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NISTIR 8271 DRAFT SUPPLEMENT

Face Recognition Technology Evaluation (FRTE)

Part 2: Identification

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Information Access Division
Information Technology Laboratory

This document is a draft supplement of NIST Interagency Report 8271

2026/10/06

National Institute of Standards and Technology
U.S. Department of Commerce

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October, 2026

U.S. Department of Commerce
Howard Lutnick, Secretary

National Institute of Standards and Technology
Arvind Raman, Under Secretary of Commerce for Standards and Technology and NIST Director

RELEASE NOTES

2026-10-06: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty-fifth draft update to NIST Interagency Report 8271. It contains results for three new developer: Cogniti Technology, IDBIO, and Security Pitch Co Ltd.
- ▷ The document also includes results for algorithms for fifteen returning developers: Akiyama S.A, Cognitec Systems GmbH, Corsight AI, FacePhi, Innovatrics, Lishui Shuyan Huirong Technology, Metsakuur, Mobbeel Solutions, Paravision, RealSense, Toshiba, Toss, Viettel Artificial Intelligence, Verigram, and Xailient.

2026-09-04: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty-fourth draft update to NIST Interagency Report 8271. It contains results for seven new developer: Innov8tif Solutions, Integrated Software and Technologies Ltd, Virtisha AI, Intpixel Labs Private Ltd, Thakaa Advanced Company for Information Technology, TrueSight Laboratories Tech LLC, and UniverseAI.
- ▷ The document also includes results for algorithms for eight returning developers: Iris ID Systems, Miaxis Biometrics, Nominder, Panasonic R+D Center Singapore, ROC, Sensetime Group, Shanghai Jiao Tong University, and Vsoft.

2026-08-04: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty-third draft update to NIST Interagency Report 8271. It contains results for three new developer: Intozi Tech Pvt Ltd, Techainer, and Youverse.
- ▷ The document also includes results for algorithms for six returning developers: Compal Electronics, Dermalog, Idemia, NEC, Onfido, and Technology Control Company.

2026-07-09: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty-second draft update to NIST Interagency Report 8271. It contains results for four new developer: AI Implementation Group, Facehawk Ltd, Fissionsoft CO Ltd, Lebentech Biometrics, NextMeta Labs, STCON LLC, Stellar Technologies, and Xailient.
- ▷ The document also includes results for algorithms for three returning developers: Hyundai Motor Company, IDENTY, Viante.AI, and WINK.

2026-05-12: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty-first draft update to NIST Interagency Report 8271. It contains results for three new developer: Naver Cloudi, Shufti, and Sparktek.
- ▷ The document also includes results for algorithms for three returning developers: DAON, Panasonic R+D Center Singapore, and Tripleize.

2026-03-30: The 1:N track of the FRTE remains open.

- ▷ This document is the fifty draft update to NIST Interagency Report 8271. It contains results for six new developer: Iris ID Systems, Lishui Shuyan Huirong Technology, Miaxis Biometrics, Onfido, Technology Control Company, and Shanghai Jiao Tong University.
- ▷ The document also includes results for algorithms for six returning developers: AllGoVision, Dermalog, Paravision, PT Ice House, Sensetime Group, and ROC.

2026-02-13: The 1:N track of the FRTE remains open.

- ▷ This document is the forty ninth draft update to NIST Interagency Report 8271. It contains results for one new developer: RealSense.

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- ▷ The document also includes results for algorithms for eight returning developers: Adiance Technologies Pvt Ltd, GTEL OTS JSC, IDENTY, Protenet Soluções em Segurança, Toshiba, STCON LLC, Viante.AI, Viettel Artificial Intelligence.

2026-01-09: The 1:N track of the FRTE remains open.

- ▷ This document is the forty eighth draft update to NIST Interagency Report 8271. It contains results for three new developer: Biometrik Systems Private Limited, FAST-NUCES Lahore Campus, and Sparktek International Pte Ltd.
- ▷ The document also includes results for algorithms for seven returning developers: Clearview AI Inc, Compal Electronics, Hyundai Motor Company, Innovatrics, NEC, Panasonic R+D Center Singapore, and VDF Software.

2025-09-30: The 1:N track of the FRTE remains open.

- ▷ This document is the forty seventh draft update to NIST Interagency Report 8271. It contains results for three new developer: LightVision Inc, Protenet Soluções em Segurança, and Toss.
- ▷ The document also includes results for algorithms for one returning developers: Dermalog.

2025-08-27: The 1:N track of the FRTE remains open.

- ▷ This document is the forty sixth draft update to NIST Interagency Report 8271. It contains results for one new developer: Private Identity LLC.
- ▷ The document also includes results for algorithms for thirteen returning developers: AllGoVision, DAON, Incode Technologies Inc, Metsakuur, NextMedia Labs, Neurotechnology, Omnigarde, Panasonic R+D Center Singapore, Pentacomp Systemy Informatyczne, ROC, TigerIT Americas LLC, Toshiba, and VinBigData.

2025-07-17: The 1:N track of the FRTE remains open.

- ▷ This document is the forty fifth draft update to NIST Interagency Report 8271. It contains results for one new developers: IDENTY.
- ▷ The document also includes results for algorithms for six returning developers: Hyundai Motor Company, Idemia, Innovatrics, Paravision, STCON LLC, and Universidade de Coimbra.

2025-05-28: The 1:N track of the FRTE remains open.

- ▷ This document is the forty fourth draft update to NIST Interagency Report 8271. It contains results for three new developers: Adiance Technologies, RNIT Solutions and Services, and VDF Software.
- ▷ The document also includes results for algorithms for four returning developers: Cubox, Dermalog, QazSmartVision.AI, and Samsung S1.

2025-04-25: The 1:N track of the FRTE remains open.

- ▷ This document is the forty third draft update to NIST Interagency Report 8271. It contains results for five new developers: Compal Electronics, FCI Vietnam, GTEL OTS JSC, i2V Systems, and ioNetworks.
- ▷ The document also includes results for algorithms for five returning developers: Intel Research Group, Mobbeel Solutions, Pyramid Cyber Security + Forensic (P), Viante.AI, and Viettel Cyberspace Center.

2025-03-18: The 1:N track of the FRTE remains open.

- ▷ This document is the forty second draft update to NIST Interagency Report 8271. It contains results for two new developers: Algorithms Inc, and Maverick Research.
- ▷ The document also includes results for algorithms for four returning developers: AFR Engine, Akuvox, EI Networks, and VinBigData.

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2025-03-05: The 1:N track of the FRTE remains open.

- ▷ This document is the forty first draft update to NIST Interagency Report 8271. It contains results for three new developers: Awidit Systems, Hyundai Motor Company, and Kloudspot.
- ▷ The document also includes results for algorithms for seven returning developers: Innovatrics, Inspur (Beijing) Electronic Information Industry, NEC, Nominder, ROC, STCON, and Viettel Group

2025-02-05: The 1:N track of the FRTE remains open.

- ▷ This document is the fortieth draft update to NIST Interagency Report 8271. It contains results for three new developers: Keyless Technologies, Pentacomp Systemy Informatyczne, and TTI Group.
- ▷ The document also includes results for algorithms for four returning developers: DAON, Hangzhuo AIlu Network Information Technology, Unissey and useB.

2024-12-18: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty nine draft update to NIST Interagency Report 8271. It contains results for two new developers: Pyramid Cyber Security + Forensic, and Shennona.
- ▷ The document also includes results for algorithms for nine returning developers: Alchera, Cubox, IMDS Software, Paravision, QazSmartVision.AI, Samsung S1, Sensetime, Vietnam Posts and Telecommunications Group, and Vision-Box.

2024-11-08: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty eighth draft update to NIST Interagency Report 8271. It contains results for six new developers: FacePhi, Metsakuur, Nextmeta Labs, Scylla Technologies, Seventh Sense Artificial Intelligence, and Vietnam Payment Solution.
- ▷ The document also includes results for algorithms for four returning developers: EI Networks, Innovatrics, Nominder, STCON, Viettel Group, VinBigData, and WINK.

2024-09-18: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty seventh draft update to NIST Interagency Report 8271. It contains results for three new developers: Antheus Technologia, Beijing Hisign Technology, and PT Ice House.
- ▷ The document also includes results for algorithms for four returning developers: Cognitec Systems, Neurotechnology, Qnap Security, and useB.
- ▷ In addition, many graphs for algorithms submitted between June 2023 and August 2024 have been included for the first time. They had incorrectly been omitted from several reports issued in that timeframe.

2024-08-09: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty sixth draft update to NIST Interagency Report 8271. It contains results for one new developer: NextMedia Labs, OptiExacta and Universidade de Coimbra.
- ▷ The document also includes results for algorithms for seven returning developers: AFR Engine, Incode, Panasonic R+D Center Singapore and QazSmartVision.AI.

2024-07-03: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty fifth draft update to NIST Interagency Report 8271. It contains results for one new developer: Sansap Technology.
- ▷ The document also includes results for algorithms for seven returning developers: Alchera, Clearview AI, Cyberlink Corp, Dermalog, Nominder, Omnigarde, and ROC.

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2024-05-30: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty forth draft update to NIST Interagency Report 8271. It contains results for seven new developers: ALTTEKGlobal, Aratek Biometrics, BioID Technologies, EI Networks Private, Kogniza Technology, STCON, and VinBigData.
- ▷ The document also includes results for algorithms for five returning developers: AadiSysTech, Cognitec Systems, Hangzhuo AIlu Network Information Technology, Innovatrics, and Mobbeel Solutions.

2024-04-29: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty third draft update to NIST Interagency Report 8271. It contains results for one new developer: FPT Smart Cloud
- ▷ The document also includes results for algorithms for one returning developer: Vision-Box

2024-03-26: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty second draft update to NIST Interagency Report 8271. It contains results for one new developer: QazSmartVision.AI
- ▷ The document also includes results for algorithms for eight returning developers: AFR Engine, Corsight AI, Dactionable Technologies, Idemia, IMDS Software, Megvii/Face++, Panasonic R+D Center, and Unissey.

2024-02-21: The 1:N track of the FRTE remains open.

- ▷ This document is the thirty first draft update to NIST Interagency Report 8271. It contains results for one new developer: Yuan High-Tech Development.
- ▷ The document also includes results for algorithms for two returning developers: Clearview AI, and useB.

2024-01-22: The 1:N track of the FRTE remains open.

- ▷ This document is the thirtieth draft update to NIST Interagency Report 8271. It contains results for two new developer: Sparsh CCTV, and VNIS Joint Stock.
- ▷ The document also includes results for algorithms for ten returning developer: Dermalog, ICM Airport Technics, Innovatrics, Maxvision, Viettel Cyberspace Center, NEC, Neurotechnology, Nominder, ROC, and WINK.

2023-12-18: The 1:N track of the FRTE remains open.

- ▷ This document is the twenty nineth draft update to NIST Interagency Report 8271. It contains results for one new developer: Mobbeel Solutions.
- ▷ The document also includes results for algorithms for seven returning developer: AadiSysTech, Alchera, Aware, Cognitec Systems, Inspur (Beijing) Electronic Information, Qnap Security, and Vietnam Posts and Telecommunications Group.

2023-11-20: The 1:N track of the FRTE remains open.

- ▷ This document is the twenty eighth draft update to NIST Interagency Report 8271. It contains results for four new developer: Dactionable Technologies, IMDS Software Inc, Luxand Inc, and Unissey.
- ▷ The document also includes results for algorithms for seven returning developer: AllGoVision, Hangzhuo AIlu Network Information Technology, Kakao Brain, Megvii/Face++, Recognito, Verigram, and Viante.AI.

2023-09-29: The 1:N track of the FRTE remains open.

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- ▷ This document is the twenty seventh draft update to NIST Interagency Report 8271. It contains results for one new developer: Nominder.
- ▷ The document also includes results for algorithms for one returning developer: useB.

2023-09-09: The 1:N track of the FRTE remains open.

- ▷ We have renamed and split FRVT into two sets of technology evaluations: The Face Recognition Technology Evaluation (FRTE) is the umbrella for the ongoing 1:1, 1:N and twins disambiguation tracks; the Face Analysis Technology Evaluation (FATE) holds the morph and attacked detection tracks, the quality assessment tracks, and the new age estimation tracks.
- ▷ This document is the twenty sixth draft update to NIST Interagency Report 8271. It contains results for two new developers: ICM Airport Technics and Papilon Savunma.
- ▷ The document also includes results for algorithms for two returning developers: Canon and Tripleize.

2023-08-16: The 1:N track of the FRVT remains open.

- ▷ This document is the twenty fifth draft update to NIST Interagency Report 8271. It contains results for two new developers: Panasonic and Sumsub.
- ▷ The document also includes results for algorithms for two returning developers: Maxvision and Innovatrics.

2023-07-28: The 1:N track of the FRVT remains open.

- ▷ This document is the twenty fourth draft update to NIST Interagency Report 8271. It contains results for four new developers: AFR Engine, Corsight, Wink and Viettel Cyberspace Center.
- ▷ The document also includes results for algorithms for seven returning developers: Alchera, Cubox (cubox.aero), NEC, Rank One Computing, Samsung S-1, Tech5 SA, and Verihubs Inteligensia.

2023-07-05: The 1:N track of the FRVT remains open.

- ▷ This document corrects mugshot-mugshot results for algorithms submitted to FRVT 2023-02 to 2023-06. The prior report's FNIR values were uniformly approximately 0.003 higher due to a bug introduced into a script by changes to the FRVT API.

2023-07-03: The 1:N track of the FRVT remains open.

- ▷ This document is the twenty third draft update to NIST Interagency Report 8271. It contains results for eight first-time participants: Advance.AI, AFIS and Biometrics Consulting, Inspur (Beijing) Electronic Information Industry, Know U Tech, Omnigarde, Serendipity, useB, and Verigram.
- ▷ The document also includes results for algorithms for four returning developers: Hangzhuo AIlu Network Information Technology, Megvii/Face++, Paravision, and Qnap Security.

2023-04-20: The 1:N track of the FRVT is closed until 2023-05-01.

- ▷ This document is the twenty third draft update to NIST Interagency Report 8271. It contains results for one first-time participant: Armatura LLC.
- ▷ The document also includes results for algorithms for two returning developers: HyperVerge, Maxvision.

2023-03-29: The 1:N track of the FRVT remains open.

- ▷ This document is the twenty third draft update to NIST Interagency Report 8271. It contains results for one first-time participant: Recognito.
- ▷ The document also includes results for algorithms for one returning developer: AllGoVision Technologies.

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2023-03-14: The 1:N track of the FRVT remains open.

- This document is the twenty second draft update to NIST Interagency Report 8271. It contains results for no first-time participants.
- The document also includes results for algorithms from nine returning developers: Cloudwalk - Moontime Smart Technology, Line Corporation, Intema-LGL Group Neurotechnology, NEC, Paravision, Samsung S1, Veridas Digital Authentication Solutions, and Thales Group

2023-02-06: The 1:N track of the FRVT remains open.

- This document is the twenty first draft update to NIST Interagency Report 8271. It contains results for no first-time participants.
- The document also includes results for algorithms from seven returning developers: Dermalog, DiluSense Technology, Hangzhuo AIlu Network Information Technology, Idemia, Innovatrics, Rank One Computing, and Sensetime Group.

2022-12-15: The 1:N track of the FRVT remains open.

- This document is the twentieth draft update to NIST Interagency Report 8271. It contains results for one first-time participant: First Credit Bureau Kazakhstan.
- The document also includes results for algorithms from five returning developers: Gorilla Technology, Pangiam, Qnap Scurity, SQIsoft, Vixvizion (formerly known as Imagus).

2022-11-09: The 1:N track of the FRVT remains open.

- This document is the nineteenth draft update to NIST Interagency Report 8271. It contains results for four first-time participant: Mukh, Turing Technology VIP, Verijelas and Verihubs Inteligensia
- The document also includes results for algorithms from two returning developers: Maxvision and Samsung S1.

2022-09-23: The 1:N track of the FRVT remains open.

- This document is the eighteenth draft update to NIST Interagency Report 8271. It contains results for two first-time participants: Intema-LGL Group and T4iSB.
- The document also includes results for algorithms from two returning developers: Cloudwalk - Moontime Smart Technology, Dermalog, Griaule, Hangzhuo AIlu Network Information Technology, Intellivision, Line Corporation, NEC, Sensetime Group, Realnetworks Inc and Vietnam Posts and Telecommunications Group.

2022-07-28: The 1:N track of the FRVT remains open.

- This document is the seventeenth draft update to NIST Interagency Report 8271. It contains results for one first-time participant: Maxvision.
- The document also includes results for algorithms from two returning developers: Rank One Computing, and Viettel Group.
- We have replaced the probe set used in the visa-border benchmark. It was previously comprised of 80 000 images; it now has size 1 212 892 - see amended entries in Table 1. False negative identification rates have increased.
- We have added images to the probe set used in the visa-kiosk benchmark. It was previously comprised of 21 016 mates and the same number of non-mates; it now has 31 579 mates and 45 460 non-mates - see amended and entries in Table 1. False negative identification rates are improved (reduced) slightly.

2022-06-08: The 1:N track of the FRVT remains open.

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- This document is the seventeenth draft update to NIST Interagency Report 8271. It includes results for algorithms submitted by three first-time participants: Digidata, DiluSense Technology, and Vietnam Posts and Telecommunications Group.
- The document also includes results for algorithms from five returning developers: Canon Inc, Imagus Technology, Neurotechnology, Thales, and Samsung S1.

2022-04-28: The 1:N track of the FRVT remains open.

- This document is the sixteenth draft update to NIST Interagency Report 8271. It includes results for algorithms submitted by one first-time participants: Hangzhuo AIlu Network Information Technology.
- The document also includes results for algorithms from three returning developers: HyperVerge Inc, Qnap Security, and Realnetworks Inc.
- The 1:N results page has been updated.

2022-03-30: The 1:N track of the FRVT remains open.

- This document is the sixteenth draft update to NIST Interagency Report 8271. It includes results for algorithms submitted by two first-time participants: Intellivision, and Pangiam.
- The document also includes results for algorithms from three returning developers: Fujitsu Research and Development Center, Idemia, and Gorilla Technology.
- The 1:N results page has been updated.

2022-02-23: The 1:N track of the FRVT remains open.

- This document is the fifteenth draft update to NIST Interagency Report 8271. It includes results for algorithms submitted by four first-time participants: Cloudwalk - Moontime Smart Technology, Decatur Industries Inc, NotionTag Technologies Private Limited, and Reveal Media Ltd.
- The document also includes results for algorithms from three returning developers: Cognitec Systems GmbH, Sensetime Group, and Viettel Group
- The 1:N results page has been updated.

2022-01-20: The 1:N track of the FRVT remains open.

- This document is the fourteenth draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by two first-time participants: Daon and SQIsoft.
- The document also includes results for algorithms from five returning developers: Cyberlink Corp, NEC, Neurotechnology, Paravision, and Rank One Computing.
- The 1:N results page has been updated.

2021-12-16: The 1:N track of the FRVT remains open.

- This document is the thirteenth draft update to NIST Interagency Report 8271. It includes results for algorithms from six returning developers: Dahua Technology, Imagus Technology, Line Corporation, N-Tech Lab, Qnap Security, and Realnetworks Inc.
- The 1:N results page has been updated.

2021-11-22: The 1:N track of the FRVT remains open.

- This document is the twelfth draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by three first-time participants Clearview AI, Griaule, and Mantra Softech India.

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- ▷ This document and the 1:N results page also include results for algorithms from six returning developers: Acer Incorporated, Canon, Dermalog, Samsung S1, VisionLabs, and Veridas Digital Authentication.

2021-10-28: The 1:N track of the FRVT remains open.

- ▷ This document is the eleventh draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by three first-time participants (20Face, Fujitsu Research and Development Center, and Vision-Box), and five returning participants (Alchera, Gorilla Technology, Tevian, Thales-Cogent, and Visidon). Visidon
- ▷ Both the main 1:N results page and the small-gallery paperless travel page have been updated.

2021-09-21: The 1:N track of the FRVT remains open. Three news items:

- ▷ This document is the tenth draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by six first-time developers: Cubox, Fincore, HyperVerge, Qnap Security, Staqu Technologies, and Tripleize (Aize, 3-ize).
- ▷ It includes results also for four returning developers: Cognitec Systems, Incode Technologies, Innovatrics, Neurotechnology, and Rank One Computing.

2021-08-02: The 1:N track of the FRVT remains open. Three news items:

- ▷ This document is the nineth draft update to NIST Interagency Report 8271. It includes results for algorithms recently submitted by eight participants: Cyberlink Corp, NEC Corp, N-Tech Lab, Realnetworks Inc., Sensetime Group, Veridas Digital, Viettel Group, and Vigilant Solutions.
- ▷ Algorithms submitted since July 24 will be included in the next update scheduled for September 9, 2021.
- ▷ A new report, NIST Interagency Report 8381 - FRVT Part 7: Identification for Paperless Travel and Immigration, has been released [PDF, webpage]. It documents the use of FRVT 1:N algorithms in positive access control and immigration status update travel applications where the enrolled population size is as low as 420 people for aircraft boarding, and 42 000 for an airport security line. These population sizes are much smaller than those used in the main 1:N evaluation. Going forward, we will update the report and webpage with results for new algorithms.

2021-07-07: The 1:N track of the FRVT remains open. One update:

- ▷ This document is the eighth draft update to NIST Interagency Report 8271. It include results for an algorithm from one participant: Kakao Enterprises.

2021-06-22: The 1:N track of the FRVT remains open. Three updates:

- ▷ This is the seventh draft of the update to NIST Interagency Report 8271. It includes results for algorithms from three new participants: Line Corporation, Rendip, and Samsung S1 Corp.
- ▷ We have also added results for algorithms from five returning developers: Imagus Technology, Kneron, Tevian, Visidon, and Xforward AI Technology.
- ▷ The algorithm-specific report cards (examples: 1, 2, and 3) now include figures showing how low threshold values can be used to reduce candidate list lengths for human review, while (usually) elevating miss rates (FNIR) only modestly. The reports also feature some minor additions and clarifications.

2021-03-26: The 1:N track of the FRVT remains open. Three updates:

- ▷ This is the sixth draft of the update to NIST Interagency Report 8271. It includes results for algorithms from three returning developers: Neurotechnology, Guangzhou Pixel Solutions, and Tech5 SA.

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- ▷ We have added results on the webpage and in the report for a new ageing dataset in which border crossing photos are searched against a gallery of border crossing photos collected between 10 and 15 years prior to the mated search photos. See section 2 for a description of the images. Table 1 has a new entry describing the experiment.
- ▷ We will mostly discontinue running the mugshot ageing test, reserving it for algorithms that show high accuracy on the new border-crossing set.

2021-03-26: Regarding the fifth draft of the update to NIST Interagency Report 8271:

- ▷ In addition have added results for first algorithms from two new participants: Viettel Group and Veridas Digital Authentication Solutions.
- ▷ We have added results for algorithms from two returning developers: Idemia and Cognitec Systems.
- ▷ In addition to the report, the results page and its hyperlinked report cards have been updated.

2021-02-08: Regarding the fourth draft of the update to NIST Interagency Report 8271:

- ▷ We have added results for eight algorithms submitted by eight developers: Cyberlink, Dermalog, Imagus, Paravision, Sensetime, Trueface, Vigilant Solutions, and X-Forward AI. With the exception of Trueface, all of these developers have participated previously.
- ▷ We anticipate updating this report again in the first week of March 2021.
- ▷ The main results page has been revised with tabs for the investigative and lights-out identification tables, and a new tab dedicated to speed and resource consumption.
- ▷ The report cards (example here) hyperlinked from the results page have been revised to improve content and format.

2020-12-14: Regarding third draft of the update to NIST Interagency Report 8271:

- ▷ We have added results for fifteen algorithms submitted by thirteen developers. The four first-time participants are: Acer, Akurat Satu Indonesia, Canon, and Xforward AI Technology. The ten returning developers are: AllGoVision, Cyberlink Corp, Dahua Technology, Deepglint, Guangzhou Pixel Solutions, IIT Vision, Innovatrics, Rank One Computing, Scanovate, Sensetime Group, Synesis, and VisionLabs.
- ▷ We have added two new datasets to the evaluation: First a set of "visa-border" photos, representing search of an airport immigration lane photo against a database of closely ISO standard portraits; second a "visa-kiosk" set representing search of a photo collected in a registered traveller kiosk against the same ISO portrait gallery. The images are described in section 2.1.
- ▷ As in previous reports, we include results for searching mugshots against a mugshot gallery containing a single image of each of 12 million people. However we have suspending running searches against a gallery in which multiple lifetime photos per person are present, because this is computationally expensive. We retain a N = 3 million search test dedicated to ageing in which mugshots taken up to 18 years after the first photograph are searched - see Table 14.
- ▷ Tables containing computational resource information, Table 2. . ., now include duration of the finalization step, in which search algorithms can, at their option, build fast-search data structures.
- ▷ We have linked revised per-algorithm PDF report cards from the main results page.
- ▷ We have regenerated all figures and tables to drop algorithms submitted before June 2018. Results for prior algorithms appear in archived editions of this report.
- ▷ Going forward, we anticipate producing more frequent updates to this report. Developers may submit one algorithm to this evaluation every four calendar months.

2020-03-24: Regarding the second draft of the update to NIST Interagency Report 8271:

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- ▷ Adds results for three algorithms from three developers, Dermalog, Innovatrics, and Synesis.
- ▷ Adds Table 14 on ageing showing the increase in false negative rates with time elapsed between two photos. Some of the results were contained in graphs in prior editions of this report, but the table adds results for some newly submitted algorithms.
- ▷ Adjusts frontal mugshot results (for recent and lifetime consolidated galleries) to include the effect of removing some images that should not have been included in image test sets. These images were mostly profile views, images of tattoos containing faces, images of faces on tee shirts, and images of photographs on walls behind the intended subject. This affects many tables and reduces false negative identification rates for all algorithms. The reduction is larger for "recent" enrollments than for "lifetime consolidated" ones with the consequence that accuracy on recent images is now superior.

2020-02-26: Regarding the first draft of the update to NIST Interagency Report 8271:

- ▷ Adds results for 38 algorithms from 31 different developers, eleven of whom are entirely new to the 1:N track of FRVT. These are Allgovision, Cyberlink, Deepsea Tencent, Farbar F8, Imperial College London, Intsys MSU, Kedacom, Kneron, Pixelall, and Scanovate.

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Specific hardware and software products identified in this report were used in order to perform the evaluations described in this document. In no case does identification of any commercial product, trade name, or vendor, imply recommendation or endorsement by the National Institute of Standards and Technology, nor does it imply that the products and equipment identified are necessarily the best available for the purpose.

INSTITUTIONAL REVIEW BOARD

The National Institute of Standards and Technology's Research Protections Office reviewed the protocol for this project and determined it is not human subjects research as defined in Department of Commerce Regulations, 15 CFR 27, also known as the Common Rule for the Protection of Human Subjects (45 CFR 46, Subpart A).

ACKNOWLEDGMENTS

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Executive Summary

This document is a draft revision of the September 2019 report NIST Interagency Report 8271. That report gave extensive documentation of face recognition applied to mugshots. This report extends that by adding more two more challenging datasets containing images with serious departures from canonical frontal image standards. The report also adds results for algorithms submitted to NIST since in 2019 and 2020. The algorithms, which implement one-to-many identification of faces appearing in two-dimensional images, are prototypes from the research and development laboratories of mostly commercial suppliers, and are submitted to NIST as compiled black-box libraries implementing a NIST-specified C++ test interface. The report therefore does not describe how algorithms operate. The report lists accuracy results alongside developer names and will therefore be useful for comparison of face recognition algorithms and assessment of absolute capability. The report is accompanied by a webpage with sortable results.

The evaluation uses six datasets: frontal mugshots, profile view mugshots, desktop webcam photos, visa-like immigration application photos, immigration lane photos, and registered traveler kiosk photos. These datasets are sequestered at NIST, meaning that developers do not have access to them for training or testing. This aspect is important because face recognition algorithms are very often deployed without the developer having access to the customers image data. A possible exception to this would be in a cloud-based application where the operational image data is uploaded to a cloud operated by a face recognition developer.

The major result in NIST IR 8271 was that massive gains in accuracy have been achieved in the years 2013 to 2018 and these far exceed improvements made in the prior period, 2010 to 2013. While the industry gains were broad - at least 30 developers' algorithms outperformed the most accurate algorithm from late 2013, there remains a wide range of capability. While this report shows accuracy gains only over the period 2018-2020, the most accurate algorithm reported here is substantially more accurate than anything reported in NIST IR 8271. This is evidence that face recognition development continues apace, and that FRVT reports are but a snapshot of contemporary capability.

From discussion with developers, the accuracy gains stem from the adoption of deep convolutional neural networks. As such, face recognition has undergone an industrial revolution, with algorithms increasingly tolerant of poorly illuminated and other low quality images, and poorly posed subjects. One related result is that a few algorithms correctly match side-view photographs to galleries of frontal photos, with search accuracy approaching that of the best c. 2010 algorithms operating on purely frontal images. The capability to recognize under a 90-degree change in viewpoint - pose invariance - has been a long-sought milestone in face recognition research.

With good quality portrait photos, the most accurate algorithms will find matching entries, when present, in galleries containing 12 million individuals, with rank one miss rates of approaching 0.1%. The remaining errors are in large part attributable to long-run ageing, facial injury and poor image quality. Given this impressive achievement - close to perfect recognition - an advocate might claim that cooperative face recognition is a solved problem, a statement that can be refuted with the following context and caveats:

- **Mugshots vs. less constrained captures:** The low error rates reported here are attained using mostly excellent cooperative live-capture mugshot images collected with an attendant present. Recognition in other circumstances, particularly those without a dedicated photographic environment and human or automated quality control checks, will lead to declines in accuracy. This is documented here for side-view images, poorer quality webcam images, and, particularly, for newly introduced ATM-style kiosk photos that were not originally intended for automated face recognition. In this case, recognition error rates are much higher, often in excess of 20% even with the more accurate algorithms which variously remain intolerant of face cropping (at image edge) and of large downward head pitch.
- **Algorithm accuracy spectrum:** Recognition accuracy is very strongly dependent on the algorithm and, more

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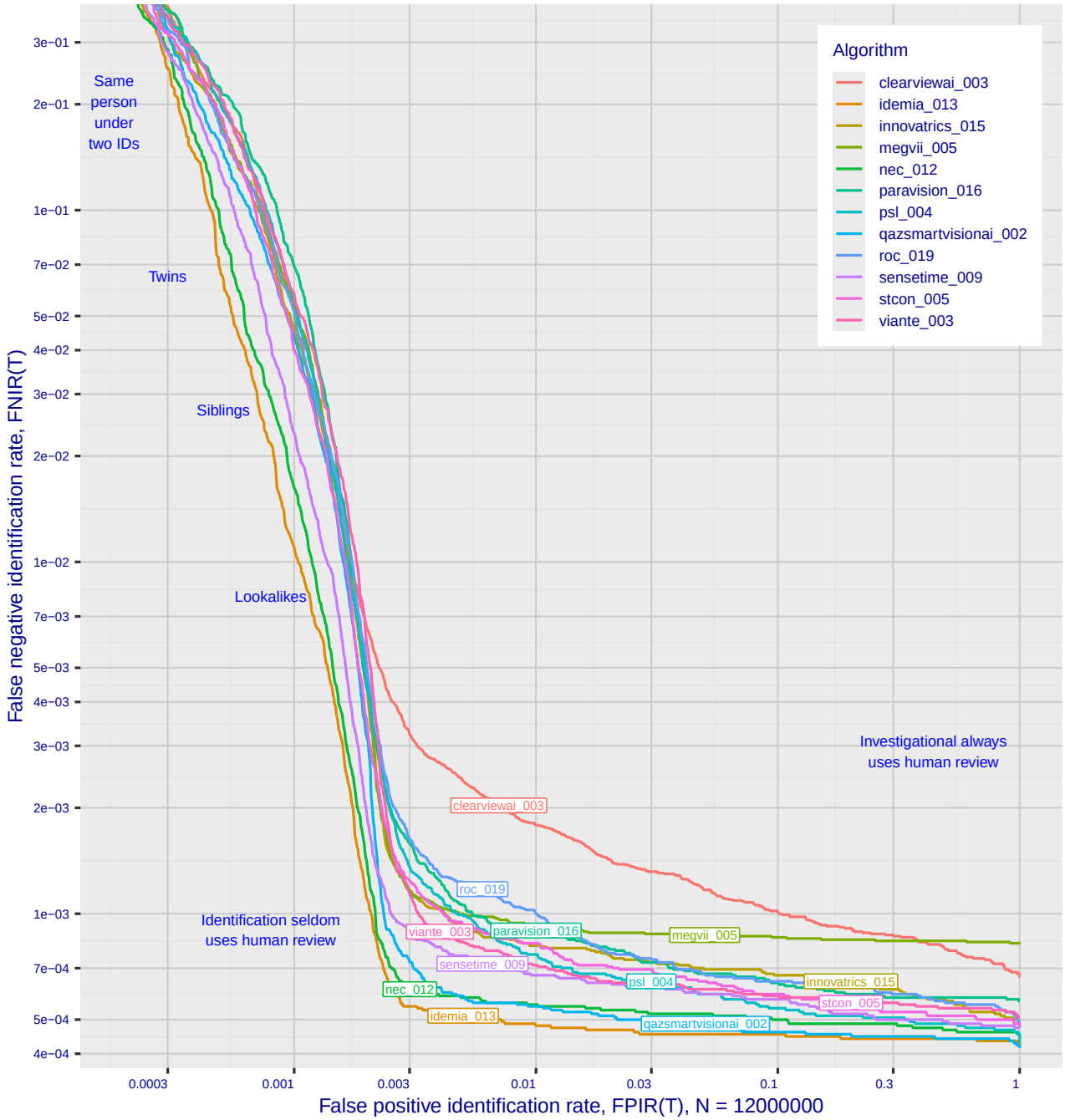

Figure 1: Identification miss rates across the false positive range. N = 12 million individuals are enrolled with one recent image.

generally, on the developer of the algorithm. False negative error rates in a particular scenario range from a few tenths of one percent to beyond fifty percent. This is tabulated exhaustively later: For example Table 24 shows accuracy across datasets. Figure 1 here compares algorithms on mugshot searches in a consolidated gallery of 12 million subjects and 12 million photos. Many algorithms do not achieve the low error rates noted above, and while many of those may still be useful and valuable to end-users, only the most accurate excel on poor quality images and those collected long after the initial enrollment sample.

▷ **Versioning:** While results for up to ten algorithms from each developer are reported here, the intra-provider

accuracy variations are usually smaller than the inter-provider variations. That said different versions give an order of magnitude fewer misses. Some developers demonstrate speed-accuracy tradeoffs[1]. See Figs. 18, 19.

- **Low similarity scores:** In thousands of mugshot cases the correct gallery image is returned at rank 1 but its similarity score is nevertheless low, below some operationally required score threshold. This is not so important when face recognition is used for "lead generation" in investigational applications because human reviewers are specifically required to review potentially long candidate lists and the threshold is effectively 0. In applications where search volumes are higher and labor is not available to review the results from searches, a higher threshold must be applied. This reduces the length of candidate lists and false positive identification rates at the expense of increased false negative miss rates. The tradeoff between the two error rates is reported extensively later.

- **Population size:** As the number of enrolled subjects grows, some mates are displaced from rank one, decreasing accuracy. As tabulated later for N up to 12 million, false negative rates generally rise slowly with population size. This enables use of face recognition in very large populations. However in most positive and negative identification applications[2], a score threshold is set to limit the rate at which non-mate searches produce false positives. This has the consequence that some mated searches will report the mate below threshold, i.e. a miss, even if it is at rank 1. The utility of this is that many non-mated searches will return no candidate identities at all. As the error-tradeoff characteristic shows, investigational miss rates on the right side are very low but then rise steadily (in the center region) as threshold is increased to support "lights-out" applications, and ultimately rise quickly (left side) as discussed below. Thus, if we demand that just one in one thousand non-mate searches produce any false positives, the most accurate algorithms there (Sensetime-004 and NEC-3) would fail on between 3 and 5% of mated searches. Even though the graph shows results for the most accurate algorithms, all but two would fail to find the mate in more than 8% of mated searches. While the two most accurate algorithms produce a relatively flat error tradeoff until the threshold is raised to limit false positives to about 1 in 400 non-mated searches[3]

  Thereafter, as the threshold is raised to further reduce false positives, miss rates rise rapidly. This means that low false positive identification rates are inaccesible with these algorithms, a result that does not apply for ten-finger identification algorithms. The rapid rise occurs because the lower mate scores are mixed with very high non-mate scores, the low scores from poor image quality and ageing, the high non-mates from the presence of lookalikes persons (doppelgangers), twins (discussed next) and, ultimately, the presence of a few unconsolidated subjects i.e. persons present under multiple IDs.

- **False negatives from ageing:** A large source of error in long-run applications where subjects are not re-enrolled on a set schedule is ageing. Changes in facial appearance increase with the time elapsed between photographs. These will depress similarity scores and eventually cause false negatives. All faces age and while this usually proceeds in a graceful and progressive manner, drug use can accelerate this [28]. Elective surgery may be effective in delaying it although this has not been formally quantified with face recognition. As ageing is essentially unavoidable, it can only be mitigated by scheduled re-capture, as in passport re-issuance. To quantify ageing effects, we used the more accurate algorithms to enroll the earliest image of 3.1 million adults and then search

[1] For example, NEC-0 prepares templates much faster than NEC-2 but gives twenty times more misses. Dermalog-5 executes a template search much more quickly than Dermalog-6 but is also much less accurate.

[2] In a positive identification application such as a registered traveler system, a user is making an implicit claim to be enrolled in the system - most users will be. In a negative application, such as with deportees, the implicit claim is that the subejct is not enrolled - most will not be.

[3] The gallery size here is 12 million people, one image per person. Given 331 201 non-mated searches, an exhaustive implementation of one-too-many search would execute almost 4 trillion comparisons. At a false positive identification rate of 0.0025 the number of false positives is, to first order, 828 corresponding to single-comparison false match rate of 828 / 4 trillion = 2.1 10^{-10} i.e. about 1 in 5 billion. Strictly this FMR computation is meaningful only for algorithms that implement 1:N search using N 1:1 comparisons, which is not always the case.

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with 10.3 million newer photos taken up to 18 years after the the initial enrollment photo. Figure 2 puts ageing into context by contrasting it with the increase in false negatives that occurs when the number of individuals in an enrollment database becomes larger and the chance of a false positive increases such that higher thresholds may become necessary[4].

The Figure shows, from to bottom, increases in false negative identification rates (FNIR) with the algorithm being tested. This applies to increases due to N on the left side, and increases due to ageing on the right side. The relative spacing of the dots shows that for all algorithms the dependency of FNIR on N (up to 12 million) is considerably less than on ΔT (up to 18 years). The figure additionally shows the most accurate nine algorithms for each year in which they were submitted, limiting to only one per developer.

In the inset table, accuracy is seen to degrade progressively with time, as mate scores decline and non-mates displace mates from rank 1 position. More accurate algorithms tend to be less sensitive to ageing. The more accurate algorithms give fewer errors after 18 years of ageing than middle tier algorithms give after four. Note also we do not quantify an ageing rate - more formal methods [2] borrowed from the longitudinal analysis literature have been published for doing so (given suitable repeated measures data). See Figures 148, 218 and 253.

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[4]Some algorithms implement stragegies to automatically adjust scores to account for increased population size. This relieves the system owner of having to increase thresholds as N increases.

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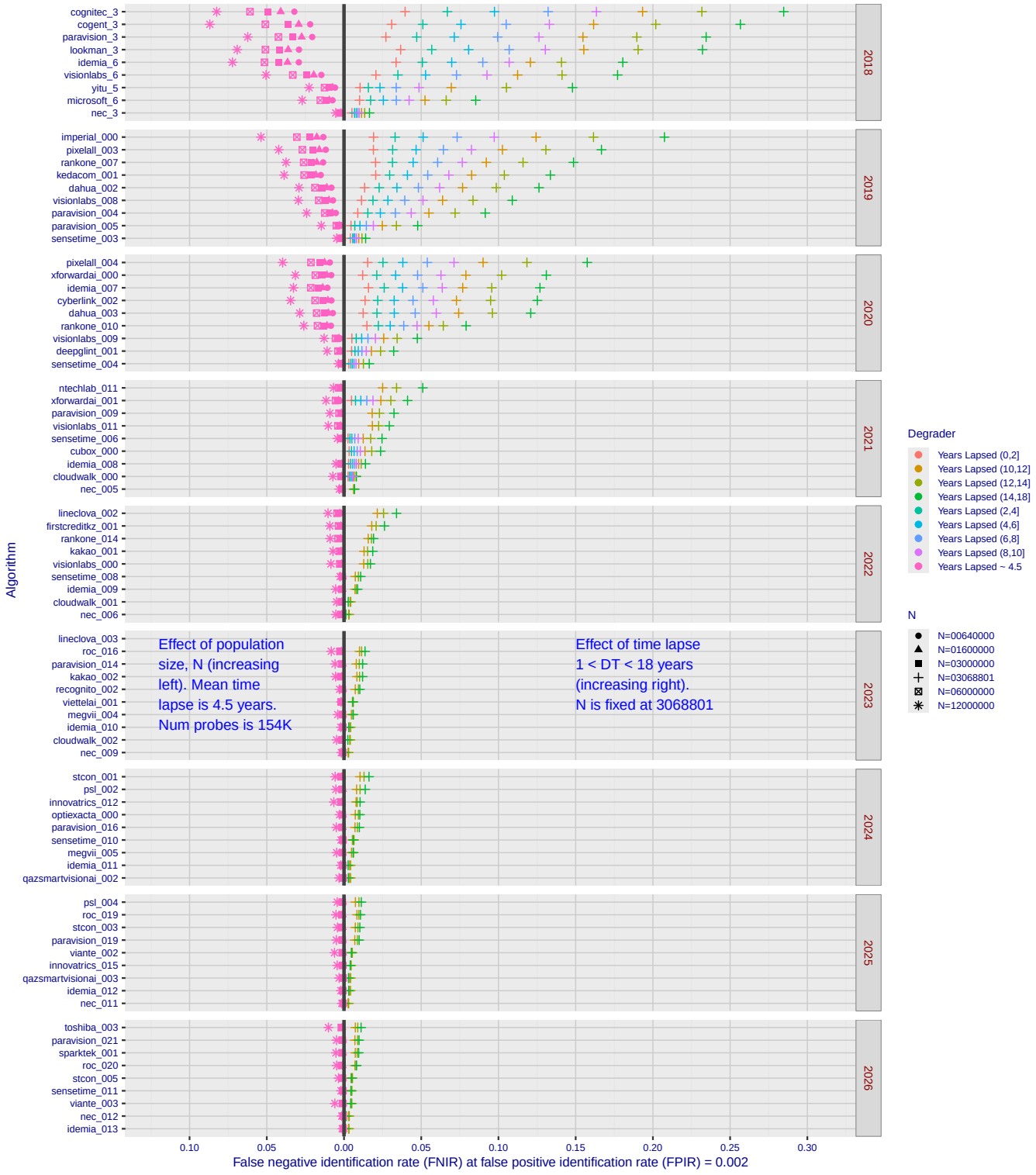

Figure 2: Identification miss rates as a function of enrolled population size, N, and time-lapse, ΔT.

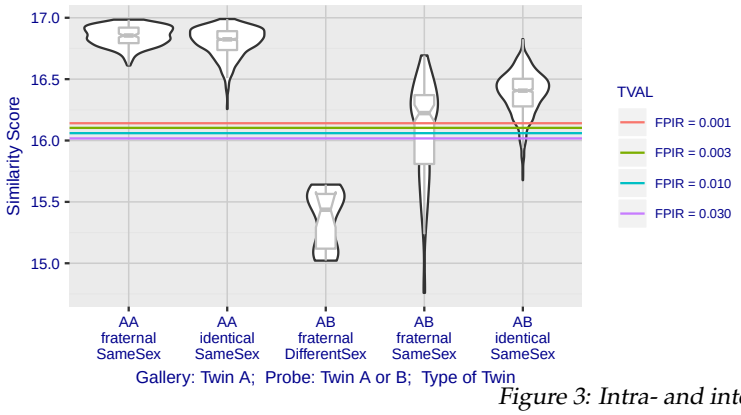

Figure 3: Intra- and inter-twin scores

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- **False positives from twins:** By enrolling 640 000 mugshots, adding photos of one twin, and then searching photos of those subjects and their twin the inset figure shows, for one typical algorithm, the similarity is generally greater when searching twins against themselves (A) than when searching twins against their sibling (B) but very often still above even stringent thresholds i.e. those corresponding to one in one thousand searches producing a false positive. Thus twins will very often produce a high-scoring non-match on a candidate list and a false alarm in an online identification system. The plot of Fig. 3 shows that fraternal twins are sometimes correctly rejected at those thresholds - including most different sex twins (at center). Figure **??** shows substantially similar behavior for all algorithms tested. In an investigative search, a twin would typically appear at rank 1, or rank 2 if their sibling happened to also be the gallery. Twins (and triplets etc.) constituted 3.3% of all live births [17] in recent years[5], and because that number is higher today than when the individuals in current adult databases were born, the false positives that arise from twins are now, and will increasingly be, an operational problem. Relative to the United States, twins are born with considerable regional variation. For example they are much less common in East Asia, and much more common in Sub-Saharan Africa [21].

  The presence of twins in the mugshot database is inevitable given its size, around 12.3 million people. As this is not an insignificant sample of the domestic United States population, people with other familial ties will be present also. The data was collected over an extended period and because location information is not available, we are unable to estimate the proportion of the domestic population that is present in the dataset. However, if we assume twins are neither more or less disposed to arrest than the general population, we can estimate that hundreds of thousands of individuals in the dataset are twins. This will affect false positive rates because we randomly set aside 331 201 individuals for nonmate searches, and some proportion of those will be twins with siblings in the gallery.

- **Database integrity:** An operational error rate should be added to all false negative rates in this report reflecting the proportion of images in a real database that are un-matchable. Such anomalies arise from images that: do not contain a face; include multiple persons; cannot be decoded; are rotated by 90° or 180°; depict a face on clothing; and others introduced by a long tail of various clerical errors. While the mugshot trials in this report have been constructed to minimize such effects, they are a real problem in actual operations.

This report is being updated continuously as new algorithms are submitted to FRVT, and run on new datasets. Participation in the one-to-many identification track is independent of participation in the one-to-one verification track of FRVT.

[5]See the CDC's National Vital Statistics Report for 2017: https://www.cdc.gov/nchs/data/nvsr/nvsr67/nvsr67_08-508.pdf

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Scope and Context

Audience: This report is intended for developers, integrators, end users, policy makers and others who have some familiarity with biometrics applications. The methods and metrics documented here will be of interest to organizations engaged in tests of face recognition algorithms. Some of these have been incorporated in the ISO/IEC 19795 Part 1 Biometric Testing and Reporting Framework standard, now nearing publication.

Prior benchmarks: Automated face recognition accuracy has improved massively in the two decades since initial commercialization of the various technologies. NIST has tracked that improvement through its conduct of regular independent, free, open, and public evaluations. These have fostered improvements in the state of the art. This report serves as an update to the NIST Interagency Report 8271 on performance of face identification algorithms, published in September 2019.

Demographics: In December 2019, NIST published a first report on demographic dependencies in face recognition, NIST Interagency Report 8280 that documented age, sex and race differentials in one-to-one and one-to-many false positive and false negative rates.

Scope: NIST IR 8271 documented recognition results for four databases containing in excess of 30.2 million still photographs of 14.4 million individuals. That constituted the largest public and independent evaluation of face recognition ever conducted. It includes results for accuracy, speed, investigative vs. identification applications, scalability to large populations, use of multiple images per person, images of cooperative and non-cooperative subjects.

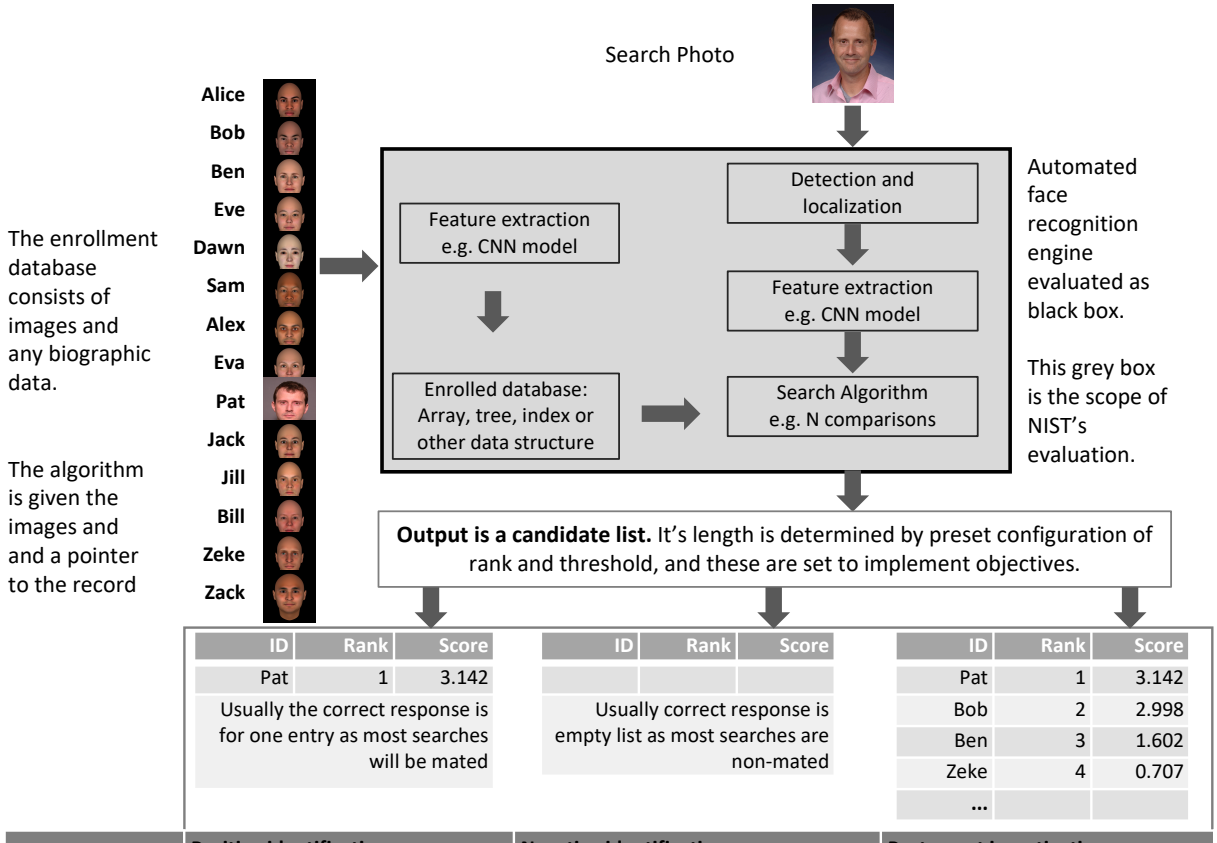
The report also includes results for ageing, recognition of twins, and recognition of profile-view images against frontal galleries. It otherwise does not address causes of recognition failure, neither image-specific problems nor subject-specific factors including demographics. Separate reports on demographic dependencies in face recognition will be published in the future. Additionally out of scope are: performance of live human-in-the-loop transactional systems like automated border control gates; human recognition accuracy as used in forensic applications; and recognition of persons in video sequences (which NIST evaluated separately [9]). Some of those applications share core matching technologies that *are* tested in this report.

Images: Five kinds of images are employed; these are either compared with images of the same kind, or against others from different capture environments as follows. The primary dataset is a set of law enforcement mugshot images (Fig. 5) which are enrolled and then searched with three kinds of images: other mugshots (i.e. within-domain); profile-view photographs (90 degree cross-view); and lower quality webcam images (Fig. 6) collected in similar detention operations (cross-domain). Additionally we compare high quality visa-like photos collected in immigration offices, with: medium quality border crossing images collected in primary immigration lanes; poor quality images collected in ATM-like registered traveller kiosks.

Participation and industry coverage: The report includes performance figures for prototype algorithms from the research laboratories of commercial developers and a few universities. This represents a substantial majority of the face recognition industry, but only a tiny minority of the academic community. Participation was open worldwide. While there is no charge for participation, developers incur some software engineering expense in implementing their algorithms behind the NIST application programming interface (API). The test is a black-box test where the function of the algorithm, and the intellectual property associated with it, is hidden inside pre-compiled libraries.

Recent technology development: Most face recognition research with deep convolutional neural networks (CNNs) has been aimed at achieving invariance to pose, illumination and expression variations that characterize photojournalism and social media images. The initial research [18, 22] employed large numbers of images of relatively few ($\sim 10^4$) individuals to learn invariance. Inevitably much larger populations ($\sim 10^7$) were employed for training [11, 20] but the benchmark, Labeled Faces in the Wild with (essentially) an equal error rate metric [12], represents an easy task,

one-to-one verification at very high false match rates. While a larger scale identification benchmark duly followed, Megaface [15], its primary metric, rank one hit rate, contrasts with the high threshold discrimination task required in most large-population applications of face recognition, namely credential de-duplication, and background checks. There, identification in galleries containing up to 10^8 individuals must be performed using a) very few images per individual and b) stringent thresholds to afford very low false positive identification rates. This track of FRVT was launched to measure the capability of the new technologies, including in these two cases. FRVT has included open-set identification tests since 2002, reporting both false negative and positive identification rates [7].



	Positive identification	Negative identification	Post-event investigation
Example	Access to a gym or cruise ship	Watchlist e.g. Detection of deportee or duplicate drivers license applications	Crime scene photos, or of detainee without ID documents.
Claim of identity	Implicit claim to be enrolled	Implicit claim to not be enrolled	No claim: Inquiry
Threshold	High, to implement security objective	High, to limit false positives	Zero
Num. candidates	1	0	L, set by request to algorithm
Human role	Review candidate to assist user in resolution of false negatives, or to detect impostor	Review candidate to determine false positive or correct hit	Review multiple candidates, refer possible hits to examiners see [26]
Intended human involvement frequency	Rare – approx. the false negative identification rate plus prior probability of impostor	Rare – approx. the false positive identification rate plus prior probability of an actual mate	Always
Performance metric of interest	**FNIR at low FPIR. See sec. 3.1, 3.2 and Tables 10, 19**	**FNIR at low FPIR. See sec. 3.1, 3.2 and Tables 10, 19**	**FNIR at ranks 1... 50, say. FPIR = 1. See sec. 3.2 and Table 12, 14, 16**

Performance metrics for applications: This report documents the performance of one-to-many face recognition algorithms. The word "performance" here refers to recognition accuracy and computational resource usage, as measured

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by executing those algorithms on massive sequestered datasets.

This report includes extensive tabulation of recognition error rates germane to the main use-cases for face search technology. The Figure below, inspired by the Figure 1 in [23] differentiates different applications of the technolgy. The last row directs readers to the main tables relevant to those applications, respectively threshold-based and rank-based metrics that are special cases of the metrics given in section 3. The terms negative identification and positive identification are taken from the ISO/IEC 2382-37:2017 standardized biometrics vocabulary.

The algorithms are specifically configured for these applications by setting thresholds and candidate list lengths. Both rank-based metrics and threshold-based metrics include tradeoffs. In investigation, overall accuracy will be reduced if labor is only available to review a few candidates from the automated system. Note that when a fixed number of candidates are returned, the false positive identification rate of the automated face recognition engine will be 100%, because a probe image of anyone not enrolled will still return candidates. In identification applications where false positives must be limited to satisfy reviewer labor availability or a security objective, higher false negative rates are implied. This report includes extensive quantification of this threshold-based tradeoff. See Sec. 3

Template diversity: The FRVT is designed to evaluate black-box technologies with the consequence that the templates that hold features extracted from face images are entirely proprietary opaque binary data that embed considerable intellectual property of the developer. Despite migration to CNN-based technologies there is no consensus on the optimal feature vector dimension. This is evidenced by template sizes ranging from below 100 bytes to more than four kilobytes. This diversity of approaches, suggests there is no prospect of a standard template something that would require a common feature set to be extracted from faces. Interoperability in automated face recognition remains solidly based on images and documentary standards for those, in particular the ICAO portrait [27] specification deriving from the ISO/IEC 19794-5 Token frontal [24] standard, which are similar to certain ANSI/NIST Type 10 [26] formats.

Training: The algorithms submitted to NIST have been developed using image datasets that developers do not disclose. The development will often include application of machine learning techniques and will additionally involve iterative training and testing cycles. NIST itself does not perform any training and does not refine or alter the algorithm in any way. Thus the model, data files, and libraries that define an algorithm are fixed for the duration of the tests. This reflects typical operational reality where recognition software, once installed, is fixed and constant until upgraded. This situation persists because on-site training of algorithms on customer data is atypical essentially because training is not a turnkey process.

Automated search and human review: Virtually all applications using automated face search require human review of the outputs at some frequency: Always for investigational applications; rarely in positive identification applications, after rejection (false or otherwise); and rarely in negative identification applications, after an alarm (false or otherwise). The human role is usually to compare a reference image with the query image or the live-subject if present, to render either a definitive decision on "exclusion" (different subjects), or "identification" (same subject), or a declaration that one or both images have "no value" and that no decision can be made. Note that automated face recognition algorithms are not built to do exclusion - low scores from a face comparison arise from different faces *and* poor quality images of the same face.

Human reviewers make recognition errors [5, 19, 25] and are sensitive to image acquisition and quality. Accurate human review is supported by high resolution - as specified in the Type 50, 51 acquisition profiles of the ANSI/NIST Type 10 record [26], and by multiple non-frontal views as specified in the same standard. These often afford views of the ear. Organizations involved in image collection should consider supporting human adjudication by collecting high-resolution frontal and non-frontal views, preparing low resolution versions for automated face recognition [24], and retaining both for any subsequent resolution of candidate matches. Along these lines, the ISO/IEC Joint Technical

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Committee 1 subcommittee 37 on biometrics has just initiated projects on image quality assessment and face-aware capture.

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Release Notes

FRVT Activities: Since February 2017, NIST has been evaluating one-to-one verification algorithms on an ongoing basis. NIST then restarted FRVT's one-to-many track in February 2018, inviting participants to send up to prototype algorithms. Both tracks allows developers to submit updated algorithms to NIST at any time but no more frequently than four calendar months. This more closely aligns development and evaluation schedules. Results are posted to the web within a few weeks of submission. Details and full report are linked from the Ongoing FRVT site.

FRVT Reports: The results of the FRVT appear in the series NIST Interagency Reports tabulated below. The reports were developed separately and released on different schedules. In prior years NIST has mostly reported FRVT results as a single report; this had the disadvantage that results from completed sub-studies were not published until all other studies were complete.

Date	Link	Title	No.
2014-03-20	PDF	FRVT Performance of Automated Age Estimation Algorithms	7995
2015-04-20	PDF	Face Recognition Vendor Test (FRVT) Performance of Automated Gender Classification Algorithms	8052
2014-05-21	PDF	FRVT Performance of face identification algorithms	8009
2017-03-07	PDF	Face In Video Evaluation (FIVE) Face Recognition of Non-Cooperative Subjects	8173
2017-11-23	PDF	The 2017 IARPA Face Recognition Prize Challenge (FRPC)	8197
2018-11-27	PDF	Face Recognition Vendor Test - Part 2: Identification	8271
2019-09-11	PDF	Face Recognition Vendor Test - Part 2: Identification	8271
2019-12-11	PDF	Face Recognition Vendor Test - Part 3: Demographic Effects	8280
2020-01-03	WWW	Face Recognition Vendor Test (FRVT) - Part 1 Verification	Draft

Details appear on pages linked from `https://www.nist.gov/programs-projects/face-projects`.

Appendices: This report is accompanied by appendices which present exhaustive results on a per-algorithm basis. These are machine-generated and are included because the authors believe that visualization of such data is broadly informative and vital to understanding the context of the report.

Typesetting: Virtually all of the tabulated content in this report was produced automatically. This involved the use of scripting tools to generate directly type-settable LaTeX content. This improves timeliness, flexibility, maintainability, and reduces transcription errors.

Graphics: Many of the Figures in this report were produced using the ggplot2 package running under R, the capabilities of which extend beyond those evident in this document.

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Contents

Release Notes 1

Disclaimer 11

Institutional Review Board 11

Acknowledgments 11

Executive Summary 12

Scope and Context 18

Release Notes 22

1 Introduction 24

2 Evaluation datasets 25

3 Performance metrics 31

4 Results 47

Appendices 125

A Accuracy on large-population FRVT 2018 mugshots 125

B Effect of time-lapse: Accuracy after face ageing 258

C Effect of enrolling multiple images 576

D Accuracy with poor quality webcam images 583

E Accuracy for profile-view to frontal recognition 593

F Search duration 597

G Gallery Insertion Timing 632

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1 Introduction

One-to-many identification represents the largest market for face recognition technology. Algorithms are used across the world in a diverse range of biometric applications: detection of duplicates in databases, detection of fraudulent applications for credentials such as passports and driving licenses, token-less access control, surveillance, social media tagging, lookalike discovery, criminal investigation, and forensic clustering.

This report contains a breadth of performance measurements relevant to many applications. Performance here refers to accuracy and resource consumption. In most applications, the core accuracy of a facial recognition algorithm is the most important performance variable. Resource consumption will be important also as it drives the amount of hardware, power, and cooling necessary to accommodate high volume workflows. Algorithms consume processing time, they require computer memory, and their static template data requires storage space. This report documents these variables.

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1.1 Open-set searches

FRVT tested open-set identification algorithms. Real-world applications are almost always "open-set", meaning that some searches have an enrolled mate, but some do not. For example, some subjects have truly not been issued a visa or drivers license before; some law enforcement searches are from first-time arrestees[6]. In an "open-set" application, algorithms make no prior assumption about whether or not to return a high-scoring result, and for a mated search, the ideal behaviour is that the search produces the correct mate at high score and first rank. For a non-mate search, the ideal behavior is that the search produces zero high-scoring candidates.

Many academic benchmarks execute only closed-set searches. The proportion of mates found in the rank one position is the default accuracy metric. This hit rate metric ignores the score with which a mate is found; weak hits count as much as strong hits. This ignores the real-world imperative that in many applications it is necessary to elevate a threshold to reduce the number of false positives.

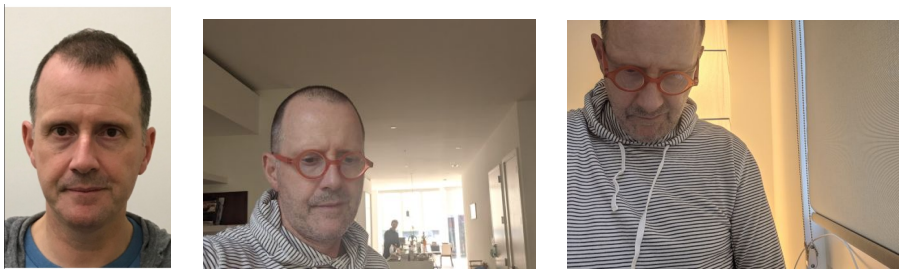
[6]Operationally closed-set applications are rare because it is usually not the case that all searches have an enrolled mate. One counter-example, however, is a cruise ship in which all passengers are enrolled and all searches should produce exactly one identity. Another example is forensic identification of dental records from an aircraft crash.

2 Evaluation datasets

This report documents accuracy for four kinds of images - mugshots, webcam, profiles and wild - as described in the following sections.

2.1 Immigration-related images

This report includes benchmark tests sharing a common enrollment of high quality frontal portrait images collected while subject make applications for various immigration benefits. We then search that with two kinds of images, webcam images collected during in-bound immigration and also images collected from registered travelers using a ATM-style kiosk. These are described below and depicted in Figure 4.



a) Application b) Immigration Lane c) Kiosk

Figure 4: Example photos.

- **Application reference photos**: The images are collected in an attended interview setting using dedicated capture equipment and lighting. The images, at size 300x300 pixels, are smaller than normally indicated by ISO. The images are all high-quality frontal portraits collected in immigration offices and with a white background. As such, potential quality related drivers of high false match rates (such as blur) can be expected to be absent. The images are encoded as ISO/IEC 10918-1 i.e. JPEG. Older images had a compression ration of about 16:1, while newer images, since 2010, are more lightly compressed at 4:1. When these images are provided as input into the algorithm, they are labeled with the type "iso". This report enrols 1 600 000 application images, one per person.

- **Border crossing photos**: Most images are have width 320 and height 240 pixels. They are JPEG compressed at 16:1 i.e. filesize just below 15KB. The images present challenges for face recognition in that subjects often exhibit non-zero yaw and pitch (associated with the rotational degrees of freedom of the camera mount), low contrast (due to varying and intense background lights), and poor spatial resolution (due to inexpensive cameras). There are often subjects standing in the background, usually at very low resolution (see Figure 4b). In such cases, algorithms should detect all faces and determine which is the largest and most centered. When these images are provided as input into the algorithm, they are labeled with the type "wild".

- **Kiosk photos**: These photos were collected from subjects whose attention was focused on interaction with an immigration kiosk. They images were not intended for use with automated face recognition. The camera is situated above a display which the user touches, and is triggered either without directing the subject to look at it, or without waiting for the subject to comply. The images are therefore characterized by pitch-down pose, sometimes exceeding 45 degrees, as in Figure 4c. Yaw-angle variation is mild, with most images close to frontal. The images

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have width 320 pixels and height 240 pixels and therefore tall individuals are sometimes cropped. This is often just above the eyes and can occur at the nose or mouth. Conversely, short individuals are sometimes cropped such that only the top part of the face is visible. In a quite small number of cases, there other subjects standing just behind the primary subject such that algoriths should detect all faces and determine which is the largest and most centered. Background ceiling lighting is often visible and this sometimes leads to under-exposure of the face When these images are provided as input into the algorithm, they are labeled with the type "wild".

2.2 Law enforcement images

The main mugshot dataset used is referred to as the FRVT 2018 set. This set was collected over the period 2002 to 2017 in routine United States law enforcement operations. This set yields three subsets

- **Mugshots**: Mugshots comprise about 86% of the database. They have reasonable compliance with the ANSI/NIST ITL1-2011 Type 10 standard's subject acquisition profiles levels 10-20 for frontal images [26]. The most common departure from the standard's requirements is the presence of mild pose variations around frontal - the images of Figure 5 are typical. The images vary in size, with many being 480x600 pixels with JPEG compression applied to produce filesizes of between 18 and 36KB with many images outside this range, implying that about 0.5 bits are being encoded per pixel. When these images are provided as input into the algorithm, they are labeled with the type "mugshot". Example images appear in Fig. 5

  NIST Interagency Report 8238 includes a comparison of this set of mugshots with the smaller and easier sets of mugshots used in tests run in 2010 and 2014.

- **Profile images**: Profile-view images have been collected in law enforcement for more than 100 years, as human capability is improved with orthogonal information. The profile images used in this report were collected during the same session as the frontal mugshot photograph, in the same standardized photographic setup. These would not therefore be used with automated face recognition. A small subset, 200 000 images, were set aside for testing. When these images are provided as input into the algorithm, they are labeled with the type "wild". Example images appear in Fig. 7

- **Webcam images**: The remaining 14% of the images were collected using an inexpensive webcam attached to a flexible operator-directed mount. These images are all of size 240x240 pixels, that are in considerable violation of most quality-related clauses of all face recognition standards. As evident in the figure, the most common defects are non-frontal pose (associated with the rotational degrees of freedom of the camera mount), low contrast (due to varying and intense background lights), and poor spatial resolution (due to inexpensive camera optics) - see examples in Fig 6. The images are overly JPEG compressed, to between 4 and 7KB, implying that only 0.5 to 1 bits are being encoded per color pixel. When these images are provided as input into the algorithm, they are labeled with the type "wild". Example images appear in Fig. 6

These are drawn from NIST Special Database 32 which may be downloaded here.

These images were partitioned in galleries and probesets for the various experiment listed in Table 1.

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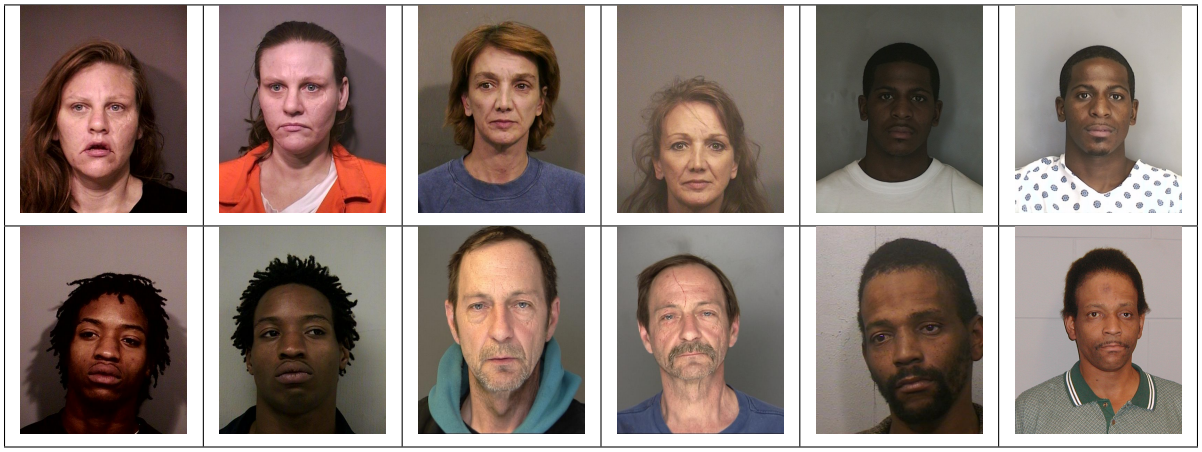

Figure 5: Six mated mugshot pairs representative of the FRVT-2014 (LEO) and FRVT-2018 datasets. The images are collected live, i.e. not scanned from paper. Image source: NIST Special Database 32 the Multiple Encounter Deceased Subjects dataset.

Figure 6: Twelve webcam images representative of probes against the FRVT-2018 mugshot gallery. The first eight images are four mated pairs. Such images present challenges to recognition including pose, non-uniform illumination, low contrast, compression, cropping, and low spatial sampling rate. Image source: NIST Special Database 32 the Multiple Encounter Deceased Subjects dataset.

Figure 7: **[Profile views]** *The three images are a frontal enrollment, subsequent frontal probe, and same-session ninety degree profile view. While collection of both frontal and profile views has been typical in law enforcement for more than a century, the recognition of profile to frontal views has essentially been impossible. However, reasonbly high accuracy results is now possible - see section E.*

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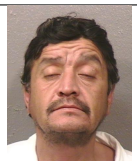
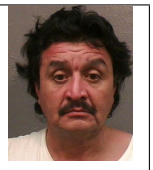
Image				
Encounter	1	...	$K_i - 1$	K_i
Capture Time	T_1	...	T_{K_i-1}	T_{K_i}
Role RECENT	Not used	Not used	Enrolled	Search
Role LIFETIME	Enrolled	Enrolled	Enrolled	Search

Figure 8: Depiction of the "recent" and "lifetime" enrollment types. Image source: NIST Special Database 32

2.3 Enrollment strategies

Many operational applications include collection and enrollment of biometric data from subjects on more than one occasion. This might be done on a regular basis, as might occur in credential (re-)issuance, or irregularly, as might happen in a criminal recidivist situation [4]. The number of images per person will depend on the application area. In civil identity credentialing (e.g. passports, driver's licenses), the images will be acquired approximately uniformly over time (e.g. ten years for a passport). While the distribution of dates for such images of a person might be assumed uniform, a number of factors might undermine this assumption[7]. In criminal applications, the number of images would depend on the number of arrests. The distribution of dates for arrest records for a person (i.e. the recidivism distribution) has been modeled using the exponential distribution but is recognized to be more complicated[8].

In any case, the 2010 NIST evaluation of face recognition showed that considerable accuracy benefits accrue with retention and use of *all* historical images [6].

To this end, the FRVT API document provides $K \geq 1$ images of an individual to the enrollment software. The software is tasked with producing a single proprietary undocumented "black-box" template[9] from the K images. This affords the algorithm an ability to generate a *model* of the individual, rather than to simply extract features from each image on a sequential basis.

As depicted in Figure 8, the i-th individual in the FRVT 2018 dataset has K_i images. These are labelled as x_k for $k = 1 \ldots K_i$ in chronological order of capture date. To measure the utility of having multiple enrollment images, this report evaluates three kinds of enrollment:

- ▷ **Recent**: Only the second most recent image, x_{K_i-1} is enrolled. This strategy of enrollment mimics the operational policy of retaining the imagery from the most recent encounter. This might be done operationally to ameliorate the effects of face ageing. Obviously retaining only the most recent image should only be done if the identity of the person is trusted to be correct. For example, in an access control situation retention of the most recent successful *authentication* image would be hazardous if it could be a false positive.

- ▷ **Lifetime-consolidated**: All but the most recent image are enrolled, $x_1 \ldots x_{K_i-1}$. This subject-centric strategy might be adopted if quality variations exist where an older image might be more suitable for matching, despite the ageing effect.

[7] For example, a person might skip applying for a passport for one cycle, letting it expire. In addition, a person might submit identical images (from the same photography session) to consecutive passport applications at five year intervals.
[8] A number of distributions have been considered to model recidivism, see for example [3].
[9] There are no formal face template standards. Template standards only exist for fingerprint minutiae - see ISO/IEC 19794-2:2011.

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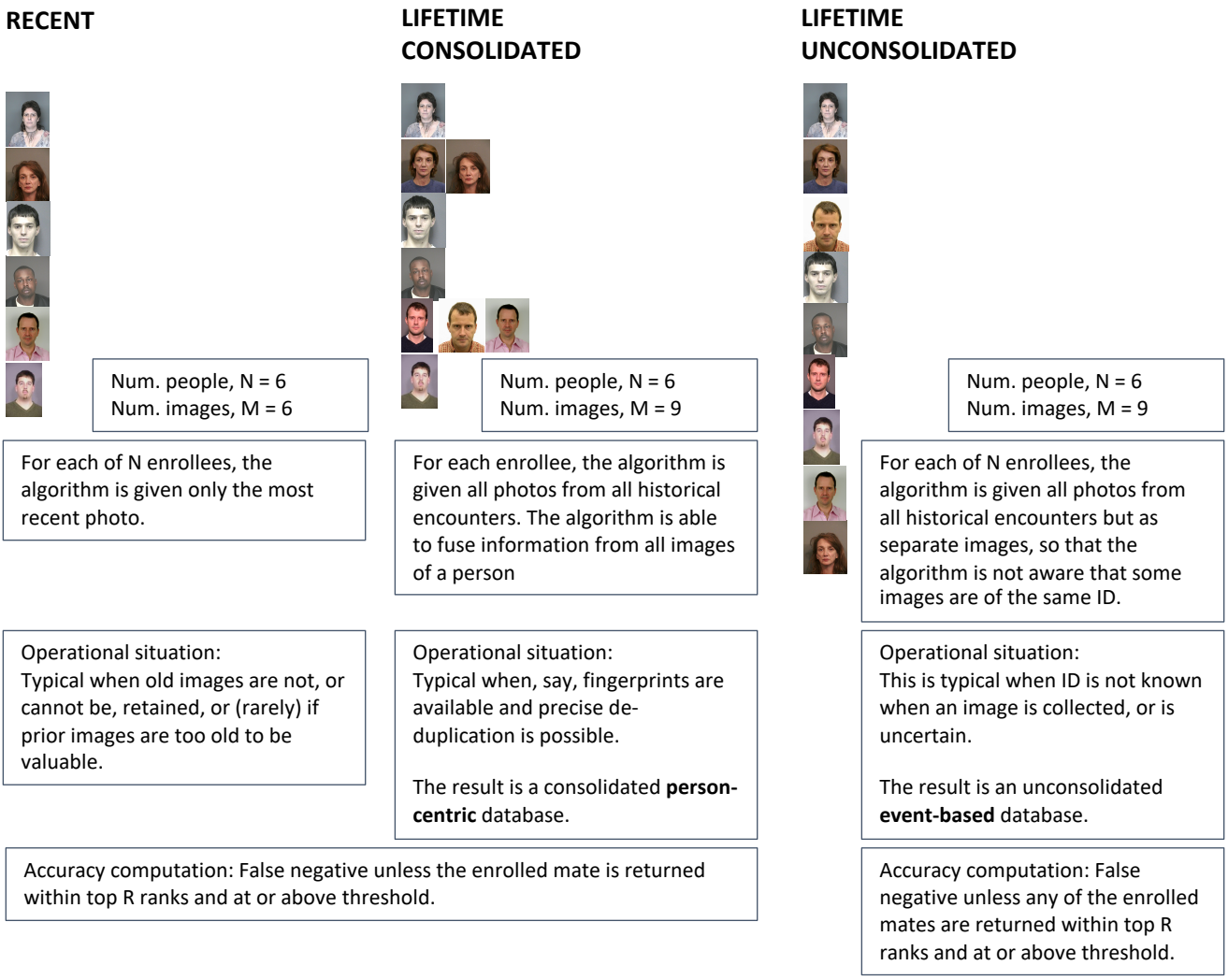

Figure 9: **Enrollment strategies**. *The figure shows the three kinds of enrollment databases examined in this report. Image source: NIST Special Database 32*

	ENROLLMENT				SEARCH			
	TYPE SEE	POPULATION			MATE		NON-MATE	
	SECTION 2.3	FILTER	N-SUBJECTS	N-IMAGES	N-SUBJECTS	N-IMAGES	N-SUBJECTS	N-IMAGES
Mugshot trials from enrollment of single images								
1	RECENT	NATURAL	640 000	640 000	154 549	154 549	331 254	331 254
2	RECENT	NATURAL	1 600 000	1 600 000				
3	RECENT	NATURAL	3 000 000	3 000 000				
4	RECENT	NATURAL	6 000 000	6 000 000				
5	RECENT	NATURAL	12 000 000	12 000 000				
Cross-domain								
13	MUGSHOTS AS ON ROW 2				82 106 WEBCAM	82 106 WEBCAM	331 254 WEBCAM	331 254 WEBCAM
Cross-view								
14	MUGSHOTS AS ON ROW 2				100 000 PROFILE	100 000 PROFILE	100 000 PROFILE	100 000 PROFILE
Mugshot ageing								
17	OLDEST	NATURAL	3 068 801	3 068 801	2 853 221	10 951 064	0	0
Border crossing ageing								
18	OLDEST	NATURAL	1 600 000	1 600 000	903 655	1 922 393	1 393 076	1 680 000
Visa-border								
19	PRIOR	NATURAL	1 600 000 VISA	1 600 000 VISA	577 444 BORDER	1 212 892 BORDER	79 769 BORDER	80 000 BORDER
20	VISA AS ON ROW 18				14 004 BORDER	31 579 BORDER	42 474 BORDER	45 460 BORDER

Table 1: **Enrollment and search sets**. *Each row summarizes one identification trial. Unless stated otherwise, all entries refer to mugshot images. The term "natural" means that subjects were selected without heed to demographics, i.e. in the distribution native to this dataset. The probe images were collected in a different calendar year to the enrollment image. Missing values in rows 2-12 are the same as in row 1.*

- **Lifetime-unconsolidated**: Again all but the most recent image are enrolled $x_1 \ldots x_{K_i-1}$ but now separately, with different identifiers, such that the algorithm is not aware that the images are from the same face. This kind of event- or encounter-centric enrollment is very common when operational constraints preclude reliable consolidation of the historical encounters into a single identity. This aspect also prevents the recognition algorithm from a) building a holistic model of identity (as is common in speaker recognition systems) and b) implementing fusion, for example template-level fusion of feature vectors, or post-search score-level fusion. The result is that searches will typically yield more than one image of a person in the top ranks. This has consequences for appropriate metrics, as detailed in section 3.2.1

  NIST first evaluated this kind of enrollment in mid 2018, and the results tables include some comparison of accuracy available from all three enrollment styles.

In all cases, the most recent image, x_{K_i}, is reserved as the search image. For the 1.6 million subject enrollment partition of the FRVT 2018 data, $1 \leq K_i \leq 33$ with $K_i = 1$ in 80.1% of the individuals, $K_i = 2$ in 13.4%, $K_i = 3$ in 3.7%, $K_i = 4$ in 1.4%, $K_i = 5$ in 0.6%, $K_i = 6$ in 0.3%, and $K_i > 6$ is 0.2% for everyone else. This distribution is substantially dependent on United States recidivism rates.

We did not evaluate the case of retaining only the highest quality image, since automated quality assessment is out of scope for this report. We do not anticipate that such strategies will prove beneficial when the quality assessment apparatus is imperfect and unvalidated.

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3 Performance metrics

This section gives specific definitions for accuracy and timing metrics. Tests of open-set biometric algorithms must quantify frequency of two error conditions:

- **False positives**: Type I errors occur when search data from a person who has never been seen before is incorrectly associated with one or more enrollees' data.
- **Misses**: Type II errors arise when a search of an enrolled person's biometric does not return the correct identity.

Many practitioners prefer to talk about "hit rates" instead of "miss rates" - the first is simply one minus the other as detailed below. Sections 3.1 and 3.2 define metrics for the Type I and Type II performance variables.

Additionally, because recognition algorithms sometimes fail to produce a template from an image, or fail to execute a one-to-many search, the occurrence of such events must be recorded. Further because algorithms might elect to not produce a template from, for example, a poor quality image, these failure rates must be combined with the recognition error rates to support algorithm comparison. This is addressed in section 3.5.

Finally, section 3.7 discusses measurement of computation duration, and section 3.8 addresses the uncertainty associated with various measurements. Template size measurement is included with the results.

3.1 Quantifying false positives

It is typical for a search to be conducted into an enrolled population of N identities, and for the algorithm to be configured to return the closest L candidate identities. These candidates are ranked by their score, in descending order, with all scores required to be greater than or equal to zero. A human analyst might examine either all L candidates, or just the top $R \leq L$ identities, or only those with score greater than threshold, T. The workload associated with such examination is discussed later, in 3.6.

False alarm performance is quantified in two related ways. These express how many searches produces false positives, and then, how many false positives are produced in a search.

False positive identification rate: The first quantity, FPIR, is the proportion of non-mate searches that produce an adverse outcome:

$$\text{FPIR}(N,T) = \frac{\text{Num. non-mate searches where one or more enrolled candidates are returned with score at or above threshold}}{\text{Num. non-mate searches attempted.}} \tag{1}$$

Under this definition, FPIR can be computed from the highest non-mate candidate produced in a search - it is not necessary to consider candidates at rank 2 and above. FPIR is the primary measure of Type I errors in this report.

Selectivity: However, note that in any given search, several non-mate may be returned above threshold. In order to quantify such events, a second quantity, selectivity (SEL), is defined as the *number* of non-mates returned on a candidate list, averaged over all searches.

$$\text{SEL}(N,T) = \frac{\text{Num. non-mate enrolled candidates returned with score at or above threshold}}{\text{Num. non-mate searches attempted.}} \tag{2}$$

where $0 \leq$ SEL(N, T) $\leq$L. Both of these metrics are useful operationally. FPIR is useful for targeting how often an

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adverse false positive outcome can occur, while SEL as a number is related to workload associated with adjudicating candidate lists. The relationship between the two quantities is complicated - it depends on whether an algorithm concentrates the false alarms in the results of a few searches or whether it disburses them across many. This was detailed in FRVT 2014, NISTIR 8009. It has not yet been detailed in FRVT 2018.

3.2 Quantifying hits and misses

If L candidates are returned in a search, a shorter candidate list can be prepared by taking the top $R \leq L$ candidates for which the score is above some threshold, $T \geq 0$. This reduction of the candidate list is done because thresholds may be applied, and only short lists might be reviewed (according to policy or labor availability, for example). It is useful then to state accuracy in terms of R and T, so we define a "miss rate" with the general name **false negative identification rate** (FNIR), as follows:

$$\text{FNIR}(N, R, T) = \frac{\text{Num. mate searches with enrolled mate found outside top R ranks or score below threshold}}{\text{Num. mate searches attempted.}} \quad (3)$$

This formulation is simple for evaluation in that it does not distinguish between causes of misses. Thus a mate that is not reported on a candidate list is treated the same as a miss arising from face finding failure, algorithm intolerance of poor quality, or software crashes. Thus if the algorithm fails to produce a candidate list, either because the search failed, or because a search template was not made, the result is regarded as a miss, adding to FNIR.

Hit rates, and true positive identification rates: While FNIR states the "miss rate" as how often the correct candidate is either not above threshold or not at good rank, many communities prefer to talk of "hit rates". This is simply the **true positive identification rate**(TPIR) which is the complement of FNIR giving a positive statement of how often mated searches are successful:

$$\text{TPIR}(N, R, T) = 1 - \text{FNIR}(N, R, T) \quad (4)$$

This report does not report true positive "hit" rates, preferring false negative miss rates for two reasons. First, costs rise linearly with error rates. For example, if we double FNIR in an access control system, then we double user inconvenience and delay. If we express that as decrease of TPIR from, say 98.5% to 97%, then we mentally have to invert the scale to see a doubling in costs. More subtly, readers don't perceive differences in numbers near 100% well, becoming inured to the "high nineties" effect where numbers close to 100 are perceived indifferently.

Reliability is a corresponding term, typically being identical to TPIR, and often cited in automated (fingerprint) identification system (AFIS) evaluations.

An important special case is the **cumulative match characteristic**(CMC) which summarizes accuracy of mated-searches only. It ignores similarity scores by relaxing the threshold requirement, and just reports the fraction of mated searches returning the mate at rank R or better.

$$\text{CMC}(N, R) = 1 - \text{FNIR}(N, R, 0) \quad (5)$$

We primarily cite the complement of this quantity, $\text{FNIR}(N, R, 0)$, the fraction of mates *not* in the top R ranks.

The **rank one hit rate** is the fraction of mated searches yielding the correct candidate at best rank, i.e. CMC(N, 1). While this quantity is the most common summary indicator of an algorithm's efficacy, it is not dependent on similarity scores, so it does not distinguish between strong (high scoring) and weak hits. It also ignores that an adjudicating reviewer is often willing to look at many candidates.

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3.2.1 False negative rates for unconsolidated galleries

As detailed in section 2.3 a common type of gallery, here referred to as the lifetime unconsolidate type, is populated with all images of an individual without any association between them. That is, the gallery construction algorithm is not provided with any ID labels that would support processing of a person's images jointly. This constrasts with the lifetime consolidate type where an algorithm may explicitly fuse features from multiple images of a person, or select a best image. In such cases, where the number of enrolled images is a random variable, we define two false negative rates as follows.

The first demands that the algorithm place any of the K_i mates in the top $R \geq 1$ ranks. The proportion of searches for which this does not occur forms a false negative identification rate:

$$\text{FNIR}_{\text{any}}(N, R, T) = 1 - \frac{\text{Num. mate searches where any enrolled mate is found in the top R ranks and at-or-above threshold}}{\text{Num. mate searches attempted.}} \tag{6}$$

The second demands that the algorithm place all K_i mates in the top $R \geq K_i$ ranks. The proportion of searches for which this does not occur forms a false negative identification rate:

$$\text{FNIR}_{\text{all}}(N, R, T) = 1 - \frac{\text{Num. mate searches where all enrolled mates are found in the top R ranks and at-or-above threshold}}{\text{Num. mate searches attempted.}} \tag{7}$$

Placing all mates in the top ranks is a more difficult task than correctly retrieving any image, so it holds that: $\text{FNIR}_{\text{all}} \geq \text{FNIR}_{\text{any}}$. This is evident in the results presented for November 2018 algorithms in Tables starting at **??**.

The information retrieval community might prefer to compute and plot *precision* and *recall*; this is a valid approach, but we advance the two metrics above because they relate to our normal definition of consolidated FNIR, and they cover the two extreme use-cases of wanting any hit vs. all hits.

3.3 DET interpretation

In biometrics, a false negative occurs when an algorithm fails to match two samples of one person – a Type II error. Correspondingly, a false positive occurs when samples from two persons are improperly associated – a Type I error.

Matches are declared by a biometric system when the native comparison score from the recognition algorithm meets some threshold. Comparison scores can be either similarity scores, in which case higher values indicate that the samples are more likely to come from the same person, or dissimilarity scores, in which case higher values indicate different people. Similarity scores are traditionally computed by fingerprint and face recognition algorithms, while dissimilarities are used in iris recognition. In some cases, the dissimilarity score is a distance possessing metric properties. In any case, scores can be either mate scores, coming from a comparison of one person's samples, or nonmate scores, coming from comparison of different persons' samples.

The words "genuine" or "authentic" are synonyms for mate, and the word "impostor" is used as a synonym for nonmate. The words "mate" and "nonmate" are traditionally used in identification applications (such as law enforcement search, or background checks) while genuine and impostor are used in verification applications (such as access control).

An error tradeoff characteristic represents the tradeoff between Type II and Type I classification errors. For identification this plots false negative vs. false positive identification rates i.e. FNIR vs. FPIR parametrically with T. Such plots

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are often called detection error tradeoff (DET) characteristics or receiver operating characteristic (ROC). These serve the same function − to show error tradeoff − but differ, for example, in plotting the complement of an error rate (e.g. TPIR = 1 − FNIR) and in transforming the axes, most commonly using logarithms, to show multiple decades of FPIR. More rarely, the function might be the inverse of the Gaussian cumulative distribution function.

The slides of Figures 10 through 15 discuss presentation and interpretation of DETs used in this document for reporting face identification accuracy. Further detail is provided in formal biometrics testing standards, see the various parts of ISO/IEC 19795 Biometrics Testing and Reporting. More terms, including and beyond those to do with accuracy, appear in ISO/IEC 2382-37 Information technology – Vocabulary – Part 37: Harmonized biometric vocabulary.

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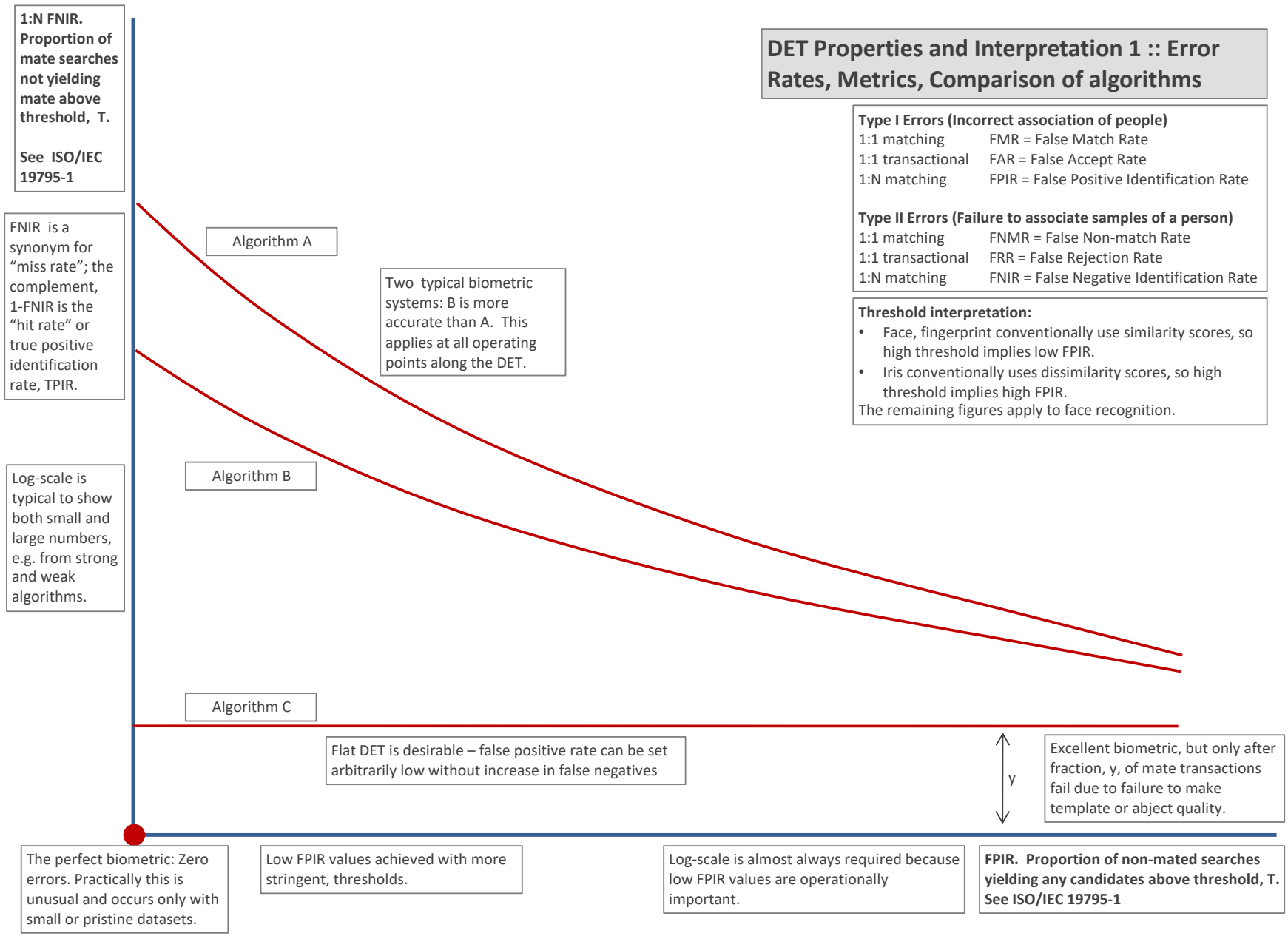

Figure 10: **DET as the primary performance reporting mechanism**.

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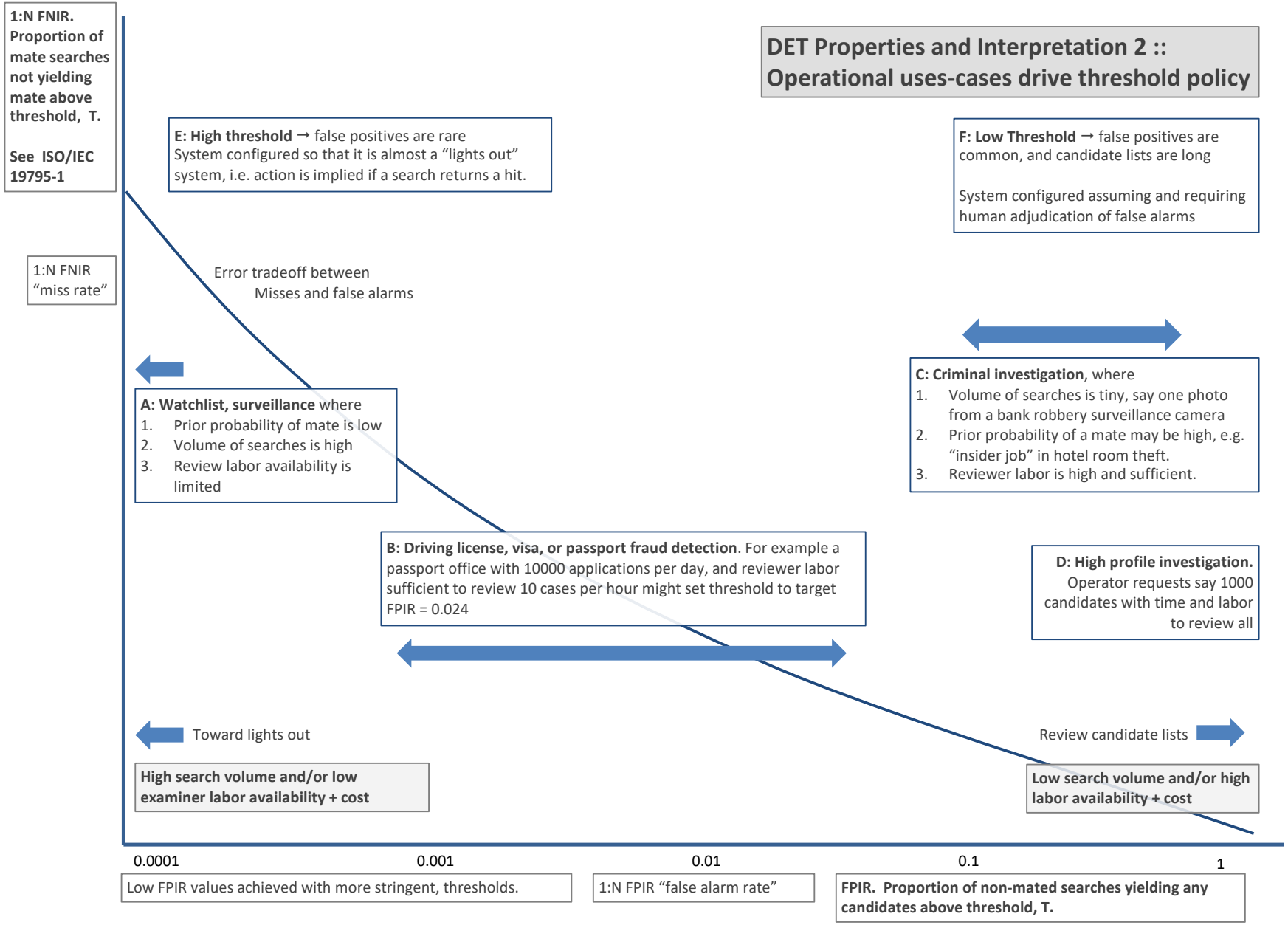

Figure 11: **DET as the primary performance reporting mechanism**.

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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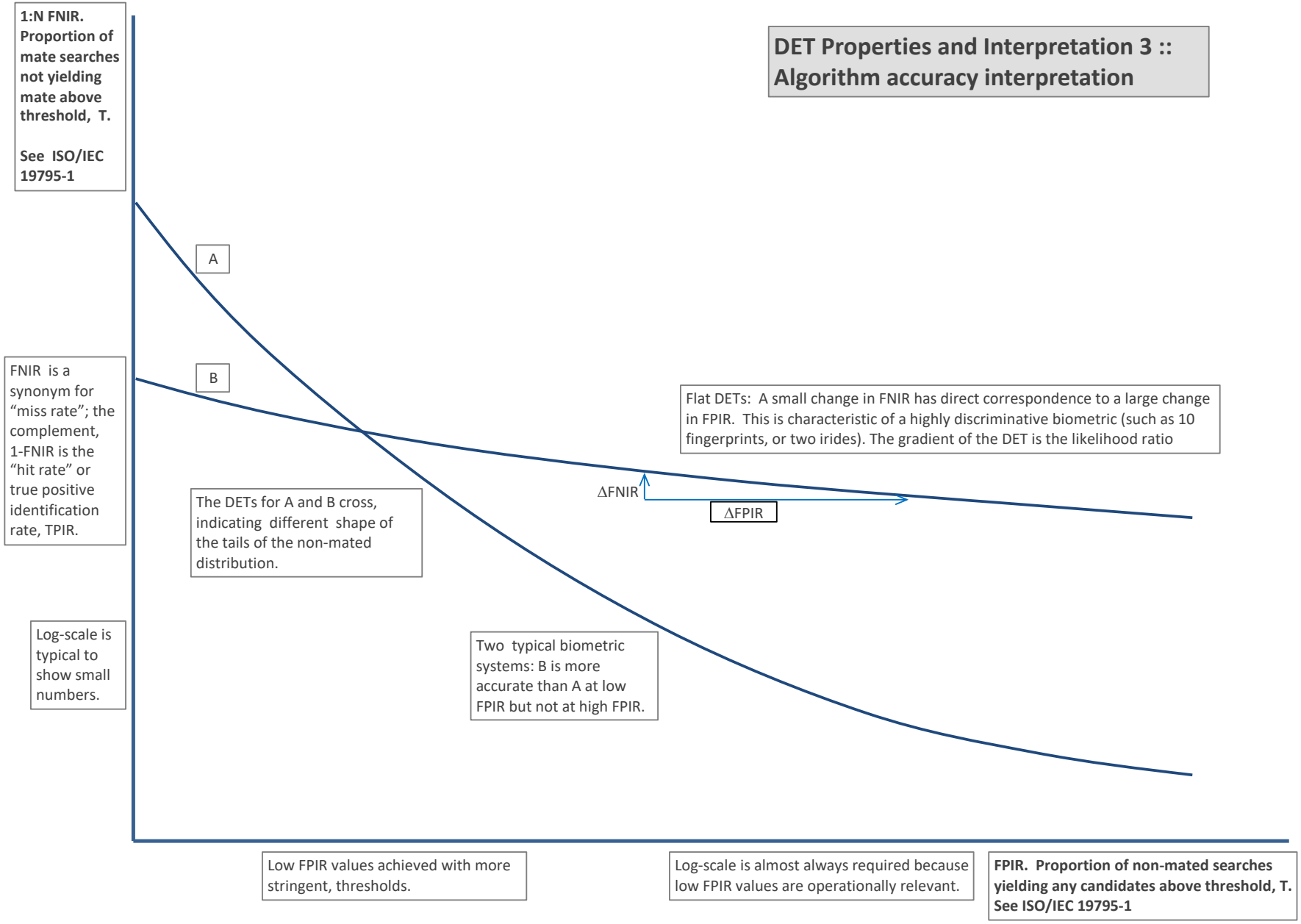

Figure 12: **DET as the primary performance reporting mechanism**.

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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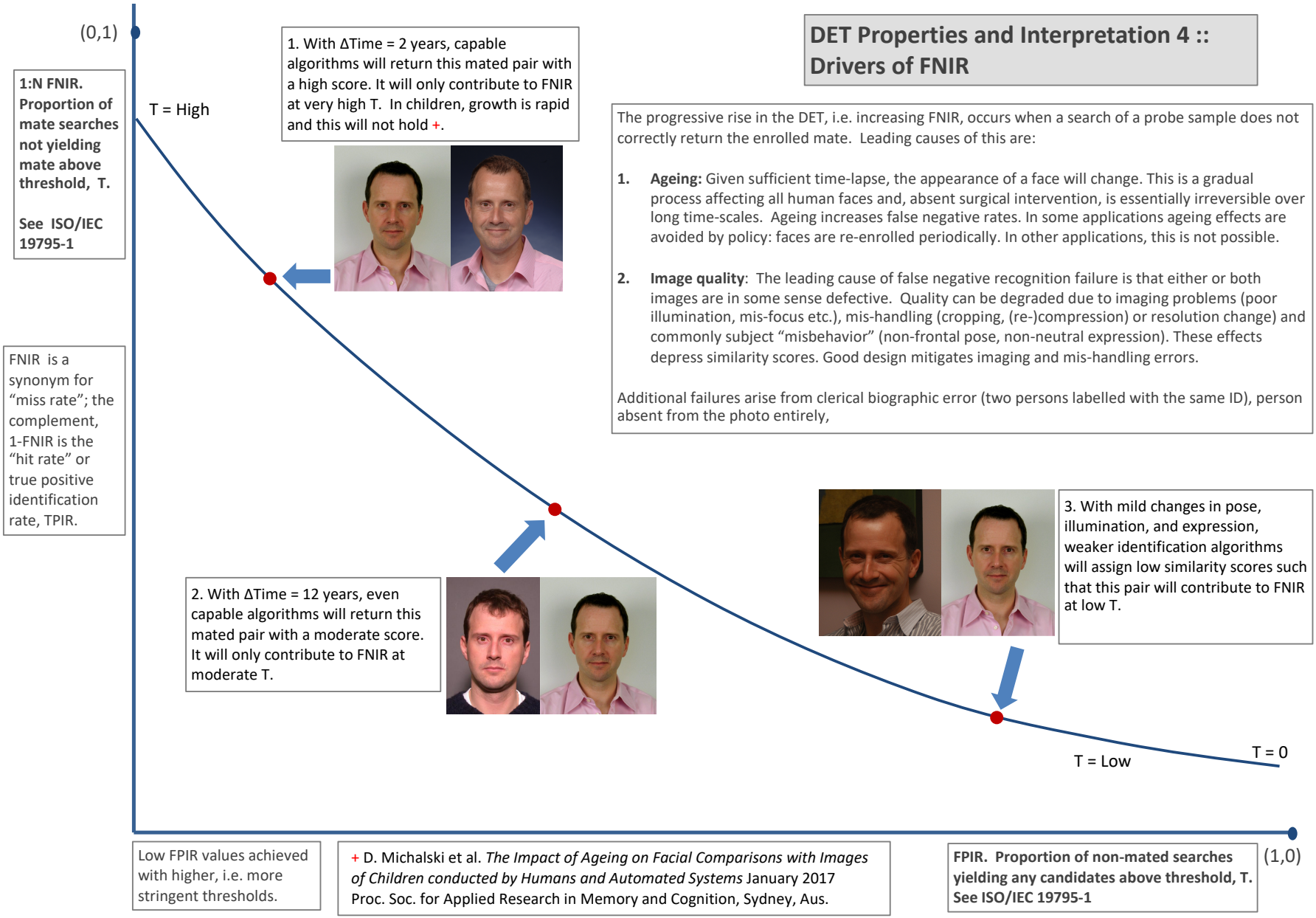

Figure 13: **DET as the primary performance reporting mechanism**.

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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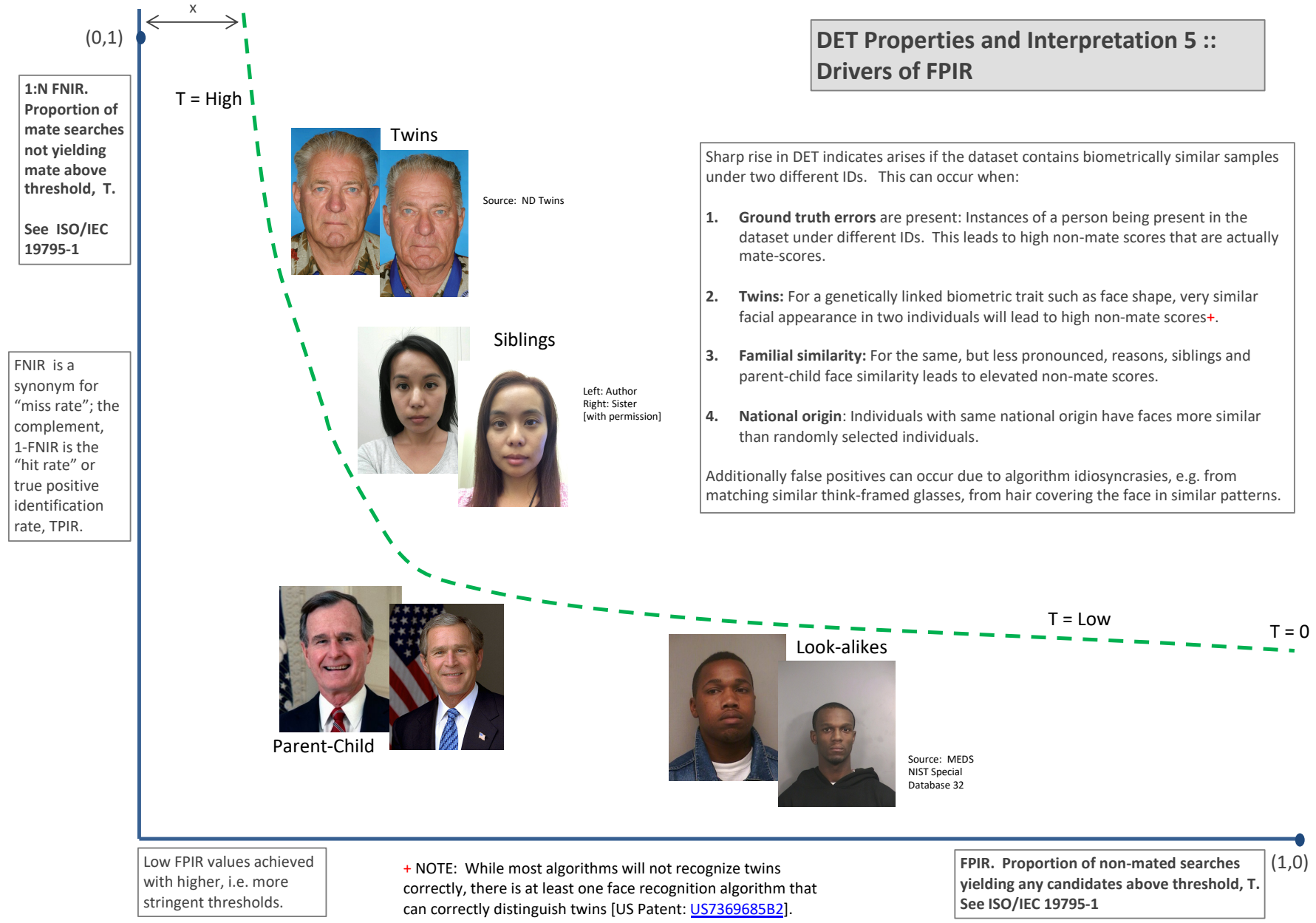

Figure 14: **DET as the primary performance reporting mechanism**.

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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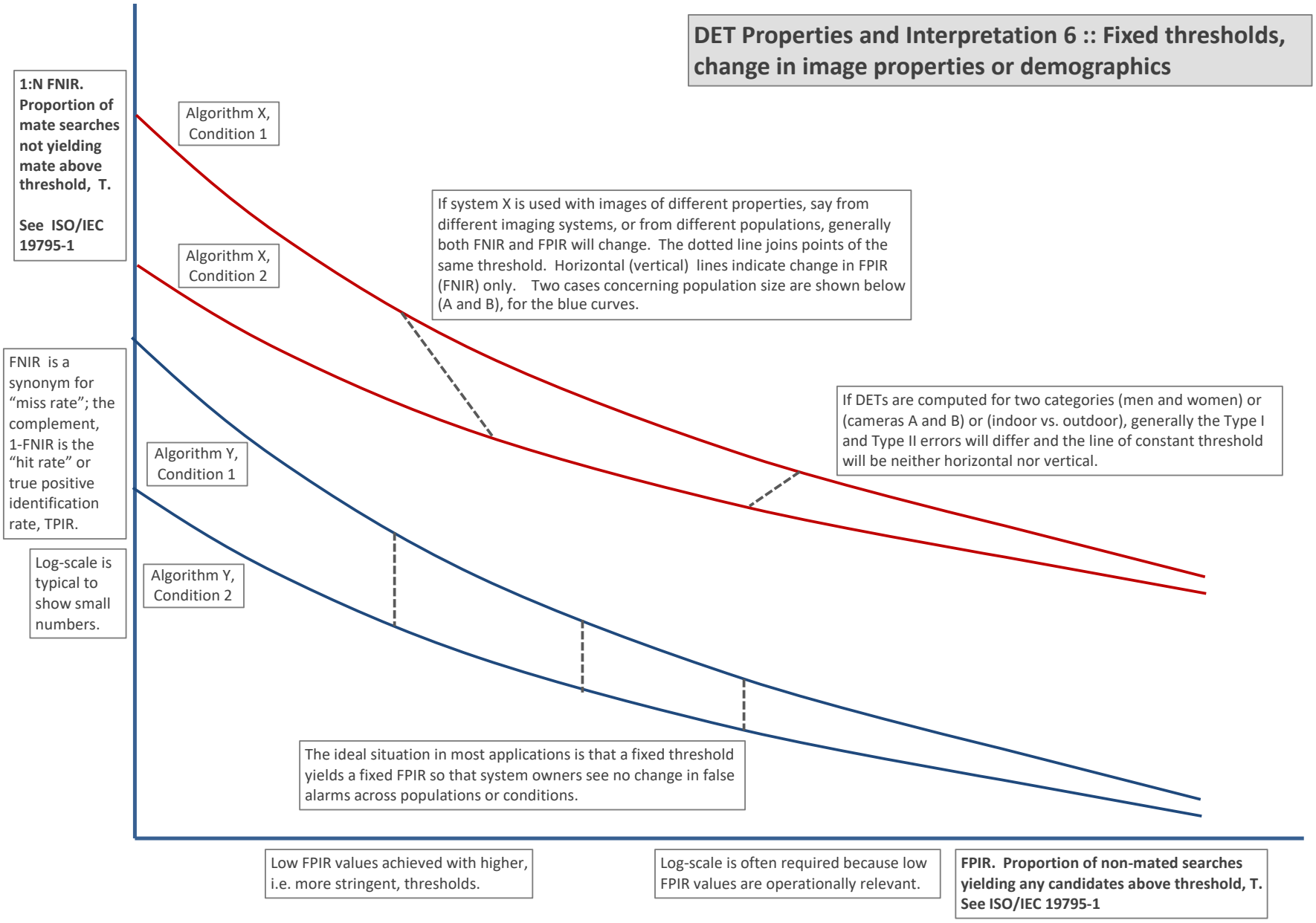

Figure 15: **DET as the primary performance reporting mechanism**.

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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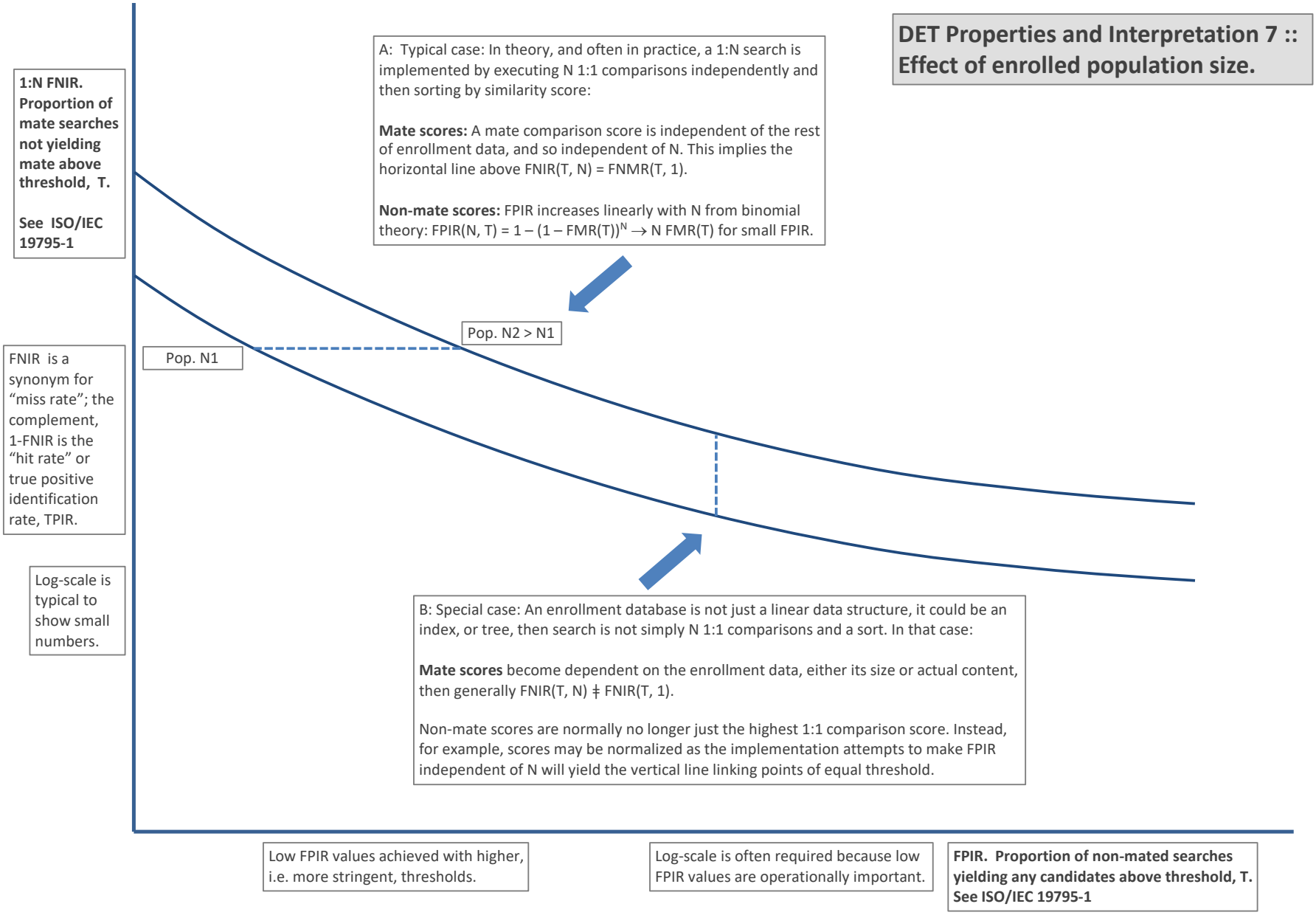

Figure 16: **DET as the primary performance reporting mechanism**.

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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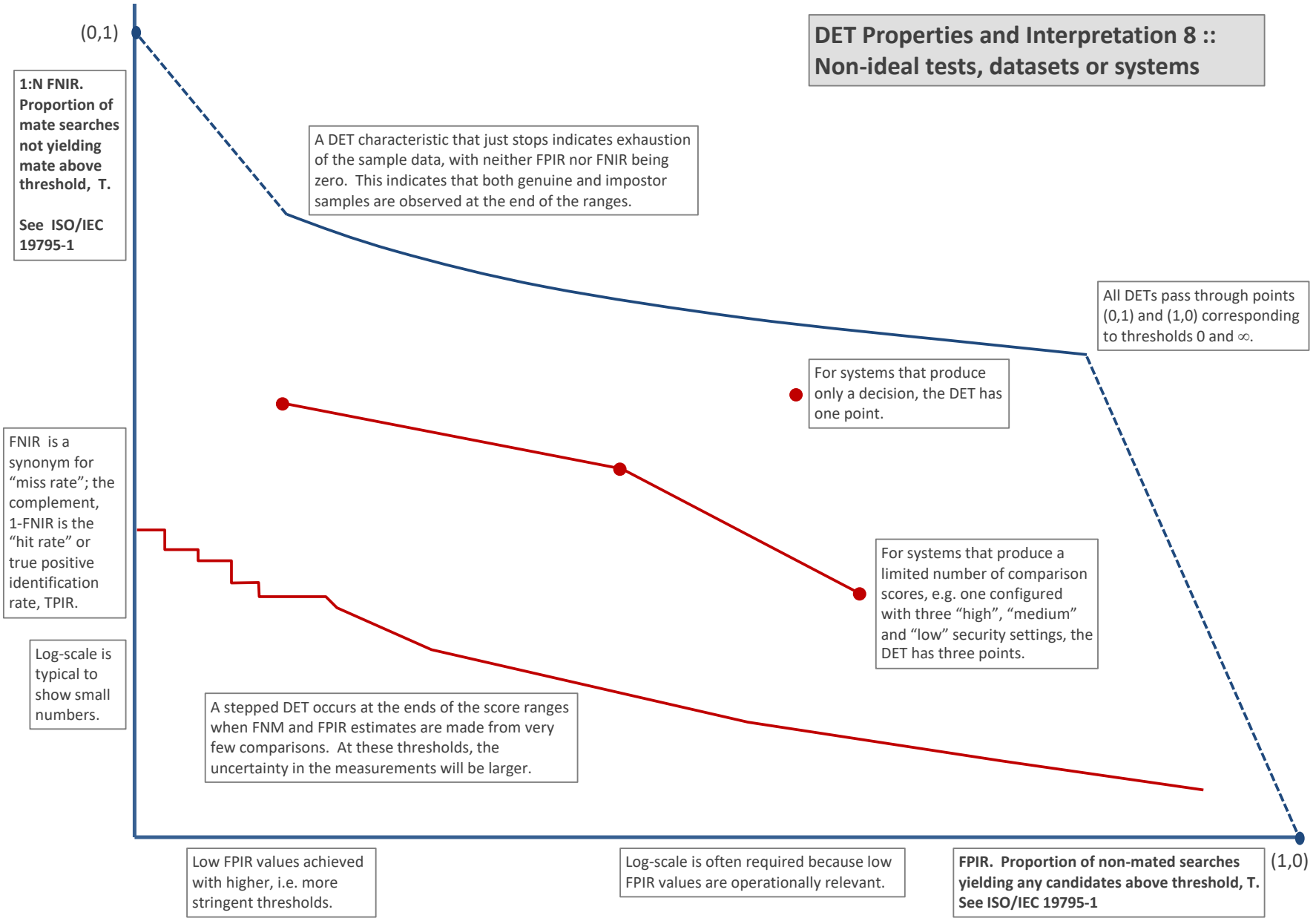

Figure 17: **DET as the primary performance reporting mechanism**.

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

3.4 Best practice testing requires execution of searches with and without mates

FRVT embeds 1:N searches of two kinds: Those for which there is an enrolled mate, and those for which there is not. The respective numbers for these types of searches appear in Table 1. However, it is common to conduct only mated searches[10]. The cumulative match characteristic is computed from candidate lists produced in mated searches. Even if the CMC is the only metric of interest, the actual trials executed in a test should nevertheless include searches for which no mate exists. As detailed in Table 1 the FRVT reserved disjoint populations of subjects for executing true non-mate searches.

3.5 Failure to extract features

During enrollment some algorithms fail to convert a face image to a template. The proportion of failures is the failure-to-enroll rate, denoted by FTE. Similarly, some search images are not converted to templates. The corresponding proportion is termed failure-to-extract, denoted by FTX.

We do not report FTX because we assume that the same underlying algorithm is used for template generation for enrollment and search.

Failure to extract rates are incorporated into FNIR and FPIR measurements as follows.

- **Enrollment templates**: Any failed enrollment is regarded as producing a zero length template. Algorithms are required by the API [10] to transparently process zero length templates. The effect of template generation failure on search accuracy depends on whether subsequent searches are mated, or non-mated: Mated searches will fail giving elevated FNIR; non-mated searches will not produce false positives so, to first order, FPIR will be reduced by a factor of $1-$FTE.

- **Search templates and 1:N search**: In cases where the algorithm fails to produce a search template from input imagery, the result is taken to be a candidate list whose entries have no hypothesized identities and zero score. The effect of template generation failure on search accuracy depends on whether searches are mated, or non-mated: Mated searches will fail giving elevated FNIR; Non-mated searches will not produce false positives, so FPIR will be reduced. Thus given a measurement of false negative and positive rates made over only those where failures-to-extract did not occur, those rates - call them $\text{FNIR}^\dagger$ and $\text{FPIR}^\dagger$ - could be adjusted by an explicit measurement of FTX as follows

$$\text{FNIR} = \text{FTX} + (1 - \text{FTX})\text{FNIR}^\dagger \tag{8}$$

$$\text{FPIR} = (1 - \text{FTX})\text{FPIR}^\dagger \tag{9}$$

This approach is the correct treatment for positive-identification applications such as access control where cooperative users are enrolled and make attempts at recognition. This approach is not appropriate to negative identification applications, such as visa fraud detection, in which hostile individuals may attempt to evade detection by submitting poor quality samples. In those cases, template generation failures should be investigated as though a false alarm had occurred.

[10]For example, the Megaface benchmark. This is bad practice for several reasons: First, if a developer knows, or can reasonably assume, that a mate always exists, then unrealistic gaming of the test is possible. A second reason is that it does not put FPIR on equal footing with FNIR and that matters because in most applications, not all searches have mates - not everyone has been previously enrolled in a driving license issuance or a criminal justice system - so addressing between-class separation becomes necessary.

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3.6 Fixed length candidate lists, threshold independent workload

Suppose an automated face identification algorithm returns L candidates, and a human reviewer is retained to examine up to R candidates, where $R \leq L$ might be set by policy, preference or labor availability. For now, assume also that the reviewer is not provided with, or ignores, similarity scores, and thresholds are not applied. Given the algorithm typically places mates at low (good) ranks, the number of candidates a reviewer can be expected to review can be derived as follows. Note that the reviewer will:

- ▷ Always inspect the first ranked image — Frac. reviewed = 1
- ▷ Then inspect those candidates where mate not confirmed at rank 1 — Frac. reviewed = 1-CMC(1)
- ▷ Then inspect those candidates where mate not confirmed at rank 1 or 2 — Frac. reviewed = 1-CMC(2)

etc. Thus if the reviewer will stop after a maximum of R candidates, the expected number of candidate reviews is

$$M(R) = 1 + (1 - CMC(1)) + (1 - CMC(2)) + \ldots + (1 - CMC(R-1)) \tag{10}$$

$$= R - \sum_{r=1}^{R-1} CMC(r) \tag{11}$$

A recognition algorithm that front-loads the cumulative match characteristic will offer reduced workload for the reviewer. This workload is defined only over the searches for which a mate exists. In the cases where there truly is no mate, the reviewer would review all R candidates. Thus, if the proportion of searches for which a mate does exist is β, which in the law enforcement context would be the recidivism rate [3], the full expression for workload becomes:

$$M(R) = \beta \left(R - \sum_{r=1}^{R-1} CMC(r) \right) \quad + \quad (1 - \beta)R \tag{12}$$

$$= R - \beta \sum_{r=1}^{R-1} CMC(r) \tag{13}$$

3.7 Timing measurement

Algorithms were submitted to NIST as implementations of the application programming interface(API) specified by NIST in the Evaluation Plan [10]. The API includes functions for initialization, template generation, finalization, search, gallery insert, and gallery delete. Two template generation functions are required, one for the preparation of an enrollment template, and one for a search template.

In NIST's test harness, all functions were wrapped by calls to the C++ std::chrono::high resolution clock which on the dedicated timing machine counts 1ns clock ticks. Precision is somewhat worse than that however.

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3.8 Uncertainty estimation

3.8.1 Random error

This study leverages operational datasets for measurement of recognition error rates. This affords several advantages. First, large numbers of searches are conducted (see Table 1) giving precision to the measurements. Moreover, for the two mugshot datasets, these do not involve reuse of individuals so binomial statistics can be expected to apply to recognition error counts. In that case, an observed count of a particular recognition outcome (i.e. a false negative or false positive) in M trials will sustain 95% confidence that the actual error rate is no larger than some value.

As an example, the minimum number of mugshot searches conducted in this report is M =154 549, and for an observed FNIR around 0.002, the measurement supports a conclusion that the actual FNIR is no higher than 0.00228 at 99% confidence level. On the false positive side, we tabulate FNIR at FPIR values as low as 0.001. Given estimates based on 331 254 non-mate trials, the actual FPIR values will be below 0.00115 at 99% confidence. In conclusion, large scale evaluation, without reuse of subjects, supports tight uncertainty bounds on the measured error rates.

3.8.2 Systematic error

The FRVT 2018 dataset includes anomalies discovered as a result of inspecting images involved in recognition failures from the most accurate algorithms. Two kinds of failure occur: False negatives (which, for the purpose here, include failures to make templates) and false positives.

False negative errors: We reviewed 600 false negative pairs for which either or both of the leading two algorithms did not put the correct mate in the top 50 candidates. Given 154 549 searches, this number represents 0.39% of the total, resulting in FNIR ~ 0.0039. Of the 600 pairs:

- **A: Poor quality**: About 20% of the pairs included images of very low quality, often greyscale, low resolution, blurred, low contrast, partially cropped, interlaced, or noisy scans of paper images. Additionally, in a few cases, the face is injured or occluded by bandages or heavy cosmetics.
- **B: Ground truth identity label bugs**: About 15% of the pairs are not actually mated. We only assigned this outcome when a pair is clearly not mated.
- **C: Profile views**: About 35% included an image of a profile (side) view of the face, or, more rarely, an image that was rotated 90 degrees in-plane (roll).
- **D: Tattoos**: About 30% included an image of a tattoo that contained a face image. These arise from mis-labelling in the parent dataset metadata.
- **E: Ageing**: There is considerable time-lapse between the two captures.

All these estimates are approximate. Of these, the tattoo and mislabled images can never be matched. These constitute an accuracy floor in the sample implying that FNIR cannot be below 0.0018[11]. The profile-views, low-quality images, and images with considerable ageing can, in principle, be successfully matched - indeed some algorithms do so - so are not part of the accuracy floor.

[11] This value is the sum of two partial false negative rates: FNIR_B = 0.15 * 0.0039 plus FNIR_D = 0.3 * 0.0039

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For the microsoft-4 algorithm the lowest miss rate from (recent entry in Table 52) is FNIR(640 000, 50, 0) = 0.0018. This is close to the value estimated from the inspection of misses. It is below the 0.0039 figure because the algorithm does match some profile and poor quality images, that the yitu-2 algorithm does not.

For many tables (e.g. Table 52), the FNIR values obtained for the FRVT-2018 mugshots could be corrected by reducing them by 0.0018. The best values would then be indistinct from zero. The results in this report *were not* adjusted to account for this systematic error.

False positive errors: As shown in Figure 1 and discussed in Figure 14 many of the DET characteristics in this report exhibit a pronounced turn upward at low false positive rates. The shape can be caused by identity labelling errors in the ground truth of a dataset, specifically persons present in the database under two IDs such that some proportion of non-mate pairs are actually mated. To look for such possibilities, we merged the highest 1000 non-mate pairs produced by three different algorithms which resulted in 1839 unique pairs. This constitutes 0.56% of all non-mate searches. We assert that it is *very* difficult for human reviewers to assign the pairs into the following three categories: twins; doppelgangers; or ground-truth errors (instances of the same person under two IDs). Given this difficulty we made no attempt to correct any possible ground truth errors except by removing 57 pairs in the following categories:

- **A: Profile views**: Thirteen pairs included one or two profile-view images. As described in Figure 478, these can cause false positives.
- **B: Same-session photographs**: For twelve pairs, the images were identical or trivially altered (e.g. cropped) versions of the same photo. These were present under a different ID likely due to some clerical or procedural mistake.
- **C: Tattoos of faces**: There were fourteen instances of tattoo photographs that contained faces causing false matches.
- **D: T-shirt faces**: There were six instances of T-shirt photographs (of Bob Marley and Che Guevara) being detected instead of the face and causing false positives.
- **E: Background faces**: There were twelve instances of one subject appearing in the background of two otherwise correct portrait photos.

Note we did not remove any images where there was a chance that the pair was actually a different person.

In any case, the results in this report have not been adjusted for this systematic error.

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4 Results

This section gives extensive results for algorithms submitted to FRVT 2018. Three page "report cards" for each algorithm are contained in a separate supplement. Performance metrics were described in section 3. The main results are summarized in tabular form with more exhaustive data included as DET, CMC and related graphs in appendices as follows:

- The three tables 2-4 list algorithms alongside full developer names, acceptance date, size of the provided configuration data, template size and generation time, and search duration data.
  - The **template generation duration** is most important to applications that require fast response. For example, an eGate taking more than two seconds to produce a template might be unacceptable. Note that GPUs may be of utility in expediting this operation for some algorithms, though at additional expense. Two additional factors should be considered[12][13].
  - The **search duration** is the time taken for a search of a search template into a gallery of N enrollment templates. This performance variable, together with the volume of searches, is influential on the amount of hardware needed to sustain an operational deployment. This is measured here with the algorithm running on a single core of a contemporary CPU. Search is most simply implemented as N computations of a distance metric followed by a sort operation to find the closest enrollments. However, considerable optimization of this process is possible, up to and including fast-search algorithms that, by various means, avoid computation of all N distances.
  - The **template size** is the size of the extracted feature vector (or vectors) and any needed header information. Large template sizes may be influential on bus or network bandwidth, storage requirements, and on search duration. While the template itself is an opaque data blob, the feature dimensionality might be estimated by assuming a four-bytes-per-float encoding. There is a wide range of encodings. For the more accurate algorithm, sizes range from 256 bytes to about 2KB bytes, indicating essentially no consensus on face modeling and template design.
  - The **template size multiplier** column shows how, given k input images, the size of the template grows. Most implementations internally extract features from each image and concatenate them, and implement some score-level fusion logic during search. Other implementations, including many of the most accurate algorithms, produce templates whose size does not grow with k. This could be achieved via selection of the best quality image - but this is not optimal in handling ageing where the oldest image could be the best quality. Another mechanism would be feature-level fusion where information is fused from all k inputs. In any case, as a black-box test, the fusion scheme is proprietary and unknown.
  - The size of the **configuration data** is the total size of all files resident in a vendor-provided directory that contains arbitrary read-only files such as parameters, recognition models (e.g caffe). Generally a large value for this quantity may prohibit the use of the algorithm on a resource-constrained device.

[12] The FRVT 2018 API prohibited threading, so some gains from parallelism may be available on multiple-cores or multiple processors, if the feature extraction code could be distributed across them.

[13] Note also that factors of two or more may be realizable by exploiting modern vector processing instructions on CPUs. It is not clear in our measurements whether all developers exploited Intel's AVX2 instructions, for example. Our machine was so equipped, but we insisted that the same compiled library should also run on older machines lacking that instruction. The more sophisticated implementations may have detected AVX2 presence and branched accordingly. The less sophisticated may be defaulted to the reduced instruction set. Readers should see the FRVT 2018 API document for the specific chip details.

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▷ Tables 52-53 report core rank-based accuracy for mugshot images. The population size is limited to N = 1.6 million identities because this is the largest gallery size on which all algorithms were executed. Notable observations from these tables are as follows:

- **Accuracy gains since 2018**: NIST Interagency Report 8238 documented massive gains over those reported in the FRVT 2014 report, NIST Interagency Report 8009. Further gains are documented in this report. Comparing the most accurate algorithm in November 2018, NEC-3, the value of FNIR(N, L, T) reduced from 0.0031 to 0.0024 for the Sensetime-004 algorithm with N = 12 million recent images. The tables show broader gains: many developers have made advances since 2018 with between two and five-fold reduction in errors.
- **Wide range in accuracy**: The rank-1 miss rates vary from FNIR(N, 1, 0) = 0.0012 for sensetime-004 up to about 0.5 for the very fast but inaccurate microfocus-x algorithms. Among the developers who are superior to NEC in 2013, the range is from 0.002 to 0.035 for camvi-3. This large accuracy range is consistent with the buyer-beware maxim, and indicates that face recognition software is far from being commoditized.

▷ Tables 61-62 report threshold-based error rates, FNIR(N, L, T), for N = 1.6 million for mugshot-mugshot accuracy on FRVT 2014, FRVT 2018, and also (in pink) mugshot-webcam accuracy using FRVT 2018 enrollments. Notable observations from these tables are as follows:

- **Order of magnitude accuracy gains since 2014**: As with rank-based results, the gains in accuracy are substantial, though somewhat reduced. At FPIR = 0.01, the best improvement over NEC in 2014 is a 27 fold reduction in FNIR using the NEC_2 algorithm. At FPIR = 0.001, the largest gain is a six-fold reduction in FNIR via the NEC_3 algorithm.
- **Broad gains across the industry**: About 19 companies realize accuracy better than the NEC benchmark from 2014. This is somewhat lower than the 28 developers who succeeded on the rank-1 metric. This may be due to the ubiquity of, and emphasis on, the rank-1 metric in many published algorithm development papers.
- **Webcam images**: Searches of webcam images give FNIR(N, T) values around 2 to 3 times higher than mugshot searches. Notably the leading developers with mugshots are approximately the same with poorer quality webcams. But some developers e.g. Camvi, Megvii, TongYi, and Neurotechnology do improve their relative rankings on webcams, perhaps indicating their algorithms were tailored to less constrained images.

▷ Tables 38, 45, 46 and show, respectively, high-threshold, rank 1, and rank 50 FNIR values for all algorithms performing searches into five different gallery sizes, N = 640 000, N = 1 600 000, N = 3 000 000, N = 6 000 000 and 12 000 000. The FPIR = 0.001 table is included to inform high-volume duplicate detection applications. The Rank-1 table is included as a primary accuracy indicator. The Rank-50 table is included to inform agencies who routinely produce 50 candidates for human-review. The notable results are:

- **Slow growth in rank-based miss rates:** FNIR(N, R) generally grows as a power law, aN^b. From the straight lines of many graphs of Figure 20 this is clearly a reasonable model for most, but not all, algorithms. The coefficient a can be interpreted as FNIR in a gallery of size 1. The more important coefficient b indicates scalability, and often, $b \ll 1$, implies very benign growth in FNIR. The coefficients of the models appear in the Tables 45 and 46.
- **Slow growth in threshold-based miss rates:** FNIR(N, T) also generally grows as a power law, aN^b except at the high threshold values corresponding to low FPIR values. This is visible in the plots of Figure 84 which

show straight lines except for FPIR = 0.001, which increase more rapidly with N above 3 000 000. Each trace in those figures shows FNIR(N, T) at fixed FPIR with both N and T varying. Thus at large N, it is usually necessary to elevate T to maintain fixed FPIR. This causes increased FNIR. Why that would no-longer obey a power-law is not known. However, if we expect large galleries to contain individuals with familial relations to the non-mate search images - in the most extreme case, twins - then suppression of false positives becomes more difficult. This is discussed in the Figures starting at Fig. 10

▷ Figure **??** shows false positives from twins against their enrolled siblings, broken out by type of twin: fraternal or identical. The Figure is based on the enrollment of 104 single images on one of a pair of twins, and then the search of 2354 second images. Note that the dataset is heavily skewed towards identical twins which is not representative of the true population. There is also a skew towards same sex fraternal twin pairs compared to different sex fraternal twin pairs again not representative of the true population.

The notable results are:

- For all algorithms tested, the 1087 mated searches (Twin A vs. Twin A) produce scores almost always above typical operational thresholds, with (not shown) matches at rank 1. The images are of good quality, so this is the result expected from the rest of this report.
- For the 1066 identical twin searches (AB), almost all produce the twin at rank 1, with a few producing the mate at further down the candidate lists rank and low score.
- For the 169 fraternal searches (AB) from same sex pairs, most algorithms give a large number of very high scores, implying false positives at all thresholds. However, there there are long tails containing lower scores that are correctly below threshold. In general, scores that are higher in this distribution are all rank 1 whereas the lower scores have much higher ranks.
- (Not shown) Of the 169, there are 24 fraternal searches (AB) involving different sex twins. Here most algorithms correctly report scores well below the lowest threshold, and usually not on the candidate list at all.

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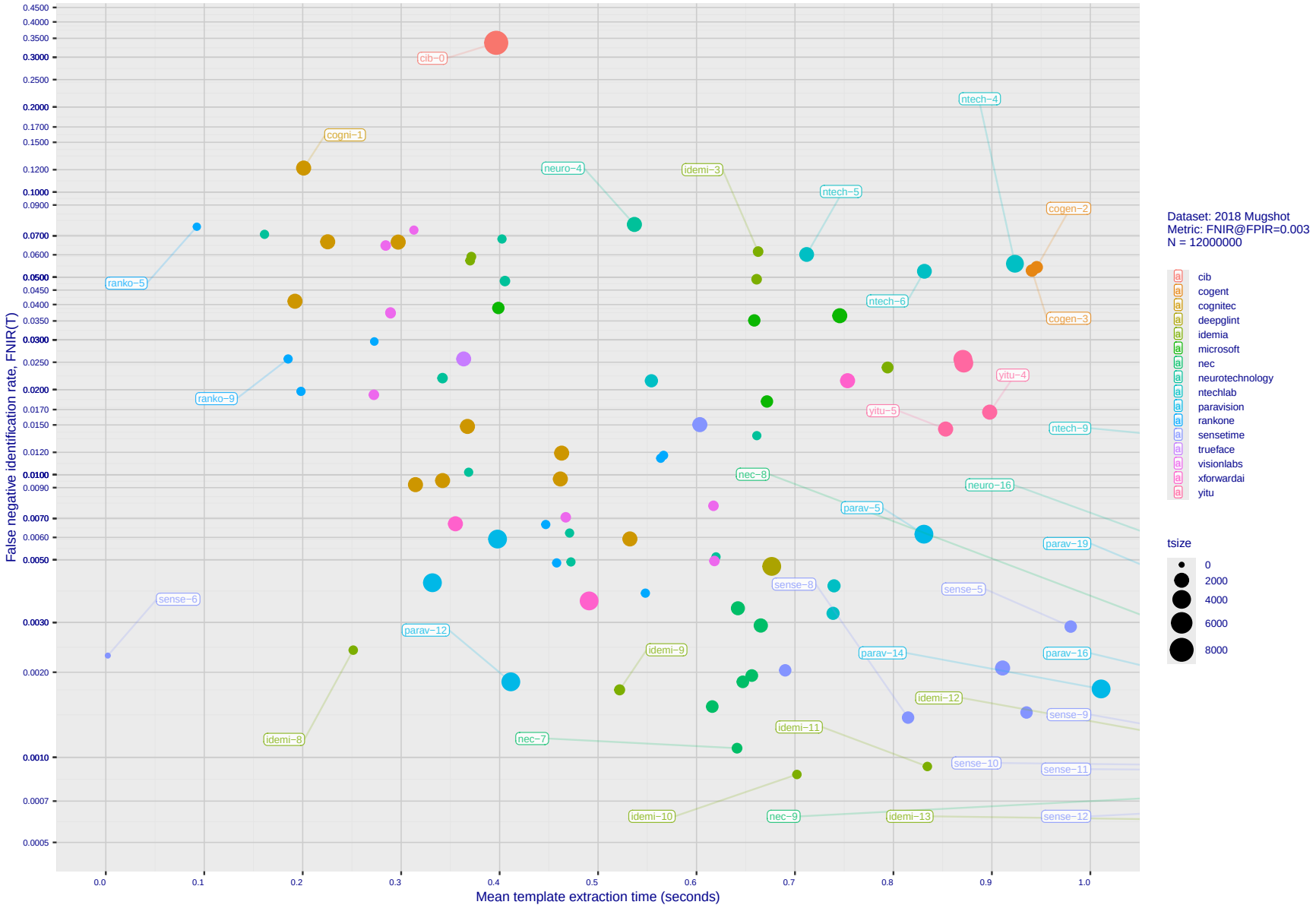

Figure 18: **[Mugshot Dataset] Speed-accuracy tradeoff**. *For developers of the more accurate algorithms the plot shows the tradeoff of high-threshold recognition miss-rates, FNIR(N, N, T) for FPIR(N, T) = 0.003, and template generation time. Developers are coded by color. Template size is encoded by the size of the circle. Some labels are quite distant from the respective point, to avoid superposing text. Without any other influences, the assumption would be that taking time to localize the face, and extract features, would lead to better accuracy. The most notable result, for NEC, is that their slower algorithms are much more accurate than the version that extract features in fewer than 90 milliseconds.*

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
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T = 0 → Investigation
T > 0 → Identification

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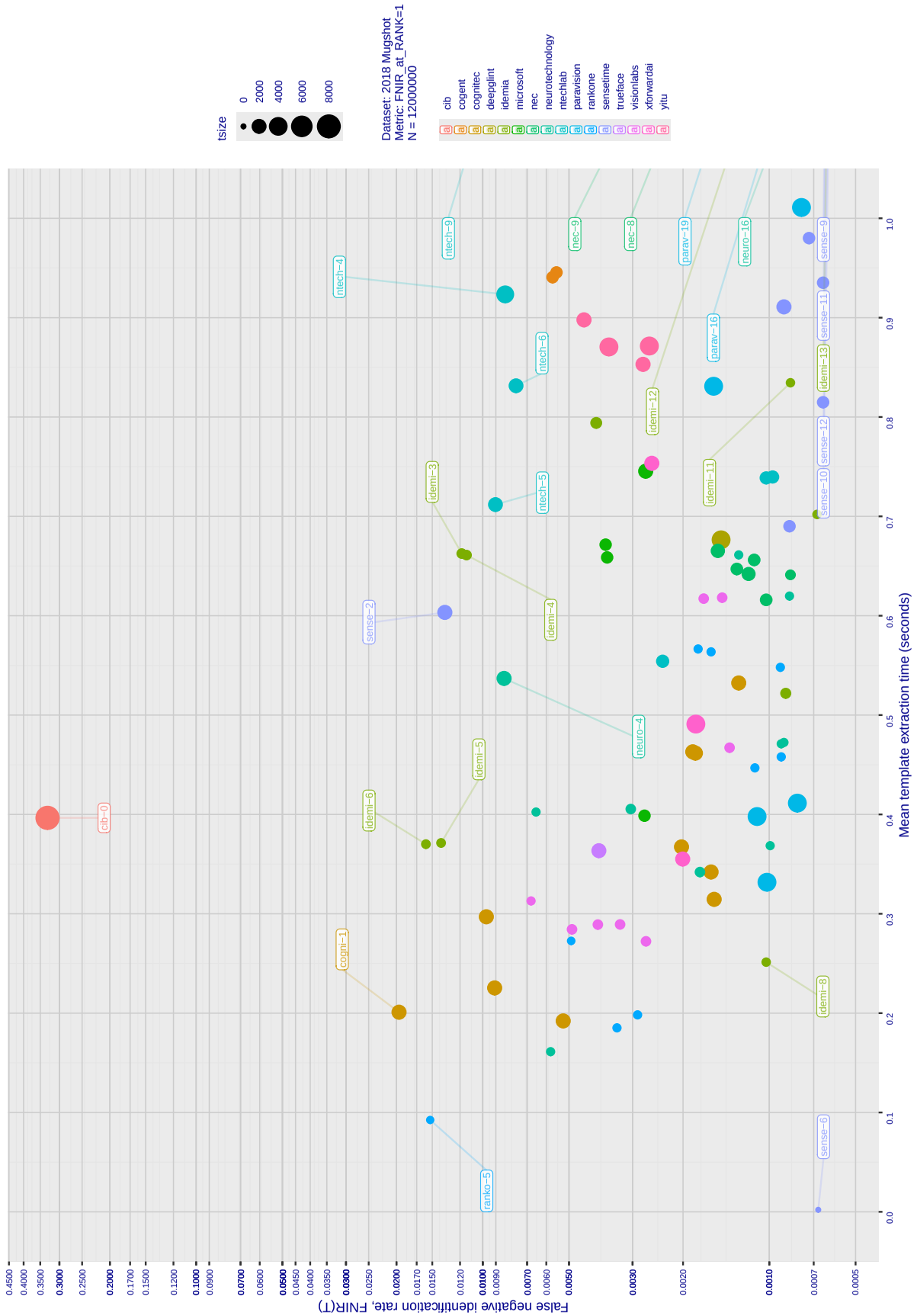

Figure 19: **[Mugshot Dataset] Speed-accuracy tradeoff**. *For developers of the more accurate algorithms the plot shows the tradeoff of rank-one recognition miss-rates, FNIR(N, 1, 0), and template generation time. Developers are coded by color. Template size is encoded by the size of the circle. Some labels are quite distant from the respective point, to avoid superposing text. Without any other influences, the assumption would be that taking time to localize the face, and extract features, would lead to better accuracy. This occurs for NEC with their slower algorithm being much accurate than the version that extract features in fewer than 90 milliseconds.*

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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
												N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
1	20Face	NL	20face	000	2021-10-01	112	319	[190]2048	-	[28]184	[125]9	[502]5679	[521]6341	-	-	-	
2	3Divi	RU	3divi	5	2018-10-26	186	51	[502]4096	k	[282]638	[427]28	[202]538	[202]537	[198]1377	[188]2614	[181]5530	$417\,0.07N^{1.1}$
3	3Divi	RU	3divi	6	2018-10-26	187	51	[86]528	k	[283]640	[56]5	[28]33	[25]33	-	-	-	
4	AFIS and Biometrics Consulting	US	afisbiometrics	000	2023-06-07	533	34	[38]265	-	[290]650	[8]2	[375]1718	[362]1563	[317]3063	[322]6541	[321]13588	$330\,0.46N^{1.1}$
5	AFR Engine	US	afrengine	000	2023-07-14	566	24	[314]2048	-	[379]818	[194]12	[484]4542	[492]4973	[423]9473	[424]20213	[419]45388	$341\,1.38N^{1.1}$
6	AFR Engine	US	afrengine	001	2024-03-15	565	24	[320]2048	-	[428]973	[240]13	[480]4505	[482]4444	[415]8555	[418]18109	[416]42844	$377\,0.89N^{1.1}$
7	AFR Engine	US	afrengine	002	2024-07-22	566	24	[241]2048	-	[451]1110	[215]12	[472]4324	[477]4120	[417]8653	[416]17832	[408]35825	$397\,0.61N^{1.1}$
8	AFR Engine	US	afrengine	003	2025-02-18	678	24	[360]2048	-	[433]1006	[244]13	[462]3803	[471]4034	[400]7389	[415]17642	[407]35554	$358\,0.90N^{1.1}$
9	AI Implementation Group	UZ	aigroup	000	2026-05-12	368	387	[456]2078	-	[185]479	[379]19	[320]1153	[320]1164	[273]2227	[270]4525	[258]8959	$204\,0.56N^{1.0}$
10	ALTTEKGlobal	VN	alttekglobal	000	2024-04-12	634	2988	[338]2048	-	[449]1108	[218]13	[105]255	[105]257	-	-	-	
11	AadiSysTech	IN	aadi	000	2023-06-27	95	83	[59]512	-	[91]346	[33]3	[399]2024	[379]1754	-	-	-	
12	AadiSysTech	IN	aadi	001	2023-12-08	307	787	[348]2048	-	[15]127	[144]10	[443]3144	[439]2850	-	-	-	
13	AadiSysTech	IN	aadi	002	2024-04-17	95	567	[372]2048	-	[10]96	[241]13	[435]2992	[438]2794	-	-	-	
14	Acer Incorporated	TW	acer	000	2020-08-12	35	67	[70]512	-	[35]198	[45]4	[121]295	[120]295	[103]623	[173]2302	[169]4915	$463\,0.00N^{1.3}$
15	Acer Incorporated	TW	acer	001	2021-11-08	42	610	[201]2048	-	[9]92	[122]9	[176]444	[215]575	-	-	-	
16	Adiance Technologies Pvt Ltd	IN	adiance	001	2025-04-28	261	332	[270]2048	-	[407]887	[401]22	[405]2096	[402]2048	-	-	-	
17	Adiance Technologies Pvt Ltd	IN	adiance	002	2026-01-13	265	367	[275]2048	-	[525]1470	[498]47	[426]2618	[433]2610	[374]4872	[370]9811	[370]19756	$115\,1.54N^{1.0}$
18	Advance.AI	SG	advance	000	2023-05-30	271	825	[178]2048	-	[247]573	[173]12	[410]2268	[415]2271	[360]4338	[354]8690	[355]17842	$183\,1.15N^{1.0}$
19	Akiyama S.A	br	akiyama	001	2026-08-25	0	765	[532]4124	-	[245]568	[459]33	[459]3663	[465]3855	-	-	-	
20	Akurat Satu Indonesia	ID	ptakuratsatu	000	2020-10-23	0	572	[87]538	-	[415]929	[565]28633	[17]18	[14]16	[7]17	[6]17	[5]17	$3\,6827.74N^{0.1}$
21	Akuvox	CN	akuvox	001	2025-02-14	0	213	[548]4843	-	[14]114	[439]30	[416]2434	[434]2613	[371]4738	[371]10025	[369]19681	$158\,1.39N^{1.0}$
22	Alchera Inc	KR	alchera	2	2018-10-30	7	14	[294]2048	k	[13]114	[514]63	[432]2923	[442]2929	-	-	-	
23	Alchera Inc	KR	alchera	3	2018-10-30	251	14	[311]2048	k	[224]531	[515]63	[434]2955	[443]2956	[390]6546	[398]15013	[406]35262	$455\,0.10N^{1.2}$
24	Alchera Inc	KR	alchera	004	2021-09-17	476	24	[226]2048	-	[365]786	[466]35	[507]5801	[529]6851	-	-	-	
25	Alchera Inc	KR	alchera	005	2023-07-18	977	482	[249]2048	-	[221]527	[468]35	[511]6123	[511]6035	-	-	-	
26	Alchera Inc	KR	alchera	006	2023-11-22	977	482	[173]2048	-	[234]546	[172]12	[530]8046	[531]6865	[435]11609	[442]27328	[439]59209	$200\,3.40N^{1.0}$
27	Alchera Inc	KR	alchera	007	2024-06-17	978	482	[188]2048	-	[240]558	[188]12	[558]20981	[568]20564	-	-	-	
28	Alchera Inc	KR	alchera	008	2024-10-25	2037	443	[271]2048	-	[515]1429	[168]12	[520]6350	[513]6134	[438]11940	[435]25274	[436]56159	$386\,1.03N^{1.1}$
29	Algorithms Inc	AE	algorithmsinc	000	2025-02-26	255	79	[264]2048	-	[166]452	[269]14	[271]903	[274]924	[269]2153	[251]4199	[248]8654	$440\,0.06N^{1.2}$
30	Alivia / Innovation Sys	RU	isystems	3	2018-10-30	350	784	[183]2048	1	[382]825	[343]16	[151]385	[161]389	[157]979	[150]1822	[265]9348	$465\,0.00N^{1.3}$
31	AllGoVision	IN	allgovision	000	2019-07-30	168	150	[313]2048	k	[134]404	[199]12	[447]3226	[451]3193	[384]6129	[384]12449	[380]25835	$230\,1.40N^{1.0}$
32	AllGoVision	IN	allgovision	001	2020-07-14	283	126	[206]2048	-	[360]777	[231]13	[445]3174	[450]3183	[382]6073	[381]12284	[378]25701	$225\,1.42N^{1.0}$
33	AllGoVision	IN	allgovision	002	2023-03-13	284	797	[240]2048	-	[172]459	[251]13	[439]3112	[458]3454	-	-	-	
34	AllGoVision	IN	allgovision	003	2023-09-19	264	799	[195]2048	-	[455]1128	[204]12	[444]3171	[447]3100	[380]5906	[387]13318	-	$317\,0.97N^{1.1}$
35	AllGoVision	IN	allgovision	004	2025-07-11	704	34	[273]2048	-	[304]669	[198]12	[351]1480	[352]1492	-	-	-	
36	AllGoVision	IN	allgovision	005	2026-03-12	455	83	[367]2048	-	[460]1138	[295]14	[370]1666	[385]1827	[100]605	[141]1734	[89]2408	$17\,10459.37N^{0.3}$
37	Anke Investments	CN	anke	0	2018-10-30	779	27	[454]2072	k	[154]429	[333]16	[230]675	[247]748	[207]1483	[203]2968	[199]6148	$332\,0.21N^{1.1}$
38	Anke Investments	CN	anke	1	2018-10-30	779	27	[452]2072	k	[155]430	[311]15	[236]707	[251]769	-	-	-	
39	Anke Investments	CN	anke	002	2019-06-27	341	401	[436]2056	k	[272]623	[264]13	[222]624	[233]682	[189]1306	[177]2403	[176]5082	$209\,0.30N^{1.0}$
40	Antheus Technologia	BR	antheus	000	2024-08-19	264	185	[196]2048	-	[494]1273	[210]12	[381]1781	[383]1788	-	-	-	
41	Aratek Biometrics Co Ltd	CN	aratek	001	2024-04-18	248	15	[177]2048	-	[43]232	[183]12	[534]8371	[541]8210	[465]16858	[460]33012	[447]68839	$319\,2.54N^{1.1}$
42	Armatura LLC	US	armatura	000	2023-04-13	0	935	[554]6144	-	[295]655	[516]66	[515]6211	[514]6138	[433]11533	[429]23189	[423]46629	$121\,3.59N^{1.0}$
43	Aware	US	aware	5	2018-10-30	368	27	[476]3100	k	[368]792	[462]34	[40]95	[48]98	[40]203	[36]371	[18]252	$22\,4.13N^{0.7}$
44	Aware	US	aware	6	2018-10-30	368	27	[5]124	k	[367]789	[13]2	[73]158	[72]162	-	-	-	
45	Aware	US	aware	007	2023-11-27	132	18	[54]480	-	[338]719	[147]11	[59]131	[58]131	[57]320	[56]655	[52]1324	$419\,0.02N^{1.1}$
46	Awidit Systems	IN	awiros	000	2025-02-10	478	584	[383]2048	-	[210]508	[243]13	[367]1628	[391]1931	-	-	-	
47	Ayonix	JP	ayonix	1	2018-10-29	74	2	[133]1036	k	[5]12	[155]11	[116]279	[113]279	-	-	-	
48	Ayonix	JP	ayonix	2	2018-10-30	74	2	[134]1036	1	[4]11	[284]14	[115]279	[112]276	[87]535	[89]1087	[86]2284	$257\,0.11N^{1.0}$
49	Beijing Hisign Technology	CN	hisign	001	2024-08-09	466	152	[458]2080	-	[271]620	[322]15	[276]926	[321]1179	[236]1855	[239]3910	[231]7639	$42\,2.25N^{0.9}$
50	BioID Technologies SA	CH	bioidtechswiss	001	2024-04-19	0	345	[68]512	-	[265]614	[42]4	[113]278	[119]294	[80]470	[80]944	[83]2141	$274\,0.10N^{1.0}$
51	Biometrik Systems Private Limited	IN	biometrik	001	2025-11-24	167	82	[331]2048	-	[526]1492	[306]15	[359]1543	[350]1486	[304]2813	[300]5694	[303]11840	$249\,0.59N^{1.0}$
52	Camvi Technologies	US	camvitech	4	2018-10-30	233	220	[115]1024	1	[316]686	[452]31	[29]33	[24]32	[15]38	[14]40	[9]48	$4\,8492.66N^{0.1}$

Notes	
1	Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas).
2	Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates.
3	This multiplier expresses the increase in template size when k images are passed to the template generation function.
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Table 2: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
											N=1.6M	N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
53	Camvi Technologies	US	camvitech	5	2018-10-30	257	220	[111]1024	1	[352]751	[449]31	[27]31	[22]30	-	-	-	
54	Canon Inc	CN	cib	000	2020-10-19	426	127	[568]8196	-	[126]396	[531]113	[450]3316	[463]3604	[393]6738	[391]13495	[383]27114	$[86]2.33N^{1.0}$
55	Canon Inc	JP	canon	001	2021-10-27	1139	91	[494]4096	-	[186]480	[396]21	[501]5489	[526]6789	[447]12741	[438]25650	[429]51922	$[140]3.82N^{1.0}$
56	Canon Inc	JP	canon	002	2022-04-26	1231	111	[556]6200	-	[243]565	[510]58	[528]7673	[524]6416	[440]12178	[433]24806	[427]49653	$[178]3.28N^{1.0}$
57	Canon Inc	JP	canon	003	2023-09-05	2462	78	[539]4148	-	[479]1211	[435]29	[396]2012	[394]1966	[344]3819	[341]7679	[348]16096	$[251]0.80N^{1.0}$
58	Clearview AI Inc	US	clearviewai	000	2021-11-12	358	316	[507]4096	-	[125]393	[445]30	[224]639	[229]657	[176]1134	[157]1939	[150]3889	$[34]1.59N^{0.9}$
59	Clearview AI Inc	US	clearviewai	001	2024-02-16	696	499	[238]2048	-	[348]743	[388]20	[158]394	[158]384	[125]722	[118]1451	[110]2919	$[125]0.22N^{1.0}$
60	Clearview AI Inc	US	clearviewai	002	2024-06-18	696	497	[202]2048	-	[364]786	[392]20	[198]517	[195]516	[155]969	[158]1946	[153]3931	$[130]0.29N^{1.0}$
61	Clearview AI Inc	US	clearviewai	003	2025-11-24	696	527	[245]2048	-	[450]1108	[385]20	[188]498	[143]362	[128]743	[148]1780	[144]3720	$[420]0.04N^{1.1}$
62	Cloudwalk - Hengrui AI Technology	CN	hr	000	2021-02-10	501	392	[300]2048	-	[325]701	[300]15	[168]420	[111]276	[89]539	[101]1268	[122]3177	$[427]0.03N^{1.1}$
63	Cloudwalk Moontime Smart Technology	CN	cloudwalk	000	2022-01-31	716	573	[352]2048	-	[402]869	[143]10	[174]440	[152]371	[91]547	[87]1065	[109]2902	$[47]0.53N^{0.9}$
64	Cloudwalk Moontime Smart Technology	CN	cloudwalk	001	2022-07-27	797	574	[219]2048	-	[223]529	[290]14	[109]273	[137]349	[109]658	[122]1492	[119]3026	$[339]0.10N^{1.1}$
65	Cloudwalk Moontime Smart Technology	CN	cloudwalk	002	2023-02-24	431	514	[212]2048	-	[225]531	[297]14	[247]767	[255]826	[214]1586	[216]3311	[198]6070	$[235]0.35N^{1.0}$
66	Cognitec Systems GmbH	DE	cognitec	2	2018-10-30	463	26	[402]2052	k	[42]225	[417]27	[377]1733	[382]1763	[338]3660	[333]7279	[324]13895	$[221]0.83N^{1.0}$
67	Cognitec Systems GmbH	DE	cognitec	3	2018-10-30	465	26	[399]2052	k	[67]297	[330]16	[376]1719	[384]1791	[336]3638	[332]7277	[333]14904	$[282]0.66N^{1.0}$
68	Cognitec Systems GmbH	DE	cognitec	004	2021-03-08	384	60	[404]2052	-	[34]192	[260]13	[372]1673	[377]1727	[307]2904	[302]5801	[300]11707	$[76]1.15N^{1.0}$
69	Cognitec Systems GmbH	DE	cognitec	005	2021-07-30	460	61	[412]2052	-	[105]367	[128]9	[361]1556	[360]1551	[309]2916	[323]6561	[325]13958	$[351]0.38N^{1.1}$
70	Cognitec Systems GmbH	DE	cognitec	006	2022-02-10	689	61	[401]2052	-	[177]463	[140]10	[294]1006	[292]1002	[260]2097	[260]4312	[230]7624	$[323]0.30N^{1.1}$
71	Cognitec Systems GmbH	DE	cognitec	007	2023-12-01	319	194	[417]2052	-	[73]314	[281]14	[304]1069	[304]1051	[274]2232	[255]4244	[237]7943	$[179]0.55N^{1.0}$
72	Cognitec Systems GmbH	DE	cognitec	008	2024-04-29	359	39	[324]2048	-	[89]342	[268]14	[329]1258	[300]1017	[284]2433	[279]4887	[284]9888	$[404]0.15N^{1.1}$
73	Cognitec Systems GmbH	DE	cognitec	009	2024-08-29	359	39	[182]2048	-	[175]461	[249]13	[301]1045	[302]1025	[262]2117	[256]4267	[240]8147	$[252]0.42N^{1.0}$
74	Cognitec Systems GmbH	DE	cognitec	010	2026-09-24	398	38	[356]2048	-	[226]532	[380]19	[286]969	[283]973	[252]2021	[232]3787	[228]7576	$[283]0.34N^{1.0}$
75	Cogniti Technology	PO	cogniti	000	2026-08-19	1002	33	[475]3096	-	[431]1004	[430]29	[219]606	[228]650	[171]1117	[183]2451	[170]4953	$[138]0.34N^{1.0}$
76	Compal Electronics	TW	compal	000	2025-03-20	980	182	[109]1024	-	[491]1256	[75]7	[491]4960	[503]5503	[395]6929	[329]6915	[217]6911	$[18]17496.15N^{0.4}$
77	Compal Electronics	TW	compal	001	2025-11-25	605	199	[392]2049	-	[413]910	[364]17	[560]22511	[569]22297	-	-	-	
78	Compal Electronics	TW	compal	002	2026-07-09	605	198	[390]2049	-	[397]855	[342]16	[427]2619	[432]2581	-	-	-	
79	Corsight AI	IL	corsight	000	2023-07-13	1277	31	[459]2080	-	[233]546	[167]12	[284]960	[330]1299	[266]2145	[235]3828	[285]9998	$[349]0.26N^{1.1}$
80	Corsight AI	IL	corsight	001	2024-03-14	1172	30	[457]2080	-	[229]543	[250]13	[295]1011	[298]1011	[248]1951	[243]4004	[270]9391	$[356]0.23N^{1.1}$
81	Corsight AI	IL	corsight	002	2026-09-07	3259	29	[455]2076	-	[524]1469	[328]16	[238]730	[265]878	[204]1451	[209]3072	[206]6415	$[220]0.37N^{1.0}$
82	Cubox	KR	cubox	000	2021-08-24	529	298	[335]2048	-	[157]440	[138]10	[452]3410	[475]4076	[403]7605	[403]15871	-	$[328]1.16N^{1.1}$
83	Cubox	KR	cubox	001	2023-07-05	1654	342	[252]2048	-	[273]624	[190]12	[448]3268	[459]3476	[383]6097	[382]12320	[382]27074	$[164]1.78N^{1.0}$
84	Cubox	KR	cubox	002	2024-11-29	1273	342	[333]2048	-	[425]962	[182]12	[463]3809	[466]3859	[398]7095	[393]13774	[389]28747	$[69]3.02N^{1.0}$
85	Cubox	KR	cubox	003	2025-05-14	219	342	[368]2048	-	[108]369	[223]13	[465]3920	[467]3915	[399]7262	[399]15013	[393]29855	$[208]1.80N^{1.0}$
86	Cyberlink Corp	TW	cyberlink	000	2019-06-12	217	93	[409]2052	1	[294]654	[440]30	[232]696	[236]701	[199]1379	[189]2639	[201]6214	$[265]0.28N^{1.0}$
87	Cyberlink Corp	TW	cyberlink	001	2019-10-07	459	102	[408]2052	1	[149]423	[428]28	[233]698	[235]700	[194]1350	[296]5524	[307]12031	$[462]0.00N^{1.3}$
88	Cyberlink Corp	TW	cyberlink	002	2020-07-31	333	109	[536]4140	-	[341]724	[552]6875	[340]1353	[452]3198	[385]6138	[380]12205	[314]13106	$[28]16.71N^{0.8}$
89	Cyberlink Corp	TW	cyberlink	003	2021-01-05	333	100	[559]6212	-	[110]370	[470]35	[269]886	[242]723	[203]1415	[200]2886	[182]5643	$[383]0.12N^{1.1}$
90	Cyberlink Corp	TW	cyberlink	004	2021-07-16	371	100	[558]6212	-	[120]386	[403]23	[275]924	[193]504	[151]923	[117]1448	[127]3350	$[46]0.73N^{0.9}$
91	Cyberlink Corp	TW	cyberlink	005	2022-01-07	371	100	[557]6212	-	[344]733	[447]30	[190]501	[189]498	[183]1193	[190]2672	[186]5693	$[450]0.03N^{1.2}$
92	Cyberlink Corp	TW	cyberlink	006	2024-06-12	1163	99	[571]8260	-	[339]722	[507]54	[401]2068	[354]1498	[343]3804	[343]7796	[343]15790	$[434]0.13N^{1.1}$
93	DAON	IE	daon	000	2021-12-23	274	2	[448]2069	-	[220]526	[99]8	[298]1029	[223]625	[205]1454	[210]3097	[204]6316	$[451]0.03N^{1.2}$
94	DAON	IE	daon	001	2025-01-03	289	64	[359]2048	-	[298]661	[179]12	[328]1242	[325]1211	-	-	-	
95	DAON	IE	daon	002	2025-07-07	288	65	[334]2048	-	[351]751	[169]12	[336]1318	[361]1552	[287]2462	[293]5452	[292]11007	$[262]0.53N^{1.0}$
96	DAON	IE	daon	003	2026-04-14	909	65	[503]4096	-	[565]2055	[426]28	[456]3584	[462]3588	[392]6731	[389]13471	[392]29819	$[240]1.50N^{1.0}$
97	Dactionable Technologies	VN	datech	000	2023-10-24	603	178	[489]4096	-	[468]1176	[541]1282	[88]194	[109]262	[84]499	[84]998	[80]2029	$[365]0.05N^{1.1}$
98	Dactionable Technologies	VN	datech	001	2024-03-14	709	178	[515]4096	-	[422]953	[521]75	[242]753	[248]751	[210]1537	[201]2946	[192]5802	$[184]0.38N^{1.0}$
99	Dahua Technology Co	CN	dahua	0	2018-10-29	276	167	[250]2048	k	[115]374	[399]22	-	[106]258	-	-	-	
100	Dahua Technology Co	CN	dahua	1	2018-10-29	276	167	[247]2048	k	[109]369	[422]28	-	[104]257	[98]602	[97]1202	[114]3007	$[437]0.02N^{1.2}$
101	Dahua Technology Co	CN	dahua	002	2019-12-02	607	137	[169]2048	k	[315]685	[383]19	[100]243	[110]269	[181]1189	[202]2950	[211]6732	$[470]0.00N^{1.5}$
102	Dahua Technology Co	CN	dahua	003	2020-11-18	889	154	[215]2048	-	[340]723	[367]18	[117]283	[103]249	[79]468	[79]935	[76]1871	$[89]0.16N^{1.0}$
103	Dahua Technology Co	CN	dahua	004	2021-11-18	812	116	[175]2048	-	[355]758	[146]11	[169]423	[167]411	[145]871	[129]1568	[121]3174	$[239]0.17N^{1.0}$
104	Decatur Industries Inc	US	decatur	000	2022-02-09	411	383	[414]2052	-	[401]863	[130]9	[379]1761	[398]2023	[332]3361	[334]7283	[330]14592	$[147]1.06N^{1.0}$

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Table 3: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

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T > 0 → Identification

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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
											N=1.6M	N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
105	Deepglint	CN	deepglint	001	2019-11-15	448	265	[482]4096	-	[308]676	[465]35	(231)677	(353)1495	(228)1724	(194)2747	(202)6246	[25]$25.27N^{0.8}$
106	Dermalog	DE	dermalog	5	2018-10-26	0	440	[7]128	1	[222]528	[546]3155	(1)0	(1)0	(1)0	(1)0	(1)0	[5]$66.21N^{0.2}$
107	Dermalog	DE	dermalog	6	2018-10-26	0	453	[18]256	1	[207]507	[17]2	(66)142	(64)144	(49)269	(48)531	(50)1294	[292]$0.05N^{1.0}$
108	Dermalog	DE	dermalog	007	2020-02-12	0	424	[11]128	1	[139]410	[3]1	(49)98	(44)96	(43)218	(39)429	(41)1013	[402]$0.01N^{1.1}$
109	Dermalog	DE	dermalog	008	2021-01-25	0	531	[67]512	-	[36]198	[47]4	(97)227	(99)246	(77)462	(78)924	(75)1849	[96]$0.15N^{1.0}$
110	Dermalog	DE	dermalog	009	2021-11-09	0	318	[58]512	-	[26]180	[31]3	(98)227	(100)246	(76)461	(77)923	(74)1846	[91]$0.16N^{1.0}$
111	Dermalog	DE	dermalog	010	2022-07-25	0	514	[72]512	-	[79]322	[41]3	(99)241	(90)210	(68)394	(68)789	(65)1580	[94]$0.13N^{1.0}$
112	Dermalog	DE	dermalog	011	2022-12-12	0	272	[6]128	-	[49]248	[4]1	(39)87	(38)88	(28)165	(27)331	(27)664	[108]$0.05N^{1.0}$
113	Dermalog	DE	dermalog	012	2023-12-19	0	287	[9]128	-	[25]177	[2]1	(44)95	(47)97	(30)177	(29)353	(29)709	[77]$0.07N^{1.0}$
114	Dermalog	DE	dermalog	013	2024-06-18	0	287	[10]128	-	[27]182	[1]1	(43)95	(43)95	(33)180	(31)361	(33)724	[131]$0.05N^{1.0}$
115	Dermalog	DE	dermalog	014	2025-05-09	0	289	[8]128	-	[24]177	[547]3293	(2)0	(2)0	(2)0	(2)1	(2)1	[13]$26.41N^{0.2}$
116	Dermalog	DE	dermalog	015	2025-09-12	0	683	[120]1024	-	[437]1033	[118]9	(96)217	(95)218	(70)410	(71)823	(68)1647	[114]$0.13N^{1.0}$
117	Dermalog	DE	dermalog	016	2026-02-04	0	888	[113]1024	-	[559]1908	[70]7	(81)181	(83)190	(61)341	(69)817	(62)1526	[326]$0.06N^{1.1}$
118	Dermalog	DE	dermalog	017	2026-06-25	0	889	[117]1024	-	[558]1906	[92]8	(80)180	(75)181	(60)340	(73)842	(64)1579	[359]$0.04N^{1.1}$
119	Digidata	IN	digidata	000	2022-06-03	248	33	[349]2048	-	[77]320	[544]2444	(3)0	(86)198	-	-	-	
120	DiluSense Technology	CN	dilusense	000	2022-05-26	311	56	[237]2048	-	[40]219	[414]26	(387)1904	(355)1498	(321)3116	(311)6126	(309)12046	[258]$0.61N^{1.0}$
121	DiluSense Technology	CN	dilusense	001	2022-12-24	599	5	[487]4096	-	[399]860	[475]38	(436)3026	(445)3042	(372)4738	(364)9473	(363)19026	[43]$5.36N^{0.9}$
122	EI Networks Private Ltd	IN	einetworksindia	000	2024-04-17	444	128	[181]2048	-	[92]347	[377]19	(181)470	(134)342	(150)915	(152)1844	(143)3719	[439]$0.03N^{1.2}$
123	EI Networks Private Ltd	IN	einetworksindia	001	2024-10-24	444	197	[214]2048	-	[87]341	[372]18	(161)402	(135)343	(108)656	(139)1714	(141)3659	[428]$0.03N^{1.1}$
124	EI Networks Private Ltd	IN	einetworksindia	002	2025-02-21	1186	141	[228]2048	-	[443]1076	[391]20	(127)327	(159)387	(106)651	(104)1310	(106)2872	[199]$0.17N^{1.0}$
125	FAST-NUCES Lahore Campus	PK	fastlahore	000	2025-09-26	186	138	[185]2048	-	[57]278	[303]15	(414)2432	(416)2272	-	-	-	
126	FCI Vietnam	VN	fci	000	2025-04-02	1452	444	[239]2048	-	[523]1466	[217]12	(553)13182	(561)13125	-	-	-	
127	FPT Smart Cloud	VN	fpt	000	2024-03-28	595	444	[216]2048	-	[394]850	[279]14	(552)13085	(560)13114	-	-	-	
128	FacePhi	ES	facephi	000	2024-10-03	134	182	[73]512	-	[56]275	[63]6	(94)214	(409)2087	(347)3854	(342)7745	(339)15540	[460]$0.01N^{1.3}$
129	FacePhi	ES	facephi	001	2026-09-23	882	111	[329]2048	-	[538]1726	[373]18	(285)967	(293)1004	(211)1539	(220)3399	(208)6622	[51]$1.20N^{1.0}$
130	Facehawk Ltd	GB	facehawk	001	2026-06-17	169	35	[449]2070	-	[100]360	[502]49	(339)1348	(340)1353	-	-	-	
131	FarBar Inc	TW	f8	001	2019-10-03	266	19	[288]2048	k	[374]810	[277]14	-	-	-	-	-	
132	Fincore	GB	fincore	000	2021-08-18	250	224	[323]2048	-	[159]443	[114]9	(287)971	(209)560	-	-	-	
133	First Credit Bureau Kazakhstan	KZ	firstcreditkz	001	2022-11-22	548	24	[39]288	-	[201]495	[14]2	(32)46	(29)38	(20)72	(20)150	(20)297	[151]$0.02N^{1.0}$
134	Fissionsoft CO Ltd	TH	fissionsfr	001	2026-04-24	138	244	[269]2048	-	[534]1704	[538]883	(4)1	(3)1	-	-	-	
135	Fujitsu Research and Development Center	CN	fujitsulab	000	2021-10-12	497	337	[128]1032	-	[264]613	[59]5	(385)1864	(370)1657	(323)3140	(317)6320	(312)12723	[207]$0.78N^{1.0}$
136	Fujitsu Research and Development Center	CN	fujitsulab	001	2022-03-15	675	386	[125]1032	-	[228]535	[115]9	(384)1854	(378)1740	(331)3324	(327)6730	(335)14943	[344]$0.45N^{1.1}$
137	GTEL OTS JSC	VN	gtelots	000	2025-04-14	246	58	[64]512	-	[366]788	[563]21132	(8)3	(6)3	-	-	-	
138	GTEL OTS JSC	VN	gtelots	001	2026-02-02	432	57	[140]1088	-	[423]955	[480]39	(7)2	(5)3	-	-	-	
139	Gorilla Technology	TW	gorilla	2	2018-10-29	91	1252	[144]1132	k	[85]338	[408]24	(67)145	(65)146	(51)293	(53)612	(60)1509	[396]$0.02N^{1.1}$
140	Gorilla Technology	TW	gorilla	3	2018-10-26	94	1252	[462]2156	k	[241]559	[557]12020	-	(400)2047	-	-	-	
141	Gorilla Technology	TW	gorilla	004	2020-01-06	182	1244	[463]2192	k	[122]388	[482]41	(118)286	(116)285	(182)1191	(179)2416	(175)5036	[461]$0.00N^{1.3}$
142	Gorilla Technology	TW	gorilla	005	2021-02-22	306	1420	[560]6288	-	[69]303	[522]78	(305)1070	(253)799	(209)1514	(267)4454	(253)8820	[443]$0.05N^{1.2}$
143	Gorilla Technology	TW	gorilla	006	2021-09-30	377	691	[572]8336	-	[148]422	[530]99	(356)1504	(366)1612	(283)2422	(264)4422	(266)9363	[206]$0.59N^{1.0}$
144	Gorilla Technology	TW	gorilla	007	2022-02-16	392	322	[561]6290	-	[75]315	[529]89	(245)765	(246)745	(202)1408	(198)2823	(189)5764	[143]$0.42N^{1.0}$
145	Gorilla Technology	TW	gorilla	008	2022-10-31	321	290	[543]4242	-	[217]521	[506]54	(193)513	(282)972	(223)1699	(185)2469	(212)6768	[271]$0.30N^{1.0}$
146	Griaule	US	griaule	000	2021-11-01	0	584	[416]2052	-	[48]247	[91]8	(488)4868	(516)6150	(432)11473	(428)22952	(421)46070	[87]$3.89N^{1.0}$
147	Griaule	US	griaule	001	2022-07-26	0	615	[413]2052	-	[310]684	[164]12	(509)5866	(510)6010	(436)11627	(431)24207	(422)46343	[191]$3.02N^{1.0}$
148	Guangzhou Pixel Solutions Co	CN	pixelall	002	2019-07-01	0	165	[468]2560	k	[32]190	[305]15	(335)1296	(337)1334	(293)2526	(286)5136	(293)11045	[260]$0.52N^{1.0}$
149	Guangzhou Pixel Solutions Co	CN	pixelall	003	2019-11-05	0	690	[465]2560	k	[328]703	[400]22	(331)1273	(332)1307	(289)2474	(287)5198	(295)11141	[296]$0.46N^{1.0}$
150	Guangzhou Pixel Solutions Co	CN	pixelall	004	2020-07-02	0	538	[467]2560	k	[163]449	[362]17	(330)1259	(331)1300	(288)2465	(294)5492	(296)11443	[345]$0.34N^{1.1}$
151	Guangzhou Pixel Solutions Co	CN	pixelall	005	2021-03-23	0	717	[466]2560	-	[390]840	[151]11	(365)1606	(357)1528	(296)2609	(281)4926	(302)11770	[176]$0.73N^{1.0}$
152	Hangzhuo Allu Network Information Technology	CN	hzailu	000	2022-03-18	855	97	[119]1024	-	[169]457	[152]11	(425)2609	(414)2251	(364)4612	(372)10075	(352)17156	[234]$1.01N^{1.0}$
153	Hangzhuo Allu Network Information Technology	CN	hzailu	001	2022-08-18	273	162	[235]2048	-	[116]377	[175]12	(483)4537	(486)4511	(414)8462	(421)19463	(415)42527	[375]$0.92N^{1.1}$
154	Hangzhuo Allu Network Information Technology	CN	hzailu	002	2023-01-12	302	293	[343]2048	-	[239]555	[201]12	(478)4481	(485)4509	(412)8413	(408)16568	(413)39517	[287]$1.59N^{1.0}$
155	Hangzhuo Allu Network Information Technology	CN	hzailu	003	2023-05-22	1434	293	[490]4096	-	[307]675	[406]24	(544)10145	(548)9351	(468)17922	(466)40619	(454)83432	[315]$3.03N^{1.0}$
156	Hangzhuo Allu Network Information Technology	CN	hzailu	004	2023-10-13	1902	293	[552]5120	-	[424]956	[432]29	(527)7508	(536)7639	(453)13673	(437)25457	(446)66435	[346]$1.83N^{1.1}$

Notes	
1	Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas).
2	Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates.
3	This multiplier expresses the increase in template size when k images are passed to the template generation function.
4	All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however.
5	Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 479. However in certain cases the model is not correct and should not be used numerically.

Table 4: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.

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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
											N=1.6M	N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
157	Hangzhuo Allu Network Information Technology	CN	hzailu	005	2024-04-26	2201	293	[565]7168	-	[513]1389	[485]41	[546]11242	[553]10996	-	-	-	
158	Hangzhuo Allu Network Information Technology	CN	hzailu	006	2025-01-03	2866	293	[566]7168	-	[518]1455	[477]38	[550]12514	[556]11557	-	-	-	
159	Hikvision Research Institute	CN	hikvision	5	2018-10-29	593	9	[152]1408	1	[262]607	[324]16	[268]883	[267]895	[244]1908	[233]3792	[269]9387	[418]$0.10N^{1.1}$
160	Hikvision Research Institute	CN	hikvision	6	2018-10-29	593	9	[151]1408	1	[256]598	[331]16	[263]871	[264]877	-	-	-	
161	HyperVerge Inc	IN	hyperverge	001	2021-08-11	1791	212	[108]1024	-	[196]490	[54]5	[234]703	[232]681	[193]1346	[191]2681	[185]5680	[224]$0.32N^{1.0}$
162	HyperVerge Inc	IN	hyperverge	002	2022-04-13	1140	1118	[116]1024	-	[158]440	[121]9	[184]484	[196]518	[169]1099	[137]1690	[77]1953	[26]$6.15N^{0.8}$
163	HyperVerge Inc	IN	hyperverge	003	2023-04-04	2089	1118	[325]2048	-	[165]452	[233]13	[246]766	[273]921	[243]1900	[211]3198	[197]6065	[148]$0.48N^{1.0}$
164	Hyundai Motor Company	KR	hyundai	000	2025-02-03	352	104	[388]2048	-	[104]366	[256]13	[469]4258	[474]4068	[405]7815	[401]15618	[403]33716	[275]$1.52N^{1.0}$
165	Hyundai Motor Company	KR	hyundai	001	2025-06-23	176	104	[267]2048	-	[95]351	[196]12	[473]4329	[480]4382	[406]7827	[406]16459	[401]33035	[172]$2.22N^{1.0}$
166	Hyundai Motor Company	KR	hyundai	002	2025-11-14	472	104	[353]2048	-	[543]1802	[318]15	[535]8376	[540]8199	[458]15300	[455]31477	[460]214557	[466]$0.04N^{1.4}$
167	Hyundai Motor Company	KR	hyundai	003	2026-06-17	472	104	[256]2048	-	[554]1860	[416]27	[487]4817	[490]4839	[420]9006	[417]18000	[409]36450	[116]$2.83N^{1.0}$
168	ICM Airport Technics	AU	icm	000	2023-08-14	991	1	[192]2048	-	[31]186	[197]12	[529]7990	[538]7943	-	-	-	
169	ICM Airport Technics	AU	icm	001	2023-12-12	858	1	[191]2048	-	[19]164	[192]12	[525]7124	[533]7219	[459]15370	[450]29562	[449]70229	[395]$1.10N^{1.1}$
170	IDBIO	US	idbio	000	2026-09-08	0	317	[84]522	-	[392]848	[474]37	[142]369	[150]370	[117]694	[127]1544	[132]3431	[360]$0.08N^{1.1}$
171	IDENTY	GB	identy	000	2025-06-18	376	38	[378]2048	-	[300]662	[473]37	[140]366	[131]330	-	-	-	
172	IDENTY	GB	identy	001	2026-01-22	1109	43	[472]3072	-	[519]1459	[442]30	[204]546	[227]638	[86]511	[82]971	[79]2024	[21]$53.38N^{0.6}$
173	IDENTY	GB	identy	002	2026-06-03	1216	77	[473]3072	-	[537]1720	[448]30	[216]587	[214]571	[170]1100	[170]2285	[165]4761	[269]$0.22N^{1.0}$
174	IMDS Software Inc	CA	imds	000	2023-10-20	364	344	[336]2048	-	[65]296	[189]12	[542]10077	[552]10387	-	-	-	
175	IMDS Software Inc	CA	imds	001	2024-03-14	362	344	[172]2048	-	[418]943	[316]15	[185]485	[186]489	[152]924	[162]1970	[157]4100	[291]$0.17N^{1.0}$
176	IMDS Software Inc	CA	imds	002	2024-11-01	364	344	[207]2048	-	[71]311	[214]12	[221]622	[222]622	[175]1132	[160]1962	[156]4000	[49]$0.86N^{0.9}$
177	Idemia	FR	idemia	5	2018-10-29	417	48	[47]352	1	[114]371	[55]5	[61]137	[61]138	[73]437	[63]724	[66]1630	[452]$0.01N^{1.2}$
178	Idemia	FR	idemia	6	2018-10-29	417	48	[48]352	1	[112]370	[51]4	[62]137	[60]138	[74]442	[72]827	[67]1646	[454]$0.01N^{1.2}$
179	Idemia	FR	idemia	007	2020-01-17	738	113	[107]860	1	[369]794	[278]14	[68]151	[66]152	[113]683	[121]1481	[117]3022	[468]$0.00N^{1.4}$
180	Idemia	FR	idemia	008	2021-03-15	378	65	[44]300	-	[50]251	[32]3	[60]137	[57]131	[46]247	[45]501	[42]1013	[149]$0.07N^{1.0}$
181	Idemia	FR	idemia	009	2022-03-01	735	68	[102]636	-	[216]521	[81]7	[92]211	[96]221	[81]473	[88]1070	[87]2352	[429]$0.02N^{1.1}$
182	Idemia	FR	idemia	010	2023-01-11	942	71	[46]300	-	[327]701	[24]2	[90]197	[87]200	[66]380	[66]766	[63]1561	[216]$0.09N^{1.0}$
183	Idemia	FR	idemia	011	2024-03-05	776	63	[45]300	-	[388]834	[22]2	[89]196	[84]193	[65]368	[65]746	[61]1525	[241]$0.08N^{1.0}$
184	Idemia	FR	idemia	012	2025-07-02	758	66	[145]1212	-	[490]1255	[86]7	[130]343	[132]335	[104]643	[128]1544	[147]3849	[438]$0.02N^{1.2}$
185	Idemia	FR	idemia	013	2026-07-01	678	68	[103]716	-	[564]2022	[67]6	[172]430	[173]427	[136]810	[164]2033	[151]3894	[381]$0.08N^{1.1}$
186	Imagus Technology Pty	AU	imagus	005	2021-01-15	222	311	[277]2048	-	[198]491	[274]14	[124]302	[125]313	[107]651	[109]1361	[93]2461	[415]$0.03N^{1.1}$
187	Imagus Technology Pty	AU	imagus	006	2021-05-27	248	369	[309]2048	-	[204]505	[131]9	[125]303	[97]234	[83]499	[102]1273	[100]2727	[449]$0.01N^{1.2}$
188	Imagus Technology Pty	AU	imagus	007	2021-11-16	248	366	[299]2048	-	[573]32200	[108]9	[123]301	[98]238	[75]442	[76]881	[73]1765	[78]$0.16N^{1.0}$
189	Imagus Technology Pty	AU	imagus	008	2022-05-26	204	335	[193]2048	-	[58]284	[360]17	[210]560	[177]441	-	-	-	
190	Imperial College London	GB	imperial	000	2019-08-28	461	15	[344]2048	1	[249]577	[226]13	[136]360	[157]379	[218]1626	[244]4057	[289]10291	[471]$0.00N^{1.5}$
191	Incode Technologies Inc	US	incode	2	2018-10-29	71	31	[205]2048	1	[61]289	[314]15	[166]411	[163]404	-	-	-	
192	Incode Technologies Inc	US	incode	3	2018-10-29	133	31	[223]2048	1	[324]697	[298]15	[165]408	[168]412	[141]847	[133]1608	[163]4486	[408]$0.05N^{1.1}$
193	Incode Technologies Inc	US	incode	004	2019-06-24	254	50	[257]2048	1	[184]475	[170]12	[139]365	[155]378	[206]1482	[135]1660	[113]2954	[338]$0.12N^{1.1}$
194	Incode Technologies Inc	US	incode	005	2021-07-29	259	21	[251]2048	-	[59]285	[136]10	[170]427	[179]454	[146]890	[151]1843	[138]3640	[382]$0.07N^{1.1}$
195	Incode Technologies Inc	US	incode	006	2024-07-15	2309	41	[567]8196	-	[441]1071	[496]47	[374]1706	[374]1713	[276]2241	[304]5891	[275]9528	[39]$3.19N^{0.9}$
196	Incode Technologies Inc	US	incode	007	2025-07-30	1766	69	[555]6148	-	[440]1047	[467]35	[321]1172	[328]1256	[240]1873	[268]4462	[238]7970	[192]$0.53N^{1.0}$
197	Innov8tif Solutions	MY	innov8tif	000	2026-07-28	115	1007	[497]4096	-	[176]461	[434]29	[390]1925	[386]1853	-	-	-	
198	Innovatrics	SK	innovatrics	4	2018-10-30	0	400	[139]1076	k	[129]399	[554]10902	[14]8	[9]8	[5]11	[3]9	[4]13	[11]$668.38N^{0.2}$
199	Innovatrics	SK	innovatrics	005	2019-09-30	0	455	[89]538	1	[383]827	[556]11897	[13]8	[10]8	[4]9	[4]9	[3]9	[1]$4055.65N^{0.1}$
200	Innovatrics	SK	innovatrics	007	2021-08-16	175	58	[88]538	-	[251]577	[283]14	[53]105	[50]100	[37]188	[38]378	[37]788	[65]$0.09N^{1.0}$
201	Innovatrics	SK	innovatrics	008	2023-01-12	609	102	[533]4136	-	[287]647	[487]42	[256]823	[259]864	[216]1610	[215]3271	[221]6943	[186]$0.43N^{1.0}$
202	Innovatrics	SK	innovatrics	009	2023-08-01	590	108	[534]4136	-	[182]474	[512]60	[274]924	[284]975	[229]1759	[225]3496	[219]6926	[66]$0.81N^{1.0}$
203	Innovatrics	SK	innovatrics	010	2024-01-09	797	42	[535]4136	-	[255]595	[509]56	[261]850	[271]907	[227]1724	[223]3444	[225]7470	[226]$0.41N^{1.0}$
204	Innovatrics	SK	innovatrics	011	2024-05-14	628	48	[540]4209	-	[258]600	[526]86	[543]10112	[551]10087	[471]18636	[462]37591	[452]75468	[137]$5.61N^{1.0}$
205	Innovatrics	SK	innovatrics	012	2024-09-20	0	1938	[546]4779	-	[520]1462	[421]28	[334]1295	[334]1312	[277]2250	[274]4635	[281]9711	[61]$1.14N^{1.0}$
206	Innovatrics	SK	innovatrics	013	2025-01-28	0	2784	[562]6845	-	[457]1133	[478]38	[354]1495	[364]1581	[310]2929	[303]5827	[299]11658	[62]$1.40N^{1.0}$
207	Innovatrics	SK	innovatrics	014	2025-05-28	0	3178	[564]6870	-	[507]1359	[479]39	[341]1405	[342]1405	[294]2556	[282]4929	[272]9479	[54]$1.49N^{1.0}$
208	Innovatrics	SK	innovatrics	015	2025-09-29	0	3057	[563]6870	-	[568]2124	[503]50	[363]1580	[363]1568	[312]2944	[305]5942	[305]11998	[135]$0.89N^{1.0}$

Notes	
1	Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas).
2	Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates.
3	This multiplier expresses the increase in template size when k images are passed to the template generation function.
4	All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however.
5	Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 479. However in certain cases the model is not correct and should not be used numerically.

Table 5: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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T = 0 → Investigation
T > 0 → Identification

	DEVELOPER / FULL NAME	DEVELOPER LOCATION	SHORT NAME	SEQ. NUM.	VALIDATION DATE	CONFIG[1] DATA (MB)	LIB[1] DATA (MB)	TEMPLATE GENERATION SIZE (B)	MULT[3]	TEMPLATE GENERATION TIME (MS)[4]	FINALIZE[2] TIME (S)	SEARCH DURATION[5] MILLISEC L=1 N=1.6M	L=50 N=1.6M	L=50 N=3M	L=50 N=6M	L=50 N=12M	POWER LAW (μs)
209	Innovatrics	SK	innovatrics	016	2026-08-24	2399	27	[547]4804	-	[529]1567	[471]36	(350)1477	(348)1456	(300)2732	(297)5527	(294)11096	[144]$0.81N^{1.0}$
210	Inspur (Beijing) Electronic Information Industry Co	CN	inspur	000	2023-05-30	356	88	[248]2048	-	[411]898	[347]16	(455)3574	(453)3248	(391)6610	(388)13468	(384)27332	[312]$1.04N^{1.0}$
211	Inspur (Beijing) Electronic Information Industry Co	CN	inspur	001	2023-12-07	437	88	[382]2048	-	[435]1025	[341]16	(454)3483	(454)3254	(389)6520	(386)13141	(379)25818	[243]$1.36N^{1.0}$
212	Inspur (Beijing) Electronic Information Industry Co	CN	inspur	002	2025-02-11	434	88	[268]2048	-	[484]1222	[340]16	(449)3313	(455)3259	(386)6169	(394)13863	(385)27477	[325]$0.98N^{1.1}$
213	Integrated Software and Technologies Ltd	BD	istl	001	2026-08-07	264	26	[445]2060	-	[257]598	[398]22	(311)1106	(310)1107	(270)2157	(252)4215	(249)8701	[185]$0.56N^{1.0}$
214	Intel Research Group	IS	intelresearch	001	2025-03-31	309	118	[244]2048	-	[480]1219	[254]13	(270)886	(261)866	(219)1652	(218)3339	(216)6862	[187]$0.44N^{1.0}$
215	Intellivision	IN	intellivision	001	2022-03-08	62	130	[431]2056	-	[81]325	[390]20	(152)388	(566)18764	-	-	-	
216	Intellivision	IN	intellivision	002	2022-07-28	114	128	[429]2056	-	[68]301	[238]13	(557)20542	(567)18784	-	-	-	
217	Intema-LGL Group	LU	intema	000	2022-08-24	1042	20	[62]512	-	[277]629	[561]13809	(25)27	(26)35	(16)39	(16)46	(10)51	[8]$3478.46N^{0.2}$
218	Intema-LGL Group	LU	intema	001	2023-02-22	723	20	[75]512	-	[372]802	[550]4808	(21)23	(19)24	(12)31	(10)38	(13)54	[14]$567.10N^{0.3}$
219	Intozi Tech Pvt Ltd	IN	intozi	001	2026-07-08	168	92	[203]2048	-	[533]1615	[460]34	(312)1111	(314)1121	(263)2133	(258)4295	(247)8626	[196]$0.55N^{1.0}$
220	Intpixel Labs Private Ltd	IN	vogic	000	2026-07-30	231	22	[523]4104	-	[203]501	[505]52	(411)2292	(419)2331	(356)4282	(355)8743	(354)17693	[154]$1.25N^{1.0}$
221	IrexAI	US	irex	000	2021-02-09	724	46	[474]3080	-	[227]532	[382]19	(265)873	(217)600	(174)1120	(186)2477	(194)5863	[364]$0.13N^{1.1}$
222	Iris ID Systems	US	irisid	001	2026-02-20	469	98	[397]2049	-	[426]968	[257]13	(408)2230	(417)2277	(357)4284	(356)8811	(356)17877	[261]$0.87N^{1.0}$
223	Iris ID Systems	US	irisid	002	2026-08-05	718	98	[391]2049	-	[539]1732	[293]14	(545)11134	(554)11093	(472)21213	(468)42047	(455)84414	[119]$6.56N^{1.0}$
224	Kakao Brain	KR	kakao	000	2021-06-23	404	124	[418]2052	-	[130]399	[94]8	(266)876	(92)215	(85)510	(81)971	(78)1955	[370]$0.05N^{1.1}$
225	Kakao Brain	KR	kakao	001	2022-06-08	615	102	[246]2048	-	[212]509	[369]18	(180)469	(169)418	(156)977	(136)1661	(152)3898	[324]$0.13N^{1.1}$
226	Kakao Brain	KR	kakao	002	2023-11-07	749	102	[374]2048	-	[254]594	[181]12	(183)478	(182)466	(165)1055	(181)2443	(168)4801	[416]$0.06N^{1.1}$
227	Kedacom International Pte	SG	kedacom	001	2019-09-16	239	36	[41]292	1	[206]507	[19]2	(244)764	(249)760	(246)1940	(204)2983	(210)6623	[268]$0.31N^{1.0}$
228	Keyless Technologies	GB	keyless	000	2025-01-16	538	35	[265]2048	-	[259]600	[160]11	(131)355	(139)356	(137)811	-	-	[401]$0.05N^{1.1}$
229	Kloudspot Inc	US	kloudspot	001	2025-01-21	575	237	[406]2052	-	[459]1138	[225]13	(453)3435	(457)3438	(388)6477	(392)13716	(388)28709	[310]$1.11N^{1.0}$
230	Kneron	US	kneron	000	2020-03-03	366	13	[291]2048	k	[218]523	[237]13	(424)2535	(429)2506	(373)4752	(369)9696	(372)20926	[270]$0.95N^{1.0}$
231	Kneron	US	kneron	001	2021-06-10	270	69	[210]2048	-	[63]291	[116]9	(442)3141	(435)2642	-	-	-	
232	Know U Tech LLC	US	knowutech	000	2023-06-05	789	34	[105]781	-	[387]831	[57]5	(461)3773	(456)3411	(387)6413	(385)12847	(386)27685	[227]$1.50N^{1.0}$
233	Kogniza Technology	US	kogniza	000	2024-05-14	0	385	[154]1536	-	[502]1336	[120]9	(526)7385	(534)7272	(454)13740	(446)28092	(437)57099	[219]$3.33N^{1.0}$
234	Lebentech Biometrics	MX	lebentech	000	2026-06-15	0	144	[464]2560	-	[44]236	[491]44	(47)96	(59)134	-	-	-	
235	LightVision Inc	KR	lightvision	000	2025-09-11	260	61	[308]2048	-	[400]862	[368]18	(497)5166	(498)5161	(425)9698	(422)19472	(411)39233	[132]$2.93N^{1.0}$
236	Line Corporation	JP	line	000	2021-06-02	138	397	[350]2048	-	[76]317	[101]8	(486)4637	(501)5418	(427)10144	-	-	[82]$3.65N^{1.0}$
237	Line Corporation	JP	line	001	2021-11-21	471	396	[230]2048	-	[187]485	[102]8	(382)1815	(393)1934	(337)3647	(340)7675	-	[304]$0.64N^{1.0}$
238	Line Corporation	JP	lineclova	002	2022-07-29	560	72	[371]2048	-	[150]425	[248]13	(107)262	(127)321	(99)604	(124)1534	(118)3023	[389]$0.05N^{1.1}$
239	Line Corporation	JP	lineclova	003	2023-01-19	574	397	[221]2048	-	[143]412	[82]7	(69)156	(85)195	(163)1018	(159)1956	(130)3405	[406]$0.05N^{1.1}$
240	Lishui Shuyan Huirong Technology	CN	lssytech	000	2026-03-02	1583	193	[488]4096	-	[512]1382	[457]33	(229)667	(266)892	(233)1811	(196)2780	(195)5907	[278]$0.29N^{1.0}$
241	Lishui Shuyan Huirong Technology	CN	lssytech	001	2026-09-17	1606	226	[542]4224	-	[517]1434	[443]30	(314)1124	(313)1121	(261)2101	(253)4218	(246)8565	[141]$0.63N^{1.0}$
242	Lomonosov Moscow State University	RU	intsysmsu	000	2019-08-19	375	168	[234]2048	1	[266]614	[252]13	(173)430	(175)431	(144)860	(140)1730	(179)5353	[435]$0.03N^{1.1}$
243	Lookman Electroplast Industries	IN	lookman	3	2018-10-28	203	24	[42]292	1	[84]336	[30]3	(240)739	(245)745	(201)1394	(197)2817	(241)8286	[388]$0.13N^{1.1}$
244	Lookman Electroplast Industries	IN	lookman	4	2018-10-28	184	24	[93]548	1	[78]320	[50]4	(289)981	(290)998	-	-	-	
245	Lookman Electroplast Industries	IN	lookman	005	2019-09-16	239	36	[94]548	1	[205]506	[46]4	(293)1005	(294)1008	(295)2597	(292)5446	(256)8939	[385]$0.19N^{1.1}$
246	Luxand Inc	US	luxand	001	2023-10-17	83	70	[236]2048	-	[152]425	[180]12	(423)2523	(437)2713	(368)4690	(366)9543	(365)19507	[68]$2.11N^{1.0}$
247	Mantra Softech India	IN	mantra	000	2021-10-28	460	61	[411]2052	-	[141]412	[135]10	(273)916	(272)910	(225)1714	(222)3411	(214)6841	[92]$0.57N^{1.0}$
248	Maverick Research	AE	maverickresearch	000	2025-02-25	250	96	[375]2048	-	[430]995	[176]12	(148)372	(178)443	-	-	-	
249	Maxvision	CN	maxvision	000	2022-06-17	167	60	[284]2048	-	[16]139	[573]-	(493)5044	(322)1195	(271)2163	(262)4380	(255)8914	[205]$0.55N^{1.0}$
250	Maxvision	CN	maxvision	001	2022-10-28	228	63	[283]2048	-	[94]351	[246]13	(322)1173	(308)1061	(275)2233	(259)4311	(260)9084	[267]$0.42N^{1.0}$
251	Maxvision	CN	maxvision	002	2023-04-12	391	63	[259]2048	-	[275]626	[320]15	(323)1180	(305)1052	(272)2213	(254)4234	(264)9310	[305]$0.37N^{1.0}$
252	Maxvision	CN	maxvision	003	2023-08-11	228	62	[346]2048	-	[93]348	[166]12	(302)1049	(306)1052	(251)2006	(250)4177	(267)9366	[297]$0.37N^{1.0}$
253	Maxvision	CN	maxvision	004	2023-12-12	308	62	[358]2048	-	[475]1202	[235]13	(308)1094	(309)1088	(254)2044	(245)4067	(254)8836	[210]$0.51N^{1.0}$
254	Megvii/Face++	CN	megvii	1	2018-10-28	1703	41	[504]4096	1	[278]631	[453]32	(207)552	(210)561	(186)1222	(175)2321	(196)5968	[405]$0.08N^{1.1}$
255	Megvii/Face++	CN	megvii	2	2018-10-28	1735	42	[498]4096	1	[279]635	[451]31	(208)553	(207)558	-	-	-	
256	Megvii/Face++	CN	megvii	003	2023-05-11	1335	43	[396]2049	-	[492]1263	[334]16	(392)1972	(387)1854	(311)2942	(328)6803	(327)14061	[314]$0.52N^{1.0}$
257	Megvii/Face++	CN	megvii	004	2023-10-18	1605	43	[394]2049	-	[500]1303	[282]14	(378)1735	(372)1694	(328)3217	(320)6495	(332)14865	[309]$0.56N^{1.0}$
258	Megvii/Face++	CN	megvii	005	2024-02-20	1605	43	[395]2049	-	[506]1348	[294]14	(380)1764	(376)1721	(330)3261	(324)6584	(319)13376	[171]$0.90N^{1.0}$
259	Metsakuur	KR	metsakuurcompany	000	2024-10-24	0	1011	[430]2056	-	[385]830	[162]12	(437)3068	(528)6828	(469)18336	(463)38444	(456)89809	[457]$0.13N^{1.2}$
260	Metsakuur	KR	metsakuurcompany	001	2025-07-30	0	299	[296]2048	-	[195]490	[216]12	(532)8191	(539)8027	(457)15224	(454)31411	(448)69959	[348]$1.96N^{1.1}$

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1	Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas).
2	Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates.
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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
											N=1.6M	N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
261	Metsakuur	KR	metsakuurcompany	002	2026-09-17	0	519	[310]2048	-	[289]650	[366]18	[195]515	[185]488	[154]966	[156]1934	[149]3886	$^{117}0.29N^{1.0}$
262	Miaxis Biometrics	CN	miaxis	001	2025-02-02	712	87	[389]2049	-	[563]1953	[365]18	[536]8566	[544]8573	[463]16097	[459]32220	[445]65943	$^{139}4.83N^{1.0}$
263	Miaxis Biometrics	CN	miaxis	002	2026-07-24	721	168	[393]2049	-	[541]1771	[356]17	[508]5855	[506]5740	[431]10637	[427]21716	[417]43897	$^{155}3.10N^{1.0}$
264	MicroFocus	GB	microfocus	5	2018-10-29	94	26	[24]256	k	[52]262	[25]2	[82]182	[81]186	[64]354	[61]708	[57]1425	$^{124}0.11N^{1.0}$
265	MicroFocus	GB	microfocus	6	2018-10-29	94	26	[17]256	k	[53]262	[27]2	[83]183	[80]186	-	-	-	
266	Microsoft	US	microsoft	5	2018-10-29	381	155	[114]1024	1	[297]658	[159]11	[364]1606	[371]1673	[319]3076	[316]6302	[316]13160	$^{203}0.79N^{1.0}$
267	Microsoft	US	microsoft	6	2018-10-29	478	155	[121]1024	1	[305]671	[304]15	[368]1642	[369]1618	[339]3710	[318]6401	[313]12892	$^{248}0.68N^{1.0}$
268	Mobbeel Solutions	ES	mobbl	000	2023-11-30	168	157	[366]2048	-	[248]577	[567]34082	[391]1958	[424]2444	[422]9181	[374]11242	[226]7502	$^{35}4.56N^{0.9}$
269	Mobbeel Solutions	ES	mobbl	001	2024-05-14	168	158	[220]2048	-	[246]568	[564]22755	[466]3993	[488]4689	-	-	-	
270	Mobbeel Solutions	ES	mobbl	002	2025-04-09	344	555	[362]2048	-	[448]1088	[178]12	[15]11	[11]11	-	-	-	
271	Mobbeel Solutions	ES	mobbl	003	2026-08-18	344	554	[339]2048	-	[549]1828	[275]14	[235]703	[237]704	-	-	-	
272	Mukh Technologies	US	mukh	002	2022-09-16	693	442	[293]2048	-	[361]780	[548]4261	[10]5	[37]82	[27]137	[26]319	[26]612	$^{106}0.05N^{1.0}$
273	N-Tech Lab	RU	ntech	5	2018-10-30	1685	113	[165]1940	k	[334]711	[508]55	[102]243	[102]246	[88]538	[90]1100	[105]2867	$^{430}0.02N^{1.1}$
274	N-Tech Lab	RU	ntech	6	2018-10-30	1686	117	[166]1940	k	[386]831	[513]63	[101]243	[101]246	[90]546	[91]1104	[107]2873	$^{432}0.02N^{1.1}$
275	N-Tech Lab	RU	ntechlab	007	2019-06-25	2450	51	[481]3348	k	[370]795	[519]73	[157]393	[172]427	[132]780	[146]1768	[135]3499	$^{266}0.16N^{1.0}$
276	N-Tech Lab	RU	ntechlab	008	2020-01-06	1111	51	[149]1300	k	[238]554	[472]36	[79]179	[78]184	[62]341	[59]683	[56]1395	$^{107}0.11N^{1.0}$
277	N-Tech Lab	RU	ntechlab	009	2021-03-01	1208	42	[150]1300	-	[561]1936	[469]35	[87]193	[79]184	[58]336	[58]676	[70]1704	$^{340}0.05N^{1.1}$
278	N-Tech Lab	RU	ntechlab	010	2021-06-24	351	213	[146]1280	-	[346]738	[62]6	[228]666	[176]435	[138]821	[134]1645	[126]3337	$^{181}0.22N^{1.0}$
279	N-Tech Lab	RU	ntechlab	011	2021-12-07	679	208	[148]1280	-	[347]739	[66]6	[257]827	[184]483	[149]912	[154]1869	[172]5003	$^{394}0.07N^{1.1}$
280	NEC	JP	nec	2	2018-10-30	705	35	[160]1616	k	[285]642	[376]18	[162]405	[166]409	[167]1072	[144]1755	[161]4255	$^{411}0.06N^{1.1}$
281	NEC	JP	nec	3	2018-10-30	774	110	[162]1712	k	[301]665	[394]21	[12]7	[8]7	[6]14	[13]40	[15]82	$^{444}0.00N^{1.2}$
282	NEC	JP	nec	004	2021-07-19	971	63	[143]1104	-	[286]647	[69]7	[179]468	[138]351	[110]662	[106]1330	[97]2685	$^{133}0.20N^{1.0}$
283	NEC	JP	nec	005	2021-12-13	922	88	[141]1104	-	[267]615	[76]7	[164]407	[205]551	[162]1017	[166]2091	[159]4242	$^{173}0.28N^{1.0}$
284	NEC	JP	nec	006	2022-08-10	701	54	[142]1104	-	[296]656	[113]9	[132]358	[165]409	[130]765	[126]1543	[148]3878	$^{353}0.10N^{1.1}$
285	NEC	JP	nec	007	2023-03-03	632	56	[97]560	-	[284]641	[64]6	[95]216	[91]214	[72]423	[74]853	[71]1720	$^{399}0.03N^{1.1}$
286	NEC	JP	nec	008	2023-07-03	1380	50	[161]1648	-	[511]1381	[153]11	[201]532	[204]548	[200]1384	[85]999	[164]4757	$^{36}1.39N^{0.9}$
287	NEC	JP	nec	009	2023-12-21	1688	51	[99]560	-	[521]1462	[60]5	[64]138	[62]139	[47]261	[46]523	[45]1056	$^{145}0.08N^{1.0}$
288	NEC	JP	nec	010	2025-01-24	2015	53	[98]560	-	[510]1373	[61]5	[63]138	[73]177	[50]270	[62]712	[58]1455	$^{272}0.06N^{1.0}$
289	NEC	JP	nec	011	2025-10-16	1966	52	[100]560	-	[570]2362	[71]7	[84]184	[77]184	[63]346	[60]692	[55]1389	$^{136}0.10N^{1.0}$
290	NEC	JP	nec	012	2026-07-01	1966	52	[101]560	-	[571]2560	[68]6	[74]158	[71]159	[56]299	[64]738	[51]1308	$^{336}0.05N^{1.1}$
291	Naver Cloud	KR	navercloud	000	2026-03-30	1079	210	[290]2048	-	[498]1300	[355]17	[167]415	[174]428	[133]783	[130]1570	[139]3647	$^{288}0.15N^{1.0}$
292	Neurotechnology	LT	neurotech	5	2018-10-30	266	53	[21]256	k	[133]402	[26]2	[259]835	[257]839	[220]1690	[214]3219	[257]8955	$^{366}0.19N^{1.1}$
293	Neurotechnology	LT	neurotech	6	2018-10-30	564	53	[15]256	k	[342]726	[23]2	[260]839	[258]842	-	-	-	
294	Neurotechnology	LT	neurotech	007	2019-10-03	57	51	[25]256	k	[18]161	[20]2	[313]1118	[312]1110	[265]2143	[263]4397	[259]9045	$^{202}0.55N^{1.0}$
295	Neurotechnology	LT	neurotechnology	008	2021-03-22	355	49	[80]514	-	[135]405	[49]4	[310]1100	[319]1149	[278]2266	[273]4573	[277]9586	$^{214}0.55N^{1.0}$
296	Neurotechnology	LT	neurotechnology	009	2021-09-01	246	82	[77]513	-	[88]341	[28]3	[279]941	[303]1049	[249]1977	[257]4270	[250]8756	$^{318}0.32N^{1.1}$
297	Neurotechnology	LT	neurotechnology	010	2022-01-07	247	83	[19]256	-	[299]661	[7]2	[290]988	[287]984	[241]1897	[241]3977	[239]8048	$^{281}0.36N^{1.0}$
298	Neurotechnology	LT	neurotechnology	012	2022-06-07	247	84	[22]256	-	[107]368	[29]3	[300]1036	[280]945	[234]1828	[231]3787	[235]7838	$^{302}0.32N^{1.0}$
299	Neurotechnology	LT	neurotechnology	013	2023-02-03	364	87	[23]256	-	[179]471	[10]2	[490]4935	[489]4812	[430]10461	[452]30539	[451]73417	$^{464}0.03N^{1.3}$
300	Neurotechnology	LT	neurotechnology	014	2023-12-27	362	95	[20]256	-	[180]472	[5]2	[93]212	[93]215	[71]414	[75]854	[72]1763	$^{286}0.08N^{1.0}$
301	Neurotechnology	LT	neurotechnology	015	2024-08-27	357	100	[16]256	-	[270]619	[18]2	[91]203	[89]207	[69]398	[70]821	[69]1690	$^{284}0.07N^{1.0}$
302	Neurotechnology	LT	neurotechnology	016	2025-08-01	788	100	[57]512	-	[461]1146	[43]4	[120]293	[117]287	[96]594	[95]1187	[90]2408	$^{321}0.09N^{1.1}$
303	Newland Computer Co	CN	newland	2	2018-10-30	96	27	[367]2048	-	[396]855	[307]15	[538]8741	[547]8854	[467]17892	[465]39356	-	$^{398}1.32N^{1.1}$
304	NextMeta Labs	IN	nextmetalabs	001	2024-09-20	353	8204	[281]2048	-	[421]948	[253]13	[505]5732	[508]5897	-	-	-	
305	NextMeta Labs	IN	nextmetalabs	002	2025-07-07	1079	6593	[384]2048	-	[472]1197	[222]13	[521]6479	[519]6259	[437]11828	[440]26060	[430]52936	$^{329}1.84N^{1.1}$
306	NextMeta Labs	IN	nextmetalabs	003	2026-06-18	1079	4076	[218]2048	-	[352]1852	[313]15	[561]28540	[570]28460	-	-	-	
307	Noblis	US	noblis	1	2018-10-30	114	176	[266]2048	1	[38]206	[301]15	[332]1273	[329]1272	-	-	-	
308	Noblis	US	noblis	2	2018-10-30	153	176	[553]6144	1	[215]517	[488]43	[422]2513	[430]2522	[376]5649	[383]12432	[418]44262	$^{458}0.04N^{1.3}$
309	Nominder	LT	nominder	000	2023-09-07	534	100	[28]258	-	[118]384	[6]2	[35]48	[33]49	[23]92	[23]186	[22]376	$^{190}0.02N^{1.0}$
310	Nominder	LT	nominder	001	2024-01-08	243	114	[26]258	-	[123]390	[15]2	[34]48	[32]48	[22]92	[22]186	[21]375	$^{195}0.02N^{1.0}$
311	Nominder	LT	nominder	002	2024-06-05	243	111	[30]258	-	[121]388	[21]2	[37]61	[30]47	[24]107	[25]234	[24]483	$^{424}0.00N^{1.1}$
312	Nominder	LT	nominder	003	2024-10-07	244	111	[27]258	-	[323]697	[12]2	[33]46	[31]47	[21]89	[21]180	[23]400	$^{306}0.02N^{1.0}$

Notes	
1	Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas).
2	Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates.
3	This multiplier expresses the increase in template size when k images are passed to the template generation function.
4	All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however.
5	Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 479. However in certain cases the model is not correct and should not be used numerically.

Table 7: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
												N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
313	Nominder	LT	nominder	004	2025-02-10	312	111	[29]258	-	[496]1294	[16]2	[36]50	[34]50	[26]116	[24]193	[25]507	[379]$0.01N^{1.1}$
314	Nominder	LT	nominder	005	2026-08-07	472	99	[79]514	-	[560]1912	[52]4	[54]110	[52]110	[41]208	[43]476	[40]990	[354]$0.03N^{1.1}$
315	NotionTag Technologies Private Limited	IN	notiontag	000	2022-01-14	265	945	[461]2120	-	[167]453	[141]10	[537]8619	[545]8705	[464]16652	[464]38794	[457]90607	[409]$1.15N^{1.1}$
316	Omnigarde	US	omnigarde	000	2023-05-22	417	34	[81]517	-	[329]705	[39]3	[393]1983	[392]1933	[333]3430	[331]7224	[340]15597	[254]$0.75N^{1.0}$
317	Omnigarde	US	omnigarde	001	2024-06-25	971	34	[82]517	-	[476]1205	[37]3	[398]2017	[407]2079	[351]3941	[348]7908	[347]15974	[128]$1.20N^{1.0}$
318	Omnigarde	US	omnigarde	002	2025-07-28	1285	34	[85]525	-	[467]1168	[38]3	[421]2512	[431]2534	[370]4734	[367]9553	[364]19203	[81]$1.72N^{1.0}$
319	Onfido	GB	onfido	000	2026-03-03	832	247	[486]4096	-	[473]1199	[441]30	[523]7075	[535]7357	[452]13562	[448]28651	[438]58683	[293]$2.49N^{1.0}$
320	Onfido	GB	onfido	001	2026-07-21	832	332	[514]4096	-	[320]690	[444]30	[451]3368	[461]3566	[394]6862	[395]13992	[387]28042	[300]$1.18N^{1.0}$
321	OptiExacta	IN	optiexacta	000	2024-07-23	505	2446	[469]2564	-	[463]1149	[308]15	[281]954	[279]944	[230]1780	[227]3583	[223]7238	[246]$0.38N^{1.0}$
322	PT Ice House	ID	icehouse	000	2024-08-30	62	58	[170]2048	-	[7]46	[187]12	[555]15996	[564]16117	-	-	-	
323	PT Ice House	ID	icehouse	001	2026-02-27	280	174	[354]2048	-	[138]408	[363]17	[146]371	[170]423	[134]792	[131]1585	[123]3202	[295]$0.14N^{1.0}$
324	Panasonic R+D Center Singapore	SG	psl	000	2023-08-08	795	607	[570]8248	-	[446]1085	[490]43	[460]3678	[464]3669	[397]7080	[390]13494	[381]26559	[197]$1.74N^{1.0}$
325	Panasonic R+D Center Singapore	SG	psl	001	2024-02-22	894	607	[537]4144	-	[452]1114	[413]25	[409]2250	[418]2288	[359]4316	[352]8577	[367]19561	[273]$0.86N^{1.0}$
326	Panasonic R+D Center Singapore	SG	psl	002	2024-07-26	992	607	[538]4148	-	[499]1300	[404]23	[428]2731	[436]2704	[375]4953	[365]9531	[371]20015	[75]$1.95N^{1.0}$
327	Panasonic R+D Center Singapore	SG	psl	003	2025-07-07	1585	607	[477]3124	-	[478]1207	[375]18	[406]2146	[421]2399	[363]4575	[361]9364	[353]17474	[237]$0.98N^{1.0}$
328	Panasonic R+D Center Singapore	SG	psl	004	2025-11-28	1088	607	[478]3124	-	[547]1824	[410]24	[446]3213	[449]3150	[381]5926	[379]12019	[394]31019	[355]$0.74N^{1.1}$
329	Panasonic R+D Center Singapore	SG	psl	005	2026-03-30	1005	607	[479]3124	-	[542]1773	[405]24	[494]5063	[496]5122	[421]9094	[412]17200	[402]33623	[30]$17.94N^{0.9}$
330	Panasonic R+D Center Singapore	SG	psl	006	2026-07-30	1005	607	[480]3124	-	[548]1826	[411]25	[492]5007	[494]5084	[419]8952	[414]17421	[404]33895	[31]$17.46N^{0.9}$
331	Pangiam	US	pangiam	000	2022-02-22	453	23	[217]2048	-	[280]636	[361]17	[111]276	[126]319	[97]601	[98]1210	[92]2443	[126]$0.18N^{1.0}$
332	Pangiam	US	pangiam	001	2022-11-17	991	36	[208]2048	-	[250]577	[271]14	[110]275	[156]378	[127]724	[116]1436	[142]3676	[362]$0.08N^{1.1}$
333	Papilon Savunma	TR	papsav1923	000	2023-08-22	923	24	[40]288	-	[358]767	[9]2	[31]37	[28]38	[18]72	[19]143	[19]286	[88]$0.02N^{1.0}$
334	Paravision (EverAI)	US	everai	2	2018-10-30	224	304	[370]2048	1	[103]366	[446]30	[114]278	[115]283	-	-	-	
335	Paravision (EverAI)	US	everai	3	2018-10-30	438	304	[289]2048	1	[337]717	[425]28	[112]278	[114]281	[93]572	[92]1146	[85]2278	[253]$0.12N^{1.0}$
336	Paravision (EverAI)	US	everai-paravision	004	2019-06-19	527	128	[491]4096	1	[306]672	[494]45	[209]559	[208]559	[297]2611	[319]6445	[329]14519	[469]$0.00N^{1.5}$
337	Paravision (EverAI)	US	paravision	005	2019-12-11	543	154	[501]4096	1	[384]830	[500]48	[211]561	[211]564	[166]1056	[171]2298	[171]4966	[334]$0.16N^{1.1}$
338	Paravision (EverAI)	US	paravision	007	2021-02-01	529	235	[509]4096	-	[128]397	[501]48	[249]786	[206]558	[168]1086	[167]2111	[160]4254	[44]$1.11N^{0.9}$
339	Paravision	US	paravision	009	2021-12-14	672	300	[519]4100	-	[82]331	[523]82	[479]4495	[478]4230	[410]8037	[407]16532	[395]31422	[256]$1.62N^{1.0}$
340	Paravision	US	paravision	012	2023-02-07	762	182	[522]4100	-	[140]411	[525]85	[471]4313	[487]4534	[416]8651	[409]16582	[398]32114	[85]$2.87N^{1.0}$
341	Paravision	US	paravision	014	2023-06-08	1501	182	[516]4100	-	[434]1010	[528]86	[481]4524	[483]4454	[408]7993	[405]16169	[396]31691	[222]$1.88N^{1.0}$
342	Paravision	US	paravision	016	2024-12-04	1546	250	[521]4100	-	[474]1200	[524]84	[477]4436	[470]4006	[407]7861	[410]16711	[400]32464	[213]$1.94N^{1.0}$
343	Paravision	US	paravision	019	2025-07-03	1558	250	[518]4100	-	[514]1415	[527]86	[470]4304	[479]4301	[404]7639	[400]15454	[397]31978	[228]$1.78N^{1.0}$
344	Paravision	US	paravision	021	2026-03-12	1607	342	[520]4100	-	[567]2070	[532]119	[524]7086	[532]7065	[451]13368	[444]27551	[433]54944	[280]$2.52N^{1.0}$
345	Paravision	US	paravision	022	2026-09-21	1607	342	[517]4100	-	[566]2057	[533]220	[522]6823	[530]6865	[449]13015	[445]27811	[434]55064	[299]$2.31N^{1.0}$
346	Pentacomp Systemy Informatyczne	PL	pentacomp	000	2025-01-10	255	187	[254]2048	-	[181]473	[242]13	[192]511	[200]534	-	-	-	
347	Pentacomp Systemy Informatyczne	PL	pentacomp	001	2025-07-28	255	187	[253]2048	-	[171]459	[280]14	[353]1485	[345]1447	[282]2408	[276]4708	[273]9479	[40]$2.83N^{0.9}$
348	Private Identity LLC	US	privid	000	2025-08-01	0	975	[326]2048	-	[497]1296	[534]226	[18]20	[15]19	-	-	-	
349	Protenet Soluções em Segurança	BR	protenet	000	2025-09-17	182	173	[233]2048	-	[350]748	[536]389	[78]178	[74]178	-	-	-	
350	Protenet Soluções em Segurança	BR	protenet	001	2026-01-27	182	134	[286]2048	-	[532]1613	[418]27	[307]1082	[311]1107	[258]2069	[248]4146	[242]8327	[122]$0.63N^{1.0}$
351	Pyramid Cyber Security + Forensic (P)	IN	pyramid	000	2024-11-19	1016	141	[302]2048	-	[389]837	[384]20	[182]473	[187]489	[140]844	[132]1587	[131]3408	[56]$0.50N^{1.0}$
352	Pyramid Cyber Security + Forensic (P)	IN	pyramid	001	2025-04-01	1023	140	[317]2048	-	[398]857	[387]20	[128]336	[136]345	[105]649	[103]1303	[116]3008	[301]$0.12N^{1.0}$
353	QazSmartVision.AI	KZ	qazsmartvisionai	000	2024-03-15	547	95	[63]512	-	[326]701	[72]7	[72]157	[68]157	[54]296	[50]597	[49]1220	[201]$0.07N^{1.0}$
354	QazSmartVision.AI	KZ	qazsmartvisionai	001	2024-07-16	1383	98	[91]544	-	[488]1245	[79]7	[57]124	[55]125	[44]234	[42]473	[39]957	[152]$0.07N^{1.0}$
355	QazSmartVision.AI	KZ	qazsmartvisionai	002	2024-12-03	1333	98	[90]544	-	[482]1221	[74]7	[70]156	[69]157	[53]294	[49]591	[54]1369	[303]$0.05N^{1.0}$
356	QazSmartVision.AI	KZ	qazsmartvisionai	003	2025-04-24	1358	98	[92]544	-	[487]1238	[73]7	[71]156	[67]156	[55]297	[51]598	[47]1204	[166]$0.08N^{1.0}$
357	Qnap Security	TW	qnap	000	2021-07-28	182	15	[318]2048	-	[131]401	[123]9	[344]1427	[381]1763	-	-	-	
358	Qnap Security	TW	qnap	001	2021-12-09	191	13	[198]2048	-	[189]487	[95]8	[347]1449	[344]1429	[334]3472	[335]7375	[337]15159	[441]$0.11N^{1.2}$
359	Qnap Security	TW	qnap	002	2022-04-15	338	32	[179]2048	-	[194]489	[354]17	[283]958	[277]938	[237]1866	[261]4359	[251]8792	[410]$0.12N^{1.1}$
360	Qnap Security	TW	qnap	003	2022-12-09	239	60	[261]2048	-	[46]244	[266]13	[371]1671	[399]2039	[352]4010	[350]8379	[334]14911	[232]$0.87N^{1.0}$
361	Qnap Security	TW	qnap	004	2023-05-05	362	60	[381]2048	-	[102]365	[158]11	[386]1885	[410]2096	[327]3191	[325]6601	[328]14490	[63]$1.66N^{1.0}$
362	Qnap Security	TW	qnap	005	2023-11-16	954	60	[363]2048	-	[427]971	[186]12	[355]1499	[412]2151	[329]3232	[349]7969	[338]15499	[174]$1.02N^{1.0}$
363	Qnap Security	TW	qnap	006	2024-08-09	1178	60	[419]2052	-	[432]1005	[221]13	[389]1908	[388]1894	[320]3098	[313]6263	[326]14003	[238]$0.72N^{1.0}$
364	Quantasoft	CZ	quantasoft	1	2018-10-30	276	452	[197]2048	k	[119]385	[65]6	[554]15422	[562]14858	[456]14717	-	[357]18323	

Notes	
1	Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas).
2	Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates.
3	This multiplier expresses the increase in template size when k images are passed to the template generation function.
4	All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however.
5	Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 479. However in certain cases the model is not correct and should not be used numerically.

Table 8: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.

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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
											N=1.6M	N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
365	RNIT Solutions and Servies Ltd	IN	rnit	000	2025-05-08	490	33	[43]296	-	[481]1220	[11]2	(154)389	(160)387	(129)752	(123)1524	(120)3098	$^{250}0.16N^{1.0}$
366	ROC	US	rankone	4	2018-10-09	0	101	[4]85	k	[6]36	[77]7	(50)101	(51)101	(39)190	-	-	$^{79}0.07N^{1.0}$
367	ROC	US	rankone	5	2018-10-24	0	101	[12]133	k	[8]92	[78]7	(65)140	(63)144	(48)266	(47)525	(44)1049	$^{73}0.11N^{1.0}$
368	ROC	US	rankone	006	2019-06-03	0	133	[13]165	k	[47]245	[93]8	-	-	-	-	-	
369	ROC	US	rankone	007	2019-11-12	0	137	[14]165	k	[54]272	[84]7	(55)116	(53)115	(42)215	(41)439	(38)877	$^{123}0.07N^{1.0}$
370	ROC	US	rankone	009	2020-06-26	0	105	[31]260	k	[30]185	[150]11	(41)95	(45)96	(34)181	(33)362	(34)727	$^{95}0.06N^{1.0}$
371	ROC	US	rankone	010	2020-11-05	0	135	[35]261	-	[37]198	[142]10	(42)95	(40)95	(31)178	(30)357	(31)714	$^{90}0.06N^{1.0}$
372	ROC	US	rankone	011	2021-08-27	0	175	[37]261	-	[244]566	[103]8	(46)96	(41)95	(36)183	(35)370	(30)714	$^{104}0.06N^{1.0}$
373	ROC	US	rankone	012	2021-12-27	0	257	[36]261	-	[242]563	[87]8	(45)95	(42)95	(32)179	(32)361	(32)718	$^{93}0.06N^{1.0}$
374	ROC	US	rankone	013	2022-07-21	0	223	[32]261	-	[161]446	[336]16	(51)101	(39)89	(29)167	(28)335	(28)672	$^{101}0.06N^{1.0}$
375	ROC	US	rankone	014	2022-12-21	0	237	[34]261	-	[170]457	[90]8	(52)101	(49)100	(38)188	(37)376	(36)775	$^{150}0.06N^{1.0}$
376	ROC	US	rankone	015	2023-07-20	0	282	[33]261	-	[235]548	[107]9	(48)97	(46)97	(35)183	(34)366	(35)734	$^{111}0.06N^{1.0}$
377	ROC	US	roc	016	2023-12-20	0	659	[49]392	-	[236]552	[96]8	(58)131	(56)130	(45)245	(44)491	(46)1084	$^{245}0.05N^{1.0}$
378	ROC	US	roc	017	2024-06-24	0	552	[50]396	-	[292]651	[110]9	(76)158	(70)159	(52)294	(52)601	(48)1204	$^{129}0.09N^{1.0}$
379	ROC	US	roc	018	2025-01-31	0	910	[106]792	-	[486]1235	[139]10	(119)290	(118)290	(92)550	(99)1222	(94)2527	$^{335}0.08N^{1.1}$
380	ROC	US	roc	019	2025-08-04	0	924	[52]444	-	[456]1131	[566]31194	(85)185	(76)183	(59)340	(57)669	(53)1338	$^{58}0.19N^{1.0}$
381	ROC	US	roc	020	2026-02-06	0	982	[51]444	-	[555]1893	[571]43943	(153)388	(162)389	(124)719	(115)1429	(104)2849	$^{59}0.36N^{1.0}$
382	ROC	US	roc	021	2026-08-05	0	980	[53]448	-	[535]1711	[570]37620	(141)367	(154)377	(112)681	(107)1352	(99)2708	$^{64}0.33N^{1.0}$
383	RealSense	US	realsense	001	2026-01-26	354	113	[213]2048	-	[562]1946	[317]15	(498)5187	(497)5144	(426)9698	(423)19548	(412)39427	$^{112}3.09N^{1.0}$
384	RealSense	US	realsense	002	2026-08-24	362	121	[342]2048	-	[556]1895	[323]15	(495)5093	(495)5093	(424)9625	(420)19403	(410)39076	$^{120}3.02N^{1.0}$
385	Realnetworks Inc	US	realnetworks	2	2018-10-30	105	104	[527]4104	k	[45]241	[424]28	(395)2008	(401)2048	(355)4194	(353)8642	(336)15035	$^{175}1.08N^{1.0}$
386	Realnetworks Inc	US	realnetworks	003	2019-06-12	93	102	[164]1848	k	[23]173	[220]13	(318)1145	(315)1132	(264)2142	(288)5241	(290)10495	$^{378}0.21N^{1.1}$
387	Realnetworks Inc	US	realnetworks	004	2019-10-17	94	102	[163]1848	1	[21]171	[149]11	(317)1143	(317)1137	(267)2149	(278)4740	(280)9693	$^{313}0.36N^{1.0}$
388	Realnetworks Inc	US	realnetworks	005	2021-06-23	168	209	[422]2056	-	[83]332	[112]9	(369)1654	(368)1616	(314)3030	(309)6068	(310)12134	$^{99}1.01N^{1.0}$
389	Realnetworks Inc	US	realnetworks	006	2021-12-02	250	56	[425]2056	-	[22]172	[97]8	(252)791	(198)531	(158)996	(163)1998	(155)3991	$^{98}0.33N^{1.0}$
390	Realnetworks Inc	US	realnetworks	007	2022-04-11	455	99	[435]2056	-	[90]342	[357]17	(255)815	(231)674	(215)1597	(229)3683	(188)5729	$^{380}0.13N^{1.1}$
391	Realnetworks Inc	US	realnetworks	008	2022-08-29	557	99	[441]2056	-	[219]524	[177]12	(203)538	(230)673	(188)1264	(205)3011	(191)5801	$^{347}0.18N^{1.1}$
392	Recognito	AE	recognito	000	2023-02-17	350	242	[147]1280	-	[359]775	[126]9	(248)780	(218)605	(178)1152	(174)2320	(187)5707	$^{384}0.10N^{1.1}$
393	Recognito	AE	recognito	001	2023-06-23	536	242	[471]2564	-	[447]1086	[310]15	(315)1128	(288)989	(259)2078	(242)3978	(158)4107	$^{33}2.45N^{0.9}$
394	Recognito	AE	recognito	002	2023-11-09	505	61	[470]2564	-	[442]1075	[326]16	(282)957	(281)945	(231)1791	(265)4437	(261)9161	$^{421}0.10N^{1.1}$
395	Remark Holdings	CN	remarkai	0	2018-10-30	187	847	[278]2048	k	[253]593	[287]14	(504)5685	(505)5723	-	-	-	
396	Remark Holdings	CN	remarkai	000	2019-06-12	234	1092	[345]2048	k	[291]650	[207]12	(506)5776	(504)5703	(434)11604	(458)32133	(458)91436	$^{459}0.05N^{1.3}$
397	Remark Holdings	CN	remarkai	1	2018-10-30	187	847	[308]2048	k	[153]427	[296]14	(503)5680	(507)5761	(444)12475	(449)28726	(440)59618	$^{446}0.37N^{1.2}$
398	Rendip	SG	rendip	000	2021-05-21	0	416	[305]2048	-	[263]611	[127]9	(122)299	(146)368	(119)697	(119)1452	(112)2926	$^{352}0.08N^{1.1}$
399	Reveal Media	GB	revealmedia	000	2022-02-02	287	196	[415]2052	-	[117]383	[134]10	(412)2322	(397)2019	(346)3838	(344)7816	(351)16559	$^{264}0.78N^{1.0}$
400	SQIsoft	KR	sqisoft	001	2021-12-20	271	377	[427]2056	-	[72]312	[124]9	(388)1907	(335)1319	(286)2456	(280)4906	(283)9755	$^{80}0.90N^{1.0}$
401	SQIsoft	KR	sqisoft	002	2022-10-26	354	593	[426]2056	-	[151]425	[245]13	(352)1480	(365)1596	(315)3044	(310)6111	(315)13138	$^{236}0.68N^{1.0}$
402	STCON LLC	SA	stcon	000	2024-04-10	1999	62	[513]4096	-	[373]804	[393]20	(438)3092	(460)3527	-	-	-	
403	STCON LLC	SA	stcon	001	2024-09-23	2973	49	[484]4096	-	[503]1343	[433]29	(253)794	(291)998	(208)1495	(206)3016	(233)7739	$^{50}1.36N^{0.9}$
404	STCON LLC	SA	stcon	002	2025-01-31	2973	49	[512]4096	-	[509]1369	[450]31	(251)790	(252)791	(235)1838	(228)3602	(222)7108	$^{369}0.18N^{1.1}$
405	STCON LLC	SA	stcon	003	2025-06-24	2973	49	[508]4096	-	[508]1359	[438]29	(292)1005	(285)977	(238)1870	(234)3795	(224)7362	$^{162}0.52N^{1.0}$
406	STCON LLC	SA	stcon	004	2026-01-26	1995	55	[510]4096	-	[530]1571	[486]41	(324)1221	(326)1218	(279)2285	(272)4570	(262)9203	$^{103}0.75N^{1.0}$
407	STCON LLC	SA	stcon	005	2026-05-25	2055	55	[524]4104	-	[527]1535	[495]45	(303)1050	(270)905	(253)2028	(237)3845	(229)7619	$^{333}0.27N^{1.1}$
408	Samsung S1 Corp	KR	s1	000	2021-06-03	257	196	[485]4096	-	[191]487	[386]20	(496)5149	(527)6794	(450)13032	(441)26372	(435)55723	$^{244}2.82N^{1.0}$
409	Samsung S1 Corp	KR	s1	001	2021-11-01	240	198	[351]2048	-	[199]491	[104]8	(413)2366	(428)2491	(369)4718	(368)9614	(377)24472	$^{368}0.53N^{1.1}$
410	Samsung S1 Corp	KR	s1	002	2022-05-04	244	93	[298]2048	-	[192]488	[325]16	(326)1234	(346)1450	(299)2729	(306)5945	(304)11923	$^{289}0.51N^{1.0}$
411	Samsung S1 Corp	KR	s1	003	2022-09-27	471	93	[187]2048	-	[190]487	[263]13	(366)1620	(347)1454	(301)2740	(308)6025	(298)11644	$^{263}0.57N^{1.0}$
412	Samsung S1 Corp	KR	s1	004	2023-02-03	471	58	[209]2048	-	[193]489	[239]13	(360)1554	(349)1481	(325)3184	(326)6701	(308)12044	$^{320}0.47N^{1.1}$
413	Samsung S1 Corp	KR	s1	005	2023-07-03	770	58	[355]2048	-	[391]847	[213]12	(346)1445	(351)1491	(305)2844	(299)5691	(317)13208	$^{327}0.44N^{1.1}$
414	Samsung S1 Corp	KR	s1	006	2024-11-29	768	172	[312]2048	-	[393]849	[224]13	(358)1534	(367)1613	(318)3063	(314)6271	(301)11735	$^{53}1.84N^{1.0}$
415	Samsung S1 Corp	KR	s1	007	2025-04-29	1439	172	[231]2048	-	[445]1083	[230]13	(349)1465	(356)1520	(306)2848	(298)5688	(318)13337	$^{294}0.53N^{1.0}$
416	Sansap Technology	IN	sansap	001	2024-06-07	0	357	[184]2048	-	[312]684	[247]13	(309)1098	(297)1010	(256)2066	(249)4160	(243)8363	$^{298}0.35N^{1.0}$

Notes	
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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
											N=1.6M	N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
417	Scanovate	IL	scanovate	000	2020-01-15	250	446	[279]2048	-	[330]705	[291]14	[342]1419	[343]1412	[313]3008	[378]11616	[306]12012	[447]$0.10N^{1.2}$
418	Scanovate	IL	scanovate	001	2020-09-10	250	446	[276]2048	-	[230]545	[236]13	[357]1520	[336]1320	[291]2502	[285]5047	[287]10163	[188]$0.65N^{1.0}$
419	Scylla Technologies Inc	US	scylla	001	2024-10-09	450	1789	[260]2048	-	[408]891	[537]467	[540]9267	[549]9373	[466]17883	[461]37069	[453]76927	[316]$2.84N^{1.0}$
420	Security Pitch Co Ltd	TH	securitypitch	000	2026-09-21	513	88	[453]2072	-	[381]822	[483]41	[171]429	[221]616	[159]1001	[168]2237	[162]4321	[57]$0.61N^{1.0}$
421	Sensetime Group	CN	sensetime	0	2018-10-30	525	6	[528]4104	k	[321]693	[484]41	[189]498	[190]501	[184]1212	[169]2281	[174]5032	[390]$0.09N^{1.1}$
422	Sensetime Group	CN	sensetime	1	2018-10-30	525	6	[526]4104	k	[276]628	[499]48	[196]516	[191]502	[177]1146	[172]2301	[166]4765	[387]$0.09N^{1.1}$
423	Sensetime Group	CN	sensetime	002	2019-06-03	523	6	[424]2056	k	[260]603	[371]18	[133]359	[147]370	[242]1897	[269]4508	[276]9543	[472]$0.00N^{1.5}$
424	Sensetime Group	CN	sensetime	003	2019-12-02	769	76	[433]2056	1	[414]910	[381]19	[489]4885	[493]4989	[442]12325	[432]24712	[425]49445	[414]$0.67N^{1.1}$
425	Sensetime Group	CN	sensetime	004	2020-08-10	456	29	[131]1032	-	[319]690	[205]12	[420]2490	[425]2477	[366]4654	[363]9402	[368]19651	[194]$1.22N^{1.0}$
426	Sensetime Group	CN	sensetime	005	2020-12-17	631	39	[126]1032	-	[429]980	[148]11	-	[469]3939	[401]7398	[396]14768	[362]19016	[29]$14.03N^{0.9}$
427	Sensetime Group	CN	sensetime	006	2021-07-26	526	54	[1]-	-	[3]-	[85]7	-	[423]2422	[362]4527	[358]9128	[358]18640	[146]$1.35N^{1.0}$
428	Sensetime Group	CN	sensetime	007	2022-01-15	526	37	[130]1032	-	[416]935	[105]8	[415]2432	[422]2406	[361]4513	[357]8998	[360]18796	[163]$1.28N^{1.0}$
429	Sensetime Group	CN	sensetime	008	2022-08-17	567	37	[122]1032	-	[376]814	[119]9	[417]2444	[396]2012	[342]3785	[339]7551	[344]15799	[170]$1.05N^{1.0}$
430	Sensetime Group	CN	sensetime	009	2023-01-04	883	59	[132]1032	-	[501]1320	[106]9	[400]2031	[403]2062	[341]3747	[338]7543	[342]15781	[167]$1.06N^{1.0}$
431	Sensetime Group	CN	sensetime	010	2024-12-03	1148	60	[124]1032	-	[464]1156	[89]8	[106]262	[107]259	[82]481	[83]981	[81]2031	[182]$0.13N^{1.0}$
432	Sensetime Group	CN	sensetime	011	2026-03-12	842	62	[123]1032	-	[485]1228	[137]10	[134]359	[142]361	[111]679	[108]1360	[102]2734	[109]$0.22N^{1.0}$
433	Sensetime Group	CN	sensetime	012	2026-07-31	1223	62	[135]1036	-	[531]1583	[272]14	[147]371	[148]370	[118]696	[120]1454	[128]3374	[342]$0.10N^{1.1}$
434	Serendipity	UZ	serendipity	000	2023-06-20	0	366	[189]2048	-	[409]894	[227]13	[160]400	[151]371	[139]827	[96]1188	[88]2386	[45]$0.60N^{0.9}$
435	Seventh Sense Artificial Intelligence	SG	seventhsense	001	2024-09-12	1410	633	[493]4096	-	[453]1127	[539]890	[6]2	[16]22	[9]23	[8]32	[7]40	[15]$381.33N^{0.3}$
436	Shaman Software	US	shaman	6	2018-10-26	0	200	[379]2048	k	[331]706	[288]14	[218]603	[219]612	-	-	-	
437	Shaman Software	US	shaman	7	2018-10-26	0	200	[274]2048	k	[332]707	[292]14	[217]602	[220]614	[180]1187	[182]2448	[177]5083	[255]$0.25N^{1.0}$
438	Shanghai Jiao Tong University	CN	sjtu	001	2026-02-10	1417	239	[545]4608	-	[551]1841	[458]33	[533]8200	[542]8315	[461]15765	[457]31878	[443]64177	[168]$4.36N^{1.0}$
439	Shanghai Jiao Tong University	CN	sjtu	002	2026-08-03	873	395	[544]4608	-	[550]1833	[464]34	[531]8190	[543]8349	[462]15902	[456]31717	[442]64136	[160]$4.46N^{1.0}$
440	Shanghai Yitu Technology	CN	yitu	4	2018-10-30	2119	136	[451]2070	1	[410]897	[493]45	[333]1288	[324]1203	[285]2440	[289]5241	[279]9671	[242]$0.52N^{1.0}$
441	Shanghai Yitu Technology	CN	yitu	5	2018-10-30	2043	136	[450]2070	1	[395]853	[492]44	[327]1237	[323]1199	[292]2513	[284]5013	[278]9620	[229]$0.55N^{1.0}$
442	Shennona Co Ltd	TW	shennona	000	2024-11-20	977	72	[110]1024	-	[495]1290	[80]7	[499]5286	[499]5244	-	-	-	
443	Shufti	GB	shufti	001	2026-03-30	432	2228	[400]2052	-	[303]667	[542]1302	[5]1	[4]3	-	-	-	
444	Smilart	DE	smilart	4	2018-10-30	65	89	[69]512	k	[20]167	[48]4	[556]16137	[563]15633	-	-	-	
445	Smilart	DE	smilart	5	2018-10-30	562	89	[174]2048	k	[164]450	[285]14	-	-	-	-	-	
446	Sparktek International Pte Ltd	CN	sparktek	000	2025-11-24	785	116	[332]2048	-	[288]649	[359]17	[187]495	[183]467	[164]1031	[153]1856	[140]3658	[165]$0.26N^{1.0}$
447	Sparktek International Pte Ltd	CN	sparktek	001	2026-04-15	2005	116	[541]4224	-	[536]1713	[461]34	[394]2006	[395]2003	[353]4118	[373]10935	[374]21982	[456]$0.06N^{1.2}$
448	Sparsh CCTV	IN	sparsh	001	2024-01-12	0	357	[224]2048	-	[313]684	[154]11	[299]1036	[296]1008	[245]1908	[240]3927	[234]7764	[180]$0.51N^{1.0}$
449	Staqu Technologies	IN	staqu	000	2021-08-30	1018	690	[506]4096	-	[311]684	[409]24	[482]4527	[491]4933	-	-	-	
450	Stellar Technologies	US	stellartecs	001	2026-05-01	1397	32	[505]4096	-	[505]1348	[454]32	[433]2925	[441]2923	[377]5652	[377]11313	[376]23169	[215]$1.36N^{1.0}$
451	Sum and Substance	GB	sumsub	000	2023-08-18	169	108	[297]2048	-	[64]292	[206]12	[149]380	[153]372	-	-	-	
452	Synesis	RU	synesis	3	2018-10-30	237	150	[483]4096	k	[11]99	[431]29	[250]789	[254]801	[247]1941	[238]3888	[252]8810	[431]$0.07N^{1.1}$
453	Synesis	RU	synesis	003	2019-07-04	143	17	[326]2048	k	[39]211	[195]12	[191]507	[192]502	[280]2297	[271]4564	[271]9452	[467]$0.00N^{1.4}$
454	Synesis	RU	synesis	005	2020-09-08	494	24	[529]4104	-	[354]756	[407]24	[267]877	[260]865	[324]3182	[275]4658	[282]9750	[448]$0.06N^{1.2}$
455	T4iSB	BR	t4isb	000	2022-08-17	228	15	[304]2048	-	[343]729	[258]13	[103]250	[123]311	-	-	-	
456	TTI Group	VN	ttigroup	000	2024-12-16	455	444	[347]2048	-	[405]887	[184]12	[551]13054	[559]12894	-	-	-	
457	Tech5 SA	CH	tech5	001	2019-08-19	1394	116	[153]1536	k	[406]887	[133]10	[150]383	[250]766	[302]2767	[312]6149	[200]6178	[413]$0.12N^{1.1}$
458	Tech5 SA	CH	tech5	002	2021-04-07	727	112	[78]513	-	[317]687	[44]4	[464]3816	[525]6689	[445]12541	[434]25145	[428]50239	[97]$4.18N^{1.0}$
459	Tech5 SA	CH	tech5	003	2023-07-14	0	357	[285]2048	-	[293]653	[208]12	[297]1025	[299]1013	[255]2050	[246]4126	[236]7935	[233]$0.45N^{1.0}$
460	Techainer	VN	techainer	000	2026-06-30	627	1071	[377]2048	-	[97]357	[346]16	[206]547	[244]734	[187]1245	[187]2496	-	[285]$0.17N^{1.0}$
461	Technology Control Company	SA	tcc	000	2026-02-25	54	1161	[112]1024	-	[106]367	[100]8	[205]546	[203]546	-	-	-	
462	Technology Control Company	SA	tcc	001	2026-07-15	963	1209	[272]2048	-	[144]415	[299]15	[220]608	[194]509	-	-	-	
463	Tencent Deepsea Lab	CN	deepsea	001	2019-07-29	250	323	[186]2048	1	[345]737	[203]12	[296]1021	[301]1020	[303]2774	[301]5767	[311]12341	[453]$0.06N^{1.2}$
464	Tevian	RU	tevian	5	2018-10-30	773	15	[180]2048	1	[136]405	[302]15	[163]405	[164]408	[142]854	[145]1757	[129]3380	[311]$0.14N^{1.0}$
465	Tevian	RU	tevian	006	2021-04-16	769	19	[127]1032	-	[232]546	[132]10	[129]343	[121]295	[94]578	[94]1187	[103]2741	[363]$0.06N^{1.1}$
466	Tevian	RU	tevian	007	2021-10-12	703	19	[129]1032	-	[349]747	[53]4	[144]369	[122]298	[95]579	[93]1179	[91]2418	[277]$0.11N^{1.0}$
467	Thakaa Advanced Company for Information Technology	SA	thakaait	000	2026-08-10	1232	480	[437]2056	-	[439]1047	[345]16	[468]4040	[472]4043	[409]8017	[404]15990	[399]32356	[247]$1.67N^{1.0}$
468	Thales Group	US	cogent	2	2018-10-30	681	39	[136]1043	k	[419]945	[419]27	[397]2017	[411]2144	[358]4298	[351]8472	[350]16429	[189]$1.08N^{1.0}$

Notes	
1	Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas).
2	Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates.
3	This multiplier expresses the increase in template size when k images are passed to the template generation function.
4	All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however.
5	Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 479. However in certain cases the model is not correct and should not be used numerically.

Table 10: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
												N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
469	Thales Group	US	cogent	3	2018-10-30	681	39	[137]1043	k	[417]940	[129]9	(325)1230	(333)1311	(298)2687	(290)5398	(288)10184	$^{218}0.62N^{1.0}$
470	Thales Group	US	cogent	004	2021-02-10	1376	59	[421]2053	-	[420]947	[270]14	(430)2903	(389)1911	(335)3566	(336)7498	(349)16370	$^{307}0.64N^{1.0}$
471	Thales Group	US	cogent	005	2021-09-13	1043	56	[138]1062	-	[145]418	[58]5	(226)648	(289)996	(239)1872	(236)3845	(227)7555	$^{223}0.44N^{1.0}$
472	Thales Group	US	cogent	006	2022-05-14	508	70	[96]550	-	[142]412	[88]8	(215)587	(238)706	(192)1328	(195)2767	(184)5675	$^{177}0.37N^{1.0}$
473	Thales Group	US	cogent	007	2023-01-30	597	72	[95]550	-	[188]485	[40]3	(197)516	(239)708	(191)1321	(199)2839	(190)5789	$^{211}0.34N^{1.0}$
474	TigerIT Americas LLC	US	tiger	2	2018-10-29	416	518	[405]2052	k	[173]461	[309]15	(383)1816	(390)1921	(345)3833	(337)7526	(331)14820	$^{231}0.83N^{1.0}$
475	TigerIT Americas LLC	US	tiger	3	2018-10-30	416	518	[420]2052	k	[174]461	[569]37431	(86)191	(82)189	-	-	-	
476	TigerIT Americas LLC	US	tiger	004	2025-08-04	248	194	[328]2048	-	[168]456	[289]14	(288)976	(286)982	(250)1997	(230)3773	(244)8432	$^{403}0.13N^{1.1}$
477	Toshiba	JP	toshiba	0	2018-10-30	961	105	[159]1548	k	[404]876	[163]12	(512)6153	(518)6236	(441)12221	(436)25355	(426)49448	$^{442}0.36N^{1.2}$
478	Toshiba	JP	toshiba	1	2018-10-30	961	105	[447]2060	k	[403]875	[572]44701	(510)6007	(522)6355	-	-	-	
479	Toshiba	JP	toshiba	002	2025-07-08	1080	75	[434]2056	-	[504]1348	[200]12	(373)1685	(380)1757	(326)3185	(330)6968	(322)13797	$^{290}0.60N^{1.0}$
480	Toshiba	JP	toshiba	003	2026-01-09	1216	83	[460]2084	-	[569]2164	[511]59	(467)4038	(468)3922	(402)7488	(397)14853	(390)28943	$^{67}3.32N^{1.0}$
481	Toshiba	JP	toshiba	004	2026-09-16	1062	200	[423]2056	-	[540]1748	[455]32	(440)3117	(448)3101	(379)5753	(375)11252	(373)21311	$^{52}3.62N^{1.0}$
482	Toss	KR	toss	000	2025-09-22	970	91	[232]2048	-	[436]1028	[370]18	(213)572	(213)571	(173)1119	(178)2413	(183)5647	$^{357}0.13N^{1.1}$
483	Toss	KR	toss	001	2026-08-28	982	91	[330]2048	-	[274]625	[358]17	(214)576	(216)576	(172)1117	(184)2456	(173)5014	$^{279}0.22N^{1.0}$
484	Tripleize	JP	aize	001	2021-08-06	262	150	[287]2048	-	[41]222	[117]9	(429)2870	(446)3080	-	-	-	
485	Tripleize	JP	aize	002	2023-08-30	251	32	[321]2048	-	[29]185	[209]12	(441)3119	(444)3037	(378)5662	(376)11312	(375)22897	$^{84}2.00N^{1.0}$
486	Tripleize	JP	aize	006	2026-04-08	270	27	[446]2060	-	[202]501	[319]15	(500)5346	(500)5326	(428)10144	(425)20404	(420)46059	$^{308}1.76N^{1.0}$
487	TrueSight Laboratories Tech LLC	AE	tsight	000	2026-08-03	1189	106	[118]1024	-	[553]1859	[315]15	(104)254	(108)259	(78)465	(86)1049	(84)2257	$^{331}0.07N^{1.1}$
488	Trueface.ai	US	trueface	000	2021-01-27	247	119	[167]2000	-	[101]363	[232]13	(108)271	(130)327	(101)614	(100)1239	(96)2678	$^{212}0.15N^{1.0}$
489	TuringTech.vip	CN	turingtechvip	001	2022-09-29	151	161	[337]2048	-	[314]685	[265]13	(559)22085	(565)17480	-	-	-	
490	Unissey	FR	unissey	001	2023-10-24	0	1052	[492]4096	-	[111]370	[553]7732	(16)15	(12)14	-	-	-	
491	Unissey	FR	unissey	002	2024-02-23	0	1055	[495]4096	-	[113]371	[551]5962	(56)119	(54)117	-	-	-	
492	Unissey	FR	unissey	003	2025-01-17	0	1100	[500]4096	-	[98]359	[497]47	(126)318	(124)312	(122)701	(125)1536	(108)2892	$^{391}0.05N^{1.1}$
493	UniverseAI	SG	universeai	000	2026-08-11	333	1362	[83]520	-	[80]325	[540]1125	(26)30	(36)73	(25)111	(40)432	(43)1021	$^{9}3042.36N^{0.2}$
494	Universidade de Coimbra	PT	visteam	000	2024-07-26	569	46	[373]2048	-	[470]1187	[219]13	(457)3619	(473)4053	(396)7043	(402)15723	(391)28968	$^{156}2.11N^{1.0}$
495	Universidade de Coimbra	PT	visteam	001	2025-05-28	569	46	[357]2048	-	[477]1207	[228]13	(458)3635	(476)4113	-	-	-	
496	VDF Software	CA	vdfsoftware	001	2025-04-25	5949	0	[263]2048	-	[86]339	[332]16	(431)2912	(440)2917	-	-	-	
497	VDF Software	CA	vdfsoftware	002	2025-10-01	5989	5	[398]2052	-	[438]1044	[402]23	(539)8765	(546)8737	(411)8209	(411)16739	(450)71934	$^{48}10.02N^{0.9}$
498	VNIS Joint Stock	VN	vnis	000	2024-01-05	278	2988	[295]2048	-	[156]435	[193]12	(227)648	(224)633	-	-	-	
499	Veridas Digital Authentication Solutions S.L.	ES	veridas	001	2021-03-05	347	875	[227]2048	-	[211]509	[234]13	(476)4407	(502)5469	(429)10350	(426)20655	(414)41264	$^{100}3.40N^{1.0}$
500	Veridas Digital Authentication Solutions S.L.	ES	veridas	002	2021-07-06	347	870	[229]2048	-	[208]507	[145]10	(178)459	(128)325	(116)685	(110)1365	(101)2730	$^{337}0.09N^{1.1}$
501	Veridas Digital Authentication Solutions S.L.	ES	veridas	003	2021-11-09	346	870	[315]2048	-	[209]508	[109]9	(177)457	(129)327	(120)699	(113)1401	(154)3954	$^{445}0.02N^{1.2}$
502	Veridas Digital Authentication Solutions S.L.	ES	veridas	004	2023-02-03	1074	608	[340]2048	-	[252]579	[261]13	(175)440	(171)426	(102)618	(54)620	(145)3775	$^{83}0.22N^{1.0}$
503	Verigram	KZ	verigram	000	2023-05-17	1553	20	[499]4096	-	[309]681	[517]68	(237)710	(234)688	(123)706	(105)1312	(98)2696	$^{23}16.28N^{0.7}$
504	Verigram	KZ	verigram	001	2023-10-26	1553	20	[525]4104	-	[322]696	[504]51	(277)929	(275)929	(190)1315	(192)2698	(193)5822	$^{41}1.67N^{0.9}$
505	Verigram	KZ	verigram	002	2026-09-08	1486	69	[569]8228	-	[528]1546	[549]4637	(258)832	(256)833	(213)1580	(213)3202	(207)6497	$^{157}0.45N^{1.0}$
506	Verijelas	ID	verijelas	000	2022-10-11	248	11	[361]2048	-	[70]310	[273]14	(19)20	(20)26	-	-	-	
507	Viante.AI	AE	viante	000	2023-10-02	1234	19	[440]2056	-	[444]1079	[171]12	(337)1336	(327)1249	(340)3733	(291)5411	(291)10878	$^{376}0.27N^{1.1}$
508	Viante.AI	AE	viante	001	2025-04-18	2541	19	[428]2056	-	[466]1160	[191]12	(306)1072	(295)1008	(268)2151	(266)4443	(268)9382	$^{392}0.16N^{1.1}$
509	Viante.AI	AE	viante	002	2025-12-16	1234	19	[439]2056	-	[546]1819	[351]17	(343)1425	(358)1531	(316)3052	(315)6290	(323)13812	$^{373}0.31N^{1.1}$
510	Viante.AI	AE	viante	003	2026-05-25	1234	19	[432]2056	-	[545]1812	[329]16	(348)1449	(359)1539	(322)3117	(321)6531	(320)13442	$^{361}0.34N^{1.1}$
511	Vietnam Payment Solution	VN	vnpay	001	2024-10-15	566	3948	[255]2048	-	[462]1148	[161]12	(223)638	(226)637	(185)1214	(180)2440	(205)6391	$^{372}0.13N^{1.1}$
512	Vietnam Posts and Telecommunications Group	VN	vnpt	001	2022-05-05	361	235	[194]2048	-	[200]491	[389]20	(254)813	(262)867	(217)1626	(219)3346	(213)6808	$^{193}0.43N^{1.0}$
513	Vietnam Posts and Telecommunications Group	VN	vnpt	002	2022-09-08	547	235	[176]2048	-	[281]636	[350]16	(262)857	(268)897	(224)1700	(217)3320	(209)6622	$^{118}0.52N^{1.0}$
514	Vietnam Posts and Telecommunications Group	VN	vnpt	003	2023-06-26	547	235	[258]2048	-	[378]816	[327]16	(272)904	(276)933	(226)1719	(226)3498	(220)6933	$^{113}0.55N^{1.0}$
515	Vietnam Posts and Telecommunications Group	VN	vnpt	004	2023-12-05	1088	351	[385]2048	-	[493]1263	[344]16	(280)941	(269)904	(222)1698	(224)3485	(218)6918	$^{153}0.50N^{1.0}$
516	Vietnam Posts and Telecommunications Group	VN	vnpt	005	2024-11-12	1088	351	[365]2048	-	[471]1189	[332]16	(278)940	(278)939	(221)1697	(221)3404	(215)6845	$^{70}0.69N^{1.0}$
517	Viettel Artificial Intelligence	VN	viettelai	000	2023-07-11	277	312	[551]4861	-	[74]315	[420]27	(156)392	(201)537	(161)1012	(143)1753	(133)3437	$^{37}1.28N^{0.9}$
518	Viettel Artificial Intelligence	VN	viettelai	001	2023-12-18	515	313	[550]4858	-	[333]710	[423]28	(186)494	(180)459	(160)1010	(161)1967	-	$^{161}0.27N^{1.0}$
519	Viettel Artificial Intelligence	VN	viettelai	002	2025-04-14	515	259	[549]4858	-	[335]714	[415]27	(159)397	(199)532	(148)905	(147)1771	(115)3007	$^{32}1.40N^{0.9}$
520	Viettel Artificial Intelligence	VN	viettelai	003	2026-02-02	1471	198	[410]2052	-	[572]3044	[321]15	(200)522	(181)465	(147)892	(149)1781	(137)3624	$^{74}0.35N^{1.0}$

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N = Num. enrolled subjects
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T = 0 → Investigation
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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
											N=1.6M	N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
521	Viettel Artificial Intelligence	VN	viettelai	004	2026-09-11	1495	357	[403]2052	-	[557]1902	[276]14	[(212)]565	[(212)]564	[(135)]792	[(165)]2081	[(136)]3533	[259]$0.19N^{1.0}$
522	Viettel Group	VN	vts	000	2021-03-12	250	257	[280]2048	-	[162]447	[543]2295	[(9)]3	[(7)]4	[(3)]6	[(5)]11	-	[20]$0.61N^{0.6}$
523	Viettel Group	VN	vts	001	2021-07-16	352	600	[322]2048	-	[160]444	[397]21	[(404)]2081	[(427)]2487	[(365)]4644	[(360)]9313	[(359)]18713	[102]$1.53N^{1.0}$
524	Viettel Group	VN	vts	002	2022-02-08	244	600	[319]2048	-	[412]903	[436]29	[(419)]2485	[(426)]2485	[(367)]4678	[(362)]9370	[(361)]18833	[110]$1.49N^{1.0}$
525	Viettel Group	VN	vts	003	2022-07-14	493	468	[341]2048	-	[124]393	[463]34	[(418)]2482	[(405)]2075	[(348)]3894	[(345)]7883	[(346)]15891	[134]$1.18N^{1.0}$
526	Viettel Group	VN	vts	004	2024-09-03	491	887	[307]2048	-	[146]420	[429]29	[(403)]2073	[(406)]2075	[(350)]3925	[(346)]7883	[(341)]15770	[127]$1.19N^{1.0}$
527	Viettel Group	VN	vts	005	2025-01-21	491	887	[171]2048	-	[132]402	[437]29	[(402)]2070	[(404)]2071	[(349)]3917	[(347)]7893	[(345)]15828	[142]$1.16N^{1.0}$
528	Vigilant Solutions	US	vigilant	5	2018-10-30	335	122	[156]1544	k	[356]762	[378]19	-	[(375)]1720	-	-	-	
529	Vigilant Solutions	US	vigilant	6	2018-10-30	337	122	[157]1544	k	[377]816	[395]21	-	[(373)]1713	-	-	-	
530	Vigilant Solutions	US	vigilantsolutions	007	2021-01-08	340	51	[155]1544	-	[237]552	[349]16	[(362)]1570	[(339)]1352	[(308)]2911	[(307)]5966	[(297)]11466	[374]$0.27N^{1.1}$
531	Vigilant Solutions	US	vigilantsolutions	008	2021-07-23	340	51	[158]1544	-	[99]360	[255]13	[(345)]1442	[(307)]1061	[(281)]2330	[(295)]5520	[(274)]9499	[422]$0.11N^{1.1}$
532	VinBigData	VN	vinbigdata	001	2024-05-07	1534	268	[511]4096	-	[458]1136	[412]25	[(291)]990	[(341)]1384	[(232)]1810	[(277)]4715	[(263)]9281	[71]$0.91N^{1.0}$
533	VinBigData	VN	vinbigdata	002	2024-10-16	1080	268	[211]2048	-	[483]1222	[165]12	[(243)]756	[(241)]721	[(196)]1369	[(193)]2745	[(180)]5514	[169]$0.38N^{1.0}$
534	VinBigData	VN	vinbigdata	003	2025-02-19	1080	268	[243]2048	-	[465]1157	[229]13	[(239)]738	[(243)]731	[(195)]1366	[(212)]3199	[(203)]6303	[343]$0.20N^{1.1}$
535	VinBigData	VN	vinbigdata	004	2025-07-24	1080	270	[303]2048	-	[469]1179	[259]13	[(241)]748	[(240)]717	[(197)]1377	[(207)]3024	[(232)]7726	[412]$0.09N^{1.1}$
536	Virtisha AI	IN	virtisha	000	2026-07-22	469	37	[438]2056	-	[380]822	[353]17	[(319)]1145	[(318)]1146	-	-	-	
537	Visidon	FI	visidon	1	2018-10-30	166	42	[407]2052	k	[302]667	[312]15	[(475)]4370	[(484)]4472	[(413)]8454	[(413)]17262	[(405)]34288	[159]$2.40N^{1.0}$
538	Visidon	FI	vd	002	2021-05-18	248	42	[3]-	-	[1]-	[111]9	-	[(420)]2336	-	-	-	
539	Visidon	FI	vd	003	2021-10-12	497	43	[2]-	-	[2]-	[98]8	-	[(408)]2082	-	-	-	
540	Vision-Box	PT	visionbox	000	2021-09-17	252	274	[444]2059	-	[318]687	[348]16	[(194)]515	[(140)]359	[(143)]855	[(55)]631	[(82)]2096	[27]$2.46N^{0.8}$
541	Vision-Box	PT	visionbox	001	2024-03-19	301	191	[530]4116	-	[66]297	[489]43	[(225)]640	[(225)]634	[(179)]1181	[(176)]2380	[(178)]5276	[276]$0.23N^{1.0}$
542	Vision-Box	PT	visionbox	002	2024-10-31	468	166	[531]4116	-	[127]397	[481]40	[(264)]872	[(263)]874	[(212)]1573	[(208)]3057	[(167)]4801	[38]$1.76N^{0.9}$
543	VisionLabs	RU	visionlabs	6	2018-10-30	360	17	[76]512	1	[60]289	[562]20290	[(30)]36	[(27)]36	[(17)]39	[(15)]44	[(12)]53	[10]$3211.93N^{0.2}$
544	VisionLabs	RU	visionlabs	7	2018-10-30	360	17	[60]512	1	[62]289	[568]34666	[(38)]63	[(35)]63	[(19)]72	[(18)]80	[(16)]115	[12]$2076.32N^{0.2}$
545	VisionLabs	RU	visionlabs	008	2019-06-18	348	17	[66]512	1	[55]272	[559]12747	[(22)]23	[(18)]24	[(10)]26	[(7)]29	[(6)]33	[6]$2539.61N^{0.2}$
546	VisionLabs	RU	visionlabs	009	2020-08-04	689	20	[71]512	-	[178]467	[560]13245	[(23)]23	[(21)]29	[(13)]34	[(17)]61	[(17)]145	[19]$8.88N^{0.6}$
547	VisionLabs	RU	visionlabs	010	2021-02-05	1042	20	[61]512	-	[268]617	[555]11837	[(20)]21	[(23)]32	[(14)]36	[(12)]39	[(8)]43	[7]$3183.79N^{0.2}$
548	VisionLabs	RU	visionlabs	011	2021-10-20	1042	20	[74]512	-	[269]618	[558]12255	[(24)]23	[(17)]23	[(11)]26	[(9)]34	[(11)]51	[16]$301.26N^{0.3}$
549	Vixvizon	AU	vixvizion	009	2022-11-28	580	460	[386]2048	-	[214]516	[267]14	[(155)]389	[(133)]339	[(126)]724	[(142)]1750	[(95)]2622	[322]$0.11N^{1.1}$
550	Vocord	RU	vocord	5	2018-10-30	1035	185	[104]768	k	[362]780	[83]7	[(75)]158	[(88)]204	[(67)]383	[(67)]767	[(59)]1466	[105]$0.12N^{1.0}$
551	Vocord	RU	vocord	6	2018-10-30	1035	185	[573]10240	k	[363]785	[535]243	[(77)]170	[(94)]216	-	-	-	
552	Vsoft	BR	vsoft	001	2026-03-16	9	108	[443]2057	-	[33]191	[518]72	[(547)]11496	[(558)]12598	-	-	-	
553	Vsoft	BR	vsoft	002	2026-07-31	476	108	[442]2057	-	[147]420	[520]74	[(549)]11666	[(557)]12583	-	-	-	
554	WINK	US	wink	000	2023-07-18	507	552	[65]512	-	[12]101	[36]3	[(135)]360	[(141)]359	-	-	-	
555	WINK	US	wink	001	2023-12-21	685	552	[55]512	-	[17]144	[35]3	[(137)]364	[(144)]363	[(114)]683	[(111)]1373	[(111)]2921	[217]$0.17N^{1.0}$
556	WINK	US	wink	002	2024-10-18	376	552	[56]512	-	[375]811	[34]3	[(138)]364	[(145)]363	[(115)]684	[(112)]1375	[(134)]3459	[350]$0.09N^{1.1}$
557	WINK	US	wink	003	2026-05-08	178	90	[225]2048	-	[544]1803	[335]16	[(338)]1341	[(338)]1335	[(290)]2477	[(283)]4939	[(286)]9998	[198]$0.63N^{1.0}$
558	Xailient	AU	xailient	000	2026-05-04	363	230	[242]2048	-	[336]714	[352]17	[(143)]369	[(188)]495	[(131)]775	[(138)]1711	[(124)]3244	[55]$0.50N^{1.0}$
559	Xailient	AU	xailient	001	2026-09-16	280	226	[282]2048	-	[371]799	[339]16	[(145)]369	[(149)]370	[(153)]948	[(155)]1896	[(146)]3840	[425]$0.04N^{1.1}$
560	Xforward AI Technology	CN	xforwardai	000	2020-07-24	236	171	[168]2048	-	[353]753	[262]13	[(485)]4603	[(537)]7647	[(460)]15723	[(430)]23900	[(431)]53729	[426]$0.56N^{1.1}$
561	Xforward AI Technology	CN	xforwardai	001	2021-01-21	332	50	[301]2048	-	[96]355	[338]16	[(474)]4338	[(481)]4384	[(418)]8798	[(419)]18553	[(424)]48993	[436]$0.32N^{1.1}$
562	Xforward AI Technology	CN	xforwardai	002	2021-05-24	691	50	[496]4096	-	[197]490	[374]18	[(519)]6340	[(523)]6400	[(446)]12659	[(453)]31077	[(444)]65158	[433]$0.52N^{1.1}$
563	Youverse	PT	youverse	001	2026-07-17	458	286	[292]2048	-	[454]1128	[456]32	[(316)]1129	[(316)]1136	[(257)]2068	[(247)]4138	[(245)]8464	[72]$0.84N^{1.0}$
564	Yuan High-Tech Development	TW	yuan	000	2024-01-23	356	206	[380]2048	-	[137]408	[174]12	[(199)]522	[(197)]521	[(121)]700	[(114)]1411	[(125)]3308	[60]$0.38N^{1.0}$
565	i2V Systems Pvt Ltd	IN	i2v	001	2025-04-17	331	249	[204]2048	-	[516]1433	[212]12	[(407)]2207	[(413)]2161	[(354)]4133	[(359)]9212	[(366)]19528	[367]$0.46N^{1.1}$
566	ioNetworks Inc	TW	ionetworks	001	2025-05-09	302	99	[199]2048	-	[357]766	[545]2884	[(11)]6	[(13)]14	[(8)]22	[(11)]39	[(14)]73	[24]$0.35N^{0.7}$
567	useB	KR	useb	000	2023-05-08	979	482	[387]2048	-	[261]606	[476]38	[(517)]6280	[(517)]6158	-	-	-	
568	useB	KR	useb	001	2023-09-12	977	482	[262]2048	-	[231]545	[156]11	[(513)]6159	[(509)]5936	[(439)]12125	[(447)]28466	[(441)]62969	[423]$0.58N^{1.1}$
569	useB	KR	useb	002	2024-01-19	977	482	[222]2048	-	[213]513	[157]11	[(516)]6231	[(512)]6094	[(455)]13842	[(451)]29794	[(432)]54153	[400]$0.88N^{1.1}$
570	useB	KR	useb	003	2024-08-22	2023	482	[200]2048	-	[489]1248	[211]12	[(518)]6295	[(520)]6316	[(448)]12799	[(439)]25794	-	[393]$0.94N^{1.1}$
571	useB	KR	useb	004	2024-12-23	2037	443	[316]2048	-	[522]1464	[202]12	[(514)]6168	[(515)]6139	[(443)]12391	[(443)]27431	-	[407]$0.81N^{1.1}$
572	verihubs-inteligensia	ID	verihubs-inteligensia	000	2022-09-29	204	75	[327]2048	-	[51]260	[286]14	[(541)]9715	[(550)]9677	[(470)]18417	[(467)]41160	[(459)]95915	[371]$2.05N^{1.1}$

Notes	
1	Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas).
2	Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates.
3	This multiplier expresses the increase in template size when k images are passed to the template generation function.
4	All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however.
5	Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 479. However in certain cases the model is not correct and should not be used numerically.

Table 12: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.

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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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	DEVELOPER	DEVELOPER	SHORT	SEQ.	VALIDATION	CONFIG[1]	LIB[1]	TEMPLATE GENERATION			FINALIZE[2]	SEARCH DURATION[5] MILLISEC					
	FULL NAME	LOCATION	NAME	NUM.	DATE	DATA (MB)	DATA (MB)	SIZE (B)	MULT[3]	TIME (MS)[4]	TIME (S)	L=1	L=50	L=50	L=50	L=50	POWER LAW
											N=1.6M	N=1.6M	N=1.6M	N=3M	N=6M	N=12M	(μs)
573	verihubs-inteligensia	ID	verihubs-inteligensia	001	2023-06-27	352	16	[364]2048	-	[183]475	[185]12	[(548)]11526	[(555)]11418	-	-	-	

Notes	
1	Configuration size does not capture static data present in libraries. Libraries are included but the size also includes any ancillary libraries for image processing (e.g. openCV) or numerical computation (e.g. blas).
2	Finalization is the processing of converting N = 1600000 templates into a searchable data structure an operation which can be a simple copy, or the building of an index or tree, for example. The duration of the operation may be data dependent, and may not be linear in the number of input templates.
3	This multiplier expresses the increase in template size when k images are passed to the template generation function.
4	All durations are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors. Estimates are made by wrapping the API function call in calls to std::chrono::high_resolution_clock which on the machine in (3) counts 1ns clock ticks. Precision is somewhat worse than that however.
5	Search durations are measured as in the prior note. The power-law model in the final column mostly fits the empirical results in Figure 479. However in certain cases the model is not correct and should not be used numerically.

Table 13: Summary of algorithms and properties included in this report. The blue superscripts give ranking for the quantity in that column. Missing search durations, denoted by "-", are absent because those runs were not executed, usually because we did not run on the larger galleries. Caution: The power-law model is sometimes an incorrect model. It is included here only to show broad sublinear behavior, which is flagged in green. The models should not be used for prediction.

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
1	3DIVI-005	[98] 0.0207	[97] 0.0304	[97] 0.0415	[97] 0.0533	[97] 0.0646	[390] 0.0735	[390] 0.0884	[391] 0.1148	[101] 0.1580	[98] 0.2316	[98] 0.3033	[98] 0.3740	[98] 0.4285	[387] 0.4742	[388] 0.5329	[385] 0.5975
2	ADIANCE-002						[257] 0.0075	[257] 0.0085	[259] 0.0089						[263] 0.0959	[262] 0.1174	[264] 0.1525
3	ADVANCE-000						[273] 0.0087	[277] 0.0099	[277] 0.0111						[293] 0.1275	[291] 0.1527	[286] 0.1920
4	AFISBIOMETRICS-000						[311] 0.0118	[311] 0.0146	[318] 0.0185						[340] 0.2012	[339] 0.2397	[340] 0.3004
5	AFRENGINE-000						[318] 0.0129	[316] 0.0150	[320] 0.0187						[262] 0.0958	[258] 0.1152	[256] 0.1469
6	AFRENGINE-001						[176] 0.0048	[156] 0.0050	[170] 0.0039						[104] 0.0200	[102] 0.0233	[104] 0.0313
7	AFRENGINE-002						[142] 0.0045	[130] 0.0048	[177] 0.0041						[92] 0.0168	[87] 0.0192	[92] 0.0273
8	AFRENGINE-003						[195] 0.0052	[187] 0.0056	[206] 0.0052						[116] 0.0239	[116] 0.0286	[116] 0.0398
9	AIGROUP-000						[224] 0.0058	[222] 0.0065	[227] 0.0063						[241] 0.0694	[240] 0.0850	[238] 0.1131
10	AIZE-002						[336] 0.0152	[331] 0.0174	[322] 0.0203						[353] 0.2500	[347] 0.2854	[345] 0.3438
11	AIZE-006						[359] 0.0213	[357] 0.0248	[351] 0.0312						[354] 0.2510	[349] 0.2920	[350] 0.3605
12	AKUVOX-001						[260] 0.0079	[269] 0.0093	[269] 0.0102						[271] 0.1009	[267] 0.1240	[268] 0.1609
13	ALCHERA-006						[168] 0.0046	[161] 0.0051	[172] 0.0040						[182] 0.0419	[181] 0.0535	[184] 0.0737
14	ALCHERA-008						[134] 0.0044	[139] 0.0049	[144] 0.0035						[130] 0.0280	[129] 0.0348	[130] 0.0469
15	ALGORITHMSINC-000						[327] 0.0137	[325] 0.0164	[325] 0.0205						[297] 0.1349	[297] 0.1654	[298] 0.2137
16	ALLGOVISION-005						[304] 0.0108	[299] 0.0126	[303] 0.0153						[313] 0.1524	[314] 0.1848	[311] 0.2309
17	ANKE-000	[95] 0.0162	[95] 0.0245	[95] 0.0333	[95] 0.0428	[95] 0.0515	[388] 0.0615	[388] 0.0780	[387] 0.1028	[96] 0.1132	[96] 0.1761	[96] 0.2402	[96] 0.3057	[95] 0.3640	[383] 0.4200	[383] 0.4928	[383] 0.5680
18	ANKE-002	[49] 0.0055	[50] 0.0074	[50] 0.0090	[49] 0.0103	[48] 0.0116	[325] 0.0135	[324] 0.0162	[321] 0.0202	[54] 0.0329	[54] 0.0560	[56] 0.0843	[57] 0.1169	[57] 0.1481	[333] 0.1820	[334] 0.2280	[333] 0.2831
19	ARATEK-001						[189] 0.0050	[183] 0.0055	[199] 0.0049						[237] 0.0686	[238] 0.0843	[237] 0.1082
20	ARMATURA-000						[107] 0.0041	[118] 0.0047	[154] 0.0037						[151] 0.0344	[151] 0.0428	[148] 0.0555
21	AWARE-005	[106] 0.0328	[106] 0.0519	[106] 0.0712	[104] 0.0910	[104] 0.1078	[397] 0.1235	[397] 0.1457	[398] 0.1831	[106] 0.3605	[107] 0.4949	[107] 0.5948	[107] 0.6783	[108] 0.7393	[399] 0.7905	[399] 0.8408	[400] 0.8831
22	AWARE-006	[110] 0.0702	[110] 0.1110	[110] 0.1502	[110] 0.1899	[110] 0.2253	[406] 0.2614	[405] 0.3045	[406] 0.3659								
23	AWARE-007						[343] 0.0162	[337] 0.0189	[340] 0.0257						[276] 0.1053	[269] 0.1246	[274] 0.1676
24	AYONIX-002	[113] 0.3360	[114] 0.4389	[114] 0.5144	[114] 0.5814	[114] 0.6340	[410] 0.6818	[410] 0.7297	[411] 0.7774	[110] 0.8288	[111] 0.9013	[111] 0.9375	[111] 0.9603	[111] 0.9744	[404] 0.9837	[404] 0.9893	[404] 0.9927
25	BIOIDTECHSWISS-001						[166] 0.0046	[165] 0.0051	[138] 0.0034						[156] 0.0355	[152] 0.0429	[147] 0.0548
26	BIOMETRIK-001						[372] 0.0281	[370] 0.0319	[370] 0.0413						[364] 0.2812	[360] 0.3249	[358] 0.3882
27	CAMVI-004	[109] 0.0623	[109] 0.0944	[109] 0.1243	[109] 0.1548	[108] 0.1812	[403] 0.2056	[403] 0.2344	[401] 0.2672	[91] 0.0810	[91] 0.1267	[88] 0.1721	[88] 0.2203	[88] 0.2619	[373] 0.3040	[372] 0.3543	[369] 0.4124
28	CAMVI-005	[111] 0.0849	[111] 0.1255	[111] 0.1631	[111] 0.1989	[111] 0.2298	[405] 0.2585	[404] 0.2915	[405] 0.3246								
29	CANON-001						[196] 0.0052	[191] 0.0057	[179] 0.0042						[200] 0.0491	[198] 0.0606	[204] 0.0826
30	CANON-002						[232] 0.0062	[235] 0.0070	[238] 0.0070						[192] 0.0472	[192] 0.0582	[199] 0.0792
31	CANON-003						[178] 0.0049	[177] 0.0053	[181] 0.0043						[106] 0.0209	[107] 0.0257	[109] 0.0358
32	CIB-000	[14] 0.0022	[14] 0.0030	[15] 0.0037	[15] 0.0044	[17] 0.0049	[221] 0.0057	[231] 0.0069	[226] 0.0062	[25] 0.0139	[26] 0.0240	[27] 0.0373	[28] 0.0525	[28] 0.0689	[251] 0.0859	[255] 0.1109	[255] 0.1454
33	CLEARVIEWAI-000	[4] 0.0017	[4] 0.0023	[4] 0.0028	[9] 0.0034	[11] 0.0039	[162] 0.0046	[188] 0.0056	[196] 0.0047	[16] 0.0066	[18] 0.0121	[18] 0.0194	[19] 0.0287	[19] 0.0385	[202] 0.0493	[216] 0.0662	[215] 0.0873
34	CLEARVIEWAI-001						[140] 0.0044	[126] 0.0047	[126] 0.0032						[168] 0.0389	[170] 0.0497	[171] 0.0676
35	CLEARVIEWAI-002						[136] 0.0044	[129] 0.0048	[146] 0.0036						[186] 0.0438	[187] 0.0566	[195] 0.0778
36	CLEARVIEWAI-003						[103] 0.0041	[75] 0.0042	[90] 0.0024						[113] 0.0224	[115] 0.0285	[113] 0.0370
37	CLOUDWALK-HR-000	[8] 0.0019	[7] 0.0024	[8] 0.0029	[6] 0.0032	[5] 0.0032	[46] 0.0036	[72] 0.0041	[62] 0.0020	[1] 0.0029	[1] 0.0041	[1] 0.0054	[1] 0.0064	[2] 0.0073	[37] 0.0085	[37] 0.0102	[37] 0.0112
38	CLOUDWALK-MT-000						[65] 0.0037	[33] 0.0038	[26] 0.0013						[27] 0.0065	[25] 0.0072	[25] 0.0075
39	CLOUDWALK-MT-001						[55] 0.0037	[22] 0.0037	[20] 0.0012						[16] 0.0045	[11] 0.0051	[10] 0.0042
40	CLOUDWALK-MT-002						[53] 0.0036	[29] 0.0038	[19] 0.0012						[9] 0.0042	[9] 0.0048	[9] 0.0034
41	COGENT-000	[90] 0.0128	[91] 0.0184	[93] 0.0250	[92] 0.0327	[92] 0.0407	[383] 0.0488	[381] 0.0611	[381] 0.0794	[77] 0.0559	[78] 0.0923	[76] 0.1342	[76] 0.1812	[75] 0.2243	[359] 0.2675	[358] 0.3240	[364] 0.3992
42	COGENT-001	[91] 0.0128	[90] 0.0184	[92] 0.0250	[93] 0.0327	[93] 0.0407	[382] 0.0488	[382] 0.0611	[380] 0.0794	[78] 0.0559	[79] 0.0923	[77] 0.1342	[77] 0.1812	[76] 0.2243	[358] 0.2675	[357] 0.3240	[365] 0.3992
43	COGENT-002	[69] 0.0081	[66] 0.0105	[63] 0.0123	[64] 0.0137	[62] 0.0157	[345] 0.0175	[343] 0.0215	[343] 0.0280	[69] 0.0499	[68] 0.0827	[67] 0.1207	[67] 0.1639	[67] 0.2037	[347] 0.2432	[351] 0.2972	[351] 0.3638
44	COGENT-003	[71] 0.0082	[67] 0.0108	[65] 0.0128	[67] 0.0145	[66] 0.0168	[353] 0.0191	[354] 0.0239	[352] 0.0312	[80] 0.0582	[80] 0.0971	[80] 0.1417	[80] 0.1918	[80] 0.2380	[367] 0.2836	[370] 0.3440	[372] 0.4207

Table 14: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = 1, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
45	COGENT-004	[59]0.0066	[53]0.0080	[45]0.0085	[39]0.0080	[31]0.0083	[286]0.0092	[280]0.0106	[292]0.0130	[63]0.0410	[65]0.0720	[65]0.1099	[65]0.1539	[64]0.1974	[348]0.2443	[354]0.3043	[354]0.3757
46	COGENT-006						[149]0.0045	[151]0.0049	[164]0.0038						[165]0.0370	[159]0.0448	[159]0.0602
47	COGENT-007						[138]0.0044	[141]0.0049	[147]0.0036						[148]0.0332	[148]0.0415	[152]0.0566
48	COGNITEC-000	[105]0.0265	[103]0.0423	[103]0.0588	[103]0.0757	[102]0.0894	[395]0.1014	[395]0.1169	[394]0.1381	[100]0.1522	[99]0.2330	[99]0.3051	[99]0.3751	[99]0.4300	[388]0.4779	[387]0.5307	[384]0.5913
49	COGNITEC-001	[93]0.0149	[94]0.0228	[94]0.0312	[94]0.0399	[94]0.0479	[385]0.0546	[384]0.0656	[382]0.0806	[93]0.0963	[93]0.1562	[93]0.2157	[93]0.2771	[93]0.3287	[381]0.3771	[380]0.4343	[379]0.4959
50	COGNITEC-002	[77]0.0101	[80]0.0138	[81]0.0170	[81]0.0201	[81]0.0237	[370]0.0264	[367]0.0309	[366]0.0389	[72]0.0517	[71]0.0879	[72]0.1269	[73]0.1707	[71]0.2098	[350]0.2463	[348]0.2919	[348]0.3535
51	COGNITEC-003	[78]0.0104	[81]0.0140	[82]0.0174	[82]0.0205	[82]0.0238	[371]0.0266	[368]0.0311	[368]0.0401	[71]0.0504	[70]0.0855	[69]0.1235	[69]0.1662	[68]0.2045	[346]0.2403	[346]0.2854	[346]0.3451
52	COGNITEC-004	[64]0.0073	[63]0.0099	[62]0.0118	[59]0.0130	[59]0.0147	[344]0.0163	[338]0.0189	[337]0.0239	[53]0.0325	[53]0.0548	[52]0.0798	[51]0.1074	[50]0.1325	[321]0.1591	[318]0.1952	[316]0.2414
53	COGNITEC-006						[265]0.0081	[258]0.0086	[260]0.0090						[245]0.0777	[244]0.0926	[245]0.1274
54	COGNITEC-007						[241]0.0066	[238]0.0072	[231]0.0066						[209]0.0507	[204]0.0616	[205]0.0826
55	COGNITEC-008						[247]0.0067	[241]0.0074	[241]0.0071						[210]0.0508	[200]0.0607	[208]0.0836
56	COGNITEC-009						[251]0.0069	[243]0.0075	[243]0.0073						[211]0.0510	[201]0.0609	[210]0.0838
57	COGNITEC-010						[225]0.0059	[219]0.0062	[218]0.0056						[145]0.0324	[141]0.0388	[139]0.0516
58	COGNITI-000						[293]0.0098	[297]0.0120	[307]0.0162						[304]0.1436	[306]0.1760	[318]0.2435
59	COMPAL-000						[407]0.3198	[406]0.3198	[404]0.3206						[378]0.3559	[374]0.3655	[357]0.3844
60	CORSIGHT-000						[192]0.0051	[192]0.0057	[182]0.0043						[213]0.0523	[211]0.0641	[206]0.0829
61	CORSIGHT-001						[180]0.0049	[176]0.0053	[186]0.0044						[220]0.0556	[220]0.0688	[218]0.0893
62	CORSIGHT-002						[150]0.0045	[162]0.0051	[153]0.0037						[170]0.0392	[169]0.0491	[173]0.0680
63	CUBOX-000	[7]0.0019	[5]0.0024	[5]0.0028	[4]0.0031	[4]0.0032	[56]0.0037	[102]0.0044	[100]0.0027	[6]0.0039	[6]0.0059	[7]0.0083	[8]0.0111	[8]0.0141	[101]0.0185	[105]0.0252	[108]0.0339
64	CUBOX-001						[94]0.0040	[97]0.0044	[106]0.0028						[112]0.0223	[112]0.0276	[111]0.0361
65	CUBOX-002						[86]0.0039	[99]0.0044	[132]0.0033						[115]0.0237	[117]0.0293	[115]0.0396
66	CUBOX-003						[118]0.0043	[128]0.0048	[176]0.0041						[123]0.0254	[119]0.0317	[120]0.0422
67	CYBERLINK-002	[50]0.0055	[45]0.0068	[41]0.0075	[35]0.0078	[32]0.0084	[288]0.0094	[288]0.0107	[280]0.0114	[32]0.0180	[33]0.0302	[33]0.0460	[32]0.0643	[33]0.0837	[277]0.1058	[279]0.1370	[280]0.1787
68	CYBERLINK-003	[35]0.0041	[34]0.0052	[27]0.0057	[25]0.0058	[25]0.0061	[248]0.0068	[249]0.0078	[252]0.0078	[19]0.0109	[19]0.0175	[20]0.0259	[21]0.0356	[21]0.0468	[223]0.0594	[230]0.0787	[235]0.1072
69	CYBERLINK-006						[100]0.0040	[92]0.0043	[84]0.0023						[102]0.0188	[99]0.0219	[98]0.0291
70	DAHUA-002	[30]0.0035	[28]0.0047	[28]0.0058	[27]0.0067	[28]0.0074	[267]0.0082	[278]0.0100	[274]0.0108	[30]0.0169	[32]0.0294	[31]0.0449	[30]0.0635	[30]0.0817	[272]0.1013	[274]0.1291	[271]0.1638
71	DAHUA-003	[19]0.0026	[19]0.0036	[19]0.0043	[20]0.0050	[20]0.0055	[231]0.0062	[252]0.0080	[244]0.0073	[29]0.0160	[30]0.0280	[29]0.0432	[29]0.0615	[29]0.0794	[268]0.0987	[272]0.1270	[267]0.1587
72	DAON-002						[236]0.0064	[245]0.0077	[248]0.0076						[240]0.0693	[241]0.0889	[241]0.1147
73	DAON-003						[145]0.0045	[145]0.0049	[134]0.0033						[138]0.0308	[137]0.0379	[131]0.0487
74	DATECH-001						[198]0.0052	[203]0.0059	[202]0.0050						[232]0.0641	[231]0.0794	[228]0.1011
75	DEEPGLINT-001	[17]0.0024	[16]0.0032	[14]0.0037	[13]0.0040	[13]0.0043	[182]0.0049	[208]0.0060	[207]0.0052	[12]0.0058	[10]0.0087	[11]0.0119	[11]0.0155	[11]0.0199	[120]0.0249	[124]0.0338	[128]0.0463
76	DEEPSEA-001	[70]0.0081	[70]0.0116	[73]0.0149	[76]0.0182	[76]0.0216	[368]0.0260	[372]0.0332	[372]0.0432	[66]0.0458	[66]0.0752	[64]0.1086	[63]0.1460	[63]0.1812	[342]0.2186	[342]0.2663	[341]0.3213
77	DERMALOG-006	[82]0.0113	[82]0.0142	[78]0.0163	[77]0.0183	[74]0.0200	[361]0.0218	[358]0.0251	[356]0.0329	[75]0.0545	[73]0.0889	[73]0.1271	[72]0.1697	[70]0.2090	[352]0.2498	[353]0.3028	[352]0.3670
78	DERMALOG-007	[88]0.0125	[88]0.0170	[88]0.0214	[88]0.0264	[87]0.0309	[377]0.0356	[378]0.0432	[378]0.0579	[92]0.0910	[92]0.1453	[92]0.2009	[92]0.2602	[92]0.3134	[380]0.3649	[379]0.4289	[380]0.5007
79	DERMALOG-008	[52]0.0057	[52]0.0077	[54]0.0095	[54]0.0110	[53]0.0128	[334]0.0148	[334]0.0180	[335]0.0223	[70]0.0501	[69]0.0850	[70]0.1247	[71]0.1692	[72]0.2105	[355]0.2541	[355]0.3102	[355]0.3762
80	DERMALOG-010						[217]0.0056	[202]0.0059	[180]0.0043						[212]0.0519	[212]0.0643	[212]0.0843
81	DERMALOG-011						[280]0.0089	[289]0.0107	[289]0.0128						[316]0.1543	[311]0.1833	[312]0.2314
82	DERMALOG-012						[283]0.0090	[287]0.0107	[294]0.0133						[312]0.1497	[307]0.1783	[304]0.2238
83	DERMALOG-013						[281]0.0089	[285]0.0106	[293]0.0132						[314]0.1516	[308]0.1803	[306]0.2264
84	DERMALOG-014						[363]0.0232	[363]0.0279	[362]0.0358						[317]0.1544	[310]0.1833	[309]0.2295
85	DERMALOG-015						[170]0.0047	[140]0.0049	[155]0.0037						[171]0.0392	[165]0.0481	[163]0.0634
86	DERMALOG-016						[81]0.0039	[88]0.0043	[109]0.0029						[110]0.0222	[113]0.0282	[117]0.0403
87	DERMALOG-017						[82]0.0039	[89]0.0043	[110]0.0029						[121]0.0250	[118]0.0307	[119]0.0408
88	DILUSENSE-000						[313]0.0123	[312]0.0146	[316]0.0180						[332]0.1814	[328]0.2149	[325]0.2644

Table 15: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
89	DILUSENSE-001						239 0.0065	230 0.0069	236 0.0069						230 0.0632	227 0.0778	227 0.1006
90	EINETWORKSINDIA-000						285 0.0091	284 0.0105	281 0.0115						284 0.1139	281 0.1413	279 0.1785
91	EINETWORKSINDIA-001						284 0.0090	280 0.0103	278 0.0113						288 0.1196	286 0.1476	283 0.1889
92	EINETWORKSINDIA-002						152 0.0045	157 0.0050	135 0.0033						161 0.0361	161 0.0463	162 0.0634
93	FACEPHI-000						209 0.0054	199 0.0059	204 0.0050						191 0.0468	188 0.0575	188 0.0748
94	FACEPHI-001						102 0.0041	93 0.0043	82 0.0023						87 0.0157	88 0.0192	89 0.0259
95	FIRSTCREDITKZ-001						219 0.0057	223 0.0066	215 0.0055						118 0.0240	114 0.0284	112 0.0368
96	FUJITSULAB-001						282 0.0089	274 0.0098	276 0.0111						303 0.1403	302 0.1723	299 0.2165
97	GORILLA-002	100 0.0213	100 0.0359	101 0.0528	102 0.0716	103 0.0895	396 0.1088	396 0.1367	396 0.1765	103 0.1828	104 0.2787	104 0.3654	104 0.4485	104 0.5168	392 0.5823	392 0.6508	392 0.7180
98	GORILLA-005	38 0.0044	47 0.0070	58 0.0102	62 0.0136	67 0.0170	356 0.0204	362 0.0272	365 0.0373	79 0.0566	81 0.0973	82 0.1432	81 0.1937	81 0.2398	369 0.2862	369 0.3437	370 0.4150
99	GORILLA-007						306 0.0108	302 0.0128	295 0.0145						335 0.1862	333 0.2198	329 0.2716
100	GORILLA-008						255 0.0071	253 0.0081	258 0.0089						319 0.1557	313 0.1847	314 0.2340
101	GRIAULE-001						160 0.0046	154 0.0050	161 0.0038						174 0.0402	168 0.0487	164 0.0636
102	HISIGN-001						234 0.0063	234 0.0070	237 0.0070						227 0.0613	225 0.0754	224 0.0969
103	HYPERVERGE-003						201 0.0053	197 0.0058	223 0.0059						239 0.0689	235 0.0826	236 0.1081
104	HYUNDAI-ROBOTICSLAB-000						279 0.0089	276 0.0099	283 0.0119						254 0.0872	251 0.1082	251 0.1405
105	HYUNDAI-ROBOTICSLAB-001						269 0.0084	267 0.0092	271 0.0104						257 0.0928	257 0.1151	257 0.1482
106	HYUNDAI-ROBOTICSLAB-002						155 0.0045	136 0.0048	140 0.0034						195 0.0481	197 0.0598	196 0.0784
107	HYUNDAI-ROBOTICSLAB-003						151 0.0045	133 0.0048	141 0.0034						193 0.0473	194 0.0589	192 0.0771
108	HZAILU-001						222 0.0058	200 0.0059	201 0.0049						214 0.0524	209 0.0630	198 0.0791
109	HZAILU-002						223 0.0058	221 0.0064	208 0.0052						199 0.0488	195 0.0597	183 0.0728
110	HZAILU-003						235 0.0063	236 0.0071	234 0.0067						233 0.0668	239 0.0850	239 0.1136
111	HZAILU-004						188 0.0050	184 0.0055	198 0.0049						207 0.0503	213 0.0655	220 0.0902
112	I2V-001						249 0.0068	237 0.0071	235 0.0069						386 0.4618	385 0.5191	386 0.6022
113	ICEHOUSE-001						214 0.0056	207 0.0060	224 0.0059						249 0.0826	248 0.1022	248 0.1331
114	ICM-001						312 0.0122	308 0.0138	308 0.0164						339 0.1997	338 0.2375	336 0.2940
115	IDBIO-000						301 0.0106	296 0.0120	299 0.0151						291 0.1231	288 0.1486	289 0.1997
116	IDEMIA-003	81 0.0110	86 0.0151	86 0.0196	85 0.0238	84 0.0281	375 0.0313	375 0.0368	374 0.0504	87 0.0717	86 0.1147	86 0.1614	86 0.2113	85 0.2553	372 0.2976	371 0.3537	373 0.4334
117	IDEMIA-004	80 0.0107	84 0.0148	85 0.0192	84 0.0233	83 0.0277	374 0.0312	374 0.0367	375 0.0512	58 0.0373	55 0.0587	54 0.0833	53 0.1100	52 0.1340	320 0.1580	317 0.1911	319 0.2482
118	IDEMIA-005	84 0.0118	87 0.0167	90 0.0218	89 0.0270	88 0.0317	378 0.0357	377 0.0425	377 0.0579	65 0.0440	64 0.0689	60 0.0964	59 0.1254	58 0.1513	330 0.1762	325 0.2113	328 0.2698
119	IDEMIA-006	87 0.0124	89 0.0171	89 0.0218	87 0.0263	86 0.0302	376 0.0321	373 0.0356	373 0.0471	62 0.0409	59 0.0620	57 0.0850	52 0.1097	49 0.1309	309 0.1486	304 0.1738	303 0.2200
120	IDEMIA-007	47 0.0050	48 0.0071	48 0.0089	50 0.0106	51 0.0124	329 0.0142	330 0.0171	333 0.0220	36 0.0202	36 0.0335	34 0.0491	33 0.0663	31 0.0825	270 0.0999	268 0.1240	273 0.1645
121	IDEMIA-008	5 0.0018	6 0.0024	6 0.0029	5 0.0032	7 0.0035	83 0.0039	115 0.0046	128 0.0033	3 0.0034	3 0.0051	5 0.0069	5 0.0087	5 0.0102	73 0.0123	69 0.0146	69 0.0186
122	IDEMIA-009						95 0.0040	78 0.0042	88 0.0024						41 0.0094	38 0.0103	39 0.0123
123	IDEMIA-010						18 0.0034	12 0.0036	15 0.0011						15 0.0045	14 0.0052	13 0.0043
124	IDEMIA-011						11 0.0033	8 0.0035	13 0.0011						10 0.0043	12 0.0051	15 0.0044
125	IDEMIA-012						17 0.0033	6 0.0034	12 0.0011						12 0.0043	10 0.0049	17 0.0048
126	IDEMIA-013						4 0.0031	1 0.0033	1 0.0007						7 0.0031	7 0.0034	7 0.0013
127	IDENTY-001							415 1.0000									
128	IDENTY-002						158 0.0046	155 0.0050	163 0.0038						124 0.0261	120 0.0321	122 0.0431
129	IMAGUS-005	33 0.0039	33 0.0052	31 0.0061	29 0.0067	30 0.0077	277 0.0088	281 0.0103	275 0.0109	39 0.0212	39 0.0357	40 0.0539	40 0.0755	38 0.0967	287 0.1183	287 0.1485	284 0.1893
130	IMAGUS-008						400 0.1625	398 0.1704	397 0.1823								
131	IMDS-SOFTWARE-001						199 0.0052	206 0.0060	205 0.0050						221 0.0569	221 0.0693	221 0.0903
132	IMDS-SOFTWARE-002						207 0.0054	201 0.0059	200 0.0049						218 0.0541	215 0.0661	211 0.0838

Table 16: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
133	IMPERIAL-000	34 0.0040	35 0.0054	36 0.0067	38 0.0079	40 0.0093	307 0.0112	309 0.0139	313 0.0178	49 0.0286	51 0.0503	51 0.0779	54 0.1116	56 0.1455	334 0.1844	337 0.2341	337 0.2951
134	INCODE-003	94 0.0155	96 0.0247	96 0.0348	96 0.0463	96 0.0571	389 0.0674	389 0.0856	390 0.1114	102 0.1627	102 0.2507	102 0.3322	100 0.4122	100 0.4772	391 0.5368	391 0.6059	391 0.6766
135	INCODE-004	56 0.0061	59 0.0087	59 0.0110	61 0.0136	64 0.0161	348 0.0185	352 0.0236	350 0.0309	73 0.0532	74 0.0908	75 0.1334	75 0.1809	77 0.2245	360 0.2675	361 0.3249	361 0.3932
136	INCODE-006						84 0.0039	108 0.0045	104 0.0028						89 0.0160	96 0.0212	102 0.0298
137	INCODE-007						61 0.0037	68 0.0041	73 0.0021						63 0.0114	77 0.0155	83 0.0229
138	INNOVATRICS-004	114 0.3594	113 0.3629	113 0.3688	112 0.3754	112 0.3813	408 0.3870	408 0.3960	408 0.4135	107 0.4234	106 0.4642	106 0.5073	106 0.5522	105 0.5902	395 0.6274	393 0.6736	393 0.7253
139	INNOVATRICS-005	41 0.0046	41 0.0063	42 0.0078	45 0.0092	45 0.0106	314 0.0124	315 0.0149	314 0.0178	55 0.0343	56 0.0590	58 0.0886	58 0.1222	59 0.1544	337 0.1881	336 0.2321	334 0.2874
140	INNOVATRICS-008						171 0.0047	144 0.0049	120 0.0031						139 0.0309	130 0.0353	127 0.0463
141	INNOVATRICS-009						123 0.0043	103 0.0044	99 0.0027						84 0.0154	82 0.0170	78 0.0214
142	INNOVATRICS-010						121 0.0043	110 0.0045	112 0.0030						91 0.0167	86 0.0192	90 0.0265
143	INNOVATRICS-011						110 0.0042	104 0.0044	86 0.0023						60 0.0109	46 0.0122	46 0.0153
144	INNOVATRICS-012						93 0.0040	82 0.0042	85 0.0023						46 0.0099	41 0.0110	41 0.0138
145	INNOVATRICS-013						52 0.0036	53 0.0040	42 0.0017						49 0.0100	43 0.0116	43 0.0142
146	INNOVATRICS-014						24 0.0034	20 0.0037	24 0.0013						19 0.0055	19 0.0059	19 0.0061
147	INNOVATRICS-015						20 0.0034	15 0.0036	27 0.0013						18 0.0051	18 0.0056	18 0.0058
148	INNOVATRICS-016						15 0.0033	7 0.0035	10 0.0010						17 0.0047	17 0.0052	16 0.0048
149	INSPUR-000						177 0.0048	175 0.0053	169 0.0039						217 0.0539	217 0.0665	214 0.0865
150	INSPUR-001						161 0.0046	158 0.0050	148 0.0036						194 0.0478	193 0.0588	191 0.0763
151	INSPUR-002						128 0.0043	131 0.0048	122 0.0031						201 0.0493	206 0.0619	202 0.0820
152	INTELLIVISION-002						386 0.0577	385 0.0694	384 0.0881								
153	INTELRESEARCH-001						104 0.0041	94 0.0043	91 0.0024						79 0.0132	71 0.0149	66 0.0181
154	INTEMA-000						97 0.0040	86 0.0043	93 0.0024						103 0.0193	103 0.0235	101 0.0294
155	INTEMA-001						90 0.0039	60 0.0040	63 0.0020						86 0.0157	89 0.0194	87 0.0246
156	INTOZI-001						259 0.0076	256 0.0083	254 0.0081						270 0.1061	270 0.1285	272 0.1640
157	IONETWORKS-001						369 0.0261	369 0.0313	369 0.0404						282 0.1113	277 0.1349	278 0.1723
158	IREX-000	24 0.0031	24 0.0042	25 0.0051	26 0.0060	26 0.0068	262 0.0080	272 0.0095	273 0.0107	52 0.0313	52 0.0539	53 0.0815	56 0.1137	55 0.1442	329 0.1755	332 0.2181	330 0.2718
159	IRISID-001						258 0.0076	259 0.0087	257 0.0085						269 0.0995	268 0.1189	266 0.1531
160	IRISID-002						253 0.0070	244 0.0076	233 0.0067						244 0.0770	245 0.0950	244 0.1213
161	ISTL-001						266 0.0082	270 0.0093	288 0.0127						258 0.0933	256 0.1138	258 0.1498
162	ISYSTEMS-002	76 0.0101	79 0.0135	80 0.0169	79 0.0197	80 0.0228	366 0.0256	366 0.0304	367 0.0398	90 0.0779	90 0.1258	91 0.1759	90 0.2299	90 0.2758	376 0.3204	377 0.3763	375 0.4401
163	ISYSTEMS-003	75 0.0089	69 0.0115	69 0.0139	69 0.0158	70 0.0177	355 0.0198	351 0.0234	348 0.0303	84 0.0647	84 0.1056	84 0.1502	84 0.1986	83 0.2402	365 0.2819	364 0.3351	363 0.3976
164	KAKAO-001						88 0.0039	87 0.0043	78 0.0022						98 0.0182	100 0.0220	99 0.0291
165	KAKAO-002						59 0.0037	46 0.0039	48 0.0018						68 0.0119	62 0.0141	60 0.0174
166	KEDACOM-001	83 0.0116	75 0.0130	67 0.0135	60 0.0133	57 0.0135	328 0.0141	317 0.0151	312 0.0176	41 0.0241	41 0.0360	39 0.0513	34 0.0689	34 0.0866	278 0.1060	275 0.1327	275 0.1694
167	KLOUDSPOT-001						308 0.0113	303 0.0130	297 0.0148						328 0.1730	324 0.2071	321 0.2534
168	KNOWUTECH-000						229 0.0061	238 0.0072	240 0.0070						247 0.0805	246 0.0995	247 0.1329
169	KOGNIZA-000						174 0.0048	179 0.0053	189 0.0045						198 0.0486	203 0.0615	193 0.0776
170	LIGHTVISION-000						326 0.0136	323 0.0158	323 0.0204						323 0.1646	319 0.1959	317 0.2422
171	LINECLOVA-002						156 0.0045	142 0.0049	113 0.0030						139 0.0307	135 0.0374	134 0.0497
172	LINECLOVA-003						132 0.0044	132 0.0048	116 0.0030						157 0.0356	157 0.0443	155 0.0586
173	LOOKMAN-003	86 0.0123	83 0.0144	77 0.0158	70 0.0168	71 0.0178	350 0.0188	342 0.0212	341 0.0260	64 0.0438	62 0.0687	61 0.0978	61 0.1296	60 0.1581	336 0.1879	335 0.2294	332 0.2756
174	LOOKMAN-005	85 0.0118	77 0.0134	70 0.0142	66 0.0144	61 0.0150	341 0.0160	332 0.0176	330 0.0213	51 0.0310	49 0.0480	46 0.0698	46 0.0954	46 0.1216	310 0.1491	316 0.1890	315 0.2381
175	LSSYTECH-000						79 0.0039	76 0.0042	72 0.0021						65 0.0116	67 0.0143	65 0.0180
176	LSSYTECH-001						58 0.0037	43 0.0039	38 0.0016						34 0.0079	35 0.0091	35 0.0105

Table 17: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = 1, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
177	LUXAND-001						330 0.0144	328 0.0166	327 0.0207						363 0.2802	359 0.3247	360 0.3916
178	MAXVISION-000						317 0.0128	314 0.0146	311 0.0169						326 0.1706	321 0.2023	322 0.2550
179	MAXVISION-001						163 0.0046	148 0.0049	133 0.0033						154 0.0351	154 0.0435	153 0.0581
180	MAXVISION-002						169 0.0047	146 0.0049	117 0.0030						143 0.0318	139 0.0385	135 0.0500
181	MAXVISION-003						172 0.0047	173 0.0052	166 0.0038						197 0.0485	199 0.0606	201 0.0812
182	MAXVISION-004						74 0.0038	80 0.0042	64 0.0020						99 0.0182	98 0.0217	100 0.0293
183	MEGVII-003						191 0.0051	209 0.0060	209 0.0052						127 0.0277	136 0.0376	142 0.0520
184	MEGVII-004						49 0.0036	62 0.0040	50 0.0018						25 0.0062	29 0.0078	28 0.0084
185	MEGVII-005						50 0.0036	61 0.0040	53 0.0018						29 0.0067	31 0.0084	30 0.0094
186	METSAKUURCOMPANY-000						213 0.0055	218 0.0062	220 0.0057						229 0.0632	232 0.0795	231 0.1043
187	METSAKUURCOMPANY-001						233 0.0063	228 0.0069	232 0.0067						248 0.0821	249 0.1024	250 0.1363
188	METSAKUURCOMPANY-002						276 0.0088	279 0.0100	261 0.0091						274 0.1033	271 0.1257	263 0.1523
189	MIAXIS-001						238 0.0065	225 0.0067	225 0.0062						222 0.0584	222 0.0717	222 0.0920
190	MIAXIS-002						237 0.0064	229 0.0069	228 0.0063						225 0.0611	226 0.0758	225 0.0970
191	MICROFOCUS-005	115 0.4269	115 0.5527	115 0.6355	116 0.7024	116 0.7503	412 0.7876	412 0.8234	413 0.8601	111 0.8338	112 0.9113	112 0.9468	112 0.9667	112 0.9771	403 0.9836	403 0.9880	403 0.9924
192	MICROSOFT-003	28 0.0034	32 0.0050	33 0.0064	36 0.0078	38 0.0092	302 0.0107	307 0.0135	309 0.0166	50 0.0288	50 0.0503	50 0.0763	50 0.1067	54 0.1359	324 0.1680	327 0.2116	326 0.2644
193	MICROSOFT-004	25 0.0032	27 0.0047	29 0.0060	32 0.0075	35 0.0087	296 0.0103	304 0.0131	305 0.0159	47 0.0268	48 0.0470	49 0.0716	48 0.1007	48 0.1291	322 0.1610	323 0.2052	323 0.2590
194	MICROSOFT-005	22 0.0031	29 0.0047	35 0.0066	43 0.0084	43 0.0103	321 0.0131	326 0.0164	317 0.0185	43 0.0243	44 0.0432	44 0.0658	44 0.0913	45 0.1172	307 0.1476	315 0.1874	307 0.2272
195	MICROSOFT-006	26 0.0032	31 0.0049	34 0.0065	42 0.0081	42 0.0096	310 0.0117	310 0.0144	306 0.0160	24 0.0134	24 0.0233	25 0.0346	23 0.0462	22 0.0578	242 0.0713	242 0.0903	242 0.1156
196	MOBBL-000						309 0.0116	305 0.0132	304 0.0156						327 0.1709	320 0.2020	320 0.2507
197	MUKH-002						398 0.1394	399 0.1754	399 0.2335						402 0.9761	402 0.9840	402 0.9899
198	NAVERCLOUD-000						119 0.0043	119 0.0047	108 0.0029						140 0.0311	145 0.0396	146 0.0539
199	NEC-000	97 0.0195	99 0.0316	99 0.0445	99 0.0581	98 0.0699	392 0.0817	392 0.0998	392 0.1237	89 0.0759	89 0.1245	89 0.1729	89 0.2240	89 0.2671	375 0.3117	373 0.3639	374 0.4348
200	NEC-001	104 0.0246	102 0.0382	100 0.0524	100 0.0672	101 0.0793	394 0.0904	393 0.1076	393 0.1317	94 0.1019	94 0.1623	94 0.2214	94 0.2834	94 0.3341	382 0.3844	382 0.4440	381 0.5183
201	NEC-002	27 0.0033	22 0.0041	18 0.0043	16 0.0044	15 0.0045	181 0.0049	186 0.0056	178 0.0041	15 0.0066	11 0.0090	10 0.0111	10 0.0131	9 0.0149	93 0.0171	93 0.0207	91 0.0267
202	NEC-003	31 0.0036	26 0.0046	24 0.0051	24 0.0055	24 0.0059	242 0.0067	246 0.0077	247 0.0073	9 0.0056	9 0.0076	9 0.0091	7 0.0105	6 0.0119	81 0.0137	79 0.0162	77 0.0209
203	NEC-004	32 0.0039	25 0.0045	22 0.0047	18 0.0046	14 0.0044	162 0.0046	174 0.0052	151 0.0036	7 0.0046	5 0.0057	2 0.0063	2 0.0066	1 0.0069	33 0.0076	33 0.0090	34 0.0105
204	NEC-005						67 0.0037	67 0.0041	66 0.0020						35 0.0080	34 0.0091	36 0.0107
205	NEC-006						87 0.0039	83 0.0042	75 0.0021						6 0.0030	6 0.0033	6 0.0012
206	NEC-007						23 0.0034	18 0.0036	32 0.0016						4 0.0027	4 0.0031	4 0.0010
207	NEC-008						57 0.0037	37 0.0038	49 0.0018						5 0.0030	5 0.0033	5 0.0012
208	NEC-009						5 0.0032	5 0.0034	6 0.0010						2 0.0027	2 0.0030	3 0.0007
209	NEC-010						3 0.0031	2 0.0034	9 0.0010						3 0.0027	1 0.0030	2 0.0007
210	NEC-011						2 0.0031	4 0.0034	5 0.0009						1 0.0027	3 0.0030	1 0.0007
211	NEC-012						1 0.0031	3 0.0034	4 0.0009						8 0.0031	8 0.0036	8 0.0018
212	NEUROTECHNOLOGY-003	101 0.0234	101 0.0379	102 0.0549	101 0.0682	100 0.0720	391 0.0747	391 0.0886	389 0.1066	109 0.6802	109 0.8187	110 0.8920	110 0.9355	110 0.9594	401 0.9738	401 0.9828	401 0.9885
213	NEUROTECHNOLOGY-004	79 0.0104	78 0.0134	76 0.0156	73 0.0173	72 0.0195	358 0.0212	356 0.0245	354 0.0320	83 0.0642	82 0.1015	81 0.1426	79 0.1881	78 0.2299	361 0.2722	362 0.3269	362 0.3943
214	NEUROTECHNOLOGY-005	74 0.0089	71 0.0116	68 0.0136	68 0.0152	69 0.0173	354 0.0196	350 0.0233	349 0.0306	76 0.0556	76 0.0913	74 0.1315	74 0.1766	74 0.2192	357 0.2617	356 0.3174	356 0.3843
215	NEUROTECHNOLOGY-007	66 0.0078	65 0.0103	64 0.0124	65 0.0140	63 0.0161	347 0.0185	347 0.0225	346 0.0290	82 0.0641	85 0.1069	85 0.1546	85 0.2075	86 0.2572	374 0.3081	376 0.3713	376 0.4421
216	NEUROTECHNOLOGY-010						202 0.0053	216 0.0061	211 0.0053						252 0.0863	250 0.1050	249 0.1333
217	NEUROTECHNOLOGY-012						141 0.0044	160 0.0051	157 0.0038						231 0.0638	228 0.0783	229 0.1027
218	NEUROTECHNOLOGY-013						117 0.0043	122 0.0047	123 0.0032						178 0.0406	171 0.0498	166 0.0654
219	NEUROTECHNOLOGY-014						122 0.0043	111 0.0046	98 0.0027						141 0.0312	138 0.0380	136 0.0501
220	NEUROTECHNOLOGY-015						109 0.0042	100 0.0044	81 0.0023						144 0.0322	143 0.0391	137 0.0502

Table 18: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
221	NEUROTECHNOLOGY-016						108 0.0041	98 0.0044	102 0.0028						153 0.0349	156 0.0438	154 0.0584
222	NEXTMETALABS-002						124 0.0043	138 0.0049	137 0.0033						169 0.0391	174 0.0500	177 0.0691
223	NOBLIS-002	112 0.1520	112 0.2419	112 0.3296	113 0.4114	113 0.4856	409 0.5528	409 0.6061	409 0.6532	113 0.9984	113 0.9996	113 0.9998	113 0.9999	113 0.9999	405 1.0000	412 1.0000	408 1.0000
224	NOMINDER-000						184 0.0050	194 0.0057	165 0.0038						189 0.0443	184 0.0545	182 0.0727
225	NOMINDER-001						197 0.0052	190 0.0056	187 0.0044						185 0.0436	183 0.0544	179 0.0709
226	NOMINDER-002						175 0.0048	182 0.0055	175 0.0041						183 0.0425	179 0.0521	176 0.0690
227	NOMINDER-003						147 0.0045	164 0.0051	159 0.0038						173 0.0401	173 0.0499	167 0.0657
228	NOMINDER-004						143 0.0045	166 0.0051	174 0.0041						177 0.0406	176 0.0505	168 0.0662
229	NOMINDER-005						129 0.0043	152 0.0049	160 0.0038						132 0.0283	133 0.0357	132 0.0488
230	NTECHLAB-003	65 0.0078	76 0.0131	87 0.0202	90 0.0295	91 0.0405	384 0.0543	387 0.0761	388 0.1035	68 0.0491	72 0.0881	79 0.1384	83 0.1985	87 0.2594	327 0.3270	378 0.4065	378 0.4891
231	NTECHLAB-004	62 0.0068	68 0.0110	79 0.0167	86 0.0239	89 0.0330	381 0.0447	383 0.0641	385 0.0891	60 0.0379	63 0.0688	66 0.1108	66 0.1629	73 0.2192	368 0.2846	375 0.3657	377 0.4524
232	NTECHLAB-006	51 0.0056	62 0.0095	72 0.0148	83 0.0218	85 0.0301	379 0.0413	380 0.0591	383 0.0814	56 0.0349	60 0.0636	63 0.1023	64 0.1506	66 0.2024	356 0.2617	366 0.3374	371 0.4185
233	NTECHLAB-007	37 0.0044	43 0.0066	49 0.0089	57 0.0118	60 0.0150	351 0.0189	359 0.0255	359 0.0342	45 0.0256	46 0.0450	48 0.0705	49 0.1012	51 0.1334	325 0.1692	329 0.2170	331 0.2752
234	NTECHLAB-008	18 0.0025	21 0.0038	26 0.0052	31 0.0074	44 0.0104	331 0.0146	353 0.0236	360 0.0348	26 0.0143	28 0.0267	32 0.0459	37 0.0733	40 0.1062	306 0.1469	322 0.2044	327 0.2698
235	NTECHLAB-009	13 0.0022	15 0.0031	16 0.0038	17 0.0045	19 0.0055	244 0.0067	261 0.0088	268 0.0100	18 0.0073	17 0.0117	17 0.0170	17 0.0238	18 0.0319	181 0.0419	190 0.0577	207 0.0833
236	NTECHLAB-011						218 0.0056	224 0.0066	245 0.0073						155 0.0351	162 0.0475	181 0.0724
237	OMNIGARDE-000						187 0.0050	195 0.0057	197 0.0048						180 0.0413	180 0.0531	186 0.0741
238	OMNIGARDE-001						80 0.0039	91 0.0043	89 0.0024						96 0.0178	97 0.0215	93 0.0275
239	OMNIGARDE-002						99 0.0040	101 0.0044	92 0.0024						70 0.0120	68 0.0146	67 0.0182
240	ONFIDO-000						340 0.0157	341 0.0197	344 0.0285						289 0.1225	294 0.1607	308 0.2273
241	ONFIDO-001						324 0.0134	327 0.0166	329 0.0211						255 0.0896	259 0.1159	270 0.1636
242	OPTIEXACTA-000						64 0.0037	73 0.0042	71 0.0021						45 0.0099	51 0.0125	52 0.0157
243	PANGIAM-000						190 0.0051	185 0.0055	192 0.0046						206 0.0503	205 0.0617	200 0.0810
244	PANGIAM-001						322 0.0132	320 0.0153	302 0.0153						219 0.0545	219 0.0685	219 0.0894
245	PAPSAV1923-000						186 0.0050	189 0.0056	171 0.0040						131 0.0280	126 0.0344	125 0.0447
246	PARAVISION-002	53 0.0058	58 0.0083	60 0.0111	63 0.0137	65 0.0162	349 0.0187	349 0.0229	347 0.0295								
247	PARAVISION-003	44 0.0048	44 0.0067	51 0.0090	52 0.0109	54 0.0128	332 0.0148	333 0.0178	331 0.0219	57 0.0354	58 0.0618	59 0.0931	60 0.1290	61 0.1625	338 0.1964	340 0.2408	335 0.2924
248	PARAVISION-004	16 0.0024	17 0.0032	17 0.0040	19 0.0047	18 0.0053	230 0.0061	240 0.0073	242 0.0072	20 0.0118	23 0.0209	24 0.0327	24 0.0465	24 0.0613	246 0.0779	249 0.1008	246 0.1285
249	PARAVISION-005	12 0.0021	13 0.0028	13 0.0035	14 0.0041	16 0.0046	208 0.0054	226 0.0067	239 0.0070	11 0.0057	12 0.0093	12 0.0144	14 0.0207	15 0.0278	163 0.0368	177 0.0508	180 0.0715
250	PARAVISION-007	6 0.0019	8 0.0025	7 0.0029	8 0.0033	8 0.0036	113 0.0042	149 0.0049	118 0.0030	10 0.0057	13 0.0094	14 0.0144	13 0.0206	14 0.0275	158 0.0357	167 0.0485	165 0.0652
251	PARAVISION-009						105 0.0041	112 0.0046	96 0.0026						133 0.0283	134 0.0371	144 0.0525
252	PARAVISION-012						62 0.0037	70 0.0041	61 0.0020						80 0.0137	80 0.0167	81 0.0219
253	PARAVISION-014						48 0.0036	54 0.0040	51 0.0018						62 0.0119	70 0.0148	73 0.0195
254	PARAVISION-016						43 0.0036	47 0.0039	41 0.0016						50 0.0100	53 0.0125	49 0.0156
255	PARAVISION-019						40 0.0035	42 0.0039	30 0.0015						51 0.0100	54 0.0125	50 0.0157
256	PARAVISION-021						38 0.0035	40 0.0039	28 0.0014						47 0.0099	49 0.0124	48 0.0156
257	PARAVISION-022						34 0.0035	45 0.0039	39 0.0016						43 0.0097	44 0.0118	57 0.0166
258	PENTACOMP-000						414 0.9396	414 0.9391	414 0.9364								
259	PENTACOMP-001						287 0.0094	291 0.0108	291 0.0130						296 0.1323	295 0.1612	292 0.2032
260	PIXELALL-002	72 0.0085	73 0.0119	71 0.0147	72 0.0172	73 0.0198	362 0.0225	361 0.0270	361 0.0349	97 0.1193	97 0.1900	97 0.2601	97 0.3332	97 0.3955	385 0.4565	386 0.5268	387 0.6030
261	PIXELALL-003	46 0.0050	42 0.0063	39 0.0072	34 0.0077	33 0.0085	289 0.0095	293 0.0113	284 0.0119	44 0.0248	43 0.0418	43 0.0622	43 0.0861	43 0.1104	299 0.1364	301 0.1723	300 0.2167
262	PIXELALL-004	45 0.0049	40 0.0063	40 0.0072	37 0.0079	36 0.0089	298 0.0103	300 0.0127	296 0.0146	38 0.0211	40 0.0360	42 0.0553	42 0.0792	39 0.1045	295 0.1317	300 0.1700	305 0.2246
263	PROTENET-001						294 0.0099	295 0.0113	298 0.0149						300 0.1375	296 0.1652	295 0.2095
264	PSL-000						68 0.0038	84 0.0042	83 0.0023						122 0.0252	122 0.0324	124 0.0436

Table 19: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
265	PSL-001						47 0.0036	44 0.0039	59 0.0019						105 0.0202	104 0.0248	107 0.0337
266	PSL-002						36 0.0035	36 0.0038	44 0.0017						75 0.0124	78 0.0160	80 0.0218
267	PSL-003						30 0.0035	30 0.0038	37 0.0016						66 0.0118	75 0.0152	76 0.0197
268	PSL-004						27 0.0034	34 0.0038	45 0.0017						57 0.0105	66 0.0142	71 0.0190
269	PSL-005						54 0.0037	57 0.0040	55 0.0018						74 0.0123	72 0.0149	75 0.0197
270	PSL-006						22 0.0034	28 0.0037	34 0.0016						38 0.0088	42 0.0114	44 0.0149
271	PTAKURATSATU-000	54 0.0061	55 0.0082	55 0.0097	53 0.0109	49 0.0120	320 0.0131	313 0.0146	315 0.0180	59 0.0375	57 0.0596	55 0.0842	55 0.1116	53 0.1357	318 0.1553	309 0.1820	313 0.2326
272	PYRAMID-000						153 0.0045	170 0.0052	168 0.0039						146 0.0325	149 0.0415	157 0.0588
273	PYRAMID-001						173 0.0047	178 0.0053	162 0.0038						167 0.0384	166 0.0482	172 0.0676
274	QAZSMARTVISIONAI-000						26 0.0034	25 0.0037	29 0.0015						52 0.0102	47 0.0123	42 0.0142
275	QAZSMARTVISIONAI-001						13 0.0033	14 0.0036	11 0.0010						28 0.0066	28 0.0077	27 0.0081
276	QAZSMARTVISIONAI-002						6 0.0032	9 0.0035	2 0.0009						11 0.0043	13 0.0051	11 0.0042
277	QAZSMARTVISIONAI-003						8 0.0033	10 0.0035	7 0.0010						14 0.0045	15 0.0052	12 0.0043
278	QNAP-004						299 0.0105	306 0.0133	310 0.0167						292 0.1243	292 0.1556	293 0.2050
279	QNAP-005						272 0.0084	266 0.0091	279 0.0113						226 0.0612	223 0.0724	230 0.1031
280	QNAP-006						254 0.0071	255 0.0082	255 0.0084						234 0.0676	236 0.0838	240 0.1137
281	RANKONE-002	99 0.0212	98 0.0313	98 0.0431	98 0.0562	99 0.0712	393 0.0881	394 0.1130	395 0.1543	95 0.1111	95 0.1707	95 0.2305	95 0.2968	96 0.3646	384 0.4345	384 0.5172	388 0.6110
282	RANKONE-004	108 0.0424	107 0.0643	107 0.0875	107 0.1127	107 0.1364	399 0.1579	400 0.1914	400 0.2378	104 0.1855	103 0.2681	103 0.3431	101 0.4155	101 0.4785	390 0.5350	390 0.5980	390 0.6722
283	RANKONE-005	92 0.0136	93 0.0192	91 0.0246	91 0.0303	90 0.0362	380 0.0422	329 0.0521	379 0.0694	81 0.0582	75 0.0910	71 0.1260	68 0.1645	65 0.2005	345 0.2353	345 0.2816	347 0.3522
284	RANKONE-007	67 0.0078	64 0.0099	61 0.0113	58 0.0123	58 0.0139	339 0.0156	340 0.0191	338 0.0242	42 0.0242	42 0.0376	41 0.0542	38 0.0737	37 0.0935	283 0.1130	282 0.1416	282 0.1811
285	RANKONE-009	48 0.0054	49 0.0072	46 0.0085	47 0.0098	47 0.0113	319 0.0130	329 0.0169	334 0.0220	37 0.0208	38 0.0345	37 0.0504	36 0.0706	36 0.0930	286 0.1174	290 0.1504	290 0.2002
286	RANKONE-010	42 0.0047	38 0.0061	38 0.0070	33 0.0076	34 0.0087	292 0.0098	294 0.0113	285 0.0120	31 0.0177	29 0.0269	26 0.0368	26 0.0479	23 0.0590	238 0.0688	233 0.0803	228 0.0991
287	RANKONE-011	23 0.0031	23 0.0041	23 0.0047	23 0.0053	22 0.0058	245 0.0067	245 0.0077	246 0.0073	23 0.0127	20 0.0194	21 0.0265	20 0.0345	20 0.0422	204 0.0499	203 0.0611	190 0.0756
288	RANKONE-012						240 0.0065	232 0.0069	210 0.0053						189 0.0460	182 0.0540	170 0.0672
289	RANKONE-013						194 0.0051	167 0.0051	145 0.0035						136 0.0306	131 0.0355	118 0.0405
290	RANKONE-014						133 0.0044	117 0.0047	107 0.0029						111 0.0222	106 0.0255	97 0.0287
291	RANKONE-015						106 0.0041	96 0.0044	82 0.0024						90 0.0162	85 0.0187	79 0.0218
292	REALNETWORKS-002	107 0.0381	108 0.0687	108 0.1062	108 0.1495	109 0.1963	404 0.2513	407 0.3206	407 0.3927	105 0.2153	105 0.3323	105 0.4444	105 0.5485	106 0.6355	397 0.7132	398 0.7855	398 0.8437
293	REALNETWORKS-003	103 0.0245	105 0.0437	105 0.0686	106 0.0975	106 0.1312	402 0.1719	402 0.2294	403 0.2907	98 0.1468	100 0.2370	101 0.3313	103 0.4269	103 0.5142	394 0.5979	395 0.6815	395 0.7567
294	REALNETWORKS-004	102 0.0244	104 0.0428	104 0.0663	105 0.0939	105 0.1251	401 0.1634	401 0.2170	402 0.2785	99 0.1484	101 0.2377	100 0.3303	102 0.4249	102 0.5106	393 0.5924	394 0.6758	394 0.7534
295	REALNETWORKS-006						252 0.0069	248 0.0077	253 0.0080						273 0.1022	270 0.1253	269 0.1622
296	REALNETWORKS-008						183 0.0049	181 0.0054	194 0.0047						190 0.0462	191 0.0577	187 0.0745
297	REALSENSE-001						131 0.0044	114 0.0046	156 0.0037						114 0.0233	110 0.0269	106 0.0333
298	REALSENSE-002						73 0.0038	50 0.0039	69 0.0021						36 0.0084	36 0.0093	33 0.0102
299	RECOGNITO-000						114 0.0042	125 0.0047	111 0.0030						94 0.0176	101 0.0221	103 0.0306
300	RECOGNITO-001						66 0.0037	71 0.0041	67 0.0021						56 0.0105	59 0.0135	59 0.0172
301	RECOGNITO-002						63 0.0037	74 0.0042	70 0.0021						44 0.0099	52 0.0125	51 0.0157
302	RNIT-000						210 0.0054	215 0.0061	191 0.0046						305 0.1456	305 0.1739	302 0.2192
303	ROC-016						75 0.0038	79 0.0042	47 0.0018						77 0.0130	74 0.0152	70 0.0187
304	ROC-017						98 0.0040	90 0.0043	97 0.0027						97 0.0181	94 0.0209	86 0.0238
305	ROC-018						71 0.0038	51 0.0039	56 0.0018						78 0.0131	73 0.0151	58 0.0171
306	ROC-019						35 0.0035	31 0.0038	40 0.0016						71 0.0121	64 0.0142	56 0.0164
307	ROC-020						32 0.0035	26 0.0037	22 0.0012						42 0.0095	40 0.0108	38 0.0120
308	ROC-021						21 0.0034	19 0.0036	3 0.0009						26 0.0064	26 0.0073	23 0.0071

Table 20: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
309	S1-002						164 0.0046	163 0.0051	167 0.0038						196 0.0482	196 0.0597	197 0.0788
310	S1-003						220 0.0057	220 0.0063	219 0.0056						235 0.0681	237 0.0839	234 0.1061
311	S1-004						111 0.0042	109 0.0045	125 0.0032						160 0.0360	158 0.0448	158 0.0598
312	S1-005						115 0.0042	106 0.0045	115 0.0030						175 0.0403	178 0.0515	178 0.0695
313	S1-006						72 0.0038	64 0.0041	79 0.0022						95 0.0176	95 0.0211	96 0.0280
314	S1-007						70 0.0038	59 0.0040	68 0.0021						83 0.0152	83 0.0177	85 0.0230
315	SANSAP-001						211 0.0054	213 0.0061	203 0.0050						172 0.0395	164 0.0479	161 0.0632
316	SCANOVATE-001	68 0.0079	72 0.0117	75 0.0151	78 0.0185	78 0.0221	367 0.0259	371 0.0321	371 0.0427	88 0.0727	88 0.1169	87 0.1650	87 0.2115	84 0.2528	371 0.2925	368 0.3437	368 0.4084
317	SCYLLA-001						291 0.0097	292 0.0112	287 0.0124						294 0.1292	293 0.1600	294 0.2093
318	SECURITYPITCH-000						357 0.0210	355 0.0242	353 0.0317						370 0.2900	365 0.3362	366 0.4005
319	SENSETIME-002	96 0.0186	92 0.0191	84 0.0183	75 0.0179	68 0.0173	323 0.0133	263 0.0089	222 0.0059	40 0.0220	25 0.0236	19 0.0237	18 0.0240	12 0.0245	109 0.0219	90 0.0195	82 0.0222
320	SENSETIME-003	11 0.0021	12 0.0028	11 0.0031	7 0.0033	6 0.0035	91 0.0040	124 0.0047	129 0.0033	8 0.0046	8 0.0064	6 0.0076	4 0.0086	4 0.0101	72 0.0122	76 0.0155	74 0.0196
321	SENSETIME-004	3 0.0016	3 0.0022	3 0.0025	3 0.0028	3 0.0030	39 0.0035	95 0.0043	95 0.0025	4 0.0036	4 0.0052	3 0.0066	3 0.0081	3 0.0099	76 0.0126	81 0.0169	84 0.0230
322	SENSETIME-005	2 0.0015	2 0.0020	2 0.0024	2 0.0026	2 0.0029	33 0.0035	85 0.0043	103 0.0028	5 0.0036	7 0.0059	8 0.0089	9 0.0128	10 0.0177	117 0.0240	127 0.0345	133 0.0493
323	SENSETIME-006	1 0.0015	1 0.0019	1 0.0022	1 0.0025	1 0.0027	9 0.0033	63 0.0040	76 0.0021	2 0.0031	2 0.0049	4 0.0068	6 0.0097	7 0.0132	100 0.0184	109 0.0262	110 0.0359
324	SENSETIME-007						31 0.0035	32 0.0038	31 0.0015						62 0.0112	61 0.0140	61 0.0176
325	SENSETIME-008						29 0.0034	38 0.0039	43 0.0017						55 0.0103	56 0.0127	55 0.0163
326	SENSETIME-010						10 0.0033	11 0.0036	17 0.0012						32 0.0071	32 0.0086	32 0.0099
327	SENSETIME-011						16 0.0033	16 0.0036	14 0.0011						21 0.0056	20 0.0065	20 0.0065
328	SENSETIME-012						44 0.0036	39 0.0039	46 0.0017						40 0.0092	57 0.0128	62 0.0177
329	SERENDIPITY-000						270 0.0084	268 0.0092	265 0.0094						266 0.0981	263 0.1180	262 0.1519
330	SEVENTHSENSE-001						316 0.0128	322 0.0156	326 0.0206						166 0.0383	163 0.0476	175 0.0688
331	SIAT-002	117 0.8309	117 0.8310	117 0.8311	117 0.8306	117 0.8296	413 0.8302	413 0.8300	412 0.8301	112 0.8340	110 0.8368	109 0.8404	109 0.8445	109 0.8480	400 0.8532	400 0.8595	399 0.8691
332	SJTU-001						51 0.0036	56 0.0040	65 0.0020						69 0.0120	65 0.0142	63 0.0178
333	SJTU-002						25 0.0034	27 0.0037	25 0.0013						30 0.0068	27 0.0076	29 0.0087
334	SPARKTEK-000						228 0.0060	227 0.0068	213 0.0054						129 0.0279	128 0.0344	126 0.0451
335	SPARKTEK-001						77 0.0039	65 0.0041	60 0.0019						53 0.0102	45 0.0121	45 0.0149
336	SPARSH-001						157 0.0046	147 0.0049	127 0.0032						162 0.0362	155 0.0435	150 0.0557
337	STCON-001						60 0.0037	69 0.0041	52 0.0018						82 0.0146	84 0.0186	88 0.0254
338	STCON-002						37 0.0035	49 0.0039	35 0.0016						64 0.0115	63 0.0142	68 0.0182
339	STCON-003						41 0.0035	41 0.0039	36 0.0016						48 0.0100	50 0.0124	47 0.0154
340	STCON-004						12 0.0033	24 0.0037	16 0.0011						22 0.0057	22 0.0068	22 0.0070
341	STCON-005						14 0.0033	23 0.0037	21 0.0012						23 0.0059	23 0.0070	26 0.0076
342	STELLARTECS-001						278 0.0088	275 0.0099	272 0.0107						280 0.1104	278 0.1354	277 0.1723
343	SYNESIS-003	89 0.0125	85 0.0151	83 0.0174	80 0.0199	79 0.0223	364 0.0240	364 0.0279	357 0.0331	85 0.0658	83 0.1052	83 0.1483	82 0.1968	82 0.2399	366 0.2834	367 0.3405	367 0.4046
344	SYNESIS-005	40 0.0044	37 0.0058	37 0.0070	40 0.0080	37 0.0091	297 0.0103	298 0.0125	300 0.0152	46 0.0262	45 0.0444	45 0.0666	45 0.0923	44 0.1156	302 0.1399	303 0.1736	301 0.2185
345	T4ISB-000						387 0.0606	386 0.0748	386 0.0970								
346	TECH5-001	57 0.0061	61 0.0093	66 0.0128	71 0.0171	77 0.0221	373 0.0289	376 0.0412	376 0.0560	86 0.0660	87 0.1156	90 0.1733	91 0.2385	91 0.2998	379 0.3629	381 0.4424	382 0.5284
347	TECH5-003						165 0.0046	153 0.0050	124 0.0032						152 0.0345	147 0.0414	145 0.0529
348	TECHAINER-000						261 0.0079	262 0.0089	264 0.0093						267 0.0985	266 0.1197	259 0.1514
349	THAKAAIT-000						263 0.0080	265 0.0090	263 0.0091						290 0.1231	289 0.1490	285 0.1911
350	TIGER-004						300 0.0106	301 0.0127	301 0.0152						298 0.1350	298 0.1654	297 0.2136
351	TOSHIBA-001	73 0.0086	74 0.0119	74 0.0150	74 0.0178	75 0.0209	365 0.0241	365 0.0292	364 0.0365								
352	TOSHIBA-002						85 0.0039	55 0.0040	74 0.0021						58 0.0105	58 0.0130	64 0.0179

Table 21: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
353	TOSHIBA-003						[78] 0.0039	[58] 0.0040	[57] 0.0018						[61] 0.0110	[60] 0.0135	[72] 0.0193
354	TOSHIBA-004						[69] 0.0038	[52] 0.0040	[33] 0.0016						[54] 0.0102	[48] 0.0123	[53] 0.0158
355	TOSS-000						[45] 0.0036	[48] 0.0039	[58] 0.0019						[59] 0.0109	[55] 0.0126	[54] 0.0163
356	TOSS-001						[76] 0.0038	[77] 0.0042	[94] 0.0025						[88] 0.0158	[92] 0.0199	[94] 0.0279
357	TRUEFACE-000	[36] 0.0043	[36] 0.0057	[30] 0.0061	[28] 0.0067	[27] 0.0073	[271] 0.0084	[273] 0.0097	[267] 0.0099	[35] 0.0200	[37] 0.0338	[38] 0.0504	[35] 0.0705	[35] 0.0904	[281] 0.1112	[280] 0.1401	[281] 0.1792
358	TSIGHT-000						[7] 0.0032	[13] 0.0036	[8] 0.0010						[13] 0.0044	[16] 0.0052	[14] 0.0044
359	UNISSEY-003						[268] 0.0083	[264] 0.0090	[262] 0.0091						[264] 0.0963	[264] 0.1181	[260] 0.1516
360	UNIVERSEAI-000						[144] 0.0045	[125] 0.0048	[121] 0.0031						[142] 0.0317	[144] 0.0393	[140] 0.0516
361	USEB-001						[137] 0.0044	[134] 0.0048	[143] 0.0035						[135] 0.0305	[142] 0.0389	[141] 0.0516
362	USEB-002						[148] 0.0045	[143] 0.0049	[139] 0.0034						[147] 0.0328	[150] 0.0417	[149] 0.0556
363	USEB-003						[135] 0.0044	[121] 0.0047	[130] 0.0033						[134] 0.0305	[140] 0.0387	[143] 0.0522
364	USEB-004						[116] 0.0043	[123] 0.0047	[142] 0.0035						[128] 0.0279	[128] 0.0347	[129] 0.0467
365	VDFSOFTWARE-002						[346] 0.0182	[344] 0.0221	[342] 0.0276						[343] 0.2342	[343] 0.2744	[343] 0.3298
366	VERIDAS-001	[58] 0.0063	[56] 0.0083	[56] 0.0099	[56] 0.0113	[56] 0.0132	[333] 0.0148	[335] 0.0184	[332] 0.0219	[61] 0.0403	[61] 0.0684	[62] 0.1012	[62] 0.1386	[62] 0.1741	[341] 0.2113	[341] 0.2611	[342] 0.3233
367	VERIDAS-004						[200] 0.0052	[205] 0.0059	[184] 0.0043						[208] 0.0506	[210] 0.0634	[209] 0.0836
368	VERIGRAM-001						[193] 0.0051	[180] 0.0053	[221] 0.0058						[150] 0.0336	[146] 0.0413	[151] 0.0561
369	VERIGRAM-002						[92] 0.0040	[81] 0.0042	[80] 0.0022						[39] 0.0091	[39] 0.0106	[40] 0.0127
370	VIANTE-000						[101] 0.0040	[105] 0.0045	[101] 0.0028						[107] 0.0210	[108] 0.0259	[105] 0.0315
371	VIANTE-001						[42] 0.0036	[35] 0.0038	[54] 0.0018						[31] 0.0070	[30] 0.0082	[31] 0.0095
372	VIANTE-002						[19] 0.0034	[17] 0.0036	[18] 0.0012						[24] 0.0061	[24] 0.0071	[24] 0.0074
373	VIANTE-003						[28] 0.0034	[21] 0.0037	[23] 0.0012						[20] 0.0056	[21] 0.0066	[21] 0.0066
374	VIETTELAI-000						[256] 0.0072	[242] 0.0074	[251] 0.0077						[389] 0.5122	[389] 0.5689	[389] 0.6493
375	VIETTELAI-001						[215] 0.0056	[210] 0.0060	[214] 0.0055								
376	VIETTELAI-002						[216] 0.0056	[211] 0.0060	[217] 0.0055						[256] 0.0902	[254] 0.1106	[253] 0.1438
377	VIETTELAI-003						[125] 0.0043	[137] 0.0048	[136] 0.0033						[188] 0.0447	[189] 0.0575	[194] 0.0778
378	VIETTELAI-004						[96] 0.0040	[107] 0.0045	[105] 0.0028						[119] 0.0249	[121] 0.0324	[121] 0.0428
379	VINBIGDATA-001						[250] 0.0068	[251] 0.0079	[250] 0.0077						[301] 0.1383	[299] 0.1670	[296] 0.2134
380	VINBIGDATA-002						[122] 0.0043	[116] 0.0047	[114] 0.0030						[108] 0.0213	[111] 0.0271	[114] 0.0372
381	VINBIGDATA-003						[126] 0.0043	[120] 0.0047	[119] 0.0031						[126] 0.0273	[132] 0.0357	[138] 0.0503
382	VINBIGDATA-004						[89] 0.0039	[66] 0.0041	[77] 0.0022						[85] 0.0156	[91] 0.0197	[95] 0.0280
383	VISIONBOX-001						[295] 0.0099	[282] 0.0103	[282] 0.0117						[396] 0.6689	[396] 0.7251	[396] 0.7917
384	VISIONBOX-002						[264] 0.0080	[260] 0.0088	[266] 0.0097						[259] 0.0937	[261] 0.1170	[265] 0.1531
385	VISIONLABS-004	[43] 0.0048	[46] 0.0069	[52] 0.0091	[55] 0.0111	[55] 0.0130	[337] 0.0152	[336] 0.0187	[339] 0.0242	[74] 0.0540	[77] 0.0916	[78] 0.1358	[78] 0.1855	[79] 0.2303	[362] 0.2745	[363] 0.3312	[359] 0.3913
386	VISIONLABS-005	[39] 0.0044	[39] 0.0063	[43] 0.0081	[46] 0.0095	[46] 0.0109	[315] 0.0125	[318] 0.0151	[319] 0.0187	[67] 0.0479	[67] 0.0812	[68] 0.1212	[70] 0.1664	[69] 0.2078	[351] 0.2473	[352] 0.2999	[349] 0.3577
387	VISIONLABS-006	[29] 0.0035	[30] 0.0048	[32] 0.0061	[30] 0.0069	[29] 0.0077	[274] 0.0087	[283] 0.0105	[286] 0.0120	[48] 0.0273	[47] 0.0465	[47] 0.0702	[47] 0.0970	[47] 0.1228	[308] 0.1486	[312] 0.1847	[310] 0.2295
388	VISIONLABS-008	[21] 0.0028	[20] 0.0037	[21] 0.0047	[22] 0.0053	[23] 0.0058	[243] 0.0067	[254] 0.0081	[256] 0.0085	[27] 0.0143	[27] 0.0241	[28] 0.0373	[27] 0.0519	[27] 0.0677	[250] 0.0850	[253] 0.1104	[254] 0.1444
389	VISIONLABS-009	[10] 0.0020	[10] 0.0026	[10] 0.0030	[10] 0.0034	[10] 0.0038	[139] 0.0044	[171] 0.0052	[190] 0.0046	[14] 0.0065	[15] 0.0105	[15] 0.0156	[15] 0.0217	[16] 0.0289	[164] 0.0368	[172] 0.0499	[174] 0.0681
390	VISIONLABS-010	[9] 0.0020	[9] 0.0025	[9] 0.0030	[11] 0.0034	[9] 0.0036	[120] 0.0043	[159] 0.0051	[195] 0.0047	[17] 0.0069	[16] 0.0113	[16] 0.0170	[16] 0.0238	[17] 0.0316	[179] 0.0411	[186] 0.0557	[185] 0.0740
391	VISIONLABS-011						[112] 0.0042	[113] 0.0046	[150] 0.0036						[125] 0.0270	[123] 0.0337	[123] 0.0432
392	VIXVIZION-009						[342] 0.0161	[339] 0.0190	[336] 0.0238						[331] 0.1787	[326] 0.2116	[324] 0.2595
393	VNPAY-001						[203] 0.0053	[196] 0.0058	[185] 0.0043						[205] 0.0501	[207] 0.0619	[203] 0.0822
394	VNPT-002						[204] 0.0053	[204] 0.0059	[188] 0.0044						[216] 0.0534	[218] 0.0670	[216] 0.0882
395	VNPT-003						[185] 0.0050	[193] 0.0057	[173] 0.0041						[215] 0.0532	[214] 0.0658	[217] 0.0883
396	VNPT-004						[179] 0.0049	[172] 0.0052	[158] 0.0038						[184] 0.0432	[185] 0.0549	[189] 0.0754

Table 22: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISS RATES		INVESTIGATION, FNIR(N, R = 1, T = 0)								IDENTIFICATION, FNIR(N, R = L, T $\geq$ 0) FOR FPIR = 0.001							
#	ALGORITHM	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]	(0, 2]	(2, 4]	(4, 6]	(6, 8]	(8, 10]	(10, 12]	(12, 14]	(14, 18]
397	VNPT-005						146 0.0045	135 0.0048	131 0.0033						159 0.0357	160 0.0456	160 0.0623
398	VOGIC-000						290 0.0096	290 0.0108	290 0.0128						285 0.1162	285 0.1425	288 0.1932
399	VTS-000	116 0.5878	116 0.6312	116 0.6602	115 0.6863	115 0.7073	411 0.7246	411 0.7458	410 0.7747	108 0.5929	108 0.6397	108 0.6729	108 0.7034	107 0.7279	398 0.7493	397 0.7739	397 0.8076
400	VTS-003						205 0.0054	198 0.0059	212 0.0054						224 0.0597	224 0.0731	223 0.0950
401	VTS-004						159 0.0046	168 0.0051	152 0.0037						176 0.0405	175 0.0505	169 0.0671
402	VTS-005						130 0.0044	150 0.0049	149 0.0036						203 0.0497	208 0.0625	213 0.0849
403	WINK-001						352 0.0190	345 0.0223	345 0.0286						344 0.2344	344 0.2754	344 0.3370
404	WINK-002						275 0.0087	271 0.0095	270 0.0103						265 0.0966	260 0.1168	261 0.1518
405	WINK-003						227 0.0059	217 0.0062	230 0.0066						243 0.0738	243 0.0911	243 0.1190
406	XAILIENT-000						360 0.0214	360 0.0265	363 0.0360						349 0.2457	350 0.2925	353 0.3694
407	XAILIENT-001						212 0.0054	212 0.0060	193 0.0046						236 0.0683	234 0.0803	233 0.1047
408	XFORWARDAI-000	20 0.0027	18 0.0034	20 0.0044	21 0.0052	21 0.0058	246 0.0067	250 0.0079	249 0.0076	28 0.0157	31 0.0281	30 0.0443	31 0.0635	32 0.0834	275 0.1050	276 0.1330	276 0.1714
409	XFORWARDAI-001	15 0.0023	11 0.0028	12 0.0034	12 0.0037	12 0.0039	154 0.0045	169 0.0052	183 0.0043	13 0.0060	14 0.0096	13 0.0144	12 0.0200	13 0.0260	149 0.0334	153 0.0435	156 0.0586
410	YITU-002	60 0.0066	57 0.0083	53 0.0094	48 0.0101	50 0.0121	335 0.0150	346 0.0223	355 0.0328	33 0.0189	34 0.0317	35 0.0494	39 0.0750	41 0.1066	311 0.1494	330 0.2171	338 0.2958
411	YITU-003	63 0.0072	60 0.0089	57 0.0100	51 0.0107	52 0.0125	338 0.0153	348 0.0226	358 0.0334	34 0.0194	35 0.0321	36 0.0500	41 0.0756	42 0.1071	313 0.1500	331 0.2177	339 0.2964
412	YITU-004	55 0.0061	51 0.0075	44 0.0081	41 0.0081	39 0.0092	303 0.0107	321 0.0154	328 0.0207	22 0.0125	22 0.0204	23 0.0314	25 0.0469	26 0.0671	261 0.0955	284 0.1421	291 0.2006
413	YITU-005	61 0.0067	54 0.0080	47 0.0087	44 0.0085	41 0.0094	305 0.0108	319 0.0151	324 0.0204	21 0.0124	21 0.0198	22 0.0308	22 0.0462	25 0.0667	260 0.0953	283 0.1418	287 0.1930
414	YOUVERSE-001						206 0.0054	214 0.0061	216 0.0055						228 0.0620	229 0.0786	232 0.1046
415	YUAN-000						226 0.0059	233 0.0070	229 0.0063						253 0.0864	252 0.1083	252 0.1406

Table 23: **Accuracy for the FRVT 2018 mugshot sets under ageing**. *The second row shows the time lapse between gallery and subsequent probe images, in years. The first two columns identify the algorithm The next 8 values give rank-based FNIR with R = 1, T = 0 and FPIR = 1. All these are relevant to investigational uses where candidates from all searches would need human review. The second 8 values give threshold-based FNIR with T $\geq$ 0, FPIR = 0.001 and no rank criterion. The shaded cells indictate indicate the three most accurate algorithms for that elapsed time. The gallery size is 3068801. The total number of searches is 10951064.*

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
1	20FACE-000	[579] 0.055	[556] 0.085	[476] 0.733	[494] 0.056	[415] 0.239	[489] 0.243	[578] 0.347	[536] 0.450	[547] 1.000	[484] 0.424	[326] 0.772	[484] 0.938	0.000	0.000	0.000		0.000	
2	3DIVI-003	[589] 0.083	[587] 0.206		[517] 0.141		[527] 0.474	[587] 0.400	[559] 0.626		[505] 0.605		[449] 0.821	0.002	0.005				
3	3DIVI-004	[536] 0.017	[544] 0.062		[479] 0.035		[495] 0.279	[545] 0.168	[521] 0.343		[460] 0.277		[404] 0.607	0.002	0.005				
4	3DIVI-005	[537] 0.017	[543] 0.062	[530] 0.929	[549] 0.821		[496] 0.279	[542] 0.166	[519] 0.339	[425] 0.996	[525] 0.864		[402] 0.597	0.002	0.005	0.442			
5	3DIVI-006	[548] 0.024	[551] 0.074		[487] 0.047		[509] 0.312	[544] 0.167	[520] 0.342		[461] 0.283		[408] 0.615	0.002	0.005				
6	AADI-000	[633] 0.999	[630] 1.000	[603] 1.000	[606] 1.000	[452] 1.000	[569] 1.000	[629] 1.000	[629] 1.000	[581] 1.000	[624] 1.000	[525] 1.000	[543] 1.000	0.028	0.154	0.990		0.274	
7	AADI-001	[629] 0.789	[625] 0.944	[588] 0.999	[553] 0.967	[447] 0.988	[554] 0.985			[506] 1.000				0.160	0.278	0.991		0.473	
8	AADI-002	[628] 0.699	[624] 0.915	[587] 0.999	[551] 0.898	[444] 0.955	[553] 0.982	[617] 0.939	[612] 0.992	[469] 0.999	[543] 0.995	[374] 0.999	[515] 0.998	0.008	0.047	0.919		0.047	
9	ACER-000	[506] 0.010	[497] 0.036	[508] 0.825	[459] 0.025		[473] 0.209	[532] 0.145	[490] 0.246	[349] 0.981	[450] 0.201		[377] 0.490	0.000	0.000	0.042			
10	ACER-001	[447] 0.005	[421] 0.020	[403] 0.416	[399] 0.008	[346] 0.050	[327] 0.098	[461] 0.056	[402] 0.109	[480] 0.999	[383] 0.068	[289] 0.406	[375] 0.479	0.001	0.001	0.041		0.000	
11	ADIANCE-001	[634] 1.000	[629] 1.000	[611] 1.000	[550] 0.823	[450] 1.000	[547] 0.837	[634] 1.000	[618] 1.000	[575] 1.000	[521] 0.833	[385] 1.000	[481] 0.931	0.001	0.000	0.043		0.000	
12	ADIANCE-002	[250] 0.001	[487] 0.031	[258] 0.098	[484] 0.045	[430] 0.441	[467] 0.197	[294] 0.012	[602] 0.955	[492] 1.000	[535] 0.974	[446] 1.000	[511] 0.996	0.000	0.000	0.031		0.000	
13	ADVANCE-000	[314] 0.002	[325] 0.013	[317] 0.177	[279] 0.003	[358] 0.060	[267] 0.081	[352] 0.020	[504] 0.270	[331] 0.973	[436] 0.163	[338] 0.884	[325] 0.326	0.000	0.000	0.043		0.000	
14	AFISBIOMETRICS-000	[256] 0.001	[185] 0.009	[275] 0.115	[282] 0.003	[249] 0.015	[279] 0.083	[405] 0.029	[357] 0.081	[199] 0.511	[327] 0.034	[217] 0.154	[244] 0.172	0.000	0.000	0.041		0.000	
15	AFRENGINE-000	[404] 0.003	[508] 0.038	[274] 0.114	[478] 0.035	[308] 0.028	[395] 0.121	[347] 0.019	[505] 0.271	[158] 0.360	[452] 0.206	[320] 0.743	[314] 0.308	0.001	0.000	0.042		0.000	
16	AFRENGINE-001	[273] 0.001	[314] 0.013	[174] 0.060	[406] 0.010	[265] 0.018	[232] 0.073	[136] 0.003	[209] 0.031	[123] 0.227	[259] 0.018	[185] 0.113	[175] 0.120	0.001	0.001	0.042		0.009	
17	AFRENGINE-002	[274] 0.001	[301] 0.012	[155] 0.058	[187] 0.001	[178] 0.008	[113] 0.054	[125] 0.003	[177] 0.026	[153] 0.344	[189] 0.008	[346] 0.926	[147] 0.102	0.001	0.000	0.042		0.000	
18	AFRENGINE-003	[380] 0.002	[320] 0.013	[175] 0.061	[191] 0.002	[169] 0.007	[162] 0.060	[183] 0.004	[224] 0.033	[92] 0.183	[215] 0.011	[177] 0.097	[165] 0.111	0.002	0.000	0.044		0.001	
19	AIGROUP-000	[226] 0.001	[528] 0.051	[229] 0.081	[504] 0.076	[431] 0.444	[501] 0.288	[270] 0.010	[611] 0.991	[432] 0.997	[515] 0.693	[575] 1.000	[479] 0.929	0.000	0.000	0.033		0.000	
20	AIZE-001	[453] 0.005	[440] 0.022	[463] 0.679	[440] 0.016	[348] 0.050	[445] 0.165	[485] 0.076	[435] 0.143	[398] 0.994	[406] 0.101	[278] 0.364	[341] 0.387	0.001	0.001	0.047		0.000	
21	AIZE-002	[412] 0.003	[410] 0.019	[376] 0.311	[382] 0.007	[341] 0.043	[320] 0.096	[447] 0.051	[608] 0.981	[556] 1.000	[474] 0.355	[400] 1.000	[633] 1.000	0.000	0.000	0.000		0.000	
22	AIZE-006	[424] 0.004	[482] 0.030	[379] 0.319	[402] 0.008	[396] 0.144	[383] 0.114	[436] 0.045	[614] 0.996	[451] 0.998	[468] 0.304	[453] 1.000	[463] 0.885	0.000	0.000	0.036		0.000	
23	AKIYAMA-000	[475] 0.007	[479] 0.029	[368] 0.297	[462] 0.027	[398] 0.152	[446] 0.167	[426] 0.041	[566] 0.664	[261] 0.793	[524] 0.862	[467] 1.000	[497] 0.972	0.003	0.006	0.042		0.000	
24	AKIYAMA-001	[553] 0.026	[574] 0.136	[280] 0.122	[457] 0.023	[400] 0.157	[301] 0.091	[416] 0.036	[450] 0.162	[194] 0.484	[337] 0.037	[356] 0.968	[248] 0.177	0.014	0.105	0.046		0.001	
25	AKUVOX-001	[280] 0.001	[304] 0.012	[278] 0.118	[230] 0.002	[283] 0.022	[203] 0.069	[301] 0.013	[282] 0.050	[315] 0.964	[267] 0.019	[317] 0.737	[206] 0.141	0.001	0.000	0.044		0.000	
26	ALCHERA-000	[532] 0.016	[525] 0.047	[516] 0.869	[486] 0.046		[504] 0.292	[529] 0.138	[474] 0.216	[455] 0.999	[442] 0.176		[445] 0.803	0.006	0.014	0.328			
27	ALCHERA-001	[632] 0.987	[631] 1.000		[556] 1.000		[585] 1.000	[625] 0.999	[637] 1.000		[604] 1.000		[564] 1.000	0.006	0.013	0.324			
28	ALCHERA-002	[592] 0.095	[580] 0.166	[548] 0.954	[543] 0.668		[524] 0.446	[596] 0.485	[554] 0.591	[488] 1.000	[520] 0.827		[446] 0.811	0.001	0.002	0.106			
29	ALCHERA-003	[505] 0.010	[495] 0.035	[478] 0.738	[441] 0.016		[471] 0.206	[535] 0.155	[486] 0.239	[471] 0.999	[441] 0.172		[368] 0.464	0.001	0.002	0.106			
30	ALCHERA-004	[510] 0.011	[501] 0.038	[388] 0.338	[442] 0.017	[375] 0.088	[428] 0.144	[586] 0.394	[549] 0.529	[386] 0.991	[485] 0.424	[312] 0.708	[393] 0.546	0.001	0.001	0.046		0.000	
31	ALCHERA-005	[476] 0.007	[469] 0.027	[353] 0.265	[422] 0.012	[370] 0.081	[390] 0.118	[574] 0.314	[534] 0.439	[363] 0.985	[469] 0.305	[342] 0.908	[364] 0.455	0.000	0.001	0.042		0.000	
32	ALCHERA-006	[168] 0.001	[204] 0.009	[253] 0.096	[81] 0.001	[174] 0.008	[107] 0.052	[195] 0.005	[163] 0.025	[352] 0.982	[124] 0.006	[298] 0.493	[120] 0.094	0.000	0.001	0.042		0.000	
33	ALCHERA-007	[164] 0.001	[201] 0.009	[248] 0.095	[108] 0.001	[177] 0.008	[104] 0.052	[152] 0.004	[145] 0.023	[355] 0.982	[105] 0.005	[321] 0.743	[95] 0.087	0.000	0.001	0.042		0.000	
34	ALCHERA-008	[191] 0.001	[196] 0.009	[93] 0.049	[60] 0.001	[379] 0.096	[100] 0.052	[143] 0.003	[133] 0.022	[87] 0.177	[212] 0.010	[388] 1.000	[261] 0.186	0.000	0.000	0.040		0.000	
35	ALGORITHMSINC-000	[306] 0.002	[252] 0.011	[320] 0.181	[291] 0.003	[380] 0.107	[281] 0.083	[330] 0.016	[424] 0.130	[197] 0.507	[408] 0.108	[404] 1.000	[358] 0.433	0.000	0.000	0.041		0.000	
36	ALLGOVISION-000	[513] 0.011	[491] 0.033	[519] 0.893	[452] 0.021		[499] 0.282	[499] 0.088	[453] 0.166	[382] 0.990	[411] 0.117		[387] 0.526	0.002	0.003	0.122			
37	ALLGOVISION-001	[491] 0.009	[507] 0.038	[457] 0.657	[449] 0.021		[486] 0.241	[506] 0.101	[478] 0.221	[365] 0.986	[429] 0.150		[378] 0.491	0.001	0.001	0.042			
38	ALLGOVISION-002	[426] 0.004	[449] 0.023	[447] 0.580	[401] 0.008	[401] 0.161	[416] 0.132	[476] 0.065	[548] 0.520	[303] 0.950	[454] 0.232	[411] 1.000	[427] 0.716	0.000	0.000	0.041		0.000	
39	ALLGOVISION-003	[335] 0.002	[335] 0.014	[316] 0.175	[277] 0.003	[250] 0.015	[278] 0.082	[334] 0.017	[311] 0.059	[218] 0.566	[280] 0.021	[207] 0.147	[374] 0.478	0.001	0.000	0.042		0.000	
40	ALLGOVISION-004	[502] 0.010	[465] 0.026	[382] 0.321	[427] 0.013	[357] 0.059	[405] 0.127	[511] 0.114	[466] 0.190	[254] 0.747	[412] 0.121	[291] 0.418	[351] 0.412	0.001	0.000	0.042		0.000	
41	ALLGOVISION-005	[330] 0.002	[321] 0.013	[230] 0.090	[239] 0.002	[361] 0.064	[240] 0.074	[358] 0.020	[320] 0.064	[166] 0.393	[348] 0.043	[392] 1.000	[324] 0.324	0.001	0.000	0.042		0.000	
42	ALTTEKGLOBAL-000	[614] 0.274	[593] 0.255	[433] 0.526	[533] 0.380	[428] 0.419	[536] 0.632	[568] 0.278	[501] 0.267	[226] 0.617	[479] 0.383	[302] 0.536	[413] 0.641	0.000	0.001	0.048		0.001	
43	ANKE-000	[524] 0.013	[502] 0.038	[534] 0.930	[570] 1.000		[613] 1.000	[513] 0.117	[476] 0.220	[399] 0.994	[587] 1.000		[599] 1.000	0.000	0.001	0.080			
44	ANKE-001	[525] 0.013	[503] 0.038	[542] 0.945	[600] 1.000		[572] 1.000	[516] 0.119	[477] 0.220	[406] 0.994	[629] 1.000		[545] 1.000	0.000	0.001	0.080			
45	ANKE-002	[389] 0.002	[382] 0.016	[428] 0.516	[344] 0.005		[391] 0.119	[410] 0.031	[353] 0.079	[302] 0.947	[329] 0.034		[289] 0.245	0.001	0.001	0.049			
46	ARATEK-001	[134] 0.001	[133] 0.008	[306] 0.152	[231] 0.002	[188] 0.009	[187] 0.065	[250] 0.008	[279] 0.050	[552] 1.000	[399] 0.086	[357] 0.971	[613] 1.000	0.000	0.000	0.000		0.000	

Table 24: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
47	ARMATURA-000	299 0.002	317 0.013	169 0.060	202 0.002	116 0.005	168 0.062	170 0.004	167 0.025	112 0.214	127 0.006	102 0.048	109 0.090	0.001	0.001	0.056		0.001	
48	AWARE-003	561 0.030	558 0.090	561 0.966	531 0.316		502 0.290	523 0.128	513 0.298	360 0.984	486 0.428		388 0.530	0.004	0.003	0.874			
49	AWARE-004	584 0.068	583 0.176	571 0.975	513 0.122		520 0.414	566 0.268	545 0.509	490 1.000	480 0.397		447 0.816	0.003	0.003	0.776			
50	AWARE-005	562 0.031	545 0.067	572 0.978	490 0.048		508 0.308	581 0.363	493 0.253	501 1.000	458 0.255		471 0.916	0.001	0.002	0.189			
51	AWARE-006	586 0.069	571 0.128	575 0.983	512 0.111		522 0.421	567 0.275	527 0.398	482 0.999	476 0.368		433 0.749	0.001	0.002	0.189			
52	AWARE-007	366 0.002	384 0.016	389 0.339	375 0.007	335 0.038	385 0.116	329 0.016	333 0.070	390 0.992	352 0.044	360 0.982	292 0.250	0.000	0.000	0.043		0.000	
53	AWIROS-000	480 0.007	486 0.031	533 0.930	446 0.018	383 0.109	444 0.165	570 0.291	541 0.471	516 1.000	536 0.976	468 1.000	520 0.999	0.000	0.000	0.000		0.000	
54	AYONIX-000	621 0.450	619 0.685	586 0.996	541 0.607		548 0.867	610 0.811	600 0.939	441 0.998	532 0.954		500 0.982	0.010	0.031	0.939			
55	AYONIX-001	616 0.341	605 0.527	580 0.993	555 0.994		543 0.778	612 0.824	594 0.920	479 0.999	546 0.999		494 0.969	0.010	0.031	0.939			
56	AYONIX-002	615 0.341	606 0.527	581 0.993	536 0.464		544 0.778	611 0.824	595 0.920	481 0.999	528 0.915		495 0.969	0.010	0.031	0.939			
57	BIOIDTECHSWISS-001	139 0.001	392 0.017	181 0.062	357 0.005	273 0.018	448 0.168	164 0.004	205 0.030	370 0.987	208 0.010	145 0.067	254 0.179	0.000	0.009	0.042		0.004	
58	BIOMETRIK-001	439 0.004	371 0.015	477 0.734	387 0.007	333 0.038	340 0.101	449 0.052	389 0.098	497 1.000	366 0.054	275 0.341	409 0.618	0.000	0.000	0.000		0.000	
59	CAMVI-003	578 0.052	559 0.090	523 0.910	507 0.093		515 0.360	480 0.071	426 0.132	323 0.970	410 0.114		346 0.402	0.006	0.013	0.675			
60	CAMVI-004	576 0.047	552 0.077	480 0.741	502 0.072		506 0.296	481 0.071	428 0.136	477 0.999	405 0.100		441 0.787	0.000	0.000	0.000			
61	CAMVI-005	583 0.065	567 0.103	482 0.743	510 0.098		514 0.341	504 0.099	463 0.179	489 1.000	432 0.156		516 0.999	0.000	0.000	0.000			
62	CANON-001	113 0.001	72 0.006	221 0.077	180 0.001	144 0.007	174 0.062	204 0.005	154 0.023	156 0.357	172 0.008	150 0.068	200 0.139	0.001	0.000	0.042		0.000	
63	CANON-002	144 0.001	75 0.006	251 0.095	146 0.001	151 0.007	160 0.059	180 0.004	122 0.020	168 0.400	235 0.013	159 0.075	264 0.188	0.001	0.000	0.042		0.000	
64	CANON-003	59 0.001	151 0.008	197 0.067	307 0.004	139 0.006	94 0.051	97 0.002	74 0.012	102 0.199	123 0.006	56 0.030	55 0.076	0.000	0.003	0.041		0.003	
65	CIB-000	247 0.001	162 0.008	237 0.090	233 0.002	220 0.011	210 0.069	291 0.012	270 0.045	514 1.000	257 0.017	199 0.141	466 0.894	0.000	0.000	0.000		0.000	
66	CLEARVIEWAI-000	114 0.001	111 0.007	111 0.051	125 0.001	127 0.006	140 0.056	213 0.005	171 0.025	332 0.974	173 0.008	126 0.057	302 0.268	0.000	0.000	0.037		0.000	
67	CLEARVIEWAI-001	118 0.001	123 0.007	46 0.044	84 0.001	54 0.004	123 0.054	149 0.003	121 0.020	76 0.167	114 0.005	91 0.042	73 0.079	0.000	0.000	0.040		0.000	
68	CLEARVIEWAI-002	124 0.001	125 0.007	48 0.044	86 0.001	55 0.004	115 0.054	162 0.004	131 0.022	83 0.170	115 0.006	90 0.041	67 0.078	0.000	0.000	0.040		0.000	
69	CLEARVIEWAI-003	119 0.001	115 0.007	38 0.043	79 0.001	50 0.003	127 0.055	96 0.002	93 0.015	49 0.128	99 0.005	78 0.038	70 0.079	0.000	0.000	0.040		0.000	
70	CLOUDWALK-HR-000	236 0.001	227 0.010	123 0.053	222 0.002	138 0.006	146 0.057	84 0.002	82 0.013	46 0.123	100 0.005	64 0.033	136 0.099	0.001	0.000	0.042		0.000	
71	CLOUDWALK-MT-000	285 0.001	268 0.011	62 0.047	122 0.001	61 0.004	91 0.051	82 0.002	81 0.013	30 0.099	53 0.002	19 0.018	42 0.072	0.001	0.000	0.042		0.000	
72	CLOUDWALK-MT-001	283 0.001	270 0.011	31 0.043	22 0.001	27 0.003	19 0.042	76 0.002	66 0.012	6 0.059	13 0.001	15 0.015	12 0.056	0.001	0.000	0.042		0.000	
73	CLOUDWALK-MT-002	284 0.001	267 0.011	21 0.041	20 0.001	19 0.002	17 0.041	73 0.002	62 0.011	3 0.053	8 0.001	8 0.013	10 0.055	0.001	0.000	0.042		0.000	
74	COGENT-000	504 0.010	523 0.046	559 0.965				452 0.052	432 0.140	413 0.995				0.000	0.000	0.000			
75	COGENT-001	503 0.010	522 0.046	560 0.965				451 0.052	431 0.140	412 0.995				0.000	0.000	0.000			
76	COGENT-002	416 0.003	424 0.020	528 0.924				433 0.044	387 0.098	438 0.998				0.000	0.000	0.000			
77	COGENT-003	418 0.003	431 0.021	539 0.938				437 0.045	380 0.095	442 0.998				0.000	0.000	0.000			
78	COGENT-004	315 0.002	327 0.013	527 0.921	318 0.004	277 0.019	379 0.113	411 0.033	283 0.051	436 0.997	284 0.022	191 0.126	365 0.456	0.000	0.000	0.000		0.000	
79	COGENT-005	269 0.001	246 0.010	276 0.116	234 0.002	214 0.010	393 0.120	260 0.009	243 0.037	375 0.989	217 0.011	166 0.082	469 0.905	0.000	0.000	0.000		0.000	
80	COGENT-006	167 0.001	127 0.007	150 0.056	158 0.001	161 0.007	341 0.101	173 0.004	155 0.023	126 0.229	141 0.006	292 0.422	191 0.130	0.000	0.000	0.041		0.000	
81	COGENT-007	116 0.001	161 0.008	127 0.053	148 0.001	134 0.006	347 0.103	144 0.003	129 0.021	93 0.185	145 0.006	100 0.047	212 0.144	0.000	0.000	0.043		0.000	
82	COGNITEC-000	550 0.025	539 0.059	557 0.963				539 0.160	514 0.303	388 0.992				0.003	0.002	0.924			
83	COGNITEC-001	515 0.011	493 0.034	552 0.957				505 0.101	482 0.230	569 1.000				0.003	0.002	0.924			
84	COGNITEC-002	454 0.005	459 0.025	544 0.948				454 0.053	462 0.178	515 1.000				0.003	0.002	0.924			
85	COGNITEC-003	460 0.006	458 0.025	532 0.929				450 0.052	451 0.162	523 1.000				0.004	0.002	0.878			
86	COGNITEC-004	403 0.003	380 0.016	505 0.811	425 0.013	335 0.057	427 0.143	409 0.031	384 0.097	379 0.990	382 0.068	269 0.316	307 0.288	0.002	0.001	0.635		0.006	
87	COGNITEC-005	265 0.001	237 0.010	472 0.710	453 0.021	332 0.037	384 0.115	266 0.009	258 0.041	590 1.000	346 0.041	220 0.157	253 0.179	0.002	0.001	0.614		0.017	
88	COGNITEC-006	249 0.001	221 0.010	468 0.699	385 0.007	297 0.024	373 0.111	244 0.007	254 0.040	636 1.000	314 0.030	227 0.171	421 0.681	0.002	0.001	0.568		0.003	
89	COGNITEC-007	208 0.001	178 0.009	357 0.275	358 0.005	286 0.022	328 0.098	211 0.005	229 0.034	230 0.621	297 0.026	244 0.210	232 0.160	0.000	0.001	0.100		0.002	
90	COGNITEC-008	189 0.001	197 0.009	337 0.216	337 0.004	272 0.018	316 0.095	217 0.006	234 0.035	217 0.564	298 0.026	202 0.142	224 0.155	0.000	0.001	0.091		0.001	
91	COGNITEC-009	241 0.001	207 0.009	339 0.219	332 0.004	257 0.016	343 0.102	221 0.006	235 0.035	216 0.554	293 0.025	189 0.124	226 0.156	0.001	0.001	0.106		0.001	
92	COGNITEC-010	157 0.001	254 0.011	307 0.153	303 0.004	264 0.017	271 0.082	159 0.004	203 0.030	191 0.479	247 0.016	167 0.086	179 0.125	0.000	0.003	0.059		0.002	

Table 25: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
93	COGNITI-000	303 0.002	297 0.012	319 0.179	325 0.004	419 0.267	257 0.078	350 0.019	351 0.078	242 0.693	434 0.160	567 1.000	381 0.502	0.000	0.000	0.040		0.000	
94	COMPAL-000	281 0.001	409 0.019	180 0.062	500 0.065	360 0.063	400 0.125	247 0.008	250 0.038		386 0.072	180 0.104	214 0.147	0.000	0.012	0.000		0.067	
95	COMPAL-001	472 0.007	349 0.014	173 0.060	373 0.006	211 0.010	188 0.065	295 0.012	223 0.033	174 0.422	232 0.013	171 0.090	170 0.117	0.004	0.002	0.049		0.005	
96	COMPAL-002	471 0.007	350 0.014	171 0.060	374 0.006	210 0.010	185 0.065	296 0.012	219 0.033	173 0.419	234 0.013	170 0.090	172 0.118	0.004	0.002	0.049		0.005	
97	CORSIGHT-000	120 0.001	110 0.007	192 0.065	149 0.001	143 0.007	144 0.057	222 0.006	196 0.029	201 0.517	199 0.009	146 0.067	401 0.594	0.000	0.000	0.000		0.000	
98	CORSIGHT-001	123 0.001	109 0.007	172 0.060	140 0.001	128 0.006	133 0.055	230 0.006	225 0.034	186 0.460	183 0.008	132 0.062	199 0.138	0.000	0.000	0.000		0.000	
99	CORSIGHT-002	194 0.001	177 0.009	110 0.051	101 0.001	103 0.005	129 0.055	187 0.004	165 0.025	114 0.218	137 0.006	108 0.049	105 0.089	0.000	0.000	0.041		0.000	
100	CUBOX-000	216 0.001	230 0.010	73 0.048	192 0.002	84 0.004	75 0.049	112 0.002	112 0.019	70 0.158	89 0.004	52 0.028	48 0.073	0.001	0.000	0.042		0.000	
101	CUBOX-001	174 0.001	166 0.008	94 0.049	130 0.001	92 0.004	59 0.047	119 0.003	126 0.021	111 0.212	152 0.007	152 0.068	86 0.085	0.001	0.001	0.041		0.000	
102	CUBOX-002	238 0.001	227 0.011	105 0.050	132 0.001	74 0.004	44 0.046	134 0.003	140 0.022	88 0.181	92 0.004	80 0.038	61 0.077	0.001	0.001	0.040		0.000	
103	CUBOX-003	253 0.001	294 0.012	142 0.056	142 0.001	95 0.005	71 0.048	140 0.003	146 0.023	116 0.219	129 0.006	113 0.051	81 0.082	0.001	0.001	0.040		0.000	
104	CYBERLINK-000	421 0.004	420 0.020	473 0.713	391 0.007		418 0.134	462 0.056	406 0.116	415 0.995	378 0.063		330 0.339	0.001	0.001	0.063			
105	CYBERLINK-001	414 0.003	401 0.018	474 0.728	379 0.007		417 0.133	455 0.053	403 0.109	410 0.995	375 0.062		416 0.652	0.000	0.000	0.040			
106	CYBERLINK-002	327 0.002	296 0.012	445 0.572	311 0.004		362 0.107	317 0.014	297 0.053	374 0.988	291 0.024		308 0.288	0.001	0.000	0.042			
107	CYBERLINK-003	252 0.001	194 0.009	417 0.468	270 0.003	221 0.012	275 0.082	245 0.007	233 0.035	329 0.972	224 0.012	179 0.100	336 0.368	0.000	0.000	0.039		0.000	
108	CYBERLINK-004	268 0.001	287 0.011	404 0.416	266 0.003	215 0.011	352 0.104	237 0.007	237 0.036	533 1.000	230 0.013	181 0.109	490 0.954	0.000	0.000	0.011		0.000	
109	CYBERLINK-005	290 0.001	258 0.011	329 0.200	244 0.002	204 0.010	329 0.098	272 0.010	255 0.041	498 1.000	239 0.014	169 0.089	478 0.926	0.000	0.000	0.034		0.000	
110	CYBERLINK-006	137 0.001	103 0.007	232 0.085	90 0.001	77 0.004	216 0.071	109 0.002	108 0.018	366 0.986	93 0.004	157 0.073	260 0.186	0.000	0.000	0.034		0.000	
111	DAHUA-000	496 0.009	464 0.026					495 0.086	422 0.135					0.004	0.003				
112	DAHUA-001	466 0.006	452 0.024	469 0.699				483 0.072	414 0.122	345 0.980				0.002	0.002	0.346			
113	DAHUA-002	293 0.001	292 0.012	369 0.296	265 0.003		282 0.084	318 0.015	274 0.046	231 0.634	254 0.017		230 0.159	0.001	0.000	0.099			
114	DAHUA-003	153 0.001	116 0.007	327 0.197	224 0.002	195 0.009	231 0.073	307 0.014	257 0.041	220 0.574	227 0.013	164 0.081	193 0.134	0.000	0.000	0.000		0.000	
115	DAHUA-004	107 0.001	133 0.008	290 0.134	194 0.002	150 0.007	205 0.069	235 0.007	175 0.026	192 0.479	185 0.008	114 0.051	169 0.113	0.000	0.000	0.000		0.000	
116	DAON-000	425 0.004	393 0.017	431 0.525	345 0.005	281 0.020	401 0.125	370 0.022	315 0.061	496 1.000	294 0.025	229 0.173	455 0.846	0.002	0.002	0.108		0.001	
117	DAON-001	551 0.025	536 0.057	545 0.951	483 0.040	344 0.046	280 0.083	428 0.043	430 0.140	304 0.952	379 0.064	216 0.154	234 0.161	0.014	0.042	0.950		0.034	
118	DAON-002	177 0.001	209 0.009	269 0.109	220 0.002	236 0.013	195 0.067	238 0.007	267 0.044		245 0.015	197 0.141	631 1.000	0.000	0.001	0.000		0.000	
119	DAON-003	106 0.001	175 0.009	162 0.059	155 0.001	216 0.011	161 0.060	139 0.003	150 0.023	148 0.319	143 0.006	147 0.067	128 0.096	0.000	0.001	0.040		0.000	
120	DATECH-000	193 0.001	167 0.008	313 0.168	171 0.001	342 0.045	126 0.055	246 0.007	248 0.038	424 0.996	250 0.016	345 0.926	316 0.311	0.000	0.000	0.039		0.000	
121	DATECH-001	128 0.001	160 0.008	289 0.132	138 0.001	351 0.052	141 0.056	231 0.006	317 0.062	395 0.993	342 0.039	373 0.998	334 0.360	0.000	0.000	0.038		0.000	
122	DECATUR-000	332 0.002	291 0.011	340 0.220	326 0.004	276 0.019	369 0.109	373 0.023	324 0.066	237 0.671	302 0.027	228 0.173	285 0.239	0.001	0.000	0.044		0.001	
123	DEEPGLINT-001	229 0.001	107 0.007	323 0.190	254 0.002		233 0.073	133 0.003	84 0.014	486 1.000	147 0.006		229 0.159	0.000	0.000	0.038			
124	DEEPSEA-001	430 0.004	378 0.016	506 0.812	405 0.010		425 0.140	439 0.046	391 0.101	362 0.984	393 0.077		326 0.326	0.000	0.001	0.047			
125	DERMALOG-003	598 0.126	590 0.217		530 0.296		534 0.560	595 0.482	563 0.655		512 0.677		460 0.870	0.002	0.002	0.103			
126	DERMALOG-004	597 0.125	588 0.215	531 0.929	516 0.135		526 0.467	594 0.479	564 0.657	416 0.995	503 0.603		457 0.856	0.001	0.002	0.107			
127	DERMALOG-005	529 0.015	500 0.037	467 0.697	524 0.242		518 0.384	498 0.088	442 0.154	380 0.990	466 0.300		407 0.614	0.001	0.002	0.102			
128	DERMALOG-006	486 0.008	457 0.024	454 0.615	409 0.010		439 0.155	448 0.051	395 0.105	348 0.981	373 0.059		321 0.318	0.003	0.006	0.181			
129	DERMALOG-007	493 0.009	467 0.027	461 0.671	435 0.014		449 0.170	496 0.086	440 0.152	381 0.990	404 0.099		396 0.557	0.001	0.002	0.102			
130	DERMALOG-008	390 0.003	361 0.015	425 0.511	376 0.007	316 0.029	424 0.139	435 0.045	377 0.094	542 1.000	369 0.057	284 0.382	485 0.940	0.000	0.000	0.002		0.000	
131	DERMALOG-009	388 0.002	348 0.014	310 0.157	386 0.007	449 0.999	355 0.106	356 0.020	325 0.066	526 1.000	316 0.031	377 0.999	452 0.840	0.001	0.001	0.018		0.003	
132	DERMALOG-010	342 0.002	284 0.011	143 0.056	481 0.038	391 0.124	380 0.113	234 0.007	301 0.055	464 0.999	401 0.089	384 1.000	385 0.522	0.001	0.001	0.018		0.003	
133	DERMALOG-011	255 0.001	239 0.010	234 0.086	290 0.003	318 0.031	303 0.092	363 0.021	365 0.087	449 0.998	418 0.129	364 0.991	436 0.764	0.000	0.000	0.013		0.000	
134	DERMALOG-012	327 0.002	325 0.013	254 0.096	274 0.003	242 0.014	222 0.071	362 0.021	327 0.067	195 0.505	286 0.023	208 0.147	350 0.409	0.001	0.000	0.040		0.000	
135	DERMALOG-013	308 0.002	319 0.013	255 0.096	284 0.003	243 0.014	224 0.071	361 0.021	330 0.068	293 0.908	287 0.024	215 0.153	390 0.541	0.000	0.000	0.038		0.000	
136	DERMALOG-014	410 0.003	377 0.016	283 0.123	404 0.009	290 0.023	363 0.107	365 0.021	331 0.069	292 0.907	306 0.028	219 0.156	357 0.432	0.000	0.000	0.038		0.000	
137	DERMALOG-015	254 0.001	265 0.011	149 0.056	219 0.002	155 0.007	137 0.056	197 0.005	210 0.031	145 0.313	192 0.009	139 0.065	228 0.158	0.001	0.000	0.040		0.000	
138	DERMALOG-016	221 0.001	244 0.010	79 0.048	195 0.002	99 0.005	86 0.050	90 0.002	119 0.020	140 0.288	84 0.004	72 0.035	391 0.544	0.001	0.000	0.040		0.000	

Table 26: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
139	DERMALOG-017	[225] 0.001	[243] 0.010	[80] 0.048	[193] 0.002	[98] 0.005	[87] 0.050	[129] 0.003	[151] 0.023	[118] 0.220	[126] 0.006	[86] 0.041	[277] 0.220	0.001	0.000	0.040		0.000	
140	DIGIDATA-000	[625] 0.590	[607] 0.548	[520] 0.894	[542] 0.642	[440] 0.707	[546] 0.813	[601] 0.610	[553] 0.577	[403] 0.994	[508] 0.646	[329] 0.789	[450] 0.824	0.002	0.001	0.070		0.001	
141	DILUSENSE-000	[344] 0.002	[302] 0.012	[363] 0.289	[394] 0.008	[312] 0.028	[333] 0.099	[407] 0.030	[352] 0.078	[233] 0.651	[340] 0.039	[310] 0.664	[271] 0.203	0.001	0.001	0.219		0.006	
142	DILUSENSE-001	[230] 0.001	[216] 0.010	[347] 0.241	[366] 0.006	[255] 0.016	[294] 0.088	[254] 0.008	[230] 0.034	[180] 0.450	[246] 0.015	[300] 0.511	[198] 0.137	0.001	0.001	0.219		0.007	
143	EINETWORKSINDIA-000	[286] 0.001	[182] 0.009	[400] 0.386	[298] 0.004	[238] 0.013	[350] 0.104	[322] 0.015	[268] 0.044	[435] 0.997	[271] 0.020	[296] 0.455	[231] 0.159	0.001	0.001	0.055		0.002	
144	EINETWORKSINDIA-001	[277] 0.001	[181] 0.009	[396] 0.375	[273] 0.003	[244] 0.014	[321] 0.096	[328] 0.016	[276] 0.048	[433] 0.997	[387] 0.072	[354] 0.957	[440] 0.782	0.001	0.000	0.043		0.000	
145	EINETWORKSINDIA-002	[135] 0.001	[87] 0.007	[98] 0.050	[67] 0.001	[324] 0.033	[85] 0.050	[158] 0.004	[136] 0.022	[96] 0.188	[170] 0.007	[352] 0.953	[185] 0.127	0.000	0.000	0.041		0.000	
146	EYEDEA-003	[588] 0.080	[577] 0.148	[555] 0.960	[511] 0.101		[517] 0.379	[584] 0.387	[551] 0.543	[407] 0.994	[500] 0.570		[443] 0.792	0.001	0.003	0.161			
147	F8-001	[519] 0.012		[459] 0.665	[582] 1.000		[633] 1.000	[541] 0.166		[453] 0.998				0.004	1.000	0.158			
148	FACEHAWK-001	[560] 0.030	[581] 0.168	[399] 0.382	[495] 0.056	[424] 0.337	[482] 0.228	[519] 0.123	[562] 0.654	[273] 0.851	[522] 0.844	[470] 1.000	[505] 0.990	0.013	0.122	0.045		0.001	
149	FACEPHI-000	[295] 0.002	[354] 0.014	[242] 0.093	[283] 0.003	[198] 0.009	[351] 0.104	[229] 0.006	[231] 0.034	[159] 0.367	[205] 0.009	[142] 0.066	[195] 0.135	0.001	0.007	0.049		0.001	
150	FACEPHI-001	[112] 0.001	[156] 0.008	[177] 0.061	[157] 0.001	[123] 0.005	[315] 0.095	[81] 0.002	[98] 0.015	[182] 0.452	[71] 0.003	[107] 0.048	[201] 0.139	0.000	0.000	0.033		0.000	
151	FASTLAHORE-000	[555] 0.026	[579] 0.154	[538] 0.938	[525] 0.277	[453] 1.000	[529] 0.499	[556] 0.214	[623] 1.000	[523] 1.000	[560] 1.000	[579] 1.000	[621] 1.000	0.000	0.064	0.000		0.000	
152	FCI-000	[296] 0.002	[309] 0.012	[176] 0.061	[141] 0.001	[356] 0.057	[92] 0.051	[225] 0.006	[178] 0.026	[307] 0.957	[190] 0.008	[327] 0.777	[149] 0.103	0.000	0.000	0.000		0.000	
153	FINCORE-000	[509] 0.010	[494] 0.034	[489] 0.764	[471] 0.032	[389] 0.117	[464] 0.191	[528] 0.133	[478] 0.217	[504] 1.000	[444] 0.187	[305] 0.598	[366] 0.458	0.000	0.001	0.043		0.000	
154	FIRSTCREDITKZ-001	[156] 0.001	[165] 0.008	[231] 0.084	[226] 0.002	[206] 0.010	[183] 0.065	[114] 0.003	[113] 0.019	[138] 0.282	[166] 0.007	[131] 0.061	[132] 0.097	0.000	0.001	0.047		0.001	
155	FISSIONSFR-001	[593] 0.098	[573] 0.131	[458] 0.659	[520] 0.157	[420] 0.267	[519] 0.408	[517] 0.121	[461] 0.177	[275] 0.859	[447] 0.193	[350] 0.950	[376] 0.490	0.001	0.000	0.040		0.000	
156	FPT-000	[305] 0.002	[356] 0.014	[201] 0.068	[348] 0.005	[411] 0.210	[336] 0.100	[264] 0.009	[523] 0.372	[454] 0.999	[475] 0.360	[592] 1.000	[482] 0.935	0.000	0.000	0.000		0.000	
157	FUJITSULAB-000	[346] 0.002	[341] 0.014	[409] 0.434	[317] 0.004	[288] 0.023	[330] 0.098	[357] 0.020	[304] 0.056		[288] 0.024	[231] 0.177	[286] 0.240	0.000	0.001	0.016		0.000	
158	FUJITSULAB-001	[297] 0.002	[333] 0.013	[412] 0.449	[320] 0.004	[301] 0.026	[358] 0.106	[342] 0.018	[306] 0.058	[392] 0.992	[290] 0.024	[318] 0.739	[291] 0.247	0.000	0.003	0.150		0.002	
159	GLORY-000	[602] 0.178	[597] 0.320	[585] 0.993	[523] 0.228		[537] 0.678	[582] 0.367	[552] 0.547	[409] 0.995	[490] 0.453		[451] 0.839	0.011	0.013	0.985			
160	GLORY-001	[599] 0.126	[594] 0.267	[579] 0.992	[521] 0.178		[535] 0.594	[571] 0.305	[550] 0.537	[396] 0.993	[482] 0.408		[448] 0.819	0.011	0.013	0.988			
161	GORILLA-001	[580] 0.060	[563] 0.095	[536] 0.936	[501] 0.069		[512] 0.329	[588] 0.406	[538] 0.453	[538] 1.000	[491] 0.468		[563] 1.000	0.001	0.001	0.069			
162	GORILLA-002	[543] 0.019	[518] 0.044	[485] 0.751	[461] 0.027		[479] 0.214	[550] 0.188	[502] 0.268	[550] 1.000	[457] 0.250		[525] 1.000	0.001	0.001	0.069			
163	GORILLA-003	[565] 0.036	[547] 0.070	[509] 0.818	[488] 0.048		[492] 0.265	[575] 0.317	[533] 0.434	[605] 1.000	[481] 0.407		[580] 1.000	0.001	0.001	0.069			
164	GORILLA-004	[461] 0.006	[453] 0.024	[465] 0.693	[417] 0.012		[443] 0.162	[502] 0.089	[448] 0.160	[308] 0.958	[423] 0.135		[360] 0.438	0.000	0.001	0.042			
165	GORILLA-005	[402] 0.003	[402] 0.018	[328] 0.199	[365] 0.006		[398] 0.124	[467] 0.058	[434] 0.142	[243] 0.697	[400] 0.088		[319] 0.315	0.000	0.000	0.040			
166	GORILLA-006	[282] 0.001	[299] 0.012	[270] 0.112	[288] 0.003	[266] 0.018	[354] 0.105	[393] 0.027	[370] 0.089	[209] 0.525	[304] 0.028	[225] 0.166	[276] 0.218	0.000	0.000	0.041		0.000	
167	GORILLA-007	[271] 0.001	[255] 0.011	[262] 0.103	[250] 0.002	[259] 0.016	[293] 0.088	[390] 0.027	[350] 0.077	[211] 0.529	[296] 0.026	[255] 0.264	[251] 0.178	0.000	0.000	0.041		0.000	
168	GORILLA-008	[237] 0.001	[224] 0.010	[219] 0.075	[206] 0.002	[228] 0.012	[276] 0.082	[374] 0.024	[360] 0.083	[183] 0.457	[313] 0.030	[272] 0.319	[249] 0.178	0.000	0.000	0.041		0.000	
169	GRIAULE-000	[369] 0.002	[334] 0.014	[380] 0.319	[412] 0.011	[321] 0.031	[403] 0.126	[353] 0.020	[319] 0.063	[411] 0.995	[323] 0.033	[236] 0.185	[268] 0.198	0.000	0.002	0.090		0.001	
170	GRIAULE-001	[150] 0.001	[148] 0.008	[281] 0.123	[112] 0.001	[292] 0.023	[184] 0.065	[181] 0.004	[189] 0.028	[276] 0.863	[160] 0.007	[367] 0.995	[135] 0.099	0.000	0.000	0.000		0.000	
171	GTELOTS-000	[606] 0.194	[589] 0.217	[444] 0.558	[526] 0.278	[432] 0.446	[533] 0.560	[554] 0.206	[495] 0.256	[247] 0.727	[464] 0.292	[566] 1.000	[403] 0.603	0.001	0.000	0.043		0.000	
172	GTELOTS-001	[531] 0.015	[416] 0.020	[315] 0.171	[482] 0.039	[354] 0.055	[468] 0.197	[440] 0.046	[368] 0.088	[241] 0.680	[388] 0.072	[262] 0.287	[311] 0.294	0.001	0.001	0.043		0.000	
173	HIK-003	[516] 0.011	[471] 0.027	[464] 0.685	[420] 0.012		[436] 0.151	[507] 0.102	[444] 0.158	[321] 0.968	[426] 0.142		[363] 0.445	0.000	0.000	0.048			
174	HIK-004	[514] 0.011	[468] 0.027	[479] 0.740	[418] 0.012		[438] 0.152	[503] 0.098	[441] 0.153	[333] 0.976	[424] 0.137		[359] 0.434	0.000	0.000	0.048			
175	HIK-005	[438] 0.004	[386] 0.017	[435] 0.530	[383] 0.007		[372] 0.111	[430] 0.043	[349] 0.077	[483] 0.999	[380] 0.068		[389] 0.541	0.000	0.000	0.000			
176	HIK-006	[437] 0.004	[385] 0.017	[436] 0.530				[441] 0.047	[364] 0.086	[545] 1.000				0.000	0.000	0.000			
177	HISIGN-001	[160] 0.001	[168] 0.008	[305] 0.150	[216] 0.002	[260] 0.016	[248] 0.076	[242] 0.007	[242] 0.037	[376] 0.990	[236] 0.013	[334] 0.824	[211] 0.143	0.000	0.000	0.041		0.000	
178	HYPERVERGE-001	[210] 0.001	[275] 0.011	[144] 0.056	[189] 0.002	[148] 0.007	[166] 0.061	[175] 0.004	[207] 0.031	[109] 0.210	[162] 0.007	[119] 0.053	[142] 0.101	0.001	0.000	0.041		0.000	
179	HYPERVERGE-002	[204] 0.001	[266] 0.011	[113] 0.052	[160] 0.001	[132] 0.006	[155] 0.058	[147] 0.003	[183] 0.027	[103] 0.201	[122] 0.006	[104] 0.048	[119] 0.093	0.001	0.000	0.041		0.000	
180	HYPERVERGE-003	[224] 0.001	[290] 0.011	[194] 0.065	[214] 0.002	[173] 0.008	[171] 0.062	[241] 0.007	[260] 0.042	[134] 0.274	[219] 0.012	[138] 0.065	[154] 0.106	0.001	0.000	0.041		0.000	
181	HYUNDAI-ROBOTICSLAB-000	[382] 0.002	[369] 0.015	[326] 0.195	[447] 0.020	[395] 0.133	[507] 0.298	[305] 0.013	[532] 0.452	[238] 0.672	[494] 0.492	[396] 1.000	[459] 0.870	0.001	0.001	0.045		0.009	
182	HYUNDAI-ROBOTICSLAB-001	[278] 0.001	[433] 0.021	[324] 0.192	[463] 0.029	[416] 0.252	[459] 0.178	[297] 0.013	[576] 0.774	[271] 0.839	[516] 0.699	[474] 1.000	[483] 0.937	0.000	0.001	0.041		0.001	
183	HYUNDAI-ROBOTICSLAB-002	[105] 0.001	[97] 0.007	[100] 0.050	[167] 0.001	[385] 0.112	[182] 0.064	[207] 0.005	[465] 0.189	[125] 0.228	[381] 0.068	[399] 1.000	[362] 0.440	0.000	0.000	0.040		0.000	
184	HYUNDAI-ROBOTICSLAB-003	[90] 0.001	[70] 0.006	[104] 0.050	[133] 0.001	[350] 0.051	[167] 0.062	[203] 0.005	[228] 0.034	[120] 0.221	[266] 0.019	[370] 0.995	[284] 0.236	0.000	0.000	0.041		0.000	

Table 27: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T → FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
185	HZAILU-000	[343] 0.002	[332] 0.013	[345] 0.235	[271] 0.003	[252] 0.015	[298] 0.090	[351] 0.019	[284] 0.051	[318] 0.967	[275] 0.020	[268] 0.316	[219] 0.153	0.001	0.001	0.054		0.001	
186	HZAILU-001	[312] 0.002	[271] 0.011	[252] 0.095	[243] 0.002	[386] 0.113	[302] 0.092	[255] 0.008	[464] 0.183	[369] 0.986	[449] 0.196	[635] 1.000	[420] 0.679	0.000	0.000	0.039		0.000	
187	HZAILU-002	[316] 0.002	[273] 0.011	[271] 0.112	[213] 0.002	[207] 0.010	[247] 0.076	[249] 0.008	[251] 0.039	[244] 0.701	[220] 0.012	[144] 0.066	[328] 0.330	0.001	0.000	0.041		0.000	
188	HZAILU-003	[263] 0.001	[260] 0.011	[245] 0.094	[154] 0.001	[156] 0.007	[228] 0.072	[251] 0.008	[247] 0.038	[213] 0.545	[204] 0.009	[134] 0.062	[174] 0.119	0.001	0.000	0.041		0.000	
189	HZAILU-004	[182] 0.001	[240] 0.010	[225] 0.079	[116] 0.001	[106] 0.005	[220] 0.071	[174] 0.004	[211] 0.031	[165] 0.390	[117] 0.006	[97] 0.045	[146] 0.102	0.001	0.000	0.041		0.000	
190	HZAILU-005	[181] 0.001	[233] 0.010	[195] 0.065	[98] 0.001	[64] 0.004	[196] 0.067	[141] 0.003	[162] 0.025	[150] 0.329	[79] 0.004	[63] 0.033	[121] 0.094	0.001	0.000	0.041		0.000	
191	HZAILU-006	[176] 0.001	[232] 0.010	[184] 0.063	[92] 0.001	[58] 0.004	[193] 0.066	[138] 0.003	[137] 0.022	[149] 0.325	[76] 0.004	[57] 0.030	[113] 0.091	0.001	0.000	0.041		0.000	
192	I2V-001	[393] 0.003	[370] 0.015	[241] 0.092	[354] 0.005	[262] 0.017	[192] 0.066	[444] 0.049	[604] 0.966	[493] 1.000	[527] 0.905	[442] 1.000	[501] 0.983	0.000	0.001	0.040		0.000	
193	ICEHOUSE-000	[440] 0.004	[476] 0.028	[495] 0.793	[407] 0.010	[392] 0.125	[441] 0.158	[501] 0.089	[588] 0.852	[495] 1.000	[409] 0.111	[504] 1.000	[371] 0.470	0.001	0.002	0.053		0.000	
194	ICEHOUSE-001	[201] 0.001	[202] 0.009	[204] 0.069	[261] 0.002	[196] 0.009	[175] 0.063	[267] 0.009	[259] 0.041	[155] 0.348	[231] 0.013	[178] 0.098	[255] 0.179	0.000	0.000	0.000		0.000	
195	ICM-000	[449] 0.005	[484] 0.031	[549] 0.957	[415] 0.012	[338] 0.040	[452] 0.174	[383] 0.026	[376] 0.092	[316] 0.965	[333] 0.036	[252] 0.241	[287] 0.242	0.005	0.014	0.949		0.006	
196	ICM-001	[399] 0.003	[381] 0.016	[397] 0.375	[353] 0.005	[313] 0.028	[377] 0.112	[404] 0.029	[491] 0.251	[510] 1.000	[470] 0.307	[402] 1.000	[461] 0.878	0.001	0.002	0.096		0.001	
197	IDBIO-000	[323] 0.002	[331] 0.013	[322] 0.187	[302] 0.004	[304] 0.027	[266] 0.081	[354] 0.020	[386] 0.097	[227] 0.618	[338] 0.037	[308] 0.636	[256] 0.180	0.000	0.000	0.042		0.000	
198	IDEMIA-003	[470] 0.007	[492] 0.034	[550] 0.957	[445] 0.018		[474] 0.210	[442] 0.047	[452] 0.165		[416] 0.123		[437] 0.766	0.000	0.000	0.041			
199	IDEMIA-004	[465] 0.006	[488] 0.032	[543] 0.947	[444] 0.018		[475] 0.210	[423] 0.037	[410] 0.118	[330] 0.973	[415] 0.123		[438] 0.766	0.000	0.000	0.041			
200	IDEMIA-005	[485] 0.008	[510] 0.039	[547] 0.953	[455] 0.021		[480] 0.217	[432] 0.044	[439] 0.150	[332] 0.977	[419] 0.130		[462] 0.879	0.000	0.000	0.041			
201	IDEMIA-006	[499] 0.009	[549] 0.072	[564] 0.969	[467] 0.030		[490] 0.253	[429] 0.043	[480] 0.226	[353] 0.982	[427] 0.144		[430] 0.733	0.000	0.000	0.041			
202	IDEMIA-007	[373] 0.002	[368] 0.015	[620] 1.000	[367] 0.006	[328] 0.036	[413] 0.131	[341] 0.018	[300] 0.055	[630] 1.000	[361] 0.052	[234] 0.182	[632] 1.000	0.000	0.000	0.040		0.000	
203	IDEMIA-008	[99] 0.001	[88] 0.007	[205] 0.069	[184] 0.001	[152] 0.007	[246] 0.075	[75] 0.002	[79] 0.013	[100] 0.195	[98] 0.005	[73] 0.036	[155] 0.106	0.000	0.000	0.040		0.000	
204	IDEMIA-009	[75] 0.001	[78] 0.006	[137] 0.055	[117] 0.001	[105] 0.005	[98] 0.051	[57] 0.001	[51] 0.011	[52] 0.131	[60] 0.003	[44] 0.027	[51] 0.074	0.000	0.000	0.040		0.000	
205	IDEMIA-010	[46] 0.001	[73] 0.006	[25] 0.048	[69] 0.001	[59] 0.004	[76] 0.049	[13] 0.001	[23] 0.008	[41] 0.120	[40] 0.002	[47] 0.028	[42] 0.070	0.000	0.000	0.037		0.000	
206	IDEMIA-011	[97] 0.001	[135] 0.008	[27] 0.042	[65] 0.001	[45] 0.003	[25] 0.043	[21] 0.001	[43] 0.009	[11] 0.070	[22] 0.002	[35] 0.025	[16] 0.061	0.000	0.000	0.036		0.000	
207	IDEMIA-012	[104] 0.001	[122] 0.007	[15] 0.041	[46] 0.001	[25] 0.003	[15] 0.041	[19] 0.001	[30] 0.009	[9] 0.069	[38] 0.002	[18] 0.016	[18] 0.061	0.000	0.000	0.036		0.000	
208	IDEMIA-013	[2] 0.000	[10] 0.005	[3] 0.039	[19] 0.001	[8] 0.002	[11] 0.039	[1] 0.000	[1] 0.006	[137] 0.282	[5] 0.001	[5] 0.010	[9] 0.055	0.000	0.000	0.030		0.000	
209	IDENTY-000	[635] 1.000	[628] 0.999	[607] 1.000	[554] 0.991	[451] 1.000	[555] 0.992	[636] 1.000	[616] 0.999	[580] 1.000	[539] 0.991	[386] 1.000	[508] 0.992	0.001	0.001	0.052		0.000	
210	IDENTY-001	[203] 0.001	[236] 0.010	[166] 0.060	[121] 0.001	[129] 0.006	[217] 0.071	[122] 0.003	[118] 0.020	[73] 0.162	[96] 0.005	[105] 0.048	[126] 0.094	0.001	0.001	0.052		0.000	
211	IDENTY-002	[95] 0.001	[90] 0.007	[103] 0.050	[106] 0.001	[102] 0.005	[88] 0.050	[117] 0.003	[101] 0.016	[69] 0.152	[94] 0.005	[99] 0.047	[76] 0.080	0.000	0.000	0.045		0.000	
212	IMAGUS-002	[611] 0.220	[595] 0.301	[577] 0.988				[606] 0.749	[582] 0.816	[534] 1.000				0.004	0.008	0.550			
213	IMAGUS-003	[618] 0.356	[603] 0.513	[582] 0.993				[609] 0.807	[592] 0.909	[519] 1.000				0.004	0.008	0.550			
214	IMAGUS-005	[311] 0.002	[298] 0.012	[325] 0.311	[364] 0.006	[284] 0.022	[415] 0.132	[345] 0.018	[323] 0.066	[270] 0.836	[307] 0.029	[224] 0.161	[281] 0.231	0.000	0.000	0.000		0.000	
215	IMAGUS-006	[319] 0.002	[338] 0.014	[362] 0.285	[321] 0.004	[279] 0.019	[376] 0.112	[349] 0.019	[332] 0.069	[288] 0.896	[305] 0.028	[223] 0.161	[296] 0.260	0.000	0.000	0.000		0.000	
216	IMAGUS-007	[321] 0.002	[328] 0.013	[327] 0.313	[312] 0.004	[285] 0.022	[388] 0.117	[372] 0.022	[341] 0.073	[286] 0.892	[317] 0.031	[226] 0.169	[301] 0.265	0.000	0.000	0.000		0.000	
217	IMAGUS-008	[591] 0.086	[561] 0.093	[369] 0.297	[454] 0.021	[371] 0.081	[393] 0.119	[619] 0.974	[577] 0.774	[419] 0.996	[497] 0.520	[387] 1.000	[384] 0.518	0.000	0.000	0.000		0.000	
218	IMDS-SOFTWARE-000	[130] 0.001	[119] 0.007	[212] 0.071	[182] 0.001	[181] 0.008	[235] 0.073	[227] 0.006	[221] 0.033	[546] 1.000	[228] 0.013	[245] 0.216	[595] 1.000	0.000	0.000	0.000		0.000	
219	IMDS-SOFTWARE-001	[136] 0.001	[95] 0.007	[186] 0.064	[176] 0.001	[200] 0.009	[322] 0.096	[232] 0.006	[226] 0.034	[557] 1.000	[269] 0.019	[293] 0.424	[634] 1.000	0.000	0.000	0.000		0.000	
220	IMDS-SOFTWARE-002	[131] 0.001	[120] 0.007	[211] 0.071	[183] 0.001	[182] 0.008	[234] 0.073	[228] 0.006	[222] 0.033	[529] 1.000	[229] 0.013	[246] 0.216	[526] 1.000	0.000	0.000	0.000		0.000	
221	IMPERIAL-000	[365] 0.002	[363] 0.015	[355] 0.272	[336] 0.004		[323] 0.097	[381] 0.026	[328] 0.068	[458] 0.999	[347] 0.042		[290] 0.245	0.000	0.000	0.000			
222	INCODE-000	[527] 0.049	[565] 0.100	[546] 0.951				[572] 0.310	[531] 0.420	[446] 0.998				0.001	0.004	0.173			
223	INCODE-001	[534] 0.016	[524] 0.046	[486] 0.759				[555] 0.211	[509] 0.296	[554] 1.000				0.001	0.004	0.173			
224	INCODE-002	[538] 0.017	[526] 0.048	[510] 0.841				[547] 0.184	[503] 0.269	[397] 0.993				0.000	0.001	0.066			
225	INCODE-003	[523] 0.013	[512] 0.040	[487] 0.762				[543] 0.167	[498] 0.264	[478] 0.999				0.000	0.001	0.066			
226	INCODE-004	[415] 0.003	[395] 0.017	[418] 0.468	[400] 0.008		[420] 0.135	[458] 0.053	[413] 0.120	[408] 0.994	[377] 0.063		[317] 0.313	0.000	0.001	0.066			
227	INCODE-005	[267] 0.001	[285] 0.011	[293] 0.138	[255] 0.002	[234] 0.013	[264] 0.079	[283] 0.011	[265] 0.043	[205] 0.522	[256] 0.017	[205] 0.145	[225] 0.155	0.000	0.000	0.042		0.000	
228	INCODE-006	[108] 0.001	[157] 0.008	[128] 0.053	[235] 0.002	[107] 0.005	[156] 0.058	[60] 0.001	[54] 0.011	[71] 0.159	[68] 0.003	[39] 0.026	[72] 0.079	0.000	0.000	0.041		0.000	
229	INCODE-007	[40] 0.001	[34] 0.006	[135] 0.054	[89] 0.001	[69] 0.004	[142] 0.056	[53] 0.001	[31] 0.009	[65] 0.146	[36] 0.002	[53] 0.028	[104] 0.089	0.000	0.000	0.043		0.000	
230	INNOV8TIF-000	[590] 0.086	[573] 0.130	[521] 0.903	[514] 0.127	[429] 0.426	[523] 0.426	[590] 0.414	[546] 0.517	[387] 0.991	[501] 0.581	[344] 0.922	[442] 0.792	0.000	0.002	0.089		0.001	

Table 28: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T → FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
231	INNOVATRICS-002	574 0.045	550 0.074	514 0.851				564 0.233	515 0.310	549 1.000				0.000	0.001	0.046			
232	INNOVATRICS-003	554 0.026	534 0.055	512 0.843				559 0.221	510 0.297	494 1.000				0.000	0.001	0.046			
233	INNOVATRICS-004	521 0.012	514 0.040	551 0.957				525 0.131	479 0.222	343 0.980				0.000	0.001	0.046			
234	INNOVATRICS-005	363 0.002	347 0.014	402 0.400	342 0.005		368 0.109	412 0.034	369 0.089	272 0.845	356 0.047		293 0.251	0.000	0.001	0.041			
235	INNOVATRICS-007	272 0.001	282 0.011	346 0.239	258 0.002	237 0.013	250 0.077	293 0.012	287 0.051	250 0.740	255 0.017	174 0.093	221 0.154	0.000	0.001	0.041		0.000	
236	INNOVATRICS-008	183 0.001	172 0.008	216 0.072	205 0.002	282 0.021	229 0.072	196 0.005	238 0.036	551 1.000	233 0.013	339 0.886	202 0.139	0.000	0.000	0.004		0.000	
237	INNOVATRICS-009	200 0.001	242 0.010	209 0.070	200 0.002	137 0.006	226 0.072	123 0.003	164 0.025	151 0.338	159 0.007	112 0.050	156 0.106	0.000	0.001	0.040		0.001	
238	INNOVATRICS-010	215 0.001	245 0.010	218 0.074	190 0.002	135 0.006	222 0.072	135 0.003	166 0.025	154 0.347	165 0.007	122 0.054	164 0.110	0.000	0.001	0.040		0.001	
239	INNOVATRICS-011	122 0.001	112 0.007	207 0.070	127 0.001	117 0.005	84 0.050	87 0.002	88 0.014	206 0.523	83 0.004	82 0.039	94 0.087	0.000	0.000	0.012		0.000	
240	INNOVATRICS-012	117 0.001	69 0.006	141 0.056	68 0.001	78 0.004	70 0.048	77 0.002	56 0.011	80 0.168	61 0.003	42 0.027	83 0.083	0.000	0.000	0.016		0.000	
241	INNOVATRICS-013	70 0.001	61 0.006	99 0.050	120 0.001	80 0.004	37 0.045	63 0.001	96 0.015	67 0.147	133 0.006	106 0.048	101 0.088	0.000	0.000	0.016		0.000	
242	INNOVATRICS-014	52 0.001	14 0.005	57 0.045	30 0.001	22 0.003	68 0.048	39 0.001	19 0.008	35 0.106	29 0.002	16 0.015	29 0.066	0.000	0.000	0.041		0.000	
243	INNOVATRICS-015	51 0.001	12 0.005	47 0.044	17 0.001	17 0.002	52 0.046	25 0.001	15 0.007	48 0.126	23 0.002	14 0.015	68 0.078	0.000	0.000	0.039		0.000	
244	INNOVATRICS-016	25 0.000	9 0.005	24 0.042	13 0.001	11 0.002	46 0.046	10 0.001	9 0.007	15 0.082	14 0.001	7 0.012	323 0.321	0.000	0.000	0.040		0.000	
245	INSPUR-000	79 0.001	613 0.606	112 0.052	466 0.030	427 0.395	466 0.195	224 0.006	593 0.915	500 1.000	531 0.936	506 1.000	514 0.998	0.000	0.000	0.000		0.000	
246	INSPUR-001	72 0.001	65 0.006	132 0.054	161 0.001	124 0.006	135 0.056	202 0.005	181 0.027	517 1.000	182 0.008	127 0.057	480 0.931	0.000	0.000	0.000		0.000	
247	INSPUR-002	63 0.001	59 0.006	86 0.049	164 0.001	112 0.005	119 0.054	209 0.005	199 0.029	467 0.999	176 0.008	118 0.052	492 0.957	0.000	0.000	0.000		0.000	
248	INTELIGENSIA-000	317 0.002	311 0.012	330 0.201	333 0.004	325 0.033	396 0.124	375 0.024	348 0.077	260 0.784	365 0.053	250 0.235	294 0.255	0.001	0.000	0.046		0.001	
249	INTELIGENSIA-001	100 0.001	92 0.007	152 0.057	204 0.002	179 0.008	348 0.103	208 0.005	187 0.028	181 0.451	174 0.008	149 0.067	190 0.129	0.001	0.000	0.046		0.001	
250	INTELLIVISION-001	566 0.036	566 0.102	566 0.971	496 0.057	413 0.222	513 0.333	569 0.279	528 0.404	502 1.000	472 0.328	322 0.749	423 0.685	0.001	0.000	0.044		0.000	
251	INTELLIVISION-002	507 0.010	485 0.031	540 0.942	443 0.018	369 0.080	469 0.200	534 0.154	467 0.196	462 0.999	422 0.134	295 0.437	367 0.460	0.001	0.000	0.043		0.000	
252	INTELRESEARCH-001	84 0.001	99 0.007	198 0.067	159 0.001	120 0.005	110 0.053	71 0.001	72 0.012	135 0.276	102 0.005	58 0.030	78 0.080	0.000	0.000	0.041		0.000	
253	INTEMA-000	132 0.001	170 0.008	68 0.047	124 0.001	122 0.005	90 0.051	93 0.002	104 0.017	541 1.000	101 0.005	263 0.288	79 0.081	0.000	0.000	0.040		0.000	
254	INTEMA-001	13 0.000	610 0.601	25 0.042	71 0.001	222 0.012	48 0.046	41 0.001	556 0.603	24 0.092	80 0.004	313 0.715	59 0.076	0.001	0.000	0.041		0.000	
255	INTOZI-001	251 0.001	456 0.024	309 0.156	499 0.064	421 0.277	487 0.242	323 0.015	599 0.938	535 1.000	544 0.998	519 1.000	544 1.000	0.000	0.000	0.000		0.000	
256	INTSYSMSU-000	600 0.145	450 0.023	443 0.557	503 0.072		414 0.132	623 0.998	617 1.000	491 1.000	545 0.999		518 0.999	0.000	0.000	0.050			
257	IONETWORKS-001	417 0.003	310 0.012	229 0.118	322 0.004	278 0.019	270 0.082	313 0.014	272 0.045	499 1.000	248 0.016	192 0.129	310 0.294	0.000	0.000	0.000		0.000	
258	IREX-000	433 0.004	212 0.010	462 0.677	252 0.002	224 0.012	273 0.082	399 0.028	314 0.060	306 0.956	351 0.044	267 0.302	243 0.170	0.000	0.000	0.042		0.000	
259	IRISID-001	219 0.001	179 0.009	260 0.103	276 0.003	253 0.016	319 0.096	306 0.013	285 0.051	175 0.422	295 0.025	209 0.148	237 0.162	0.000	0.001	0.046		0.001	
260	IRISID-002	173 0.001	173 0.008	257 0.097	268 0.003	246 0.014	309 0.093	278 0.010	271 0.045	169 0.407	274 0.020	194 0.133	239 0.164	0.000	0.001	0.044		0.001	
261	ISTL-001	368 0.002	400 0.017	259 0.098	272 0.003	376 0.089	215 0.070	299 0.013	340 0.073	161 0.378	358 0.048	383 1.000	349 0.407	0.001	0.000	0.041		0.000	
262	ISYSTEMS-002	464 0.006	463 0.026	511 0.842				487 0.078	419 0.126	437 0.998				0.002	0.002	0.142			
263	ISYSTEMS-003	450 0.005	446 0.023	493 0.789				468 0.058	401 0.107	503 1.000				0.002	0.002	0.142			
264	KAKAO-000	232 0.001	251 0.011	267 0.109	256 0.002	231 0.013	258 0.078	320 0.015	303 0.056	187 0.462	270 0.019	198 0.141	227 0.158	0.000	0.000	0.041		0.000	
265	KAKAO-001	209 0.001	198 0.009	61 0.047	87 0.001	89 0.004	61 0.047	104 0.002	107 0.017	68 0.150	88 0.004	92 0.042	50 0.074	0.000	0.000	0.040		0.000	
266	KAKAO-002	197 0.001	184 0.009	55 0.045	53 0.001	53 0.004	36 0.045	85 0.002	89 0.014	53 0.134	63 0.003	77 0.037	45 0.070	0.000	0.000	0.040		0.000	
267	KEDACOM-001	481 0.007	496 0.036	568 0.972	474 0.034		483 0.237	371 0.022	338 0.072	368 0.986	367 0.055		313 0.305	0.000	0.000	0.000			
268	KEYLESS-000	103 0.001	58 0.006	196 0.066	137 0.001	326 0.034	236 0.073	216 0.006	194 0.029	371 0.987	218 0.011	366 0.994	216 0.149	0.000	0.000	0.041		0.000	
269	KLOUDSPOT-001	358 0.002	422 0.021	301 0.145	395 0.008	402 0.170	334 0.100	376 0.024		214 0.550				0.001	0.000	0.042		0.000	
270	KNERON-000	458 0.006	470 0.027	441 0.547	463 0.028		465 0.195							0.000	0.000	0.000			
271	KNERON-001	559 0.029	615 0.621	344 0.228	518 0.144	410 0.207	498 0.280							0.000	0.000	0.000		0.000	
272	KNOWUTECH-000	126 0.001	100 0.007	206 0.069	197 0.002	184 0.008	214 0.070	262 0.009	249 0.038	143 0.310	225 0.012	172 0.091	189 0.129	0.000	0.000	0.041		0.000	
273	KOGNIZA-000	260 0.001	250 0.010	224 0.078	110 0.001	70 0.004	139 0.056	218 0.006	188 0.028	179 0.444	110 0.005	111 0.050	96 0.087	0.001	0.001	0.041		0.000	
274	LEBENTECH-000	623 0.540	616 0.644	583 0.993	529 0.294	436 0.517	532 0.560	628 1.000	590 0.882	562 1.000	498 0.527	395 1.000	491 0.954	0.007	0.012	0.076		0.007	
275	LIGHTVISION-000	397 0.003	474 0.028	449 0.585	340 0.005	394 0.129	326 0.098	392 0.027	578 0.797	507 1.000	324 0.033	434 1.000	274 0.214	0.000	0.000	0.000		0.000	
276	LINE-000	347 0.002	339 0.014	336 0.214	355 0.005	314 0.029	360 0.107	408 0.030	381 0.095		354 0.046	258 0.278	581 1.000	0.000	0.000	0.000		0.000	

Table 29: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate False pos. identification rate
N = Num. enrolled subjects R = Num. candidates examined
T = Threshold
T = 0 → Investigation T > 0 → Identification

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
277	LINE-001	129 0.001	124 0.007	119 0.053	215 0.002	185 0.008	289 0.085	186 0.004	180 0.027	560 1.000	197 0.009	153 0.072	629 1.000	0.000	0.000	0.000		0.000	
278	LINECLOVA-002	186 0.001	130 0.008	165 0.059	199 0.002	217 0.011	153 0.058	155 0.004	421 0.130	347 0.980	344 0.040	393 1.000	425 0.700	0.000	0.001	0.040		0.001	
279	LINECLOVA-003	77 0.001	612 0.601	236 0.089	118 0.001	189 0.009	287 0.085	107 0.002	557 0.606		116 0.006	358 0.974	163 0.110	0.000	0.000	0.024		0.000	
280	LOOKMAN-003	490 0.008	506 0.038		472 0.035		485 0.239	431 0.043	405 0.112		398 0.084		333 0.355	0.000	0.000				
281	LOOKMAN-004	492 0.009	511 0.039	570 0.973				434 0.044	398 0.105	335 0.977				0.000	0.000	0.000			
282	LOOKMAN-005	484 0.008	499 0.036	569 0.972	476 0.035		484 0.237	406 0.030	362 0.086	338 0.977	376 0.062		315 0.308	0.000	0.000	0.000			
283	LSSYTECH-000	68 0.001	86 0.007	164 0.059	77 0.001	71 0.004	150 0.057	51 0.001	41 0.009	78 0.167	56 0.003	38 0.026	75 0.080	0.000	0.000	0.040		0.000	
284	LSSYTECH-001	53 0.001	79 0.006	147 0.056	61 0.001	46 0.003	29 0.043	18 0.001	22 0.008	77 0.167	44 0.002	29 0.022	37 0.068	0.000	0.000	0.040		0.000	
285	LUXAND-001	401 0.003	423 0.020	373 0.305	414 0.012	363 0.068	407 0.127	464 0.057	525 0.385	418 0.995	509 0.662	343 0.909	468 0.903	0.001	0.002	0.088		0.001	
286	MANTRA-000	276 0.001	251 0.010	471 0.705	327 0.007	298 0.024	374 0.112	273 0.010	256 0.041	614 1.000	309 0.029	212 0.152	524 1.000	0.002	0.001	0.591		0.003	
287	MAVERICKRESEARCH-000	508 0.010	460 0.025	406 0.419	431 0.014	343 0.045	429 0.144	491 0.084	459 0.176	389 0.992	413 0.122	285 0.394	392 0.546	0.004	0.001	0.055		0.005	
288	MAXVISION-000	360 0.002	357 0.015	381 0.319	325 0.004	349 0.051	337 0.101	400 0.028	483 0.237	257 0.764	428 0.149	371 0.997	395 0.557	0.000	0.000	0.042		0.000	
289	MAXVISION-001	155 0.001	136 0.008	131 0.054	134 0.001	270 0.018	151 0.057	169 0.004	169 0.025	108 0.210	154 0.007	351 0.951	141 0.100	0.000	0.000	0.042		0.000	
290	MAXVISION-002	163 0.001	106 0.007	121 0.053	156 0.001	199 0.009	208 0.069	150 0.004	139 0.022	95 0.188	150 0.007	340 0.895	151 0.104	0.001	0.000	0.044		0.000	
291	MAXVISION-003	172 0.001	134 0.008	133 0.054	170 0.001	232 0.013	157 0.059	215 0.006	252 0.039	127 0.237	216 0.011	276 0.348	166 0.111	0.000	0.000	0.042		0.000	
292	MAXVISION-004	154 0.001	193 0.009	44 0.044	62 0.001	331 0.037	134 0.055	93 0.002	117 0.020	43 0.122	78 0.004	362 0.990	82 0.082	0.001	0.000	0.042		0.000	
293	MEGVII-001	518 0.012	397 0.017		620 1.000			482 0.072	385 0.097					0.002	0.000				
294	MEGVII-002	520 0.012	399 0.017	410 0.444	557 1.000			486 0.077	383 0.096	452 0.998				0.002	0.000	0.033			
295	MEGVII-003	170 0.001	154 0.008	107 0.050	109 0.001	85 0.004	142 0.057	102 0.002	59 0.011	64 0.145	67 0.003	46 0.027	66 0.078	0.001	0.000	0.044		0.000	
296	MEGVII-004	152 0.001	143 0.008	42 0.044	83 0.001	30 0.003	120 0.054	35 0.001	26 0.008	12 0.072	11 0.001	4 0.009	28 0.065	0.001	0.000	0.044		0.000	
297	MEGVII-005	151 0.001	145 0.008	43 0.044	50 0.001	142 0.007	24 0.042	36 0.001	42 0.009	89 0.181	120 0.006	405 1.000	171 0.117	0.001	0.000	0.044		0.000	
298	METSAKUURCOMPANY-000	138 0.001	101 0.007	182 0.063	196 0.002	323 0.083	159 0.059	240 0.007	236 0.036	465 0.999	395 0.081	369 0.995	422 0.683	0.000	0.000	0.000		0.000	
299	METSAKUURCOMPANY-001	205 0.001	140 0.008	256 0.097	211 0.002	183 0.008	230 0.072	263 0.009	208 0.031	484 1.000	209 0.010	155 0.073	454 0.841	0.000	0.000	0.000		0.000	
300	METSAKUURCOMPANY-002	192 0.001	114 0.007	264 0.105	186 0.001	336 0.039	200 0.068	287 0.012	217 0.033	221 0.577	242 0.015	311 0.684	182 0.127	0.000	0.000	0.038		0.000	
301	MIAXIS-001	264 0.001	223 0.010	338 0.218	247 0.002	225 0.012	366 0.108	248 0.008	218 0.033	305 0.954	241 0.014	214 0.153	247 0.176	0.000	0.000	0.000		0.000	
302	MIAXIS-002	266 0.001	289 0.011	304 0.148	249 0.002	230 0.013	346 0.102	252 0.008	239 0.036	212 0.536	253 0.017	237 0.192	238 0.163	0.000	0.000	0.000		0.000	
303	MICROFOCUS-003	627 0.594	623 0.781		548 0.708		551 0.907	616 0.931	607 0.979		538 0.982		507 0.991	0.001	0.005				
304	MICROFOCUS-004	624 0.576	622 0.758		547 0.701		550 0.904	624 0.999	605 0.975		534 0.974		503 0.989	0.001	0.005				
305	MICROFOCUS-005	619 0.424	611 0.601		538 0.494		542 0.777	613 0.835	597 0.928		530 0.935		502 0.985	0.001	0.005				
306	MICROFOCUS-006	620 0.427	609 0.583		537 0.490		545 0.782	620 0.978	596 0.923		529 0.923		496 0.971	0.001	0.005				
307	MICROSOFT-003	298 0.002	305 0.012		309 0.004		370 0.109	395 0.027	373 0.091		334 0.036		283 0.233	0.000	0.001				
308	MICROSOFT-004	287 0.001	303 0.012		297 0.004		371 0.109	382 0.026	366 0.087		325 0.033		278 0.222	0.000	0.001				
309	MICROSOFT-005	345 0.002	264 0.011	291 0.134	281 0.003		331 0.099	379 0.025	335 0.070	222 0.582	299 0.027		257 0.180	0.000	0.001	0.049			
310	MICROSOFT-006	352 0.002	288 0.011	296 0.140	305 0.004		335 0.100	289 0.012	240 0.037	162 0.379	320 0.032		250 0.178	0.000	0.001	0.049			
311	MOBBL-000	381 0.002	329 0.013	408 0.431	335 0.004	300 0.026	382 0.114	398 0.028	346 0.077	361 0.984	332 0.036	306 0.621	280 0.228	0.000	0.002	0.088		0.001	
312	MOBBL-001	517 0.012	413 0.019	411 0.446	464 0.029	345 0.047	457 0.178	414 0.035	356 0.080	357 0.983	372 0.058	307 0.622	305 0.275	0.000	0.002	0.088		0.001	
313	MOBBL-002	609 0.212	586 0.193	470 0.701	527 0.282	435 0.500	531 0.550	558 0.215	472 0.207	378 0.990	462 0.285	372 0.997	397 0.558	0.000	0.000	0.042		0.000	
314	MOBBL-003	477 0.007	481 0.029	341 0.220	439 0.016	388 0.115	434 0.148	521 0.124	603 0.960	521 1.000	467 0.302	614 1.000	429 0.728	0.000	0.000	0.042		0.000	
315	MUKH-002	552 0.025	498 0.036	455 0.633	416 0.012	368 0.079	409 0.129	599 0.594	489 0.242	513 1.000	439 0.170	319 0.741	342 0.389	0.000	0.000	0.042		0.000	
316	NAVERCLOUD-000	171 0.001	84 0.007	156 0.058	177 0.001	97 0.005	181 0.064	146 0.003	123 0.020	326 0.971	104 0.005	81 0.039	122 0.094	0.000	0.001	0.041		0.001	
317	NEC-000	535 0.017	516 0.041	554 0.958	458 0.025		488 0.243	489 0.079	433 0.140	341 0.979			373 0.474	0.001	0.002	0.890			
318	NEC-001	544 0.021	535 0.056	562 0.967	472 0.033		494 0.277	509 0.105	469 0.197	367 0.986	421 0.133		370 0.468	0.005	0.003	0.934			
319	NEC-002	179 0.001	188 0.009	392 0.355	293 0.003		386 0.117	108 0.002	120 0.020	475 0.999	179 0.008		419 0.676	0.000	0.001	0.041			
320	NEC-003	262 0.001	228 0.010	391 0.345	304 0.004	229 0.013	394 0.120	103 0.002	106 0.017	266 0.822	184 0.008	74 0.036	418 0.668	0.000	0.001	0.041		0.001	
321	NEC-004	220 0.001	180 0.009	437 0.533	278 0.003	166 0.007	245 0.075	66 0.001	78 0.013	228 0.618	90 0.004	21 0.019	139 0.100	0.000	0.001	0.041		0.001	
322	NEC-005	149 0.001	137 0.008	213 0.071	208 0.002	101 0.005	238 0.073	62 0.001	67 0.012	236 0.669	70 0.003	20 0.019	134 0.099	0.000	0.001	0.040		0.001	

Table 30: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T → FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
323	NEC-006	178 0.001	163 0.008	138 0.055	221 0.002	110 0.005	190 0.065	80 0.002	110 0.018	185 0.457	77 0.004	41 0.026	125 0.094	0.000	0.001	0.040		0.001	
324	NEC-007	93 0.001	60 0.006	82 0.048	114 0.001	29 0.003	79 0.049	38 0.001	33 0.009	58 0.137	31 0.002	6 0.010	27 0.065	0.000	0.000	0.040		0.000	
325	NEC-008	111 0.001	82 0.006	45 0.044	104 0.001	21 0.003	58 0.047	48 0.001	29 0.009	21 0.089	26 0.002	3 0.008	21 0.062	0.000	0.000	0.040		0.000	
326	NEC-009	74 0.001	57 0.006	12 0.040	80 0.001	10 0.002	33 0.044	15 0.001	11 0.007	10 0.070	9 0.001	2 0.005	7 0.052	0.000	0.000	0.040		0.000	
327	NEC-010	30 0.000	13 0.005	4 0.039	74 0.001	7 0.002	27 0.043	7 0.001	5 0.006	8 0.067	7 0.001	1 0.004	2 0.050	0.000	0.000	0.040		0.000	
328	NEC-011	15 0.000	2 0.005	8 0.039	7 0.001	226 0.012	9 0.037	6 0.001	7 0.006	59 0.140	2 0.001	328 0.789	5 0.052	0.000	0.000	0.039		0.000	
329	NEC-012	14 0.000	3 0.005	9 0.039	10 0.001	2 0.002	8 0.037	4 0.001	2 0.006	5 0.056	1 0.001	95 0.044	1 0.049	0.000	0.000	0.039		0.000	
330	NEUROTECHNOLOGY-003	546 0.022	517 0.042	556 0.960				603 0.631	500 0.266	592 1.000				0.000	0.001	0.131			
331	NEUROTECHNOLOGY-004	452 0.005	419 0.020	563 0.970				475 0.062	407 0.117	402 0.994				0.000	0.001	0.131			
332	NEUROTECHNOLOGY-005	431 0.004	455 0.024	518 0.892				459 0.054	422 0.130	443 0.998				0.000	0.000	0.030			
333	NEUROTECHNOLOGY-006	539 0.018	521 0.045	452 0.602				565 0.249	529 0.418					0.000	0.000				
334	NEUROTECHNOLOGY-007	420 0.004	429 0.021	497 0.794	403 0.009		460 0.180	474 0.062	457 0.173	511 1.000	473 0.339		594 1.000	0.001	0.001	0.041			
335	NEUROTECHNOLOGY-008	340 0.002	346 0.014	414 0.451	314 0.004	294 0.023	342 0.101	453 0.053	355 0.080	540 1.000	331 0.035	265 0.293	270 0.203	0.000	0.001	0.052		0.001	
336	NEUROTECHNOLOGY-009	227 0.001	261 0.011	314 0.169	241 0.002	235 0.013	263 0.079	321 0.015	291 0.052	223 0.583	273 0.020	213 0.153	240 0.165	0.001	0.000	0.046		0.000	
337	NEUROTECHNOLOGY-010	165 0.001	211 0.009	167 0.060	168 0.001	163 0.007	201 0.068	271 0.010	244 0.037	132 0.269	210 0.010	161 0.075	180 0.126	0.000	0.000	0.041		0.000	
338	NEUROTECHNOLOGY-012	91 0.001	146 0.008	116 0.052	82 0.001	118 0.005	148 0.057	233 0.006	216 0.032	310 0.959	191 0.008	130 0.061	472 0.916	0.000	0.000	0.039		0.000	
339	NEUROTECHNOLOGY-013	92 0.001	141 0.008	72 0.047	75 0.001	91 0.004	130 0.056	168 0.004	148 0.023	146 0.316	144 0.006	98 0.046	412 0.641	0.000	0.000	0.039		0.000	
340	NEUROTECHNOLOGY-014	61 0.001	71 0.006	88 0.049	93 0.001	94 0.005	130 0.055	131 0.003	138 0.022	113 0.215	134 0.006	116 0.052	124 0.094	0.000	0.000	0.039		0.000	
341	NEUROTECHNOLOGY-015	45 0.001	49 0.006	70 0.047	45 0.001	79 0.004	64 0.048	137 0.003	158 0.024	82 0.170	135 0.006	110 0.049	93 0.086	0.000	0.000	0.039		0.000	
342	NEUROTECHNOLOGY-016	35 0.001	30 0.006	59 0.046	34 0.001	68 0.004	72 0.048	130 0.003	128 0.021	90 0.182	161 0.007	120 0.054	118 0.093	0.000	0.000	0.039		0.000	
343	NEWLAND-002	587 0.078	570 0.117	535 0.935				591 0.437	539 0.466	466 0.999				0.007	0.012	0.200			
344	NEXTMETALABS-001	563 0.031	560 0.090	423 0.510	519 0.154	438 0.550	521 0.419	561 0.233	619 1.000	548 1.000	586 1.000	612 1.000	602 1.000	0.000	0.000	0.000		0.000	
345	NEXTMETALABS-002	80 0.001	104 0.007	101 0.050	242 0.002	406 0.182	252 0.077	160 0.004	418 0.125	101 0.198	420 0.131	397 1.000	382 0.510	0.000	0.000	0.041		0.000	
346	NEXTMETALABS-003	78 0.001	53 0.006	102 0.050	111 0.001	119 0.005	209 0.069	161 0.004	141 0.022	99 0.194	132 0.006	117 0.052	158 0.107	0.000	0.000	0.041		0.000	
347	NOBLIS-001	613 0.249	604 0.522	584 0.993				627 1.000	620 1.000	543 1.000				0.000	0.000	0.000			
348	NOBLIS-002	603 0.179	600 0.392	574 0.981				622 0.997	628 1.000	537 1.000				0.000	0.000	0.000			
349	NOMINDER-000	94 0.001	174 0.009	297 0.142	323 0.004	399 0.155	297 0.090	188 0.004	506 0.280	299 0.929	477 0.370	599 1.000	444 0.799	0.000	0.000	0.040		0.000	
350	NOMINDER-001	240 0.001	274 0.011	263 0.104	95 0.001	280 0.020	154 0.058	201 0.005	363 0.086	239 0.675	177 0.008	394 1.000	220 0.154	0.000	0.000	0.040		0.000	
351	NOMINDER-002	242 0.001	262 0.011	109 0.051	102 0.001	158 0.007	67 0.048	199 0.005	198 0.029	121 0.223	146 0.006	337 0.867	107 0.089	0.001	0.000	0.040		0.000	
352	NOMINDER-003	243 0.001	256 0.011	89 0.049	96 0.001	133 0.006	40 0.045	192 0.005	185 0.027	105 0.208	130 0.006	333 0.816	89 0.085	0.001	0.000	0.040		0.000	
353	NOMINDER-004	222 0.001	234 0.010	91 0.049	103 0.001	100 0.005	66 0.048	185 0.004	174 0.026	106 0.208	139 0.006	273 0.331	106 0.089	0.001	0.000	0.040		0.000	
354	NOMINDER-005	234 0.001	269 0.011	74 0.048	113 0.001	108 0.005	41 0.045	127 0.003	132 0.022	81 0.169	121 0.006	277 0.356	103 0.088	0.001	0.000	0.040		0.000	
355	NOTIONTAG-000	364 0.002	306 0.012	325 0.195	315 0.004	254 0.016	314 0.095	336 0.017	309 0.059	225 0.606	281 0.021	211 0.150	246 0.176	0.000	0.000	0.000		0.000	
356	NTECHLAB-003	463 0.006	443 0.023	420 0.498				457 0.053	408 0.118	269 0.835				0.000	0.000	0.040			
357	NTECHLAB-004	445 0.005	411 0.019	421 0.500	393 0.008		410 0.129	425 0.040	396 0.105	268 0.831	364 0.053		298 0.263	0.000	0.000	0.040			
358	NTECHLAB-005	443 0.005	403 0.018	393 0.360	398 0.008		389 0.118	427 0.042	393 0.102	258 0.769	389 0.073		309 0.294	0.000	0.000	0.040			
359	NTECHLAB-006	432 0.004	390 0.017	390 0.340	389 0.007		381 0.113	419 0.036	378 0.094	256 0.751	370 0.057		297 0.260	0.000	0.000	0.040			
360	NTECHLAB-007	378 0.002	312 0.012	378 0.318	334 0.004		359 0.107	378 0.025	326 0.067	253 0.747	321 0.032		279 0.223	0.000	0.000	0.042			
361	NTECHLAB-008	275 0.001	213 0.010	302 0.147	292 0.003		283 0.084	311 0.014	273 0.045	207 0.524	326 0.033		258 0.183	0.000	0.000	0.044			
362	NTECHLAB-009	180 0.001	147 0.008	285 0.128	237 0.002	239 0.013	241 0.074	194 0.005	142 0.022	176 0.423	243 0.015	182 0.109	207 0.142	0.000	0.000	0.041		0.001	
363	NTECHLAB-010	121 0.001	164 0.008	220 0.075	210 0.002	175 0.008	147 0.057	101 0.002	91 0.015	129 0.243	151 0.007	129 0.059	130 0.098	0.001	0.001	0.043		0.000	
364	NTECHLAB-011	96 0.001	98 0.007	128 0.062	173 0.001	170 0.007	99 0.051	115 0.003	90 0.015	115 0.219	196 0.009	158 0.074	112 0.091	0.000	0.000	0.040		0.000	
365	OMNIGARDE-000	102 0.001	89 0.007	183 0.063	165 0.001	142 0.007	198 0.068	165 0.004	149 0.023	128 0.243	167 0.007	123 0.055	196 0.136	0.000	0.000	0.041		0.000	
366	OMNIGARDE-001	48 0.001	48 0.006	65 0.047	76 0.001	62 0.004	128 0.055	86 0.002	95 0.015	56 0.135	73 0.003	59 0.030	65 0.078	0.000	0.000	0.041		0.000	
367	OMNIGARDE-002	66 0.001	51 0.006	67 0.047	63 0.001	75 0.004	138 0.056	61 0.001	50 0.011	37 0.111	50 0.002	25 0.022	71 0.079	0.000	0.000	0.041		0.000	
368	ONFIDO-000	336 0.002	215 0.010	483 0.748	295 0.003	293 0.023	313 0.094	303 0.013	232 0.035	327 0.971	311 0.029	253 0.244	266 0.194	0.000	0.000	0.040		0.000	

Table 31: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
369	ONFIDO-001	270 0.001	132 0.008	389 0.338	236 0.002	267 0.018	268 0.081	280 0.010	153 0.023	252 0.744	238 0.014	203 0.142	218 0.152	0.000	0.000	0.040		0.000	
370	OPTIEXACTA-000	43 0.001	56 0.006	95 0.049	131 0.001	62 0.004	80 0.049	31 0.001	25 0.008	54 0.135	64 0.003	66 0.033	44 0.070	0.000	0.000	0.040		0.000	
371	PANGIAM-000	146 0.001	138 0.008	185 0.064	223 0.002	168 0.007	186 0.065	219 0.006	201 0.030	144 0.310	203 0.009	196 0.136	152 0.105	0.000	0.001	0.044		0.001	
372	PANGIAM-001	467 0.007	324 0.013	199 0.067	139 0.001	201 0.009	180 0.064	284 0.011	200 0.030	160 0.376	195 0.009	336 0.860	205 0.141	0.003	0.000	0.040		0.000	
373	PAPSAV1923-000	142 0.001	150 0.008	214 0.071	341 0.005	208 0.010	177 0.063	126 0.003	125 0.021	122 0.226	206 0.009	109 0.049	123 0.094	0.000	0.001	0.052		0.003	
374	PARAVISION-000	540 0.019	505 0.038	434 0.529	535 0.423		530 0.529	500 0.088	454 0.170	420 0.999	492 0.470		475 0.926	0.000	0.000	0.000			
375	PARAVISION-001	422 0.004	425 0.020	383 0.322	534 0.414		528 0.484	443 0.049	420 0.128	457 0.999	489 0.444		431 0.739	0.000	0.000	0.000			
376	PARAVISION-002	429 0.004	436 0.022	385 0.328	437 0.015		454 0.175	445 0.050	411 0.119	356 0.983	394 0.080		379 0.497	0.000	0.000	0.032			
377	PARAVISION-003	411 0.003	414 0.019	348 0.244	438 0.015		447 0.167	413 0.035	382 0.096	404 0.994	371 0.058		312 0.296	0.000	0.000	0.032			
378	PARAVISION-004	259 0.001	238 0.010	244 0.094	362 0.006		375 0.112	274 0.010	245 0.038	527 1.000	262 0.018		470 0.908	0.000	0.000	0.032			
379	PARAVISION-005	246 0.001	222 0.010	203 0.068	380 0.007		352 0.106	163 0.004	159 0.024	344 0.980	214 0.011		192 0.132	0.000	0.000	0.038			
380	PARAVISION-007	145 0.001	159 0.008	139 0.055	350 0.005	209 0.010	338 0.101	156 0.004	168 0.025	559 1.000	193 0.009	184 0.113	627 1.000	0.000	0.000	0.000		0.000	
381	PARAVISION-009	89 0.001	118 0.007	145 0.056	166 0.001	76 0.004	114 0.054	106 0.002	114 0.019	249 0.732	65 0.003	65 0.033	49 0.073	0.000	0.001	0.025		0.000	
382	PARAVISION-012	69 0.001	108 0.007	106 0.050	163 0.001	63 0.004	102 0.052	55 0.001	73 0.012	188 0.469	55 0.002	34 0.025	39 0.068	0.000	0.001	0.025		0.000	
383	PARAVISION-014	64 0.001	105 0.007	83 0.048	72 0.001	48 0.003	54 0.047	45 0.001	61 0.011	210 0.527	37 0.002	67 0.033	20 0.061	0.000	0.001	0.025		0.000	
384	PARAVISION-016	67 0.001	81 0.006	56 0.045	52 0.001	42 0.003	93 0.051	34 0.001	58 0.011	38 0.112	24 0.002	48 0.028	24 0.065	0.000	0.000	0.040		0.000	
385	PARAVISION-019	29 0.000	41 0.006	53 0.045	39 0.001	32 0.003	34 0.044	28 0.001	46 0.010	86 0.176	21 0.002	54 0.028	19 0.061	0.000	0.000	0.003		0.000	
386	PARAVISION-021	16 0.000	42 0.006	58 0.046	37 0.001	31 0.003	31 0.044	30 0.001	49 0.011	47 0.124	18 0.001	51 0.028	17 0.061	0.000	0.000	0.024		0.000	
387	PARAVISION-022	12 0.000	33 0.006	54 0.045	27 0.001	26 0.003	30 0.043	32 0.001	44 0.010	57 0.135	15 0.001	37 0.026	23 0.063	0.000	0.000	0.024		0.000	
388	PENTACOMP-000	631 0.975	627 0.978	567 0.972	552 0.948	446 0.980	552 0.951	621 0.996	615 0.998	384 0.990	533 0.955	496 1.000	493 0.960	0.000	0.000	0.041		0.000	
389	PENTACOMP-001	288 0.001	295 0.012	298 0.142	251 0.002	390 0.123	204 0.069	332 0.016	455 0.172	200 0.517	392 0.076	423 1.000	356 0.430	0.000	0.000	0.041		0.000	
390	PIXELALL-002	435 0.004	439 0.022	503 0.808	411 0.011		462 0.187	508 0.105	526 0.388	539 1.000	502 0.602		561 1.000	0.000	0.000	0.000			
391	PIXELALL-003	338 0.002	345 0.014	424 0.510	372 0.006		435 0.151	366 0.021	339 0.073	487 1.000	339 0.037		394 0.554	0.000	0.000	0.000			
392	PIXELALL-004	324 0.002	358 0.015	429 0.518	359 0.005		432 0.152	344 0.018	354 0.079	522 1.000	359 0.051		509 0.994	0.000	0.000	0.000			
393	PIXELALL-005	302 0.002	263 0.011	351 0.256	419 0.012	309 0.028	432 0.146	286 0.012	280 0.050	530 1.000	300 0.027	239 0.203	521 1.000	0.000	0.000	0.000		0.000	
394	PRIVID-000	545 0.021	430 0.021	233 0.086	480 0.037	425 0.369	461 0.181	387 0.027	584 0.825	281 0.882	504 0.603	615 1.000	498 0.972	0.000	0.000	0.039		0.000	
395	PROTENET-000	582 0.063	585 0.190	450 0.597	508 0.096	433 0.448	525 0.462	607 0.794	585 0.834	342 0.980	519 0.804	523 1.000	464 0.890	0.015	0.111	0.216		0.015	
396	PROTENET-001	356 0.002	373 0.016	282 0.123	289 0.003	407 0.187	339 0.101	360 0.021	392 0.105	215 0.551	414 0.123	581 1.000	405 0.610	0.001	0.002	0.040		0.000	
397	PSL-000	32 0.000	47 0.006	118 0.053	91 0.001	83 0.004	109 0.053	105 0.002	127 0.021	104 0.208	111 0.005	121 0.054	131 0.097	0.000	0.000	0.040		0.000	
398	PSL-001	21 0.000	22 0.005	20 0.041	9 0.001	28 0.003	35 0.045	94 0.002	102 0.017	44 0.122	69 0.003	50 0.028	52 0.075	0.000	0.000	0.040		0.000	
399	PSL-002	9 0.000	6 0.005	16 0.041	15 0.001	14 0.002	47 0.046	42 0.001	60 0.011	27 0.095	42 0.002	31 0.023	69 0.079	0.000	0.000	0.040		0.000	
400	PSL-003	8 0.000	16 0.005	33 0.043	31 0.001	15 0.002	57 0.047	46 0.001	55 0.011	33 0.102	39 0.002	28 0.022	148 0.102	0.000	0.000	0.040		0.000	
401	PSL-004	7 0.000	11 0.005	30 0.043	32 0.001	18 0.002	62 0.047	33 0.001	38 0.009	32 0.101	25 0.002	30 0.022	58 0.076	0.000	0.000	0.040		0.000	
402	PSL-005	11 0.000	38 0.006	35 0.043	25 0.001	56 0.004	43 0.046	43 0.001	35 0.009	17 0.083	30 0.002	33 0.024	26 0.065	0.000	0.000	0.040		0.000	
403	PSL-006	6 0.000	7 0.005	23 0.041	16 0.001	16 0.002	39 0.045	22 0.001	24 0.008	20 0.085	17 0.001	22 0.019	31 0.067	0.000	0.000	0.040		0.000	
404	PTAKURATSATU-000	398 0.003	389 0.017	451 0.600	356 0.005	305 0.027	353 0.105	418 0.036	417 0.124	296 0.923	355 0.046	242 0.206	282 0.232	0.000	0.001	0.039		0.000	
405	PYRAMID-000	115 0.001	76 0.006	117 0.052	94 0.001	164 0.007	131 0.055	153 0.004	124 0.021	97 0.190	155 0.007	294 0.433	129 0.095	0.001	0.000	0.042		0.000	
406	PYRAMID-001	148 0.001	490 0.032	120 0.053	456 0.021	287 0.022	176 0.063	182 0.004	273 0.047	98 0.193	301 0.027	160 0.075	138 0.100	0.000	0.037	0.041		0.018	
407	QAZSMARTVISIONAI-000	18 0.000	26 0.005	26 0.042	24 0.001	38 0.003	23 0.042	47 0.001	47 0.010	26 0.094	54 0.002	87 0.041	32 0.067	0.000	0.000	0.035		0.000	
408	QAZSMARTVISIONAI-001	3 0.000	17 0.005	10 0.040	11 0.001	9 0.002	10 0.039	9 0.001	20 0.008	7 0.065	28 0.002	40 0.026	13 0.058	0.000	0.000	0.031		0.000	
409	QAZSMARTVISIONAI-002	1 0.000	5 0.005	1 0.039	4 0.001	4 0.002	5 0.037	3 0.001	6 0.006	2 0.052	6 0.001	10 0.014	5 0.052	0.000	0.000	0.031		0.000	
410	QAZSMARTVISIONAI-003	5 0.000	1 0.005	5 0.039	6 0.001	5 0.002	4 0.036	5 0.001	4 0.006	4 0.053	4 0.001	17 0.016	4 0.052	0.000	0.000	0.031		0.000	
411	QNAP-000	482 0.007	473 0.027	427 0.516	429 0.013	352 0.054	440 0.158	524 0.129	488 0.238	564 1.000	445 0.191	303 0.539	513 0.998	0.001	0.000	0.054		0.000	
412	QNAP-001	427 0.004	437 0.022	419 0.492	371 0.006	339 0.041	378 0.112	456 0.053	429 0.137	297 0.927	396 0.081	280 0.368	329 0.331	0.000	0.000	0.004		0.000	
413	QNAP-002	442 0.005	428 0.021	312 0.163	324 0.004	320 0.031	402 0.125	384 0.026	399 0.106	259 0.769	363 0.052	260 0.281	303 0.272	0.001	0.004	0.057		0.001	
414	QNAP-003	387 0.002	395 0.017	299 0.142	396 0.008	359 0.061	308 0.093	348 0.019	586 0.835	391 0.992	496 0.502	535 1.000	458 0.865	0.000	0.001	0.002		0.001	

Table 32: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BORi10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BORi10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BORi10YR	KIOSK
415	QNAP-004	304 0.002	353 0.014	268 0.109	347 0.005	340 0.041	291 0.086	308 0.014	591 0.890	426 0.996	495 0.501	451 1.000	474 0.922	0.000	0.001	0.002		0.001	
416	QNAP-005	405 0.003	461 0.026	239 0.091	388 0.007	274 0.019	286 0.085	276 0.010	322 0.066	157 0.359	289 0.024	183 0.110	183 0.127	0.005	0.015	0.054		0.005	
417	QNAP-006	235 0.001	276 0.011	249 0.095	225 0.002	245 0.014	199 0.068	256 0.008	263 0.042	164 0.388	268 0.019	193 0.131	197 0.137	0.000	0.000	0.046		0.000	
418	QUANTASOFT-001	610 0.217	621 0.727					604 0.638						0.000	0.000				
419	RANKONE-002	541 0.019	548 0.071					514 0.117	496 0.261					0.000	0.000				
420	RANKONE-003	542 0.019	546 0.068					515 0.117	494 0.255					0.000	0.000				
421	RANKONE-004	572 0.041	575 0.141					551 0.193	532 0.426					0.000	0.000				
422	RANKONE-005	495 0.009	515 0.041	576 0.986				470 0.059	458 0.173	444 0.998				0.000	0.000	0.489			
423	RANKONE-006	446 0.005		499 0.795				420 0.037		336 0.977				0.002		0.167			
424	RANKONE-007	407 0.003	406 0.019	496 0.793				369 0.022	379 0.095	317 0.967				0.001	0.001	0.102			
425	RANKONE-009	355 0.002	315 0.013	438 0.544	361 0.006		419 0.134	339 0.017	344 0.076	322 0.969	374 0.062		327 0.328	0.000	0.000	0.000			
426	RANKONE-010	348 0.002	225 0.010	394 0.367	346 0.005	302 0.027	404 0.126	304 0.013	307 0.058	264 0.799	362 0.052	243 0.208	295 0.259	0.000	0.000	0.000		0.000	
427	RANKONE-011	239 0.001	286 0.011	335 0.214	301 0.004	275 0.019	277 0.082	257 0.008	277 0.048		336 0.037	233 0.182	499 0.977	0.000	0.000	0.000		0.000	
428	RANKONE-012	206 0.001	248 0.010	277 0.117	280 0.003	241 0.014	207 0.069	243 0.007	298 0.053		310 0.029	204 0.144	369 0.465	0.000	0.000	0.000		0.000	
429	RANKONE-013	101 0.001	93 0.007	193 0.065	145 0.001	186 0.008	112 0.054	176 0.004	227 0.034	422 0.996	263 0.018	200 0.141	208 0.142	0.000	0.000	0.033		0.000	
430	RANKONE-014	76 0.001	68 0.006	153 0.057	99 0.001	121 0.005	81 0.050	128 0.003	160 0.024		200 0.009	165 0.081	592 1.000	0.000	0.000	0.006		0.000	
431	RANKONE-015	83 0.001	63 0.006	115 0.052	48 0.001	115 0.005	118 0.054	110 0.002	111 0.018		156 0.007	93 0.042	162 0.109	0.000	0.000	0.000		0.000	
432	REALNETWORKS-000	571 0.040	555 0.078					562 0.233	518 0.319					0.001	0.000				
433	REALNETWORKS-001	570 0.040	554 0.078					563 0.233	517 0.319					0.001	0.000				
434	REALNETWORKS-002	567 0.039	553 0.078					560 0.231	516 0.315					0.001	0.000				
435	REALNETWORKS-003	549 0.024	542 0.062	490 0.768	470 0.031		472 0.209	538 0.159	499 0.266	450 0.998	437 0.164		380 0.500	0.001	0.000	0.009			
436	REALNETWORKS-004	547 0.023	540 0.059	498 0.794	469 0.031		478 0.213	537 0.158	497 0.263	472 0.999	440 0.170		406 0.613	0.001	0.000	0.009			
437	REALNETWORKS-005	354 0.002	330 0.013	407 0.427	330 0.004	289 0.023	344 0.102	394 0.027	342 0.074	324 0.970	335 0.037	248 0.223	275 0.215	0.000	0.000	0.006		0.000	
438	REALNETWORKS-006	213 0.001	226 0.010	359 0.279	260 0.002	212 0.010	259 0.078	312 0.014	294 0.053	346 0.980	249 0.016	188 0.120	222 0.154	0.000	0.000	0.009		0.000	
439	REALNETWORKS-007	195 0.001	208 0.009	352 0.258	201 0.002	190 0.009	225 0.072	268 0.009	266 0.043	340 0.979	221 0.012	297 0.463	204 0.140	0.000	0.000	0.009		0.000	
440	REALNETWORKS-008	141 0.001	139 0.008	222 0.078	209 0.002	160 0.007	299 0.091	220 0.006	197 0.029	319 0.967	187 0.008	151 0.068	187 0.129	0.000	0.000	0.042		0.000	
441	REALSENSE-001	87 0.001	96 0.007	187 0.064	144 0.001	111 0.005	96 0.051	116 0.003	80 0.013	119 0.221	85 0.004	49 0.028	63 0.077	0.000	0.000	0.041		0.000	
442	REALSENSE-002	55 0.001	85 0.007	122 0.053	128 0.001	96 0.005	89 0.050	54 0.001	57 0.011	63 0.144	75 0.004	36 0.025	54 0.075	0.000	0.000	0.041		0.000	
443	RECOGNITO-000	62 0.001	74 0.006	125 0.053	64 0.001	306 0.028	38 0.045	79 0.002	64 0.012	85 0.175	168 0.007	316 0.730	115 0.092	0.000	0.000	0.040		0.000	
444	RECOGNITO-001	58 0.001	50 0.006	81 0.048	152 0.001	66 0.004	83 0.050	50 0.001	32 0.009	290 0.899	72 0.003	68 0.033	53 0.075	0.000	0.000	0.032		0.000	
445	RECOGNITO-002	42 0.001	55 0.006	96 0.049	36 0.001	263 0.017	32 0.044	29 0.001	27 0.008	55 0.135	59 0.003	299 0.510	36 0.068	0.000	0.000	0.040		0.000	
446	REMARKAI-000	489 0.008	483 0.030					522 0.127	470 0.203					0.000	0.001				
447	REMARKAI-000	409 0.003	405 0.018	456 0.656	392 0.008		433 0.148	460 0.055	412 0.120	468 0.999	384 0.069		428 0.717	0.000	0.000	0.000			
448	REMARKAI-002	487 0.008	480 0.029	501 0.800				520 0.123	468 0.196	385 0.991				0.000	0.001	0.017			
449	RENDIP-000	248 0.001	362 0.015	405 0.417	368 0.006	307 0.028	284 0.084	285 0.012	308 0.059	287 0.892	282 0.022	235 0.185	241 0.167	0.000	0.000	0.041		0.000	
450	REVEALMEDIA-000	294 0.001	220 0.010	354 0.266	238 0.002	227 0.012	239 0.074	288 0.012	262 0.042	240 0.677	279 0.021	175 0.093	210 0.143	0.000	0.000	0.041		0.000	
451	RNIT-000	233 0.001	142 0.008	273 0.114	172 0.001	146 0.007	143 0.056	281 0.011	176 0.026	198 0.510	223 0.012	251 0.239	179 0.123	0.000	0.001	0.076		0.000	
452	ROC-016	86 0.001	60 0.006	97 0.050	38 0.001	113 0.005	55 0.047	99 0.002	105 0.017		171 0.008	96 0.045	159 0.108	0.000	0.000	0.001		0.000	
453	ROC-017	71 0.001	67 0.006	87 0.049	58 0.001	82 0.004	45 0.046	121 0.003	135 0.022	84 0.174	149 0.006	83 0.039	114 0.092	0.000	0.000	0.032		0.000	
454	ROC-018	49 0.001	24 0.005	51 0.045	29 0.001	51 0.003	28 0.043	91 0.002	109 0.018	62 0.142	118 0.006	76 0.037	85 0.085	0.000	0.000	0.039		0.000	
455	ROC-019	31 0.000	15 0.005	28 0.042	23 0.001	39 0.003	18 0.042	52 0.001	36 0.009	29 0.098	47 0.002	45 0.027	43 0.070	0.000	0.000	0.041		0.000	
456	ROC-020	41 0.001	18 0.005	29 0.042	14 0.001	40 0.003	13 0.040	56 0.001	77 0.013	31 0.101	62 0.003	89 0.041	35 0.067	0.000	0.000	0.041		0.000	
457	ROC-021	26 0.000	8 0.005	19 0.041	18 0.001	24 0.003	12 0.040	16 0.001	17 0.007	16 0.083	27 0.002	156 0.073	15 0.060	0.000	0.000	0.003		0.000	
458	S1-000	359 0.002	388 0.017	350 0.250	360 0.005	299 0.025	296 0.090	397 0.028	361 0.085	563 1.000	357 0.047	461 1.000	574 1.000	0.000	0.000	0.040		0.000	
459	S1-001	400 0.003	342 0.014	333 0.206	269 0.003	268 0.018	249 0.077	325 0.015	290 0.052	364 0.985	265 0.019	195 0.136	213 0.148	0.001	0.000	0.035		0.000	
460	S1-002	218 0.001	205 0.009	230 0.083	135 0.001	213 0.010	132 0.055	210 0.005	206 0.031	94 0.186	163 0.007	330 0.792	453 0.841	0.000	0.000	0.028		0.000	

Table 33: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
461	S1-003	231 0.001	214 0.010	261 0.103	178 0.001	157 0.007	163 0.060	258 0.008	241 0.037	509 1.000	240 0.014	286 0.396	539 1.000	0.000	0.000	0.033		0.000	
462	S1-004	196 0.001	200 0.009	126 0.053	97 0.001	109 0.005	78 0.049	157 0.004	161 0.025	430 0.997	138 0.006	137 0.064	537 1.000	0.000	0.000	0.033		0.000	
463	S1-005	207 0.001	191 0.009	78 0.048	55 0.001	90 0.004	65 0.048	171 0.004	143 0.022	405 0.994	125 0.006	141 0.065	467 0.895	0.000	0.000	0.033		0.000	
464	S1-006	188 0.001	189 0.009	64 0.047	125 0.001	93 0.005	77 0.049	92 0.002	100 0.016	476 0.999	106 0.005	85 0.040	355 0.423	0.000	0.000	0.033		0.000	
465	S1-007	159 0.001	144 0.008	63 0.047	115 0.001	72 0.004	50 0.046	78 0.002	86 0.014	75 0.165	87 0.004	69 0.034	60 0.077	0.000	0.001	0.039		0.000	
466	SANSAP-001	143 0.001	218 0.010	200 0.068	174 0.001	374 0.087	165 0.061	172 0.004	543 0.480	309 0.958	292 0.024	390 1.000	306 0.280	0.000	0.000	0.041		0.000	
467	SCANOVATE-000	444 0.005	519 0.045	442 0.555	475 0.035		476 0.211	479 0.066	488 0.240	285 0.892	453 0.215		344 0.400	0.000	0.001	0.057			
468	SCANOVATE-001	451 0.005	513 0.040	448 0.580	468 0.031		458 0.178	490 0.081	481 0.227	294 0.910	446 0.192		348 0.404	0.000	0.001	0.044			
469	SCYLLA-001	341 0.002	475 0.028	266 0.107	451 0.021	414 0.231	431 0.146	333 0.016	615 0.995	184 0.457	487 0.431		488 0.953	0.000	0.000	0.041		0.000	
470	SECURITYPITCH-000	419 0.004	404 0.018	432 0.526	381 0.007	327 0.034	430 0.145	471 0.060	416 0.123	431 0.997	385 0.071	266 0.302	322 0.320	0.001	0.001	0.073		0.001	
471	SENSETIME-000	371 0.002	376 0.016	430 0.523				359 0.020	318 0.063	635 1.000				0.004	0.000	0.042			
472	SENSETIME-001	372 0.002	375 0.016					367 0.022	321 0.064					0.004	0.000				
473	SENSETIME-002	526 0.013	415 0.020	398 0.377	410 0.011		349 0.104	316 0.014	190 0.028	401 0.994	319 0.032		386 0.523	0.009	0.000	0.040			
474	SENSETIME-003	85 0.001	91 0.007	294 0.140	264 0.003		300 0.091	70 0.001	70 0.012	189 0.471	180 0.008		193 0.133	0.000	0.000	0.041			
475	SENSETIME-004	81 0.001	102 0.007	179 0.062	248 0.002		285 0.084	65 0.001	75 0.013	117 0.220	128 0.006		168 0.113	0.000	0.000	0.041			
476	SENSETIME-005	34 0.001	62 0.006	85 0.048	246 0.002	153 0.007	272 0.082	89 0.002	85 0.014	74 0.163	153 0.007	115 0.051	150 0.104	0.000	0.000	0.041		0.000	
477	SENSETIME-006	33 0.001	45 0.006	49 0.044	88 0.001	52 0.004	179 0.064	67 0.001	71 0.012	447 0.998	82 0.004	70 0.034	116 0.093	0.000	0.000	0.025		0.000	
478	SENSETIME-007	24 0.000	32 0.006	22 0.041	54 0.001	36 0.003	170 0.062	40 0.001	40 0.009	474 0.999	58 0.003	32 0.024	87 0.085	0.000	0.000	0.025		0.000	
479	SENSETIME-008	23 0.000	27 0.006	36 0.043	35 0.001	35 0.003	194 0.067	37 0.001	34 0.009	167 0.398	48 0.002	24 0.021	77 0.080	0.000	0.000	0.039		0.000	
480	SENSETIME-009	20 0.000	20 0.005	17 0.041	43 0.001	33 0.003	254 0.078	11 0.001	13 0.007	277 0.866	19 0.001	12 0.015	90 0.085	0.000	0.000	0.039		0.000	
481	SENSETIME-010	10 0.000	23 0.005	11 0.040	44 0.001	37 0.003	256 0.078	14 0.001	21 0.008	28 0.096	32 0.002	23 0.020	100 0.087	0.000	0.000	0.039		0.000	
482	SENSETIME-011	19 0.000	21 0.005	18 0.041	42 0.001	34 0.003	255 0.078	12 0.001	12 0.007	278 0.866	20 0.001	13 0.015	91 0.085	0.000	0.000	0.039		0.000	
483	SENSETIME-012	17 0.000	19 0.005	39 0.043	41 0.001	43 0.003	261 0.078	8 0.001	8 0.006	22 0.090	12 0.001	26 0.022	88 0.085	0.000	0.000	0.039		0.000	
484	SERENDIPITY-000	244 0.001	359 0.015	365 0.291	329 0.004	381 0.108	237 0.073	292 0.012	473 0.213	253 0.745	260 0.018	406 1.000	209 0.142	0.000	0.000	0.041		0.000	
485	SEVENTHSENSE-001	322 0.002	192 0.009	191 0.064	220 0.002	323 0.032	213 0.070	184 0.004	264 0.043	124 0.228	207 0.010	309 0.643	184 0.127	0.000	0.000	0.039		0.000	
486	SHAMAN-003	595 0.124	582 0.172					593 0.451	555 0.597					0.020	0.011				
487	SHAMAN-004	612 0.222	596 0.319					602 0.615	574 0.754					0.020	0.011				
488	SHAMAN-006	569 0.039	538 0.058	537 0.938				530 0.141	484 0.237	328 0.971				0.020	0.011	0.869			
489	SHAMAN-007	568 0.039	537 0.057					531 0.141	487 0.240					0.020	0.010				
490	SHENNONA-000	522 0.012	557 0.086	235 0.088	430 0.014	418 0.257	312 0.094	368 0.022	449 0.160	163 0.383	435 0.160	495 1.000	398 0.558	0.007	0.063	0.041		0.000	
491	SHUFTI-001	511 0.011	489 0.032	349 0.246	473 0.033	362 0.065	450 0.171	401 0.028	394 0.104	232 0.650	368 0.056	283 0.382	288 0.243	0.000	0.000	0.041		0.000	
492	SIAT-001	320 0.002	598 0.333		331 0.004		332 0.099	338 0.017	522 0.365		315 0.031			0.000	0.000				
493	SIAT-002	325 0.002	601 0.446		532 0.348		345 0.102	364 0.021	542 0.478		478 0.372		475 0.923	0.000	0.000				
494	SJTU-001	27 0.000	39 0.006	151 0.057	126 0.001	187 0.009	164 0.060	69 0.001	65 0.012	142 0.306	119 0.006	186 0.114	144 0.101	0.000	0.000	0.030		0.000	
495	SJTU-002	22 0.000	25 0.005	108 0.050	123 0.001	202 0.009	158 0.059	20 0.001	32 0.009	295 0.911	108 0.005	301 0.516	181 0.126	0.000	0.000	0.030		0.000	
496	SMILART-004	630 0.965	626 0.974					618 0.968	606 0.976					0.011	0.013				
497	SMILART-005													0.011	0.013				
498	SPARKTEK-000	127 0.001	126 0.007	308 0.154	212 0.002	162 0.007	206 0.069	124 0.003	92 0.015	171 0.414	136 0.006	103 0.048	157 0.106	0.000	0.000	0.062		0.001	
499	SPARKTEK-001	60 0.001	80 0.006	190 0.064	66 0.001	47 0.003	121 0.054	26 0.001	28 0.009	109 0.210	45 0.002	27 0.022	57 0.076	0.000	0.000	0.040		0.000	
500	SPARSH-001	110 0.001	249 0.010	157 0.058	217 0.002	372 0.082	189 0.065	167 0.004	575 0.759	334 0.976	438 0.166	595 1.000	410 0.633	0.000	0.000	0.041		0.000	
501	SQISOFT-001	428 0.004	412 0.019	356 0.273	352 0.005	303 0.027	324 0.097	526 0.132	492 0.252	262 0.794	343 0.040	271 0.317	353 0.420	0.000	0.000	0.039		0.000	
502	SQISOFT-002	185 0.001	462 0.026	226 0.080	349 0.005	422 0.282	262 0.078	403 0.029		291 0.903	507 0.621		489 0.953	0.000	0.000	0.039		0.000	
503	STAQU-000	473 0.007	422 0.020	453 0.609	448 0.020	353 0.055	442 0.159	472 0.061	538 0.443	505 1.000	499 0.535	355 0.961	550 1.000	0.000	0.000	0.000		0.000	
504	STCON-000	605 0.183	618 0.675	361 0.284	545 0.674	439 0.704	539 0.709	548 0.184	568 0.678	147 0.318	511 0.674	348 0.941	426 0.709	0.000	0.000	0.041		0.000	
505	STCON-001	47 0.001	37 0.006	52 0.045	33 0.001	194 0.009	20 0.042	68 0.001	68 0.012	51 0.131	57 0.003	281 0.370	40 0.069	0.000	0.000	0.041		0.000	
506	STCON-002	38 0.001	31 0.006	41 0.043	21 0.001	205 0.010	16 0.041	58 0.001	53 0.011	39 0.115	51 0.002	290 0.415	34 0.067	0.000	0.000	0.041		0.000	

Table 34: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
507	STCON-003	[39] 0.001	[36] 0.006	[40] 0.043	[12] 0.001	[20] 0.003	[14] 0.041	[49] 0.001	[45] 0.010	[36] 0.110	[43] 0.002	[60] 0.031	[25] 0.065	0.000	0.000	0.041		0.000	
508	STCON-004	[36] 0.001	[28] 0.006	[13] 0.041	[5] 0.001	[6] 0.002	[7] 0.037	[23] 0.001	[18] 0.008	[19] 0.083	[16] 0.001	[11] 0.014	[11] 0.055	0.000	0.000	0.041		0.000	
509	STCON-005	[37] 0.001	[29] 0.006	[14] 0.041	[3] 0.000	[3] 0.002	[6] 0.037	[24] 0.001	[16] 0.007	[18] 0.083	[10] 0.001	[55] 0.029	[6] 0.052	0.000	0.000	0.041		0.000	
510	STELLARTECS-001	[337] 0.002	[293] 0.012	[358] 0.277	[207] 0.002	[256] 0.016	[242] 0.075	[314] 0.014	[286] 0.051	[298] 0.927	[277] 0.021	[323] 0.750	[223] 0.155	0.001	0.001	0.042		0.000	
511	SUMSUB-000	[575] 0.045	[568] 0.106	[578] 0.989	[509] 0.097	[448] 0.999	[510] 0.313	[592] 0.445	[572] 0.742	[528] 1.000	[523] 0.852	[477] 1.000	[512] 0.996	0.000	0.000	0.000		0.000	
512	SYNESIS-003	[533] 0.016	[442] 0.023	[509] 0.825	[423] 0.013		[422] 0.136	[477] 0.065	[415] 0.123	[311] 0.960	[391] 0.075		[318] 0.314	0.000	0.001	0.063			
513	SYNESIS-003	[601] 0.170	[591] 0.235					[598] 0.582	[561] 0.646					0.006	0.015				
514	SYNESIS-005	[488] 0.008	[318] 0.013	[481] 0.741	[287] 0.003		[304] 0.092	[377] 0.024	[336] 0.072	[358] 0.983	[322] 0.032		[273] 0.214	0.001	0.000	0.135			
515	T4ISB-000	[501] 0.010	[451] 0.023	[415] 0.456	[262] 0.003	[382] 0.115	[269] 0.081	[326] 0.015	[295] 0.053	[196] 0.505	[276] 0.021	[324] 0.759	[235] 0.161	0.000	0.000	0.000		0.000	
516	TCC-000	[573] 0.045	[602] 0.448	[541] 0.944	[539] 0.595	[445] 0.978	[549] 0.879	[614] 0.868	[610] 0.990	[520] 1.000	[541] 0.992	[584] 1.000	[523] 1.000	0.000	0.000	0.002		0.000	
517	TCC-001	[459] 0.006	[520] 0.045	[426] 0.515	[492] 0.054	[405] 0.178	[477] 0.212	[540] 0.161	[609] 0.987	[518] 1.000	[540] 0.991	[428] 1.000	[522] 1.000	0.001	0.001	0.033		0.000	
518	TECH5-001	[423] 0.004	[387] 0.017	[446] 0.579	[378] 0.007		[361] 0.107	[463] 0.057	[598] 0.935	[561] 1.000	[456] 0.244		[510] 0.994	0.000	0.000	0.006			
519	TECH5-002	[379] 0.002	[259] 0.011	[322] 0.304	[285] 0.003	[315] 0.029	[295] 0.089	[388] 0.027	[334] 0.070	[265] 0.803	[341] 0.039	[240] 0.205	[361] 0.440	0.001	0.000	0.041		0.000	
520	TECH5-003	[109] 0.001	[272] 0.011	[158] 0.058	[227] 0.002	[377] 0.089	[191] 0.066	[154] 0.004	[579] 0.801	[354] 0.982	[443] 0.179	[591] 1.000	[417] 0.656	0.000	0.000	0.041		0.000	
521	TECHAINER-000	[351] 0.002	[360] 0.015	[240] 0.092	[306] 0.004	[233] 0.013	[212] 0.070	[302] 0.013	[296] 0.053	[193] 0.479	[258] 0.018	[218] 0.155	[213] 0.145	0.001	0.007	0.042		0.002	
522	TEVIAN-003	[528] 0.015	[529] 0.052					[546] 0.177	[511] 0.298					0.001	0.002				
523	TEVIAN-004	[512] 0.011	[504] 0.038					[512] 0.117	[460] 0.176					0.001	0.002				
524	TEVIAN-005	[478] 0.007	[477] 0.028	[416] 0.460				[497] 0.087	[436] 0.144	[314] 0.961				0.001	0.002	0.116			
525	TEVIAN-006	[361] 0.002	[280] 0.011	[272] 0.113	[267] 0.003	[240] 0.013	[219] 0.071	[269] 0.009	[213] 0.032	[172] 0.418	[251] 0.016	[173] 0.093	[487] 0.951	0.001	0.000	0.062		0.000	
526	TEVIAN-007	[289] 0.001	[206] 0.009	[229] 0.082	[228] 0.002	[203] 0.009	[197] 0.067	[200] 0.005	[134] 0.022	[141] 0.293	[201] 0.009	[140] 0.065	[176] 0.122	0.000	0.000	0.062		0.000	
527	THAKAAIT-000	[223] 0.001	[190] 0.009	[324] 0.307	[299] 0.004	[251] 0.015	[310] 0.093	[324] 0.015	[292] 0.052	[440] 0.998	[264] 0.018	[259] 0.279	[233] 0.160	0.000	0.001	0.067		0.001	
528	TIGER-000	[581] 0.061	[564] 0.095					[585] 0.390	[544] 0.500					0.000	0.000				
529	TIGER-002	[456] 0.005	[444] 0.023	[422] 0.509				[494] 0.086	[445] 0.158	[461] 0.999				0.000	0.000	0.056			
530	TIGER-003	[457] 0.005	[445] 0.023					[493] 0.086	[446] 0.158					0.000	0.000				
531	TIGER-004	[374] 0.002	[391] 0.017	[265] 0.106	[390] 0.007	[367] 0.076	[243] 0.075	[335] 0.017	[337] 0.072	[170] 0.410	[402] 0.095	[401] 1.000	[345] 0.402	0.001	0.009	0.041		0.004	
532	TONGYITRANS-000	[468] 0.007	[441] 0.022					[484] 0.074	[404] 0.112					0.003	0.001				
533	TONGYITRANS-001	[469] 0.007	[442] 0.022					[478] 0.066	[392] 0.101					0.003	0.001				
534	TOSHIBA-000	[436] 0.004	[434] 0.022	[488] 0.764				[473] 0.062	[409] 0.118	[414] 0.995				0.000	0.000	0.070			
535	TOSHIBA-001	[441] 0.005	[438] 0.022					[466] 0.057	[374] 0.092					0.000	0.000				
536	TOSHIBA-002	[184] 0.001	[253] 0.011	[37] 0.043	[49] 0.001	[23] 0.003	[56] 0.047	[72] 0.001	[94] 0.015	[25] 0.094	[34] 0.002	[61] 0.031	[30] 0.066	0.001	0.000	0.040		0.000	
537	TOSHIBA-003	[125] 0.001	[157] 0.008	[34] 0.043	[40] 0.001	[44] 0.003	[74] 0.049	[64] 0.001	[69] 0.012	[45] 0.123	[41] 0.002	[230] 0.176	[92] 0.085	0.000	0.000	0.039		0.000	
538	TOSHIBA-004	[28] 0.000	[35] 0.006	[32] 0.043	[28] 0.001	[86] 0.004	[21] 0.042	[44] 0.001	[52] 0.011	[23] 0.092	[33] 0.002	[304] 0.573	[186] 0.128	0.000	0.000	0.041		0.000	
539	TOSS-000	[198] 0.001	[183] 0.009	[50] 0.045	[56] 0.001	[57] 0.004	[42] 0.046	[83] 0.002	[87] 0.014	[42] 0.121	[74] 0.003	[75] 0.037	[41] 0.069	0.000	0.000	0.040		0.000	
540	TOSS-001	[211] 0.001	[186] 0.009	[71] 0.047	[70] 0.001	[65] 0.004	[53] 0.047	[100] 0.002	[103] 0.017	[61] 0.141	[81] 0.004	[94] 0.043	[64] 0.078	0.000	0.000	0.040		0.000	
541	TRUEFACE-000	[406] 0.003	[336] 0.014	[342] 0.221	[384] 0.007	[296] 0.024	[305] 0.092	[343] 0.018	[316] 0.062	[280] 0.881	[312] 0.030	[238] 0.194	[263] 0.188	0.001	0.001	0.047		0.003	
542	TSIGHT-000	[4] 0.000	[4] 0.005	[2] 0.039	[2] 0.000	[1] 0.002	[3] 0.036	[2] 0.001	[3] 0.006	[1] 0.051	[3] 0.001	[9] 0.013	[3] 0.050	0.000	0.000	0.031		0.000	
543	TTIGROUP-000	[307] 0.002	[351] 0.014	[202] 0.068	[316] 0.004	[412] 0.213	[307] 0.093	[259] 0.009	[512] 0.298	[448] 0.998	[463] 0.286	[403] 1.000	[456] 0.855	0.000	0.000	0.000		0.000	
544	TURINGTECHVIP-001	[497] 0.009	[466] 0.026	[208] 0.070	[485] 0.045	[409] 0.199	[481] 0.220	[576] 0.345	[587] 0.850	[393] 0.993	[537] 0.978	[492] 1.000	[519] 0.999	0.001	0.003	0.044		0.000	
545	UNISSEY-001	[608] 0.210	[620] 0.692	[413] 0.449	[546] 0.690	[443] 0.731	[540] 0.737	[557] 0.214	[570] 0.700	[229] 0.619	[514] 0.693	[379] 0.999	[432] 0.748	0.000	0.000	0.041		0.000	
546	UNISSEY-002	[604] 0.181	[617] 0.673	[366] 0.291	[544] 0.674	[441] 0.710	[538] 0.703	[549] 0.187	[569] 0.685	[219] 0.571	[518] 0.724	[380] 1.000	[465] 0.891	0.000	0.000	0.041		0.000	
547	UNISSEY-003	[413] 0.003	[372] 0.016	[288] 0.131	[432] 0.014	[291] 0.023	[244] 0.075	[315] 0.014	[281] 0.050	[208] 0.524	[308] 0.029	[241] 0.205	[245] 0.173	0.001	0.006	0.042		0.011	
548	UNIVERSEAI-000	[98] 0.001	[83] 0.007	[92] 0.049	[286] 0.003	[180] 0.008	[169] 0.062	[145] 0.003	[115] 0.020	[79] 0.168	[186] 0.008	[176] 0.094	[145] 0.101	0.001	0.001	0.043		0.002	
549	USEB-000	[564] 0.034	[532] 0.054	[395] 0.373	[450] 0.021	[403] 0.172	[421] 0.135	[605] 0.715	[571] 0.722	[320] 0.968	[493] 0.490	[331] 0.804	[414] 0.648	0.000	0.001	0.042		0.000	
550	USEB-001	[162] 0.001	[205] 0.009	[246] 0.094	[107] 0.001	[172] 0.008	[106] 0.052	[151] 0.004	[147] 0.023	[350] 0.982	[103] 0.005	[314] 0.728	[97] 0.087	0.000	0.001	0.042		0.000	
551	USEB-002	[217] 0.001	[217] 0.010	[250] 0.095	[105] 0.001	[171] 0.008	[108] 0.052	[166] 0.004	[156] 0.023	[351] 0.982	[113] 0.005	[315] 0.728	[102] 0.088	0.000	0.001	0.042		0.000	
552	USEB-003	[161] 0.001	[199] 0.009	[247] 0.094	[85] 0.001	[145] 0.007	[97] 0.051	[148] 0.003	[130] 0.021	[325] 0.971	[112] 0.005	[288] 0.398	[98] 0.087	0.000	0.001	0.042		0.000	

Table 35: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T → FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
553	USEB-004	[187]0.001	[195]0.009	[90]0.049	[57]0.001	[378]0.096	[101]0.052	[142]0.003	[144]0.022	[91]0.182	[211]0.010	[389]1.000	[262]0.186	0.000	0.000	0.040		0.000	
554	VD-000	[622]0.474	[608]0.551					[615]0.917	[601]0.946					0.011	0.013				
555	VD-001	[557]0.027	[531]0.053					[552]0.201	[507]0.281					0.005	0.001				
556	VD-002	[498]0.009	[472]0.027	[517]0.892	[428]0.013	[347]0.050	[455]0.176	[488]0.078	[438]0.148	[417]0.995	[403]0.095	[279]0.367	[338]0.372	0.004	0.003	0.156		0.002	
557	VD-003	[479]0.007	[435]0.022	[491]0.770	[397]0.008	[317]0.030	[423]0.137	[438]0.045	[390]0.100	[463]0.999	[360]0.051	[254]0.244	[320]0.315	0.003	0.003	0.144		0.002	
558	VDFSOFTWARE-001	[607]0.206	[584]0.179	[500]0.799	[408]0.010	[393]0.128	[387]0.117	[626]1.000	[547]0.518	[544]1.000	[513]0.680	[381]1.000	[486]0.948	0.000	0.000	0.000		0.000	
559	VDFSOFTWARE-002	[334]0.002	[257]0.011	[303]0.148	[319]0.004	[365]0.071	[292]0.087	[424]0.038	[400]0.107	[245]0.713	[390]0.075	[398]1.000	[354]0.423	0.000	0.000	0.040		0.000	
560	VERIDAS-001	[385]0.002	[344]0.014	[440]0.545	[370]0.006	[310]0.028	[411]0.131	[421]0.037	[359]0.082	[372]0.987	[349]0.044	[256]0.266	[299]0.264	0.000	0.002	0.093		0.001	
561	VERIDAS-002	[384]0.002	[343]0.014	[439]0.545	[369]0.006	[311]0.028	[412]0.131	[422]0.037	[358]0.082	[373]0.987	[350]0.044	[257]0.266	[300]0.264	0.000	0.002	0.093		0.001	
562	VERIDAS-003	[291]0.001	[278]0.011	[364]0.289	[313]0.004	[258]0.016	[365]0.108	[331]0.016	[302]0.055	[427]0.997	[272]0.020	[210]0.150	[252]0.178	0.000	0.002	0.093		0.001	
563	VERIDAS-004	[199]0.001	[152]0.008	[318]0.177	[263]0.003	[197]0.009	[325]0.097	[212]0.005	[170]0.025	[339]0.978	[178]0.008	[128]0.058	[173]0.118	0.000	0.002	0.094		0.001	
564	VERIGRAM-000	[370]0.002	[374]0.016	[130]0.053	[218]0.002	[167]0.007	[103]0.052	[191]0.005	[191]0.028	[110]0.212	[142]0.006	[168]0.088	[84]0.083	0.001	0.004	0.046		0.001	
565	VERIGRAM-001	[367]0.002	[366]0.015	[124]0.053	[203]0.002	[159]0.007	[60]0.047	[193]0.005	[253]0.039	[224]0.584	[140]0.006	[361]0.988	[80]0.082	0.001	0.003	0.045		0.001	
566	VERIGRAM-002	[375]0.002	[219]0.010	[66]0.047	[47]0.001	[49]0.003	[26]0.043	[120]0.003	[195]0.029	[40]0.116	[49]0.002	[43]0.027	[62]0.077	0.000	0.000	0.041		0.000	
567	VERIJELAS-000	[617]0.354	[599]0.369	[563]0.968	[506]0.086	[408]0.191	[505]0.292	[608]0.799	[581]0.813	[459]0.999	[471]0.324	[347]0.933	[400]0.589	0.002	0.001	0.070		0.001	
568	VIANTE-000	[175]0.001	[158]0.008	[215]0.072	[153]0.001	[81]0.004	[125]0.055	[113]0.003	[83]0.014	[66]0.147	[148]0.006	[79]0.038	[99]0.087	0.001	0.000	0.073		0.001	
569	VIANTE-001	[166]0.001	[149]0.008	[210]0.071	[59]0.001	[41]0.003	[73]0.049	[59]0.001	[39]0.009	[34]0.105	[52]0.002	[264]0.293	[33]0.067	0.001	0.000	0.073		0.001	
570	VIANTE-002	[50]0.001	[54]0.006	[7]0.039	[1]0.000	[12]0.002	[2]0.036	[27]0.001	[14]0.007	[14]0.078	[46]0.002	[245]0.211	[22]0.063	0.000	0.000	0.041		0.000	
571	VIANTE-003	[44]0.001	[52]0.006	[6]0.039	[8]0.001	[13]0.002	[1]0.036	[17]0.001	[10]0.007	[13]0.076	[35]0.002	[187]0.117	[14]0.060	0.000	0.000	0.041		0.000	
572	VIETTELAI-000	[434]0.004	[408]0.019	[217]0.073	[294]0.003	[322]0.032	[223]0.071	[469]0.058	[573]0.753	[553]1.000	[488]0.435	[368]0.995	[619]1.000	0.000	0.000	0.000		0.000	
573	VIETTELAI-001	[329]0.002	[283]0.011	[159]0.058	[433]0.014	[330]0.036	[290]0.086							0.000	0.000	0.040		0.001	
574	VIETTELAI-002	[326]0.002	[478]0.028	[160]0.058	[351]0.005	[337]0.040	[306]0.092	[279]0.010	[560]0.631	[133]0.269	[483]0.421	[378]0.999	[434]0.750	0.000	0.000	0.040		0.001	
575	VIETTELAI-003	[82]0.001	[64]0.006	[76]0.048	[51]0.001	[104]0.005	[82]0.050	[179]0.004	[157]0.023	[267]0.823	[131]0.006	[222]0.160	[111]0.090	0.000	0.000	0.000		0.000	
576	VIETTELAI-004	[57]0.001	[46]0.006	[60]0.047	[26]0.001	[60]0.004	[22]0.042	[111]0.002	[99]0.015	[536]1.000	[97]0.005	[136]0.063	[74]0.080	0.000	0.000	0.000		0.000	
577	VIGILANTSOLUTIONS-003	[585]0.069	[578]0.151	[553]0.957				[589]0.408	[565]0.660	[456]0.999				0.000	0.001	0.127			
578	VIGILANTSOLUTIONS-004	[596]0.125	[592]0.244	[558]0.965				[597]0.549	[583]0.817	[421]0.996				0.000	0.001	0.127			
579	VIGILANTSOLUTIONS-005	[494]0.009		[525]0.919				[583]0.386		[531]1.000				0.000	0.001	0.127			
580	VIGILANTSOLUTIONS-006	[500]0.010		[526]0.920				[579]0.349		[558]1.000				0.000	0.001	0.127			
581	VIGILANTSOLUTIONS-007	[408]0.003	[394]0.017	[529]0.924	[424]0.013	[364]0.068	[453]0.175	[402]0.028	[367]0.088	[420]0.996	[397]0.081	[282]0.371	[343]0.391	0.000	0.001	0.127		0.001	
582	VIGILANTSOLUTIONS-008	[394]0.003	[396]0.017	[524]0.912	[434]0.014	[366]0.072	[456]0.178	[355]0.020	[347]0.077	[460]0.999	[407]0.104	[287]0.398	[383]0.511	0.000	0.001	0.127		0.001	
583	VINBIGDATA-001	[158]0.001	[117]0.007	[243]0.093	[185]0.001	[218]0.011	[152]0.057	[327]0.016	[312]0.059	[300]0.931	[261]0.018	[349]0.944	[304]0.274	0.000	0.000	0.041		0.000	
584	VINBIGDATA-002	[65]0.001	[40]0.006	[134]0.054	[100]0.001	[126]0.006	[95]0.051	[88]0.002	[63]0.012	[72]0.160	[95]0.005	[270]0.317	[56]0.076	0.000	0.000	0.040		0.000	
585	VINBIGDATA-003	[54]0.001	[44]0.006	[84]0.048	[78]0.001	[87]0.004	[51]0.046	[98]0.002	[76]0.013	[60]0.141	[86]0.004	[84]0.039	[46]0.070	0.000	0.000	0.040		0.000	
586	VINBIGDATA-004	[56]0.001	[43]0.006	[69]0.047	[73]0.001	[73]0.004	[47]0.046	[74]0.002	[48]0.010	[50]0.130	[66]0.003	[62]0.032	[38]0.068	0.000	0.000	0.041		0.000	
587	VIRTISHA-000	[530]0.015	[569]0.111	[286]0.128	[489]0.048	[417]0.255	[451]0.173	[389]0.027	[569]0.674	[202]0.517	[526]0.899	[608]1.000	[504]0.990	0.009	0.083	0.042		0.000	
588	VISIONBOX-000	[309]0.002	[281]0.011	[484]0.750	[339]0.005	[261]0.017	[260]0.078	[340]0.017	[305]0.057	[383]0.990	[285]0.023	[206]0.146	[236]0.162	0.000	0.001	0.043		0.001	
589	VISIONBOX-001	[395]0.003	[365]0.015	[513]0.848	[296]0.003	[271]0.018	[426]0.140	[527]0.133	[313]0.060	[532]1.000	[430]0.152	[375]0.999	[476]0.924	0.002	0.000	0.264		0.001	
590	VISIONBOX-002	[292]0.001	[241]0.010	[300]0.143	[232]0.002	[219]0.011	[173]0.062	[282]0.011	[261]0.042	[248]0.730	[244]0.015	[190]0.125	[203]0.140	0.001	0.001	0.056		0.001	
591	VISIONLABS-004	[396]0.003	[418]0.020	[386]0.335				[465]0.057	[447]0.159	[284]0.888				0.001	0.001	0.046			
592	VISIONLABS-005	[376]0.002	[407]0.019	[384]0.326				[446]0.050	[437]0.147	[283]0.886				0.001	0.001	0.046			
593	VISIONLABS-006	[333]0.002	[367]0.015	[332]0.202	[308]0.004		[318]0.096	[386]0.026	[372]0.090	[234]0.668				0.001	0.001	0.051			
594	VISIONLABS-007	[331]0.002	[364]0.015	[331]0.202	[300]0.004		[317]0.095	[385]0.026	[371]0.090	[235]0.668	[318]0.031		[259]0.185	0.001	0.001	0.051			
595	VISIONLABS-008	[318]0.002	[337]0.014	[287]0.131	[245]0.002		[265]0.081	[298]0.013	[289]0.051	[190]0.475	[252]0.017		[217]0.151	0.001	0.000	0.075			
596	VISIONLABS-009	[133]0.001	[171]0.008	[228]0.081	[169]0.001		[218]0.071	[177]0.004	[172]0.025	[263]0.797	[188]0.008		[167]0.113	0.000	0.000	0.060			
597	VISIONLABS-010	[214]0.001	[247]0.010	[161]0.058	[162]0.001	[131]0.006	[211]0.069	[198]0.005	[186]0.027		[175]0.008	[124]0.055	[160]0.109	0.000	0.000	0.040		0.000	
598	VISIONLABS-011	[147]0.001	[187]0.009	[129]0.053	[150]0.001	[88]0.004	[178]0.063	[132]0.003	[116]0.020		[91]0.004	[71]0.034	[110]0.090	0.000	0.000	0.032		0.000	

Table 36: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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FNIR(N, R, T) = FPIR(N, T) =

False neg. identification rate
False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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#	ALGORITHM	INVESTIGATION MODE						IDENTIFICATION MODE						FAILURE TO EXTRACT					
		RANK ONE MISS RATE, FNIR(N, 0, 1)						HIGH T ⟶ FPIR = 0.001, FNIR(N, T, L)						FEATURES					
		N=1.6M						N=1.6M											
	GALLERY	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	VISA	MUGSHOT	MUGSHOT	MUGSHOT	VISA	BORDER	KIOSK
	PROBE	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK	MUGSHOT	WEBCAM	PROFILE	BORDER	BOR¿10YR	KIOSK
599	VISTEAM-000	448 0.005	417 0.020	370 0.301	413 0.011	382 0.108	397 0.124	492 0.085	471 0.205	282 0.886	417 0.128	391 1.000	332 0.345	0.000	0.000	0.040		0.000	
600	VISTEAM-001	455 0.005	432 0.021	371 0.302	426 0.013	384 0.110	408 0.127	510 0.109	540 0.469	279 0.878	431 0.154	502 1.000	339 0.374	0.000	0.000	0.040		0.000	
601	VIXVIZION-009	349 0.002	316 0.013	334 0.210	338 0.005	319 0.031	364 0.107	391 0.027	345 0.077	251 0.742	345 0.041	261 0.286	372 0.472	0.000	0.000	0.000		0.000	
602	VNIS-000	594 0.105	576 0.143	460 0.671	515 0.127	434 0.448	497 0.279	573 0.311	530 0.418	394 0.993	517 0.701	335 0.838	424 0.696	0.012	0.019	0.098		0.062	
603	VNPAY-001	228 0.001	131 0.008	188 0.064	343 0.005	225 0.012	288 0.085	214 0.006	179 0.026	136 0.280	222 0.012	143 0.066	161 0.109	0.001	0.001	0.048		0.006	
604	VNPT-001	339 0.002	352 0.014	292 0.135	259 0.002	295 0.023	221 0.071	310 0.014	329 0.068	246 0.715	330 0.035	363 0.990	335 0.362	0.001	0.000	0.042		0.000	
605	VNPT-002	301 0.002	307 0.012	154 0.058	143 0.001	141 0.006	116 0.054	226 0.006	212 0.032	139 0.284	169 0.007	154 0.072	129 0.096	0.001	0.000	0.042		0.000	
606	VNPT-003	257 0.001	129 0.007	170 0.060	142 0.001	136 0.006	124 0.054	223 0.006	192 0.028	312 0.960	164 0.007	133 0.062	130 0.096	0.001	0.000	0.041		0.000	
607	VNPT-004	258 0.001	128 0.007	163 0.059	136 0.001	125 0.006	122 0.054	205 0.005	175 0.026	289 0.899	158 0.007	125 0.057	117 0.093	0.001	0.000	0.041		0.000	
608	VNPT-005	261 0.001	121 0.007	140 0.056	119 0.001	114 0.005	105 0.052	178 0.004	152 0.023	274 0.853	109 0.005	101 0.047	108 0.090	0.001	0.000	0.041		0.000	
609	VOCORD-003	462 0.006	454 0.024	502 0.802	498 0.061		463 0.188	518 0.122	443 0.155	445 0.998	433 0.157		347 0.404	0.001	0.011	0.425			
610	VOCORD-004	483 0.008	426 0.021	494 0.789	421 0.012		406 0.127	580 0.353	456 0.173	512 1.000	448 0.193		506 0.991	0.000	0.000	0.000			
611	VOCORD-005	474 0.007	448 0.023	504 0.810	493 0.055		470 0.206	536 0.156	423 0.130	434 0.997	425 0.138		340 0.381	0.001	0.009	0.554			
612	VOCORD-006	637 1.000	637 1.000	599 1.000	637 1.000		589 1.000	631 1.000	634 1.000	587 1.000	599 1.000		568 1.000	0.001	0.009	0.554			
613	VOGIC-000	383 0.002	509 0.039	321 0.185	505 0.081	437 0.539	511 0.317	309 0.014	589 0.855	428 0.997	542 0.993	554 1.000	517 0.999	0.000	0.000	0.032		0.000	
614	VSOFT-001	527 0.014	530 0.052	475 0.729	460 0.026	399 0.149	493 0.265	533 0.153	508 0.288	377 0.990	451 0.202	326 0.999	399 0.573	0.000	0.002	0.094		0.001	
615	VSOFT-002	353 0.002	308 0.012	401 0.393	328 0.004	329 0.036	367 0.108	396 0.028	388 0.098	429 0.997	328 0.034	365 0.992	269 0.201	0.000	0.002	0.094		0.001	
616	VTS-000	626 0.594	614 0.608	522 0.908	540 0.607	442 0.724	541 0.739	600 0.598	558 0.619	473 0.999	506 0.613	325 0.760	435 0.761	0.000	0.001	0.047		0.000	
617	VTS-001	245 0.001	229 0.010	311 0.158	363 0.006	269 0.018	253 0.077	300 0.013	288 0.051	400 0.994	283 0.022	201 0.141	265 0.192	0.000	0.000	0.040		0.000	
618	VTS-002	310 0.002	326 0.013	343 0.224	436 0.014	334 0.038	399 0.125	380 0.026	343 0.075	485 1.000	353 0.045	249 0.231	352 0.417	0.000	0.000	0.029		0.000	
619	VTS-003	140 0.001	113 0.007	189 0.064	198 0.002	192 0.009	111 0.053	239 0.007	220 0.033	524 1.000	237 0.014	353 0.954	411 0.635	0.000	0.001	0.029		0.000	
620	VTS-004	88 0.001	93 0.007	136 0.055	151 0.001	165 0.007	69 0.048	189 0.004	184 0.027	555 1.000	194 0.009	382 1.000	272 0.203	0.000	0.001	0.029		0.000	
621	VTS-005	73 0.001	77 0.006	114 0.052	129 0.001	130 0.006	63 0.047	206 0.005	204 0.030	439 0.998	157 0.007	332 0.811	140 0.100	0.000	0.001	0.029		0.000	
622	WINK-000	558 0.029	562 0.095	573 0.979	491 0.049	404 0.176	516 0.366	553 0.203	524 0.377	359 0.983	459 0.261	601 1.000	415 0.649	0.007	0.001	0.979		0.000	
623	WINK-001	391 0.003	527 0.049	492 0.785	522 0.206	423 0.296	491 0.253	417 0.036	425 0.131	423 0.996	455 0.237	341 0.903	337 0.372	0.001	0.036	0.043		0.169	
624	WINK-002	190 0.001	533 0.055	466 0.696	528 0.285	426 0.386	503 0.291	290 0.012	375 0.092	313 0.961	465 0.294	359 0.978	331 0.342	0.000	0.048	0.041		0.247	
625	WINK-003	357 0.002	383 0.016	360 0.281	240 0.002	191 0.009	356 0.106	261 0.009	246 0.038	508 1.000	213 0.010	162 0.077	439 0.774	0.001	0.002	0.058		0.001	
626	XAILIENT-000	386 0.002	279 0.011	295 0.140	253 0.002	247 0.015	202 0.068	415 0.035	310 0.059	204 0.521	303 0.027	232 0.181	267 0.196	0.000	0.000	0.000		0.000	
627	XAILIENT-001	212 0.001	210 0.009	148 0.056	188 0.002	140 0.006	145 0.057	253 0.008	214 0.032	130 0.251	198 0.009	148 0.067	153 0.105	0.000	0.000	0.000		0.000	
628	XFORWARDAI-000	350 0.002	340 0.014	223 0.078	310 0.004	248 0.015	311 0.094	319 0.015	299 0.053	177 0.434	278 0.021	221 0.159	242 0.169	0.000	0.000	0.000		0.000	
629	XFORWARDAI-001	328 0.002	313 0.013	146 0.056	275 0.003	193 0.009	274 0.082	190 0.005	193 0.028	178 0.441	181 0.008	135 0.062	178 0.123	0.000	0.000	0.000		0.000	
630	XFORWARDAI-002	313 0.002	300 0.012	77 0.048	257 0.002	149 0.007	251 0.077	118 0.003	99 0.016	203 0.519	102 0.005	88 0.041	137 0.099	0.000	0.000	0.000		0.000	
631	YISHENG-001	556 0.026	541 0.060		497 0.058		500 0.287	577 0.346	580 0.808		510 0.666		473 0.919	0.002	0.005				
632	YITU-002	300 0.002	235 0.010					337 0.017	278 0.049					0.000	0.000				
633	YITU-003	392 0.003	329 0.016					346 0.018	293 0.052					0.003	0.001				
634	YITU-004	202 0.001	169 0.008	515 0.864				265 0.009	182 0.027	301 0.936				0.000	0.000	0.000			
635	YITU-005	362 0.002	355 0.014					275 0.010	215 0.032					0.003	0.001				
636	YOUVERSE-001	169 0.001	126 0.009	168 0.060	181 0.001	154 0.007	117 0.054	236 0.007	202 0.030	131 0.253	202 0.009	163 0.081	143 0.101	0.000	0.001	0.041		0.000	
637	YUAN-000	279 0.001	322 0.013	284 0.128	179 0.001	176 0.008	172 0.062	277 0.010	269 0.045	152 0.342	226 0.012	274 0.333	188 0.129	0.001	0.006	0.071		0.000	

Table 37: **Miss rates by dataset:** *At left, rank 1 miss rates relevant to investigations; at right, with threshold set to target FPIR = 0.01 for higher volume, low prior, uses. Yellow indicates most accurate algorithm. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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MISSES BELOW THRESHOLD, T FNIR(N, T> 0, R >L)		ENROL MOST RECENT DATASET: FRVT 2018 MUGSHOTS				
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M
1	3DIVI-005	542 0.1353	542 0.1661	478 0.1907	462 0.2339	452 0.2995
2	ACER-000	533 0.1182	532 0.1452	472 0.1711	456 0.2071	445 0.2534
3	ADIANCE-002	290 0.0089	294 0.0124	282 0.0160	274 0.0291	274 0.1106
4	ADVANCE-000	349 0.0138	352 0.0195	333 0.0243	319 0.0401	321 0.1278
5	AFISBIOMETRICS-000	401 0.0201	405 0.0295	382 0.0378	371 0.0559	300 0.1173
6	AFRENGINE-000	345 0.0134	345 0.0185	334 0.0245	340 0.0461	391 0.1717
7	AFRENGINE-001	145 0.0025	136 0.0030	131 0.0040	149 0.0115	199 0.0919
8	AFRENGINE-002	134 0.0023	125 0.0028	116 0.0034	157 0.0117	186 0.0901
9	AFRENGINE-003	207 0.0036	183 0.0044	166 0.0054	148 0.0115	201 0.0921
10	AIGROUP-000	266 0.0062	270 0.0095	258 0.0127	259 0.0269	311 0.1239
11	AIZE-002	449 0.0374	445 0.0508	415 0.0653	414 0.1022	430 0.2232
12	AIZE-006	442 0.0354	436 0.0453	410 0.0588	399 0.0867	398 0.1786
13	AKUVOX-001	296 0.0093	301 0.0130	286 0.0168	286 0.0317	308 0.1234
14	ALCHERA-003	531 0.1172	535 0.1550	474 0.1834	463 0.2389	461 0.3457
15	ALCHERA-006	195 0.0034	195 0.0046	184 0.0060	207 0.0166	249 0.1034
16	ALCHERA-008	142 0.0025	143 0.0032	137 0.0042	125 0.0098	148 0.0815
17	ALGORITHMSINC-000	340 0.0132	330 0.0160	326 0.0235	320 0.0404	292 0.1156
18	ALLGOVISION-000	499 0.0684	499 0.0878	453 0.1081	440 0.1386	425 0.2126
19	ALLGOVISION-001	506 0.0782	506 0.1014	461 0.1215	448 0.1581	433 0.2270
20	ALLGOVISION-005	363 0.0147	358 0.0204	341 0.0263	324 0.0410	286 0.1144
21	ANKE-000	515 0.0939	513 0.1166	467 0.1394	453 0.1767	443 0.2486
22	ANKE-002	409 0.0225	410 0.0315	388 0.0402	374 0.0602	360 0.1463
23	ARATEK-001	245 0.0053	250 0.0078	245 0.0109	305 0.0367	353 0.1433
24	ARMATURA-000	180 0.0031	170 0.0039	155 0.0051	121 0.0095	77 0.0613
25	AWARE-003	528 0.1094	523 0.1279	468 0.1438	451 0.1754	436 0.2325
26	AWARE-005	582 0.3375	581 0.3633	486 0.3963	475 0.4470	444 0.2519
27	AWARE-007	328 0.0117	329 0.0158	308 0.0199	288 0.0319	288 0.1146
28	AYONIX-002	610 0.7859	611 0.8241	493 0.8500	481 0.8697	474 0.8927
29	BIOIDTECHSWISS-001	164 0.0028	164 0.0038	160 0.0053	167 0.0125	233 0.0991
30	BIOMETRIK-001	451 0.0385	449 0.0520	425 0.0675	413 0.0984	420 0.2062
31	CAMVI-004	445 0.0363	481 0.0713	447 0.0976	467 0.2503	450 0.2679
32	CANON-001	206 0.0036	204 0.0051	202 0.0070	197 0.0154	200 0.0920
33	CANON-002	187 0.0033	180 0.0043	177 0.0058	163 0.0121	144 0.0805
34	CANON-003	89 0.0015	97 0.0021	99 0.0028	81 0.0068	93 0.0660
35	CIB-000	284 0.0082	291 0.0121	270 0.0156	279 0.0299	463 0.3606
36	CLEARVIEWAI-000	210 0.0037	213 0.0055	208 0.0074	201 0.0155	215 0.0968
37	CLEARVIEWAI-001	139 0.0024	149 0.0035	152 0.0049	124 0.0096	59 0.0556
38	CLEARVIEWAI-002	155 0.0027	162 0.0038	167 0.0054	135 0.0102	76 0.0605
39	CLEARVIEWAI-003	97 0.0016	96 0.0021	97 0.0027	71 0.0065	50 0.0522
40	CLOUDWALK-HR-000	91 0.0015	84 0.0017	76 0.0020	82 0.0068	109 0.0698
41	CLOUDWALK-MT-000	94 0.0016	82 0.0017	71 0.0018	37 0.0046	35 0.0463
42	CLOUDWALK-MT-001	85 0.0015	76 0.0016	62 0.0017	44 0.0048	58 0.0551
43	CLOUDWALK-MT-002	86 0.0015	73 0.0015	58 0.0016	43 0.0047	70 0.0583
44	COGENT-000	466 0.0426	452 0.0523	426 0.0688	425 0.1106	406 0.1872
45	COGENT-001	467 0.0426	451 0.0523	427 0.0688	426 0.1106	405 0.1872
46	COGENT-002	429 0.0318	433 0.0441	409 0.0587	420 0.1077	423 0.2080
47	COGENT-003	431 0.0324	435 0.0453	420 0.0667	432 0.1241	437 0.2341
48	COGENT-004	406 0.0206	411 0.0327	404 0.0523	427 0.1135	424 0.2116
49	COGENT-005	259 0.0061	260 0.0087	251 0.0119	281 0.0299	307 0.1230
50	COGENT-006	169 0.0029	173 0.0040	164 0.0053	155 0.0117	158 0.0827
51	COGENT-007	144 0.0025	144 0.0033	147 0.0046	140 0.0107	131 0.0753
52	COGNITEC-000	545 0.1374	539 0.1602	477 0.1858	458 0.2154	451 0.2784
53	COGNITEC-001	509 0.0803	505 0.1013	460 0.1204	443 0.1495	426 0.2159
54	COGNITEC-002	459 0.0402	454 0.0527	416 0.0654	403 0.0910	403 0.1810
55	COGNITEC-003	456 0.0395	450 0.0520	413 0.0642	400 0.0874	381 0.1648
56	COGNITEC-004	408 0.0219	409 0.0310	384 0.0385	364 0.0537	271 0.1100
57	COGNITEC-005	257 0.0060	266 0.0093	268 0.0140	268 0.0284	213 0.0964
58	COGNITEC-006	238 0.0049	244 0.0073	249 0.0113	249 0.0251	195 0.0916
59	COGNITEC-007	204 0.0036	211 0.0053	215 0.0079	200 0.0155	147 0.0812
60	COGNITEC-008	209 0.0036	217 0.0056	212 0.0077	195 0.0152	153 0.0822
61	COGNITEC-009	212 0.0038	221 0.0059	217 0.0080	204 0.0158	161 0.0829
62	COGNITEC-010	163 0.0028	159 0.0037	151 0.0049	115 0.0090	106 0.0689
63	COGNITI-000	338 0.0130	350 0.0191	346 0.0271	342 0.0469	268 0.1094
64	COMPAL-000	246 0.0053	247 0.0075	546 1.0000	581 1.0000	504 1.0000
65	CORSIGHT-000	221 0.0042	222 0.0059	218 0.0080	211 0.0176	250 0.1039
66	CORSIGHT-001	228 0.0045	230 0.0064	221 0.0084	213 0.0180	207 0.0945
67	CORSIGHT-002	182 0.0031	187 0.0045	182 0.0059	178 0.0133	236 0.0994
68	CUBOX-001	119 0.0020	119 0.0027	113 0.0034	88 0.0071	66 0.0575
69	CUBOX-002	130 0.0023	134 0.0030	124 0.0037	89 0.0072	43 0.0493
70	CUBOX-003	140 0.0024	140 0.0032	133 0.0040	97 0.0074	44 0.0506
71	CYBERLINK-000	461 0.0411	462 0.0561	432 0.0704	416 0.1028	419 0.2047
72	CYBERLINK-001	452 0.0389	455 0.0532	429 0.0691	411 0.0969	399 0.1791

Table 38: **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with* $N \geq 3\,000\,000$. *Throughout blue superscripts indicate the rank of the algorithm for that column.*

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MISSES BELOW THRESHOLD, T		ENROL MOST RECENT				
FNIR(N, T> 0, R >L)		DATASET: FRVT 2018 MUGSHOTS				
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M
73	CYBERLINK-002	[311] 0.0102	[317] 0.0145	[307] 0.0198	[318] 0.0395	[315] 0.1252
74	CYBERLINK-003	[244] 0.0052	[245] 0.0074	[234] 0.0097	[243] 0.0232	[309] 0.1234
75	CYBERLINK-004	[235] 0.0048	[237] 0.0068	[237] 0.0099	[338] 0.0458	[365] 0.1509
76	CYBERLINK-005	[267] 0.0063	[272] 0.0096	[266] 0.0134	[314] 0.0391	[375] 0.1563
77	CYBERLINK-006	[108] 0.0018	[109] 0.0023	[108] 0.0030	[101] 0.0078	[65] 0.0571
78	DAHUA-001	[485] 0.0566	[483] 0.0724	[441] 0.0868	[428] 0.1138	[396] 0.1773
79	DAHUA-002	[318] 0.0105	[318] 0.0148	[300] 0.0188	[273] 0.0288	[289] 0.1150
80	DAHUA-003	[302] 0.0097	[307] 0.0135	[294] 0.0177	[276] 0.0293	[281] 0.1127
81	DAHUA-004	[227] 0.0044	[235] 0.0065	[223] 0.0086	[206] 0.0160	[165] 0.0850
82	DAON-000	[370] 0.0157	[370] 0.0223	[355] 0.0289	[370] 0.0558	[387] 0.1699
83	DAON-002	[236] 0.0048	[238] 0.0069	[231] 0.0095	[232] 0.0212	[258] 0.1072
84	DAON-003	[133] 0.0023	[139] 0.0031	[139] 0.0042	[129] 0.0100	[97] 0.0673
85	DATECH-000	[242] 0.0051	[246] 0.0074	[238] 0.0099	[269] 0.0284	[314] 0.1244
86	DATECH-001	[232] 0.0046	[231] 0.0064	[232] 0.0096	[250] 0.0253	[287] 0.1145
87	DECATUR-000	[373] 0.0170	[373] 0.0226	[358] 0.0302	[339] 0.0460	[352] 0.1430
88	DEEPGLINT-001	[135] 0.0023	[133] 0.0030	[132] 0.0040	[158] 0.0118	[198] 0.0919
89	DEEPSEA-001	[440] 0.0343	[439] 0.0459	[408] 0.0583	[397] 0.0799	[390] 0.1705
90	DERMALOG-005	[503] 0.0696	[498] 0.0876	[455] 0.1135	[447] 0.1550	[438] 0.2356
91	DERMALOG-006	[454] 0.0392	[448] 0.0514	[414] 0.0651	[409] 0.0951	[383] 0.1664
92	DERMALOG-007	[500] 0.0687	[496] 0.0859	[454] 0.1104	[444] 0.1501	[435] 0.2296
93	DERMALOG-008	[436] 0.0334	[435] 0.0452	[412] 0.0622	[418] 0.1057	[434] 0.2273
94	DERMALOG-009	[359] 0.0144	[356] 0.0202	[342] 0.0265	[327] 0.0413	[340] 0.1371
95	DERMALOG-010	[237] 0.0048	[234] 0.0065	[222] 0.0084	[226] 0.0203	[216] 0.0968
96	DERMALOG-011	[361] 0.0145	[363] 0.0212	[352] 0.0276	[336] 0.0457	[303] 0.1188
97	DERMALOG-012	[352] 0.0141	[362] 0.0207	[347] 0.0272	[330] 0.0425	[263] 0.1083
98	DERMALOG-013	[354] 0.0142	[361] 0.0207	[351] 0.0275	[331] 0.0435	[272] 0.1105
99	DERMALOG-014	[358] 0.0144	[365] 0.0214	[356] 0.0291	[344] 0.0472	[297] 0.1168
100	DERMALOG-015	[193] 0.0033	[197] 0.0047	[190] 0.0062	[168] 0.0125	[175] 0.0871
101	DERMALOG-016	[100] 0.0016	[90] 0.0020	[90] 0.0026	[96] 0.0074	[84] 0.0632
102	DERMALOG-017	[132] 0.0023	[129] 0.0029	[123] 0.0037	[103] 0.0079	[91] 0.0656
103	DILUSENSE-000	[404] 0.0205	[407] 0.0301	[381] 0.0374	[366] 0.0540	[350] 0.1426
104	DILUSENSE-001	[251] 0.0057	[254] 0.0082	[242] 0.0106		[293] 0.1158
105	EINETWORKSINDIA-000	[321] 0.0107	[322] 0.0150	[310] 0.0203	[311] 0.0385	[347] 0.1402
106	EINETWORKSINDIA-001	[325] 0.0110	[328] 0.0157	[316] 0.0218	[335] 0.0451	[376] 0.1569
107	EINETWORKSINDIA-002	[150] 0.0026	[158] 0.0037	[156] 0.0052	[139] 0.0107	[142] 0.0797
108	FACEPHI-000	[229] 0.0045	[229] 0.0063	[216] 0.0079	[183] 0.0141	[92] 0.0657
109	FACEPHI-001	[76] 0.0013	[81] 0.0017	[82] 0.0021	[75] 0.0066	[83] 0.0631
110	FIRSTCREDITKZ-001	[114] 0.0019	[114] 0.0026	[120] 0.0036	[114] 0.0090	[132] 0.0756
111	FUJITSULAB-000	[360] 0.0144	[357] 0.0203	[350] 0.0274	[365] 0.0538	[394] 0.1736
112	FUJITSULAB-001	[331] 0.0122	[342] 0.0178	[335] 0.0247	[377] 0.0642	[422] 0.2076
113	GORILLA-002	[548] 0.1534	[550] 0.1877	[482] 0.2172	[469] 0.2586	[460] 0.3357
114	GORILLA-004	[502] 0.0696	[502] 0.0889	[450] 0.1045	[439] 0.1367	[413] 0.1967
115	GORILLA-005	[472] 0.0450	[467] 0.0579	[431] 0.0701	[412] 0.0971	[361] 0.1471
116	GORILLA-006	[391] 0.0192	[393] 0.0271	[365] 0.0328	[355] 0.0513	[278] 0.1110
117	GORILLA-007	[388] 0.0187	[390] 0.0267	[372] 0.0345	[358] 0.0516	[280] 0.1126
118	GORILLA-008	[372] 0.0167	[374] 0.0235	[359] 0.0305	[341] 0.0465	[279] 0.1117
119	GRIAULE-000	[353] 0.0141	[353] 0.0198	[336] 0.0250	[321] 0.0404	[354] 0.1437
120	GRIAULE-001	[175] 0.0030	[181] 0.0043	[188] 0.0061	[192] 0.0150	[190] 0.0907
121	HIK-003	[510] 0.0824	[507] 0.1023	[459] 0.1194	[446] 0.1513	[440] 0.2401
122	HIK-004	[507] 0.0792	[503] 0.0984	[456] 0.1142	[442] 0.1460	[441] 0.2403
123	HIK-005	[426] 0.0309	[430] 0.0432	[407] 0.0555	[402] 0.0894	[415] 0.2000
124	HISIGN-001	[241] 0.0051	[242] 0.0073	[229] 0.0093	[222] 0.0189	[239] 0.1011
125	HYPERVERGE-001	[173] 0.0030	[175] 0.0041	[172] 0.0056	[146] 0.0113	[172] 0.0869
126	HYPERVERGE-002	[146] 0.0025	[147] 0.0034	[140] 0.0043	[64] 0.0060	[52] 0.0535
127	HYPERVERGE-003	[243] 0.0052	[241] 0.0071	[233] 0.0096	[218] 0.0185	[227] 0.0983
128	HYUNDAI-ROBOTICSLAB-000	[303] 0.0097	[305] 0.0135	[287] 0.0169	[267] 0.0284	[243] 0.1016
129	HYUNDAI-ROBOTICSLAB-001	[293] 0.0091	[297] 0.0125	[289] 0.0171	[270] 0.0284	[256] 0.1071
130	HYUNDAI-ROBOTICSLAB-002	[196] 0.0034	[207] 0.0052	[201] 0.0070	[174] 0.0130	[122] 0.0720
131	HYUNDAI-ROBOTICSLAB-003	[190] 0.0033	[203] 0.0051	[198] 0.0068	[170] 0.0127	[111] 0.0700
132	HZAILU-000	[351] 0.0140	[351] 0.0194	[337] 0.0252	[322] 0.0408	[299] 0.1170
133	HZAILU-001	[264] 0.0062	[255] 0.0082	[241] 0.0105	[227] 0.0204	[251] 0.1049
134	HZAILU-002	[253] 0.0057	[249] 0.0076	[236] 0.0098	[216] 0.0184	[192] 0.0911
135	HZAILU-003	[250] 0.0057	[251] 0.0078	[244] 0.0108	[212] 0.0179	[214] 0.0966
136	HZAILU-004	[174] 0.0030	[174] 0.0041	[179] 0.0058	[144] 0.0111	[112] 0.0702
137	I2V-001	[369] 0.0157	[444] 0.0488	[457] 0.1152	[464] 0.2377	[462] 0.3600
138	ICEHOUSE-001	[272] 0.0066	[267] 0.0093	[259] 0.0128	[237] 0.0223	[221] 0.0976
139	ICM-001	[405] 0.0206	[404] 0.0287	[389] 0.0405	[396] 0.0792	[432] 0.2254
140	IDBIO-000	[357] 0.0143	[354] 0.0199	[338] 0.0254	[315] 0.0391	[275] 0.1107
141	IDEMIA-003	[438] 0.0343	[442] 0.0468	[442] 0.0869	[470] 0.2689	[468] 0.4184
142	IDEMIA-004	[425] 0.0296	[423] 0.0370	[393] 0.0441	[375] 0.0607	[373] 0.1545
143	IDEMIA-005	[444] 0.0356	[432] 0.0436	[406] 0.0532	[392] 0.0745	[401] 0.1797
144	IDEMIA-006	[441] 0.0348	[429] 0.0429	[403] 0.0520	[387] 0.0722	[421] 0.2062

Table 39: **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR *= 0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N \geq 3\,000\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column.*

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MISSES BELOW THRESHOLD, T		ENROL MOST RECENT				
FNIR(N, T> 0, R >L)		DATASET: FRVT 2018 MUGSHOTS				
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M
145	IDEMIA-007	[341]0.0133	[341]0.0177	[319]0.0224	[301]0.0353	[346]0.1399
146	IDEMIA-008	[73]0.0013	[75]0.0016	[79]0.0020	[47]0.0049	[37]0.0466
147	IDEMIA-009	[55]0.0010	[57]0.0012	[49]0.0014	[58]0.0058	[55]0.0547
148	IDEMIA-010	[15]0.0007	[13]0.0007	[12]0.0008	[12]0.0023	[6]0.0190
149	IDEMIA-011	[34]0.0008	[21]0.0009	[17]0.0009	[11]0.0022	[13]0.0258
150	IDEMIA-012	[30]0.0008	[19]0.0008	[16]0.0009	[9]0.0021	[15]0.0266
151	IDEMIA-013	[1]0.0005	[1]0.0005	[1]0.0005	[1]0.0012	[1]0.0111
152	IDENTY-001	[121]0.0021	[122]0.0027		[483]0.9606	[477]0.9688
153	IDENTY-002	[111]0.0019	[117]0.0026	[122]0.0037	[134]0.0102	[188]0.0906
154	IMAGUS-005	[343]0.0133	[345]0.0181	[324]0.0234	[304]0.0364	[254]0.1064
155	IMAGUS-006	[344]0.0134	[349]0.0187	[330]0.0241	[316]0.0392	[290]0.1156
156	IMAGUS-007	[367]0.0157	[372]0.0224	[353]0.0280	[332]0.0441	[301]0.1176
157	IMDS-SOFTWARE-001	[230]0.0045	[232]0.0064	[226]0.0091	[228]0.0204	[304]0.1190
158	IMDS-SOFTWARE-002	[226]0.0044	[228]0.0063	[224]0.0087	[223]0.0192	[262]0.1083
159	IMPERIAL-000	[382]0.0183	[381]0.0256	[378]0.0354	[390]0.0730	[400]0.1791
160	INCODE-003	[540]0.1321	[543]0.1668	[479]0.1951	[460]0.2319	[453]0.3041
161	INCODE-004	[457]0.0400	[458]0.0535	[419]0.0658	[405]0.0914	[380]0.1616
162	INCODE-005	[282]0.0079	[283]0.0109	[269]0.0141	[247]0.0244	[191]0.0909
163	INCODE-006	[58]0.0011	[60]0.0013	[64]0.0017	[33]0.0041	[40]0.0485
164	INCODE-007	[39]0.0009	[53]0.0011	[47]0.0014	[25]0.0036	[56]0.0549
165	INNOVATRICS-007	[291]0.0090	[293]0.0122	[278]0.0156	[251]0.0256	[264]0.1089
166	INNOVATRICS-008	[194]0.0034	[196]0.0046	[191]0.0062	[225]0.0203	[266]0.1090
167	INNOVATRICS-009	[137]0.0023	[123]0.0028	[119]0.0035	[133]0.0102	[120]0.0714
168	INNOVATRICS-010	[136]0.0023	[135]0.0030	[125]0.0037	[132]0.0102	[141]0.0783
169	INNOVATRICS-011	[88]0.0015	[87]0.0018	[86]0.0023	[87]0.0071	[102]0.0685
170	INNOVATRICS-012	[78]0.0014	[77]0.0016	[75]0.0019	[72]0.0065	[86]0.0649
171	INNOVATRICS-013	[66]0.0012	[63]0.0014	[67]0.0018	[68]0.0062	[57]0.0550
172	INNOVATRICS-014	[36]0.0008	[39]0.0010	[29]0.0011	[49]0.0050	[41]0.0486
173	INNOVATRICS-015	[32]0.0008	[25]0.0009	[21]0.0010	[42]0.0047	[38]0.0467
174	INNOVATRICS-016	[11]0.0006	[10]0.0007	[10]0.0008	[30]0.0037	[49]0.0522
175	INSPUR-000	[222]0.0042	[224]0.0060	[220]0.0083	[215]0.0181	[222]0.0977
176	INSPUR-001	[202]0.0035	[202]0.0050	[200]0.0070	[177]0.0132	[166]0.0851
177	INSPUR-002	[201]0.0034	[209]0.0052	[199]0.0069	[173]0.0129	[128]0.0742
178	INTELRESEARCH-001	[64]0.0011	[71]0.0015	[78]0.0020	[98]0.0074	[135]0.0765
179	INTEMA-000	[96]0.0016	[93]0.0021	[102]0.0028	[119]0.0094	[129]0.0741
180	INTEMA-001	[57]0.0010	[41]0.0010	[73]0.0019	[80]0.0068	[90]0.0675
181	INTOZI-001	[315]0.0103	[323]0.0150	[304]0.0193	[289]0.0319	[328]0.1310
182	INTSYSMSU-000	[625]0.9982	[623]0.9984	[497]0.9985	[485]0.9987	[478]0.9988
183	IONETWORKS-001	[301]0.0095	[313]0.0142	[312]0.0207	[339]0.0457	[374]0.1556
184	IREX-000	[389]0.0187	[399]0.0277	[385]0.0388	[380]0.0674	[362]0.1476
185	IRISID-001	[297]0.0094	[306]0.0135	[292]0.0175	[280]0.0299	[261]0.1081
186	IRISID-002	[277]0.0071	[278]0.0099	[260]0.0129	[244]0.0235	[224]0.0981
187	ISTL-001	[299]0.0094	[299]0.0129	[283]0.0161	[260]0.0272	[230]0.0987
188	ISYSTEMS-002	[487]0.0579	[487]0.0779	[446]0.0963	[438]0.1360	[429]0.2221
189	ISYSTEMS-003	[469]0.0434	[468]0.0585	[436]0.0792	[431]0.1224	[431]0.2243
190	KAKAO-000	[319]0.0105	[320]0.0148	[302]0.0192	[292]0.0321	[238]0.1007
191	KAKAO-001	[103]0.0017	[104]0.0022	[103]0.0029	[104]0.0081	[107]0.0689
192	KAKAO-002	[80]0.0014	[85]0.0018	[83]0.0022	[56]0.0057	[60]0.0559
193	KEDACOM-001	[377]0.0177	[371]0.0223	[339]0.0262	[328]0.0419	[333]0.1337
194	KLOUDSPOT-001	[375]0.0176	[376]0.0243	[371]0.0343	[382]0.0685	[330]0.1319
195	KNOWUTECH-000	[262]0.0061	[262]0.0089	[250]0.0118	[233]0.0214	[225]0.0982
196	KOGNIZA-000	[220]0.0041	[218]0.0056	[207]0.0073	[186]0.0144	[123]0.0723
197	LIGHTVISION-000	[392]0.0193	[392]0.0270	[379]0.0355	[373]0.0591	[395]0.1742
198	LINECLOVA-002	[143]0.0025	[155]0.0036	[146]0.0046	[156]0.0117	[150]0.0821
199	LINECLOVA-003	[128]0.0022	[107]0.0022	[145]0.0045	[198]0.0154	[228]0.0986
200	LOOKMAN-003	[439]0.0343	[431]0.0433	[401]0.0509	[386]0.0712	[367]0.1530
201	LOOKMAN-005	[412]0.0237	[406]0.0298	[377]0.0352	[352]0.0509	[331]0.1331
202	LSSYTECH-000	[44]0.0009	[51]0.0011	[48]0.0014	[571]1.0000	[85]0.0633
203	LSSYTECH-001		[18]0.0008	[27]0.0011	[61]0.0059	[51]0.0533
204	LUXAND-001	[478]0.0465	[464]0.0573	[424]0.0674	[410]0.0954	[385]0.1681
205	MANTRA-000	[263]0.0062	[273]0.0097	[271]0.0147	[283]0.0304	[246]0.1032
206	MAXVISION-000	[402]0.0202	[400]0.0278	[375]0.0351	[356]0.0514	[334]0.1337
207	MAXVISION-001	[162]0.0028	[169]0.0039	[154]0.0051	[160]0.0119	[181]0.0891
208	MAXVISION-002	[159]0.0027	[150]0.0035	[148]0.0047	[162]0.0120	[208]0.0946
209	MAXVISION-003	[216]0.0039	[215]0.0055	[203]0.0071	[188]0.0145	[105]0.0688
210	MAXVISION-004	[99]0.0016	[95]0.0021	[93]0.0027	[109]0.0084	[129]0.0743
211	MEGVII-001	[483]0.0558	[482]0.0719	[440]0.0861	[434]0.1281	[449]0.2600
212	MEGVII-003	[104]0.0017	[102]0.0022	[101]0.0028	[105]0.0082	[139]0.0781
213	MEGVII-004	[45]0.0009	[35]0.0010	[25]0.0011	[32]0.0040	[48]0.0521
214	MEGVII-005	[46]0.0009	[36]0.0010	[26]0.0011	[67]0.0061	[53]0.0540
215	METSAKUURCOMPANY-000	[239]0.0049	[240]0.0070	[228]0.0093	[231]0.0209	[255]0.1065
216	METSAKUURCOMPANY-001	[265]0.0062	[263]0.0089	[256]0.0126	[349]0.0498	[392]0.1724

Table 40: **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with* $N \geq 3\,000\,000$. *Throughout blue superscripts indicate the rank of the algorithm for that column.*

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MISSES BELOW THRESHOLD, T FNIR(N, T> 0, R >L)		ENROL MOST RECENT DATASET: FRVT 2018 MUGSHOTS				
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M
217	METSAKUURCOMPANY-002	[285]0.0084	[287]0.0118	[274]0.0153	[265]0.0274	[265]0.1089
218	MIAXIS-001	[248]0.0055	[248]0.0076	[239]0.0099	[219]0.0188	[218]0.0970
219	MIAXIS-002	[256]0.0060	[252]0.0080	[240]0.0105	[221]0.0189	[223]0.0978
220	MICROFOCUS-005	[620]0.9732	[613]0.8352	[494]0.8550	[482]0.8748	[475]0.8941
221	MICROSOFT-003	[395]0.0196	[395]0.0274	[374]0.0351	[362]0.0526	[356]0.1440
222	MICROSOFT-004	[380]0.0182	[382]0.0256	[366]0.0329	[353]0.0509	[349]0.1415
223	MICROSOFT-005	[378]0.0180	[379]0.0254	[363]0.0317	[350]0.0499	[339]0.1368
224	MICROSOFT-006	[292]0.0090	[289]0.0119	[280]0.0158	[275]0.0292	[345]0.1395
225	MOBBL-000	[393]0.0195	[398]0.0276	[380]0.0364	[329]0.0424	[28]0.0424
226	MUKH-002	[598]0.5039	[599]0.5941	[491]0.6673	[479]0.7313	[473]0.8275
227	NAVERCLOUD-000	[147]0.0025	[146]0.0033	[143]0.0044	[131]0.0101	[126]0.0740
228	NEC-000	[491]0.0634	[489]0.0786	[445]0.0928	[429]0.1150	[404]0.1862
229	NEC-001	[511]0.0859	[509]0.1050	[462]0.1237	[445]0.1511	[428]0.2196
230	NEC-002	[106]0.0018	[108]0.0023	[107]0.0030	[161]0.0119	[73]0.0594
231	NEC-003	[113]0.0019	[103]0.0022	[89]0.0026	[53]0.0053	[42]0.0492
232	NEC-004	[77]0.0013	[66]0.0015	[61]0.0016	[22]0.0033	[20]0.0326
233	NEC-005	[65]0.0012	[62]0.0013	[57]0.0015	[65]0.0061	[18]0.0304
234	NEC-006	[84]0.0015	[80]0.0017	[84]0.0022	[128]0.0099	[64]0.0570
235	NEC-007	[48]0.0009	[38]0.0010	[28]0.0011	[13]0.0025	[14]0.0265
236	NEC-008	[56]0.0010	[48]0.0011	[38]0.0012	[18]0.0030	[17]0.0292
237	NEC-009	[21]0.0007	[15]0.0008	[15]0.0008	[10]0.0022	[7]0.0191
238	NEC-010	[7]0.0006	[7]0.0006	[7]0.0007	[3]0.0019	[5]0.0167
239	NEC-011	[6]0.0005	[6]0.0006	[6]0.0006	[4]0.0018	[3]0.0164
240	NEC-012	[5]0.0005	[4]0.0006	[2]0.0006	[2]0.0013	[4]0.0164
241	NEUROTECHNOLOGY-003	[601]0.5681	[603]0.6312	[492]0.6951	[480]0.7551	[472]0.8014
242	NEUROTECHNOLOGY-004	[474]0.0462	[475]0.0625	[434]0.0772	[424]0.1101	[417]0.2008
243	NEUROTECHNOLOGY-005	[453]0.0390	[459]0.0535	[422]0.0672	[406]0.0934	[407]0.1874
244	NEUROTECHNOLOGY-007	[468]0.0433	[474]0.0619	[437]0.0799	[436]0.1317	[439]0.2391
245	NEUROTECHNOLOGY-008	[432]0.0335	[453]0.0527	[443]0.0889	[452]0.1766	[459]0.3286
246	NEUROTECHNOLOGY-009	[317]0.0104	[321]0.0148	[303]0.0192	[290]0.0320	[270]0.1099
247	NEUROTECHNOLOGY-010	[271]0.0066	[271]0.0096	[267]0.0135	[334]0.0445	[393]0.1724
248	NEUROTECHNOLOGY-012	[224]0.0044	[233]0.0065	[230]0.0094	[255]0.0261	[336]0.1340
249	NEUROTECHNOLOGY-013	[149]0.0026	[168]0.0039	[168]0.0054	[229]0.0205	[305]0.1199
250	NEUROTECHNOLOGY-014	[120]0.0020	[131]0.0029	[138]0.0042	[138]0.0103	[173]0.0870
251	NEUROTECHNOLOGY-015	[116]0.0020	[137]0.0030	[135]0.0041	[123]0.0096	[211]0.0951
252	NEUROTECHNOLOGY-016	[118]0.0020	[130]0.0029	[129]0.0039	[110]0.0087	[145]0.0807
253	NEXTMETALABS-002	[161]0.0027	[160]0.0037	[163]0.0053	[151]0.0116	[149]0.0818
254	NOMINDER-000	[184]0.0032	[188]0.0045	[186]0.0060	[196]0.0152	[189]0.0906
255	NOMINDER-001	[205]0.0036	[201]0.0049	[194]0.0064	[187]0.0145	[209]0.0947
256	NOMINDER-002	[203]0.0035	[199]0.0047	[192]0.0063	[184]0.0141	[193]0.0913
257	NOMINDER-003	[186]0.0033	[192]0.0045	[185]0.0060	[166]0.0125	[185]0.0900
258	NOMINDER-004	[183]0.0031	[185]0.0044	[180]0.0058	[176]0.0130	[206]0.0943
259	NOMINDER-005	[126]0.0022	[127]0.0029	[128]0.0039	[141]0.0109	[163]0.0833
260	NOTIONTAG-000	[333]0.0124	[336]0.0171	[320]0.0224	[302]0.0354	[319]0.1267
261	NTECHLAB-003	[463]0.0417	[457]0.0534	[421]0.0667	[401]0.0893	[364]0.1508
262	NTECHLAB-004	[427]0.0309	[425]0.0402	[402]0.0514	[385]0.0705	[355]0.1440
263	NTECHLAB-005	[434]0.0331	[427]0.0420	[405]0.0530	[393]0.0748	[359]0.1459
264	NTECHLAB-006	[422]0.0285	[419]0.0364	[396]0.0464	[379]0.0660	[358]0.1445
265	NTECHLAB-007	[385]0.0184	[378]0.0252	[361]0.0314	[347]0.0492	[327]0.1303
266	NTECHLAB-008	[314]0.0103	[311]0.0141	[298]0.0184	[266]0.0282	[234]0.0991
267	NTECHLAB-009	[191]0.0033	[194]0.0045	[181]0.0059	[164]0.0121	[125]0.0731
268	NTECHLAB-010	[98]0.0016	[101]0.0022	[95]0.0027	[94]0.0074	[115]0.0707
269	NTECHLAB-011	[110]0.0019	[115]0.0026	[117]0.0034	[90]0.0072	[80]0.0622
270	OMNIGARDE-000	[166]0.0028	[165]0.0039	[158]0.0052	[171]0.0127	[177]0.0875
271	OMNIGARDE-001	[79]0.0014	[86]0.0018	[87]0.0024	[74]0.0066	[118]0.0711
272	OMNIGARDE-002	[60]0.0011	[61]0.0013	[60]0.0016	[62]0.0060	[116]0.0708
273	ONFIDO-000	[308]0.0100	[303]0.0131	[293]0.0175	[282]0.0300	[306]0.1225
274	ONFIDO-001	[278]0.0074	[280]0.0104	[265]0.0134	[258]0.0267	[320]0.1271
275	OPTIEXACTA-000	[27]0.0008	[31]0.0009	[32]0.0011	[14]0.0027	[25]0.0372
276	PANGIAM-000	[215]0.0039	[219]0.0057	[211]0.0076	[202]0.0156	[176]0.0873
277	PANGIAM-001	[300]0.0094	[284]0.0110	[261]0.0130	[240]0.0229	[170]0.0862
278	PAPSAV1923-000	[125]0.0021	[126]0.0029	[127]0.0038	[109]0.0087	[114]0.0705
279	PARAVISION-003	[415]0.0257	[413]0.0347	[392]0.0441	[378]0.0646	[370]0.1539
280	PARAVISION-004	[274]0.0070	[274]0.0098	[264]0.0133	[256]0.0263	[316]0.1253
281	PARAVISION-005	[162]0.0029	[163]0.0038	[162]0.0053	[210]0.0171	[248]0.1034
282	PARAVISION-007	[151]0.0026	[156]0.0036	[157]0.0052	[230]0.0207	[269]0.1094
283	PARAVISION-009	[101]0.0016	[106]0.0022	[114]0.0034	[120]0.0094	[167]0.0854
284	PARAVISION-012	[52]0.0009	[55]0.0012	[56]0.0015	[66]0.0061	[136]0.0767
285	PARAVISION-014	[42]0.0009	[45]0.0011	[43]0.0013	[55]0.0056	[121]0.0717
286	PARAVISION-016	[40]0.0009	[34]0.0010	[41]0.0013	[46]0.0049	[110]0.0699
287	PARAVISION-019	[29]0.0008	[28]0.0009	[34]0.0012	[41]0.0047	[108]0.0696
288	PARAVISION-021	[25]0.0007	[30]0.0009	[33]0.0011	[39]0.0046	[99]0.0678

Table 41: **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with* $N \geq 3\,000\,000$. *Throughout blue superscripts indicate the rank of the algorithm for that column.*

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MISSES BELOW THRESHOLD, T FNIR(N, T> 0, R >L)		ENROL MOST RECENT DATASET: FRVT 2018 MUGSHOTS				
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M
289	PARAVISION-022	[20] 0.0007	[32] 0.0009	[31] 0.0011	[50] 0.0050	[88] 0.0653
290	PENTACOMP-001	[348] 0.0136	[332] 0.0162	[329] 0.0238	[326] 0.0412	[291] 0.1156
291	PIXELALL-002	[504] 0.0713	[508] 0.1049	[470] 0.1472	[466] 0.2487	[466] 0.3902
292	PIXELALL-003	[366] 0.0155	[366] 0.0215	[354] 0.0285	[343] 0.0470	[284] 0.1135
293	PIXELALL-004	[336] 0.0126	[344] 0.0179	[331] 0.0242	[372] 0.0573	[384] 0.1678
294	PIXELALL-005	[286] 0.0084	[286] 0.0117	[285] 0.0168	[310] 0.0380	[323] 0.1290
295	PROTENET-001	[362] 0.0146	[360] 0.0206	[348] 0.0272	[419] 0.1060	[338] 0.1359
296	PSL-000	[82] 0.0014	[105] 0.0022	[106] 0.0030	[102] 0.0078	[204] 0.0937
297	PSL-001	[90] 0.0015	[94] 0.0021	[100] 0.0028	[100] 0.0075	[95] 0.0663
298	PSL-002	[33] 0.0008	[42] 0.0011	[44] 0.0013	[59] 0.0058	[104] 0.0687
299	PSL-003	[41] 0.0009	[46] 0.0011	[55] 0.0015	[63] 0.0060	[101] 0.0685
300	PSL-004	[23] 0.0007	[33] 0.0009	[36] 0.0012	[38] 0.0046	[61] 0.0560
301	PSL-005	[50] 0.0009	[43] 0.0011	[46] 0.0014	[78] 0.0067	[164] 0.0846
302	PSL-006	[17] 0.0007	[22] 0.0009	[22] 0.0010	[29] 0.0037	[79] 0.0618
303	PTAKURATSATU-000	[419] 0.0272	[418] 0.0363	[395] 0.0455	[359] 0.0520	[47] 0.0520
304	PYRAMID-000	[153] 0.0026	[153] 0.0036	[150] 0.0047	[117] 0.0092	[103] 0.0686
305	PYRAMID-001	[178] 0.0031	[182] 0.0044	[178] 0.0058	[145] 0.0111	[138] 0.0780
306	QAZSMARTVISIONAI-000	[38] 0.0009	[47] 0.0011	[42] 0.0013	[35] 0.0043	[69] 0.0582
307	QAZSMARTVISIONAI-001	[8] 0.0006	[9] 0.0007	[9] 0.0008	[23] 0.0035	[34] 0.0453
308	QAZSMARTVISIONAI-002	[4] 0.0005	[3] 0.0006	[5] 0.0006	[21] 0.0033	[29] 0.0444
309	QAZSMARTVISIONAI-003	[3] 0.0005	[5] 0.0006	[4] 0.0006	[19] 0.0032	[33] 0.0453
310	QNAP-001	[458] 0.0400	[456] 0.0533	[418] 0.0658	[404] 0.0912	[378] 0.1592
311	QNAP-002	[396] 0.0197	[384] 0.0261	[364] 0.0324	[346] 0.0487	[335] 0.1338
312	QNAP-003	[347] 0.0136	[348] 0.0186	[327] 0.0235	[307] 0.0376	[348] 0.1411
313	QNAP-004	[295] 0.0092	[308] 0.0136	[296] 0.0180	[299] 0.0342	[329] 0.1318
314	QNAP-005	[280] 0.0075	[276] 0.0098	[253] 0.0124	[236] 0.0219	[277] 0.1108
315	QNAP-006	[254] 0.0059	[256] 0.0083	[246] 0.0111	[238] 0.0224	[325] 0.1300
316	QUANTASOFT-001	[603] 0.6384	[604] 0.6384	[490] 0.6384		[470] 0.6384
317	RANKONE-002	[520] 0.0970	[514] 0.1170	[463] 0.1349	[450] 0.1697	[447] 0.2553
318	RANKONE-003	[521] 0.0970	[515] 0.1170	[464] 0.1349	[449] 0.1697	[448] 0.2553
319	RANKONE-005	[476] 0.0469	[470] 0.0588	[428] 0.0690	[407] 0.0934	[409] 0.1921
320	RANKONE-007	[371] 0.0164	[369] 0.0218	[340] 0.0262	[309] 0.0378	[282] 0.1129
321	RANKONE-009	[337] 0.0129	[339] 0.0174	[322] 0.0226	[298] 0.0341	[197] 0.0918
322	RANKONE-010	[312] 0.0102	[304] 0.0133	[288] 0.0170	[254] 0.0261	[140] 0.0782
323	RANKONE-011	[255] 0.0059	[257] 0.0084	[247] 0.0112	[257] 0.0266	[283] 0.1132
324	RANKONE-012	[247] 0.0054	[243] 0.0073	[235] 0.0097	[234] 0.0216	[276] 0.1107
325	RANKONE-013	[177] 0.0030	[176] 0.0042	[173] 0.0056	[165] 0.0124	[174] 0.0871
326	RANKONE-014	[122] 0.0021	[128] 0.0029	[130] 0.0039	[136] 0.0102	[87] 0.0652
327	RANKONE-015	[109] 0.0018	[110] 0.0024	[110] 0.0031	[85] 0.0071	[82] 0.0628
328	REALNETWORKS-002	[559] 0.1939	[560] 0.2308	[485] 0.2647	[474] 0.3110	[457] 0.3185
329	REALNETWORKS-003	[539] 0.1296	[538] 0.1591	[476] 0.1855	[459] 0.2243	[454] 0.3074
330	REALNETWORKS-004	[538] 0.1276	[537] 0.1578	[475] 0.1854	[461] 0.2326	[456] 0.3177
331	REALNETWORKS-005	[397] 0.0198	[394] 0.0273	[376] 0.0352	[369] 0.0557	[351] 0.1428
332	REALNETWORKS-006	[298] 0.0094	[312] 0.0142	[295] 0.0178	[284] 0.0305	[232] 0.0988
333	REALNETWORKS-007	[269] 0.0064	[268] 0.0093	[252] 0.0121	[241] 0.0230	[194] 0.0913
334	REALNETWORKS-008	[219] 0.0041	[220] 0.0058	[214] 0.0079	[179] 0.0135	[151] 0.0821
335	REALSENSE-001	[107] 0.0018	[116] 0.0026	[118] 0.0035	[142] 0.0110	[205] 0.0941
336	REALSENSE-002	[54] 0.0010	[54] 0.0012	[53] 0.0014	[76] 0.0066	[94] 0.0662
337	RECOGNITO-000	[74] 0.0013	[79] 0.0017	[85] 0.0023	[51] 0.0053	[45] 0.0516
338	RECOGNITO-001	[47] 0.0009	[50] 0.0011	[45] 0.0014	[17] 0.0029	[22] 0.0368
339	RECOGNITO-002	[26] 0.0008	[29] 0.0009	[30] 0.0011	[15] 0.0027	[24] 0.0372
340	REMARKAI-000	[460] 0.0402	[460] 0.0549	[423] 0.0673	[415] 0.1025	[416] 0.2000
341	RENDIP-000	[283] 0.0081	[285] 0.0117	[273] 0.0153	[264] 0.0274	[302] 0.1179
342	REVEALMEDIA-000	[287] 0.0086	[288] 0.0118	[275] 0.0154	[263] 0.0274	[242] 0.1016
343	RNIT-000	[258] 0.0061	[281] 0.0109	[357] 0.0292	[430] 0.1179	[446] 0.2553
344	ROC-016	[95] 0.0016	[99] 0.0021	[94] 0.0027	[111] 0.0088	[81] 0.0626
345	ROC-017	[115] 0.0020	[121] 0.0027	[121] 0.0036	[113] 0.0088	[90] 0.0655
346	ROC-018	[93] 0.0015	[91] 0.0020	[91] 0.0026	[86] 0.0071	[75] 0.0599
347	ROC-019	[35] 0.0008	[52] 0.0011	[52] 0.0014	[69] 0.0063	[46] 0.0519
348	ROC-020	[51] 0.0009	[56] 0.0012	[63] 0.0017	[45] 0.0048	[36] 0.0465
349	ROC-021	[16] 0.0007	[16] 0.0008	[19] 0.0010	[36] 0.0045	[27] 0.0408
350	S1-000	[400] 0.0200	[397] 0.0276	[383] 0.0378	[376] 0.0627	[389] 0.1704
351	S1-001	[326] 0.0112	[325] 0.0153	[305] 0.0195	[312] 0.0388	[317] 0.1253
352	S1-002	[208] 0.0036	[210] 0.0053	[205] 0.0073	[253] 0.0261	[322] 0.1282
353	S1-003	[252] 0.0057	[258] 0.0084	[248] 0.0112	[261] 0.0273	[324] 0.1295
354	S1-004	[156] 0.0027	[157] 0.0037	[161] 0.0053	[205] 0.0159	[229] 0.0986
355	S1-005	[165] 0.0028	[171] 0.0040	[165] 0.0053	[185] 0.0143	[182] 0.0896
356	S1-006	[92] 0.0015	[92] 0.0020	[96] 0.0027	[147] 0.0114	[156] 0.0824
357	S1-007	[75] 0.0013	[78] 0.0016	[81] 0.0021	[79] 0.0067	[74] 0.0598
358	SANSAP-001	[170] 0.0029	[172] 0.0040	[170] 0.0055	[203] 0.0157	[220] 0.0975
359	SCANOVATE-000	[479] 0.0494	[479] 0.0664	[438] 0.0801	[422] 0.1094	[273] 0.1105
360	SCANOVATE-001	[490] 0.0626	[490] 0.0812	[448] 0.0989	[435] 0.1289	[411] 0.1958

Table 42: **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with* $N \geq 3\,000\,000$. *Throughout blue superscripts indicate the rank of the algorithm for that column.*

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MISSES BELOW THRESHOLD, T		ENROL MOST RECENT				
FNIR(N, T> 0, R >L)		DATASET: FRVT 2018 MUGSHOTS				
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M
361	SCYLLA-001	332 0.0123	333 0.0164	313 0.0211	296 0.0340	318 0.1261
362	SECURITYPITCH-000	471 0.0442	471 0.0603	433 0.0768	417 0.1052	410 0.1927
363	SENSETIME-000	365 0.0155	359 0.0204	343 0.0266	313 0.0389	285 0.1135
364	SENSETIME-001	368 0.0157	367 0.0216	349 0.0273	325 0.0410	310 0.1238
365	SENSETIME-002	355 0.0142	316 0.0145	272 0.0149	242 0.0230	89 0.0654
366	SENSETIME-003	70 0.0012	70 0.0015	66 0.0018	48 0.0050	31 0.0448
367	SENSETIME-004	68 0.0012	65 0.0014	65 0.0017	28 0.0037	21 0.0350
368	SENSETIME-005	72 0.0013	89 0.0019	98 0.0027	108 0.0085	32 0.0451
369	SENSETIME-006	59 0.0011	67 0.0015	77 0.0020	34 0.0043	23 0.0369
370	SENSETIME-007	43 0.0009	40 0.0010	40 0.0012	20 0.0032	19 0.0312
371	SENSETIME-008	31 0.0008	37 0.0010	37 0.0012	16 0.0027	16 0.0285
372	SENSETIME-009	12 0.0006	11 0.0007	13 0.0008	7 0.0020	10 0.0234
373	SENSETIME-010	10 0.0006	14 0.0007	11 0.0008	6 0.0019	12 0.0241
374	SENSETIME-011	13 0.0006	12 0.0007	14 0.0008	8 0.0020	11 0.0234
375	SENSETIME-012	9 0.0006	8 0.0007	8 0.0007	3 0.0014	8 0.0195
376	SERENDIPITY-000	289 0.0088	293 0.0121	277 0.0156	262 0.0274	240 0.1013
377	SEVENTHSENSE-001	172 0.0029	184 0.0044	197 0.0067	193 0.0150	159 0.0828
378	SHAMAN-007	535 0.1209	531 0.1410	471 0.1577	455 0.1864	442 0.2404
379	SIAT-001	342 0.0133	338 0.0173	321 0.0225	293 0.0337	210 0.0950
380	SIAT-002	364 0.0151	364 0.0213	345 0.0270	317 0.0392	298 0.1169
381	SJTU-001	63 0.0011	69 0.0015	69 0.0018	57 0.0058	54 0.0543
382	SPARKTEK-000	123 0.0021	124 0.0028	126 0.0037	118 0.0093	162 0.0831
383	SPARKTEK-001	28 0.0008	26 0.0009	39 0.0012	73 0.0066	63 0.0567
384	SPARSH-001	168 0.0029	167 0.0039	171 0.0055	175 0.0130	180 0.0890
385	SQISOFT-001	513 0.0918	526 0.1319	473 0.1778	463 0.2345	455 0.3174
386	SQISOFT-002	374 0.0173	403 0.0286	390 0.0411	391 0.0736	337 0.1347
387	STCON-001	69 0.0012	68 0.0015	72 0.0019	60 0.0058	96 0.0670
388	STCON-002	53 0.0010	58 0.0012	59 0.0016	40 0.0046	67 0.0579
389	STCON-003	49 0.0009	49 0.0011	51 0.0014	31 0.0040	72 0.0590
390	STCON-004	24 0.0007	23 0.0009	20 0.0010	27 0.0036	30 0.0445
391	STCON-005	19 0.0007	24 0.0009	24 0.0010	26 0.0036	26 0.0400
392	STELLARTECS-001	316 0.0104	314 0.0144	306 0.0196	297 0.0341	295 0.1161
393	SYNESIS-003	480 0.0496	477 0.0649	439 0.0801	421 0.1092	408 0.1913
394	SYNESIS-003	599 0.5339	598 0.5817	489 0.6099	478 0.6473	471 0.6801
395	SYNESIS-005	376 0.0177	377 0.0245	362 0.0316	357 0.0515	377 0.1577
396	TECH5-001	462 0.0417	463 0.0571	444 0.0907	457 0.2103	464 0.3723
397	TECH5-002	390 0.0190	388 0.0266	370 0.0343	363 0.0534	379 0.1604
398	TECH5-003	152 0.0026	154 0.0036	153 0.0050	169 0.0125	203 0.0931
399	TECHAINER-000	305 0.0099	302 0.0131	291 0.0172	278 0.0297	252 0.1049
400	TEVIAN-005	501 0.0687	492 0.0870	451 0.1053	433 0.1279	402 0.1798
401	TEVIAN-006	279 0.0074	269 0.0095	257 0.0127	252 0.0258	326 0.1302
402	TEVIAN-007	200 0.0034	200 0.0048	189 0.0062	194 0.0151	212 0.0954
403	THAKAAIT-000	324 0.0107	324 0.0151	311 0.0207	306 0.0376	332 0.1331
404	TIGER-002	493 0.0643	494 0.0857	449 0.1027	437 0.1317	427 0.2175
405	TIGER-004	350 0.0140	335 0.0168	332 0.0242	323 0.0410	294 0.1159
406	TOSHIBA-000	473 0.0457	473 0.0616	435 0.0776	423 0.1097	418 0.2014
407	TOSHIBA-002	71 0.0012	72 0.0015	70 0.0018	91 0.0073	124 0.0727
408	TOSHIBA-003	67 0.0012	64 0.0014	68 0.0018	137 0.0102	184 0.0898
409	TOSHIBA-004	37 0.0008	44 0.0011	50 0.0014	84 0.0069	134 0.0763
410	TOSS-000	81 0.0014	83 0.0017	80 0.0020	54 0.0056	39 0.0476
411	TOSS-001	102 0.0017	100 0.0021	92 0.0026	83 0.0068	71 0.0587
412	TRUEFACE-000	339 0.0130	343 0.0178	325 0.0235	308 0.0377	342 0.1382
413	TSIGHT-000	2 0.0005	2 0.0005	3 0.0006	491 1.0000	595 1.0000
414	UNISSEY-003	323 0.0107	315 0.0144	297 0.0183	285 0.0306	267 0.1090
415	UNIVERSEAI-000	138 0.0024	145 0.0033	141 0.0044	126 0.0099	155 0.0823
416	USEB-001	158 0.0027	151 0.0036	149 0.0047	159 0.0118	154 0.0822
417	USEB-002	171 0.0029	166 0.0039	159 0.0052	199 0.0154	187 0.0901
418	USEB-003	154 0.0026	148 0.0034	142 0.0044	143 0.0111	168 0.0854
419	USEB-004	141 0.0025	142 0.0032	136 0.0042	122 0.0095	137 0.0774
420	VD-001	550 0.1639	552 0.2010	484 0.2344	471 0.2719	458 0.3253
421	VDFSOFTWARE-002	418 0.0267	424 0.0376	398 0.0482	383 0.0685	366 0.1523
422	VERIDAS-001	420 0.0275	421 0.0369	399 0.0488	395 0.0749	368 0.1538
423	VERIDAS-002	421 0.0275	422 0.0369	400 0.0488	394 0.0749	369 0.1538
424	VERIDAS-003	327 0.0114	331 0.0162	315 0.0215	333 0.0443	371 0.1540
425	VERIDAS-004	214 0.0038	212 0.0054	206 0.0073	93 0.0073	9 0.0229
426	VERIGRAM-000	197 0.0034	191 0.0045	144 0.0045	52 0.0053	2 0.0111
427	VERIGRAM-001	198 0.0034	193 0.0045	169 0.0054	172 0.0129	146 0.0811
428	VERIGRAM-002	148 0.0025	120 0.0027	105 0.0029	106 0.0082	117 0.0710
429	VIANTE-000	112 0.0019	113 0.0025	115 0.0034	154 0.0116	133 0.0763
430	VIANTE-001	61 0.0011	59 0.0012	54 0.0015	77 0.0067	78 0.0614
431	VIANTE-002	22 0.0007	27 0.0009	35 0.0012	92 0.0073	62 0.0567
432	VIANTE-003	14 0.0007	17 0.0008	18 0.0009	95 0.0074	68 0.0581

Table 43: **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with* $N \geq 3\,000\,000$. *Throughout blue superscripts indicate the rank of the algorithm for that column.*

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MISSES BELOW THRESHOLD, T FNIR(N, T> 0, R >L)		ENROL MOST RECENT DATASET: FRVT 2018 MUGSHOTS				
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M
433	VIETTELAI-000	[410] 0.0227	[469] 0.0585	[466] 0.1394	[468] 0.2506	[465] 0.3742
434	VIETTELAI-002	[275] 0.0070	[279] 0.0102	[276] 0.0156	[389] 0.0724	[412] 0.1964
435	VIETTELAI-003	[185] 0.0032	[179] 0.0043	[183] 0.0059	[220] 0.0188	[244] 0.1019
436	VIETTELAI-004	[105] 0.0018	[111] 0.0024	[112] 0.0033	[116] 0.0092	[119] 0.0712
437	VIGILANTSOLUTIONS-008	[356] 0.0143	[353] 0.0202	[344] 0.0266	[345] 0.0485	[296] 0.1161
438	VINBIGDATA-001	[320] 0.0106	[327] 0.0157	[314] 0.0213	[295] 0.0339	[226] 0.0983
439	VINBIGDATA-002	[83] 0.0015	[88] 0.0019	[88] 0.0024	[130] 0.0101	[157] 0.0824
440	VINBIGDATA-003	[87] 0.0015	[98] 0.0021	[104] 0.0029	[127] 0.0099	[169] 0.0860
441	VINBIGDATA-004	[62] 0.0011	[74] 0.0015	[74] 0.0019	[112] 0.0088	[130] 0.0744
442	VISIONBOX-000	[329] 0.0119	[340] 0.0174	[328] 0.0236		[476] 0.9538
443	VISIONBOX-001	[505] 0.0757	[527] 0.1326	[480] 0.2095	[473] 0.3064	[467] 0.4116
444	VISIONBOX-002	[281] 0.0078	[282] 0.0109	[270] 0.0145	[246] 0.0242	[259] 0.1079
445	VISIONLABS-004	[465] 0.0424	[465] 0.0575	[430] 0.0697	[408] 0.0940	[397] 0.1785
446	VISIONLABS-005	[447] 0.0366	[446] 0.0498	[411] 0.0620	[398] 0.0831	[388] 0.1701
447	VISIONLABS-006	[387] 0.0186	[386] 0.0262	[368] 0.0331	[361] 0.0525	[343] 0.1392
448	VISIONLABS-007	[386] 0.0186	[385] 0.0262	[367] 0.0330	[360] 0.0523	[344] 0.1393
449	VISIONLABS-008	[294] 0.0092	[298] 0.0127	[284] 0.0162	[272] 0.0288	[313] 0.1244
450	VISIONLABS-009	[181] 0.0031	[177] 0.0042	[175] 0.0056	[180] 0.0136	[178] 0.0878
451	VISIONLABS-010	[199] 0.0034	[198] 0.0047	[196] 0.0066	[189] 0.0145	[196] 0.0917
452	VISIONLABS-011	[124] 0.0021	[132] 0.0030	[134] 0.0040	[153] 0.0116	[152] 0.0822
453	VISTEAM-000	[496] 0.0656	[492] 0.0852	[452] 0.1072	[441] 0.1387	[414] 0.1987
454	VIXVIZION-009	[399] 0.0199	[391] 0.0270	[373] 0.0345	[367] 0.0541	[341] 0.1374
455	VNPAY-001	[217] 0.0039	[214] 0.0055	[209] 0.0075	[191] 0.0149	[160] 0.0829
456	VNPT-001	[307] 0.0100	[310] 0.0140	[299] 0.0186	[277] 0.0293	[245] 0.1025
457	VNPT-002	[234] 0.0047	[226] 0.0062	[219] 0.0080	[209] 0.0169	[237] 0.1002
458	VNPT-003	[231] 0.0046	[223] 0.0060	[213] 0.0078	[208] 0.0168	[231] 0.0987
459	VNPT-004	[213] 0.0038	[205] 0.0051	[193] 0.0064	[190] 0.0148	[179] 0.0885
460	VNPT-005	[188] 0.0033	[178] 0.0042	[174] 0.0056	[150] 0.0115	[171] 0.0866
461	VOCORD-005	[532] 0.1174	[536] 0.1559	[481] 0.2152	[472] 0.3062	[469] 0.4354
462	VOGIC-000	[304] 0.0098	[309] 0.0136	[309] 0.0200	[348] 0.0498	[363] 0.1492
463	VTS-001	[306] 0.0099	[300] 0.0130	[290] 0.0172	[287] 0.0319	[312] 0.1240
464	VTS-002	[379] 0.0181	[380] 0.0255	[369] 0.0341	[368] 0.0546	[357] 0.1444
465	VTS-003	[240] 0.0050	[239] 0.0069	[227] 0.0092	[217] 0.0184	[241] 0.1014
466	VTS-004	[179] 0.0031	[189] 0.0045	[187] 0.0061	[182] 0.0140	[183] 0.0896
467	VTS-005	[192] 0.0033	[206] 0.0051	[204] 0.0071	[181] 0.0138	[100] 0.0680
468	WINK-001	[413] 0.0244	[417] 0.0359	[397] 0.0481	[388] 0.0724	[386] 0.1692
469	WINK-002	[288] 0.0086	[290] 0.0121	[281] 0.0159	[271] 0.0285	[217] 0.0968
470	WINK-003	[261] 0.0061	[261] 0.0088	[255] 0.0126	[354] 0.0511	[372] 0.1544
471	XAILIENT-000	[416] 0.0260	[415] 0.0354	[394] 0.0454	[381] 0.0678	[382] 0.1650
472	XAILIENT-001	[249] 0.0057	[253] 0.0082	[243] 0.0107	[224] 0.0197	[202] 0.0923
473	XFORWARDAI-000	[313] 0.0103	[319] 0.0148	[301] 0.0192	[291] 0.0321	[253] 0.1054
474	XFORWARDAI-001	[189] 0.0033	[190] 0.0045	[176] 0.0057	[152] 0.0116	[143] 0.0797
475	XFORWARDAI-002	[131] 0.0023	[118] 0.0027	[111] 0.0031	[99] 0.0074	[113] 0.0702
476	YITU-002	[335] 0.0125	[337] 0.0173	[318] 0.0223	[294] 0.0337	[257] 0.1071
477	YITU-003	[346] 0.0134	[346] 0.0181	[323] 0.0232	[300] 0.0345	[260] 0.1079
478	YITU-004	[268] 0.0064	[265] 0.0092	[254] 0.0124	[235] 0.0219	[219] 0.0973
479	YITU-005	[276] 0.0071	[275] 0.0098	[262] 0.0131	[245] 0.0242	[235] 0.0993
480	YOUVERSE-001	[233] 0.0047	[236] 0.0067	[225] 0.0090	[214] 0.0180	[247] 0.1033
481	YUAN-000	[273] 0.0067	[277] 0.0099	[263] 0.0132	[239] 0.0227	[579] 1.0000

Table 44: **Identification-mode: Effect of N on FNIR at high threshold.** *Values are threshold-based miss rates i.e.* FNIR *at* FPIR = *0.001 for five enrollment population sizes, N. The right six columns apply for enrollment of one image. Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with* $N \geq 3\,000\,000$. *Throughout blue superscripts indicate the rank of the algorithm for that column.*

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MISSES AT GIVEN RANK		ENROL MOST RECENT											
FNIR(N, T= 0, R)		RANK 1						RANK 50					
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b
1	3DIVI-005	539 0.0134	537 0.0173	473 0.0207	462 0.0250	456 0.0298	396 0.0003 $N^{0.275}$ 411	509 0.0038	508 0.0046	461 0.0054	452 0.0065	446 0.0079	78 0.0001 $N^{0.251}$ 435
2	ACER-000	499 0.0077	506 0.0102	461 0.0124	454 0.0153	446 0.0191	267 0.0001 $N^{0.308}$ 452	428 0.0016	458 0.0022	435 0.0027	430 0.0034	426 0.0042	23 0.0000 $N^{0.324}$ 457
3	ADIANCE-002	248 0.0011	250 0.0012	242 0.0013	247 0.0015	251 0.0018	206 0.0001 $N^{0.181}$ 284	241 0.0008	230 0.0008	231 0.0009	227 0.0009	232 0.0010	223 0.0004 $N^{0.056}$ 296
4	ADVANCE-000	305 0.0014	314 0.0016	310 0.0018	304 0.0022	303 0.0025	192 0.0001 $N^{0.206}$ 334	328 0.0010	322 0.0011	314 0.0011	312 0.0012	308 0.0013	267 0.0004 $N^{0.064}$ 321
5	AFISBIOMETRICS-000	208 0.0010	256 0.0012	279 0.0016	292 0.0020	302 0.0025	18 0.0000 $N^{0.336}$ 462	97 0.0005	104 0.0006	136 0.0006	137 0.0007	140 0.0007	83 0.0001 $N^{0.097}$ 367
6	AFRENGINE-000	400 0.0023	404 0.0029	394 0.0034	384 0.0041	384 0.0052	99 0.0001 $N^{0.275}$ 414	395 0.0013	391 0.0014	370 0.0015	361 0.0016	354 0.0017	263 0.0004 $N^{0.084}$ 348
7	AFRENGINE-001	288 0.0013	273 0.0013	263 0.0014	254 0.0016	245 0.0018	386 0.0003 $N^{0.106}$ 102	384 0.0012	361 0.0013	345 0.0013	329 0.0013	311 0.0013	431 0.0011 $N^{0.007}$ 60
8	AFRENGINE-002	289 0.0013	274 0.0013	256 0.0014	242 0.0015	234 0.0017	420 0.0004 $N^{0.087}$ 60	385 0.0012	363 0.0013	346 0.0013	330 0.0013	310 0.0013	436 0.0012 $N^{0.005}$ 43
9	AFRENGINE-003	397 0.0023	380 0.0023	355 0.0024	335 0.0025	313 0.0027	453 0.0010 $N^{0.063}$ 24	471 0.0022	454 0.0022	421 0.0022	408 0.0022	392 0.0022	459 0.0021 $N^{0.002}$ 14
10	AIGROUP-000	222 0.0010	226 0.0011	222 0.0012	218 0.0014	222 0.0016	238 0.0001 $N^{0.161}$ 242	200 0.0007	200 0.0007	188 0.0007	177 0.0008	174 0.0008	305 0.0005 $N^{0.029}$ 215
11	AIZE-002	410 0.0025	412 0.0031	398 0.0036	391 0.0043	386 0.0052	191 0.0001 $N^{0.251}$ 391	396 0.0013	400 0.0014	380 0.0015	370 0.0017	363 0.0018	157 0.0003 $N^{0.106}$ 371
12	AIZE-006	423 0.0029	424 0.0036	414 0.0044	404 0.0054	398 0.0065	138 0.0001 $N^{0.278}$ 421	387 0.0013	381 0.0013	378 0.0015	374 0.0017	373 0.0019	94 0.0002 $N^{0.146}$ 396
13	AKUVOX-001	285 0.0013	280 0.0014	273 0.0016	271 0.0018	265 0.0020	304 0.0002 $N^{0.154}$ 222	320 0.0010	320 0.0011	307 0.0011	285 0.0011	278 0.0011	361 0.0007 $N^{0.031}$ 223
14	ALCHERA-003	498 0.0077	505 0.0102	459 0.0120	453 0.0145	445 0.0177	320 0.0002 $N^{0.282}$ 429	485 0.0025	484 0.0030	445 0.0033	437 0.0040	428 0.0046	89 0.0002 $N^{0.206}$ 427
15	ALCHERA-006	178 0.0009	168 0.0009	168 0.0010	159 0.0011	171 0.0013	281 0.0001 $N^{0.132}$ 178	197 0.0007	194 0.0007	187 0.0007	181 0.0008	170 0.0008	290 0.0005 $N^{0.032}$ 231
16	ALCHERA-008	203 0.0009	191 0.0010	174 0.0010	174 0.0011	163 0.0012	367 0.0003 $N^{0.095}$ 81	267 0.0009	252 0.0009	238 0.0009	223 0.0009	213 0.0009	402 0.0008 $N^{0.004}$ 34
17	ALGORITHMSINC-000	272 0.0012	306 0.0016	306 0.0018	313 0.0022	326 0.0029	30 0.0000 $N^{0.293}$ 439	344 0.0011	376 0.0013	382 0.0015	386 0.0018	395 0.0023	30 0.0000 $N^{0.256}$ 437
18	ALLGOVISION-000	518 0.0097	513 0.0111	460 0.0124	451 0.0142	444 0.0162	451 0.0009 $N^{0.176}$ 275	533 0.0059	528 0.0064	468 0.0067	455 0.0072	445 0.0078	454 0.0017 $N^{0.091}$ 359
19	ALLGOVISION-001	483 0.0066	491 0.0086	454 0.0103	446 0.0124	441 0.0154	286 0.0001 $N^{0.287}$ 436	459 0.0020	465 0.0023	436 0.0028	423 0.0032	423 0.0040	56 0.0001 $N^{0.238}$ 431
20	ALLGOVISION-005	335 0.0015	330 0.0017	321 0.0020	315 0.0023	308 0.0026	248 0.0001 $N^{0.191}$ 313	349 0.0011	339 0.0011	328 0.0012	331 0.0013	321 0.0014	209 0.0004 $N^{0.079}$ 342
21	ANKE-000	521 0.0099	524 0.0129	469 0.0152	459 0.0185	451 0.0222	365 0.0003 $N^{0.275}$ 412	494 0.0029	498 0.0037	456 0.0044	443 0.0053	436 0.0063	61 0.0001 $N^{0.261}$ 440
22	ANKE-002	384 0.0021	389 0.0024	378 0.0029	371 0.0034	366 0.0040	208 0.0001 $N^{0.228}$ 360	380 0.0012	370 0.0013	356 0.0014	349 0.0014	346 0.0016	232 0.0004 $N^{0.085}$ 349
23	ARATEK-001	137 0.0007	134 0.0008	126 0.0008	132 0.0010	138 0.0011	187 0.0001 $N^{0.159}$ 233	143 0.0006	133 0.0006	129 0.0006	123 0.0006	118 0.0006	229 0.0004 $N^{0.031}$ 227
24	ARMATURA-000	332 0.0015	299 0.0015	277 0.0016	263 0.0017	249 0.0018	437 0.0007 $N^{0.061}$ 21	410 0.0015	401 0.0015	373 0.0015	352 0.0015	333 0.0015	448 0.0014 $N^{0.004}$ 29
25	AWARE-003	559 0.0235	561 0.0303	485 0.0358	474 0.0427	470 0.0503	444 0.0007 $N^{0.260}$ 399	526 0.0052	538 0.0072	479 0.0088	468 0.0109	465 0.0140	45 0.0001 $N^{0.335}$ 462
26	AWARE-005	562 0.0241	562 0.0308	486 0.0363	475 0.0431	460 0.0309	475 0.0055 $N^{0.120}$ 139	531 0.0058	543 0.0079	481 0.0098	471 0.0125	448 0.0086	349 0.0006 $N^{0.176}$ 416
27	AWARE-007	354 0.0017	366 0.0021	364 0.0025	360 0.0031	361 0.0038	52 0.0000 $N^{0.282}$ 428	214 0.0008	238 0.0009	273 0.0010	294 0.0011	316 0.0013	46 0.0001 $N^{0.187}$ 420
28	AYONIX-002	615 0.2933	615 0.3412	496 0.3734	484 0.4099	477 0.4463	480 0.0439 $N^{0.143}$ 205	608 0.0947	610 0.1271	496 0.1522	483 0.1825	475 0.2147	464 0.0023 $N^{0.279}$ 447
29	BIOIDTECHSWISS-001	150 0.0008	139 0.0008	129 0.0009	118 0.0009	117 0.0011	297 0.0002 $N^{0.118}$ 135	163 0.0006	152 0.0006	147 0.0007	141 0.0007	143 0.0007	278 0.0004 $N^{0.027}$ 199
30	BIOMETRIK-001	432 0.0033	439 0.0044	425 0.0052	410 0.0062	406 0.0075	181 0.0001 $N^{0.278}$ 420	370 0.0012	390 0.0014	386 0.0016	382 0.0018	382 0.0021	64 0.0001 $N^{0.191}$ 423
31	CAMVI-004	532 0.0122	576 0.0466	490 0.0717	483 0.2360	475 0.2365	9 0.0000 $N^{1.059}$ 476	570 0.0116	593 0.0462	492 0.0713	485 0.2358	477 0.2362	10 0.0000 $N^{1.075}$ 477
32	CANON-001	118 0.0007	113 0.0007	119 0.0008	125 0.0010	122 0.0011	175 0.0001 $N^{0.159}$ 234	103 0.0005	108 0.0006	107 0.0006	108 0.0006	106 0.0006	146 0.0003 $N^{0.042}$ 267
33	CANON-002	130 0.0007	144 0.0008	160 0.0009	166 0.0011	168 0.0012	84 0.0001 $N^{0.196}$ 319	106 0.0005	103 0.0006	98 0.0006	93 0.0006	100 0.0006	226 0.0004 $N^{0.025}$ 191
34	CANON-003	56 0.0005	59 0.0006	71 0.0007	73 0.0008	67 0.0009	89 0.0001 $N^{0.173}$ 260	42 0.0005	45 0.0005	31 0.0005	32 0.0005	36 0.0005	201 0.0004 $N^{0.017}$ 144
35	CIB-000	246 0.0011	247 0.0012	244 0.0013	251 0.0016	476 0.3301	6 0.0000 $N^{1.598}$ 478	231 0.0008	217 0.0008	219 0.0008	224 0.0009	479 0.3295	6 0.0000 $N^{1.644}$ 478
36	CLEARVIEWAI-000	104 0.0007	114 0.0007	116 0.0008	133 0.0010	133 0.0011	96 0.0001 $N^{0.183}$ 295	115 0.0006	102 0.0006	103 0.0006	109 0.0006	104 0.0006	194 0.0004 $N^{0.031}$ 222
37	CLEARVIEWAI-001	128 0.0007	118 0.0007	106 0.0008	103 0.0009	99 0.0010	279 0.0001 $N^{0.121}$ 148	176 0.0007	156 0.0007	148 0.0007	143 0.0007	137 0.0007	334 0.0006 $N^{0.011}$ 90
38	CLEARVIEWAI-002	126 0.0007	124 0.0008	117 0.0008	117 0.0009	111 0.0010	255 0.0001 $N^{0.132}$ 176	175 0.0007	159 0.0007	153 0.0007	142 0.0007	132 0.0007	338 0.0006 $N^{0.008}$ 71
39	CLEARVIEWAI-003	135 0.0007	119 0.0008	110 0.0008	100 0.0009	94 0.0010	307 0.0002 $N^{0.107}$ 107	167 0.0006	162 0.0007	146 0.0007	132 0.0007	124 0.0007	348 0.0006 $N^{0.003}$ 21
40	CLOUDWALK-HR-000	256 0.0011	236 0.0011	216 0.0012	194 0.0012	188 0.0014	423 0.0004 $N^{0.069}$ 30	324 0.0010	313 0.0010	298 0.0011	283 0.0011	265 0.0011	400 0.0008 $N^{0.016}$ 139
41	CLOUDWALK-MT-000	313 0.0014	285 0.0014	265 0.0015	241 0.0015	223 0.0016	446 0.0008 $N^{0.045}$ 12	402 0.0014	389 0.0014	359 0.0014	342 0.0014	325 0.0014	444 0.0014 $N^{0.002}$ 15
42	CLOUDWALK-MT-001	309 0.0014	283 0.0014	259 0.0014	235 0.0015	216 0.0016	449 0.0009 $N^{0.036}$ 8	401 0.0014	388 0.0014	360 0.0014	344 0.0014	326 0.0014	448 0.0013 $N^{0.003}$ 25
43	CLOUDWALK-MT-002	310 0.0014	284 0.0014	258 0.0014	236 0.0015	212 0.0016	450 0.0009 $N^{0.034}$ 6	403 0.0014	387 0.0014	362 0.0014	345 0.0014	327 0.0014	445 0.0014 $N^{0.002}$ 16
44	COGENT-000	519 0.0098	504 0.0101	455 0.0106	441 0.0112	434 0.0122	471 0.0036 $N^{0.073}$ 33	446 0.0018	450 0.0021	428 0.0024	427 0.0033	451 0.0092	16 0.0000 $N^{0.507}$ 469
45	COGENT-001	520 0.0098	503 0.0101	456 0.0106	442 0.0112	433 0.0122	472 0.0036 $N^{0.073}$ 34	445 0.0018	449 0.0021	427 0.0024	426 0.0033	452 0.0092	15 0.0000 $N^{0.507}$ 468
46	COGENT-002	411 0.0025	416 0.0033	402 0.0038	394 0.0046	390 0.0055	156 0.0001 $N^{0.264}$ 405	357 0.0011	362 0.0013	355 0.0014	359 0.0016	360 0.0018	74 0.0001 $N^{0.163}$ 410
47	COGENT-003	417 0.0027	418 0.0035	405 0.0040	397 0.0048	393 0.0057	216 0.0001 $N^{0.249}$ 388	369 0.0012	386 0.0014	383 0.0015	371 0.0017	371 0.0019	80 0.0001 $N^{0.165}$ 411
48	COGENT-004	315 0.0014	315 0.0016	307 0.0018	303 0.0021	295 0.0025	234 0.0001 $N^{0.190}$ 312	302 0.0010	311 0.0010	308 0.0011	298 0.0011	293 0.0012	237 0.0004 $N^{0.067}$ 327
49	COGENT-005	270 0.0012	269 0.0013	266 0.0015	258 0.0016	246 0.0018	333 0.0002 $N^{0.134}$ 183	306 0.0010	293 0.0010	278 0.0010	263 0.0010	257 0.0010	371 0.0007 $N^{0.023}$ 184
50	COGENT-006	168 0.0008	167 0.0009	161 0.0009	154 0.0011	146 0.0012	306 0.0002 $N^{0.120}$ 140	187 0.0007	192 0.0007	184 0.0007	168 0.0007	163 0.0008	303 0.0005 $N^{0.028}$ 205
51	COGENT-007	112 0.0007	116 0.0007	134 0.0009	131 0.0010	131 0.0011	129 0.0001 $N^{0.172}$ 255	138 0.0006	132 0.0006	127 0.0006	121 0.0006	120 0.0006	198 0.0004 $N^{0.035}$ 245
52	COGNITEC-000	551 0.0192	550 0.0249	481 0.0294	471 0.0349	464 0.0414	439 0.0006 $N^{0.262}$ 404	521 0.0046	526 0.0062	475 0.0074	466 0.0093	460 0.0118	48 0.0001 $N^{0.319}$ 455
53	COGNITEC-001	512 0.0087	515 0.0114	466 0.0136	457 0.0163	448 0.0196	340 0.0002 $N^{0.278}$ 419	488 0.0027	480 0.0031	451 0.0036	441 0.0043	431 0.0051	79 0.0001 $N^{0.224}$ 430
54	COGNITEC-002	459 0.0045	454 0.0053	432 0.0063	420 0.0075	417 0.0091	309 0.0002 $N^{0.244}$ 383	463 0.0021	463 0.0023	431 0.0025	420 0.0027	410 0.0030	186 0.0004 $N^{0.131}$ 386
55	COGNITEC-003	463 0.0049	460 0.0059	437 0.0069	427 0.0082	420 0.0097	347 0.0002 $N^{0.232}$ 365	484 0.0025	475 0.0028	437 0.0029	422 0.0032	416 0.0034	347 0.0006 $N^{0.105}$ 370
56	COGNITEC-004	399 0.0023	403 0.0028	393 0.0034	386 0.0042	385 0.0052	92 0.0001 $N^{0.280}$ 424	301 0.0010	315 0.0010	322 0.0011	333 0.0013	337 0.0015	71 0.0001 $N^{0.160}$ 407
57	COGNITEC-005	252 0.0011	265 0.0013	268 0.0015	265 0.0017	267 0.0020	124 0.0001 $N^{0.210}$ 340	208 0.0007	209 0.0008	205 0.0008	212 0.0008	200 0.0009	202 0.0004 $N^{0.053}$ 293
58	COGNITEC-006	241 0.0011	249 0.0012	245 0.0013	250 0.0016	253 0.0018	167 0.0001 $N^{0.194}$ 317	210 0.0007	207 0.0008	203 0.0008	206 0.0008	202 0.0009	205 0.0004 $N^{0.052}$ 292
59	COGNITEC-007	189 0.0009	208 0.0010	200 0.0011	206 0.0013	215 0.0016	145 0.0001 $N^{0.186}$ 303	160 0.0006	165 0.0007	157 0.0007	155 0.0007	156 0.0007	144 0.0003 $N^{0.056}$ 298
60	COGNITEC-008	177 0.0008	189 0.0010	194 0.0011	205 0.0013	219 0.0016	67 0.0000 $N^{0.218}$ 347	150 0.0006	149 0.0006	143 0.0006	133 0.0007	128 0.0007	258 0.0004 $N^{0.028}$ 207
61	COGNITEC-009	236 0.0010	241 0.0012	236 0.0013	240 0.0015	250 0.0018	179 0.0001 $N^{0.189}$ 308	236 0.0008	225 0.0008	218 0.0008	211 0.0008	203 0.0009	343 0.0006 $N^{0.022}$ 176
62	COGNITEC-010	151 0.0008	157 0.0008	158 0.0009	155 0.0011	175 0.0013	148 0.0001 $N^{0.173}$ 263	157 0.0006	148 0.0006	137 0.0006	134 0.0007	130 0.0007	289 0.0005 $N^{0.022}$ 172
63	COGNITI-000	298 0.0013	303 0.0015	295 0.0017	289 0.0020	287 0.0023	239 0.0001 $N^{0.185}$ 299	335 0.0011	326 0.0011	311 0.0011	299 0.0011	289 0.0012	355 0.0007 $N^{0.035}$ 244
64	COMPAL-000	290 0.0013	281 0.0014	613 1.0000	507 1.0000	606 1.0000	3 0.0000 $N^{2.726}$ 482	355 0.0011	344 0.0011	513 1.0000	516 1.0000	488 1.0000	3 0.0000 $N^{2.800}$ 481
65	CORSIGHT-000	122 0.0007	120 0.0008	129 0.0008	139 0.0010	140 0.0011	126 0.0001 $N^{0.173}$ 267	149 0.0006	145 0.0006	134 0.0006	122 0.0006	110 0.0006	335 0.0006 $N^{0.006}$ 59
66	CORSIGHT-001	129 0.0007	123 0.0008	133 0.0009	128 0.0010	129 0.0011	196 0.0001 $N^{0.155}$ 225	142 0.0006	142 0.0006	131 0.0006	129 0.0006	129 0.0007	188 0.0004 $N^{0.038}$ 257
67	CORSIGHT-002	194 0.0009	194 0.0010	185 0.0011	180 0.0011	179 0.0013	322 0.0002 $N^{0.120}$ 141	254 0.0009	242 0.0009	225 0.0009	216 0.0009	201 0.0009	394 0.0008 $N^{0.005}$ 47
68	CUBOX-001	185 0.0009	174 0.0009	159 0.0009	161 0.0011	149 0.0012	334 0.0002 $N^{0.107}$ 106	242 0.0008	224 0.0008	215 0.0008	200 0.0008	189 0.0008	389 0.0008 $N^{0.004}$ 36
69	CUBOX-002	259 0.0011	238 0.0011	226 0.0012	209 0.0013	191 0.0014	404 0.0004 $N^{0.082}$ 48	333 0.0010	317 0.0011	302 0.0011	282 0.0011	263 0.0011	421 0.0010 $N^{0.007}$ 62
70	CUBOX-003	274 0.0012	253 0.0012	237 0.0013	221 0.0014	213 0.0016	412 0.0004 $N^{0.085}$ 57	354 0.0011	340 0.0011	325 0.0011	306 0.0012	287 0.0012	415 0.0009 $N^{0.014}$ 123
71	CYBERLINK-000	428 0.0031	421 0.0036	411 0.0042	400 0.0050	395 0.0059	292 0.0001 $N^{0.227}$ 358	439 0.0017	435 0.0019	414 0.0020	406 0.0022	398 0.0024	239 0.0004 $N^{0.108}$ 373
72	CYBERLINK-001	414 0.0026	414 0.0032	403 0.0038	396 0.0047	392 0.0057	147 0.0001 $N^{0.266}$ 406	383 0.0012	380 0.0013	366 0.0014	365 0.0016	365 0.0018	106 0.0002 $N^{0.135}$ 390

Table 45: **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES AT GIVEN RANK FNIR(N, T= 0, R)		ENROL MOST RECENT											
		RANK 1						RANK 50					
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b
73	CYBERLINK-002	385 0.0021	377 0.0023	360 0.0025	344 0.0027	336 0.0031	393 0.0003 $N^{0.138\ 192}$	430 0.0017	422 0.0017	401 0.0018	385 0.0018	368 0.0019	417 0.0009 $N^{0.043\ 270}$
74	CYBERLINK-003	253 0.0011	252 0.0012	246 0.0013	233 0.0015	229 0.0017	311 0.0002 $N^{0.139\ 197}$	218 0.0008	215 0.0008	212 0.0008	219 0.0009	222 0.0009	148 0.0003 $N^{0.067\ 328}$
75	CYBERLINK-004	282 0.0012	268 0.0013	261 0.0014	257 0.0016	239 0.0018	360 0.0002 $N^{0.122\ 151}$	321 0.0010	319 0.0011	299 0.0011	288 0.0011	279 0.0011	358 0.0007 $N^{0.032\ 229}$
76	CYBERLINK-005	297 0.0013	290 0.0015	270 0.0015	267 0.0017	260 0.0019	368 0.0003 $N^{0.121\ 149}$	331 0.0010	327 0.0011	316 0.0011	303 0.0011	290 0.0012	339 0.0006 $N^{0.042\ 265}$
77	CYBERLINK-006	145 0.0008	137 0.0008	118 0.0008	112 0.0009	103 0.0010	323 0.0002 $N^{0.104\ 97}$	178 0.0007	168 0.0007	156 0.0007	149 0.0007	138 0.0007	333 0.0006 $N^{0.012\ 102}$
78	DAHUA-001	470 0.0051	466 0.0064	443 0.0077	432 0.0090	425 0.0109	301 0.0002 $N^{0.261\ 400}$	482 0.0025	476 0.0028	439 0.0030	425 0.0033	418 0.0036	296 0.0005 $N^{0.124\ 383}$
79	DAHUA-002	294 0.0013	293 0.0015	289 0.0017	285 0.0019	286 0.0023	225 0.0001 $N^{0.188\ 307}$	293 0.0009	292 0.0010	284 0.0010	276 0.0011	274 0.0011	259 0.0004 $N^{0.059\ 305}$
80	DAHUA-003	113 0.0007	153 0.0008	183 0.0011	197 0.0012	205 0.0015	25 0.0000 $N^{0.270\ 410}$	88 0.0005	79 0.0005	80 0.0005	91 0.0006	94 0.0006	136 0.0003 $N^{0.044\ 273}$
81	DAHUA-004	110 0.0007	107 0.0007	114 0.0008	107 0.0009	124 0.0011	162 0.0001 $N^{0.161\ 239}$	96 0.0005	98 0.0005	92 0.0005	97 0.0006	95 0.0006	155 0.0003 $N^{0.037\ 254}$
82	DAON-000	442 0.0035	425 0.0037	404 0.0039	385 0.0042	379 0.0045	456 0.0011 $N^{0.084\ 55}$	499 0.0032	491 0.0033	443 0.0033	429 0.0033	415 0.0034	466 0.0027 $N^{0.014\ 121}$
83	DAON-002	156 0.0008	177 0.0009	193 0.0011	204 0.0013	221 0.0016	38 0.0000 $N^{0.239\ 377}$	139 0.0006	144 0.0006	139 0.0006	139 0.0007	136 0.0007	153 0.0003 $N^{0.047\ 281}$
84	DAON-003	102 0.0006	106 0.0007	101 0.0008	106 0.0009	101 0.0010	166 0.0001 $N^{0.158\ 231}$	95 0.0005	84 0.0005	87 0.0005	84 0.0005	84 0.0006	196 0.0004 $N^{0.026\ 197}$
85	DATECH-000	158 0.0008	193 0.0010	192 0.0011	203 0.0013	218 0.0016	60 0.0000 $N^{0.223\ 354}$	155 0.0006	186 0.0007	168 0.0007	161 0.0007	158 0.0007	150 0.0003 $N^{0.055\ 294}$
86	DATECH-001	120 0.0007	128 0.0008	124 0.0008	134 0.0010	130 0.0011	165 0.0001 $N^{0.163\ 247}$	127 0.0006	119 0.0006	120 0.0006	117 0.0006	111 0.0006	179 0.0003 $N^{0.036\ 251}$
87	DECATUR-000	325 0.0015	332 0.0017	329 0.0021	329 0.0024	327 0.0029	117 0.0001 $N^{0.236\ 371}$	273 0.0009	281 0.0009	276 0.0010	289 0.0011	295 0.0012	123 0.0002 $N^{0.097\ 365}$
88	DEEPGLINT-001	240 0.0011	229 0.0011	213 0.0012	202 0.0013	203 0.0015	346 0.0002 $N^{0.115\ 129}$	290 0.0009	274 0.0009	263 0.0010	245 0.0010	244 0.0010	359 0.0007 $N^{0.023\ 185}$
89	DEEPSEA-001	424 0.0029	430 0.0040	417 0.0048	411 0.0062	408 0.0077	41 0.0000 $N^{0.335\ 461}$	268 0.0009	303 0.0010	333 0.0012	336 0.0014	349 0.0016	44 0.0001 $N^{0.204\ 426}$
90	DERMALOG-005	530 0.0112	529 0.0147	471 0.0198	467 0.0287	468 0.0445	22 0.0000 $N^{0.475\ 469}$	560 0.0092	563 0.0120	487 0.0169	476 0.0252	471 0.0403	18 0.0000 $N^{0.510\ 470}$
91	DERMALOG-006	493 0.0073	486 0.0079	449 0.0084	433 0.0091	423 0.0101	465 0.0017 $N^{0.110\ 113}$	534 0.0061	525 0.0062	465 0.0063	450 0.0064	438 0.0066	472 0.0042 $N^{0.027\ 203}$
92	DERMALOG-007	497 0.0076	493 0.0089	453 0.0099	444 0.0115	437 0.0136	434 0.0005 $N^{0.197\ 320}$	522 0.0048	512 0.0050	460 0.0053	444 0.0055	435 0.0059	455 0.0018 $N^{0.072\ 334}$
93	DERMALOG-008	382 0.0021	390 0.0025	382 0.0030	379 0.0036	377 0.0045	98 0.0001 $N^{0.268\ 408}$	352 0.0011	350 0.0012	343 0.0012	335 0.0014	338 0.0015	119 0.0002 $N^{0.113\ 377}$
94	DERMALOG-009	396 0.0022	388 0.0024	368 0.0027	356 0.0029	345 0.0033	399 0.0004 $N^{0.137\ 188}$	444 0.0018	436 0.0019	406 0.0019	391 0.0019	378 0.0020	430 0.0012 $N^{0.034\ 238}$
95	DERMALOG-010	364 0.0018	342 0.0018	319 0.0019	293 0.0020	277 0.0021	445 0.0008 $N^{0.064\ 25}$	432 0.0017	420 0.0017	395 0.0017	373 0.0017	356 0.0017	452 0.0015 $N^{0.010\ 87}$
96	DERMALOG-011	249 0.0011	255 0.0012	264 0.0015	264 0.0017	266 0.0020	97 0.0001 $N^{0.219\ 348}$	185 0.0007	183 0.0007	194 0.0008	193 0.0008	187 0.0008	139 0.0003 $N^{0.065\ 325}$
97	DERMALOG-012	316 0.0014	327 0.0017	313 0.0019	295 0.0021	290 0.0024	290 0.0001 $N^{0.172\ 258}$	350 0.0011	338 0.0011	321 0.0011	310 0.0012	303 0.0012	369 0.0007 $N^{0.032\ 230}$
98	DERMALOG-013	293 0.0013	308 0.0016	296 0.0018	287 0.0020	283 0.0023	251 0.0001 $N^{0.182\ 290}$	314 0.0010	310 0.0010	294 0.0010	279 0.0011	277 0.0011	352 0.0006 $N^{0.034\ 240}$
99	DERMALOG-014	369 0.0019	410 0.0030	421 0.0050	437 0.0095	449 0.0197	11 0.0000 $N^{0.807\ 473}$	425 0.0016	470 0.0026	458 0.0045	465 0.0089	467 0.0190	11 0.0000 $N^{0.855\ 473}$
100	DERMALOG-015	255 0.0011	254 0.0012	231 0.0012	219 0.0014	211 0.0015	375 0.0003 $N^{0.105\ 98}$	325 0.0010	308 0.0010	295 0.0010	272 0.0010	260 0.0011	416 0.0009 $N^{0.008\ 68}$
101	DERMALOG-016	244 0.0011	221 0.0011	197 0.0011	189 0.0012	180 0.0013	414 0.0004 $N^{0.075\ 38}$	319 0.0010	304 0.0010	282 0.0010	268 0.0010	253 0.0010	420 0.0010 $N^{0.004\ 40}$
102	DERMALOG-017	247 0.0011	225 0.0011	206 0.0011	186 0.0012	183 0.0013	411 0.0004 $N^{0.076\ 39}$	315 0.0010	305 0.0010	280 0.0010	265 0.0010	252 0.0010	419 0.0010 $N^{0.004\ 39}$
103	DILUSENSE-000	338 0.0016	344 0.0018	334 0.0021	330 0.0025	329 0.0030	185 0.0001 $N^{0.219\ 350}$	249 0.0009	254 0.0009	255 0.0009	261 0.0010	269 0.0011	130 0.0003 $N^{0.086\ 352}$
104	DILUSENSE-001	218 0.0010	230 0.0011	227 0.0012		225 0.0016	211 0.0001 $N^{0.172\ 256}$	221 0.0008	213 0.0008	207 0.0008		199 0.0009	300 0.0005 $N^{0.036\ 250}$
105	EINETWORKSINDIA-000	284 0.0013	286 0.0014	286 0.0016	280 0.0019	279 0.0022	245 0.0001 $N^{0.180\ 281}$	334 0.0011	323 0.0011	306 0.0011	290 0.0011	276 0.0011	392 0.0008 $N^{0.021\ 166}$
106	EINETWORKSINDIA-001	281 0.0012	277 0.0014	271 0.0016	269 0.0017	269 0.0020	257 0.0001 $N^{0.172\ 259}$	300 0.0010	289 0.0010	277 0.0010	269 0.0010	264 0.0011	336 0.0006 $N^{0.038\ 256}$
107	EINETWORKSINDIA-002	138 0.0007	135 0.0008	121 0.0008	114 0.0009	110 0.0010	287 0.0001 $N^{0.120\ 142}$	161 0.0006	155 0.0006	144 0.0006	130 0.0006	131 0.0007	320 0.0005 $N^{0.014\ 119}$
108	FACEPHI-000	320 0.0015	295 0.0015	276 0.0016	261 0.0017	254 0.0019	432 0.0005 $N^{0.077\ 42}$	394 0.0013	383 0.0014	354 0.0014	339 0.0014	324 0.0014	429 0.0011 $N^{0.017\ 145}$
109	FACEPHI-001	140 0.0007	112 0.0007	99 0.0008	85 0.0008	77 0.0009	371 0.0003 $N^{0.073\ 32}$	177 0.0007	171 0.0007	158 0.0007	148 0.0007	146 0.0007	317 0.0005 $N^{0.018\ 152}$
110	FIRSTCREDITKZ-001	166 0.0008	156 0.0008	147 0.0009	136 0.0010	125 0.0011	336 0.0002 $N^{0.101\ 90}$	234 0.0008	222 0.0008	213 0.0008	197 0.0008	184 0.0008	382 0.0008 $N^{0.006\ 50}$
111	FUJITSULAB-000	339 0.0016	346 0.0018	338 0.0022	336 0.0026	328 0.0030	177 0.0001 $N^{0.222\ 352}$	305 0.0010	296 0.0010	303 0.0011	295 0.0011	302 0.0012	169 0.0003 $N^{0.079\ 340}$
112	FUJITSULAB-001	295 0.0013	297 0.0015	291 0.0017	284 0.0019	282 0.0023	253 0.0001 $N^{0.180\ 282}$	304 0.0010	307 0.0010	297 0.0010	291 0.0011	285 0.0012	265 0.0004 $N^{0.060\ 312}$
113	GORILLA-002	541 0.0145	543 0.0194	477 0.0235	466 0.0285	461 0.0348	373 0.0003 $N^{0.299\ 445}$	496 0.0030	503 0.0039	459 0.0047	448 0.0060	444 0.0078	35 0.0000 $N^{0.321\ 456}$
114	GORILLA-004	458 0.0045	461 0.0059	439 0.0072	431 0.0088	426 0.0111	150 0.0001 $N^{0.307\ 451}$	371 0.0012	399 0.0014	402 0.0018	405 0.0021	404 0.0026	29 0.0000 $N^{0.269\ 445}$
115	GORILLA-005	393 0.0022	402 0.0028	397 0.0035	388 0.0043	387 0.0053	58 0.0000 $N^{0.301\ 448}$	258 0.0009	275 0.0009	289 0.0010	317 0.0012	323 0.0014	60 0.0001 $N^{0.168\ 413}$
116	GORILLA-006	258 0.0011	282 0.0014	288 0.0017	290 0.0020	301 0.0025	34 0.0000 $N^{0.275\ 413}$	186 0.0007	198 0.0007	201 0.0008	203 0.0008	204 0.0009	117 0.0002 $N^{0.081\ 346}$
117	GORILLA-007	238 0.0010	271 0.0013	272 0.0016	288 0.0020	298 0.0025	26 0.0000 $N^{0.295\ 442}$	183 0.0007	179 0.0007	175 0.0007	173 0.0008	186 0.0008	140 0.0003 $N^{0.063\ 317}$
118	GORILLA-008	209 0.0010	237 0.0011	243 0.0013	253 0.0016	255 0.0019	66 0.0000 $N^{0.229\ 362}$	179 0.0007	181 0.0007	174 0.0007	165 0.0007	168 0.0008	177 0.0003 $N^{0.050\ 286}$
119	GRIAULE-000	367 0.0018	369 0.0021	356 0.0024	340 0.0027	337 0.0031	317 0.0002 $N^{0.175\ 271}$	382 0.0012	367 0.0013	365 0.0014	353 0.0015	340 0.0015	274 0.0004 $N^{0.078\ 338}$
120	GRIAULE-001	148 0.0008	150 0.0008	143 0.0009	144 0.0010	142 0.0012	231 0.0001 $N^{0.144\ 206}$	165 0.0006	172 0.0007	159 0.0007	150 0.0007	149 0.0007	276 0.0004 $N^{0.029\ 212}$
121	HIK-003	513 0.0089	516 0.0115	467 0.0137	458 0.0167	450 0.0201	343 0.0002 $N^{0.278\ 423}$	472 0.0022	481 0.0029	448 0.0034	440 0.0042	433 0.0052	41 0.0001 $N^{0.283\ 449}$
122	HIK-004	511 0.0086	514 0.0111	464 0.0132	455 0.0160	447 0.0192	341 0.0002 $N^{0.276\ 417}$	476 0.0023	477 0.0028	446 0.0033	438 0.0040	430 0.0050	47 0.0001 $N^{0.267\ 442}$
123	HIK-005	438 0.0034	438 0.0044	422 0.0051	412 0.0063	409 0.0079	170 0.0001 $N^{0.283\ 431}$	364 0.0011	392 0.0014	377 0.0015	380 0.0018	384 0.0021	57 0.0001 $N^{0.198\ 425}$
124	HISIGN-001	142 0.0007	160 0.0009	175 0.0010	181 0.0012	184 0.0013	77 0.0000 $N^{0.206\ 331}$	137 0.0006	136 0.0006	126 0.0006	127 0.0006	123 0.0006	191 0.0004 $N^{0.037\ 253}$
125	HYPERVERGE-001	215 0.0010	210 0.0010	191 0.0011	185 0.0012	187 0.0014	342 0.0002 $N^{0.111\ 120}$	292 0.0009	278 0.0009	258 0.0009	240 0.0009	227 0.0009	405 0.0009 $N^{0.006\ 57}$
126	HYPERVERGE-002	219 0.0010	204 0.0010	190 0.0011	170 0.0011	154 0.0012	421 0.0004 $N^{0.062\ 23}$	278 0.0009	268 0.0009	256 0.0009	237 0.0009	226 0.0009	390 0.0008 $N^{0.012\ 98}$
127	HYPERVERGE-003	226 0.0010	224 0.0011	211 0.0012	207 0.0013	209 0.0015	303 0.0002 $N^{0.136\ 186}$	281 0.0009	277 0.0009	259 0.0009	241 0.0009	229 0.0010	388 0.0008 $N^{0.013\ 110}$
128	HYUNDAI-ROBOTICSLAB-000	395 0.0022	382 0.0023	363 0.0025	343 0.0027	334 0.0030	433 0.0005 $N^{0.107\ 104}$	447 0.0018	438 0.0019	411 0.0019	392 0.0019	379 0.0020	438 0.0012 $N^{0.030\ 219}$
129	HYUNDAI-ROBOTICSLAB-001	279 0.0012	278 0.0014	275 0.0016	272 0.0018	270 0.0021	240 0.0001 $N^{0.178\ 278}$	259 0.0009	253 0.0009	260 0.0009	257 0.0010	261 0.0011	180 0.0003 $N^{0.067\ 329}$
130	HYUNDAI-ROBOTICSLAB-002	99 0.0006	105 0.0007	103 0.0008	105 0.0009	121 0.0011	100 0.0001 $N^{0.178\ 279}$	114 0.0006	107 0.0006	105 0.0006	101 0.0006	97 0.0006	251 0.0004 $N^{0.021\ 164}$
131	HYUNDAI-ROBOTICSLAB-003	83 0.0006	90 0.0007	96 0.0007	92 0.0009	108 0.0010	69 0.0000 $N^{0.190\ 311}$	78 0.0005	78 0.0005	76 0.0005	72 0.0005	68 0.0005	206 0.0004 $N^{0.023\ 178}$
132	HZAILU-000	343 0.0016	343 0.0018	333 0.0021	331 0.0025	332 0.0030	190 0.0001 $N^{0.216\ 344}$	337 0.0011	331 0.0011	312 0.0011	308 0.0012	297 0.0012	344 0.0006 $N^{0.042\ 264}$
133	HZAILU-001	326 0.0015	312 0.0016	290 0.0017	276 0.0018	273 0.0021	392 0.0003 $N^{0.114\ 127}$	345 0.0011	348 0.0011	335 0.0012	323 0.0012	313 0.0013	292 0.0005 $N^{0.063\ 314}$
134	HZAILU-002	323 0.0015	316 0.0016	297 0.0018	286 0.0020	281 0.0022	357 0.0002 $N^{0.137\ 190}$	346 0.0011	346 0.0011	332 0.0012	326 0.0013	314 0.0013	291 0.0005 $N^{0.063\ 315}$
135	HZAILU-003	267 0.0012	263 0.0013	257 0.0014	256 0.0016	252 0.0018	293 0.0001 $N^{0.153\ 221}$	257 0.0009	263 0.0009	272 0.0010	262 0.0010	256 0.0010	210 0.0004 $N^{0.064\ 320}$
136	HZAILU-004	191 0.0009	182 0.0009	172 0.0010	173 0.0011	165 0.0012	331 0.0002 $N^{0.111\ 119}$	252 0.0009	241 0.0009	226 0.0009	218 0.0009	206 0.0009	383 0.0008 $N^{0.009\ 80}$
137	I2V-001	406 0.0025	393 0.0026	369 0.0027	348 0.0028	330 0.0030	455 0.0011 $N^{0.059\ 20}$	478 0.0024	467 0.0024	426 0.0024	413 0.0024	400 0.0024	462 0.0022 $N^{0.006\ 56}$
138	ICEHOUSE-001	199 0.0009	201 0.0010	204 0.0011	201 0.0013	195 0.0014	264 0.0001 $N^{0.149\ 215}$	195 0.0007	201 0.0007	195 0.0008	183 0.0008	177 0.0008	261 0.0004 $N^{0.039\ 258}$
139	ICM-001	409 0.0025	399 0.0028	384 0.0030	366 0.0033	360 0.0038	409 0.0004 $N^{0.141\ 201}$	443 0.0018	440 0.0019	413 0.0020	401 0.0020	389 0.0021	404 0.0009 $N^{0.056\ 297}$
140	IDBIO-000	287 0.0013	323 0.0017	324 0.0020	326 0.0024	331 0.0030	37 0.0000 $N^{0.282\ 430}$	152 0.0006	180 0.0007	186 0.0007	180 0.0008	217 0.0009	72 0.0001 $N^{0.129\ 385}$
141	IDEMIA-003	466 0.0050	470 0.0066	446 0.0080	439 0.0098	432 0.0118	220 0.0001 $N^{0.293\ 440}$	458 0.0020	468 0.0024	434 0.0027	424 0.0033	419 0.0038	67 0.0001 $N^{0.224\ 429}$
142	IDEMIA-004	469 0.0051	465 0.0063	442 0.0076	436 0.0094	428 0.0114	250 0.0001 $N^{0.281\ 425}$	415 0.0015	426 0.0018	423 0.0022	416 0.0027	413 0.0033	33 0.0000 $N^{0.272\ 446}$
143	IDEMIA-005	477 0.0061	485 0.0077	451 0.0093	443 0.0115	438 0.0140	274 0.0001 $N^{0.287\ 435}$	456 0.0019	474 0.0027	442 0.0033	439 0.0041	432 0.0051	26 0.0000 $N^{0.329\ 460}$
144	IDEMIA-006	492 0.0072	499 0.0092	458 0.0109	448 0.0132	442 0.0158	335 0.0002 $N^{0.267\ 407}$	480 0.0025	493 0.0034	455 0.0043	445 0.0055	442 0.0073	21 0.0000 $N^{0.369\ 465}$

Table 46: **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES AT GIVEN RANK FNIR(N, T= 0, R)		ENROL MOST RECENT											
		RANK 1						RANK 50					
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	$a.N^b$	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	$a.N^b$
145	IDEMIA-007	360 0.0017	373 0.0022	367 0.0026	364 0.0032	367 0.0040	54 0.0000 $N^{0.284}$ 432	206 0.0007	228 0.0008	235 0.0009	266 0.0010	288 0.0012	66 0.0001 $N^{0.153}$ 401
146	IDEMIA-008	90 0.0006	99 0.0007	108 0.0008	102 0.0009	106 0.0010	101 0.0001 $N^{0.176}$ 273	99 0.0005	99 0.0005	91 0.0005	94 0.0006	89 0.0006	214 0.0004 $N^{0.026}$ 193
147	IDEMIA-009	79 0.0006	75 0.0006	74 0.0007	68 0.0008	68 0.0009	174 0.0001 $N^{0.146}$ 210	71 0.0005	68 0.0005	69 0.0005	66 0.0005	61 0.0005	249 0.0004 $N^{0.013}$ 107
148	IDEMIA-010	46 0.0005	46 0.0005	44 0.0006	38 0.0006	21 0.0007	285 0.0001 $N^{0.095}$ 79	72 0.0005	67 0.0005	64 0.0005	58 0.0005	54 0.0005	295 0.0005 $N^{0.004}$ 31
149	IDEMIA-011	107 0.0007	97 0.0007	86 0.0007	72 0.0008	61 0.0008	356 0.0002 $N^{0.077}$ 40	173 0.0006	157 0.0007	155 0.0007	144 0.0007	134 0.0007	332 0.0006 $N^{0.011}$ 94
150	IDEMIA-012	109 0.0007	104 0.0007	95 0.0007	76 0.0008	59 0.0008	369 0.0003 $N^{0.070}$ 31	162 0.0006	150 0.0006	145 0.0006	135 0.0007	125 0.0007	325 0.0005 $N^{0.011}$ 97
151	IDEMIA-013	4 0.0004	2 0.0004	1 0.0005	1 0.0005	1 0.0006	212 0.0001 $N^{0.106}$ 100	11 0.0004	7 0.0004	7 0.0004	3 0.0004	3 0.0004	256 0.0004 $N^{0.000}$ 8
152	IDENTY-001	214 0.0010	203 0.0010	501 1.0000	486 0.5187	479 0.5726	4 0.0000 $N^{2.612}$ 480	276 0.0009	260 0.0009	500 0.9999	484 0.1882	476 0.2328	4 0.0000 $N^{2.262}$ 480
153	IDENTY-002	97 0.0006	95 0.0007	93 0.0007	99 0.0009	96 0.0010	163 0.0001 $N^{0.156}$ 226	112 0.0006	117 0.0006	108 0.0006	102 0.0006	103 0.0006	235 0.0004 $N^{0.025}$ 192
154	IMAGUS-005	311 0.0014	311 0.0016	303 0.0018	298 0.0021	293 0.0024	232 0.0001 $N^{0.189}$ 309	272 0.0009	291 0.0010	283 0.0010	280 0.0011	298 0.0012	134 0.0003 $N^{0.089}$ 356
155	IMAGUS-006	318 0.0014	319 0.0016	304 0.0018	297 0.0021	299 0.0025	252 0.0001 $N^{0.185}$ 301	316 0.0010	329 0.0011	318 0.0011	300 0.0011	306 0.0012	257 0.0004 $N^{0.065}$ 324
156	IMAGUS-007	306 0.0014	321 0.0016	315 0.0019	319 0.0023	309 0.0026	133 0.0001 $N^{0.224}$ 356	243 0.0008	259 0.0009	246 0.0009	247 0.0010	271 0.0011	128 0.0003 $N^{0.088}$ 355
157	IMDS-SOFTWARE-001	121 0.0007	136 0.0008	153 0.0009	158 0.0011	166 0.0012	76 0.0000 $N^{0.201}$ 324	125 0.0006	121 0.0006	117 0.0006	111 0.0006	116 0.0006	172 0.0003 $N^{0.039}$ 259
158	IMDS-SOFTWARE-002	114 0.0007	131 0.0008	142 0.0009	153 0.0011	159 0.0012	75 0.0000 $N^{0.200}$ 323	109 0.0005	106 0.0006	109 0.0006	104 0.0006	107 0.0006	167 0.0003 $N^{0.037}$ 255
159	IMPERIAL-000	366 0.0018	365 0.0021	351 0.0023	342 0.0027	339 0.0031	305 0.0002 $N^{0.181}$ 287	388 0.0013	375 0.0013	358 0.0014	347 0.0014	334 0.0015	351 0.0006 $N^{0.052}$ 291
160	INCODE-003	517 0.0096	523 0.0127	468 0.0151	460 0.0188	452 0.0230	319 0.0002 $N^{0.299}$ 446	475 0.0023	482 0.0029	450 0.0035	442 0.0044	434 0.0054	37 0.0000 $N^{0.295}$ 453
161	INCODE-004	413 0.0026	415 0.0032	401 0.0038	395 0.0046	391 0.0056	140 0.0001 $N^{0.268}$ 409	405 0.0014	407 0.0015	388 0.0016	377 0.0017	366 0.0018	234 0.0004 $N^{0.094}$ 362
162	INCODE-005	264 0.0012	267 0.0013	260 0.0014	259 0.0016	262 0.0019	235 0.0001 $N^{0.174}$ 268	253 0.0009	258 0.0009	250 0.0009	250 0.0010	248 0.0010	240 0.0004 $N^{0.056}$ 299
163	INCODE-006	125 0.0007	108 0.0007	100 0.0008	83 0.0008	80 0.0009	339 0.0002 $N^{0.088}$ 62	181 0.0007	177 0.0007	166 0.0007	154 0.0007	142 0.0007	342 0.0006 $N^{0.009}$ 83
164	INCODE-007	39 0.0005	40 0.0005	40 0.0006	28 0.0006	33 0.0007	209 0.0001 $N^{0.120}$ 143	32 0.0005	29 0.0005	28 0.0005	33 0.0005	38 0.0005	162 0.0003 $N^{0.025}$ 190
165	INNOVATRICS-007	273 0.0012	272 0.0013	269 0.0015	266 0.0017	268 0.0020	237 0.0001 $N^{0.176}$ 276	251 0.0009	247 0.0009	240 0.0009	242 0.0009	242 0.0010	255 0.0004 $N^{0.052}$ 290
166	INNOVATRICS-008	184 0.0009	183 0.0009	178 0.0010	179 0.0011	177 0.0013	278 0.0001 $N^{0.137}$ 189	189 0.0007	184 0.0007	176 0.0007	169 0.0007	165 0.0008	309 0.0005 $N^{0.027}$ 200
167	INNOVATRICS-009	205 0.0010	200 0.0010	189 0.0011	187 0.0012	182 0.0013	329 0.0002 $N^{0.117}$ 133	261 0.0009	246 0.0009	232 0.0009	221 0.0009	212 0.0009	384 0.0008 $N^{0.010}$ 86
168	INNOVATRICS-010	224 0.0010	215 0.0010	202 0.0011	184 0.0012	181 0.0013	377 0.0003 $N^{0.095}$ 76	265 0.0009	248 0.0009	239 0.0009	231 0.0009	225 0.0009	350 0.0006 $N^{0.024}$ 187
169	INNOVATRICS-011	124 0.0007	122 0.0008	123 0.0008	110 0.0009	112 0.0010	247 0.0001 $N^{0.133}$ 181	140 0.0006	140 0.0006	135 0.0006	131 0.0006	135 0.0007	168 0.0003 $N^{0.043}$ 271
170	INNOVATRICS-012	115 0.0007	117 0.0007	115 0.0008	119 0.0009	116 0.0011	194 0.0001 $N^{0.153}$ 220	147 0.0006	131 0.0006	130 0.0006	126 0.0006	122 0.0006	230 0.0004 $N^{0.031}$ 228
171	INNOVATRICS-013	59 0.0005	70 0.0006	69 0.0007	78 0.0008	82 0.0009	63 0.0000 $N^{0.186}$ 302	39 0.0005	48 0.0005	49 0.0005	41 0.0005	33 0.0005	197 0.0004 $N^{0.018}$ 149
172	INNOVATRICS-014	47 0.0005	52 0.0006	52 0.0006	55 0.0007	51 0.0008	160 0.0001 $N^{0.142}$ 202	41 0.0005	35 0.0005	40 0.0005	55 0.0005	59 0.0005	129 0.0003 $N^{0.039}$ 260
173	INNOVATRICS-015	45 0.0005	51 0.0006	49 0.0006	48 0.0007	41 0.0007	221 0.0001 $N^{0.120}$ 146	40 0.0005	28 0.0005	36 0.0005	35 0.0005	47 0.0005	147 0.0003 $N^{0.028}$ 208
174	INNOVATRICS-016	25 0.0005	25 0.0005	20 0.0005	17 0.0006	19 0.0007	227 0.0001 $N^{0.109}$ 112	21 0.0004	19 0.0004	21 0.0004	20 0.0005	21 0.0005	158 0.0003 $N^{0.022}$ 173
175	INSPUR-000	71 0.0006	79 0.0006	78 0.0007	87 0.0008	105 0.0010	42 0.0000 $N^{0.205}$ 330	51 0.0005	55 0.0005	61 0.0005	59 0.0005	53 0.0005	175 0.0003 $N^{0.024}$ 189
176	INSPUR-001	72 0.0006	72 0.0006	89 0.0007	89 0.0008	88 0.0010	57 0.0000 $N^{0.195}$ 318	52 0.0005	56 0.0005	55 0.0005	50 0.0005	46 0.0005	213 0.0004 $N^{0.017}$ 147
177	INSPUR-002	57 0.0005	63 0.0006	73 0.0007	67 0.0008	69 0.0009	88 0.0001 $N^{0.173}$ 266	36 0.0005	40 0.0005	35 0.0005	26 0.0005	28 0.0005	220 0.0004 $N^{0.014}$ 116
178	INTELRESEARCH-001	82 0.0006	84 0.0006	79 0.0007	77 0.0008	83 0.0009	135 0.0001 $N^{0.158}$ 229	77 0.0005	71 0.0005	66 0.0005	68 0.0005	65 0.0005	233 0.0004 $N^{0.017}$ 148
179	INTEMA-000	129 0.0007	132 0.0008	128 0.0009	130 0.0010	133 0.0012	132 0.0001 $N^{0.173}$ 261	172 0.0006	174 0.0007	192 0.0008	182 0.0008	218 0.0009	81 0.0001 $N^{0.116}$ 379
180	INTEMA-001	16 0.0005	13 0.0005	34 0.0006	46 0.0007	81 0.0009	21 0.0000 $N^{0.237}$ 374	23 0.0004	15 0.0004	31 0.0005	61 0.0005	148 0.0007	42 0.0001 $N^{0.151}$ 400
181	INTOZI-001	235 0.0010	251 0.0012	255 0.0014	262 0.0017	263 0.0020	86 0.0001 $N^{0.224}$ 355	158 0.0006	175 0.0007	169 0.0007	162 0.0007	180 0.0008	109 0.0002 $N^{0.081}$ 345
182	INTSYSMSU-000	601 0.1392	608 0.1454	492 0.1495	480 0.1541	472 0.1588	482 0.0765 $N^{0.045}$ 13	610 0.1095	608 0.1160	495 0.1203	482 0.1248	474 0.1293	482 0.0516 $N^{0.057}$ 300
183	IONETWORKS-001	348 0.0016	417 0.0034	435 0.0066	452 0.0142	454 0.0263	8 0.0000 $N^{0.974}$ 475	406 0.0014	487 0.0031	464 0.0062	473 0.0138	469 0.0257	8 0.0000 $N^{1.013}$ 475
184	IREX-000	448 0.0040	433 0.0040	406 0.0041	387 0.0042	375 0.0044	467 0.0024 $N^{0.035}$ 7	511 0.0039	502 0.0039	454 0.0039	434 0.0039	422 0.0039	469 0.0038 $N^{0.002}$ 18
185	IRISID-001	193 0.0009	219 0.0011	228 0.0012	227 0.0015	247 0.0018	61 0.0000 $N^{0.229}$ 361	128 0.0006	127 0.0006	132 0.0006	147 0.0007	151 0.0007	96 0.0002 $N^{0.085}$ 351
186	IRISID-002	154 0.0008	173 0.0009	182 0.0010	198 0.0012	210 0.0015	44 0.0000 $N^{0.230}$ 363	108 0.0005	112 0.0006	114 0.0006	116 0.0006	121 0.0006	121 0.0002 $N^{0.060}$ 311
187	ISTL-001	378 0.0020	368 0.0021	344 0.0023	325 0.0024	312 0.0027	431 0.0005 $N^{0.102}$ 95	440 0.0017	425 0.0018	403 0.0018	383 0.0018	364 0.0018	446 0.0014 $N^{0.018}$ 150
188	ISYSTEMS-002	465 0.0050	464 0.0061	438 0.0070	425 0.0080	418 0.0093	387 0.0003 $N^{0.209}$ 337	497 0.0031	490 0.0032	447 0.0034	431 0.0035	420 0.0038	434 0.0012 $N^{0.072}$ 333
189	ISYSTEMS-003	455 0.0043	450 0.0049	424 0.0055	413 0.0063	405 0.0073	413 0.0004 $N^{0.181}$ 285	493 0.0029	485 0.0031	440 0.0031	428 0.0033	417 0.0035	439 0.0012 $N^{0.063}$ 316
190	KAKAO-000	198 0.0009	232 0.0011	235 0.0013	248 0.0016	257 0.0019	43 0.0000 $N^{0.245}$ 384	129 0.0006	141 0.0006	141 0.0006	151 0.0007	150 0.0007	107 0.0002 $N^{0.079}$ 341
191	KAKAO-001	227 0.0010	209 0.0010	196 0.0011	182 0.0012	173 0.0013	397 0.0003 $N^{0.079}$ 45	289 0.0009	280 0.0009	264 0.0010	245 0.0010	236 0.0010	377 0.0007 $N^{0.016}$ 142
192	KAKAO-002	217 0.0010	197 0.0010	187 0.0011	175 0.0011	158 0.0012	401 0.0004 $N^{0.074}$ 36	285 0.0009	269 0.0009	254 0.0009	243 0.0010	231 0.0010	376 0.0007 $N^{0.016}$ 138
193	KEDACOM-001	491 0.0072	481 0.0074	441 0.0076	424 0.0079	410 0.0083	473 0.0037 $N^{0.049}$ 15	543 0.0068	534 0.0068	472 0.0069	453 0.0069	439 0.0070	476 0.0060 $N^{0.010}$ 84
194	KLOUDSPOT-001	358 0.0017	358 0.0020	354 0.0024	353 0.0028	351 0.0035	127 0.0001 $N^{0.241}$ 380	330 0.0010	321 0.0011	315 0.0011	305 0.0011	304 0.0012	308 0.0005 $N^{0.055}$ 295
195	KNERON-000	456 0.0044	458 0.0056	433 0.0064	421 0.0075	415 0.0089	324 0.0002 $N^{0.239}$ 376	516 0.0044	518 0.0056	466 0.0064	458 0.0075	450 0.0089	99 0.0002 $N^{0.239}$ 432
196	KNOWUTECH-000	103 0.0007	126 0.0008	149 0.0009	162 0.0011	157 0.0012	48 0.0000 $N^{0.216}$ 345	67 0.0005	80 0.0005	79 0.0005	71 0.0005	77 0.0005	152 0.0003 $N^{0.034}$ 239
197	KOGNIZA-000	271 0.0012	260 0.0013	254 0.0014	244 0.0015	230 0.0017	366 0.0003 $N^{0.115}$ 130	339 0.0011	332 0.0011	317 0.0011	293 0.0011	280 0.0011	399 0.0008 $N^{0.019}$ 155
198	LIGHTVISION-000	404 0.0024	397 0.0027	383 0.0030	369 0.0034	365 0.0040	361 0.0002 $N^{0.170}$ 253	455 0.0019	442 0.0019	412 0.0019	397 0.0020	383 0.0021	447 0.0014 $N^{0.024}$ 188
199	LINECLOVA-002	195 0.0009	186 0.0009	177 0.0010	172 0.0011	170 0.0013	344 0.0002 $N^{0.106}$ 101	238 0.0008	229 0.0008	217 0.0008	204 0.0008	193 0.0009	368 0.0007 $N^{0.011}$ 93
200	LINECLOVA-003	89 0.0006	77 0.0006	94 0.0007	94 0.0009	89 0.0010	104 0.0001 $N^{0.171}$ 254	33 0.0005	30 0.0005	54 0.0005	57 0.0005	63 0.0005	116 0.0002 $N^{0.050}$ 284
201	LOOKMAN-003	503 0.0080	490 0.0084	450 0.0087	435 0.0093	422 0.0101	469 0.0027 $N^{0.079}$ 44	547 0.0069	537 0.0070	474 0.0071	456 0.0072	443 0.0074	471 0.0051 $N^{0.023}$ 180
202	LOOKMAN-005	495 0.0075	484 0.0077	445 0.0080	428 0.0082	414 0.0088	470 0.0035 $N^{0.056}$ 16	544 0.0068	535 0.0069	473 0.0069	454 0.0070	440 0.0071	475 0.0056 $N^{0.014}$ 118
203	LSSYTECH-000	70 0.0006	68 0.0006	64 0.0006	616 1.0000	56 0.0008	7 0.0000 $N^{1.127}$ 477	105 0.0005	97 0.0005	96 0.0005	607 1.0000	81 0.0006	7 0.0000 $N^{1.040}$ 476
204	LSSYTECH-001	63 0.0006	53 0.0006	50 0.0006	47 0.0007	45 0.0008	256 0.0001 $N^{0.111}$ 117	91 0.0005	91 0.0005	81 0.0005	74 0.0005	72 0.0005	285 0.0005 $N^{0.010}$ 88
205	LUXAND-001	403 0.0024	401 0.0028	390 0.0032	380 0.0037	372 0.0043	300 0.0002 $N^{0.203}$ 326	419 0.0015	416 0.0016	394 0.0017	378 0.0017	367 0.0019	354 0.0006 $N^{0.064}$ 318
206	MANTRA-000	266 0.0012	276 0.0014	274 0.0016	274 0.0018	278 0.0022	139 0.0001 $N^{0.210}$ 338	239 0.0008	234 0.0008	229 0.0009	229 0.0009	234 0.0010	208 0.0004 $N^{0.058}$ 304
207	MAXVISION-000	356 0.0017	360 0.0021	352 0.0024	350 0.0028	350 0.0035	143 0.0001 $N^{0.237}$ 372	277 0.0009	297 0.0010	300 0.0011	313 0.0012	320 0.0013	86 0.0002 $N^{0.132}$ 387
208	MAXVISION-001	159 0.0008	155 0.0008	152 0.0009	140 0.0010	139 0.0011	298 0.0002 $N^{0.122}$ 153	199 0.0007	195 0.0007	183 0.0007	176 0.0008	164 0.0008	323 0.0005 $N^{0.021}$ 170
209	MAXVISION-002	160 0.0008	163 0.0009	157 0.0009	146 0.0010	144 0.0012	296 0.0001 $N^{0.124}$ 158	222 0.0008	210 0.0008	202 0.0008	186 0.0008	173 0.0008	385 0.0008 $N^{0.002}$ 20
210	MAXVISION-003	179 0.0009	172 0.0009	167 0.0010	165 0.0011	160 0.0012	299 0.0002 $N^{0.126}$ 160	193 0.0007	193 0.0007	189 0.0007	179 0.0008	167 0.0008	302 0.0005 $N^{0.029}$ 214
211	MAXVISION-004	172 0.0008	154 0.0008	140 0.0009	120 0.0009	115 0.0011	372 0.0003 $N^{0.082}$ 49	228 0.0008	216 0.0008	204 0.0008	191 0.0008	179 0.0008	396 0.0008 $N^{0.000}$ 7
212	MEGVII-001	525 0.0104	518 0.0116	462 0.0127	450 0.0140	443 0.0158	463 0.0015 $N^{0.142}$ 204	551 0.0077	545 0.0080	478 0.0082	463 0.0085	449 0.0088	471 0.0041 $N^{0.046}$ 277
213	MEGVII-003	183 0.0009	170 0.0009	150 0.0009	135 0.0010	128 0.0011	374 0.0003 $N^{0.084}$ 54	237 0.0008	221 0.0008	216 0.0008	201 0.0008	190 0.0008	370 0.0007 $N^{0.010}$ 85
214	MEGVII-004	163 0.0008	152 0.0008	130 0.0009	109 0.0009	91 0.0010	406 0.0004 $N^{0.059}$ 19	230 0.0008	218 0.0008	209 0.0008	196 0.0008	182 0.0008	379 0.0007 $N^{0.006}$ 52
215	MEGVII-005	167 0.0008	151 0.0008	131 0.0009	108 0.0009	93 0.0010	407 0.0004 $N^{0.059}$ 18	227 0.0008	219 0.0008	206 0.0008	195 0.0008	181 0.0008	378 0.0007 $N^{0.006}$ 51
216	METSAKUURCOMPANY-000	116 0.0007	138 0.0008	137 0.0009	148 0.0010	169 0.0013	71 0.0000 $N^{0.202}$ 325	90 0.0005	89 0.0005	86 0.0005	86 0.0005	83 0.0006	182 0.0003 $N^{0.029}$ 213

Table 47: **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES AT GIVEN RANK FNIR(N, T= 0, R)		ENROL MOST RECENT											
		RANK 1						RANK 50					
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b
217	METSAKUURCOMPANY-001	192 0.0009	205 0.0010	195 0.0011	199 0.0013	207 0.0015	210 0.0001 $N^{0.166\ 250}$	198 0.0007	202 0.0008	197 0.0008	190 0.0008	183 0.0008	244 0.0004 $N^{0.042\ 266}$
218	METSAKUURCOMPANY-002	173 0.0008	192 0.0010	214 0.0012	225 0.0014	236 0.0017	35 0.0000 $N^{0.251\ 390}$	92 0.0005	100 0.0005	106 0.0006	106 0.0006	112 0.0006	112 0.0002 $N^{0.065\ 323}$
219	MIAXIS-001	262 0.0011	264 0.0013	252 0.0014	246 0.0015	232 0.0017	327 0.0002 $N^{0.135\ 188}$	294 0.0009	290 0.0010	274 0.0010	260 0.0010	254 0.0010	345 0.0006 $N^{0.033\ 235}$
220	MIAXIS-002	265 0.0012	266 0.0013	253 0.0014	249 0.0016	240 0.0018	316 0.0002 $N^{0.140\ 199}$	309 0.0010	300 0.0010	286 0.0010	271 0.0010	266 0.0011	360 0.0007 $N^{0.028\ 206}$
221	MICROFOCUS-005	619 0.3698	619 0.4240	497 0.4608	485 0.4998	478 0.5390	481 0.0673 $N^{0.128\ 166}$	614 0.1298	613 0.1721	497 0.2043	486 0.2423	478 0.2808	470 0.0039 $N^{0.263\ 441}$
222	MICROSOFT-003	269 0.0012	298 0.0015	302 0.0018	306 0.0022	314 0.0027	36 0.0000 $N^{0.276\ 418}$	117 0.0006	139 0.0006	172 0.0007	188 0.0008	198 0.0009	50 0.0001 $N^{0.158\ 405}$
223	MICROSOFT-004	263 0.0011	287 0.0014	293 0.0017	294 0.0021	311 0.0027	31 0.0000 $N^{0.287\ 437}$	113 0.0006	120 0.0006	142 0.0006	156 0.0007	185 0.0008	59 0.0001 $N^{0.139\ 393}$
224	MICROSOFT-005	317 0.0014	345 0.0018	348 0.0023	357 0.0030	352 0.0037	23 0.0000 $N^{0.326\ 458}$	130 0.0006	146 0.0006	167 0.0007	175 0.0008	194 0.0009	63 0.0001 $N^{0.137\ 392}$
225	MICROSOFT-006	331 0.0015	352 0.0019	361 0.0025	358 0.0030	358 0.0037	29 0.0000 $N^{0.310\ 453}$	134 0.0006	160 0.0007	173 0.0007	205 0.0008	249 0.0010	38 0.0000 $N^{0.188\ 421}$
226	MOBBL-000	329 0.0020	381 0.0023	366 0.0026	347 0.0027	315 0.0027	430 0.0005 $N^{0.107\ 105}$	376 0.0012	364 0.0013	348 0.0013	337 0.0014	322 0.0014	356 0.0007 $N^{0.046\ 278}$
227	MUKH-002	555 0.0200	552 0.0255	482 0.0301	472 0.0358	467 0.0427	436 0.0006 $N^{0.258\ 397}$	525 0.0051	533 0.0066	476 0.0079	467 0.0097	461 0.0120	70 0.0001 $N^{0.293\ 452}$
228	NAVERCLOUD-000	162 0.0008	171 0.0009	162 0.0009	143 0.0010	148 0.0012	302 0.0002 $N^{0.121\ 150}$	209 0.0007	203 0.0008	191 0.0008	174 0.0008	162 0.0008	364 0.0007 $N^{0.006\ 58}$
229	NEC-000	535 0.0128	535 0.0167	472 0.0199	461 0.0241	455 0.0291	384 0.0003 $N^{0.281\ 426}$	486 0.0026	495 0.0035	457 0.0044	446 0.0056	441 0.0071	27 0.0000 $N^{0.344\ 463}$
230	NEC-001	548 0.0177	544 0.0206	476 0.0230	465 0.0262	457 0.0300	464 0.0016 $N^{0.181\ 286}$	568 0.0106	557 0.0109	482 0.0113	469 0.0118	462 0.0125	473 0.0049 $N^{0.057\ 301}$
231	NEC-002	175 0.0008	179 0.0009	170 0.0010	164 0.0011	147 0.0012	314 0.0002 $N^{0.117\ 132}$	190 0.0007	190 0.0007	185 0.0007	171 0.0008	172 0.0008	280 0.0004 $N^{0.034\ 241}$
232	NEC-003	277 0.0012	262 0.0013	248 0.0013	223 0.0014	208 0.0015	426 0.0005 $N^{0.074\ 35}$	360 0.0011	343 0.0011	326 0.0011	309 0.0012	292 0.0012	418 0.0009 $N^{0.013\ 111}$
233	NEC-004	234 0.0010	220 0.0011	207 0.0011	188 0.0012	178 0.0013	400 0.0004 $N^{0.078\ 43}$	286 0.0009	273 0.0009	262 0.0009	253 0.0010	238 0.0010	362 0.0007 $N^{0.023\ 179}$
234	NEC-005	155 0.0008	149 0.0008	141 0.0009	121 0.0009	107 0.0010	349 0.0002 $N^{0.092\ 69}$	192 0.0007	185 0.0007	180 0.0007	170 0.0007	161 0.0008	324 0.0005 $N^{0.020\ 159}$
235	NEC-006	182 0.0009	178 0.0009	163 0.0009	156 0.0011	136 0.0011	359 0.0002 $N^{0.095\ 80}$	217 0.0008	211 0.0008	210 0.0008	202 0.0008	195 0.0009	299 0.0005 $N^{0.036\ 248}$
236	NEC-007	98 0.0006	93 0.0007	81 0.0007	71 0.0008	60 0.0008	315 0.0002 $N^{0.095\ 78}$	136 0.0006	128 0.0006	123 0.0006	112 0.0006	109 0.0006	273 0.0004 $N^{0.021\ 169}$
237	NEC-008	133 0.0007	111 0.0007	104 0.0008	96 0.0009	73 0.0009	350 0.0002 $N^{0.085\ 56}$	151 0.0006	147 0.0006	140 0.0006	125 0.0006	119 0.0006	313 0.0005 $N^{0.016\ 135}$
238	NEC-009	84 0.0006	74 0.0006	66 0.0006	56 0.0007	46 0.0008	318 0.0002 $N^{0.088\ 63}$	135 0.0006	126 0.0006	112 0.0006	103 0.0006	96 0.0006	329 0.0006 $N^{0.003\ 26}$
239	NEC-010	35 0.0005	30 0.0005	29 0.0005	19 0.0006	17 0.0007	263 0.0001 $N^{0.100\ 89}$	29 0.0004	37 0.0005	39 0.0005	48 0.0005	40 0.0005	143 0.0003 $N^{0.031\ 221}$
240	NEC-011	11 0.0004	15 0.0005	12 0.0005	7 0.0005	7 0.0006	217 0.0001 $N^{0.110\ 116}$	8 0.0004	12 0.0004	10 0.0004	11 0.0004	14 0.0004	160 0.0003 $N^{0.019\ 157}$
241	NEC-012	10 0.0004	14 0.0005	13 0.0005	9 0.0006	9 0.0006	204 0.0001 $N^{0.115\ 128}$	5 0.0004	14 0.0004	9 0.0004	10 0.0004	12 0.0004	161 0.0003 $N^{0.019\ 158}$
242	NEUROTECHNOLOGY-003	547 0.0175	546 0.0222	478 0.0259	468 0.0302	462 0.0358	443 0.0007 $N^{0.242\ 381}$	512 0.0039	517 0.0054	471 0.0069	464 0.0087	458 0.0109	32 0.0000 $N^{0.348\ 464}$
243	NEUROTECHNOLOGY-004	451 0.0043	452 0.0052	431 0.0061	418 0.0071	413 0.0084	330 0.0002 $N^{0.231\ 364}$	451 0.0019	456 0.0022	429 0.0025	421 0.0028	411 0.0031	103 0.0002 $N^{0.173\ 414}$
244	NEUROTECHNOLOGY-005	431 0.0033	431 0.0040	415 0.0046	405 0.0054	399 0.0065	282 0.0001 $N^{0.235\ 370}$	452 0.0019	447 0.0021	419 0.0022	410 0.0023	403 0.0025	316 0.0005 $N^{0.097\ 368}$
245	NEUROTECHNOLOGY-007	420 0.0028	420 0.0036	407 0.0041	398 0.0049	394 0.0058	236 0.0001 $N^{0.242\ 382}$	434 0.0017	427 0.0018	408 0.0019	398 0.0020	393 0.0022	310 0.0005 $N^{0.091\ 357}$
246	NEUROTECHNOLOGY-008	330 0.0015	340 0.0018	330 0.0021	334 0.0025	333 0.0030	112 0.0001 $N^{0.240\ 379}$	298 0.0009	283 0.0010	275 0.0010	275 0.0010	275 0.0011	262 0.0004 $N^{0.058\ 303}$
247	NEUROTECHNOLOGY-009	207 0.0010	227 0.0011	223 0.0012	234 0.0015	238 0.0017	103 0.0001 $N^{0.207\ 335}$	204 0.0007	204 0.0008	196 0.0008	192 0.0008	188 0.0008	247 0.0004 $N^{0.043\ 268}$
248	NEUROTECHNOLOGY-010	152 0.0008	165 0.0009	166 0.0010	171 0.0011	174 0.0013	173 0.0001 $N^{0.170\ 251}$	171 0.0006	167 0.0007	154 0.0007	146 0.0007	144 0.0007	304 0.0005 $N^{0.021\ 171}$
249	NEUROTECHNOLOGY-012	101 0.0006	91 0.0007	85 0.0007	84 0.0008	97 0.0010	183 0.0001 $N^{0.149\ 214}$	133 0.0006	125 0.0006	113 0.0006	105 0.0006	105 0.0006	301 0.0005 $N^{0.014\ 117}$
250	NEUROTECHNOLOGY-013	96 0.0006	92 0.0007	82 0.0007	79 0.0008	76 0.0009	246 0.0001 $N^{0.124\ 159}$	146 0.0006	134 0.0006	125 0.0006	119 0.0006	115 0.0006	270 0.0004 $N^{0.023\ 183}$
251	NEUROTECHNOLOGY-014	64 0.0006	61 0.0006	63 0.0006	63 0.0007	71 0.0009	109 0.0001 $N^{0.162\ 244}$	76 0.0005	77 0.0005	83 0.0005	80 0.0005	90 0.0006	132 0.0003 $N^{0.045\ 275}$
252	NEUROTECHNOLOGY-015	41 0.0005	45 0.0005	51 0.0006	53 0.0007	63 0.0009	74 0.0000 $N^{0.175\ 270}$	49 0.0005	54 0.0005	74 0.0005	89 0.0006	99 0.0006	88 0.0002 $N^{0.080\ 344}$
253	NEUROTECHNOLOGY-016	42 0.0005	35 0.0005	38 0.0006	44 0.0006	52 0.0008	111 0.0001 $N^{0.154\ 223}$	43 0.0005	61 0.0005	65 0.0005	77 0.0005	71 0.0005	110 0.0002 $N^{0.058\ 302}$
254	NEXTMETALABS-002	69 0.0006	80 0.0006	72 0.0007	64 0.0007	70 0.0009	142 0.0001 $N^{0.152\ 219}$	48 0.0005	46 0.0005	57 0.0005	51 0.0005	55 0.0005	164 0.0003 $N^{0.026\ 198}$
255	NOMINDER-000	94 0.0006	94 0.0007	97 0.0007	98 0.0009	109 0.0010	105 0.0001 $N^{0.174\ 269}$	69 0.0005	72 0.0005	71 0.0005	73 0.0005	73 0.0005	159 0.0003 $N^{0.031\ 224}$
256	NOMINDER-001	257 0.0011	240 0.0011	220 0.0012	215 0.0013	204 0.0015	382 0.0003 $N^{0.096\ 84}$	329 0.0010	316 0.0010	296 0.0010	274 0.0010	259 0.0011	424 0.0010 $N^{0.005\ 48}$
257	NOMINDER-002	260 0.0011	242 0.0012	224 0.0012	213 0.0013	196 0.0014	405 0.0004 $N^{0.082\ 50}$	327 0.0010	314 0.0010	293 0.0010	278 0.0011	262 0.0011	412 0.0009 $N^{0.009\ 81}$
258	NOMINDER-003	261 0.0011	243 0.0012	221 0.0012	217 0.0013	198 0.0014	402 0.0004 $N^{0.084\ 53}$	342 0.0011	324 0.0011	305 0.0011	284 0.0011	268 0.0011	427 0.0010 $N^{0.003\ 24}$
259	NOMINDER-004	237 0.0010	222 0.0011	198 0.0011	193 0.0012	186 0.0014	383 0.0003 $N^{0.091\ 68}$	303 0.0010	288 0.0010	268 0.0010	256 0.0010	241 0.0010	408 0.0009 $N^{0.007\ 64}$
260	NOMINDER-005	250 0.0011	234 0.0011	217 0.0012	208 0.0013	197 0.0014	376 0.0003 $N^{0.100\ 88}$	318 0.0010	306 0.0010	285 0.0010	270 0.0010	255 0.0010	422 0.0010 $N^{0.004\ 42}$
261	NOTIONTAG-000	374 0.0019	364 0.0021	341 0.0022	333 0.0025	322 0.0029	388 0.0003 $N^{0.135\ 184}$	423 0.0016	414 0.0016	390 0.0016	368 0.0016	353 0.0017	426 0.0010 $N^{0.032\ 233}$
262	NTECHLAB-003	457 0.0045	463 0.0060	440 0.0074	434 0.0092	427 0.0112	125 0.0001 $N^{0.315\ 454}$	378 0.0012	406 0.0015	396 0.0017	402 0.0021	402 0.0025	39 0.0000 $N^{0.241\ 433}$
263	NTECHLAB-004	441 0.0035	445 0.0046	427 0.0056	416 0.0069	411 0.0083	130 0.0001 $N^{0.296\ 443}$	332 0.0010	359 0.0012	368 0.0014	362 0.0016	374 0.0019	49 0.0001 $N^{0.207\ 428}$
264	NTECHLAB-005	434 0.0034	443 0.0046	428 0.0057	419 0.0071	416 0.0090	51 0.0000 $N^{0.337\ 463}$	223 0.0008	301 0.0010	327 0.0011	350 0.0015	361 0.0018	20 0.0000 $N^{0.288\ 450}$
265	NTECHLAB-006	422 0.0029	432 0.0040	419 0.0049	409 0.0061	407 0.0077	45 0.0000 $N^{0.330\ 460}$	196 0.0007	245 0.0009	290 0.0010	318 0.0012	342 0.0015	24 0.0000 $N^{0.260\ 439}$
266	NTECHLAB-007	365 0.0018	378 0.0023	370 0.0027	368 0.0034	368 0.0041	64 0.0000 $N^{0.278\ 422}$	216 0.0008	235 0.0008	249 0.0009	277 0.0011	294 0.0012	69 0.0001 $N^{0.150\ 399}$
267	NTECHLAB-008	245 0.0011	275 0.0014	284 0.0016	291 0.0020	289 0.0024	32 0.0000 $N^{0.275\ 415}$	166 0.0006	173 0.0007	171 0.0007	187 0.0008	192 0.0008	95 0.0002 $N^{0.096\ 364}$
268	NTECHLAB-009	161 0.0008	180 0.0009	179 0.0010	183 0.0012	190 0.0014	120 0.0001 $N^{0.187\ 305}$	132 0.0006	130 0.0006	133 0.0006	138 0.0007	139 0.0007	124 0.0002 $N^{0.064\ 319}$
269	NTECHLAB-010	123 0.0007	121 0.0008	113 0.0008	115 0.0009	104 0.0010	258 0.0001 $N^{0.130\ 173}$	170 0.0006	169 0.0007	152 0.0007	140 0.0007	133 0.0007	337 0.0006 $N^{0.008\ 75}$
270	NTECHLAB-011	95 0.0006	96 0.0007	88 0.0007	91 0.0008	92 0.0010	178 0.0001 $N^{0.151\ 217}$	94 0.0005	94 0.0005	94 0.0005	90 0.0006	88 0.0006	189 0.0004 $N^{0.028\ 210}$
271	OMNIGARDE-000	87 0.0006	102 0.0007	105 0.0008	116 0.0009	123 0.0011	56 0.0000 $N^{0.203\ 327}$	54 0.0005	66 0.0005	75 0.0005	75 0.0005	74 0.0005	126 0.0002 $N^{0.049\ 283}$
272	OMNIGARDE-001	54 0.0005	48 0.0005	56 0.0006	62 0.0007	58 0.0008	118 0.0001 $N^{0.157\ 227}$	26 0.0004	26 0.0004	27 0.0005	36 0.0005	39 0.0005	135 0.0003 $N^{0.035\ 242}$
273	OMNIGARDE-002	60 0.0005	66 0.0006	65 0.0006	61 0.0007	65 0.0009	136 0.0001 $N^{0.151\ 218}$	18 0.0004	24 0.0004	32 0.0005	47 0.0005	52 0.0005	111 0.0002 $N^{0.052\ 288}$
274	ONFIDO-000	312 0.0014	336 0.0018	332 0.0021	338 0.0026	340 0.0032	40 0.0000 $N^{0.281\ 427}$	168 0.0006	187 0.0007	214 0.0008	228 0.0009	246 0.0010	58 0.0001 $N^{0.155\ 403}$
275	ONFIDO-001	232 0.0010	270 0.0013	292 0.0017	309 0.0022	321 0.0028	16 0.0000 $N^{0.348\ 464}$	101 0.0005	113 0.0006	122 0.0006	136 0.0007	147 0.0007	84 0.0001 $N^{0.097\ 366}$
276	OPTIEXACTA-000	43 0.0005	43 0.0005	39 0.0006	34 0.0006	25 0.0007	271 0.0001 $N^{0.101\ 93}$	65 0.0005	62 0.0005	56 0.0005	49 0.0005	41 0.0005	287 0.0005 $N^{0.004\ 33}$
277	PANGIAM-000	153 0.0008	146 0.0008	154 0.0009	149 0.0010	164 0.0012	188 0.0001 $N^{0.162\ 243}$	159 0.0006	170 0.0007	161 0.0007	153 0.0007	152 0.0007	204 0.0004 $N^{0.041\ 263}$
278	PANGIAM-001	482 0.0065	467 0.0065	436 0.0066	415 0.0068	402 0.0069	474 0.0049 $N^{0.021\ 4}$	538 0.0064	529 0.0064	467 0.0064	451 0.0064	437 0.0064	477 0.0063 $N^{0.002\ 13}$
279	PAPSAV1923-000	147 0.0008	142 0.0008	122 0.0008	104 0.0009	100 0.0010	338 0.0002 $N^{0.095\ 77}$	205 0.0007	197 0.0007	182 0.0007	163 0.0007	154 0.0007	374 0.0007 $N^{-0.000\ 3}$
280	PARAVISION-003	412 0.0026	411 0.0031	395 0.0034	382 0.0041	381 0.0047	294 0.0001 $N^{0.212\ 342}$	424 0.0016	419 0.0017	400 0.0018	389 0.0019	380 0.0020	306 0.0005 $N^{0.087\ 353}$
281	PARAVISION-004	268 0.0012	259 0.0013	251 0.0014	243 0.0015	241 0.0018	321 0.0002 $N^{0.139\ 193}$	291 0.0009	282 0.0009	265 0.0010	251 0.0010	250 0.0010	353 0.0006 $N^{0.027\ 201}$
282	PARAVISION-005	254 0.0011	246 0.0012	230 0.0012	222 0.0014	217 0.0016	345 0.0002 $N^{0.120\ 145}$	297 0.0009	285 0.0010	270 0.0010	254 0.0010	247 0.0010	372 0.0007 $N^{0.020\ 163}$
283	PARAVISION-007	146 0.0008	145 0.0008	139 0.0009	122 0.0010	127 0.0011	277 0.0001 $N^{0.126\ 161}$	169 0.0006	161 0.0007	150 0.0007	152 0.0007	145 0.0007	277 0.0004 $N^{0.028\ 204}$
284	PARAVISION-009	93 0.0006	89 0.0006	92 0.0007	93 0.0009	102 0.0010	108 0.0001 $N^{0.172\ 257}$	118 0.0006	124 0.0006	116 0.0006	114 0.0006	117 0.0006	137 0.0003 $N^{0.051\ 287}$
285	PARAVISION-012	80 0.0006	69 0.0006	55 0.0006	54 0.0007	54 0.0008	265 0.0001 $N^{0.111\ 118}$	123 0.0006	105 0.0006	100 0.0006	96 0.0006	87 0.0006	319 0.0005 $N^{0.005\ 45}$
286	PARAVISION-014	76 0.0006	64 0.0006	54 0.0006	59 0.0007	48 0.0008	270 0.0001 $N^{0.109\ 111}$	110 0.0005	101 0.0006	97 0.0006	92 0.0006	85 0.0006	307 0.0005 $N^{0.008\ 73}$
287	PARAVISION-016	75 0.0006	67 0.0006	61 0.0006	52 0.0007	47 0.0008	284 0.0001 $N^{0.101\ 94}$	122 0.0006	111 0.0006	102 0.0006	95 0.0006	86 0.0006	321 0.0005 $N^{0.003\ 27}$
288	PARAVISION-019	31 0.0005	29 0.0005	30 0.0005	26 0.0006	28 0.0007	205 0.0001 $N^{0.121\ 147}$	34 0.0005	38 0.0005	42 0.0005	37 0.0005	32 0.0005	178 0.0003 $N^{0.021\ 168}$

Table 48: **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES AT GIVEN RANK		ENROL MOST RECENT											
FNIR(N, T= 0, R)		RANK 1						RANK 50					
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b
289	PARAVISION-021	17 0.0005	16 0.0005	14 0.0005	23 0.0006	22 0.0007	131 0.0001 $N^{0.139\ 195}$	24 0.0004	22 0.0004	19 0.0004	17 0.0004	16 0.0004	246 0.0004 $N^{0.004\ 41}$
290	PARAVISION-022	13 0.0005	12 0.0005	8 0.0005	12 0.0006	14 0.0006	197 0.0001 $N^{0.119\ 138}$	19 0.0004	18 0.0004	14 0.0004	12 0.0004	9 0.0004	272 0.0004 $N^{0.000\ 6}$
291	PENTACOMP-001	280 0.0012	288 0.0014	281 0.0016	279 0.0019	284 0.0023	155 0.0001 $N^{0.208\ 336}$	248 0.0008	251 0.0009	243 0.0009	255 0.0010	258 0.0011	151 0.0003 $N^{0.073\ 335}$
292	PIXELALL-002	435 0.0034	435 0.0041	418 0.0048	408 0.0059	404 0.0071	230 0.0001 $N^{0.256\ 394}$	398 0.0014	413 0.0016	397 0.0017	399 0.0020	396 0.0023	73 0.0001 $N^{0.185\ 419}$
293	PIXELALL-003	337 0.0015	338 0.0018	327 0.0021	324 0.0024	323 0.0029	198 0.0001 $N^{0.213\ 343}$	317 0.0010	318 0.0011	309 0.0011	301 0.0011	295 0.0012	281 0.0004 $N^{0.060\ 310}$
294	PIXELALL-004	296 0.0013	324 0.0017	318 0.0019	322 0.0023	317 0.0027	87 0.0001 $N^{0.245\ 385}$	295 0.0009	294 0.0010	287 0.0010	286 0.0011	283 0.0011	211 0.0004 $N^{0.069\ 330}$
295	PIXELALL-005	307 0.0014	302 0.0015	285 0.0016	275 0.0018	271 0.0021	358 0.0002 $N^{0.132\ 179}$	368 0.0012	357 0.0012	341 0.0012	322 0.0012	309 0.0013	398 0.0008 $N^{0.026\ 195}$
296	PROTENET-001	363 0.0018	356 0.0020	340 0.0022	478 0.0685	325 0.0029	12 0.0000 $N^{0.620\ 472}$	400 0.0014	396 0.0014	371 0.0015	479 0.0676	342 0.0016	13 0.0000 $N^{0.569\ 472}$
297	PSL-000	18 0.0005	32 0.0005	33 0.0005	43 0.0006	49 0.0008	68 0.0000 $N^{0.173\ 262}$	13 0.0004	6 0.0004	15 0.0004	13 0.0004	10 0.0004	190 0.0004 $N^{0.012\ 105}$
298	PSL-001	19 0.0005	21 0.0005	27 0.0005	35 0.0006	35 0.0007	114 0.0001 $N^{0.149\ 213}$	12 0.0004	10 0.0004	12 0.0004	16 0.0004	17 0.0004	156 0.0003 $N^{0.021\ 165}$
299	PSL-002	6 0.0004	9 0.0004	15 0.0005	21 0.0006	31 0.0007	82 0.0000 $N^{0.161\ 241}$	10 0.0004	4 0.0004	4 0.0004	5 0.0004	7 0.0004	193 0.0004 $N^{0.010\ 89}$
300	PSL-003	7 0.0004	8 0.0004	11 0.0005	14 0.0006	13 0.0006	161 0.0001 $N^{0.128\ 167}$	9 0.0004	9 0.0004	13 0.0004	9 0.0004	8 0.0004	212 0.0004 $N^{0.009\ 78}$
301	PSL-004	9 0.0004	7 0.0004	9 0.0005	6 0.0005	6 0.0006	202 0.0001 $N^{0.115\ 131}$	15 0.0004	13 0.0004	8 0.0004	6 0.0004	5 0.0004	264 0.0004 $N^{0.000\ 2}$
302	PSL-005	15 0.0005	11 0.0005	21 0.0005	29 0.0006	38 0.0007	80 0.0000 $N^{0.164\ 249}$	7 0.0004	11 0.0004	11 0.0004	8 0.0004	13 0.0004	170 0.0003 $N^{0.017\ 146}$
303	PSL-006	8 0.0004	6 0.0004	5 0.0005	5 0.0005	5 0.0006	207 0.0001 $N^{0.112\ 123}$	6 0.0004	8 0.0004	5 0.0004	7 0.0004	6 0.0004	195 0.0004 $N^{0.011\ 92}$
304	PTAKURATSATU-000	391 0.0021	398 0.0027	392 0.0033	377 0.0036	353 0.0036	326 0.0002 $N^{0.188\ 306}$	366 0.0012	366 0.0013	374 0.0015	364 0.0016	350 0.0016	118 0.0002 $N^{0.121\ 380}$
305	PYRAMID-000	111 0.0007	115 0.0007	120 0.0008	123 0.0010	118 0.0011	172 0.0001 $N^{0.160\ 236}$	145 0.0006	138 0.0006	128 0.0006	120 0.0006	113 0.0006	288 0.0005 $N^{0.019\ 154}$
306	PYRAMID-001	149 0.0008	148 0.0008	136 0.0009	127 0.0010	126 0.0011	288 0.0001 $N^{0.123\ 156}$	174 0.0007	166 0.0007	149 0.0007	145 0.0007	141 0.0007	312 0.0005 $N^{0.020\ 160}$
307	QAZSMARTVISIONAI-000	27 0.0005	18 0.0005	23 0.0005	24 0.0006	27 0.0007	184 0.0001 $N^{0.128\ 163}$	14 0.0004	20 0.0004	18 0.0004	23 0.0005	20 0.0005	142 0.0003 $N^{0.028\ 209}$
308	QAZSMARTVISIONAI-001	2 0.0004	3 0.0004	7 0.0005	13 0.0006	10 0.0006	113 0.0001 $N^{0.141\ 200}$	2 0.0004	5 0.0004	6 0.0004	4 0.0004	2 0.0004	187 0.0004 $N^{0.011\ 95}$
309	QAZSMARTVISIONAI-002	1 0.0004	1 0.0004	2 0.0005	2 0.0005	3 0.0006	164 0.0001 $N^{0.123\ 155}$	1 0.0004	3 0.0004	1 0.0004	1 0.0004	1 0.0004	225 0.0004 $N^{0.004\ 38}$
310	QAZSMARTVISIONAI-003	5 0.0004	5 0.0004	4 0.0005	3 0.0005	2 0.0006	203 0.0001 $N^{0.112\ 124}$	4 0.0004	1 0.0004	3 0.0004	2 0.0004	4 0.0004	185 0.0004 $N^{0.011\ 91}$
311	QNAP-001	430 0.0031	427 0.0038	412 0.0043	399 0.0050	396 0.0059	312 0.0002 $N^{0.216\ 346}$	450 0.0019	444 0.0020	418 0.0021	409 0.0022	399 0.0024	340 0.0006 $N^{0.085\ 350}$
312	QNAP-002	452 0.0043	442 0.0045	416 0.0048	401 0.0051	389 0.0055	460 0.0014 $N^{0.085\ 58}$	506 0.0038	500 0.0038	453 0.0039	436 0.0039	428 0.0040	467 0.0028 $N^{0.021\ 167}$
313	QNAP-003	390 0.0021	387 0.0024	371 0.0027	362 0.0031	353 0.0036	325 0.0002 $N^{0.182\ 292}$	348 0.0011	356 0.0012	347 0.0013	341 0.0014	351 0.0016	98 0.0002 $N^{0.133\ 389}$
314	QNAP-004	302 0.0014	304 0.0016	299 0.0018	296 0.0021	297 0.0025	193 0.0001 $N^{0.204\ 329}$	312 0.0010	302 0.0010	288 0.0010	287 0.0011	282 0.0011	318 0.0005 $N^{0.047\ 280}$
315	QNAP-005	421 0.0028	405 0.0029	381 0.0030	363 0.0032	349 0.0034	454 0.0011 $N^{0.069\ 28}$	489 0.0027	473 0.0027	433 0.0027	419 0.0027	407 0.0027	465 0.0025 $N^{0.006\ 53}$
316	QNAP-006	242 0.0011	235 0.0011	232 0.0012	229 0.0015	237 0.0017	218 0.0001 $N^{0.173\ 265}$	240 0.0008	227 0.0008	224 0.0009	215 0.0009	211 0.0009	322 0.0005 $N^{0.030\ 218}$
317	QUANTASOFT-001	614 0.2174	610 0.2174	495 0.2174		473 0.2174	483 0.2174 $N^{0.000\ 1}$	611 0.1113	607 0.1113	494 0.1113		473 0.1113	483 0.1113 $N^{0.000\ 4}$
318	RANKONE-002	543 0.0151	541 0.0191	475 0.0221	463 0.0259	458 0.0300	440 0.0007 $N^{0.233\ 368}$	518 0.0045	519 0.0057	470 0.0068	461 0.0081	456 0.0098	76 0.0001 $N^{0.268\ 443}$
319	RANKONE-003	542 0.0151	542 0.0191	474 0.0221	464 0.0259	459 0.0300	441 0.0007 $N^{0.233\ 367}$	519 0.0045	520 0.0057	469 0.0068	462 0.0081	455 0.0098	75 0.0001 $N^{0.268\ 444}$
320	RANKONE-005	490 0.0072	495 0.0090	457 0.0106	447 0.0129	440 0.0153	348 0.0002 $N^{0.259\ 398}$	477 0.0023	478 0.0028	444 0.0033	435 0.0039	429 0.0047	62 0.0001 $N^{0.242\ 434}$
321	RANKONE-007	408 0.0025	407 0.0030	396 0.0035	383 0.0041	383 0.0049	233 0.0001 $N^{0.233\ 366}$	361 0.0011	369 0.0013	363 0.0014	360 0.0016	358 0.0018	82 0.0001 $N^{0.154\ 402}$
322	RANKONE-009	350 0.0016	355 0.0020	350 0.0023	352 0.0028	348 0.0034	94 0.0001 $N^{0.253\ 393}$	307 0.0010	312 0.0010	313 0.0011	307 0.0012	305 0.0012	176 0.0003 $N^{0.079\ 339}$
323	RANKONE-010	353 0.0017	348 0.0019	335 0.0022	332 0.0025	324 0.0029	266 0.0001 $N^{0.192\ 315}$	336 0.0011	333 0.0011	331 0.0012	316 0.0012	317 0.0013	218 0.0004 $N^{0.076\ 336}$
324	RANKONE-011	225 0.0010	239 0.0011	239 0.0013	237 0.0015	244 0.0018	169 0.0001 $N^{0.191\ 314}$	224 0.0008	214 0.0008	211 0.0008	208 0.0008	197 0.0009	315 0.0005 $N^{0.033\ 234}$
325	RANKONE-012	196 0.0009	206 0.0010	215 0.0012	210 0.0013	220 0.0016	146 0.0001 $N^{0.187\ 304}$	207 0.0007	206 0.0008	198 0.0008	184 0.0008	176 0.0008	326 0.0006 $N^{0.023\ 181}$
326	RANKONE-013	92 0.0006	101 0.0007	112 0.0008	124 0.0010	132 0.0011	53 0.0000 $N^{0.206\ 332}$	87 0.0005	85 0.0005	85 0.0005	82 0.0005	82 0.0006	181 0.0003 $N^{0.028\ 211}$
327	RANKONE-014	73 0.0006	76 0.0006	75 0.0007	74 0.0008	75 0.0009	115 0.0001 $N^{0.164\ 248}$	85 0.0005	93 0.0005	82 0.0005	76 0.0005	70 0.0005	250 0.0004 $N^{0.016\ 134}$
328	RANKONE-015	86 0.0006	83 0.0006	77 0.0007	80 0.0008	78 0.0009	186 0.0001 $N^{0.146\ 208}$	98 0.0005	95 0.0005	90 0.0005	85 0.0005	80 0.0006	253 0.0004 $N^{0.017\ 143}$
329	REALNETWORKS-002	572 0.0297	567 0.0391	488 0.0467	477 0.0560	471 0.0577	458 0.0013 $N^{0.237\ 373}$	527 0.0053	539 0.0075	480 0.0095	470 0.0124	463 0.0130	53 0.0001 $N^{0.324\ 458}$
330	REALNETWORKS-003	549 0.0179	549 0.0238	480 0.0287	470 0.0349	466 0.0420	410 0.0004 $N^{0.291\ 438}$	507 0.0038	511 0.0050	463 0.0061	459 0.0077	454 0.0098	40 0.0000 $N^{0.325\ 459}$
331	REALNETWORKS-004	546 0.0172	547 0.0233	479 0.0280	469 0.0344	463 0.0413	394 0.0003 $N^{0.299\ 447}$	504 0.0036	509 0.0047	462 0.0058	457 0.0074	453 0.0095	34 0.0000 $N^{0.335\ 461}$
332	REALNETWORKS-005	349 0.0016	354 0.0019	343 0.0022	337 0.0026	346 0.0033	121 0.0001 $N^{0.240\ 378}$	229 0.0008	240 0.0009	261 0.0009	267 0.0010	273 0.0011	97 0.0002 $N^{0.111\ 375}$
333	REALNETWORKS-006	200 0.0009	213 0.0010	225 0.0012	228 0.0015	233 0.0017	93 0.0001 $N^{0.210\ 341}$	153 0.0006	163 0.0007	163 0.0007	172 0.0008	178 0.0008	100 0.0002 $N^{0.091\ 358}$
334	REALNETWORKS-007	190 0.0009	195 0.0010	188 0.0011	191 0.0012	199 0.0014	215 0.0001 $N^{0.163\ 245}$	141 0.0006	151 0.0006	160 0.0007	158 0.0007	166 0.0008	101 0.0002 $N^{0.087\ 354}$
335	REALNETWORKS-008	119 0.0007	141 0.0008	146 0.0009	141 0.0010	152 0.0012	110 0.0001 $N^{0.182\ 293}$	80 0.0005	81 0.0005	99 0.0006	98 0.0006	101 0.0006	114 0.0002 $N^{0.061\ 313}$
336	REALSENSE-001	81 0.0006	87 0.0006	87 0.0007	88 0.0008	98 0.0010	79 0.0000 $N^{0.185\ 300}$	75 0.0005	74 0.0005	78 0.0005	79 0.0005	75 0.0005	149 0.0003 $N^{0.035\ 243}$
337	REALSENSE-002	52 0.0005	55 0.0006	47 0.0006	49 0.0007	50 0.0008	199 0.0001 $N^{0.131\ 174}$	66 0.0005	63 0.0005	60 0.0005	62 0.0005	67 0.0005	174 0.0003 $N^{0.026\ 196}$
338	RECOGNITO-000	53 0.0005	62 0.0006	58 0.0006	60 0.0007	57 0.0008	141 0.0001 $N^{0.148\ 212}$	57 0.0005	50 0.0005	47 0.0005	45 0.0005	44 0.0005	227 0.0004 $N^{0.015\ 124}$
339	RECOGNITO-001	61 0.0006	58 0.0006	48 0.0006	45 0.0007	39 0.0007	291 0.0001 $N^{0.097\ 86}$	84 0.0005	82 0.0005	73 0.0005	70 0.0005	64 0.0005	284 0.0005 $N^{0.009\ 77}$
340	RECOGNITO-002	44 0.0005	42 0.0005	43 0.0006	31 0.0006	26 0.0007	272 0.0001 $N^{0.101\ 92}$	62 0.0005	64 0.0005	53 0.0005	42 0.0005	37 0.0005	286 0.0005 $N^{0.004\ 32}$
341	REMARKAI-000	402 0.0024	409 0.0030	400 0.0037	393 0.0045	388 0.0055	90 0.0001 $N^{0.286\ 434}$	341 0.0011	354 0.0012	344 0.0013	346 0.0014	348 0.0016	92 0.0002 $N^{0.137\ 391}$
342	RENDIP-000	243 0.0011	248 0.0012	250 0.0014	252 0.0016	256 0.0019	144 0.0001 $N^{0.198\ 321}$	226 0.0008	233 0.0008	221 0.0008	214 0.0009	215 0.0009	314 0.0005 $N^{0.035\ 246}$
343	REVEALMEDIA-000	292 0.0013	294 0.0015	283 0.0016	283 0.0019	275 0.0021	295 0.0001 $N^{0.163\ 246}$	244 0.0008	243 0.0009	236 0.0009	232 0.0009	233 0.0010	269 0.0004 $N^{0.048\ 282}$
344	RNIT-000	212 0.0010	233 0.0011	234 0.0013	226 0.0014	226 0.0016	195 0.0001 $N^{0.180\ 283}$	260 0.0009	276 0.0009	279 0.0010	273 0.0010	270 0.0011	145 0.0003 $N^{0.079\ 343}$
345	ROC-016	77 0.0006	86 0.0006	80 0.0007	82 0.0008	74 0.0009	123 0.0001 $N^{0.161\ 240}$	89 0.0005	83 0.0005	95 0.0005	83 0.0005	76 0.0005	215 0.0004 $N^{0.024\ 186}$
346	ROC-017	67 0.0006	71 0.0006	84 0.0007	95 0.0009	95 0.0010	39 0.0000 $N^{0.210\ 339}$	79 0.0005	87 0.0005	84 0.0005	81 0.0005	78 0.0006	163 0.0003 $N^{0.033\ 236}$
347	ROC-018	50 0.0005	49 0.0005	59 0.0006	69 0.0008	66 0.0009	65 0.0000 $N^{0.181\ 288}$	68 0.0005	65 0.0005	62 0.0005	54 0.0005	56 0.0005	275 0.0004 $N^{0.007\ 67}$
348	ROC-019	33 0.0005	31 0.0005	35 0.0006	39 0.0006	44 0.0008	122 0.0001 $N^{0.147\ 211}$	53 0.0005	43 0.0005	37 0.0005	31 0.0005	27 0.0005	293 0.0005 $N^{0.000\ 9}$
349	ROC-020	40 0.0005	41 0.0005	36 0.0006	41 0.0006	43 0.0008	171 0.0001 $N^{0.136\ 187}$	50 0.0005	41 0.0005	38 0.0005	29 0.0005	35 0.0005	254 0.0004 $N^{0.007\ 66}$
350	ROC-021	20 0.0005	26 0.0005	24 0.0005	27 0.0006	29 0.0007	157 0.0001 $N^{0.134\ 182}$	38 0.0005	27 0.0005	26 0.0005	24 0.0005	22 0.0005	252 0.0004 $N^{0.006\ 55}$
351	S1-000	357 0.0017	359 0.0021	359 0.0024	354 0.0029	347 0.0034	154 0.0001 $N^{0.234\ 369}$	347 0.0011	342 0.0011	334 0.0012	324 0.0012	318 0.0013	248 0.0004 $N^{0.071\ 332}$
352	S1-001	416 0.0027	400 0.0028	385 0.0030	365 0.0033	356 0.0037	439 0.0007 $N^{0.103\ 96}$	457 0.0019	443 0.0020	415 0.0020	403 0.0021	388 0.0021	443 0.0013 $N^{0.027\ 202}$
353	S1-002	229 0.0010	218 0.0011	209 0.0011	195 0.0012	194 0.0014	353 0.0002 $N^{0.110\ 114}$	296 0.0009	286 0.0010	266 0.0010	246 0.0010	240 0.0010	386 0.0008 $N^{0.015\ 131}$
354	S1-003	228 0.0010	231 0.0011	219 0.0012	214 0.0013	206 0.0015	315 0.0002 $N^{0.131\ 175}$	255 0.0009	257 0.0009	241 0.0009	238 0.0009	228 0.0009	328 0.0006 $N^{0.033\ 237}$
355	S1-004	213 0.0010	196 0.0010	180 0.0010	176 0.0011	172 0.0013	378 0.0003 $N^{0.089\ 66}$	270 0.0009	256 0.0009	248 0.0009	234 0.0009	221 0.0009	375 0.0007 $N^{0.015\ 126}$
356	S1-005	216 0.0010	207 0.0010	201 0.0011	190 0.0012	185 0.0013	354 0.0002 $N^{0.107\ 108}$	284 0.0009	272 0.0009	253 0.0009	236 0.0009	224 0.0009	397 0.0008 $N^{0.009\ 76}$
357	S1-006	197 0.0009	188 0.0010	176 0.0010	167 0.0011	150 0.0012	380 0.0003 $N^{0.086\ 59}$	263 0.0009	250 0.0009	237 0.0009	222 0.0009	214 0.0009	387 0.0008 $N^{0.009\ 79}$
358	S1-007	174 0.0008	159 0.0009	145 0.0009	129 0.0010	120 0.0011	362 0.0002 $N^{0.088\ 65}$	219 0.0008	208 0.0008	200 0.0008	185 0.0008	175 0.0008	365 0.0007 $N^{0.008\ 72}$
359	SANSAP-001	141 0.0007	143 0.0008	144 0.0009	150 0.0010	161 0.0012	128 0.0001 $N^{0.176\ 274}$	121 0.0006	123 0.0006	118 0.0006	107 0.0006	114 0.0006	166 0.0003 $N^{0.039\ 261}$
360	SCANOVATE-000	440 0.0035	444 0.0046	426 0.0055	417 0.0069	403 0.0070	269 0.0001 $N^{0.252\ 392}$	322 0.0010	353 0.0012	352 0.0014	369 0.0017	353 0.0017	65 0.0001 $N^{0.180\ 417}$

Table 49: **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES AT GIVEN RANK FNIR(N, T= 0, R)		ENROL MOST RECENT											
		RANK 1						RANK 50					
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	$a.N^b$	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	$a.N^b$
361	SCANOVATE-001	445 0.0037	451 0.0050	430 0.0060	422 0.0076	419 0.0095	91 0.0001 $N^{0.317\ 455}$	311 0.0010	351 0.0012	353 0.0014	376 0.0017	386 0.0021	28 0.0000 $N^{0.259\ 438}$
362	SCYLLA-001	336 0.0015	341 0.0018	323 0.0020	316 0.0023	318 0.0028	243 0.0001 $N^{0.194\ 316}$	308 0.0010	309 0.0010	304 0.0011	292 0.0011	300 0.0012	241 0.0004 $N^{0.066\ 326}$
363	SECURITYPITCH-000	418 0.0028	419 0.0035	408 0.0041	402 0.0051	397 0.0062	132 0.0001 $N^{0.275\ 416}$	392 0.0013	402 0.0015	389 0.0016	381 0.0018	376 0.0020	102 0.0002 $N^{0.145\ 395}$
364	SENSETIME-000	381 0.0020	371 0.0022	357 0.0024	339 0.0027	335 0.0031	390 0.0003 $N^{0.139\ 196}$	421 0.0015	415 0.0016	392 0.0016	375 0.0017	362 0.0018	363 0.0007 $N^{0.059\ 308}$
365	SENSETIME-001	380 0.0020	372 0.0022	353 0.0024	345 0.0027	352 0.0036	308 0.0002 $N^{0.183\ 296}$	414 0.0015	410 0.0015	387 0.0016	372 0.0017	394 0.0023	127 0.0003 $N^{0.128\ 384}$
366	SENSETIME-002	537 0.0133	526 0.0133	465 0.0133	449 0.0134	436 0.0136	478 0.0120 $N^{0.007\ 2}$	575 0.0132	565 0.0132	484 0.0132	422 0.0132	464 0.0132	479 0.0131 $N^{0.001\ 11}$
367	SENSETIME-003	91 0.0006	85 0.0006	76 0.0007	75 0.0008	72 0.0009	229 0.0001 $N^{0.127\ 162}$	126 0.0006	122 0.0006	110 0.0006	110 0.0006	102 0.0006	268 0.0004 $N^{0.020\ 162}$
368	SENSETIME-004	88 0.0006	81 0.0006	67 0.0006	70 0.0008	64 0.0009	242 0.0001 $N^{0.122\ 152}$	60 0.0005	70 0.0005	67 0.0005	65 0.0005	66 0.0005	154 0.0003 $N^{0.031\ 225}$
369	SENSETIME-005	30 0.0005	34 0.0005	37 0.0006	37 0.0006	40 0.0007	158 0.0001 $N^{0.137\ 191}$	31 0.0005	34 0.0005	24 0.0005	18 0.0005	19 0.0005	266 0.0004 $N^{0.004\ 30}$
370	SENSETIME-006	32 0.0005	33 0.0005	26 0.0005	25 0.0006	20 0.0007	224 0.0001 $N^{0.112\ 125}$	35 0.0005	39 0.0005	29 0.0005	34 0.0005	31 0.0005	183 0.0003 $N^{0.019\ 156}$
371	SENSETIME-007	26 0.0005	24 0.0005	19 0.0005	15 0.0006	15 0.0006	254 0.0001 $N^{0.101\ 91}$	37 0.0005	36 0.0005	30 0.0005	30 0.0005	29 0.0005	219 0.0004 $N^{0.014\ 115}$
372	SENSETIME-008	24 0.0005	23 0.0005	22 0.0005	22 0.0006	16 0.0006	223 0.0001 $N^{0.111\ 121}$	17 0.0004	33 0.0005	25 0.0005	22 0.0005	24 0.0005	165 0.0003 $N^{0.022\ 177}$
373	SENSETIME-009	28 0.0005	20 0.0005	18 0.0005	11 0.0006	12 0.0006	275 0.0001 $N^{0.093\ 73}$	27 0.0004	31 0.0005	33 0.0005	28 0.0005	26 0.0005	200 0.0004 $N^{0.016\ 136}$
374	SENSETIME-010	14 0.0005	10 0.0005	6 0.0005	8 0.0005	8 0.0006	222 0.0001 $N^{0.108\ 109}$	16 0.0004	17 0.0004	17 0.0004	15 0.0004	15 0.0004	222 0.0004 $N^{0.008\ 74}$
375	SENSETIME-011	29 0.0005	19 0.0005	17 0.0005	10 0.0006	11 0.0006	276 0.0001 $N^{0.093\ 72}$	28 0.0004	32 0.0005	34 0.0005	27 0.0005	25 0.0005	199 0.0004 $N^{0.016\ 137}$
376	SENSETIME-012	12 0.0004	17 0.0005	10 0.0005	4 0.0005	4 0.0006	262 0.0001 $N^{0.095\ 82}$	25 0.0004	16 0.0004	16 0.0004	14 0.0004	11 0.0004	271 0.0004 $N^{0.000\ 5}$
377	SERENDIPITY-000	230 0.0010	244 0.0012	238 0.0013	239 0.0015	242 0.0018	189 0.0001 $N^{0.184\ 297}$	233 0.0008	226 0.0008	228 0.0009	225 0.0009	219 0.0009	245 0.0004 $N^{0.050\ 285}$
378	SEVENTHSENSE-001	251 0.0011	322 0.0016	388 0.0031	407 0.0058	430 0.0117	10 0.0000 $N^{0.832\ 474}$	280 0.0009	408 0.0015	438 0.0030	447 0.0056	459 0.0115	9 0.0000 $N^{0.880\ 474}$
379	SHAMAN-007	527 0.0368	568 0.0393	487 0.0413	476 0.0439	469 0.0470	477 0.0120 $N^{0.083\ 52}$	594 0.0306	587 0.0311	490 0.0316	477 0.0323	470 0.0334	480 0.0205 $N^{0.029\ 216}$
380	SIAT-001	328 0.0015	320 0.0016	309 0.0018	302 0.0021	306 0.0026	261 0.0001 $N^{0.185\ 298}$	283 0.0009	295 0.0010	292 0.0010	296 0.0011	296 0.0012	133 0.0003 $N^{0.091\ 360}$
381	SIAT-002	319 0.0014	325 0.0017	311 0.0018	300 0.0021	296 0.0025	268 0.0001 $N^{0.182\ 291}$	313 0.0010	325 0.0011	319 0.0011	311 0.0012	301 0.0012	260 0.0004 $N^{0.064\ 322}$
382	SJTU-001	21 0.0005	27 0.0005	25 0.0005	33 0.0006	24 0.0007	168 0.0001 $N^{0.133\ 180}$	20 0.0004	23 0.0004	22 0.0004	19 0.0005	18 0.0005	184 0.0004 $N^{0.016\ 140}$
383	SPARKTEK-000	134 0.0007	127 0.0008	138 0.0009	126 0.0010	141 0.0011	180 0.0001 $N^{0.161\ 237}$	131 0.0006	129 0.0006	124 0.0006	113 0.0006	108 0.0006	243 0.0004 $N^{0.026\ 194}$
384	SPARKTEK-001	62 0.0006	60 0.0006	57 0.0006	50 0.0007	53 0.0008	219 0.0001 $N^{0.124\ 157}$	93 0.0005	86 0.0005	77 0.0005	69 0.0005	69 0.0005	297 0.0005 $N^{0.007\ 61}$
385	SPARSH-001	105 0.0007	110 0.0007	111 0.0008	111 0.0009	113 0.0011	176 0.0001 $N^{0.157\ 228}$	124 0.0006	114 0.0006	104 0.0006	100 0.0006	93 0.0006	311 0.0005 $N^{0.009\ 82}$
386	SQISOFT-001	407 0.0025	428 0.0038	425 0.0055	426 0.0081	431 0.0118	14 0.0000 $N^{0.537\ 471}$	148 0.0006	176 0.0007	190 0.0008	209 0.0008	235 0.0010	52 0.0001 $N^{0.160\ 408}$
387	SQISOFT-002	157 0.0008	185 0.0009	210 0.0012	238 0.0015	258 0.0019	20 0.0000 $N^{0.301\ 449}$	116 0.0006	110 0.0006	119 0.0006	118 0.0006	126 0.0007	125 0.0002 $N^{0.060\ 309}$
388	STCON-001	49 0.0005	47 0.0005	53 0.0006	51 0.0007	42 0.0007	201 0.0001 $N^{0.129\ 170}$	59 0.0005	51 0.0005	45 0.0005	38 0.0005	43 0.0005	236 0.0004 $N^{0.013\ 109}$
389	STCON-002	38 0.0005	38 0.0005	42 0.0006	30 0.0006	32 0.0007	214 0.0001 $N^{0.119\ 137}$	56 0.0005	58 0.0005	44 0.0005	52 0.0005	50 0.0005	207 0.0004 $N^{0.018\ 153}$
390	STCON-003	37 0.0005	39 0.0005	41 0.0006	32 0.0006	30 0.0007	226 0.0001 $N^{0.114\ 126}$	63 0.0005	59 0.0005	43 0.0005	44 0.0005	49 0.0005	238 0.0004 $N^{0.013\ 106}$
391	STCON-004	36 0.0005	36 0.0005	32 0.0005	18 0.0006	23 0.0007	249 0.0001 $N^{0.105\ 99}$	45 0.0005	44 0.0005	46 0.0005	39 0.0005	30 0.0005	216 0.0004 $N^{0.015\ 132}$
392	STCON-005	34 0.0005	37 0.0005	31 0.0005	20 0.0006	18 0.0007	273 0.0001 $N^{0.097\ 85}$	47 0.0005	42 0.0005	50 0.0005	40 0.0005	34 0.0005	217 0.0004 $N^{0.015\ 133}$
393	STELLARTECS-001	345 0.0016	337 0.0018	322 0.0020	308 0.0022	304 0.0026	329 0.0002 $N^{0.158\ 230}$	393 0.0013	384 0.0014	351 0.0014	340 0.0014	329 0.0014	425 0.0010 $N^{0.023\ 182}$
394	SYNESIS-003	545 0.0157	533 0.0159	420 0.0160	456 0.0162	453 0.0250	468 0.0026 $N^{0.129\ 168}$	580 0.0156	573 0.0156	486 0.0156	475 0.0156	468 0.0242	407 0.0009 $N^{0.196\ 424}$
395	SYNESIS-003	602 0.1453	601 0.1697	493 0.1873	481 0.2085	474 0.2315	479 0.0176 $N^{0.158\ 232}$	604 0.0825	603 0.0866	493 0.0917	481 0.0995	472 0.1101	481 0.0216 $N^{0.098\ 369}$
396	SYNESIS-005	505 0.0081	488 0.0082	447 0.0082	429 0.0083	412 0.0084	476 0.0069 $N^{0.012\ 3}$	555 0.0081	546 0.0081	477 0.0081	460 0.0081	447 0.0081	478 0.0081 $N^{0.000\ 10}$
397	TECH5-001	419 0.0028	423 0.0036	413 0.0044	403 0.0053	400 0.0067	95 0.0001 $N^{0.294\ 441}$	379 0.0012	377 0.0013	367 0.0014	363 0.0016	369 0.0019	90 0.0002 $N^{0.148\ 397}$
398	TECH5-002	351 0.0017	379 0.0023	374 0.0028	367 0.0034	374 0.0044	28 0.0000 $N^{0.324\ 456}$	111 0.0006	164 0.0007	178 0.0007	198 0.0008	237 0.0010	36 0.0000 $N^{0.190\ 422}$
399	TECH5-003	106 0.0007	109 0.0007	109 0.0008	113 0.0009	114 0.0011	159 0.0001 $N^{0.161\ 238}$	119 0.0006	115 0.0006	101 0.0006	99 0.0006	92 0.0006	294 0.0005 $N^{0.013\ 108}$
400	TECHAINER-000	359 0.0017	351 0.0019	326 0.0020	320 0.0023	300 0.0025	389 0.0003 $N^{0.128\ 165}$	408 0.0014	397 0.0014	372 0.0015	354 0.0015	344 0.0016	411 0.0009 $N^{0.032\ 232}$
401	TEVIAN-005	473 0.0055	478 0.0071	448 0.0082	440 0.0103	435 0.0128	260 0.0001 $N^{0.285\ 433}$	453 0.0019	453 0.0022	430 0.0025	418 0.0027	412 0.0033	91 0.0002 $N^{0.181\ 418}$
402	TEVIAN-006	371 0.0019	361 0.0021	347 0.0023	328 0.0024	316 0.0027	408 0.0004 $N^{0.123\ 154}$	381 0.0012	368 0.0013	349 0.0013	338 0.0014	332 0.0014	346 0.0006 $N^{0.052\ 289}$
403	TEVIAN-007	300 0.0014	289 0.0014	267 0.0015	260 0.0016	243 0.0018	419 0.0004 $N^{0.090\ 67}$	269 0.0009	270 0.0009	257 0.0009	249 0.0010	243 0.0010	331 0.0006 $N^{0.035\ 247}$
404	THAKAAIT-000	204 0.0010	223 0.0011	229 0.0012	232 0.0015	235 0.0017	116 0.0001 $N^{0.203\ 328}$	184 0.0007	178 0.0007	170 0.0007	166 0.0007	169 0.0008	203 0.0004 $N^{0.045\ 276}$
405	TIGER-002	453 0.0043	456 0.0054	434 0.0066	430 0.0084	424 0.0102	153 0.0001 $N^{0.302\ 450}$	375 0.0012	395 0.0014	393 0.0016	394 0.0020	405 0.0026	31 0.0000 $N^{0.256\ 436}$
406	TIGER-004	372 0.0019	374 0.0022	358 0.0024	341 0.0027	338 0.0031	351 0.0002 $N^{0.160\ 235}$	433 0.0017	429 0.0018	407 0.0019	396 0.0020	390 0.0022	327 0.0006 $N^{0.083\ 347}$
407	TOSHIBA-000	433 0.0033	436 0.0043	420 0.0050	406 0.0058	439 0.0151	15 0.0000 $N^{0.462\ 468}$	409 0.0014	418 0.0016	399 0.0017	393 0.0019	457 0.0103	12 0.0000 $N^{0.568\ 471}$
408	TOSHIBA-002	202 0.0009	184 0.0009	165 0.0010	142 0.0010	135 0.0011	418 0.0004 $N^{0.062\ 22}$	279 0.0009	264 0.0009	242 0.0009	230 0.0009	216 0.0009	409 0.0009 $N^{0.002\ 17}$
409	TOSHIBA-003	144 0.0008	125 0.0008	107 0.0008	97 0.0009	87 0.0010	363 0.0002 $N^{0.080\ 47}$	203 0.0007	199 0.0007	181 0.0007	167 0.0007	155 0.0007	367 0.0007 $N^{0.004\ 28}$
410	TOSHIBA-004	22 0.0005	28 0.0005	28 0.0005	36 0.0006	37 0.0007	119 0.0001 $N^{0.145\ 207}$	30 0.0004	25 0.0004	23 0.0004	25 0.0005	23 0.0005	221 0.0004 $N^{0.012\ 99}$
411	TOSS-000	211 0.0010	198 0.0010	181 0.0010	168 0.0011	148 0.0012	415 0.0004 $N^{0.069\ 29}$	287 0.0009	266 0.0009	247 0.0009	235 0.0009	220 0.0009	406 0.0009 $N^{0.005\ 46}$
412	TOSS-001	221 0.0010	211 0.0010	186 0.0011	177 0.0011	167 0.0012	403 0.0004 $N^{0.074\ 37}$	282 0.0009	261 0.0009	244 0.0009	239 0.0009	230 0.0010	380 0.0008 $N^{0.014\ 122}$
413	TRUEFACE-000	415 0.0027	406 0.0029	389 0.0032	374 0.0035	363 0.0039	429 0.0005 $N^{0.128\ 164}$	468 0.0022	452 0.0022	424 0.0023	412 0.0024	401 0.0024	437 0.0012 $N^{0.044\ 274}$
414	TSIGHT-000	3 0.0004	4 0.0004	3 0.0005	581 1.0000	525 1.0000	1 0.0000 $N^{3.173\ 483}$	3 0.0004	2 0.0004	2 0.0004	596 1.0000	553 1.0000	1 0.0000 $N^{3.187\ 483}$
415	UNISSEY-003	426 0.0029	413 0.0031	391 0.0032	374 0.0034	359 0.0038	452 0.0009 $N^{0.083\ 51}$	487 0.0026	471 0.0026	432 0.0026	417 0.0027	408 0.0027	461 0.0022 $N^{0.014\ 114}$
416	UNIVERSEAI-000	171 0.0008	98 0.0007	98 0.0007	101 0.0009	134 0.0011	310 0.0002 $N^{0.108\ 110}$	215 0.0008	135 0.0006	121 0.0006	128 0.0006	157 0.0007	366 0.0007 $N^{-0.003\ 1}$
417	USEB-001	169 0.0008	162 0.0009	156 0.0009	160 0.0011	156 0.0012	280 0.0001 $N^{0.132\ 177}$	201 0.0007	191 0.0007	177 0.0007	160 0.0007	159 0.0007	357 0.0007 $N^{0.007\ 63}$
418	USEB-002	231 0.0010	217 0.0011	203 0.0011	200 0.0013	193 0.0014	352 0.0002 $N^{0.110\ 115}$	288 0.0009	267 0.0009	245 0.0009	233 0.0009	223 0.0009	403 0.0009 $N^{0.006\ 54}$
419	USEB-003	165 0.0008	161 0.0009	155 0.0009	147 0.0010	155 0.0012	283 0.0001 $N^{0.129\ 171}$	191 0.0007	188 0.0007	179 0.0007	164 0.0007	160 0.0008	330 0.0006 $N^{0.018\ 151}$
420	USEB-004	201 0.0009	187 0.0010	173 0.0010	169 0.0011	162 0.0012	364 0.0002 $N^{0.096\ 83}$	262 0.0009	249 0.0009	234 0.0009	220 0.0009	209 0.0009	401 0.0008 $N^{0.004\ 35}$
421	VD-001	558 0.0227	557 0.0272	483 0.0312	473 0.0360	465 0.0414	461 0.0014 $N^{0.206\ 333}$	572 0.0117	564 0.0127	485 0.0137	474 0.0151	466 0.0167	463 0.0022 $N^{0.123\ 382}$
422	VDFSOFTWARE-002	301 0.0014	334 0.0018	346 0.0023	355 0.0029	362 0.0038	19 0.0000 $N^{0.358\ 465}$	104 0.0005	137 0.0006	165 0.0007	178 0.0008	207 0.0009	43 0.0001 $N^{0.168\ 412}$
423	VERIDAS-001	373 0.0019	385 0.0024	376 0.0028	372 0.0034	370 0.0041	106 0.0001 $N^{0.261\ 401}$	326 0.0010	335 0.0011	330 0.0012	327 0.0013	331 0.0014	122 0.0002 $N^{0.107\ 372}$
424	VERIDAS-002	370 0.0019	384 0.0024	375 0.0028	370 0.0034	369 0.0041	102 0.0001 $N^{0.261\ 402}$	323 0.0010	334 0.0011	329 0.0012	325 0.0013	330 0.0014	120 0.0002 $N^{0.108\ 374}$
425	VERIDAS-003	286 0.0013	291 0.0015	282 0.0016	278 0.0019	280 0.0022	241 0.0001 $N^{0.181\ 289}$	271 0.0009	279 0.0009	267 0.0010	259 0.0010	267 0.0011	242 0.0004 $N^{0.059\ 307}$
426	VERIDAS-004	210 0.0010	199 0.0010	184 0.0011	152 0.0011	143 0.0012	424 0.0004 $N^{0.058\ 17}$	247 0.0008	244 0.0009	230 0.0009	217 0.0009	208 0.0009	373 0.0007 $N^{0.012\ 101}$
427	VERIGRAM-000	389 0.0021	370 0.0022	337 0.0022	310 0.0022	285 0.0023	462 0.0015 $N^{0.025\ 5}$	464 0.0021	448 0.0021	417 0.0021	404 0.0021	385 0.0021	458 0.0020 $N^{0.002\ 19}$
428	VERIGRAM-001	386 0.0021	367 0.0021	336 0.0022	318 0.0023	288 0.0023	457 0.0012 $N^{0.042\ 11}$	462 0.0020	446 0.0020	416 0.0020	400 0.0020	381 0.0020	457 0.0019 $N^{0.003\ 22}$
429	VERIGRAM-002	394 0.0022	375 0.0022	349 0.0023	323 0.0024	294 0.0025	459 0.0014 $N^{0.036\ 9}$	470 0.0022	452 0.0022	420 0.0022	407 0.0022	391 0.0022	460 0.0021 $N^{0.001\ 12}$
430	VIANTE-000	186 0.0009	175 0.0009	164 0.0009	145 0.0010	137 0.0011	370 0.0003 $N^{0.088\ 64}$	246 0.0008	236 0.0008	222 0.0008	213 0.0009	196 0.0009	391 0.0008 $N^{0.005\ 49}$
431	VIANTE-001	181 0.0009	166 0.0009	151 0.0009	138 0.0010	119 0.0011	381 0.0003 $N^{0.079\ 46}$	245 0.0008	231 0.0008	223 0.0008	210 0.0008	191 0.0008	395 0.0008 $N^{0.003\ 23}$
432	VIANTE-002	48 0.0005	50 0.0005	46 0.0006	42 0.0006	36 0.0007	244 0.0001 $N^{0.111\ 122}$	74 0.0005	73 0.0005	70 0.0005	67 0.0005	60 0.0005	283 0.0004 $N^{0.008\ 70}$

Table 50: **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES AT GIVEN RANK FNIR(N, T= 0, R)		ENROL MOST RECENT											
		RANK 1						RANK 50					
#	ALGORITHM	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b	N=0.64M	N=1.6M	N=3.0M	N=6.0M	N=12.0M	aN^b
433	VIANTE-003	[51] 0.0005	[44] 0.0005	[45] 0.0006	[40] 0.0006	[34] 0.0007	[259] 0.0001 N[0.107 103]	[70] 0.0005	[76] 0.0005	[68] 0.0005	[63] 0.0005	[57] 0.0005	[282] 0.0004 N[0.008 69]
434	VIETTELAI-000	[449] 0.0040	[434] 0.0041	[410] 0.0042	[390] 0.0043	[378] 0.0045	[466] 0.0024 N[0.039 10]	[508] 0.0038	[499] 0.0038	[452] 0.0038	[433] 0.0039	[421] 0.0039	[468] 0.0031 N[0.013 112]
435	VIETTELAI-001	[347] 0.0016	[329] 0.0017	[300] 0.0018	[282] 0.0019	[274] 0.0021	[428] 0.0005 N[0.093 71]	[413] 0.0015	[403] 0.0015	[376] 0.0015	[356] 0.0015	[343] 0.0015	[441] 0.0013 N[0.012 104]
436	VIETTELAI-002	[346] 0.0016	[326] 0.0017	[301] 0.0018	[281] 0.0019	[276] 0.0021	[425] 0.0004 N[0.094 75]	[412] 0.0015	[404] 0.0015	[375] 0.0015	[355] 0.0015	[341] 0.0015	[440] 0.0013 N[0.012 103]
437	VIETTELAI-003	[78] 0.0006	[82] 0.0006	[90] 0.0007	[90] 0.0008	[86] 0.0010	[85] 0.0001 N[0.179 280]	[83] 0.0005	[75] 0.0005	[72] 0.0005	[78] 0.0005	[79] 0.0006	[171] 0.0003 N[0.030 217]
438	VIETTELAI-004	[58] 0.0005	[57] 0.0006	[60] 0.0006	[58] 0.0007	[55] 0.0008	[182] 0.0001 N[0.139 194]	[73] 0.0005	[69] 0.0005	[63] 0.0005	[60] 0.0005	[58] 0.0005	[229] 0.0004 N[0.007 65]
439	VIGILANTSOLUTIONS-008	[388] 0.0021	[394] 0.0026	[386] 0.0031	[381] 0.0037	[373] 0.0044	[149] 0.0001 N[0.250 389]	[232] 0.0008	[271] 0.0009	[301] 0.0011	[315] 0.0012	[319] 0.0013	[55] 0.0001 N[0.175 415]
440	VINBIGDATA-001	[139] 0.0007	[158] 0.0009	[171] 0.0010	[178] 0.0011	[192] 0.0014	[49] 0.0000 N[0.223 353]	[81] 0.0005	[92] 0.0005	[88] 0.0005	[87] 0.0006	[91] 0.0006	[138] 0.0003 N[0.043 272]
441	VINBIGDATA-002	[65] 0.0006	[65] 0.0006	[68] 0.0006	[65] 0.0008	[85] 0.0009	[70] 0.0000 N[0.183 294]	[55] 0.0005	[53] 0.0005	[52] 0.0005	[43] 0.0005	[42] 0.0005	[231] 0.0004 N[0.015 127]
442	VINBIGDATA-003	[55] 0.0005	[54] 0.0006	[70] 0.0007	[66] 0.0008	[79] 0.0009	[59] 0.0000 N[0.190 310]	[64] 0.0005	[60] 0.0005	[58] 0.0005	[56] 0.0005	[48] 0.0005	[224] 0.0004 N[0.016 141]
443	VINBIGDATA-004	[66] 0.0006	[56] 0.0006	[62] 0.0006	[57] 0.0007	[62] 0.0008	[151] 0.0001 N[0.146 209]	[61] 0.0005	[52] 0.0005	[41] 0.0005	[53] 0.0005	[51] 0.0005	[192] 0.0004 N[0.020 161]
444	VISIONBOX-000	[304] 0.0014	[309] 0.0016	[308] 0.0018	[596] 1.0000	[480] 0.9526	[2] 0.0000 N[2.661 481]	[264] 0.0009	[265] 0.0009	[269] 0.0010	[488] 1.0000	[480] 0.9525	[2] 0.0000 N[2.857 482]
445	VISIONBOX-001	[405] 0.0025	[395] 0.0026	[373] 0.0027	[359] 0.0030	[342] 0.0033	[442] 0.0007 N[0.094 74]	[473] 0.0023	[462] 0.0023	[425] 0.0023	[411] 0.0023	[397] 0.0024	[456] 0.0019 N[0.015 129]
446	VISIONBOX-002	[303] 0.0014	[292] 0.0015	[278] 0.0016	[273] 0.0018	[272] 0.0021	[332] 0.0002 N[0.142 203]	[362] 0.0011	[355] 0.0012	[339] 0.0012	[319] 0.0012	[309] 0.0012	[381] 0.0008 N[0.031 226]
447	VISIONLABS-004	[387] 0.0021	[396] 0.0026	[387] 0.0031	[389] 0.0043	[401] 0.0068	[17] 0.0000 N[0.395 467]	[374] 0.0012	[378] 0.0013	[391] 0.0016	[414] 0.0024	[427] 0.0044	[17] 0.0000 N[0.440 466]
448	VISIONLABS-005	[368] 0.0019	[376] 0.0023	[372] 0.0027	[376] 0.0035	[382] 0.0049	[27] 0.0000 N[0.328 459]	[367] 0.0012	[360] 0.0013	[379] 0.0015	[388] 0.0019	[409] 0.0029	[22] 0.0000 N[0.303 454]
449	VISIONLABS-006	[334] 0.0015	[333] 0.0018	[328] 0.0021	[346] 0.0027	[364] 0.0040	[24] 0.0000 N[0.324 457]	[365] 0.0011	[358] 0.0012	[364] 0.0014	[384] 0.0018	[406] 0.0026	[25] 0.0000 N[0.281 448]
450	VISIONLABS-007	[329] 0.0015	[331] 0.0017	[316] 0.0019	[314] 0.0023	[343] 0.0033	[72] 0.0000 N[0.256 396]	[359] 0.0011	[352] 0.0012	[337] 0.0012	[328] 0.0013	[370] 0.0019	[77] 0.0001 N[0.156 404]
451	VISIONLABS-008	[327] 0.0015	[318] 0.0016	[298] 0.0018	[305] 0.0022	[310] 0.0027	[213] 0.0001 N[0.199 322]	[389] 0.0013	[382] 0.0013	[357] 0.0014	[366] 0.0016	[375] 0.0020	[104] 0.0002 N[0.140 394]
452	VISIONLABS-009	[136] 0.0007	[133] 0.0008	[135] 0.0009	[151] 0.0011	[189] 0.0014	[47] 0.0000 N[0.219 349]	[154] 0.0006	[154] 0.0006	[162] 0.0007	[180] 0.0008	[251] 0.0010	[51] 0.0001 N[0.158 406]
453	VISIONLABS-010	[223] 0.0010	[214] 0.0010	[212] 0.0012	[216] 0.0013	[231] 0.0017	[200] 0.0001 N[0.176 272]	[274] 0.0009	[255] 0.0009	[271] 0.0010	[281] 0.0011	[315] 0.0013	[93] 0.0002 N[0.121 381]
454	VISIONLABS-011	[143] 0.0008	[147] 0.0008	[148] 0.0009	[157] 0.0011	[201] 0.0015	[50] 0.0000 N[0.219 351]	[182] 0.0007	[182] 0.0007	[193] 0.0008	[207] 0.0008	[281] 0.0011	[54] 0.0001 N[0.162 409]
455	VISTEAM-000	[436] 0.0034	[448] 0.0048	[429] 0.0060	[423] 0.0077	[421] 0.0097	[33] 0.0000 N[0.361 466]	[213] 0.0008	[269] 0.0010	[342] 0.0012	[348] 0.0014	[359] 0.0018	[19] 0.0000 N[0.288 451]
456	VIXVIZION-009	[341] 0.0016	[349] 0.0019	[345] 0.0023	[351] 0.0028	[344] 0.0033	[78] 0.0000 N[0.262 403]	[188] 0.0007	[232] 0.0008	[251] 0.0009	[258] 0.0010	[272] 0.0011	[68] 0.0001 N[0.148 398]
457	VNPAY-001	[239] 0.0011	[228] 0.0011	[218] 0.0012	[211] 0.0013	[202] 0.0015	[337] 0.0002 N[0.118 134]	[275] 0.0009	[262] 0.0009	[252] 0.0009	[252] 0.0010	[239] 0.0010	[341] 0.0006 N[0.030 220]
458	VNPT-001	[355] 0.0017	[339] 0.0018	[317] 0.0019	[299] 0.0021	[291] 0.0024	[395] 0.0003 N[0.119 136]	[404] 0.0014	[398] 0.0014	[369] 0.0015	[351] 0.0015	[335] 0.0015	[428] 0.0010 N[0.022 174]
459	VNPT-002	[321] 0.0015	[301] 0.0015	[280] 0.0016	[270] 0.0017	[259] 0.0019	[427] 0.0005 N[0.087 61]	[399] 0.0014	[385] 0.0014	[361] 0.0014	[343] 0.0014	[328] 0.0014	[435] 0.0012 N[0.012 100]
460	VNPT-003	[276] 0.0012	[257] 0.0013	[249] 0.0014	[231] 0.0015	[227] 0.0016	[391] 0.0003 N[0.100 87]	[356] 0.0011	[341] 0.0011	[323] 0.0011	[302] 0.0011	[291] 0.0012	[413] 0.0009 N[0.015 128]
461	VNPT-004	[275] 0.0012	[258] 0.0013	[241] 0.0013	[224] 0.0014	[224] 0.0016	[398] 0.0003 N[0.092 70]	[358] 0.0011	[345] 0.0011	[324] 0.0011	[304] 0.0011	[286] 0.0012	[423] 0.0010 N[0.011 96]
462	VNPT-005	[278] 0.0012	[261] 0.0013	[240] 0.0013	[220] 0.0014	[214] 0.0016	[422] 0.0004 N[0.077 41]	[353] 0.0011	[336] 0.0011	[320] 0.0011	[297] 0.0011	[284] 0.0011	[414] 0.0009 N[0.014 113]
463	VOCORD-005	[474] 0.0057	[474] 0.0068	[444] 0.0079	[438] 0.0095	[429] 0.0114	[353] 0.0002 N[0.238 375]	[498] 0.0031	[492] 0.0033	[449] 0.0034	[432] 0.0038	[428] 0.0041	[410] 0.0009 N[0.092 361]
464	VOGIC-000	[392] 0.0022	[383] 0.0024	[365] 0.0026	[349] 0.0028	[341] 0.0032	[417] 0.0004 N[0.129 169]	[442] 0.0018	[432] 0.0018	[405] 0.0019	[390] 0.0019	[377] 0.0020	[430] 0.0011 N[0.036 249]
465	VTS-001	[220] 0.0010	[245] 0.0012	[247] 0.0013	[255] 0.0016	[261] 0.0019	[81] 0.0000 N[0.226 357]	[156] 0.0006	[158] 0.0007	[164] 0.0007	[159] 0.0007	[171] 0.0008	[113] 0.0002 N[0.077 337]
466	VTS-002	[291] 0.0013	[310] 0.0016	[312] 0.0019	[317] 0.0023	[320] 0.0028	[62] 0.0000 N[0.256 395]	[194] 0.0007	[205] 0.0008	[220] 0.0008	[226] 0.0009	[245] 0.0010	[87] 0.0002 N[0.113 376]
467	VTS-003	[132] 0.0007	[140] 0.0008	[132] 0.0009	[137] 0.0010	[151] 0.0012	[134] 0.0001 N[0.173 264]	[86] 0.0005	[90] 0.0005	[93] 0.0005	[88] 0.0006	[98] 0.0006	[141] 0.0003 N[0.043 269]
468	VTS-004	[85] 0.0006	[88] 0.0006	[91] 0.0007	[86] 0.0008	[90] 0.0010	[107] 0.0001 N[0.170 252]	[44] 0.0005	[57] 0.0005	[59] 0.0005	[64] 0.0005	[62] 0.0005	[131] 0.0003 N[0.041 262]
469	VTS-005	[74] 0.0006	[73] 0.0006	[83] 0.0007	[81] 0.0008	[84] 0.0009	[83] 0.0001 N[0.177 277]	[58] 0.0005	[49] 0.0005	[48] 0.0005	[46] 0.0005	[45] 0.0005	[228] 0.0004 N[0.015 125]
470	WINK-001	[383] 0.0021	[391] 0.0026	[380] 0.0030	[375] 0.0035	[371] 0.0043	[152] 0.0001 N[0.249 387]	[340] 0.0011	[347] 0.0011	[340] 0.0012	[332] 0.0013	[336] 0.0015	[115] 0.0002 N[0.114 378]
471	WINK-002	[176] 0.0008	[190] 0.0010	[208] 0.0011	[212] 0.0013	[228] 0.0016	[55] 0.0000 N[0.227 359]	[102] 0.0005	[116] 0.0006	[115] 0.0006	[115] 0.0006	[127] 0.0007	[108] 0.0002 N[0.069 331]
472	WINK-003	[377] 0.0020	[357] 0.0020	[331] 0.0021	[311] 0.0022	[292] 0.0024	[447] 0.0008 N[0.067 27]	[449] 0.0018	[433] 0.0018	[404] 0.0019	[387] 0.0019	[372] 0.0019	[453] 0.0015 N[0.015 130]
473	XAILIENT-000	[375] 0.0019	[386] 0.0024	[379] 0.0030	[378] 0.0036	[380] 0.0046	[46] 0.0000 N[0.298 444]	[266] 0.0009	[287] 0.0010	[291] 0.0010	[314] 0.0012	[312] 0.0013	[85] 0.0001 N[0.132 388]
474	XAILIENT-001	[206] 0.0010	[212] 0.0010	[199] 0.0011	[192] 0.0012	[200] 0.0014	[289] 0.0001 N[0.140 198]	[225] 0.0008	[212] 0.0008	[208] 0.0008	[199] 0.0008	[205] 0.0009	[298] 0.0005 N[0.036 252]
475	XFORWARDAI-000	[362] 0.0018	[350] 0.0019	[325] 0.0020	[321] 0.0023	[305] 0.0026	[385] 0.0003 N[0.130 172]	[422] 0.0016	[412] 0.0016	[385] 0.0016	[367] 0.0016	[352] 0.0017	[432] 0.0011 N[0.022 175]
476	XFORWARDAI-001	[352] 0.0017	[328] 0.0017	[294] 0.0017	[277] 0.0018	[264] 0.0020	[438] 0.0007 N[0.066 26]	[420] 0.0015	[411] 0.0015	[384] 0.0015	[358] 0.0015	[345] 0.0016	[451] 0.0014 N[0.005 44]
477	XFORWARDAI-002	[342] 0.0016	[313] 0.0016	[287] 0.0017	[268] 0.0017	[248] 0.0018	[448] 0.0008 N[0.047 14]	[417] 0.0015	[409] 0.0015	[381] 0.0015	[357] 0.0015	[339] 0.0015	[450] 0.0014 N[0.004 37]
478	YITU-002	[283] 0.0013	[300] 0.0015	[305] 0.0018	[301] 0.0021	[307] 0.0026	[73] 0.0000 N[0.247 386]	[180] 0.0007	[196] 0.0007	[199] 0.0008	[194] 0.0008	[210] 0.0009	[105] 0.0002 N[0.095 363]
479	YITU-003	[398] 0.0023	[392] 0.0026	[377] 0.0028	[361] 0.0031	[354] 0.0036	[379] 0.0003 N[0.154 224]	[441] 0.0018	[430] 0.0018	[409] 0.0019	[395] 0.0020	[387] 0.0021	[393] 0.0008 N[0.059 306]
480	YITU-004	[180] 0.0009	[202] 0.0010	[233] 0.0012	[230] 0.0015	[376] 0.0044	[13] 0.0000 N[0.500 470]	[144] 0.0006	[143] 0.0006	[138] 0.0006	[124] 0.0006	[414] 0.0033	[14] 0.0000 N[0.470 467]
481	YITU-005	[376] 0.0019	[362] 0.0021	[342] 0.0022	[327] 0.0024	[319] 0.0028	[416] 0.0004 N[0.120 144]	[436] 0.0017	[423] 0.0017	[398] 0.0017	[379] 0.0017	[357] 0.0018	[449] 0.0014 N[0.014 120]
482	YOUVERSE-001	[164] 0.0008	[169] 0.0009	[169] 0.0010	[163] 0.0011	[176] 0.0013	[228] 0.0001 N[0.150 216]	[164] 0.0006	[153] 0.0006	[151] 0.0007	[157] 0.0007	[153] 0.0007	[173] 0.0003 N[0.047 279]
483	YUAN-000	[299] 0.0014	[279] 0.0014	[262] 0.0014	[245] 0.0015	[520] 1.0000	[5] 0.0000 N[1.807 479]	[397] 0.0013	[379] 0.0013	[350] 0.0013	[334] 0.0013	[481] 1.0000	[5] 0.0000 N[1.793 479]

Table 51: **Investigation-mode: Effect of N on FNIR on recent images** *For five enrollment population sizes, N, with T = 0 and FPIR = 1. The left five columns are rank 1 miss rates The right five columns are rank 50 miss rates Missing entries usually apply because another algorithm from the same developer was run instead. Some developers are missing because less accurate algorithms were not run on galleries with $N > 1\,600\,000$. Throughout blue superscripts indicate the rank of the algorithm for that column, and yellow highlighting indicates the most accurate value. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES OUTSIDE RANK R		RESOURCE USAGE		ENROL MOST RECENT, N = 1.6M					
FNIR(N, T=0, R)		TEMPLATE		FRVT 2018 MUGSHOTS					
#	ALGORITHM	BYTES	MSEC	R=1	R=5	R=10	R=20	R=50	WORK-10
1	20FACE-000	637 -	637 -	579 0.0549	571 0.0265	566 0.0194	560 0.0143	550 0.0095	573 1.272
2	3DIVI-003	637 -	637 -	589 0.0830	583 0.0441	582 0.0346	578 0.0267	575 0.0188	585 1.444
3	3DIVI-004	637 -	637 -	536 0.0172	522 0.0088	519 0.0072	513 0.0058	507 0.0046	527 1.089
4	3DIVI-005	637 -	637 -	537 0.0173	523 0.0088	518 0.0071	511 0.0058	508 0.0046	528 1.090
5	3DIVI-006	637 -	637 -	548 0.0237	556 0.0168	562 0.0157	563 0.0151	569 0.0145	554 1.160
6	AADI-000	637 -	637 -	633 0.9991	633 0.9987	633 0.9984	633 0.9979	633 0.9969	633 9.988
7	AADI-001	637 -	637 -	629 0.7887	629 0.7331	629 0.7098	630 0.6856	630 0.6528	629 7.655
8	AADI-002	637 -	637 -	628 0.6994	628 0.6287	628 0.5993	626 0.5688	626 0.5255	628 6.734
9	ACER-000	637 -	637 -	506 0.0102	482 0.0048	476 0.0037	470 0.0030	458 0.0022	484 1.050
10	ACER-001	637 -	637 -	447 0.0047	449 0.0029	444 0.0024	440 0.0021	434 0.0019	445 1.028
11	ADIANCE-001	637 -	637 -	634 1.0000	635 1.0000	636 1.0000	636 1.0000	634 1.0000	636 10.000
12	ADIANCE-002	637 -	637 -	250 0.0012	249 0.0010	237 0.0009	235 0.0009	230 0.0008	247 1.009
13	ADVANCE-000	637 -	637 -	314 0.0016	326 0.0013	328 0.0012	320 0.0011	322 0.0011	319 1.012
14	AFISBIOMETRICS-000	637 -	637 -	256 0.0012	164 0.0008	139 0.0007	136 0.0006	104 0.0006	183 1.007
15	AFRENGINE-000	637 -	637 -	404 0.0029	390 0.0019	388 0.0017	393 0.0015	391 0.0014	393 1.018
16	AFRENGINE-001	637 -	637 -	273 0.0013	323 0.0013	339 0.0013	352 0.0013	361 0.0013	311 1.012
17	AFRENGINE-002	637 -	637 -	274 0.0013	321 0.0013	336 0.0013	350 0.0013	363 0.0013	309 1.012
18	AFRENGINE-003	637 -	637 -	380 0.0023	420 0.0022	433 0.0022	443 0.0022	454 0.0022	411 1.020
19	AIGROUP-000	637 -	637 -	226 0.0011	195 0.0008	185 0.0008	191 0.0008	200 0.0007	200 1.008
20	AIZE-001	637 -	637 -	453 0.0053	453 0.0033	458 0.0030	459 0.0027	464 0.0023	453 1.032
21	AIZE-002	637 -	637 -	412 0.0031	395 0.0019	395 0.0017	399 0.0016	400 0.0014	399 1.019
22	AIZE-006	637 -	637 -	424 0.0036	413 0.0021	402 0.0018	396 0.0016	381 0.0013	417 1.021
23	AKIYAMA-000	637 -	637 -	475 0.0069	497 0.0060	505 0.0059	512 0.0058	522 0.0057	492 1.055
24	AKIYAMA-001	637 -	637 -	553 0.0256	568 0.0254	574 0.0253	577 0.0253	582 0.0253	568 1.229
25	AKUVOX-001	637 -	637 -	280 0.0014	294 0.0012	300 0.0011	308 0.0011	320 0.0011	285 1.011
26	ALCHERA-000	637 -	637 -	532 0.0159	539 0.0122	544 0.0114	549 0.0108	555 0.0102	538 1.114
27	ALCHERA-001	637 -	637 -	632 0.9869	632 0.9782	632 0.9735	632 0.9679	631 0.9589	632 9.811
28	ALCHERA-002	637 -	637 -	592 0.0946	590 0.0552	588 0.0440	586 0.0351	581 0.0251	590 1.542
29	ALCHERA-003	637 -	637 -	505 0.0102	488 0.0051	487 0.0043	485 0.0036	484 0.0030	490 1.053
30	ALCHERA-004	637 -	637 -	510 0.0106	480 0.0045	471 0.0035	463 0.0028	455 0.0022	482 1.048
31	ALCHERA-005	637 -	637 -	476 0.0070	450 0.0030	441 0.0023	426 0.0019	405 0.0015	452 1.032
32	ALCHERA-006	637 -	637 -	168 0.0009	178 0.0008	188 0.0008	194 0.0008	194 0.0007	176 1.007
33	ALCHERA-007	637 -	637 -	164 0.0009	166 0.0008	169 0.0007	171 0.0007	189 0.0007	165 1.007
34	ALCHERA-008	637 -	637 -	191 0.0010	220 0.0009	227 0.0009	239 0.0009	252 0.0009	213 1.008
35	ALGORITHMSINC-000	637 -	637 -	306 0.0016	333 0.0013	345 0.0013	357 0.0013	376 0.0013	329 1.012
36	ALLGOVISION-000	637 -	637 -	513 0.0111	516 0.0080	522 0.0074	523 0.0069	528 0.0064	516 1.076
37	ALLGOVISION-001	637 -	637 -	491 0.0086	478 0.0045	474 0.0036	469 0.0030	465 0.0023	477 1.045
38	ALLGOVISION-002	637 -	637 -	426 0.0037	421 0.0022	404 0.0018	397 0.0016	374 0.0013	420 1.022
39	ALLGOVISION-003	637 -	637 -	335 0.0018	329 0.0013	329 0.0012	345 0.0012	349 0.0011	333 1.012
40	ALLGOVISION-004	637 -	637 -	502 0.0101	477 0.0044	461 0.0033	451 0.0025	439 0.0019	481 1.046
41	ALLGOVISION-005	637 -	637 -	330 0.0017	337 0.0013	332 0.0012	334 0.0012	339 0.0011	331 1.012
42	ALTTEKGLOBAL-000	637 -	637 -	614 0.2740	617 0.2740	620 0.2740	621 0.2740	625 0.2740	617 3.466
43	ANKE-000	637 -	637 -	524 0.0129	506 0.0070	500 0.0057	500 0.0047	498 0.0037	511 1.070
44	ANKE-001	637 -	637 -	525 0.0129	507 0.0070	502 0.0058	501 0.0047	497 0.0037	512 1.070
45	ANKE-002	637 -	637 -	389 0.0024	377 0.0016	371 0.0015	375 0.0014	370 0.0013	378 1.016
46	ARATEK-001	637 -	637 -	134 0.0008	127 0.0007	128 0.0006	129 0.0006	133 0.0006	129 1.006
47	ARMATURA-000	637 -	637 -	299 0.0015	357 0.0015	370 0.0015	383 0.0015	401 0.0015	353 1.013
48	AWARE-003	637 -	637 -	561 0.0303	554 0.0158	549 0.0124	543 0.0097	538 0.0072	555 1.160
49	AWARE-004	637 -	637 -	584 0.0676	579 0.0345	576 0.0271	572 0.0205	568 0.0142	579 1.351
50	AWARE-005	637 -	637 -	562 0.0308	555 0.0164	551 0.0130	546 0.0104	543 0.0079	557 1.164
51	AWARE-006	637 -	637 -	586 0.0694	581 0.0366	577 0.0284	574 0.0220	572 0.0155	581 1.368
52	AWARE-007	637 -	637 -	366 0.0021	315 0.0012	291 0.0011	265 0.0010	238 0.0009	327 1.012
53	AWIROS-000	637 -	637 -	480 0.0073	483 0.0048	488 0.0044	495 0.0041	501 0.0039	479 1.046
54	AYONIX-000	637 -	637 -	621 0.4503	622 0.3538	622 0.3174	622 0.2831	621 0.2379	622 4.286
55	AYONIX-001	637 -	637 -	616 0.3412	614 0.2335	612 0.1974	609 0.1649	609 0.1271	614 3.223
56	AYONIX-002	637 -	637 -	615 0.3412	615 0.2336	613 0.1975	608 0.1649	610 0.1271	615 3.224
57	BIOIDTECHSWISS-001	637 -	637 -	139 0.0008	149 0.0007	157 0.0007	153 0.0007	152 0.0006	144 1.006
58	BIOMETRIK-001	637 -	637 -	439 0.0044	423 0.0023	416 0.0019	405 0.0016	390 0.0014	427 1.023
59	CAMVI-003	637 -	637 -	578 0.0517	589 0.0514	590 0.0514	593 0.0514	594 0.0514	587 1.463
60	CAMVI-004	637 -	637 -	576 0.0466	586 0.0463	589 0.0463	590 0.0462	593 0.0462	583 1.417
61	CAMVI-005	637 -	637 -	583 0.0650	591 0.0646	595 0.0646	596 0.0645	599 0.0645	591 1.582
62	CANON-001	637 -	637 -	113 0.0007	120 0.0006	113 0.0006	109 0.0006	108 0.0006	118 1.006
63	CANON-002	637 -	637 -	144 0.0008	113 0.0006	107 0.0006	99 0.0006	103 0.0006	113 1.006
64	CANON-003	637 -	637 -	59 0.0006	49 0.0005	42 0.0005	40 0.0005	45 0.0005	50 1.005
65	CIB-000	637 -	637 -	247 0.0012	236 0.0010	228 0.0009	220 0.0008	217 0.0008	241 1.009
66	CLEARVIEWAI-000	637 -	637 -	114 0.0007	116 0.0006	117 0.0006	104 0.0006	102 0.0006	116 1.006
67	CLEARVIEWAI-001	637 -	637 -	118 0.0007	132 0.0007	144 0.0007	154 0.0007	156 0.0007	137 1.006
68	CLEARVIEWAI-002	637 -	637 -	124 0.0008	136 0.0007	146 0.0007	151 0.0007	159 0.0007	136 1.006
69	CLEARVIEWAI-003	637 -	637 -	119 0.0008	138 0.0007	141 0.0007	146 0.0007	162 0.0007	135 1.006
70	CLOUDWALK-HR-000	637 -	637 -	236 0.0011	282 0.0011	292 0.0011	301 0.0011	313 0.0010	272 1.010
71	CLOUDWALK-MT-000	637 -	637 -	285 0.0014	349 0.0014	358 0.0014	372 0.0014	389 0.0014	338 1.013
72	CLOUDWALK-MT-001	637 -	637 -	283 0.0014	347 0.0014	357 0.0014	371 0.0014	388 0.0014	337 1.013

Table 52: **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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MISSES OUTSIDE RANK R		RESOURCE USAGE		ENROL MOST RECENT, N = 1.6M					
FNIR(N, T=0, R)		TEMPLATE		FRVT 2018 MUGSHOTS					
#	ALGORITHM	BYTES	MSEC	R=1	R=5	R=10	R=20	R=50	WORK-10
73	CLOUDWALK-MT-002	637 -	637 -	284 0.0014	348 0.0014	355 0.0014	373 0.0014	382 0.0014	339 1.013
74	COGENT-000	637 -	637 -	504 0.0101	529 0.0093	532 0.0092	464 0.0028	450 0.0021	522 1.085
75	COGENT-001	637 -	637 -	503 0.0101	528 0.0093	533 0.0092	465 0.0028	449 0.0021	523 1.085
76	COGENT-002	637 -	637 -	416 0.0033	394 0.0019	390 0.0017	384 0.0015	362 0.0013	398 1.019
77	COGENT-003	637 -	637 -	418 0.0035	411 0.0021	399 0.0018	400 0.0016	386 0.0014	415 1.020
78	COGENT-004	637 -	637 -	315 0.0016	311 0.0012	305 0.0011	314 0.0011	311 0.0010	307 1.011
79	COGENT-005	637 -	637 -	269 0.0013	277 0.0011	272 0.0010	284 0.0010	293 0.0010	274 1.010
80	COGENT-006	637 -	637 -	162 0.0009	176 0.0008	172 0.0007	183 0.0007	192 0.0007	171 1.007
81	COGENT-007	637 -	637 -	116 0.0007	121 0.0006	126 0.0006	131 0.0006	132 0.0006	125 1.006
82	COGNITEC-000	637 -	637 -	550 0.0249	545 0.0133	541 0.0104	535 0.0082	526 0.0062	546 1.133
83	COGNITEC-001	637 -	637 -	515 0.0114	495 0.0058	493 0.0047	492 0.0039	480 0.0031	496 1.059
84	COGNITEC-002	637 -	637 -	454 0.0053	455 0.0033	452 0.0028	452 0.0026	463 0.0023	454 1.032
85	COGNITEC-003	637 -	637 -	460 0.0059	461 0.0037	462 0.0033	471 0.0030	475 0.0028	463 1.036
86	COGNITEC-004	637 -	637 -	403 0.0028	379 0.0017	354 0.0014	335 0.0012	315 0.0010	383 1.016
87	COGNITEC-005	637 -	637 -	265 0.0013	224 0.0009	217 0.0009	210 0.0008	209 0.0008	231 1.009
88	COGNITEC-006	637 -	637 -	249 0.0012	222 0.0009	212 0.0008	207 0.0008	207 0.0008	220 1.008
89	COGNITEC-007	637 -	637 -	208 0.0010	170 0.0008	168 0.0007	166 0.0007	165 0.0007	178 1.007
90	COGNITEC-008	637 -	637 -	189 0.0010	150 0.0007	142 0.0007	147 0.0007	149 0.0006	156 1.007
91	COGNITEC-009	637 -	637 -	241 0.0012	219 0.0009	220 0.0009	222 0.0008	225 0.0008	221 1.008
92	COGNITEC-010	637 -	637 -	157 0.0008	144 0.0007	148 0.0007	143 0.0006	148 0.0006	149 1.006
93	COGNITI-000	637 -	637 -	303 0.0015	302 0.0012	310 0.0011	315 0.0011	326 0.0011	298 1.011
94	COMPAL-000	637 -	637 -	281 0.0014	309 0.0012	324 0.0012	332 0.0012	344 0.0011	302 1.011
95	COMPAL-001	637 -	637 -	472 0.0067	504 0.0066	512 0.0066	519 0.0065	532 0.0065	499 1.059
96	COMPAL-002	637 -	637 -	471 0.0067	503 0.0066	511 0.0065	518 0.0065	531 0.0065	498 1.059
97	CORSIGHT-000	637 -	637 -	120 0.0008	118 0.0006	123 0.0006	132 0.0006	145 0.0006	120 1.006
98	CORSIGHT-001	637 -	637 -	123 0.0008	134 0.0007	143 0.0007	140 0.0006	142 0.0006	133 1.006
99	CORSIGHT-002	637 -	637 -	194 0.0010	209 0.0009	221 0.0009	232 0.0009	242 0.0009	204 1.008
100	CUBOX-000	637 -	637 -	216 0.0011	259 0.0010	274 0.0010	286 0.0010	298 0.0010	253 1.009
101	CUBOX-001	637 -	637 -	174 0.0009	196 0.0008	203 0.0008	216 0.0008	224 0.0008	192 1.008
102	CUBOX-002	637 -	637 -	238 0.0011	272 0.0011	290 0.0011	300 0.0011	317 0.0011	268 1.010
103	CUBOX-003	637 -	637 -	253 0.0012	297 0.0012	316 0.0012	330 0.0011	340 0.0011	288 1.011
104	CYBERLINK-000	637 -	637 -	421 0.0036	432 0.0024	436 0.0022	439 0.0021	435 0.0019	431 1.023
105	CYBERLINK-001	637 -	637 -	414 0.0032	401 0.0020	392 0.0017	380 0.0015	380 0.0013	405 1.019
106	CYBERLINK-002	637 -	637 -	327 0.0023	397 0.0019	407 0.0019	414 0.0018	422 0.0017	392 1.018
107	CYBERLINK-003	637 -	637 -	252 0.0012	239 0.0010	235 0.0009	221 0.0008	215 0.0008	244 1.009
108	CYBERLINK-004	637 -	637 -	268 0.0013	290 0.0011	302 0.0011	305 0.0011	319 0.0011	283 1.011
109	CYBERLINK-005	637 -	637 -	290 0.0015	310 0.0012	318 0.0012	324 0.0011	327 0.0011	304 1.011
110	CYBERLINK-006	637 -	637 -	137 0.0008	157 0.0007	151 0.0007	155 0.0007	168 0.0007	151 1.007
111	DAHUA-000	637 -	637 -	496 0.0091	500 0.0064	504 0.0059	509 0.0056	516 0.0052	502 1.061
112	DAHUA-001	637 -	637 -	466 0.0064	464 0.0039	469 0.0034	474 0.0031	476 0.0028	467 1.038
113	DAHUA-002	637 -	637 -	293 0.0015	287 0.0011	296 0.0011	294 0.0010	292 0.0010	293 1.011
114	DAHUA-003	637 -	637 -	153 0.0008	100 0.0006	96 0.0006	87 0.0005	79 0.0005	111 1.006
115	DAHUA-004	637 -	637 -	107 0.0007	101 0.0006	100 0.0006	97 0.0006	98 0.0005	104 1.006
116	DAON-000	637 -	637 -	425 0.0037	456 0.0034	465 0.0033	478 0.0033	491 0.0033	451 1.031
117	DAON-001	637 -	637 -	551 0.0252	567 0.0249	572 0.0249	576 0.0248	580 0.0248	567 1.225
118	DAON-002	637 -	637 -	177 0.0009	148 0.0007	140 0.0007	144 0.0006	144 0.0006	153 1.007
119	DAON-003	637 -	637 -	106 0.0007	93 0.0006	90 0.0006	84 0.0005	84 0.0005	94 1.005
120	DATECH-000	637 -	637 -	193 0.0010	174 0.0008	173 0.0007	180 0.0007	186 0.0007	175 1.007
121	DATECH-001	637 -	637 -	128 0.0008	122 0.0006	119 0.0006	117 0.0006	119 0.0006	122 1.006
122	DECATUR-000	637 -	637 -	332 0.0017	312 0.0012	306 0.0011	297 0.0011	281 0.0009	318 1.012
123	DEEPGLINT-001	637 -	637 -	229 0.0011	257 0.0010	260 0.0010	266 0.0010	274 0.0009	251 1.009
124	DEEPSEA-001	637 -	637 -	430 0.0040	389 0.0019	367 0.0015	346 0.0012	303 0.0010	406 1.019
125	DERMALOG-003	637 -	637 -	598 0.1256	595 0.0741	593 0.0600	592 0.0477	590 0.0345	595 1.729
126	DERMALOG-004	637 -	637 -	597 0.1248	594 0.0737	592 0.0595	591 0.0473	589 0.0341	594 1.725
127	DERMALOG-005	637 -	637 -	529 0.0147	542 0.0127	548 0.0123	554 0.0121	563 0.0120	539 1.116
128	DERMALOG-006	637 -	637 -	486 0.0079	505 0.0067	509 0.0064	515 0.0063	525 0.0062	504 1.062
129	DERMALOG-007	637 -	637 -	493 0.0089	499 0.0063	499 0.0057	506 0.0054	512 0.0050	497 1.059
130	DERMALOG-008	637 -	637 -	390 0.0025	376 0.0016	366 0.0014	355 0.0013	350 0.0012	377 1.016
131	DERMALOG-009	637 -	637 -	388 0.0024	407 0.0020	418 0.0020	425 0.0019	436 0.0019	403 1.019
132	DERMALOG-010	637 -	637 -	342 0.0018	384 0.0017	391 0.0017	407 0.0017	420 0.0017	376 1.016
133	DERMALOG-011	637 -	637 -	255 0.0012	208 0.0009	199 0.0008	195 0.0008	183 0.0007	210 1.008
134	DERMALOG-012	637 -	637 -	327 0.0017	316 0.0012	325 0.0012	325 0.0011	338 0.0011	316 1.012
135	DERMALOG-013	637 -	637 -	308 0.0016	292 0.0012	298 0.0011	295 0.0010	310 0.0010	295 1.011
136	DERMALOG-014	637 -	637 -	410 0.0030	442 0.0027	448 0.0026	458 0.0026	470 0.0026	437 1.025
137	DERMALOG-015	637 -	637 -	254 0.0012	270 0.0011	282 0.0011	293 0.0010	308 0.0010	270 1.010
138	DERMALOG-016	637 -	637 -	221 0.0011	262 0.0010	273 0.0010	285 0.0010	304 0.0010	254 1.009
139	DERMALOG-017	637 -	637 -	225 0.0011	264 0.0010	275 0.0010	287 0.0010	305 0.0010	255 1.009
140	DIGIDATA-000	637 -	637 -	625 0.5896	626 0.5890	626 0.5889	627 0.5889	628 0.5889	626 6.302
141	DILUSENSE-000	637 -	637 -	344 0.0018	305 0.0012	286 0.0011	277 0.0010	254 0.0009	308 1.011
142	DILUSENSE-001	637 -	637 -	230 0.0011	215 0.0009	213 0.0008	217 0.0008	213 0.0008	214 1.008
143	EINETWORKSINDIA-000	637 -	637 -	286 0.0014	289 0.0011	301 0.0011	311 0.0011	323 0.0011	290 1.011
144	EINETWORKSINDIA-001	637 -	637 -	277 0.0014	273 0.0011	281 0.0010	282 0.0010	289 0.0010	278 1.010

Table 53: **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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MISSES OUTSIDE RANK R		RESOURCE USAGE		ENROL MOST RECENT, N = 1.6M					
FNIR(N, T=0, R)		TEMPLATE		FRVT 2018 MUGSHOTS					
#	ALGORITHM	BYTES	MSEC	R=1	R=5	R=10	R=20	R=50	WORK-10
145	EINETWORKSINDIA-002	637 -	637 -	135 0.0008	139 0.0007	134 0.0006	142 0.0006	155 0.0006	132 1.006
146	EYEDEA-003	637 -	637 -	588 0.0797	584 0.0447	583 0.0359	579 0.0286	577 0.0208	586 1.445
147	F8-001	637 -	637 -	519 0.0116	531 0.0101	536 0.0099	542 0.0097	549 0.0095	530 1.093
148	FACEHAWK-001	637 -	637 -	560 0.0300	569 0.0257	573 0.0249	575 0.0244	579 0.0239	569 1.236
149	FACEPHI-000	637 -	637 -	295 0.0015	351 0.0014	361 0.0014	366 0.0014	383 0.0014	344 1.013
150	FACEPHI-001	637 -	637 -	112 0.0007	156 0.0007	156 0.0007	161 0.0007	171 0.0007	145 1.006
151	FASTLAHORE-000	637 -	637 -	555 0.0261	560 0.0178	559 0.0155	558 0.0138	560 0.0115	559 1.169
152	FCI-000	637 -	637 -	296 0.0015	339 0.0013	344 0.0013	354 0.0013	365 0.0013	332 1.012
153	FINCORE-000	637 -	637 -	509 0.0105	485 0.0049	477 0.0038	472 0.0030	461 0.0023	486 1.051
154	FIRSTCREDITKZ-001	637 -	637 -	156 0.0008	190 0.0008	202 0.0008	214 0.0008	222 0.0008	186 1.007
155	FISSIONSFR-001	637 -	637 -	593 0.0980	599 0.0976	600 0.0975	602 0.0975	604 0.0975	598 1.879
156	FPT-000	637 -	637 -	305 0.0016	345 0.0014	353 0.0014	363 0.0013	373 0.0013	343 1.013
157	FUJITSULAB-000	637 -	637 -	346 0.0018	324 0.0013	319 0.0012	310 0.0011	296 0.0010	325 1.012
158	FUJITSULAB-001	637 -	637 -	297 0.0015	299 0.0012	308 0.0011	298 0.0011	307 0.0010	299 1.011
159	GLORY-000	637 -	637 -	602 0.1778	604 0.1388	604 0.1263	604 0.1151	605 0.1004	603 2.295
160	GLORY-001	637 -	637 -	599 0.1265	598 0.0964	598 0.0866	599 0.0775	600 0.0670	599 1.900
161	GORILLA-001	637 -	637 -	580 0.0600	574 0.0300	570 0.0227	568 0.0171	559 0.0114	574 1.306
162	GORILLA-002	637 -	637 -	543 0.0194	524 0.0090	513 0.0068	504 0.0052	503 0.0039	532 1.094
163	GORILLA-003	637 -	637 -	565 0.0359	548 0.0144	540 0.0103	531 0.0076	514 0.0052	552 1.156
164	GORILLA-004	637 -	637 -	461 0.0059	448 0.0028	439 0.0023	427 0.0019	399 0.0014	450 1.030
165	GORILLA-005	637 -	637 -	402 0.0028	359 0.0015	347 0.0013	319 0.0011	275 0.0009	373 1.015
166	GORILLA-006	637 -	637 -	282 0.0014	213 0.0009	209 0.0008	205 0.0008	198 0.0007	239 1.009
167	GORILLA-007	637 -	637 -	271 0.0013	204 0.0009	196 0.0008	179 0.0007	179 0.0007	218 1.008
168	GORILLA-008	637 -	637 -	237 0.0011	188 0.0008	182 0.0008	175 0.0007	181 0.0007	195 1.008
169	GRIAULE-000	637 -	637 -	369 0.0021	370 0.0016	374 0.0015	376 0.0014	369 0.0013	369 1.015
170	GRIAULE-001	637 -	637 -	150 0.0008	153 0.0007	159 0.0007	160 0.0007	172 0.0007	154 1.007
171	GTELOTS-000	637 -	637 -	606 0.1940	610 0.1938	611 0.1938	614 0.1938	618 0.1938	610 2.745
172	GTELOTS-001	637 -	637 -	531 0.0155	549 0.0151	557 0.0151	562 0.0150	570 0.0150	548 1.136
173	HIK-003	637 -	637 -	516 0.0115	494 0.0058	491 0.0046	489 0.0038	481 0.0029	500 1.059
174	HIK-004	637 -	637 -	514 0.0111	492 0.0057	490 0.0045	483 0.0035	477 0.0028	494 1.058
175	HIK-005	637 -	637 -	438 0.0044	428 0.0024	411 0.0019	401 0.0016	392 0.0014	430 1.023
176	HIK-006	637 -	637 -	437 0.0044	427 0.0024	412 0.0019	402 0.0016	393 0.0014	429 1.023
177	HISIGN-001	637 -	637 -	160 0.0009	131 0.0007	133 0.0006	128 0.0006	136 0.0006	141 1.006
178	HYPERVERGE-001	637 -	637 -	210 0.0010	231 0.0009	246 0.0009	258 0.0009	278 0.0009	230 1.009
179	HYPERVERGE-002	637 -	637 -	204 0.0010	233 0.0009	248 0.0009	255 0.0009	268 0.0009	229 1.009
180	HYPERVERGE-003	637 -	637 -	224 0.0011	244 0.0010	247 0.0009	259 0.0009	277 0.0009	236 1.009
181	HYUNDAI-ROBOTICSLAB-000	637 -	637 -	382 0.0023	405 0.0020	420 0.0020	428 0.0019	438 0.0019	397 1.019
182	HYUNDAI-ROBOTICSLAB-001	637 -	637 -	278 0.0014	280 0.0011	271 0.0010	273 0.0010	253 0.0009	280 1.010
183	HYUNDAI-ROBOTICSLAB-002	637 -	637 -	105 0.0007	103 0.0006	106 0.0006	106 0.0006	107 0.0006	105 1.006
184	HYUNDAI-ROBOTICSLAB-003	637 -	637 -	90 0.0007	77 0.0005	79 0.0005	80 0.0005	78 0.0005	83 1.005
185	HZAILU-000	637 -	637 -	343 0.0018	319 0.0013	321 0.0012	318 0.0011	331 0.0011	322 1.012
186	HZAILU-001	637 -	637 -	312 0.0016	340 0.0014	341 0.0013	347 0.0012	348 0.0011	336 1.012
187	HZAILU-002	637 -	637 -	316 0.0016	335 0.0013	337 0.0013	340 0.0012	346 0.0011	335 1.012
188	HZAILU-003	637 -	637 -	263 0.0013	269 0.0011	272 0.0010	283 0.0010	263 0.0009	269 1.010
189	HZAILU-004	637 -	637 -	182 0.0009	211 0.0009	225 0.0009	234 0.0009	241 0.0009	205 1.008
190	HZAILU-005	637 -	637 -	181 0.0009	206 0.0009	219 0.0009	229 0.0009	239 0.0009	203 1.008
191	HZAILU-006	637 -	637 -	176 0.0009	205 0.0009	218 0.0009	228 0.0009	237 0.0009	202 1.008
192	I2V-001	637 -	637 -	393 0.0026	433 0.0024	446 0.0024	447 0.0024	467 0.0024	422 1.022
193	ICEHOUSE-000	637 -	637 -	440 0.0044	440 0.0026	437 0.0022	433 0.0020	424 0.0017	443 1.026
194	ICEHOUSE-001	637 -	637 -	201 0.0010	194 0.0008	192 0.0008	198 0.0008	201 0.0007	194 1.008
195	ICM-000	637 -	637 -	449 0.0049	475 0.0043	485 0.0042	496 0.0042	505 0.0041	470 1.040
196	ICM-001	637 -	637 -	399 0.0028	417 0.0022	424 0.0021	432 0.0020	440 0.0019	412 1.020
197	IDBIO-000	637 -	637 -	323 0.0017	258 0.0010	223 0.0009	193 0.0008	180 0.0007	273 1.010
198	IDEMIA-003	637 -	637 -	470 0.0066	472 0.0041	472 0.0035	473 0.0031	468 0.0024	471 1.040
199	IDEMIA-004	637 -	637 -	465 0.0063	459 0.0035	453 0.0029	448 0.0024	426 0.0018	458 1.035
200	IDEMIA-005	637 -	637 -	485 0.0077	470 0.0041	463 0.0033	467 0.0029	474 0.0027	473 1.042
201	IDEMIA-006	637 -	637 -	499 0.0092	484 0.0048	480 0.0039	482 0.0035	493 0.0034	483 1.049
202	IDEMIA-007	637 -	637 -	373 0.0022	314 0.0012	288 0.0011	251 0.0009	228 0.0008	328 1.012
203	IDEMIA-008	637 -	637 -	99 0.0007	94 0.0006	99 0.0006	103 0.0006	99 0.0005	96 1.005
204	IDEMIA-009	637 -	637 -	75 0.0006	67 0.0005	69 0.0005	69 0.0005	68 0.0005	70 1.005
205	IDEMIA-010	637 -	637 -	46 0.0005	52 0.0005	60 0.0005	64 0.0005	67 0.0005	51 1.005
206	IDEMIA-011	637 -	637 -	97 0.0007	133 0.0007	145 0.0007	149 0.0007	157 0.0007	127 1.006
207	IDEMIA-012	637 -	637 -	104 0.0007	129 0.0007	137 0.0007	141 0.0006	150 0.0006	126 1.006
208	IDEMIA-013	637 -	637 -	2 0.0004	3 0.0004	4 0.0004	5 0.0004	7 0.0004	3 1.004
209	IDENTY-000	637 -	637 -	635 1.0000	634 1.0000	634 1.0000	634 1.0000	635 1.0000	634 10.000
210	IDENTY-001	637 -	637 -	203 0.0010	242 0.0010	241 0.0009	248 0.0009	260 0.0009	225 1.009
211	IDENTY-002	637 -	637 -	95 0.0007	111 0.0006	110 0.0006	112 0.0006	117 0.0006	106 1.006
212	IMAGUS-002	637 -	637 -	611 0.2200	603 0.1339	602 0.1087	600 0.0868	598 0.0629	604 2.305
213	IMAGUS-003	637 -	637 -	618 0.3557	616 0.2489	616 0.2129	610 0.1788	611 0.1394	616 3.360
214	IMAGUS-005	637 -	637 -	311 0.0016	317 0.0012	304 0.0011	292 0.0010	291 0.0010	310 1.012
215	IMAGUS-006	637 -	637 -	319 0.0016	328 0.0013	322 0.0012	316 0.0011	329 0.0011	320 1.012
216	IMAGUS-007	637 -	637 -	321 0.0016	291 0.0011	279 0.0010	261 0.0009	259 0.0009	296 1.011

Table 54: **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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MISSES OUTSIDE RANK R		RESOURCE USAGE		ENROL MOST RECENT, N = 1.6M					
FNIR(N, T=0, R)		TEMPLATE		FRVT 2018 MUGSHOTS					
#	ALGORITHM	BYTES	MSEC	R=1	R=5	R=10	R=20	R=50	WORK-10
217	IMAGUS-008	637 -	637 -	591 0.0857	592 0.0698	594 0.0643	594 0.0587	595 0.0514	592 1.646
218	IMDS-SOFTWARE-000	637 -	637 -	130 0.0008	114 0.0006	109 0.0006	111 0.0006	109 0.0006	114 1.006
219	IMDS-SOFTWARE-001	637 -	637 -	136 0.0008	123 0.0006	124 0.0006	125 0.0006	121 0.0006	123 1.006
220	IMDS-SOFTWARE-002	637 -	637 -	131 0.0008	112 0.0006	108 0.0006	108 0.0006	106 0.0006	115 1.006
221	IMPERIAL-000	637 -	637 -	365 0.0021	360 0.0015	369 0.0015	374 0.0014	375 0.0013	364 1.014
222	INCODE-000	637 -	637 -	577 0.0486	570 0.0258	567 0.0201	566 0.0157	558 0.0114	570 1.259
223	INCODE-001	637 -	637 -	534 0.0164	518 0.0082	510 0.0065	505 0.0053	506 0.0041	520 1.084
224	INCODE-002	637 -	637 -	538 0.0175	521 0.0088	514 0.0068	508 0.0054	504 0.0041	526 1.089
225	INCODE-003	637 -	637 -	523 0.0127	498 0.0062	494 0.0049	490 0.0038	482 0.0029	506 1.064
226	INCODE-004	637 -	637 -	415 0.0032	402 0.0020	401 0.0018	404 0.0016	407 0.0015	407 1.019
227	INCODE-005	637 -	637 -	267 0.0013	261 0.0010	268 0.0010	262 0.0009	258 0.0009	262 1.010
228	INCODE-006	637 -	637 -	108 0.0007	146 0.0007	161 0.0007	167 0.0007	177 0.0007	140 1.006
229	INCODE-007	637 -	637 -	40 0.0005	40 0.0005	34 0.0005	33 0.0005	29 0.0005	38 1.004
230	INNOV8TIF-000	637 -	637 -	590 0.0856	588 0.0479	584 0.0376	581 0.0296	578 0.0210	589 1.475
231	INNOVATRICS-002	637 -	637 -	574 0.0448	577 0.0340	579 0.0319	582 0.0306	585 0.0295	578 1.319
232	INNOVATRICS-003	637 -	637 -	554 0.0260	541 0.0124	534 0.0092	525 0.0072	513 0.0051	543 1.127
233	INNOVATRICS-004	637 -	637 -	521 0.0120	496 0.0059	492 0.0047	488 0.0037	479 0.0029	503 1.061
234	INNOVATRICS-005	637 -	637 -	363 0.0021	354 0.0015	346 0.0013	341 0.0012	330 0.0011	356 1.014
235	INNOVATRICS-007	637 -	637 -	272 0.0013	268 0.0010	258 0.0010	243 0.0009	247 0.0009	265 1.010
236	INNOVATRICS-008	637 -	637 -	183 0.0009	177 0.0008	177 0.0008	184 0.0007	184 0.0007	174 1.007
237	INNOVATRICS-009	637 -	637 -	200 0.0010	217 0.0009	222 0.0009	233 0.0009	246 0.0009	215 1.008
238	INNOVATRICS-010	637 -	637 -	215 0.0010	234 0.0009	239 0.0009	245 0.0009	248 0.0009	228 1.009
239	INNOVATRICS-011	637 -	637 -	122 0.0008	142 0.0007	136 0.0007	134 0.0006	140 0.0006	138 1.006
240	INNOVATRICS-012	637 -	637 -	117 0.0007	126 0.0007	132 0.0006	137 0.0006	131 0.0006	128 1.006
241	INNOVATRICS-013	637 -	637 -	70 0.0006	54 0.0005	38 0.0005	37 0.0005	48 0.0005	59 1.005
242	INNOVATRICS-014	637 -	637 -	52 0.0006	55 0.0005	66 0.0005	52 0.0005	35 0.0005	57 1.005
243	INNOVATRICS-015	637 -	637 -	51 0.0006	46 0.0005	46 0.0005	36 0.0005	28 0.0005	47 1.005
244	INNOVATRICS-016	637 -	637 -	25 0.0005	21 0.0005	22 0.0005	24 0.0005	19 0.0004	22 1.004
245	INSPUR-000	637 -	637 -	79 0.0006	53 0.0005	62 0.0005	61 0.0005	55 0.0005	63 1.005
246	INSPUR-001	637 -	637 -	72 0.0006	60 0.0005	50 0.0005	52 0.0005	56 0.0005	65 1.005
247	INSPUR-002	637 -	637 -	63 0.0006	57 0.0005	33 0.0005	32 0.0005	40 0.0005	54 1.005
248	INTELIGENSIA-000	637 -	637 -	317 0.0016	285 0.0011	263 0.0010	241 0.0009	223 0.0008	286 1.011
249	INTELIGENSIA-001	637 -	637 -	100 0.0007	98 0.0006	94 0.0006	98 0.0006	96 0.0005	99 1.005
250	INTELLIVISION-001	637 -	637 -	566 0.0361	564 0.0195	560 0.0156	556 0.0123	548 0.0091	563 1.196
251	INTELLIVISION-002	637 -	637 -	507 0.0103	486 0.0051	483 0.0040	479 0.0034	472 0.0027	488 1.052
252	INTELRESEARCH-001	637 -	637 -	84 0.0006	76 0.0005	75 0.0005	77 0.0005	71 0.0005	76 1.005
253	INTEMA-000	637 -	637 -	132 0.0008	154 0.0007	160 0.0007	163 0.0007	174 0.0007	147 1.006
254	INTEMA-001	637 -	637 -	13 0.0005	20 0.0005	20 0.0005	21 0.0004	15 0.0004	18 1.004
255	INTOZI-001	637 -	637 -	251 0.0012	203 0.0009	187 0.0008	169 0.0007	175 0.0007	207 1.008
256	INTSYSMSU-000	637 -	637 -	600 0.1454	602 0.1317	605 0.1269	605 0.1222	608 0.1160	602 2.200
257	IONETWORKS-001	637 -	637 -	417 0.0034	451 0.0032	460 0.0032	475 0.0031	487 0.0031	449 1.029
258	IREX-000	637 -	637 -	433 0.0040	467 0.0039	481 0.0039	494 0.0039	502 0.0039	462 1.036
259	IRISID-001	637 -	637 -	219 0.0011	163 0.0007	158 0.0007	148 0.0007	127 0.0006	168 1.007
260	IRISID-002	637 -	637 -	173 0.0009	130 0.0007	127 0.0006	123 0.0006	112 0.0006	142 1.006
261	ISTL-001	637 -	637 -	368 0.0021	391 0.0019	405 0.0018	417 0.0018	425 0.0018	388 1.017
262	ISYSTEMS-002	637 -	637 -	464 0.0061	471 0.0041	475 0.0037	480 0.0034	490 0.0032	469 1.039
263	ISYSTEMS-003	637 -	637 -	450 0.0049	460 0.0037	468 0.0034	477 0.0032	485 0.0031	457 1.034
264	KAKAO-000	637 -	637 -	232 0.0011	171 0.0008	162 0.0007	145 0.0007	141 0.0006	179 1.007
265	KAKAO-001	637 -	637 -	209 0.0010	247 0.0010	256 0.0010	270 0.0010	280 0.0009	238 1.009
266	KAKAO-002	637 -	637 -	197 0.0010	246 0.0010	254 0.0010	256 0.0009	269 0.0009	235 1.009
267	KEDACOM-001	637 -	637 -	481 0.0074	509 0.0070	515 0.0069	522 0.0069	534 0.0068	505 1.064
268	KEYLESS-000	637 -	637 -	103 0.0007	95 0.0006	89 0.0006	93 0.0005	88 0.0005	95 1.005
269	KLOUDSPOT-001	637 -	637 -	358 0.0020	332 0.0013	317 0.0012	326 0.0011	321 0.0011	340 1.013
270	KNERON-000	637 -	637 -	458 0.0056	489 0.0056	498 0.0056	510 0.0056	518 0.0056	485 1.050
271	KNERON-001	637 -	637 -	559 0.0292	573 0.0292	578 0.0292	580 0.0292	584 0.0291	571 1.262
272	KNOWUTECH-000	637 -	637 -	126 0.0008	84 0.0006	77 0.0005	78 0.0005	80 0.0005	91 1.005
273	KOGNIZA-000	637 -	637 -	260 0.0013	286 0.0011	303 0.0011	313 0.0011	332 0.0011	281 1.010
274	LEBENTECH-000	637 -	637 -	623 0.5395	625 0.5376	625 0.5371	625 0.5367	627 0.5362	625 5.841
275	LIGHTVISION-000	637 -	637 -	397 0.0027	412 0.0021	421 0.0020	430 0.0020	442 0.0019	408 1.020
276	LINE-000	637 -	637 -	347 0.0019	304 0.0012	278 0.0010	242 0.0009	220 0.0008	306 1.011
277	LINE-001	637 -	637 -	129 0.0008	117 0.0006	120 0.0006	114 0.0006	118 0.0006	119 1.006
278	LINECLOVA-002	637 -	637 -	186 0.0009	202 0.0009	207 0.0008	219 0.0008	229 0.0008	199 1.008
279	LINECLOVA-003	637 -	637 -	77 0.0006	64 0.0005	47 0.0005	56 0.0005	30 0.0005	64 1.005
280	LOOKMAN-003	637 -	637 -	490 0.0084	513 0.0075	521 0.0073	529 0.0072	537 0.0070	509 1.068
281	LOOKMAN-004	637 -	637 -	492 0.0087	514 0.0075	520 0.0073	526 0.0072	536 0.0070	510 1.069
282	LOOKMAN-005	637 -	637 -	484 0.0077	510 0.0071	517 0.0070	524 0.0069	535 0.0069	507 1.065
283	LSSYTECH-000	637 -	637 -	68 0.0006	83 0.0006	84 0.0005	90 0.0005	97 0.0005	77 1.005
284	LSSYTECH-001	637 -	637 -	53 0.0006	75 0.0005	76 0.0005	83 0.0005	91 0.0005	73 1.005
285	LUXAND-001	637 -	637 -	401 0.0028	399 0.0020	403 0.0018	408 0.0017	416 0.0016	401 1.019
286	MANTRA-000	637 -	637 -	276 0.0014	250 0.0010	250 0.0009	238 0.0009	234 0.0008	259 1.009
287	MAVERICKRESEARCH-000	637 -	637 -	508 0.0104	520 0.0085	523 0.0081	533 0.0080	542 0.0078	517 1.079
288	MAXVISION-000	637 -	637 -	360 0.0021	341 0.0014	330 0.0012	321 0.0011	297 0.0010	350 1.013

Table 55: **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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MISSES OUTSIDE RANK R		RESOURCE USAGE		ENROL MOST RECENT, N = 1.6M					
FNIR(N, T=0, R)		TEMPLATE		FRVT 2018 MUGSHOTS					
#	ALGORITHM	BYTES	MSEC	R=1	R=5	R=10	R=20	R=50	WORK-10
289	MAXVISION-001	637 -	637 -	155 0.0008	179 0.0008	181 0.0008	190 0.0008	195 0.0007	169 1.007
290	MAXVISION-002	637 -	637 -	163 0.0009	180 0.0008	189 0.0008	202 0.0008	210 0.0008	173 1.007
291	MAXVISION-003	637 -	637 -	172 0.0009	175 0.0008	184 0.0008	189 0.0008	193 0.0007	172 1.007
292	MAXVISION-004	637 -	637 -	154 0.0008	186 0.0008	192 0.0008	208 0.0008	216 0.0008	180 1.007
293	MEGVII-001	637 -	637 -	518 0.0116	526 0.0092	526 0.0086	537 0.0083	545 0.0080	524 1.085
294	MEGVII-002	637 -	637 -	520 0.0117	527 0.0092	528 0.0087	536 0.0083	544 0.0079	525 1.086
295	MEGVII-003	637 -	637 -	170 0.0009	197 0.0008	205 0.0008	215 0.0008	221 0.0008	190 1.008
296	MEGVII-004	637 -	637 -	152 0.0008	189 0.0008	200 0.0008	212 0.0008	218 0.0008	185 1.007
297	MEGVII-005	637 -	637 -	151 0.0008	192 0.0008	201 0.0008	213 0.0008	219 0.0008	184 1.007
298	METSAKUURCOMPANY-000	637 -	637 -	138 0.0008	109 0.0006	93 0.0006	88 0.0005	89 0.0005	107 1.006
299	METSAKUURCOMPANY-001	637 -	637 -	205 0.0010	187 0.0008	193 0.0008	199 0.0008	202 0.0008	196 1.008
300	METSAKUURCOMPANY-002	637 -	637 -	192 0.0010	128 0.0007	121 0.0006	110 0.0006	100 0.0005	139 1.006
301	MIAXIS-001	637 -	637 -	264 0.0013	263 0.0010	270 0.0010	281 0.0010	290 0.0010	266 1.010
302	MIAXIS-002	637 -	637 -	266 0.0013	283 0.0011	287 0.0011	288 0.0010	300 0.0010	279 1.010
303	MICROFOCUS-003	637 -	637 -	627 0.5940	624 0.4690	624 0.4202	624 0.3722	625 0.3093	624 5.359
304	MICROFOCUS-004	637 -	637 -	624 0.5762	623 0.4517	623 0.4024	623 0.3558	624 0.2954	623 5.198
305	MICROFOCUS-005	637 -	637 -	619 0.4240	619 0.3025	617 0.2604	617 0.2206	613 0.1721	619 3.859
306	MICROFOCUS-006	637 -	637 -	620 0.4266	620 0.3046	618 0.2621	619 0.2230	614 0.1743	620 3.878
307	MICROSOFT-003	637 -	637 -	298 0.0015	227 0.0009	206 0.0008	186 0.0007	139 0.0006	246 1.009
308	MICROSOFT-004	637 -	637 -	287 0.0014	214 0.0009	176 0.0008	158 0.0007	120 0.0006	226 1.009
309	MICROSOFT-005	637 -	637 -	345 0.0018	223 0.0009	198 0.0008	178 0.0007	146 0.0006	260 1.009
310	MICROSOFT-006	637 -	637 -	352 0.0019	284 0.0011	238 0.0009	197 0.0008	160 0.0007	294 1.011
311	MOBBL-000	637 -	637 -	381 0.0023	372 0.0016	364 0.0014	365 0.0014	364 0.0013	374 1.015
312	MOBBL-001	637 -	637 -	517 0.0115	536 0.0109	542 0.0108	548 0.0108	556 0.0107	533 1.099
313	MOBBL-002	637 -	637 -	609 0.2119	613 0.2119	615 0.2119	616 0.2119	620 0.2119	613 2.907
314	MOBBL-003	637 -	637 -	477 0.0071	493 0.0058	497 0.0056	507 0.0054	515 0.0052	491 1.054
315	MUKH-002	637 -	637 -	552 0.0255	546 0.0136	543 0.0109	538 0.0086	533 0.0066	547 1.136
316	NAVERCLOUD-000	637 -	637 -	171 0.0009	173 0.0008	180 0.0008	192 0.0008	203 0.0008	170 1.007
317	NEC-000	637 -	637 -	535 0.0167	519 0.0083	506 0.0063	502 0.0049	495 0.0035	521 1.084
318	NEC-001	637 -	637 -	544 0.0206	547 0.0138	550 0.0125	552 0.0115	557 0.0109	545 1.132
319	NEC-002	637 -	637 -	179 0.0009	181 0.0008	183 0.0008	188 0.0008	190 0.0007	181 1.007
320	NEC-003	637 -	637 -	262 0.0013	306 0.0012	313 0.0012	329 0.0011	343 0.0011	292 1.011
321	NEC-004	637 -	637 -	220 0.0011	253 0.0010	266 0.0010	263 0.0010	273 0.0009	250 1.009
322	NEC-005	637 -	637 -	149 0.0008	172 0.0008	178 0.0008	185 0.0007	185 0.0007	164 1.007
323	NEC-006	637 -	637 -	178 0.0009	201 0.0009	208 0.0008	209 0.0008	211 0.0008	197 1.008
324	NEC-007	637 -	637 -	93 0.0007	108 0.0006	118 0.0006	121 0.0006	128 0.0006	108 1.006
325	NEC-008	637 -	637 -	111 0.0007	135 0.0007	135 0.0006	138 0.0006	147 0.0006	130 1.006
326	NEC-009	637 -	637 -	74 0.0006	96 0.0006	105 0.0006	115 0.0006	126 0.0006	93 1.005
327	NEC-010	637 -	637 -	30 0.0005	36 0.0005	44 0.0005	41 0.0005	37 0.0005	39 1.004
328	NEC-011	637 -	637 -	15 0.0005	16 0.0004	13 0.0004	8 0.0004	12 0.0004	14 1.004
329	NEC-012	637 -	637 -	14 0.0005	15 0.0004	14 0.0004	9 0.0004	14 0.0004	13 1.004
330	NEUROTECHNOLOGY-003	637 -	637 -	546 0.0222	540 0.0123	535 0.0096	530 0.0075	517 0.0054	542 1.122
331	NEUROTECHNOLOGY-004	637 -	637 -	452 0.0052	454 0.0033	452 0.0029	457 0.0026	456 0.0022	458 1.032
332	NEUROTECHNOLOGY-005	637 -	637 -	431 0.0040	439 0.0026	445 0.0024	444 0.0022	447 0.0021	441 1.025
333	NEUROTECHNOLOGY-006	637 -	637 -	539 0.0178	515 0.0078	501 0.0058	498 0.0045	489 0.0032	519 1.082
334	NEUROTECHNOLOGY-007	637 -	637 -	420 0.0036	424 0.0023	429 0.0021	429 0.0019	427 0.0018	423 1.022
335	NEUROTECHNOLOGY-008	637 -	637 -	340 0.0018	293 0.0012	294 0.0011	289 0.0010	283 0.0010	305 1.011
336	NEUROTECHNOLOGY-009	637 -	637 -	227 0.0011	200 0.0009	204 0.0008	203 0.0008	204 0.0008	206 1.008
337	NEUROTECHNOLOGY-010	637 -	637 -	165 0.0009	145 0.0007	150 0.0007	150 0.0007	167 0.0007	152 1.007
338	NEUROTECHNOLOGY-012	637 -	637 -	91 0.0007	110 0.0006	111 0.0006	113 0.0006	125 0.0006	103 1.006
339	NEUROTECHNOLOGY-013	637 -	637 -	92 0.0007	106 0.0006	115 0.0006	124 0.0006	134 0.0006	101 1.005
340	NEUROTECHNOLOGY-014	637 -	637 -	61 0.0006	56 0.0005	68 0.0005	72 0.0005	77 0.0005	56 1.005
341	NEUROTECHNOLOGY-015	637 -	637 -	45 0.0005	30 0.0005	36 0.0005	43 0.0005	54 0.0005	37 1.004
342	NEUROTECHNOLOGY-016	637 -	637 -	35 0.0005	44 0.0005	55 0.0005	62 0.0005	61 0.0005	42 1.004
343	NEWLAND-002	637 -	637 -	587 0.0783	587 0.0477	585 0.0394	585 0.0329	583 0.0261	588 1.466
344	NEXTMETALABS-001	637 -	637 -	563 0.0312	552 0.0156	546 0.0116	539 0.0087	524 0.0061	553 1.159
345	NEXTMETALABS-002	637 -	637 -	80 0.0006	69 0.0005	57 0.0005	51 0.0005	46 0.0005	72 1.005
346	NEXTMETALABS-003	637 -	637 -	78 0.0006	66 0.0005	59 0.0005	54 0.0005	47 0.0005	71 1.005
347	NOBLIS-001	637 -	637 -	613 0.2489	607 0.1769	607 0.1539	607 0.1336	606 0.1109	609 2.676
348	NOBLIS-002	637 -	637 -	603 0.1791	600 0.1104	599 0.0900	597 0.0719	597 0.0532	600 2.074
349	NOMINDER-000	637 -	637 -	94 0.0007	78 0.0005	80 0.0005	76 0.0005	72 0.0005	84 1.005
350	NOMINDER-001	637 -	637 -	240 0.0011	271 0.0011	284 0.0011	296 0.0010	316 0.0010	267 1.010
351	NOMINDER-002	637 -	637 -	242 0.0012	276 0.0011	289 0.0011	299 0.0011	314 0.0010	271 1.010
352	NOMINDER-003	637 -	637 -	243 0.0012	279 0.0011	295 0.0011	304 0.0011	324 0.0011	275 1.010
353	NOMINDER-004	637 -	637 -	222 0.0011	254 0.0010	267 0.0010	274 0.0010	288 0.0010	248 1.009
354	NOMINDER-005	637 -	637 -	234 0.0011	266 0.0010	276 0.0010	290 0.0010	306 0.0010	258 1.009
355	NOTIONTAG-000	637 -	637 -	364 0.0021	385 0.0017	389 0.0017	406 0.0016	414 0.0016	380 1.016
356	NTECHLAB-003	637 -	637 -	463 0.0060	444 0.0028	434 0.0022	419 0.0018	406 0.0015	448 1.029
357	NTECHLAB-004	637 -	637 -	445 0.0046	418 0.0022	398 0.0018	389 0.0015	359 0.0012	424 1.023
358	NTECHLAB-005	637 -	637 -	443 0.0046	414 0.0021	386 0.0016	349 0.0013	301 0.0010	421 1.022
359	NTECHLAB-006	637 -	637 -	432 0.0040	387 0.0018	363 0.0014	317 0.0011	245 0.0009	402 1.019
360	NTECHLAB-007	637 -	637 -	378 0.0023	330 0.0013	293 0.0011	276 0.0010	235 0.0008	342 1.013

Table 56: **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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MISSES OUTSIDE RANK R		RESOURCE USAGE		ENROL MOST RECENT, N = 1.6M					
FNIR(N, T=0, R)		TEMPLATE		FRVT 2018 MUGSHOTS					
#	ALGORITHM	BYTES	MSEC	R=1	R=5	R=10	R=20	R=50	WORK-10
361	NTECHLAB-008	637 -	637 -	275 0.0014	199 0.0008	193 0.0008	187 0.0007	173 0.0007	219 1.008
362	NTECHLAB-009	637 -	637 -	180 0.0009	155 0.0007	153 0.0007	139 0.0006	130 0.0006	158 1.007
363	NTECHLAB-010	637 -	637 -	121 0.0008	137 0.0007	147 0.0007	152 0.0007	169 0.0007	134 1.006
364	NTECHLAB-011	637 -	637 -	96 0.0007	92 0.0006	92 0.0006	91 0.0005	94 0.0005	92 1.005
365	OMNIGARDE-000	637 -	637 -	102 0.0007	79 0.0005	78 0.0005	73 0.0005	66 0.0005	86 1.005
366	OMNIGARDE-001	637 -	637 -	48 0.0005	42 0.0005	45 0.0005	31 0.0005	26 0.0004	45 1.004
367	OMNIGARDE-002	637 -	637 -	66 0.0006	59 0.0005	48 0.0005	39 0.0005	24 0.0004	55 1.005
368	ONFIDO-000	637 -	637 -	336 0.0018	278 0.0011	257 0.0010	227 0.0009	187 0.0007	282 1.010
369	ONFIDO-001	637 -	637 -	270 0.0013	168 0.0008	149 0.0007	133 0.0006	113 0.0006	187 1.008
370	OPTIEXACTA-000	637 -	637 -	43 0.0005	37 0.0005	43 0.0005	53 0.0005	62 0.0005	41 1.004
371	PANGIAM-000	637 -	637 -	146 0.0008	158 0.0007	163 0.0007	162 0.0007	170 0.0007	157 1.007
372	PANGIAM-001	637 -	637 -	467 0.0065	501 0.0064	508 0.0064	516 0.0064	529 0.0064	495 1.058
373	PAPSAV1923-000	637 -	637 -	142 0.0008	161 0.0007	165 0.0007	176 0.0007	197 0.0007	160 1.007
374	PARAVISION-000	637 -	637 -	540 0.0187	558 0.0170	564 0.0166	567 0.0164	574 0.0163	551 1.155
375	PARAVISION-001	637 -	637 -	422 0.0036	425 0.0023	428 0.0021	431 0.0020	428 0.0018	425 1.023
376	PARAVISION-002	637 -	637 -	429 0.0039	434 0.0025	432 0.0022	434 0.0020	431 0.0018	432 1.024
377	PARAVISION-003	637 -	637 -	411 0.0031	416 0.0021	419 0.0020	420 0.0018	419 0.0017	413 1.020
378	PARAVISION-004	637 -	637 -	259 0.0013	265 0.0010	264 0.0010	272 0.0010	282 0.0009	264 1.010
379	PARAVISION-005	637 -	637 -	246 0.0012	260 0.0010	265 0.0010	275 0.0010	285 0.0010	256 1.009
380	PARAVISION-007	637 -	637 -	145 0.0008	159 0.0007	155 0.0007	159 0.0007	161 0.0007	150 1.006
381	PARAVISION-009	637 -	637 -	89 0.0006	102 0.0006	116 0.0006	119 0.0006	124 0.0006	98 1.005
382	PARAVISION-012	637 -	637 -	69 0.0006	90 0.0006	95 0.0006	100 0.0006	105 0.0006	88 1.005
383	PARAVISION-014	637 -	637 -	64 0.0006	88 0.0006	91 0.0006	96 0.0006	101 0.0006	82 1.005
384	PARAVISION-016	637 -	637 -	67 0.0006	89 0.0006	97 0.0006	101 0.0006	111 0.0006	87 1.005
385	PARAVISION-019	637 -	637 -	29 0.0005	35 0.0005	39 0.0005	44 0.0005	38 0.0005	33 1.004
386	PARAVISION-021	637 -	637 -	16 0.0005	12 0.0004	15 0.0004	19 0.0004	22 0.0004	12 1.004
387	PARAVISION-022	637 -	637 -	12 0.0005	10 0.0004	12 0.0004	15 0.0004	18 0.0004	10 1.004
388	PENTACOMP-000	637 -	637 -	631 0.9751	630 0.8897	630 0.7909	629 0.5980	371 0.0013	630 9.021
389	PENTACOMP-001	637 -	637 -	288 0.0014	274 0.0011	269 0.0010	267 0.0010	251 0.0009	277 1.010
390	PIXELALL-002	637 -	637 -	435 0.0041	437 0.0025	430 0.0021	415 0.0018	413 0.0016	436 1.025
391	PIXELALL-003	637 -	637 -	338 0.0018	322 0.0013	320 0.0012	309 0.0011	318 0.0011	323 1.012
392	PIXELALL-004	637 -	637 -	324 0.0017	300 0.0012	299 0.0011	303 0.0011	294 0.0010	303 1.011
393	PIXELALL-005	637 -	637 -	302 0.0015	331 0.0013	334 0.0012	348 0.0012	357 0.0012	321 1.012
394	PRIVID-000	637 -	637 -	545 0.0208	565 0.0207	568 0.0207	573 0.0207	576 0.0207	561 1.186
395	PROTENET-000	637 -	637 -	582 0.0632	585 0.0461	587 0.0426	588 0.0400	591 0.0377	584 1.436
396	PROTENET-001	637 -	637 -	356 0.0020	375 0.0016	383 0.0016	388 0.0015	396 0.0014	370 1.015
397	PSL-000	637 -	637 -	32 0.0005	11 0.0004	11 0.0004	13 0.0004	6 0.0004	17 1.004
398	PSL-001	637 -	637 -	21 0.0005	22 0.0005	18 0.0004	16 0.0004	10 0.0004	20 1.004
399	PSL-002	637 -	637 -	9 0.0004	7 0.0004	6 0.0004	3 0.0004	4 0.0004	6 1.004
400	PSL-003	637 -	637 -	8 0.0004	6 0.0004	7 0.0004	7 0.0004	9 0.0004	9 1.004
401	PSL-004	637 -	637 -	7 0.0004	8 0.0004	8 0.0004	10 0.0004	13 0.0004	8 1.004
402	PSL-005	637 -	637 -	11 0.0005	17 0.0004	17 0.0004	17 0.0004	11 0.0004	16 1.004
403	PSL-006	637 -	637 -	6 0.0004	9 0.0004	9 0.0004	11 0.0004	8 0.0004	7 1.004
404	PTAKURATSATU-000	637 -	637 -	398 0.0027	383 0.0017	382 0.0016	368 0.0014	366 0.0013	385 1.017
405	PYRAMID-000	637 -	637 -	115 0.0007	124 0.0006	129 0.0006	130 0.0006	138 0.0006	124 1.006
406	PYRAMID-001	637 -	637 -	148 0.0008	151 0.0007	154 0.0007	157 0.0007	166 0.0007	148 1.006
407	QAZSMARTVISIONAI-000	637 -	637 -	18 0.0005	26 0.0005	26 0.0005	22 0.0004	20 0.0004	26 1.004
408	QAZSMARTVISIONAI-001	637 -	637 -	3 0.0004	4 0.0004	5 0.0004	6 0.0004	5 0.0004	5 1.004
409	QAZSMARTVISIONAI-002	637 -	637 -	1 0.0004	2 0.0004	3 0.0004	2 0.0004	3 0.0004	2 1.004
410	QAZSMARTVISIONAI-003	637 -	637 -	5 0.0004	5 0.0004	2 0.0004	4 0.0004	1 0.0004	4 1.004
411	QNAP-000	637 -	637 -	482 0.0074	469 0.0040	466 0.0033	468 0.0029	469 0.0025	472 1.040
412	QNAP-001	637 -	637 -	427 0.0038	436 0.0025	440 0.0023	441 0.0021	444 0.0020	434 1.024
413	QNAP-002	637 -	637 -	442 0.0045	468 0.0040	482 0.0040	493 0.0039	500 0.0038	464 1.037
414	QNAP-003	637 -	637 -	387 0.0024	380 0.0017	375 0.0015	358 0.0013	356 0.0012	381 1.016
415	QNAP-004	637 -	637 -	304 0.0016	303 0.0012	307 0.0011	291 0.0010	302 0.0010	300 1.011
416	QNAP-005	637 -	637 -	405 0.0029	443 0.0027	450 0.0027	461 0.0027	473 0.0027	438 1.025
417	QNAP-006	637 -	637 -	235 0.0011	225 0.0009	224 0.0009	226 0.0009	227 0.0008	223 1.009
418	QUANTASOFT-001	637 -	637 -	610 0.2174	606 0.1641	606 0.1465	606 0.1309	607 0.1113	606 2.537
419	RANKONE-002	637 -	637 -	541 0.0191	533 0.0108	529 0.0089	528 0.0074	519 0.0057	536 1.108
420	RANKONE-003	637 -	637 -	542 0.0191	534 0.0108	530 0.0089	529 0.0074	520 0.0057	535 1.108
421	RANKONE-004	637 -	637 -	572 0.0412	566 0.0222	565 0.0174	559 0.0138	551 0.0098	566 1.222
422	RANKONE-005	637 -	637 -	495 0.0090	487 0.0051	486 0.0043	481 0.0035	478 0.0028	487 1.051
423	RANKONE-006	637 -	637 -	446 0.0047	441 0.0027	442 0.0023	437 0.0020	421 0.0017	444 1.027
424	RANKONE-007	637 -	637 -	407 0.0030	398 0.0020	393 0.0017	386 0.0015	369 0.0013	400 1.019
425	RANKONE-009	637 -	637 -	355 0.0020	325 0.0013	323 0.0012	323 0.0011	312 0.0010	330 1.012
426	RANKONE-010	637 -	637 -	348 0.0019	350 0.0014	340 0.0013	337 0.0012	333 0.0011	349 1.013
427	RANKONE-011	637 -	637 -	239 0.0011	210 0.0009	216 0.0009	218 0.0008	214 0.0008	216 1.008
428	RANKONE-012	637 -	637 -	206 0.0010	193 0.0008	191 0.0008	201 0.0008	206 0.0008	189 1.008
429	RANKONE-013	637 -	637 -	101 0.0007	91 0.0006	87 0.0005	85 0.0005	85 0.0005	90 1.005
430	RANKONE-014	637 -	637 -	76 0.0006	81 0.0005	83 0.0005	82 0.0005	93 0.0005	78 1.005
431	RANKONE-015	637 -	637 -	83 0.0006	85 0.0006	86 0.0005	92 0.0005	95 0.0005	89 1.005
432	REALNETWORKS-000	637 -	637 -	571 0.0399	563 0.0193	555 0.0147	550 0.0109	540 0.0075	565 1.199

Table 57: **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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MISSES OUTSIDE RANK R		RESOURCE USAGE		ENROL MOST RECENT, N = 1.6M					
FNIR(N, T=0, R)		TEMPLATE		FRVT 2018 MUGSHOTS					
#	ALGORITHM	BYTES	MSEC	R=1	R=5	R=10	R=20	R=50	WORK-10
433	REALNETWORKS-001	637 -	637 -	570 0.0399	562 0.0193	556 0.0147	551 0.0109	541 0.0075	564 1.199
434	REALNETWORKS-002	637 -	637 -	567 0.0391	561 0.0187	553 0.0140	542 0.0106	539 0.0075	562 1.193
435	REALNETWORKS-003	637 -	637 -	549 0.0238	537 0.0113	527 0.0086	521 0.0067	511 0.0050	540 1.117
436	REALNETWORKS-004	637 -	637 -	547 0.0233	535 0.0109	525 0.0084	512 0.0065	509 0.0047	537 1.113
437	REALNETWORKS-005	637 -	637 -	354 0.0019	308 0.0012	280 0.0010	278 0.0010	240 0.0009	315 1.012
438	REALNETWORKS-006	637 -	637 -	213 0.0010	191 0.0008	186 0.0008	170 0.0007	163 0.0007	191 1.008
439	REALNETWORKS-007	637 -	637 -	195 0.0010	182 0.0008	175 0.0007	165 0.0007	151 0.0006	182 1.007
440	REALNETWORKS-008	637 -	637 -	141 0.0008	99 0.0006	101 0.0006	105 0.0006	81 0.0005	109 1.006
441	REALSENSE-001	637 -	637 -	87 0.0006	80 0.0005	81 0.0005	81 0.0005	74 0.0005	81 1.005
442	REALSENSE-002	637 -	637 -	55 0.0006	73 0.0005	70 0.0005	66 0.0005	63 0.0005	68 1.005
443	RECOGNITO-000	637 -	637 -	62 0.0006	51 0.0005	51 0.0005	45 0.0005	50 0.0005	53 1.005
444	RECOGNITO-001	637 -	637 -	58 0.0006	71 0.0005	72 0.0005	74 0.0005	82 0.0005	69 1.005
445	RECOGNITO-002	637 -	637 -	42 0.0005	39 0.0005	49 0.0005	58 0.0005	64 0.0005	40 1.004
446	REMARKAI-000	637 -	637 -	489 0.0083	474 0.0042	467 0.0034	466 0.0029	466 0.0024	475 1.043
447	REMARKAI-000	637 -	637 -	409 0.0030	381 0.0017	373 0.0015	359 0.0013	354 0.0012	387 1.017
448	REMARKAI-002	637 -	637 -	487 0.0079	463 0.0037	454 0.0029	450 0.0024	437 0.0019	468 1.039
449	RENDIP-000	637 -	637 -	248 0.0012	241 0.0010	230 0.0009	225 0.0008	233 0.0008	240 1.009
450	REVEALMEDIA-000	637 -	637 -	294 0.0015	248 0.0010	253 0.0009	244 0.0009	243 0.0009	263 1.010
451	RNIT-000	637 -	637 -	233 0.0011	252 0.0010	255 0.0010	268 0.0010	276 0.0009	249 1.009
452	ROC-016	637 -	637 -	86 0.0006	87 0.0006	85 0.0005	89 0.0005	83 0.0005	85 1.005
453	ROC-017	637 -	637 -	71 0.0006	86 0.0006	88 0.0005	86 0.0005	87 0.0005	79 1.005
454	ROC-018	637 -	637 -	49 0.0005	47 0.0005	58 0.0005	60 0.0005	65 0.0005	48 1.005
455	ROC-019	637 -	637 -	31 0.0005	32 0.0005	31 0.0005	30 0.0005	43 0.0005	31 1.004
456	ROC-020	637 -	637 -	41 0.0005	31 0.0005	37 0.0005	29 0.0005	41 0.0005	34 1.004
457	ROC-021	637 -	637 -	26 0.0005	24 0.0005	24 0.0005	28 0.0005	27 0.0005	24 1.004
458	S1-000	637 -	637 -	359 0.0021	344 0.0014	342 0.0013	339 0.0012	342 0.0011	354 1.013
459	S1-001	637 -	637 -	400 0.0028	415 0.0021	426 0.0021	435 0.0020	443 0.0020	410 1.020
460	S1-002	637 -	637 -	218 0.0011	251 0.0010	259 0.0010	269 0.0010	286 0.0010	243 1.009
461	S1-003	637 -	637 -	231 0.0011	245 0.0010	245 0.0009	249 0.0009	257 0.0009	242 1.009
462	S1-004	637 -	637 -	196 0.0010	230 0.0009	243 0.0009	250 0.0009	256 0.0009	224 1.009
463	S1-005	637 -	637 -	207 0.0010	232 0.0009	251 0.0009	257 0.0009	272 0.0009	227 1.009
464	S1-006	637 -	637 -	188 0.0010	221 0.0009	233 0.0009	240 0.0009	250 0.0009	212 1.008
465	S1-007	637 -	637 -	159 0.0009	183 0.0008	190 0.0008	200 0.0008	208 0.0008	177 1.007
466	SANSAP-001	637 -	637 -	143 0.0008	141 0.0007	112 0.0006	118 0.0006	125 0.0006	131 1.006
467	SCANOVATE-000	637 -	637 -	444 0.0046	422 0.0023	408 0.0019	382 0.0015	353 0.0012	428 1.023
468	SCANOVATE-001	637 -	637 -	451 0.0050	426 0.0023	410 0.0019	385 0.0015	351 0.0012	435 1.024
469	SCYLLA-001	637 -	637 -	341 0.0018	327 0.0013	312 0.0012	307 0.0011	309 0.0010	324 1.012
470	SECURITYPITCH-000	637 -	637 -	419 0.0035	408 0.0021	414 0.0019	410 0.0017	402 0.0015	416 1.021
471	SENSETIME-000	637 -	637 -	371 0.0022	393 0.0019	400 0.0018	409 0.0017	415 0.0016	389 1.017
472	SENSETIME-001	637 -	637 -	372 0.0022	388 0.0018	396 0.0018	403 0.0016	410 0.0015	386 1.017
473	SENSETIME-002	637 -	637 -	526 0.0133	544 0.0132	552 0.0132	557 0.0132	565 0.0132	541 1.119
474	SENSETIME-003	637 -	637 -	85 0.0006	107 0.0006	114 0.0006	116 0.0006	122 0.0006	97 1.005
475	SENSETIME-004	637 -	637 -	81 0.0006	74 0.0005	74 0.0005	71 0.0005	70 0.0005	75 1.005
476	SENSETIME-005	637 -	637 -	34 0.0005	33 0.0005	27 0.0005	26 0.0005	34 0.0005	30 1.004
477	SENSETIME-006	637 -	637 -	33 0.0005	34 0.0005	32 0.0005	38 0.0005	39 0.0005	32 1.004
478	SENSETIME-007	637 -	637 -	24 0.0005	29 0.0005	28 0.0005	27 0.0005	36 0.0005	29 1.004
479	SENSETIME-008	637 -	637 -	23 0.0005	25 0.0005	25 0.0005	25 0.0005	33 0.0005	25 1.004
480	SENSETIME-009	637 -	637 -	20 0.0005	28 0.0005	30 0.0005	35 0.0005	31 0.0005	28 1.004
481	SENSETIME-010	637 -	637 -	10 0.0005	14 0.0004	10 0.0004	12 0.0004	17 0.0004	11 1.004
482	SENSETIME-011	637 -	637 -	19 0.0005	27 0.0005	29 0.0005	34 0.0005	32 0.0005	27 1.004
483	SENSETIME-012	637 -	637 -	17 0.0005	13 0.0004	16 0.0004	14 0.0004	16 0.0004	15 1.004
484	SERENDIPITY-000	637 -	637 -	244 0.0012	240 0.0010	232 0.0009	231 0.0009	226 0.0008	237 1.009
485	SEVENTHSENSE-001	637 -	637 -	322 0.0016	360 0.0016	384 0.0016	395 0.0016	408 0.0015	363 1.014
486	SHAMAN-003	637 -	637 -	595 0.1240	596 0.0819	596 0.0704	595 0.0613	596 0.0515	596 1.786
487	SHAMAN-004	637 -	637 -	612 0.2218	605 0.1470	603 0.1238	603 0.1046	602 0.0822	605 2.409
488	SHAMAN-006	637 -	637 -	569 0.0395	578 0.0341	581 0.0329	584 0.0320	588 0.0312	576 1.313
489	SHAMAN-007	637 -	637 -	568 0.0393	576 0.0339	580 0.0328	583 0.0319	587 0.0311	575 1.312
490	SHENNONA-000	637 -	637 -	522 0.0120	538 0.0118	547 0.0118	553 0.0117	561 0.0117	534 1.106
491	SHUFTI-001	637 -	637 -	511 0.0107	532 0.0103	539 0.0103	545 0.0102	554 0.0102	531 1.093
492	SIAT-001	637 -	637 -	320 0.0016	320 0.0013	314 0.0012	302 0.0011	295 0.0010	314 1.012
493	SIAT-002	637 -	637 -	325 0.0017	313 0.0012	326 0.0012	331 0.0012	325 0.0011	317 1.012
494	SJTU-001	637 -	637 -	27 0.0005	18 0.0005	21 0.0005	23 0.0005	23 0.0004	21 1.004
495	SJTU-002	637 -	637 -	22 0.0005	19 0.0005	19 0.0005	18 0.0004	21 0.0004	19 1.004
496	SMILART-004	637 -	637 -	630 0.9648	631 0.9641	631 0.9640	631 0.9639	632 0.9637	631 9.678
497	SMILART-005	637 -	637 -						635 10.000
498	SPARKTEK-000	637 -	637 -	127 0.0008	125 0.0006	122 0.0006	120 0.0006	129 0.0006	121 1.006
499	SPARKTEK-001	637 -	637 -	60 0.0006	68 0.0005	73 0.0005	79 0.0005	86 0.0005	67 1.005
500	SPARSH-001	637 -	637 -	110 0.0007	104 0.0006	104 0.0006	107 0.0006	114 0.0006	102 1.006
501	SQISOFT-001	637 -	637 -	428 0.0038	281 0.0011	234 0.0009	206 0.0008	176 0.0007	345 1.013
502	SQISOFT-002	637 -	637 -	185 0.0009	140 0.0007	131 0.0006	122 0.0006	110 0.0006	146 1.006
503	STAQU-000	637 -	637 -	473 0.0068	491 0.0057	495 0.0054	503 0.0052	510 0.0050	489 1.052
504	STCON-000	637 -	637 -	605 0.1826	609 0.1826	609 0.1826	612 0.1826	616 0.1825	608 2.643

Table 58: **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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MISSES OUTSIDE RANK R FNIR(N, T=0, R)		RESOURCE USAGE TEMPLATE		ENROL MOST RECENT, N = 1.6M FRVT 2018 MUGSHOTS					
#	ALGORITHM	BYTES	MSEC	R=1	R=5	R=10	R=20	R=50	WORK-10
505	STCON-001	[637] -	[637] -	[47] 0.0005	[45] 0.0005	[40] 0.0005	[42] 0.0005	[51] 0.0005	[44] 1.004
506	STCON-002	[637] -	[637] -	[38] 0.0005	[48] 0.0005	[53] 0.0005	[49] 0.0005	[58] 0.0005	[43] 1.004
507	STCON-003	[637] -	[637] -	[39] 0.0005	[50] 0.0005	[56] 0.0005	[50] 0.0005	[59] 0.0005	[46] 1.005
508	STCON-004	[637] -	[637] -	[36] 0.0005	[41] 0.0005	[41] 0.0005	[48] 0.0005	[44] 0.0005	[35] 1.004
509	STCON-005	[637] -	[637] -	[37] 0.0005	[38] 0.0005	[35] 0.0005	[42] 0.0005	[42] 0.0005	[36] 1.004
510	STELLARTECS-001	[637] -	[637] -	[337] 0.0018	[353] 0.0014	[360] 0.0014	[362] 0.0014	[384] 0.0014	[351] 1.013
511	SUMSUB-000	[637] -	[637] -	[575] 0.0453	[572] 0.0270	[569] 0.0221	[570] 0.0182	[566] 0.0136	[572] 1.264
512	SYNESIS-003	[637] -	[637] -	[533] 0.0159	[553] 0.0156	[561] 0.0156	[565] 0.0156	[573] 0.0156	[550] 1.141
513	SYNESIS-003	[637] -	[637] -	[601] 0.1697	[601] 0.1169	[601] 0.1045	[601] 0.0950	[603] 0.0866	[601] 2.118
514	SYNESIS-005	[637] -	[637] -	[488] 0.0082	[517] 0.0081	[524] 0.0081	[534] 0.0081	[546] 0.0081	[514] 1.073
515	T4ISB-000	[637] -	[637] -	[501] 0.0100	[530] 0.0100	[538] 0.0100	[544] 0.0100	[552] 0.0100	[529] 1.090
516	TCC-000	[637] -	[637] -	[573] 0.0448	[582] 0.0413	[586] 0.0405	[587] 0.0399	[592] 0.0393	[582] 1.376
517	TCC-001	[637] -	[637] -	[459] 0.0059	[465] 0.0039	[470] 0.0034	[476] 0.0032	[483] 0.0030	[465] 1.037
518	TECH5-001	[637] -	[637] -	[423] 0.0036	[404] 0.0020	[394] 0.0017	[387] 0.0015	[377] 0.0013	[414] 1.020
519	TECH5-002	[637] -	[637] -	[379] 0.0023	[275] 0.0011	[231] 0.0009	[196] 0.0008	[164] 0.0007	[301] 1.011
520	TECH5-003	[637] -	[637] -	[109] 0.0007	[91] 0.0006	[103] 0.0006	[102] 0.0006	[115] 0.0006	[100] 1.005
521	TECHAINER-000	[637] -	[637] -	[351] 0.0019	[369] 0.0016	[376] 0.0015	[381] 0.0015	[397] 0.0014	[366] 1.015
522	TEVIAN-003	[637] -	[637] -	[528] 0.0145	[511] 0.0072	[503] 0.0058	[499] 0.0046	[496] 0.0036	[515] 1.074
523	TEVIAN-004	[637] -	[637] -	[512] 0.0111	[490] 0.0056	[489] 0.0045	[482] 0.0036	[480] 0.0029	[493] 1.057
524	TEVIAN-005	[637] -	[637] -	[478] 0.0071	[462] 0.0037	[459] 0.0030	[455] 0.0026	[453] 0.0022	[466] 1.037
525	TEVIAN-006	[637] -	[637] -	[361] 0.0021	[355] 0.0015	[356] 0.0014	[364] 0.0013	[368] 0.0013	[357] 1.014
526	TEVIAN-007	[637] -	[637] -	[289] 0.0014	[256] 0.0010	[262] 0.0010	[271] 0.0010	[270] 0.0009	[261] 1.010
527	THAKAAIT-000	[637] -	[637] -	[223] 0.0011	[185] 0.0008	[171] 0.0007	[177] 0.0007	[178] 0.0007	[188] 1.008
528	TIGER-000	[637] -	[637] -	[581] 0.0613	[575] 0.0308	[571] 0.0234	[569] 0.0176	[562] 0.0118	[577] 1.313
529	TIGER-002	[637] -	[637] -	[456] 0.0054	[446] 0.0028	[438] 0.0023	[413] 0.0018	[395] 0.0014	[446] 1.029
530	TIGER-003	[637] -	[637] -	[457] 0.0054	[447] 0.0028	[435] 0.0022	[412] 0.0018	[394] 0.0014	[447] 1.029
531	TIGER-004	[637] -	[637] -	[374] 0.0022	[392] 0.0019	[409] 0.0019	[421] 0.0018	[429] 0.0018	[391] 1.017
532	TONGYITRANS-000	[637] -	[637] -	[468] 0.0066	[457] 0.0035	[456] 0.0029	[454] 0.0026	[460] 0.0023	[459] 1.035
533	TONGYITRANS-001	[637] -	[637] -	[469] 0.0066	[458] 0.0035	[455] 0.0029	[453] 0.0026	[459] 0.0023	[460] 1.035
534	TOSHIBA-000	[637] -	[637] -	[436] 0.0043	[431] 0.0024	[423] 0.0020	[416] 0.0018	[418] 0.0016	[433] 1.024
535	TOSHIBA-001	[637] -	[637] -	[441] 0.0045	[438] 0.0025	[427] 0.0021	[418] 0.0018	[417] 0.0016	[440] 1.025
536	TOSHIBA-002	[637] -	[637] -	[184] 0.0009	[226] 0.0009	[236] 0.0009	[247] 0.0009	[264] 0.0009	[217] 1.008
537	TOSHIBA-003	[637] -	[637] -	[125] 0.0008	[160] 0.0007	[167] 0.0007	[182] 0.0007	[199] 0.0007	[159] 1.007
538	TOSHIBA-004	[637] -	[637] -	[28] 0.0005	[23] 0.0005	[23] 0.0005	[20] 0.0004	[25] 0.0004	[23] 1.004
539	TOSS-000	[637] -	[637] -	[198] 0.0010	[228] 0.0009	[240] 0.0009	[254] 0.0009	[266] 0.0009	[222] 1.009
540	TOSS-001	[637] -	[637] -	[211] 0.0010	[237] 0.0010	[252] 0.0009	[253] 0.0009	[261] 0.0009	[232] 1.009
541	TRUEFACE-000	[637] -	[637] -	[406] 0.0029	[435] 0.0025	[447] 0.0024	[445] 0.0023	[457] 0.0022	[426] 1.023
542	TSIGHT-000	[637] -	[637] -	[4] 0.0004	[1] 0.0004	[1] 0.0004	[1] 0.0004	[2] 0.0004	[1] 1.004
543	TTIGROUP-000	[637] -	[637] -	[307] 0.0016	[343] 0.0014	[352] 0.0014	[360] 0.0013	[372] 0.0013	[341] 1.013
544	TURINGTECHVIP-001	[637] -	[637] -	[497] 0.0091	[525] 0.0090	[531] 0.0090	[541] 0.0089	[547] 0.0089	[518] 1.081
545	UNISSEY-001	[637] -	[637] -	[608] 0.2098	[612] 0.2097	[614] 0.2097	[615] 0.2097	[619] 0.2097	[612] 2.887
546	UNISSEY-002	[637] -	[637] -	[604] 0.1808	[608] 0.1805	[608] 0.1805	[611] 0.1805	[615] 0.1804	[607] 2.625
547	UNISSEY-003	[637] -	[637] -	[413] 0.0031	[445] 0.0028	[449] 0.0027	[460] 0.0027	[471] 0.0026	[442] 1.025
548	UNIVERSEAI-000	[637] -	[637] -	[98] 0.0007	[119] 0.0006	[125] 0.0006	[127] 0.0006	[135] 0.0006	[117] 1.006
549	USEB-000	[637] -	[637] -	[564] 0.0342	[551] 0.0153	[545] 0.0115	[540] 0.0089	[523] 0.0060	[556] 1.162
550	USEB-001	[637] -	[637] -	[162] 0.0009	[165] 0.0008	[170] 0.0007	[173] 0.0007	[191] 0.0007	[166] 1.007
551	USEB-002	[637] -	[637] -	[217] 0.0011	[243] 0.0010	[249] 0.0009	[252] 0.0009	[267] 0.0009	[234] 1.009
552	USEB-003	[637] -	[637] -	[161] 0.0009	[169] 0.0008	[174] 0.0007	[181] 0.0007	[188] 0.0007	[167] 1.007
553	USEB-004	[637] -	[637] -	[187] 0.0010	[218] 0.0009	[226] 0.0009	[237] 0.0009	[249] 0.0009	[209] 1.008
554	VD-000	[637] -	[637] -	[622] 0.4735	[621] 0.3202	[619] 0.2693	[618] 0.2212	[612] 0.1675	[621] 4.056
555	VD-001	[637] -	[637] -	[557] 0.0272	[559] 0.0178	[563] 0.0158	[561] 0.0143	[564] 0.0127	[560] 1.171
556	VD-002	[637] -	[637] -	[498] 0.0092	[512] 0.0073	[516] 0.0070	[520] 0.0067	[530] 0.0064	[508] 1.068
557	VD-003	[637] -	[637] -	[429] 0.0073	[502] 0.0065	[507] 0.0064	[514] 0.0063	[527] 0.0062	[501] 1.059
558	VDFSOFTWARE-001	[637] -	[637] -	[607] 0.2055	[611] 0.1958	[610] 0.1926	[613] 0.1893	[617] 0.1853	[611] 2.773
559	VDFSOFTWARE-002	[637] -	[637] -	[334] 0.0018	[235] 0.0010	[195] 0.0008	[172] 0.0007	[137] 0.0006	[257] 1.009
560	VERIDAS-001	[637] -	[637] -	[385] 0.0024	[366] 0.0016	[350] 0.0013	[338] 0.0012	[335] 0.0011	[368] 1.015
561	VERIDAS-002	[637] -	[637] -	[384] 0.0024	[364] 0.0016	[348] 0.0013	[336] 0.0012	[334] 0.0011	[367] 1.015
562	VERIDAS-003	[637] -	[637] -	[291] 0.0015	[288] 0.0011	[283] 0.0011	[279] 0.0010	[279] 0.0009	[284] 1.011
563	VERIDAS-004	[637] -	[637] -	[199] 0.0010	[216] 0.0009	[229] 0.0009	[236] 0.0009	[244] 0.0009	[211] 1.008
564	VERIGRAM-000	[637] -	[637] -	[370] 0.0022	[409] 0.0021	[425] 0.0021	[438] 0.0021	[448] 0.0021	[404] 1.019
565	VERIGRAM-001	[637] -	[637] -	[367] 0.0021	[406] 0.0020	[422] 0.0020	[436] 0.0020	[446] 0.0020	[394] 1.018
566	VERIGRAM-002	[637] -	[637] -	[375] 0.0022	[419] 0.0022	[431] 0.0022	[442] 0.0022	[452] 0.0022	[409] 1.020
567	VERIJELAS-000	[637] -	[637] -	[617] 0.3545	[618] 0.2973	[621] 0.2802	[620] 0.2652	[622] 0.2486	[618] 3.742
568	VIANTE-000	[637] -	[637] -	[175] 0.0009	[207] 0.0009	[215] 0.0009	[224] 0.0008	[236] 0.0008	[201] 1.008
569	VIANTE-001	[637] -	[637] -	[166] 0.0009	[198] 0.0008	[210] 0.0008	[223] 0.0008	[231] 0.0008	[193] 1.008
570	VIANTE-002	[637] -	[637] -	[50] 0.0005	[62] 0.0005	[65] 0.0005	[68] 0.0005	[73] 0.0005	[60] 1.005
571	VIANTE-003	[637] -	[637] -	[44] 0.0005	[58] 0.0005	[67] 0.0005	[70] 0.0005	[76] 0.0005	[52] 1.005
572	VIETTELAI-000	[637] -	[637] -	[434] 0.0041	[466] 0.0039	[478] 0.0039	[491] 0.0039	[499] 0.0038	[461] 1.036
573	VIETTELAI-001	[637] -	[637] -	[329] 0.0017	[363] 0.0016	[378] 0.0015	[390] 0.0015	[403] 0.0015	[361] 1.014
574	VIETTELAI-002	[637] -	[637] -	[326] 0.0017	[365] 0.0016	[379] 0.0015	[391] 0.0015	[404] 0.0015	[360] 1.014
575	VIETTELAI-003	[637] -	[637] -	[82] 0.0006	[82] 0.0006	[82] 0.0005	[75] 0.0005	[75] 0.0005	[80] 1.005
576	VIETTELAI-004	[637] -	[637] -	[57] 0.0006	[61] 0.0005	[61] 0.0005	[65] 0.0005	[69] 0.0005	[62] 1.005

Table 59: **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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MISSES OUTSIDE RANK R		RESOURCE USAGE		ENROL MOST RECENT, N = 1.6M					
FNIR(N, T=0, R)		TEMPLATE		FRVT 2018 MUGSHOTS					
#	ALGORITHM	BYTES	MSEC	R=1	R=5	R=10	R=20	R=50	WORK-10
577	VIGILANTSOLUTIONS-003	637 -	637 -	585 0.0691	580 0.0346	575 0.0259	571 0.0198	567 0.0137	580 1.352
578	VIGILANTSOLUTIONS-004	637 -	637 -	596 0.1246	593 0.0702	591 0.0553	589 0.0431	586 0.0302	593 1.696
579	VIGILANTSOLUTIONS-005	637 -	637 -	494 0.0089	473 0.0042	464 0.0033	456 0.0026	441 0.0019	476 1.043
580	VIGILANTSOLUTIONS-006	637 -	637 -	500 0.0096	479 0.0045	473 0.0035	462 0.0028	445 0.0020	480 1.046
581	VIGILANTSOLUTIONS-007	637 -	637 -	408 0.0030	371 0.0016	349 0.0013	333 0.0012	284 0.0010	379 1.016
582	VIGILANTSOLUTIONS-008	637 -	637 -	394 0.0026	358 0.0015	333 0.0012	312 0.0011	271 0.0009	365 1.014
583	VINBIGDATA-001	637 -	637 -	158 0.0009	105 0.0006	98 0.0006	95 0.0005	92 0.0005	110 1.006
584	VINBIGDATA-002	637 -	637 -	65 0.0006	43 0.0005	54 0.0005	59 0.0005	53 0.0005	49 1.005
585	VINBIGDATA-003	637 -	637 -	54 0.0006	63 0.0005	64 0.0005	63 0.0005	60 0.0005	58 1.005
586	VINBIGDATA-004	637 -	637 -	56 0.0006	65 0.0005	63 0.0005	46 0.0005	52 0.0005	61 1.005
587	VIRTISHA-000	637 -	637 -	530 0.0154	550 0.0153	558 0.0152	564 0.0152	571 0.0152	549 1.138
588	VISIONBOX-000	637 -	637 -	309 0.0016	295 0.0012	297 0.0011	280 0.0010	265 0.0009	297 1.011
589	VISIONBOX-001	637 -	637 -	395 0.0026	429 0.0024	443 0.0023	446 0.0023	462 0.0023	418 1.022
590	VISIONBOX-002	637 -	637 -	292 0.0015	318 0.0013	331 0.0012	342 0.0012	355 0.0012	313 1.012
591	VISIONLABS-004	637 -	637 -	396 0.0026	382 0.0017	380 0.0016	377 0.0015	378 0.0013	382 1.016
592	VISIONLABS-005	637 -	637 -	376 0.0023	373 0.0016	368 0.0015	362 0.0013	360 0.0013	371 1.015
593	VISIONLABS-006	637 -	637 -	333 0.0018	346 0.0014	343 0.0013	351 0.0013	358 0.0012	347 1.013
594	VISIONLABS-007	637 -	637 -	331 0.0017	334 0.0013	335 0.0013	344 0.0012	352 0.0012	334 1.012
595	VISIONLABS-008	637 -	637 -	318 0.0016	356 0.0015	365 0.0014	370 0.0014	382 0.0013	352 1.013
596	VISIONLABS-009	637 -	637 -	133 0.0008	152 0.0007	152 0.0007	156 0.0007	154 0.0006	143 1.006
597	VISIONLABS-010	637 -	637 -	214 0.0010	238 0.0010	244 0.0009	246 0.0009	255 0.0009	233 1.009
598	VISIONLABS-011	637 -	637 -	147 0.0008	167 0.0008	179 0.0008	174 0.0007	182 0.0007	163 1.007
599	VISTEAM-000	637 -	637 -	448 0.0048	410 0.0021	385 0.0016	356 0.0013	299 0.0010	419 1.022
600	VISTEAM-001	637 -	637 -	455 0.0053	430 0.0024	406 0.0018	378 0.0015	328 0.0011	439 1.025
601	VIXVIZION-009	637 -	637 -	349 0.0019	307 0.0012	285 0.0011	264 0.0010	232 0.0008	312 1.012
602	VNIS-000	637 -	637 -	594 0.1054	599 0.0880	597 0.0825	598 0.0771	601 0.0707	597 1.811
603	VNPAY-001	637 -	637 -	228 0.0011	255 0.0010	261 0.0010	260 0.0009	262 0.0009	245 1.009
604	VNPT-001	637 -	637 -	339 0.0018	361 0.0015	372 0.0015	379 0.0015	398 0.0014	359 1.014
605	VNPT-002	637 -	637 -	301 0.0015	352 0.0014	362 0.0014	369 0.0014	385 0.0014	346 1.013
606	VNPT-003	637 -	637 -	257 0.0013	296 0.0012	315 0.0012	327 0.0011	341 0.0011	291 1.011
607	VNPT-004	637 -	637 -	258 0.0013	301 0.0012	311 0.0011	328 0.0011	345 0.0011	289 1.011
608	VNPT-005	637 -	637 -	261 0.0013	298 0.0012	309 0.0011	322 0.0011	336 0.0011	287 1.011
609	VOCORD-003	637 -	637 -	462 0.0060	452 0.0032	451 0.0027	449 0.0024	451 0.0021	456 1.032
610	VOCORD-004	637 -	637 -	483 0.0077	481 0.0046	484 0.0040	486 0.0036	488 0.0032	478 1.045
611	VOCORD-005	637 -	637 -	474 0.0068	476 0.0044	479 0.0039	484 0.0036	492 0.0033	474 1.042
612	VOCORD-006	637 -	637 -	637 1.0000	637 1.0000	635 1.0000	635 1.0000	636 1.0000	637 10.000
613	VOGIC-000	637 -	637 -	383 0.0024	403 0.0020	417 0.0020	424 0.0019	432 0.0018	395 1.019
614	VSOFT-001	637 -	637 -	527 0.0143	508 0.0070	496 0.0055	497 0.0045	494 0.0034	513 1.072
615	VSOFT-002	637 -	637 -	353 0.0019	338 0.0013	338 0.0013	343 0.0012	337 0.0011	348 1.013
616	VTS-000	637 -	637 -	626 0.5936	627 0.5935	627 0.5935	628 0.5935	629 0.5935	627 6.341
617	VTS-001	637 -	637 -	245 0.0012	184 0.0008	166 0.0007	168 0.0007	158 0.0007	198 1.008
618	VTS-002	637 -	637 -	310 0.0016	267 0.0010	242 0.0009	230 0.0009	205 0.0008	276 1.010
619	VTS-003	637 -	637 -	140 0.0008	115 0.0006	102 0.0006	94 0.0005	90 0.0005	112 1.006
620	VTS-004	637 -	637 -	88 0.0006	70 0.0005	71 0.0005	67 0.0005	57 0.0005	74 1.005
621	VTS-005	637 -	637 -	73 0.0006	72 0.0005	52 0.0005	57 0.0005	49 0.0005	66 1.005
622	WINK-000	637 -	637 -	558 0.0291	557 0.0170	554 0.0142	555 0.0121	553 0.0100	558 1.168
623	WINK-001	637 -	637 -	391 0.0026	374 0.0016	359 0.0014	353 0.0013	347 0.0011	375 1.016
624	WINK-002	637 -	637 -	190 0.0010	143 0.0007	130 0.0006	126 0.0006	116 0.0006	155 1.007
625	WINK-003	637 -	637 -	357 0.0020	396 0.0019	413 0.0019	423 0.0019	433 0.0018	390 1.017
626	XAILIENT-000	637 -	637 -	386 0.0024	342 0.0014	327 0.0012	308 0.0011	287 0.0010	355 1.014
627	XAILIENT-001	637 -	637 -	212 0.0010	212 0.0009	211 0.0008	211 0.0008	212 0.0008	208 1.008
628	XFORWARDAI-000	637 -	637 -	350 0.0019	378 0.0017	387 0.0016	398 0.0016	412 0.0016	372 1.015
629	XFORWARDAI-001	637 -	637 -	328 0.0017	368 0.0016	381 0.0016	394 0.0015	411 0.0015	362 1.014
630	XFORWARDAI-002	637 -	637 -	313 0.0016	362 0.0015	377 0.0015	392 0.0015	409 0.0015	358 1.014
631	YISHENG-001	637 -	637 -	556 0.0262	543 0.0128	537 0.0099	532 0.0077	521 0.0057	544 1.131
632	YITU-002	637 -	637 -	300 0.0015	229 0.0009	214 0.0009	204 0.0008	196 0.0007	252 1.009
633	YITU-003	637 -	637 -	392 0.0026	400 0.0020	415 0.0019	422 0.0018	430 0.0018	396 1.019
634	YITU-004	637 -	637 -	202 0.0010	147 0.0007	138 0.0007	135 0.0006	143 0.0006	161 1.007
635	YITU-005	637 -	637 -	362 0.0021	386 0.0018	397 0.0018	411 0.0017	423 0.0017	384 1.017
636	YOUVERSE-001	637 -	637 -	169 0.0009	162 0.0007	164 0.0007	164 0.0007	153 0.0006	162 1.007
637	YUAN-000	637 -	637 -	279 0.0014	336 0.0013	351 0.0013	361 0.0013	379 0.0013	326 1.012

Table 60: **Rank-based accuracy for the FRVT 2018 mugshot sets**. *In columns 3 and 4 are template size and template generation duration. Thereafter values are rank-based FNIR with T = 0 and FPIR = 1. This is appropriate to investigational uses but not those with higher volumes where candidates from all searches would need review. The next column is a workload statistic, a small value shows an algorithm front-loads mates into the first 10 candidates. Throughout, blue superscripts indicate the rank of the algorithm for that column, and the best value is highlighted in yellow.*

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
1	20FACE-000	561 0.462	578 0.347	585 0.229	550 0.763	536 0.450	556 0.301	518 1.000	547 1.000	549 1.000	484 0.424	490 0.255	326 0.772	359 0.599	484 0.938	505 0.836
2	3DIVI-003	564 0.481	587 0.400	593 0.281	539 0.685	559 0.626	580 0.497				505 0.605	512 0.445			449 0.821	490 0.717
3	3DIVI-004	523 0.255	545 0.168	550 0.093	488 0.400	521 0.343	545 0.237				460 0.277	478 0.172			404 0.607	451 0.485
4	3DIVI-005	522 0.254	542 0.166	549 0.092	487 0.395	519 0.339	541 0.234	408 0.998	425 0.996	459 0.990	525 0.864	535 0.846			402 0.597	450 0.484
5	3DIVI-006	520 0.252	544 0.167	552 0.095	491 0.403	520 0.342	546 0.238				461 0.283	479 0.174			408 0.615	452 0.490
6	AADI-000	629 1.000	629 1.000	631 1.000	622 1.000	629 1.000	633 1.000	561 1.000	581 1.000	576 1.000	624 1.000	625 1.000	525 1.000	431 1.000	543 1.000	572 1.000
7	AADI-001								506 1.000	518 0.998						542 1.000
8	AADI-002	611 0.952	617 0.939	621 0.899	599 0.995	612 0.992	618 0.980	461 1.000	469 0.999	501 0.997	543 0.995	544 0.986	374 0.999	399 0.997	515 0.998	536 0.996
9	ACER-000	500 0.207	532 0.145	537 0.073	463 0.300	490 0.246	513 0.157	320 0.987	349 0.981	409 0.955	450 0.201	468 0.114			377 0.490	426 0.363
10	ACER-001	428 0.108	461 0.056	467 0.025	376 0.136	402 0.109	423 0.069	458 1.000	480 0.999	517 0.998	383 0.068	400 0.036	289 0.406	327 0.250	375 0.479	348 0.206
11	ADIANCE-001	636 1.000	634 1.000	632 1.000	612 1.000	618 1.000	622 1.000	570 1.000	575 1.000	609 1.000	521 0.833	532 0.828	385 1.000	412 1.000	481 0.931	507 0.852
12	ADIANCE-002	296 0.032	294 0.012	290 0.005	596 0.991	602 0.955	592 0.634	474 1.000	492 1.000	239 0.333	535 0.974	528 0.692	446 1.000	531 1.000	511 0.996	513 0.902
13	ADVANCE-000	318 0.038	352 0.020	346 0.008	482 0.371	504 0.270	486 0.112	393 0.997	331 0.973	279 0.470	436 0.163	415 0.047	338 0.884	364 0.667	325 0.326	355 0.209
14	AFISBIOMETRICS-000	366 0.053	405 0.029	385 0.010	332 0.104	357 0.081	358 0.044	178 0.619	199 0.511	244 0.340	327 0.034	326 0.015	217 0.154	250 0.080	244 0.172	266 0.127
15	AFRENGINE-000	298 0.033	347 0.019	369 0.009	481 0.364	505 0.271	505 0.149	165 0.565	158 0.360	198 0.237	452 0.206	467 0.114	320 0.743	292 0.135	314 0.308	370 0.231
16	AFRENGINE-001	135 0.008	136 0.003	193 0.002	196 0.041	209 0.031	227 0.020	160 0.550	123 0.227	122 0.119	259 0.018	310 0.013	185 0.113	209 0.051	175 0.120	205 0.097
17	AFRENGINE-002	110 0.006	125 0.003	178 0.002	217 0.046	177 0.026	192 0.018	243 0.901	153 0.344	98 0.106	189 0.008	182 0.004	346 0.926	185 0.039	147 0.102	126 0.075
18	AFRENGINE-003	142 0.008	183 0.004	247 0.003	209 0.045	224 0.033	237 0.022	116 0.333	92 0.183	108 0.113	215 0.011	203 0.005	177 0.097	184 0.039	165 0.111	157 0.083
19	AIGROUP-000	316 0.037	270 0.010	265 0.003	603 0.996	611 0.991	613 0.935	422 0.999	432 0.997	214 0.276	515 0.693	516 0.522	525 1.000	625 1.000	429 0.929	508 0.858
20	AIZE-001	442 0.126	483 0.076	486 0.033	414 0.187	435 0.143	452 0.087	372 0.995	398 0.994	443 0.983	406 0.101	424 0.052	278 0.364	320 0.216	341 0.387	399 0.289
21	AIZE-002	592 0.787	447 0.051	430 0.019	633 1.000	608 0.981	416 0.065	531 1.000	556 1.000	552 1.000	474 0.355	396 0.036	400 1.000	332 0.272	633 1.000	612 1.000
22	AIZE-006	433 0.116	436 0.045	437 0.020	611 1.000	614 0.996	572 0.413		451 0.998	330 0.708	468 0.304	426 0.055	453 1.000	526 1.000	463 0.885	460 0.544
23	AKIYAMA-000	391 0.066	426 0.041	439 0.020	555 0.805	566 0.664	460 0.093	252 0.927	261 0.793	305 0.598	524 0.862	517 0.548	467 1.000	430 1.000	492 0.972	511 0.900
24	AKIYAMA-001	346 0.046	416 0.036	476 0.029	404 0.174	450 0.162	504 0.148	161 0.555	194 0.484	245 0.340	337 0.037	384 0.029	356 0.968	372 0.767	248 0.177	278 0.134
25	AKUVOX-001	275 0.026	301 0.013	295 0.005	272 0.067	282 0.050	294 0.031	291 0.979	315 0.964	357 0.844	267 0.019	267 0.009	317 0.737	333 0.275	206 0.141	228 0.110
26	ALCHERA-000	511 0.228	529 0.138	531 0.070	450 0.259	474 0.216	502 0.146	430 0.999	455 0.999	494 0.996	442 0.176	465 0.111			445 0.803	444 0.456
27	ALCHERA-001	625 1.000	625 0.999	628 0.999	617 1.000	637 1.000	634 1.000				604 1.000	599 1.000			564 1.000	552 1.000
28	ALCHERA-002	595 0.806	596 0.485	597 0.301	538 0.685	554 0.591	574 0.442	484 1.000	488 1.000	526 0.999	520 0.827	530 0.770			446 0.811	485 0.705
29	ALCHERA-003	556 0.449	535 0.155	532 0.070	464 0.304	486 0.239	510 0.152	472 1.000	471 0.999	500 0.997	441 0.172	457 0.097			368 0.464	425 0.362
30	ALCHERA-004	569 0.519	586 0.394	582 0.211	531 0.642	549 0.529	562 0.327	373 0.995	386 0.991	350 0.811	485 0.424	488 0.232	312 0.708	351 0.515	393 0.546	438 0.398
31	ALCHERA-005	554 0.432	574 0.314	570 0.145	514 0.534	534 0.439	548 0.258	340 0.991	363 0.985	391 0.935	469 0.305	475 0.144	342 0.908	350 0.502	364 0.455	410 0.309
32	ALCHERA-006	198 0.012	195 0.005	190 0.002	162 0.033	163 0.025	178 0.017	331 0.989	352 0.982	378 0.913	124 0.006	122 0.003	298 0.493	233 0.068	120 0.094	109 0.072
33	ALCHERA-007	154 0.009	152 0.004	158 0.002	140 0.029	145 0.023	162 0.016	342 0.991	355 0.982	369 0.885	105 0.005	106 0.003	321 0.743	265 0.086	95 0.087	98 0.069
34	ALCHERA-008	130 0.008	143 0.003	146 0.002	142 0.029	133 0.022	145 0.015	90 0.272	87 0.177	95 0.105	212 0.010	90 0.002	388 1.000	390 0.975	261 0.186	186 0.090
35	ALGORITHMSINC-000	302 0.034	330 0.016	341 0.007	486 0.394	424 0.130	313 0.033	207 0.760	197 0.507	255 0.367	408 0.108	319 0.014	404 1.000	400 0.997	388 0.433	306 0.153
36	ALLGOVISION-000	454 0.138	499 0.088	510 0.045	429 0.202	453 0.166	482 0.106	349 0.993	382 0.990	441 0.982	411 0.117	437 0.066			387 0.526	437 0.396
37	ALLGOVISION-001	470 0.154	506 0.101	517 0.052	456 0.275	478 0.221	501 0.141	357 0.993	365 0.986	388 0.932	429 0.150	450 0.081			378 0.491	435 0.389
38	ALLGOVISION-002	449 0.132	476 0.065	473 0.028	548 0.753	548 0.520	487 0.113	303 0.983	303 0.950	363 0.868	454 0.232	420 0.049	411 1.000	419 1.000	427 0.716	428 0.366
39	ALLGOVISION-003	295 0.032	334 0.017	336 0.007	289 0.075	311 0.059	331 0.037	188 0.656	218 0.566	265 0.423	280 0.021	284 0.010	207 0.147	262 0.082	374 0.478	277 0.133
40	ALLGOVISION-004	476 0.164	511 0.114	525 0.059	437 0.229	466 0.190	491 0.120	221 0.804	254 0.747	313 0.627	412 0.121	436 0.066	291 0.418	331 0.271	351 0.412	387 0.262
41	ALLGOVISION-005	308 0.035	358 0.020	359 0.008	306 0.086	320 0.064	335 0.038	137 0.468	166 0.393	208 0.262	348 0.043	305 0.012	392 1.000	384 0.943	324 0.324	300 0.149
42	ALTTEKGLOBAL-000	531 0.282	568 0.278	591 0.275	455 0.272	501 0.267	549 0.259	184 0.646	226 0.617	299 0.571	479 0.383	504 0.381	302 0.536	345 0.434	413 0.641	477 0.636
43	ANKE-000	485 0.183	513 0.117	527 0.062	449 0.256	476 0.220	508 0.151	369 0.995	399 0.994	456 0.990	587 1.000	591 1.000			599 1.000	630 1.000
44	ANKE-001	484 0.183	516 0.119	528 0.062	448 0.256	477 0.220	509 0.151	374 0.995	406 0.994	471 0.992	629 1.000	627 1.000			545 1.000	577 1.000
45	ANKE-002	380 0.062	410 0.031	409 0.013	330 0.103	353 0.079	377 0.050	280 0.974	302 0.947	345 0.793	329 0.034	346 0.018			289 0.245	342 0.190
46	ARATEK-001	461 0.141	250 0.008	227 0.003	578 0.958	279 0.050	223 0.020	541 1.000	552 1.000	527 0.999	399 0.086	217 0.005	357 0.971	198 0.045	613 1.000	539 0.999

Table 61: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
47	ARMATURA-000	147 0.008	170 0.004	202 0.002	156 0.031	167 0.025	200 0.018	95 0.283	112 0.214	127 0.124	127 0.006	133 0.003	102 0.048	99 0.019	109 0.090	117 0.073
48	AWARE-003	481 0.173	523 0.128	541 0.082	477 0.351	513 0.298	535 0.204	316 0.986	360 0.984	431 0.977	486 0.428	503 0.378			388 0.530	443 0.443
49	AWARE-004	543 0.354	566 0.268	575 0.174	527 0.619	545 0.509	567 0.375	480 1.000	490 1.000	531 0.999	480 0.397	493 0.279			447 0.816	474 0.631
50	AWARE-005	576 0.606	581 0.363	544 0.085	471 0.342	493 0.253	516 0.163	476 1.000	501 1.000	529 0.999	458 0.255	471 0.122			471 0.916	488 0.714
51	AWARE-006	562 0.470	567 0.275	576 0.175	507 0.466	527 0.398	552 0.283	453 1.000	482 0.999	521 0.999	476 0.368	489 0.254			433 0.749	471 0.623
52	AWARE-007	286 0.029	329 0.016	338 0.007	422 0.197	333 0.070	332 0.038	402 0.998	390 0.992	327 0.694	352 0.044	360 0.021	360 0.982	338 0.338	292 0.250	328 0.177
53	AWIROS-000	590 0.761	570 0.291	494 0.037	563 0.909	541 0.471	484 0.109	506 1.000	516 1.000	554 1.000	536 0.976	500 0.320	468 1.000	403 0.998	520 0.999	514 0.924
54	AYONIX-000	598 0.846	610 0.811	617 0.724	576 0.956	600 0.939	610 0.892	409 0.998	441 0.998	492 0.995	532 0.954	538 0.891			500 0.982	520 0.959
55	AYONIX-001	602 0.874	612 0.824	615 0.701	570 0.946	594 0.920	606 0.845	469 1.000	479 0.999	496 0.996	546 0.999	547 0.998			494 0.969	515 0.926
56	AYONIX-002	603 0.874	611 0.824	616 0.701	571 0.946	595 0.920	605 0.845	471 1.000	481 0.999	495 0.996	528 0.915	531 0.821			495 0.969	516 0.926
57	BIOIDTECHSWISS-001	159 0.009	164 0.004	148 0.002	185 0.038	205 0.030	253 0.023	390 0.997	370 0.987	191 0.207	208 0.010	240 0.007	145 0.067	178 0.036	254 0.179	322 0.172
58	BIOMETRIK-001	432 0.115	449 0.052	455 0.023	365 0.129	389 0.098	400 0.058	529 1.000	497 1.000	499 0.997	366 0.054	381 0.027	275 0.341	314 0.183	409 0.618	339 0.188
59	CAMVI-003	415 0.094	480 0.071	522 0.057	388 0.152	426 0.132	483 0.108	292 0.979	323 0.970	396 0.940	410 0.114	459 0.100			346 0.402	432 0.377
60	CAMVI-004	426 0.106	481 0.071	520 0.054	442 0.240	428 0.136	469 0.100	456 1.000	477 0.999	505 0.998	405 0.100	449 0.081			441 0.787	454 0.507
61	CAMVI-005	455 0.138	504 0.099	540 0.076	503 0.451	463 0.179	496 0.132	462 1.000	489 1.000	515 0.998	432 0.156	466 0.112			516 0.999	529 0.983
62	CANON-001	188 0.012	204 0.005	200 0.002	149 0.031	154 0.023	146 0.015	180 0.629	156 0.357	192 0.208	172 0.008	180 0.004	150 0.068	169 0.034	200 0.139	192 0.092
63	CANON-002	161 0.009	180 0.004	186 0.002	126 0.027	122 0.020	113 0.013	141 0.481	168 0.400	203 0.245	235 0.013	186 0.004	159 0.075	199 0.046	264 0.188	219 0.106
64	CANON-003	87 0.005	97 0.002	82 0.001	65 0.015	74 0.012	77 0.010	91 0.274	102 0.199	131 0.127	123 0.006	199 0.004	56 0.030	71 0.014	55 0.076	64 0.063
65	CIB-000	335 0.044	291 0.012	281 0.005	291 0.077	270 0.045	265 0.025	512 1.000	514 1.000	543 1.000	257 0.017	249 0.008	199 0.141	234 0.068	466 0.894	456 0.521
66	CLEARVIEWAI-000	203 0.013	213 0.005	196 0.002	181 0.036	171 0.025	160 0.016	432 0.999	332 0.974	145 0.139	173 0.008	169 0.004	126 0.057	139 0.027	302 0.268	146 0.080
67	CLEARVIEWAI-001	129 0.007	149 0.003	123 0.001	118 0.026	121 0.020	111 0.013	60 0.234	76 0.167	82 0.093	114 0.005	87 0.002	91 0.042	88 0.016	73 0.079	82 0.066
68	CLEARVIEWAI-002	141 0.008	162 0.004	136 0.002	136 0.029	131 0.022	121 0.013	72 0.250	83 0.170	84 0.095	115 0.006	102 0.002	90 0.041	90 0.017	67 0.078	77 0.066
69	CLEARVIEWAI-003	88 0.005	96 0.002	91 0.001	96 0.019	93 0.015	84 0.010	27 0.172	49 0.128	47 0.074	99 0.005	78 0.002	78 0.038	77 0.015	70 0.079	75 0.065
70	CLOUDWALK-HR-000	68 0.004	84 0.002	112 0.001	72 0.015	82 0.013	99 0.012	31 0.179	46 0.123	66 0.085	100 0.005	149 0.003	64 0.033	95 0.018	136 0.099	127 0.075
71	CLOUDWALK-MT-000	48 0.003	82 0.002	128 0.001	66 0.015	81 0.013	103 0.012	22 0.160	30 0.099	41 0.066	53 0.002	68 0.002	19 0.018	33 0.009	47 0.072	65 0.063
72	CLOUDWALK-MT-001	41 0.003	76 0.002	120 0.001	51 0.013	66 0.012	98 0.011	4 0.093	6 0.059	12 0.049	13 0.001	17 0.001	15 0.015	15 0.006	12 0.056	16 0.049
73	CLOUDWALK-MT-002	39 0.003	73 0.002	117 0.001	43 0.012	62 0.011	95 0.011	1 0.070	3 0.053	8 0.045	8 0.001	7 0.001	8 0.013	8 0.005	10 0.055	9 0.047
74	COGENT-000	464 0.143	452 0.052	475 0.029	405 0.175	432 0.140	473 0.100	381 0.996	413 0.995	468 0.991						
75	COGENT-001	463 0.143	451 0.052	474 0.029	406 0.175	431 0.140	472 0.100	380 0.996	412 0.995	467 0.991						
76	COGENT-002	473 0.157	433 0.044	425 0.017	360 0.124	387 0.098	413 0.063	419 0.998	438 0.998	478 0.994						
77	COGENT-003	498 0.199	437 0.045	416 0.015	357 0.121	380 0.095	409 0.061	420 0.999	442 0.998	488 0.995						
78	COGENT-004	503 0.209	411 0.033	306 0.006	270 0.067	283 0.051	298 0.031	407 0.998	436 0.997	491 0.995	284 0.022	302 0.012	191 0.126	238 0.072	365 0.456	331 0.178
79	COGENT-005	353 0.050	260 0.009	267 0.003	230 0.050	243 0.037	255 0.023	377 0.996	375 0.989	235 0.315	217 0.011	229 0.006	166 0.082	193 0.043	469 0.905	344 0.202
80	COGENT-006	174 0.010	173 0.004	171 0.002	166 0.033	155 0.023	152 0.015	127 0.376	126 0.229	139 0.135	141 0.006	155 0.003	292 0.422	147 0.028	191 0.130	247 0.120
81	COGENT-007	136 0.008	144 0.003	137 0.002	129 0.027	129 0.021	142 0.014	65 0.241	93 0.185	115 0.116	145 0.006	151 0.003	100 0.047	110 0.022	212 0.144	269 0.129
82	COGNITEC-000	509 0.225	539 0.160	551 0.095	499 0.439	514 0.303	533 0.200	376 0.995	388 0.992	419 0.971						
83	COGNITEC-001	496 0.191	505 0.101	518 0.052	604 0.997	482 0.230	498 0.135	563 1.000	569 1.000	414 0.965						
84	COGNITEC-002	438 0.120	454 0.053	461 0.024	595 0.990	462 0.178	476 0.101	601 1.000	515 1.000	410 0.956						
85	COGNITEC-003	429 0.099	450 0.052	464 0.025	436 0.222	451 0.162	470 0.100	630 1.000	523 1.000	398 0.945						
86	COGNITEC-004	365 0.055	409 0.031	410 0.013	363 0.127	384 0.097	401 0.058	371 0.995	379 0.990	382 0.918	382 0.068	401 0.038	269 0.316	319 0.196	307 0.288	362 0.218
87	COGNITEC-005	370 0.055	266 0.009	255 0.003	252 0.058	258 0.041	247 0.022	586 1.000	590 1.000	366 0.877	346 0.041	382 0.028	220 0.157	270 0.092	253 0.179	296 0.145
88	COGNITEC-006	283 0.029	244 0.007	237 0.003	263 0.065	254 0.040	241 0.022	629 1.000	636 1.000	522 0.999	314 0.030	315 0.013	227 0.171	255 0.081	421 0.681	359 0.214
89	COGNITEC-007	208 0.013	211 0.005	206 0.002	233 0.051	229 0.034	214 0.019	202 0.735	230 0.621	281 0.483	297 0.026	300 0.012	244 0.210	264 0.085	232 0.160	260 0.124
90	COGNITEC-008	213 0.014	217 0.006	213 0.002	225 0.048	234 0.035	226 0.020	186 0.653	217 0.564	263 0.414	298 0.026	296 0.011	202 0.142	232 0.068	224 0.155	257 0.122
91	COGNITEC-009	215 0.014	221 0.006	225 0.003	223 0.048	235 0.035	225 0.020	181 0.633	216 0.554	264 0.415	293 0.025	297 0.011	189 0.124	218 0.059	226 0.156	263 0.126
92	COGNITEC-010	127 0.007	159 0.004	173 0.002	194 0.040	203 0.030	220 0.020	169 0.573	191 0.479	237 0.331	247 0.016	259 0.009	167 0.086	197 0.045	179 0.125	221 0.107

Table 62: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
93	COGNITI-000	[324] 0.040	[350] 0.019	[316] 0.006	[348] 0.116	[351] 0.078	[321] 0.035	[232] 0.858	[242] 0.693	[266] 0.426	[434] 0.160	[417] 0.048	[567] 1.000	[628] 1.000	[381] 0.502	[407] 0.300
94	COMPAL-000	[538] 0.323	[247] 0.008	[239] 0.003	[216] 0.046	[250] 0.038	[282] 0.028			[563] 1.000	[386] 0.072	[440] 0.068	[180] 0.104	[254] 0.081	[214] 0.147	[283] 0.135
95	COMPAL-001	[540] 0.366	[295] 0.012	[348] 0.008	[215] 0.046	[223] 0.033	[248] 0.023	[353] 0.993	[174] 0.422	[141] 0.136	[232] 0.013	[265] 0.009	[171] 0.090	[163] 0.033	[170] 0.117	[164] 0.084
96	COMPAL-002	[547] 0.368	[296] 0.012	[347] 0.008	[220] 0.046	[219] 0.033	[250] 0.023	[352] 0.993	[173] 0.419	[140] 0.135	[234] 0.013	[264] 0.009	[170] 0.090	[161] 0.033	[172] 0.118	[166] 0.084
97	CORSIGHT-000	[225] 0.016	[222] 0.006	[214] 0.002	[186] 0.038	[196] 0.029	[193] 0.018	[276] 0.971	[201] 0.517	[173] 0.173	[199] 0.009	[192] 0.004	[146] 0.067	[164] 0.033	[401] 0.594	[179] 0.088
98	CORSIGHT-001	[218] 0.015	[230] 0.006	[211] 0.002	[206] 0.043	[225] 0.034	[218] 0.020	[244] 0.903	[186] 0.460	[171] 0.171	[183] 0.008	[185] 0.004	[132] 0.062	[154] 0.030	[199] 0.138	[152] 0.082
99	CORSIGHT-002	[173] 0.010	[187] 0.004	[191] 0.002	[159] 0.032	[165] 0.025	[171] 0.016	[117] 0.334	[114] 0.218	[117] 0.117	[137] 0.006	[124] 0.003	[108] 0.049	[122] 0.023	[105] 0.089	[113] 0.073
100	CUBOX-000	[75] 0.004	[112] 0.002	[133] 0.002	[107] 0.022	[112] 0.019	[129] 0.014	[85] 0.267	[70] 0.158	[83] 0.093	[89] 0.004	[112] 0.003	[52] 0.028	[72] 0.014	[48] 0.073	[62] 0.062
101	CUBOX-001	[89] 0.005	[119] 0.003	[124] 0.001	[120] 0.026	[126] 0.021	[130] 0.014	[106] 0.311	[111] 0.212	[120] 0.118	[152] 0.007	[146] 0.003	[152] 0.068	[116] 0.023	[86] 0.085	[84] 0.067
102	CUBOX-002	[100] 0.006	[134] 0.003	[140] 0.002	[127] 0.027	[140] 0.022	[174] 0.016	[88] 0.270	[88] 0.181	[92] 0.103	[92] 0.004	[86] 0.002	[80] 0.038	[84] 0.016	[61] 0.077	[58] 0.061
103	CUBOX-003	[102] 0.006	[140] 0.003	[176] 0.002	[128] 0.027	[146] 0.023	[176] 0.017	[113] 0.321	[116] 0.219	[135] 0.131	[129] 0.006	[143] 0.003	[113] 0.051	[109] 0.021	[81] 0.082	[79] 0.066
104	CYBERLINK-000	[453] 0.137	[462] 0.056	[451] 0.022	[397] 0.162	[406] 0.116	[426] 0.070	[397] 0.997	[415] 0.995	[440] 0.981	[378] 0.063	[392] 0.032			[330] 0.339	[372] 0.232
105	CYBERLINK-001	[418] 0.096	[455] 0.053	[449] 0.022	[380] 0.138	[403] 0.109	[419] 0.067	[396] 0.997	[410] 0.995	[445] 0.983	[375] 0.062	[387] 0.031			[416] 0.652	[375] 0.239
106	CYBERLINK-002	[319] 0.038	[317] 0.014	[318] 0.006	[276] 0.068	[297] 0.053	[304] 0.032	[361] 0.994	[374] 0.988	[411] 0.957	[291] 0.024	[306] 0.013			[308] 0.288	[311] 0.157
107	CYBERLINK-003	[339] 0.045	[245] 0.007	[251] 0.003	[212] 0.045	[233] 0.035	[233] 0.021	[364] 0.994	[329] 0.972	[356] 0.843	[224] 0.012	[242] 0.007	[179] 0.100	[208] 0.051	[336] 0.368	[249] 0.120
108	CYBERLINK-004	[493] 0.188	[237] 0.007	[234] 0.003	[261] 0.063	[237] 0.036	[245] 0.022	[548] 1.000	[533] 1.000	[533] 0.999	[230] 0.013	[239] 0.007	[181] 0.109	[204] 0.050	[490] 0.954	[402] 0.291
109	CYBERLINK-005	[501] 0.208	[272] 0.010	[271] 0.004	[240] 0.054	[255] 0.041	[271] 0.026	[493] 1.000	[498] 1.000	[370] 0.887	[239] 0.014	[245] 0.007	[169] 0.089	[190] 0.043	[478] 0.926	[390] 0.266
110	CYBERLINK-006	[143] 0.008	[109] 0.002	[98] 0.001	[106] 0.022	[108] 0.018	[105] 0.012	[348] 0.992	[366] 0.986	[243] 0.339	[93] 0.004	[100] 0.002	[157] 0.073	[78] 0.015	[260] 0.186	[202] 0.096
111	DAHUA-000	[443] 0.127	[495] 0.086	[507] 0.044	[409] 0.179	[427] 0.135	[450] 0.083									
112	DAHUA-001	[425] 0.105	[483] 0.072	[490] 0.036	[387] 0.151	[414] 0.122	[440] 0.075	[318] 0.987	[345] 0.980	[389] 0.933						
113	DAHUA-002	[273] 0.026	[318] 0.015	[313] 0.006	[256] 0.060	[274] 0.046	[284] 0.029	[195] 0.678	[231] 0.634	[290] 0.517	[254] 0.017	[256] 0.008			[230] 0.159	[261] 0.125
114	DAHUA-003	[268] 0.025	[307] 0.014	[293] 0.005	[239] 0.054	[257] 0.041	[258] 0.024	[182] 0.643	[220] 0.574	[271] 0.441	[227] 0.013	[236] 0.006	[164] 0.081	[191] 0.043	[194] 0.134	[226] 0.109
115	DAHUA-004	[209] 0.014	[235] 0.007	[226] 0.003	[169] 0.033	[175] 0.026	[165] 0.016	[159] 0.547	[192] 0.479	[241] 0.337	[185] 0.008	[191] 0.004	[114] 0.051	[136] 0.027	[169] 0.113	[196] 0.094
116	DAON-000	[451] 0.135	[370] 0.022	[370] 0.009	[296] 0.079	[315] 0.061	[336] 0.039	[482] 1.000	[496] 1.000	[513] 0.998	[294] 0.025	[307] 0.013	[229] 0.173	[269] 0.091	[455] 0.846	[321] 0.172
117	DAON-001	[383] 0.063	[428] 0.043	[479] 0.031	[403] 0.173	[430] 0.140	[465] 0.096	[267] 0.953	[304] 0.952	[403] 0.951	[379] 0.064	[419] 0.048	[216] 0.154	[276] 0.098	[234] 0.161	[271] 0.129
118	DAON-002	[233] 0.018	[238] 0.007	[244] 0.003	[381] 0.139	[267] 0.044	[262] 0.024			[566] 1.000	[245] 0.015	[234] 0.006	[197] 0.141	[196] 0.044	[631] 1.000	[436] 0.395
119	DAON-003	[131] 0.008	[139] 0.003	[129] 0.002	[157] 0.031	[150] 0.023	[149] 0.015	[319] 0.987	[148] 0.319	[149] 0.140	[143] 0.006	[147] 0.003	[147] 0.067	[146] 0.028	[128] 0.096	[138] 0.079
120	DATECH-000	[327] 0.041	[246] 0.007	[241] 0.003	[331] 0.104	[248] 0.038	[219] 0.020	[403] 0.998	[424] 0.996	[442] 0.982	[250] 0.016	[220] 0.005	[345] 0.926	[362] 0.632	[316] 0.311	[231] 0.111
121	DATECH-001	[317] 0.037	[231] 0.006	[217] 0.002	[400] 0.168	[317] 0.062	[195] 0.018	[386] 0.996	[395] 0.993	[420] 0.971	[342] 0.039	[246] 0.007	[373] 0.998	[380] 0.875	[334] 0.360	[337] 0.184
122	DECATUR-000	[331] 0.043	[373] 0.023	[376] 0.010	[304] 0.085	[324] 0.066	[344] 0.040	[206] 0.754	[237] 0.671	[285] 0.503	[302] 0.027	[320] 0.014	[228] 0.173	[277] 0.098	[285] 0.239	[309] 0.156
123	DEEPGLINT-001	[167] 0.010	[133] 0.003	[150] 0.002	[88] 0.018	[84] 0.014	[78] 0.010	[519] 1.000	[486] 1.000	[283] 0.498	[147] 0.006	[189] 0.004			[229] 0.159	[203] 0.097
124	DEEPSEA-001	[398] 0.073	[439] 0.046	[447] 0.021	[367] 0.129	[391] 0.101	[404] 0.059	[326] 0.988	[362] 0.984	[425] 0.973	[393] 0.077	[407] 0.041			[326] 0.326	[381] 0.251
125	DERMALOG-003	[573] 0.549	[595] 0.482	[600] 0.360	[545] 0.715	[563] 0.655	[584] 0.526				[512] 0.677	[518] 0.554			[460] 0.870	[500] 0.791
126	DERMALOG-004	[574] 0.553	[594] 0.479	[599] 0.358	[543] 0.711	[564] 0.657	[582] 0.526	[388] 0.997	[416] 0.995	[470] 0.991	[503] 0.603	[514] 0.458			[457] 0.856	[495] 0.751
127	DERMALOG-005	[494] 0.189	[498] 0.088	[502] 0.043	[425] 0.201	[442] 0.154	[466] 0.096	[383] 0.996	[380] 0.990	[402] 0.949	[466] 0.300	[491] 0.267			[407] 0.614	[445] 0.459
128	DERMALOG-006	[417] 0.095	[448] 0.051	[466] 0.025	[378] 0.137	[395] 0.105	[418] 0.067	[329] 0.989	[348] 0.981	[390] 0.933	[373] 0.059	[389] 0.031			[321] 0.318	[369] 0.230
129	DERMALOG-007	[492] 0.188	[496] 0.086	[499] 0.040	[424] 0.200	[440] 0.152	[462] 0.093	[382] 0.996	[381] 0.990	[401] 0.949	[404] 0.099	[422] 0.052			[396] 0.557	[406] 0.299
130	DERMALOG-008	[525] 0.267	[435] 0.045	[424] 0.017	[439] 0.231	[377] 0.094	[391] 0.054	[528] 1.000	[542] 1.000	[546] 1.000	[369] 0.057	[376] 0.025	[284] 0.382	[307] 0.158	[485] 0.940	[482] 0.678
131	DERMALOG-009	[325] 0.041	[356] 0.020	[366] 0.009	[307] 0.086	[325] 0.066	[347] 0.040	[494] 1.000	[526] 1.000	[540] 1.000	[316] 0.031	[332] 0.016	[327] 0.999	[404] 0.999	[452] 0.840	[363] 0.222
132	DERMALOG-010	[230] 0.018	[234] 0.007	[250] 0.003	[374] 0.134	[301] 0.055	[252] 0.023	[455] 1.000	[464] 0.999	[405] 0.951	[401] 0.089	[409] 0.041	[384] 1.000	[389] 0.971	[385] 0.522	[358] 0.211
133	DERMALOG-011	[343] 0.046	[363] 0.021	[337] 0.007	[419] 0.193	[365] 0.087	[339] 0.039	[433] 0.999	[449] 0.998	[446] 0.985	[418] 0.129	[312] 0.013	[364] 0.991	[356] 0.567	[436] 0.764	[368] 0.228
134	DERMALOG-012	[328] 0.041	[362] 0.021	[343] 0.007	[310] 0.089	[327] 0.067	[342] 0.040	[256] 0.932	[195] 0.505	[227] 0.294	[286] 0.023	[290] 0.011	[208] 0.147	[243] 0.076	[350] 0.409	[258] 0.123
135	DERMALOG-013	[326] 0.041	[361] 0.021	[340] 0.007	[316] 0.095	[330] 0.068	[341] 0.040	[365] 0.995	[293] 0.908	[232] 0.306	[287] 0.024	[294] 0.011	[215] 0.153	[244] 0.076	[390] 0.541	[267] 0.128
136	DERMALOG-014	[329] 0.042	[365] 0.021	[361] 0.008	[317] 0.095	[331] 0.069	[349] 0.041	[366] 0.995	[292] 0.907	[234] 0.312	[306] 0.028	[336] 0.016	[219] 0.156	[251] 0.080	[357] 0.432	[295] 0.144
137	DERMALOG-015	[178] 0.011	[197] 0.005	[208] 0.002	[188] 0.039	[210] 0.031	[231] 0.021	[222] 0.809	[145] 0.313	[152] 0.143	[192] 0.009	[204] 0.005	[139] 0.065	[166] 0.033	[228] 0.158	[158] 0.083
138	DERMALOG-016	[97] 0.006	[90] 0.002	[115] 0.001	[123] 0.027	[119] 0.020	[131] 0.014	[266] 0.953	[140] 0.288	[69] 0.088	[84] 0.004	[111] 0.003	[72] 0.035	[92] 0.017	[391] 0.544	[112] 0.073

*Table 63: **Threshold-based accuracy**. Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
139	DERMALOG-017	111 0.006	129 0.003	142 0.002	147 0.030	151 0.023	181 0.017	176 0.608	118 0.220	99 0.106	126 0.006	142 0.003	86 0.041	107 0.020	277 0.220	115 0.073
140	DIGIDATA-000	577 0.619	601 0.610	612 0.598	523 0.588	553 0.577	587 0.560	436 0.999	403 0.994	397 0.941	508 0.646	522 0.643	329 0.789	369 0.722	450 0.824	503 0.816
141	DILUSENSE-000	361 0.053	407 0.030	403 0.012	326 0.100	352 0.078	369 0.047	229 0.850	233 0.651	280 0.482	340 0.039	366 0.022	310 0.664	324 0.242	271 0.203	308 0.154
142	DILUSENSE-001	235 0.018	254 0.008	254 0.003	218 0.046	230 0.034	239 0.022	212 0.769	180 0.450	238 0.332	246 0.015	279 0.010	300 0.511	281 0.109	198 0.137	232 0.112
143	EINETWORKSINDIA-000	369 0.055	322 0.015	312 0.006	258 0.063	268 0.044	273 0.026	417 0.998	435 0.997	464 0.991	271 0.020	272 0.010	296 0.455	216 0.057	231 0.159	274 0.132
144	EINETWORKSINDIA-001	419 0.096	328 0.016	308 0.006	298 0.080	276 0.048	272 0.026	415 0.998	433 0.997	460 0.990	387 0.072	269 0.009	354 0.957	246 0.077	440 0.782	330 0.178
145	EINETWORKSINDIA-002	139 0.008	158 0.004	143 0.002	131 0.028	136 0.022	124 0.014	67 0.243	96 0.188	112 0.114	170 0.007	127 0.003	352 0.953	368 0.701	185 0.127	121 0.074
146	EYEDEA-003	567 0.507	584 0.387	590 0.265	529 0.625	551 0.543	570 0.404	387 0.997	407 0.994	455 0.989	500 0.570	506 0.392			443 0.792	479 0.658
147	F8-001	559 0.458	541 0.166	487 0.035				431 0.999	453 0.998	493 0.995						
148	FACEHAWK-001	477 0.168	519 0.123	538 0.073	551 0.771	562 0.654	554 0.289	247 0.913	273 0.851	329 0.700	522 0.844	515 0.502	470 1.000	427 1.000	505 0.990	510 0.900
149	FACEPHI-000	194 0.012	229 0.006	240 0.003	200 0.042	231 0.034	260 0.024	142 0.487	159 0.367	197 0.236	205 0.009	230 0.006	142 0.066	153 0.030	195 0.135	254 0.121
150	FACEPHI-001	85 0.005	81 0.002	75 0.001	100 0.021	98 0.015	90 0.011	154 0.526	182 0.452	172 0.173	71 0.003	67 0.002	107 0.048	102 0.019	201 0.139	235 0.113
151	FASTLAHORE-000	558 0.457	556 0.214	554 0.098	631 1.000	623 1.000	623 1.000	537 1.000	525 1.000	539 1.000	560 1.000	563 1.000	579 1.000	616 1.000	621 1.000	599 1.000
152	FCI-000	297 0.033	225 0.006	243 0.003	164 0.033	178 0.026	206 0.019	293 0.979	307 0.957	352 0.820	190 0.008	183 0.004	327 0.777	353 0.538	149 0.103	142 0.079
153	FINCORE-000	491 0.187	528 0.133	535 0.071	454 0.267	475 0.217	499 0.140	473 1.000	504 1.000	483 0.995	444 0.187	464 0.108	305 0.598	343 0.418	366 0.458	419 0.349
154	FIRSTCREDITKZ-001	105 0.006	114 0.003	106 0.001	115 0.025	115 0.019	118 0.013	126 0.372	138 0.282	167 0.167	166 0.007	179 0.004	131 0.061	148 0.028	132 0.097	141 0.079
155	FISSIONSFR-001	460 0.141	517 0.121	557 0.105	420 0.194	461 0.177	512 0.153	240 0.893	275 0.859	343 0.789	447 0.193	477 0.169	350 0.950	371 0.748	376 0.490	442 0.441
156	FPT-000	447 0.129	264 0.009	268 0.004	532 0.643	523 0.372	327 0.036	434 0.999	454 0.999	463 0.991	475 0.360	458 0.100	592 1.000	416 1.000	482 0.935	481 0.668
157	FUJITSULAB-000	516 0.246	357 0.020	357 0.008	280 0.070	304 0.056	319 0.035				288 0.024	314 0.013	231 0.177	274 0.093	286 0.240	310 0.156
158	FUJITSULAB-001	581 0.655	342 0.018	326 0.006	343 0.112	306 0.058	306 0.033	378 0.996	392 0.992	395 0.940	290 0.024	298 0.011	318 0.739	336 0.310	291 0.247	299 0.146
159	GLORY-000	535 0.440	582 0.367	596 0.294	522 0.586	552 0.547	577 0.470	367 0.995	409 0.995	425 0.993	490 0.453	505 0.381			451 0.839	501 0.795
160	GLORY-001	542 0.354	571 0.305	587 0.236	520 0.582	550 0.537	575 0.448	358 0.994	396 0.993	462 0.991	482 0.408	501 0.336			448 0.819	496 0.753
161	GORILLA-001	589 0.743	588 0.406	588 0.246	524 0.590	538 0.453	559 0.314	513 1.000	538 1.000	574 1.000	491 0.468	497 0.299			563 1.000	487 0.710
162	GORILLA-002	524 0.265	550 0.188	558 0.105	473 0.342	502 0.268	522 0.170	522 1.000	550 1.000	476 0.993	457 0.250	473 0.137			525 1.000	449 0.466
163	GORILLA-003	586 0.690	575 0.317	572 0.157	537 0.684	533 0.434	547 0.247	614 1.000	605 1.000	551 1.000	481 0.407	483 0.213			580 1.000	464 0.562
164	GORILLA-004	450 0.135	502 0.089	503 0.043	426 0.202	448 0.160	474 0.101	276 0.972	308 0.958	373 0.902	423 0.135	443 0.072			360 0.438	409 0.309
165	GORILLA-005	411 0.086	467 0.058	468 0.026	408 0.179	434 0.142	454 0.088	211 0.768	243 0.697	296 0.548	400 0.088	405 0.040			319 0.315	364 0.223
166	GORILLA-006	345 0.046	393 0.027	386 0.010	352 0.118	370 0.089	387 0.053	173 0.597	209 0.525	250 0.361	304 0.028	309 0.013	225 0.166	273 0.093	276 0.218	307 0.154
167	GORILLA-007	341 0.046	390 0.027	380 0.010	327 0.101	350 0.077	363 0.045	179 0.622	211 0.529	251 0.361	296 0.026	299 0.012	255 0.264	280 0.108	251 0.178	287 0.138
168	GORILLA-008	333 0.043	374 0.024	360 0.008	341 0.111	360 0.083	372 0.048	157 0.536	183 0.457	222 0.286	313 0.030	295 0.011	272 0.319	268 0.090	249 0.178	275 0.132
169	GRIAULE-000	334 0.043	353 0.020	364 0.008	300 0.082	319 0.063	334 0.038	399 0.997	411 0.995	404 0.951	323 0.033	357 0.020	236 0.185	279 0.107	268 0.198	315 0.166
170	GRIAULE-001	201 0.012	181 0.004	168 0.002	232 0.051	189 0.028	170 0.016	253 0.927	276 0.863	309 0.621	160 0.007	145 0.003	367 0.995	361 0.610	135 0.099	150 0.081
171	GTELOTS-000	505 0.217	554 0.206	579 0.198	465 0.307	495 0.256	543 0.232	213 0.770	247 0.727	320 0.651	464 0.292	494 0.282	566 1.000	636 1.000	403 0.603	469 0.577
172	GTELOTS-001	395 0.069	440 0.046	469 0.026	347 0.115	368 0.088	384 0.052	238 0.883	241 0.680	268 0.432	388 0.072	421 0.051	262 0.287	302 0.150	311 0.294	376 0.239
173	HIK-003	472 0.156	507 0.102	521 0.057	418 0.190	444 0.158	479 0.105	295 0.980	321 0.968	384 0.924	426 0.142	448 0.080			363 0.445	423 0.359
174	HIK-004	471 0.155	503 0.098	519 0.054	412 0.182	441 0.153	475 0.101	306 0.983	333 0.976	399 0.946	424 0.137	447 0.077			359 0.434	421 0.353
175	HIK-005	422 0.102	430 0.043	431 0.019	321 0.098	349 0.077	373 0.048	463 1.000	483 0.999	508 0.998	380 0.068	398 0.036			389 0.541	384 0.258
176	HIK-006	462 0.141	441 0.047	435 0.020	340 0.111	364 0.086	385 0.052	530 1.000	545 1.000	536 0.999						
177	HISIGN-001	223 0.015	242 0.007	242 0.003	253 0.058	242 0.037	243 0.022	425 0.999	376 0.990	261 0.397	236 0.013	238 0.007	334 0.824	318 0.194	211 0.143	227 0.109
178	HYPERVERGE-001	158 0.009	175 0.004	189 0.002	189 0.039	207 0.031	224 0.020	84 0.267	109 0.210	142 0.136	162 0.007	178 0.004	119 0.053	141 0.027	142 0.101	159 0.083
179	HYPERVERGE-002	140 0.008	147 0.003	156 0.002	172 0.034	183 0.027	197 0.018	89 0.270	103 0.201	123 0.120	122 0.006	136 0.003	104 0.048	120 0.023	119 0.093	134 0.077
180	HYPERVERGE-003	222 0.015	241 0.007	238 0.003	237 0.053	260 0.042	277 0.026	121 0.346	134 0.274	176 0.174	219 0.012	223 0.005	138 0.065	162 0.033	154 0.106	168 0.085
181	HYUNDAI-ROBOTICSLAB-000	265 0.025	305 0.013	325 0.006	534 0.673	537 0.452	428 0.070	237 0.876	238 0.672	269 0.439	494 0.492	482 0.197	396 1.000	407 0.999	459 0.870	428 0.654
182	HYUNDAI-ROBOTICSLAB-001	277 0.027	297 0.013	303 0.005	565 0.921	576 0.774	560 0.322	299 0.981	271 0.839	273 0.444	516 0.699	508 0.404	474 1.000	428 1.000	483 0.937	502 0.807
183	HYUNDAI-ROBOTICSLAB-002	184 0.011	207 0.005	174 0.002	506 0.463	465 0.189	230 0.021	119 0.344	125 0.228	133 0.129	381 0.068	257 0.008	399 1.000	392 0.981	362 0.440	320 0.171
184	HYUNDAI-ROBOTICSLAB-003	179 0.011	203 0.005	163 0.002	273 0.067	228 0.034	180 0.017	109 0.311	120 0.221	132 0.127	266 0.019	195 0.004	370 0.995	373 0.781	284 0.236	215 0.102

Table 64: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
185	HZAILU-000	305 0.035	351 0.019	365 0.008	262 0.064	284 0.051	297 0.031	304 0.983	318 0.967	351 0.811	275 0.020	275 0.010	268 0.316	247 0.077	219 0.153	252 0.120
186	HZAILU-001	227 0.016	255 0.008	275 0.004	492 0.414	464 0.183	263 0.024	413 0.998	369 0.986	213 0.273	449 0.196	362 0.021	635 1.000	401 0.997	420 0.679	424 0.360
187	HZAILU-002	216 0.015	249 0.008	272 0.004	231 0.050	251 0.039	264 0.024	333 0.990	244 0.701	224 0.291	220 0.012	226 0.006	144 0.066	171 0.034	328 0.330	229 0.110
188	HZAILU-003	217 0.015	251 0.008	273 0.004	234 0.051	247 0.038	246 0.022	200 0.711	213 0.545	216 0.278	204 0.009	200 0.004	134 0.062	159 0.031	174 0.119	201 0.096
189	HZAILU-004	157 0.009	174 0.004	177 0.002	202 0.042	211 0.031	205 0.019	155 0.528	165 0.390	201 0.241	117 0.006	117 0.003	97 0.045	108 0.021	146 0.102	174 0.087
190	HZAILU-005	115 0.007	141 0.003	134 0.002	163 0.033	162 0.025	159 0.015	136 0.463	150 0.329	186 0.199	79 0.004	75 0.002	63 0.033	75 0.015	121 0.094	137 0.078
191	HZAILU-006	124 0.007	138 0.003	127 0.001	132 0.028	137 0.022	143 0.015	147 0.497	149 0.325	184 0.198	76 0.004	71 0.002	57 0.030	69 0.014	113 0.091	133 0.077
192	I2V-001	600 0.851	444 0.049	280 0.004	598 0.994	604 0.966	429 0.071	485 1.000	493 1.000	433 0.977	527 0.905	323 0.014	442 1.000	365 0.683	501 0.983	415 0.328
193	ICEHOUSE-000	535 0.298	501 0.089	485 0.033	564 0.917	588 0.852	542 0.223	468 1.000	495 1.000	523 0.999	409 0.111	423 0.052	504 1.000	426 1.000	371 0.470	417 0.346
194	ICEHOUSE-001	242 0.019	267 0.009	260 0.003	248 0.056	259 0.041	267 0.025	150 0.510	155 0.348	187 0.204	231 0.013	237 0.007	178 0.098	201 0.046	255 0.179	197 0.094
195	ICM-000	374 0.058	383 0.026	406 0.012	344 0.113	376 0.092	412 0.062	313 0.984	316 0.965	413 0.960	333 0.036	368 0.022	252 0.241	293 0.136	287 0.242	354 0.208
196	ICM-001	553 0.421	404 0.029	378 0.010	568 0.942	491 0.251	380 0.051	534 1.000	510 1.000	473 0.993	470 0.307	327 0.015	402 1.000	295 0.145	461 0.878	324 0.173
197	IDBIO-000	311 0.036	354 0.020	356 0.008	358 0.121	386 0.097	396 0.055	199 0.703	227 0.618	278 0.463	338 0.037	348 0.018	308 0.636	337 0.335	256 0.180	289 0.139
198	IDEMIA-003	572 0.541	442 0.047	442 0.021	614 1.000	452 0.165	444 0.079			635 1.000	416 0.123	430 0.061			437 0.766	472 0.630
199	IDEMIA-004	367 0.054	423 0.037	441 0.021	383 0.144	410 0.118	443 0.079	285 0.975	330 0.973	415 0.967	415 0.123	429 0.061			438 0.766	473 0.630
200	IDEMIA-005	389 0.065	432 0.044	465 0.025	411 0.181	439 0.150	477 0.102	294 0.979	337 0.977	424 0.973	419 0.130	441 0.070			462 0.879	492 0.743
201	IDEMIA-006	386 0.064	429 0.043	462 0.025	453 0.266	480 0.226	515 0.161	309 0.984	353 0.982	437 0.980	427 0.144	454 0.090			430 0.733	458 0.531
202	IDEMIA-007	307 0.035	341 0.018	350 0.008	286 0.073	300 0.055	307 0.033	624 1.000	630 1.000	599 1.000	361 0.052	369 0.022	234 0.182	282 0.109	632 1.000	527 0.982
203	IDEMIA-008	55 0.003	75 0.002	80 0.001	80 0.016	79 0.013	73 0.009	86 0.267	100 0.195	130 0.126	98 0.005	121 0.003	73 0.036	103 0.019	155 0.106	190 0.092
204	IDEMIA-009	53 0.003	57 0.001	56 0.001	44 0.012	51 0.011	52 0.008	42 0.193	52 0.131	71 0.088	60 0.003	69 0.002	44 0.027	63 0.013	51 0.074	70 0.064
205	IDEMIA-010	22 0.002	13 0.001	20 0.001	22 0.009	23 0.008	34 0.007	52 0.219	41 0.120	42 0.068	40 0.002	45 0.001	47 0.028	32 0.009	42 0.070	47 0.058
206	IDEMIA-011	17 0.002	21 0.001	54 0.001	31 0.010	43 0.009	58 0.008	40 0.192	11 0.070	14 0.050	22 0.002	33 0.001	35 0.025	25 0.008	16 0.061	23 0.052
207	IDEMIA-012	12 0.002	19 0.001	45 0.001	27 0.010	30 0.009	50 0.008	8 0.116	9 0.069	10 0.048	38 0.002	42 0.001	18 0.016	17 0.007	18 0.061	20 0.051
208	IDEMIA-013	1 0.001	1 0.000	1 0.000	1 0.006	1 0.006	2 0.005	168 0.569	137 0.282	7 0.044	5 0.001	6 0.001	5 0.010	3 0.004	9 0.055	11 0.048
209	IDENTY-000	634 1.000	636 1.000	635 1.000	610 0.999	616 0.999	621 0.999	554 1.000	580 1.000	613 1.000	539 0.991	546 0.991	386 1.000	417 1.000	508 0.992	534 0.992
210	IDENTY-001	121 0.007	122 0.003	131 0.002	112 0.024	118 0.020	140 0.014	82 0.264	73 0.162	89 0.103	96 0.005	92 0.002	105 0.048	123 0.023	126 0.094	154 0.082
211	IDENTY-002	118 0.007	117 0.003	107 0.001	101 0.021	101 0.016	94 0.011	63 0.240	69 0.152	79 0.092	94 0.005	88 0.002	99 0.047	113 0.022	76 0.080	78 0.066
212	IMAGUS-002	607 0.907	606 0.749	610 0.564	569 0.944	582 0.816	591 0.645	497 1.000	534 1.000	556 1.000						
213	IMAGUS-003	606 0.897	609 0.807	613 0.669	575 0.954	592 0.909	604 0.809	524 1.000	519 1.000	547 1.000						
214	IMAGUS-005	301 0.034	345 0.018	351 0.008	308 0.088	323 0.066	345 0.040	251 0.925	270 0.836	317 0.642	307 0.029	334 0.016	224 0.161	275 0.094	281 0.231	340 0.189
215	IMAGUS-006	320 0.039	349 0.019	353 0.008	313 0.093	332 0.069	354 0.042	297 0.980	288 0.896	307 0.617	305 0.028	324 0.015	223 0.161	271 0.092	296 0.260	334 0.181
216	IMAGUS-007	332 0.043	372 0.022	375 0.009	325 0.100	341 0.073	361 0.045	278 0.973	286 0.892	318 0.647	317 0.031	329 0.016	226 0.169	278 0.098	301 0.265	333 0.181
217	IMAGUS-008	618 0.995	619 0.974	608 0.523	579 0.958	577 0.774	553 0.285	465 1.000	419 0.996	328 0.697	497 0.520	442 0.071	387 1.000	354 0.540	384 0.518	379 0.246
218	IMDS-SOFTWARE-000	237 0.018	227 0.006	224 0.003	485 0.389	221 0.033	208 0.019	517 1.000	546 1.000	561 1.000	228 0.013	211 0.005	247 0.216	187 0.041	595 1.000	633 1.000
219	IMDS-SOFTWARE-001	267 0.025	232 0.006	221 0.003	557 0.834	226 0.034	196 0.018	532 1.000	557 1.000	553 1.000	269 0.019	206 0.005	293 0.424	215 0.055	634 1.000	611 1.000
220	IMDS-SOFTWARE-002	236 0.018	228 0.006	220 0.003	484 0.389	222 0.033	207 0.019	508 1.000	529 1.000	571 1.000	229 0.013	212 0.005	246 0.216	186 0.041	526 1.000	560 1.000
221	IMPERIAL-000	467 0.154	381 0.026	372 0.009	311 0.089	328 0.068	351 0.041	509 1.000	458 0.999	482 0.995	347 0.042	358 0.020			290 0.245	316 0.168
222	INCODE-000	552 0.421	572 0.310	580 0.199	512 0.486	531 0.420	558 0.304	467 1.000	446 0.998	479 0.994						
223	INCODE-001	537 0.317	555 0.211	560 0.111	476 0.348	509 0.296	531 0.198	544 1.000	554 1.000	542 1.000						
224	INCODE-002	533 0.285	547 0.184	556 0.100	469 0.333	503 0.269	525 0.176	401 0.998	397 0.993	430 0.975						
225	INCODE-003	532 0.282	543 0.167	543 0.083	483 0.372	498 0.264	518 0.164	478 1.000	478 0.999	497 0.996						
226	INCODE-004	421 0.099	458 0.053	454 0.023	399 0.167	413 0.120	427 0.070	398 0.997	408 0.994	386 0.928	377 0.063	388 0.031			317 0.313	366 0.226
227	INCODE-005	245 0.020	283 0.011	278 0.004	247 0.055	265 0.043	275 0.026	177 0.609	205 0.522	252 0.365	256 0.017	261 0.009	205 0.145	239 0.073	225 0.155	240 0.116
228	INCODE-006	51 0.003	60 0.001	66 0.001	46 0.013	54 0.011	68 0.009	56 0.230	71 0.159	81 0.093	68 0.003	93 0.002	39 0.026	61 0.013	72 0.079	87 0.067
229	INCODE-007	38 0.003	53 0.001	36 0.001	40 0.012	31 0.009	20 0.006	49 0.205	65 0.146	78 0.092	36 0.002	35 0.001	53 0.028	49 0.011	104 0.089	69 0.064
230	INNOV8TIF-000	565 0.487	590 0.414	594 0.286	521 0.584	546 0.517	568 0.378	360 0.994	387 0.991	444 0.983	501 0.581	509 0.421	344 0.922	375 0.825	442 0.792	484 0.702

Table 65: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
231	INNOVATRICS-002	[549] 0.374	[564] 0.233	[569] 0.139	[490] 0.403	[515] 0.310	[538] 0.209	[516] 1.000	[549] 1.000	[537] 0.999						
232	INNOVATRICS-003	[534] 0.296	[559] 0.221	[565] 0.131	[478] 0.351	[510] 0.297	[534] 0.203	[481] 1.000	[494] 1.000	[511] 0.998						
233	INNOVATRICS-004	[487] 0.183	[525] 0.131	[539] 0.073	[451] 0.262	[479] 0.222	[506] 0.149	[308] 0.983	[343] 0.980	[422] 0.972						
234	INNOVATRICS-005	[372] 0.056	[412] 0.034	[412] 0.014	[346] 0.114	[369] 0.089	[386] 0.052	[239] 0.889	[272] 0.845	[331] 0.720	[356] 0.047	[367] 0.022			[293] 0.251	[335] 0.182
235	INNOVATRICS-007	[256] 0.023	[293] 0.012	[294] 0.005	[264] 0.065	[287] 0.051	[300] 0.032	[220] 0.803	[250] 0.740	[297] 0.562	[255] 0.017	[271] 0.009	[174] 0.093	[214] 0.053	[221] 0.154	[251] 0.120
236	INNOVATRICS-008	[263] 0.024	[196] 0.005	[194] 0.002	[379] 0.138	[238] 0.036	[194] 0.018	[539] 1.000	[551] 1.000	[490] 0.995	[233] 0.013	[244] 0.007	[339] 0.886	[344] 0.430	[202] 0.139	[224] 0.108
237	INNOVATRICS-009	[114] 0.007	[123] 0.003	[154] 0.002	[152] 0.031	[164] 0.025	[190] 0.018	[138] 0.475	[151] 0.338	[183] 0.197	[159] 0.007	[190] 0.004	[112] 0.050	[132] 0.025	[156] 0.106	[188] 0.090
238	INNOVATRICS-010	[116] 0.007	[135] 0.003	[169] 0.002	[161] 0.032	[166] 0.025	[189] 0.018	[149] 0.500	[154] 0.347	[190] 0.206	[165] 0.007	[187] 0.004	[122] 0.054	[142] 0.027	[164] 0.110	[193] 0.093
239	INNOVATRICS-011	[86] 0.005	[87] 0.002	[96] 0.001	[86] 0.017	[88] 0.014	[89] 0.010	[196] 0.684	[206] 0.523	[188] 0.204	[83] 0.004	[107] 0.003	[82] 0.039	[93] 0.017	[94] 0.087	[100] 0.069
240	INNOVATRICS-012	[63] 0.004	[77] 0.002	[87] 0.001	[56] 0.014	[56] 0.011	[59] 0.008	[69] 0.247	[80] 0.168	[94] 0.104	[61] 0.003	[65] 0.002	[42] 0.027	[58] 0.012	[83] 0.083	[76] 0.066
241	INNOVATRICS-013	[57] 0.004	[63] 0.001	[63] 0.001	[102] 0.022	[96] 0.015	[72] 0.009	[57] 0.230	[67] 0.147	[70] 0.088	[133] 0.006	[94] 0.002	[106] 0.048	[89] 0.016	[101] 0.088	[66] 0.063
242	INNOVATRICS-014	[33] 0.002	[39] 0.001	[50] 0.001	[18] 0.009	[19] 0.008	[12] 0.006	[66] 0.241	[35] 0.106	[40] 0.066	[29] 0.002	[31] 0.001	[16] 0.015	[23] 0.007	[29] 0.066	[38] 0.056
243	INNOVATRICS-015	[30] 0.002	[25] 0.001	[41] 0.001	[15] 0.008	[15] 0.007	[10] 0.006	[175] 0.604	[48] 0.126	[39] 0.065	[23] 0.002	[18] 0.001	[14] 0.015	[18] 0.007	[68] 0.078	[36] 0.056
244	INNOVATRICS-016	[20] 0.002	[10] 0.001	[15] 0.001	[11] 0.008	[9] 0.007	[9] 0.006	[29] 0.260	[15] 0.082	[19] 0.055	[14] 0.001	[12] 0.001	[7] 0.012	[13] 0.005	[323] 0.321	[27] 0.054
245	INSPUR-000	[219] 0.015	[224] 0.006	[203] 0.002	[580] 0.963	[593] 0.915	[603] 0.740	[483] 1.000	[500] 1.000	[520] 0.999	[531] 0.936	[519] 0.584	[506] 1.000	[481] 1.000	[514] 0.998	[523] 0.966
246	INSPUR-001	[186] 0.011	[202] 0.005	[179] 0.002	[203] 0.043	[181] 0.027	[161] 0.016	[495] 1.000	[517] 1.000	[532] 0.999	[182] 0.008	[176] 0.004	[127] 0.057	[140] 0.027	[480] 0.931	[255] 0.121
247	INSPUR-002	[193] 0.012	[209] 0.005	[172] 0.002	[205] 0.043	[199] 0.029	[173] 0.016	[543] 1.000	[467] 0.999	[438] 0.980	[176] 0.008	[168] 0.004	[118] 0.052	[130] 0.024	[492] 0.957	[200] 0.095
248	INTELIGENSIA-000	[359] 0.050	[375] 0.024	[384] 0.010	[319] 0.097	[348] 0.077	[376] 0.049	[300] 0.981	[260] 0.784	[284] 0.501	[365] 0.053	[397] 0.036	[250] 0.235	[297] 0.145	[294] 0.255	[352] 0.208
249	INTELIGENSIA-001	[196] 0.012	[208] 0.005	[185] 0.002	[183] 0.037	[187] 0.028	[179] 0.017	[216] 0.787	[181] 0.451	[137] 0.132	[174] 0.008	[188] 0.004	[149] 0.067	[160] 0.031	[190] 0.129	[237] 0.115
250	INTELLIVISION-001	[568] 0.508	[569] 0.279	[573] 0.158	[505] 0.459	[528] 0.404	[557] 0.302	[492] 1.000	[502] 1.000	[534] 0.999	[472] 0.328	[485] 0.219	[322] 0.749	[358] 0.598	[423] 0.685	[465] 0.562
251	INTELLIVISION-002	[530] 0.281	[534] 0.154	[536] 0.072	[440] 0.236	[467] 0.196	[491] 0.127	[441] 0.999	[462] 0.999	[504] 0.997	[422] 0.134	[444] 0.073	[295] 0.437	[335] 0.297	[367] 0.460	[418] 0.348
252	INTELRESEARCH-001	[73] 0.004	[71] 0.001	[65] 0.001	[67] 0.015	[72] 0.012	[69] 0.009	[130] 0.396	[135] 0.276	[136] 0.132	[102] 0.005	[128] 0.003	[58] 0.030	[80] 0.015	[78] 0.080	[85] 0.067
253	INTEMA-000	[146] 0.008	[93] 0.002	[90] 0.001	[108] 0.022	[104] 0.017	[106] 0.012		[541] 1.000	[73] 0.090	[101] 0.005	[95] 0.002	[263] 0.288	[189] 0.042	[79] 0.081	[88] 0.067
254	INTEMA-001	[13] 0.002	[41] 0.001	[39] 0.001	[525] 0.604	[556] 0.603	[589] 0.602	[10] 0.127	[24] 0.092	[33] 0.062	[80] 0.004	[73] 0.002	[313] 0.715	[334] 0.294	[59] 0.076	[59] 0.062
255	INTOZI-001	[293] 0.031	[323] 0.015	[304] 0.006	[583] 0.968	[599] 0.938	[588] 0.575	[498] 1.000	[535] 1.000	[555] 1.000	[544] 0.998	[537] 0.884	[519] 1.000	[479] 1.000	[544] 1.000	[541] 0.999
256	INTSYSMSU-000	[620] 0.999	[623] 0.998	[625] 0.990	[613] 1.000	[617] 1.000	[619] 0.998	[475] 1.000	[491] 1.000	[507] 0.998	[545] 0.999	[545] 0.989			[518] 0.999	[532] 0.988
257	IONETWORKS-001	[427] 0.108	[313] 0.014	[328] 0.007	[251] 0.057	[272] 0.045	[285] 0.029	[477] 1.000	[499] 1.000	[512] 0.998	[248] 0.016	[262] 0.009	[192] 0.129	[230] 0.065	[310] 0.294	[233] 0.112
258	IREX-000	[393] 0.068	[399] 0.028	[349] 0.008	[324] 0.099	[314] 0.060	[302] 0.032	[325] 0.988	[306] 0.956	[324] 0.676	[351] 0.044	[289] 0.011	[267] 0.302	[224] 0.062	[243] 0.170	[282] 0.135
259	IRISID-001	[281] 0.028	[306] 0.013	[296] 0.005	[267] 0.066	[285] 0.051	[290] 0.030	[151] 0.511	[175] 0.422	[223] 0.287	[295] 0.025	[291] 0.011	[209] 0.148	[240] 0.075	[237] 0.162	[276] 0.132
260	IRISID-002	[246] 0.021	[278] 0.010	[277] 0.004	[255] 0.059	[271] 0.045	[281] 0.027	[139] 0.479	[169] 0.407	[211] 0.271	[274] 0.020	[278] 0.010	[194] 0.133	[231] 0.067	[239] 0.164	[273] 0.130
261	ISTL-001	[252] 0.023	[299] 0.013	[309] 0.006	[445] 0.249	[340] 0.073	[328] 0.036	[140] 0.481	[161] 0.378	[204] 0.255	[358] 0.048	[274] 0.010	[383] 1.000	[386] 0.952	[349] 0.407	[285] 0.138
262	ISYSTEMS-002	[469] 0.154	[487] 0.078	[482] 0.032	[394] 0.161	[419] 0.126	[446] 0.080	[412] 0.998	[437] 0.998	[474] 0.993						
263	ISYSTEMS-003	[499] 0.203	[468] 0.058	[457] 0.024	[375] 0.135	[401] 0.107	[421] 0.068	[490] 1.000	[503] 1.000	[503] 0.997						
264	KAKAO-000	[282] 0.028	[320] 0.015	[320] 0.006	[284] 0.071	[303] 0.056	[317] 0.034	[156] 0.534	[187] 0.462	[236] 0.319	[270] 0.019	[273] 0.010	[198] 0.141	[241] 0.075	[227] 0.158	[250] 0.120
265	KAKAO-001	[98] 0.006	[104] 0.002	[119] 0.001	[104] 0.022	[107] 0.017	[116] 0.013	[51] 0.218	[68] 0.150	[75] 0.090	[88] 0.004	[77] 0.002	[92] 0.042	[86] 0.016	[50] 0.074	[63] 0.063
266	KAKAO-002	[74] 0.004	[85] 0.002	[101] 0.001	[83] 0.017	[89] 0.014	[96] 0.011	[50] 0.217	[53] 0.134	[60] 0.083	[63] 0.003	[62] 0.002	[77] 0.037	[56] 0.012	[45] 0.070	[49] 0.058
267	KEDACOM-001	[323] 0.040	[371] 0.022	[408] 0.013	[318] 0.096	[338] 0.072	[390] 0.054	[330] 0.989	[368] 0.986	[426] 0.973	[367] 0.055	[411] 0.043			[313] 0.305	[388] 0.264
268	KEYLESS-000	[200] 0.012	[216] 0.006	[201] 0.002	[195] 0.041	[194] 0.029	[182] 0.017	[362] 0.994	[371] 0.987	[275] 0.455	[218] 0.011	[174] 0.004	[366] 0.994	[381] 0.888	[216] 0.149	[210] 0.099
269	KLOUDSPOT-001	[394] 0.069	[376] 0.024	[377] 0.010			[495] 0.128	[205] 0.752	[214] 0.550	[259] 0.381		[456] 0.094				[439] 0.406
270	KNERON-000			[484] 0.033			[468] 0.099									
271	KNERON-001			[515] 0.051												
272	KNOWUTECH-000	[230] 0.017	[262] 0.009	[259] 0.003	[228] 0.049	[249] 0.038	[249] 0.023	[129] 0.392	[143] 0.310	[182] 0.196	[225] 0.012	[235] 0.006	[172] 0.091	[202] 0.046	[189] 0.129	[218] 0.103
273	KOGNIZA-000	[192] 0.012	[218] 0.006	[223] 0.003	[179] 0.036	[188] 0.028	[204] 0.019	[197] 0.692	[179] 0.444	[205] 0.255	[110] 0.005	[115] 0.003	[111] 0.050	[101] 0.019	[96] 0.087	[95] 0.069
274	LEBENTECH-000	[628] 1.000	[628] 1.000	[630] 1.000	[600] 0.995	[590] 0.882	[595] 0.659	[550] 1.000	[562] 1.000	[572] 1.000	[498] 0.527	[499] 0.319	[395] 1.000	[387] 0.955	[491] 0.954	[480] 0.665
275	LIGHTVISION-000	[412] 0.086	[392] 0.027	[388] 0.010	[566] 0.925	[578] 0.797	[431] 0.072	[479] 1.000	[507] 1.000	[519] 0.998	[324] 0.033	[337] 0.017	[434] 1.000	[429] 1.000	[274] 0.214	[305] 0.153
276	LINE-000	[381] 0.062	[408] 0.030	[404] 0.012	[371] 0.132	[381] 0.095	[392] 0.054			[559] 1.000	[354] 0.046	[363] 0.021	[258] 0.278	[304] 0.151	[581] 1.000	[391] 0.268

Table 66: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
277	LINE-001	[289]0.030	[186]0.004	[165]0.002	[265]0.066	[180]0.027	[158]0.015	[536]1.000	[560]1.000	[568]1.000	[197]0.009	[184]0.004	[153]0.072	[172]0.034	[629]1.000	[509]0.858
278	LINECLOVA-002	[164]0.009	[155]0.004	[151]0.002	[513]0.508	[421]0.130	[134]0.014	[347]0.992	[347]0.980	[300]0.572	[344]0.040	[162]0.004	[393]1.000	[366]0.690	[425]0.700	[189]0.091
279	LINECLOVA-003	[69]0.004	[109]0.002	[89]0.001	[526]0.612	[557]0.606	[590]0.603			[564]1.000	[116]0.006	[103]0.002	[358]0.974	[308]0.158	[163]0.110	[206]0.097
280	LOOKMAN-003	[387]0.065	[431]0.043	[460]0.024	[370]0.131	[405]0.112	[447]0.082				[398]0.084	[431]0.061			[333]0.355	[408]0.304
281	LOOKMAN-004	[399]0.073	[434]0.044	[456]0.024	[359]0.123	[398]0.105	[438]0.075	[289]0.978	[335]0.977	[428]0.974						
282	LOOKMAN-005	[354]0.050	[406]0.030	[422]0.017	[328]0.102	[362]0.086	[414]0.063	[296]0.980	[338]0.977	[423]0.973	[376]0.062	[413]0.047			[315]0.308	[394]0.273
283	LSSYTECH-000	[49]0.003	[51]0.001	[53]0.001	[35]0.011	[41]0.009	[43]0.008	[58]0.231	[78]0.167	[100]0.106	[56]0.003	[58]0.001	[38]0.026	[43]0.011	[75]0.080	[101]0.069
284	LSSYTECH-001	[92]0.005	[18]0.001	[29]0.001	[23]0.009	[22]0.008	[37]0.007	[70]0.249	[77]0.167	[91]0.103	[44]0.002	[50]0.001	[29]0.022	[28]0.009	[37]0.068	[37]0.056
285	LUXAND-001	[458]0.140	[464]0.057	[429]0.019	[516]0.553	[525]0.385	[514]0.161	[602]1.000	[418]0.995	[332]0.723	[509]0.662	[482]0.222	[343]0.909	[367]0.698	[468]0.903	[466]0.562
286	MANTRA-000	[390]0.066	[273]0.010	[257]0.003	[257]0.063	[256]0.041	[244]0.022	[609]1.000	[614]1.000	[530]0.999	[309]0.029	[316]0.014	[212]0.152	[256]0.081	[524]1.000	[303]0.151
287	MAVERICKRESEARCH-000	[456]0.138	[491]0.084	[500]0.040	[443]0.245	[459]0.176	[480]0.105	[392]0.997	[389]0.992	[342]0.785	[413]0.122	[428]0.058	[285]0.394	[321]0.220	[392]0.546	[395]0.279
288	MAXVISION-000	[349]0.048	[400]0.028	[405]0.012	[508]0.468	[483]0.237	[393]0.054	[226]0.825	[257]0.764	[312]0.626	[428]0.149	[364]0.022	[371]0.997	[379]0.872	[395]0.557	[378]0.245
289	MAXVISION-001	[162]0.009	[169]0.004	[160]0.002	[208]0.044	[169]0.025	[155]0.015	[92]0.274	[108]0.210	[129]0.126	[154]0.007	[138]0.003	[351]0.951	[349]0.485	[141]0.100	[136]0.078
290	MAXVISION-002	[138]0.008	[150]0.004	[162]0.002	[138]0.029	[139]0.022	[135]0.014	[68]0.246	[95]0.188	[116]0.116	[150]0.007	[140]0.003	[340]0.895	[206]0.050	[151]0.104	[172]0.086
291	MAXVISION-003	[190]0.012	[215]0.006	[205]0.002	[235]0.051	[252]0.039	[242]0.022	[104]0.304	[127]0.237	[151]0.142	[216]0.011	[215]0.005	[276]0.348	[298]0.145	[166]0.111	[173]0.086
292	MAXVISION-004	[90]0.005	[95]0.002	[92]0.001	[116]0.025	[117]0.020	[137]0.014	[25]0.171	[43]0.122	[54]0.078	[78]0.004	[84]0.002	[362]0.990	[377]0.846	[82]0.082	[106]0.071
293	MEGVII-001	[502]0.209	[482]0.072	[489]0.036	[354]0.119	[385]0.097	[406]0.061									
294	MEGVII-002	[519]0.251	[486]0.077	[492]0.037	[355]0.120	[383]0.096	[403]0.059	[429]0.999	[452]0.998	[365]0.870						
295	MEGVII-003	[95]0.006	[102]0.002	[110]0.001	[53]0.013	[59]0.011	[70]0.009	[83]0.265	[64]0.145	[74]0.090	[67]0.003	[70]0.002	[46]0.027	[68]0.014	[66]0.078	[94]0.069
296	MEGVII-004	[16]0.002	[35]0.001	[60]0.001	[20]0.009	[26]0.008	[51]0.008	[44]0.196	[12]0.072	[16]0.053	[11]0.001	[34]0.001	[4]0.009	[11]0.005	[28]0.065	[53]0.060
297	MEGVII-005	[108]0.006	[36]0.001	[61]0.001	[85]0.017	[42]0.009	[53]0.008	[170]0.580	[89]0.181	[17]0.053	[120]0.006	[28]0.001	[405]1.000	[317]0.190	[171]0.117	[51]0.058
298	METSAKUURCOMPANY-000	[229]0.017	[240]0.007	[232]0.003	[302]0.084	[236]0.036	[202]0.018	[437]0.999	[463]0.999	[487]0.995	[395]0.081	[281]0.010	[369]0.995	[383]0.915	[422]0.683	[389]0.264
299	METSAKUURCOMPANY-001	[468]0.154	[263]0.009	[253]0.003	[254]0.059	[208]0.031	[201]0.018	[503]1.000	[484]1.000	[506]0.998	[209]0.010	[213]0.005	[155]0.073	[177]0.036	[454]0.841	[440]0.416
300	METSAKUURCOMPANY-002	[250]0.022	[287]0.012	[285]0.005	[204]0.043	[217]0.033	[217]0.020	[246]0.905	[221]0.577	[217]0.281	[242]0.015	[251]0.008	[311]0.684	[346]0.434	[182]0.127	[213]0.101
301	MIAXIS-001	[221]0.015	[248]0.008	[252]0.003	[199]0.042	[218]0.033	[234]0.021	[283]0.975	[305]0.954	[344]0.792	[241]0.014	[241]0.007	[214]0.153	[192]0.043	[247]0.176	[288]0.139
302	MIAXIS-002	[228]0.016	[252]0.008	[262]0.003	[221]0.046	[239]0.036	[261]0.024	[191]0.666	[212]0.536	[256]0.367	[253]0.017	[252]0.008	[237]0.192	[212]0.052	[238]0.163	[284]0.135
303	MICROFOCUS-003	[613]0.958	[616]0.931	[620]0.866	[593]0.988	[607]0.979	[615]0.948				[538]0.982	[541]0.945			[507]0.991	[525]0.977
304	MICROFOCUS-004	[622]0.999	[624]0.999	[629]0.999	[592]0.984	[605]0.975	[614]0.940				[534]0.974	[540]0.935			[503]0.989	[524]0.976
305	MICROFOCUS-005	[605]0.882	[613]0.835	[618]0.736	[573]0.951	[597]0.928	[608]0.865				[530]0.935	[536]0.848			[502]0.985	[522]0.965
306	MICROFOCUS-006	[616]0.983	[620]0.978	[622]0.963	[572]0.950	[596]0.923	[607]0.858				[529]0.923	[534]0.843			[496]0.971	[517]0.939
307	MICROSOFT-003	[350]0.048	[395]0.027	[397]0.011	[349]0.117	[373]0.091	[397]0.056				[334]0.036	[354]0.019			[283]0.233	[326]0.176
308	MICROSOFT-004	[344]0.046	[382]0.026	[389]0.010	[342]0.111	[366]0.087	[389]0.053				[325]0.033	[347]0.018			[278]0.222	[319]0.170
309	MICROSOFT-005	[347]0.047	[379]0.025	[383]0.010	[312]0.090	[335]0.070	[352]0.041	[446]0.999	[222]0.582	[246]0.347	[299]0.027	[311]0.013			[257]0.180	[281]0.134
310	MICROSOFT-006	[264]0.025	[289]0.012	[317]0.006	[224]0.048	[240]0.037	[259]0.024	[133]0.446	[162]0.379	[212]0.272	[320]0.032	[325]0.015			[250]0.178	[286]0.138
311	MOBBL-000	[376]0.059	[398]0.028	[396]0.011	[337]0.107	[346]0.077	[359]0.045	[336]0.990	[361]0.984	[381]0.918	[352]0.036	[345]0.017	[306]0.621	[294]0.145	[280]0.228	[318]0.170
312	MOBBL-001	[384]0.063	[414]0.035	[438]0.020	[338]0.109	[356]0.080	[374]0.049	[335]0.990	[357]0.983	[376]0.911	[372]0.058	[406]0.040	[307]0.622	[309]0.160	[305]0.275	[365]0.225
313	MOBBL-002	[507]0.218	[558]0.215	[584]0.213	[434]0.217	[472]0.207	[530]0.197	[329]0.996	[328]0.990	[333]0.729	[462]0.285	[495]0.283	[372]0.997	[378]0.866	[397]0.558	[461]0.553
314	MOBBL-003	[478]0.169	[521]0.124	[530]0.066	[594]0.988	[603]0.960	[579]0.481	[521]1.000	[521]1.000	[338]0.743	[467]0.302	[472]0.132	[614]1.000	[413]1.000	[429]0.728	[467]0.566
315	MUKH-002	[623]0.999	[599]0.594	[559]0.110	[467]0.326	[489]0.242	[511]0.153	[514]1.000	[513]1.000	[450]0.987	[439]0.170	[453]0.089	[319]0.741	[341]0.382	[342]0.389	[397]0.286
316	NAVERCLOUD-000	[122]0.007	[146]0.003	[135]0.002	[121]0.026	[123]0.020	[117]0.013	[321]0.987	[326]0.971	[242]0.337	[104]0.005	[120]0.003	[81]0.039	[100]0.019	[122]0.094	[145]0.079
317	NEC-000	[431]0.111	[489]0.079	[512]0.046	[401]0.171	[433]0.140	[458]0.093	[305]0.983	[341]0.979	[418]0.968					[373]0.474	[433]0.377
318	NEC-001	[466]0.148	[509]0.105	[526]0.060	[441]0.238	[469]0.197	[497]0.133	[341]0.991	[367]0.986	[421]0.972	[421]0.133	[451]0.082			[370]0.468	[434]0.378
319	NEC-002	[231]0.017	[108]0.002	[125]0.001	[144]0.029	[120]0.020	[122]0.013	[457]1.000	[475]0.999	[485]0.995	[179]0.008	[224]0.005			[419]0.676	[403]0.292
320	NEC-003	[72]0.004	[103]0.002	[149]0.002	[98]0.021	[106]0.017	[120]0.013	[242]0.901	[266]0.822	[311]0.623	[184]0.008	[227]0.006	[74]0.036	[115]0.023	[418]0.668	[385]0.261
321	NEC-004	[29]0.002	[66]0.001	[99]0.001	[71]0.015	[78]0.013	[85]0.010	[185]0.649	[228]0.618	[298]0.570	[90]0.004	[163]0.004	[21]0.019	[57]0.012	[139]0.100	[180]0.088
322	NEC-005	[109]0.006	[62]0.001	[78]0.001	[61]0.014	[67]0.012	[74]0.009	[241]0.900	[236]0.669	[168]0.168	[70]0.003	[101]0.002	[20]0.019	[42]0.011	[134]0.099	[177]0.087

Table 67: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
323	NEC-006	163 0.009	80 0.002	94 0.001	113 0.024	110 0.018	114 0.013	230 0.855	185 0.457	107 0.112	77 0.004	114 0.003	41 0.026	62 0.013	125 0.094	151 0.081
324	NEC-007	14 0.002	38 0.001	57 0.001	33 0.010	33 0.009	39 0.007	59 0.232	58 0.137	55 0.078	31 0.002	47 0.001	6 0.010	14 0.006	27 0.065	43 0.057
325	NEC-008	19 0.002	48 0.001	67 0.001	29 0.010	29 0.009	44 0.008	19 0.157	21 0.089	28 0.060	26 0.002	46 0.001	3 0.008	9 0.005	21 0.062	31 0.055
326	NEC-009	18 0.002	15 0.001	38 0.001	9 0.007	11 0.007	23 0.007	24 0.165	10 0.070	9 0.048	9 0.001	25 0.001	2 0.005	2 0.003	7 0.052	10 0.048
327	NEC-010	11 0.002	7 0.001	10 0.001	2 0.006	5 0.006	7 0.006	34 0.184	8 0.067	2 0.043	7 0.001	13 0.001	1 0.004	1 0.003	2 0.050	7 0.046
328	NEC-011	9 0.001	6 0.001	7 0.001	7 0.007	7 0.006	5 0.005	128 0.380	59 0.140	3 0.043	2 0.001	1 0.001	328 0.789	340 0.354	5 0.052	1 0.043
329	NEC-012	2 0.001	4 0.001	5 0.001	3 0.006	2 0.006	1 0.005	13 0.134	5 0.056	1 0.043	1 0.001	4 0.001	95 0.044	12 0.005	1 0.049	2 0.043
330	NEUROTECHNOLOGY-003	621 0.999	603 0.631	555 0.098	552 0.773	500 0.266	517 0.164	597 1.000	592 1.000	627 1.000						
331	NEUROTECHNOLOGY-004	436 0.119	475 0.062	471 0.027	384 0.146	407 0.117	433 0.073	384 0.996	402 0.994	458 0.990						
332	NEUROTECHNOLOGY-005	434 0.116	459 0.054	448 0.022	446 0.252	422 0.130	437 0.074	427 0.999	443 0.998	453 0.988						
333	NEUROTECHNOLOGY-006	617 0.987	565 0.249	562 0.121	616 1.000	529 0.418	536 0.206									
334	NEUROTECHNOLOGY-007	521 0.252	474 0.062	445 0.021	602 0.996	457 0.173	420 0.068	523 1.000	511 1.000	502 0.997	473 0.339	399 0.036			594 1.000	533 0.989
335	NEUROTECHNOLOGY-008	594 0.797	453 0.053	401 0.012	339 0.110	355 0.080	368 0.047	510 1.000	540 1.000	575 1.000	331 0.035	342 0.017	265 0.293	301 0.149	270 0.203	304 0.152
336	NEUROTECHNOLOGY-009	274 0.026	321 0.015	307 0.006	268 0.066	291 0.052	301 0.032	189 0.657	223 0.583	267 0.430	273 0.020	282 0.010	213 0.153	261 0.082	240 0.165	270 0.129
337	NEUROTECHNOLOGY-010	540 0.346	271 0.010	246 0.003	222 0.047	244 0.037	256 0.023	125 0.370	132 0.269	163 0.160	210 0.010	205 0.005	161 0.075	183 0.039	180 0.126	207 0.097
338	NEUROTECHNOLOGY-012	414 0.092	233 0.006	195 0.002	213 0.045	216 0.032	211 0.019	487 1.000	310 0.959	144 0.139	191 0.008	171 0.004	130 0.061	150 0.028	472 0.916	181 0.088
339	NEUROTECHNOLOGY-013	375 0.058	168 0.004	118 0.001	148 0.030	148 0.023	141 0.014	281 0.974	146 0.316	102 0.107	144 0.006	125 0.003	98 0.046	112 0.022	412 0.641	162 0.083
340	NEUROTECHNOLOGY-014	128 0.007	131 0.003	111 0.001	150 0.031	138 0.022	119 0.013	166 0.568	113 0.215	101 0.107	134 0.006	123 0.003	116 0.052	125 0.024	124 0.094	125 0.075
341	NEUROTECHNOLOGY-015	119 0.007	137 0.003	95 0.001	158 0.032	158 0.024	136 0.014	75 0.254	82 0.170	93 0.104	135 0.006	116 0.003	110 0.049	114 0.023	93 0.086	90 0.068
342	NEUROTECHNOLOGY-016	125 0.007	130 0.003	97 0.001	133 0.028	128 0.021	115 0.013	62 0.238	90 0.182	105 0.109	161 0.007	152 0.003	120 0.054	131 0.025	118 0.093	114 0.073
343	NEWLAND-002	570 0.522	591 0.437	595 0.294	515 0.535	539 0.466	563 0.335	443 0.999	466 0.999	510 0.998						
344	NEXTMETALABS-001	536 0.305	561 0.233	571 0.145	636 1.000	619 1.000	612 0.933	520 1.000	548 1.000	562 1.000	586 1.000	548 0.999	612 1.000	569 1.000	602 1.000	629 1.000
345	NEXTMETALABS-002	144 0.008	160 0.004	145 0.002	460 0.285	418 0.125	209 0.019	80 0.260	101 0.198	119 0.118	420 0.131	370 0.023	397 1.000	406 0.999	382 0.510	383 0.257
346	NEXTMETALABS-003	145 0.008	161 0.004	144 0.002	141 0.029	141 0.022	132 0.014	77 0.257	99 0.194	118 0.117	133 0.006	137 0.003	117 0.052	127 0.024	158 0.107	175 0.087
347	NOBLIS-001	626 1.000	627 1.000	626 0.991	637 1.000	620 1.000	624 1.000	522 1.000	543 1.000	558 1.000						
348	NOBLIS-002	624 1.000	622 0.997	605 0.488	624 1.000	628 1.000	632 1.000	500 1.000	537 1.000	573 1.000						
349	NOMINDER-000	195 0.012	188 0.004	167 0.002	510 0.481	506 0.280	286 0.029	301 0.982	299 0.929	272 0.441	475 0.370	452 0.088	599 1.000	418 1.000	444 0.799	462 0.557
350	NOMINDER-001	199 0.012	201 0.005	207 0.002	391 0.156	363 0.086	222 0.020	236 0.875	239 0.675	218 0.282	177 0.008	148 0.003	394 1.000	385 0.948	220 0.154	155 0.082
351	NOMINDER-002	187 0.011	199 0.005	199 0.002	211 0.045	198 0.029	203 0.019	111 0.317	121 0.223	124 0.121	146 0.006	135 0.003	337 0.867	223 0.062	107 0.089	99 0.069
352	NOMINDER-003	171 0.010	192 0.005	187 0.002	190 0.039	185 0.027	187 0.018	102 0.299	105 0.208	111 0.114	130 0.006	126 0.003	333 0.816	170 0.034	89 0.085	74 0.065
353	NOMINDER-004	175 0.010	185 0.004	188 0.002	176 0.035	174 0.026	186 0.017	99 0.296	106 0.208	114 0.116	139 0.006	129 0.003	273 0.331	138 0.027	106 0.089	93 0.068
354	NOMINDER-005	166 0.010	127 0.003	138 0.002	207 0.044	132 0.022	156 0.015	100 0.297	81 0.169	77 0.091	121 0.006	119 0.003	277 0.356	149 0.028	103 0.088	86 0.067
355	NOTIONTAG-000	294 0.032	336 0.017	334 0.007	290 0.076	309 0.059	324 0.036	192 0.667	225 0.606	277 0.460	281 0.021	293 0.011	211 0.150	263 0.084	246 0.176	290 0.140
356	NTECHLAB-003	403 0.079	457 0.053	422 0.028	385 0.148	408 0.118	439 0.075	234 0.872	269 0.835	341 0.749						
357	NTECHLAB-004	382 0.062	425 0.040	443 0.021	369 0.131	396 0.105	417 0.065	233 0.866	268 0.831	339 0.743	364 0.053	386 0.030			298 0.263	361 0.214
358	NTECHLAB-005	379 0.062	427 0.042	444 0.021	368 0.130	393 0.102	415 0.063	223 0.813	258 0.769	321 0.657	389 0.073	403 0.039			309 0.294	367 0.227
359	NTECHLAB-006	371 0.056	419 0.036	427 0.018	356 0.121	378 0.094	402 0.059	219 0.799	256 0.751	315 0.631	370 0.057	391 0.032			297 0.260	350 0.207
360	NTECHLAB-007	322 0.040	378 0.025	398 0.012	303 0.085	326 0.067	350 0.041	218 0.793	255 0.747	316 0.638	321 0.032	344 0.017			279 0.223	325 0.176
361	NTECHLAB-008	261 0.024	311 0.014	327 0.006	250 0.057	273 0.045	287 0.029	172 0.596	207 0.524	260 0.384	326 0.033	349 0.018			258 0.183	291 0.140
362	NTECHLAB-009	165 0.009	194 0.005	210 0.002	154 0.028	142 0.022	138 0.014	152 0.517	176 0.423	230 0.303	243 0.015	254 0.008	182 0.109	221 0.061	207 0.142	236 0.114
363	NTECHLAB-010	81 0.005	101 0.002	93 0.001	91 0.018	91 0.015	92 0.011	115 0.327	129 0.243	161 0.159	151 0.007	177 0.004	129 0.059	157 0.031	133 0.098	131 0.077
364	NTECHLAB-011	94 0.006	115 0.003	104 0.001	90 0.018	90 0.015	87 0.010	94 0.282	115 0.219	148 0.140	196 0.009	194 0.004	158 0.074	182 0.038	112 0.091	124 0.075
365	OMNIGARDE-000	170 0.010	165 0.004	166 0.002	139 0.029	149 0.023	139 0.014	114 0.324	128 0.243	156 0.150	167 0.007	160 0.003	123 0.055	145 0.027	196 0.136	191 0.092
366	OMNIGARDE-001	71 0.004	86 0.002	72 0.001	95 0.019	95 0.015	81 0.010	43 0.193	56 0.135	67 0.087	73 0.003	74 0.002	59 0.030	74 0.015	65 0.078	91 0.068
367	OMNIGARDE-002	54 0.003	61 0.001	64 0.001	54 0.013	50 0.011	54 0.008	18 0.156	37 0.111	56 0.079	50 0.002	56 0.001	25 0.022	52 0.012	71 0.079	89 0.068
368	ONFIDO-000	272 0.026	303 0.013	324 0.006	210 0.045	232 0.035	238 0.022	288 0.978	327 0.971	400 0.947	311 0.029	330 0.016	253 0.244	285 0.113	266 0.194	301 0.149

Table 68: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
369	ONFIDO-001	247 0.021	280 0.010	284 0.005	154 0.031	153 0.023	133 0.014	225 0.823	252 0.744	303 0.592	238 0.014	233 0.006	203 0.142	217 0.059	218 0.152	253 0.121
370	OPTIEXACTA-000	25 0.002	31 0.001	31 0.001	24 0.010	25 0.008	26 0.007	36 0.186	54 0.135	63 0.085	64 0.003	76 0.002	66 0.033	79 0.015	44 0.070	52 0.059
371	PANGIAM-000	206 0.013	219 0.006	204 0.002	192 0.039	201 0.030	198 0.018	279 0.974	144 0.310	165 0.165	203 0.009	201 0.005	196 0.136	168 0.033	152 0.105	161 0.083
372	PANGIAM-001	251 0.023	284 0.011	352 0.008	187 0.039	200 0.030	221 0.020	183 0.646	160 0.376	169 0.171	195 0.009	175 0.004	336 0.860	253 0.081	205 0.141	169 0.085
373	PAPSAV1923-000	117 0.007	126 0.003	105 0.001	124 0.027	125 0.021	126 0.014	103 0.300	122 0.226	147 0.140	206 0.009	231 0.006	109 0.049	124 0.024	123 0.094	130 0.076
374	PARAVISION-000	527 0.271	500 0.088	509 0.045	501 0.447	454 0.170	471 0.100	545 1.000	470 0.999	498 0.997	492 0.470	511 0.443			477 0.926	499 0.779
375	PARAVISION-001	457 0.139	443 0.049	440 0.020	431 0.207	420 0.128	436 0.074	540 1.000	457 0.999	480 0.994	489 0.444	510 0.428			431 0.739	468 0.573
376	PARAVISION-002	410 0.085	445 0.050	450 0.022	389 0.152	411 0.119	441 0.076	346 0.992	356 0.983	340 0.746	394 0.080	412 0.043			379 0.497	392 0.268
377	PARAVISION-003	383 0.063	413 0.035	417 0.015	362 0.124	382 0.096	405 0.060	395 0.997	404 0.994	334 0.730	371 0.058	394 0.034			312 0.296	371 0.232
378	PARAVISION-004	266 0.025	274 0.010	276 0.004	227 0.049	245 0.038	257 0.024	504 1.000	527 1.000	348 0.794	262 0.018	287 0.011			470 0.908	357 0.211
379	PARAVISION-005	212 0.014	163 0.004	183 0.002	153 0.031	159 0.024	172 0.016	391 0.997	344 0.980	170 0.171	214 0.011	255 0.008			192 0.132	248 0.120
380	PARAVISION-007	348 0.047	156 0.004	132 0.002	518 0.560	168 0.025	154 0.015	535 1.000	559 1.000	567 1.000	193 0.009	232 0.006	184 0.113	126 0.024	627 1.000	602 1.000
381	PARAVISION-009	113 0.007	106 0.002	77 0.001	117 0.026	114 0.019	100 0.012	214 0.775	249 0.732	295 0.545	65 0.003	63 0.002	65 0.033	81 0.015	49 0.073	57 0.061
382	PARAVISION-012	65 0.004	55 0.001	43 0.001	76 0.016	73 0.012	66 0.009	158 0.541	188 0.469	210 0.263	55 0.002	59 0.002	34 0.025	54 0.012	39 0.068	42 0.057
383	PARAVISION-014	47 0.003	45 0.001	42 0.001	60 0.014	61 0.011	57 0.008	171 0.595	210 0.527	195 0.235	37 0.002	38 0.001	67 0.033	59 0.012	20 0.061	22 0.052
384	PARAVISION-016	36 0.002	34 0.001	35 0.001	64 0.015	58 0.011	48 0.008	17 0.156	38 0.112	43 0.068	24 0.002	20 0.001	48 0.028	46 0.011	24 0.065	29 0.055
385	PARAVISION-019	43 0.003	28 0.001	17 0.001	58 0.014	46 0.010	36 0.007	54 0.226	86 0.176	104 0.108	21 0.002	14 0.001	54 0.028	44 0.011	19 0.061	14 0.049
386	PARAVISION-021	35 0.002	30 0.001	16 0.001	59 0.014	49 0.011	31 0.007	20 0.159	47 0.124	46 0.074	18 0.001	16 0.001	51 0.028	45 0.011	17 0.061	12 0.049
387	PARAVISION-022	37 0.002	32 0.001	12 0.001	50 0.013	44 0.010	33 0.007	39 0.190	57 0.135	52 0.076	15 0.001	11 0.001	37 0.026	38 0.010	23 0.063	19 0.051
388	PENTACOMP-000	619 0.996	621 0.996	627 0.996	607 0.998	615 0.998	620 0.998	337 0.991	384 0.990	457 0.990	533 0.955	542 0.955	496 1.000	422 1.000	493 0.960	519 0.958
389	PENTACOMP-001	303 0.034	332 0.016	344 0.008	498 0.436	455 0.172	325 0.036	193 0.667	200 0.517	257 0.374	392 0.076	313 0.013	423 1.000	402 0.998	356 0.430	327 0.177
390	PIXELALL-002	582 0.664	508 0.105	477 0.029	588 0.974	526 0.388	449 0.083		539 1.000	557 1.000	502 0.602	416 0.047			561 1.000	543 1.000
391	PIXELALL-003	351 0.048	366 0.021	368 0.009	329 0.102	339 0.073	355 0.043		482 1.000	509 0.998	339 0.037	355 0.020			394 0.554	382 0.255
392	PIXELALL-004	437 0.120	344 0.018	332 0.007	553 0.783	354 0.079	330 0.037		522 1.000	525 0.999	359 0.051	328 0.015			509 0.994	518 0.942
393	PIXELALL-005	402 0.078	286 0.012	279 0.004	504 0.456	280 0.050	280 0.027		530 1.000	538 0.999	300 0.027	338 0.017	239 0.203	236 0.071	521 1.000	528 0.983
394	PRIVID-000	313 0.036	387 0.027	452 0.022	585 0.968	584 0.825	537 0.207	464 1.000	281 0.882	185 0.199	504 0.603	481 0.195	615 1.000	573 1.000	498 0.972	506 0.842
395	PROTENET-000	604 0.875	607 0.794	609 0.540	561 0.884	585 0.834	599 0.682	324 0.988	342 0.980	387 0.930	519 0.804	520 0.587	523 1.000	425 1.000	464 0.890	493 0.747
396	PROTENET-001	309 0.035	360 0.021	367 0.009	415 0.187	397 0.105	356 0.043	248 0.914	215 0.551	248 0.347	414 0.123	390 0.032	581 1.000	423 1.000	405 0.610	429 0.367
397	PSL-000	101 0.006	105 0.002	74 0.001	143 0.029	127 0.021	107 0.012	98 0.295	104 0.208	113 0.115	111 0.005	91 0.002	121 0.054	134 0.026	131 0.097	119 0.074
398	PSL-001	80 0.005	94 0.002	68 0.001	105 0.022	102 0.017	91 0.011	28 0.173	44 0.122	49 0.075	69 0.003	60 0.002	50 0.028	65 0.013	52 0.075	54 0.060
399	PSL-002	50 0.003	42 0.001	24 0.001	69 0.015	60 0.011	42 0.008	16 0.155	27 0.095	26 0.059	42 0.002	26 0.001	31 0.023	41 0.010	69 0.079	56 0.060
400	PSL-003	66 0.004	46 0.001	21 0.001	57 0.014	55 0.011	35 0.007	47 0.202	33 0.102	37 0.064	39 0.002	39 0.001	28 0.022	35 0.010	148 0.102	46 0.058
401	PSL-004	52 0.003	33 0.001	14 0.001	41 0.012	38 0.009	22 0.007	29 0.174	32 0.101	38 0.065	25 0.002	19 0.001	30 0.022	39 0.010	58 0.076	44 0.058
402	PSL-005	42 0.003	43 0.001	51 0.001	37 0.011	35 0.009	30 0.007	7 0.115	17 0.083	31 0.062	30 0.002	21 0.001	33 0.024	53 0.012	26 0.065	39 0.057
403	PSL-006	34 0.002	22 0.001	11 0.001	34 0.011	24 0.008	16 0.006	9 0.123	20 0.085	20 0.058	17 0.001	9 0.001	22 0.019	30 0.009	31 0.067	28 0.054
404	PTAKURATSATU-000	373 0.057	418 0.036	423 0.017	398 0.165	417 0.124	430 0.071	264 0.947	296 0.923	362 0.867	355 0.046	365 0.022	242 0.206	287 0.120	282 0.232	332 0.179
405	PYRAMID-000	137 0.008	153 0.004	141 0.002	125 0.027	124 0.021	123 0.013	64 0.241	97 0.190	125 0.122	155 0.007	134 0.003	294 0.433	211 0.051	127 0.095	120 0.074
406	PYRAMID-001	148 0.008	182 0.004	184 0.002	242 0.054	275 0.047	337 0.039	81 0.262	98 0.193	128 0.125	301 0.027	374 0.023	160 0.075	194 0.044	138 0.100	160 0.083
407	QAZSMARTVISIONAI-000	27 0.002	47 0.001	40 0.001	47 0.013	47 0.010	47 0.008	11 0.130	26 0.094	36 0.064	54 0.002	52 0.001	87 0.041	50 0.011	32 0.067	33 0.056
408	QAZSMARTVISIONAI-001	15 0.002	9 0.001	6 0.001	19 0.009	20 0.008	13 0.006	5 0.095	7 0.065	13 0.050	28 0.002	22 0.001	40 0.026	21 0.007	13 0.058	17 0.049
409	QAZSMARTVISIONAI-002	5 0.001	3 0.001	4 0.000	6 0.007	6 0.006	6 0.005	6 0.107	2 0.052	5 0.043	6 0.001	3 0.001	10 0.014	5 0.004	8 0.052	6 0.045
410	QAZSMARTVISIONAI-003	8 0.001	5 0.001	3 0.000	4 0.006	4 0.006	4 0.005	2 0.071	4 0.053	6 0.044	4 0.001	5 0.001	17 0.016	6 0.004	4 0.052	4 0.044
411	QNAP-000	615 0.971	524 0.129	516 0.052	606 0.998	485 0.238	489 0.117	552 1.000	564 1.000	560 1.000	445 0.191	439 0.068	303 0.539	330 0.263	513 0.998	531 0.985
412	QNAP-001	409 0.083	456 0.053	459 0.024	407 0.176	429 0.137	451 0.085	259 0.942	297 0.927	364 0.869	396 0.081	408 0.041	280 0.368	323 0.227	329 0.331	380 0.248
413	QNAP-002	338 0.044	384 0.026	407 0.013	377 0.136	399 0.106	422 0.068	224 0.818	259 0.769	308 0.617	363 0.052	377 0.025	260 0.281	312 0.171	303 0.272	360 0.214
414	QNAP-003	312 0.036	348 0.019	362 0.008	590 0.980	586 0.835	398 0.057	486 1.000	391 0.992	253 0.365	496 0.502	438 0.066	535 1.000	421 1.000	458 0.865	431 0.373

Table 69: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
415	QNAP-004	299 0.034	308 0.014	299 0.005	609 0.999	591 0.890	333 0.038	511 1.000	426 0.996	228 0.295	495 0.501	427 0.056	451 1.000	409 0.999	474 0.922	457 0.530
416	QNAP-005	232 0.018	276 0.010	292 0.005	301 0.082	322 0.066	364 0.045	135 0.460	157 0.359	194 0.227	289 0.024	321 0.014	183 0.110	222 0.061	183 0.127	223 0.108
417	QNAP-006	234 0.018	256 0.008	258 0.003	244 0.055	263 0.042	276 0.026	144 0.488	164 0.388	202 0.244	268 0.019	266 0.009	193 0.131	229 0.065	197 0.137	217 0.102
418	QUANTASOFT-001	587 0.713	604 0.638	606 0.493												
419	RANKONE-002	483 0.182	514 0.117	533 0.070	466 0.308	496 0.261	529 0.190									
420	RANKONE-003	482 0.182	515 0.117	534 0.070	462 0.300	494 0.255	527 0.187									
421	RANKONE-004	518 0.250	551 0.193	564 0.124	511 0.482	532 0.426	561 0.324									
422	RANKONE-005	416 0.095	470 0.059	483 0.033	432 0.212	458 0.173	490 0.119	445 0.999	444 0.998	481 0.994						
423	RANKONE-006	378 0.060	420 0.037	433 0.019				317 0.987	336 0.977	392 0.936						
424	RANKONE-007	300 0.034	369 0.022	392 0.011	353 0.118	379 0.095	407 0.061	284 0.975	317 0.967	383 0.923						
425	RANKONE-009	290 0.030	339 0.017	355 0.008	320 0.098	344 0.076	360 0.045	302 0.982	322 0.969	360 0.857	374 0.062	385 0.029			327 0.328	349 0.206
426	RANKONE-010	253 0.023	304 0.013	333 0.007	293 0.077	307 0.058	323 0.036	245 0.904	264 0.799	319 0.648	362 0.052	380 0.027	243 0.208	286 0.119	295 0.259	343 0.194
427	RANKONE-011	429 0.109	257 0.008	270 0.004	297 0.079	277 0.048	288 0.029				336 0.037	345 0.017	233 0.182	272 0.092	499 0.977	448 0.465
428	RANKONE-012	244 0.020	243 0.007	261 0.003	285 0.072	298 0.053	291 0.030				310 0.029	318 0.014	204 0.144	237 0.072	369 0.465	268 0.128
429	RANKONE-013	172 0.010	176 0.004	180 0.002	219 0.046	227 0.034	216 0.020	410 0.998	422 0.996	189 0.205	263 0.018	250 0.008	200 0.141	207 0.050	208 0.142	208 0.097
430	RANKONE-014	132 0.008	128 0.003	116 0.001	167 0.033	160 0.024	147 0.015				200 0.009	181 0.004	165 0.081	133 0.026	592 1.000	156 0.083
431	RANKONE-015	91 0.005	110 0.002	109 0.001	114 0.024	111 0.018	108 0.012				156 0.007	154 0.003	93 0.042	105 0.020	162 0.109	178 0.088
432	REALNETWORKS-000	544 0.365	562 0.233	568 0.137	496 0.433	518 0.319	540 0.209									
433	REALNETWORKS-001	545 0.365	563 0.233	567 0.137	497 0.433	517 0.319	539 0.209									
434	REALNETWORKS-002	548 0.369	560 0.231	566 0.137	493 0.416	516 0.315	541 0.209									
435	REALNETWORKS-003	528 0.272	538 0.159	546 0.090	472 0.342	499 0.266	523 0.172	440 0.999	450 0.998	449 0.987	437 0.164	460 0.103			380 0.500	427 0.364
436	REALNETWORKS-004	514 0.242	537 0.158	545 0.090	479 0.353	497 0.263	520 0.169	459 1.000	472 0.999	472 0.992	440 0.170	461 0.103			406 0.613	430 0.370
437	REALNETWORKS-005	360 0.052	394 0.027	400 0.012	315 0.094	342 0.074	367 0.047	307 0.983	324 0.970	372 0.894	335 0.037	340 0.017	248 0.223	289 0.123	275 0.215	314 0.165
438	REALNETWORKS-006	269 0.025	312 0.014	301 0.005	278 0.068	294 0.053	305 0.032	350 0.993	346 0.980	355 0.836	249 0.016	258 0.008	188 0.120	225 0.063	222 0.154	239 0.116
439	REALNETWORKS-007	241 0.019	268 0.009	269 0.004	249 0.057	266 0.043	279 0.027	343 0.991	340 0.979	359 0.853	221 0.012	221 0.005	297 0.463	226 0.063	204 0.140	212 0.100
440	REALNETWORKS-008	197 0.012	220 0.006	229 0.003	182 0.037	197 0.029	199 0.018	327 0.988	319 0.967	209 0.262	187 0.008	196 0.004	151 0.068	174 0.035	187 0.129	230 0.110
441	REALSENSE-001	133 0.008	116 0.003	100 0.001	79 0.016	80 0.013	75 0.009	120 0.346	119 0.221	121 0.119	85 0.004	97 0.002	49 0.028	70 0.014	63 0.077	73 0.065
442	REALSENSE-002	64 0.004	54 0.001	44 0.001	52 0.013	57 0.011	62 0.009	93 0.279	63 0.144	64 0.085	75 0.004	83 0.002	36 0.025	60 0.013	54 0.075	68 0.063
443	RECOGNITO-000	60 0.004	79 0.002	76 0.001	68 0.015	64 0.012	67 0.009	76 0.257	85 0.175	109 0.113	168 0.007	159 0.003	316 0.730	347 0.437	115 0.092	105 0.070
444	RECOGNITO-001	28 0.002	50 0.001	46 0.001	32 0.010	32 0.009	32 0.007	274 0.971	290 0.899	86 0.099	72 0.003	80 0.002	68 0.033	82 0.016	53 0.075	67 0.063
445	RECOGNITO-002	26 0.002	29 0.001	32 0.001	26 0.010	27 0.008	28 0.007	37 0.186	55 0.135	62 0.085	59 0.003	61 0.002	299 0.510	326 0.248	36 0.068	32 0.055
446	REMARKAI-000	497 0.196	522 0.127	524 0.059	452 0.263	470 0.203	493 0.123									
447	REMARKAI-000	441 0.124	460 0.055	453 0.022	402 0.173	412 0.120	425 0.070	451 0.999	468 0.999	484 0.995	384 0.069	393 0.033			428 0.717	412 0.315
448	REMARKAI-002	490 0.186	520 0.123	523 0.058	444 0.248	468 0.196	492 0.122	356 0.993	385 0.991	436 0.980						
449	RENDIP-000	254 0.023	283 0.012	289 0.005	417 0.189	308 0.059	315 0.034	263 0.945	287 0.892	337 0.741	282 0.022	308 0.013	235 0.185	267 0.089	241 0.167	272 0.130
450	REVEALMEDIA-000	255 0.023	288 0.012	298 0.005	241 0.054	262 0.042	269 0.025	204 0.752	240 0.677	293 0.533	229 0.021	288 0.011	175 0.093	210 0.051	210 0.143	244 0.118
451	RNIT-000	550 0.388	281 0.011	228 0.003	201 0.042	176 0.026	164 0.016	194 0.672	198 0.510	221 0.285	223 0.012	209 0.005	251 0.239	188 0.041	177 0.123	165 0.084
452	ROC-016	96 0.006	99 0.002	86 0.001	103 0.022	105 0.017	102 0.012				171 0.008	170 0.004	96 0.045	106 0.020	159 0.108	144 0.079
453	ROC-017	106 0.006	121 0.003	108 0.001	135 0.028	135 0.022	127 0.014	74 0.253	84 0.174	97 0.105	149 0.006	132 0.003	83 0.039	83 0.016	114 0.092	111 0.073
454	ROC-018	103 0.006	91 0.002	79 0.001	110 0.023	109 0.018	97 0.011	53 0.223	62 0.142	72 0.089	118 0.006	105 0.002	76 0.037	73 0.015	85 0.085	81 0.066
455	ROC-019	59 0.004	52 0.001	34 0.001	38 0.011	36 0.009	25 0.007	48 0.203	29 0.098	32 0.062	47 0.002	43 0.001	45 0.027	36 0.010	43 0.070	40 0.057
456	ROC-020	56 0.003	56 0.001	52 0.001	82 0.017	77 0.013	60 0.008	35 0.186	31 0.101	34 0.063	62 0.003	57 0.001	89 0.041	55 0.012	35 0.067	25 0.053
457	ROC-021	31 0.002	16 0.001	23 0.001	21 0.009	17 0.007	11 0.006	12 0.132	16 0.083	18 0.054	27 0.002	27 0.001	156 0.073	66 0.013	15 0.060	15 0.049
458	S1-000	452 0.136	397 0.028	390 0.010	366 0.129	361 0.085	371 0.048	549 1.000	563 1.000	302 0.591	357 0.047	350 0.018	461 1.000	288 0.123	574 1.000	475 0.632
459	S1-001	365 0.053	325 0.015	329 0.007	266 0.066	290 0.052	310 0.033	344 0.992	364 0.985	406 0.952	265 0.019	280 0.010	195 0.136	242 0.075	215 0.148	245 0.119
460	S1-002	377 0.060	210 0.005	198 0.002	305 0.085	206 0.031	191 0.018	250 0.923	94 0.186	65 0.085	163 0.007	150 0.003	330 0.792	305 0.151	453 0.841	294 0.144

Table 70: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
461	s1-003	356 0.050	258 0.008	249 0.003	236 0.052	241 0.037	240 0.022	502 1.000	509 1.000	454 0.989	240 0.014	228 0.006	286 0.396	180 0.037	539 1.000	356 0.209
462	s1-004	258 0.024	157 0.004	161 0.002	170 0.034	161 0.025	169 0.016	489 1.000	430 0.997	177 0.182	138 0.006	130 0.003	137 0.064	128 0.024	537 1.000	170 0.086
463	s1-005	262 0.024	171 0.004	170 0.002	151 0.031	143 0.022	157 0.015	448 0.999	405 0.994	143 0.138	125 0.006	96 0.002	141 0.065	119 0.023	462 0.895	129 0.076
464	s1-006	226 0.016	92 0.002	102 0.001	99 0.021	100 0.016	101 0.012	551 1.000	476 0.999	90 0.103	106 0.005	98 0.002	85 0.040	98 0.019	355 0.423	104 0.070
465	s1-007	84 0.005	78 0.002	85 0.001	89 0.018	86 0.014	88 0.010	124 0.367	75 0.165	68 0.088	87 0.004	85 0.002	69 0.034	85 0.016	60 0.077	60 0.062
466	SANSAP-001	189 0.012	172 0.004	175 0.002	549 0.755	543 0.480	268 0.025	328 0.989	309 0.958	178 0.184	292 0.024	225 0.006	390 1.000	397 0.991	306 0.280	259 0.124
467	SCANOVATE-000	423 0.102	479 0.066	478 0.030	461 0.296	488 0.240	507 0.150	255 0.930	285 0.892	349 0.800	453 0.215	470 0.118			344 0.400	405 0.299
468	SCANOVATE-001	445 0.128	490 0.081	493 0.037	459 0.281	481 0.227	500 0.140	257 0.934	294 0.910	354 0.832	446 0.192	462 0.103			348 0.404	401 0.290
469	SCYLLA-001	288 0.030	333 0.016	342 0.007	608 0.999	613 0.995	598 0.680	190 0.664	184 0.457	229 0.299	487 0.431	492 0.279		437 1.000	488 0.953	489 0.714
470	SECURITYPITCH-000	448 0.130	471 0.060	463 0.025	392 0.158	416 0.123	434 0.074	404 0.998	431 0.997	371 0.890	385 0.071	395 0.034	266 0.302	311 0.171	322 0.320	374 0.237
471	SENSETIME-000	310 0.035	359 0.020	371 0.009	294 0.078	318 0.063	346 0.040	628 1.000	635 1.000	451 0.987						
472	SENSETIME-001	314 0.036	367 0.022	374 0.009	299 0.080	321 0.064	353 0.041									
473	SENSETIME-002	315 0.036	316 0.014	413 0.014	361 0.124	190 0.028	254 0.023	389 0.997	401 0.994	434 0.978	319 0.032	341 0.017			386 0.523	313 0.160
474	SENSETIME-003	67 0.004	70 0.001	70 0.001	63 0.014	70 0.012	71 0.009	174 0.602	189 0.471	231 0.303	180 0.008	219 0.005			193 0.133	238 0.115
475	SENSETIME-004	46 0.003	65 0.001	69 0.001	70 0.015	75 0.013	79 0.010	97 0.293	117 0.220	146 0.140	128 0.006	167 0.004			168 0.113	211 0.100
476	SENSETIME-005	177 0.010	89 0.002	62 0.001	93 0.018	85 0.014	76 0.010	71 0.250	74 0.163	80 0.092	153 0.007	172 0.004	115 0.051	121 0.023	150 0.104	194 0.093
477	SENSETIME-006	70 0.004	67 0.001	55 0.001	74 0.016	71 0.012	63 0.009	424 0.999	447 0.998	323 0.676	82 0.004	72 0.002	70 0.034	87 0.016	116 0.093	143 0.079
478	SENSETIME-007	44 0.003	40 0.001	37 0.001	39 0.012	40 0.009	38 0.007	460 1.000	474 0.999	292 0.533	58 0.003	51 0.001	32 0.024	47 0.011	87 0.085	122 0.074
479	SENSETIME-008	23 0.002	37 0.001	28 0.001	36 0.011	34 0.009	29 0.007	332 0.990	167 0.398	50 0.075	48 0.002	41 0.001	24 0.021	31 0.009	77 0.080	118 0.074
480	SENSETIME-009	6 0.001	11 0.001	18 0.001	13 0.008	13 0.007	15 0.006	439 0.999	277 0.866	23 0.058	19 0.001	23 0.001	12 0.015	19 0.007	90 0.085	148 0.081
481	SENSETIME-010	10 0.001	14 0.001	9 0.001	25 0.010	21 0.008	24 0.007	30 0.176	28 0.096	21 0.058	32 0.002	32 0.001	23 0.020	26 0.008	100 0.087	153 0.082
482	SENSETIME-011	7 0.001	12 0.001	19 0.001	14 0.008	12 0.007	14 0.006	438 0.999	278 0.866	22 0.058	20 0.001	24 0.001	13 0.015	20 0.007	91 0.085	149 0.081
483	SENSETIME-012	4 0.001	8 0.001	8 0.001	8 0.007	8 0.006	8 0.006	14 0.135	22 0.090	29 0.060	12 0.001	8 0.001	26 0.022	37 0.010	88 0.085	147 0.080
484	SERENDIPITY-000	257 0.023	292 0.012	288 0.005	509 0.474	473 0.213	326 0.036	228 0.842	253 0.745	301 0.575	260 0.018	276 0.010	406 1.000	410 0.999	209 0.142	220 0.106
485	SEVENTHSENSE-001	149 0.009	184 0.004	222 0.003	245 0.055	264 0.043	266 0.025	122 0.346	124 0.228	134 0.130	207 0.010	207 0.005	309 0.643	339 0.351	184 0.127	198 0.095
486	SHAMAN-003	566 0.506	593 0.451	598 0.346	533 0.650	555 0.597	578 0.472									
487	SHAMAN-004	584 0.678	602 0.615	604 0.488	556 0.812	574 0.754	593 0.639									
488	SHAMAN-006	488 0.184	530 0.141	548 0.092	457 0.278	484 0.237	519 0.168	287 0.977	328 0.971	412 0.959						
489	SHAMAN-007	486 0.183	531 0.141	547 0.092	458 0.280	487 0.240	521 0.169									
490	SHENNONA-000	353 0.049	368 0.022	415 0.015	435 0.221	449 0.160	478 0.104	153 0.524	163 0.383	196 0.235	435 0.160	433 0.064	495 1.000	414 1.000	398 0.558	420 0.350
491	SHUFTI-001	340 0.045	401 0.028	421 0.016		394 0.104	399 0.058	501 1.000	232 0.650	274 0.453	368 0.056	410 0.042	283 0.382	296 0.145	288 0.243	353 0.208
492	SIAT-001	444 0.127	338 0.017	330 0.007	530 0.641	522 0.365	564 0.348				315 0.031	317 0.014				
493	SIAT-002	551 0.417	364 0.021	335 0.007	567 0.942	542 0.478	576 0.460				478 0.372	502 0.356			425 0.923	317 0.169
494	SJTU-001	61 0.004	69 0.001	49 0.001	73 0.016	65 0.012	56 0.008	131 0.435	142 0.306	153 0.144	119 0.006	139 0.003	186 0.114	179 0.037	144 0.101	163 0.084
495	SJTU-002	21 0.002	20 0.001	13 0.001	45 0.012	37 0.009	27 0.007	269 0.958	295 0.911	110 0.113	108 0.005	131 0.003	301 0.516	213 0.052	181 0.126	171 0.086
496	SMILART-004	614 0.970	618 0.968	623 0.965	589 0.977	606 0.976	617 0.973									
497	SMILART-005															
498	SPARKTEK-000	120 0.007	124 0.003	126 0.001	94 0.018	92 0.015	86 0.010	143 0.487	171 0.414	226 0.294	136 0.006	158 0.003	103 0.048	111 0.022	157 0.106	176 0.087
499	SPARKTEK-001	82 0.005	26 0.001	33 0.001	28 0.010	28 0.009	40 0.007	108 0.311	107 0.210	126 0.124	45 0.002	54 0.001	27 0.022	29 0.009	57 0.076	83 0.067
500	SPARSH-001	160 0.009	167 0.004	139 0.002	574 0.952	575 0.759	340 0.039	345 0.992	334 0.976	164 0.164	438 0.166	356 0.020	595 1.000	408 0.999	410 0.633	416 0.340
501	SQISOFT-001	510 0.226	526 0.132	506 0.044	470 0.340	492 0.252	485 0.111	268 0.955	262 0.794	306 0.604	343 0.040	353 0.019	271 0.317	303 0.150	353 0.420	341 0.189
502	SQISOFT-002	400 0.074	403 0.029	311 0.006			611 0.908		291 0.903	206 0.258	507 0.621	446 0.074			489 0.953	441 0.435
503	STAQU-000	539 0.334	472 0.061	446 0.021	558 0.848	535 0.443	408 0.061	488 1.000	505 1.000	528 0.999	499 0.535	404 0.039	355 0.961	315 0.183	550 1.000	540 0.999
504	STCON-000	489 0.185	548 0.184	527 0.183	536 0.679	568 0.678	596 0.675	123 0.355	147 0.318	220 0.284	511 0.674	525 0.674	348 0.941	376 0.837	426 0.709	486 0.709
505	STCON-001	58 0.004	68 0.001	58 0.001	77 0.016	68 0.012	64 0.009	41 0.192	51 0.131	59 0.082	57 0.003	55 0.001	281 0.370	306 0.158	40 0.069	35 0.056
506	STCON-002	45 0.003	58 0.001	47 0.001	55 0.013	53 0.011	46 0.008	23 0.162	39 0.115	48 0.074	51 0.002	49 0.001	290 0.415	316 0.185	34 0.067	30 0.055

Table 71: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
507	STCON-003	[40] 0.003	[49] 0.001	[48] 0.001	[42] 0.012	[45] 0.010	[41] 0.007	[26] 0.171	[36] 0.110	[44] 0.071	[43] 0.002	[44] 0.001	[60] 0.031	[34] 0.009	[25] 0.065	[24] 0.053
508	STCON-004	[24] 0.002	[23] 0.001	[30] 0.001	[17] 0.009	[18] 0.008	[21] 0.006	[15] 0.136	[19] 0.083	[27] 0.060	[16] 0.001	[15] 0.001	[11] 0.014	[7] 0.005	[11] 0.055	[8] 0.047
509	STCON-005	[32] 0.002	[24] 0.001	[25] 0.001	[16] 0.008	[16] 0.007	[17] 0.006	[21] 0.159	[18] 0.083	[24] 0.058	[10] 0.001	[10] 0.001	[55] 0.029	[10] 0.005	[6] 0.052	[5] 0.045
510	STELLARTECS-001	[284] 0.028	[314] 0.014	[322] 0.006	[277] 0.068	[286] 0.051	[292] 0.031	[270] 0.961	[298] 0.927	[325] 0.686	[277] 0.021	[263] 0.009	[323] 0.750	[329] 0.263	[223] 0.155	[234] 0.112
511	SUMSUB-000	[583] 0.675	[592] 0.445	[574] 0.173	[562] 0.888	[572] 0.742	[565] 0.348	[505] 1.000	[528] 1.000	[569] 1.000	[523] 0.852	[513] 0.453	[427] 1.000	[453] 1.000	[512] 0.996	[512] 0.901
512	SYNESIS-003	[430] 0.111	[477] 0.065	[480] 0.031	[390] 0.155	[415] 0.123	[442] 0.078	[277] 0.973	[311] 0.960	[374] 0.910	[391] 0.075	[402] 0.039			[318] 0.314	[373] 0.235
513	SYNESIS-003	[579] 0.648	[598] 0.582	[603] 0.443	[541] 0.708	[561] 0.646	[581] 0.524									
514	SYNESIS-005	[352] 0.049	[377] 0.024	[394] 0.011	[309] 0.088	[336] 0.072	[357] 0.043	[370] 0.995	[358] 0.983	[346] 0.793	[322] 0.032	[331] 0.016			[273] 0.214	[312] 0.158
515	T4ISB-000	[276] 0.027	[326] 0.015	[387] 0.010	[275] 0.068	[295] 0.053	[316] 0.034	[163] 0.561	[196] 0.505	[276] 0.457	[276] 0.021	[283] 0.010	[324] 0.759	[313] 0.177	[235] 0.161	[262] 0.125
516	TCC-000	[609] 0.943	[614] 0.868	[601] 0.372	[601] 0.996	[610] 0.990	[616] 0.967	[525] 1.000	[520] 1.000	[535] 0.999	[541] 0.992	[543] 0.973	[584] 1.000	[548] 1.000	[523] 1.000	[538] 0.998
517	TCC-001	[588] 0.734	[540] 0.161	[491] 0.037	[605] 0.997	[609] 0.987	[586] 0.554	[515] 1.000	[518] 1.000	[544] 1.000	[540] 0.991	[539] 0.915	[428] 1.000	[432] 1.000	[522] 1.000	[535] 0.994
518	TECH5-001	[596] 0.807	[463] 0.057	[426] 0.018	[597] 0.994	[598] 0.935	[595] 0.055	[559] 1.000	[561] 1.000	[548] 1.000	[456] 0.244	[383] 0.028			[510] 0.994	[504] 0.817
519	TECH5-002	[364] 0.053	[388] 0.027	[399] 0.012	[314] 0.094	[334] 0.070	[348] 0.040	[235] 0.872	[265] 0.803	[310] 0.622	[341] 0.039	[352] 0.019	[240] 0.205	[283] 0.111	[361] 0.440	[336] 0.182
520	TECH5-003	[153] 0.009	[154] 0.004	[130] 0.002	[581] 0.964	[579] 0.801	[575] 0.049	[359] 0.994	[354] 0.982	[162] 0.160	[443] 0.179	[373] 0.023	[591] 1.000	[420] 1.000	[417] 0.656	[422] 0.358
521	TECHAINER-000	[271] 0.026	[302] 0.013	[310] 0.006	[288] 0.074	[296] 0.053	[312] 0.033	[215] 0.780	[193] 0.479	[207] 0.259	[258] 0.018	[268] 0.009	[218] 0.155	[228] 0.064	[213] 0.145	[225] 0.109
522	TEVIAN-003	[513] 0.238	[546] 0.177	[553] 0.096	[475] 0.346	[511] 0.298	[532] 0.198									
523	TEVIAN-004	[479] 0.169	[512] 0.117	[529] 0.063	[433] 0.216	[460] 0.176	[488] 0.115									
524	TEVIAN-005	[446] 0.128	[497] 0.087	[508] 0.044	[410] 0.180	[436] 0.144	[455] 0.089	[322] 0.987	[314] 0.961	[347] 0.793						
525	TEVIAN-006	[260] 0.024	[269] 0.009	[287] 0.005	[197] 0.041	[213] 0.032	[236] 0.021	[162] 0.556	[172] 0.418	[219] 0.282	[251] 0.016	[260] 0.009	[173] 0.093	[205] 0.050	[487] 0.951	[243] 0.117
526	TEVIAN-007	[181] 0.011	[200] 0.005	[216] 0.002	[130] 0.028	[134] 0.022	[144] 0.015	[148] 0.498	[141] 0.293	[175] 0.174	[201] 0.009	[208] 0.005	[140] 0.065	[167] 0.033	[176] 0.122	[216] 0.102
527	THAKAAIT-000	[330] 0.042	[324] 0.015	[300] 0.005	[281] 0.071	[292] 0.052	[293] 0.031	[423] 0.999	[440] 0.998	[439] 0.981	[264] 0.018	[270] 0.009	[259] 0.279	[235] 0.071	[233] 0.160	[256] 0.121
528	TIGER-000	[560] 0.461	[585] 0.390	[589] 0.261	[519] 0.565	[544] 0.500	[566] 0.366									
529	TIGER-002	[475] 0.157	[494] 0.086	[498] 0.039	[428] 0.202	[445] 0.158	[464] 0.095	[452] 0.999	[461] 0.999	[429] 0.975						
530	TIGER-003	[474] 0.157	[493] 0.086	[497] 0.039	[427] 0.202	[446] 0.158	[463] 0.095									
531	TIGER-004	[306] 0.035	[335] 0.017	[358] 0.008	[333] 0.105	[337] 0.072	[343] 0.040	[145] 0.489	[170] 0.410	[215] 0.278	[402] 0.095	[333] 0.016	[401] 1.000	[394] 0.990	[345] 0.402	[293] 0.143
532	TONGYITRANS-000	[424] 0.105	[484] 0.074	[496] 0.038	[382] 0.141	[404] 0.112	[424] 0.069									
533	TONGYITRANS-001	[440] 0.123	[478] 0.066	[481] 0.032	[364] 0.128	[392] 0.101	[411] 0.062									
534	TOSHIBA-000	[439] 0.122	[473] 0.062	[470] 0.026	[386] 0.150	[409] 0.118	[435] 0.074	[394] 0.997	[414] 0.995	[452] 0.988						
535	TOSHIBA-001	[508] 0.222	[466] 0.057	[428] 0.018	[373] 0.133	[374] 0.092	[394] 0.054									
536	TOSHIBA-002	[78] 0.005	[72] 0.001	[83] 0.001	[84] 0.017	[94] 0.015	[109] 0.012	[32] 0.179	[25] 0.094	[30] 0.060	[34] 0.002	[36] 0.001	[61] 0.031	[24] 0.008	[30] 0.066	[26] 0.054
537	TOSHIBA-003	[123] 0.007	[64] 0.001	[71] 0.001	[62] 0.014	[69] 0.012	[80] 0.010	[112] 0.320	[45] 0.123	[35] 0.064	[41] 0.002	[37] 0.001	[230] 0.176	[48] 0.011	[92] 0.085	[34] 0.056
538	TOSHIBA-004	[79] 0.005	[44] 0.001	[27] 0.001	[48] 0.013	[52] 0.011	[49] 0.008	[38] 0.189	[23] 0.092	[25] 0.059	[33] 0.002	[29] 0.001	[304] 0.573	[117] 0.023	[186] 0.128	[21] 0.051
539	TOSS-000	[62] 0.004	[83] 0.002	[103] 0.001	[87] 0.017	[87] 0.014	[93] 0.011	[33] 0.182	[42] 0.121	[53] 0.077	[24] 0.003	[64] 0.002	[25] 0.037	[64] 0.013	[41] 0.069	[50] 0.058
540	TOSS-001	[83] 0.005	[100] 0.002	[121] 0.001	[97] 0.020	[103] 0.017	[110] 0.012	[46] 0.202	[61] 0.141	[76] 0.091	[81] 0.004	[79] 0.002	[94] 0.043	[76] 0.015	[64] 0.078	[61] 0.062
541	TRUEFACE-000	[342] 0.046	[343] 0.018	[354] 0.008	[295] 0.079	[316] 0.062	[338] 0.039	[325] 0.995	[280] 0.881	[282] 0.493	[312] 0.030	[335] 0.016	[238] 0.194	[284] 0.111	[263] 0.188	[297] 0.145
542	TSIGHT-000	[3] 0.001	[2] 0.001	[2] 0.000	[5] 0.006	[3] 0.006	[3] 0.005	[3] 0.075	[1] 0.051	[4] 0.043	[3] 0.001	[2] 0.001	[9] 0.013	[4] 0.004	[3] 0.050	[3] 0.044
543	TTIGROUP-000	[435] 0.116	[259] 0.009	[256] 0.003	[517] 0.553	[512] 0.298	[299] 0.032	[428] 0.999	[448] 0.998	[447] 0.986	[463] 0.286	[434] 0.065	[403] 1.000	[415] 1.000	[456] 0.855	[459] 0.540
544	TURINGTECHVIP-001	[601] 0.865	[576] 0.345	[391] 0.011	[582] 0.967	[587] 0.850	[524] 0.173	[450] 0.999	[393] 0.993	[181] 0.196	[537] 0.978	[529] 0.754	[497] 1.000	[424] 1.000	[519] 0.999	[530] 0.984
545	UNISSEY-001	[506] 0.217	[557] 0.214	[583] 0.211	[542] 0.709	[570] 0.700	[600] 0.695	[210] 0.767	[229] 0.619	[291] 0.532	[514] 0.693	[527] 0.691	[329] 0.999	[396] 0.991	[432] 0.748	[491] 0.742
546	UNISSEY-002	[495] 0.190	[540] 0.187	[578] 0.183	[540] 0.703	[569] 0.685	[597] 0.677	[217] 0.792	[219] 0.571	[270] 0.441	[518] 0.724	[526] 0.678	[380] 1.000	[398] 0.992	[465] 0.891	[497] 0.769
547	UNISSEY-003	[287] 0.029	[315] 0.014	[339] 0.007	[260] 0.063	[281] 0.050	[311] 0.033	[231] 0.857	[208] 0.524	[247] 0.347	[308] 0.029	[359] 0.020	[241] 0.205	[245] 0.077	[245] 0.173	[241] 0.116
548	UNIVERSEAI-000	[150] 0.009	[145] 0.003	[122] 0.001	[119] 0.026	[115] 0.020	[112] 0.013	[87] 0.268	[79] 0.168	[88] 0.102	[186] 0.008	[216] 0.005	[176] 0.094	[135] 0.026	[145] 0.101	[140] 0.079
549	USEB-000	[593] 0.789	[605] 0.715	[607] 0.509	[554] 0.803	[571] 0.722	[585] 0.531	[298] 0.980	[320] 0.968	[379] 0.914	[493] 0.490	[498] 0.301	[331] 0.804	[363] 0.651	[414] 0.648	[453] 0.506
550	USEB-001	[155] 0.009	[151] 0.004	[157] 0.002	[137] 0.029	[147] 0.023	[163] 0.016	[338] 0.991	[350] 0.982	[367] 0.877	[103] 0.005	[108] 0.003	[314] 0.728	[257] 0.081	[97] 0.087	[92] 0.068
551	USEB-002	[240] 0.019	[166] 0.004	[182] 0.002	[146] 0.030	[156] 0.023	[168] 0.016	[339] 0.991	[351] 0.982	[368] 0.877	[113] 0.005	[109] 0.003	[315] 0.728	[258] 0.081	[102] 0.088	[97] 0.069
552	USEB-003	[152] 0.009	[148] 0.003	[155] 0.002	[122] 0.026	[130] 0.021	[151] 0.015	[314] 0.984	[325] 0.971	[361] 0.861	[112] 0.005	[110] 0.003	[288] 0.398	[195] 0.044	[98] 0.087	[102] 0.069

Table 72: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
553	USEB-004	[126]0.007	[142]0.003	[147]0.002	[155]0.031	[144]0.022	[148]0.015	[96]0.284	[91]0.182	[96]0.105	[211]0.010	[89]0.002	[389]1.000	[391]0.975	[262]0.186	[187]0.090
554	VD-000	[610]0.949	[615]0.917	[619]0.827	[584]0.968	[601]0.946	[609]0.871									
555	VD-001	[529]0.278	[552]0.201	[561]0.115	[468]0.331	[507]0.281	[528]0.188									
556	VD-002	[465]0.144	[488]0.078	[488]0.036	[416]0.188	[438]0.148	[456]0.092	[406]0.998	[417]0.995	[448]0.987	[403]0.095	[418]0.048	[279]0.367	[322]0.220	[338]0.372	[396]0.280
557	VD-003	[512]0.234	[438]0.045	[434]0.019	[372]0.133	[390]0.100	[410]0.061	[449]0.999	[463]0.999	[477]0.994	[360]0.051	[329]0.027	[254]0.244	[290]0.133	[320]0.315	[345]0.203
558	VDFSOFTWARE-001	[627]1.000	[626]1.000	[624]0.990	[547]0.746	[547]0.518	[555]0.301	[526]1.000	[544]1.000	[550]1.000	[513]0.680	[474]0.144	[381]1.000	[393]0.989	[486]0.948	[470]0.608
559	VDFSOFTWARE-002	[392]0.068	[424]0.038	[414]0.015	[438]0.229	[400]0.107	[383]0.052	[271]0.969	[245]0.713	[262]0.398	[390]0.075	[375]0.024	[398]1.000	[395]0.990	[354]0.423	[351]0.207
560	VERIDAS-001	[404]0.080	[421]0.037	[418]0.016	[336]0.106	[359]0.082	[378]0.051	[355]0.993	[372]0.987	[393]0.937	[349]0.044	[371]0.023	[256]0.266	[299]0.146	[299]0.264	[346]0.204
561	VERIDAS-002	[405]0.080	[422]0.037	[419]0.016	[335]0.106	[358]0.082	[379]0.051	[354]0.993	[373]0.987	[394]0.937	[350]0.044	[372]0.023	[257]0.266	[300]0.146	[300]0.264	[347]0.204
562	VERIDAS-003	[396]0.072	[331]0.016	[315]0.006	[282]0.071	[302]0.055	[308]0.033	[416]0.998	[427]0.997	[385]0.927	[272]0.020	[286]0.011	[210]0.150	[248]0.078	[252]0.178	[292]0.142
563	VERIDAS-004	[270]0.026	[212]0.005	[209]0.002	[171]0.034	[170]0.025	[177]0.017	[334]0.990	[339]0.978	[353]0.820	[178]0.008	[202]0.005	[128]0.058	[151]0.029	[173]0.118	[222]0.107
564	VERIGRAM-000	[169]0.010	[191]0.005	[231]0.003	[174]0.034	[191]0.028	[232]0.021	[201]0.722	[110]0.212	[103]0.108	[142]0.006	[161]0.004	[168]0.088	[129]0.024	[84]0.083	[96]0.069
565	VERIGRAM-001	[205]0.013	[193]0.005	[230]0.003	[494]0.419	[253]0.039	[229]0.021	[312]0.984	[224]0.584	[106]0.110	[140]0.006	[156]0.003	[361]0.988	[156]0.030	[80]0.082	[80]0.066
566	VERIGRAM-002	[76]0.005	[120]0.003	[212]0.002	[238]0.053	[195]0.029	[104]0.012	[45]0.201	[40]0.116	[45]0.073	[49]0.002	[53]0.001	[43]0.027	[40]0.010	[62]0.077	[55]0.060
567	VERIJELAS-000	[599]0.846	[608]0.799	[614]0.681	[560]0.868	[581]0.813	[601]0.697	[442]0.999	[459]0.999	[489]0.995	[471]0.324	[484]0.216	[347]0.933	[355]0.561	[400]0.589	[446]0.462
568	VIANTE-000	[176]0.010	[113]0.003	[113]0.001	[78]0.016	[83]0.014	[83]0.010	[107]0.311	[66]0.147	[87]0.102	[148]0.006	[141]0.003	[79]0.038	[94]0.018	[99]0.087	[107]0.071
569	VIANTE-001	[93]0.006	[59]0.001	[73]0.001	[30]0.010	[39]0.009	[61]0.008	[78]0.260	[34]0.105	[58]0.080	[52]0.002	[48]0.001	[264]0.293	[27]0.008	[33]0.067	[41]0.057
570	VIANTE-002	[77]0.005	[27]0.001	[26]0.001	[12]0.008	[14]0.007	[18]0.006	[101]0.299	[14]0.078	[15]0.051	[46]0.002	[40]0.001	[245]0.211	[22]0.007	[22]0.063	[18]0.050
571	VIANTE-003	[112]0.007	[17]0.001	[22]0.001	[10]0.008	[10]0.007	[19]0.006	[118]0.340	[13]0.076	[11]0.049	[35]0.002	[30]0.001	[187]0.117	[16]0.006	[14]0.060	[13]0.049
572	VIETTELAI-000	[591]0.784	[469]0.058	[314]0.006	[587]0.972	[573]0.753	[329]0.036	[538]1.000	[553]1.000	[541]1.000	[488]0.435	[351]0.018	[368]0.995	[360]0.601	[619]1.000	[537]0.997
573	VIETTELAI-001															
574	VIETTELAI-002	[517]0.249	[279]0.010	[263]0.003	[544]0.713	[560]0.631	[571]0.411	[132]0.438	[133]0.269	[157]0.152	[483]0.421	[480]0.176	[378]0.999	[388]0.957	[434]0.750	[455]0.514
575	VIETTELAI-003	[214]0.014	[179]0.004	[159]0.002	[165]0.033	[157]0.023	[153]0.015	[290]0.978	[267]0.823	[294]0.543	[131]0.006	[113]0.003	[222]0.160	[144]0.027	[111]0.090	[108]0.072
576	VIETTELAI-004	[156]0.009	[111]0.002	[88]0.001	[109]0.023	[97]0.015	[82]0.010	[499]1.000	[536]1.000	[572]1.000	[97]0.005	[82]0.002	[136]0.063	[97]0.019	[74]0.080	[72]0.064
577	VIGILANTSOLUTIONS-003	[563]0.481	[589]0.408	[592]0.281	[546]0.730	[565]0.660	[583]0.526	[444]0.999	[456]0.999	[486]0.995						
578	VIGILANTSOLUTIONS-004	[578]0.623	[597]0.549	[602]0.422	[559]0.858	[583]0.817	[602]0.709	[411]0.998	[421]0.996	[466]0.991						
579	VIGILANTSOLUTIONS-005	[608]0.934	[583]0.386	[501]0.042				[507]1.000	[531]1.000	[570]1.000						
580	VIGILANTSOLUTIONS-006	[612]0.955	[579]0.349	[504]0.043				[533]1.000	[558]1.000	[565]1.000						
581	VIGILANTSOLUTIONS-007	[401]0.075	[402]0.028	[393]0.011	[345]0.113	[367]0.088	[388]0.053	[400]0.997	[420]0.996	[469]0.991	[397]0.081	[414]0.047	[282]0.371	[325]0.242	[343]0.391	[404]0.295
582	VIGILANTSOLUTIONS-008	[357]0.050	[355]0.020	[373]0.009	[334]0.105	[347]0.077	[365]0.046	[454]1.000	[460]0.999	[465]0.991	[407]0.104	[425]0.054	[287]0.398	[328]0.259	[383]0.511	[413]0.316
583	VINBIGDATA-001	[280]0.028	[327]0.016	[291]0.005	[430]0.204	[312]0.059	[289]0.029	[368]0.995	[300]0.931	[225]0.293	[261]0.018	[247]0.007	[349]0.944	[220]0.061	[304]0.274	[199]0.095
584	VINBIGDATA-002	[104]0.006	[88]0.002	[84]0.001	[75]0.016	[63]0.012	[55]0.008	[61]0.237	[72]0.160	[85]0.095	[95]0.005	[99]0.002	[270]0.317	[104]0.019	[56]0.076	[71]0.064
585	VINBIGDATA-003	[134]0.008	[98]0.002	[81]0.001	[81]0.016	[76]0.013	[65]0.009	[55]0.226	[60]0.141	[57]0.080	[86]0.004	[81]0.002	[84]0.039	[67]0.014	[46]0.070	[45]0.058
586	VINBIGDATA-004	[99]0.006	[74]0.002	[59]0.001	[49]0.013	[48]0.010	[45]0.008	[73]0.251	[50]0.130	[51]0.076	[66]0.003	[66]0.002	[62]0.032	[51]0.012	[38]0.068	[48]0.058
587	VIRTISHA-000	[321]0.039	[389]0.027	[432]0.019	[586]0.970	[567]0.674	[503]0.147	[203]0.748	[202]0.517	[249]0.351	[526]0.899	[523]0.644	[608]1.000	[581]1.000	[504]0.990	[521]0.960
588	VISIONBOX-000	[397]0.073	[340]0.017	[331]0.007	[283]0.071	[305]0.057	[320]0.035	[363]0.994	[383]0.990	[427]0.973	[285]0.023	[303]0.012	[206]0.146	[252]0.081	[236]0.162	[264]0.126
589	VISIONBOX-001	[580]0.649	[527]0.133	[321]0.006	[500]0.441	[313]0.060	[295]0.031	[496]1.000	[532]1.000	[545]1.000	[430]0.152	[277]0.010	[375]0.999	[266]0.087	[476]0.924	[393]0.269
590	VISIONBOX-002	[249]0.022	[282]0.011	[283]0.005	[243]0.055	[261]0.042	[270]0.025	[310]0.984	[248]0.730	[240]0.336	[244]0.015	[248]0.007	[190]0.125	[227]0.063	[203]0.140	[209]0.099
591	VISIONLABS-004	[413]0.090	[465]0.057	[458]0.024	[423]0.199	[447]0.159	[467]0.097	[260]0.943	[284]0.888	[336]0.739						
592	VISIONLABS-005	[406]0.080	[446]0.050	[436]0.020	[413]0.183	[437]0.147	[453]0.087	[262]0.945	[283]0.886	[335]0.733						
593	VISIONLABS-006	[337]0.044	[386]0.026	[382]0.010	[351]0.117	[372]0.090	[382]0.051	[208]0.762	[234]0.668	[288]0.506						
594	VISIONLABS-007	[336]0.044	[385]0.026	[381]0.010	[350]0.117	[371]0.090	[381]0.051	[209]0.762	[235]0.668	[287]0.505	[318]0.031	[322]0.014			[259]0.185	[298]0.145
595	VISIONLABS-008	[279]0.027	[298]0.013	[297]0.005	[274]0.068	[289]0.051	[303]0.032	[167]0.569	[190]0.475	[233]0.309	[252]0.017	[253]0.008			[217]0.151	[246]0.119
596	VISIONLABS-009	[191]0.012	[177]0.004	[152]0.002	[160]0.032	[172]0.025	[183]0.017	[254]0.929	[263]0.797	[180]0.186	[188]0.008	[193]0.004			[167]0.113	[195]0.093
597	VISIONLABS-010	[211]0.014	[198]0.005	[192]0.002	[173]0.034	[186]0.027	[210]0.019			[160]0.159	[175]0.008	[173]0.004	[124]0.055	[137]0.027	[160]0.109	[184]0.089
598	VISIONLABS-011	[183]0.011	[132]0.003	[114]0.001	[111]0.024	[116]0.020	[128]0.014			[179]0.184	[91]0.004	[104]0.002	[71]0.034	[91]0.017	[110]0.090	[139]0.079

Table 73: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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MISSES BELOW THRESHOLD, T		ENROL RECENT MUGSHOT, N = 1.6M									ENROL APPLICATION PORTRAIT, N = 1.6M					
		ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: MUGSHOT			ENROL: VISA		ENROL: BORDER		ENROL: VISA	
		PROBE: MUGSHOT			PROBE: WEBCAM			PROBE: PROFILE			PROBE: BORDER		PROBE: BORDER 10+YR		PROBE: KIOSK	
#	ALGORITHM	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.0003	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01	FPIR=0.001	FPIR=0.01
599	VISTEAM-000	459 0.140	492 0.085	495 0.038	535 0.674	471 0.205	459 0.093	273 0.970	282 0.886	322 0.666	417 0.128	432 0.062	391 1.000	405 0.999	332 0.345	377 0.240
600	VISTEAM-001	480 0.172	510 0.109	511 0.046	626 1.000	540 0.469	481 0.105	261 0.944	279 0.878	326 0.691	431 0.154	445 0.074	502 1.000	411 1.000	339 0.374	386 0.262
601	VIXVIZION-009	362 0.053	391 0.027	402 0.012	322 0.098	345 0.077	370 0.048	286 0.976	251 0.742	289 0.513	345 0.041	361 0.021	261 0.286	310 0.165	372 0.472	329 0.178
602	VNIS-000	571 0.541	573 0.311	586 0.234	502 0.448	530 0.418	569 0.383	385 0.996	394 0.993	408 0.953	517 0.701	524 0.663	335 0.838	374 0.782	424 0.696	476 0.635
603	VNPAY-001	203 0.013	214 0.006	215 0.002	175 0.035	179 0.026	167 0.016	146 0.491	136 0.280	158 0.156	222 0.012	243 0.007	143 0.066	173 0.034	161 0.109	204 0.097
604	VNPT-001	278 0.027	310 0.014	319 0.006	393 0.158	329 0.068	322 0.036	249 0.921	246 0.715	254 0.365	330 0.035	292 0.011	363 0.990	352 0.537	335 0.362	279 0.134
605	VNPT-002	204 0.013	226 0.006	245 0.003	193 0.040	212 0.032	235 0.021	164 0.563	139 0.284	154 0.144	169 0.007	165 0.004	154 0.072	158 0.031	129 0.096	128 0.075
606	VNPT-003	207 0.013	223 0.006	235 0.003	180 0.036	192 0.028	188 0.018	323 0.988	312 0.960	174 0.173	164 0.007	166 0.004	133 0.062	152 0.030	130 0.096	132 0.077
607	VNPT-004	182 0.011	205 0.005	219 0.002	168 0.033	173 0.026	175 0.016	272 0.969	289 0.899	166 0.166	158 0.007	157 0.003	125 0.057	143 0.027	117 0.093	123 0.075
608	VNPT-005	151 0.009	178 0.004	197 0.002	145 0.030	152 0.023	150 0.015	258 0.936	274 0.853	150 0.141	109 0.005	118 0.003	101 0.047	118 0.023	108 0.090	110 0.072
609	VOCORD-003	541 0.350	518 0.122	513 0.047	421 0.195	443 0.155	461 0.093	426 0.999	445 0.998	461 0.990	433 0.157	463 0.105			347 0.404	400 0.289
610	VOCORD-004	597 0.825	580 0.353	514 0.051	489 0.401	456 0.173	457 0.093	546 1.000	512 1.000	524 0.999	448 0.193	435 0.065			506 0.991	498 0.776
611	VOCORD-005	585 0.687	536 0.156	505 0.043	395 0.161	423 0.130	445 0.080	435 0.999	434 0.997	416 0.968	425 0.138	455 0.090			340 0.381	398 0.287
612	VOCORD-006	631 1.000	631 1.000	637 1.000	619 1.000	634 1.000	635 1.000	593 1.000	587 1.000	624 1.000	599 1.000	603 1.000			568 1.000	548 1.000
613	VOGIC-000	407 0.080	309 0.014	305 0.006	577 0.957	589 0.855	573 0.435	421 0.999	428 0.997	304 0.597	542 0.993	533 0.834	554 1.000	589 1.000	517 0.999	526 0.977
614	VSOFT-001	504 0.213	533 0.153	542 0.083	480 0.355	508 0.288	526 0.185	351 0.993	377 0.990	417 0.968	451 0.202	469 0.118	376 0.999	382 0.915	399 0.573	447 0.462
615	VSOFT-002	358 0.050	396 0.028	395 0.011	447 0.254	388 0.098	362 0.045	418 0.998	429 0.997	380 0.916	328 0.034	339 0.017	365 0.992	357 0.596	269 0.201	302 0.150
616	VTS-000	575 0.605	600 0.598	611 0.595	528 0.624	558 0.619	591 0.613	447 0.999	475 0.999	516 0.998	506 0.613	521 0.609	325 0.760	370 0.739	435 0.761	494 0.749
617	VTS-001	304 0.034	300 0.013	302 0.005	271 0.067	288 0.051	296 0.031	405 0.998	400 0.994	286 0.504	283 0.022	304 0.012	201 0.141	249 0.079	265 0.192	265 0.126
618	VTS-002	363 0.053	380 0.026	379 0.010	323 0.098	343 0.075	366 0.046	470 1.000	485 1.000	407 0.953	353 0.045	378 0.026	249 0.231	291 0.133	352 0.417	338 0.187
619	VTS-003	220 0.015	239 0.007	233 0.003	226 0.048	220 0.033	213 0.019	542 1.000	524 1.000	314 0.628	237 0.014	210 0.005	353 0.954	219 0.060	411 0.635	185 0.089
620	VTS-004	180 0.011	189 0.004	164 0.002	246 0.055	184 0.027	166 0.016	547 1.000	555 1.000	358 0.847	194 0.009	164 0.004	382 1.000	200 0.046	272 0.203	116 0.073
621	VTS-005	185 0.011	206 0.005	153 0.002	214 0.045	204 0.030	184 0.017	466 1.000	439 0.998	199 0.238	157 0.007	144 0.003	332 0.811	181 0.037	140 0.100	103 0.070
622	WINK-000	526 0.269	553 0.203	563 0.122	495 0.432	524 0.377	551 0.274	315 0.985	359 0.983	435 0.979	459 0.261	476 0.163	601 1.000	433 1.000	415 0.649	463 0.558
623	WINK-001	408 0.082	417 0.036	411 0.014	474 0.346	425 0.131	448 0.083	414 0.998	423 0.996	432 0.977	455 0.237	486 0.220	341 0.903	342 0.412	337 0.372	411 0.312
624	WINK-002	259 0.024	290 0.012	286 0.005	396 0.162	375 0.092	432 0.072	311 0.984	313 0.961	375 0.910	465 0.294	496 0.289	359 0.978	348 0.441	331 0.342	414 0.316
625	WINK-003	515 0.242	261 0.009	266 0.003	229 0.049	246 0.038	274 0.026	491 1.000	508 1.000	514 0.998	213 0.010	218 0.005	162 0.077	176 0.036	439 0.774	323 0.173
626	XAILIENT-000	388 0.065	415 0.035	420 0.016	292 0.077	310 0.059	314 0.034	187 0.653	204 0.521	258 0.376	303 0.027	301 0.012	232 0.181	260 0.082	267 0.196	242 0.117
627	XAILIENT-001	224 0.015	253 0.008	248 0.003	198 0.041	214 0.032	228 0.021	105 0.310	130 0.251	155 0.149	198 0.009	197 0.004	148 0.067	165 0.033	153 0.105	167 0.085
628	XFORWARDAI-000	283 0.028	319 0.015	323 0.006	279 0.070	299 0.053	318 0.034	198 0.694	177 0.434	200 0.241	278 0.021	285 0.011	221 0.159	259 0.082	242 0.169	280 0.134
629	XFORWARDAI-001	165 0.010	190 0.005	218 0.002	178 0.036	193 0.028	215 0.020	227 0.836	178 0.441	138 0.133	181 0.008	214 0.005	135 0.062	155 0.030	178 0.123	214 0.102
630	XFORWARDAI-002	107 0.006	118 0.003	181 0.002	92 0.018	99 0.016	125 0.014	282 0.974	203 0.519	61 0.084	107 0.005	153 0.003	88 0.041	96 0.018	137 0.099	183 0.089
631	YISHENG-001	557 0.451	577 0.346	581 0.206	591 0.983	580 0.808	559 0.269				510 0.666	507 0.396			473 0.919	483 0.695
632	YITU-002	291 0.031	337 0.017	345 0.008	259 0.063	278 0.049	283 0.028									
633	YITU-003	292 0.031	346 0.018	363 0.008	269 0.067	293 0.052	309 0.033									
634	YITU-004	239 0.019	265 0.009	274 0.004	177 0.035	182 0.027	185 0.017	265 0.947	301 0.936	377 0.912						
635	YITU-005	248 0.021	275 0.010	282 0.005	191 0.039	215 0.032	251 0.023									
636	YOUVERSE-001	210 0.014	236 0.007	236 0.003	184 0.038	202 0.030	212 0.019	110 0.311	131 0.253	159 0.159	202 0.009	198 0.004	163 0.081	175 0.036	143 0.101	135 0.078
637	YUAN-000	243 0.019	277 0.010	264 0.003	287 0.073	269 0.045	278 0.027	134 0.454	152 0.342	193 0.210	226 0.012	222 0.005	274 0.333	203 0.048	188 0.129	182 0.088

Table 74: **Threshold-based accuracy**. *Values are FNIR(N, T, L) with N = 1.6 million with thresholds set to produce FPIR = 0.0003, 0.001, and 0.01 in non-mate searches. Throughout blue superscripts indicate the rank of the algorithm for that column. Caution: The Power-low models are mostly intended to draw attention to the kind of behavior, not as a model to be used for prediction.*

2026/10/06 08:21:15

FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

Appendices

Appendix A Accuracy on large-population FRVT 2018 mugshots

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2026/10/06 08:21:15	FNIR(N, R, T) = FPIR(N, T) =	False neg. identification rate False pos. identification rate	N = Num. enrolled subjects R = Num. candidates examined	T = Threshold	$T = 0 \rightarrow$ Investigation $T > 0 \rightarrow$ Identification

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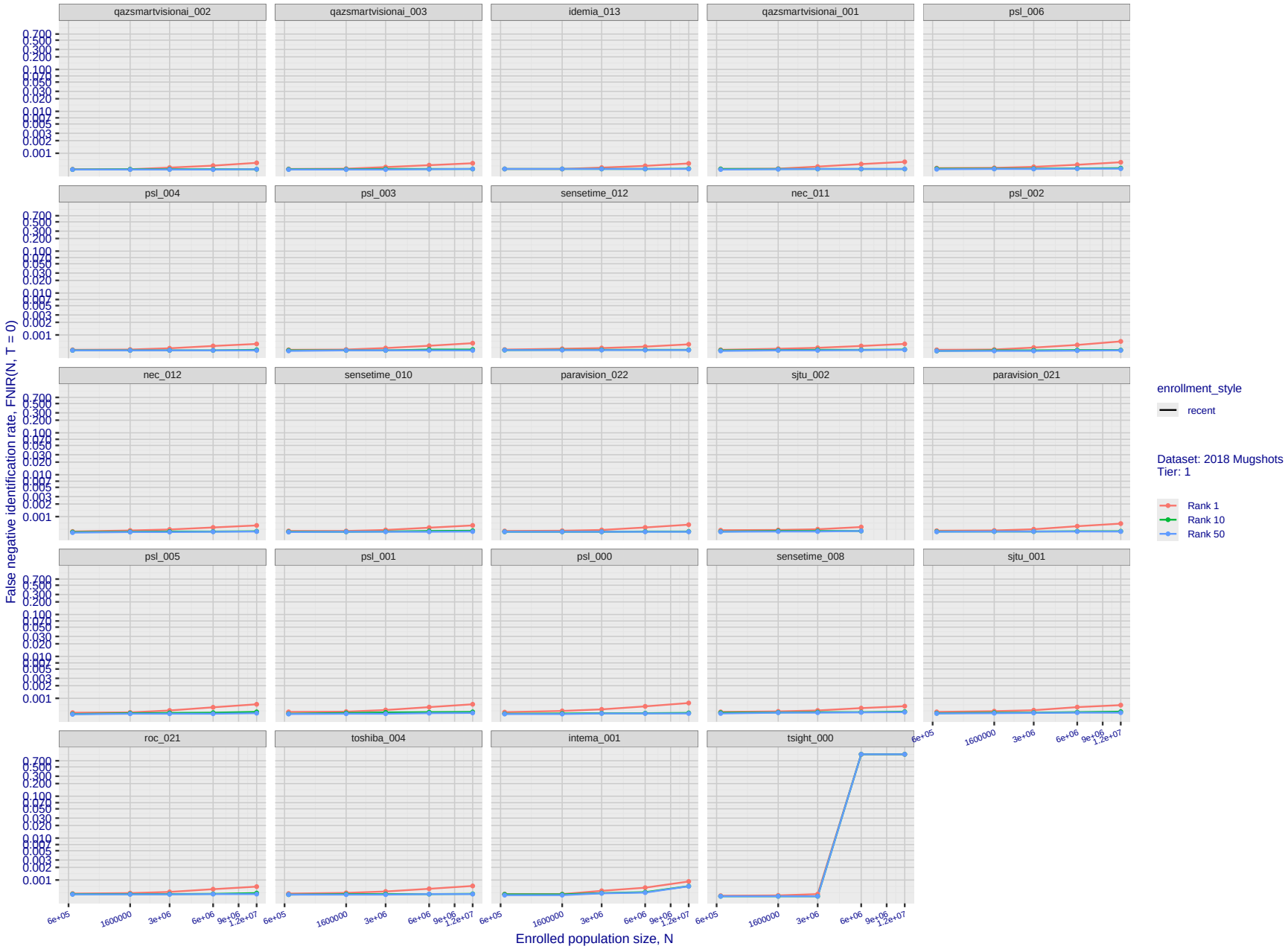

Figure 20: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

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R = Num. candidates examined

T = Threshold

T = 0 → Investigation
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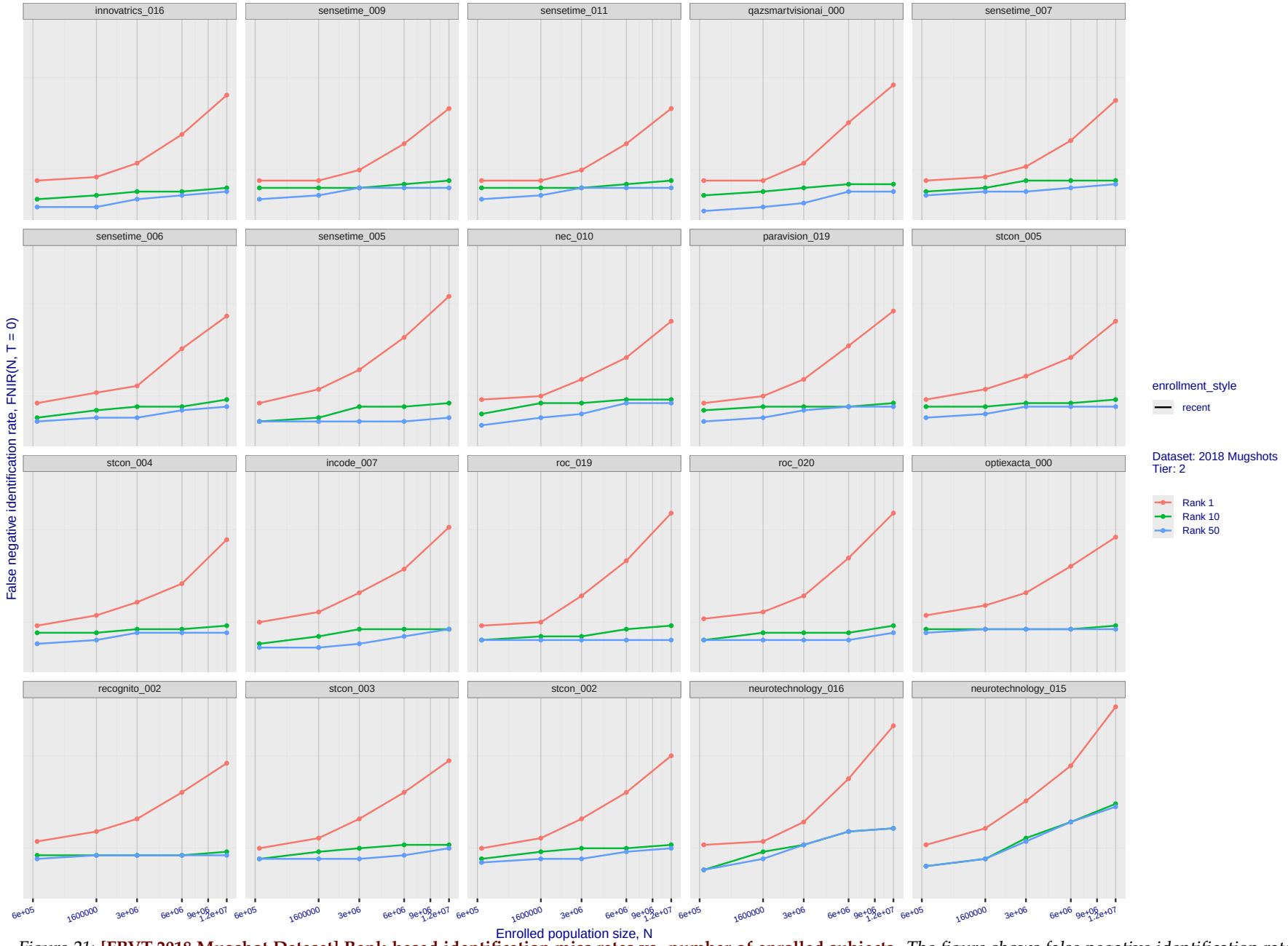

Figure 21: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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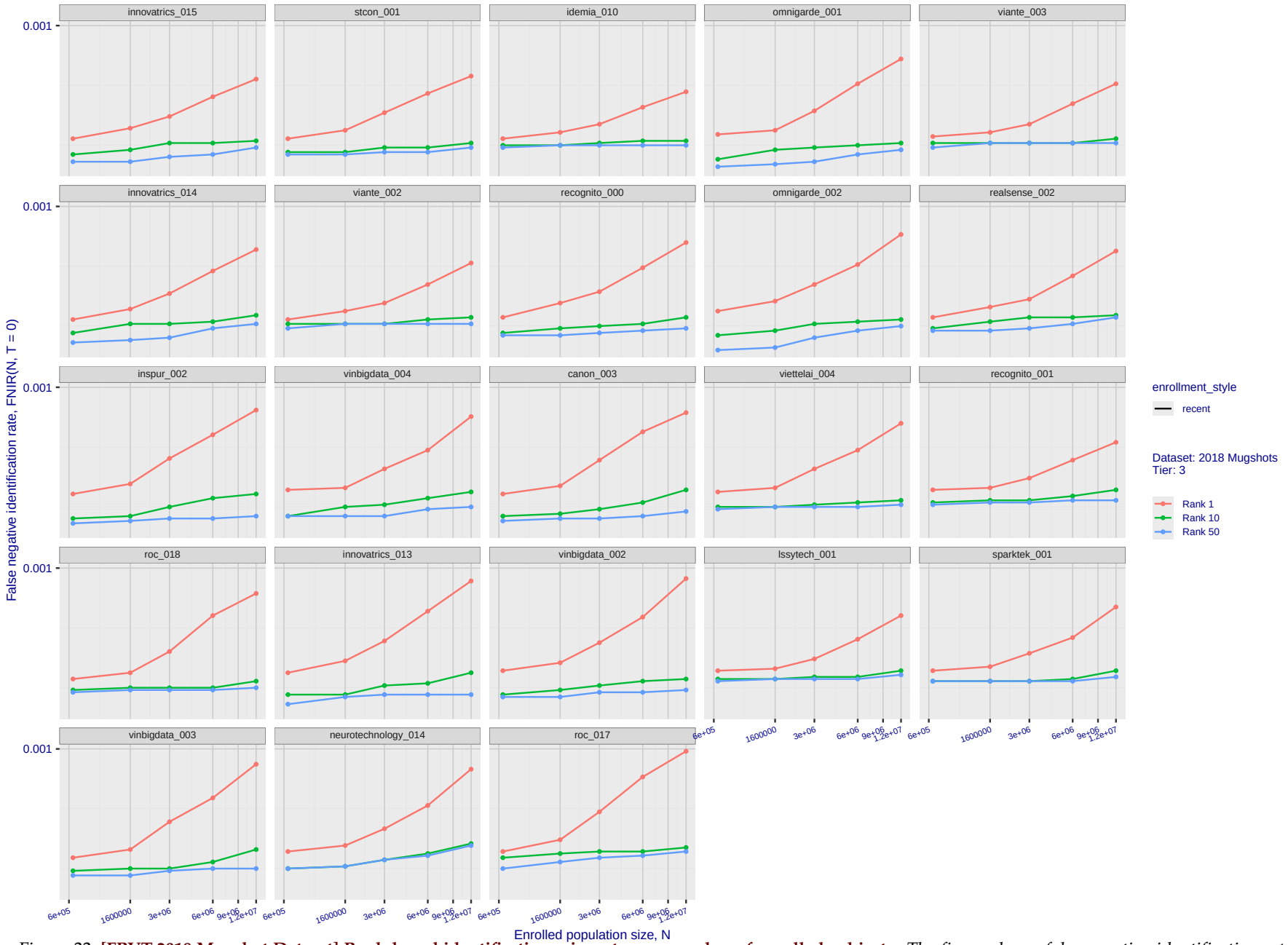

Figure 22: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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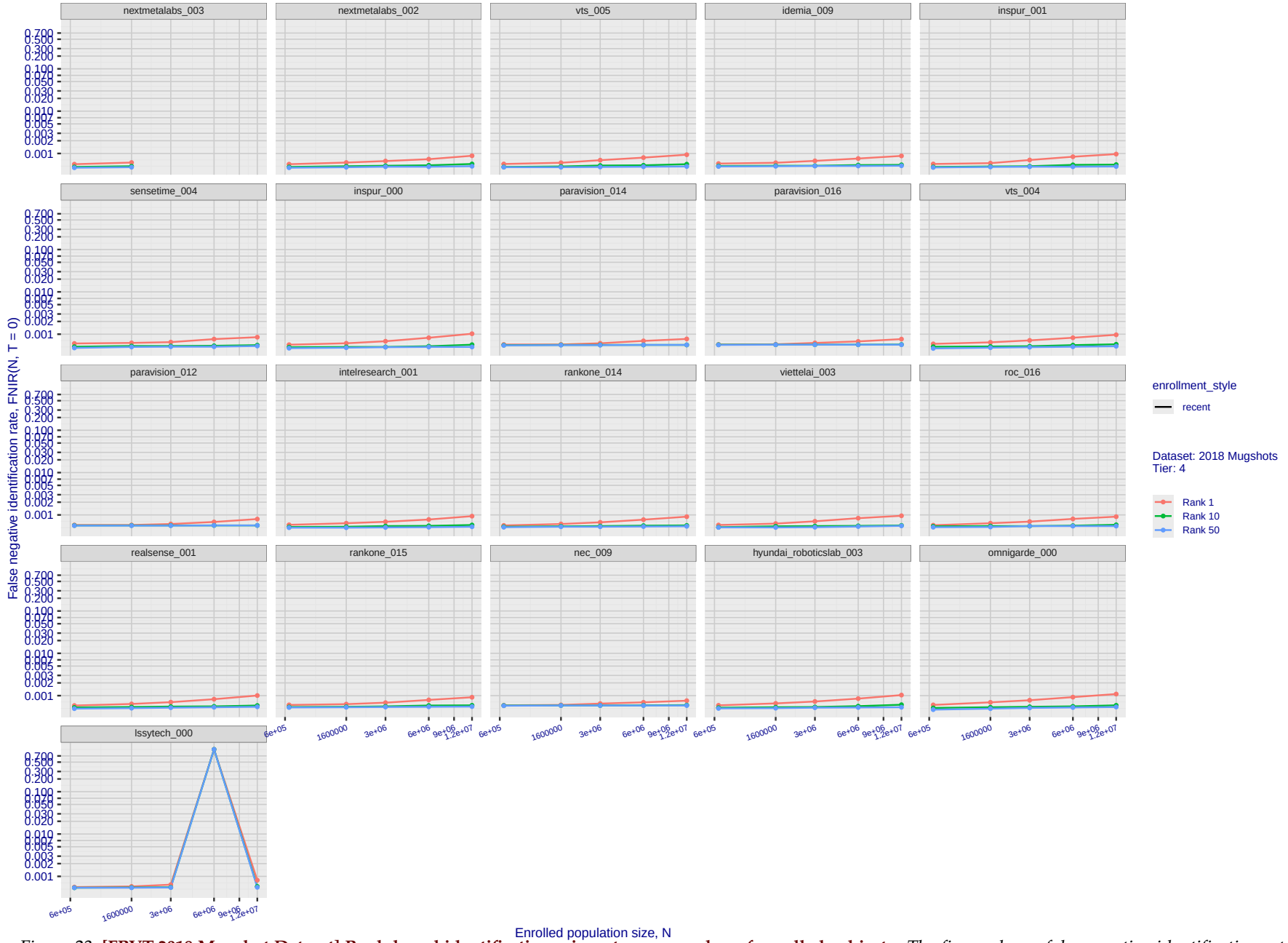

Figure 23: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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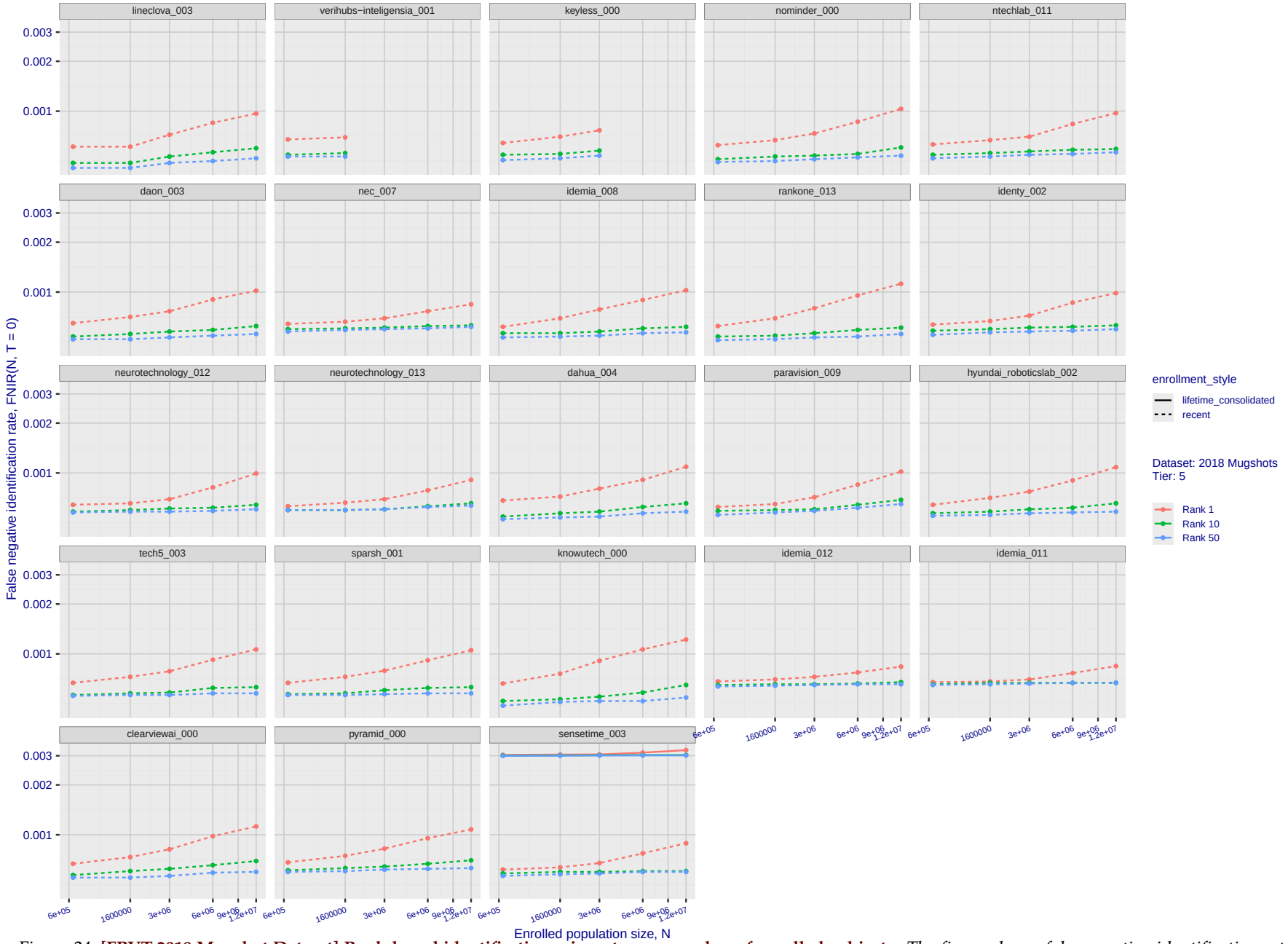

Figure 24: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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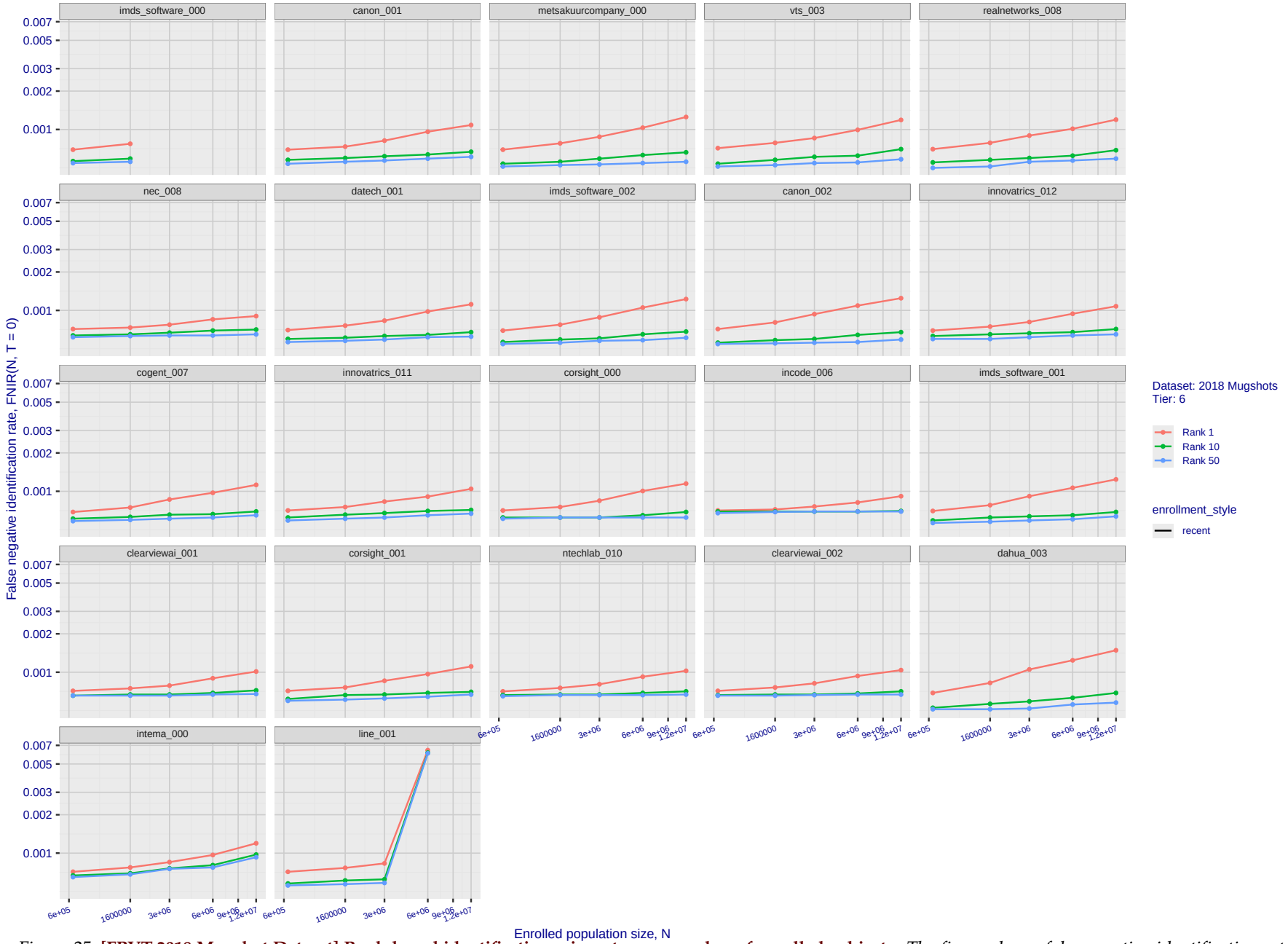

Figure 25: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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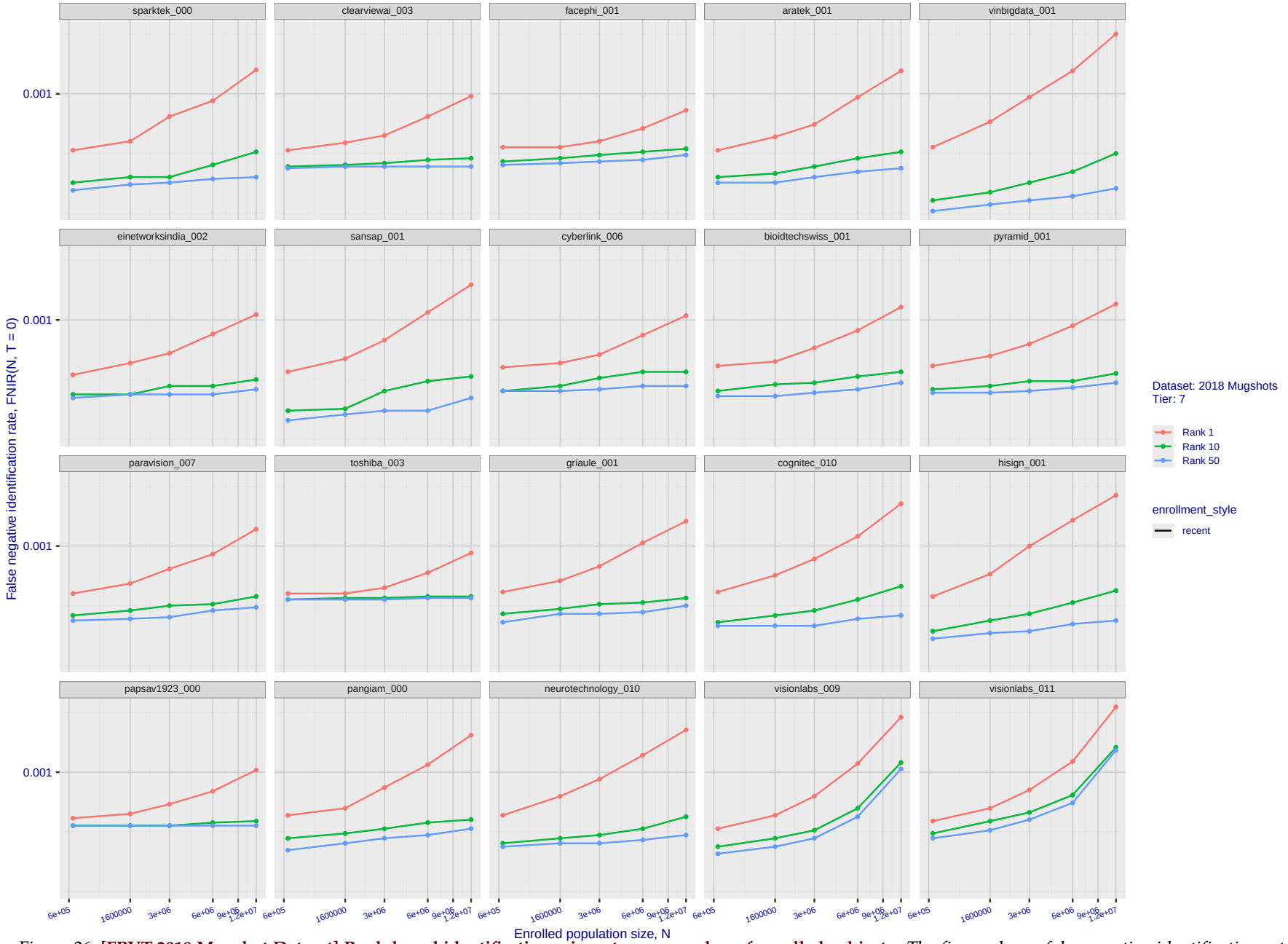

Figure 26: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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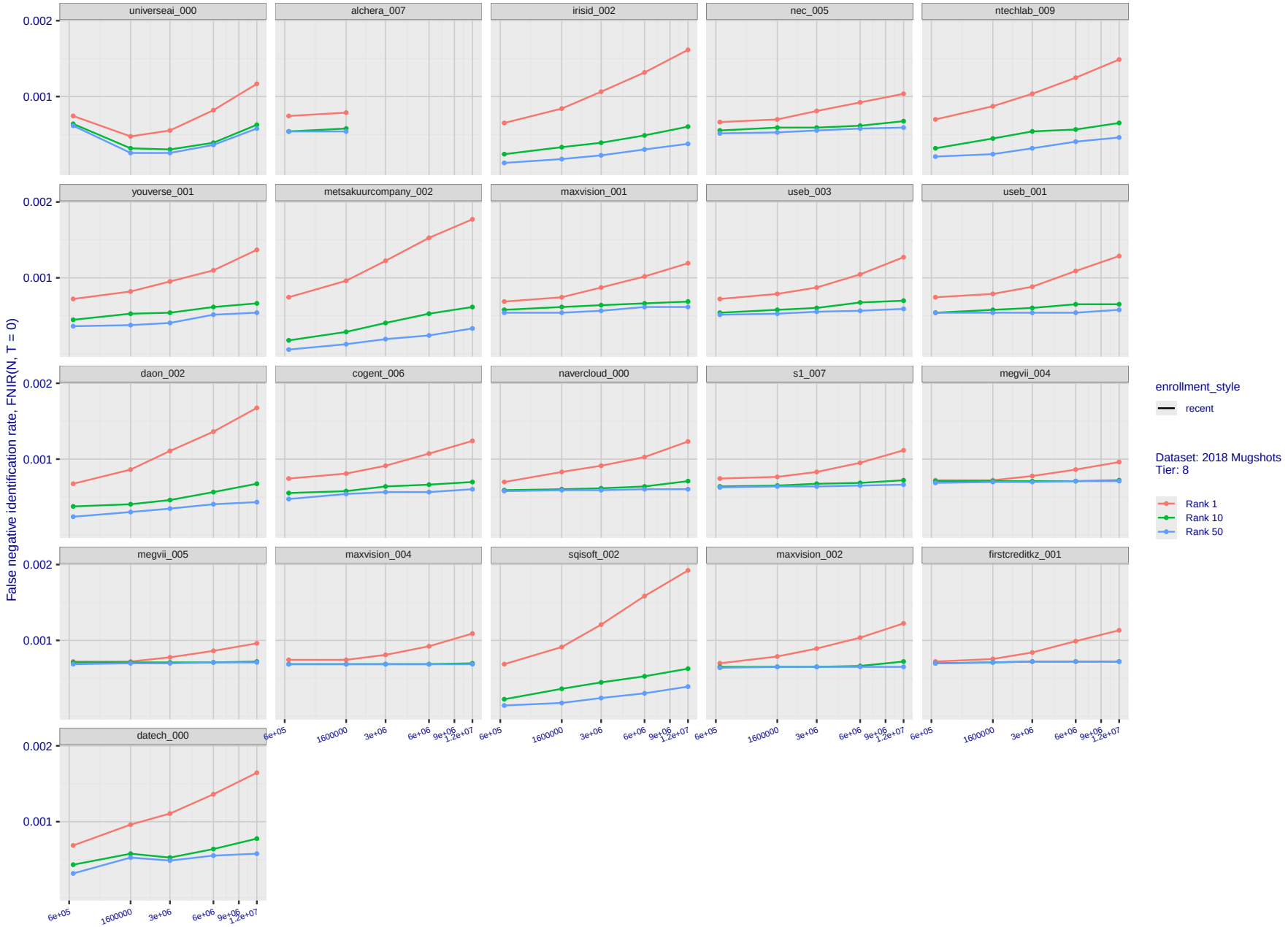

Figure 27: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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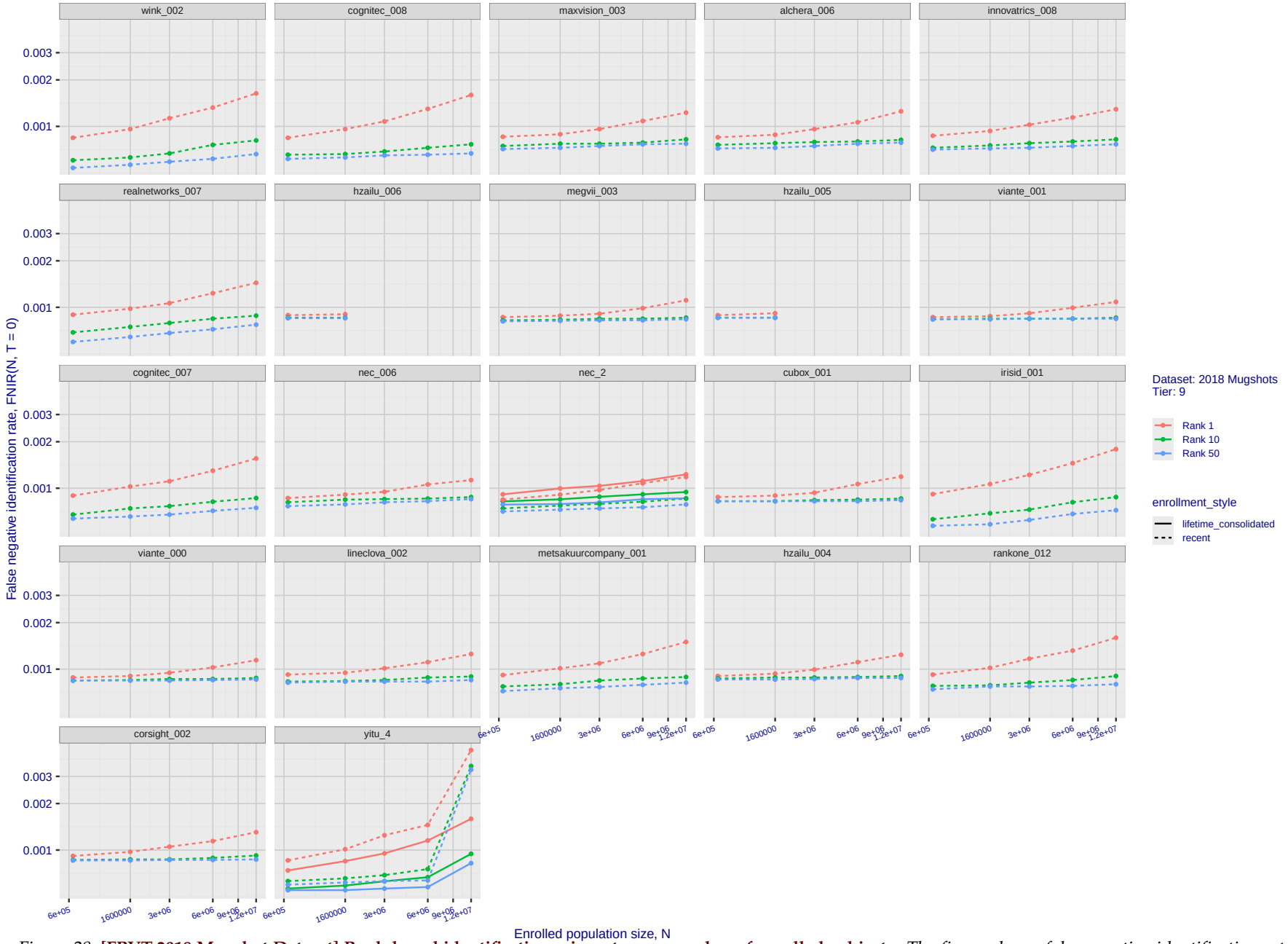

Figure 28: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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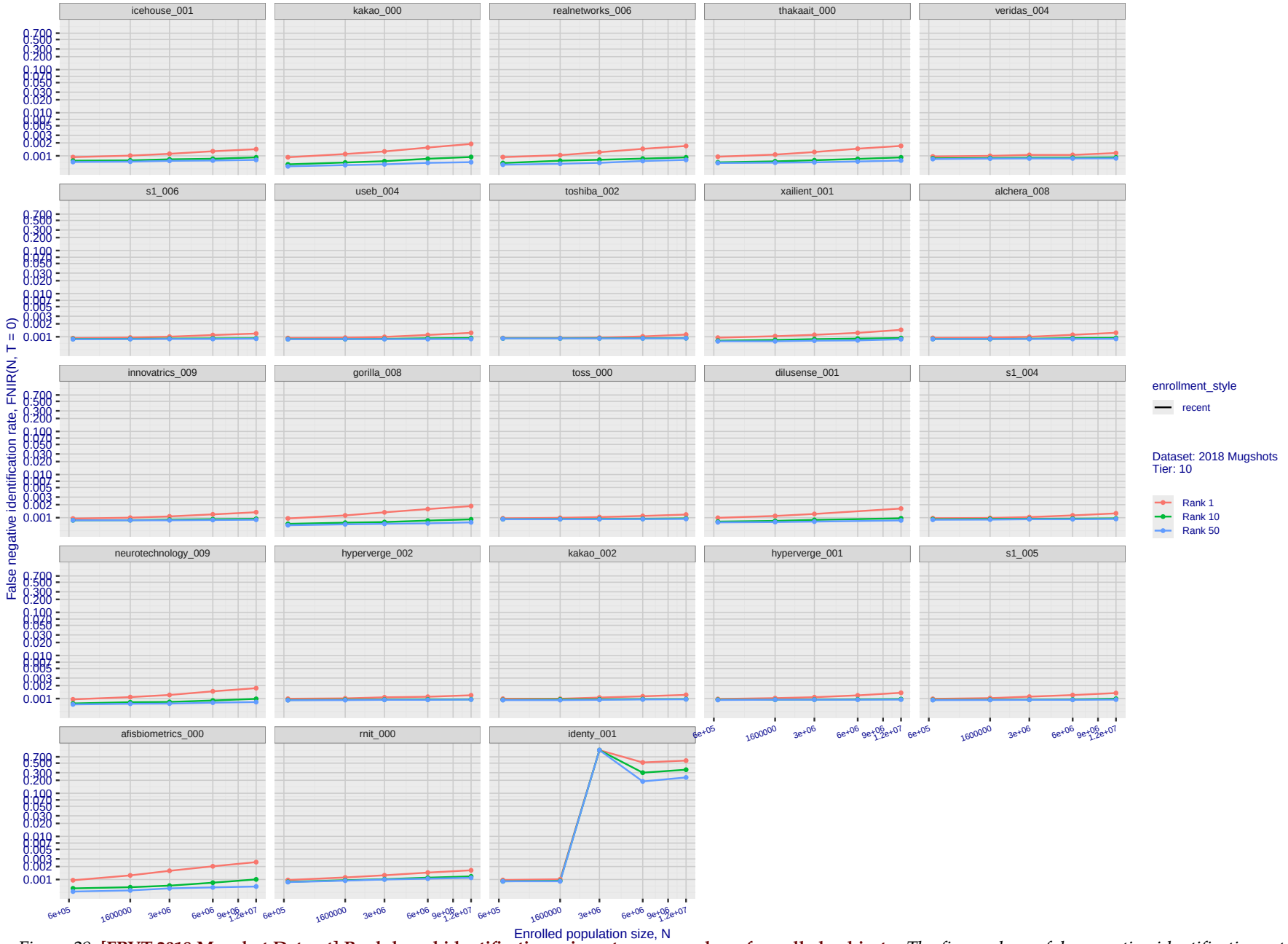

Figure 29: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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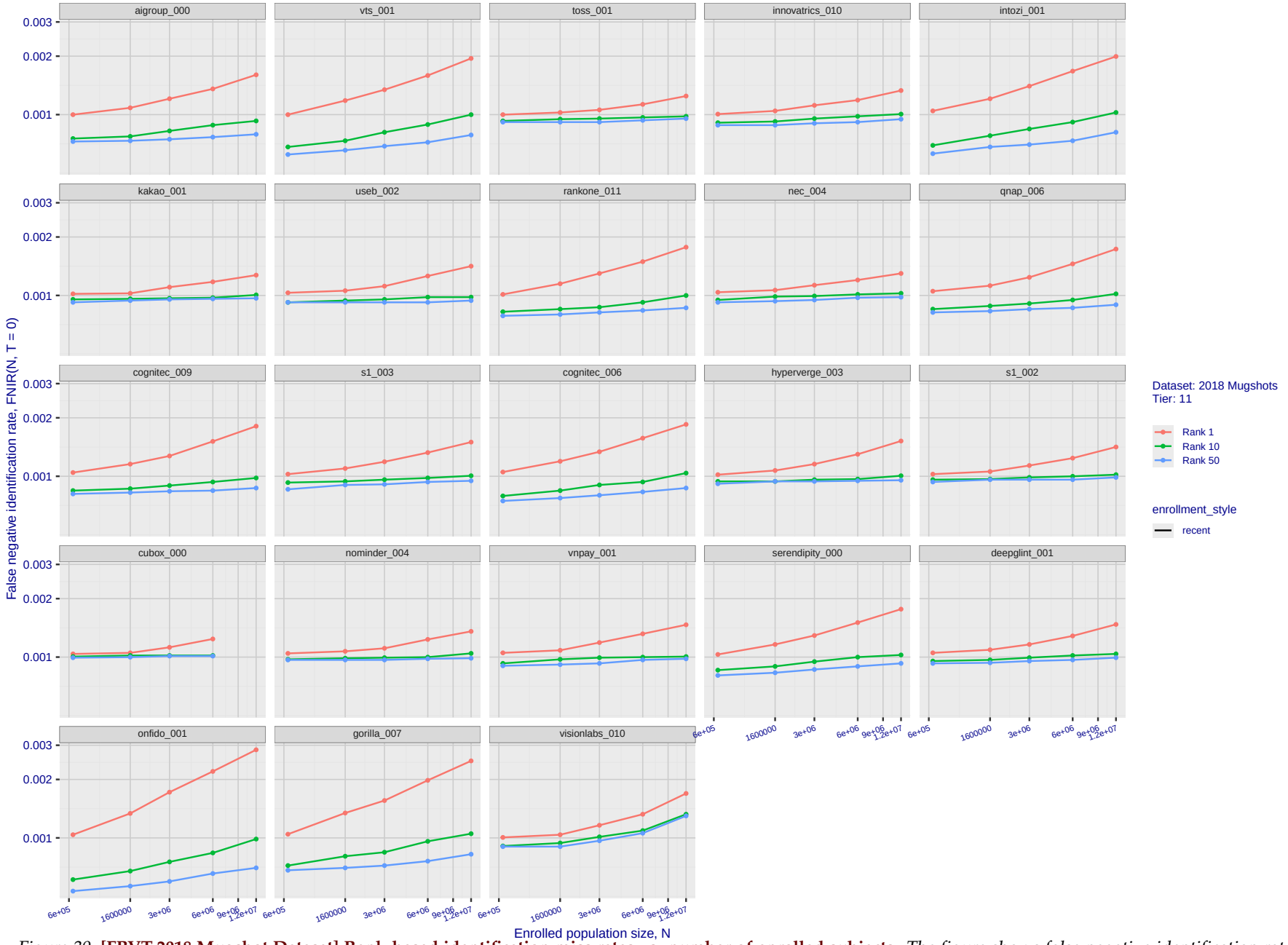

Figure 30: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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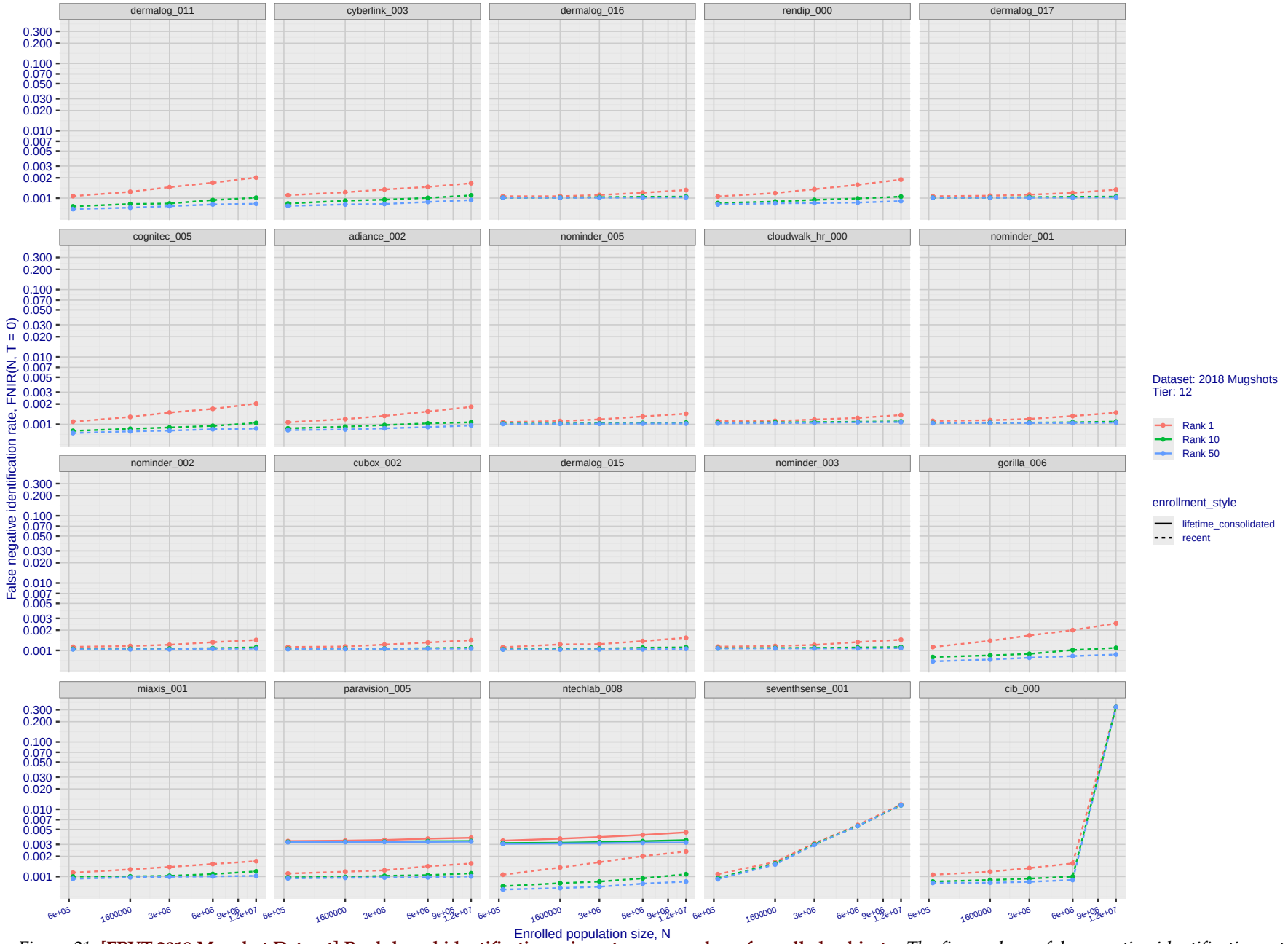

Figure 31: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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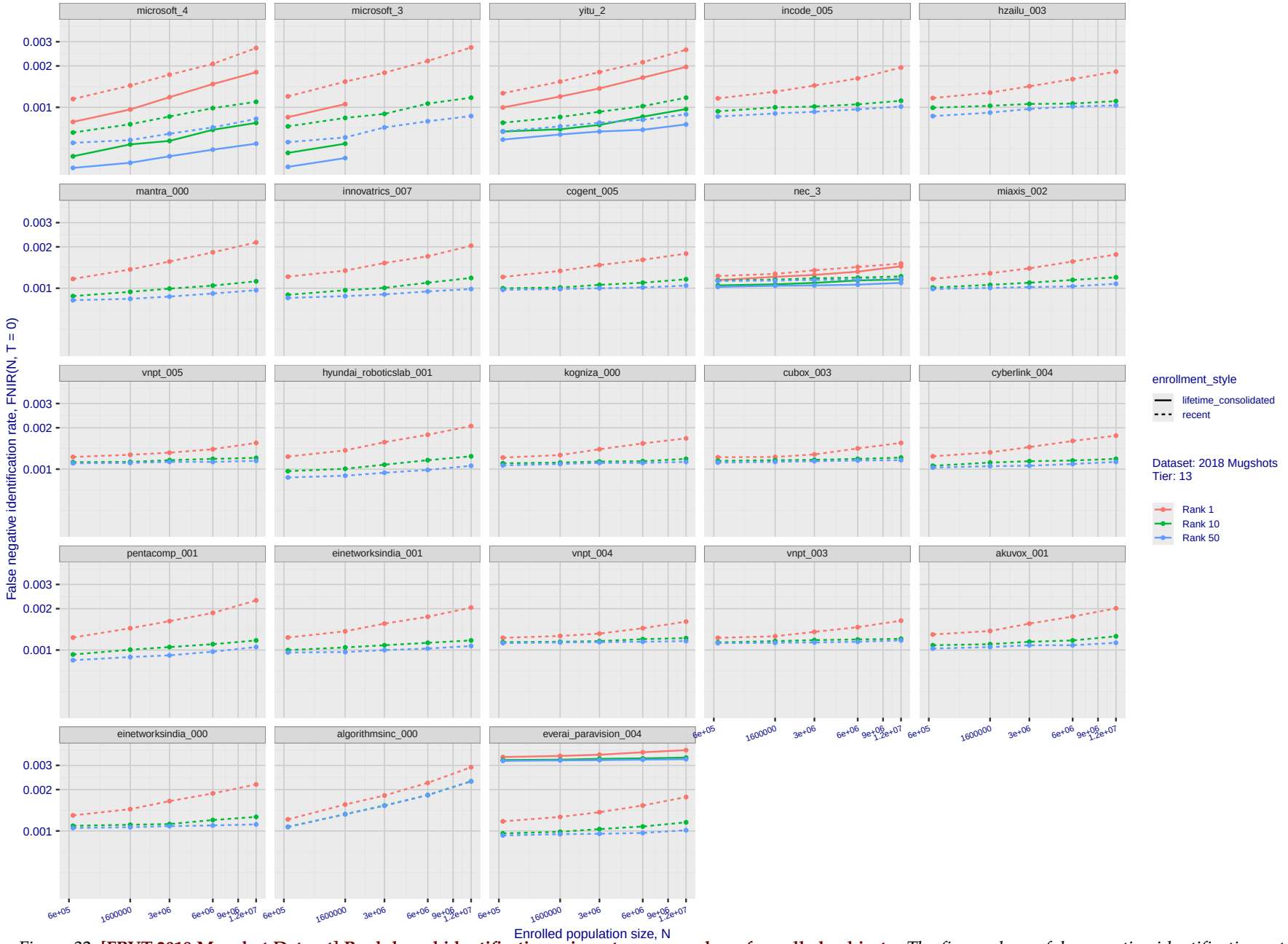

Figure 32: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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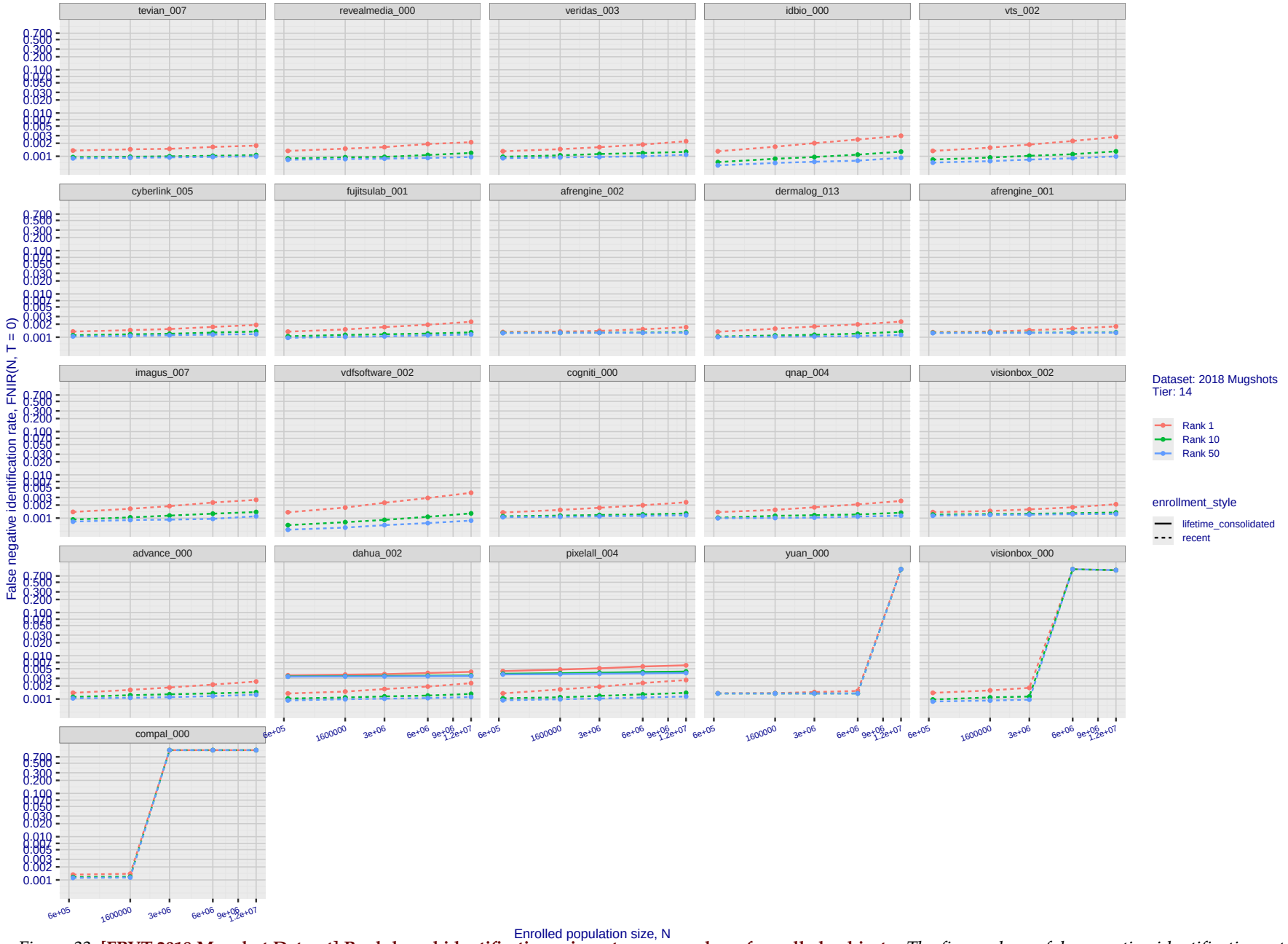

Figure 33: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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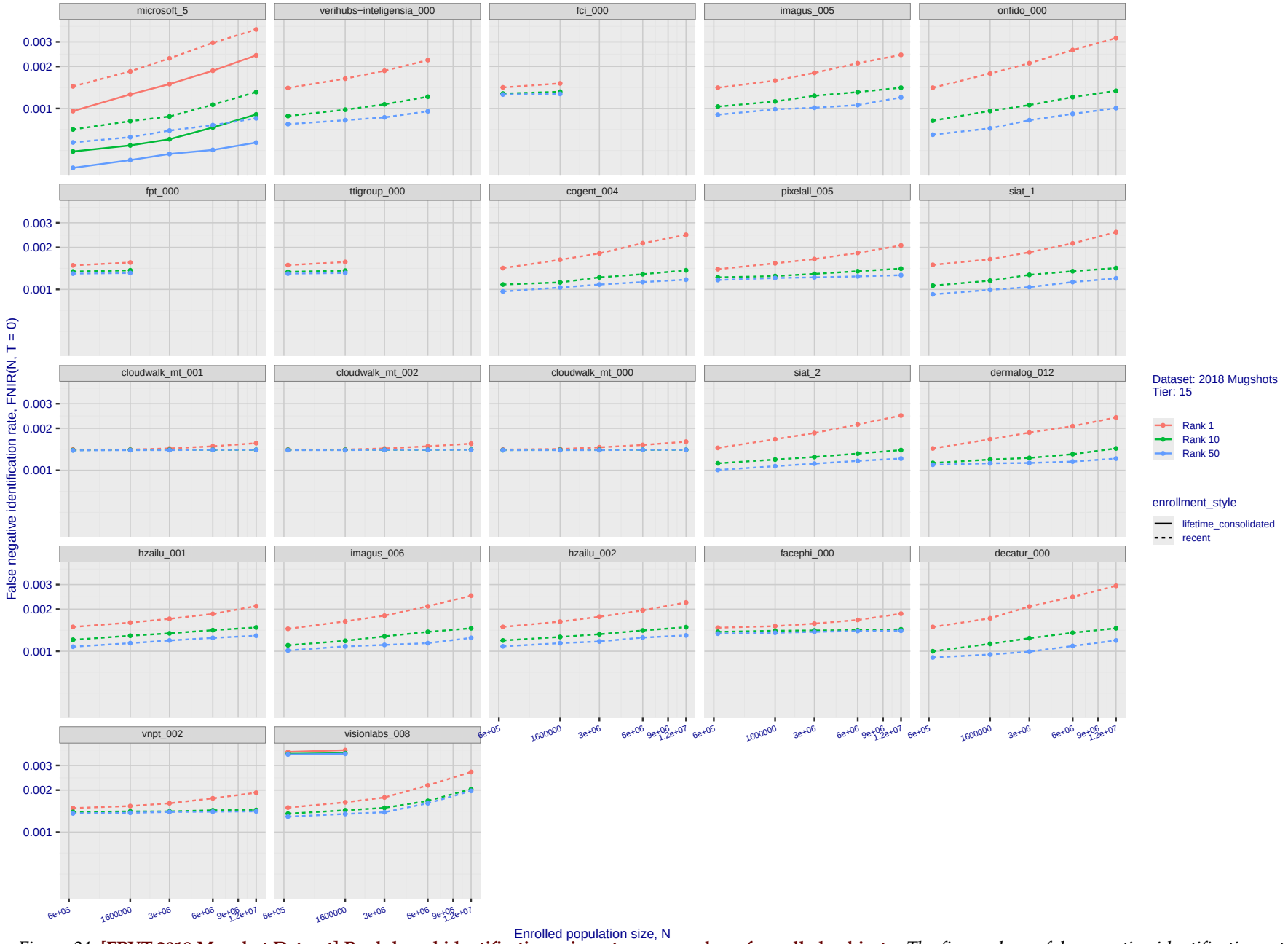

Figure 34: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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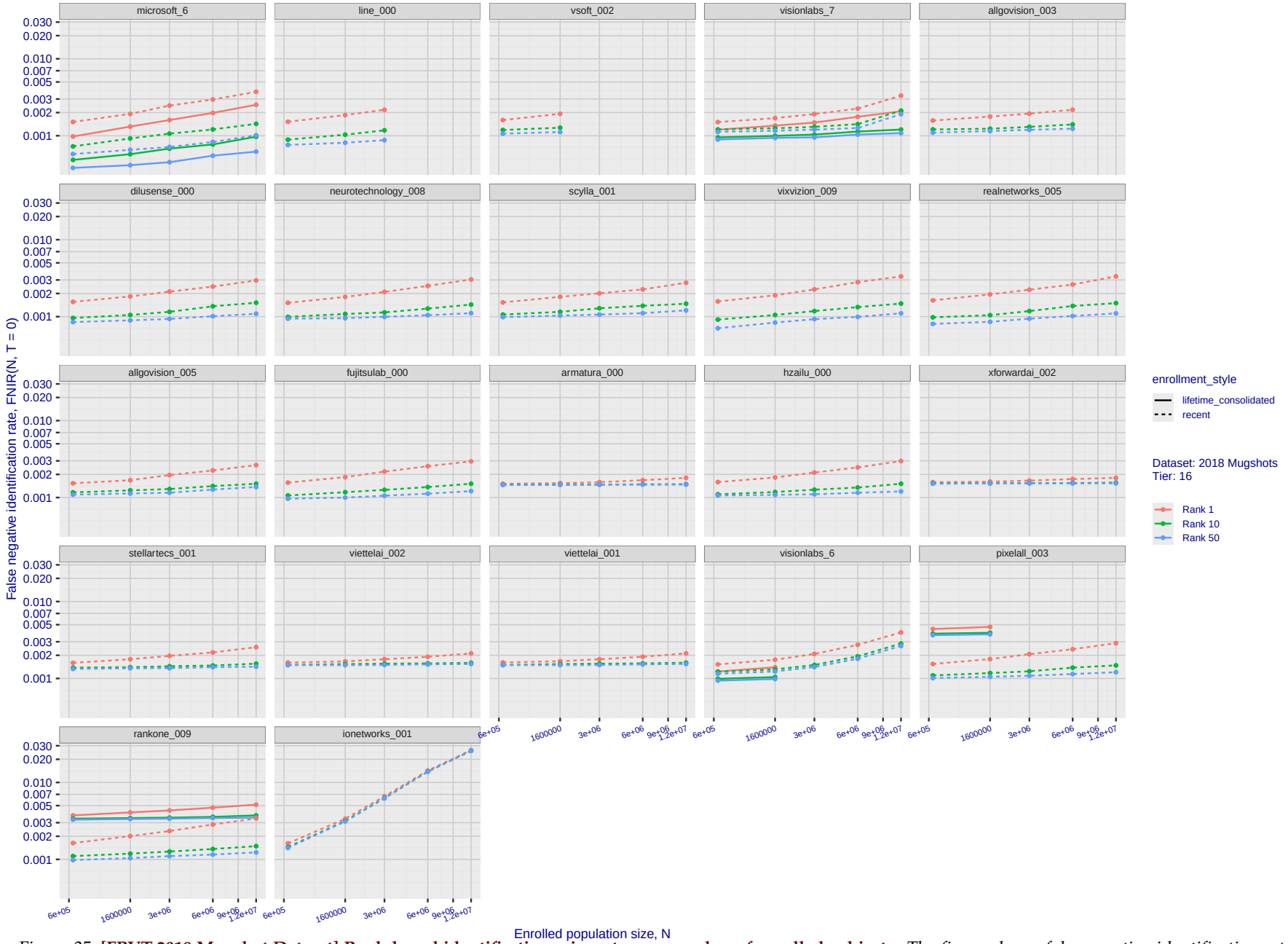

Figure 35: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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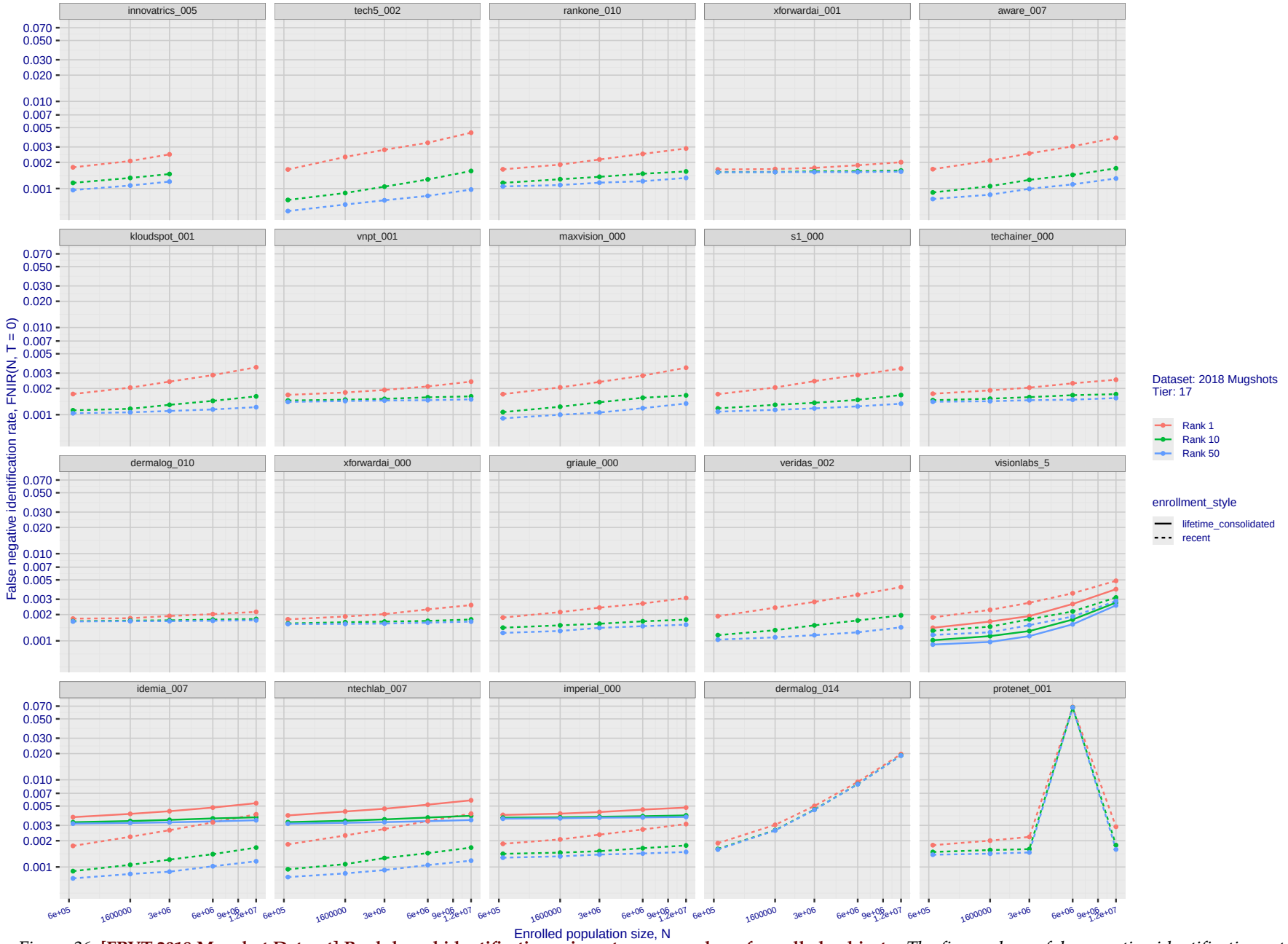

Figure 36: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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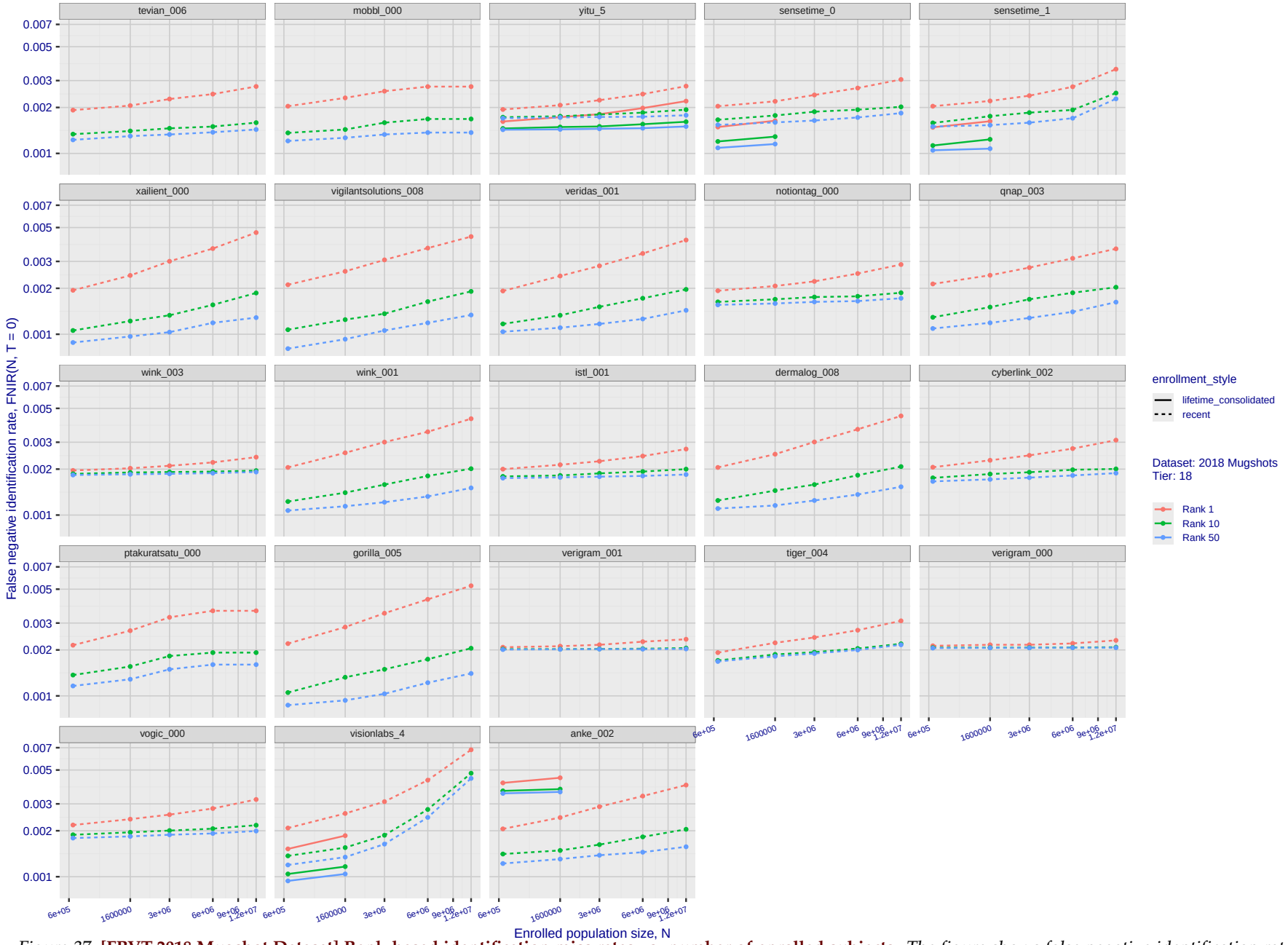

Figure 37: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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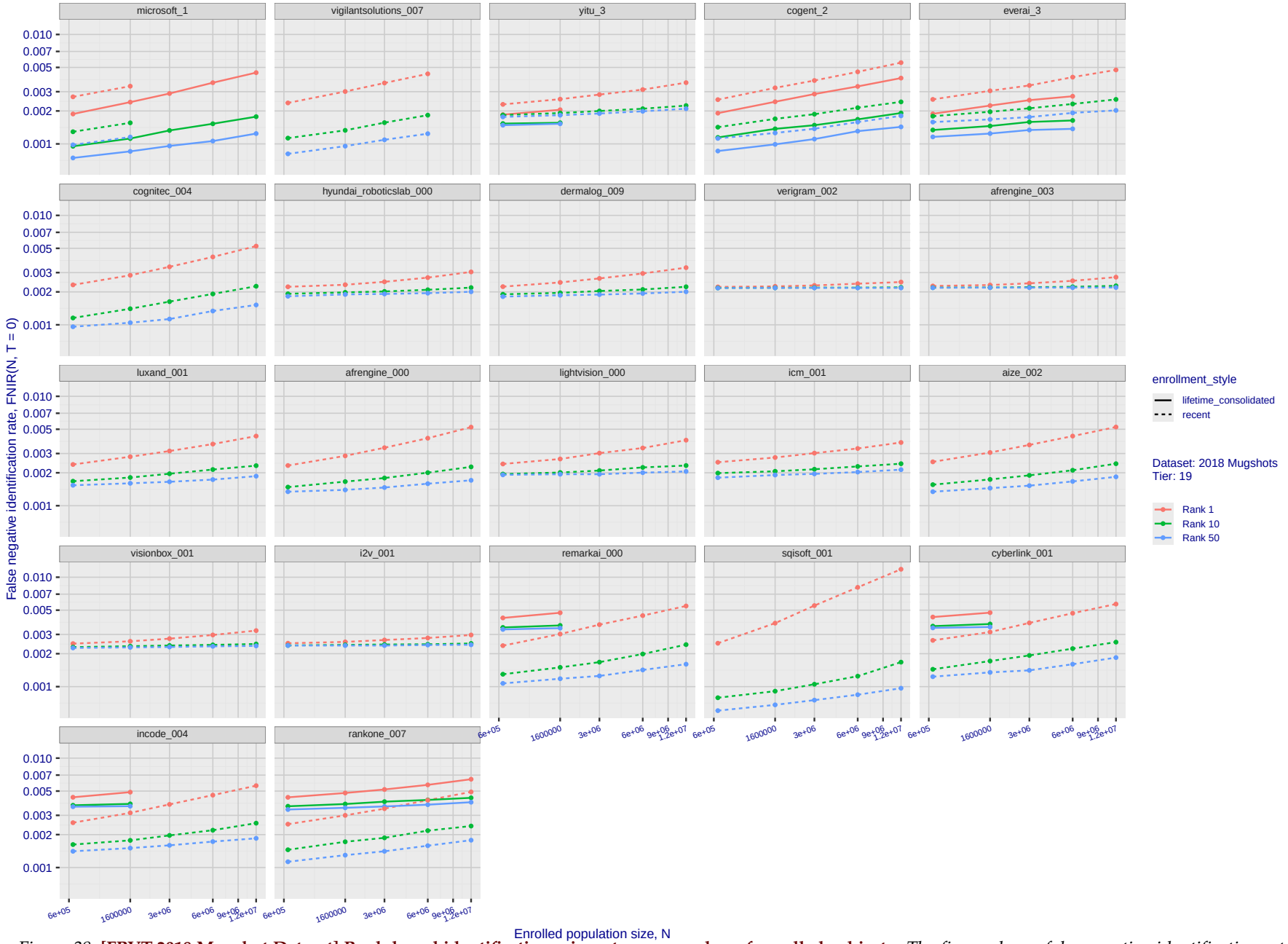

Figure 38: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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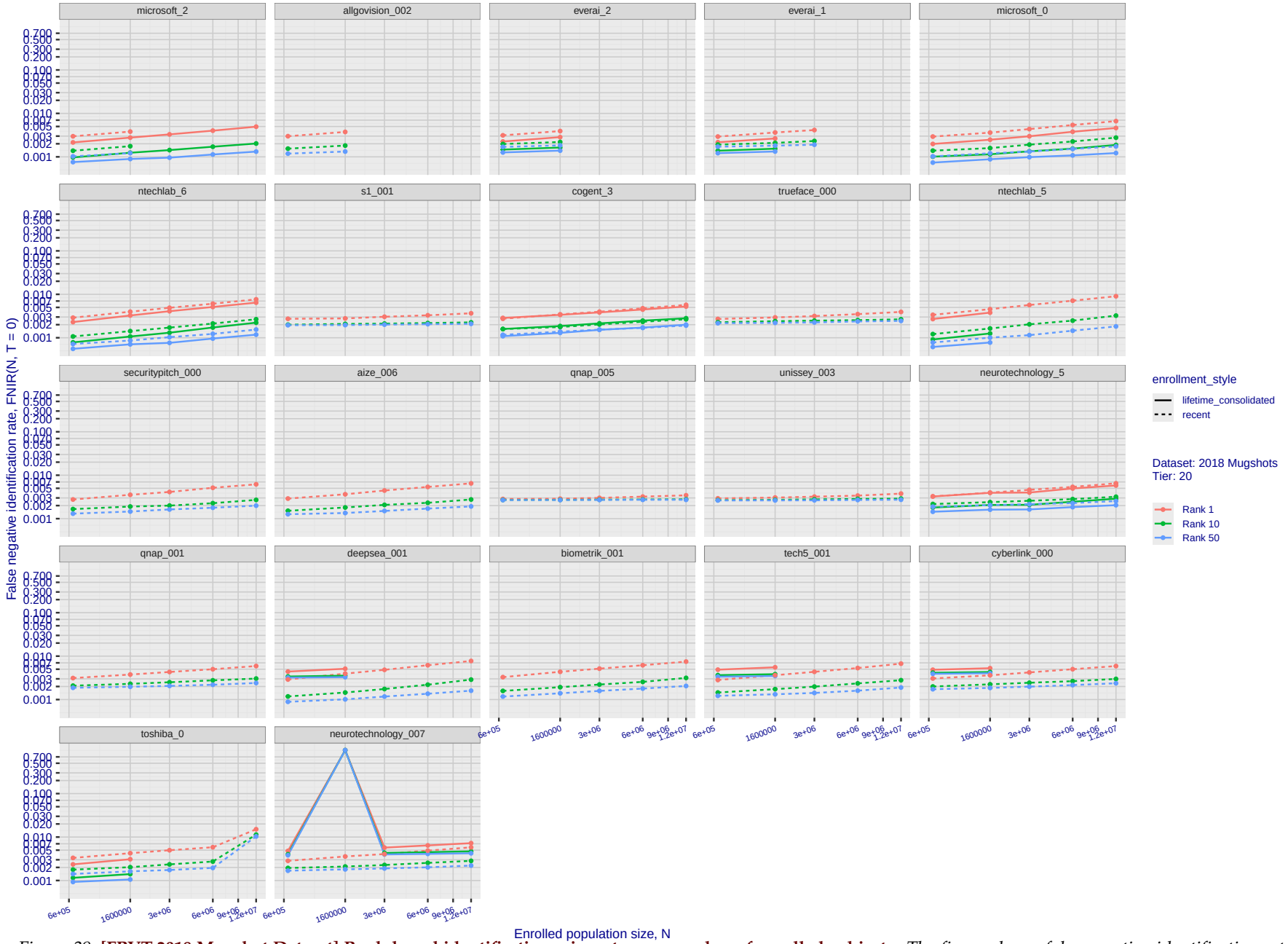

Figure 39: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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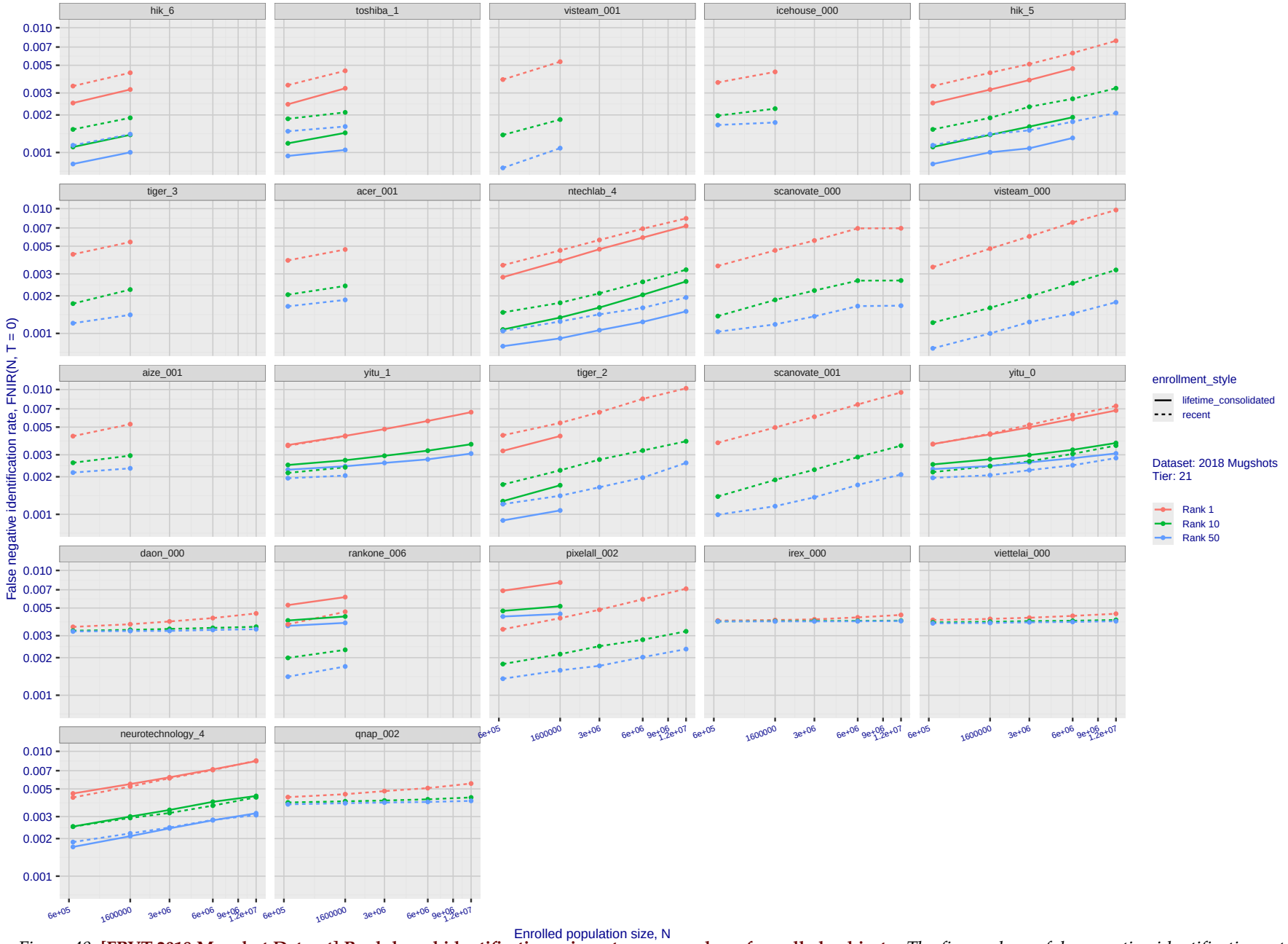

Figure 40: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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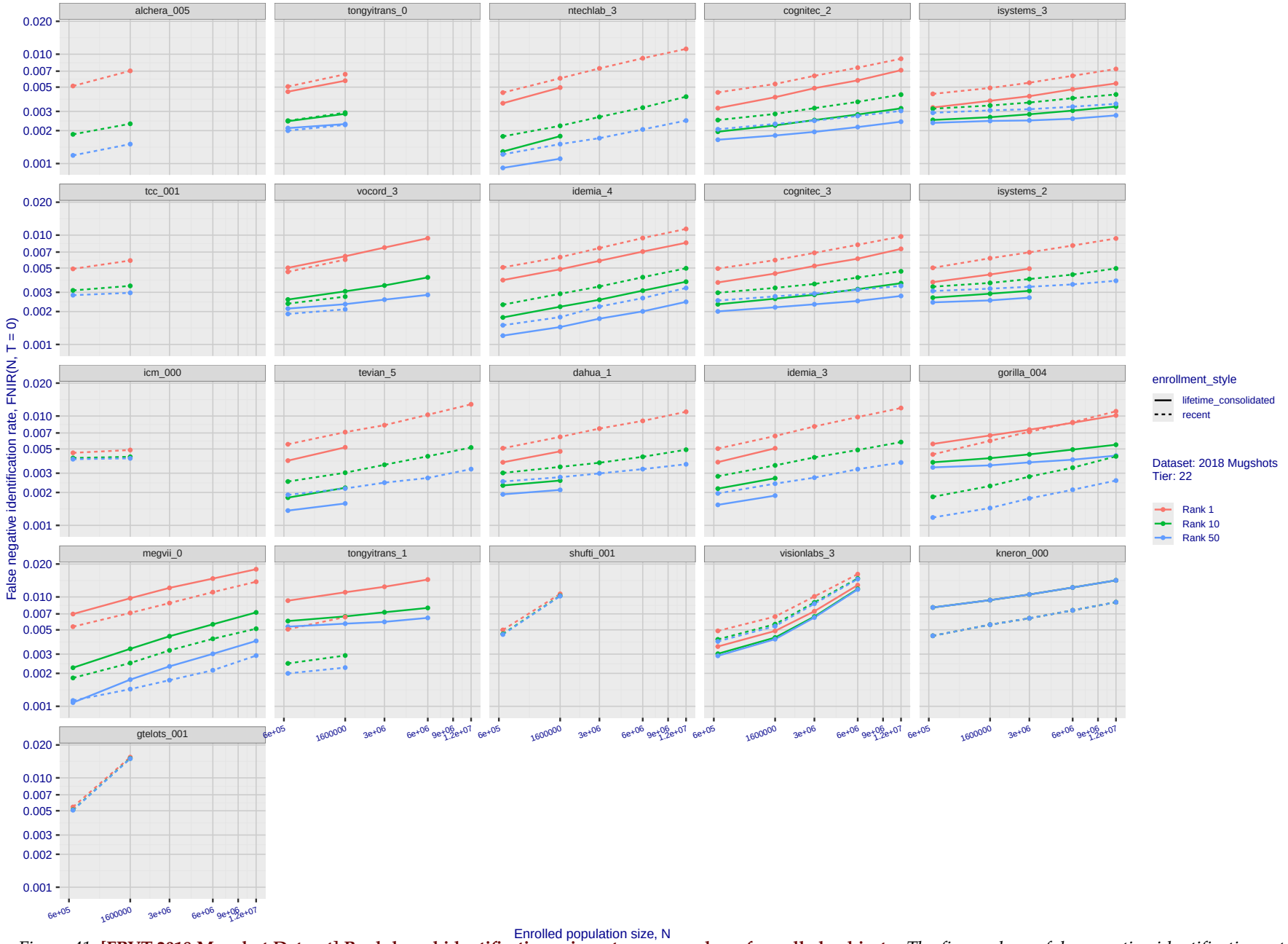

Figure 41: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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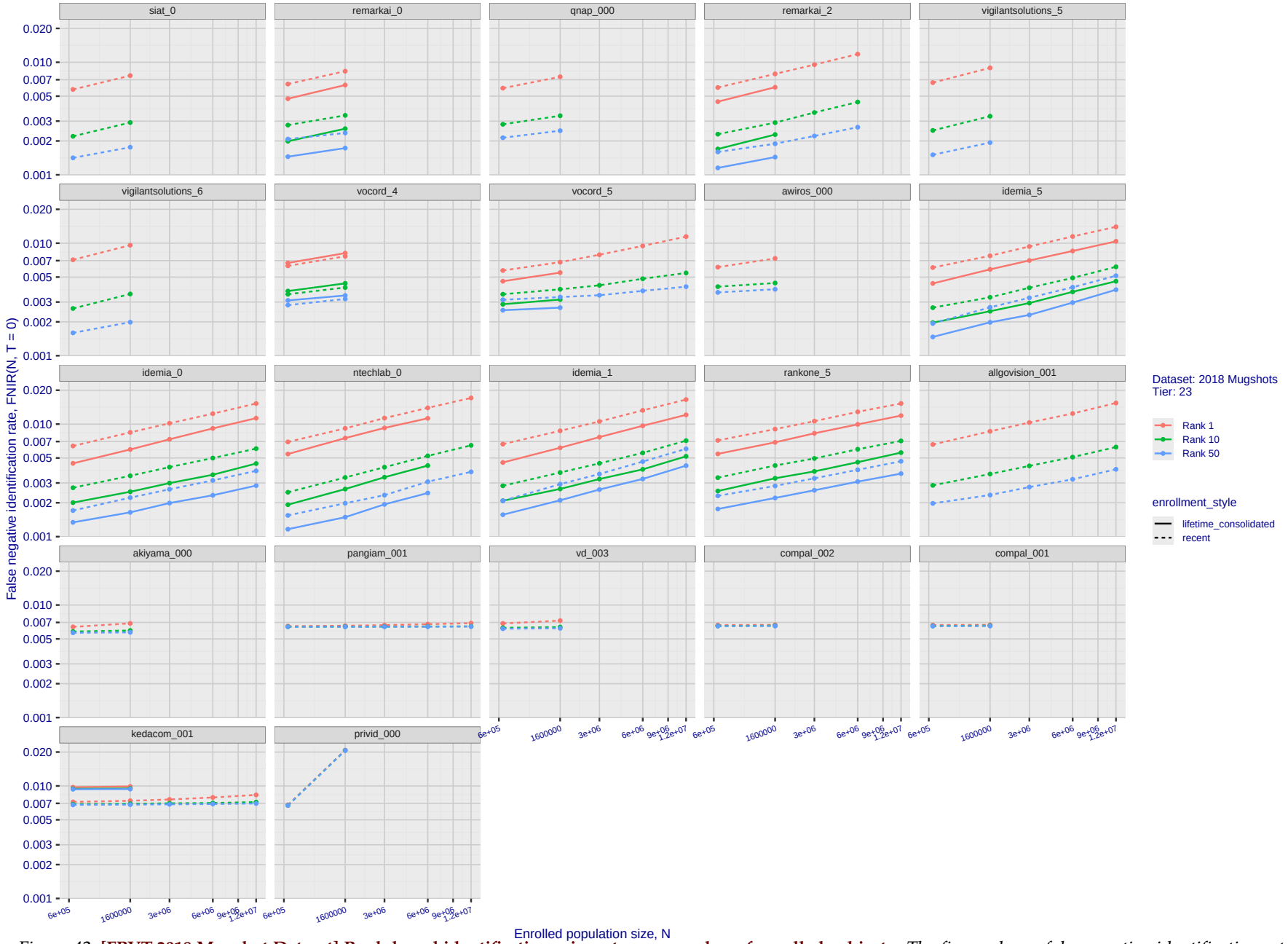

Figure 42: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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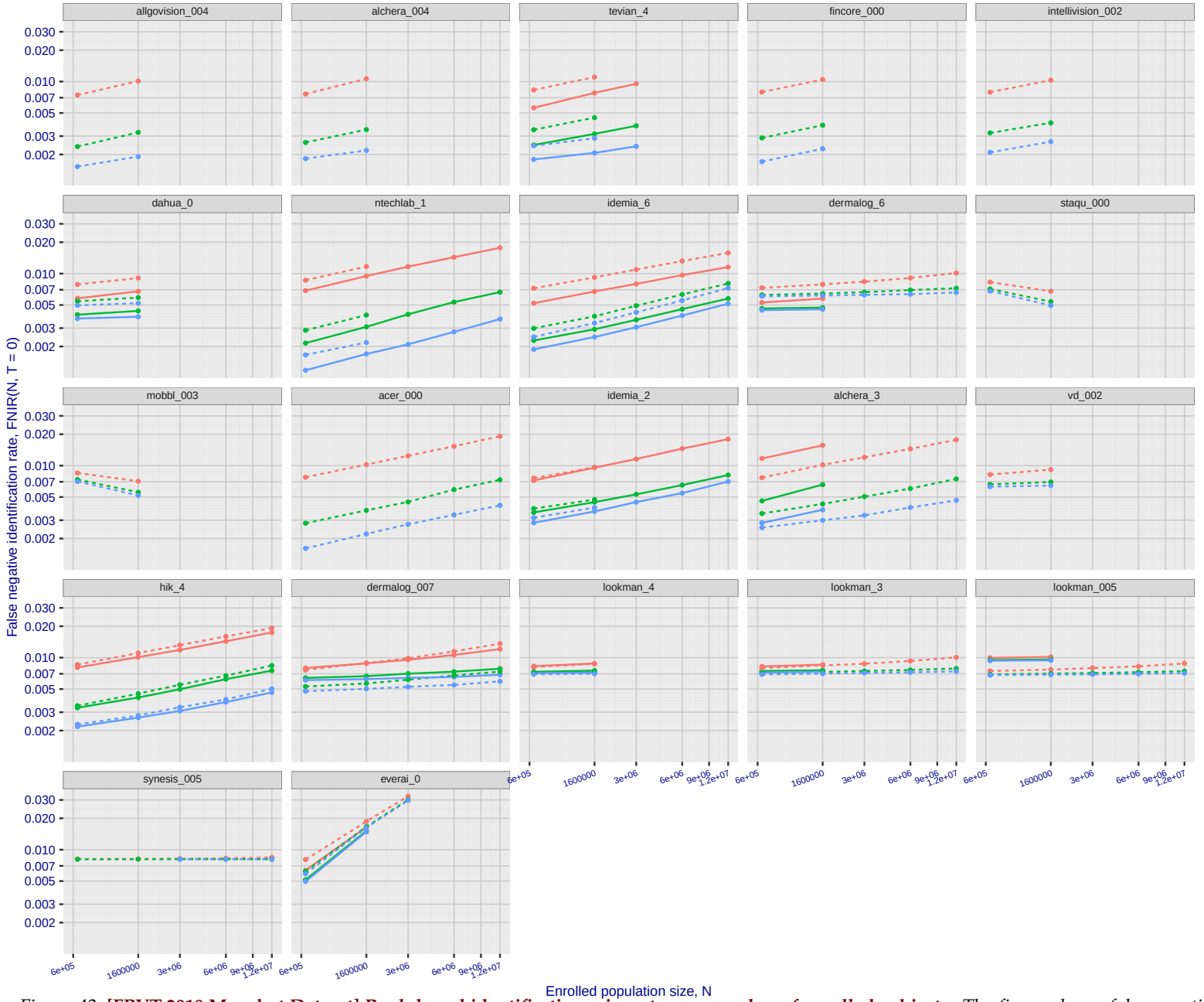

Figure 43: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
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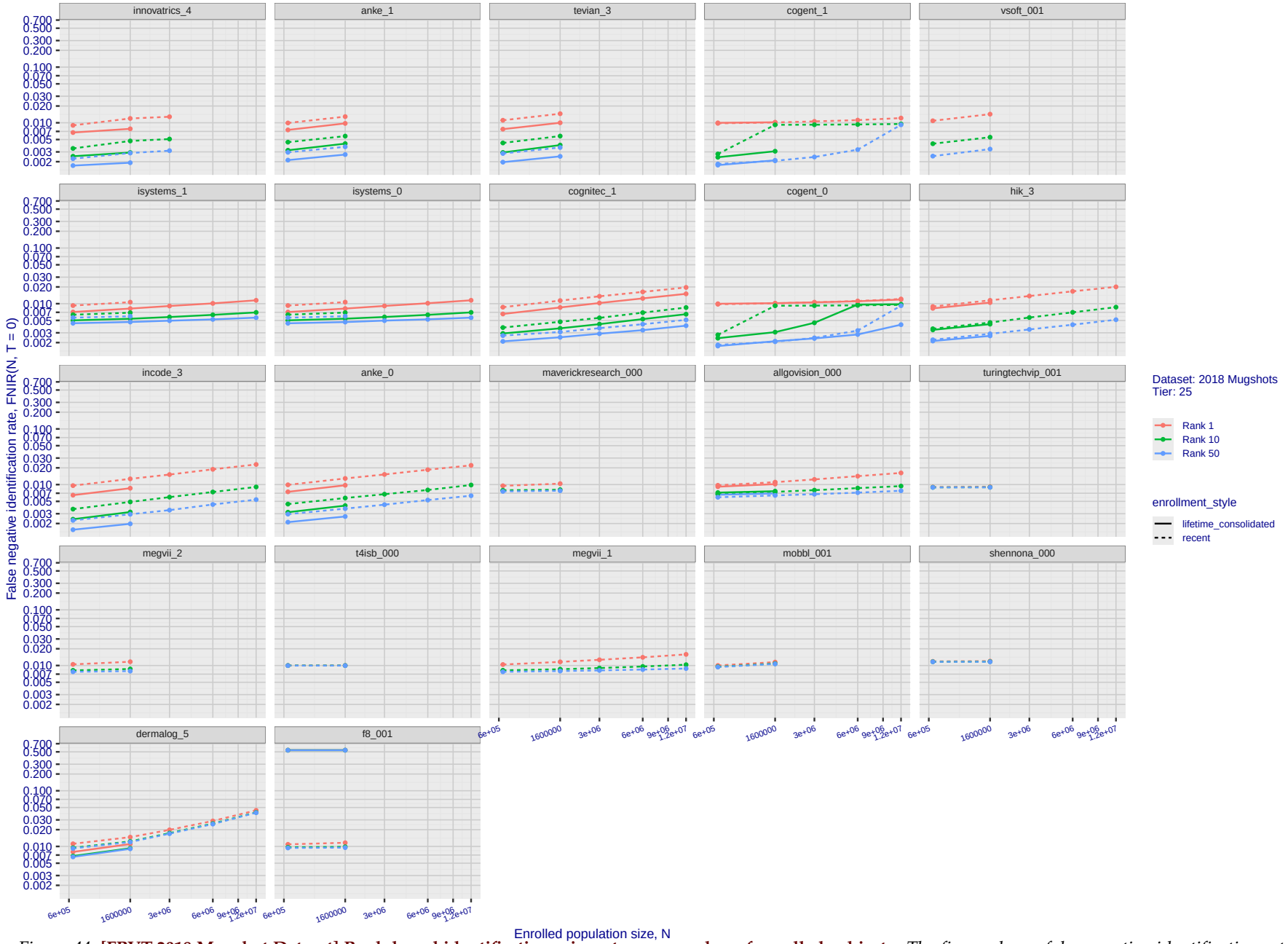

Figure 44: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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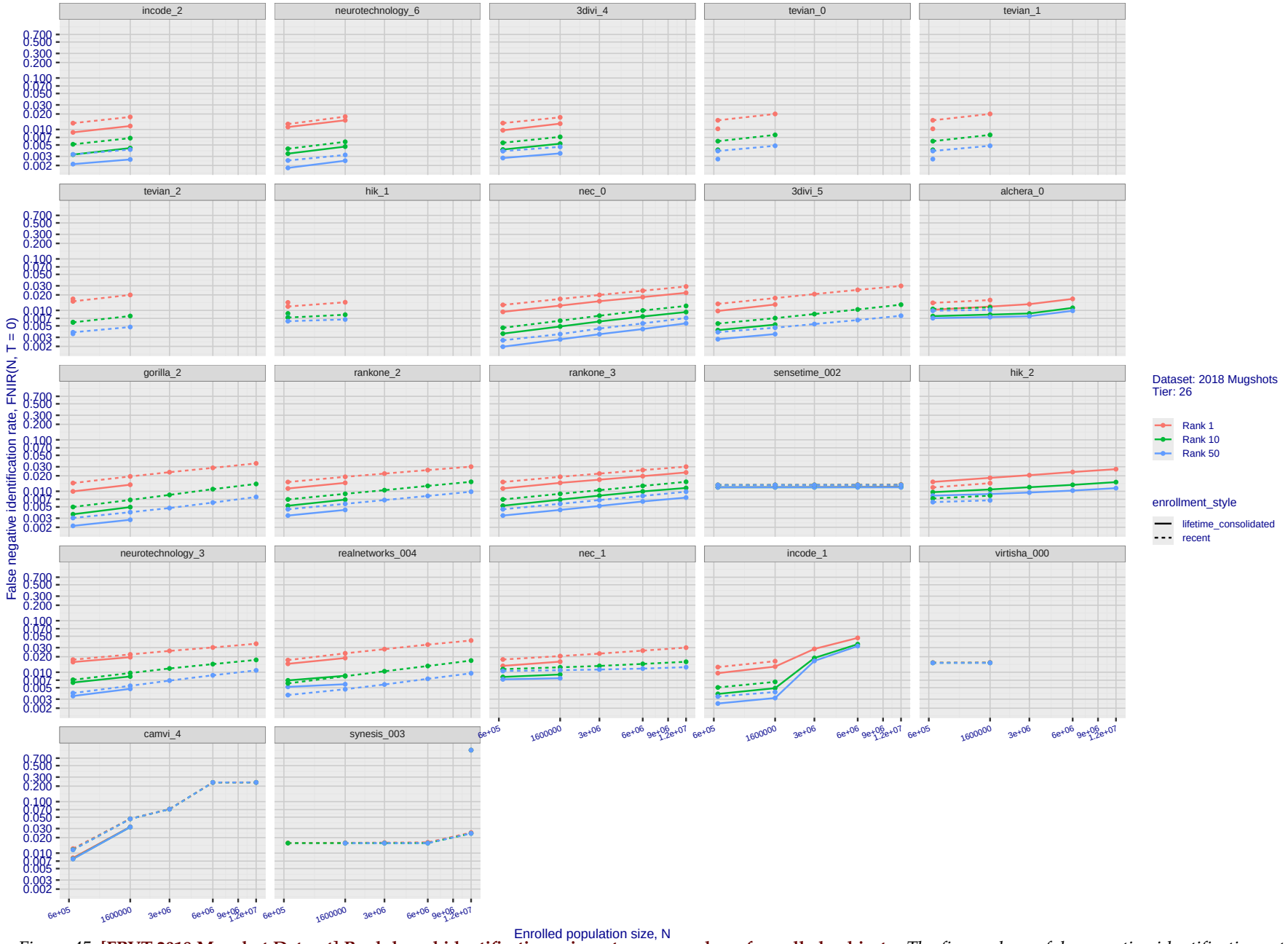

Figure 45: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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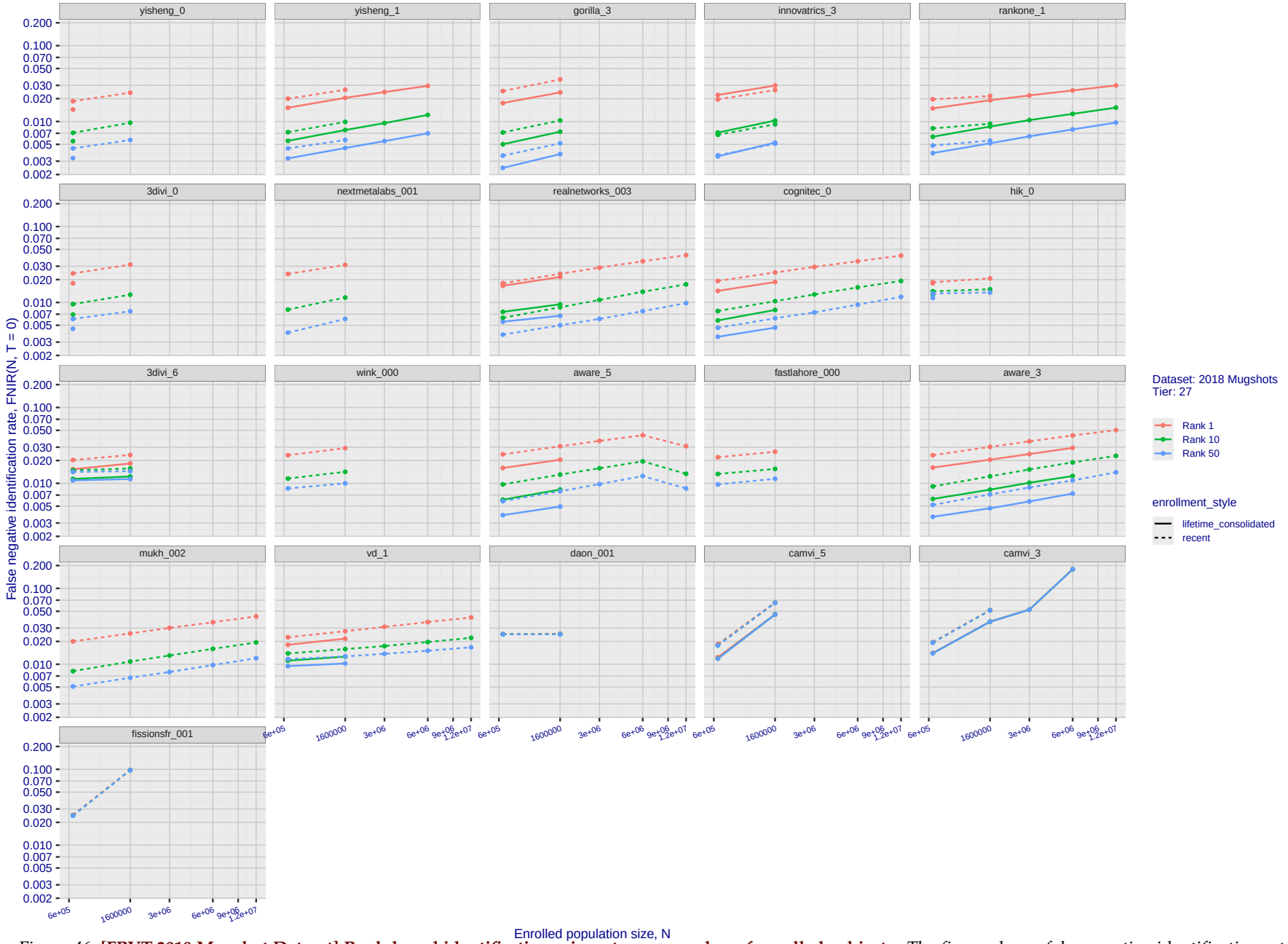

Figure 46: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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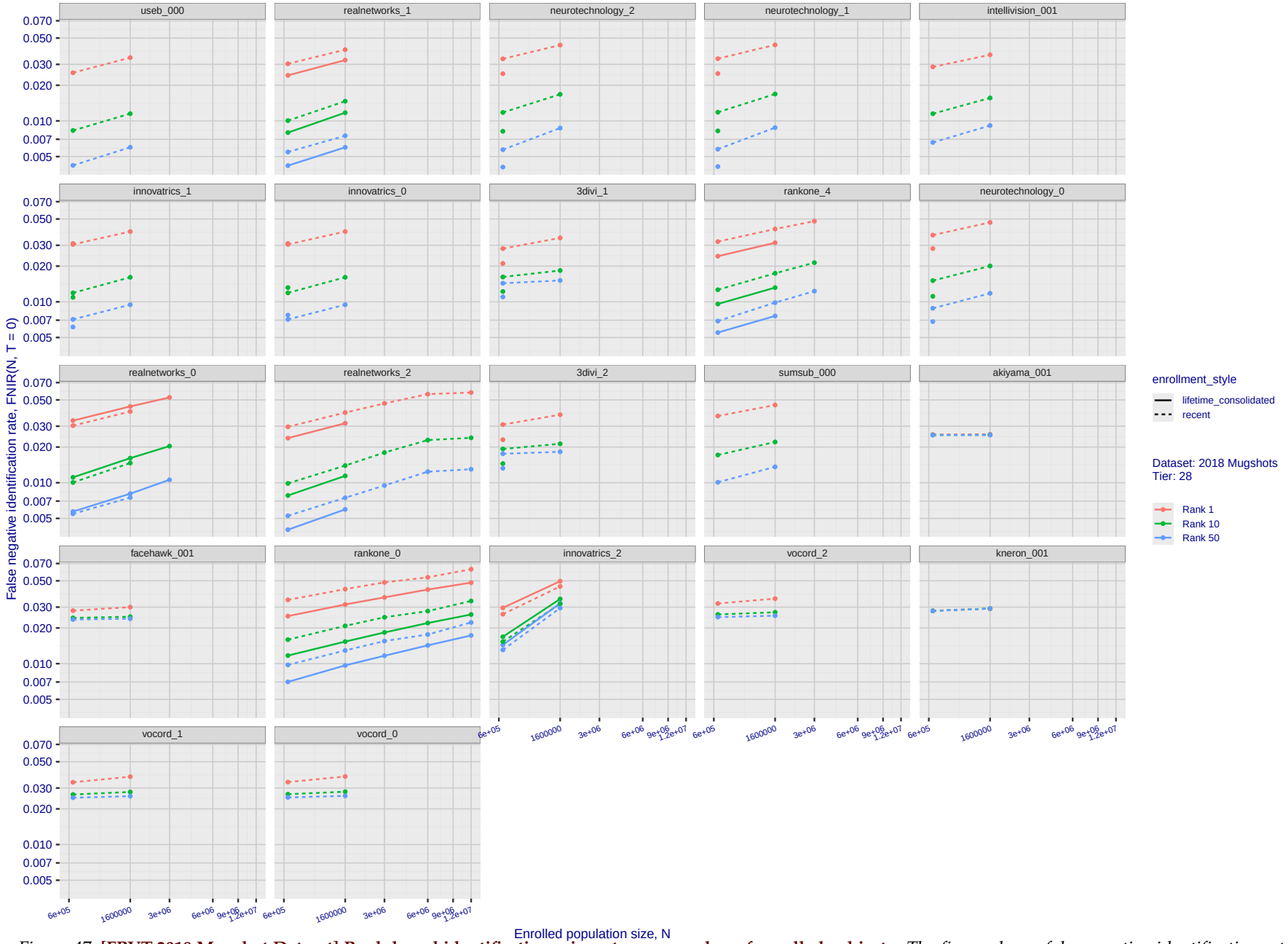

Figure 47: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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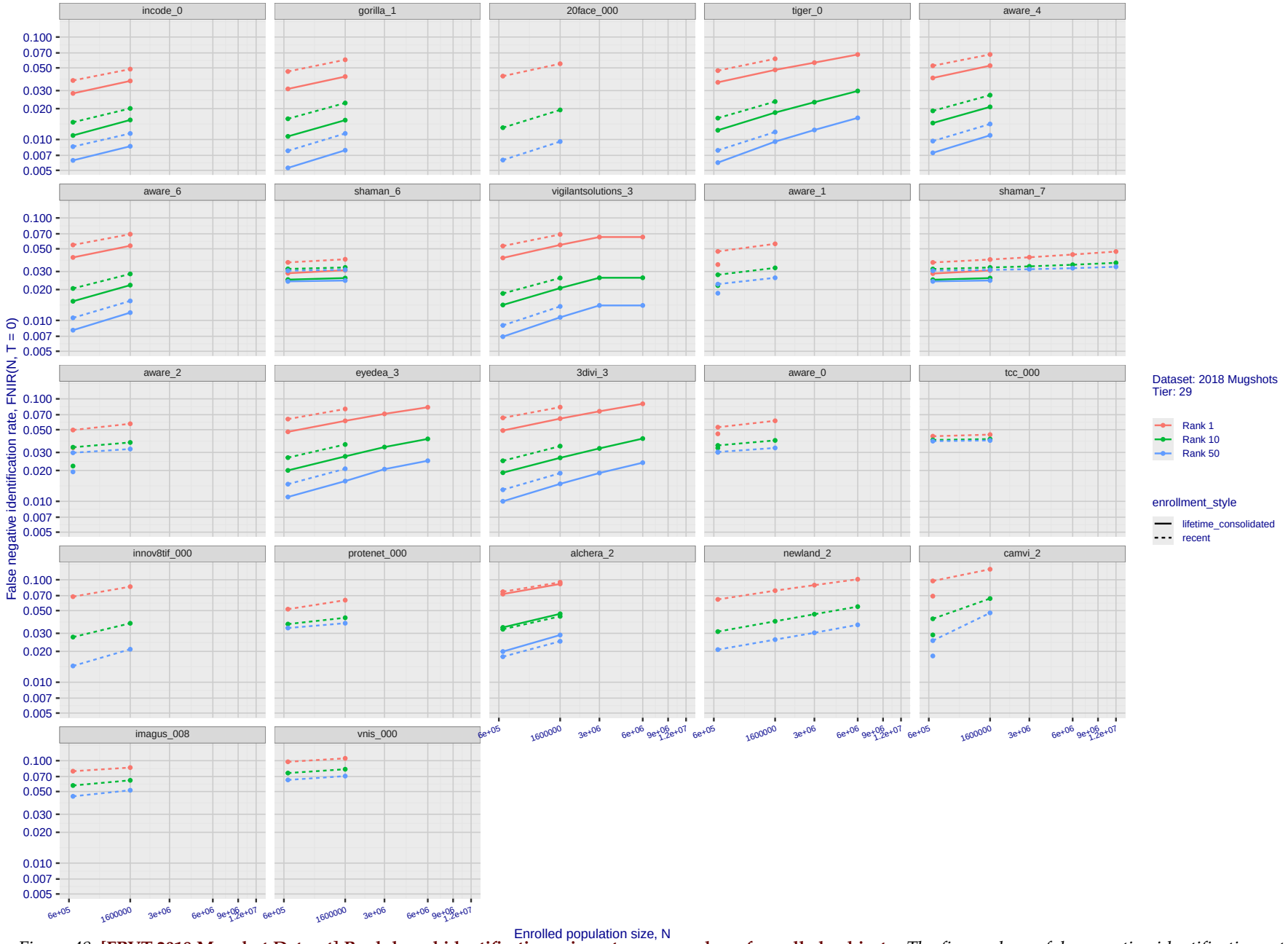

Figure 48: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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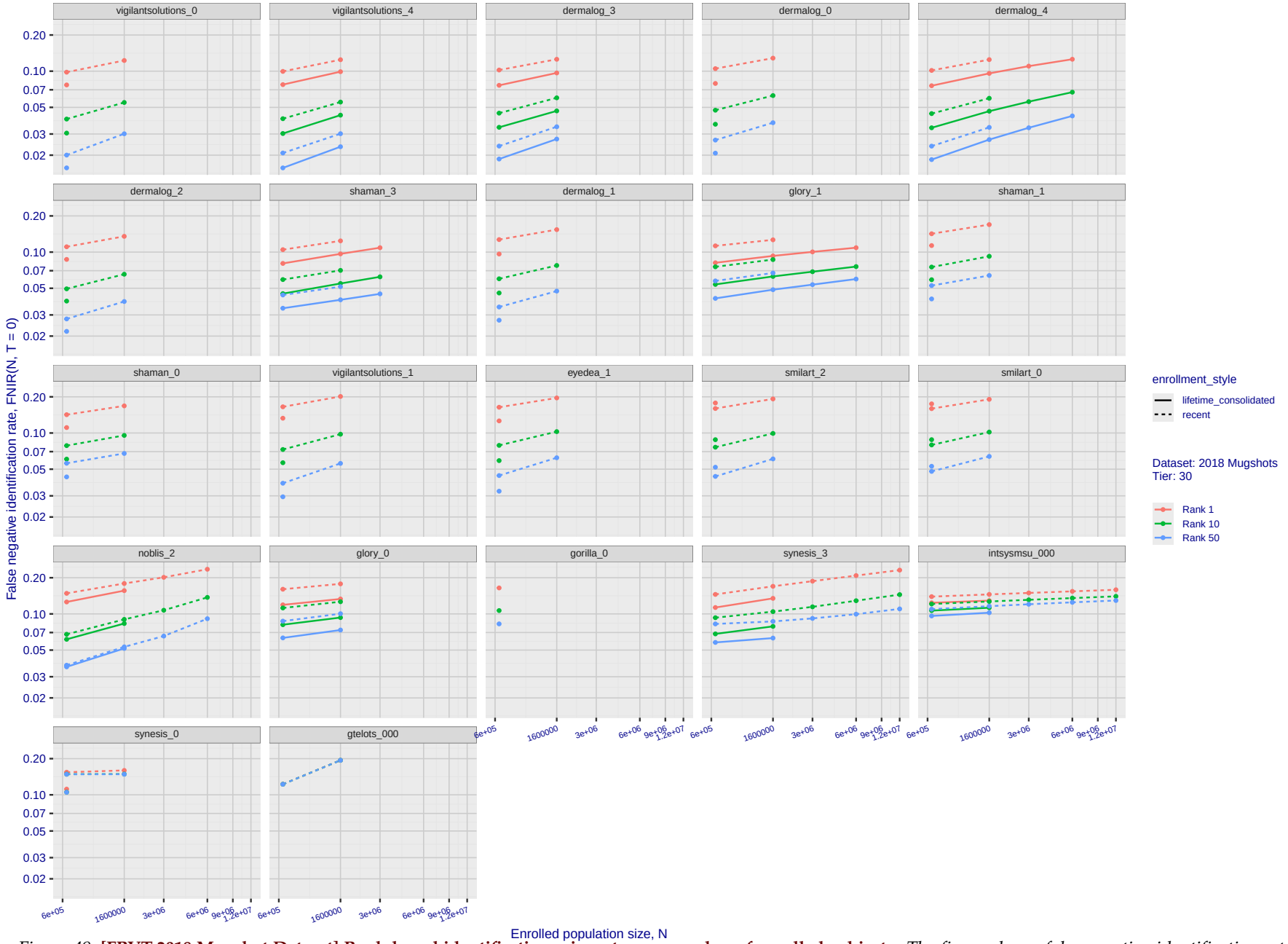

Figure 49: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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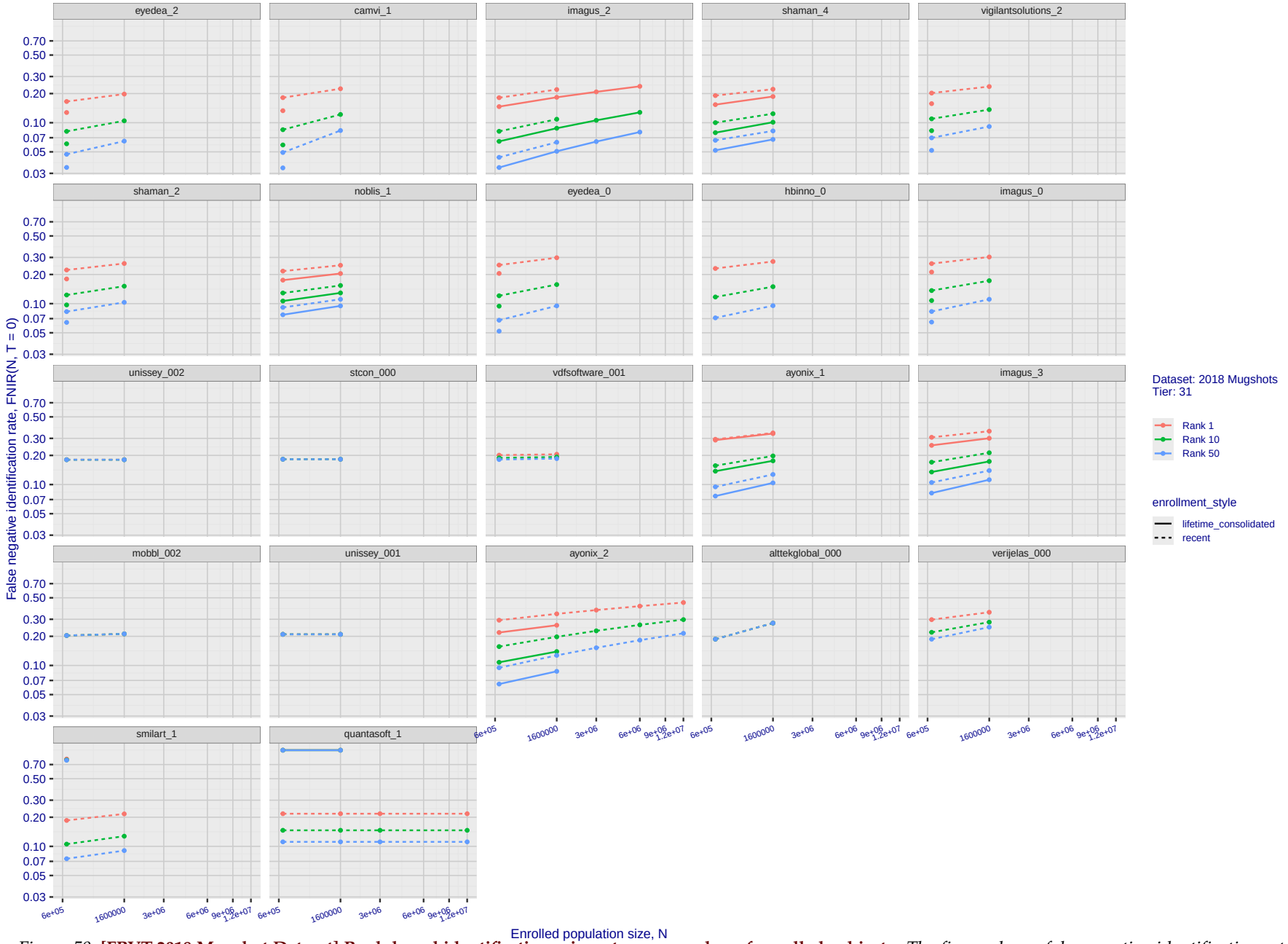

Figure 50: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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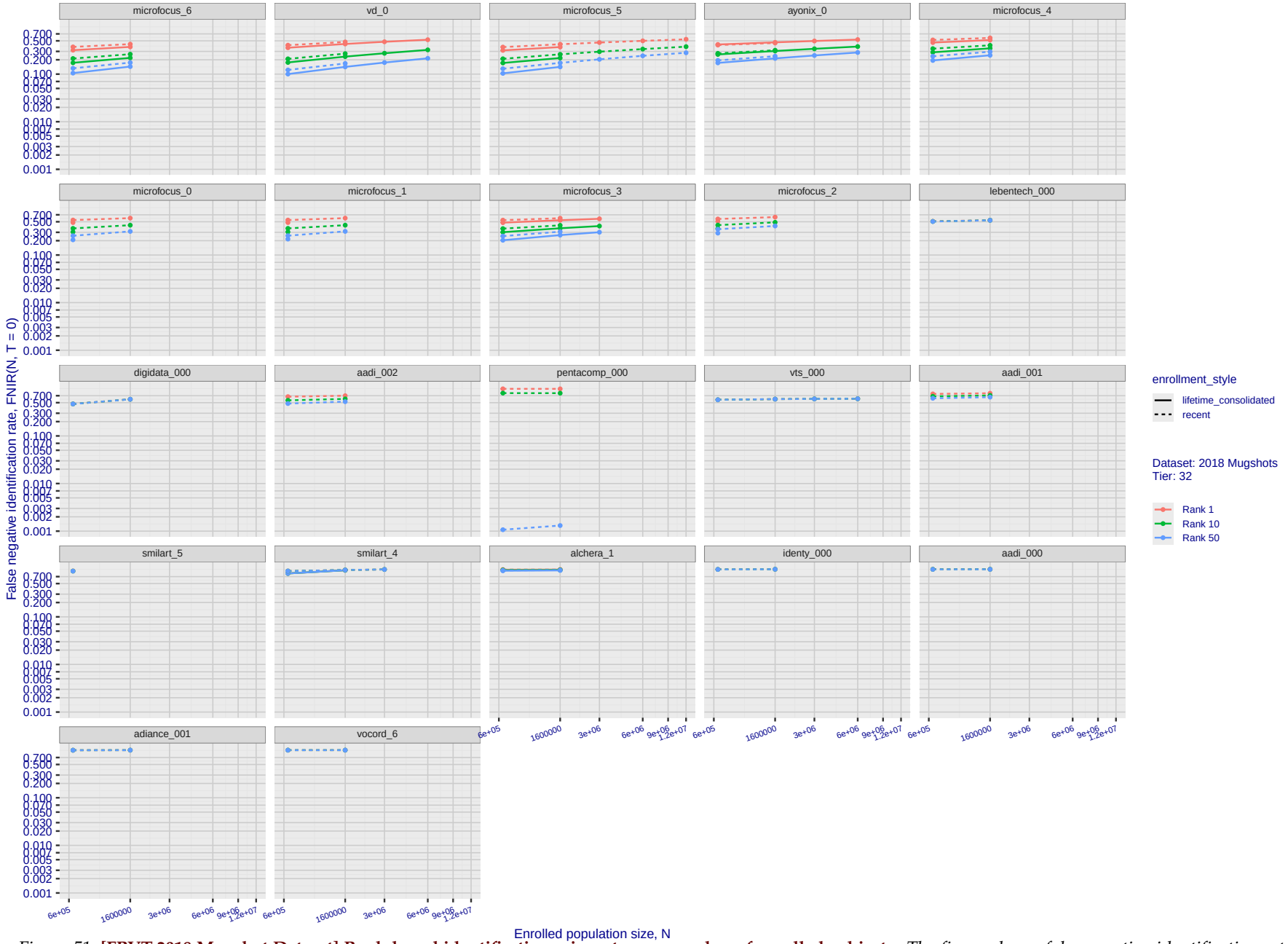

Figure 51: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. number of enrolled subjects**. *The figure shows false negative identification rates, FNIR(N, R), across various gallery sizes and ranks 1, 10 and 50. The threshold is set to zero, so this metric rewards even weak scoring rank 1 mates. This also means FPIR = 1, so any search without an enrolled mate will return non-mated candidates. For clarity, results are sorted and reported into tiers spanning multiple pages, the tiering criteria being rank 1 hit rate on a gallery size of 640 000.*

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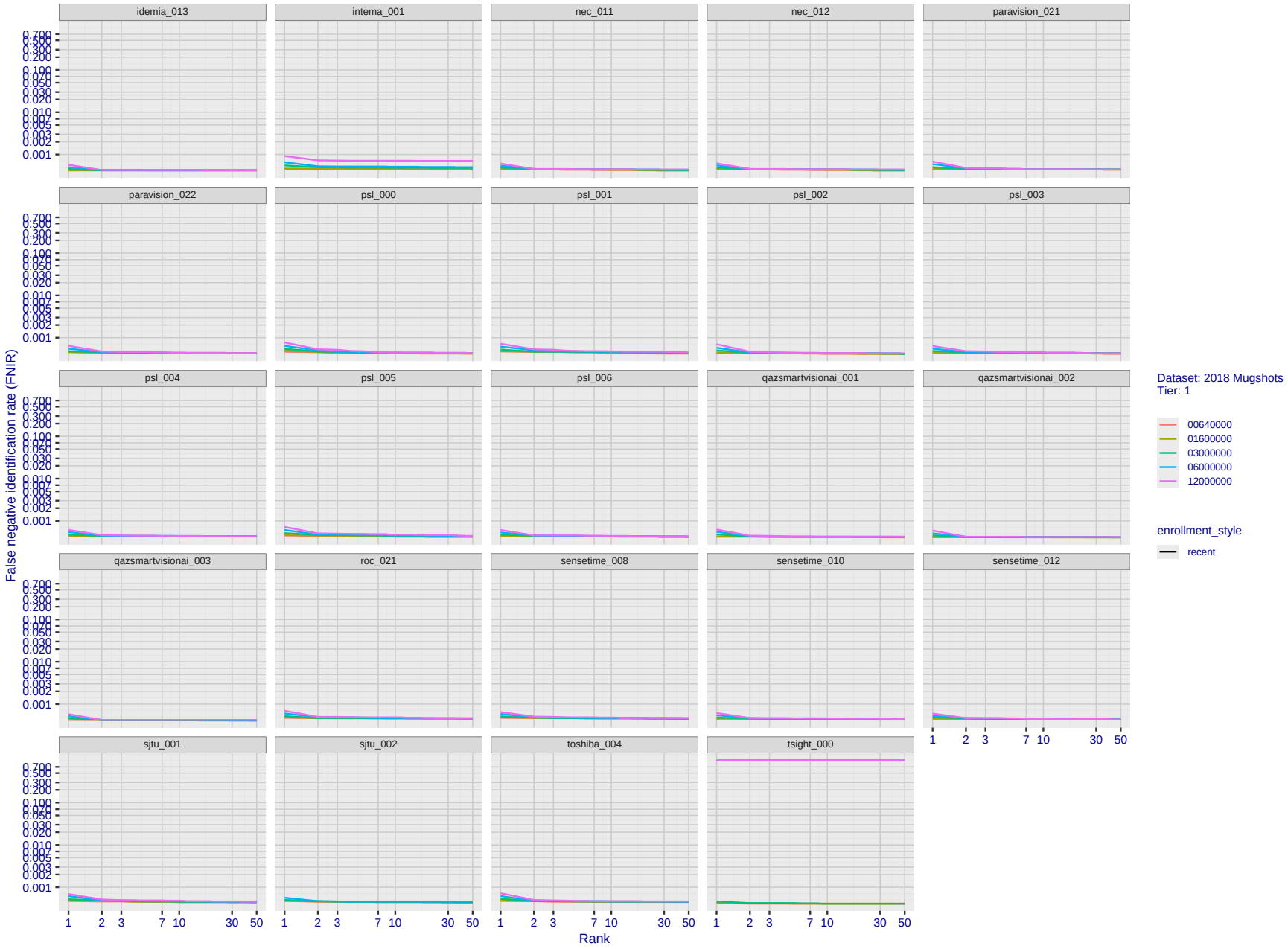

Figure 52: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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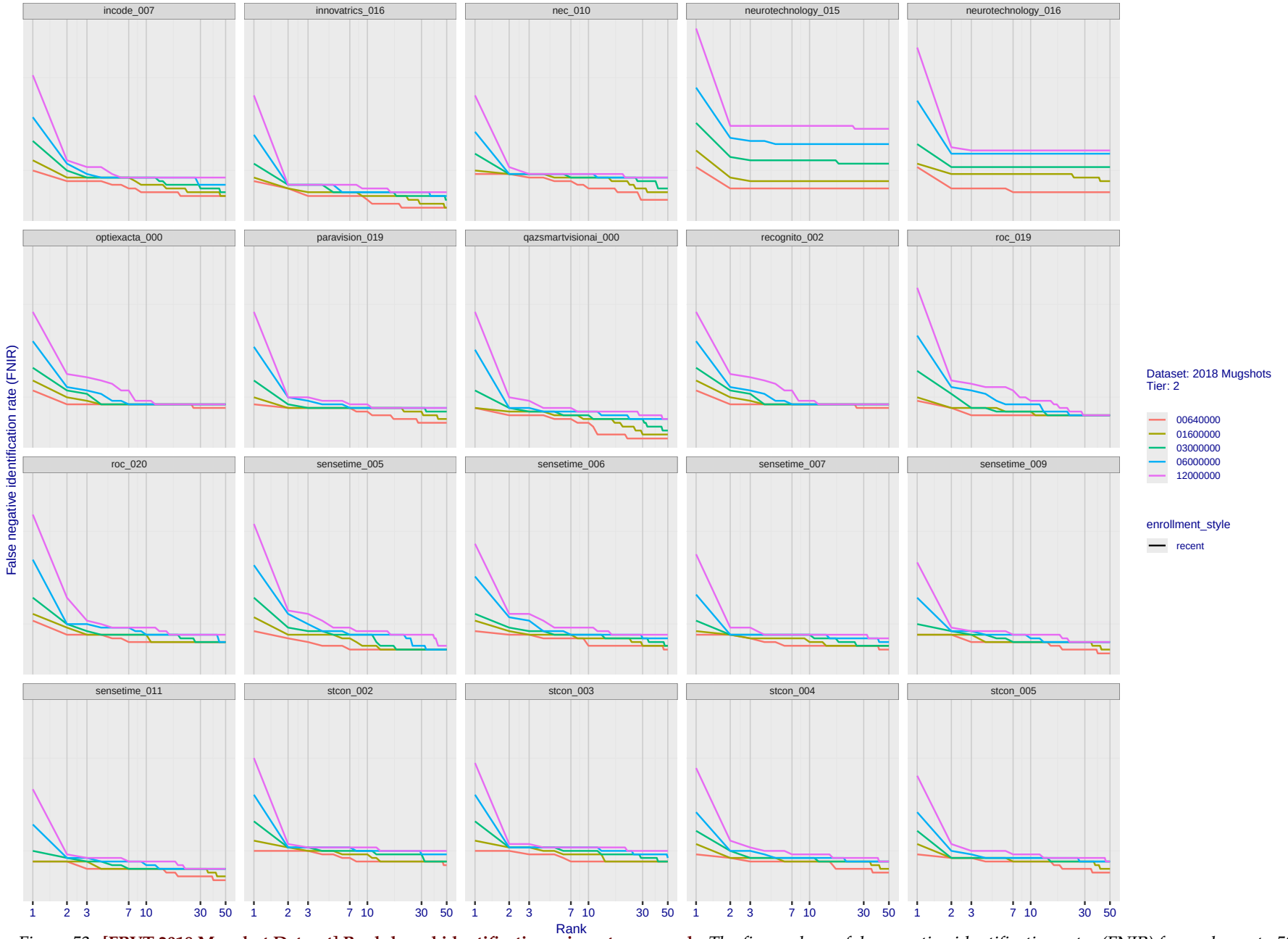

Figure 53: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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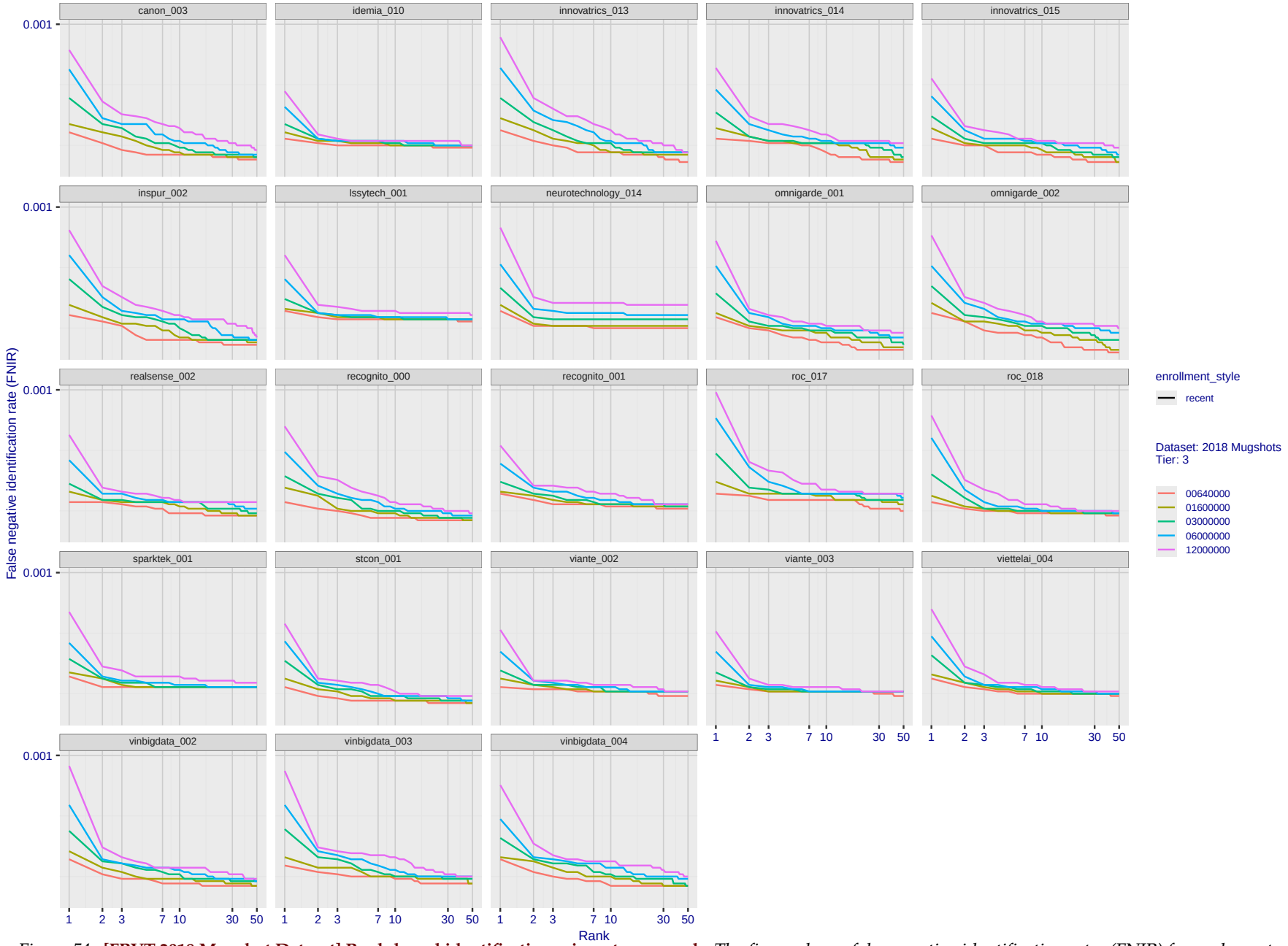

Figure 54: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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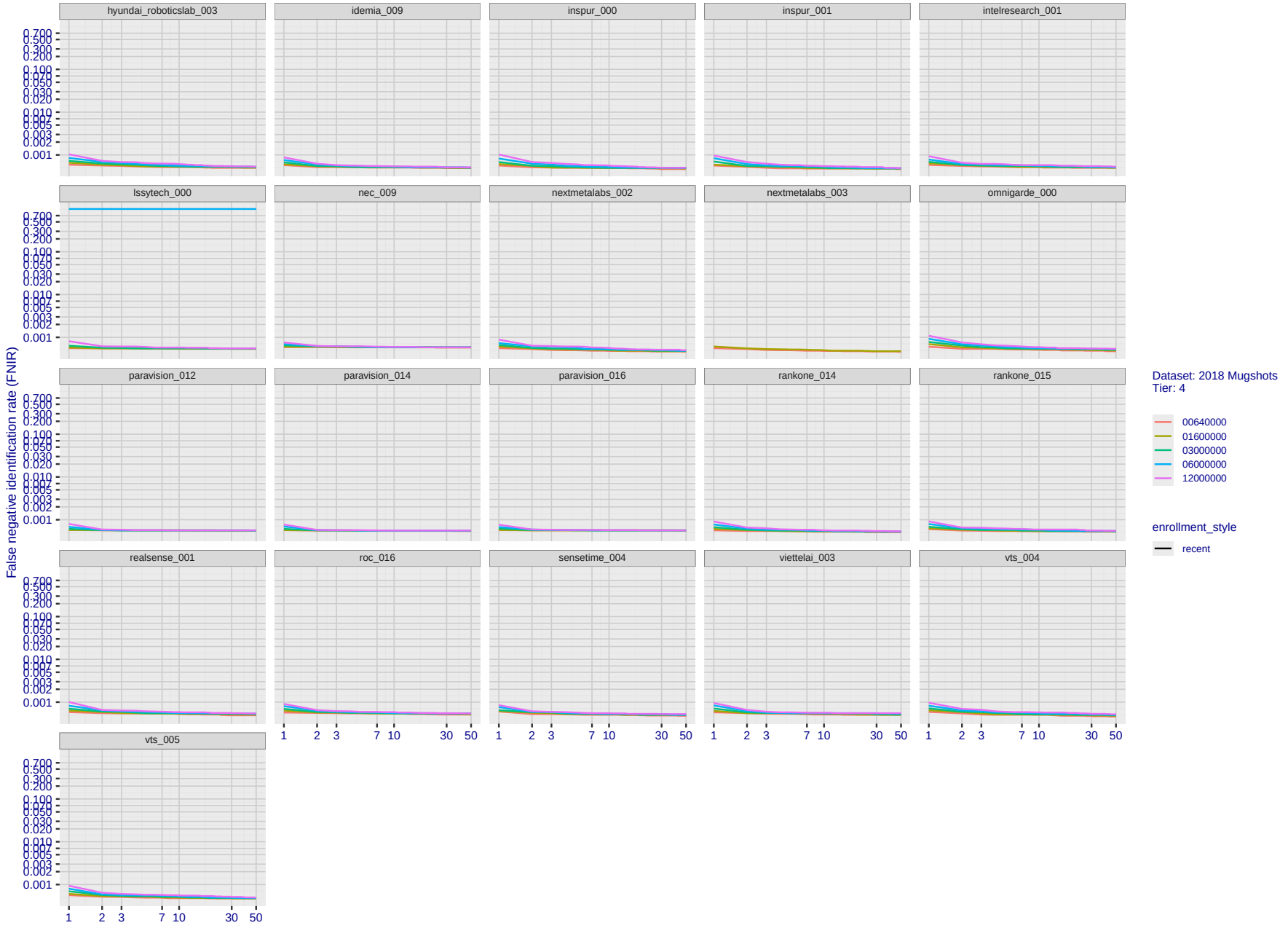

Figure 55: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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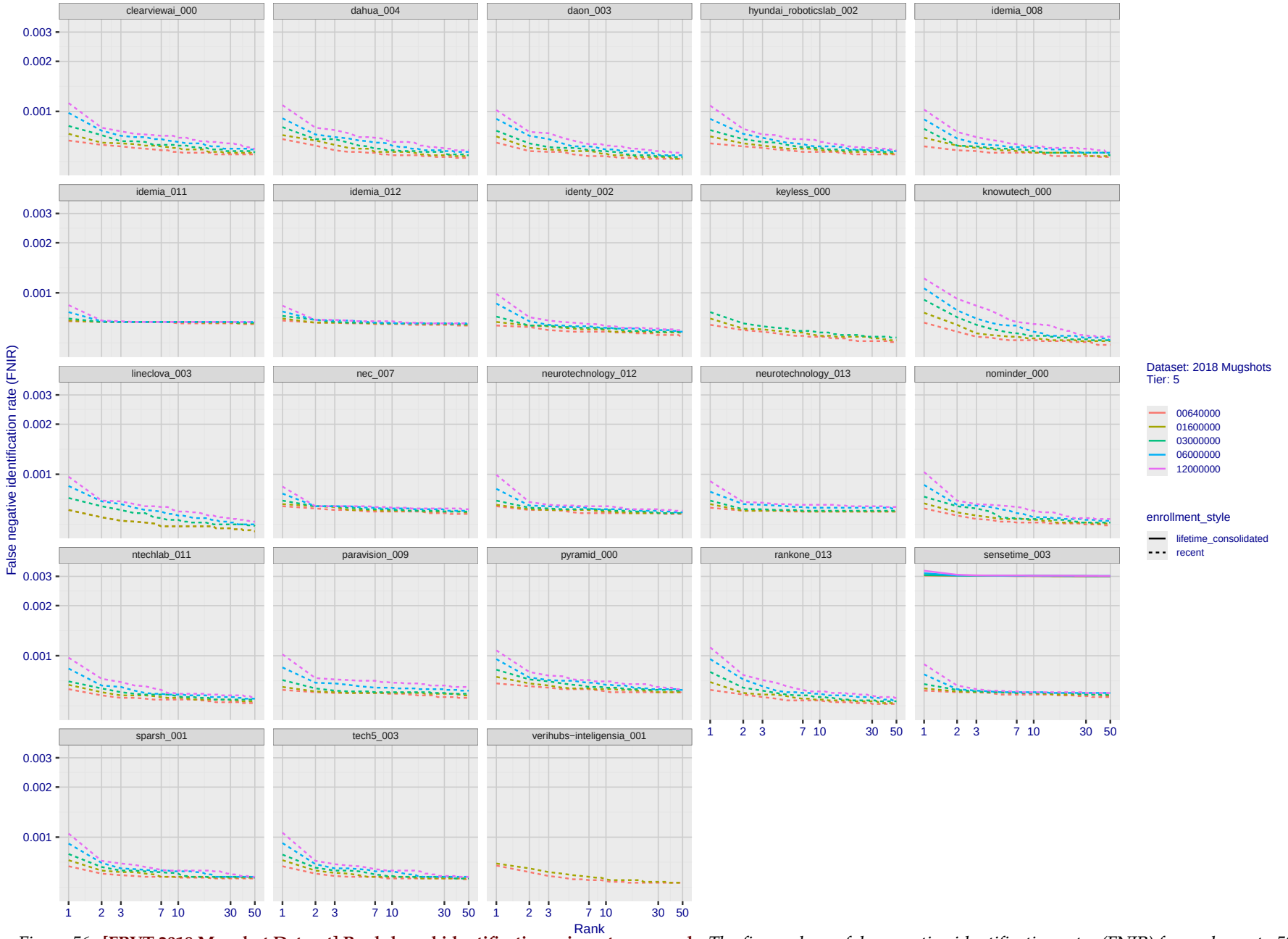

Figure 56: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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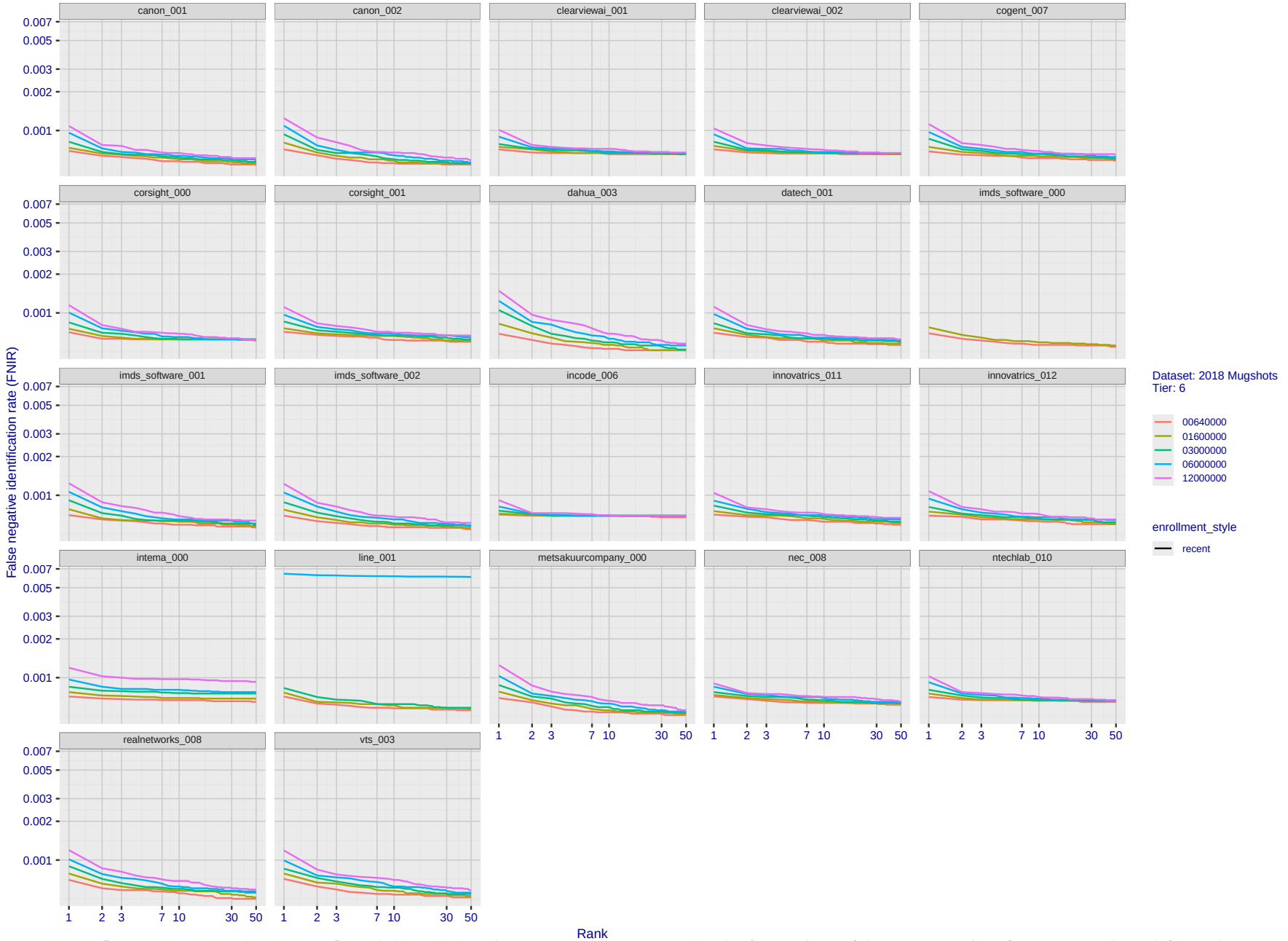

Figure 57: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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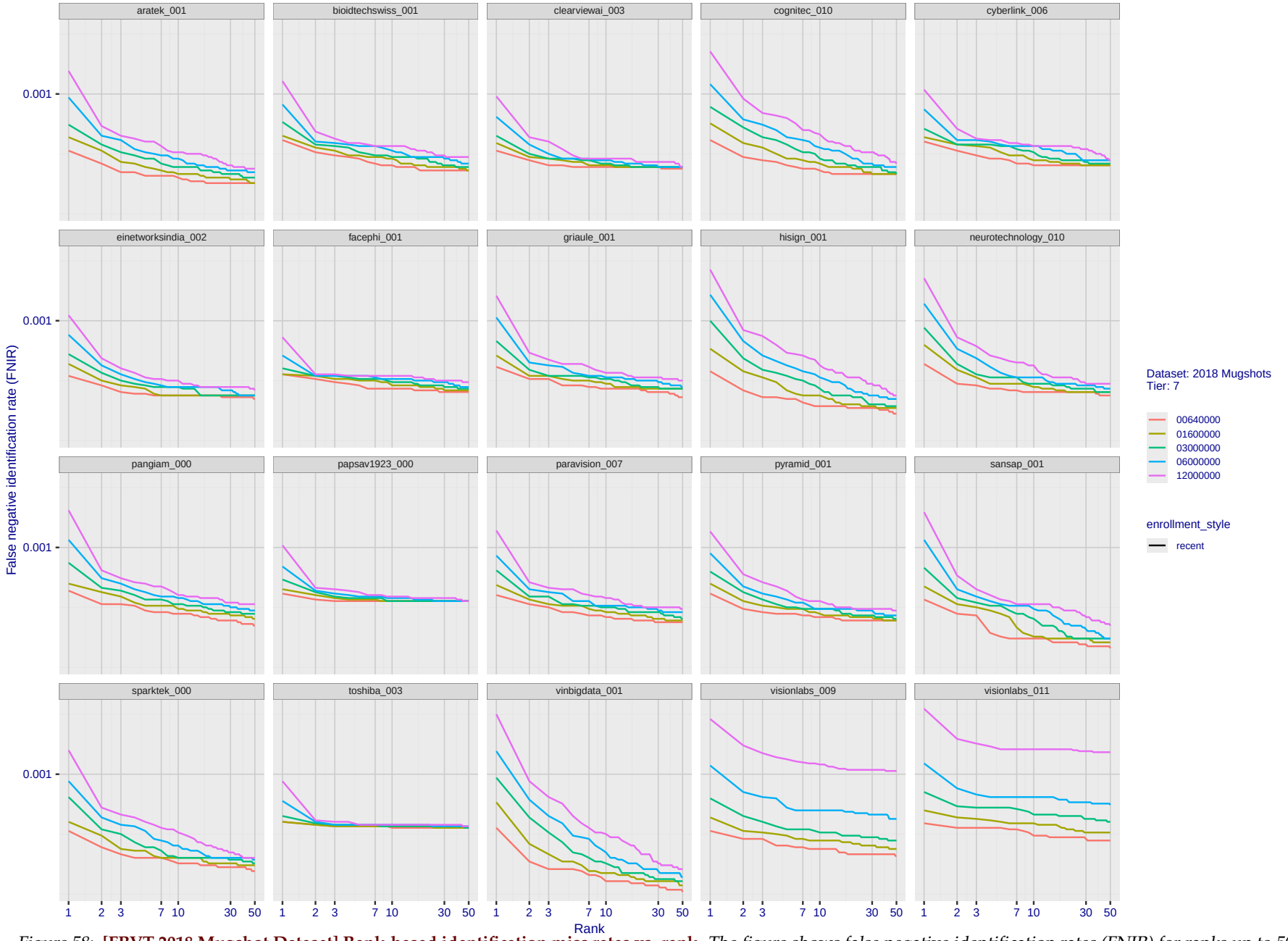

Figure 58: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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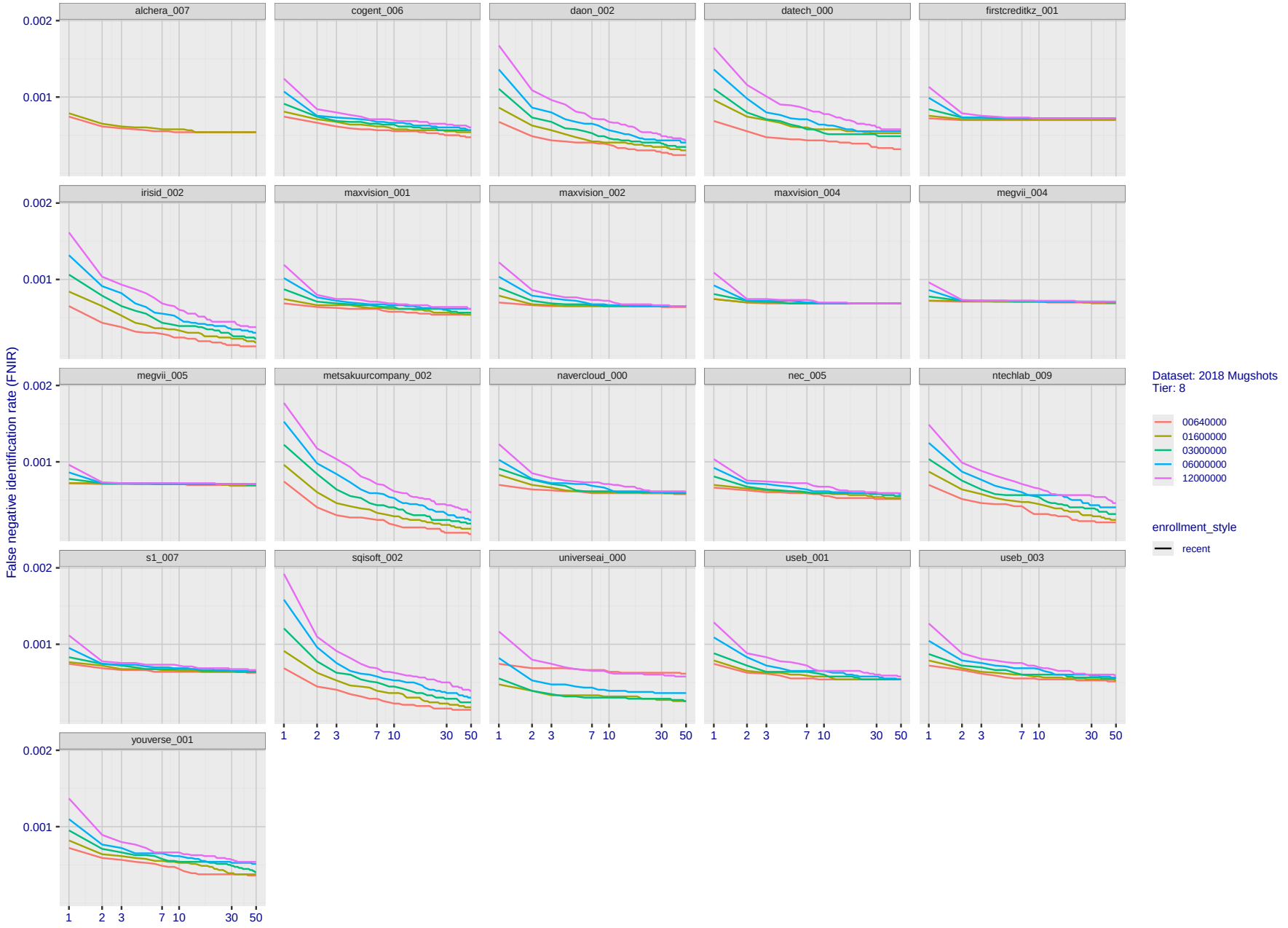

Figure 59: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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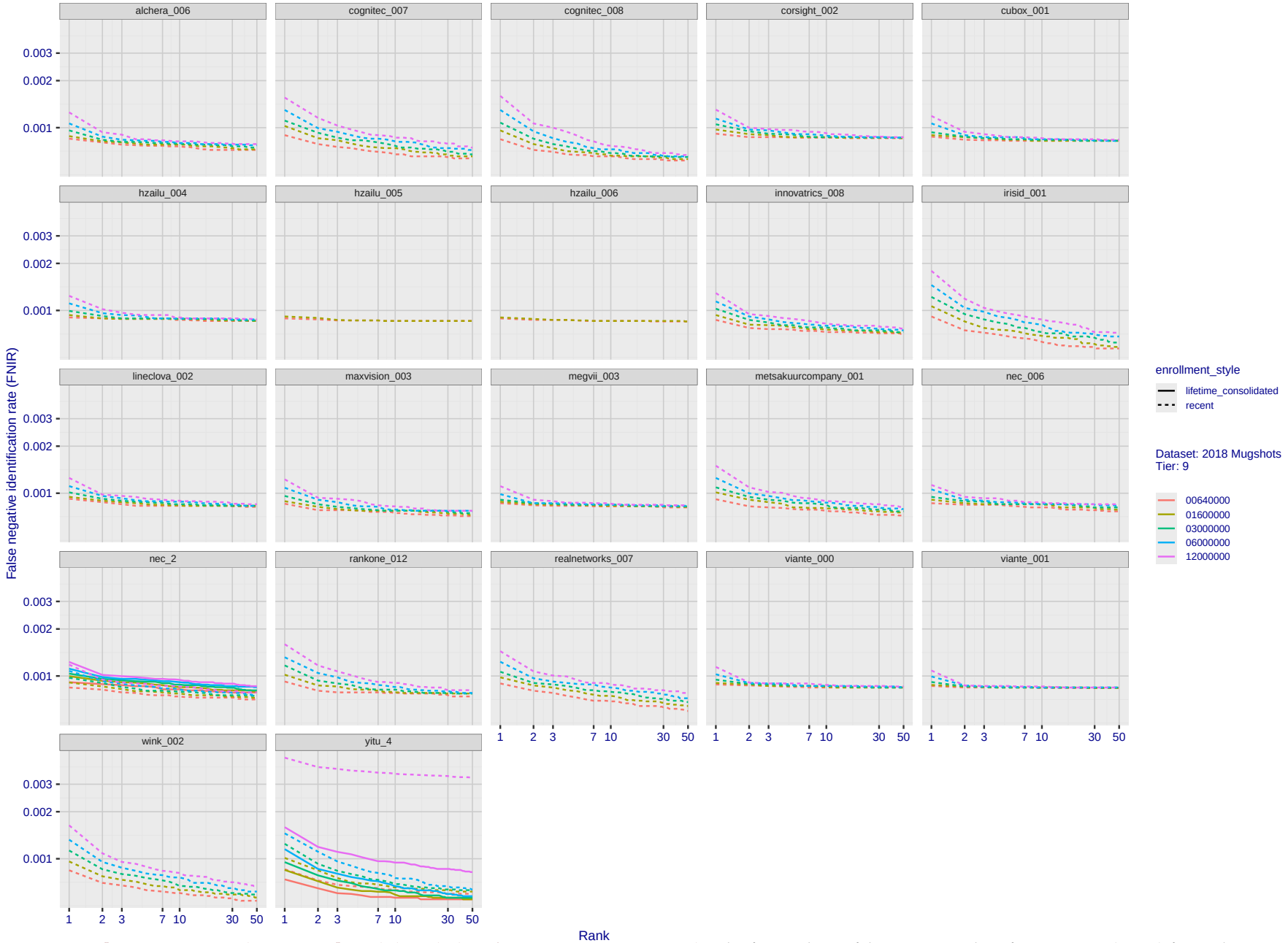

Figure 60: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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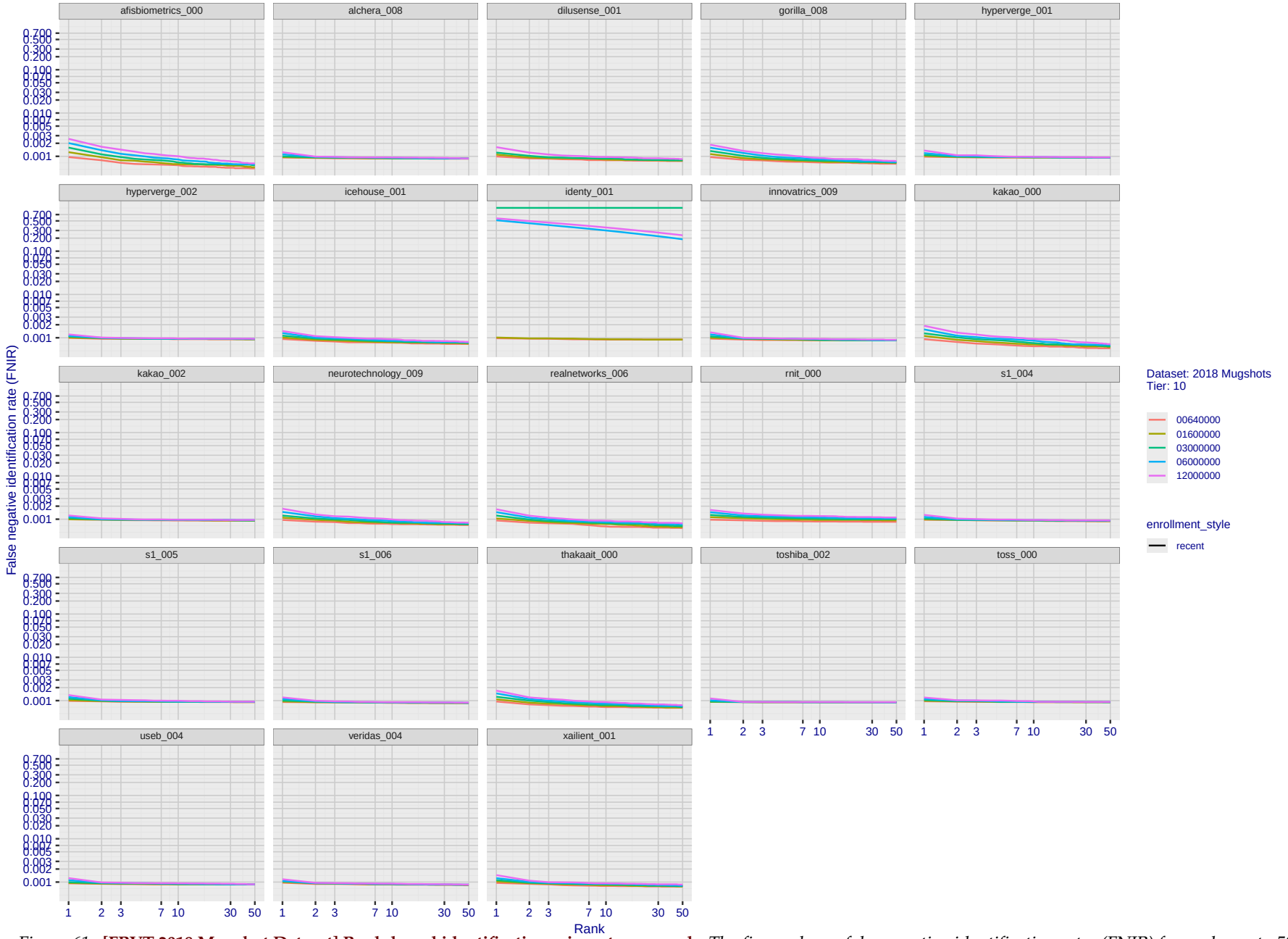

Figure 61: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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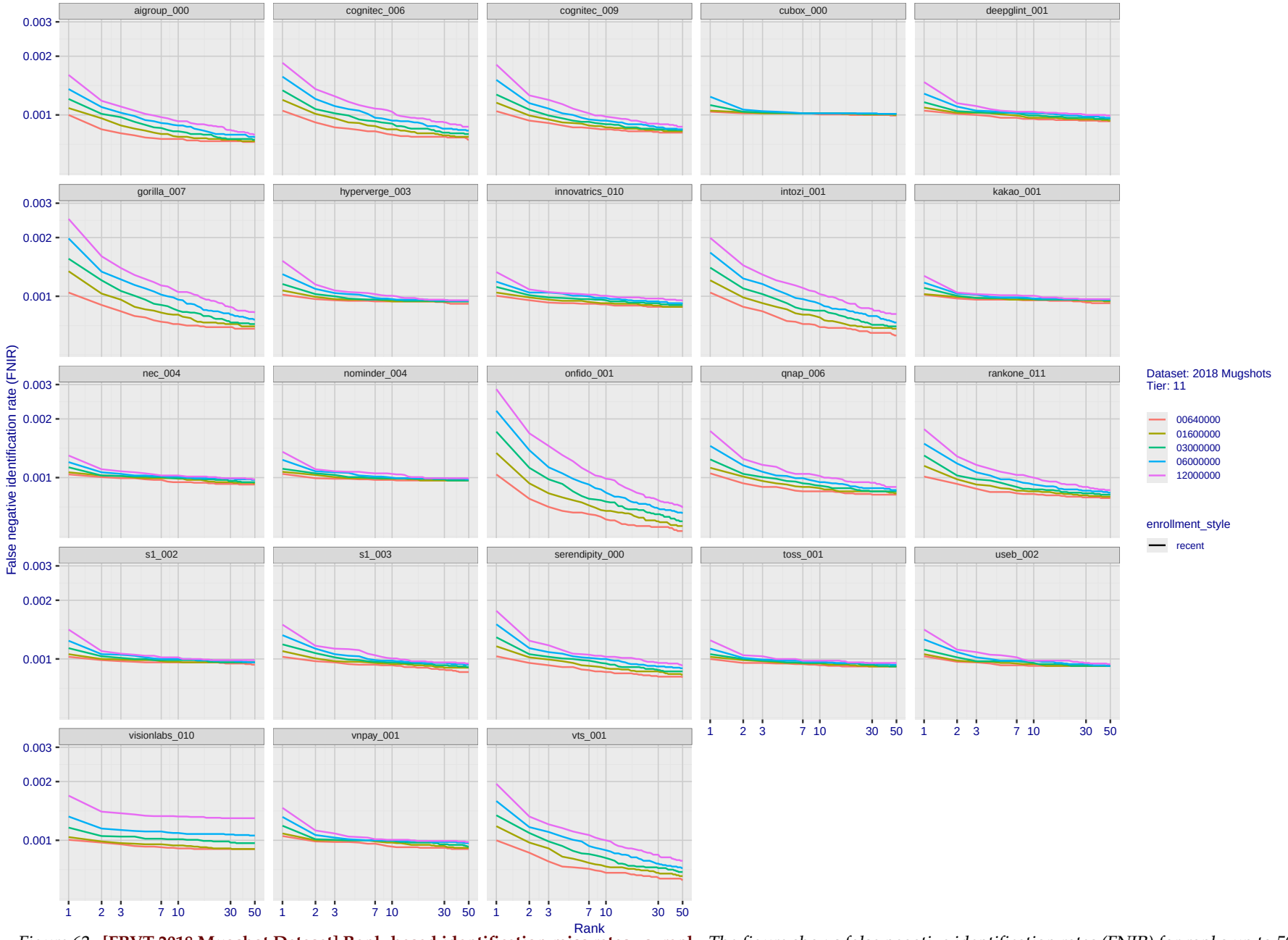

Figure 62: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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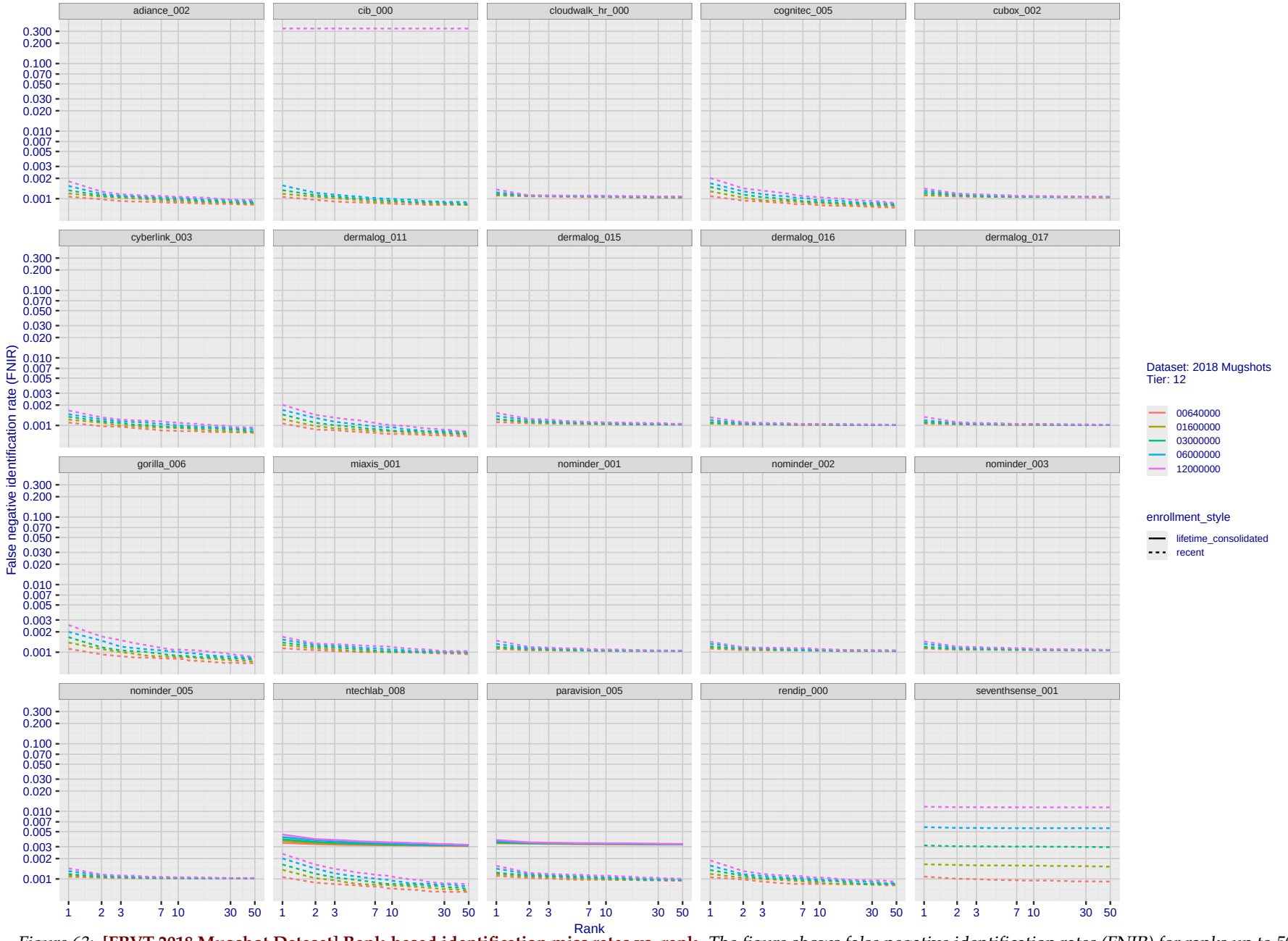

Figure 63: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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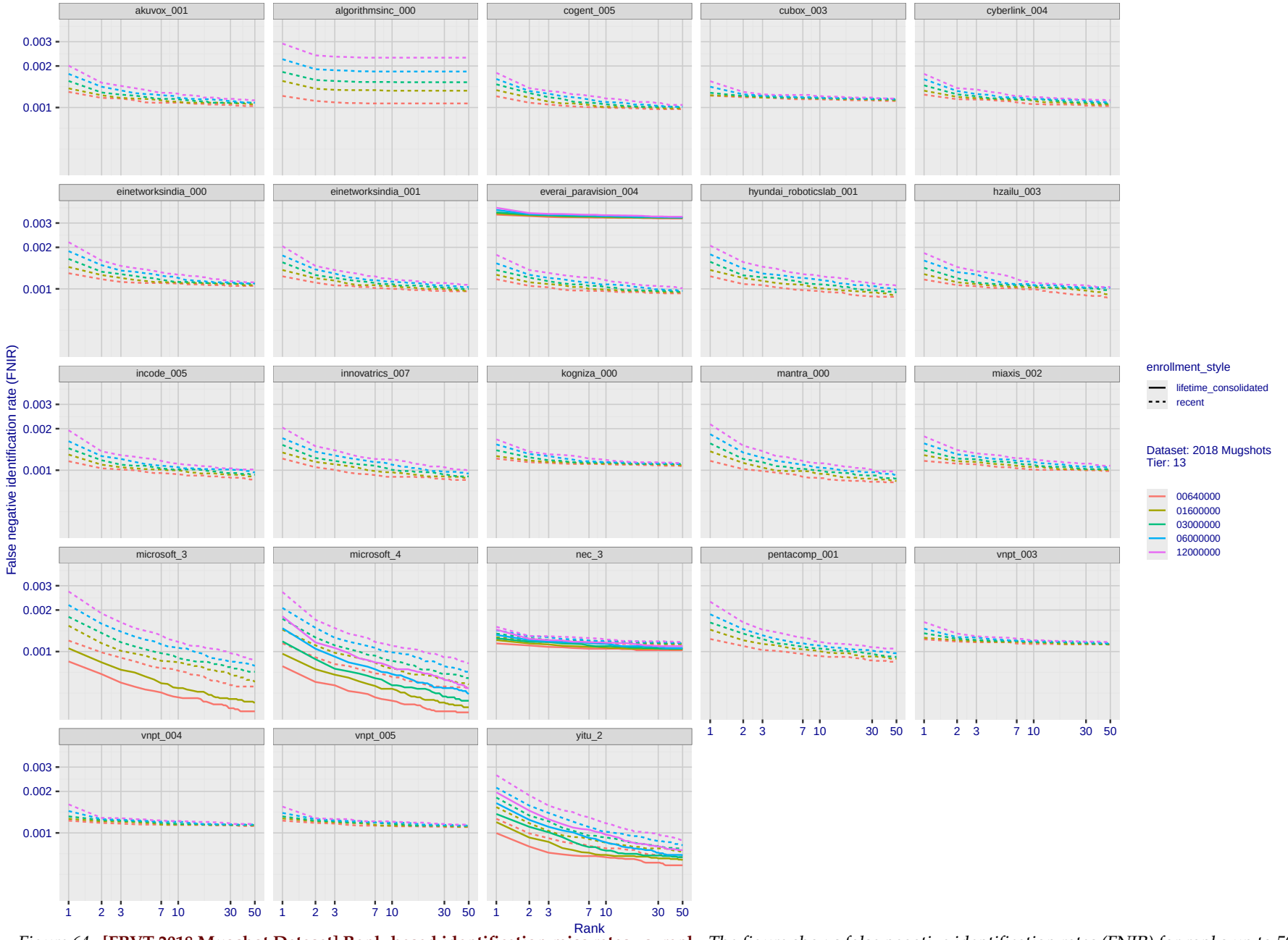

Figure 64: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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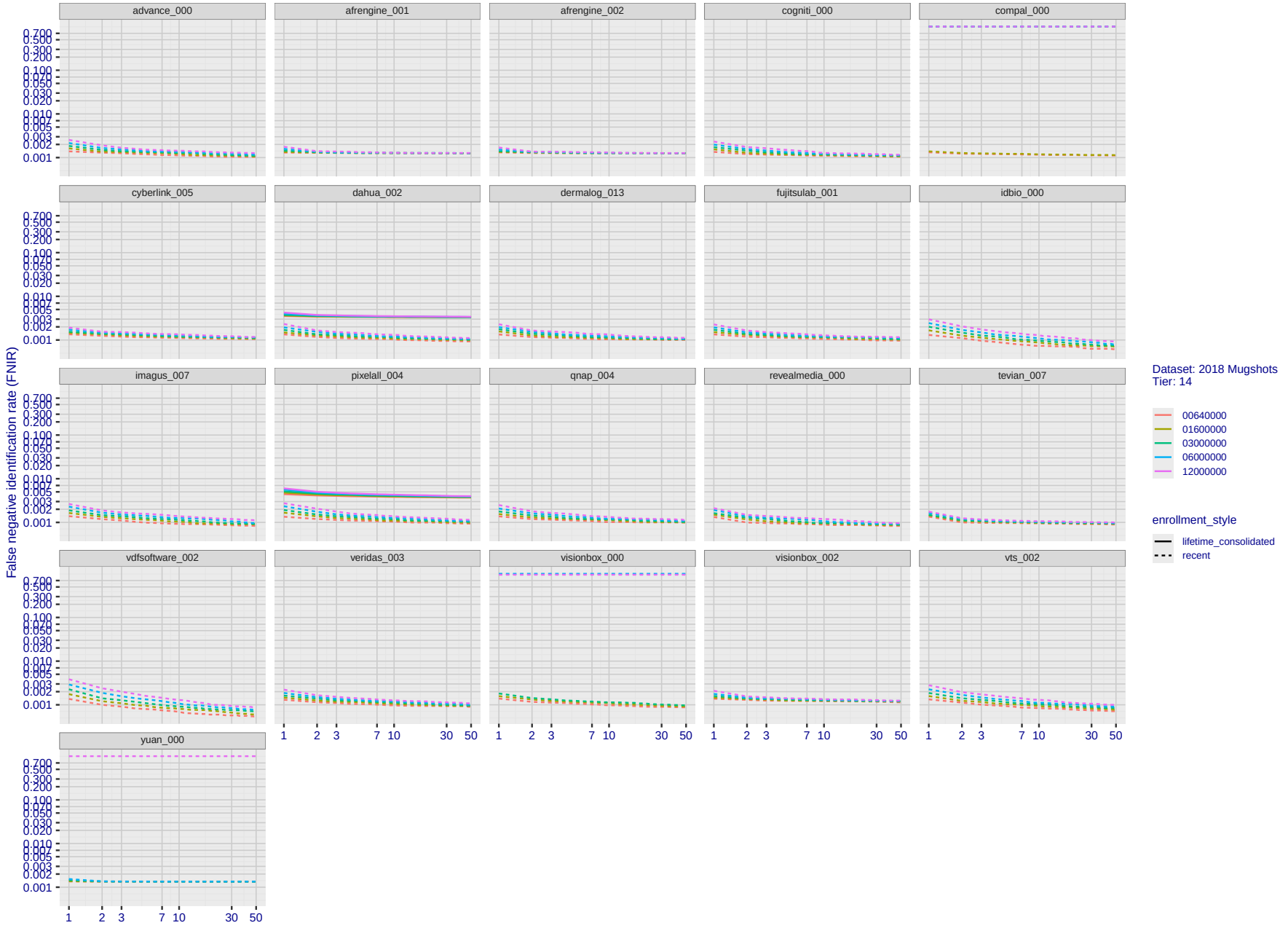

Figure 65: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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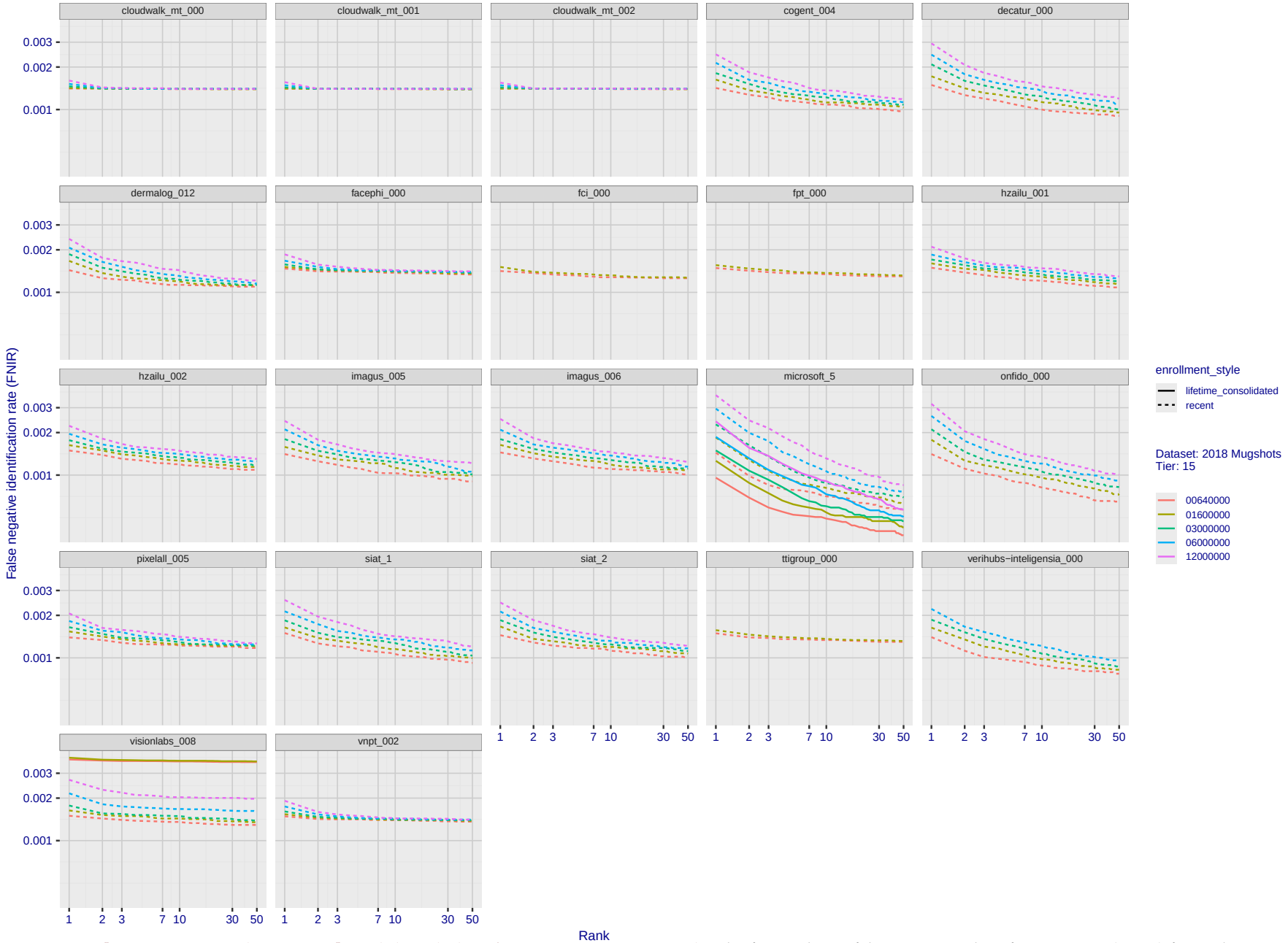

Figure 66: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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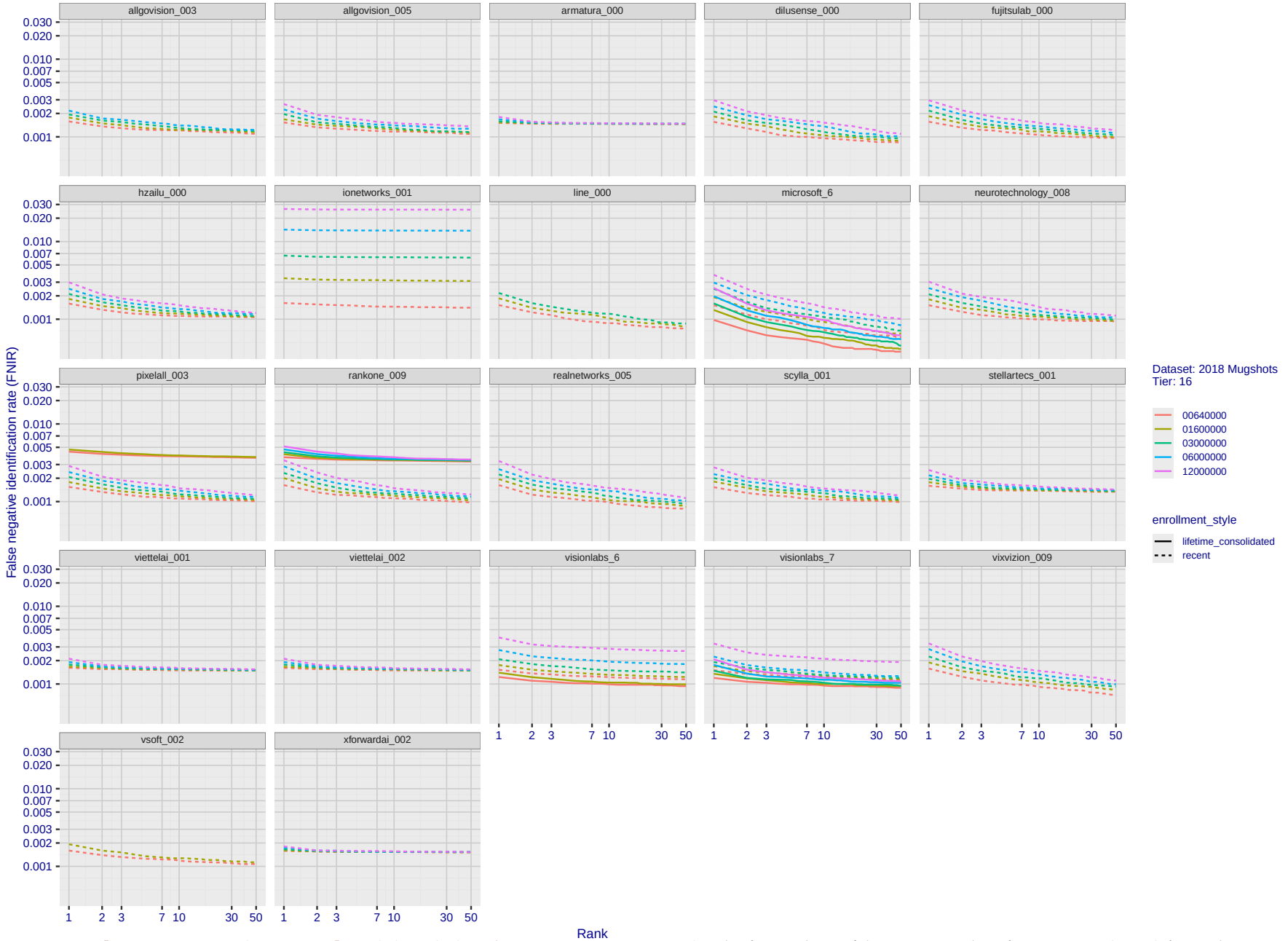

Figure 67: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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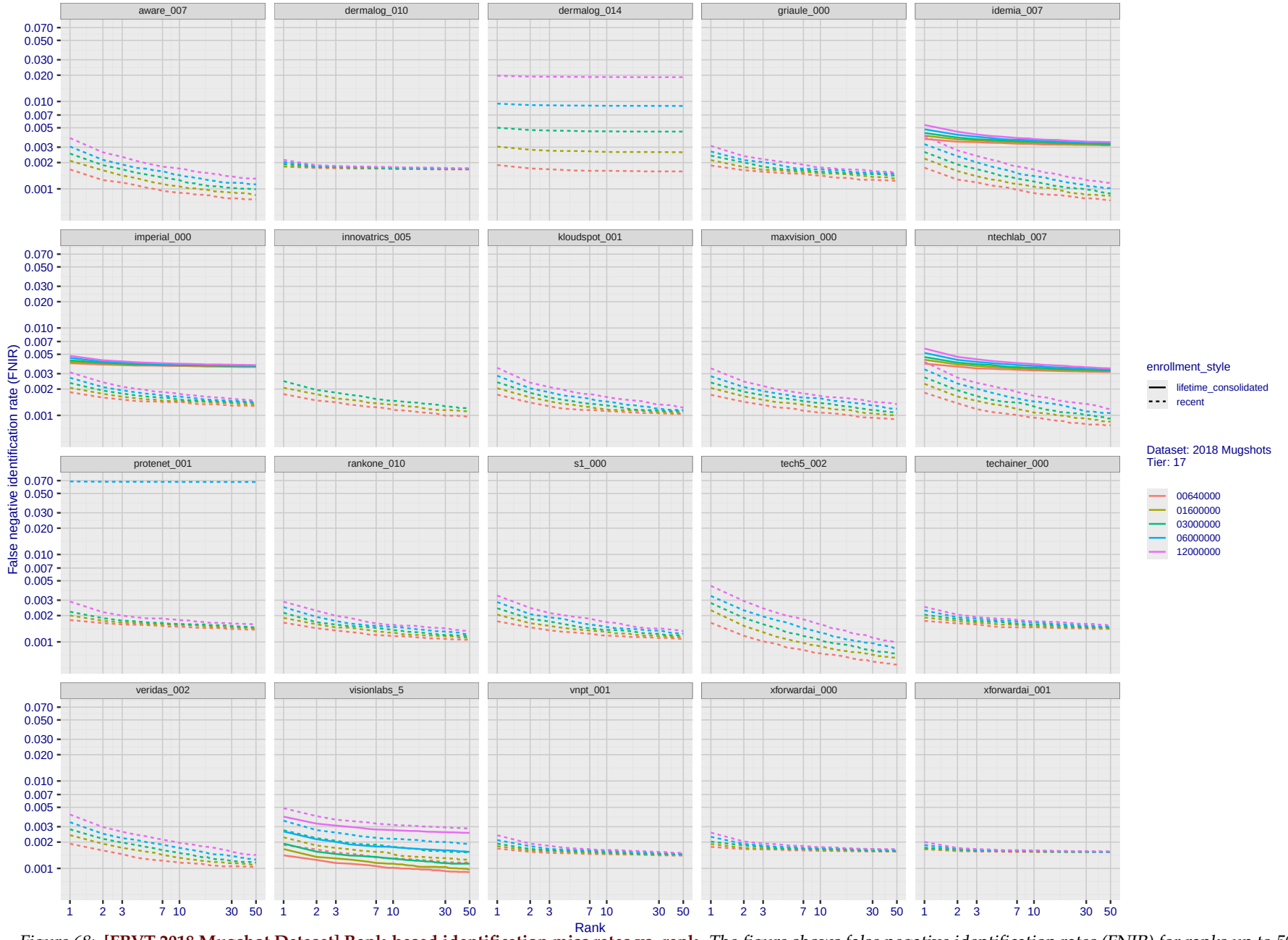

Figure 68: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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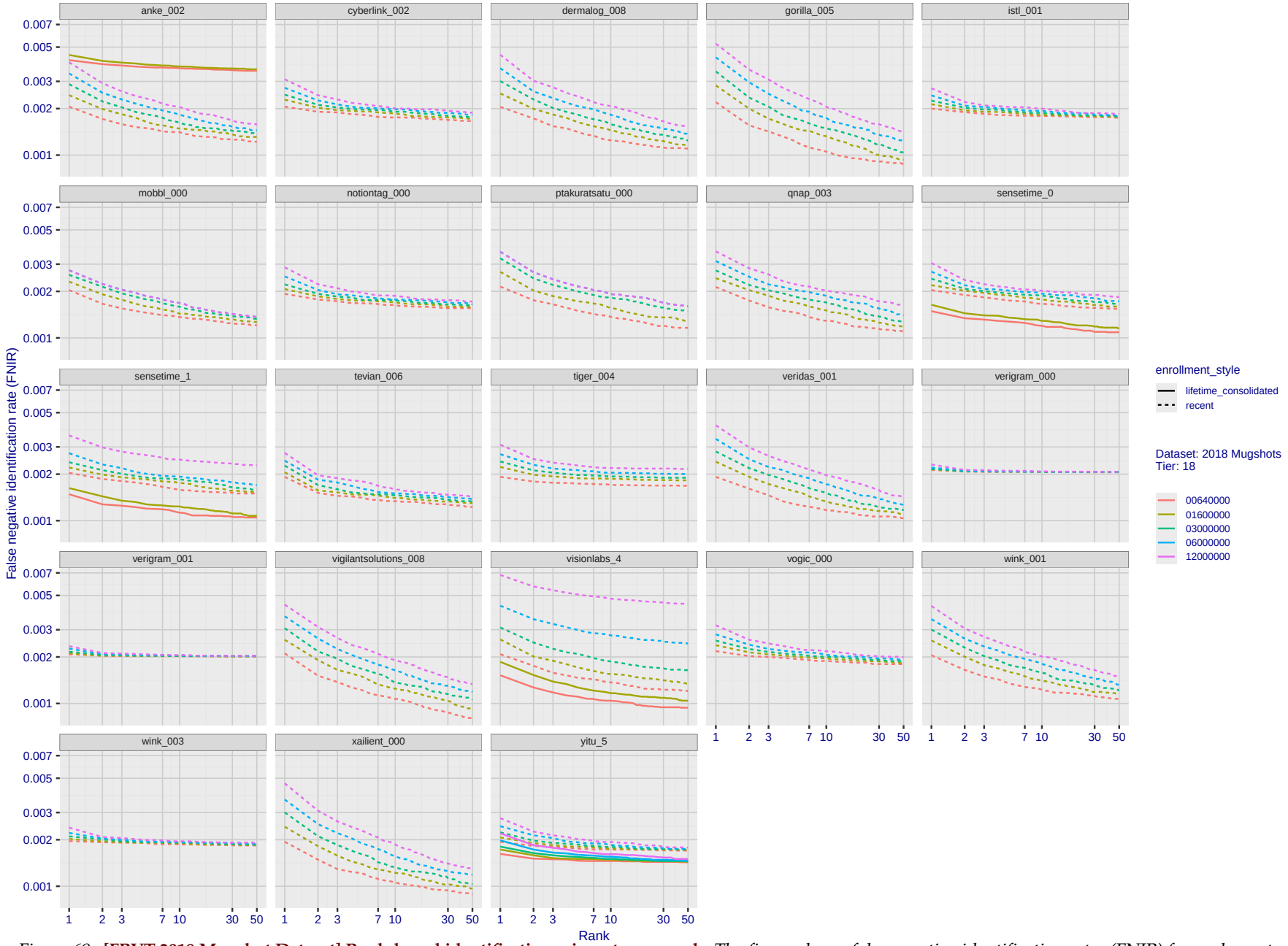

Figure 69: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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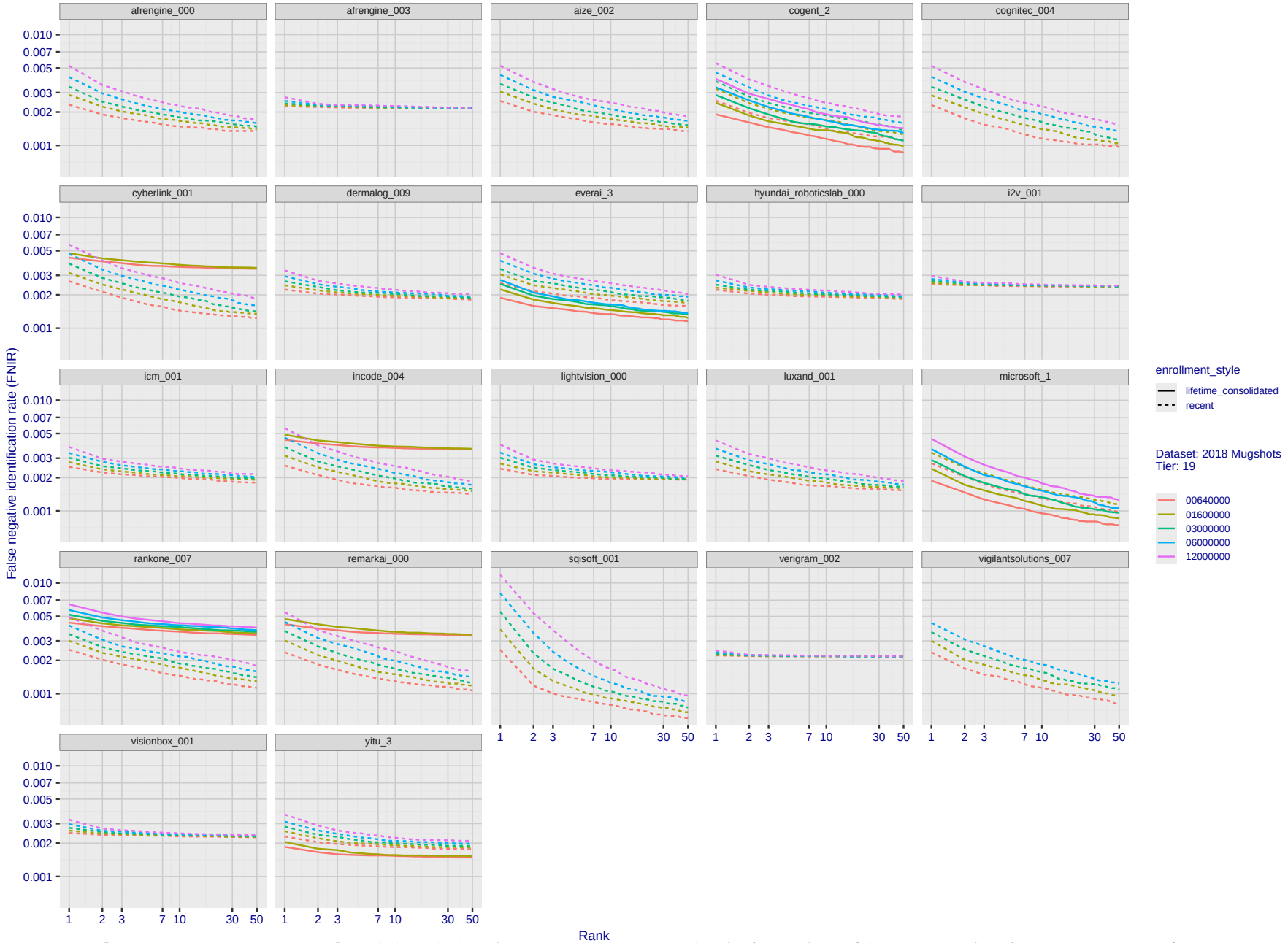

Figure 70: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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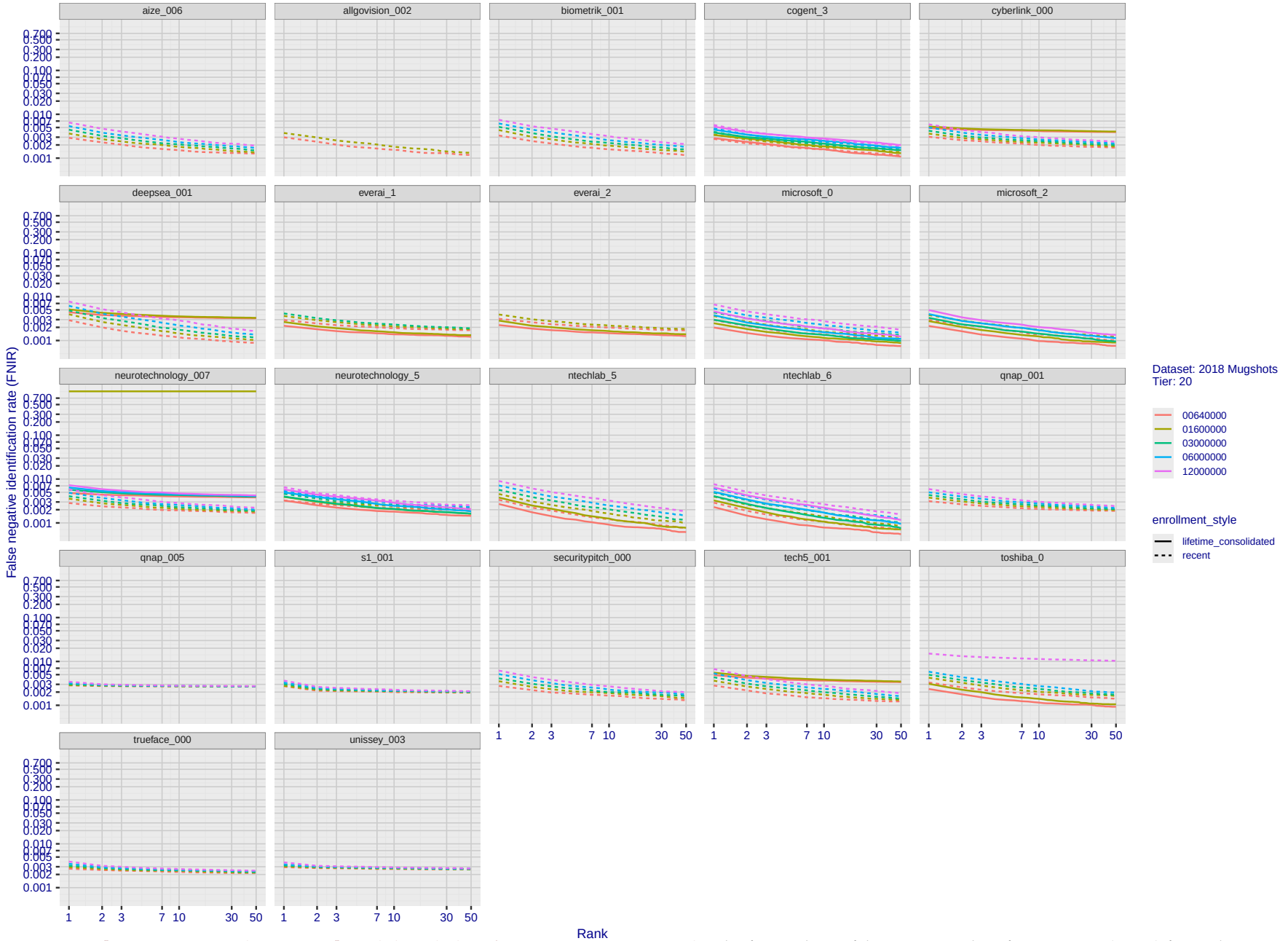

Figure 71: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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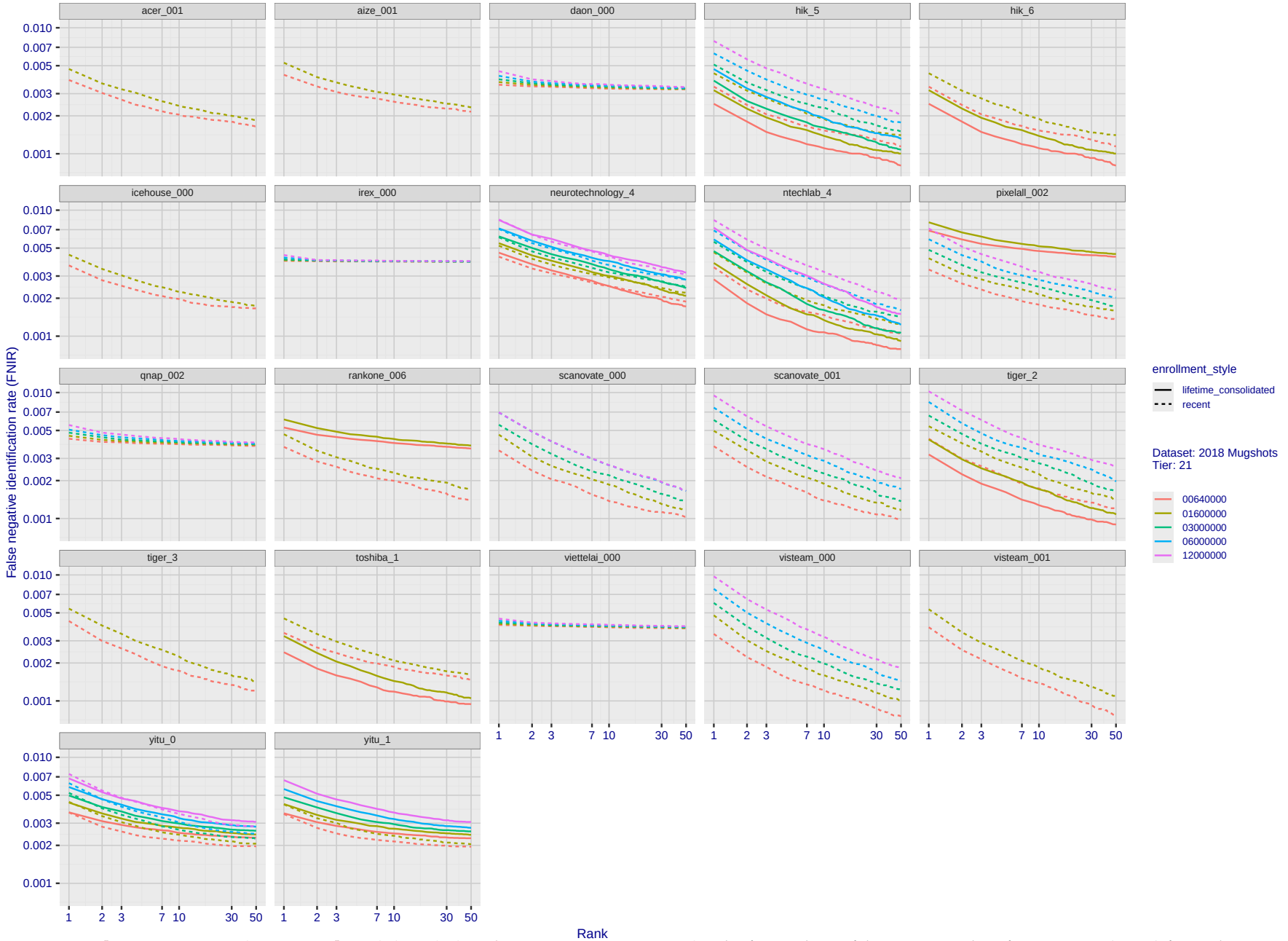

Figure 72: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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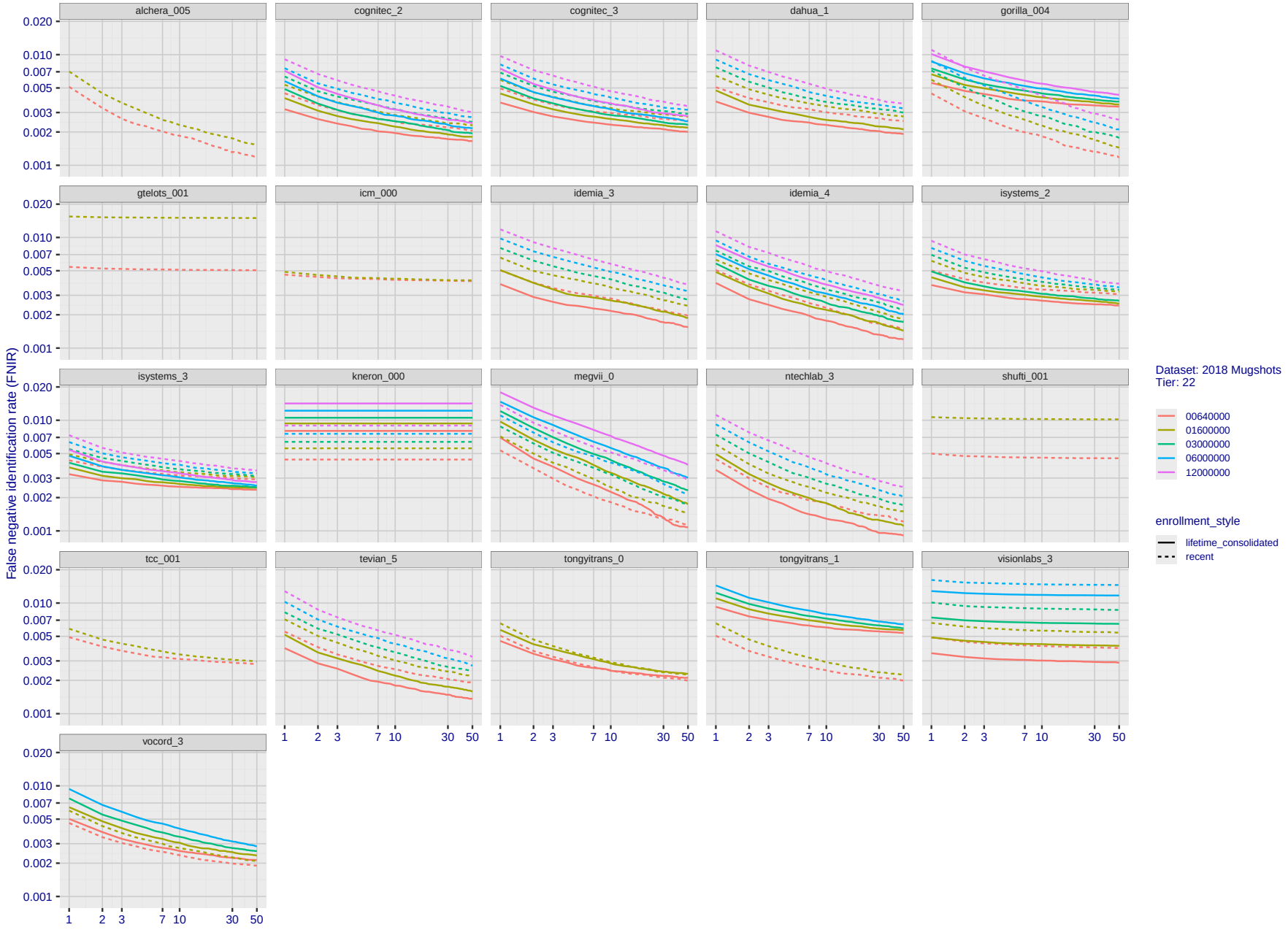

Figure 73: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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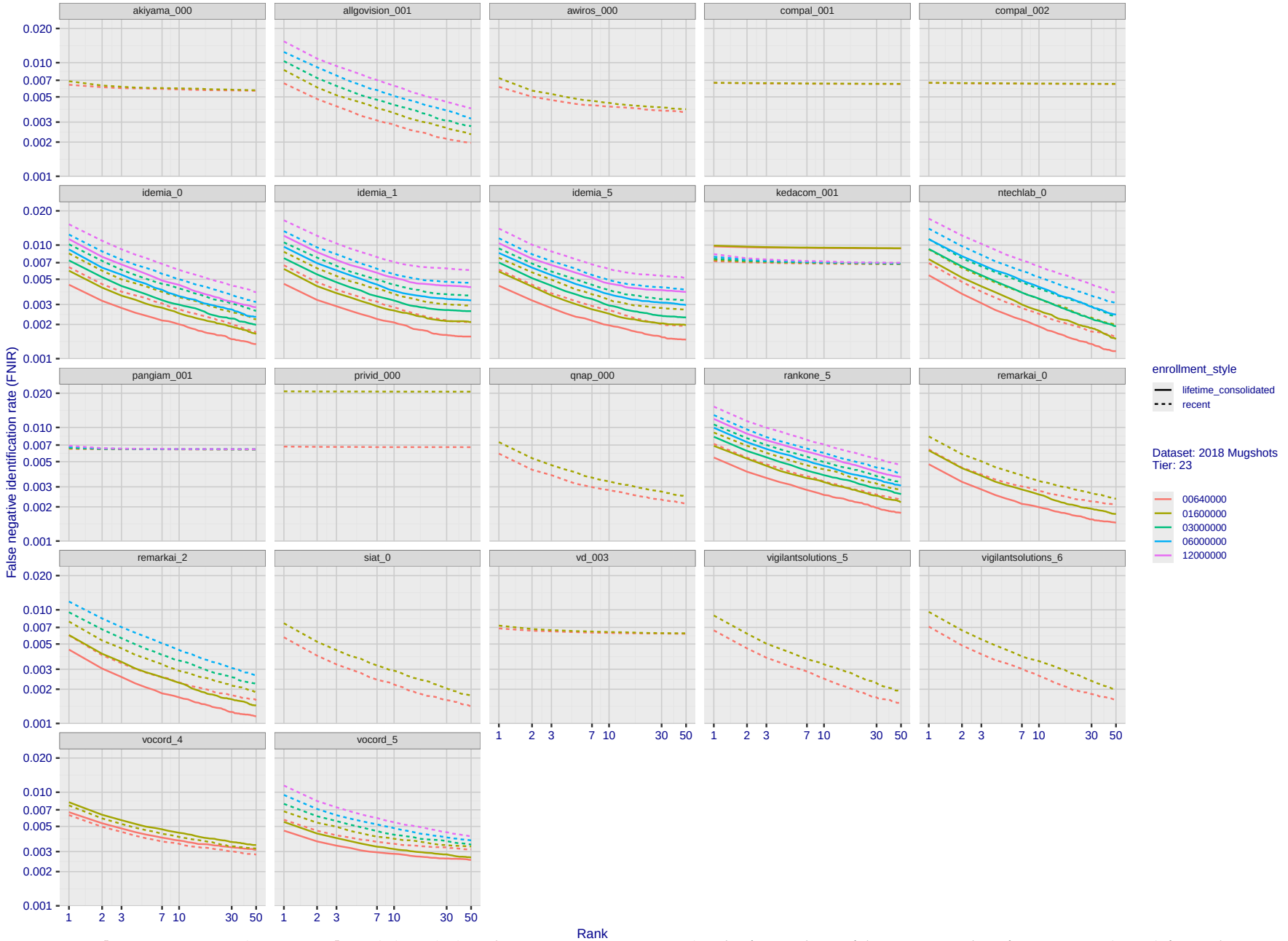

Figure 74: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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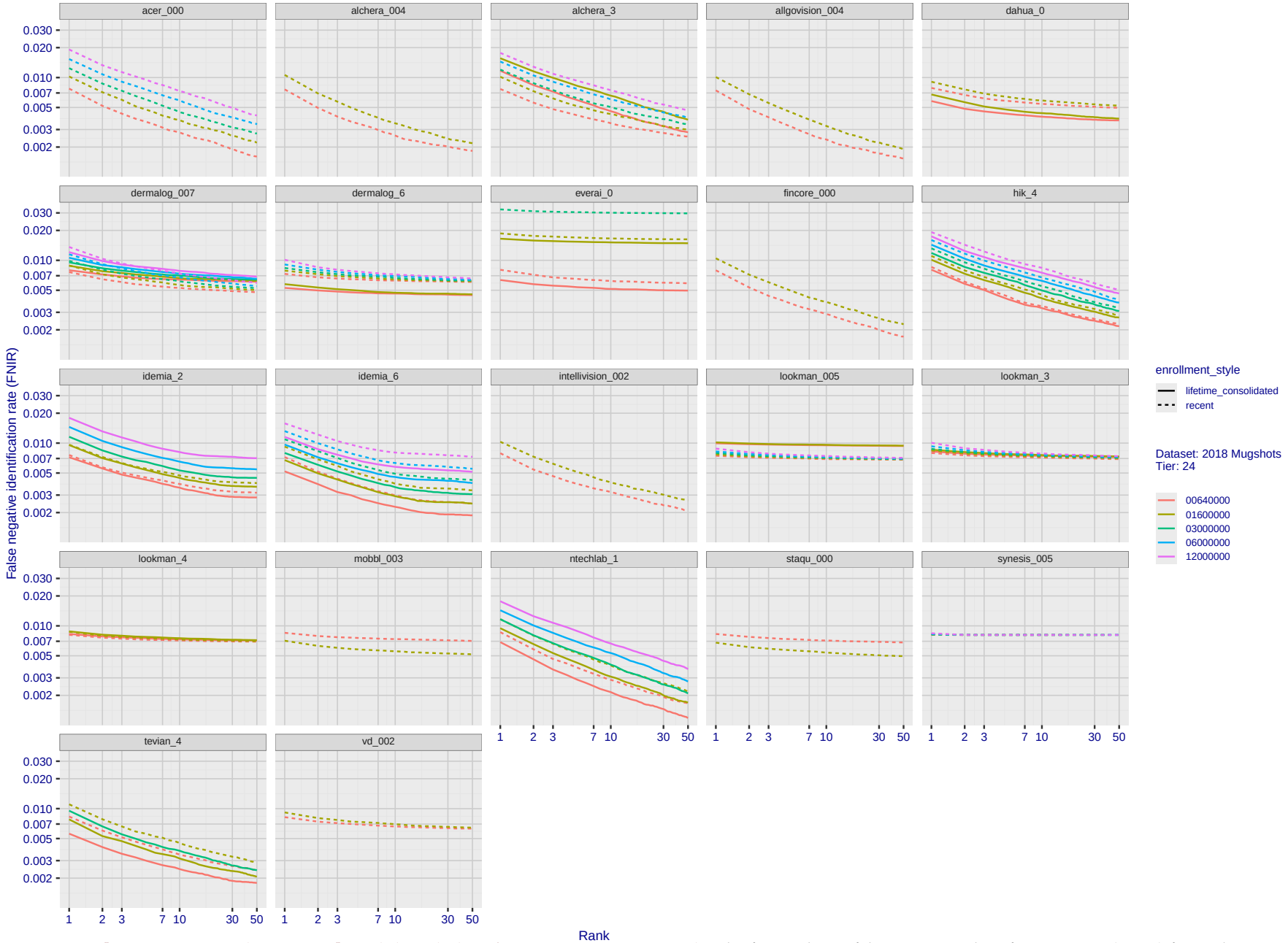

Figure 75: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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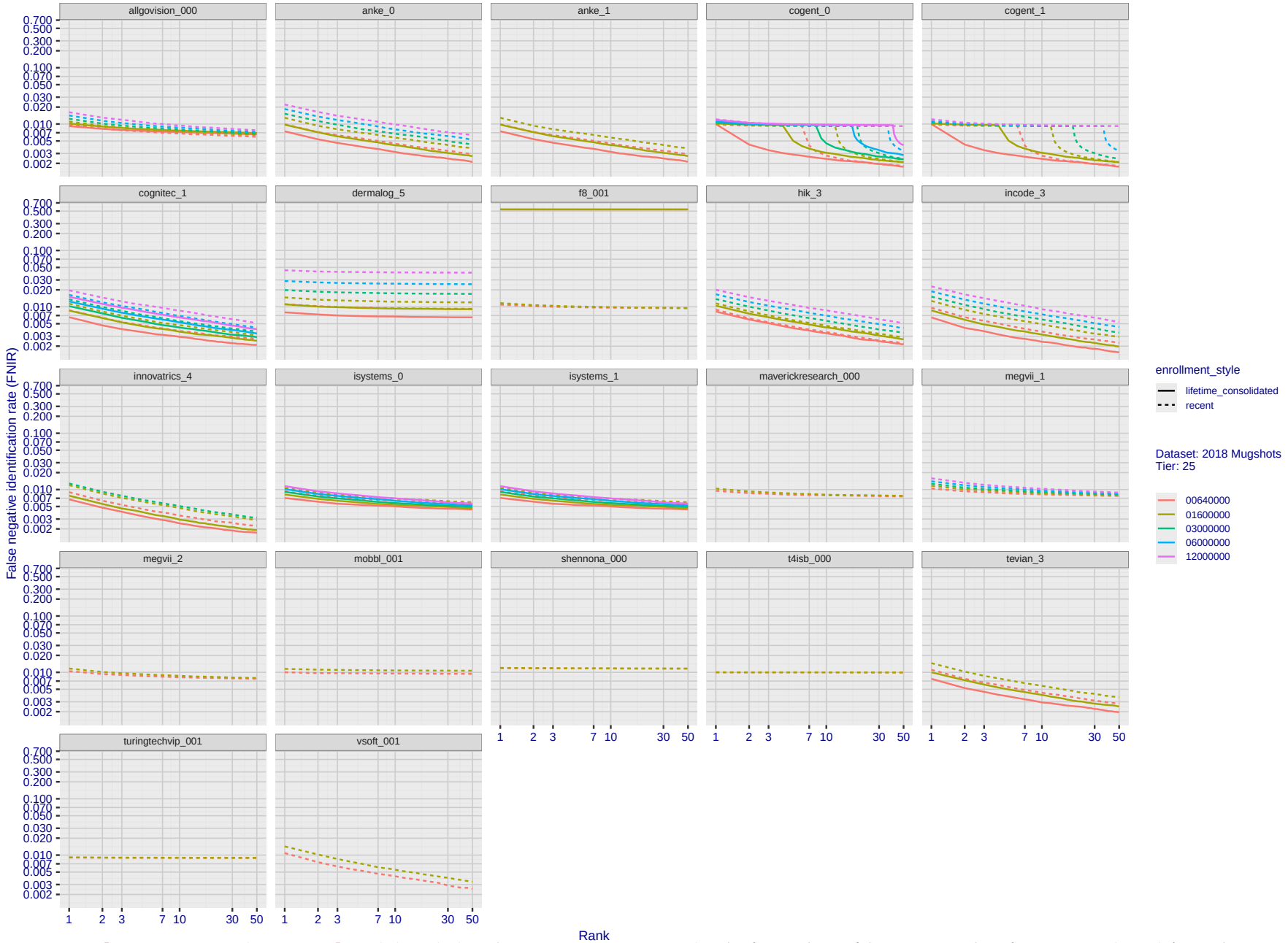

Figure 76: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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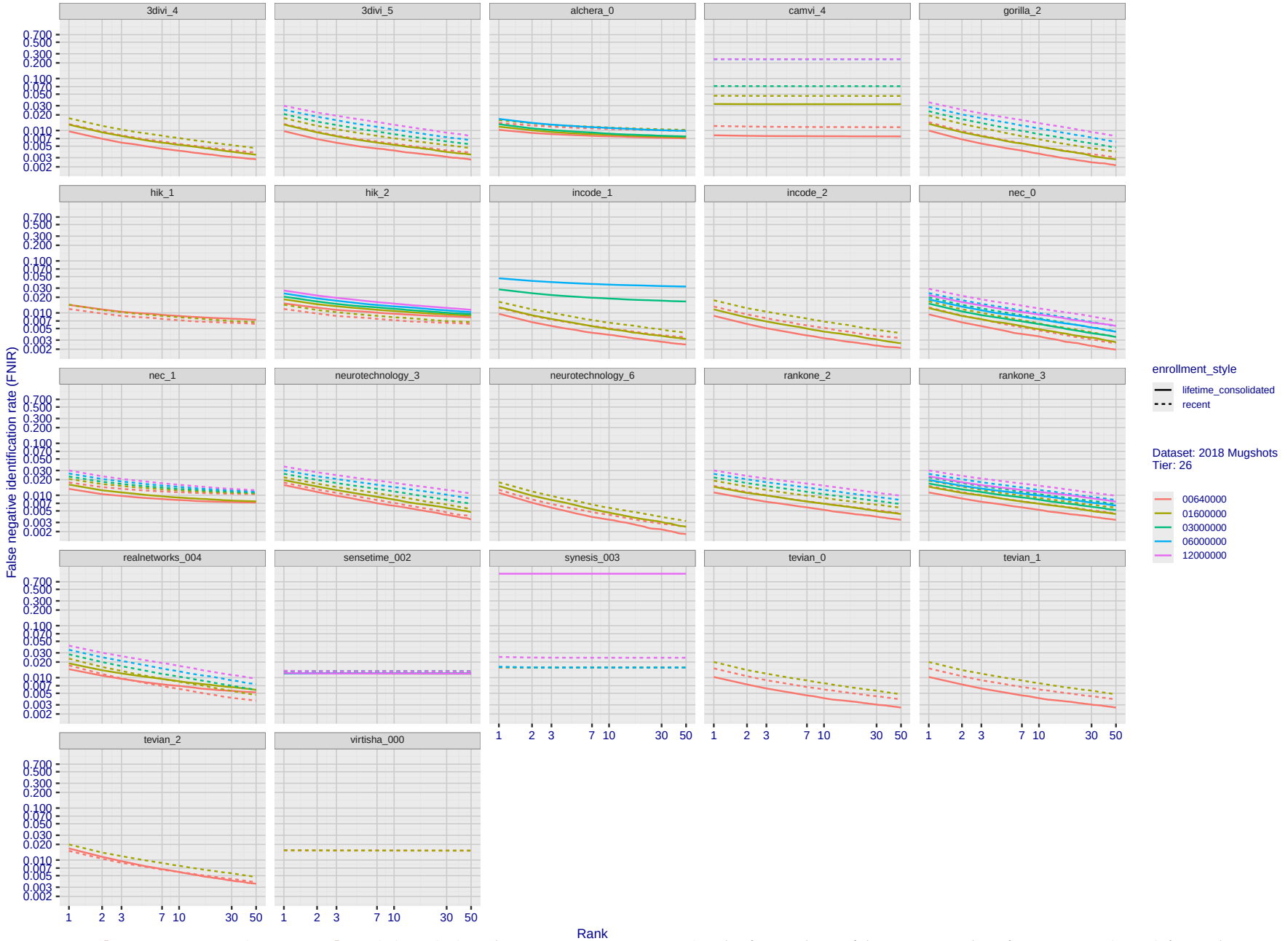

Figure 77: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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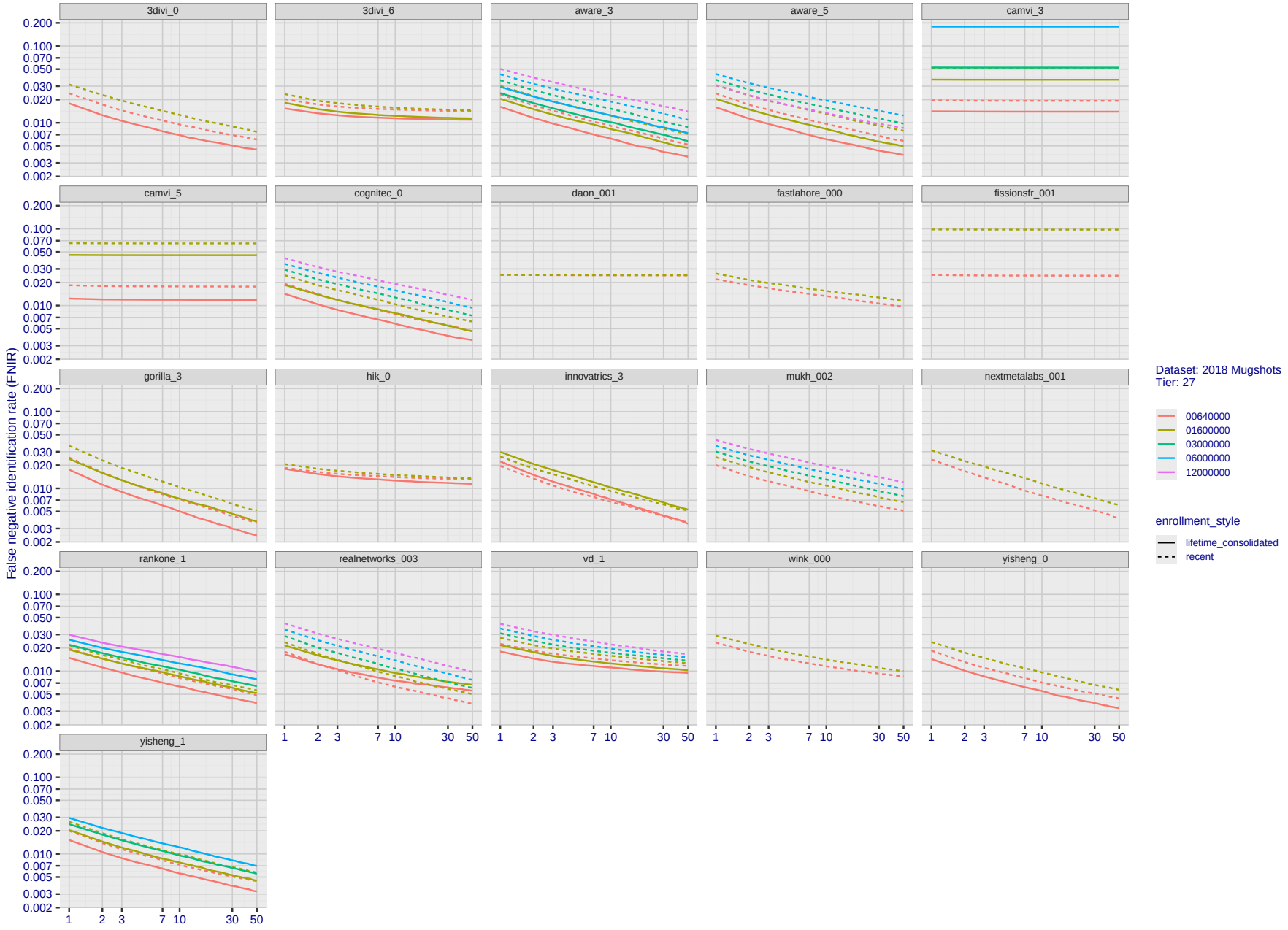

Figure 78: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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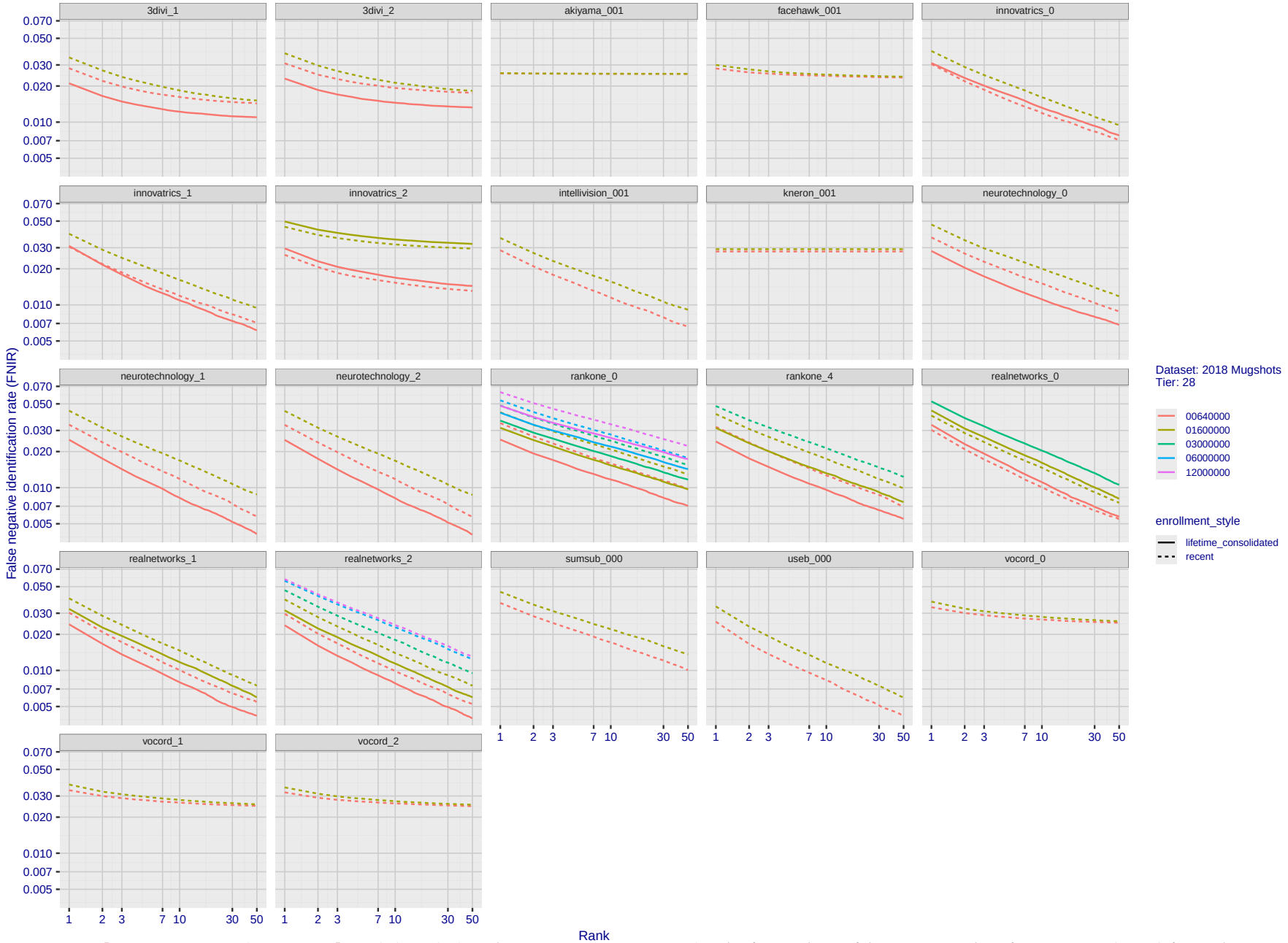

Figure 79: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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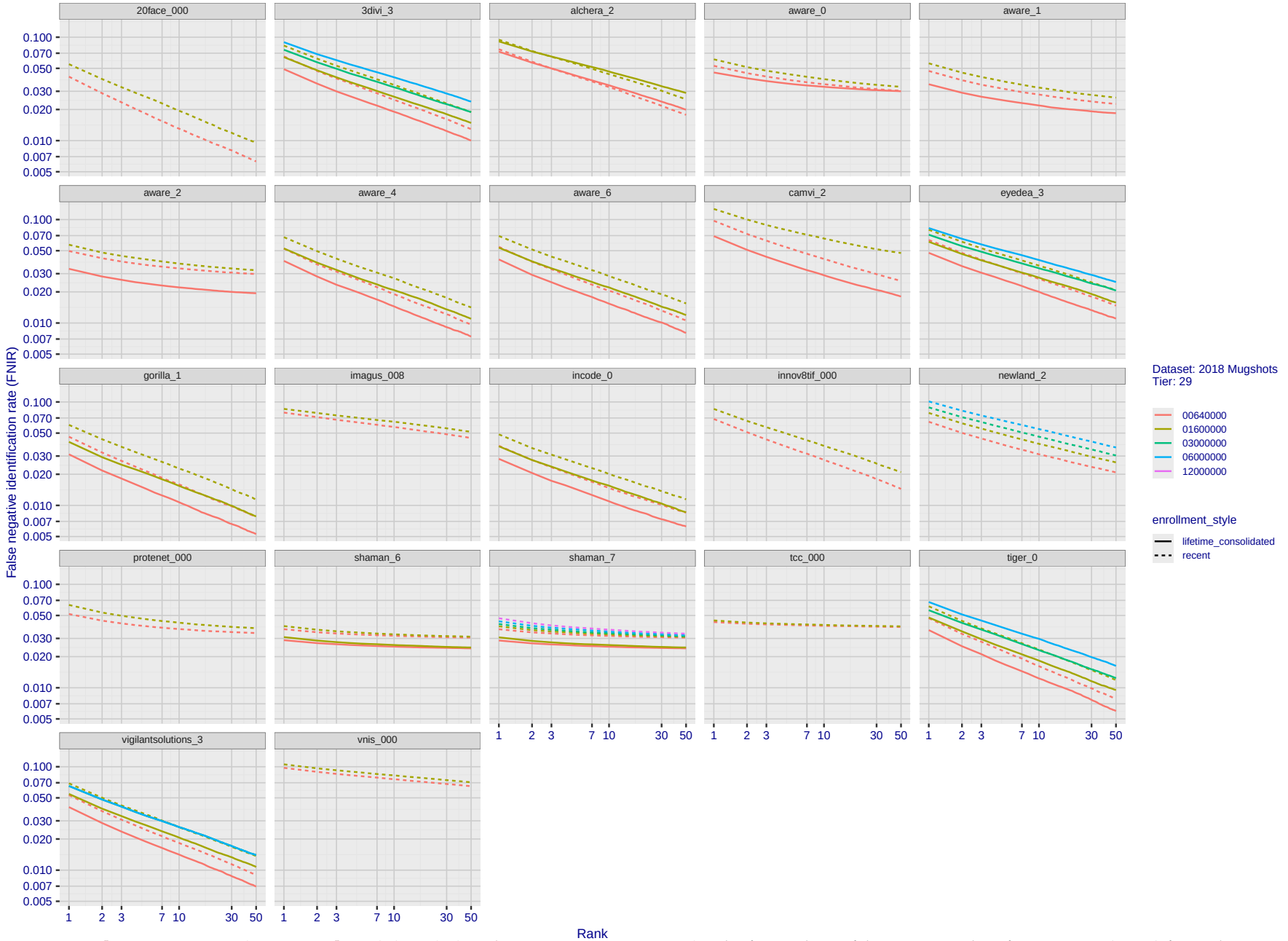

Figure 80: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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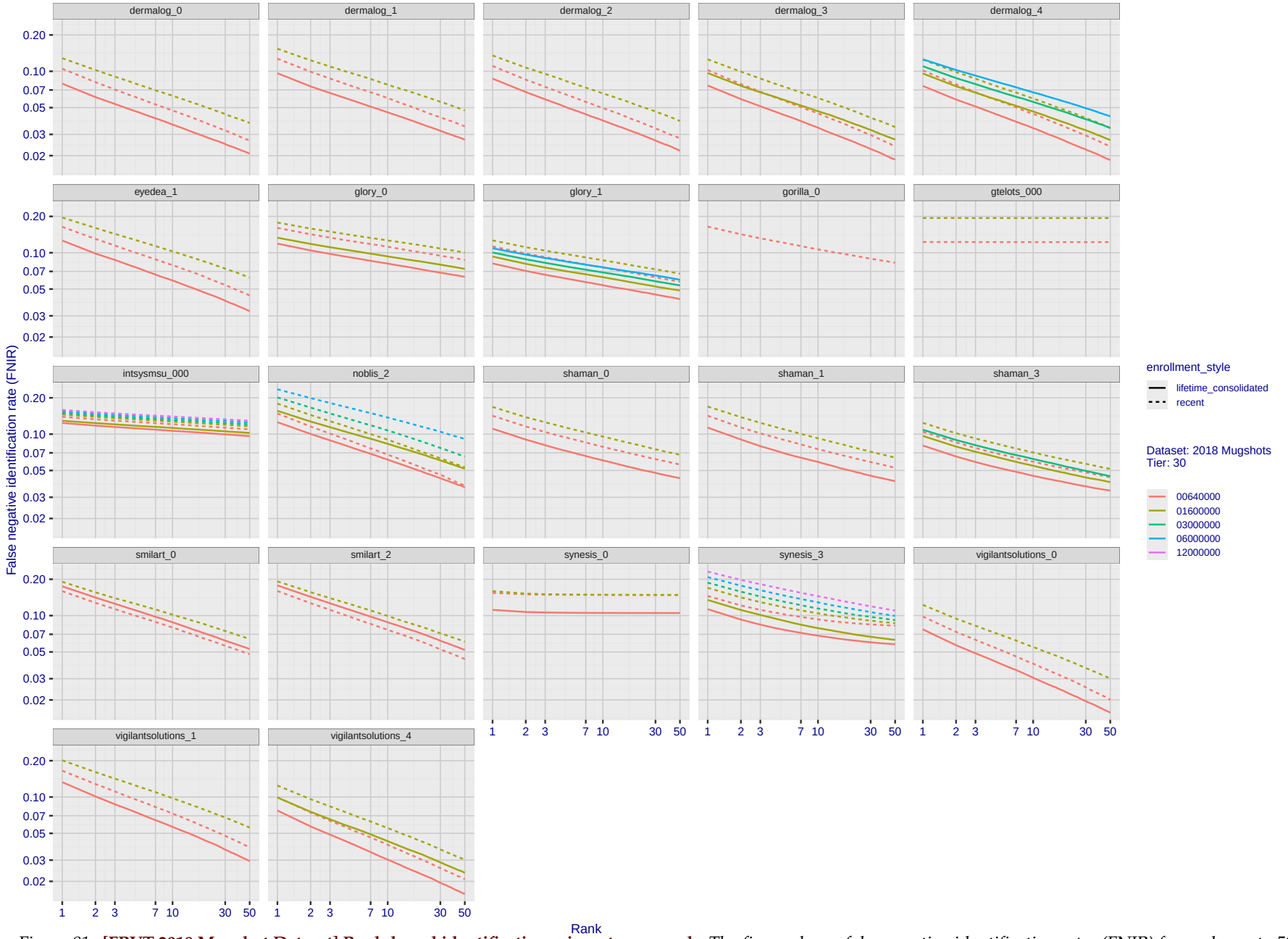

Figure 81: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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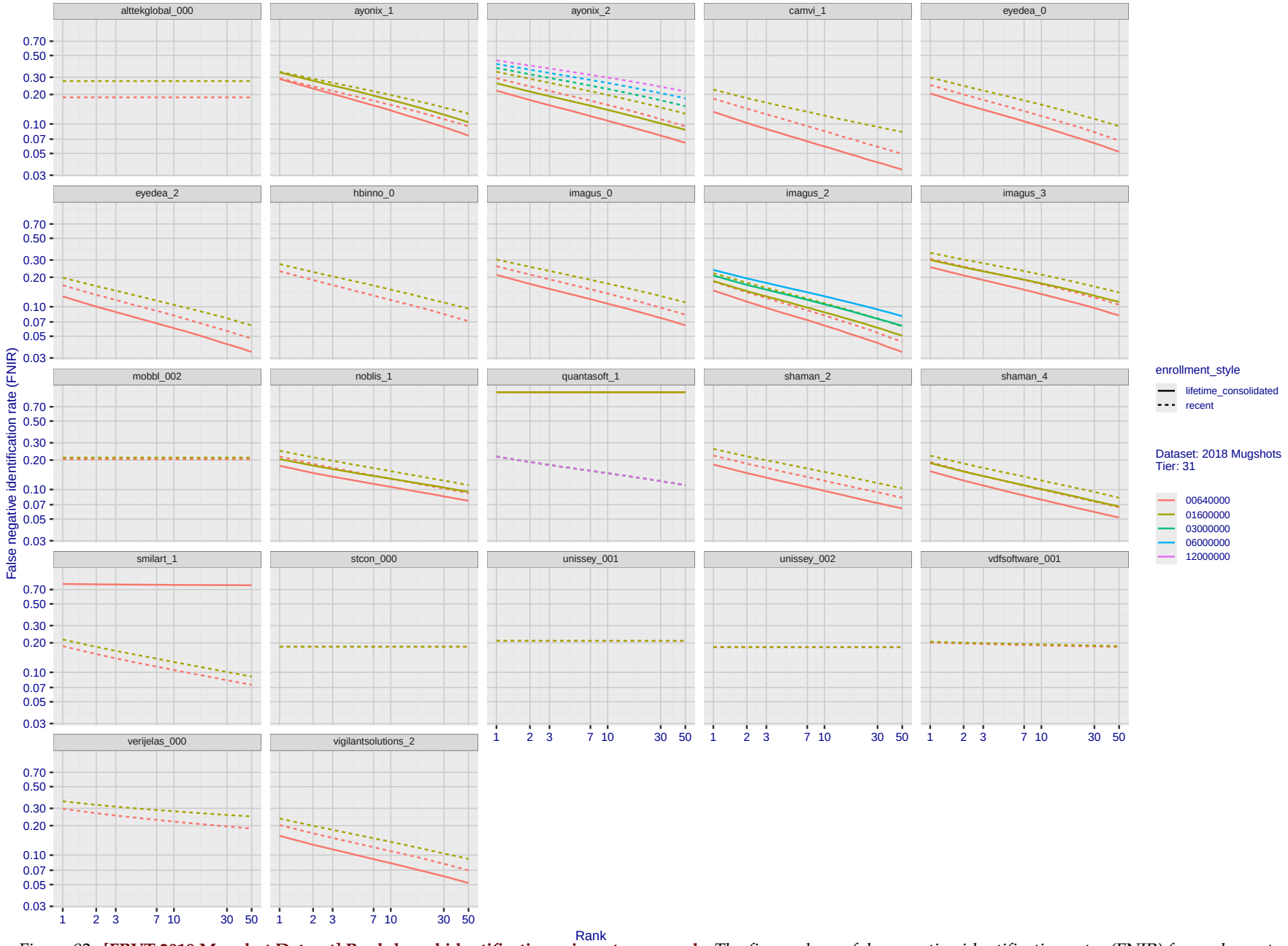

Figure 82: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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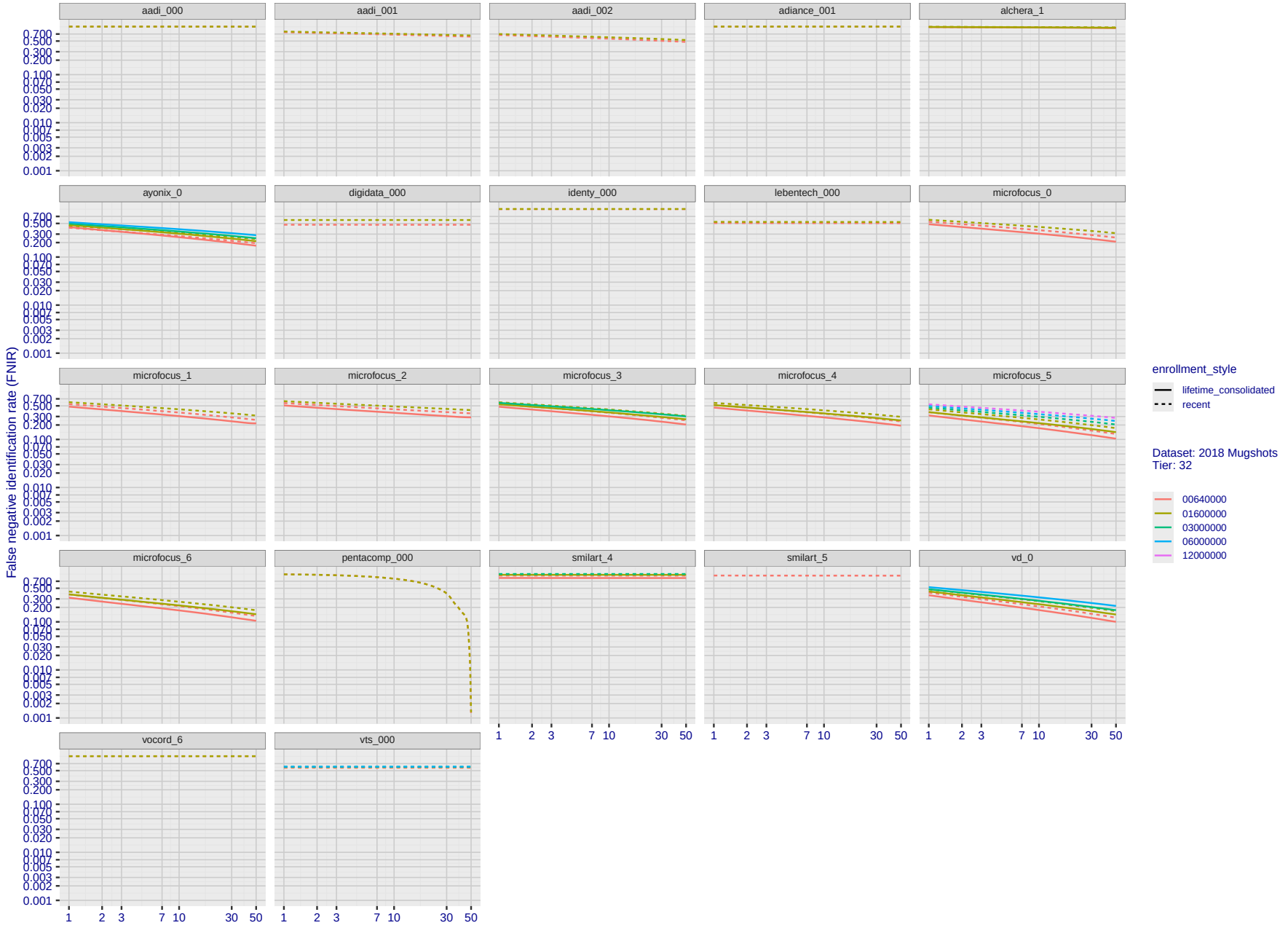

Figure 83: **[FRVT-2018 Mugshot Dataset] Rank-based identification miss rates vs. rank**. *The figure shows false negative identification rates (FNIR) for ranks up to 50. This metric is appropriate to investigational applications where human reviewers will adjudicate sorted candidate lists. Note that with threshold set to zero, FPIR = 1, i.e. any search without an enrolled mate will return non-mated candidates. Results are sorted and reported into tiers for clarity, with the tiering criteria being rank 1 hit rate on a gallery size of N = 640 000 subjects.*

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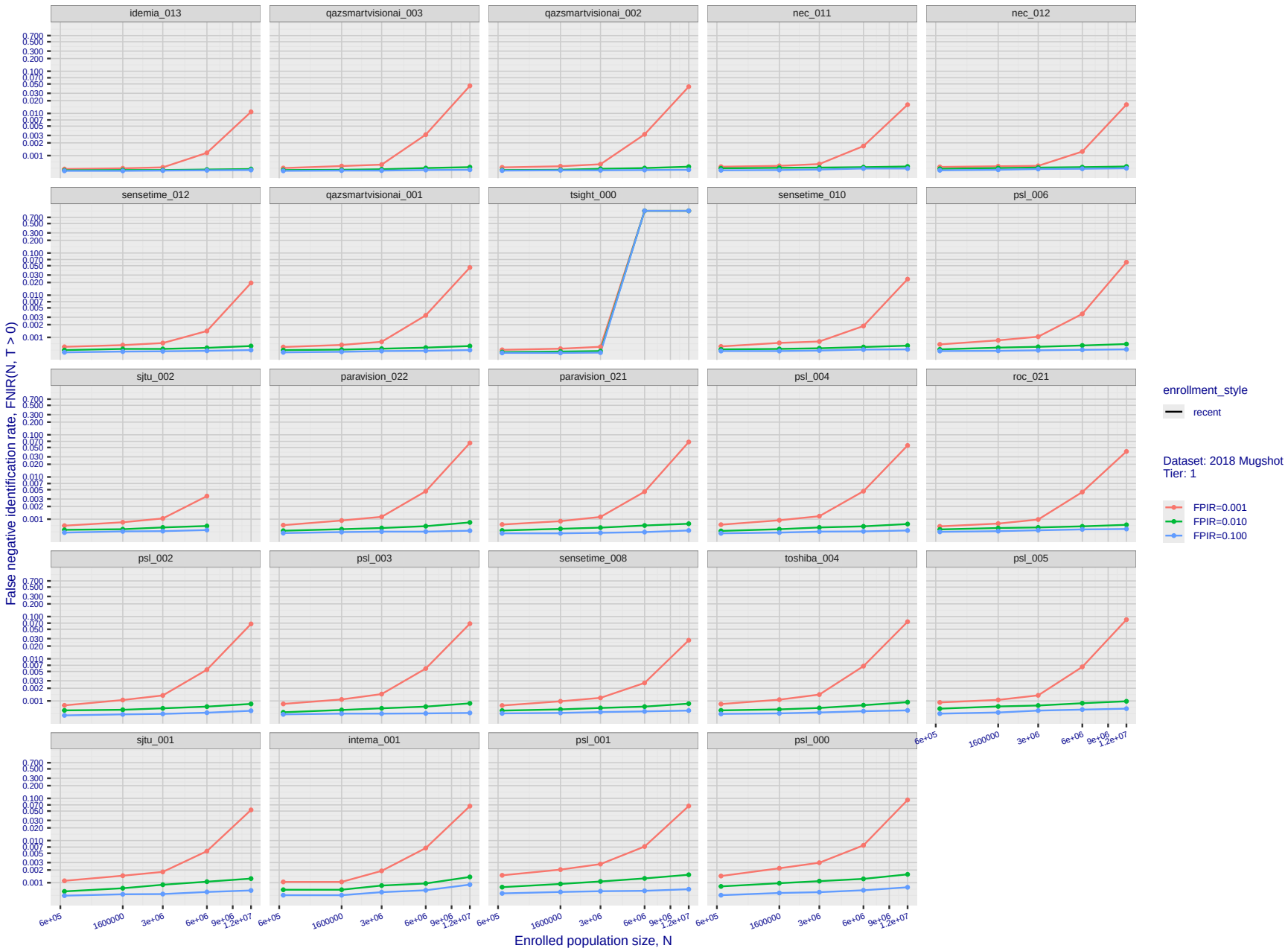

Figure 84: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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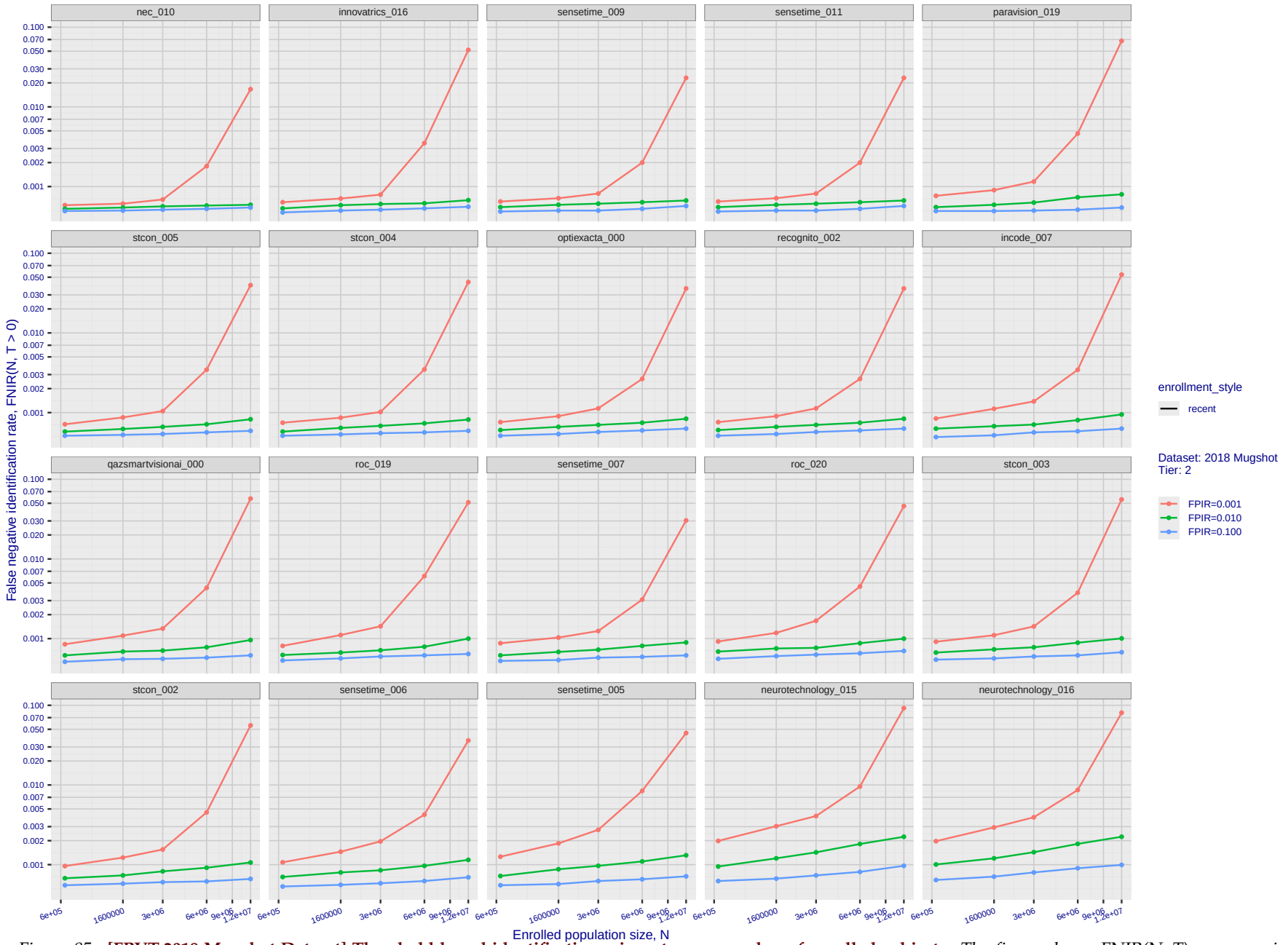

Figure 85: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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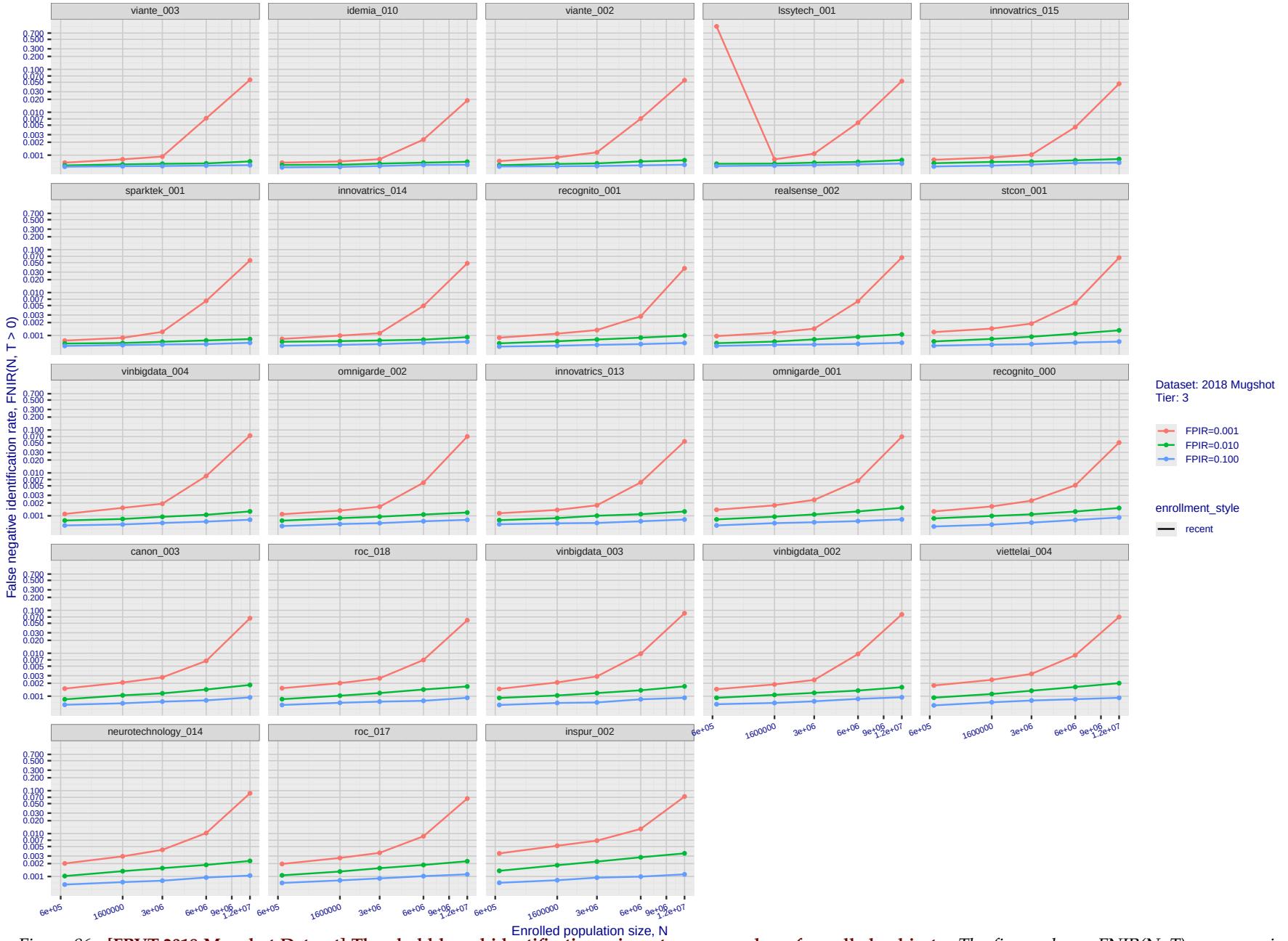

Figure 86: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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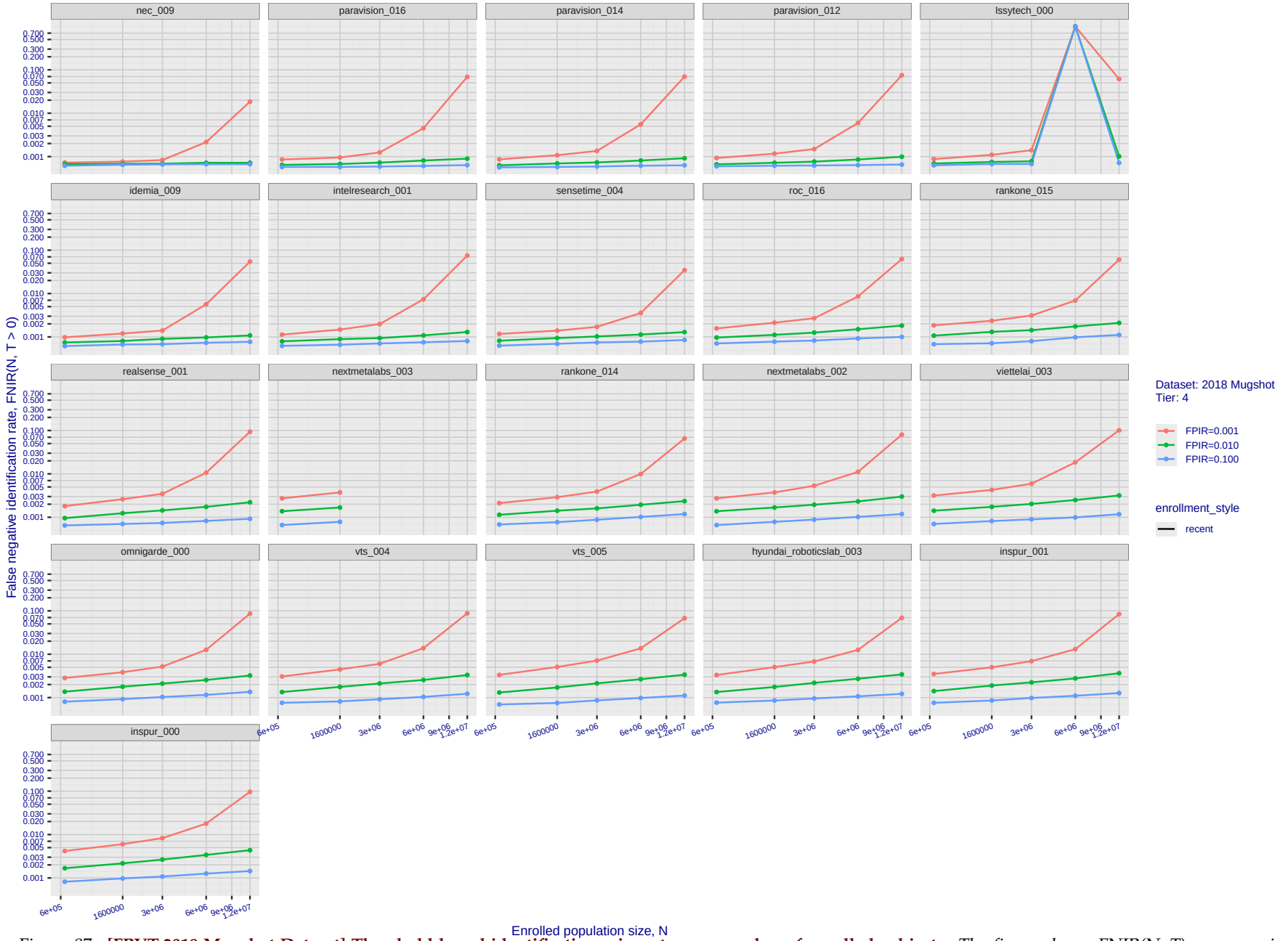

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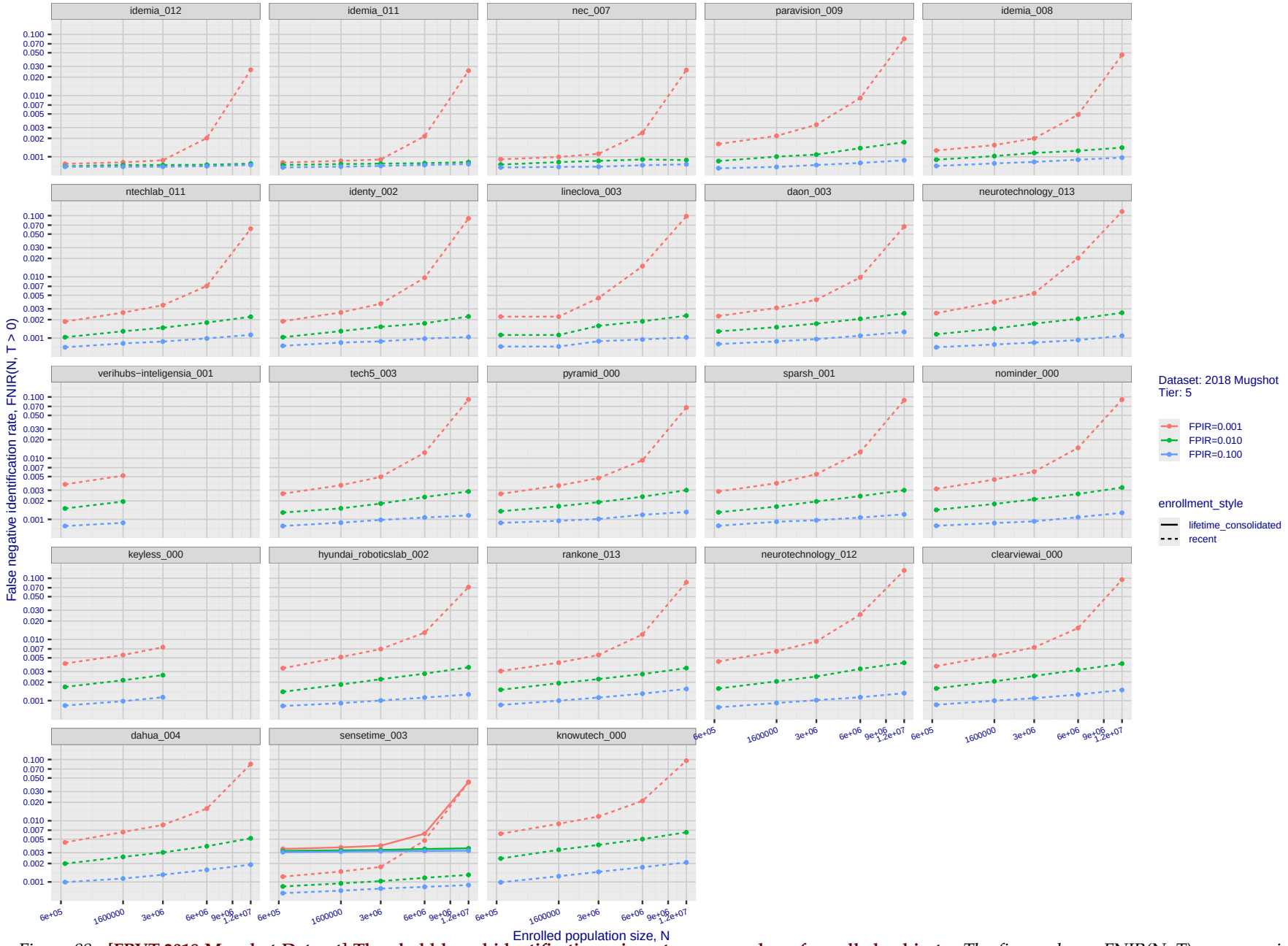

Figure 88: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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T = Threshold
T = 0 → Investigation
T > 0 → Identification

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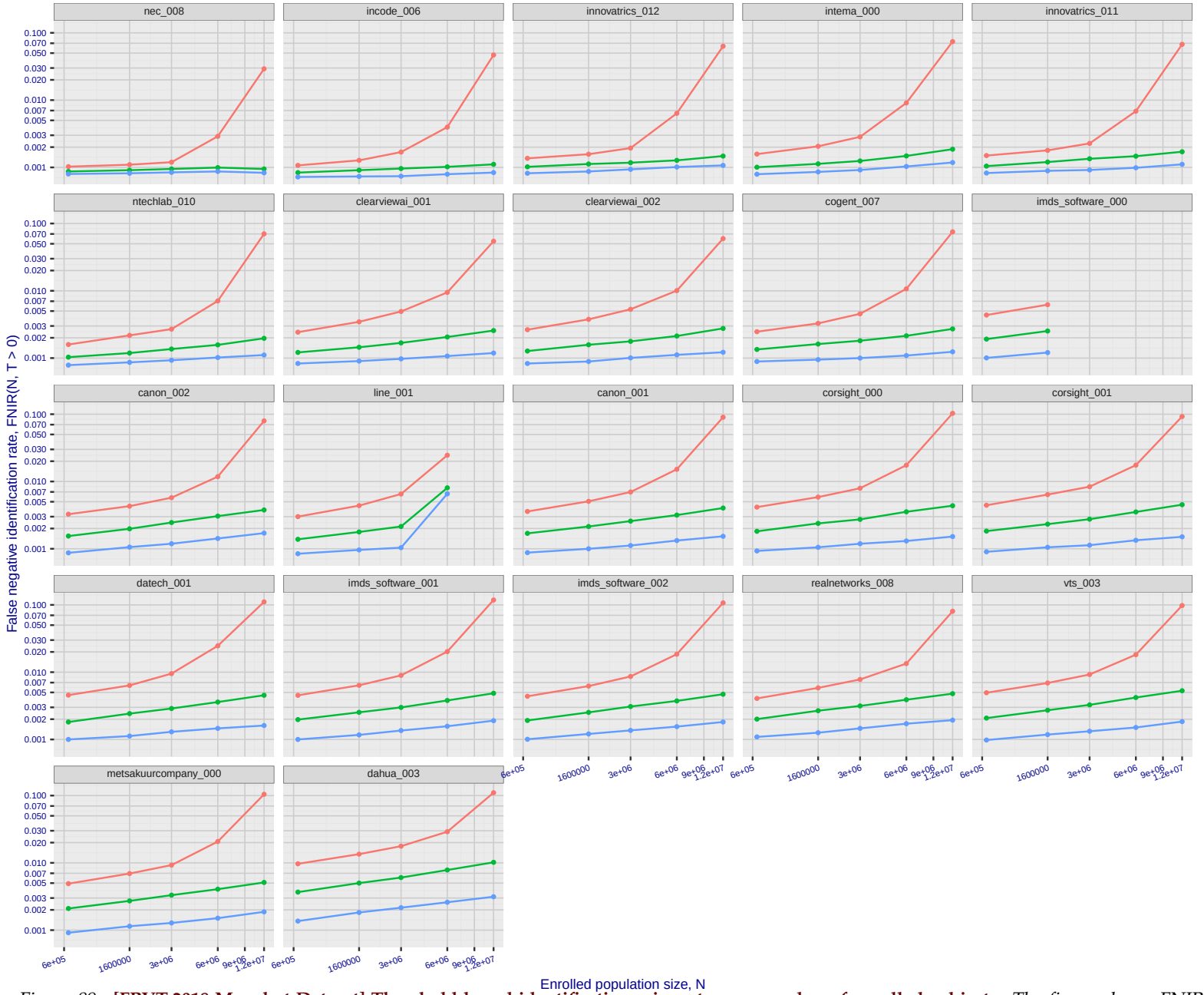

Figure 89: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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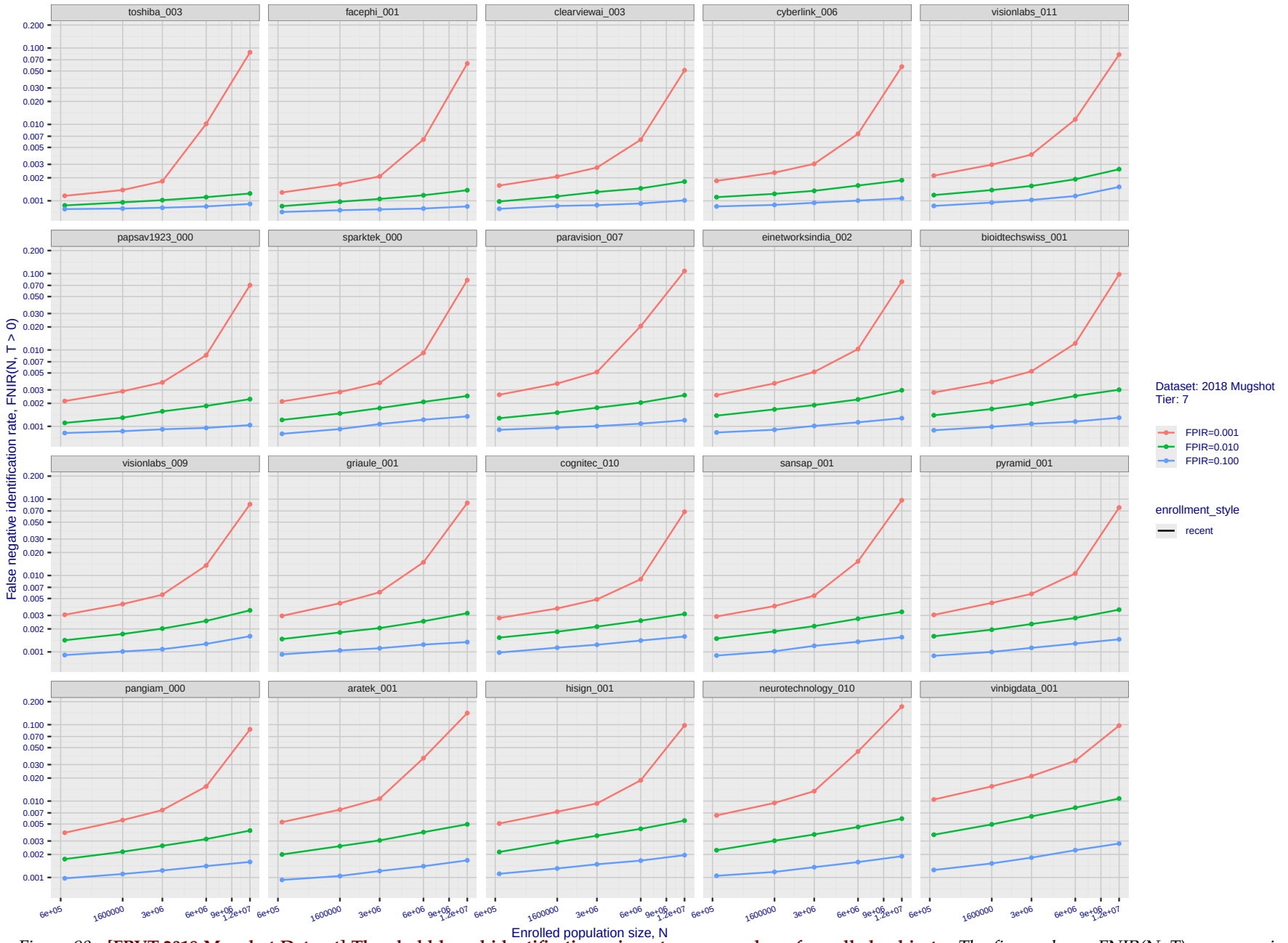

Figure 90: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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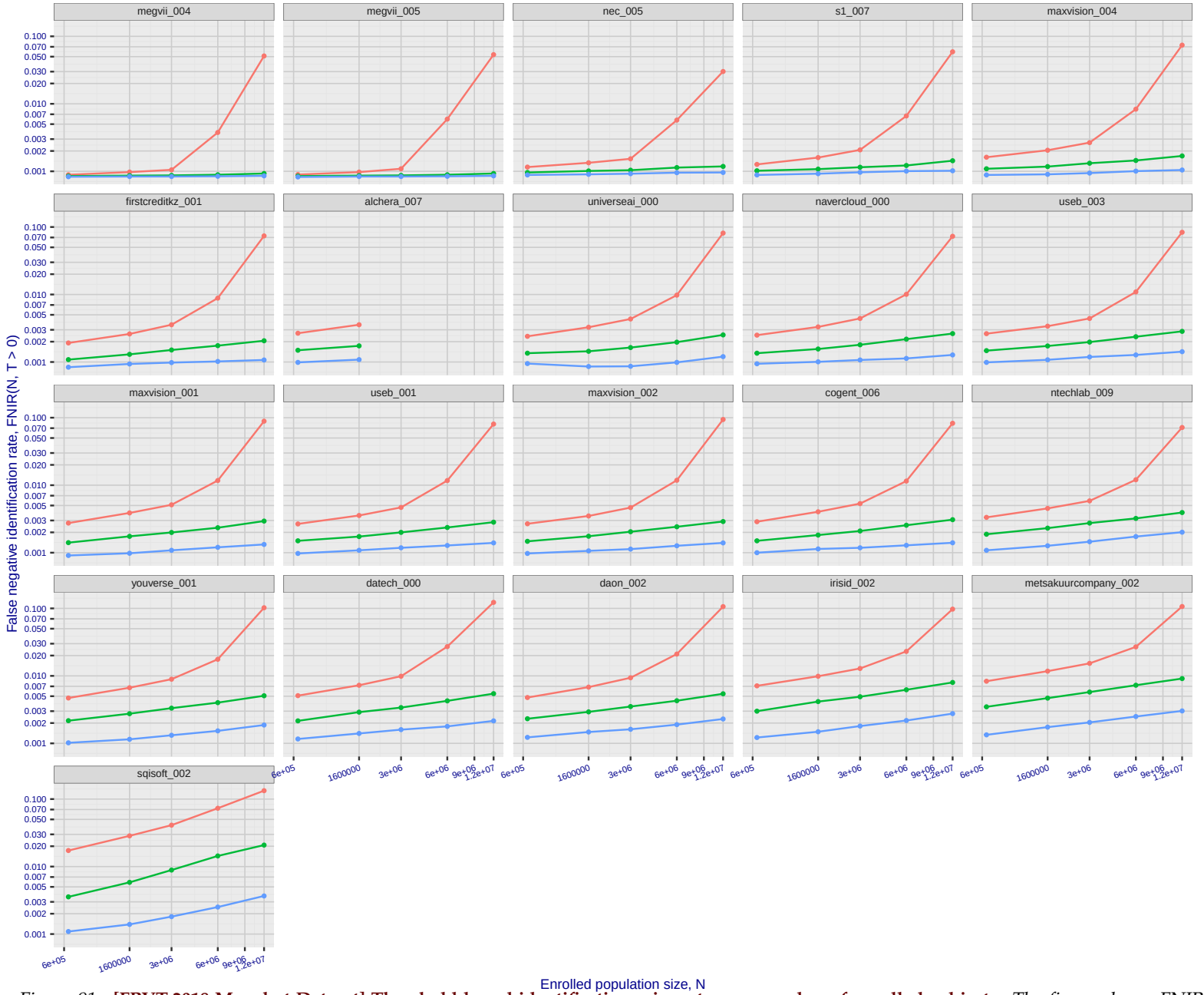

Figure 91: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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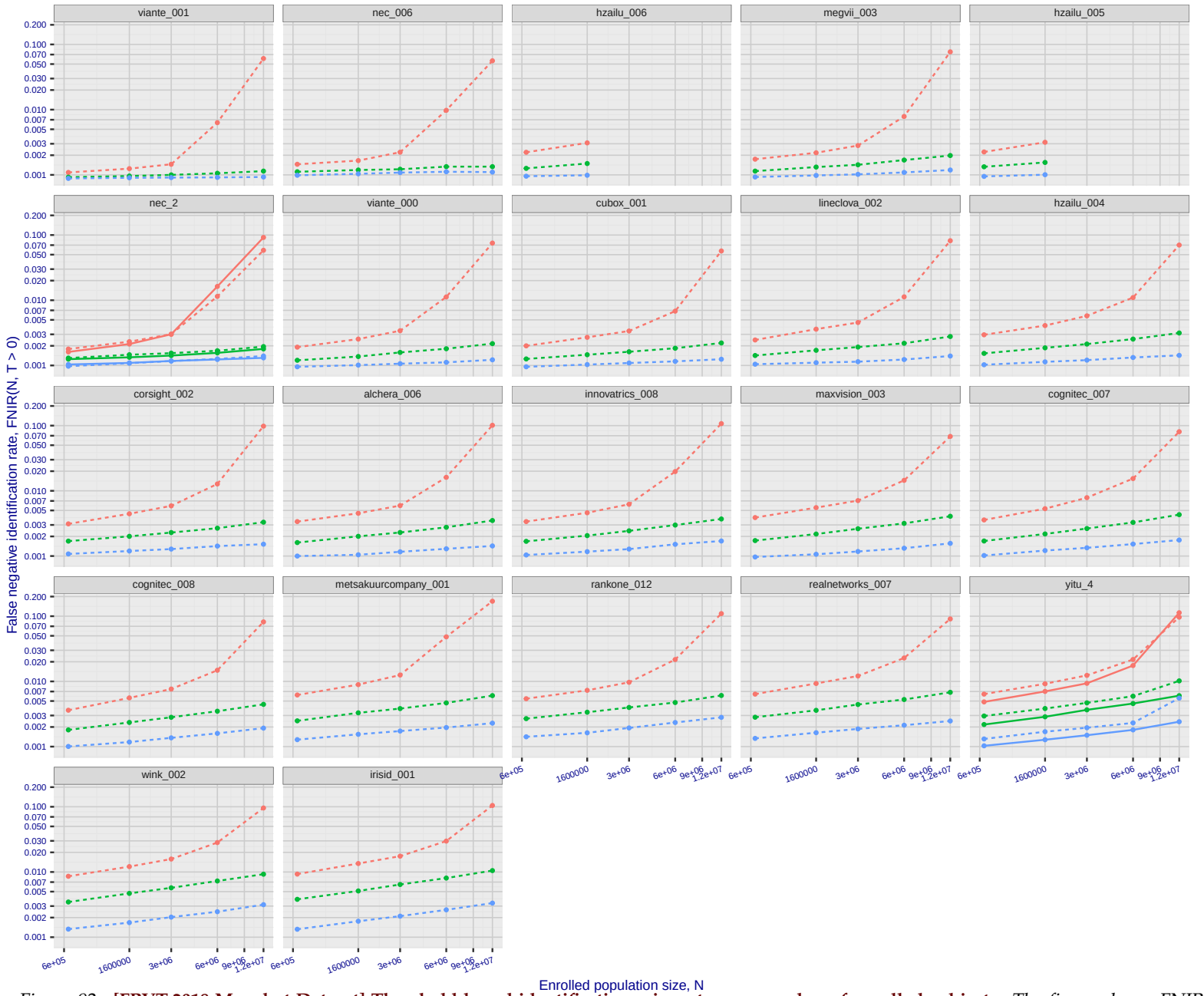

Figure 92: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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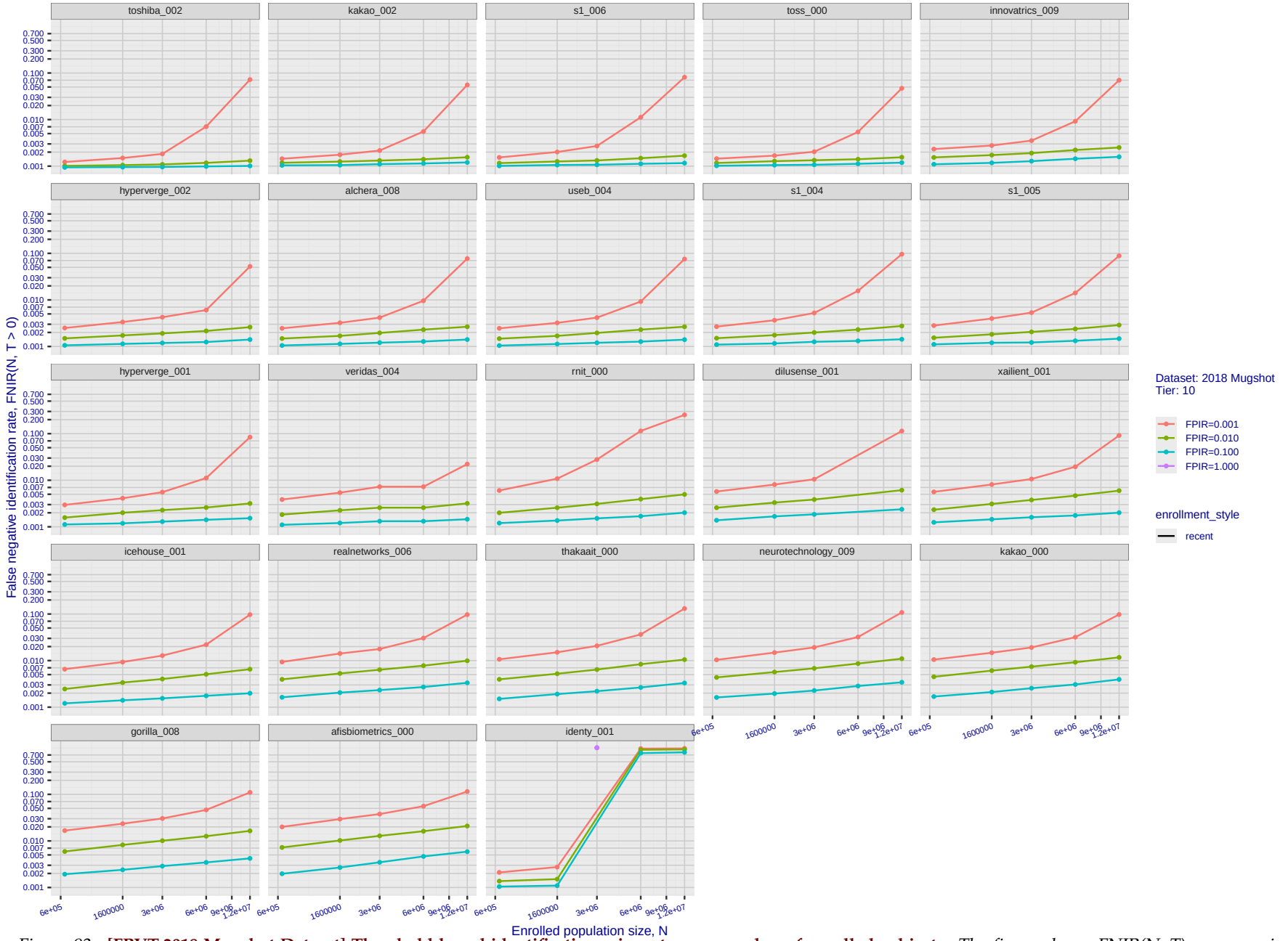

Figure 93: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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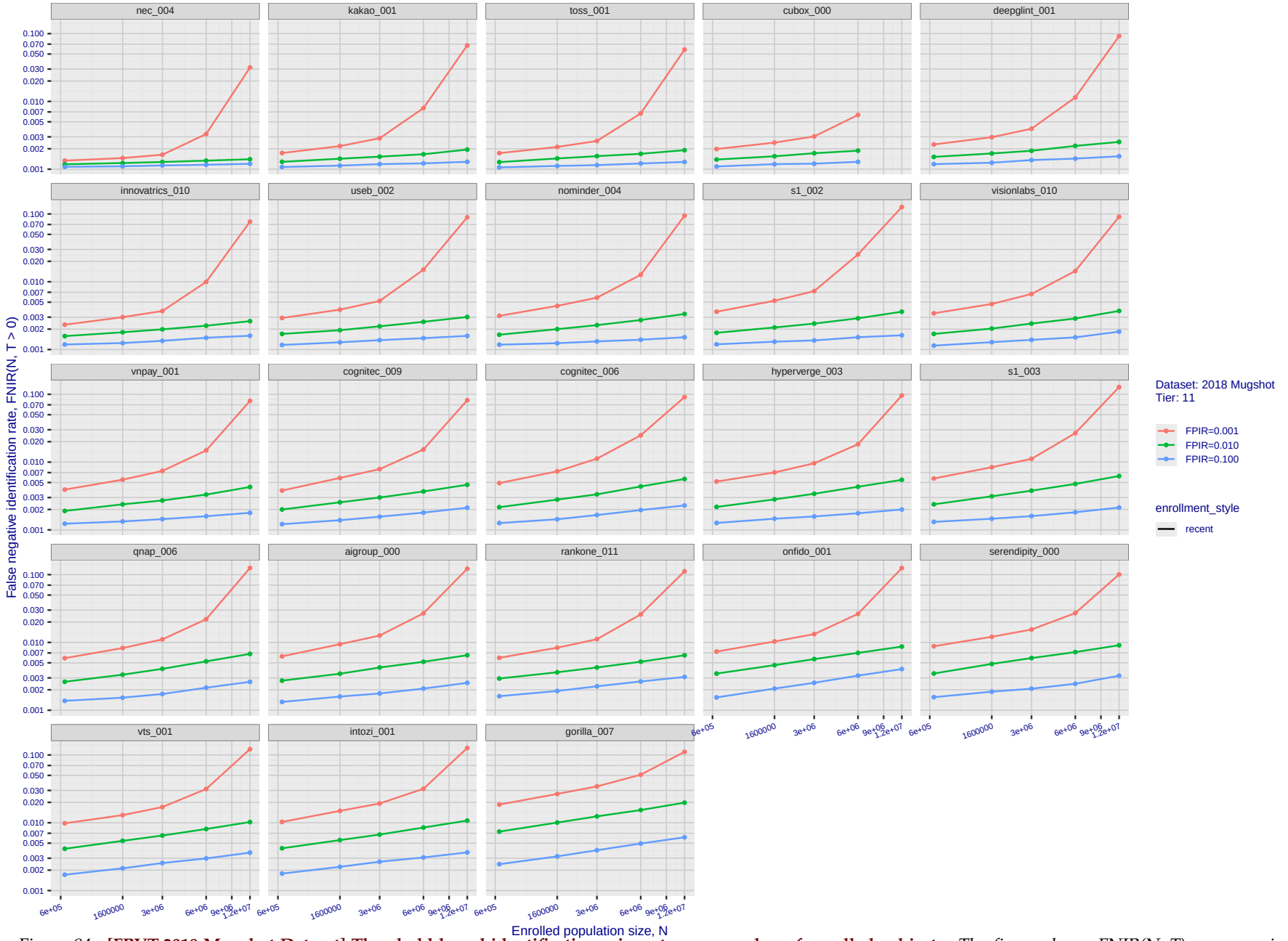

Figure 94: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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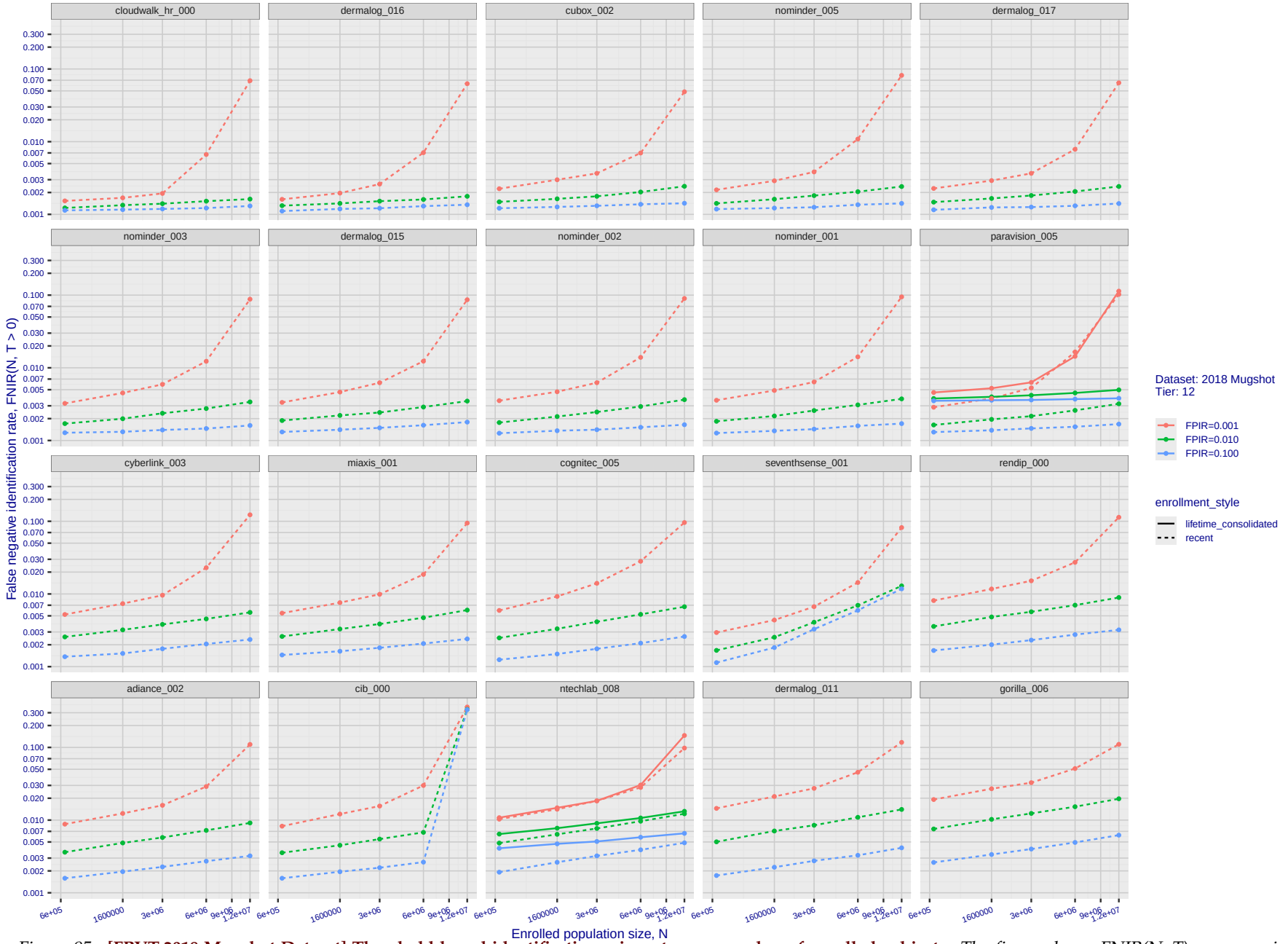

Figure 95: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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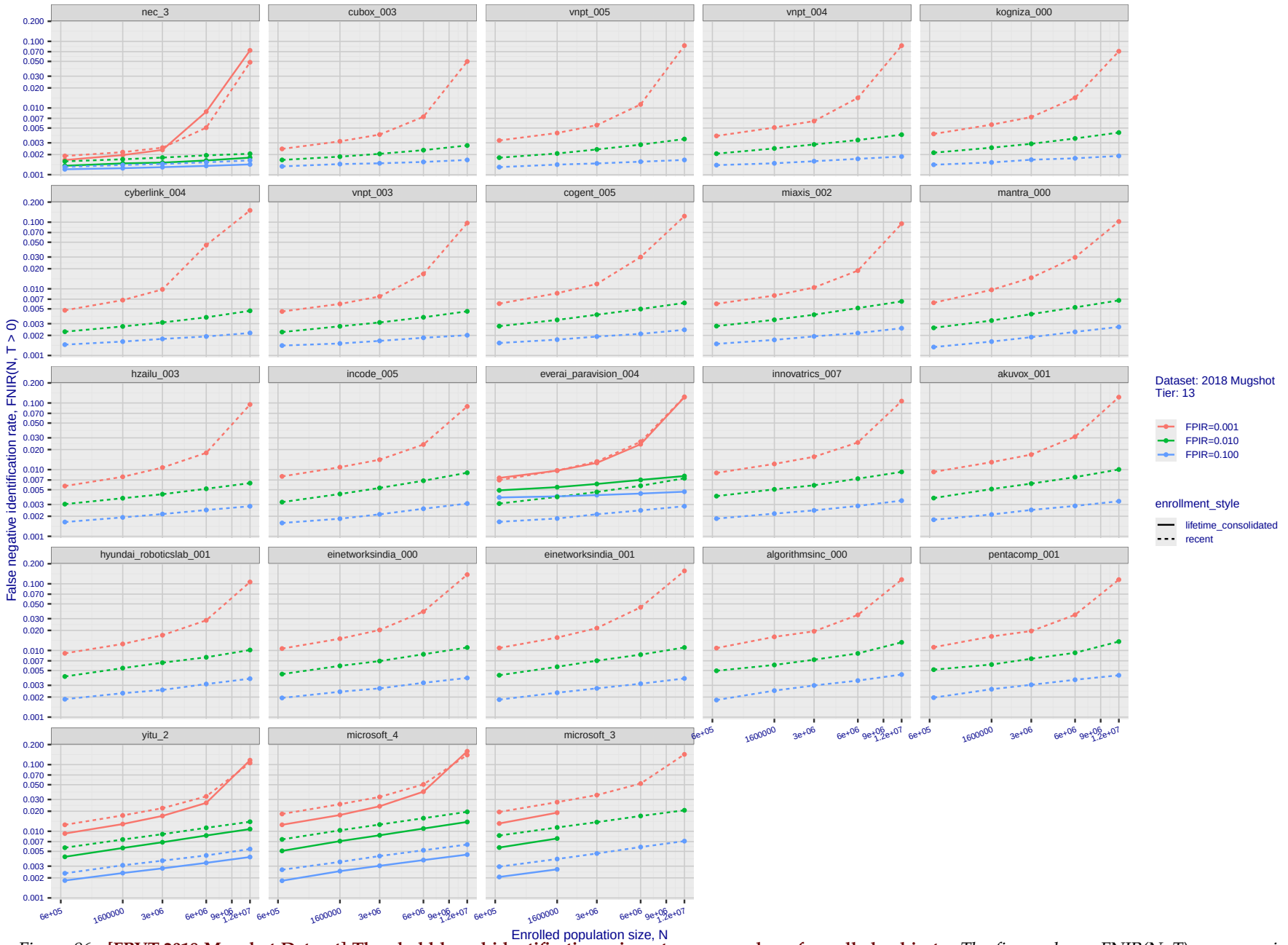

Figure 96: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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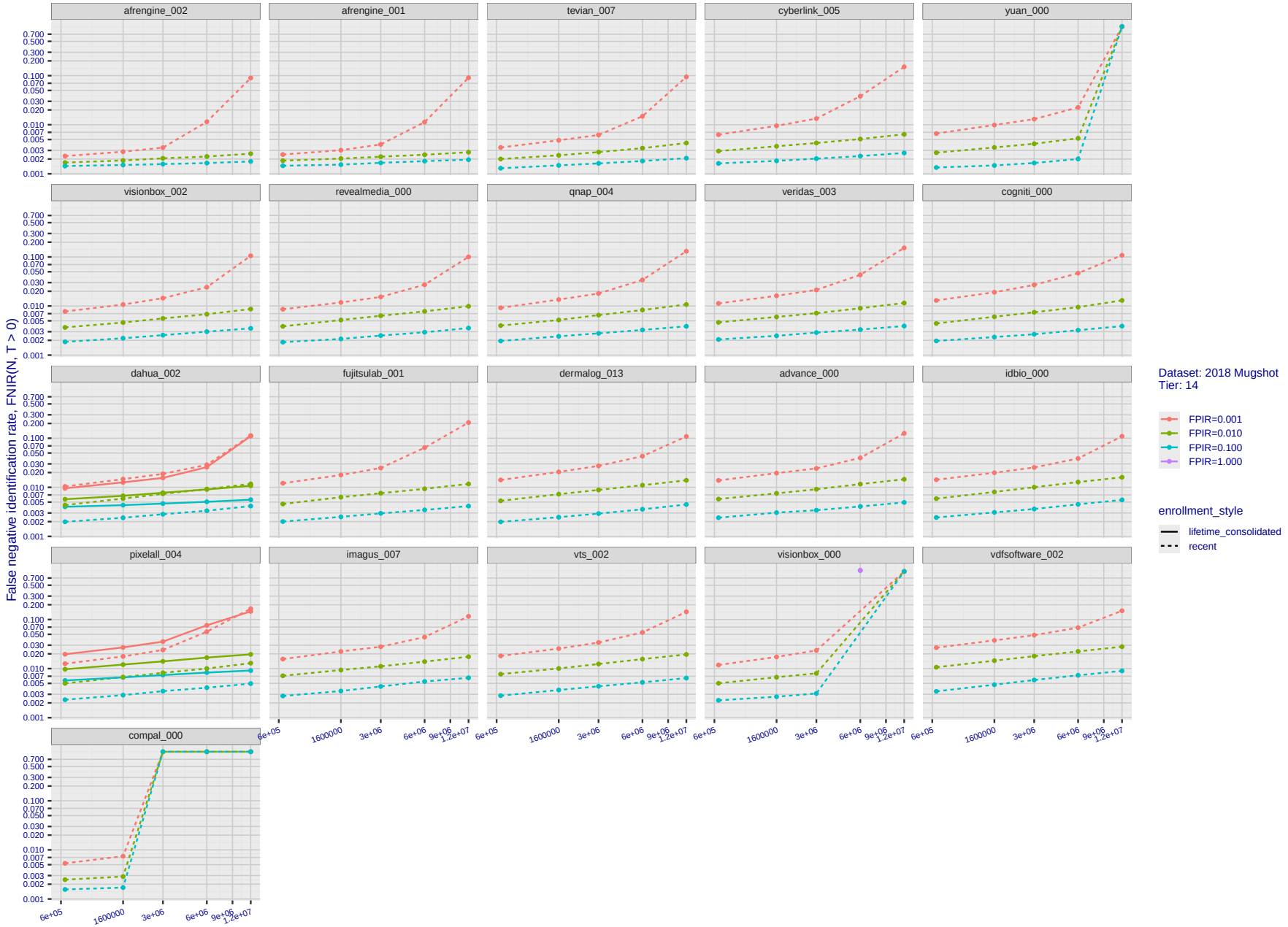

Figure 97: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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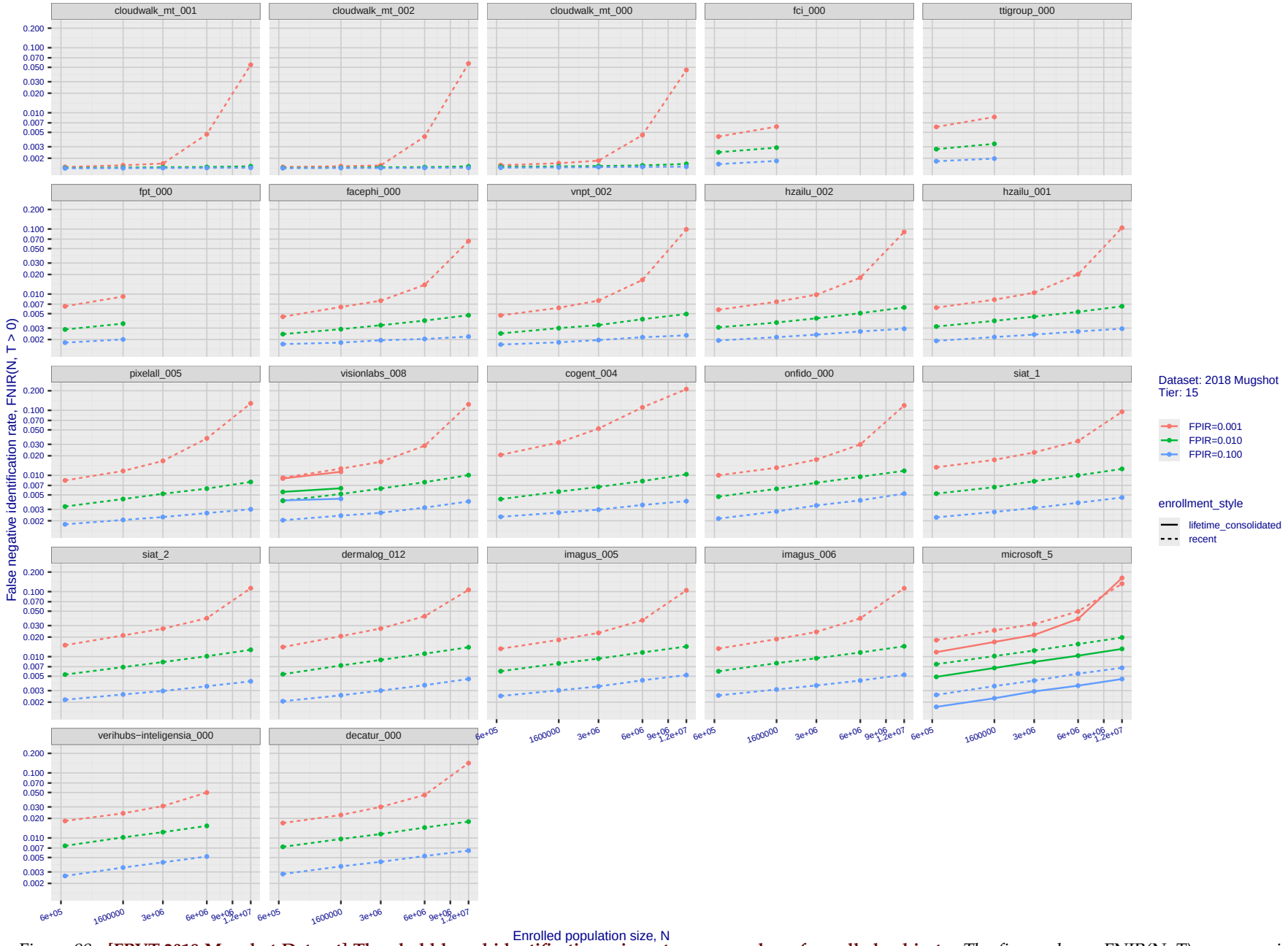

Figure 98: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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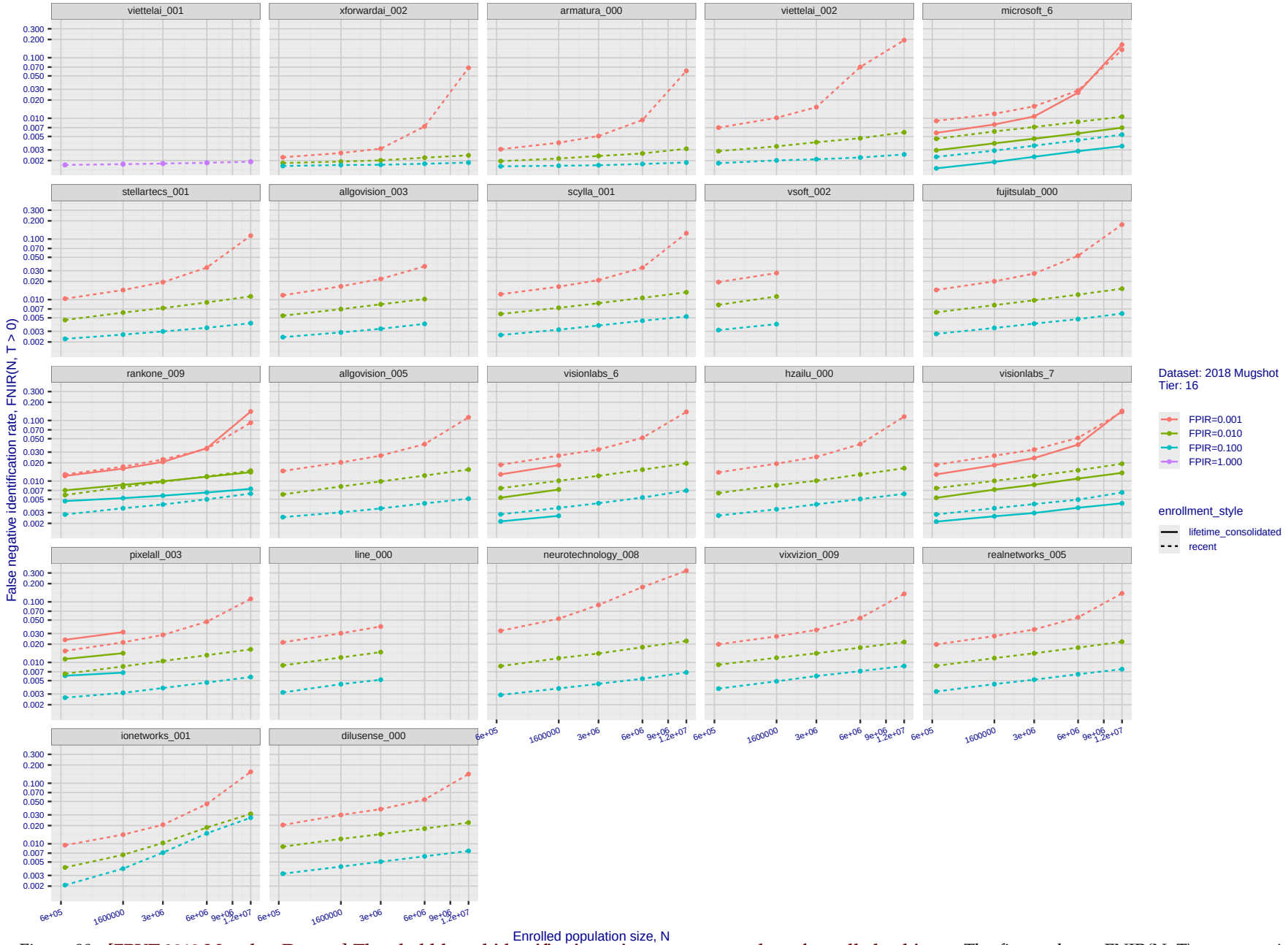

Figure 99: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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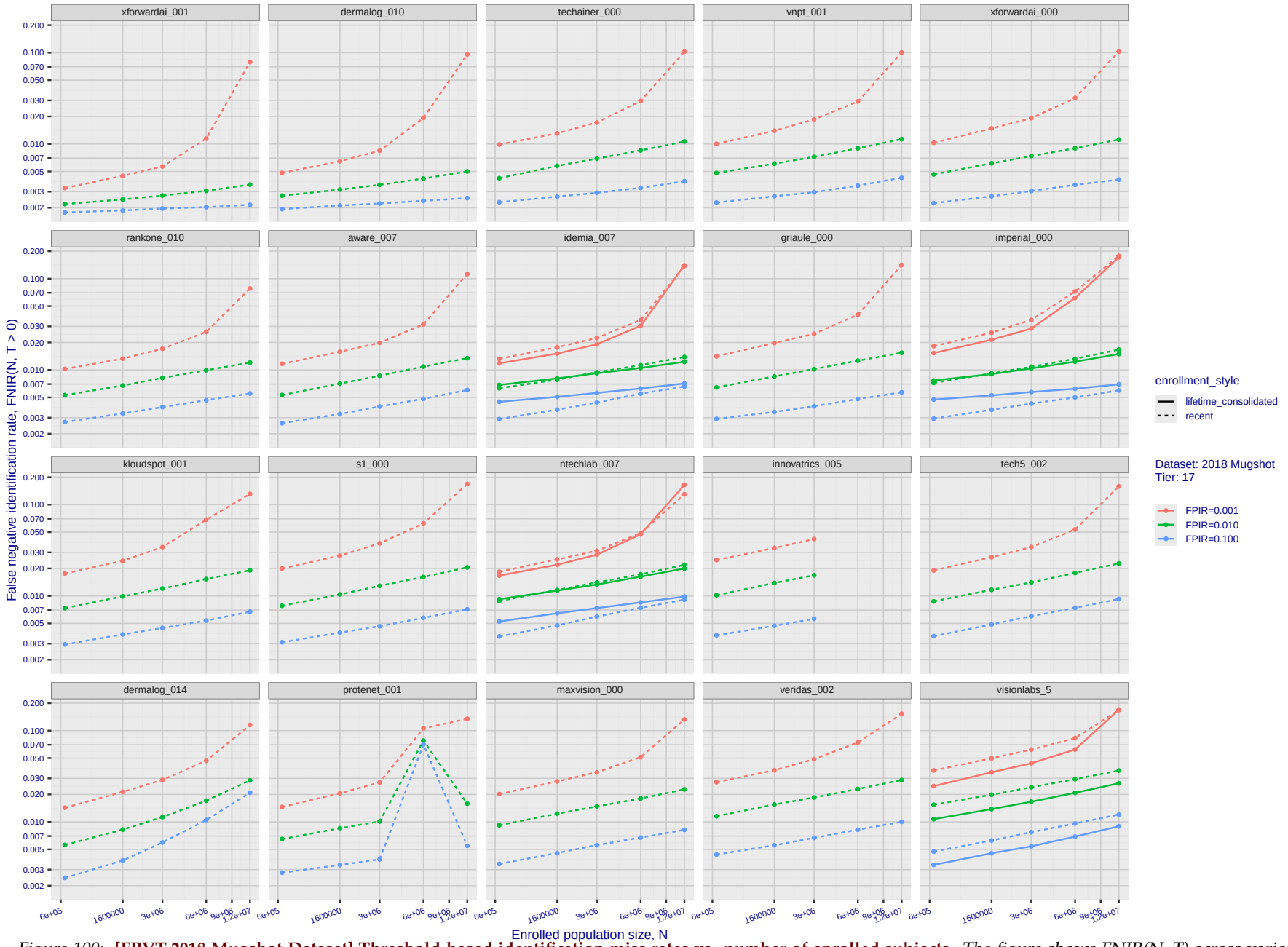

Figure 100: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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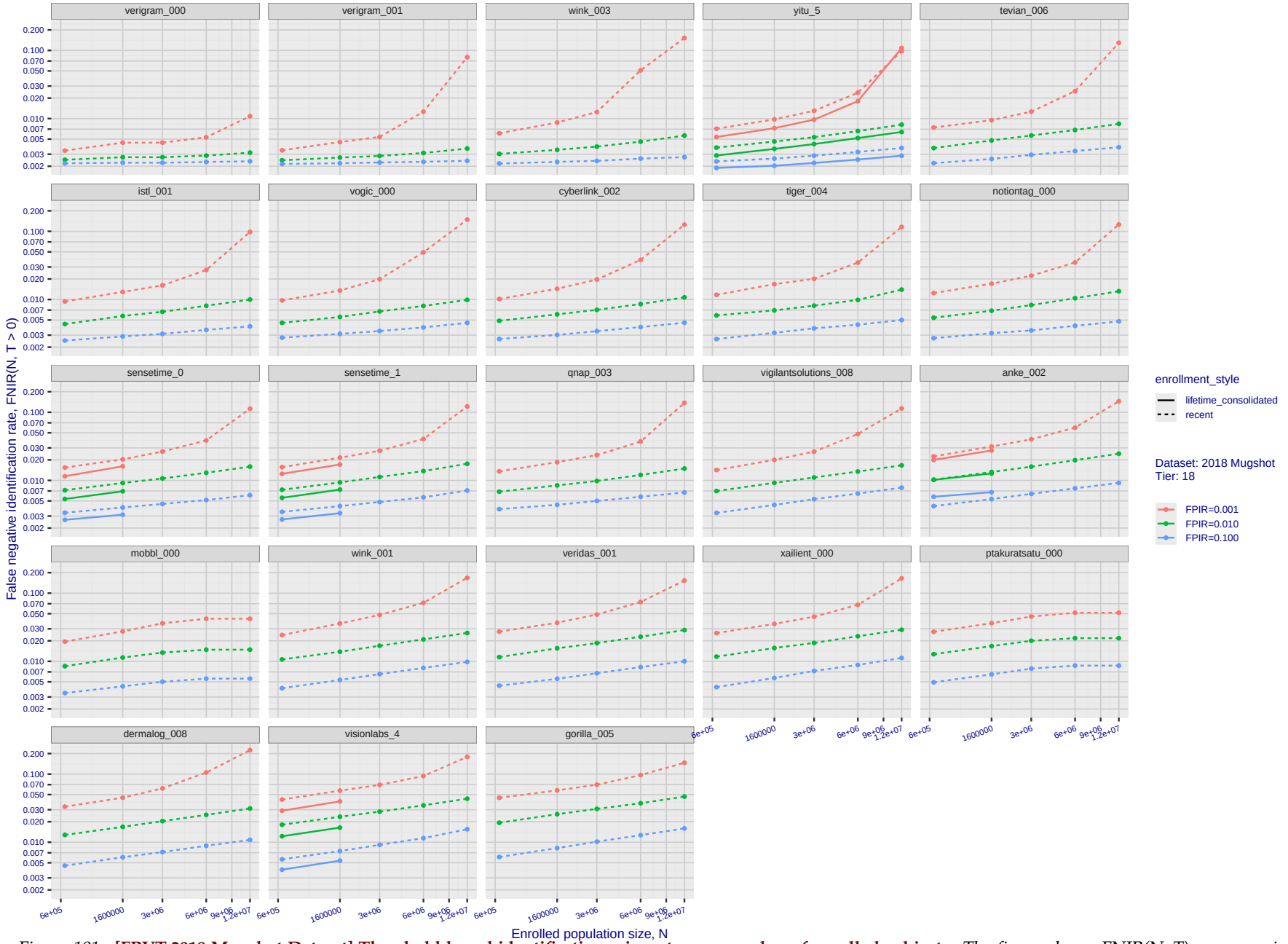

Figure 101: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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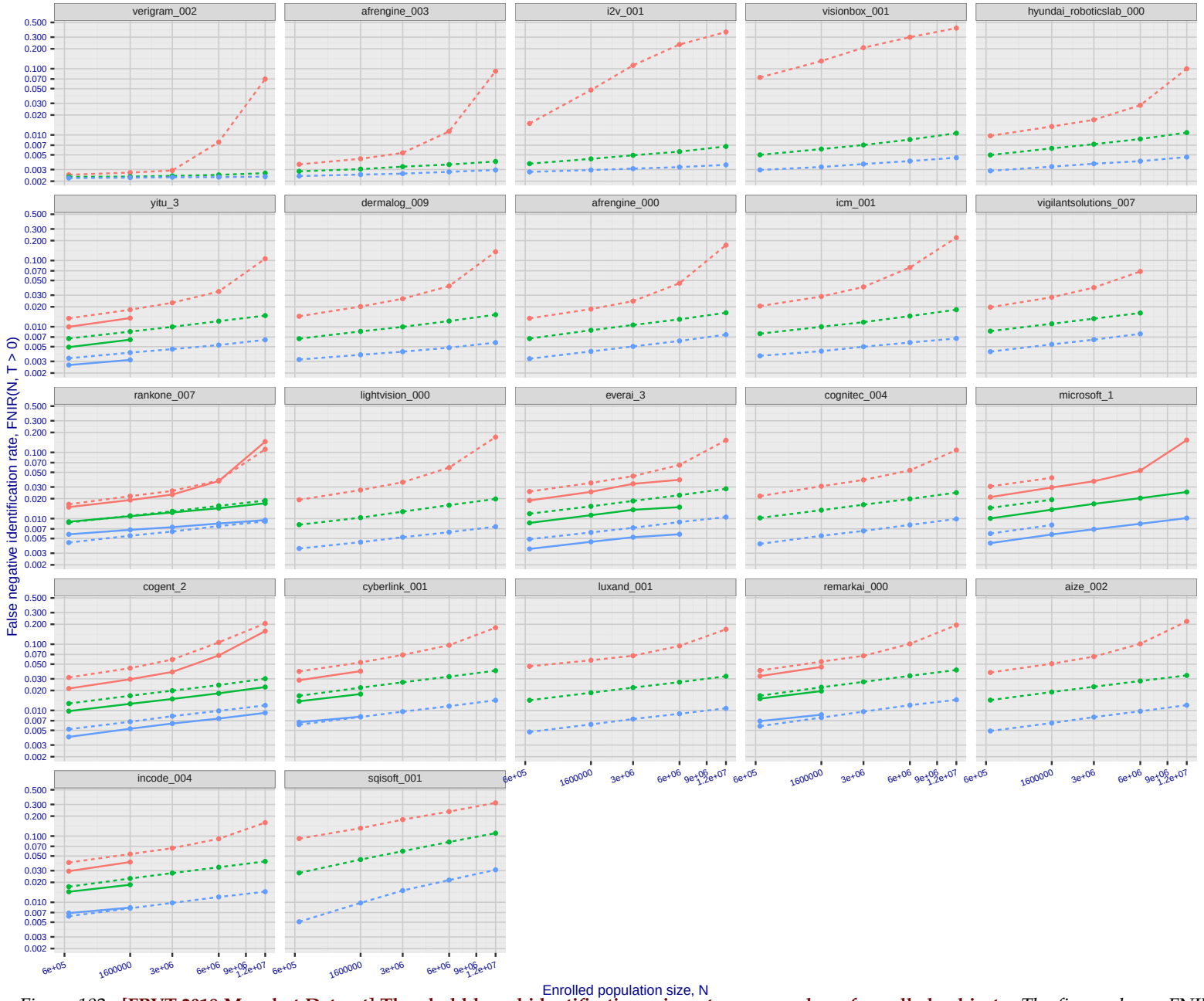

Figure 102: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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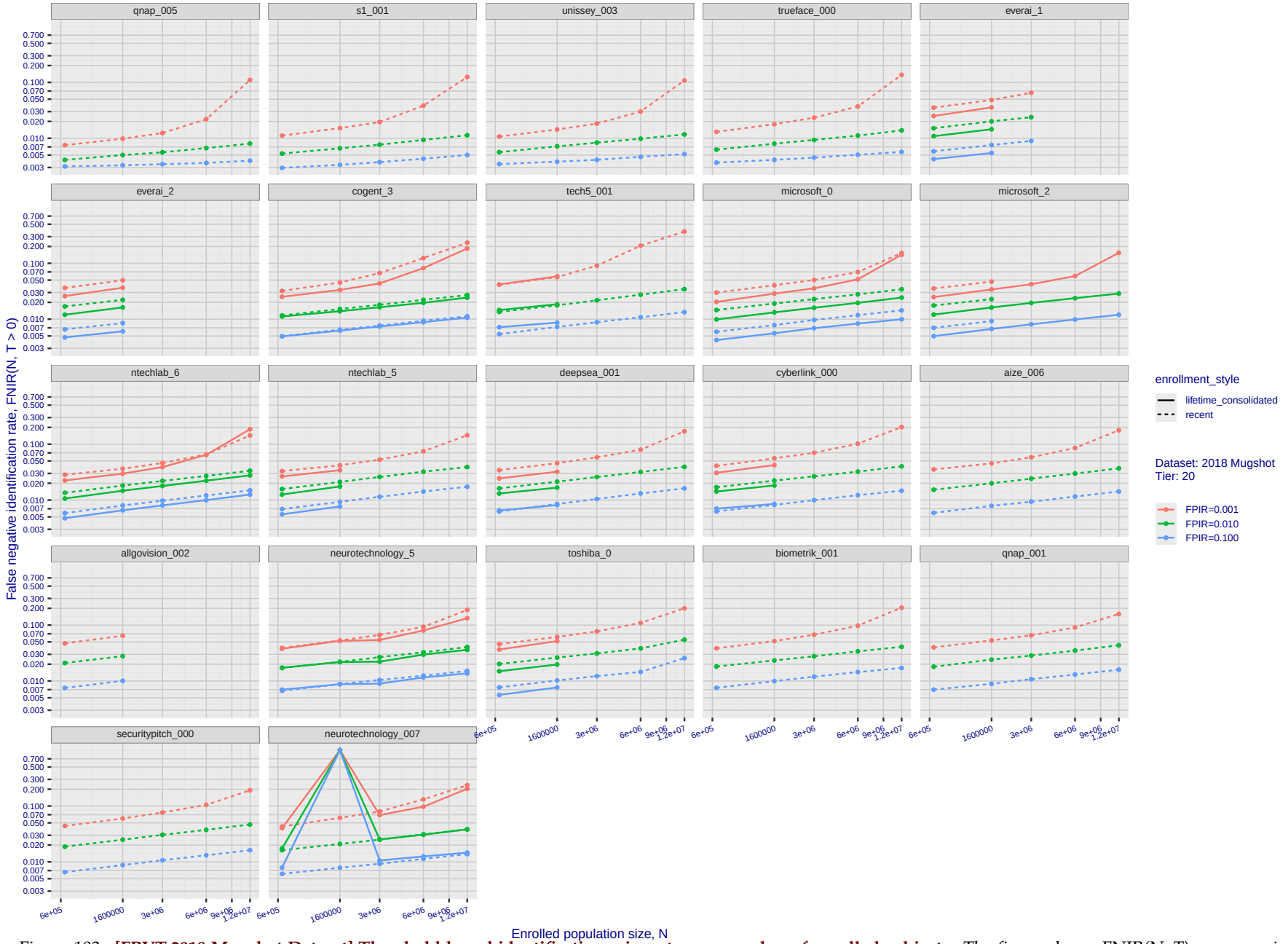

Figure 103: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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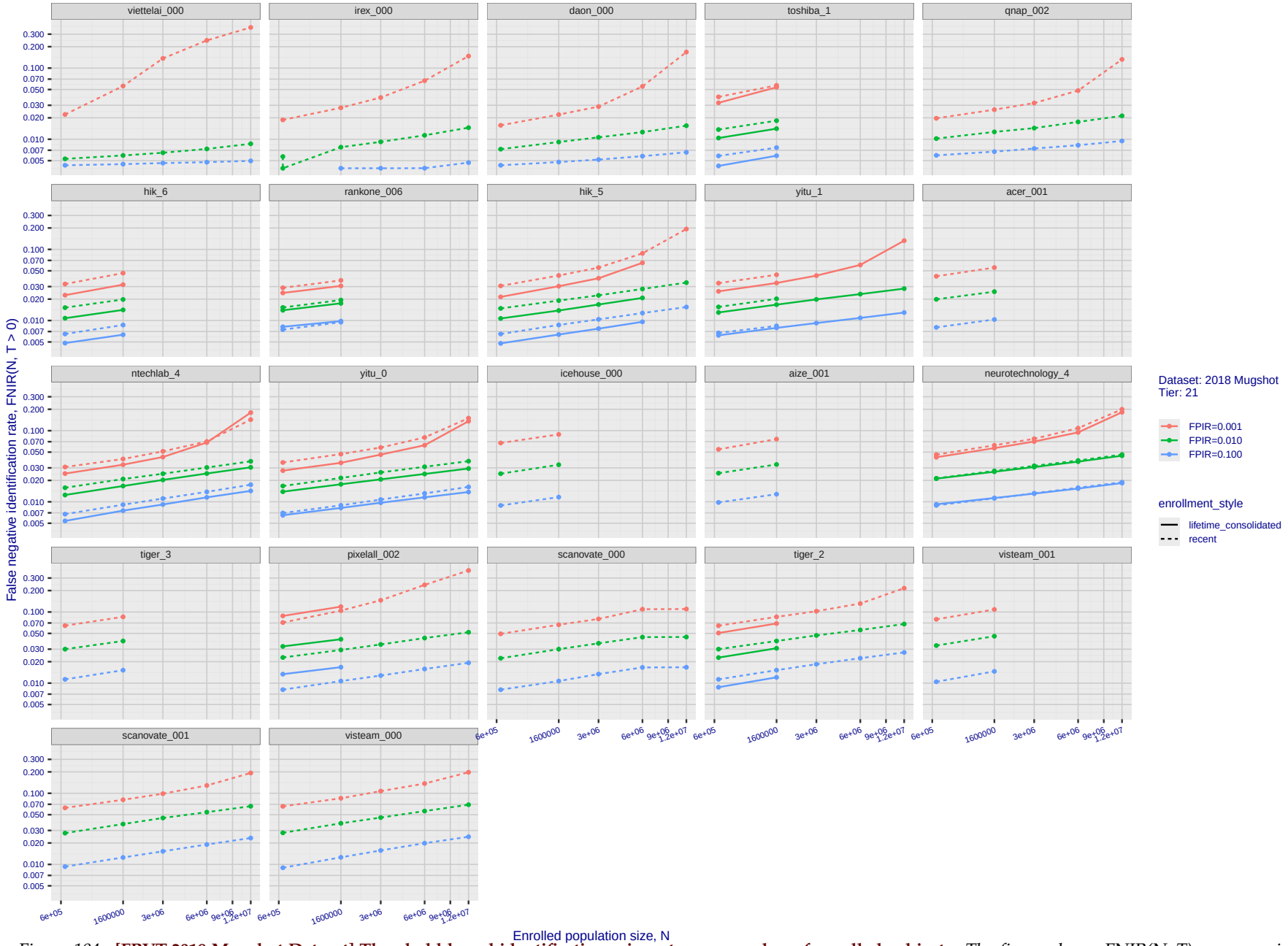

Figure 104: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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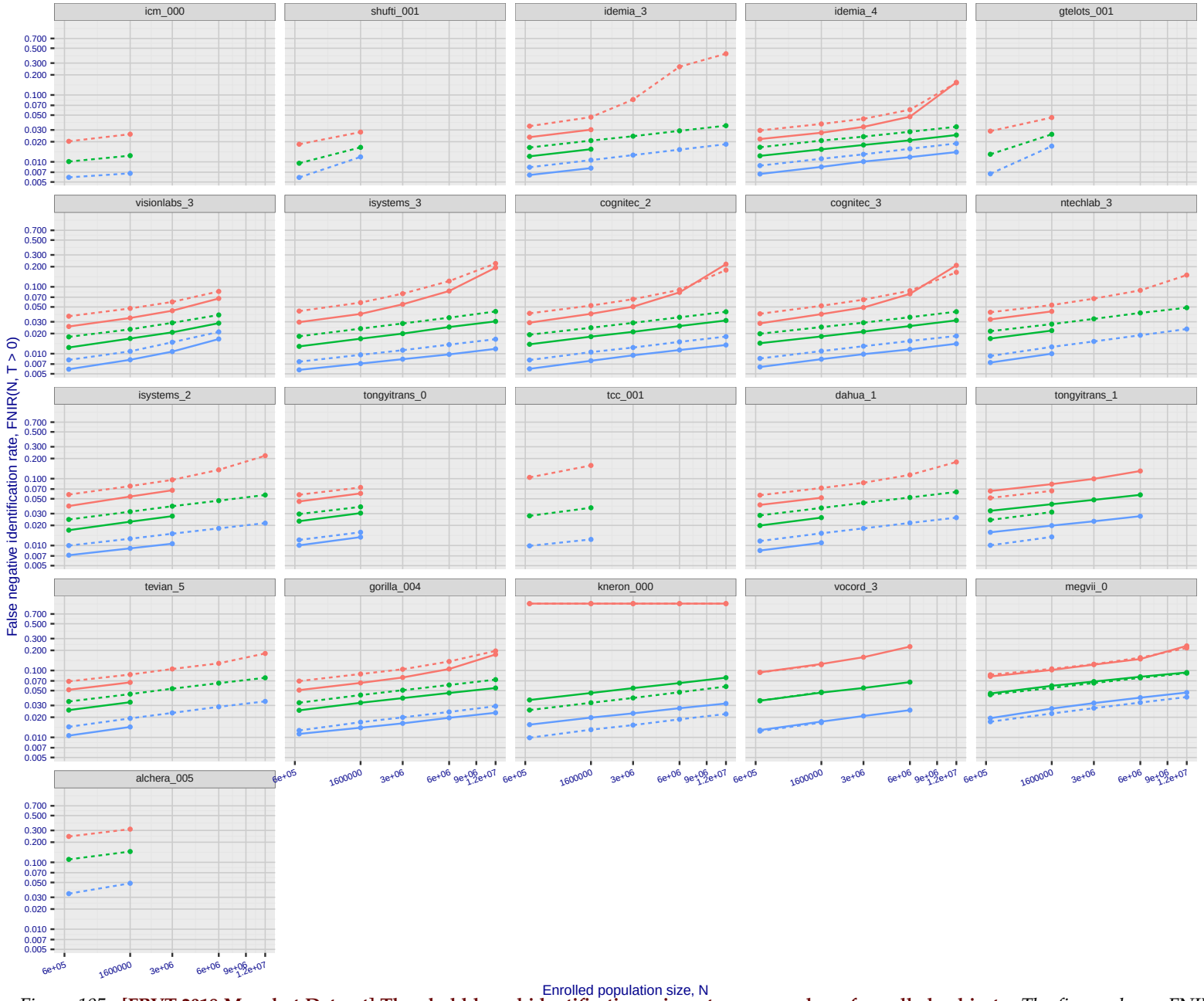

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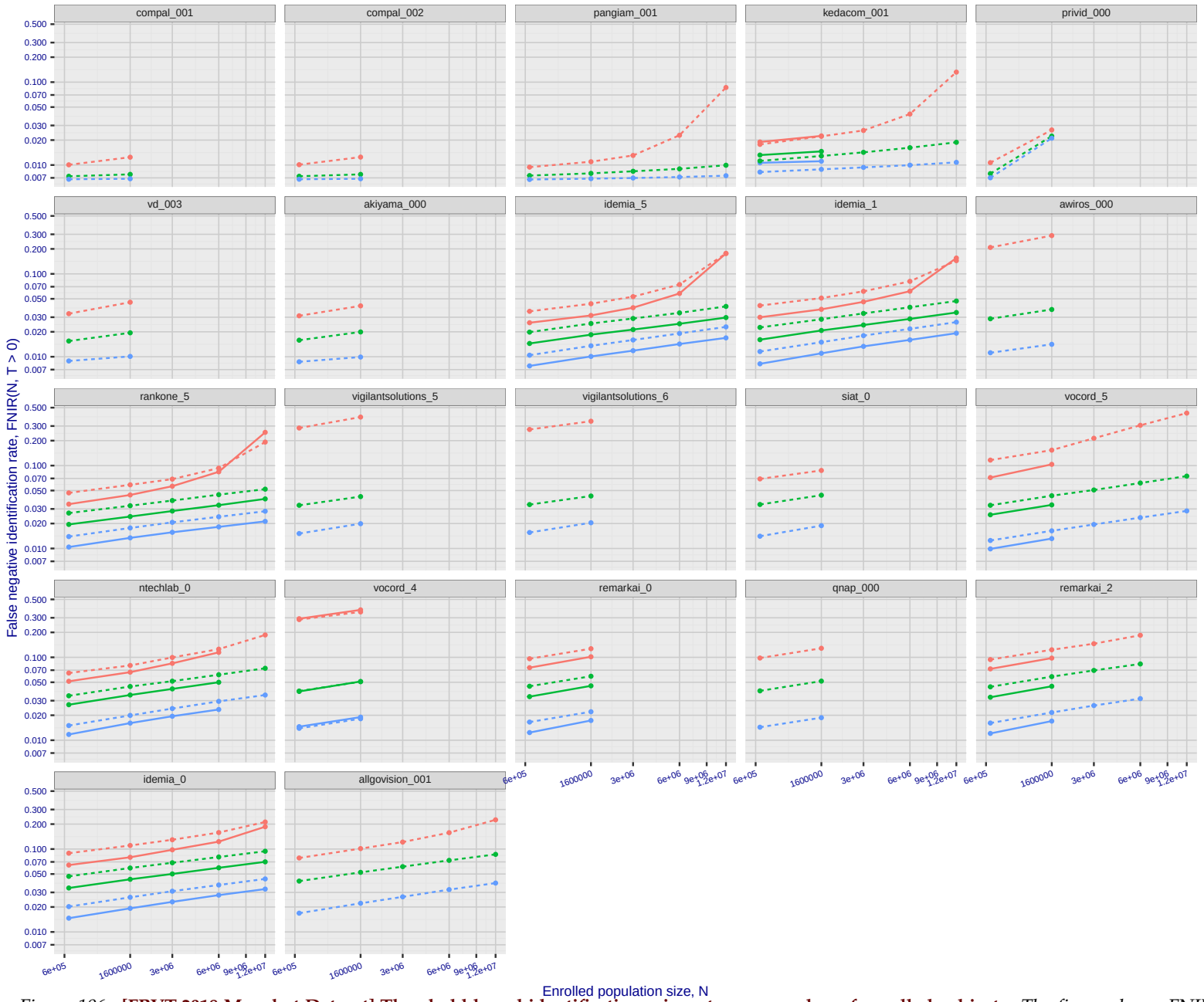

Figure 106: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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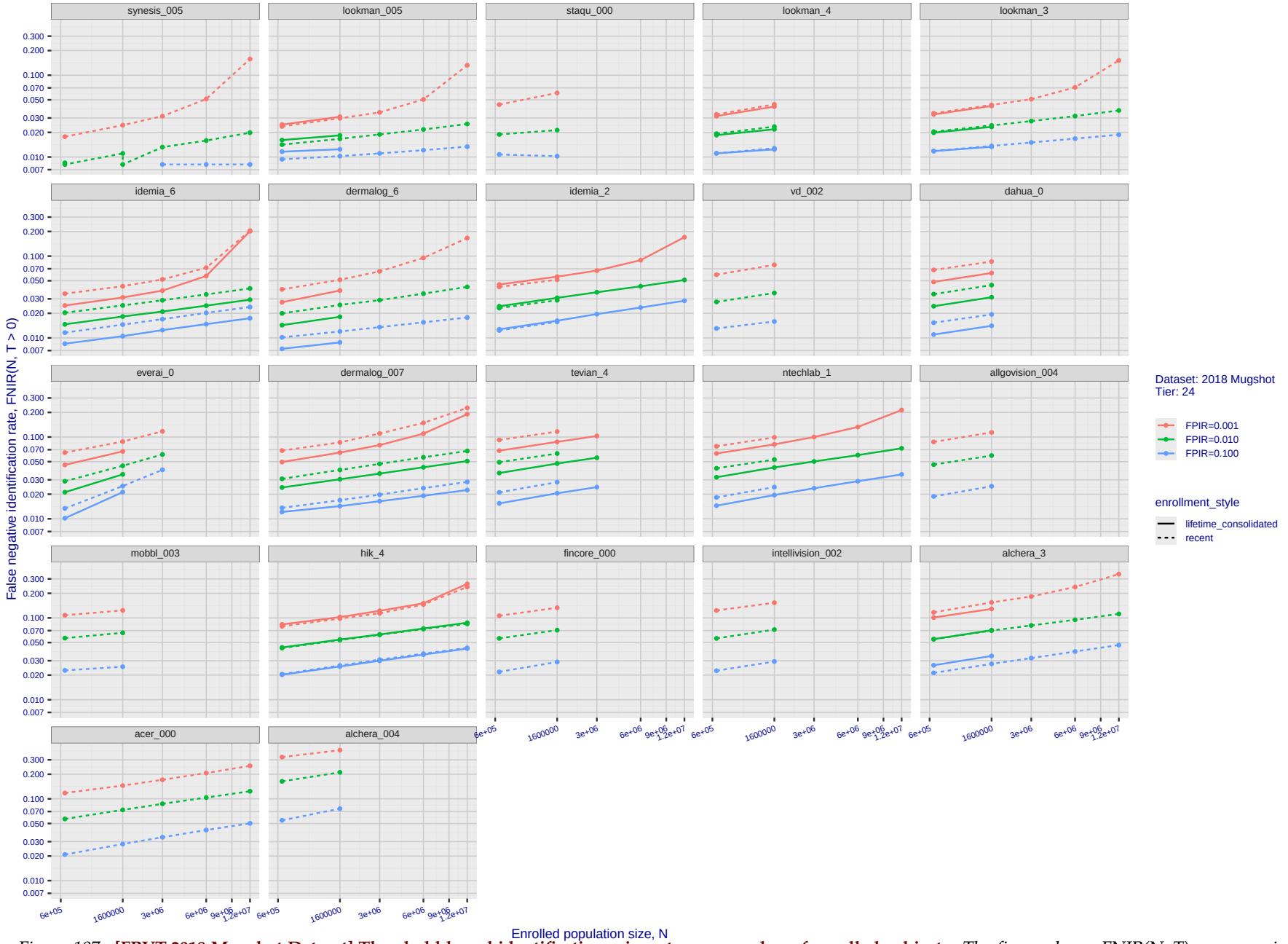

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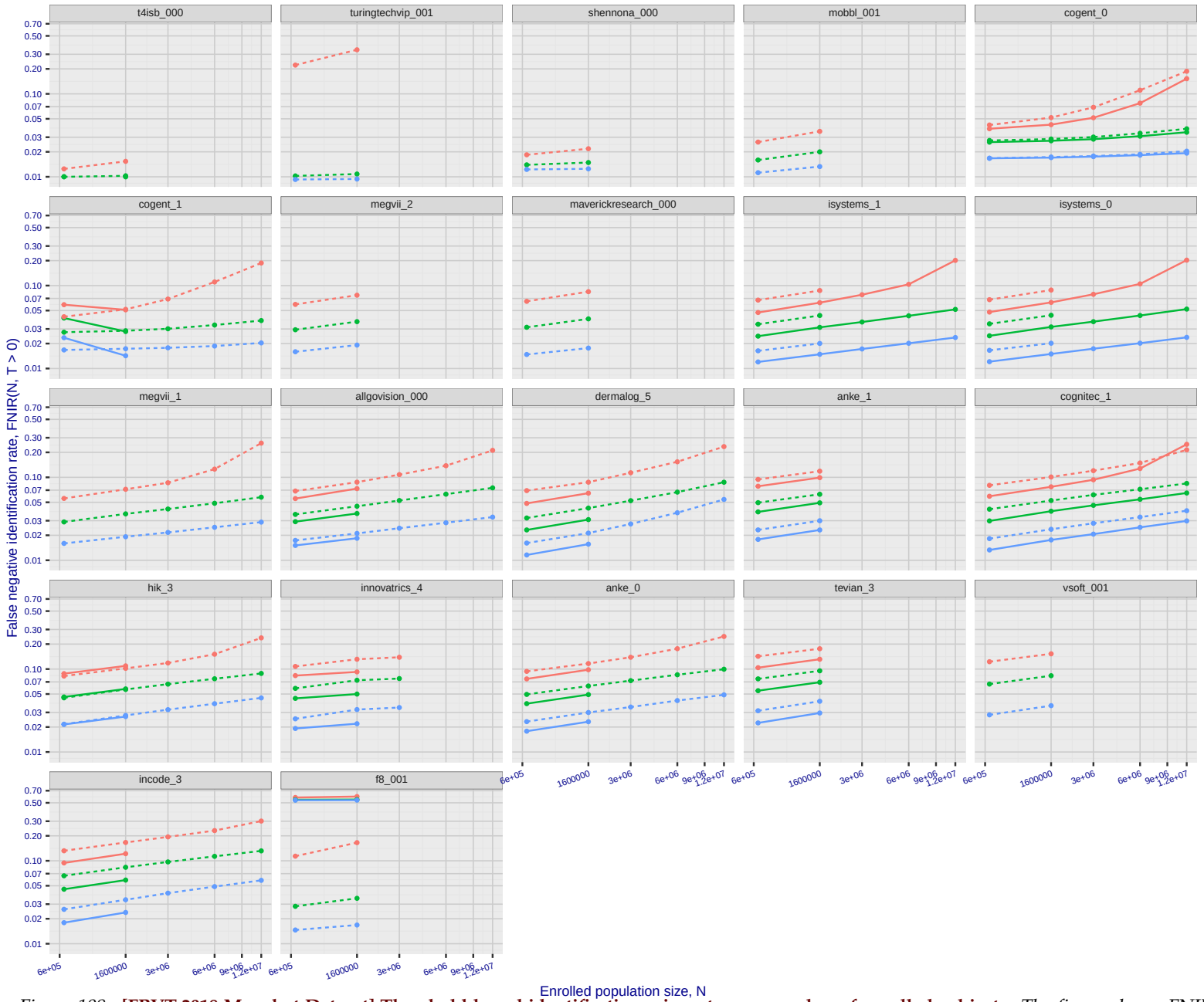

Figure 108: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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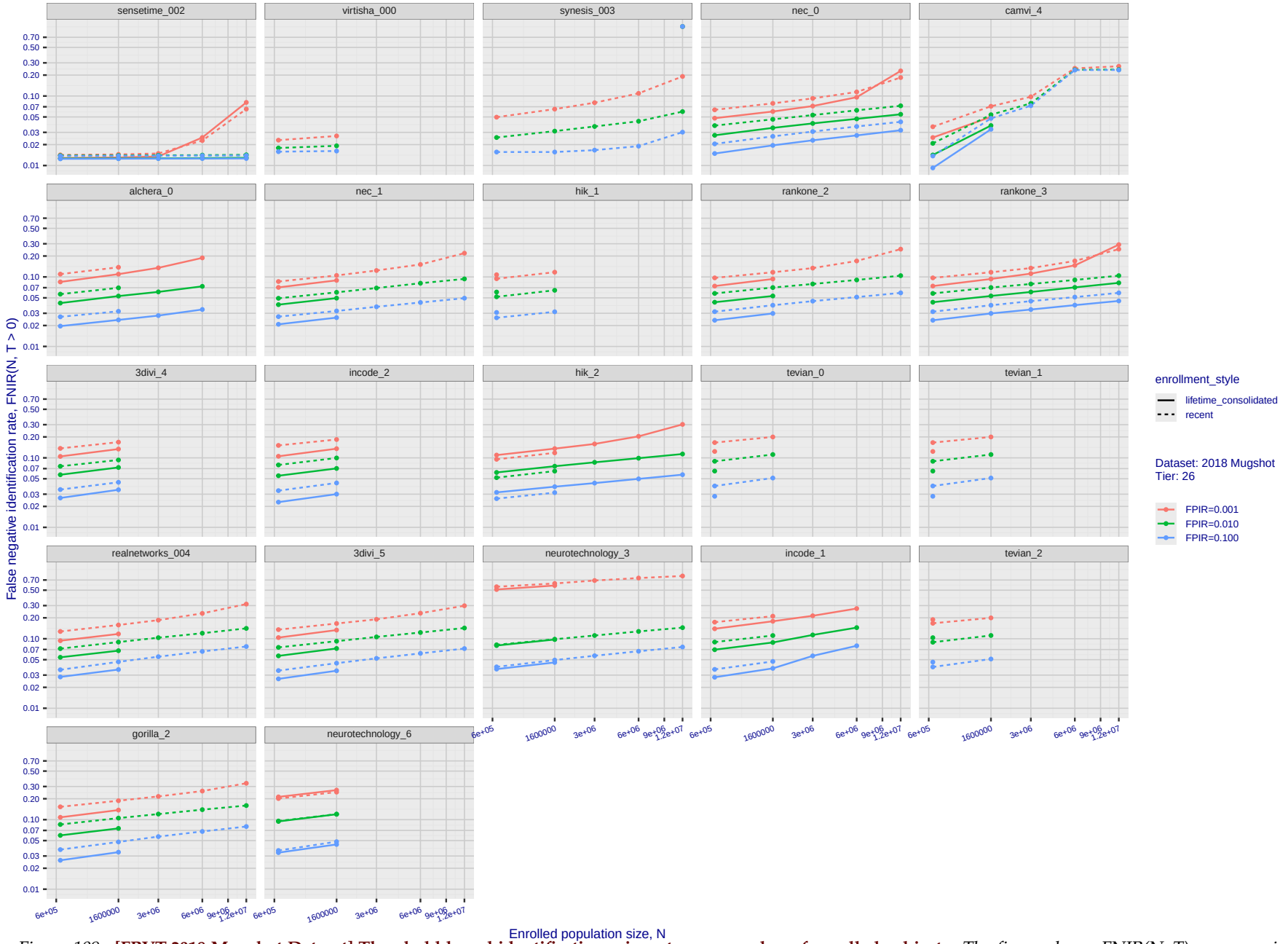

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T = Threshold
T = 0 → Investigation
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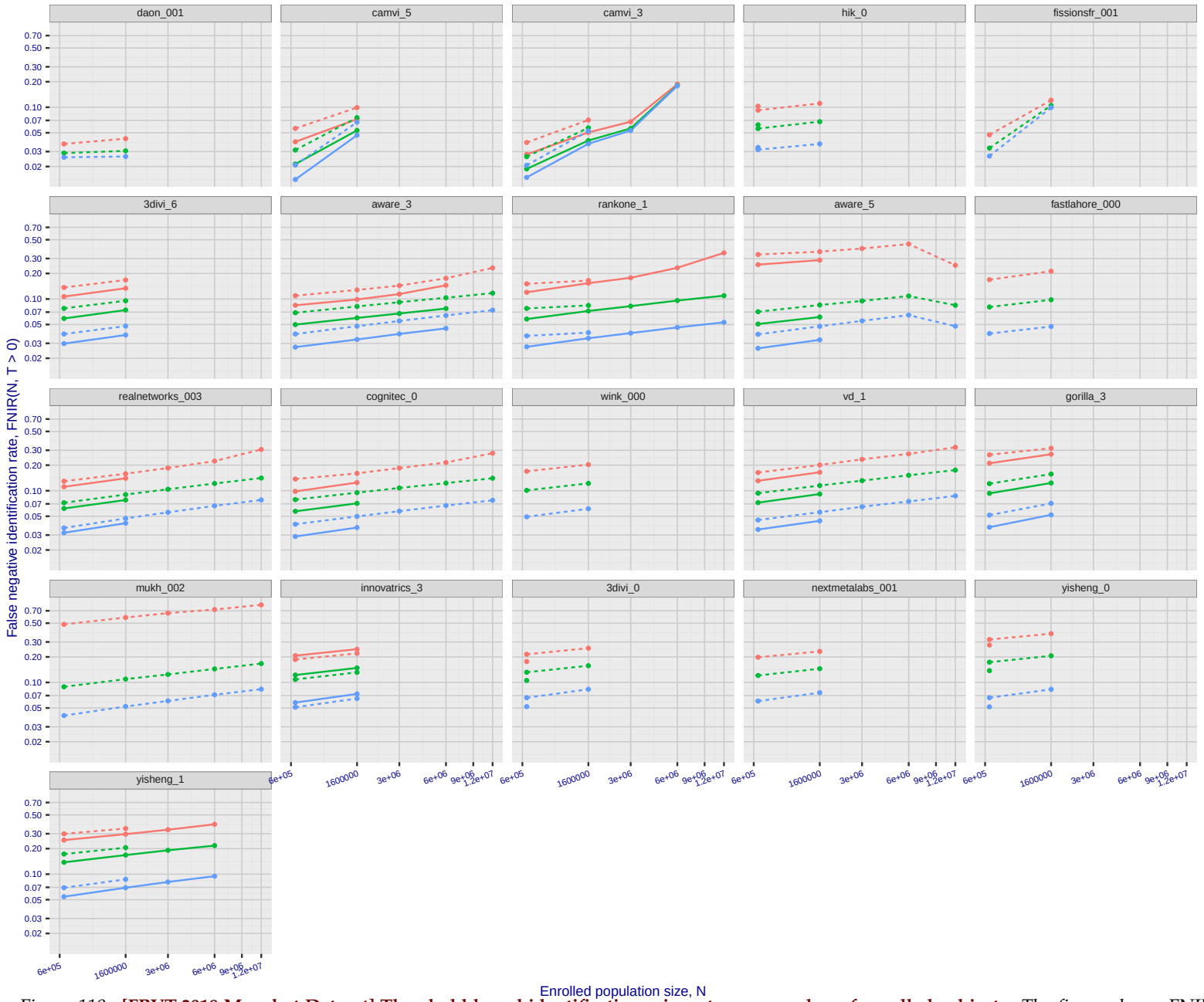

Figure 110: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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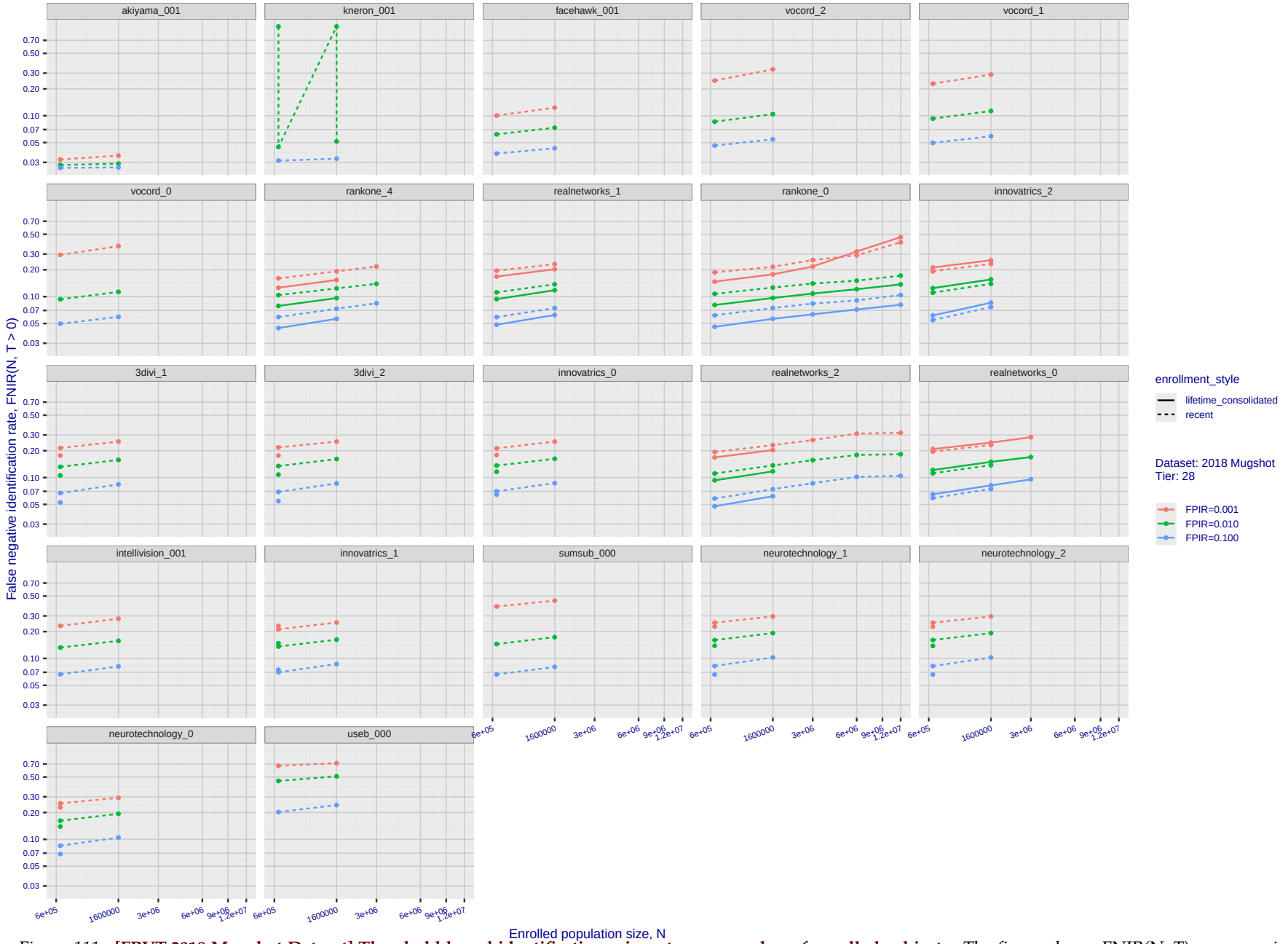

Figure 111: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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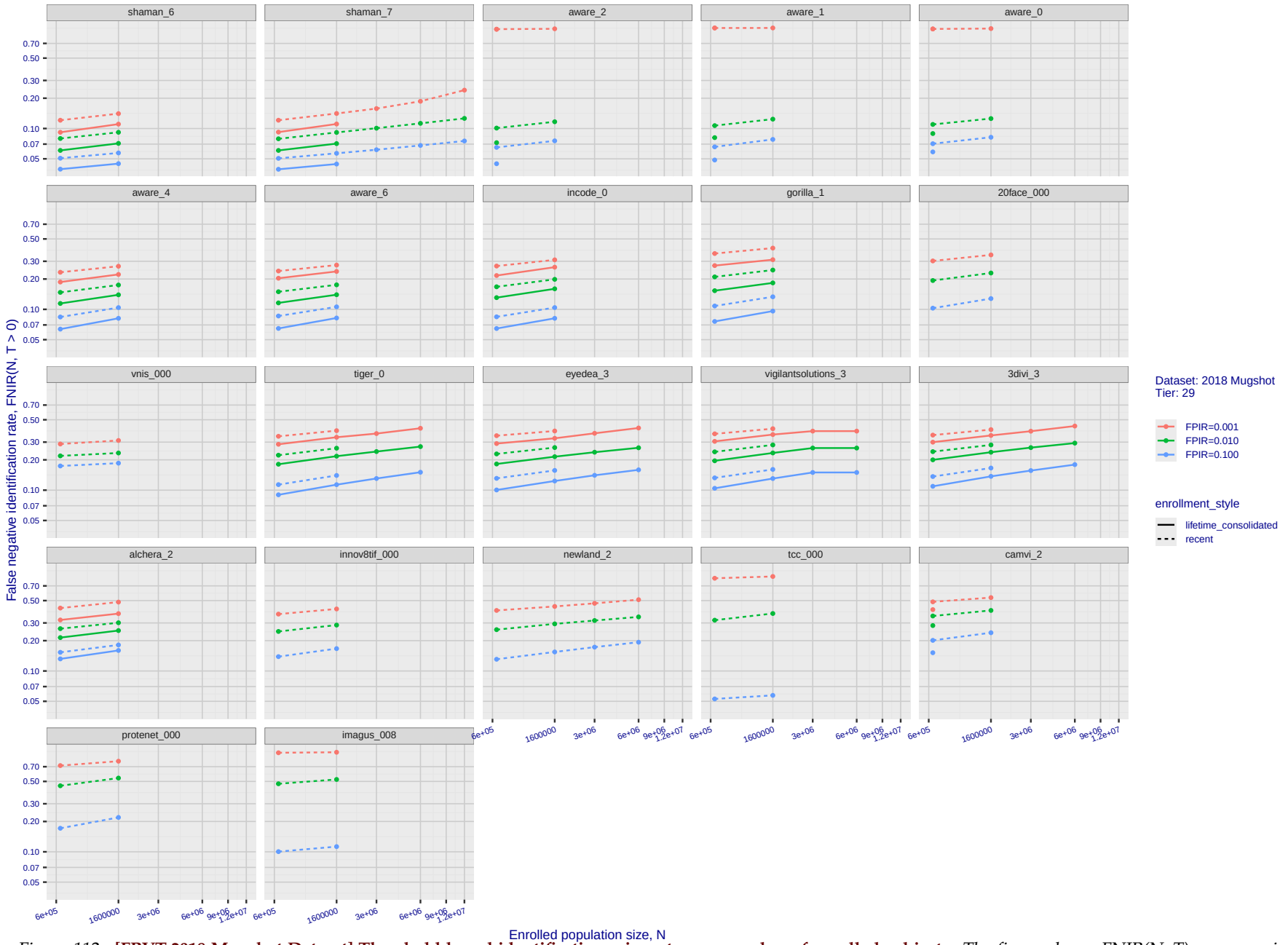

Figure 112: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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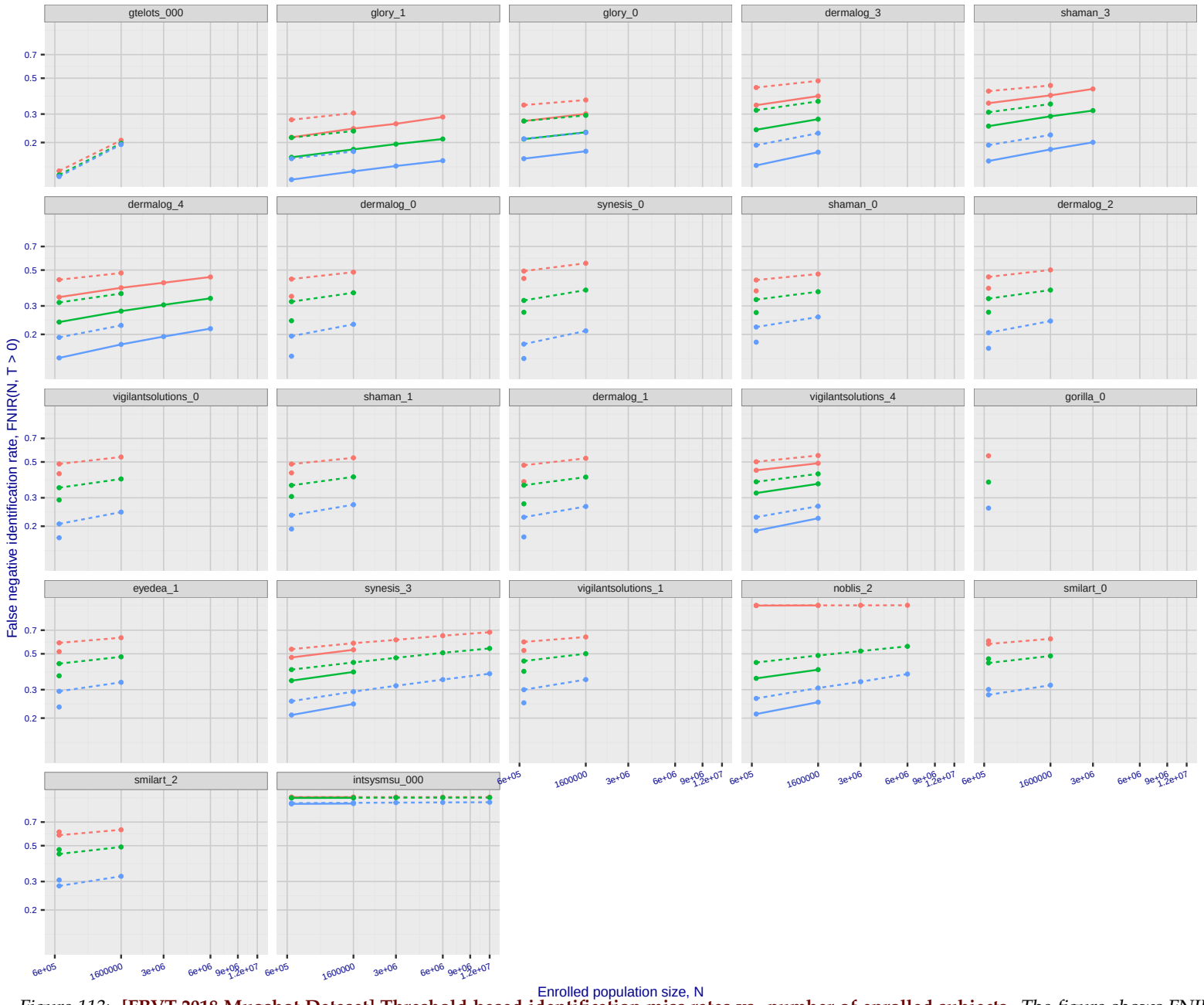

Figure 113: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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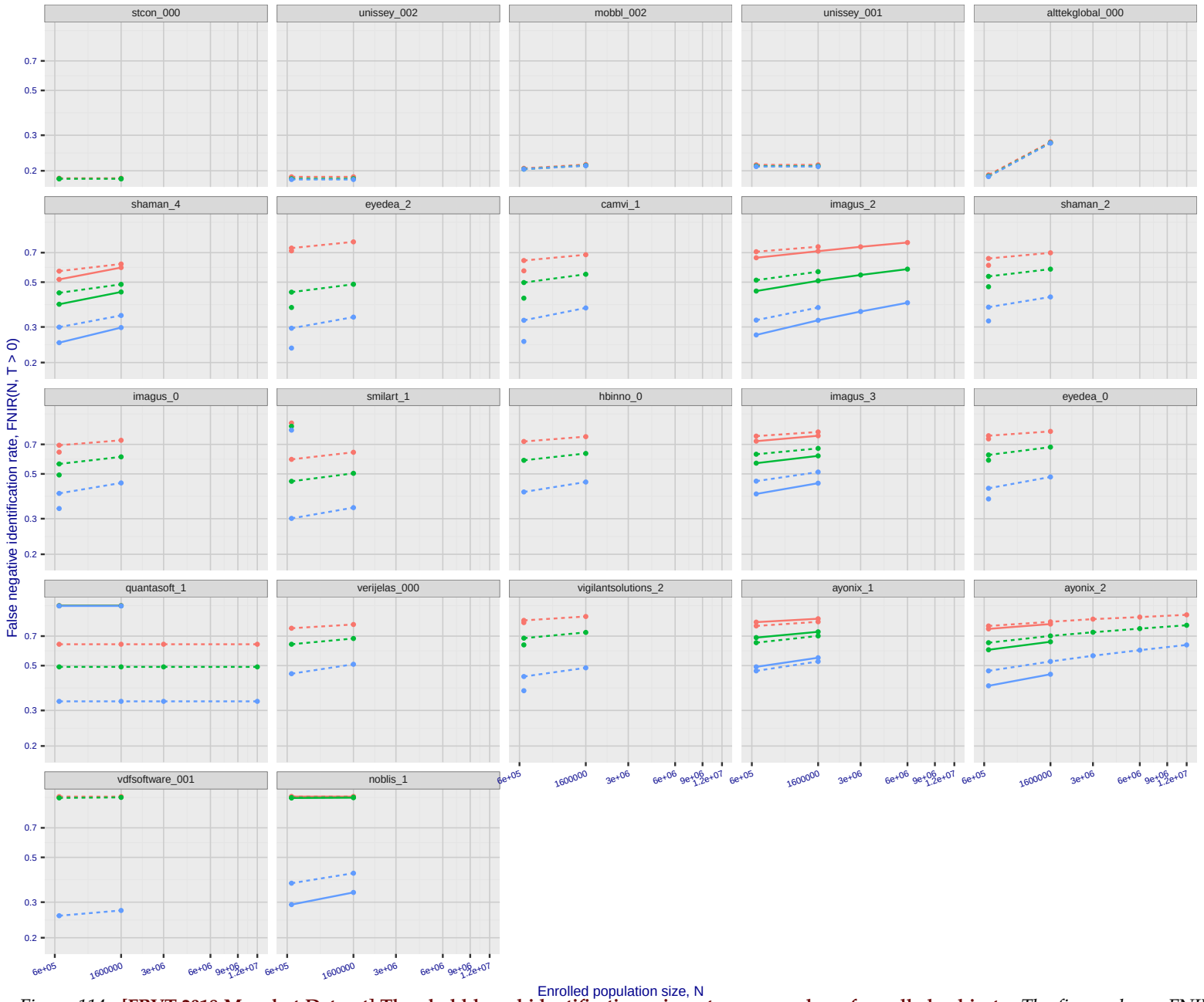

Figure 114: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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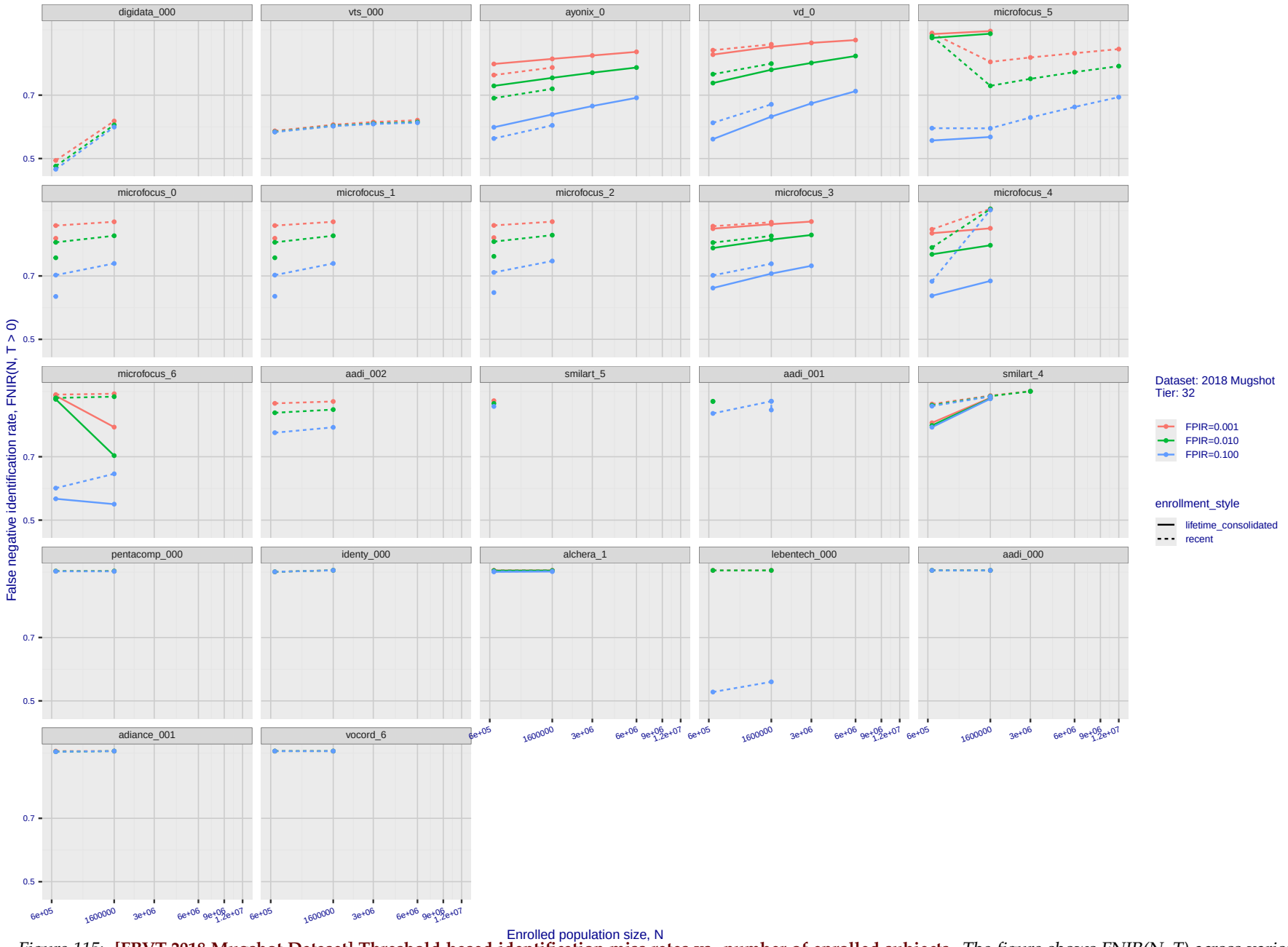

Figure 115: **[FRVT-2018 Mugshot Dataset] Threshold-based identification miss rates vs. number of enrolled subjects**. *The figure shows FNIR(N, T) across various gallery sizes when the threshold is set to achieve the given FPIRs. The rank criterion is irrelevant at high thresholds as mates are always at rank 1. The results are computed from the trials listed in rows 1-10 of Table 1. Less accurate algorithms were not run on large N, so results are missing. For clarity, results are sorted and reported into tiers spanning multiple pages. The tiering criteria is complicated: First paging by FNIR(N_b, 1, 0), then sorting by median FNIR(N_b, T), N_b = 640 000.*

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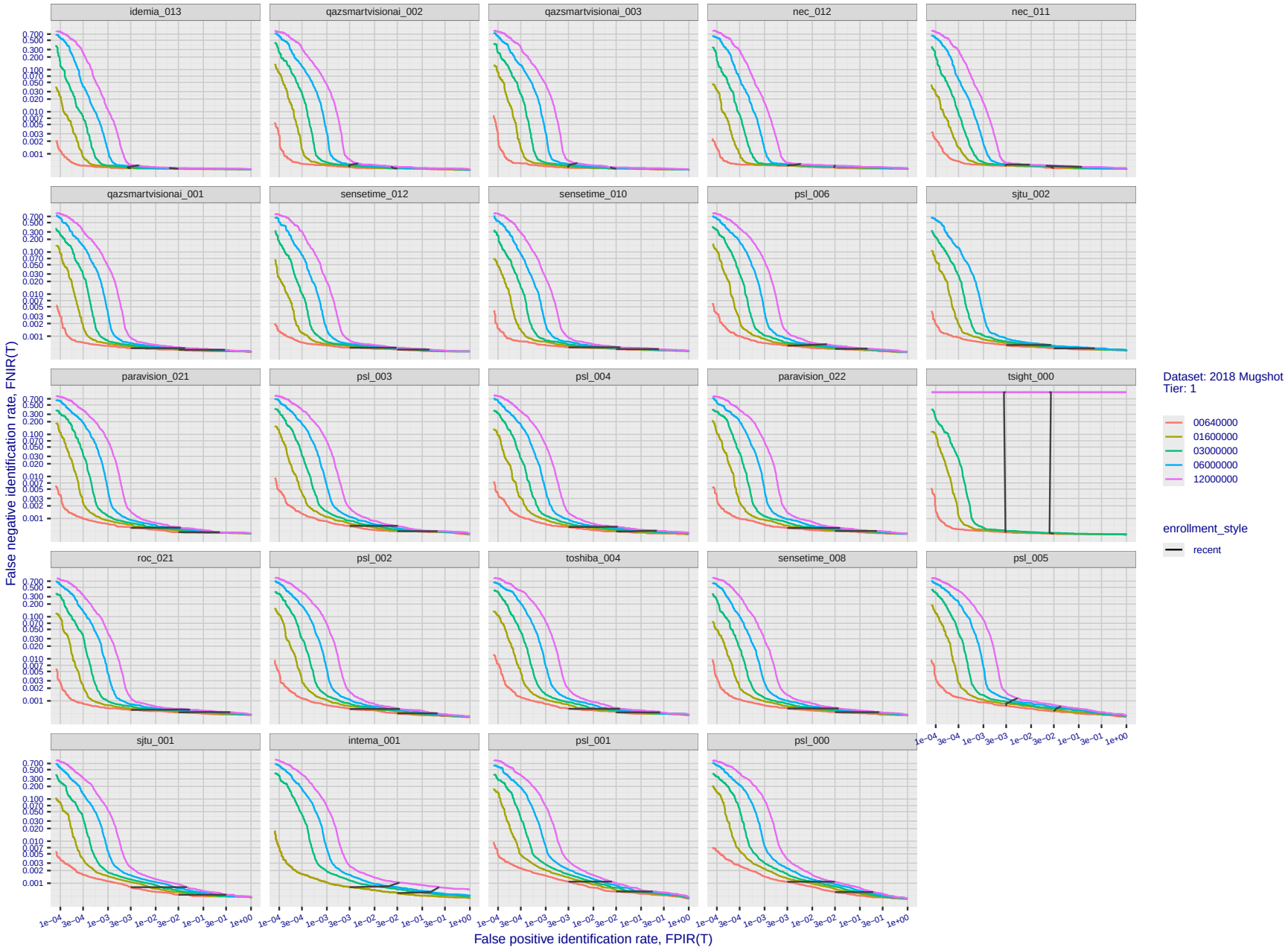

Figure 116: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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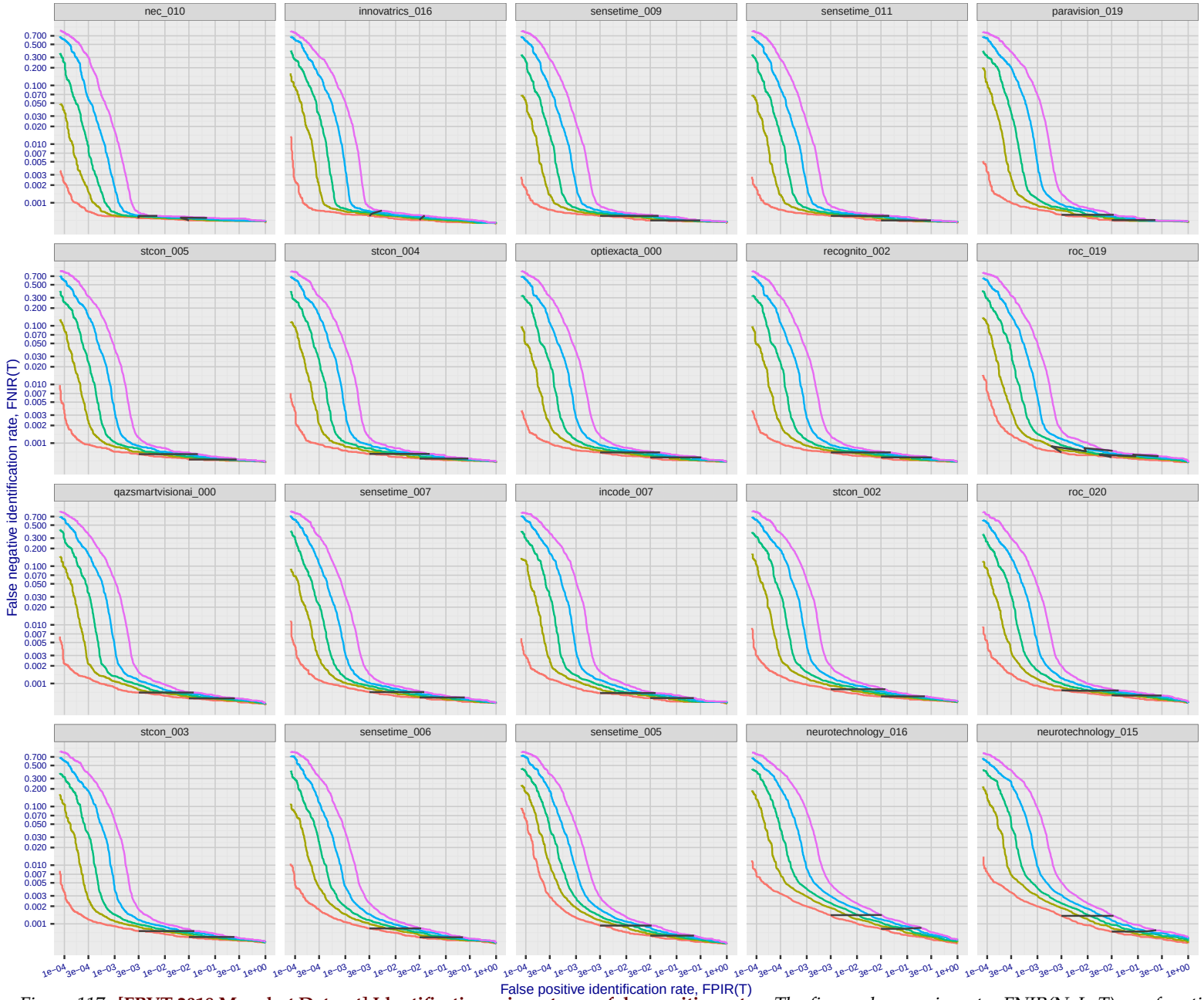

Figure 117: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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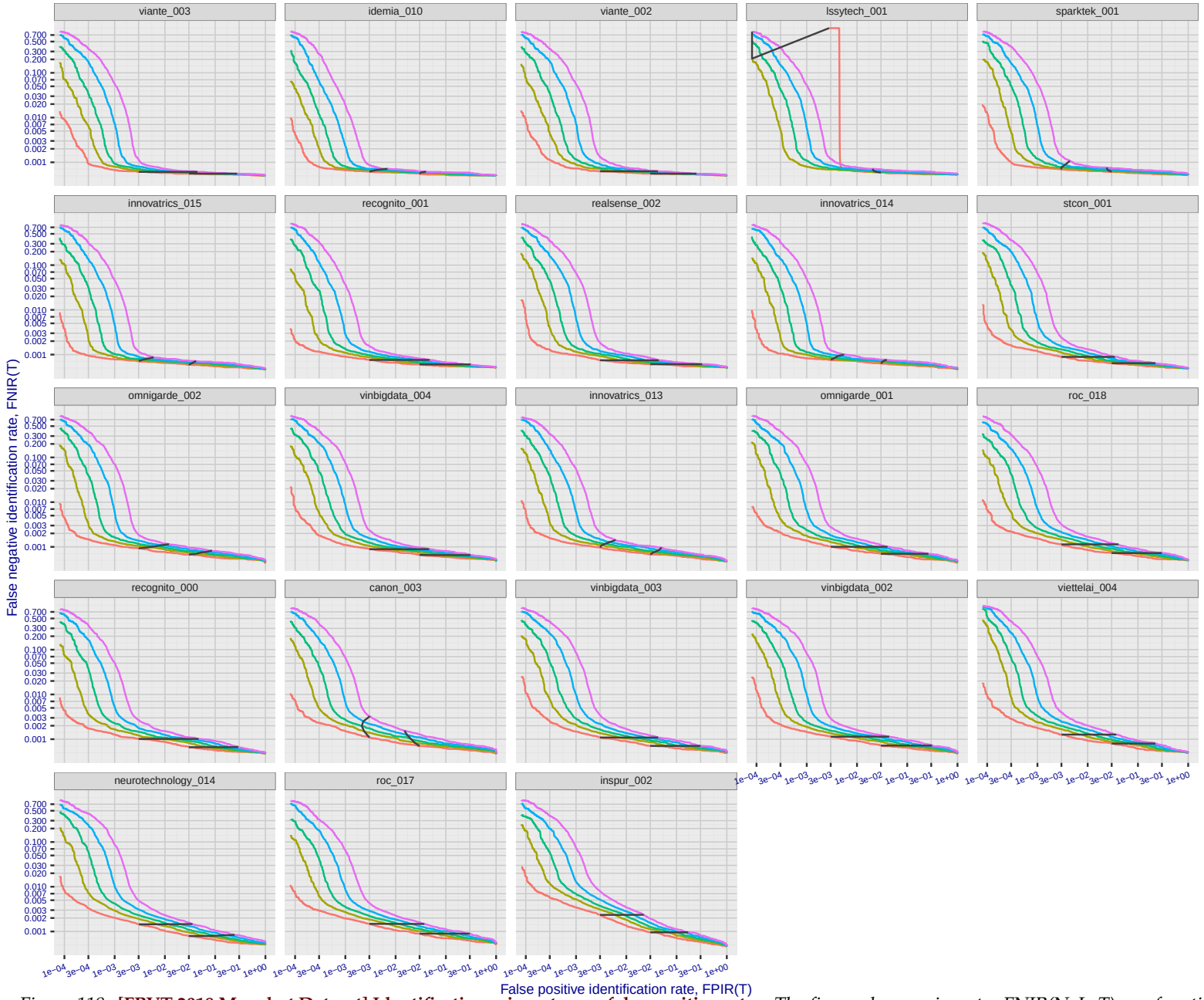

Figure 118: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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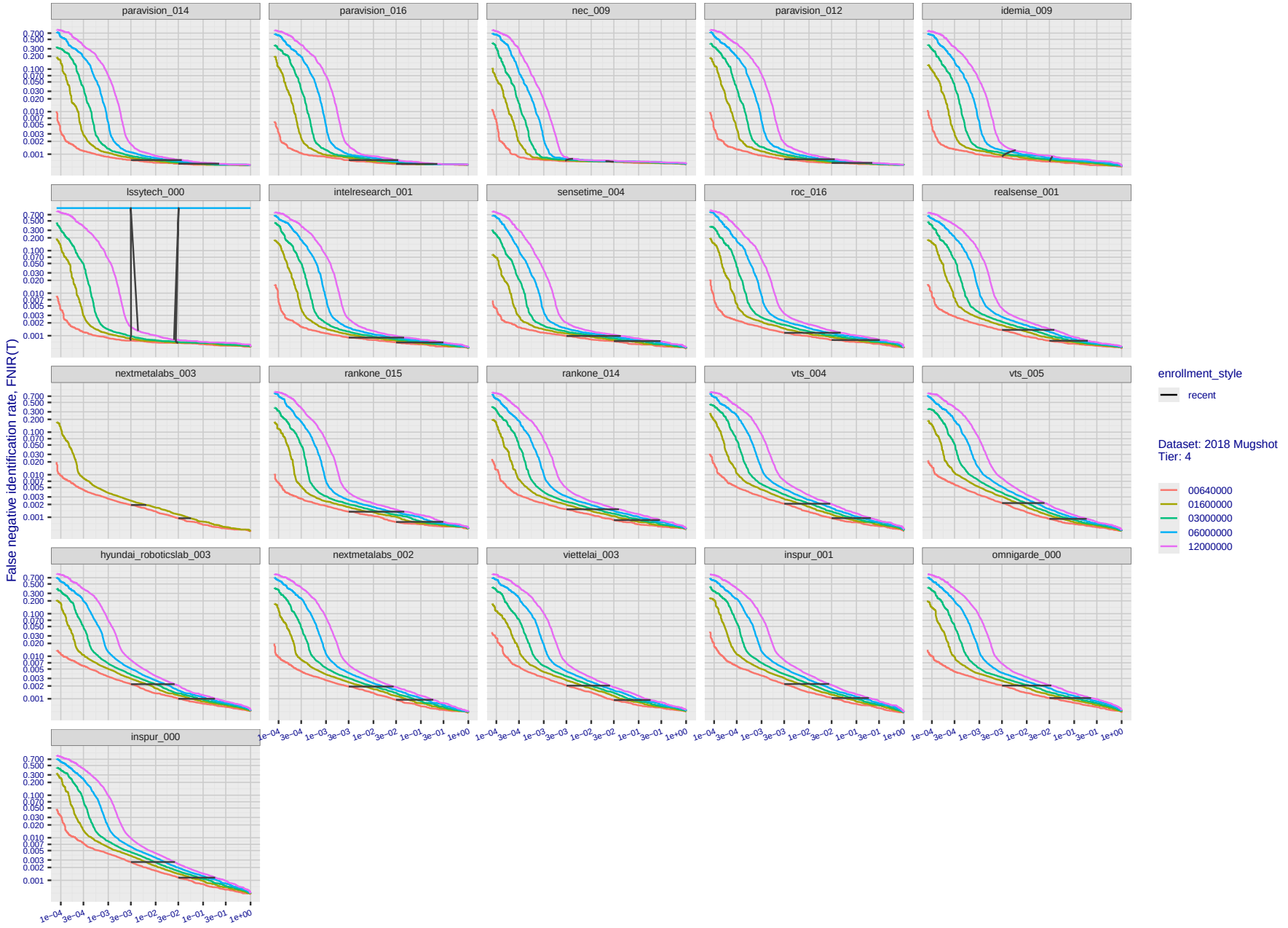

Figure 119: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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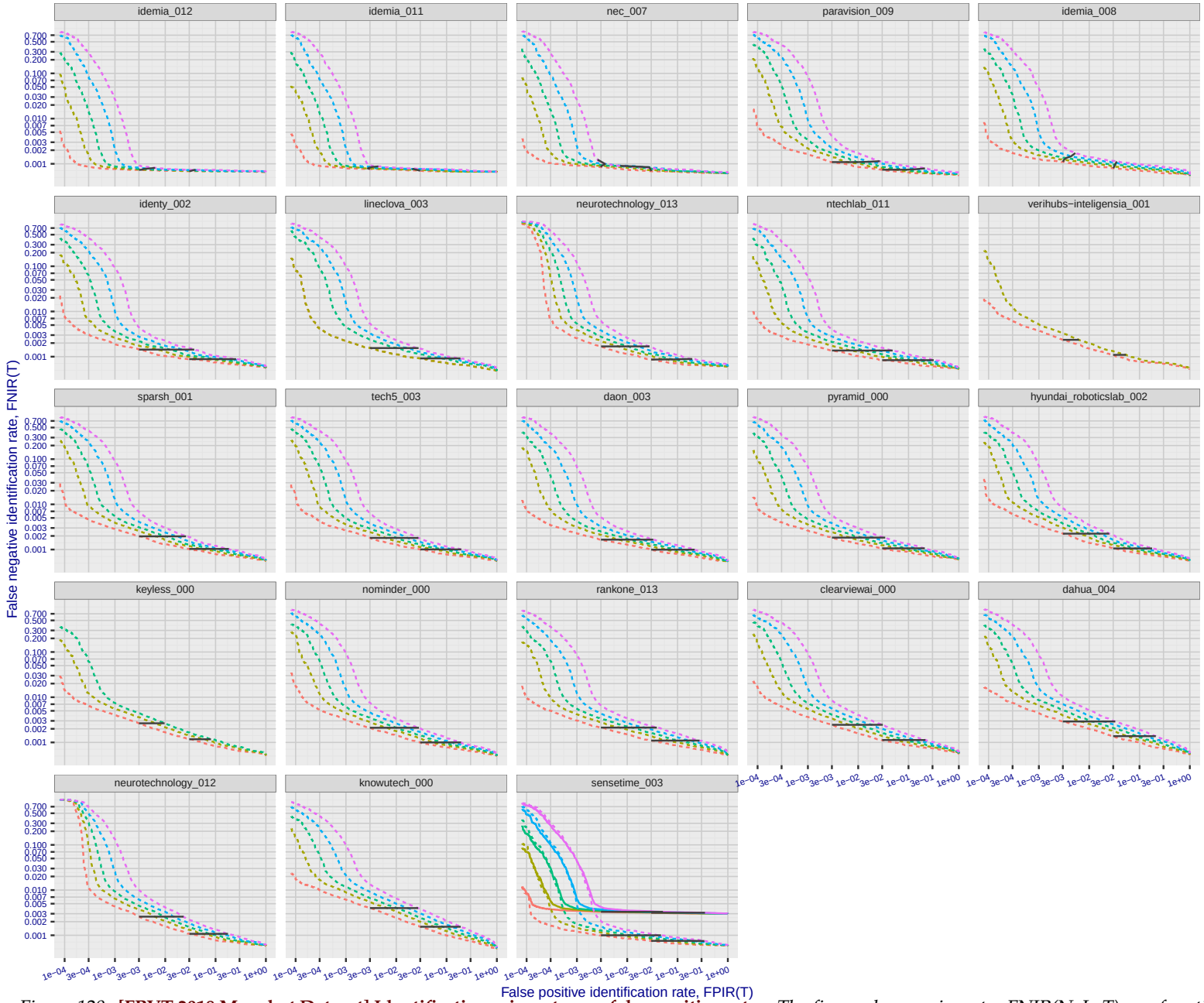

Figure 120: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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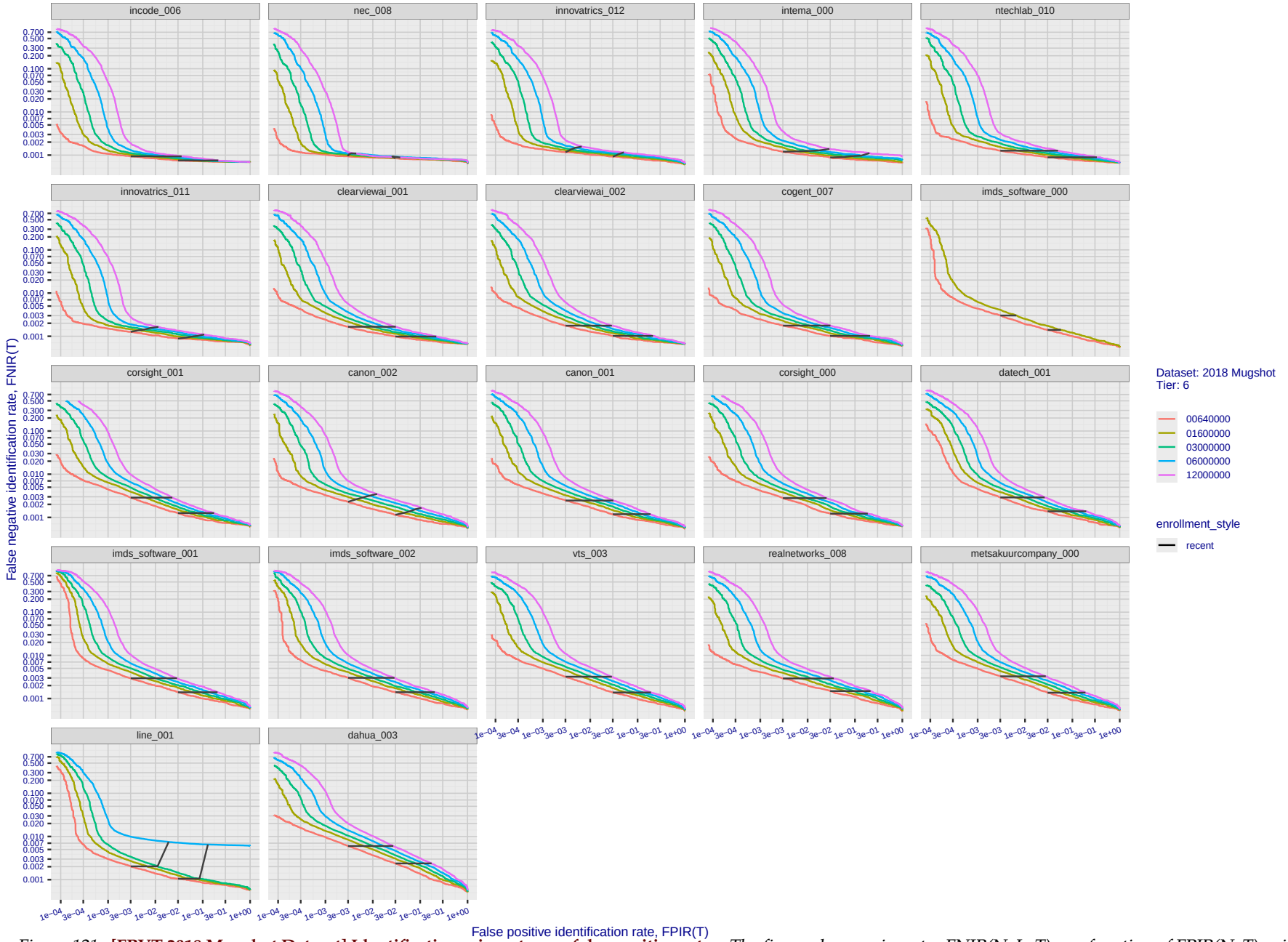

Figure 121: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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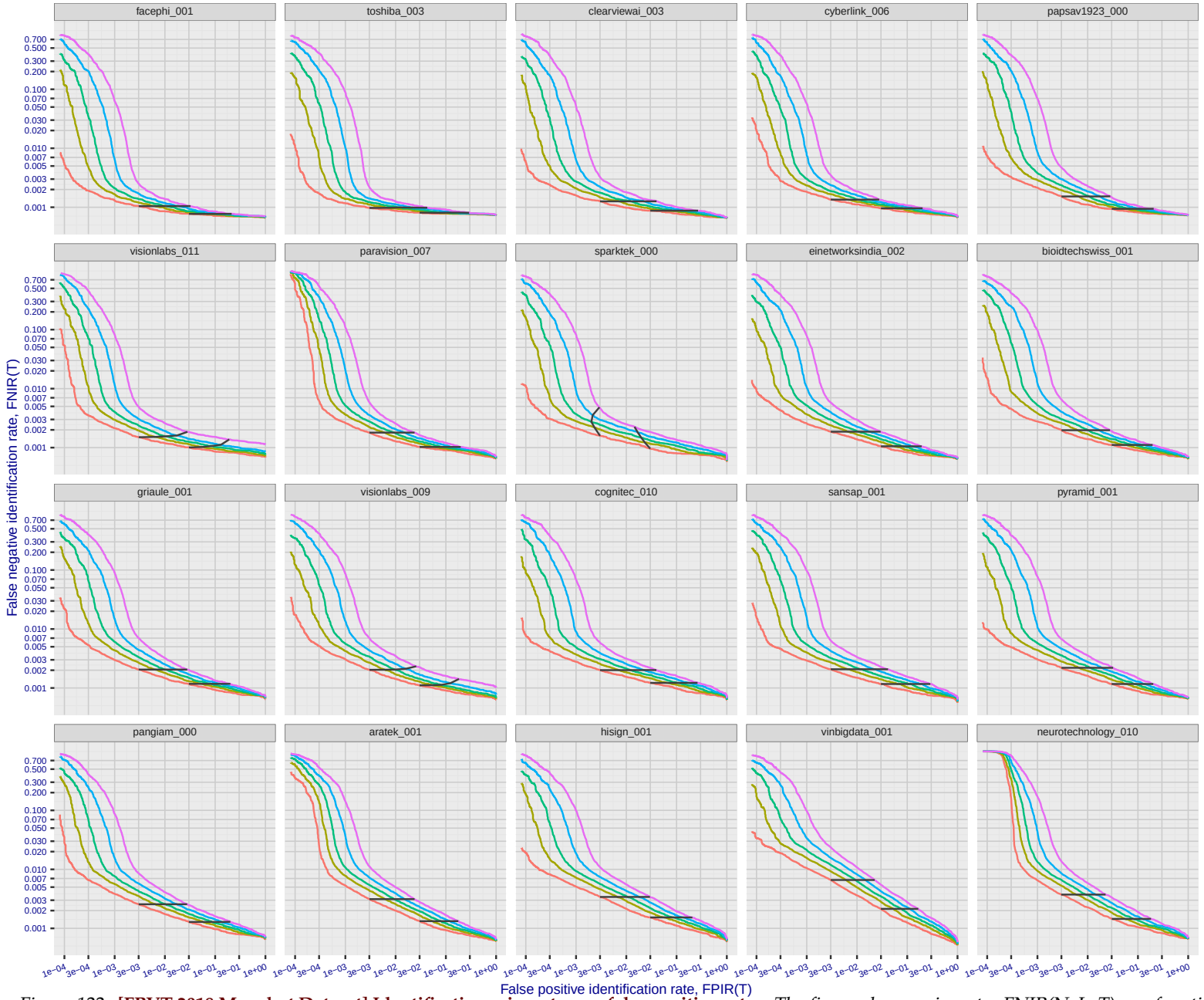

Figure 122: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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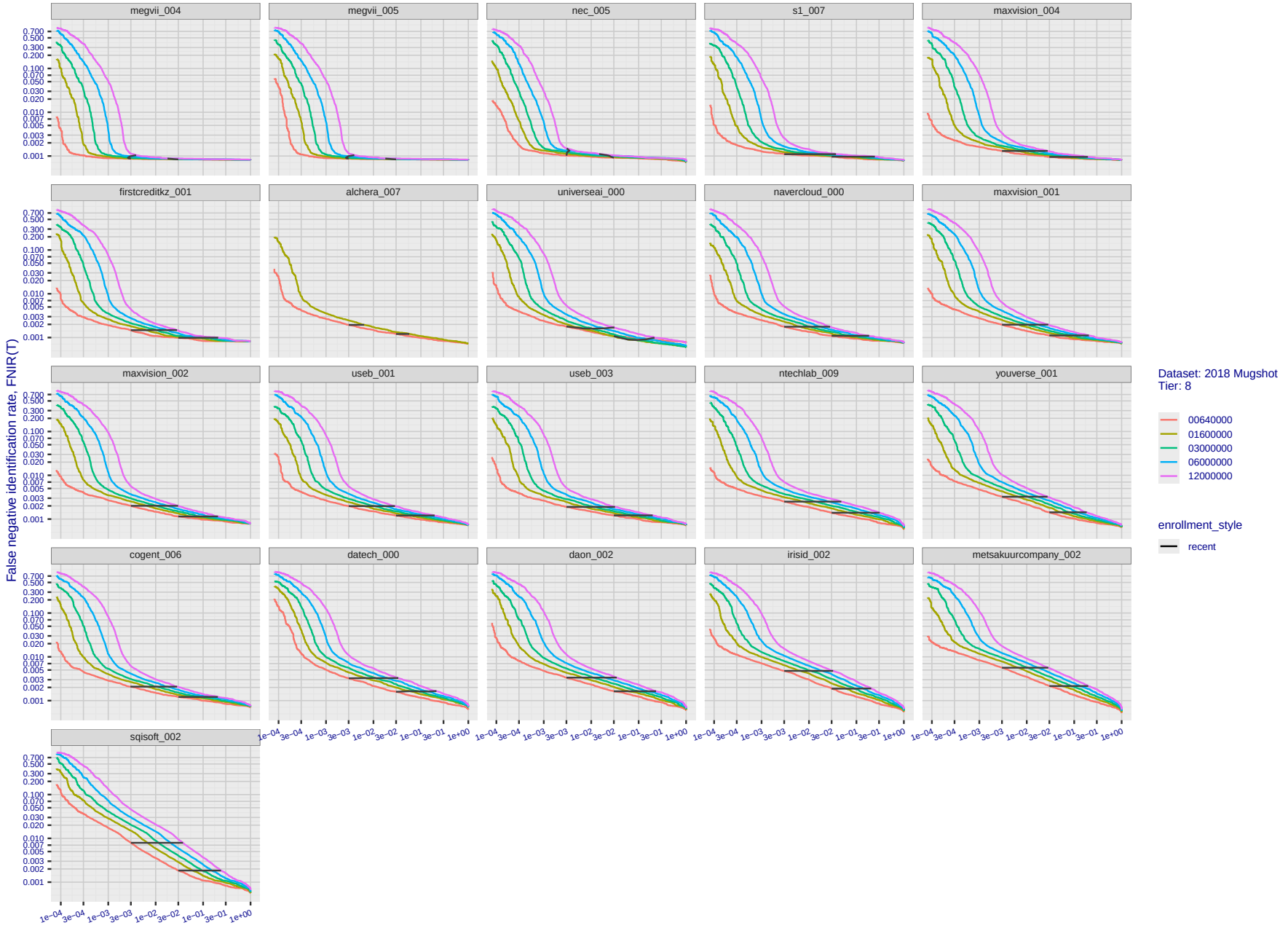

Figure 123: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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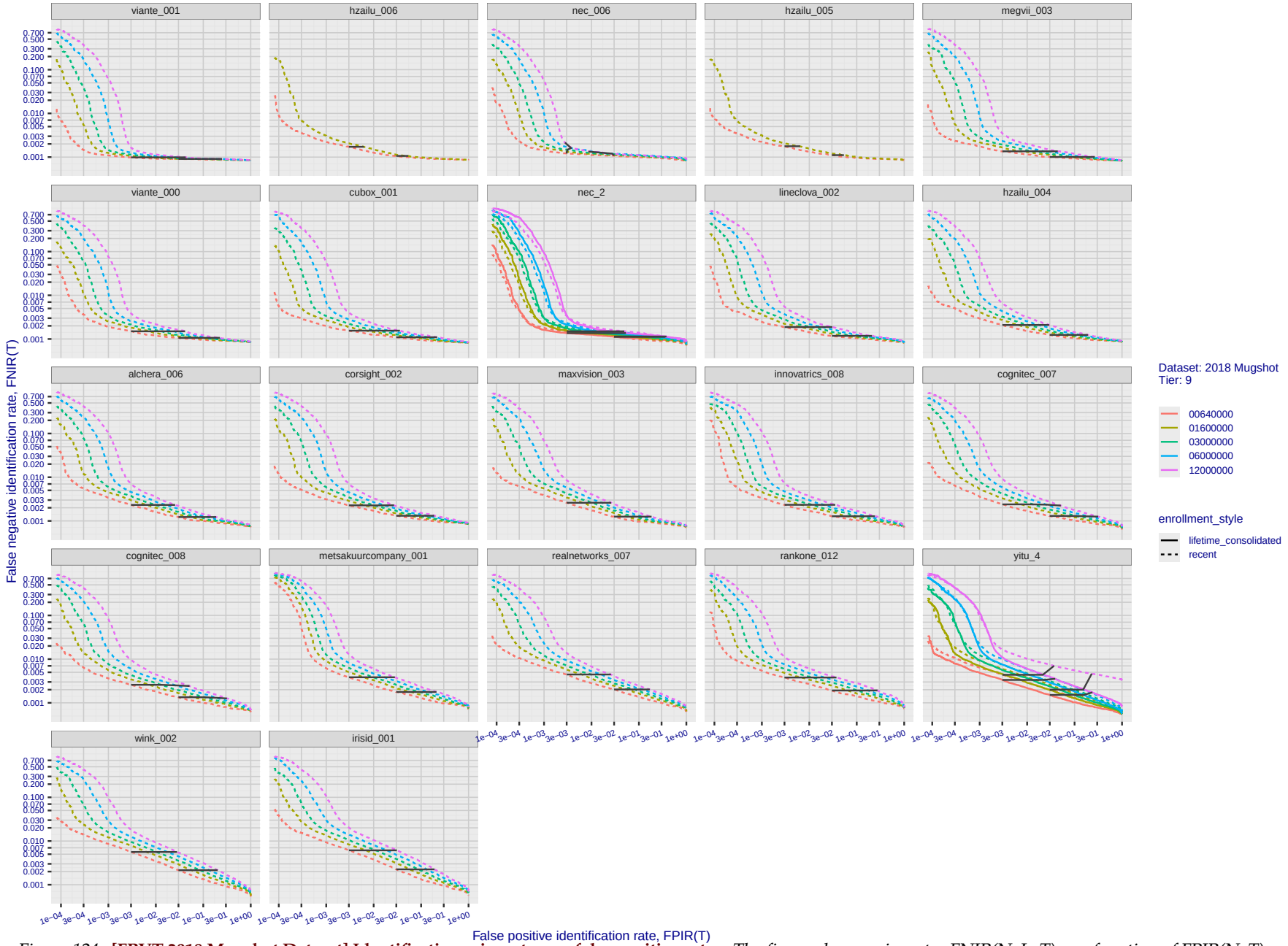

Figure 124: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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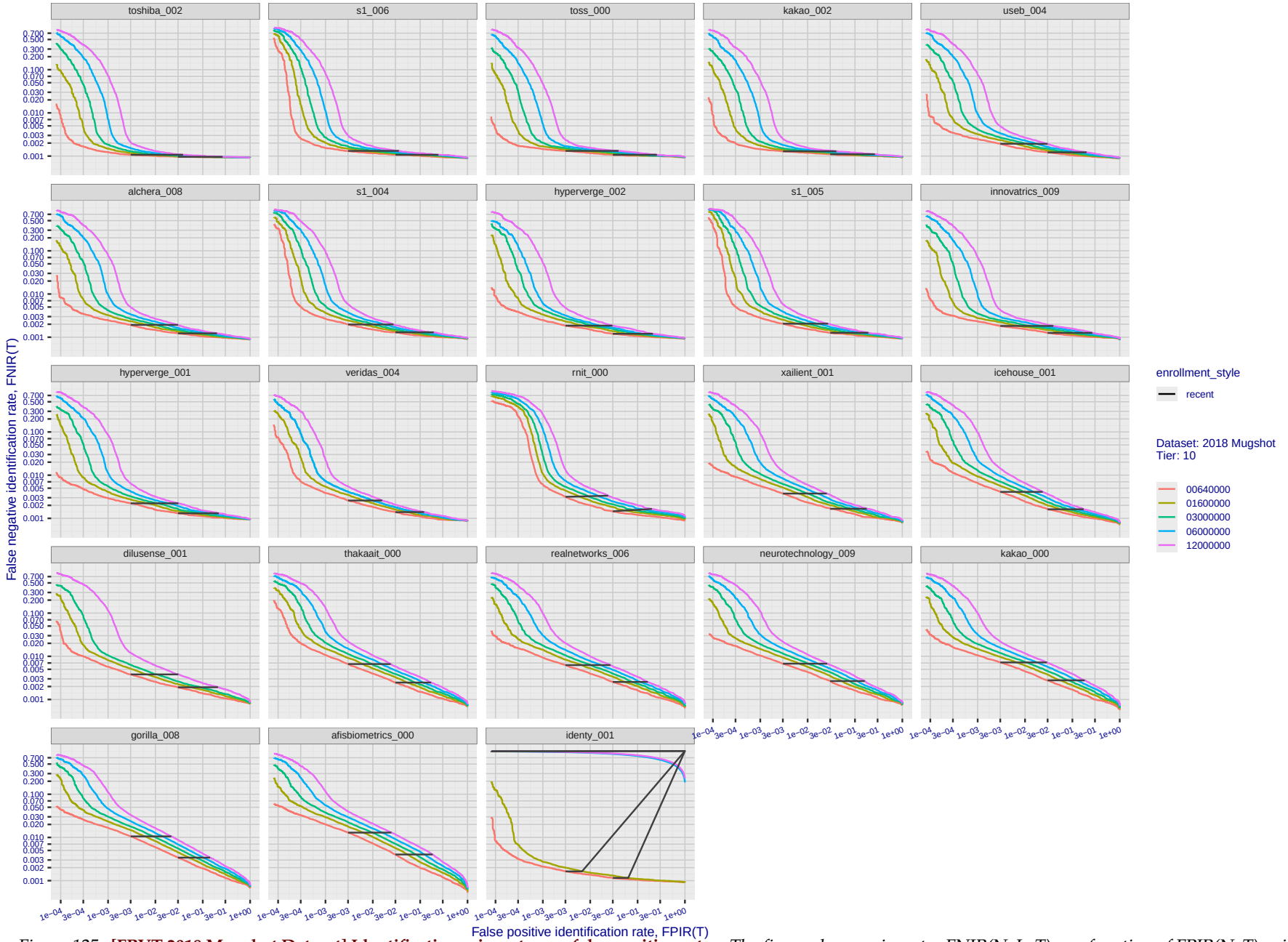

Figure 125: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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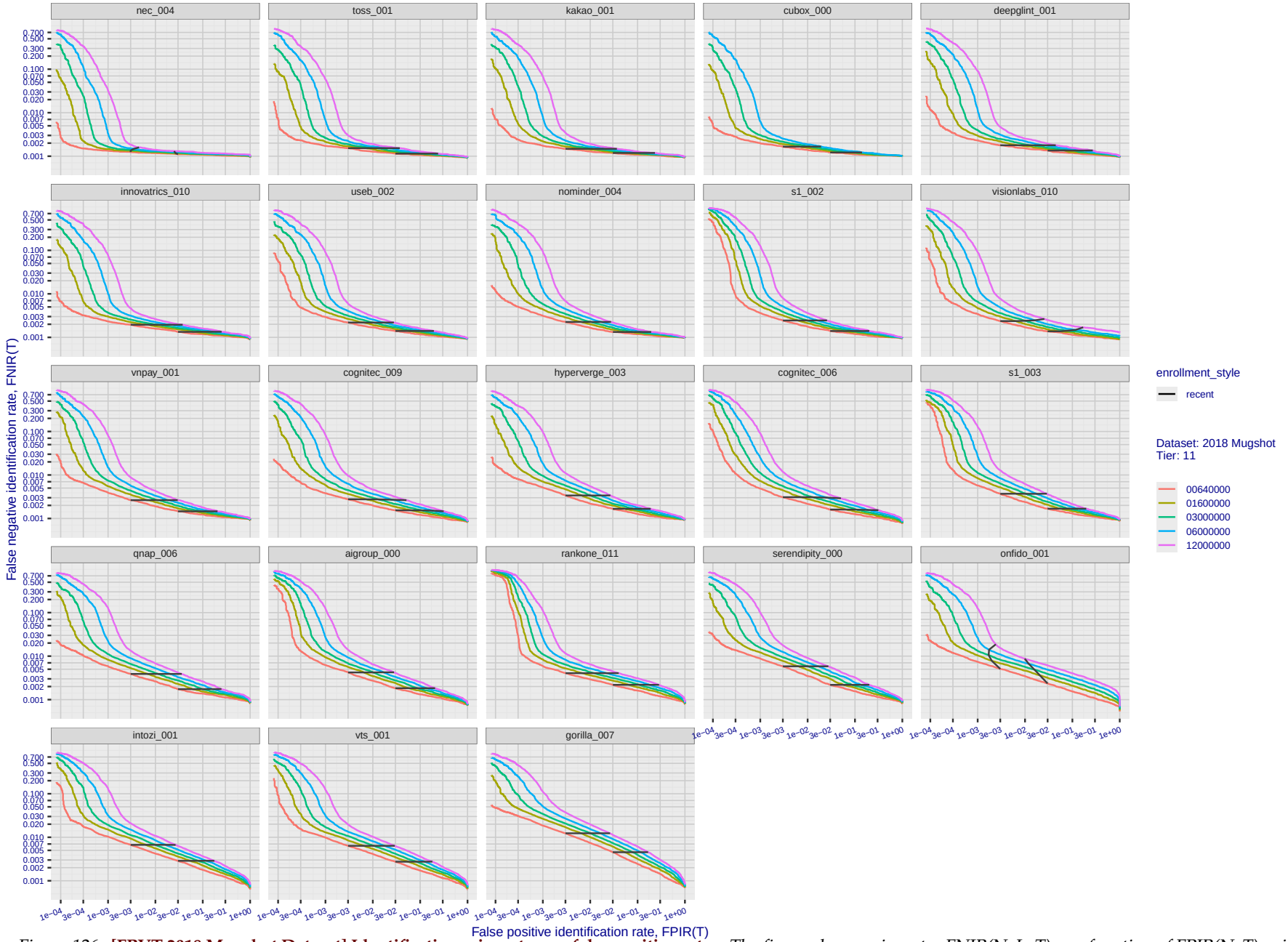

Figure 126: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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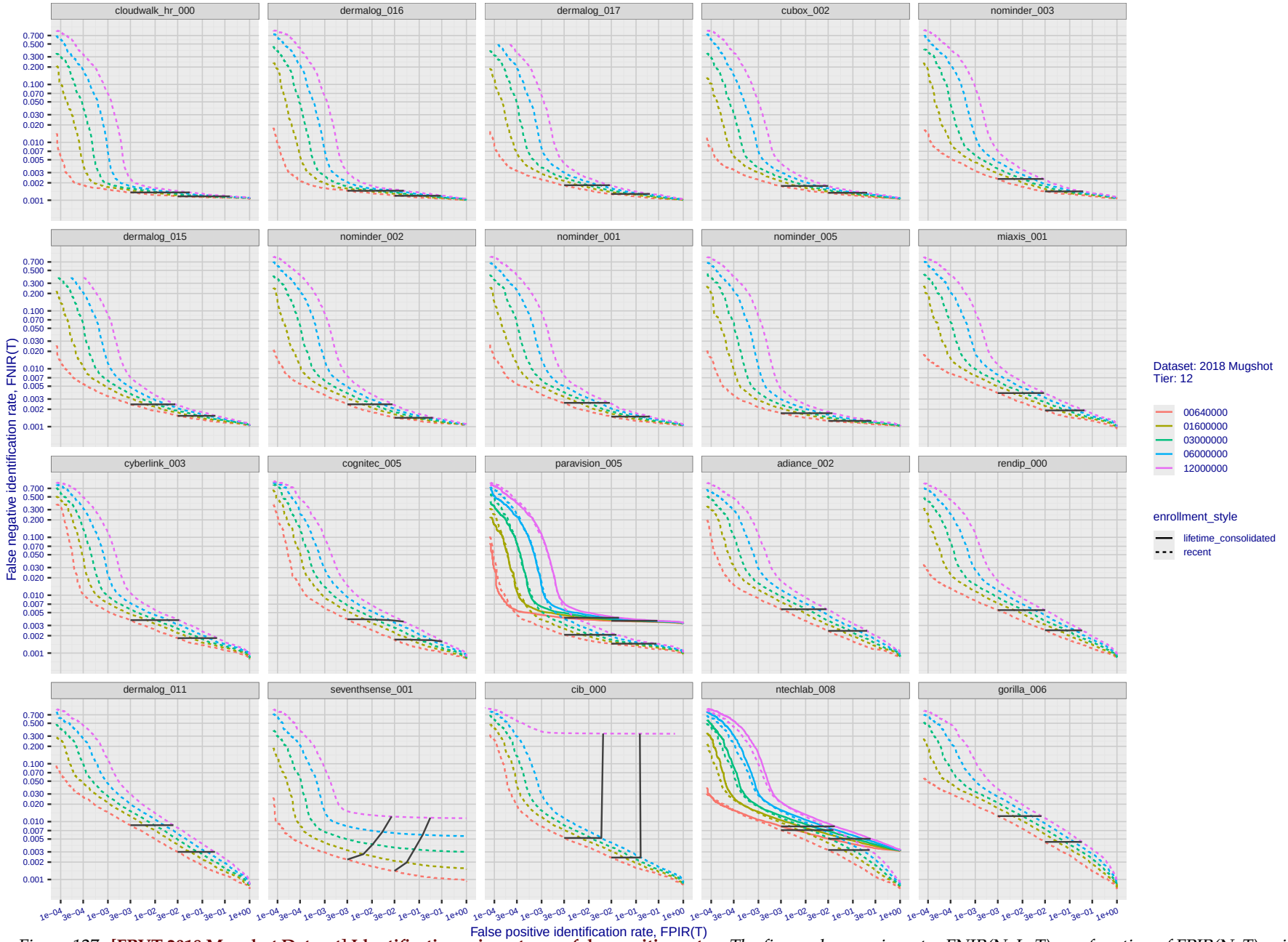

Figure 127: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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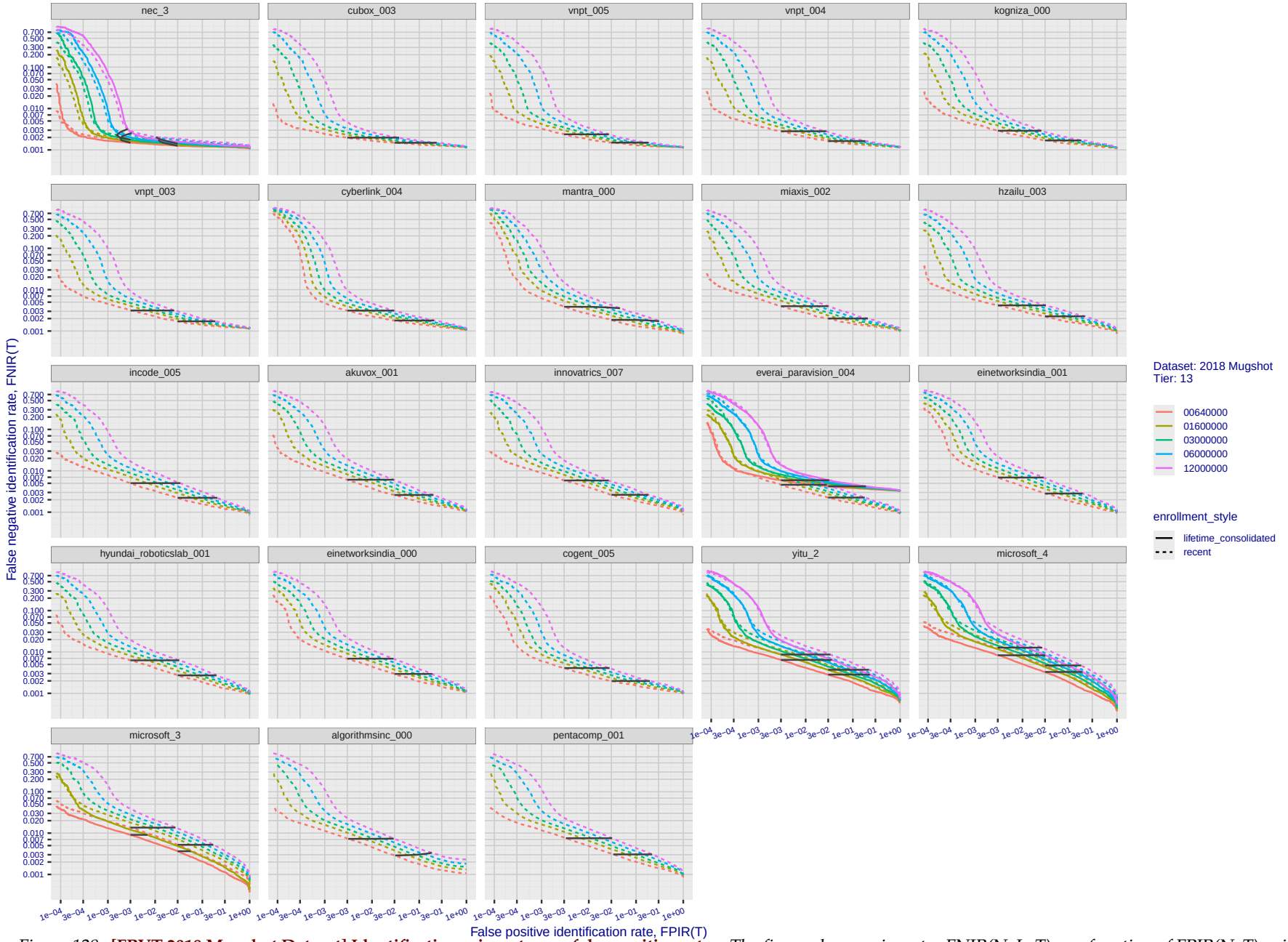

Figure 128: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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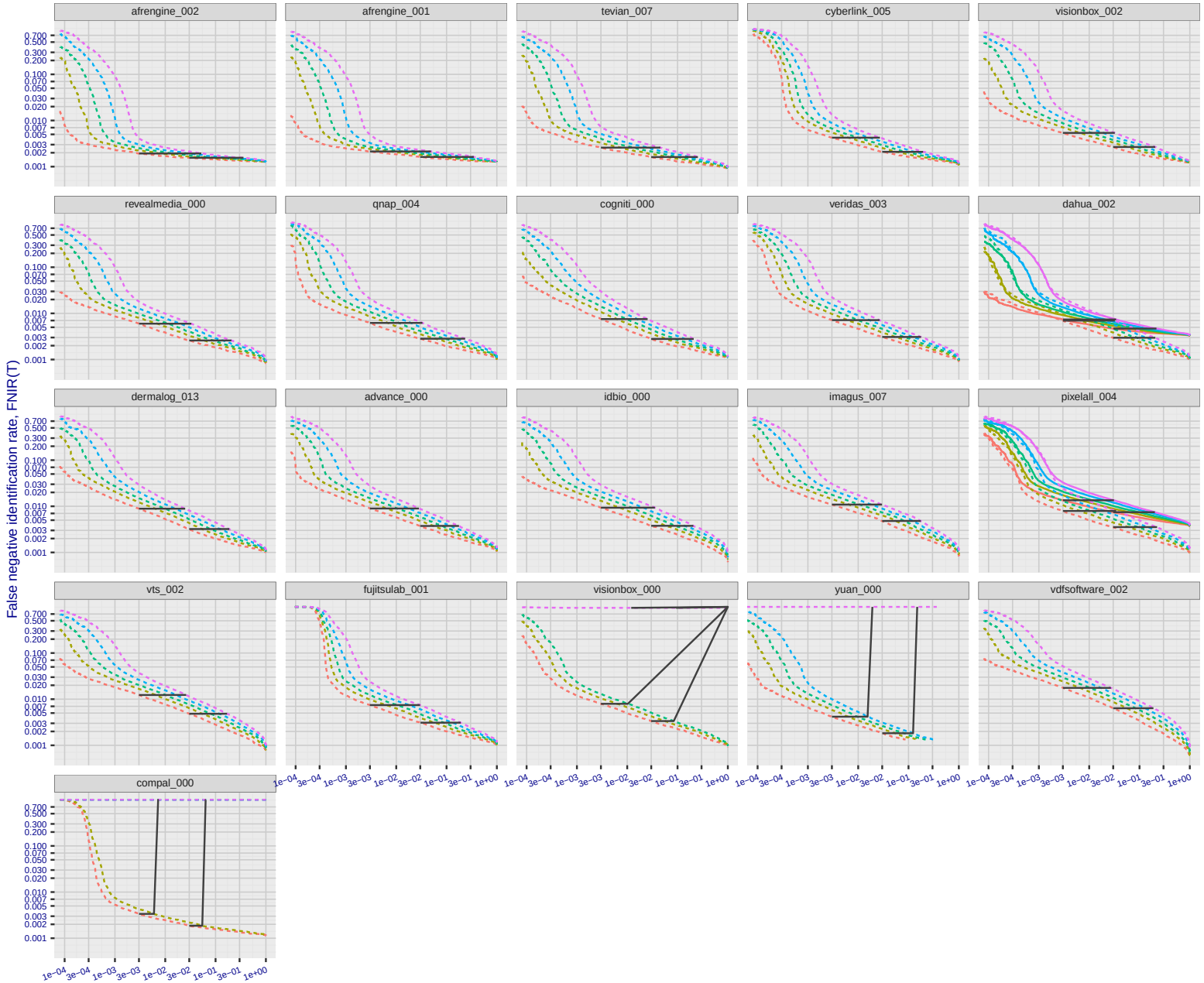

Figure 129: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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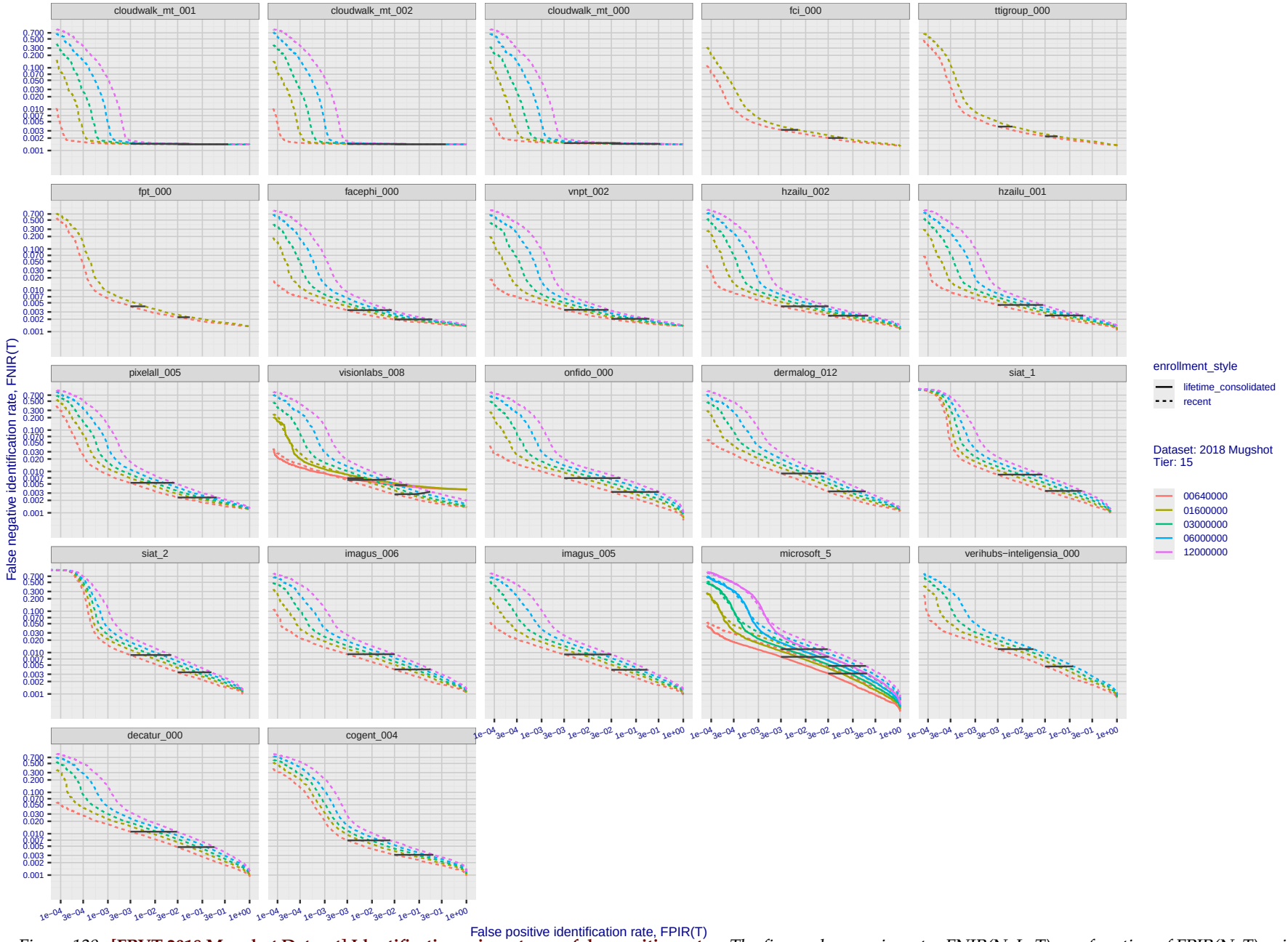

Figure 130: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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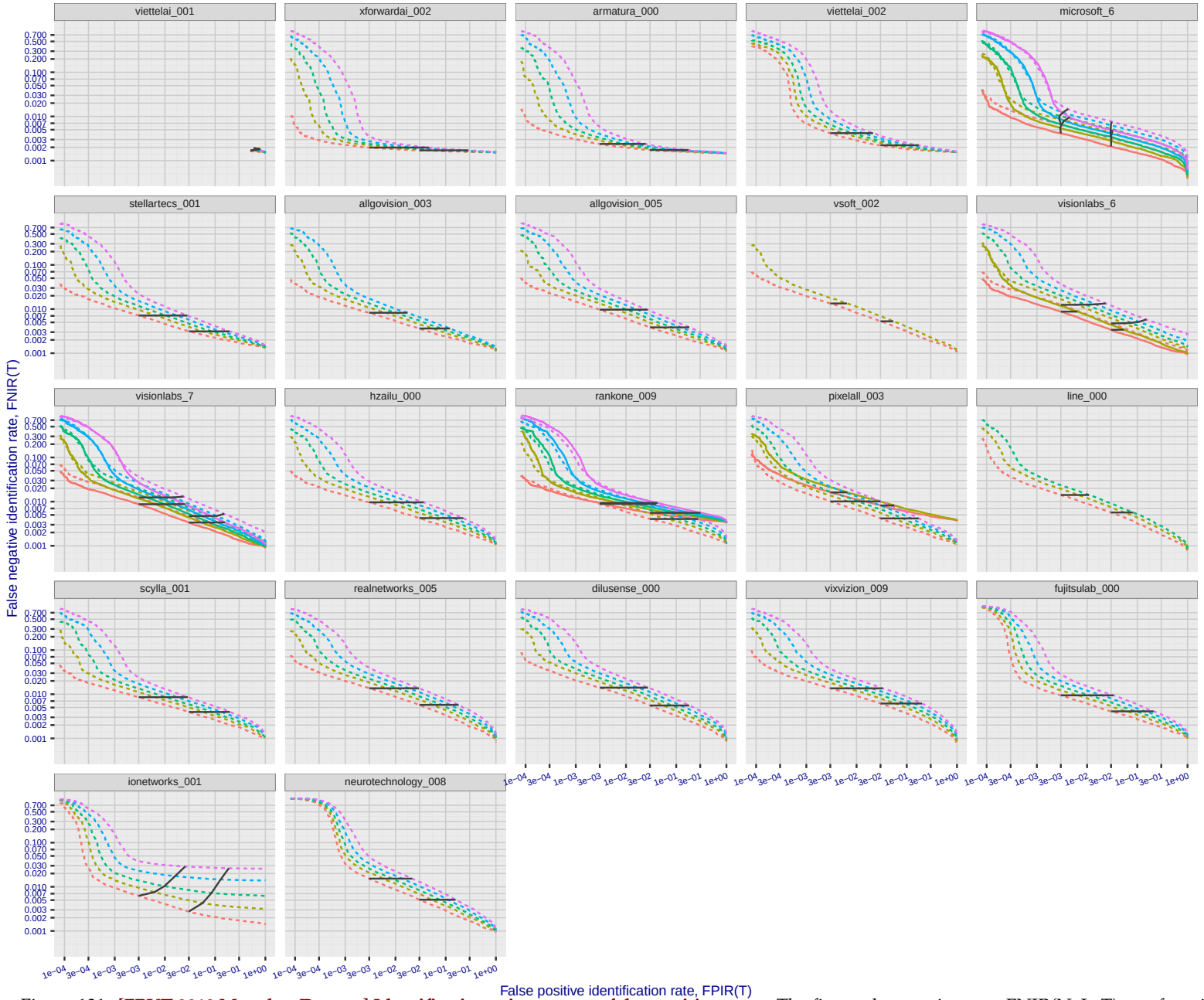

Figure 131: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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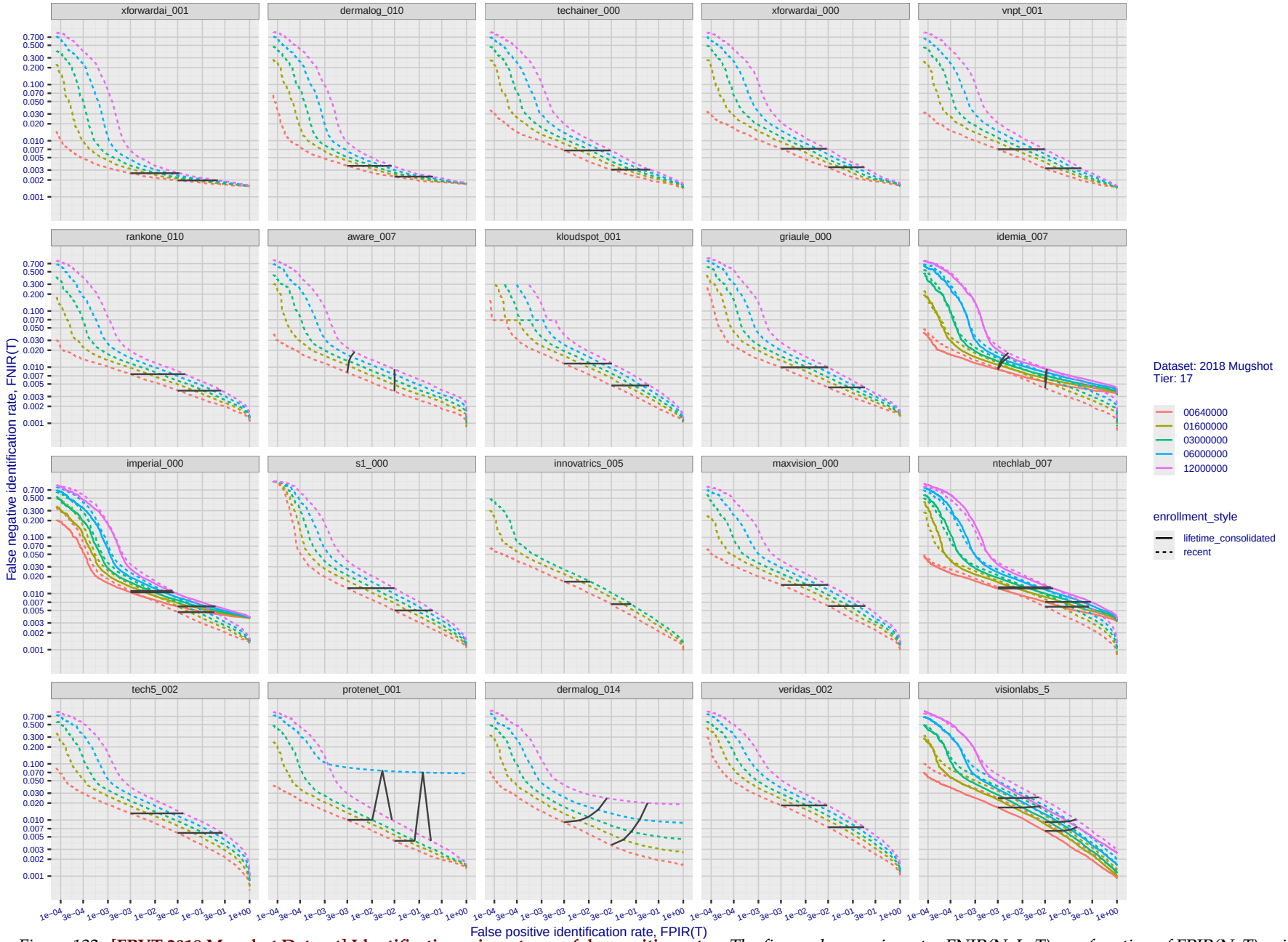

Figure 132: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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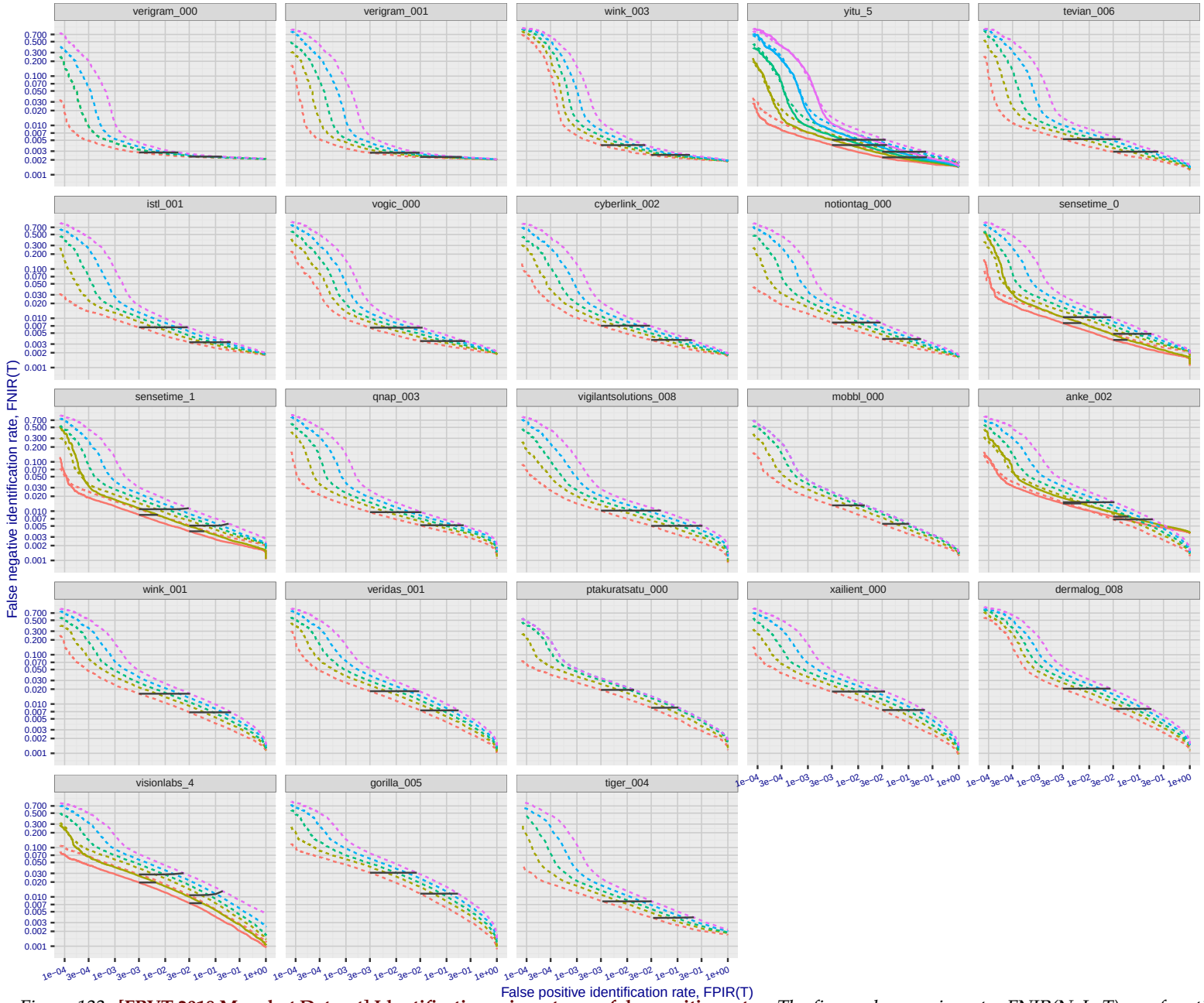

Figure 133: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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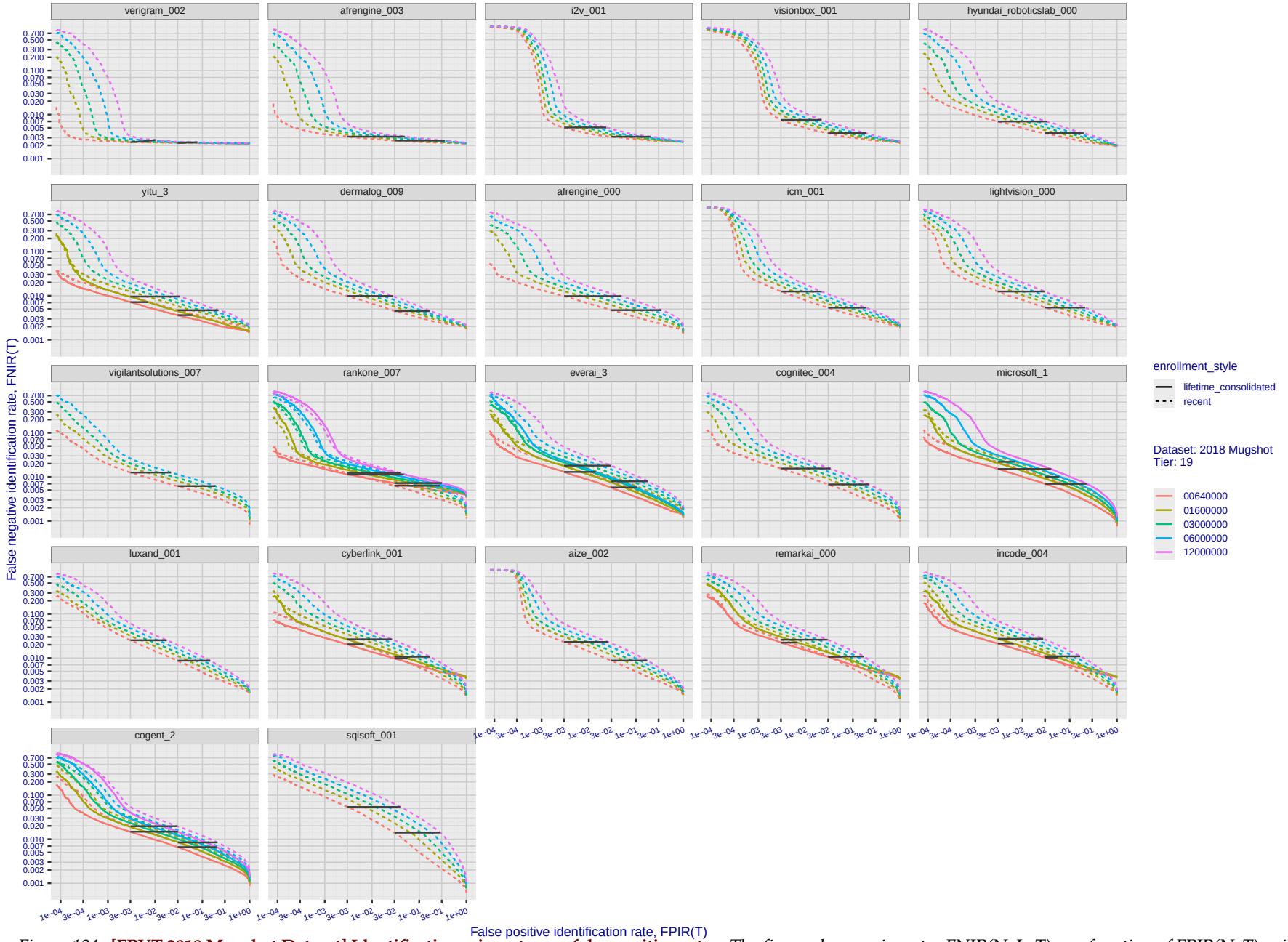

Figure 134: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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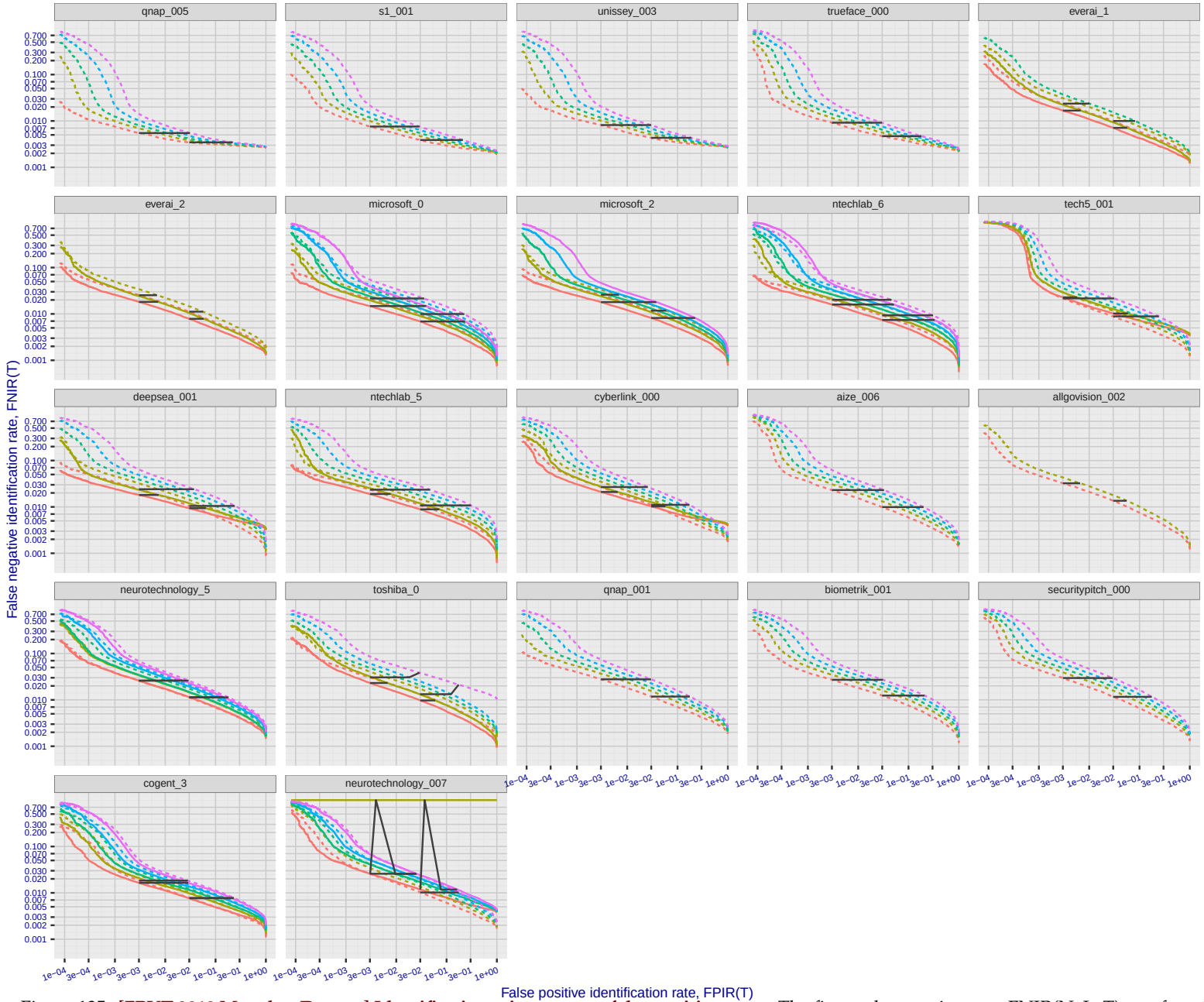

Figure 135: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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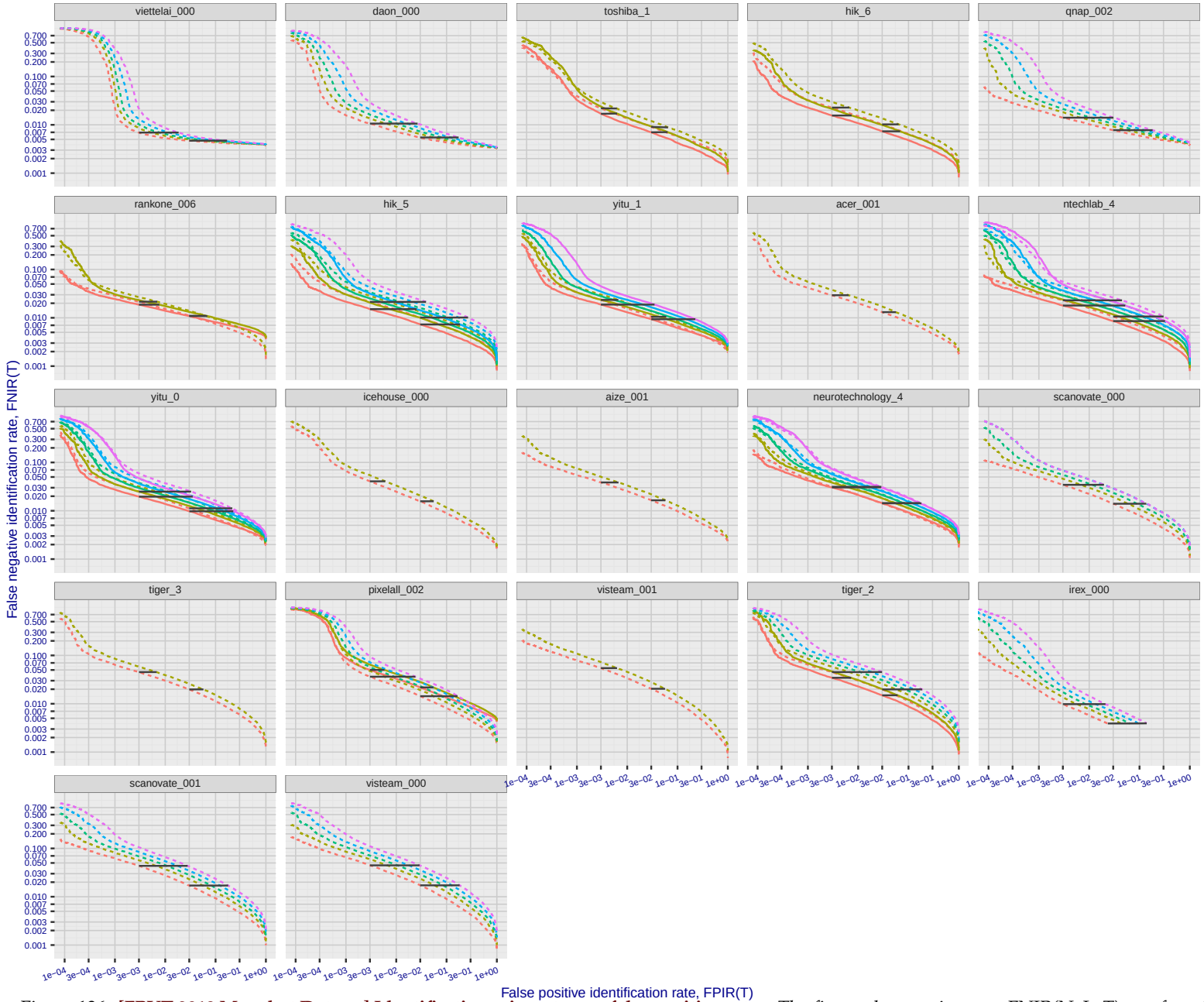

Figure 136: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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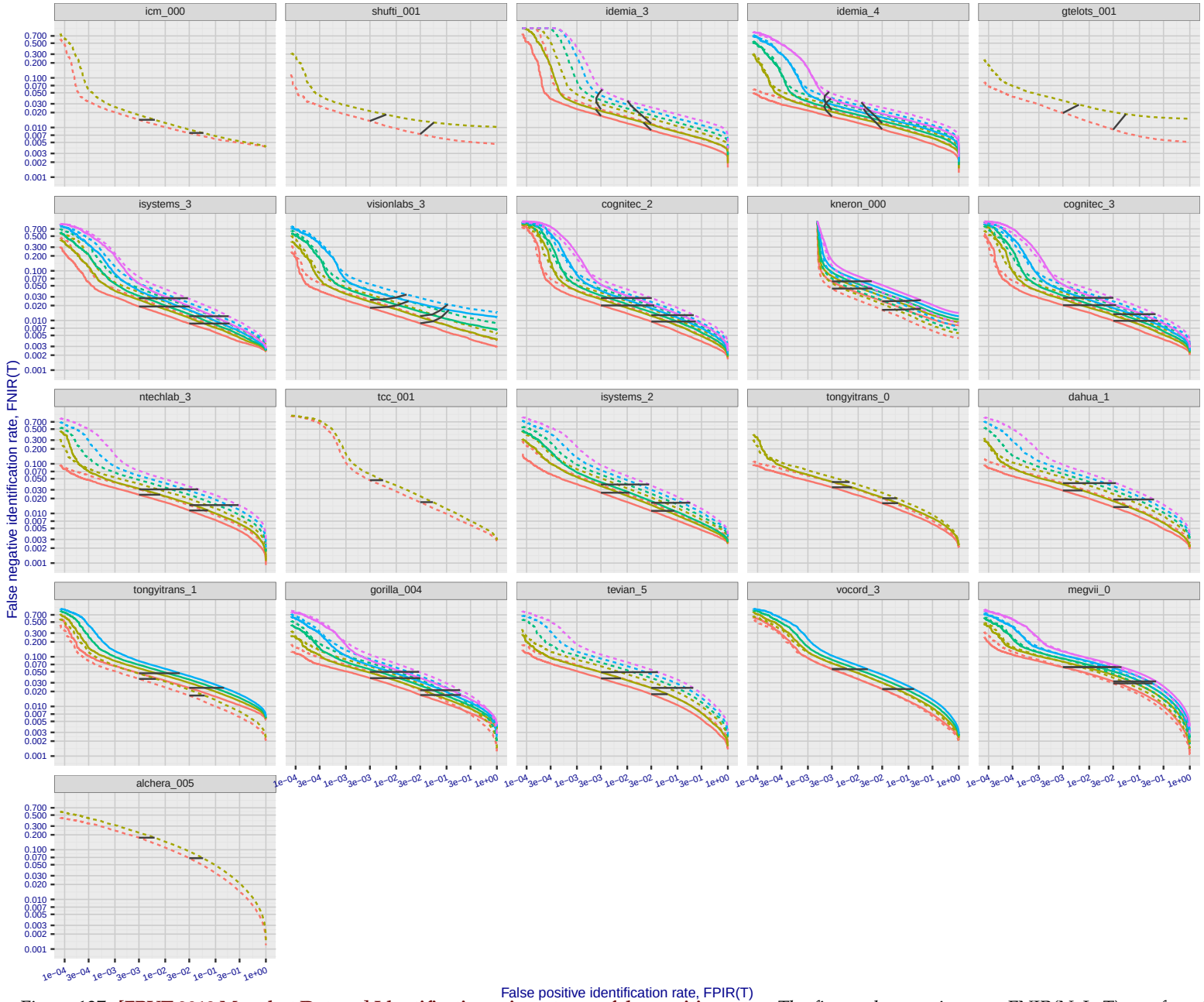

Figure 137: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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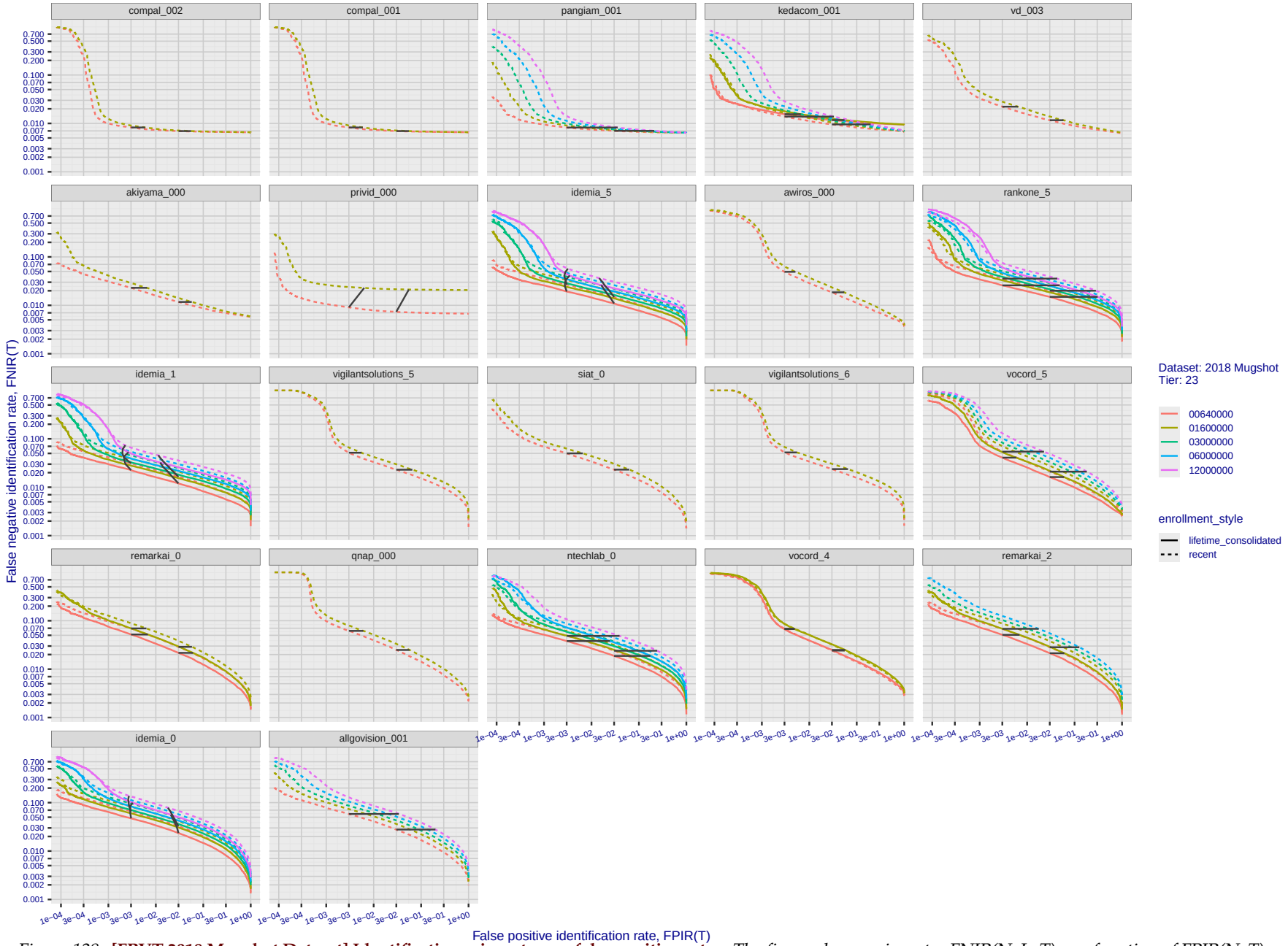

Figure 138: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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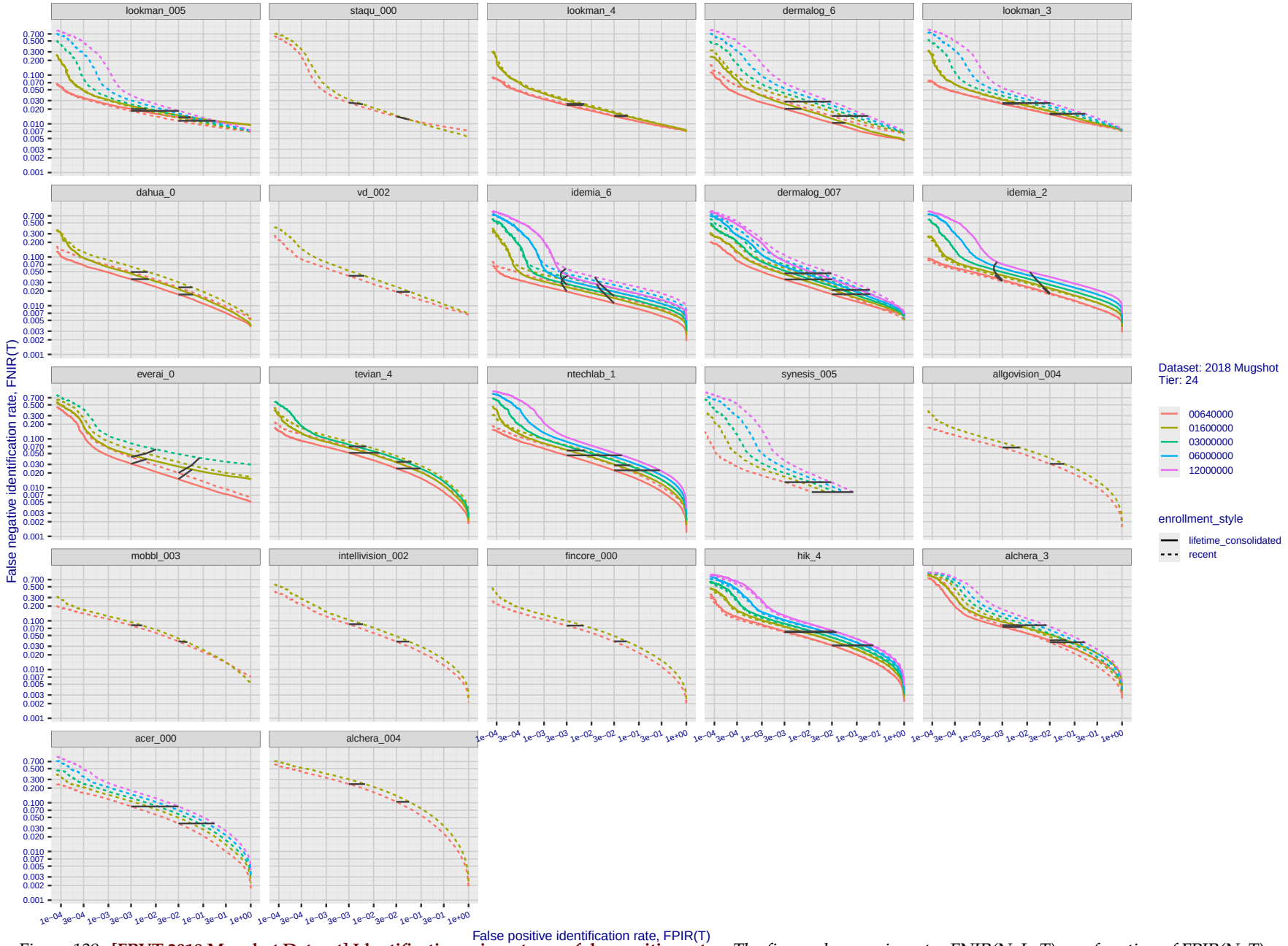

Figure 139: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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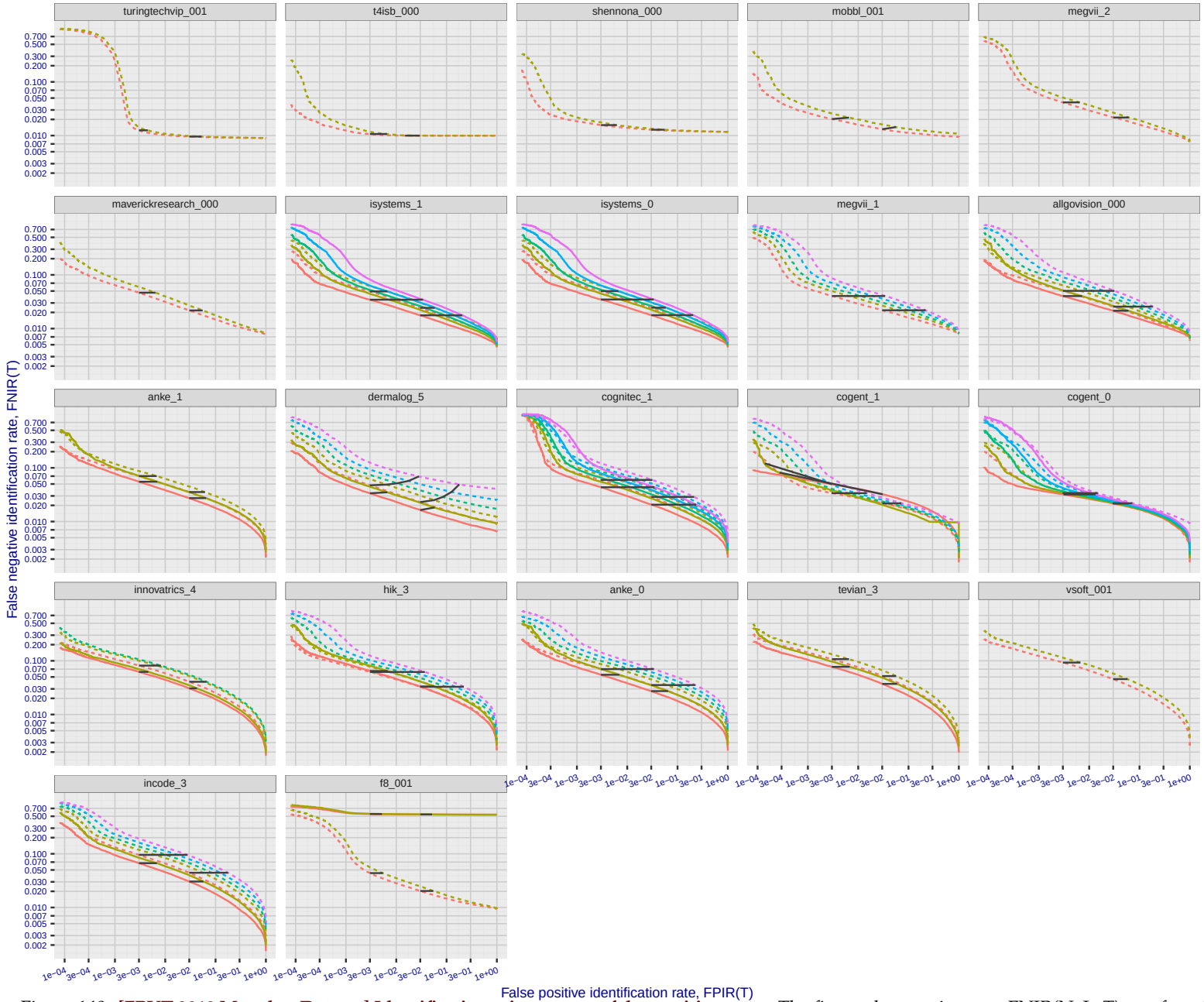

Figure 140: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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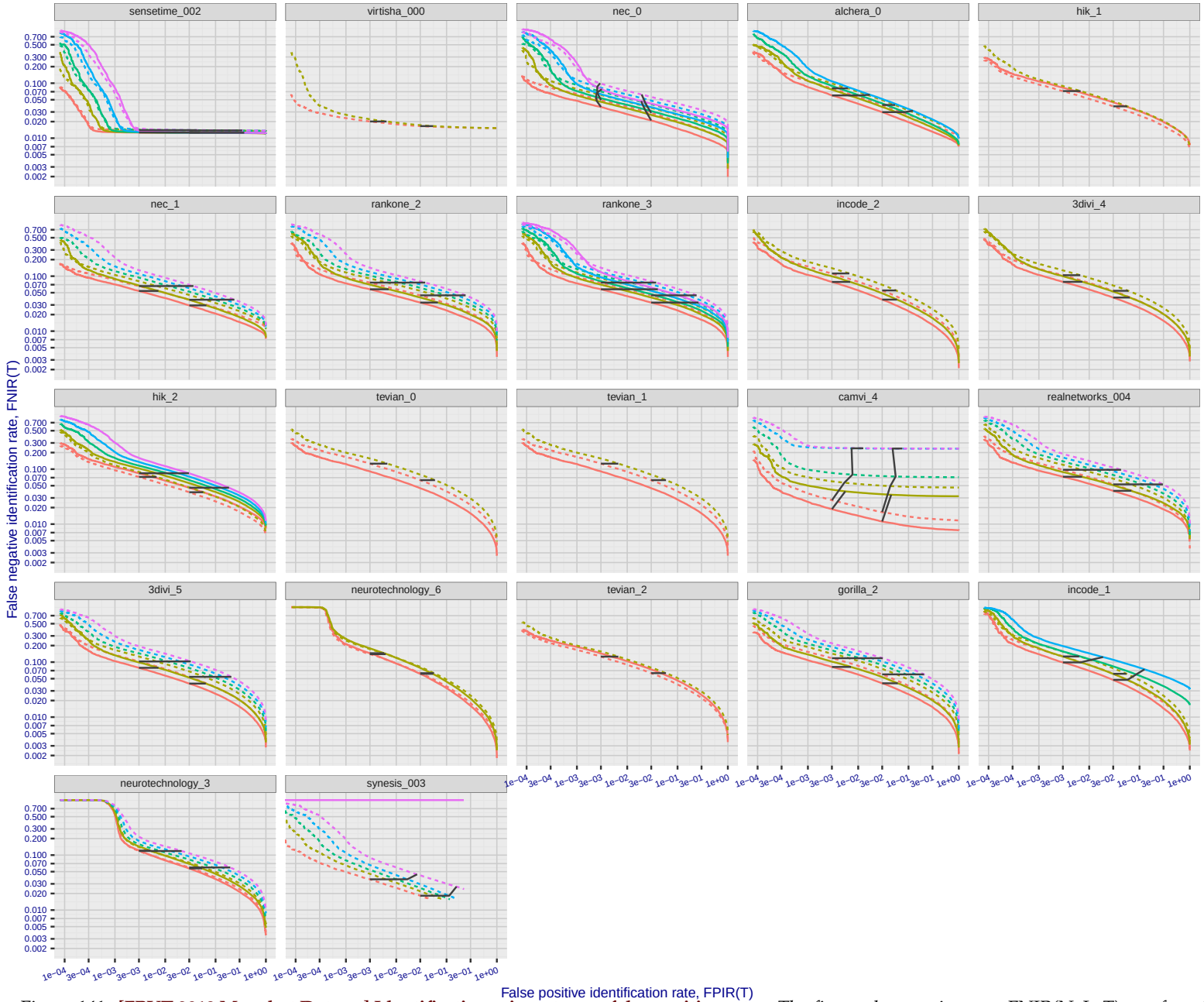

Figure 141: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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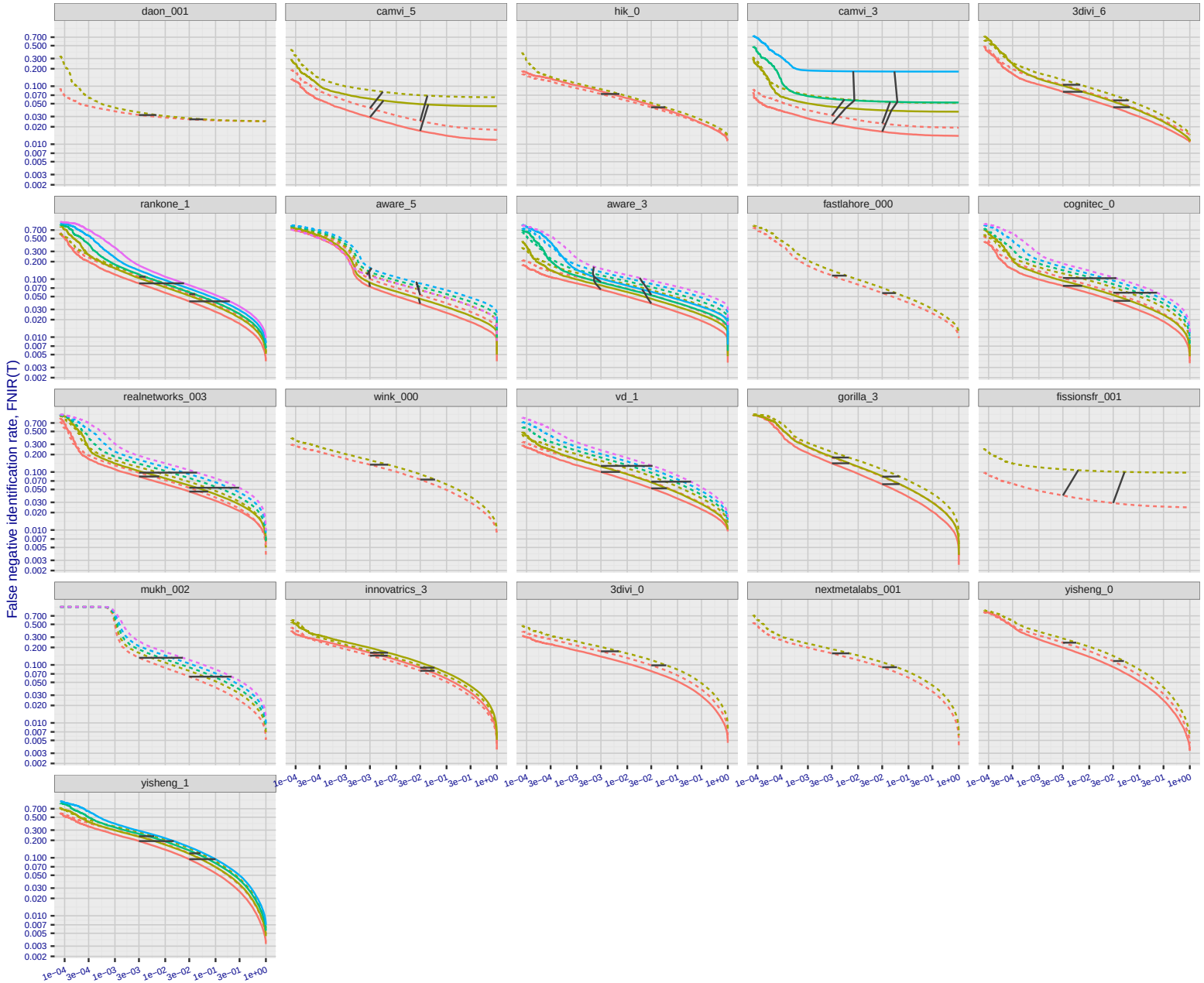

Figure 142: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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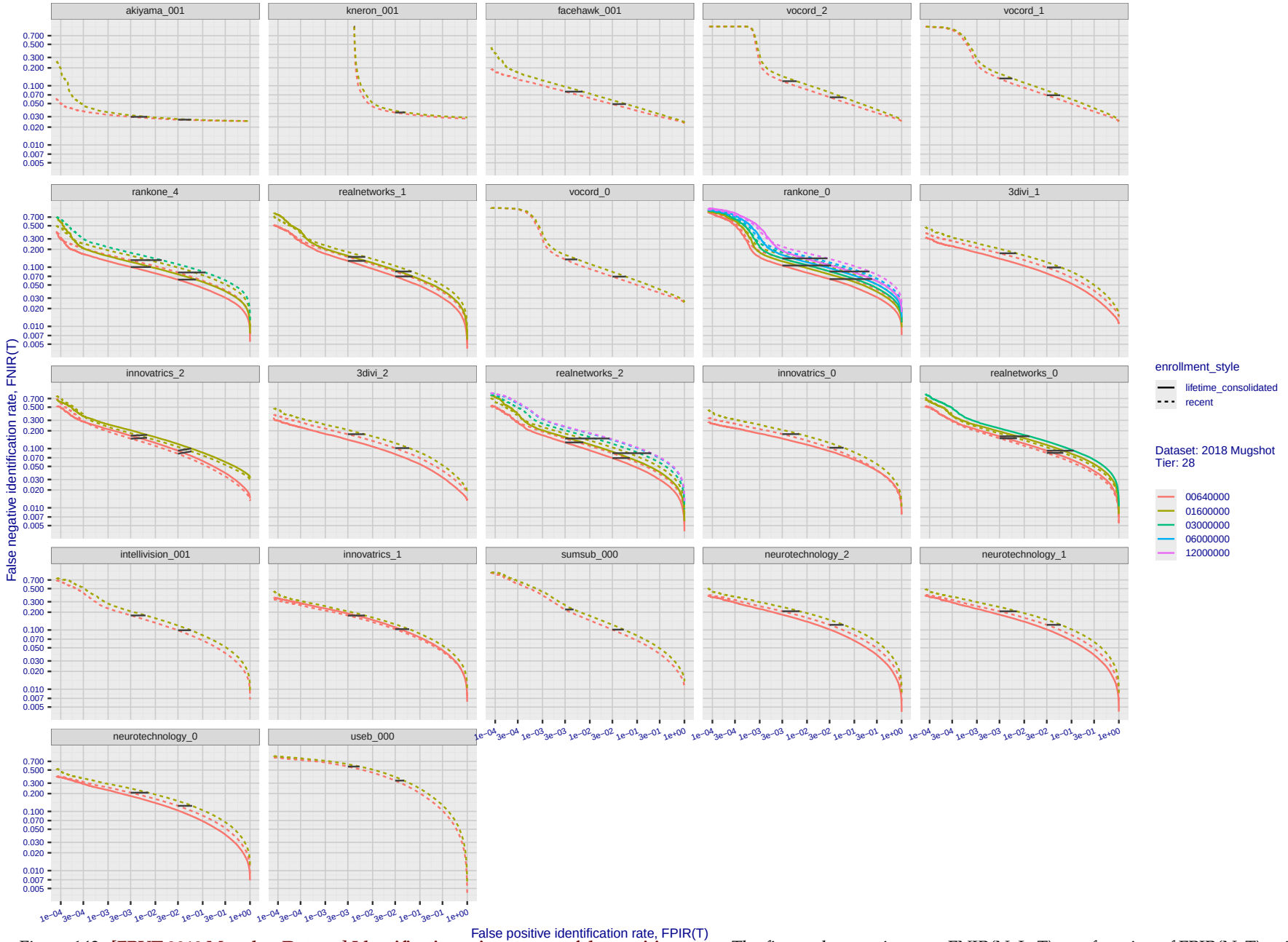

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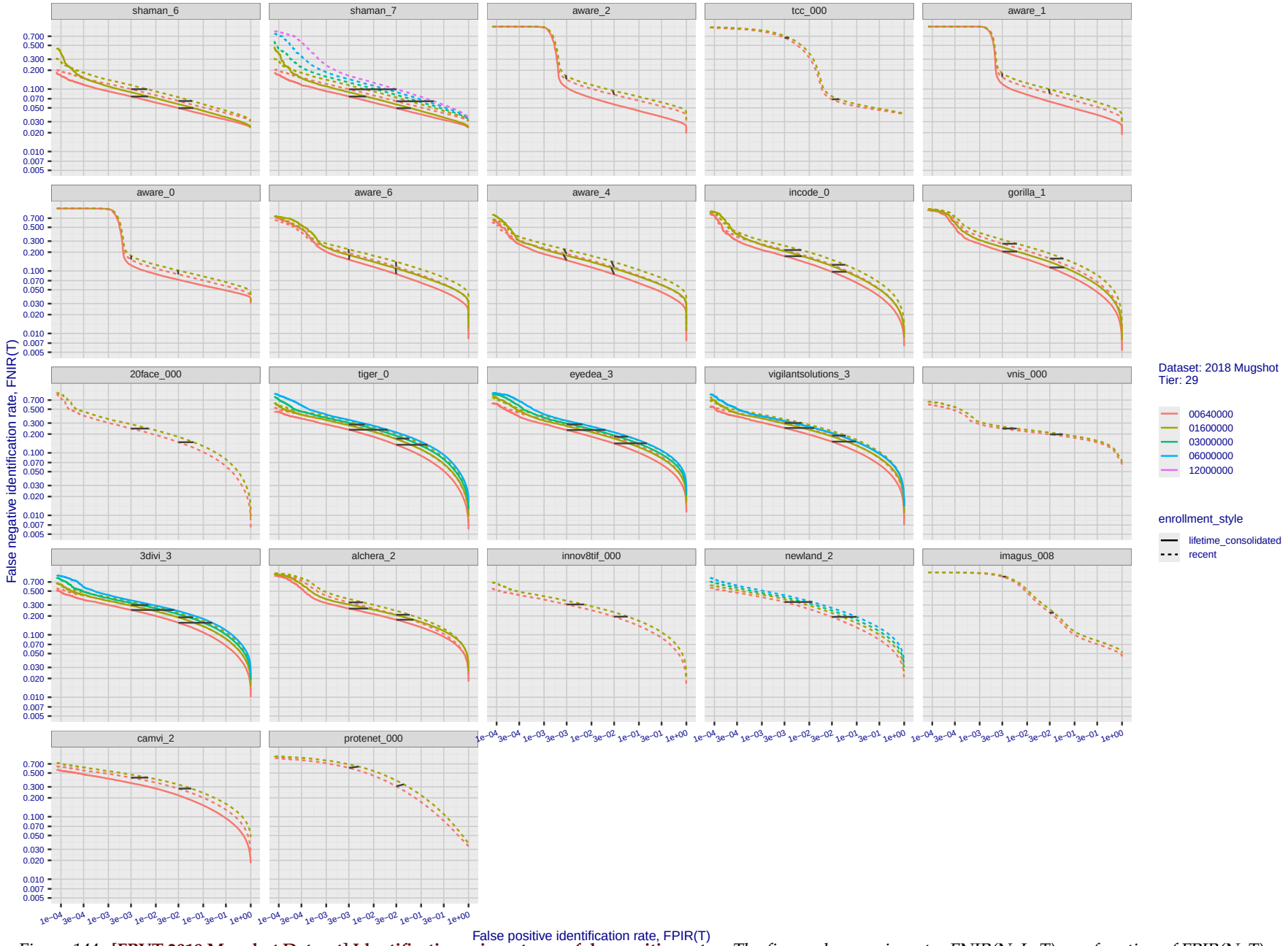

Figure 144: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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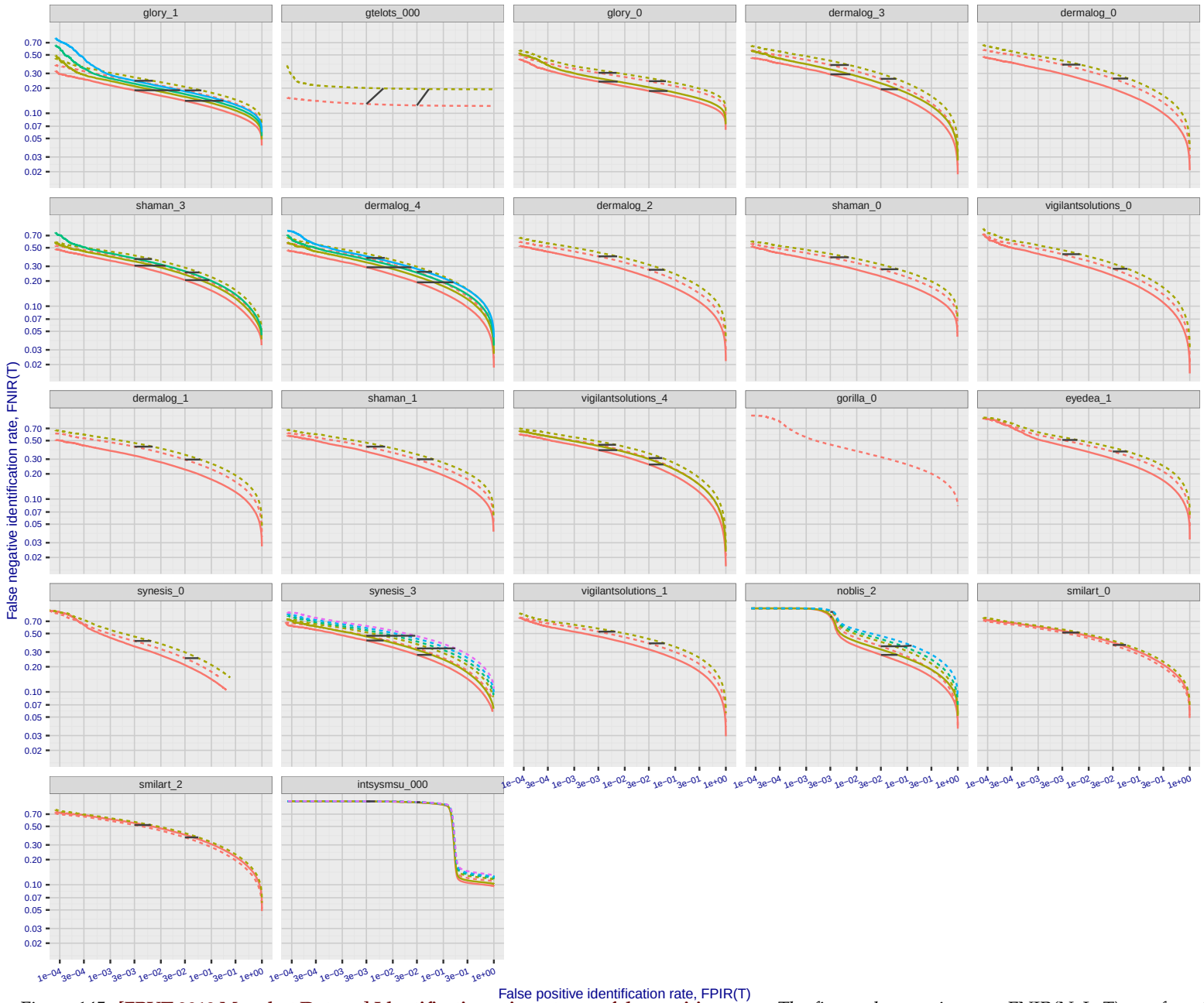

Figure 145: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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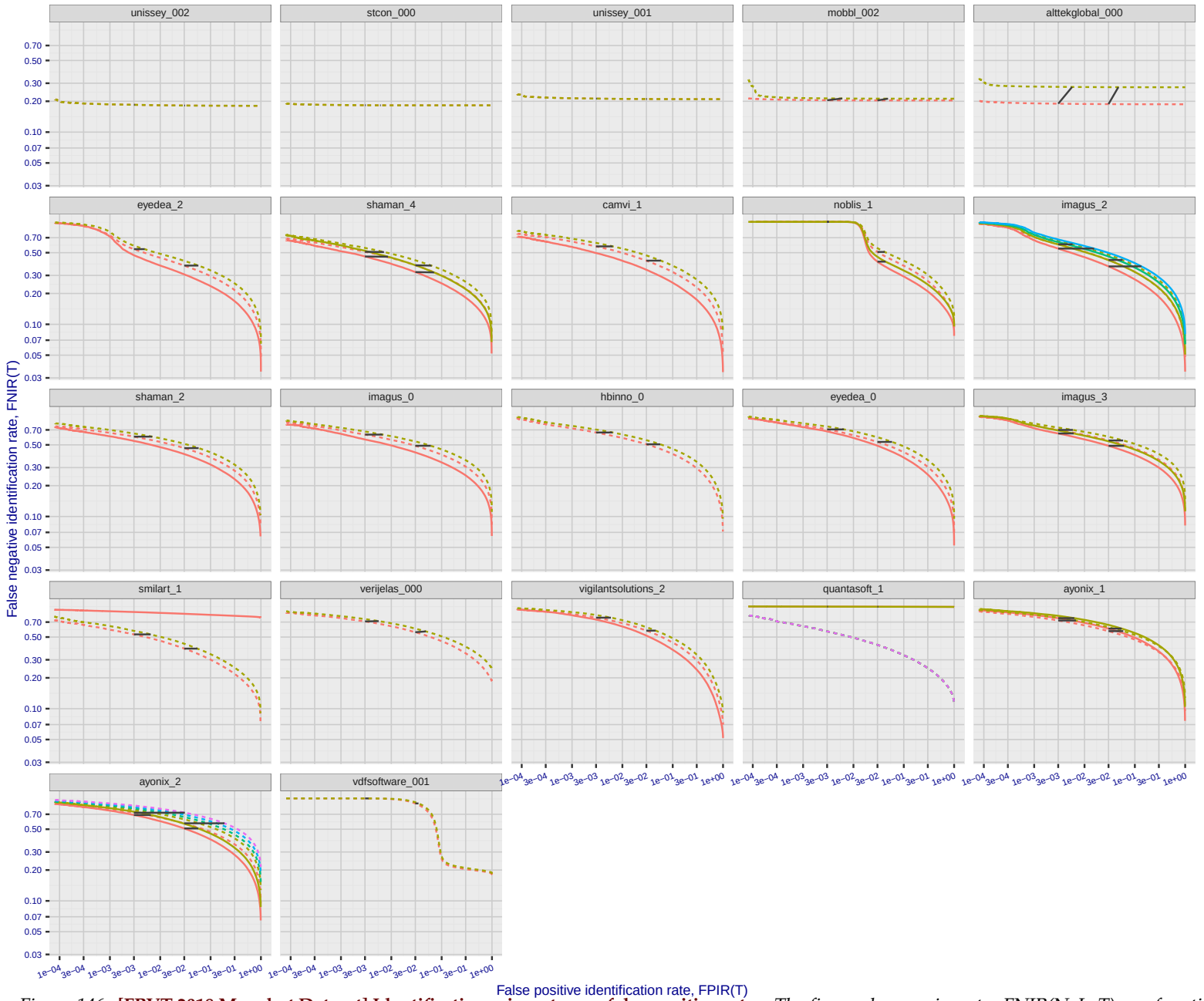

Figure 146: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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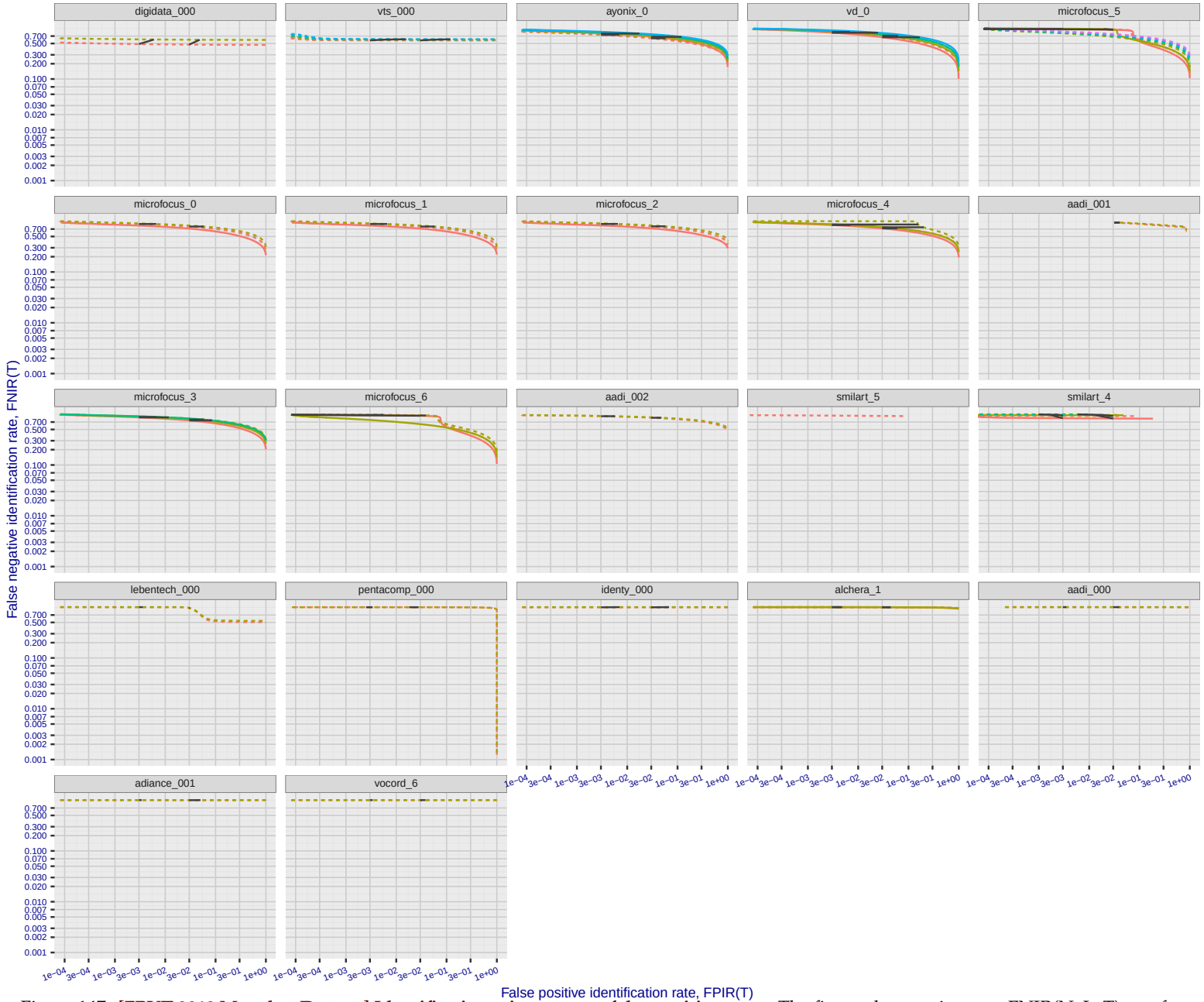

Figure 147: **[FRVT-2018 Mugshot Dataset] Identification miss rates vs. false positive rates**. *The figure shows miss rates FNIR(N, L, T) as a function of FPIR(N, T), with N ranging from 640 000 to 12 000 000 as noted in rows 1-10 of Table 1. These error tradeoff characteristics are useful for applications where a threshold must be elevated to limit false positives, such as when human reviewer labor is not matched to the volume of searches. Dark lines join points of equal threshold: If horizontal, FPIR(T) rises with N, and mate scores are independent of N. Other algorithms adjust scores in an attempt to make FPIR independent of N.*

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Appendix B Effect of time-lapse: Accuracy after face ageing

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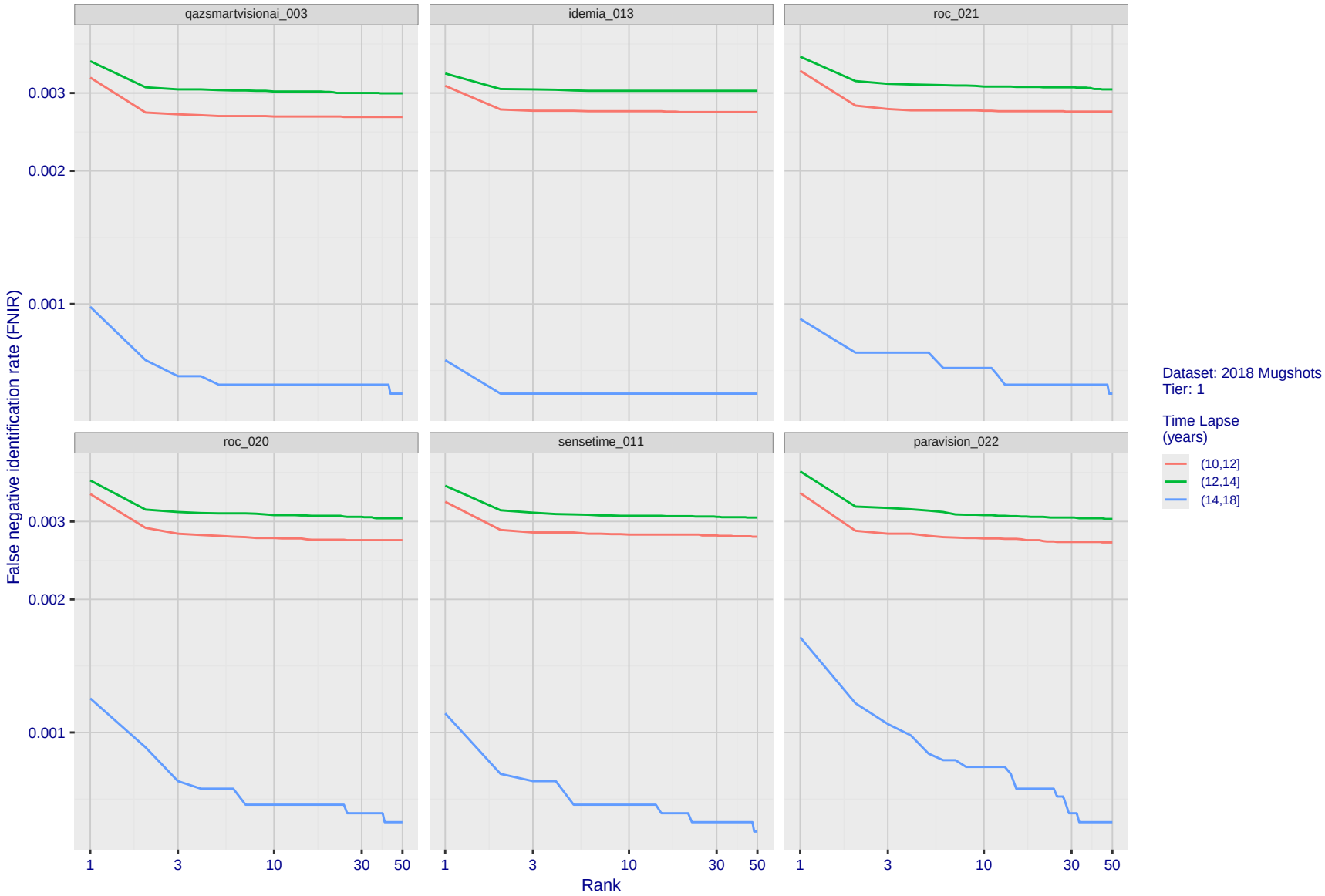

Figure 148: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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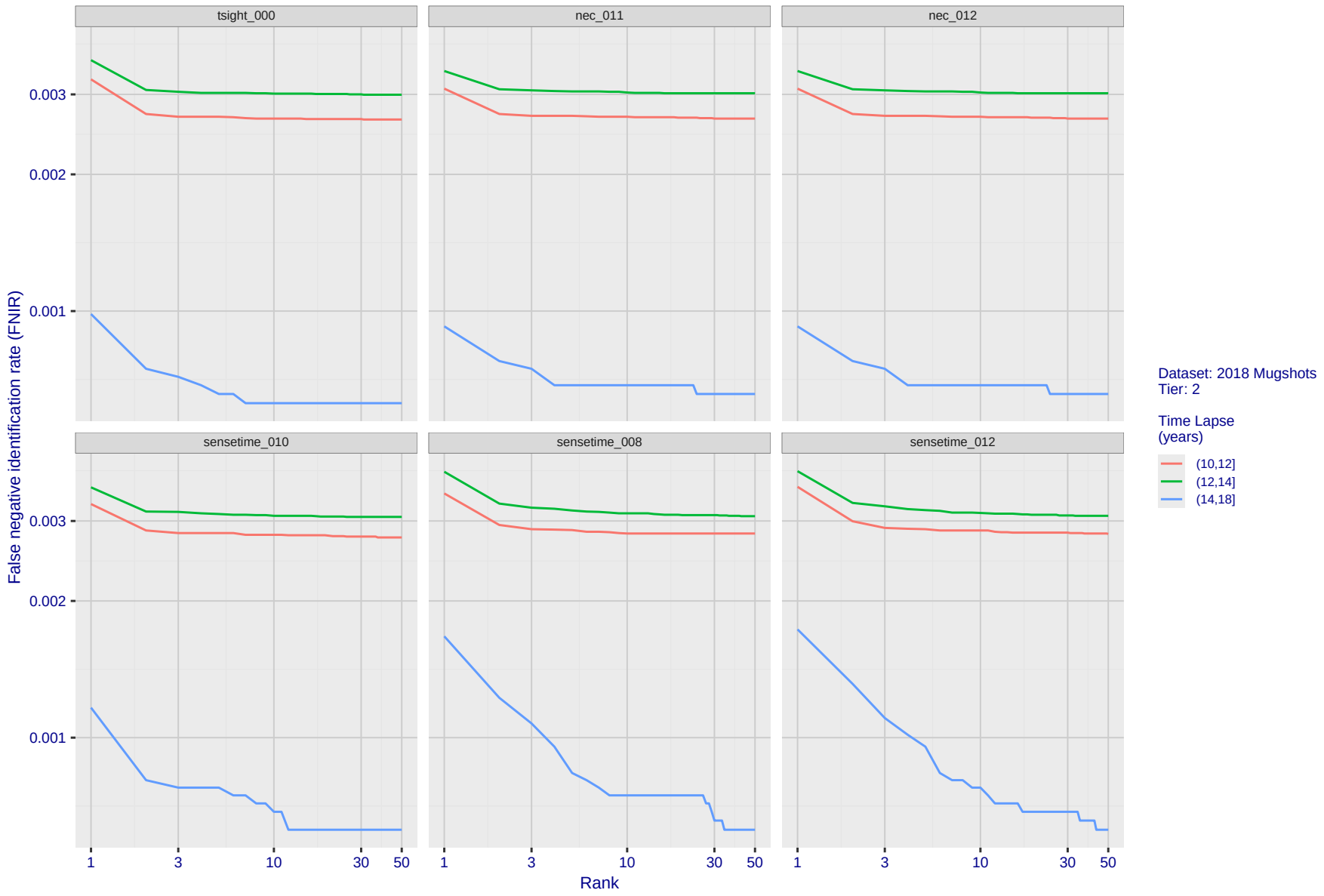

Figure 149: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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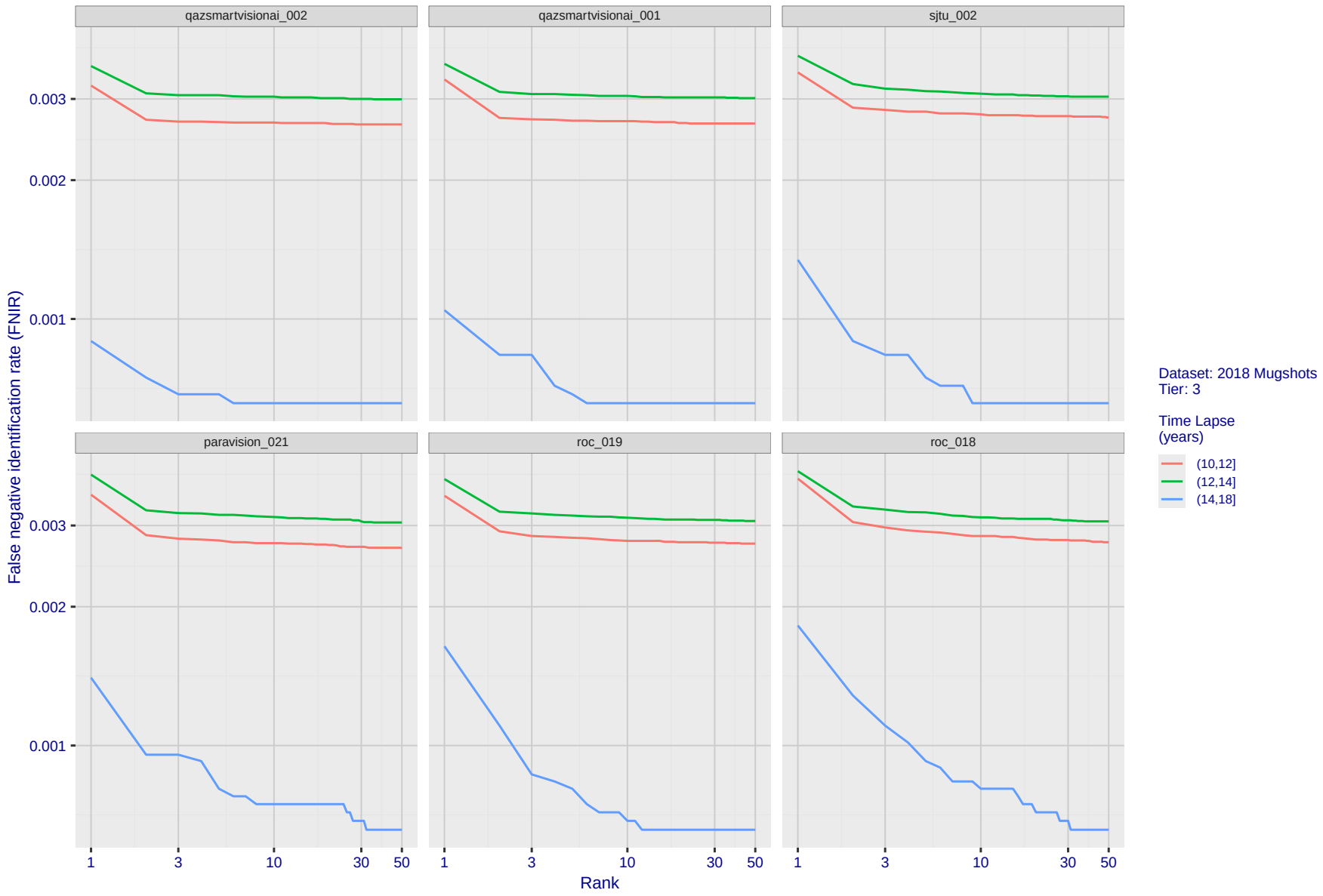

Figure 150: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
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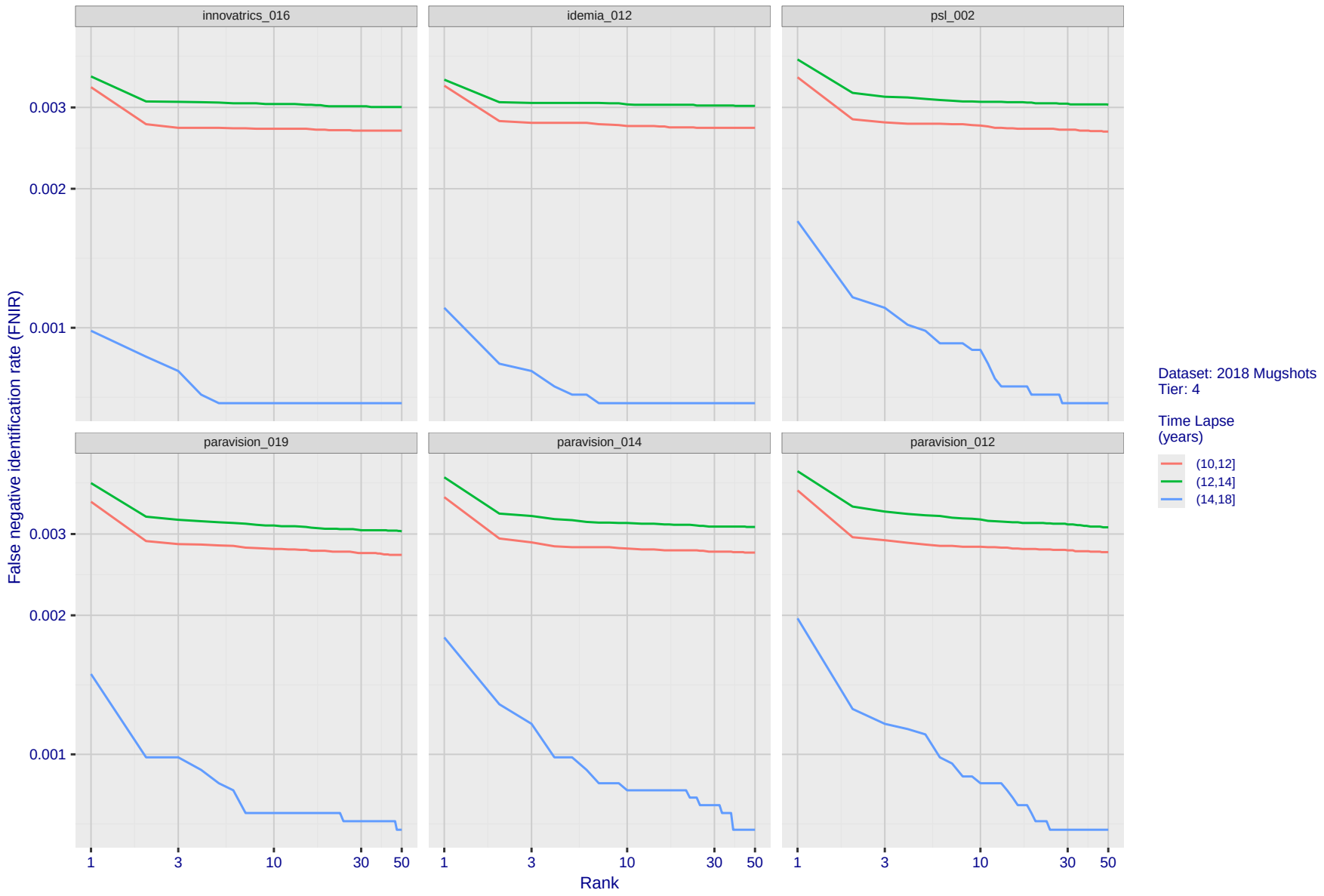

Figure 151: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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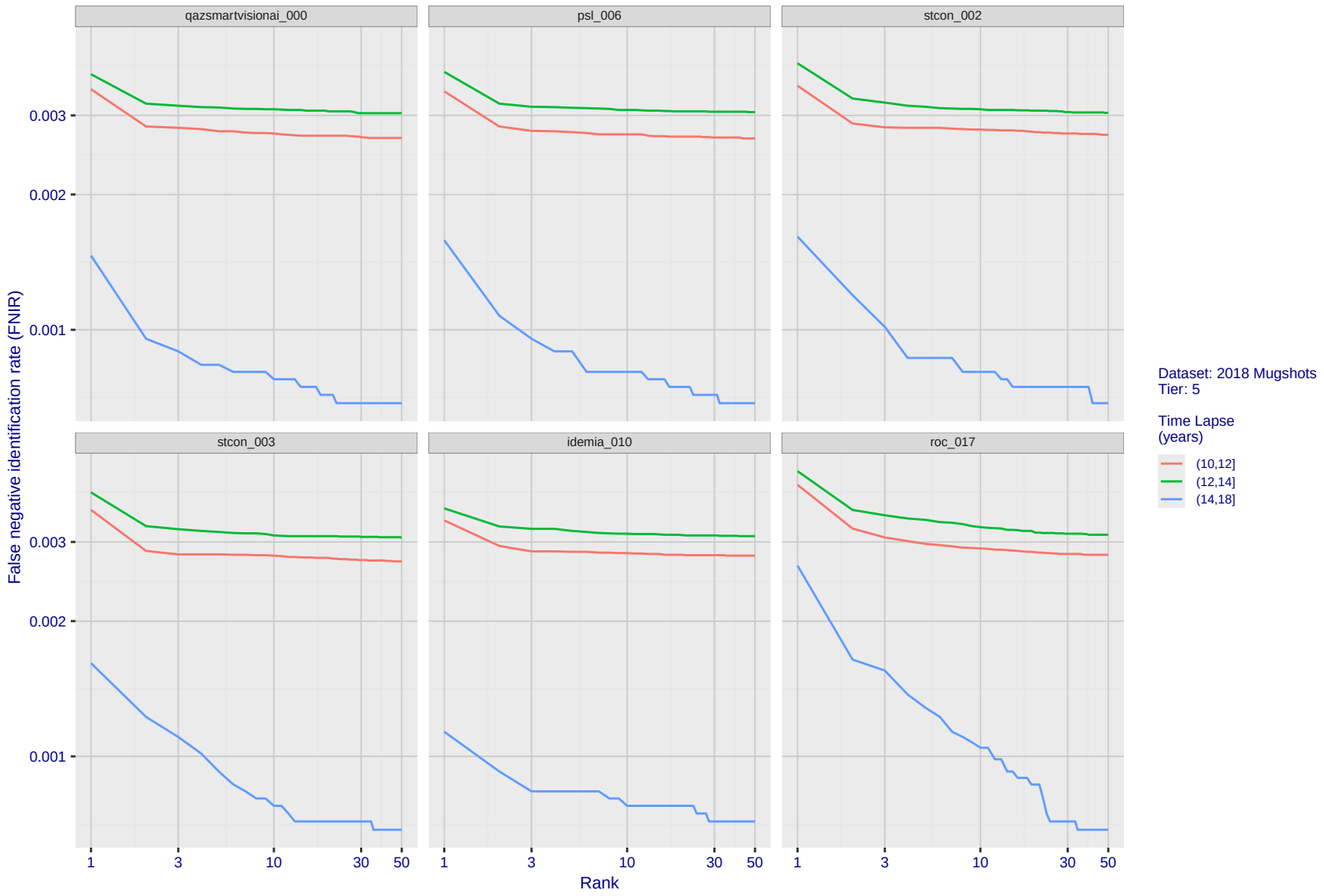

Figure 152: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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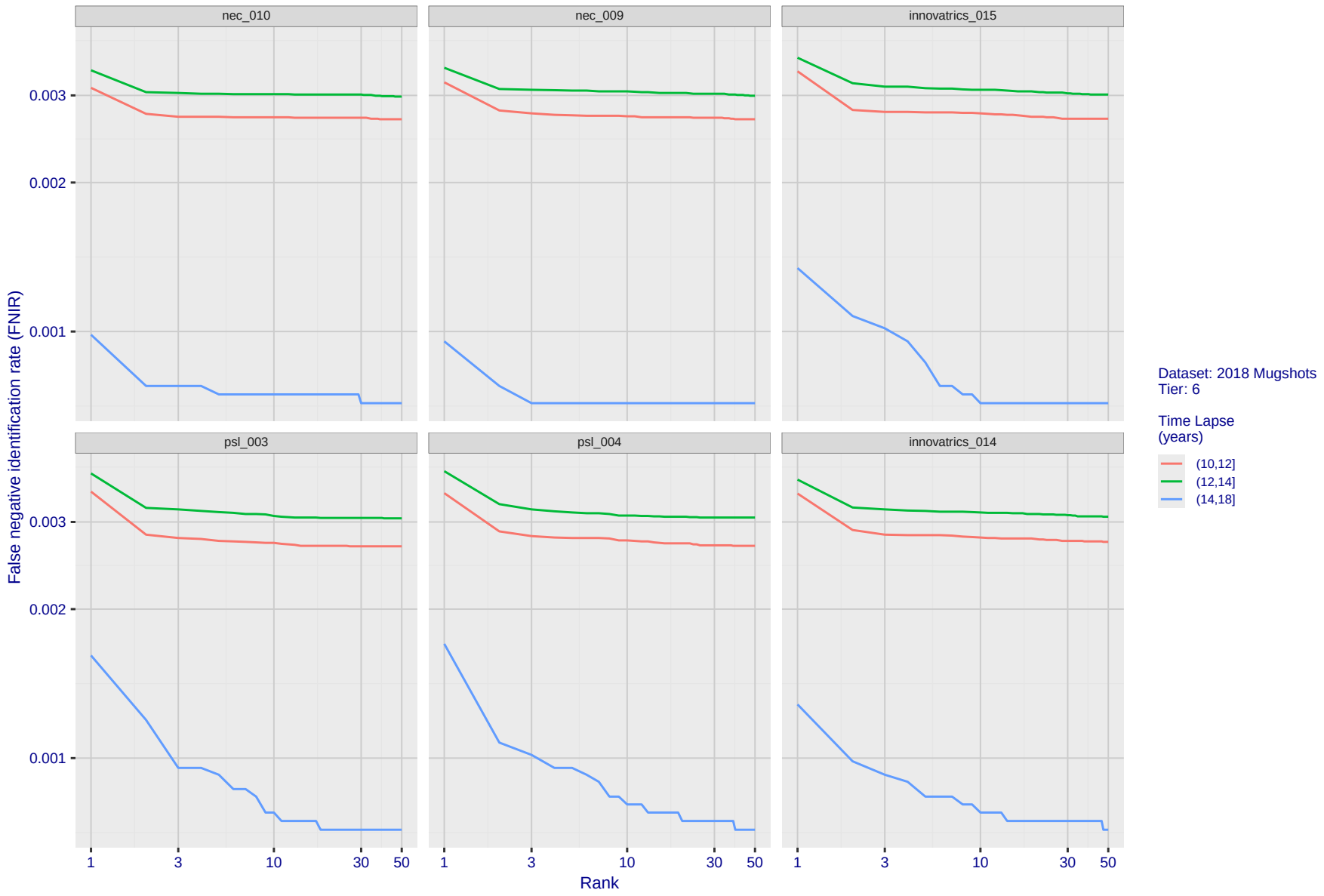

Figure 153: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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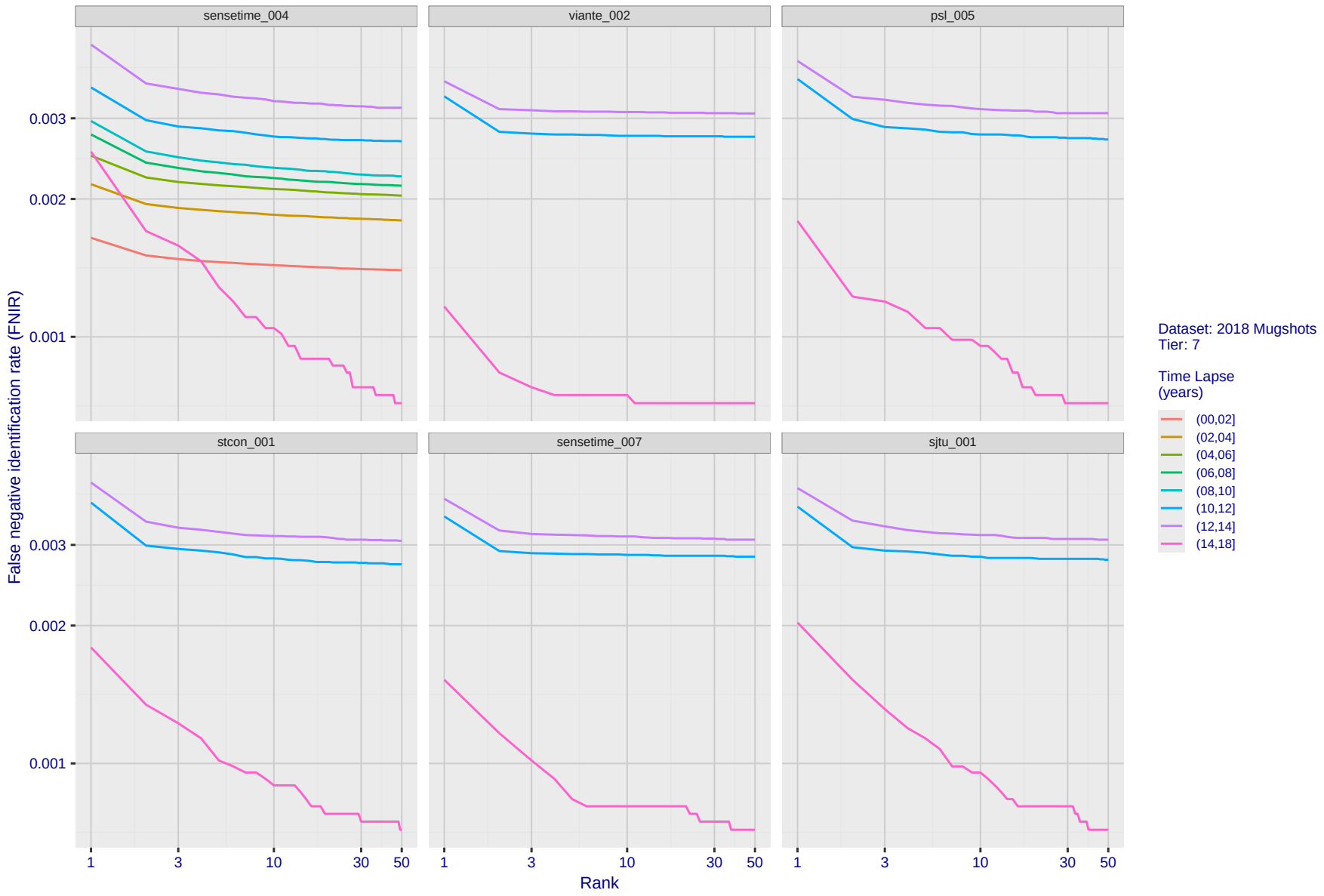

Figure 154: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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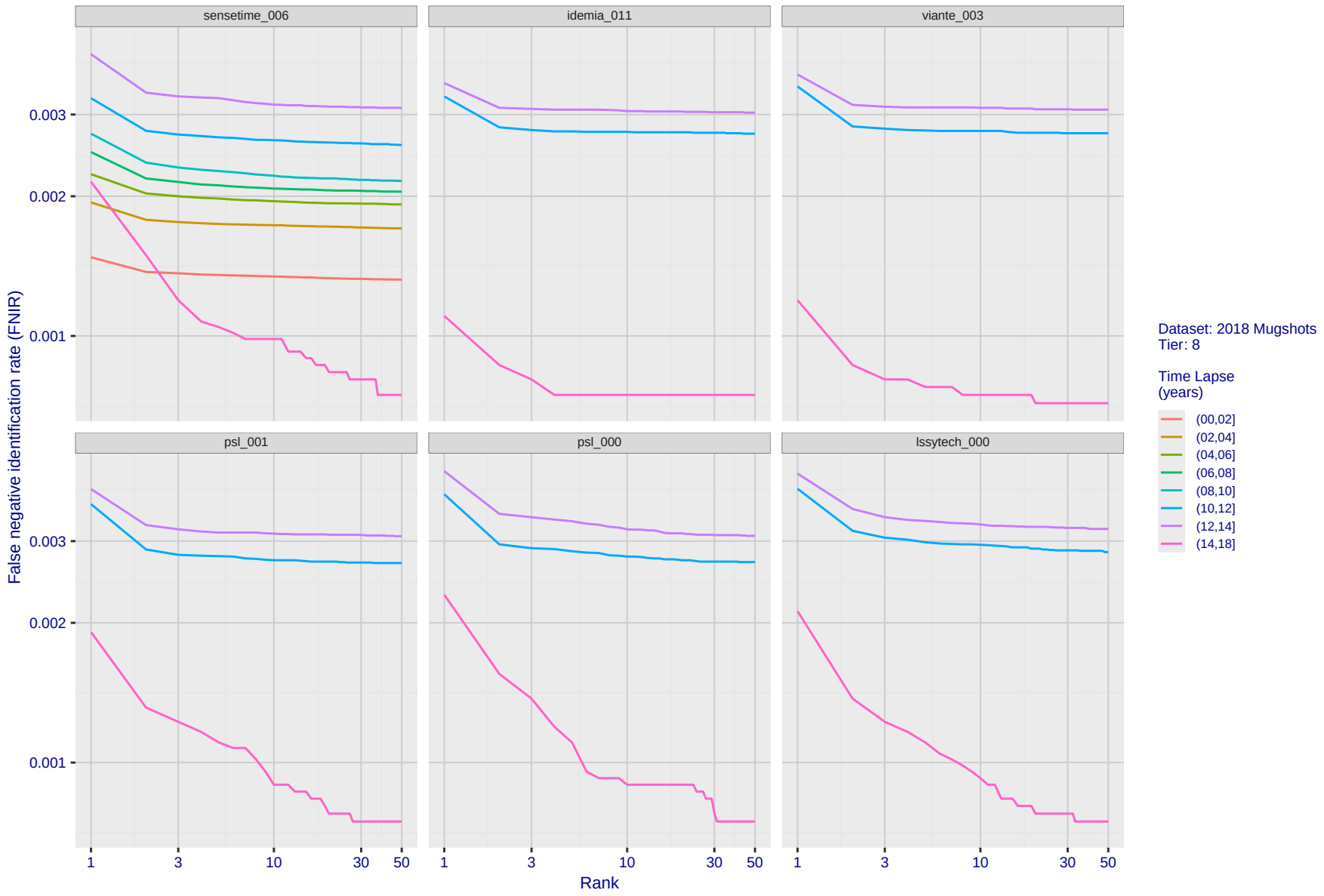

Figure 155: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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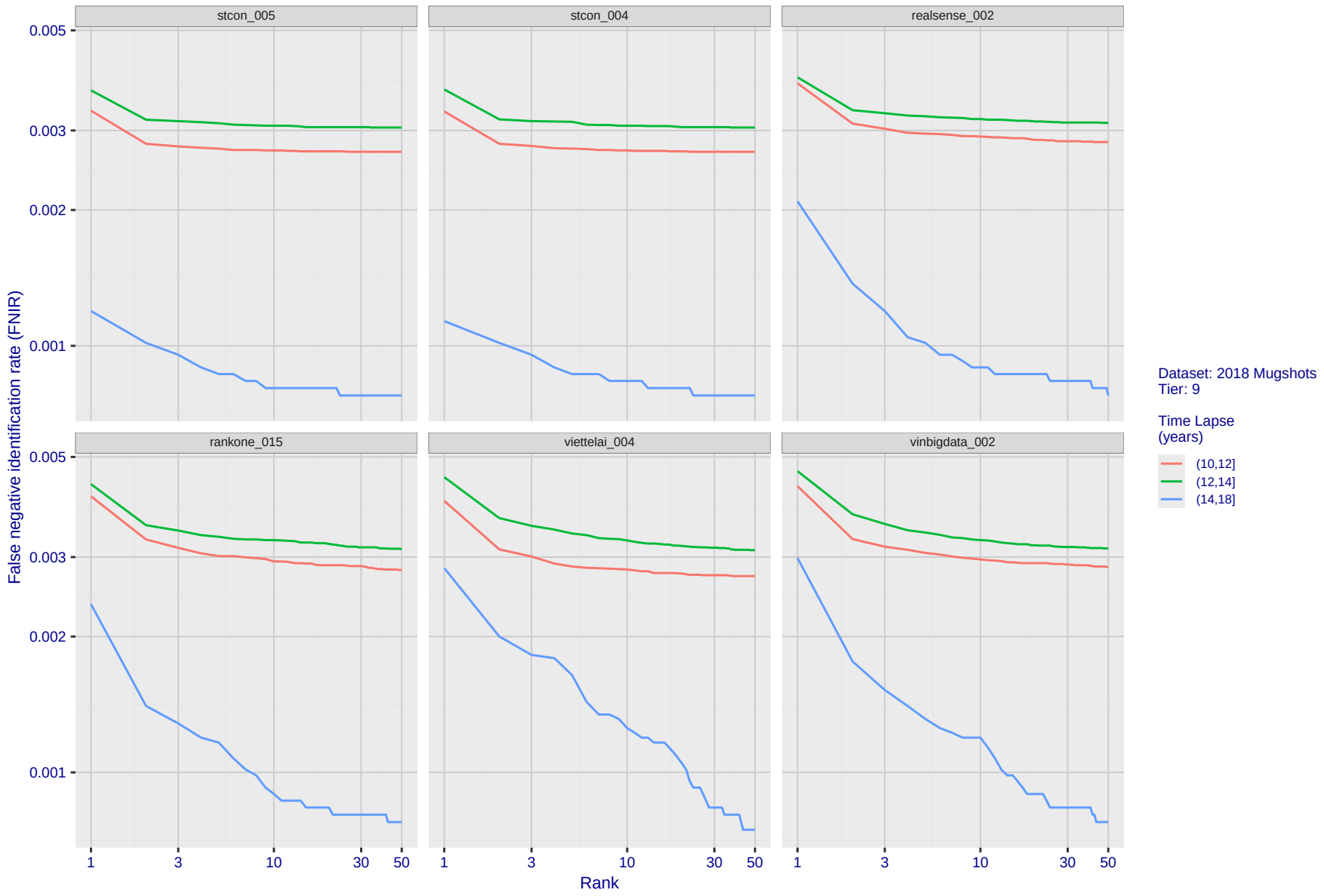

Figure 156: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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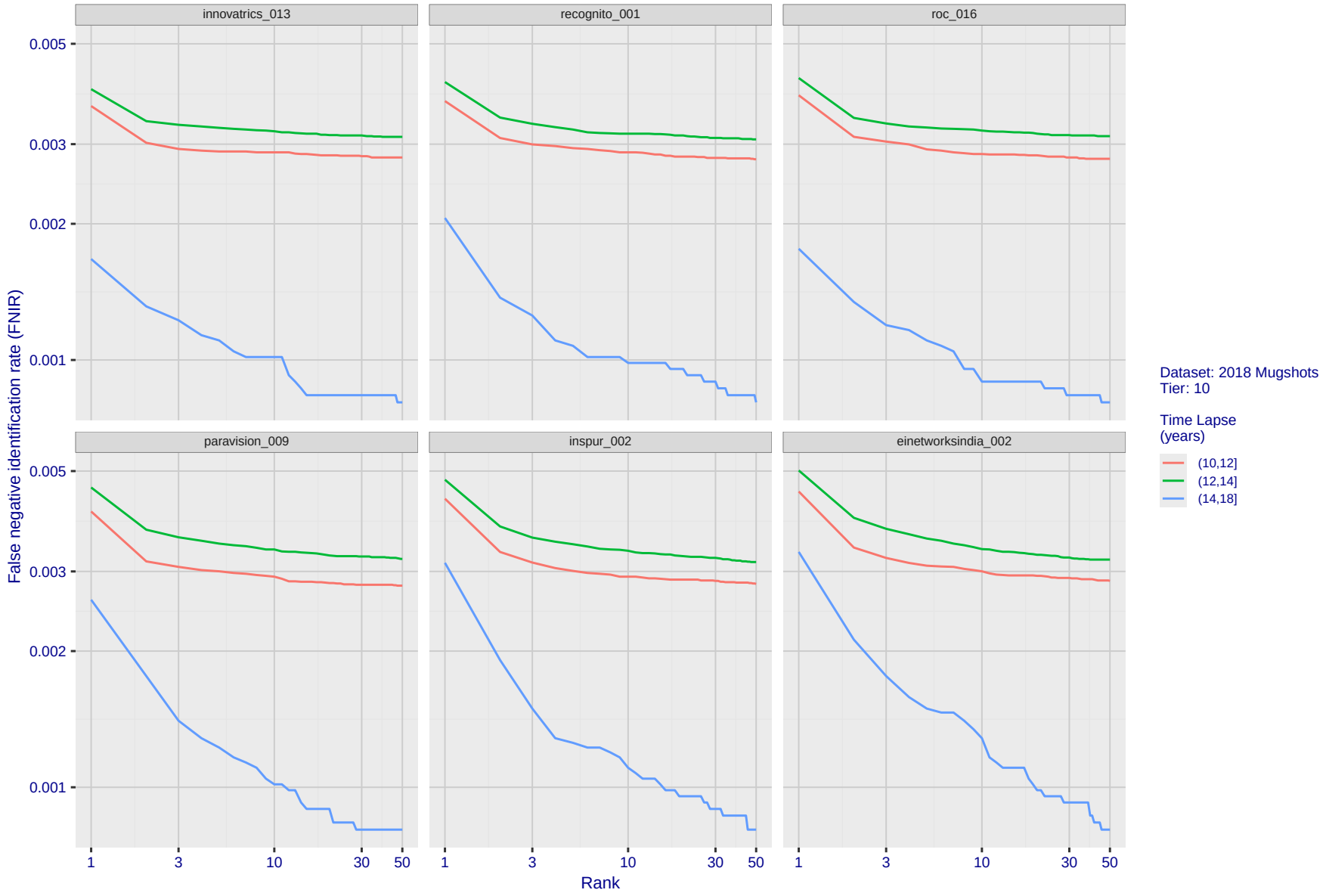

Figure 157: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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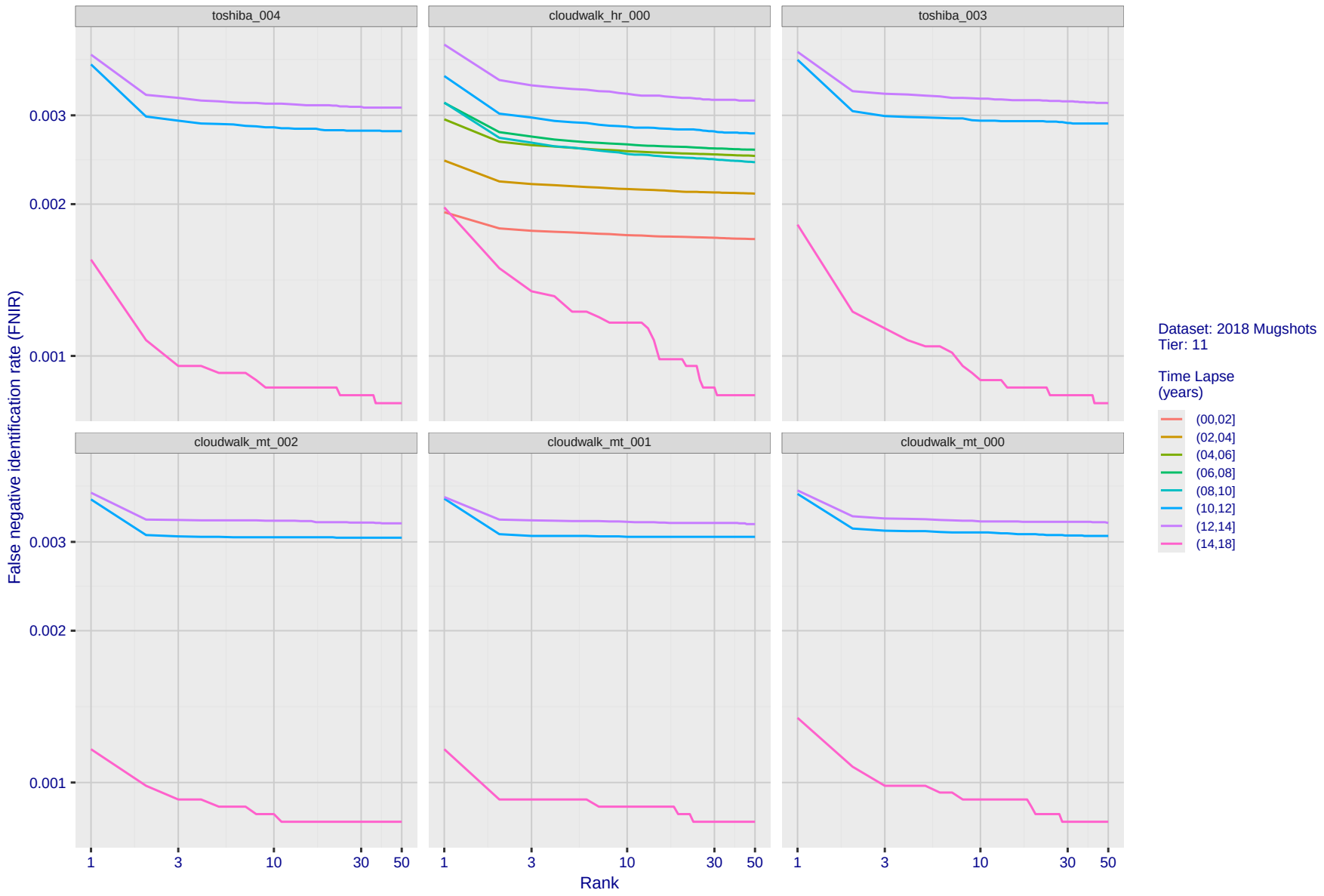

Figure 158: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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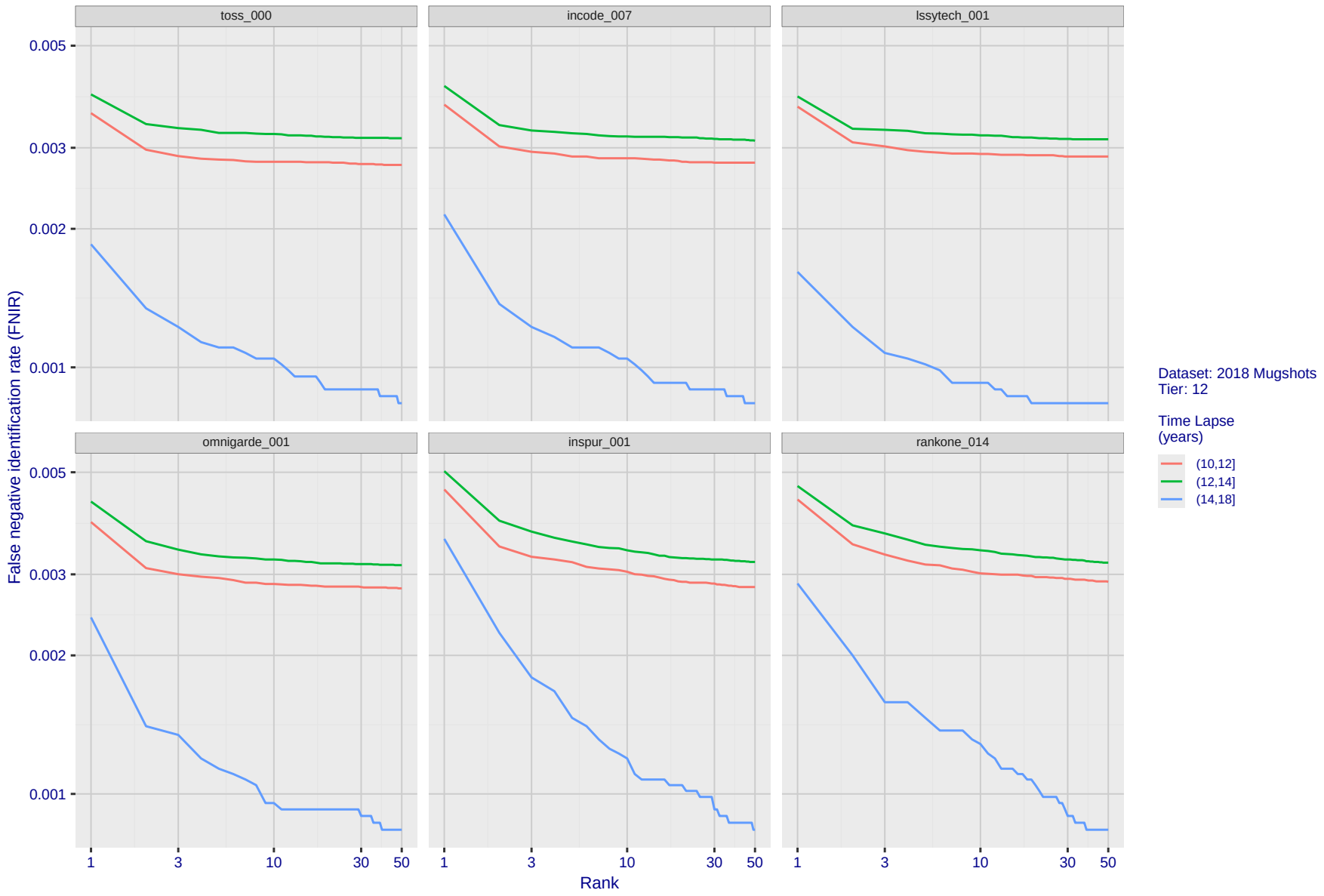

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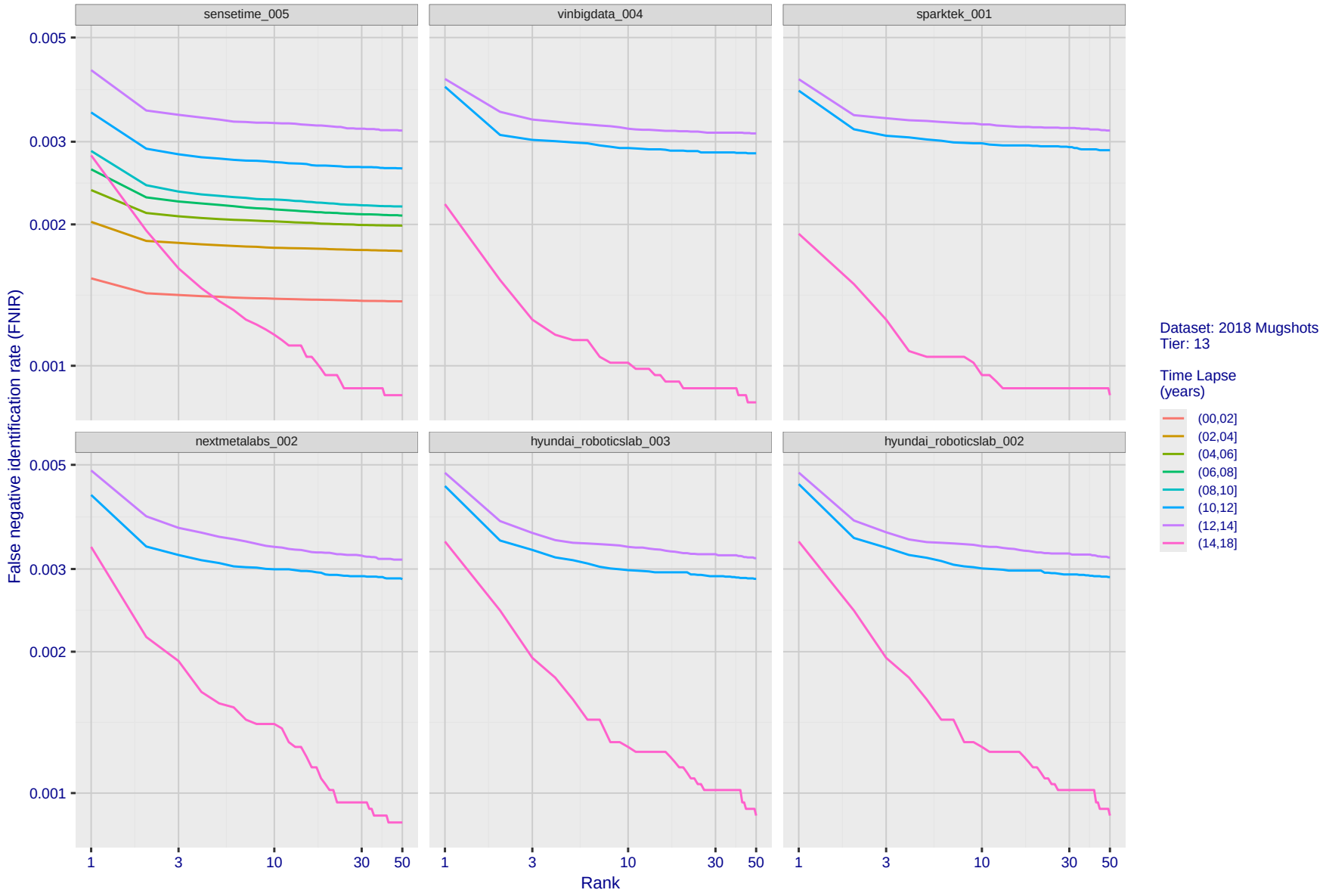

Figure 160: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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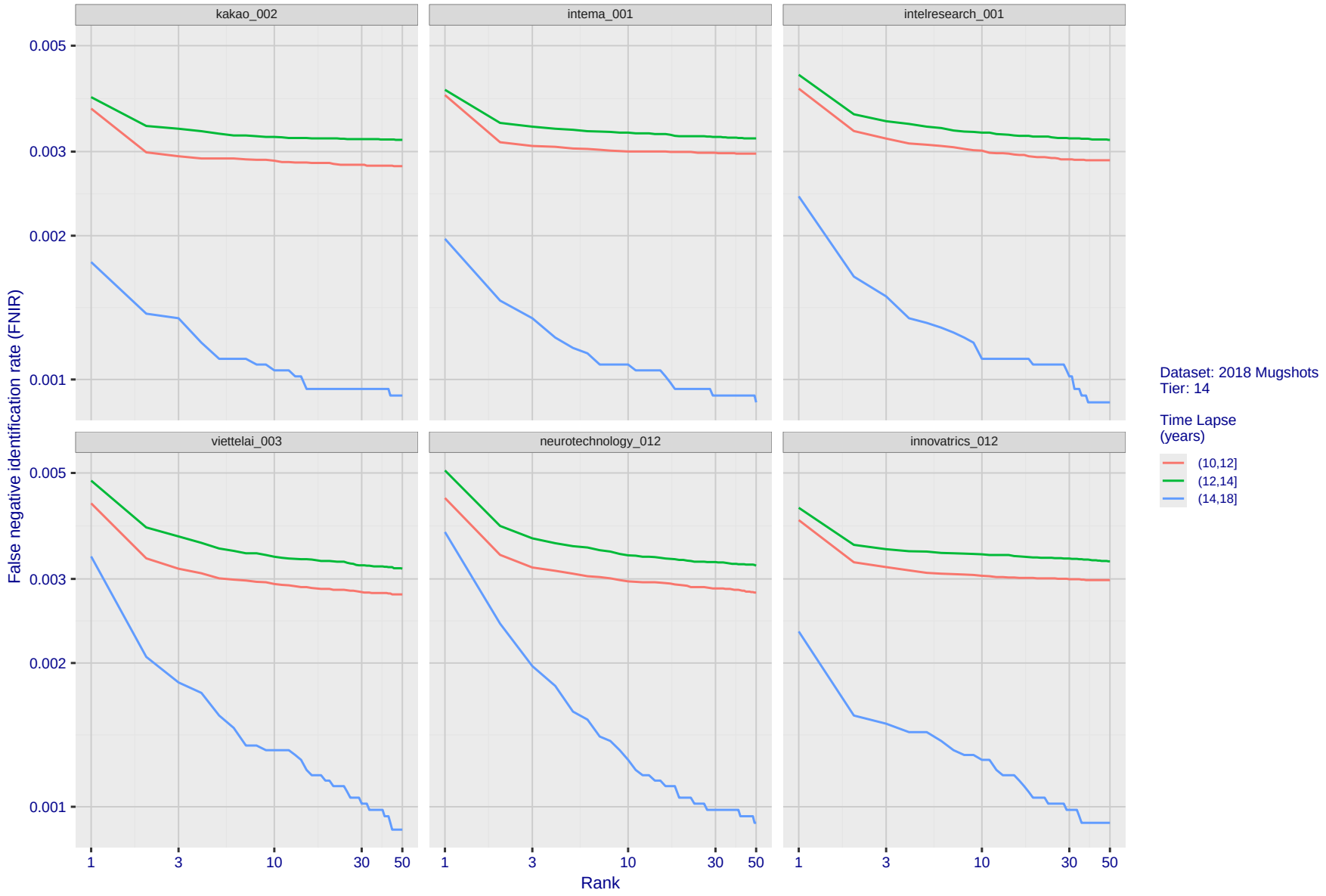

Figure 161: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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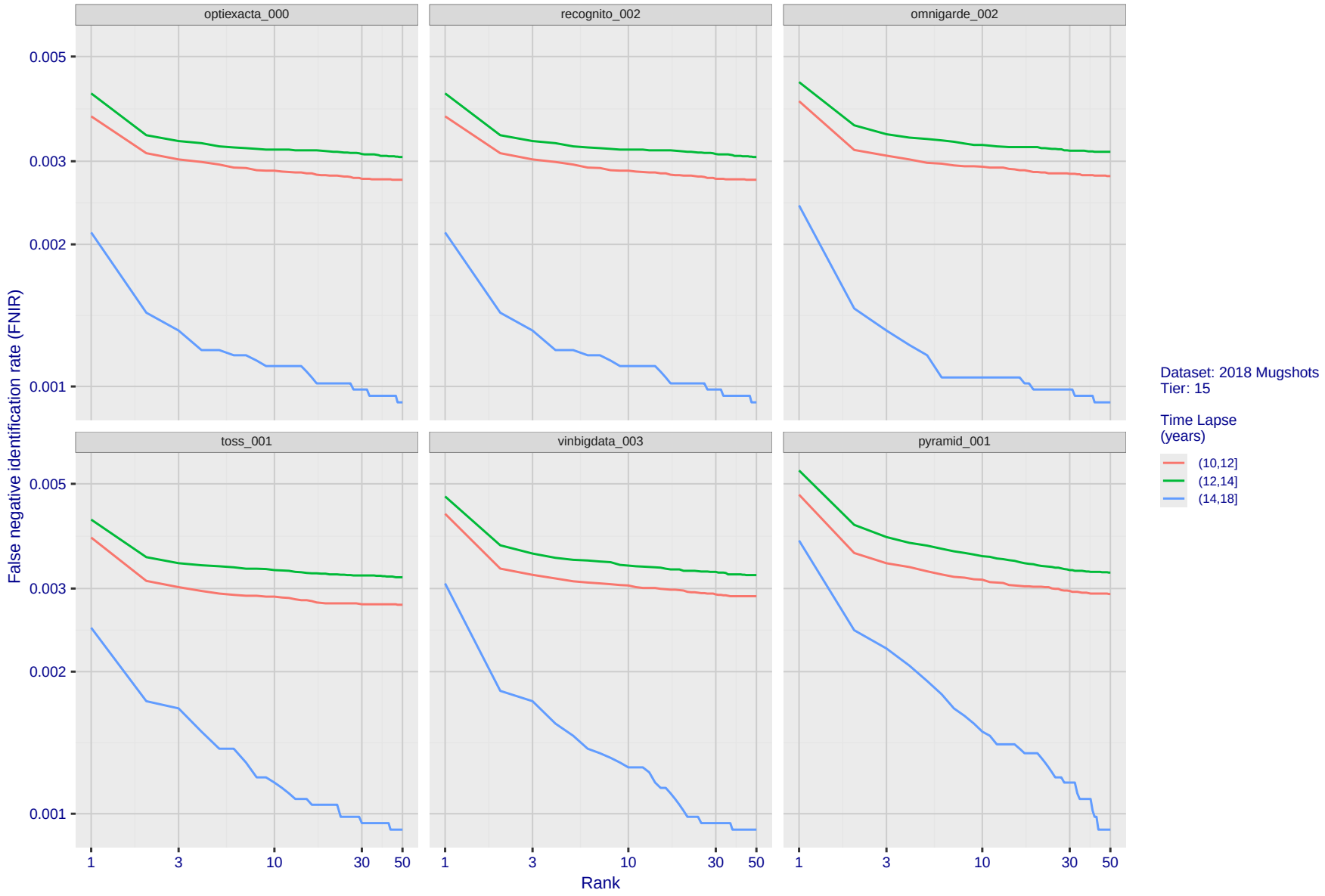

Figure 162: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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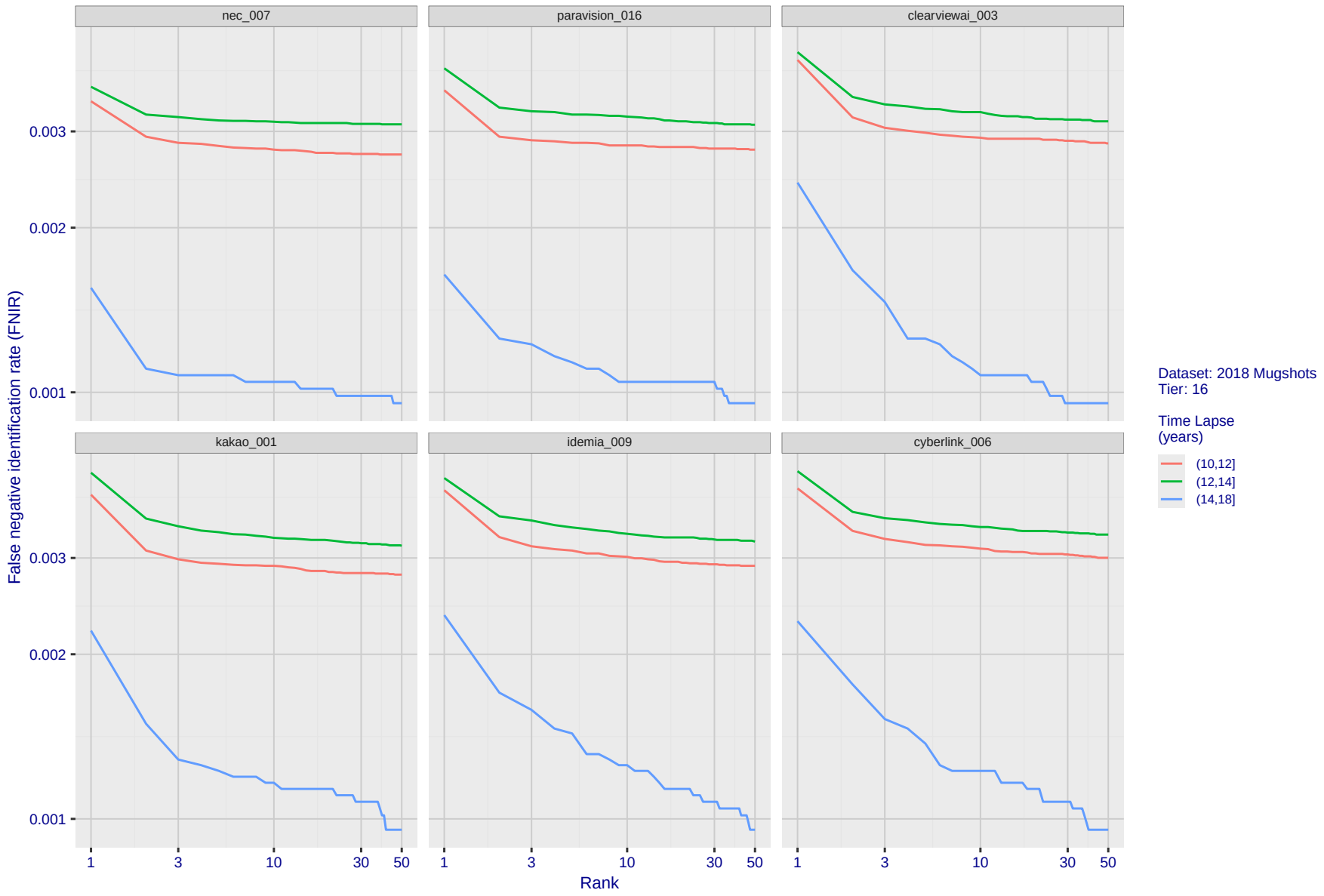

Figure 163: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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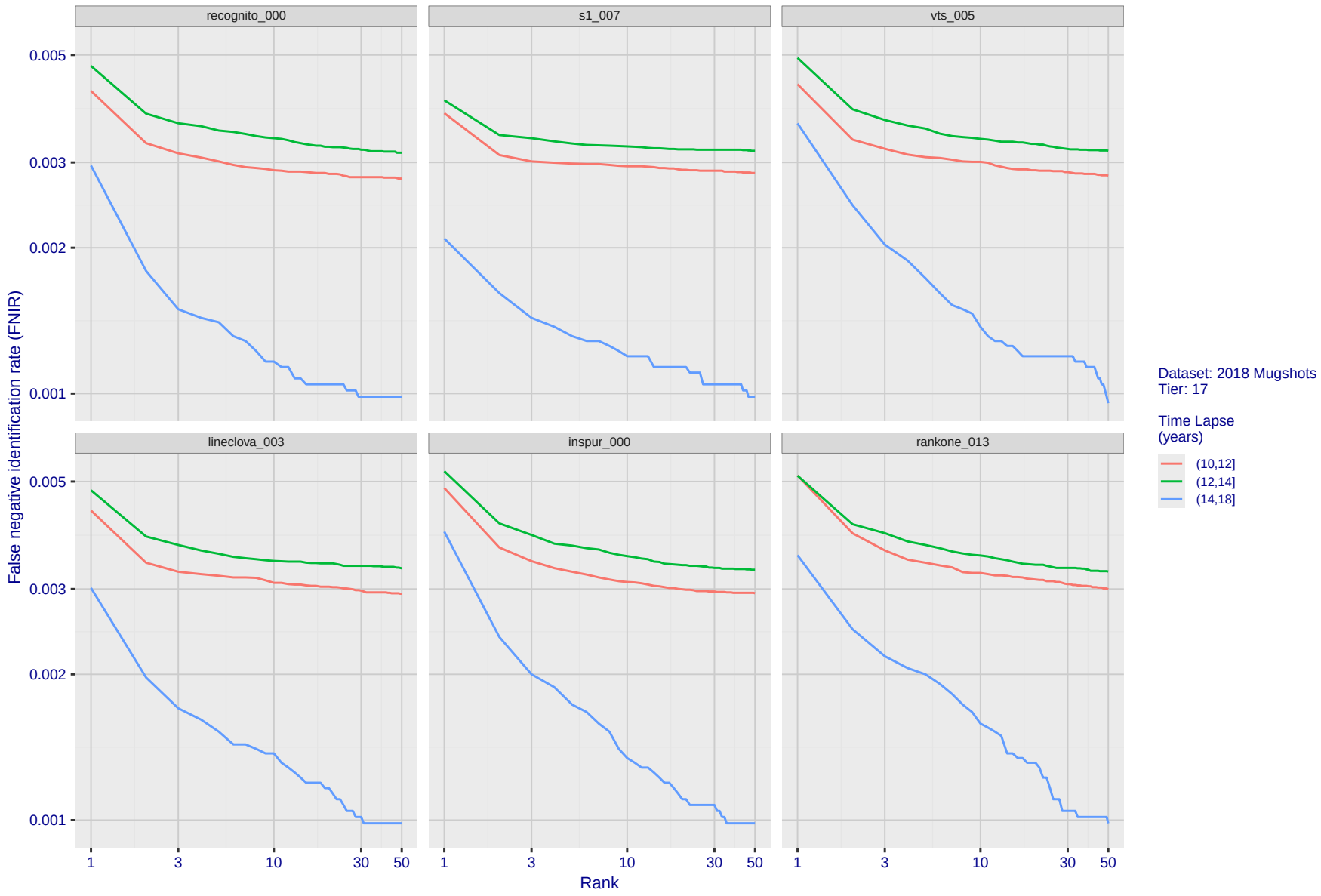

Figure 164: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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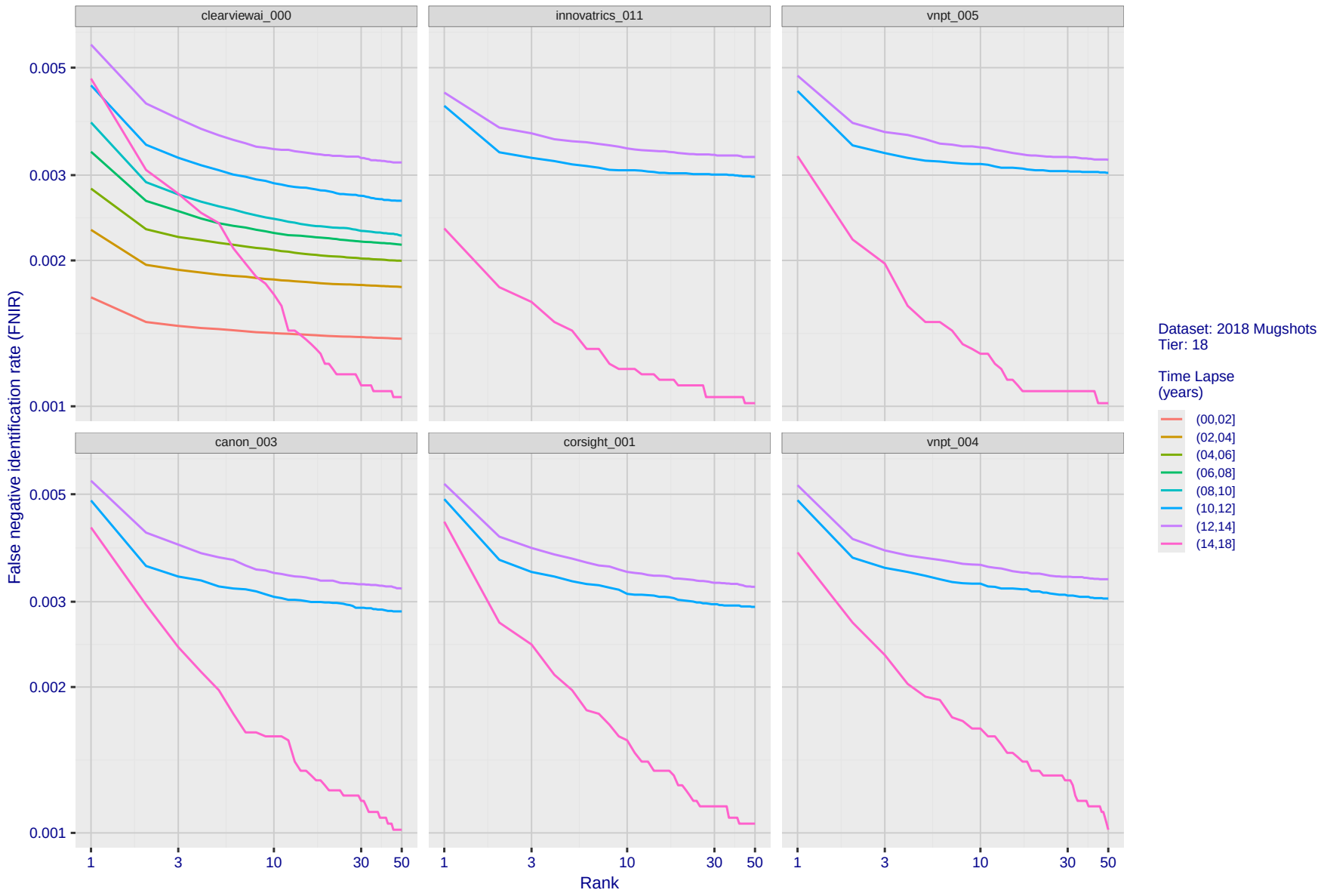

Figure 165: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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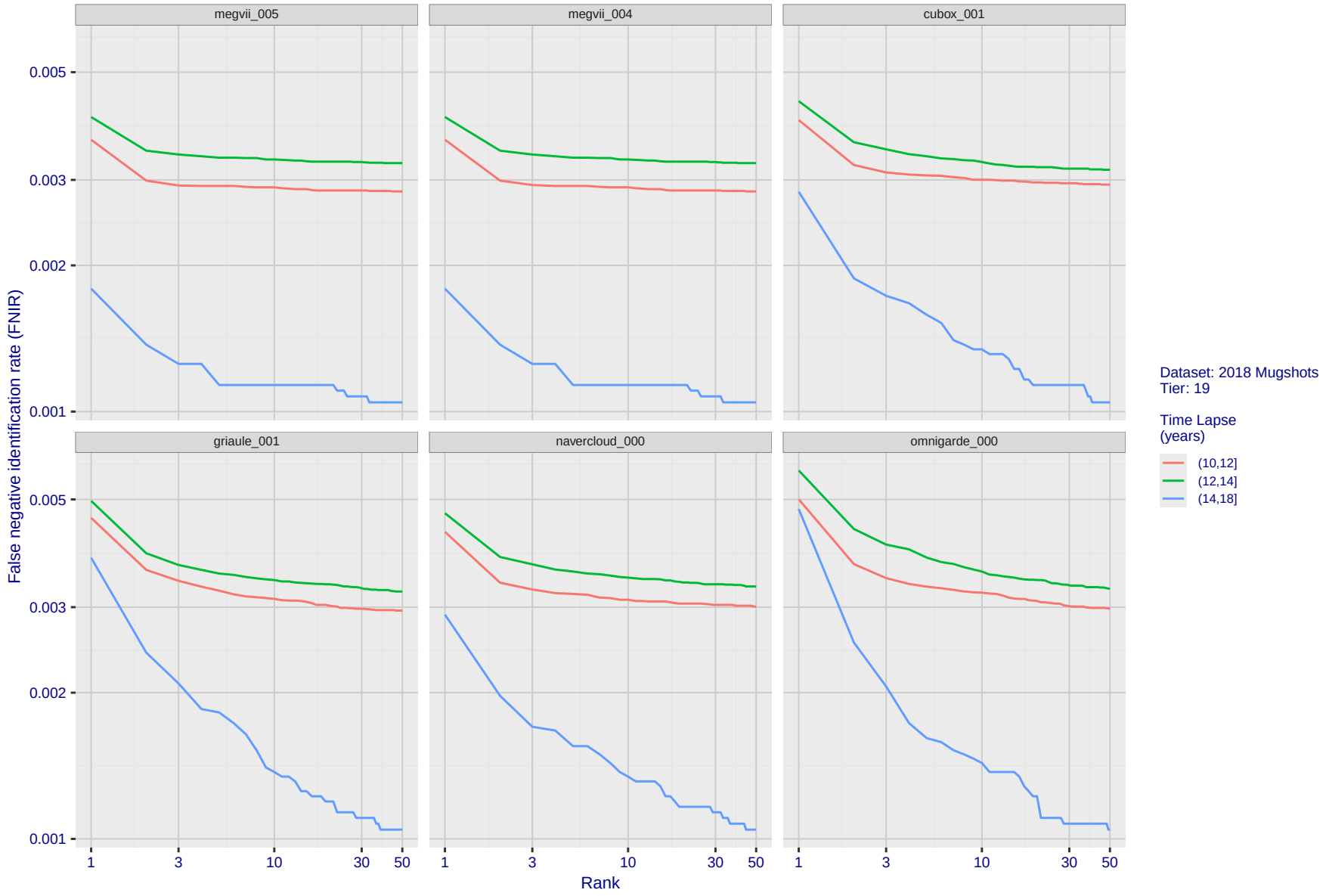

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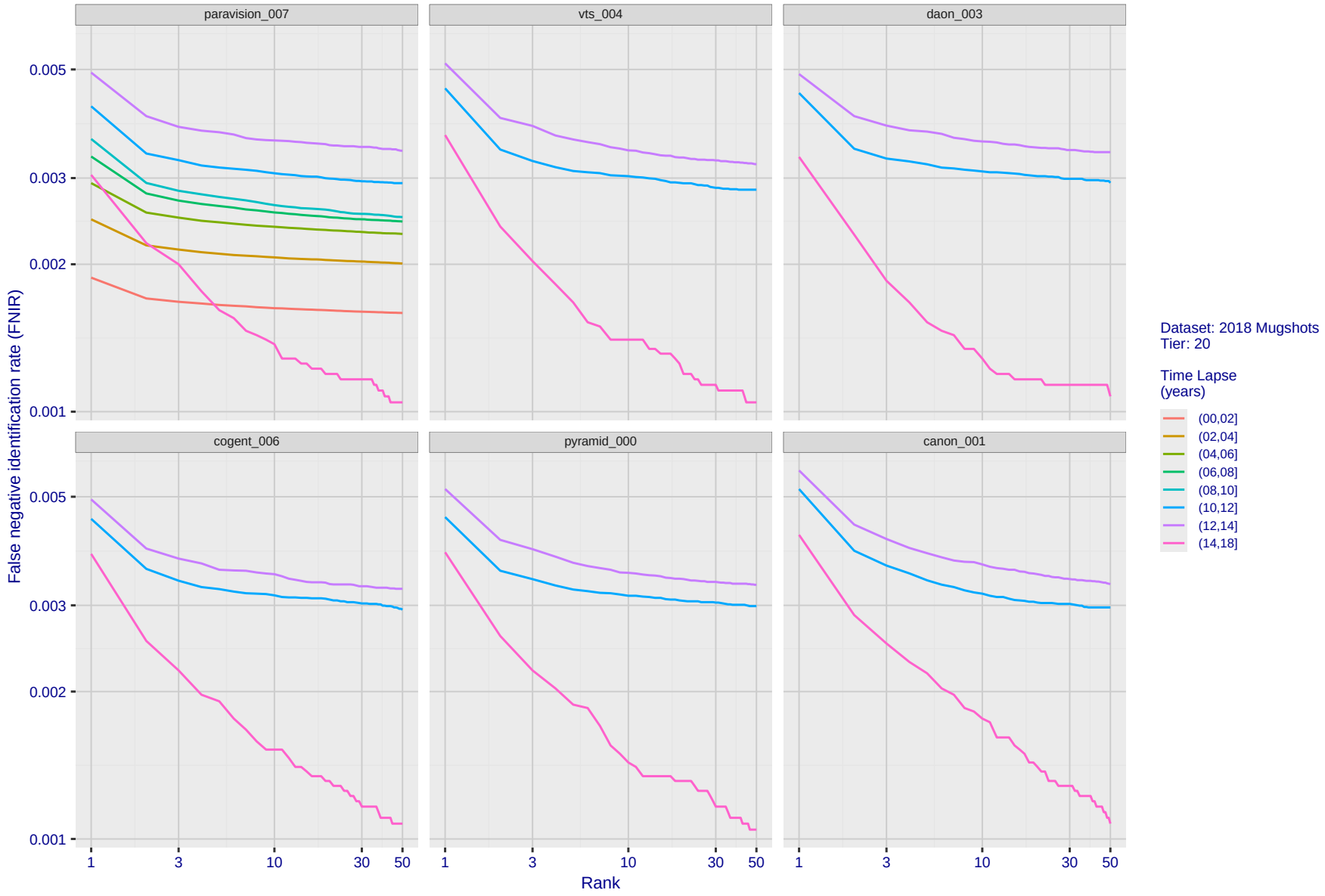

Figure 167: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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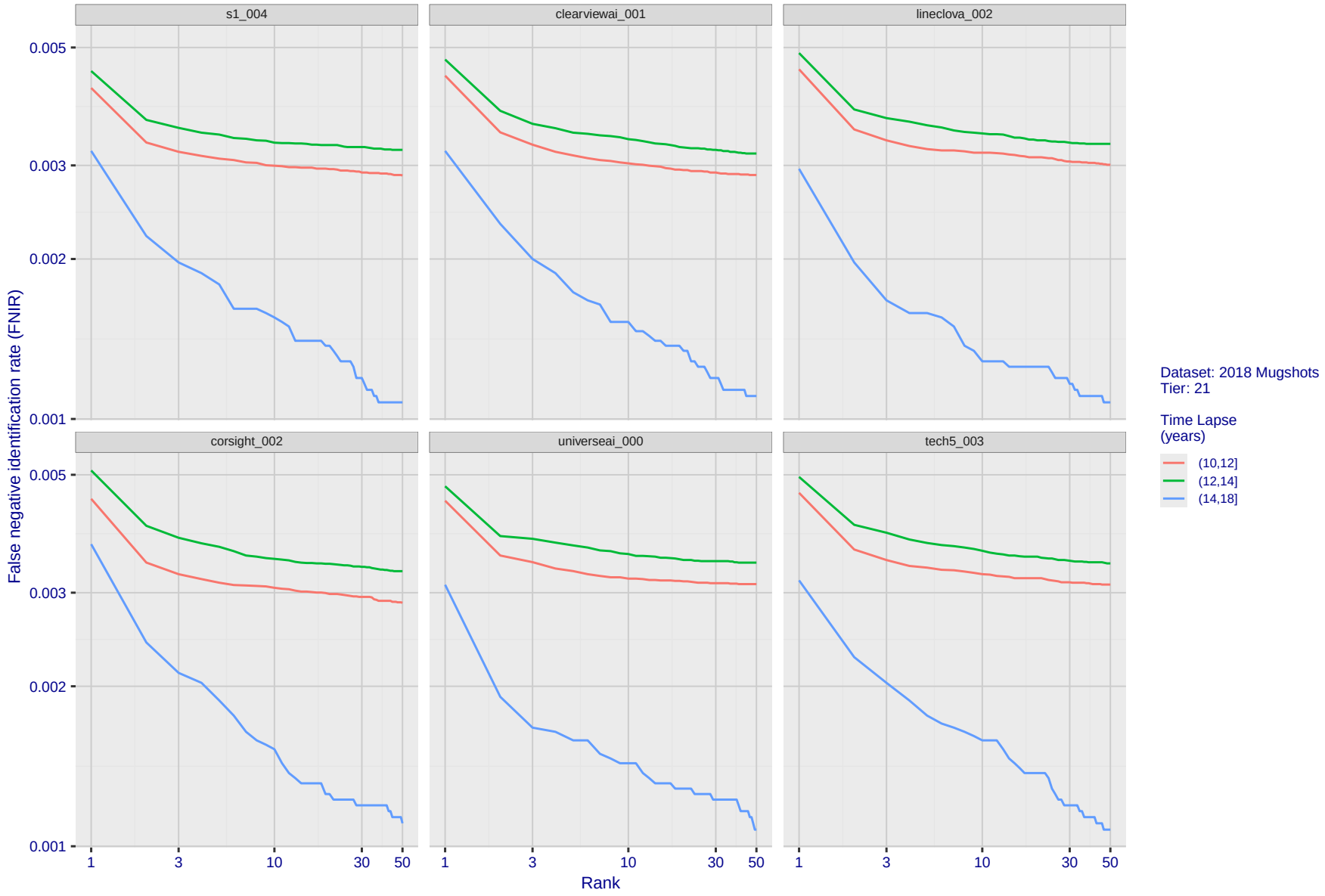

Figure 168: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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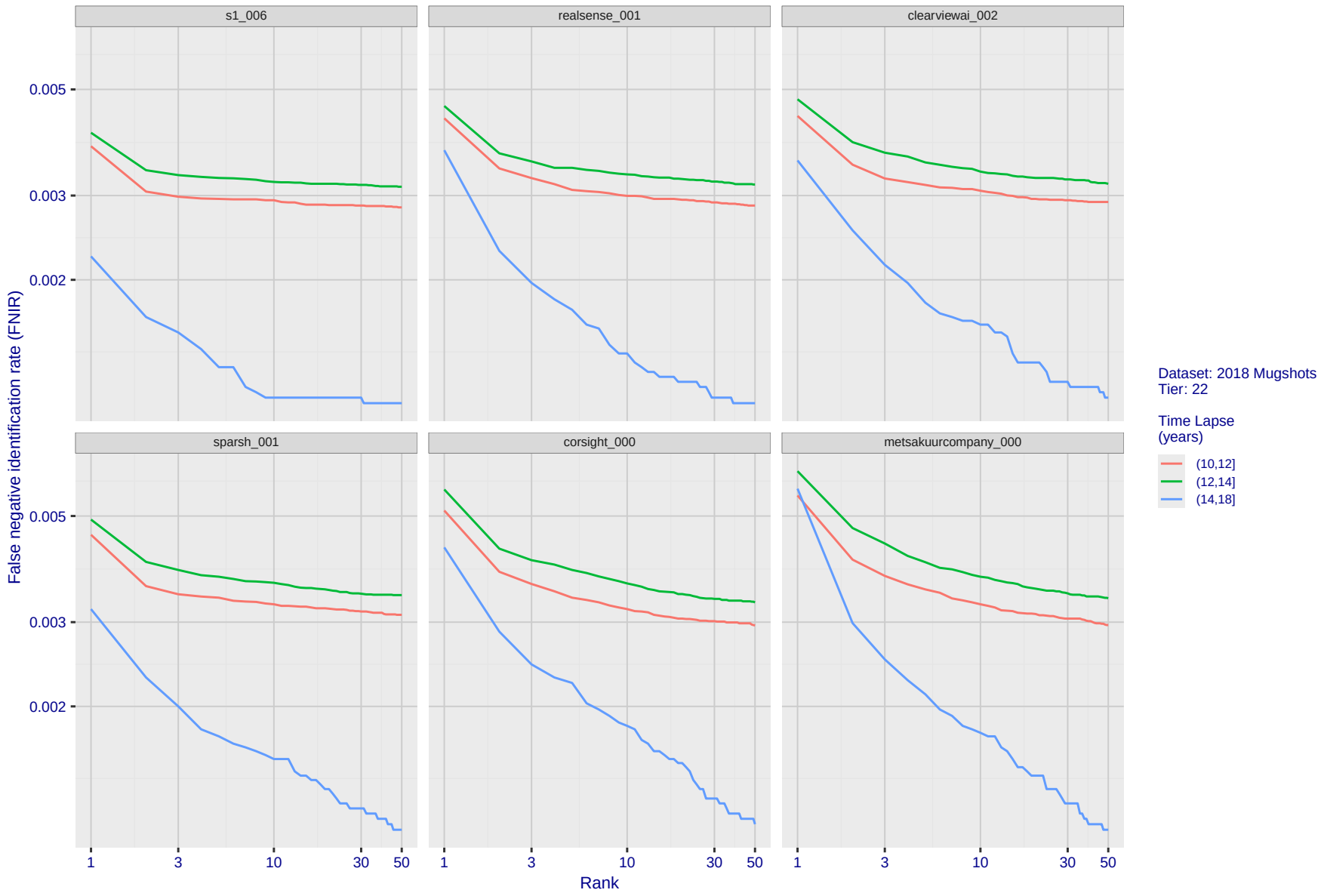

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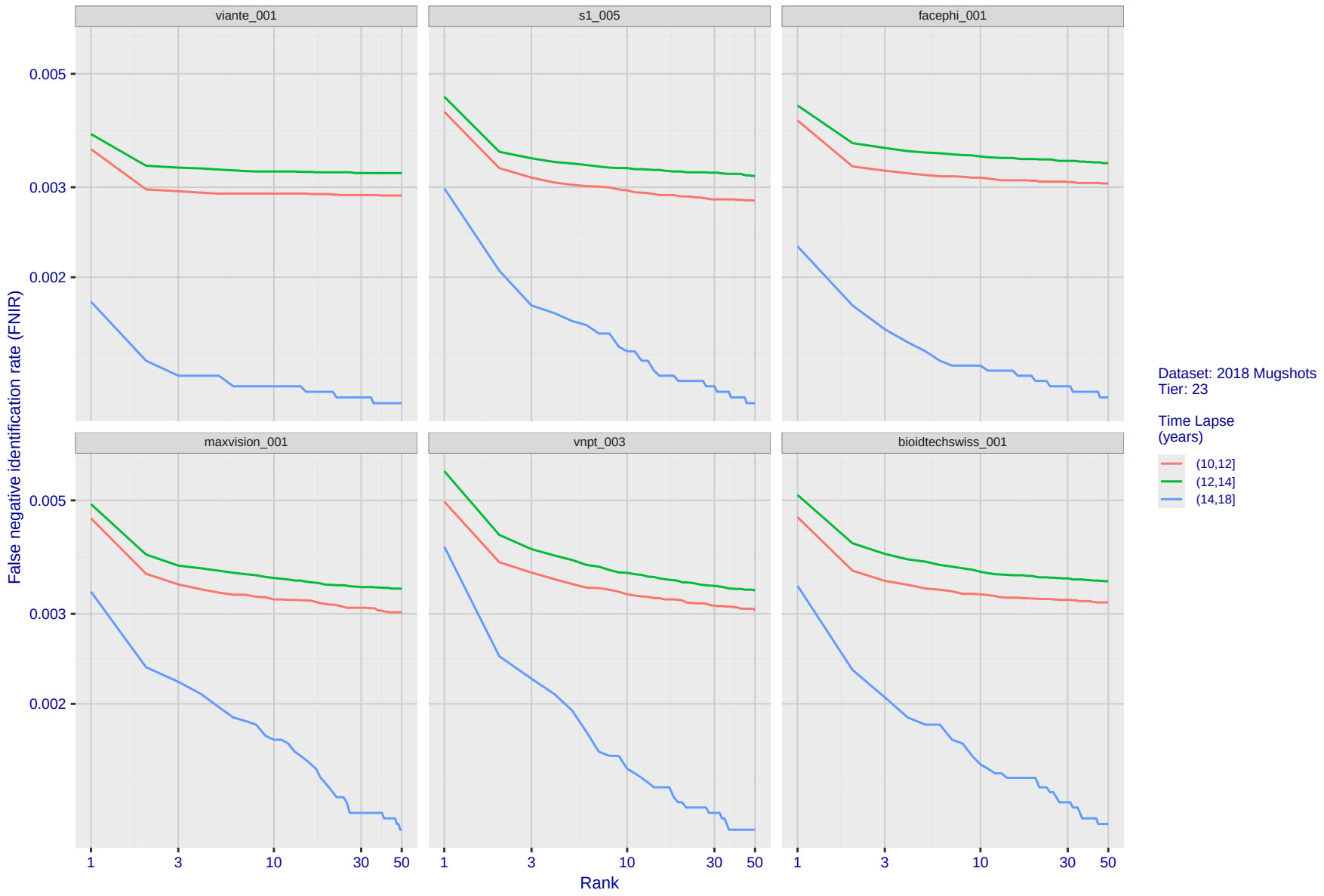

Figure 170: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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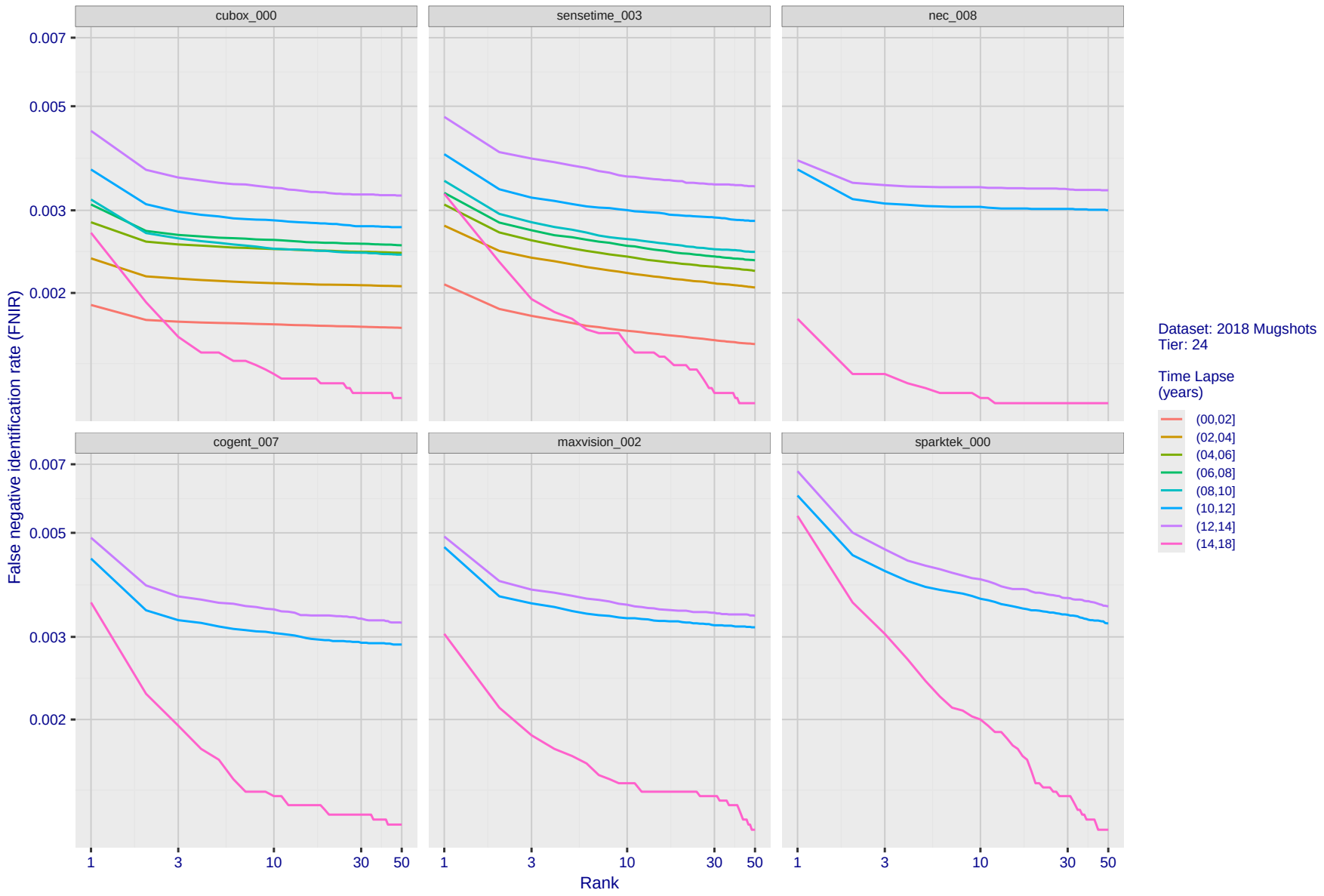

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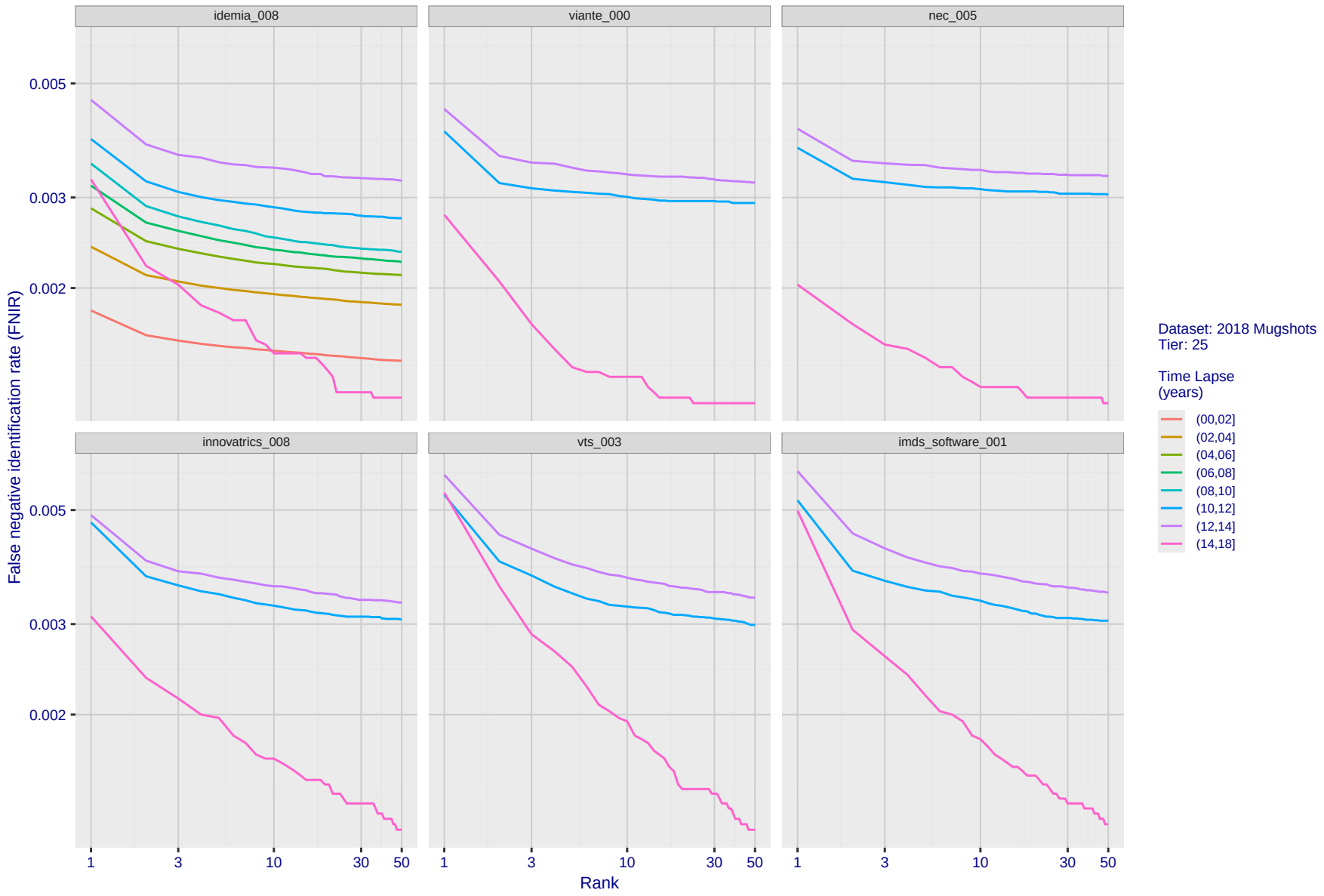

Figure 172: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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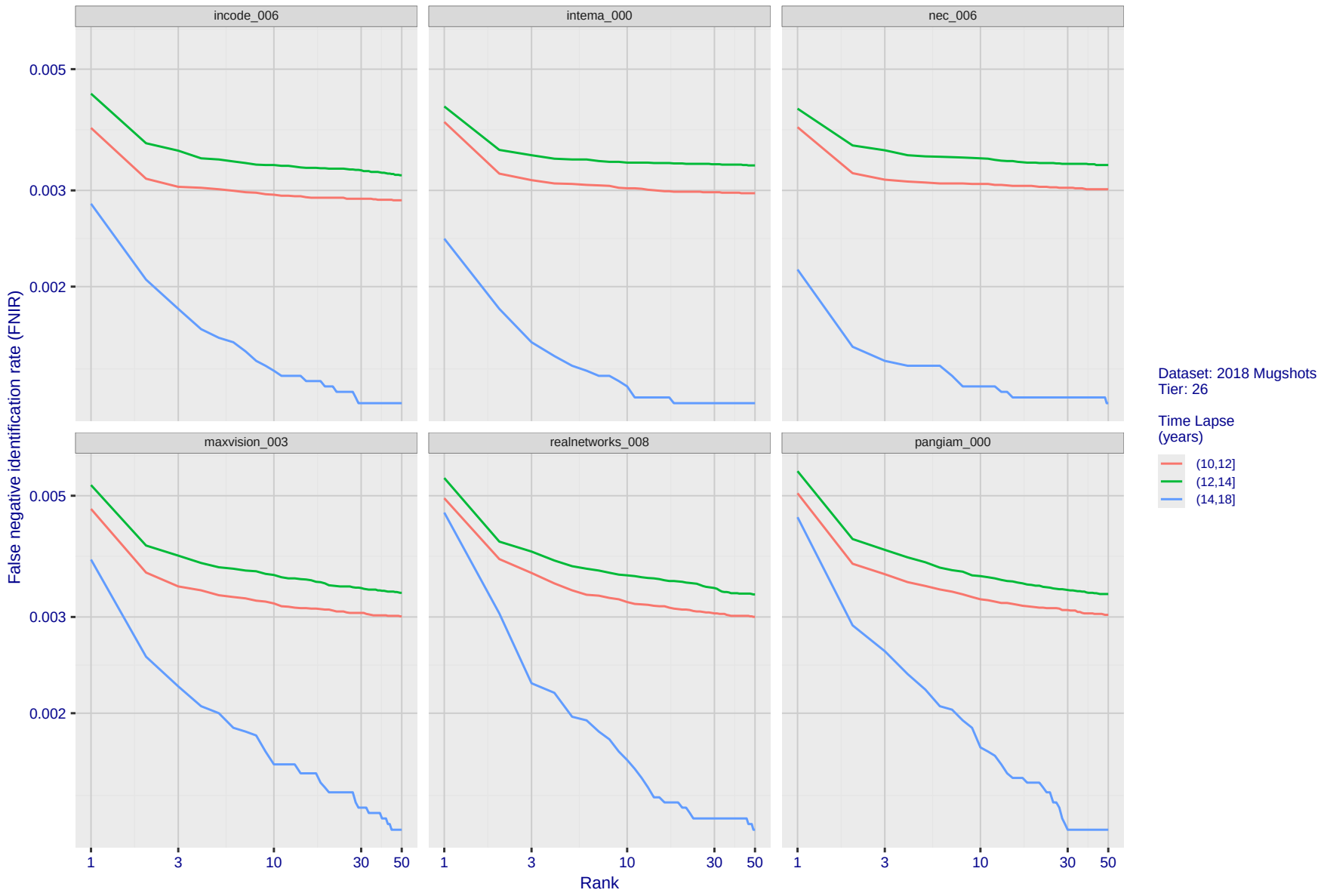

Figure 173: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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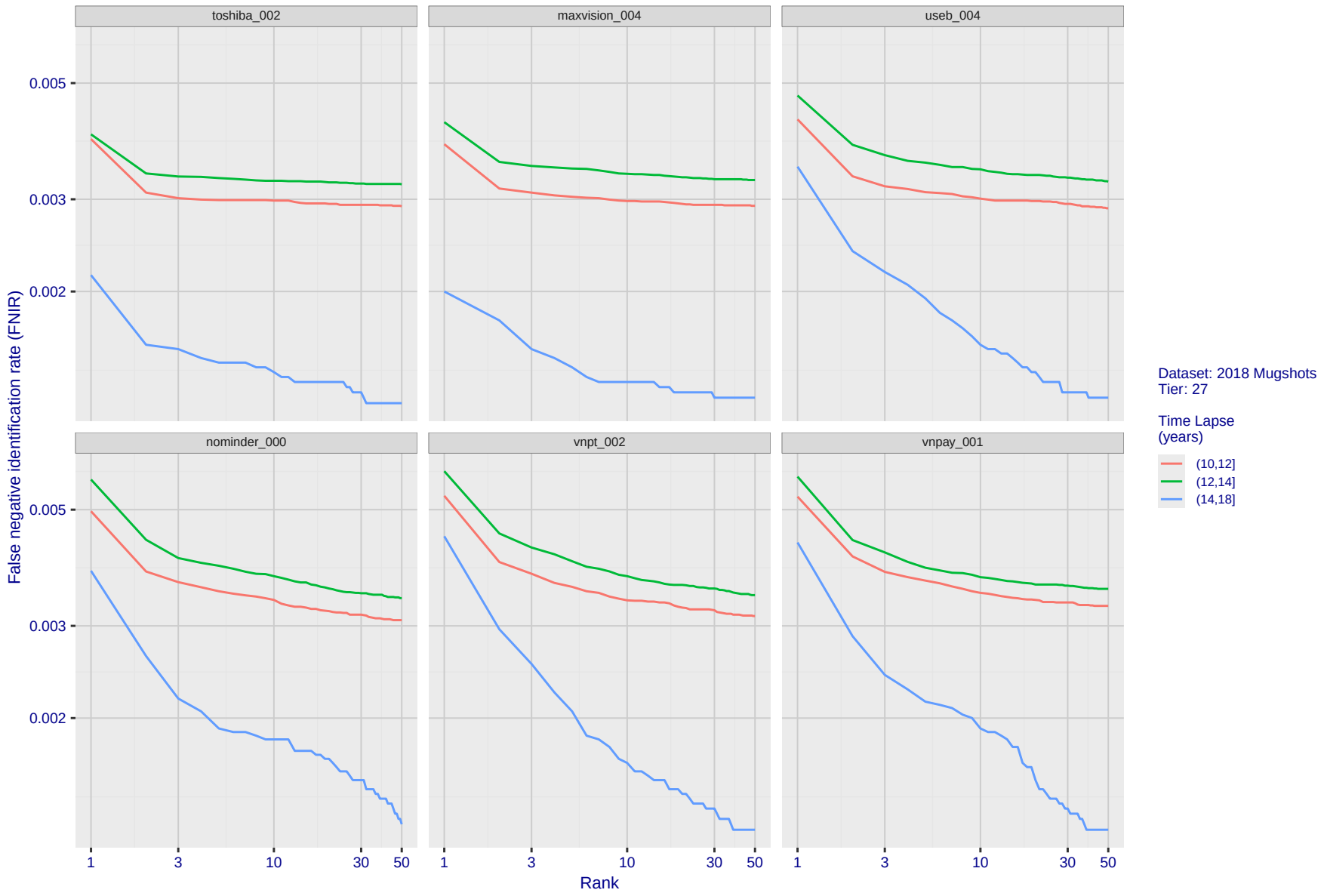

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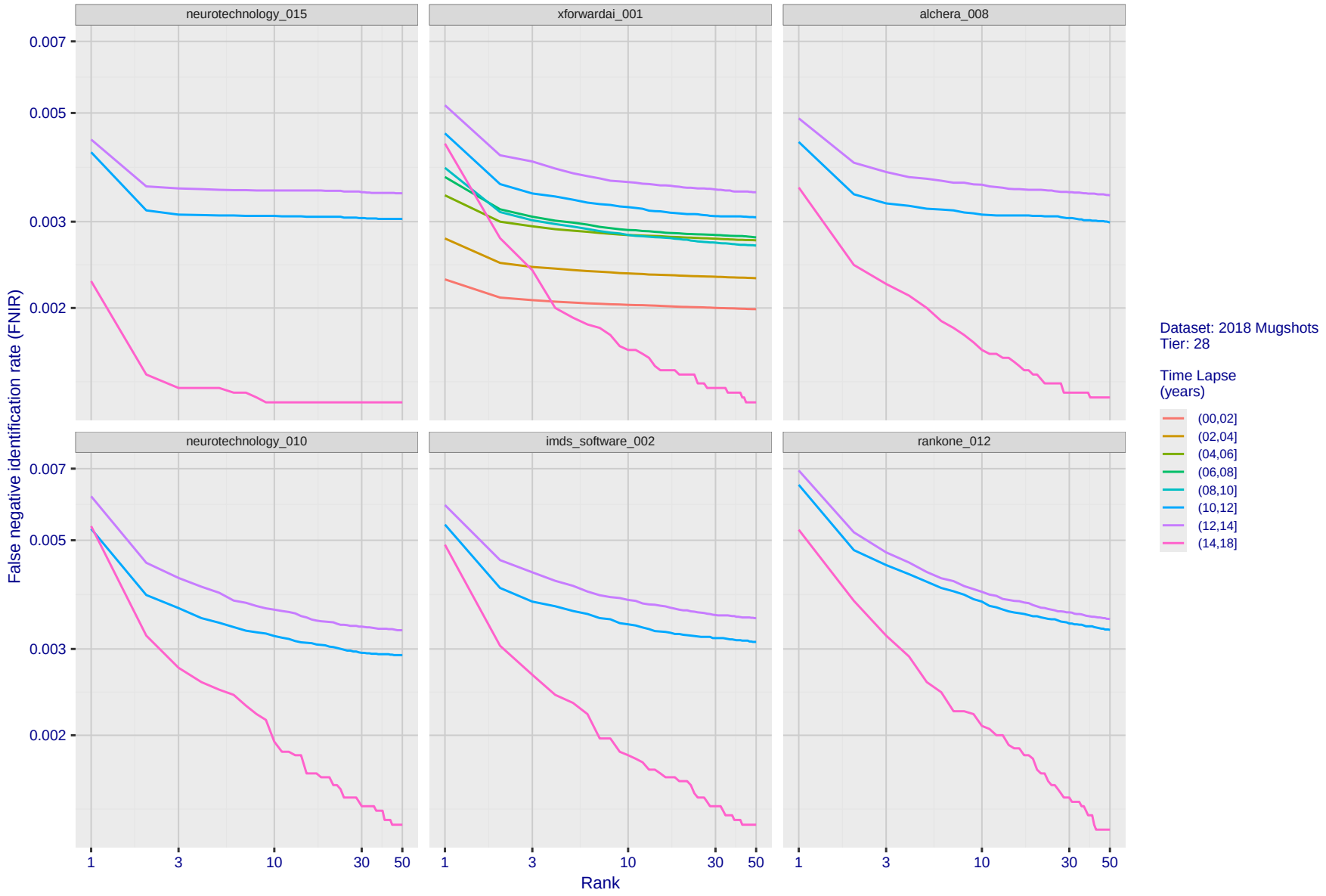

Figure 175: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
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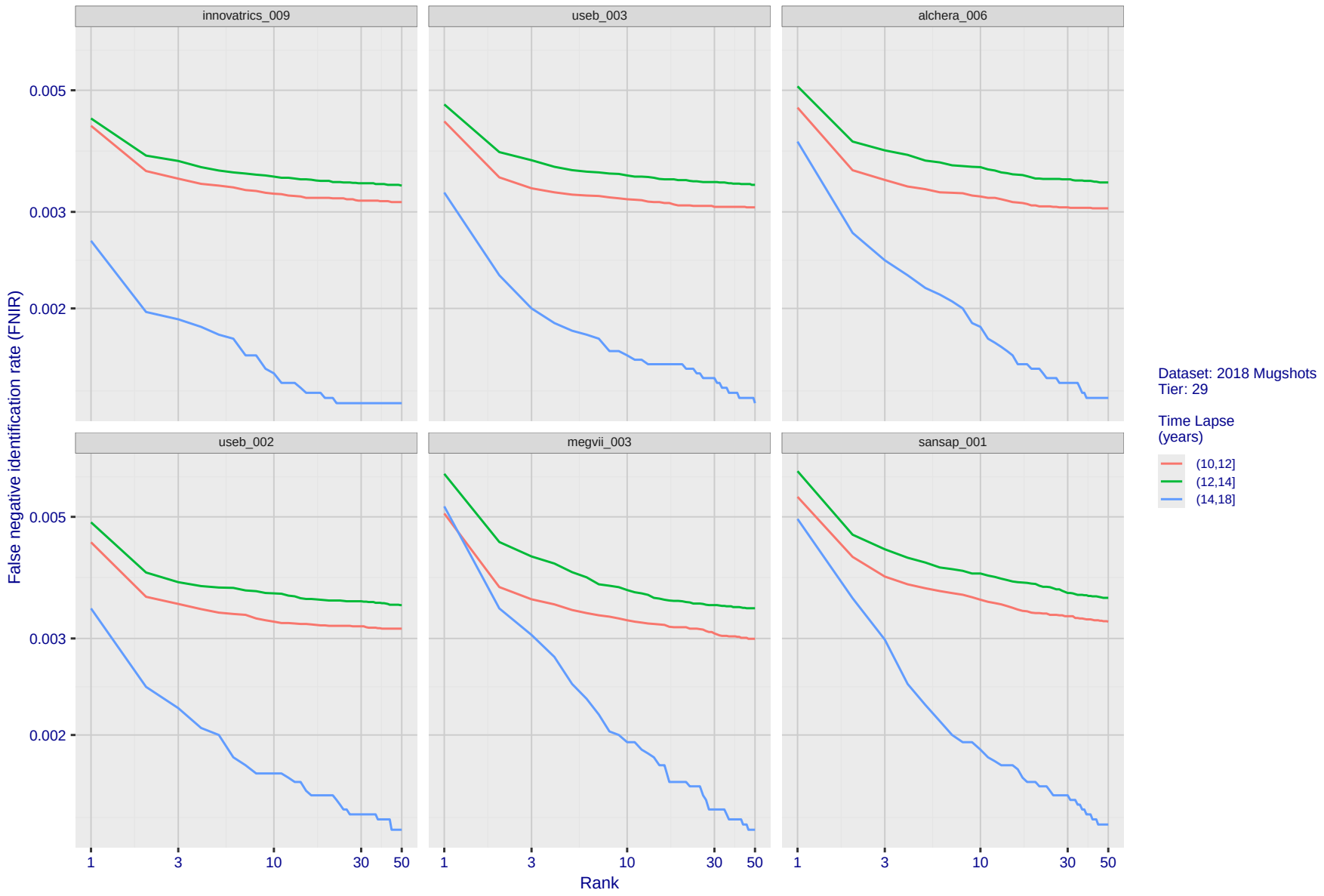

Figure 176: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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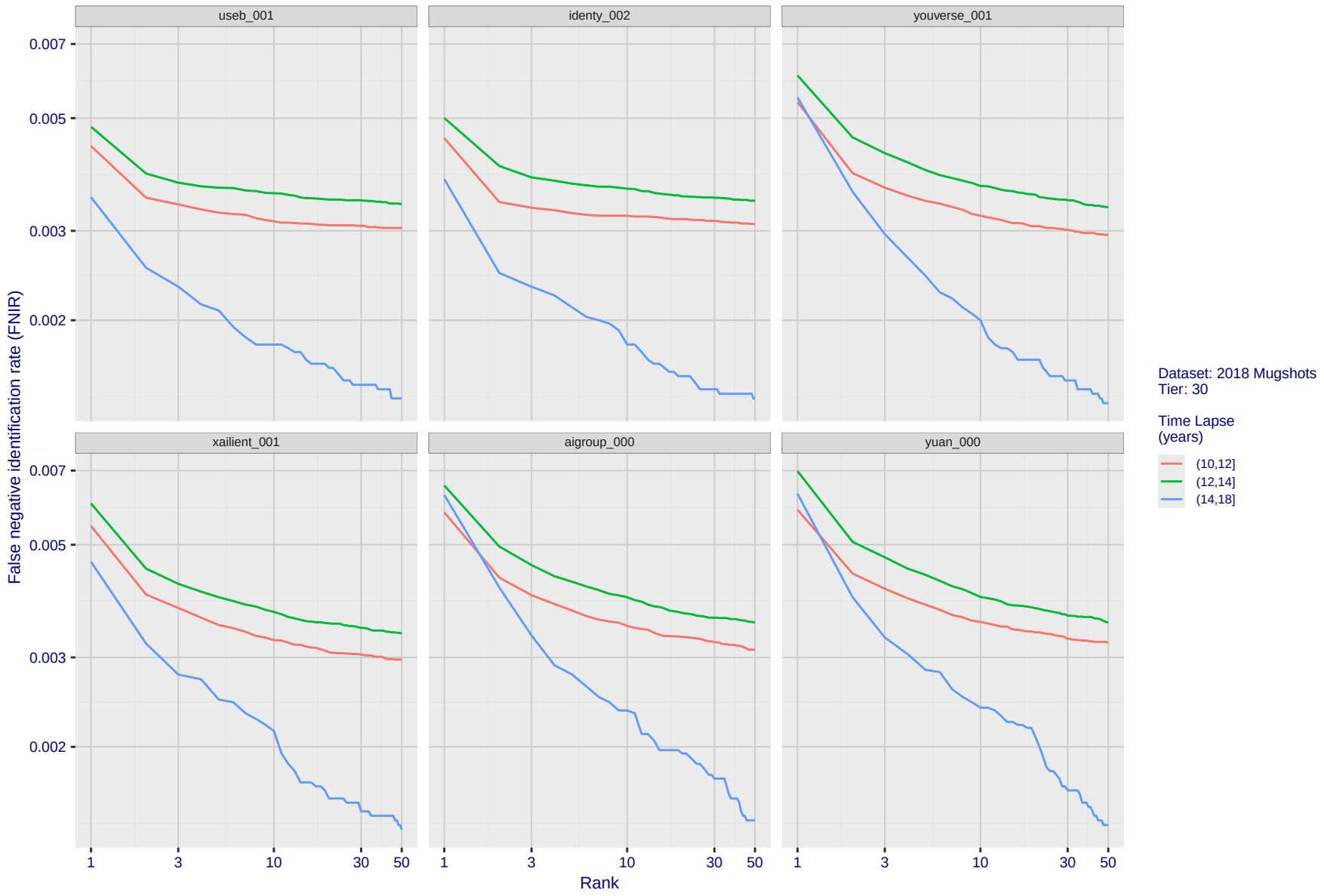

Figure 177: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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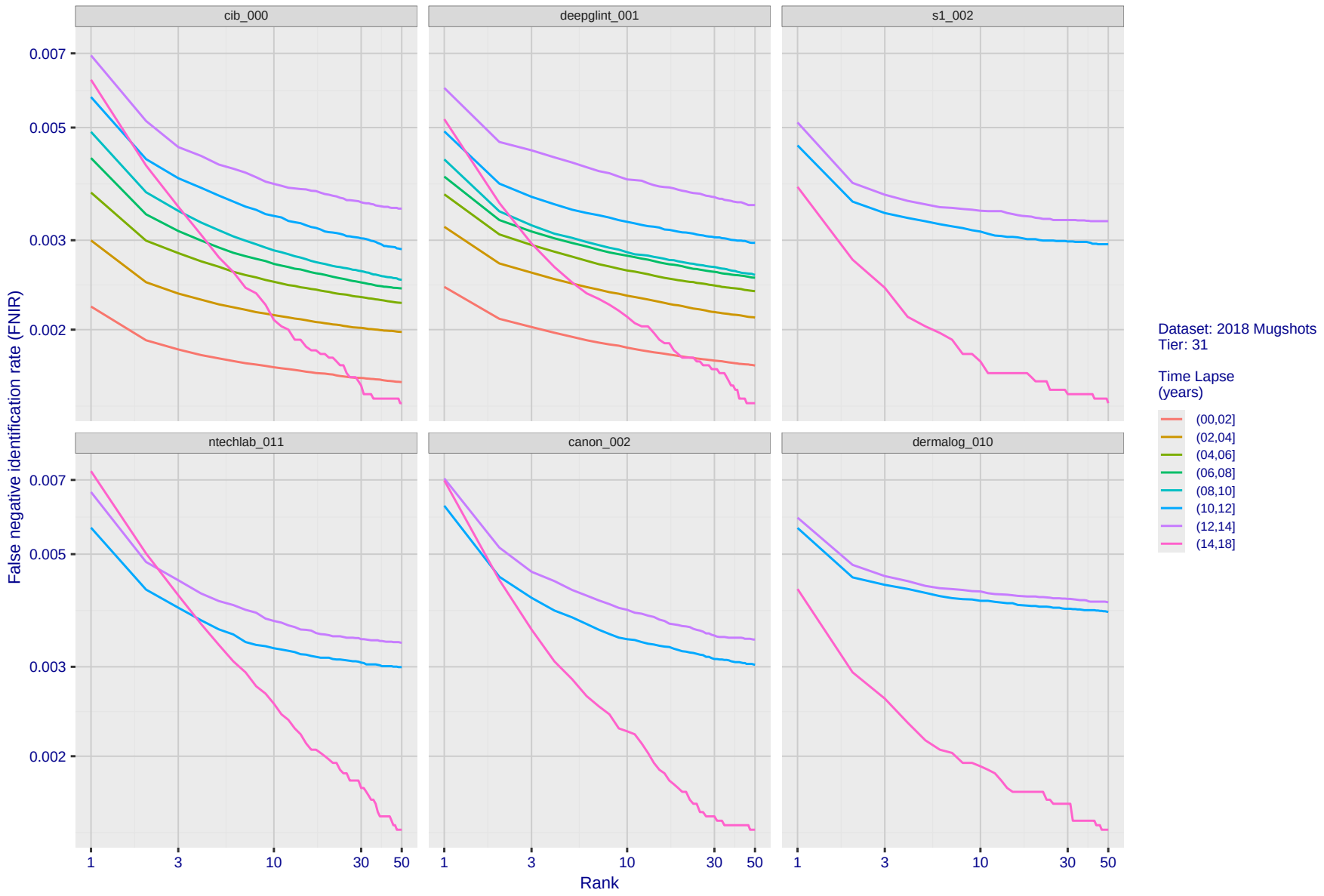

Figure 178: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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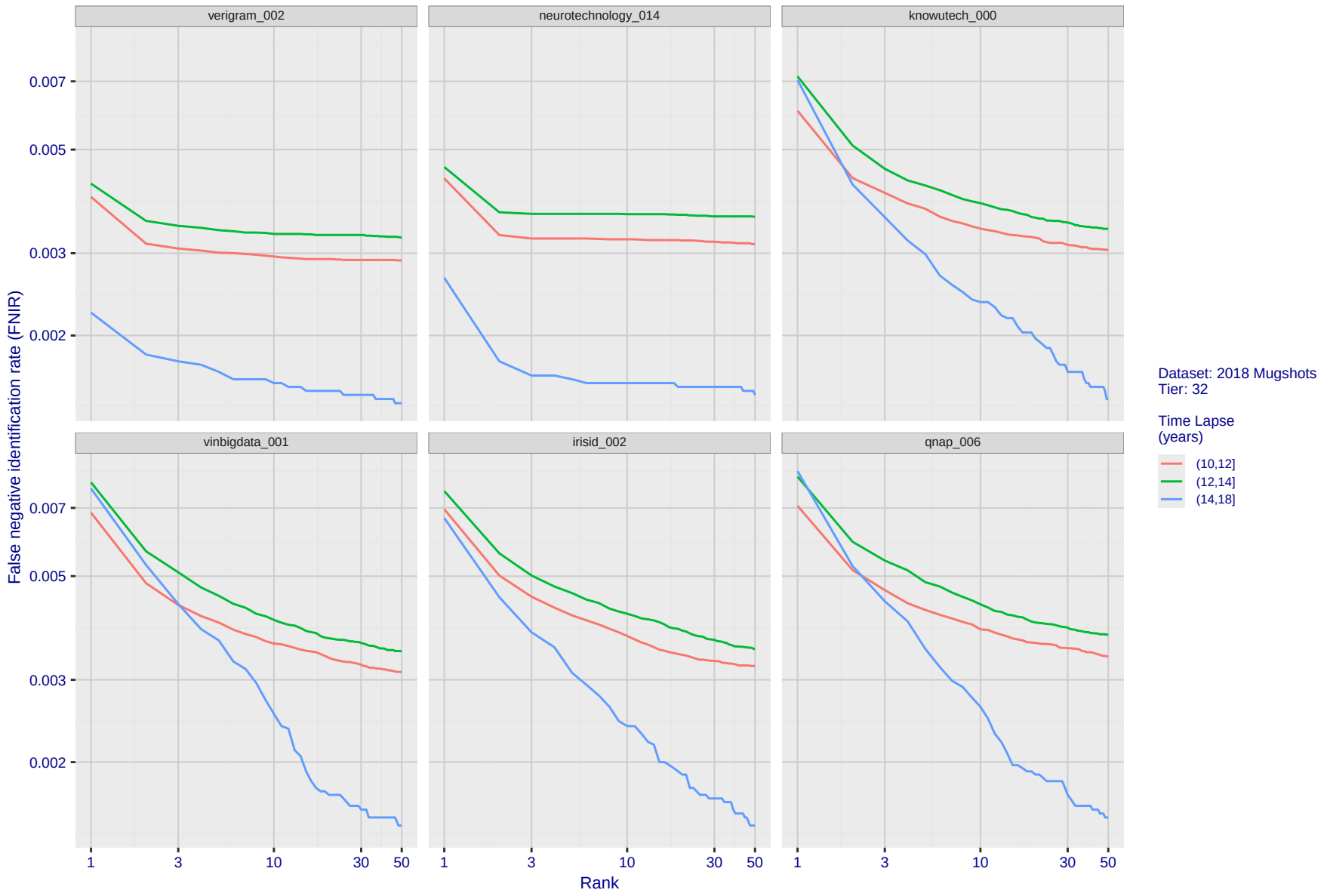

Figure 179: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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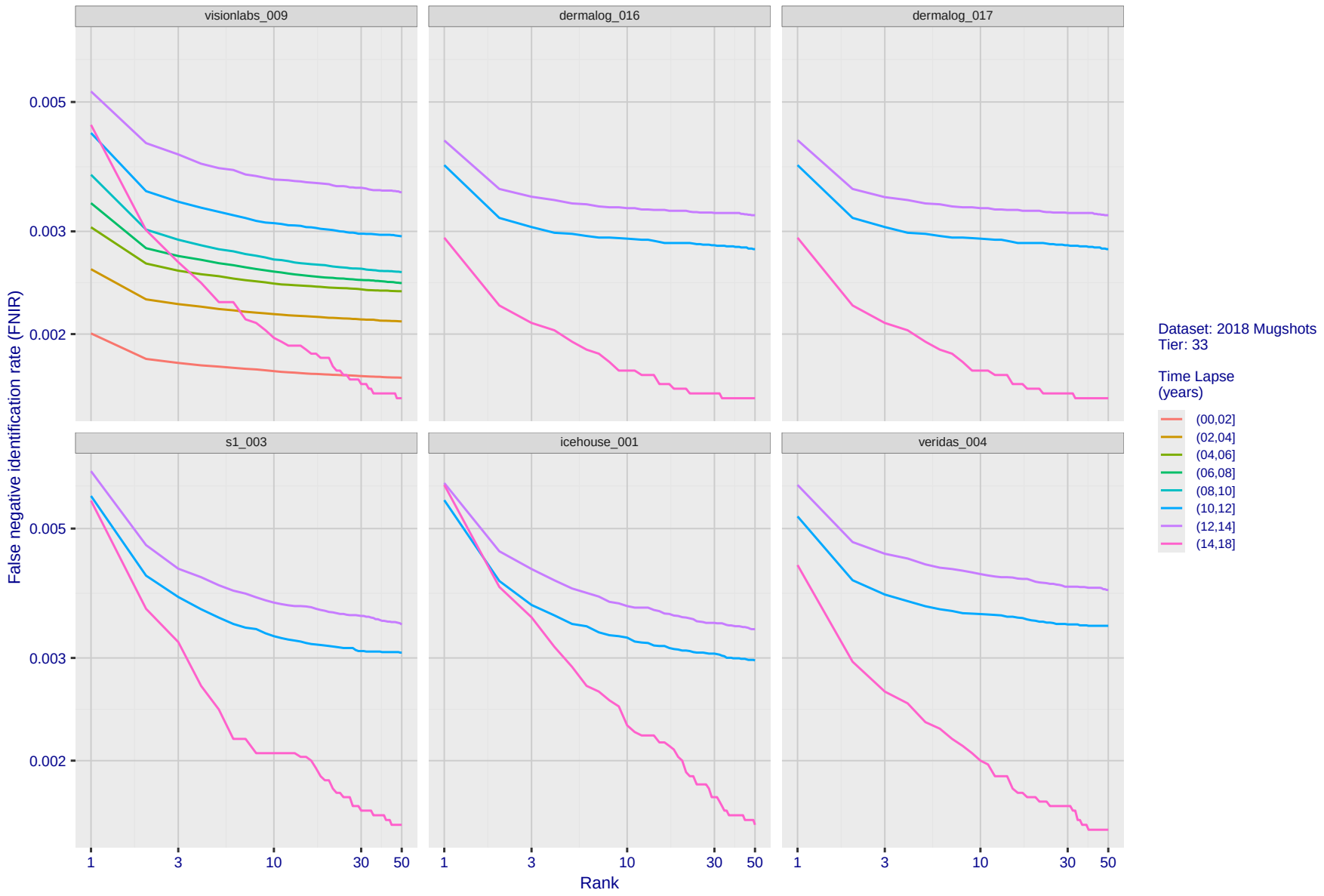

Figure 180: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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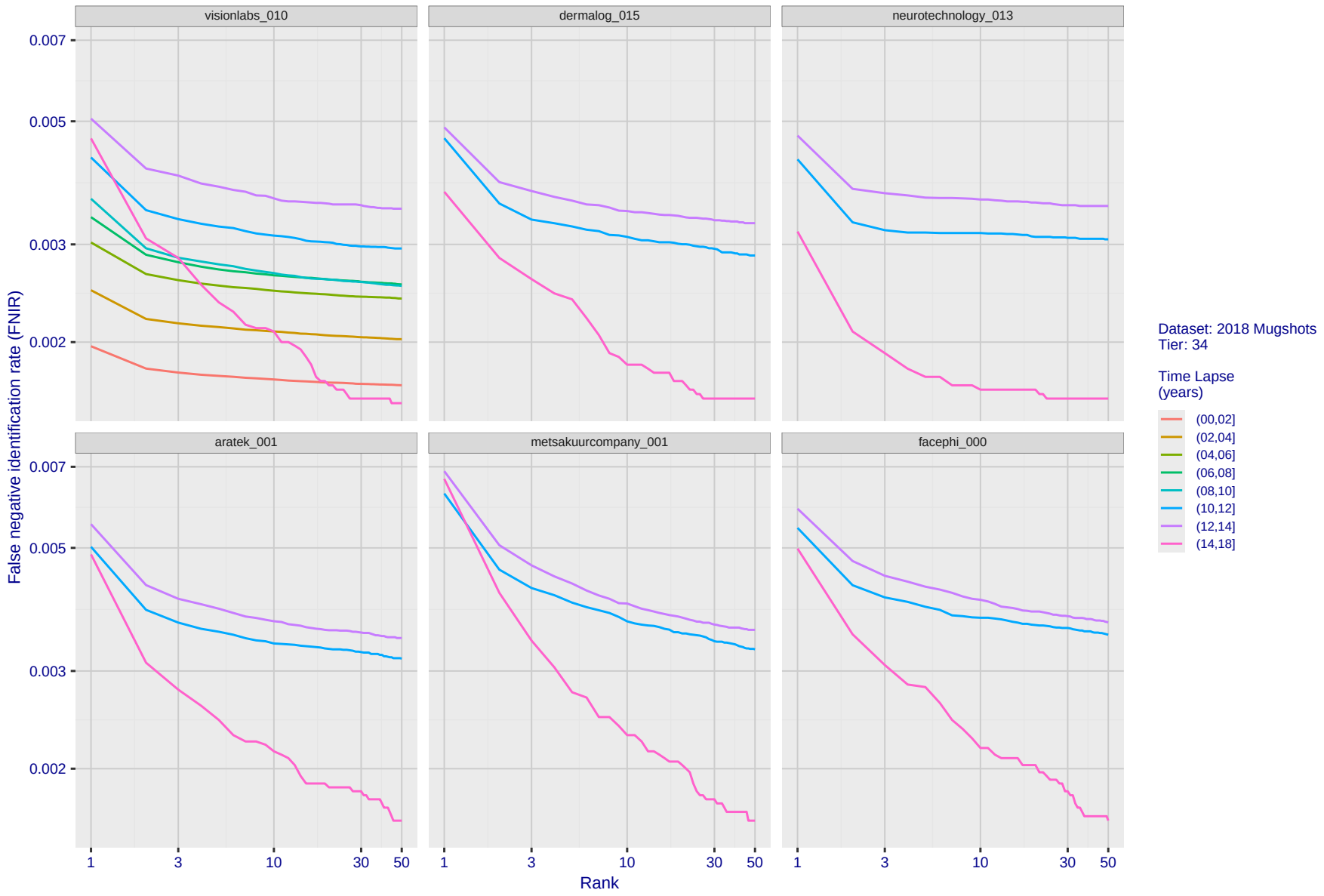

Figure 181: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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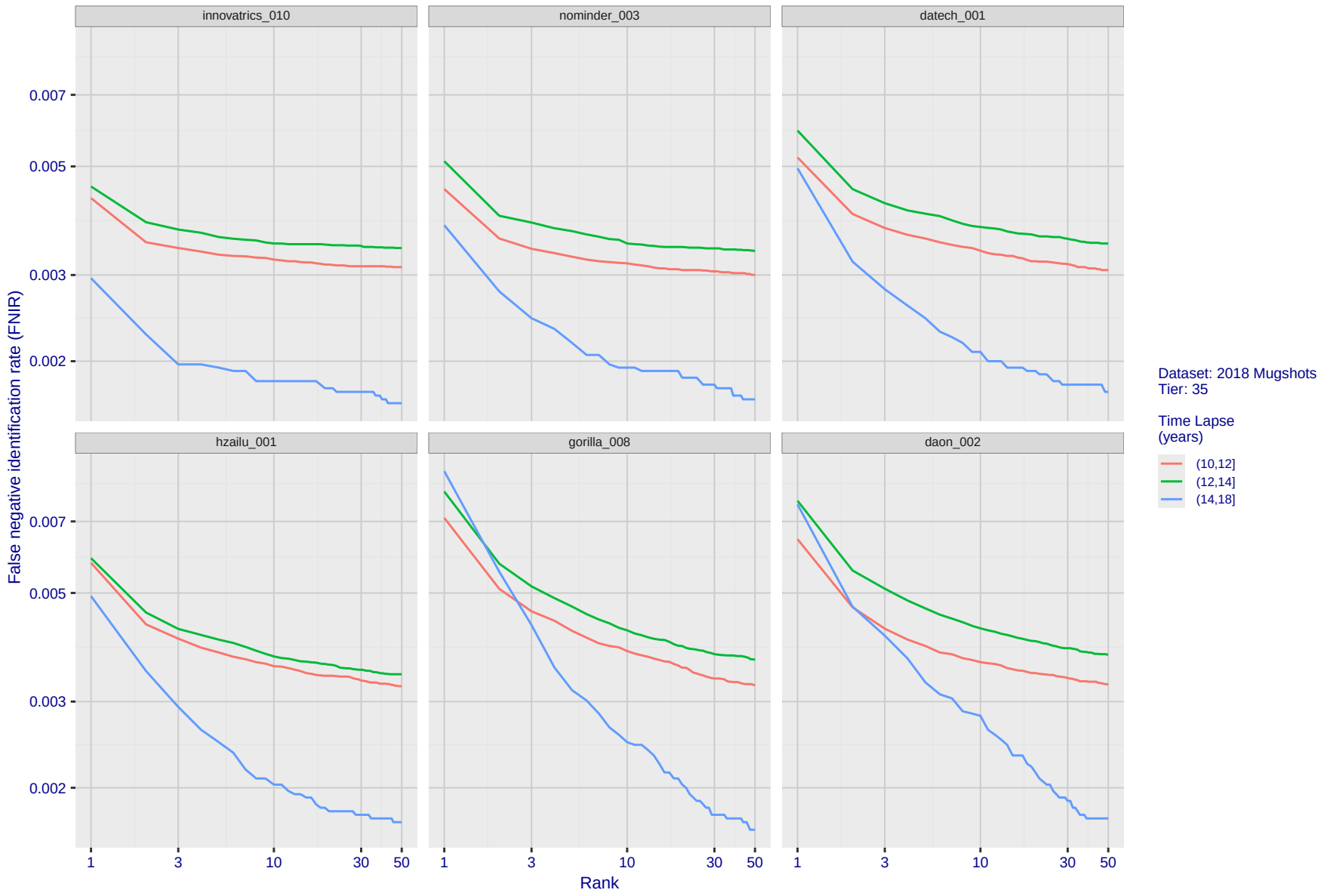

Figure 182: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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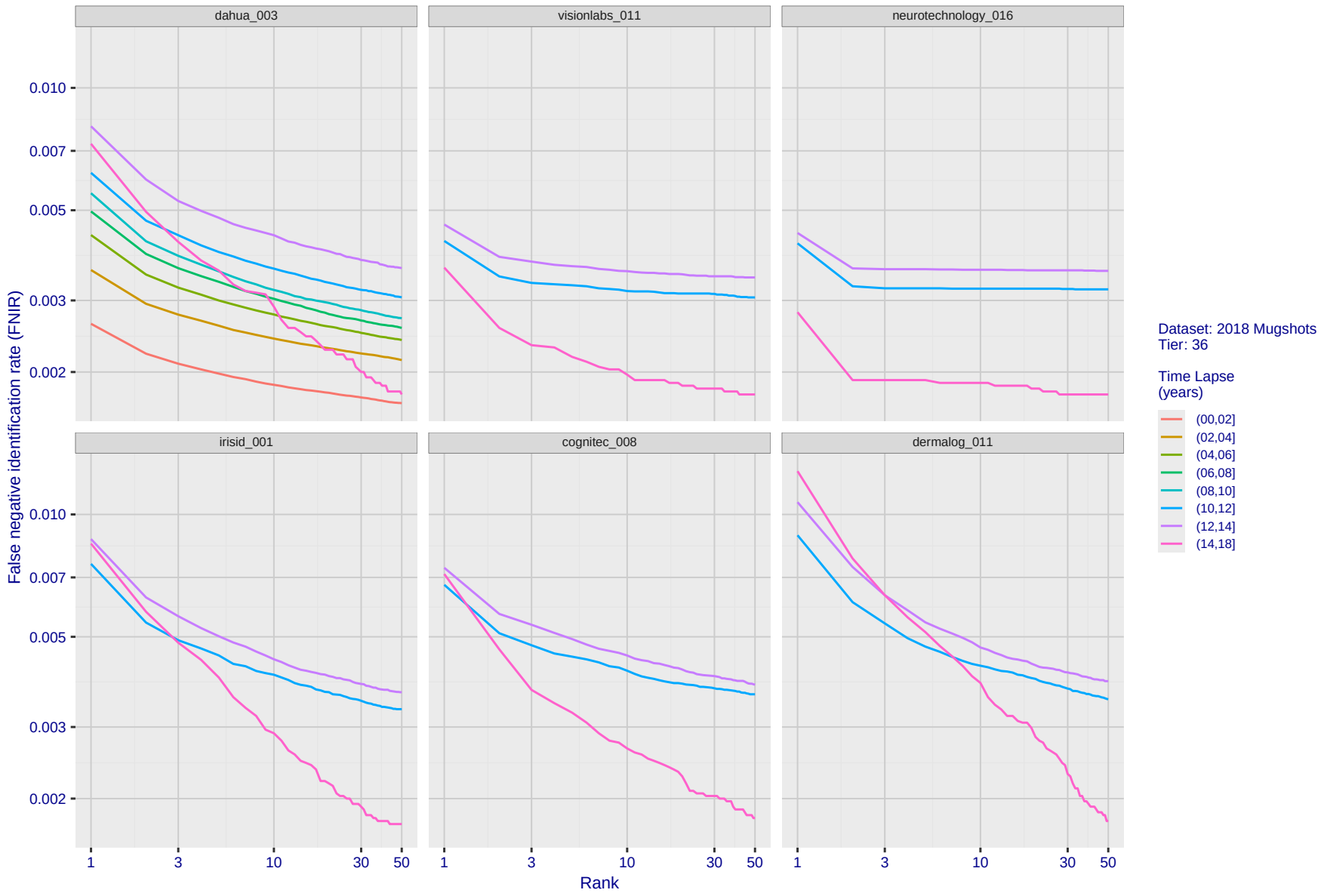

Figure 183: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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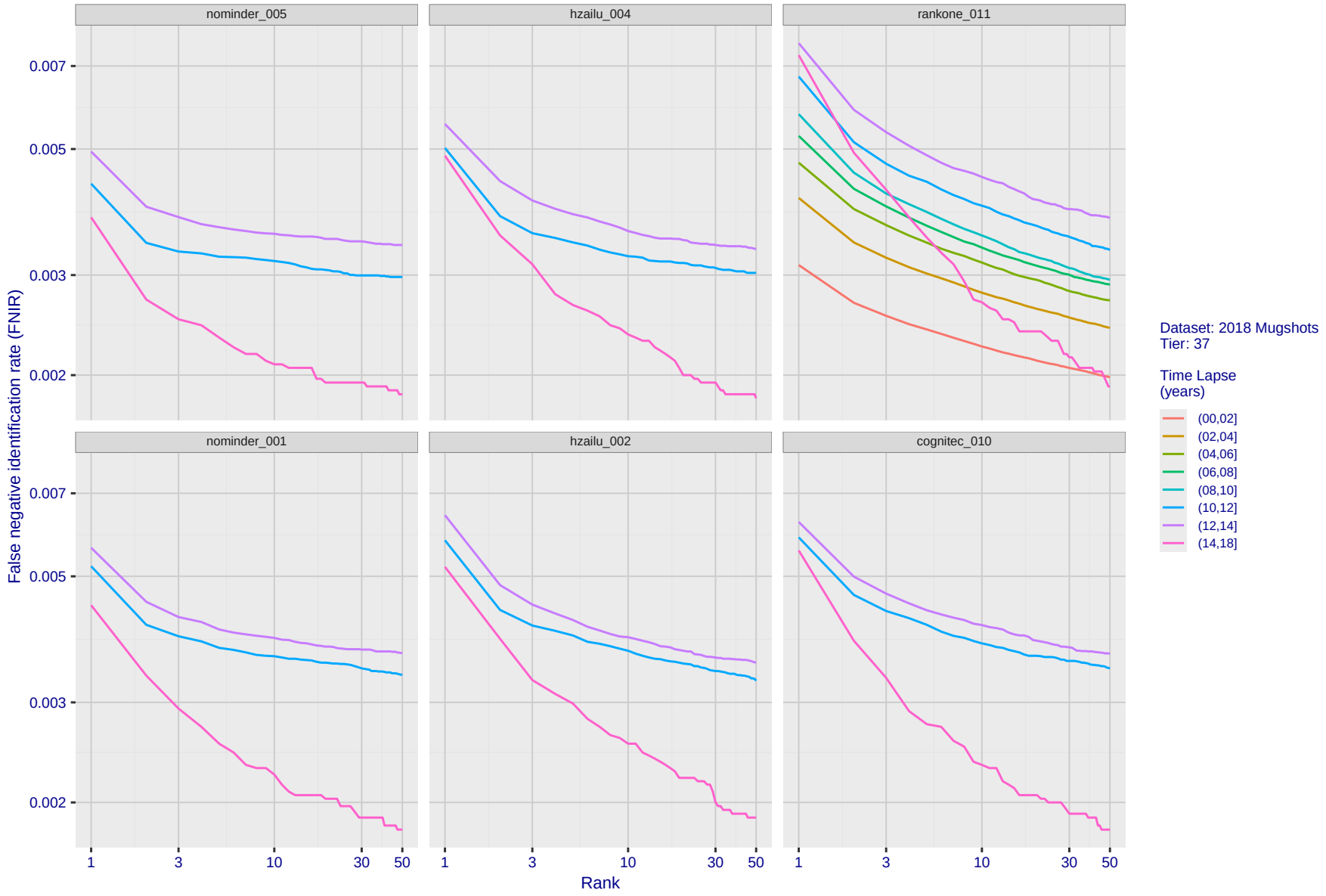

Figure 184: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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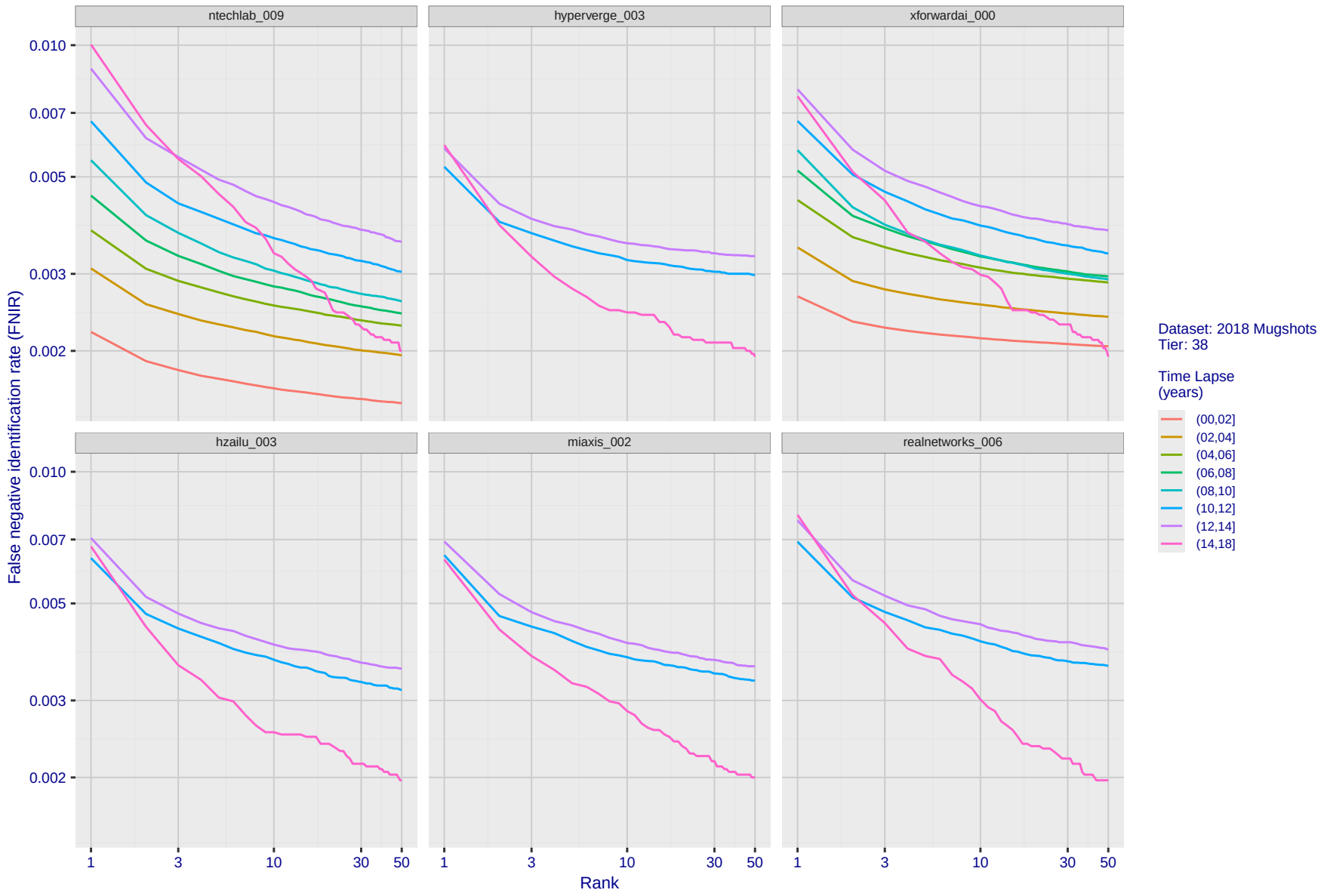

Figure 185: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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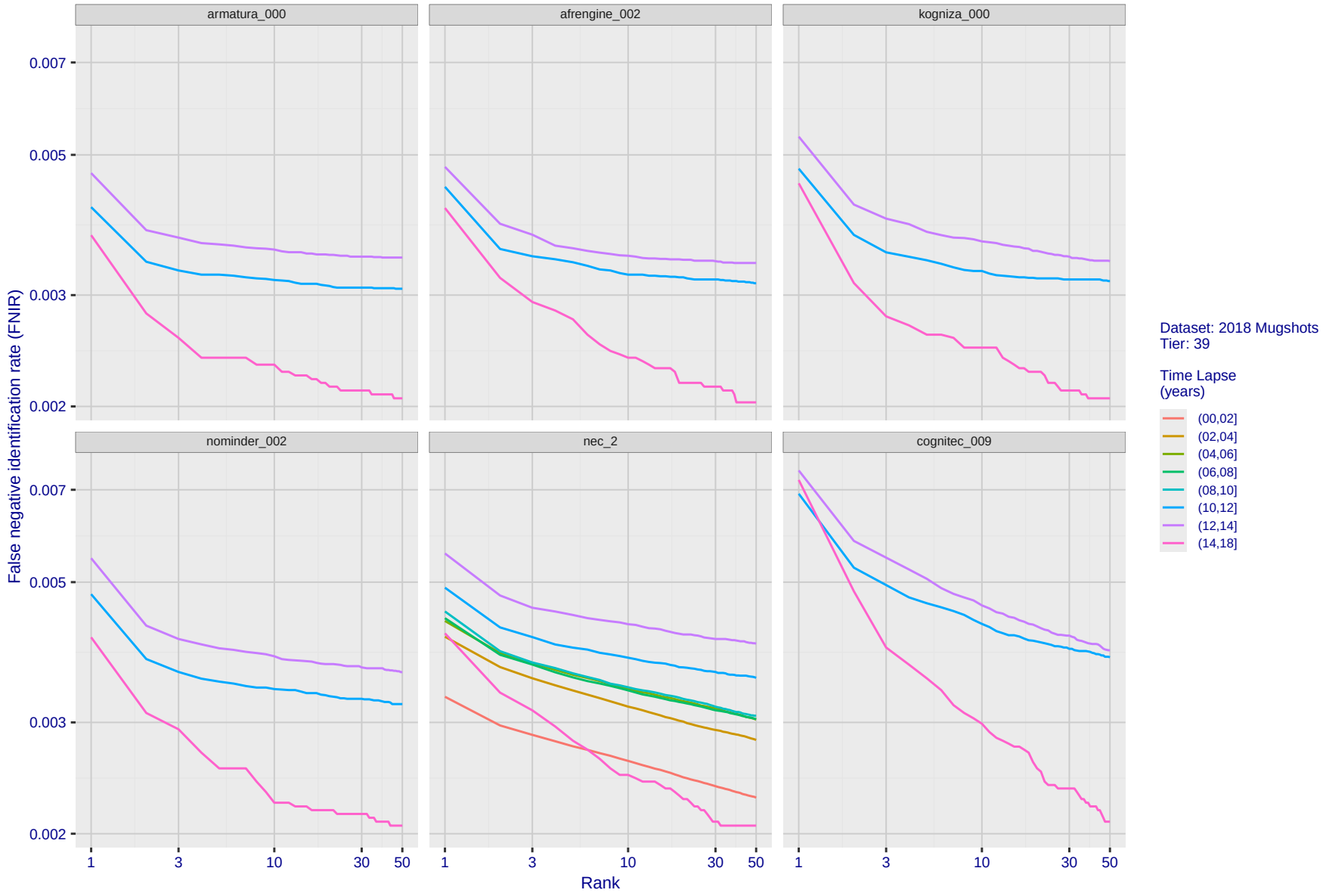

Figure 186: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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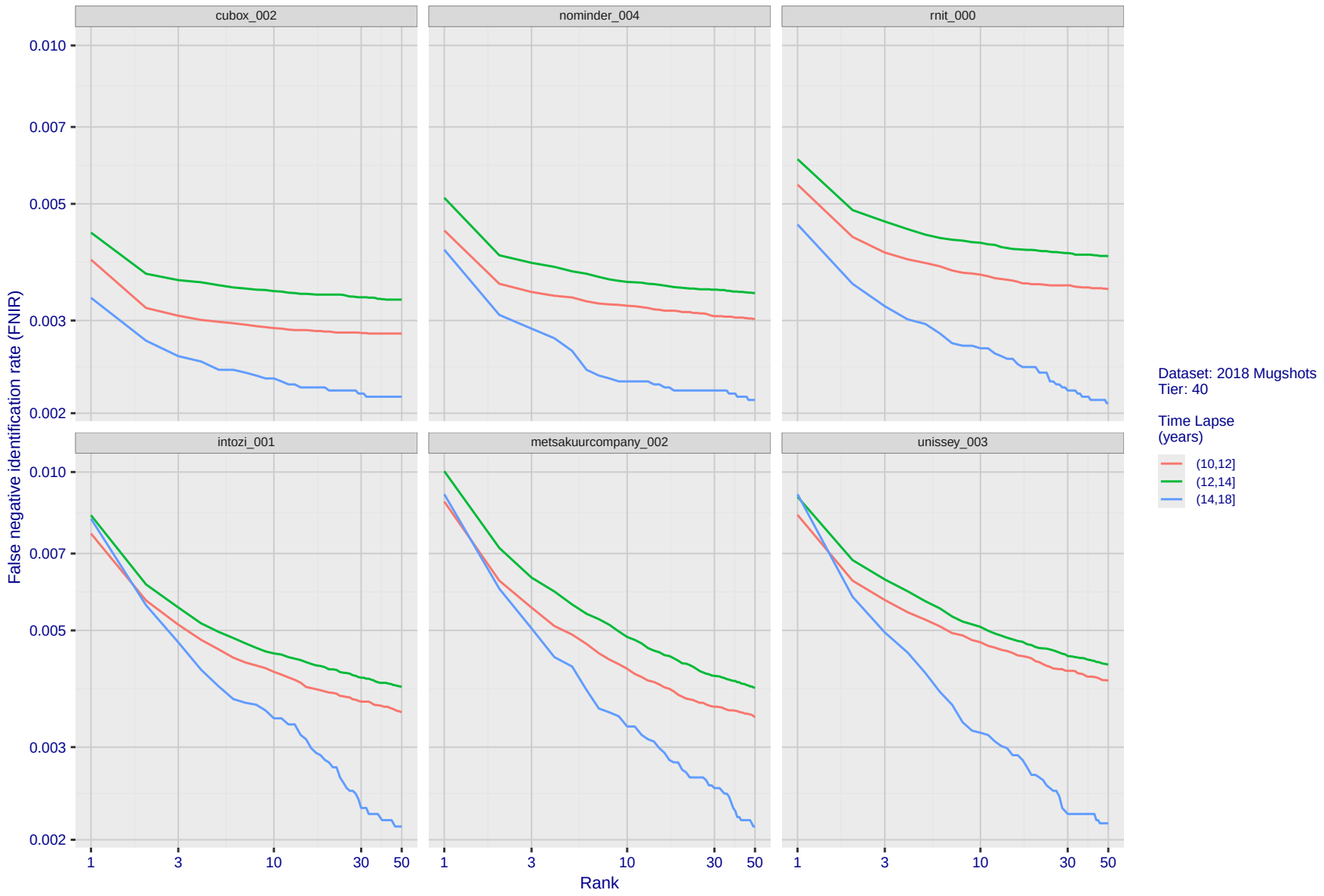

Figure 187: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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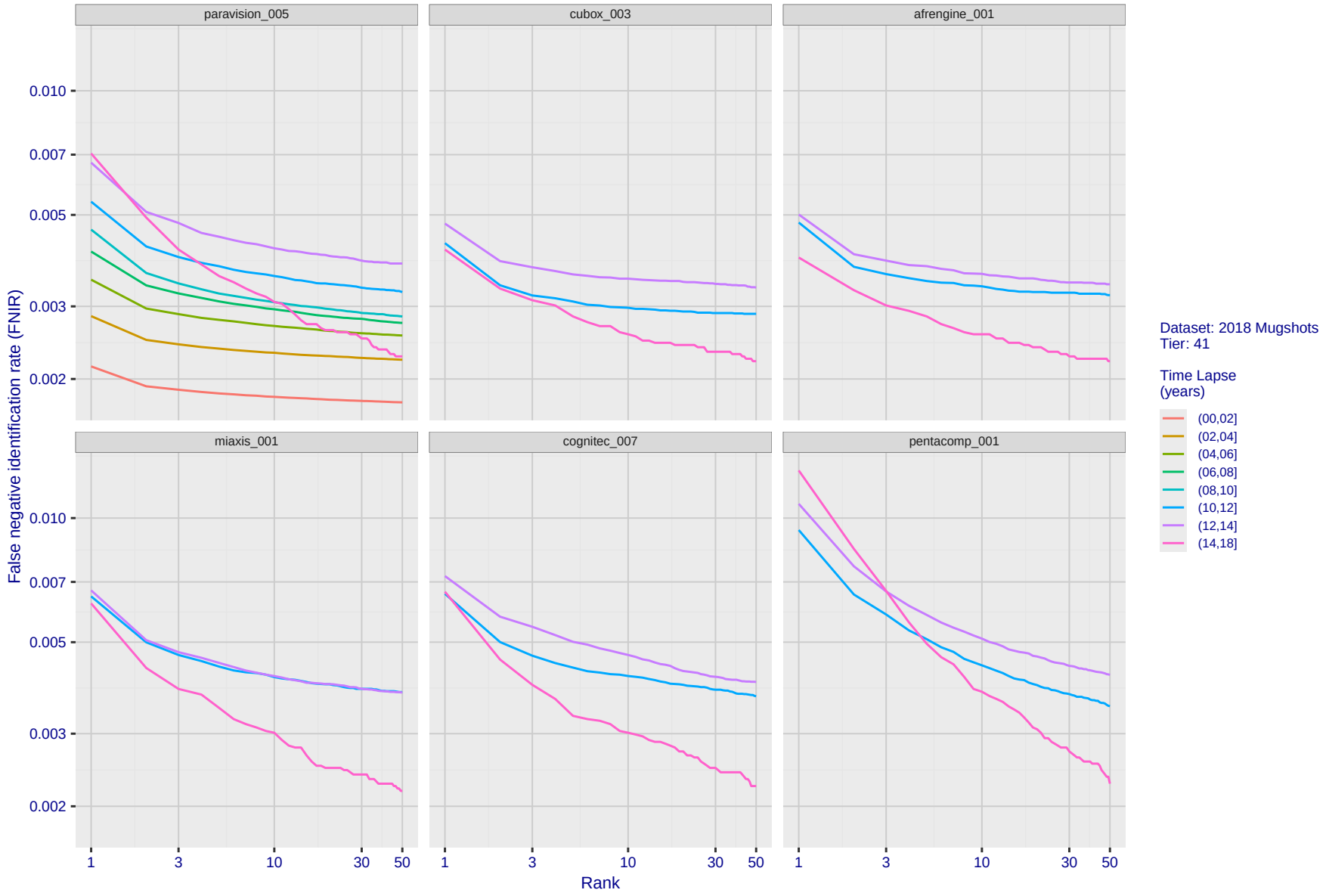

Figure 188: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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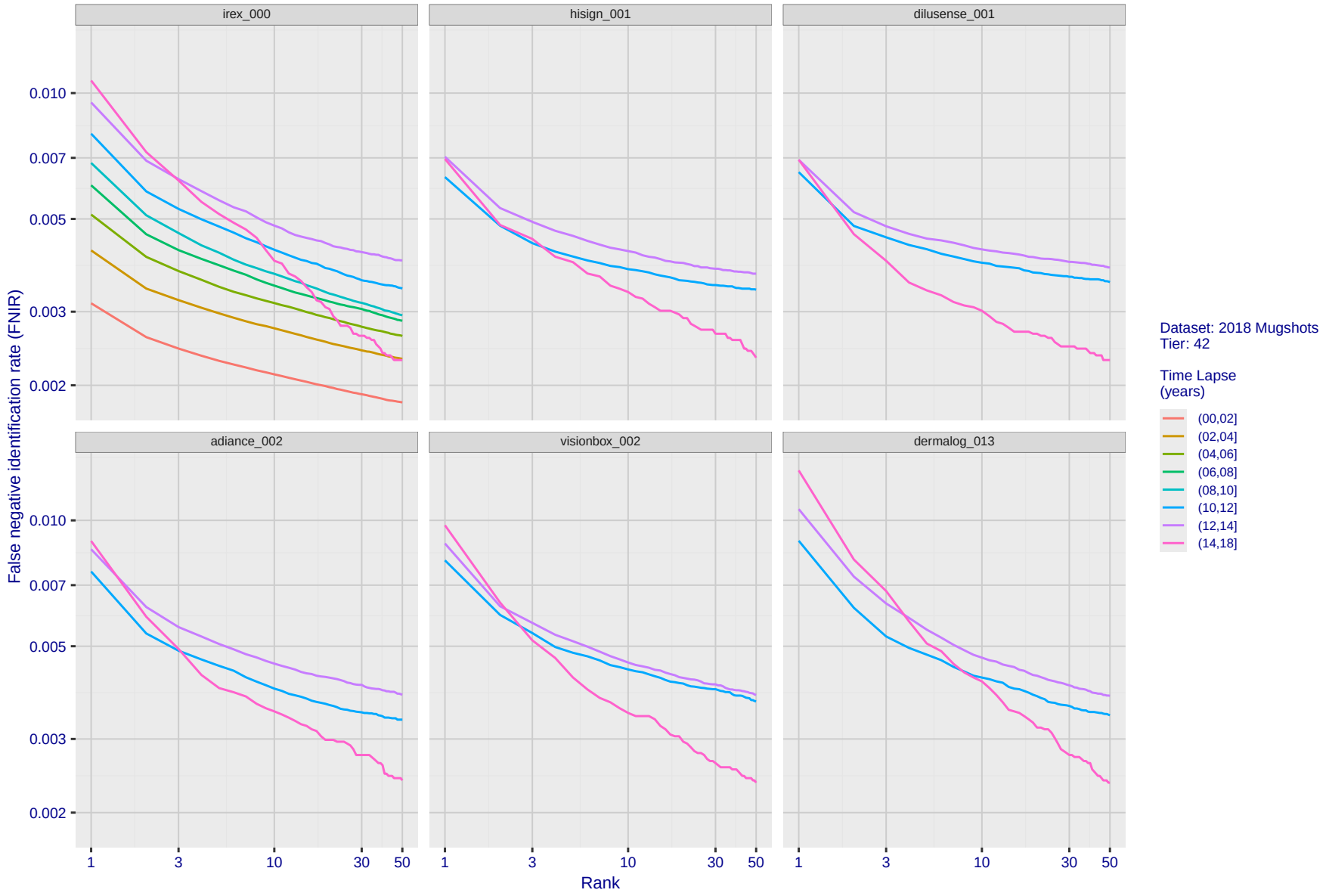

Figure 189: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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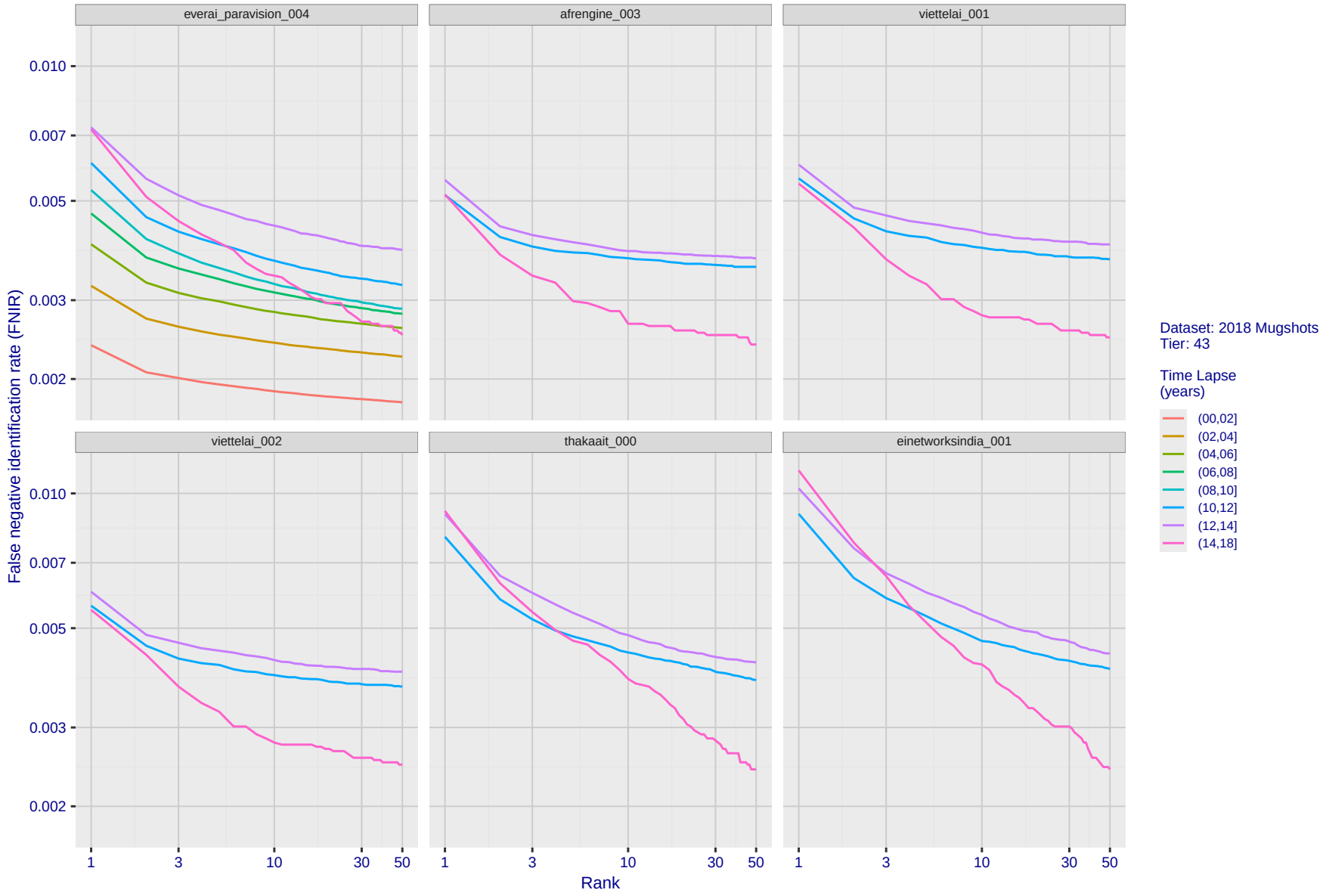

Figure 190: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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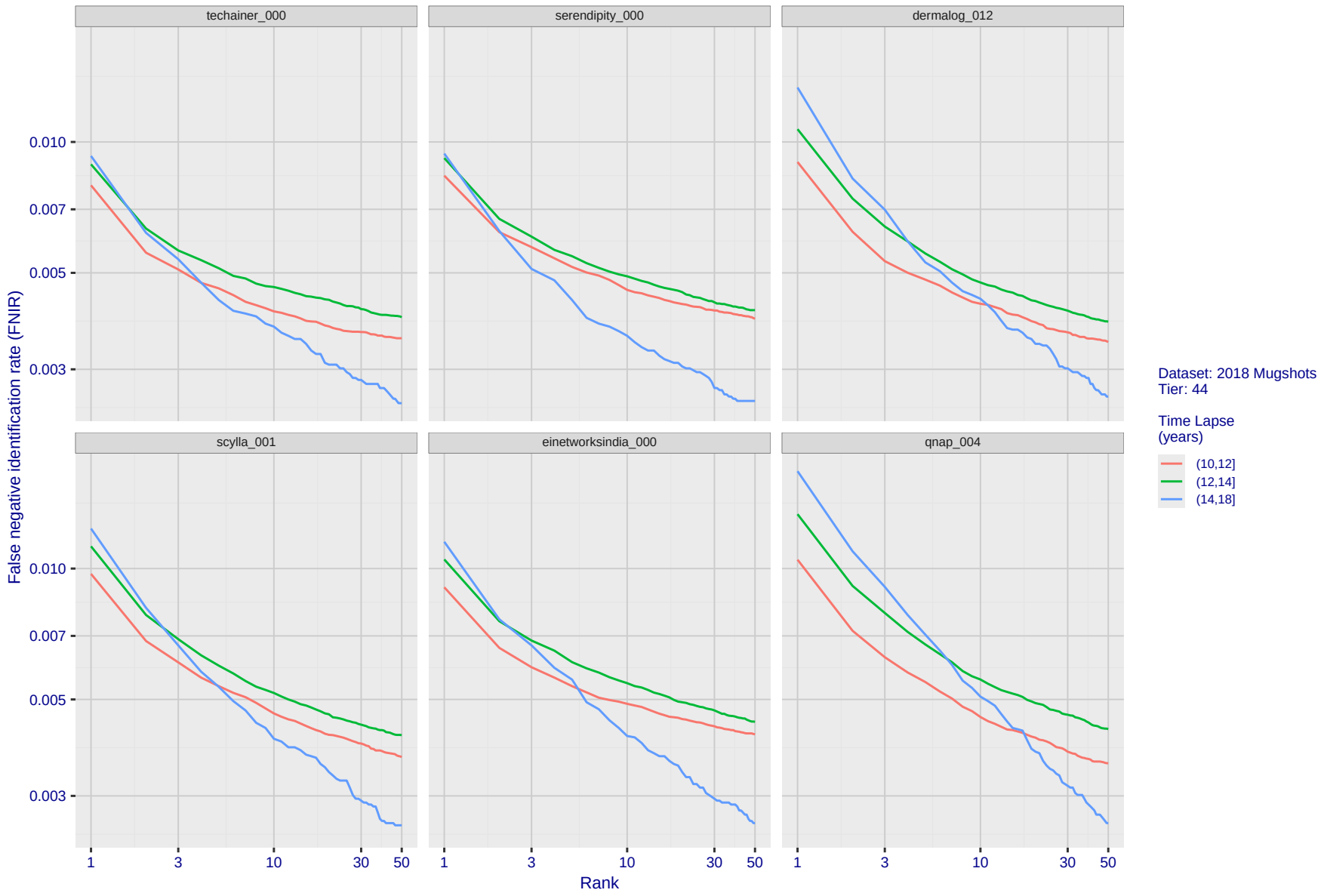

Figure 191: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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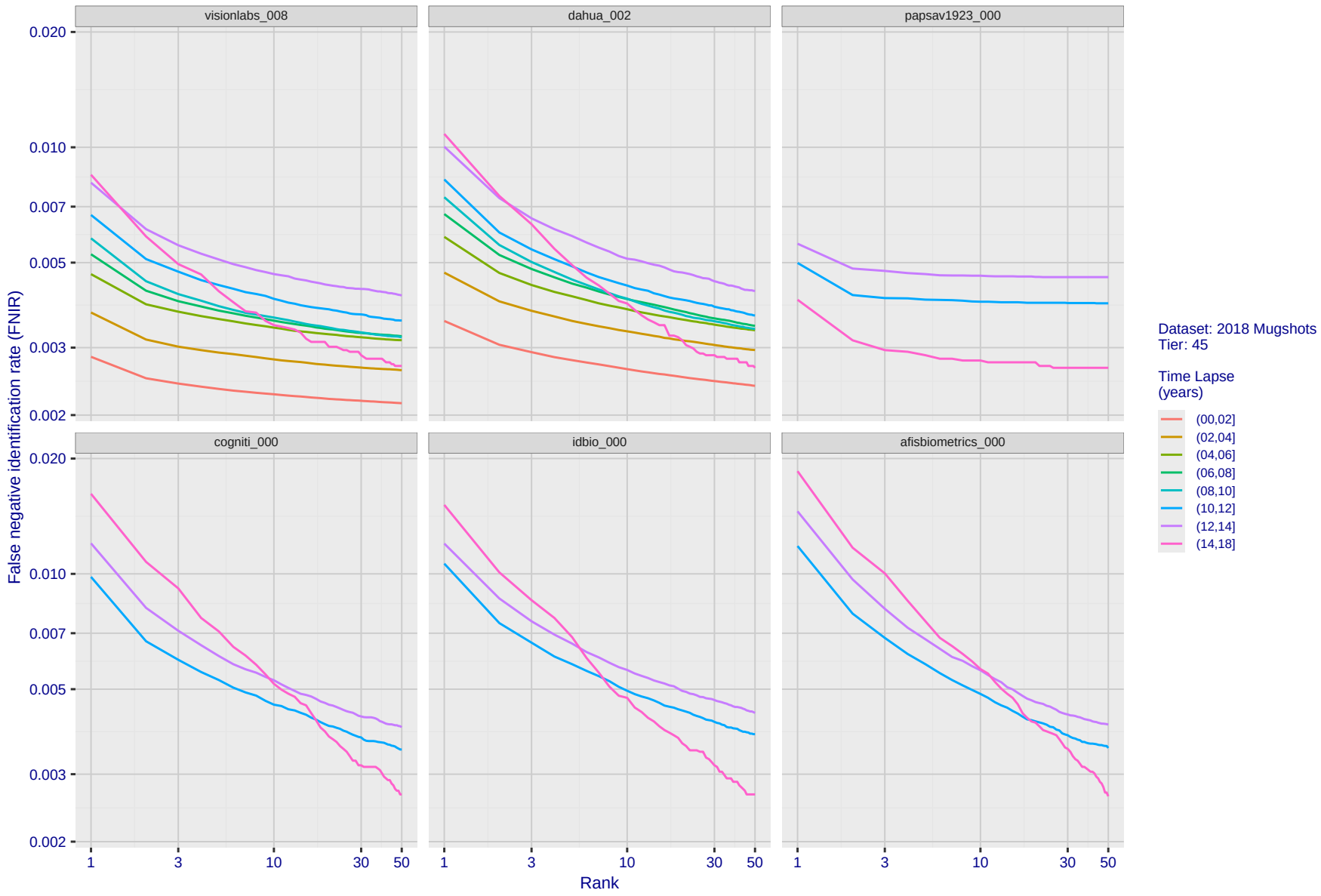

Figure 192: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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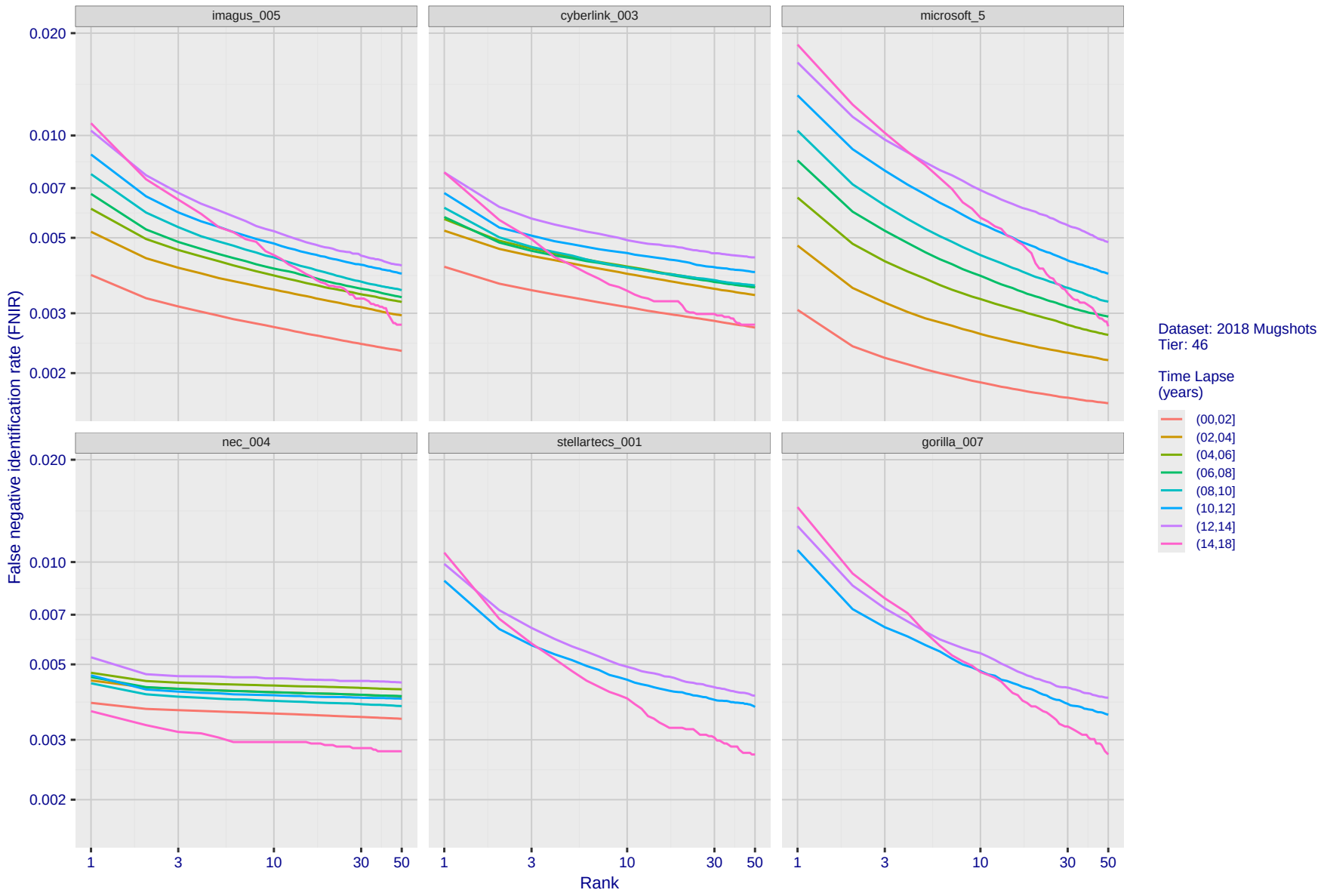

Figure 193: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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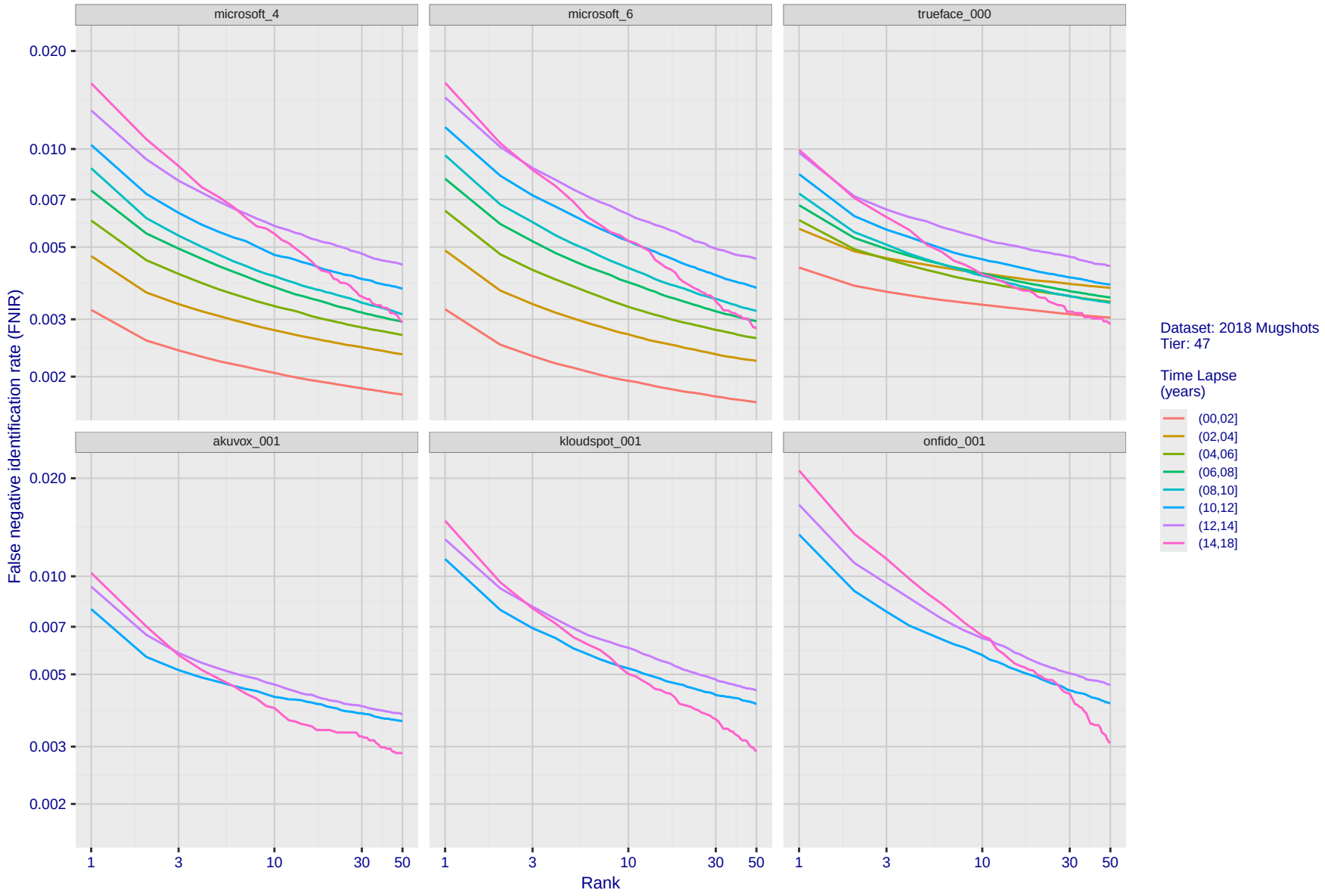

Figure 194: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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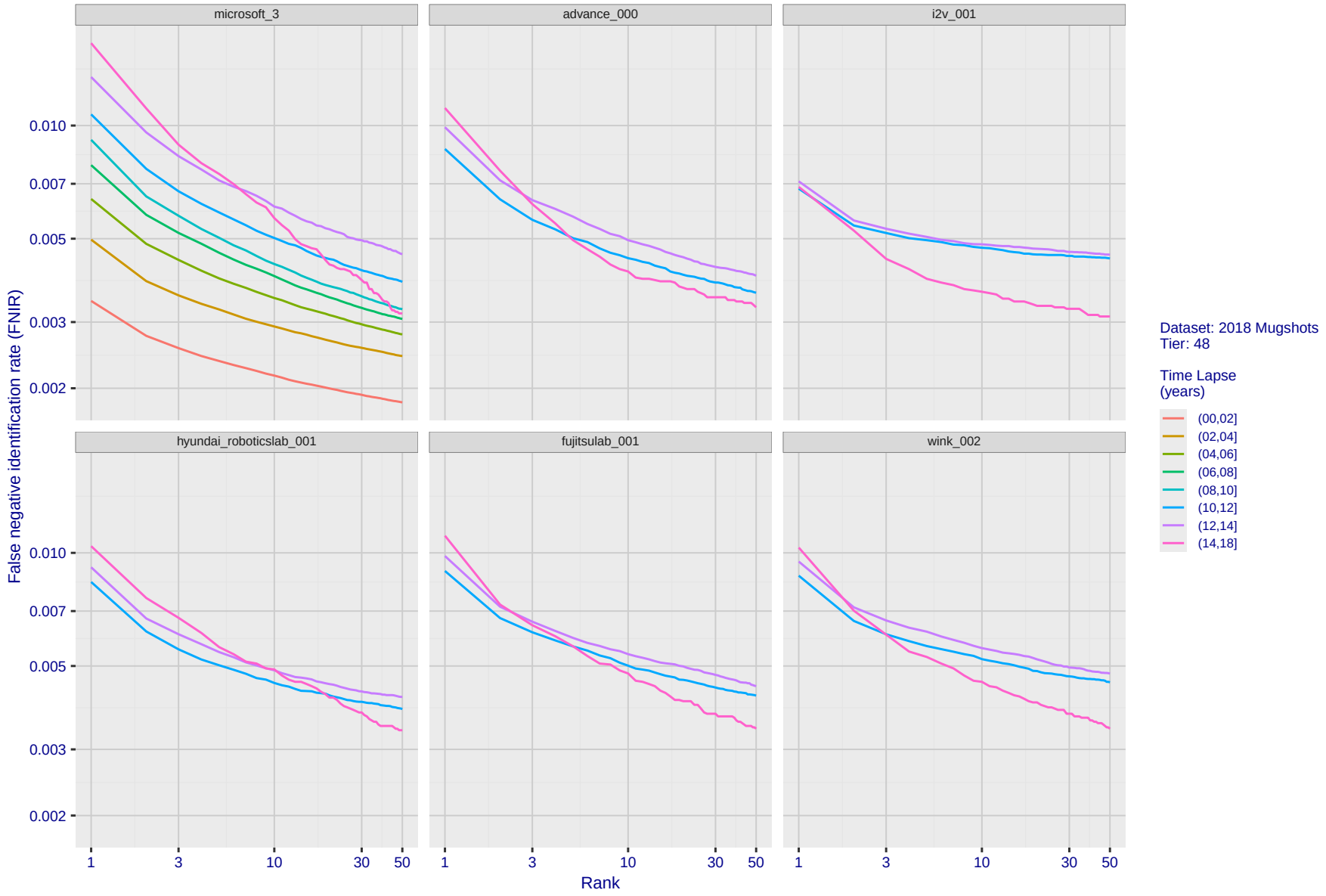

Figure 195: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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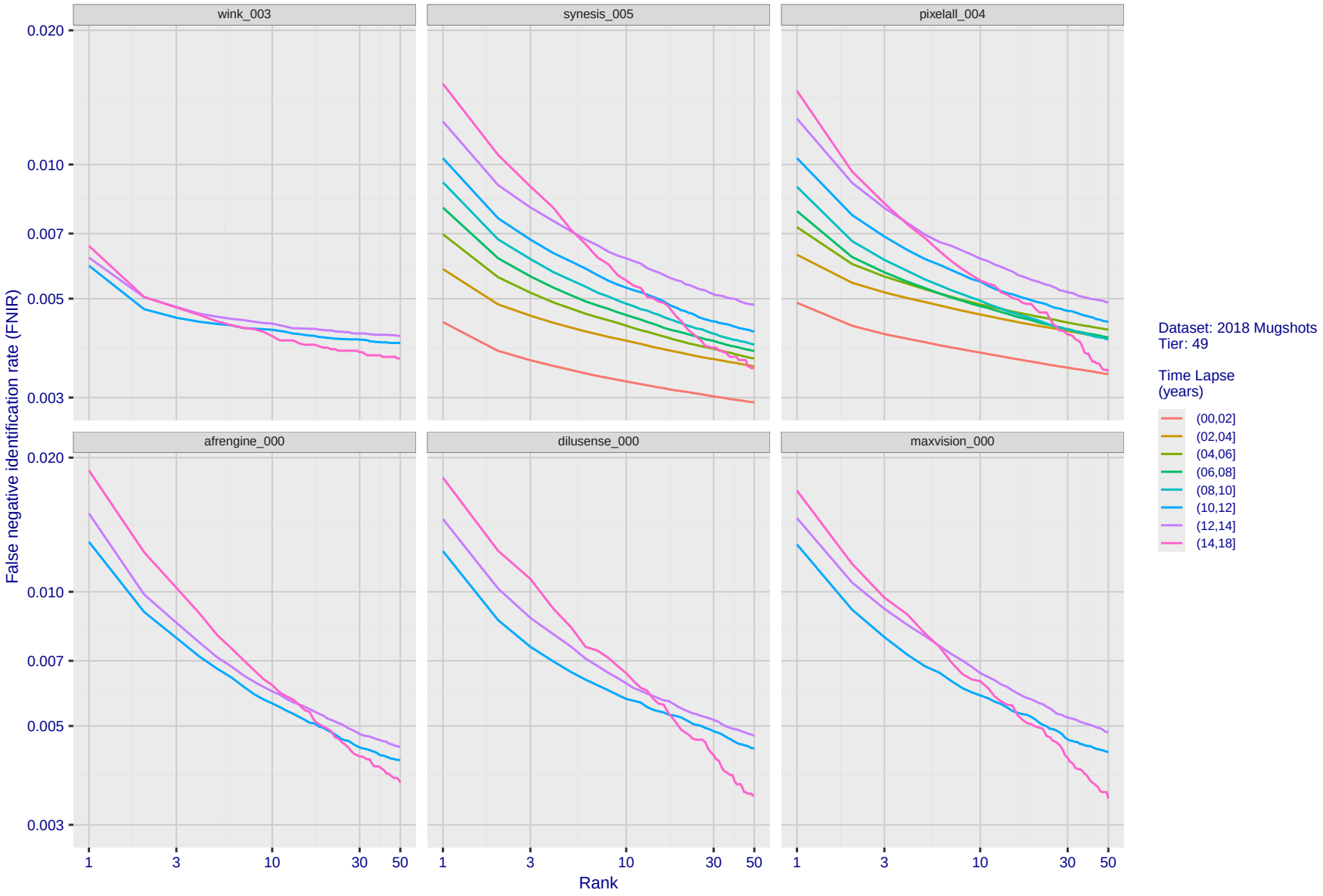

Figure 196: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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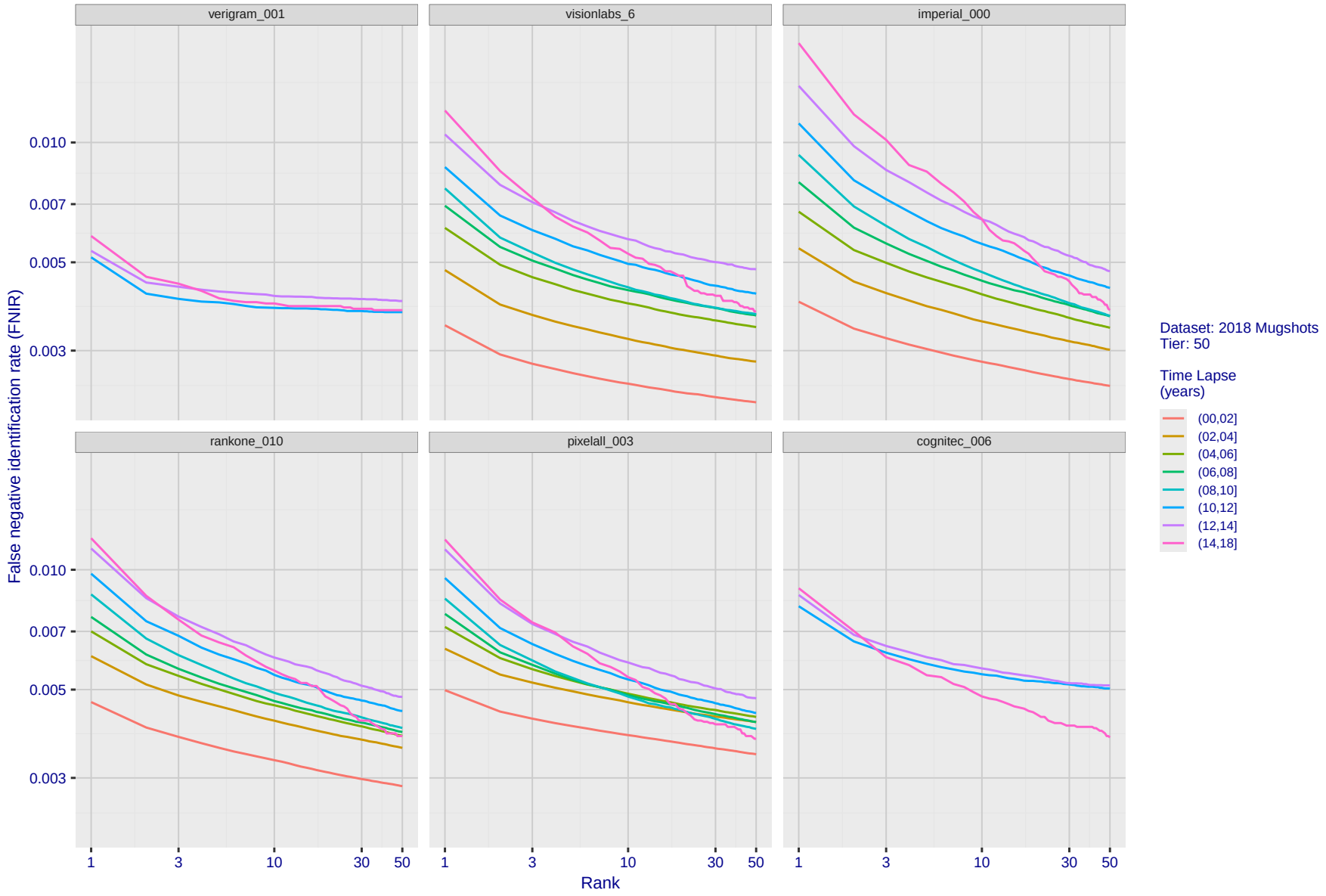

Figure 197: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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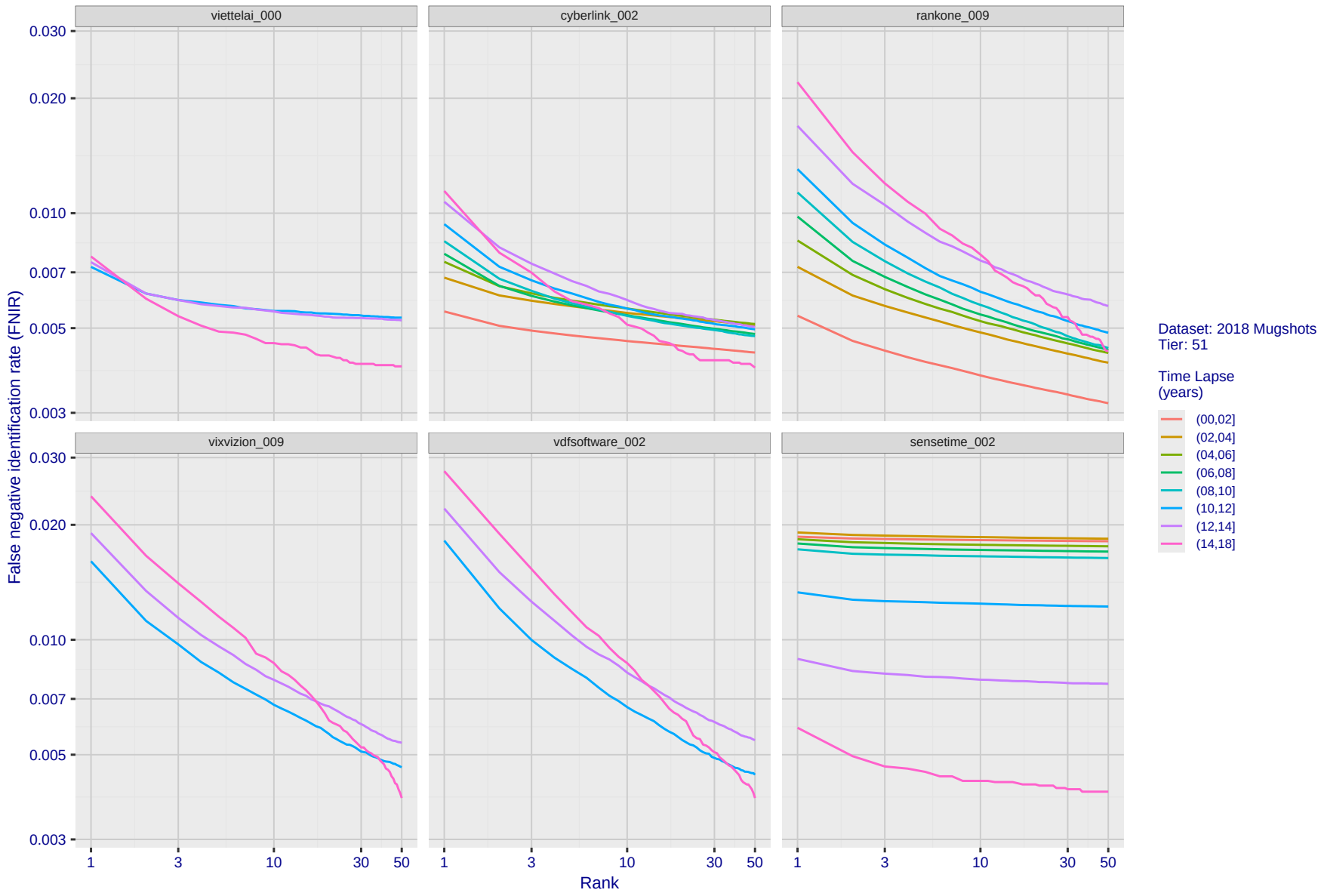

Figure 198: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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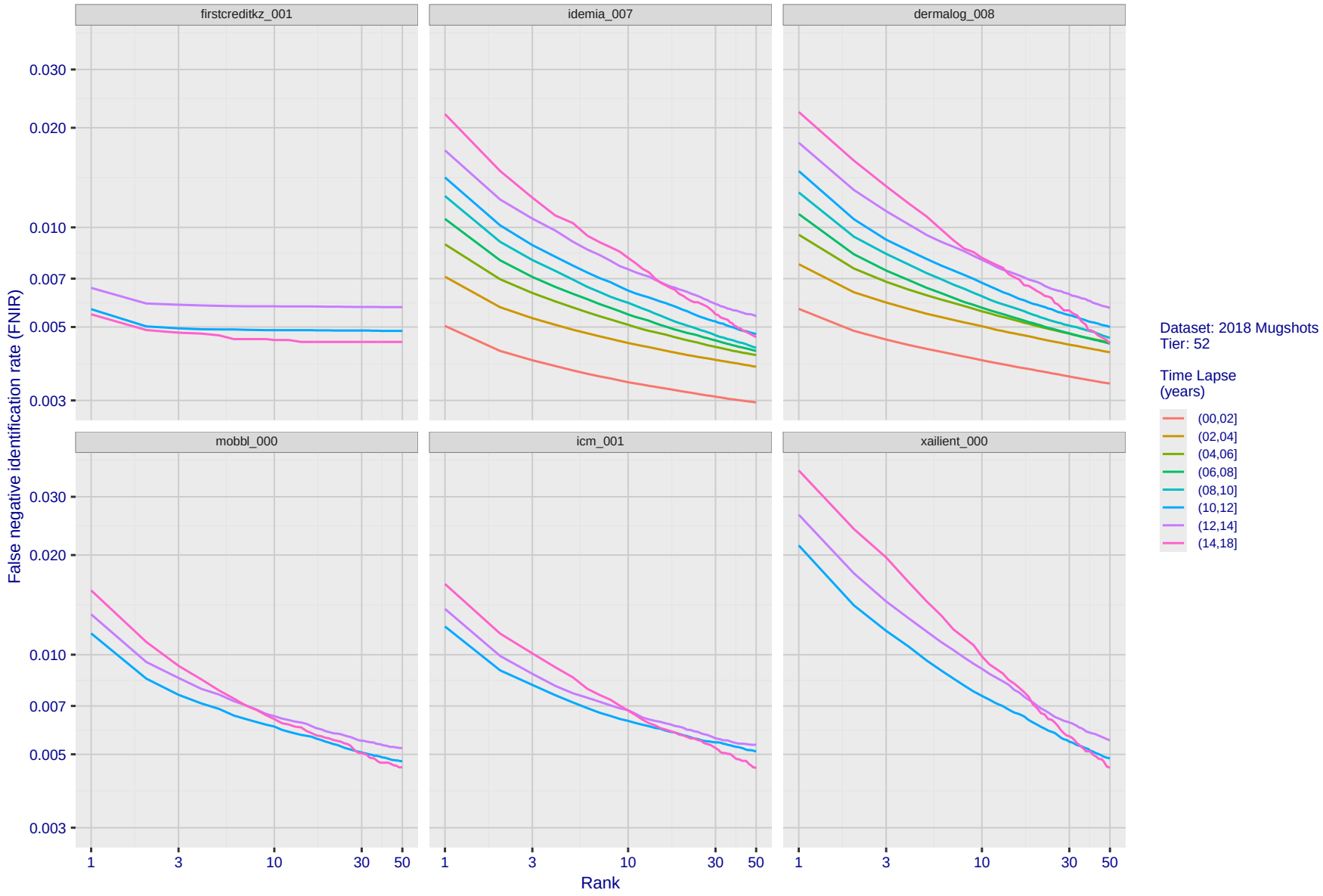

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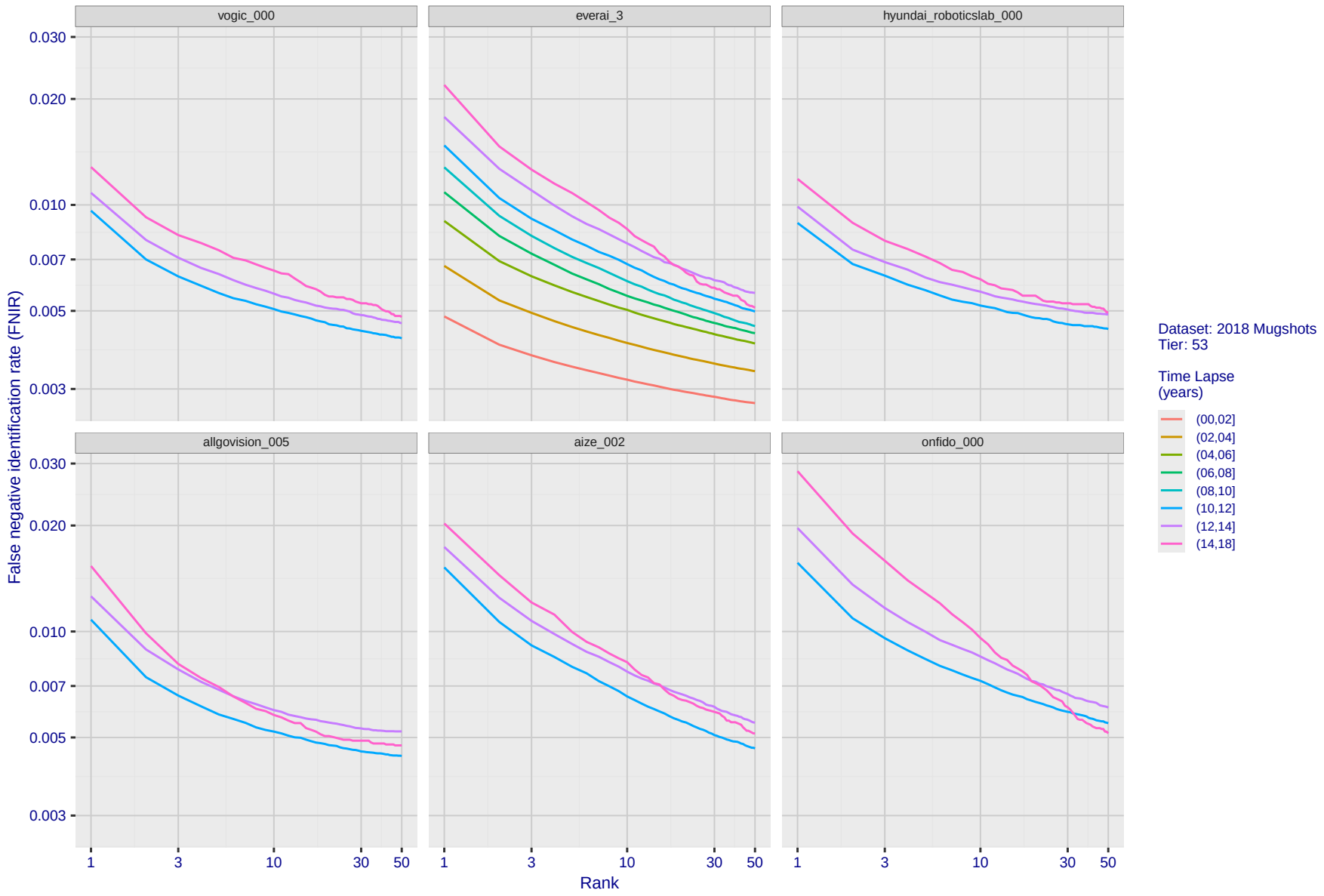

Figure 200: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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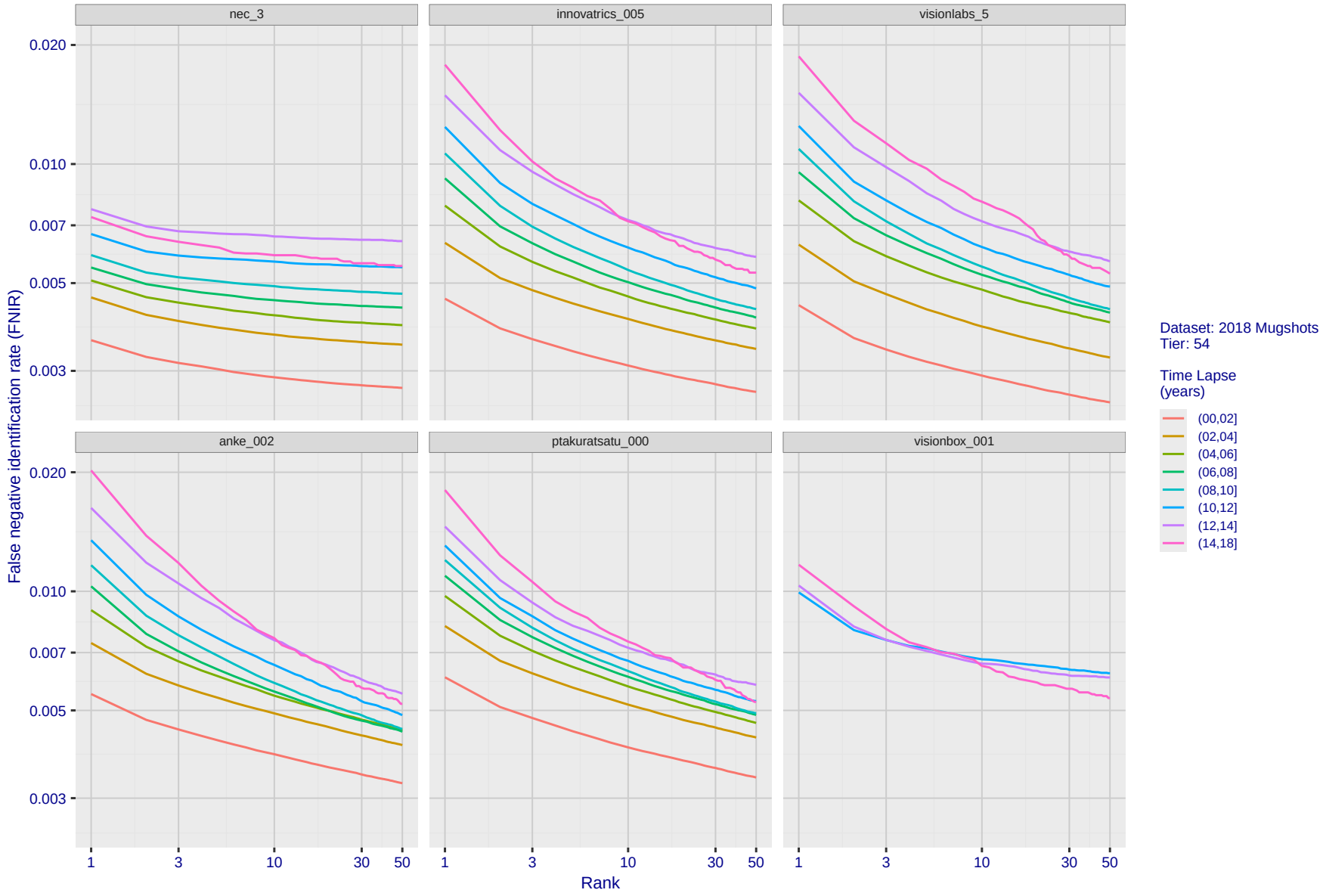

Figure 201: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
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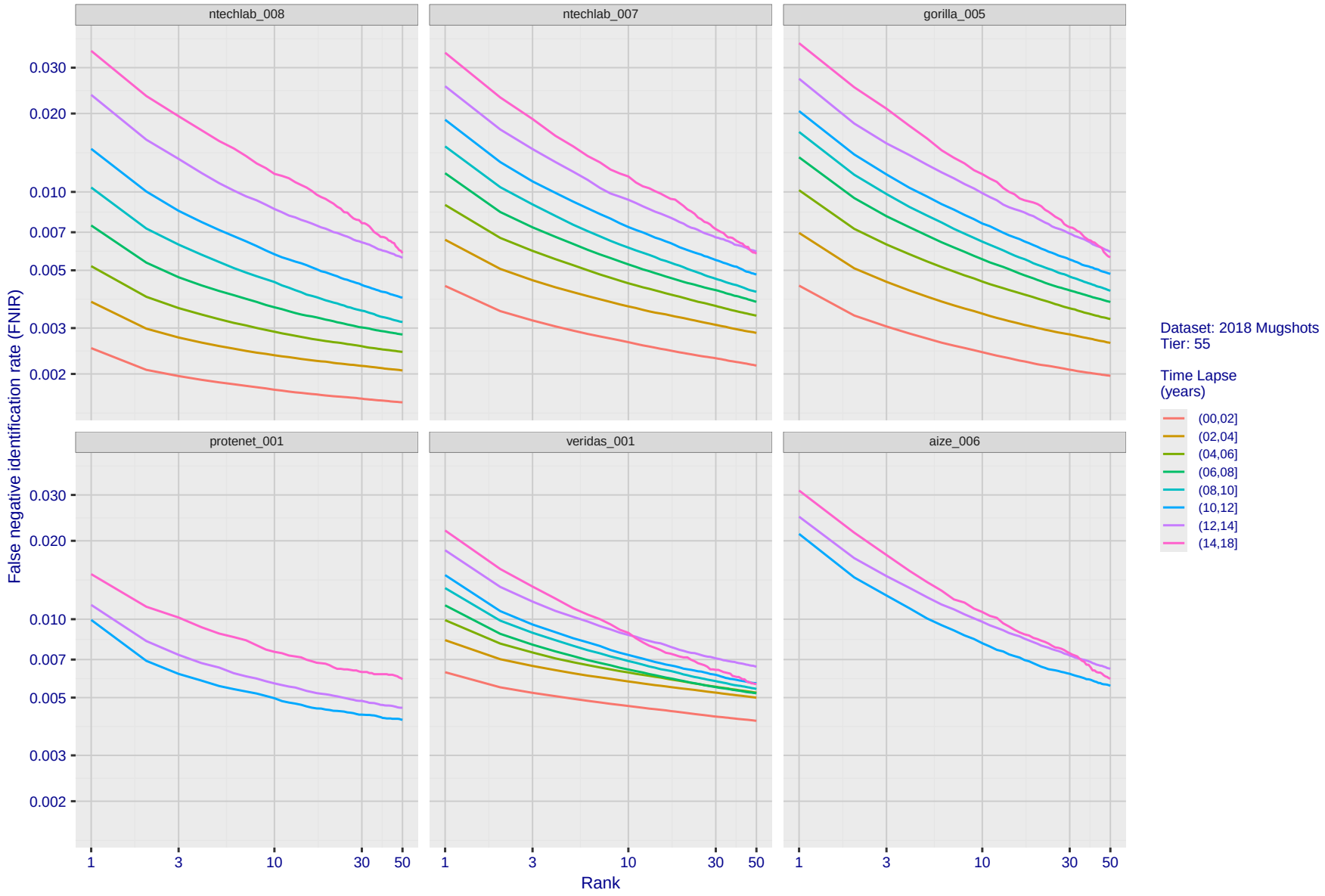

Figure 202: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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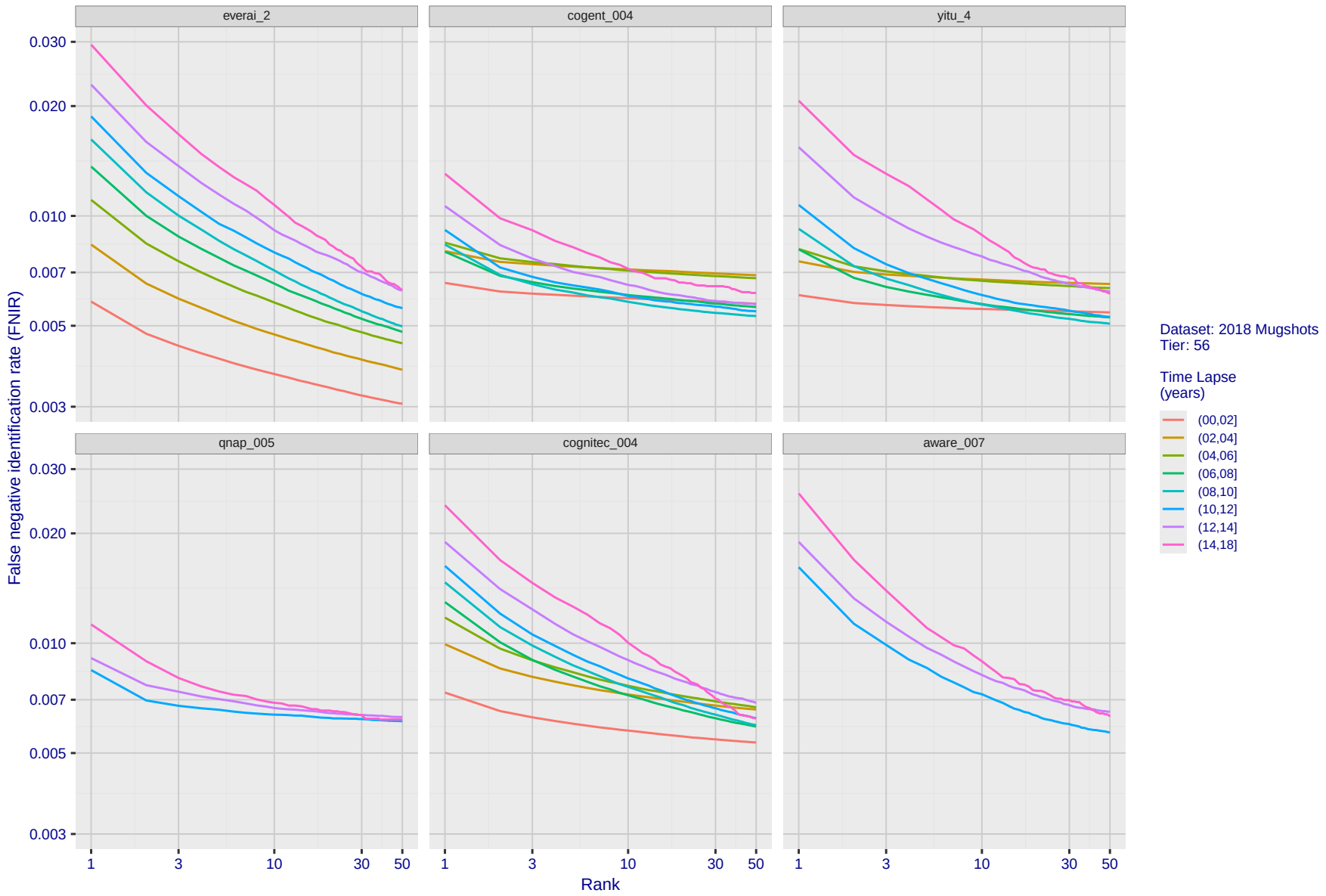

Figure 203: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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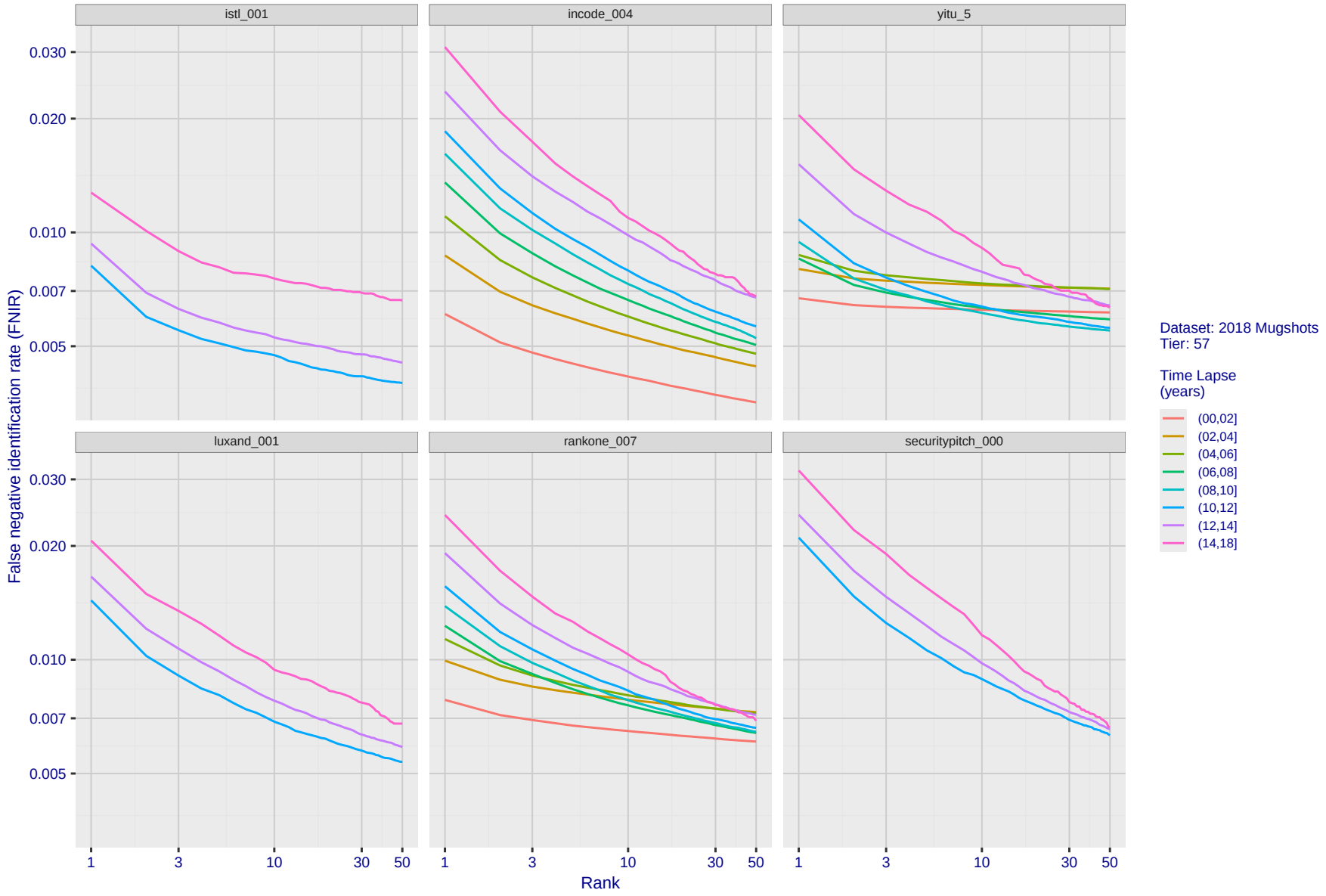

Figure 204: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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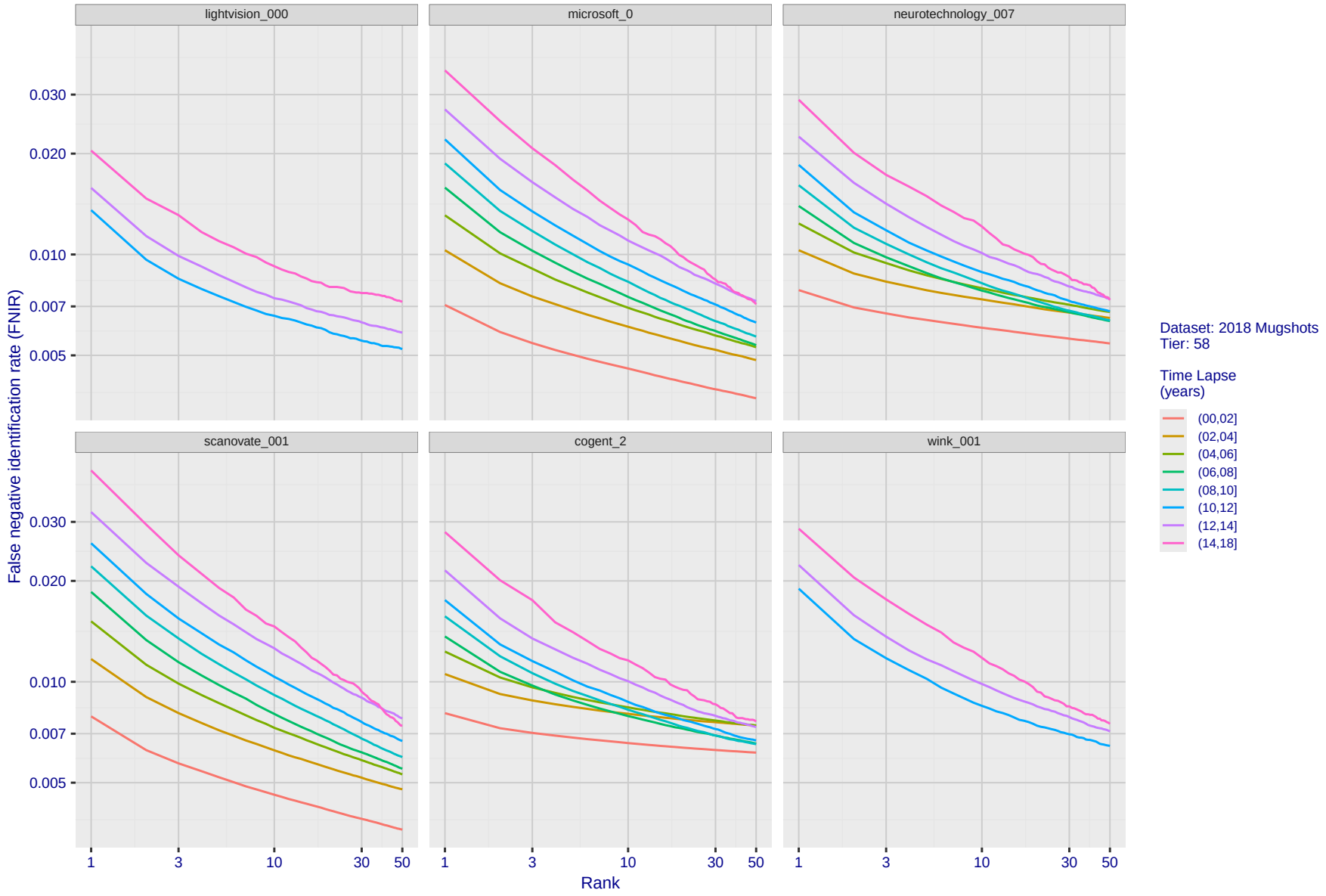

Figure 205: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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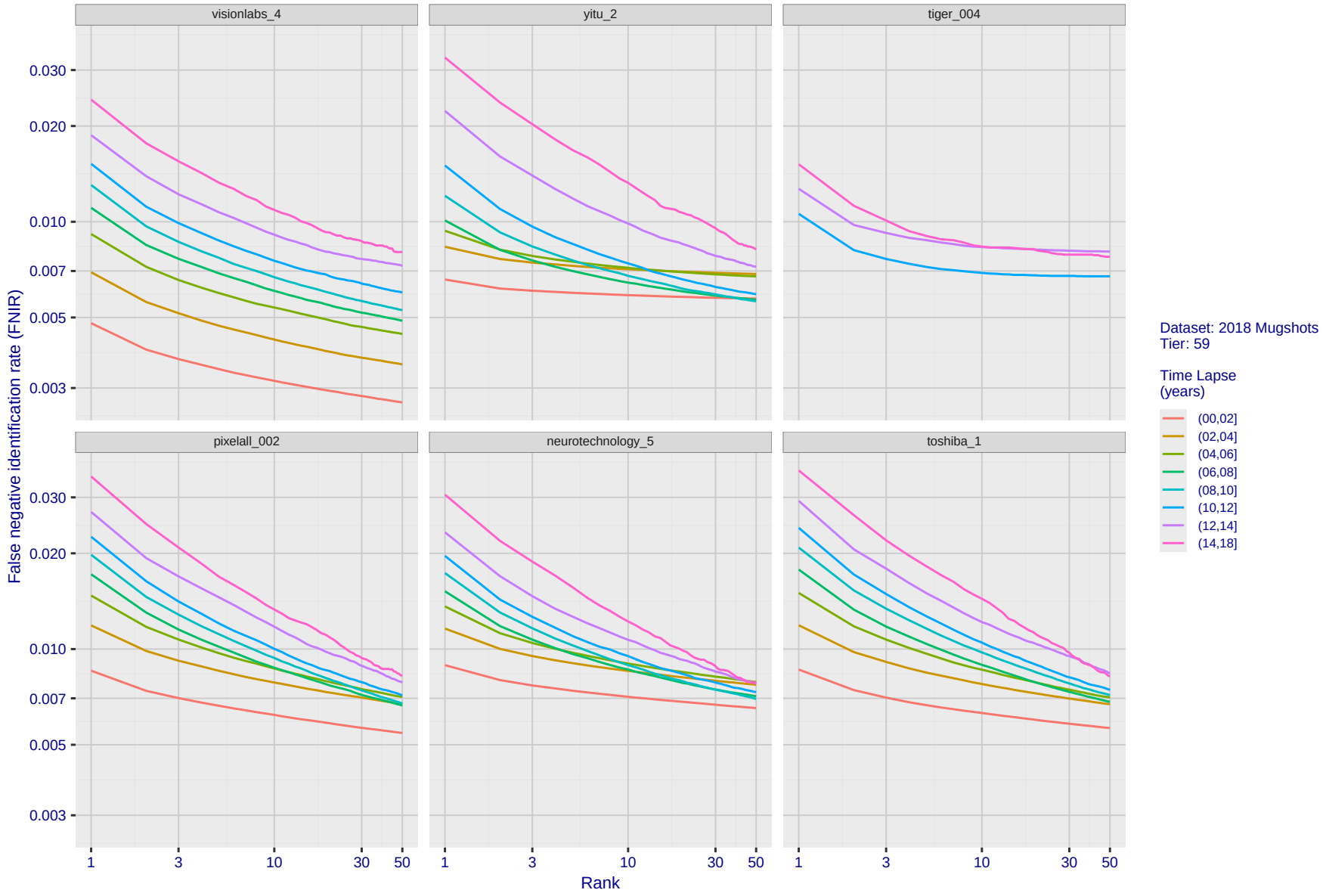

Figure 206: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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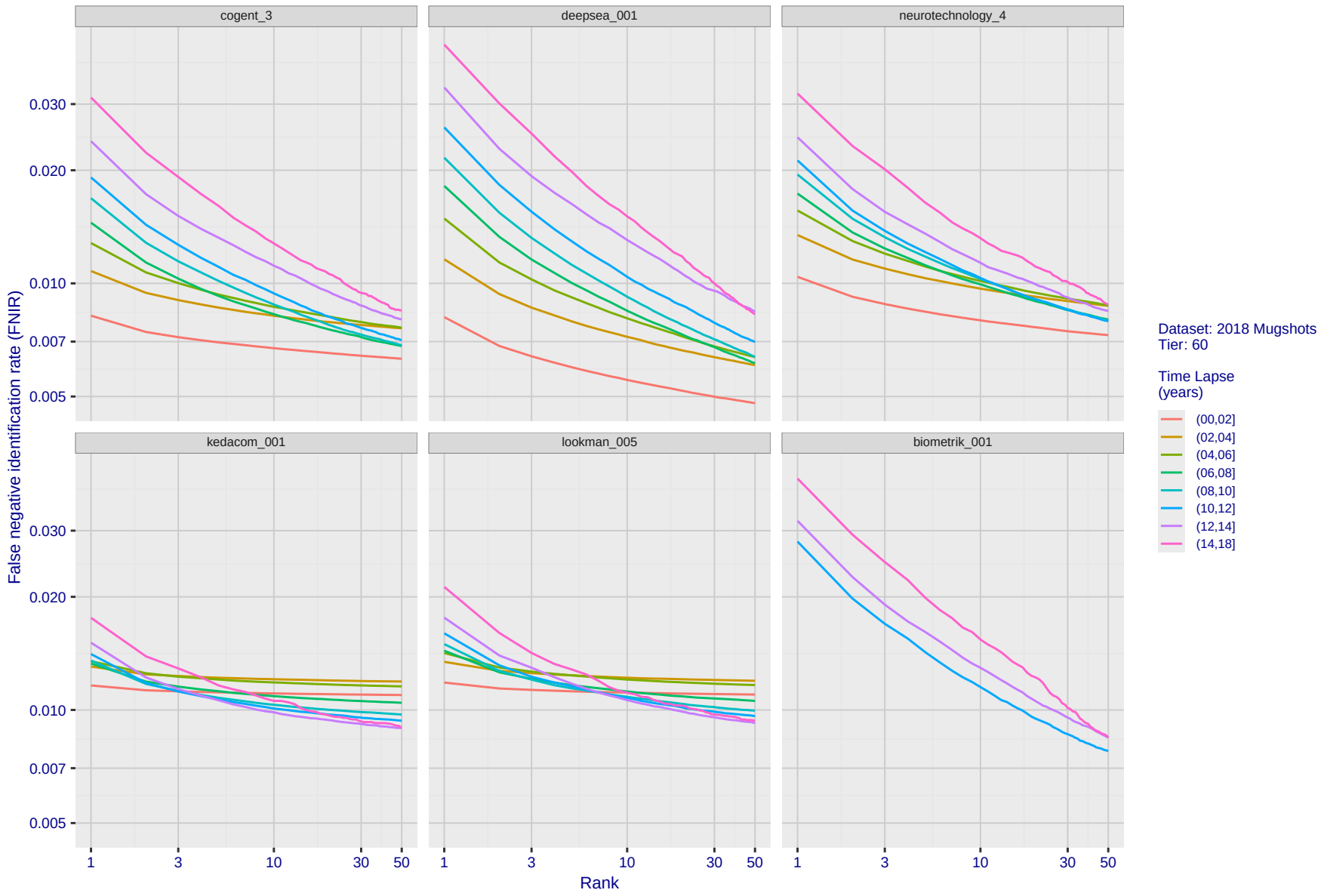

Figure 207: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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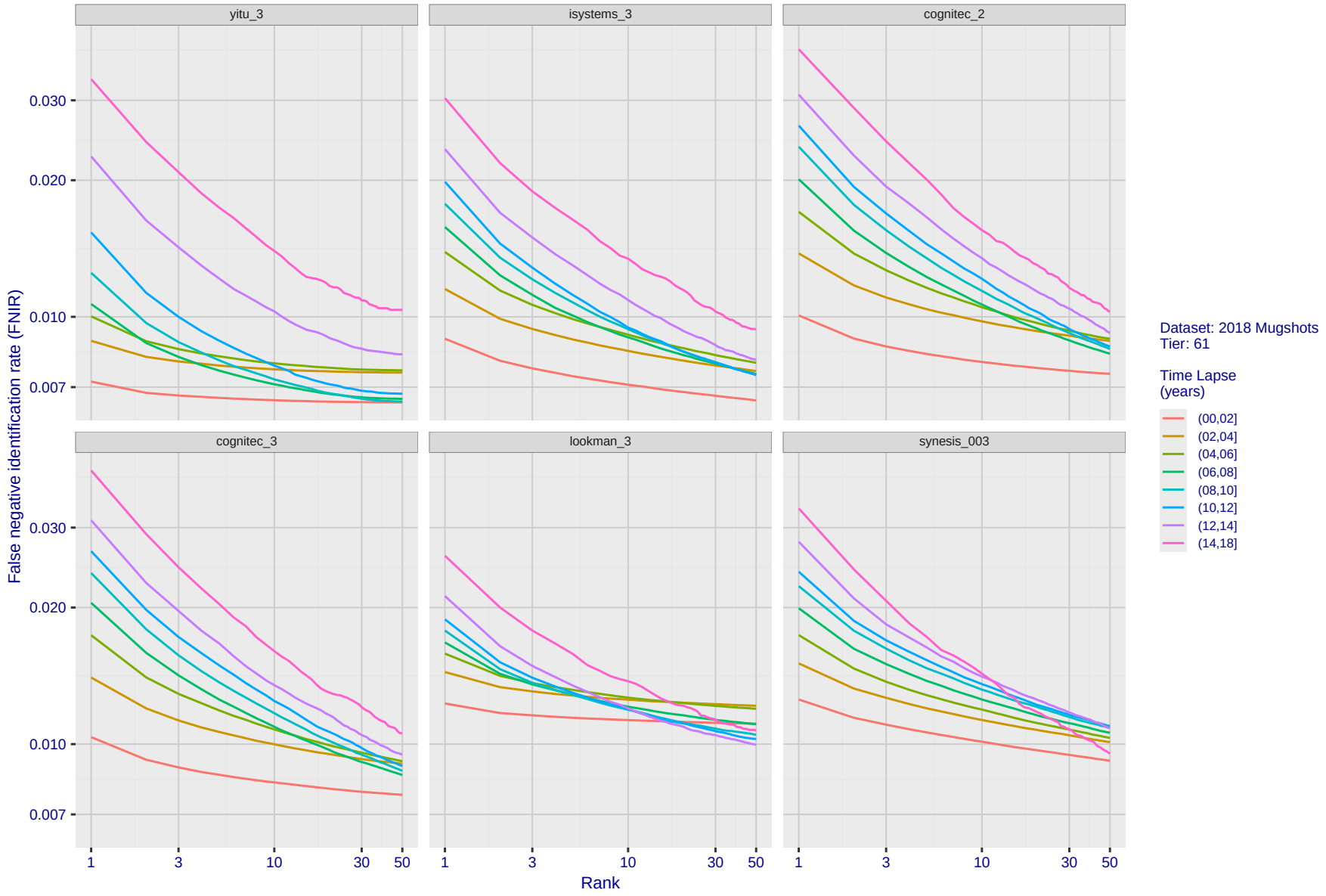

Figure 208: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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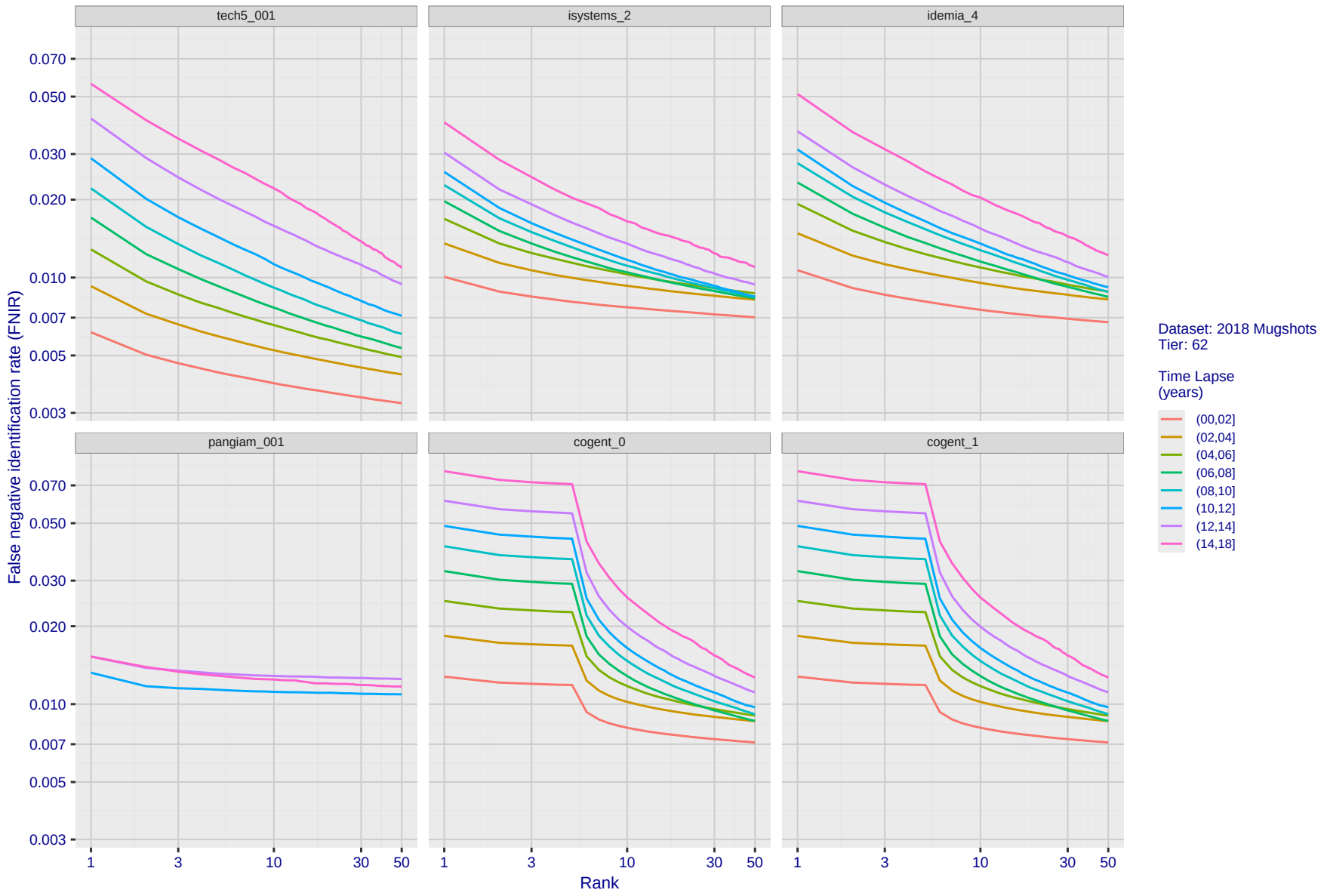

Figure 209: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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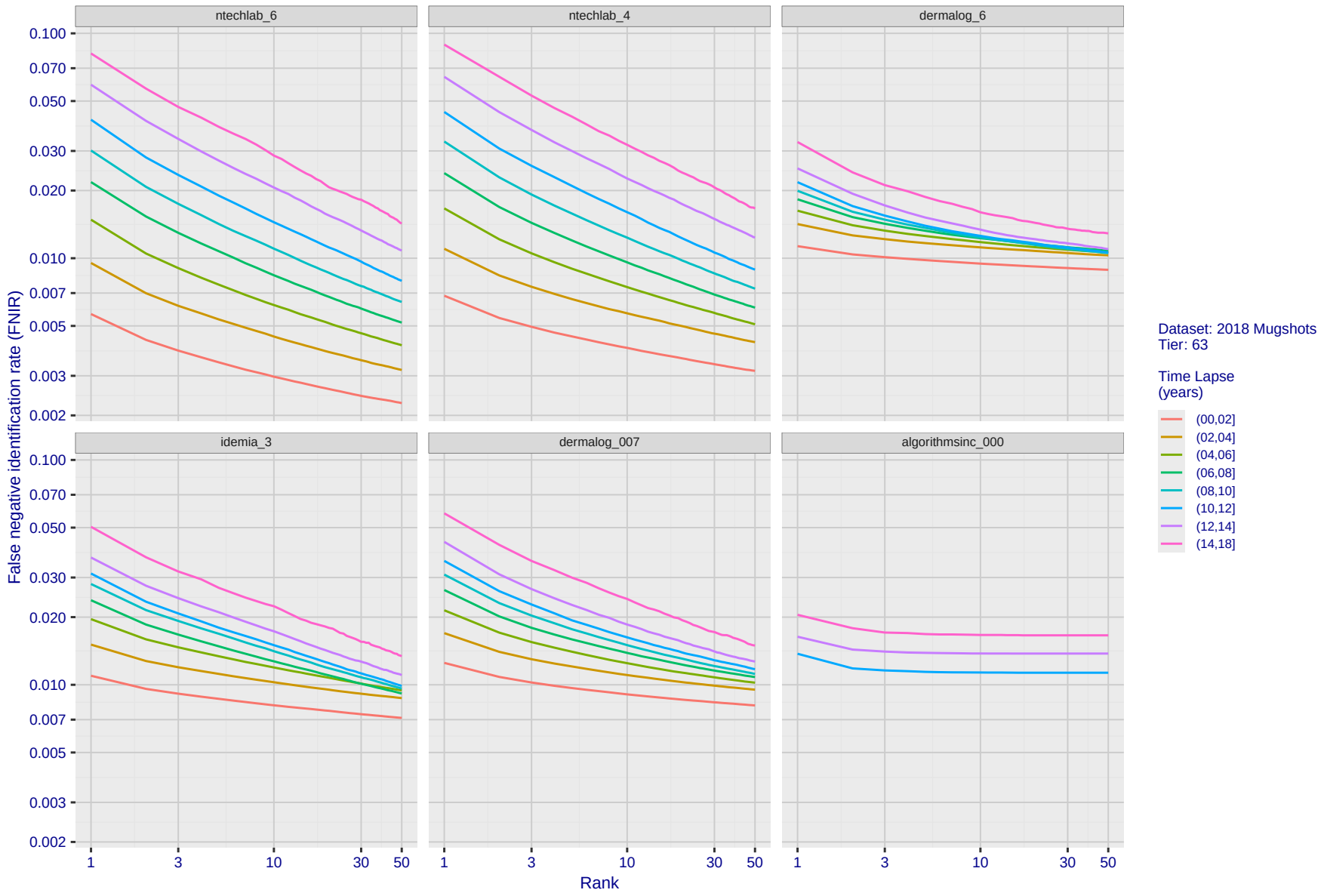

Figure 210: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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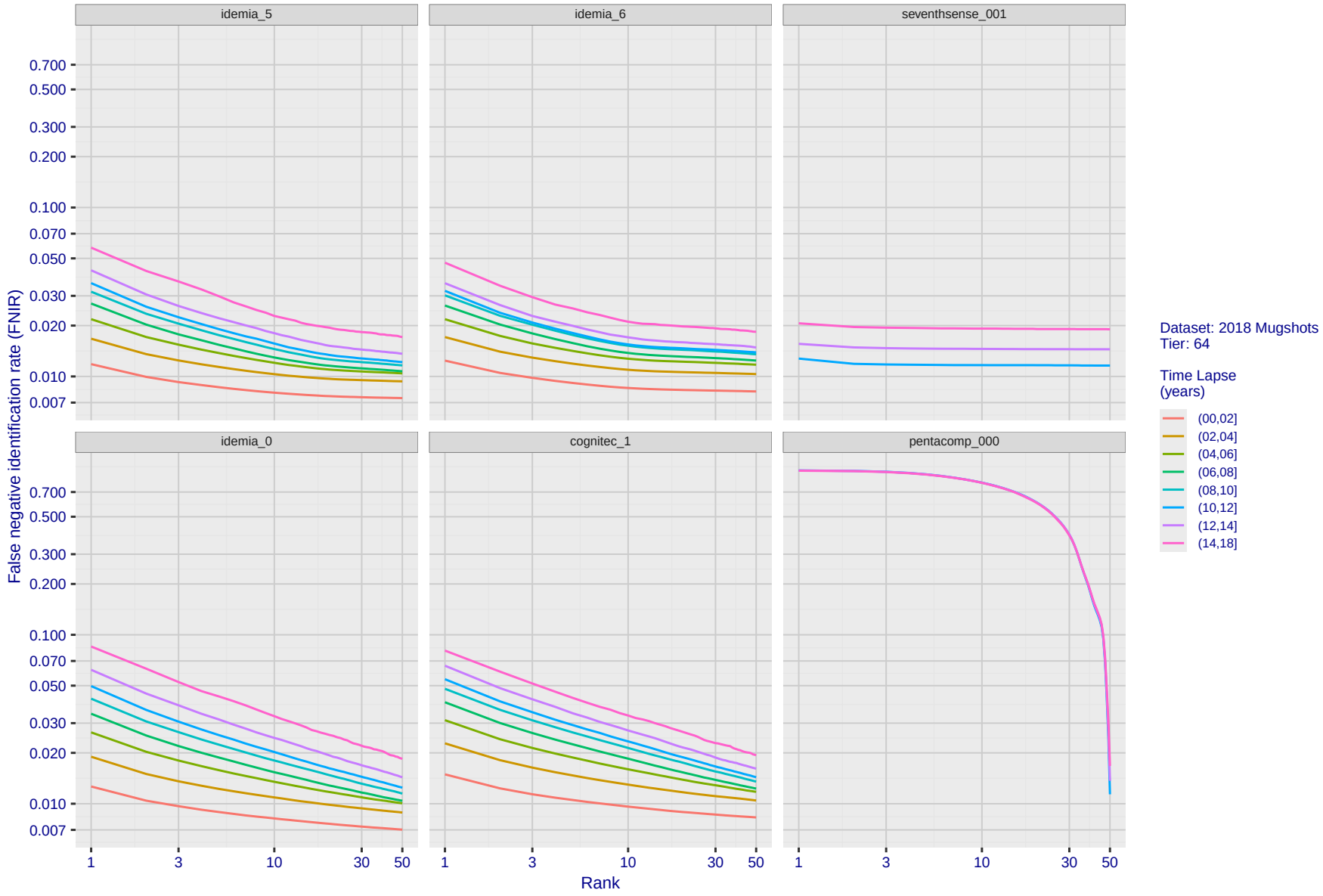

Figure 211: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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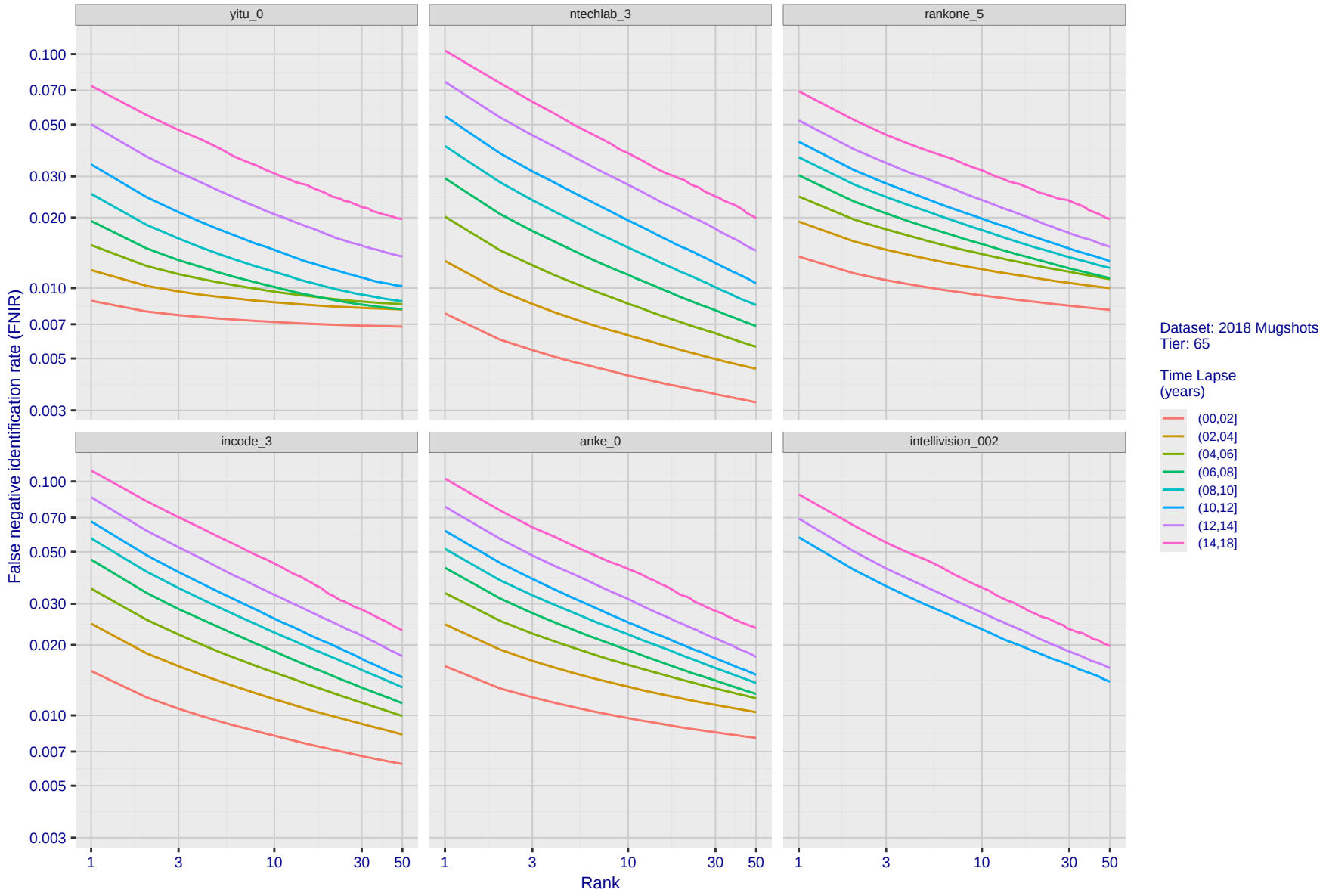

Figure 212: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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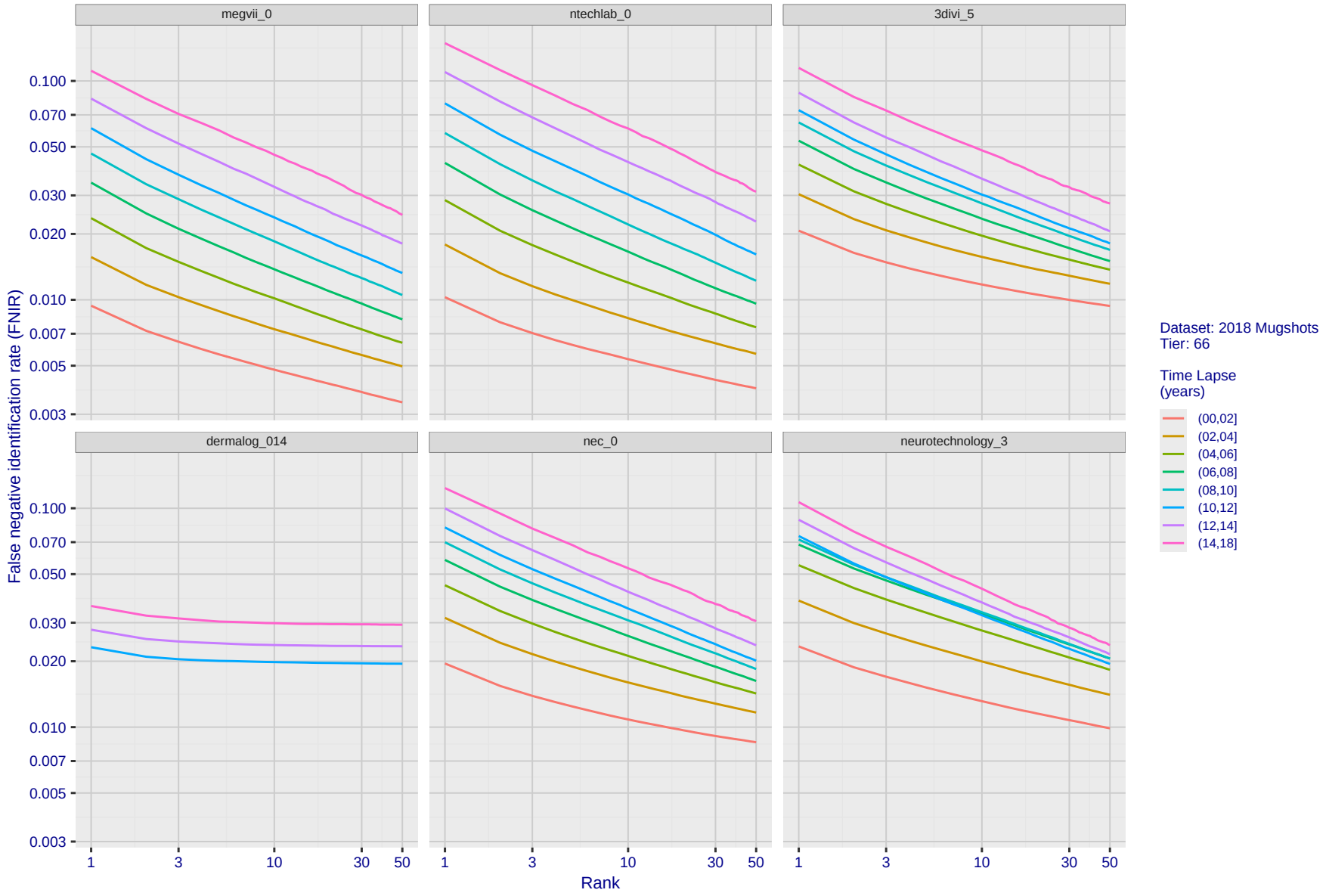

Figure 213: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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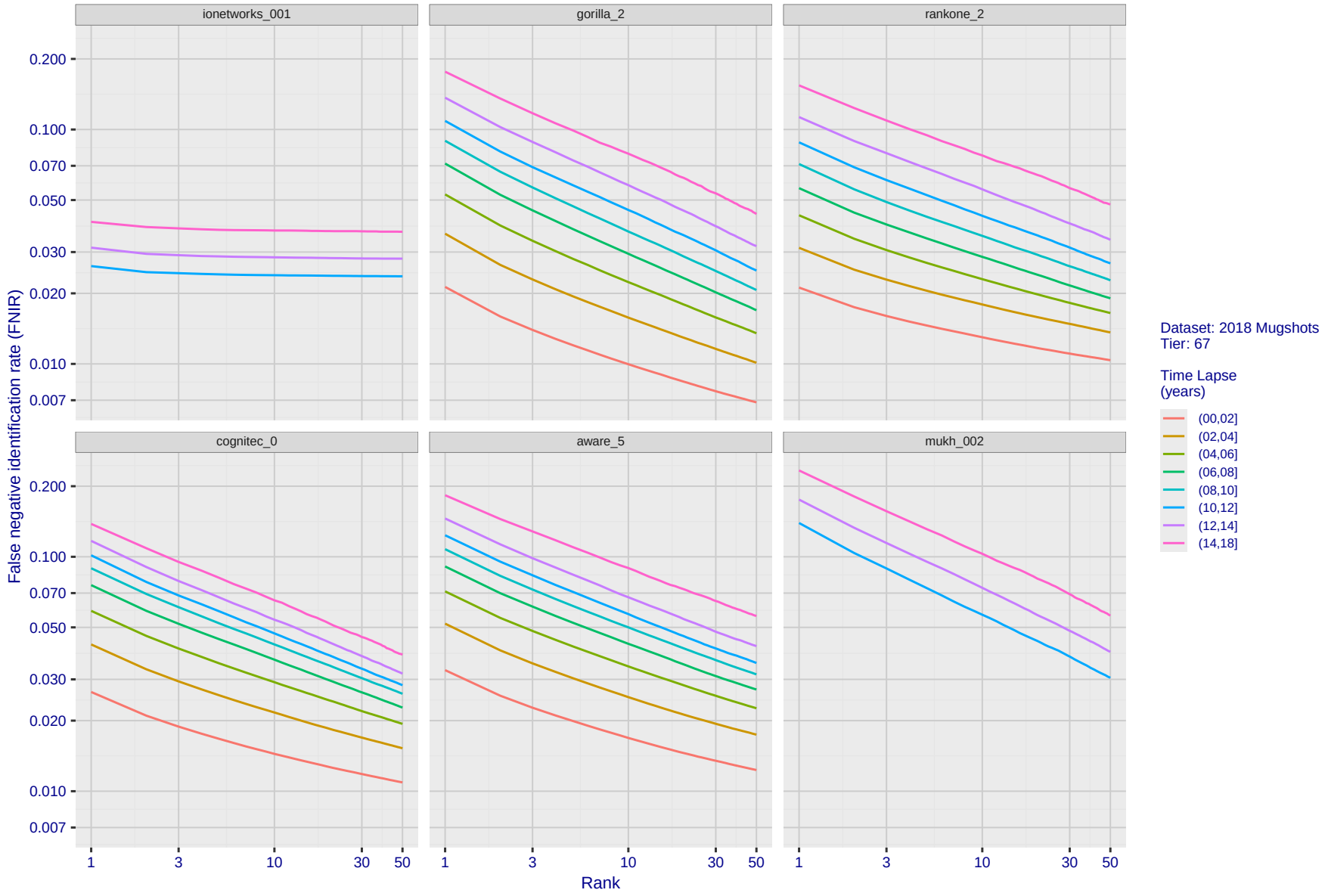

Figure 214: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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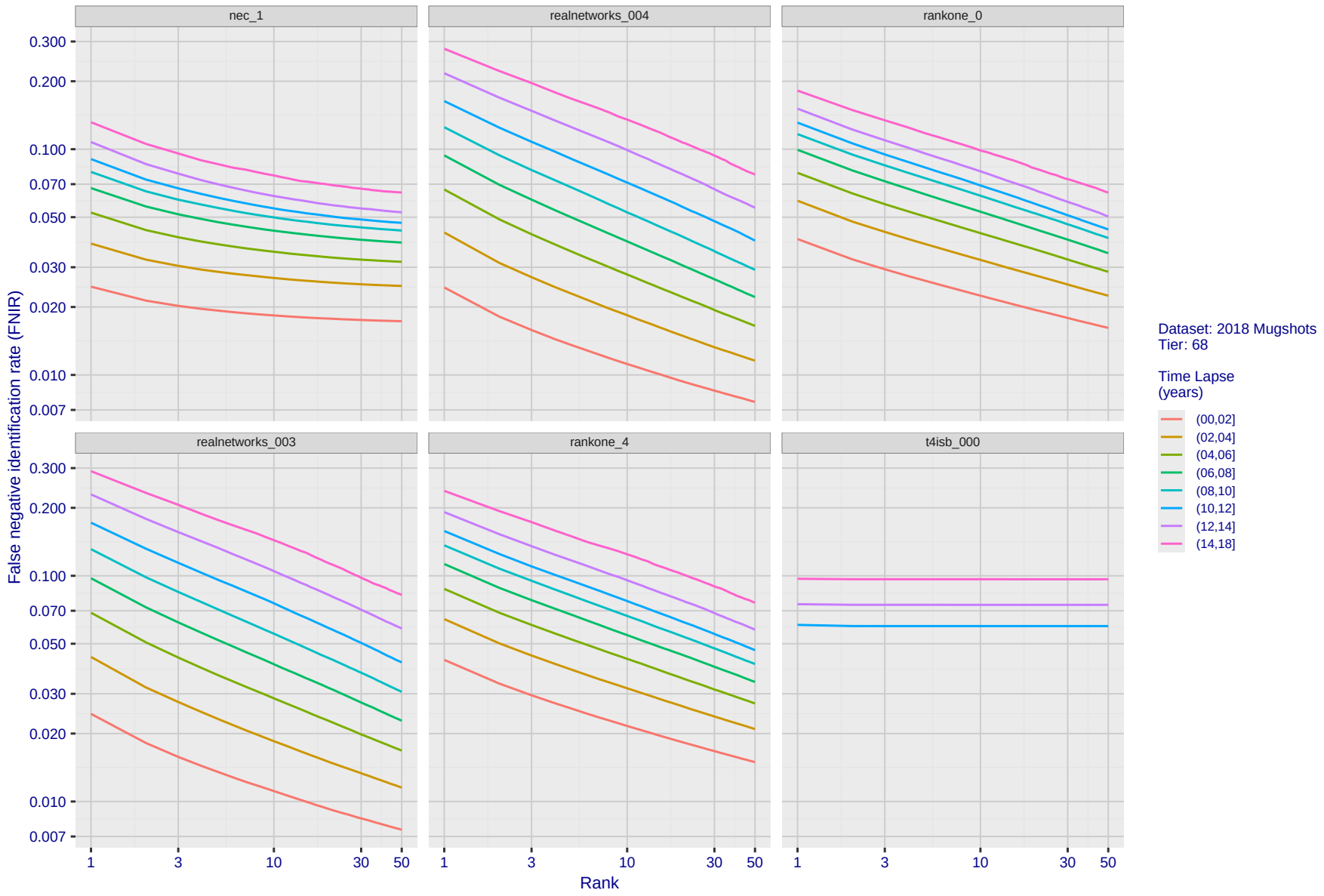

Figure 215: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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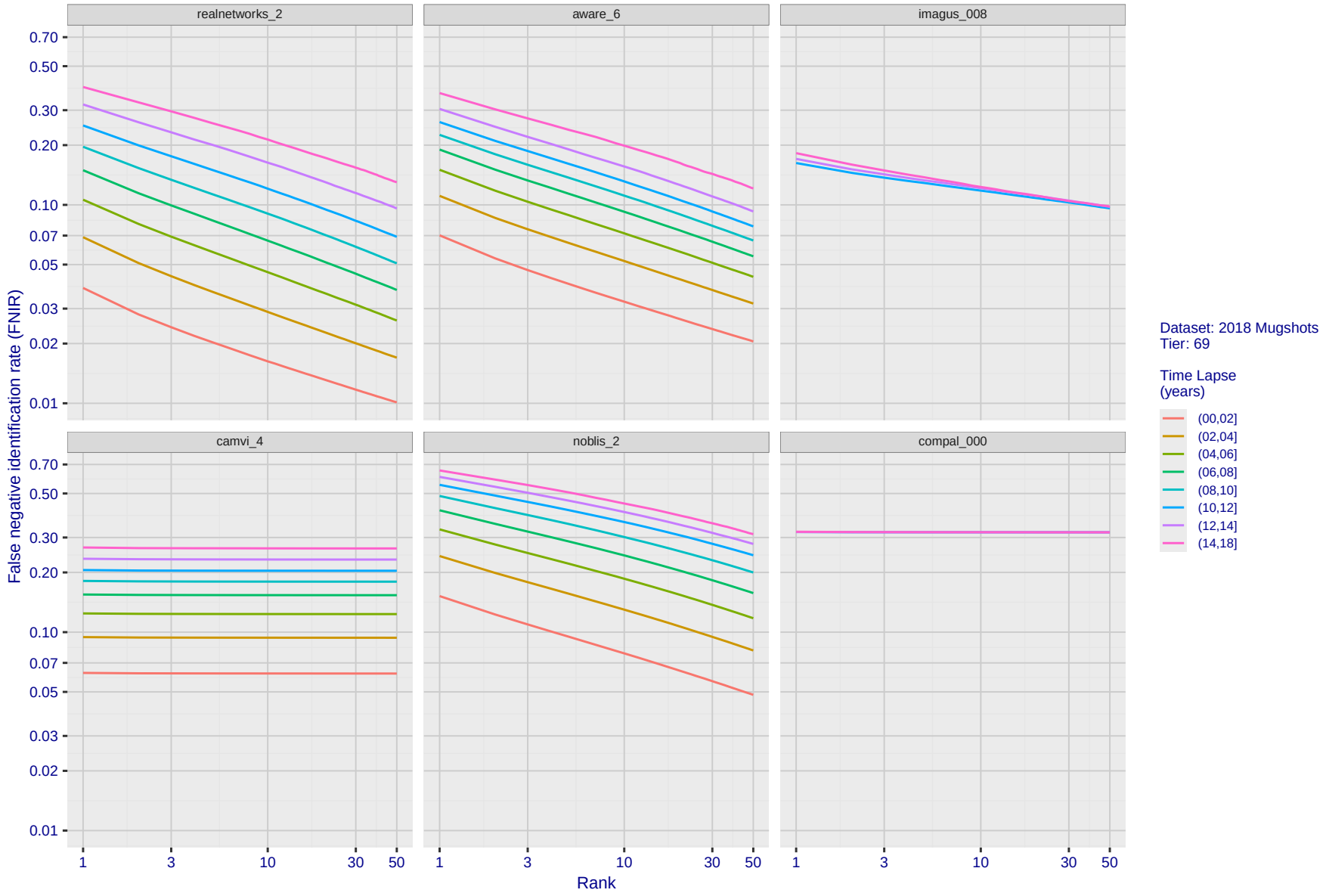

Figure 216: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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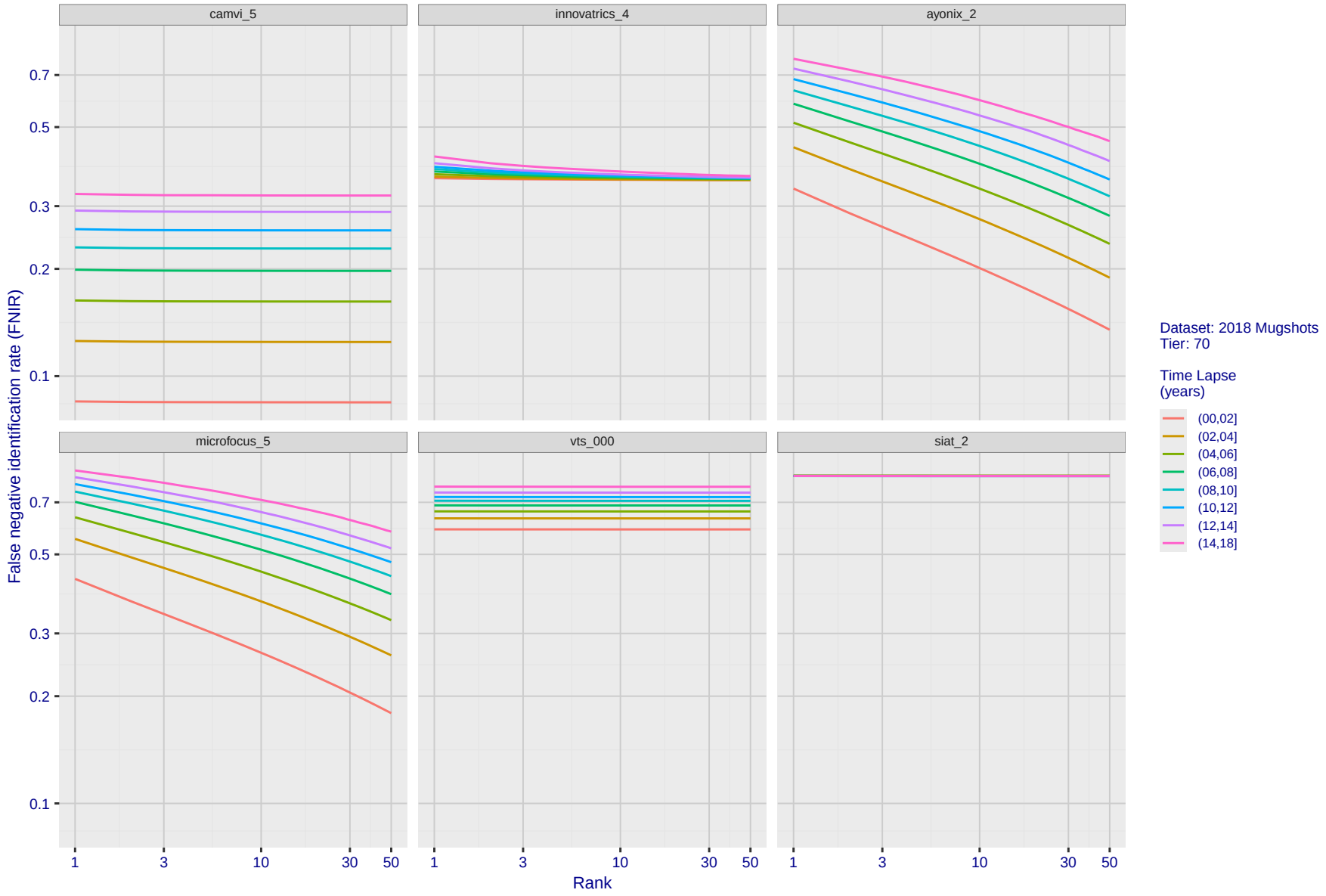

Figure 217: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. rank by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment.*

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2026/10/06 08:21:15	FNIR(N, R, T) = FPIR(N, T) =	False neg. identification rate False pos. identification rate	N = Num. enrolled subjects R = Num. candidates examined	T = Threshold	$T = 0 \rightarrow$ Investigation $T > 0 \rightarrow$ Identification

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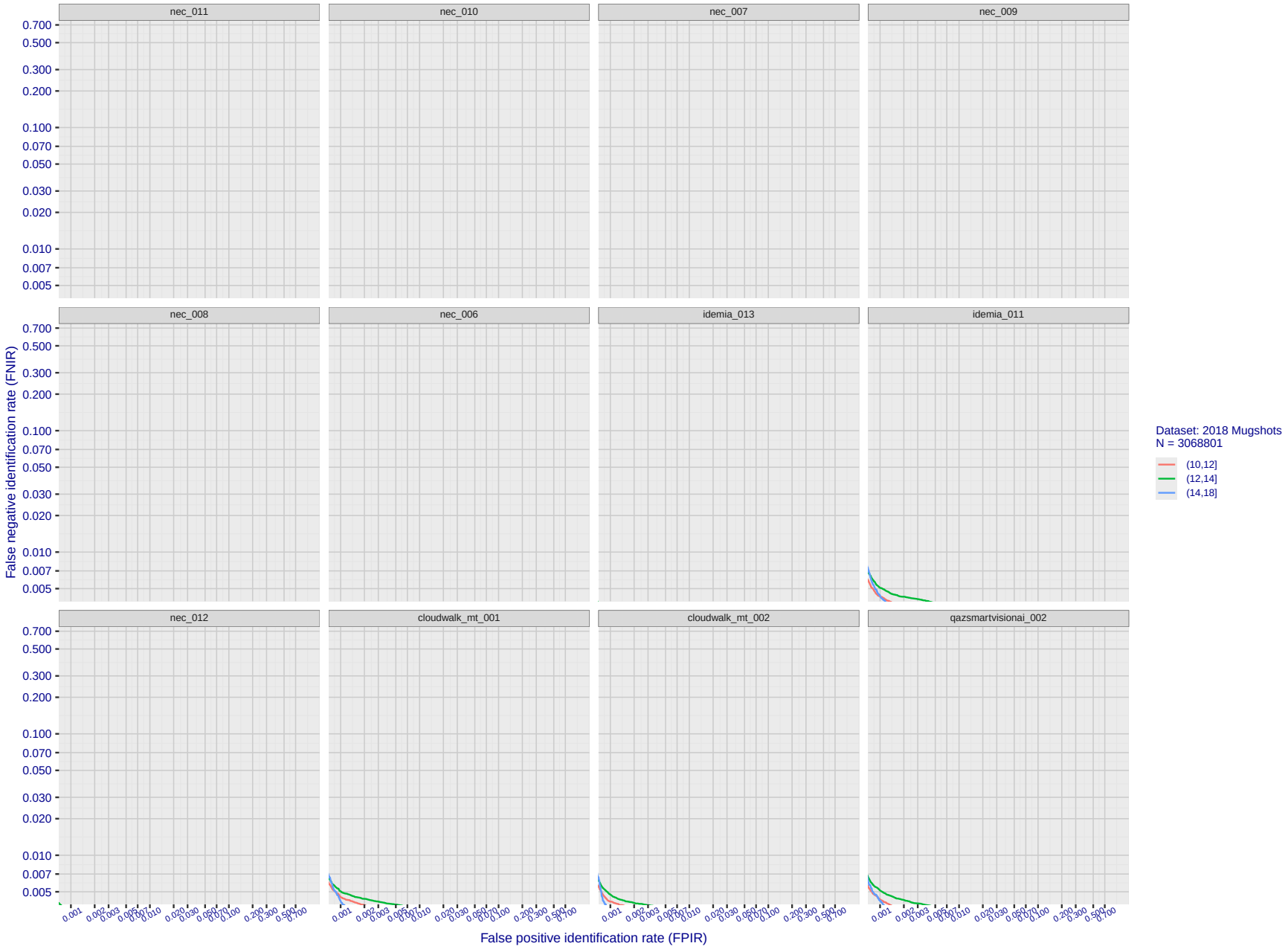

Figure 218: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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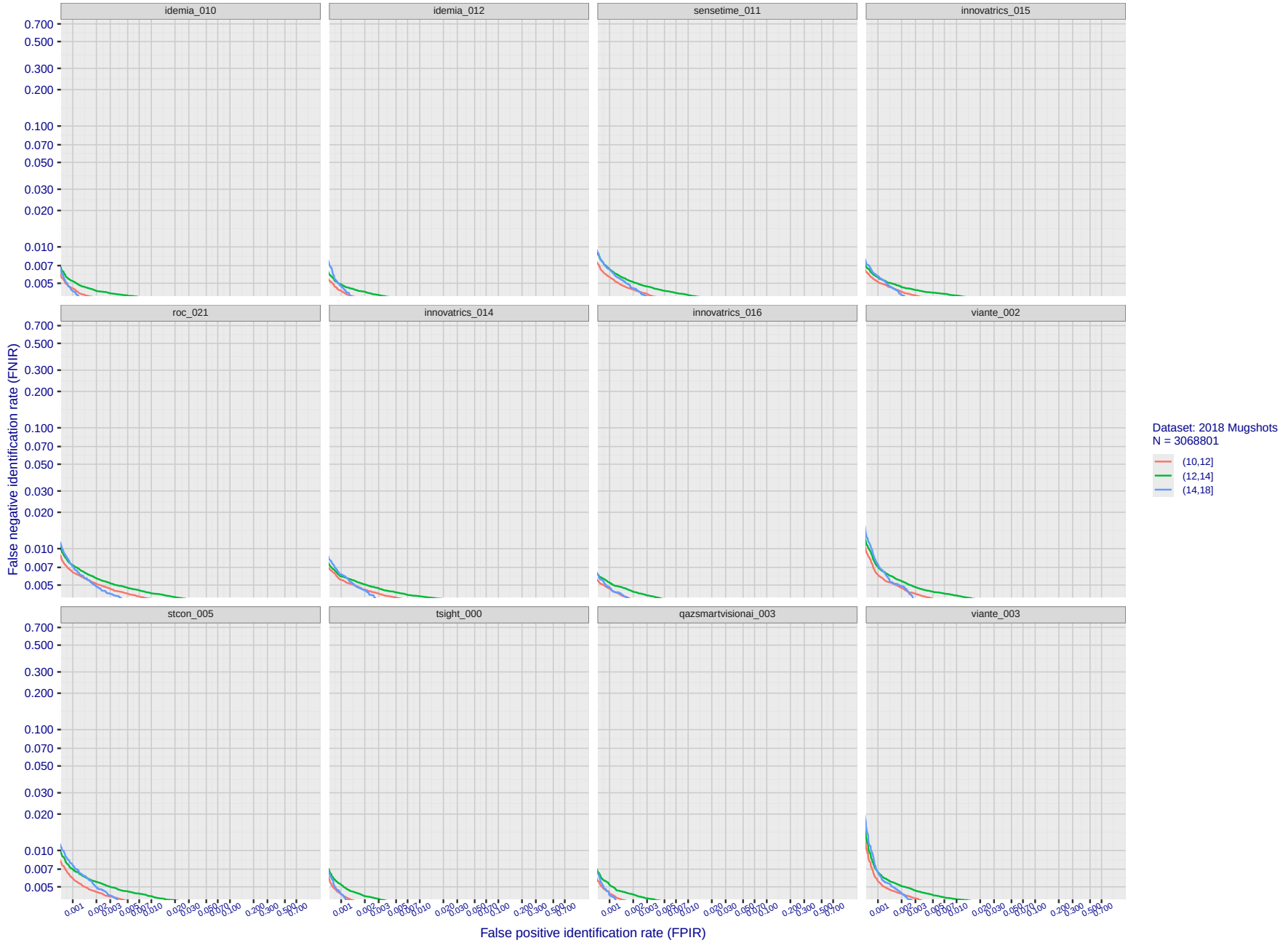

Figure 219: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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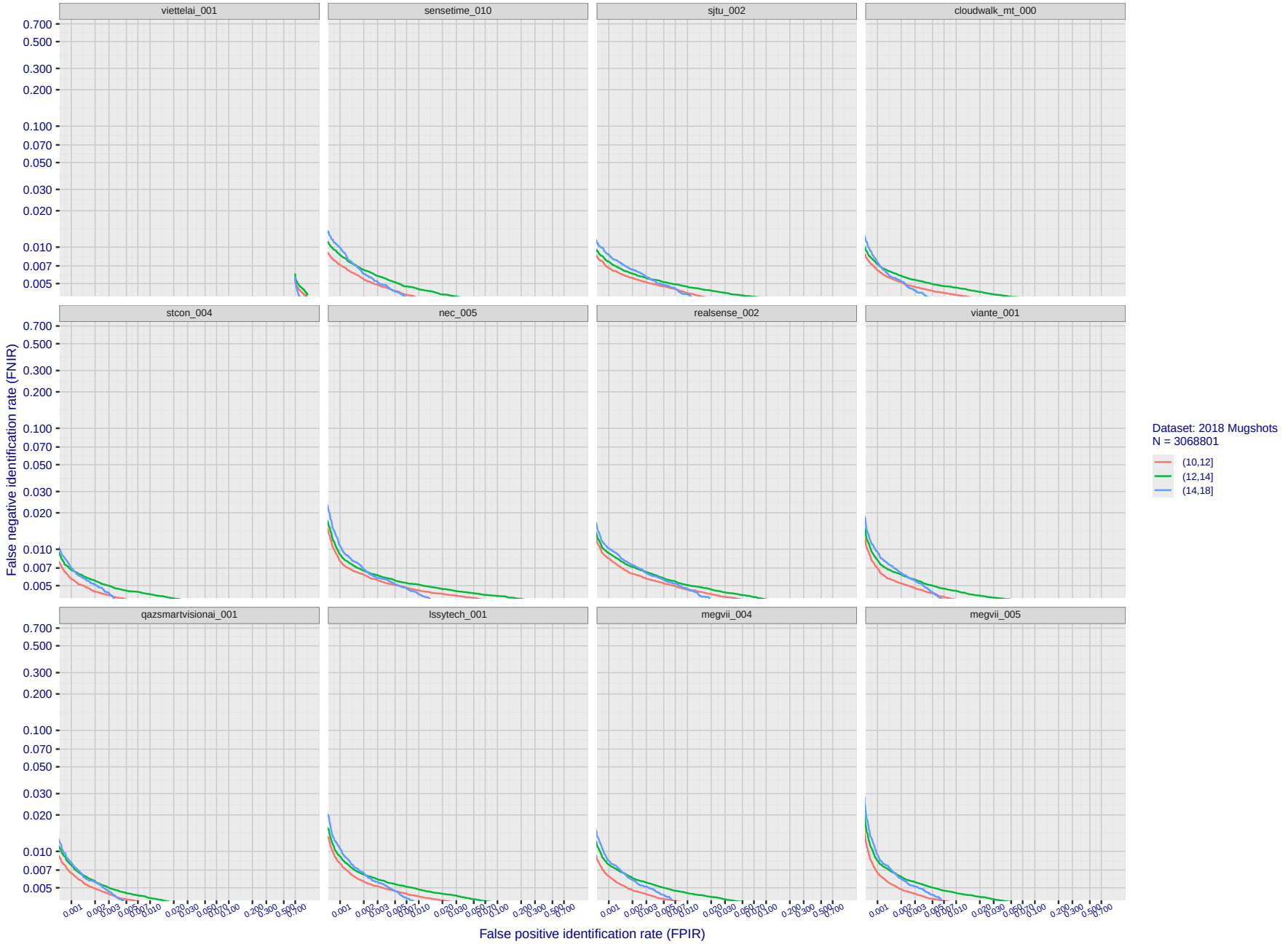

Figure 220: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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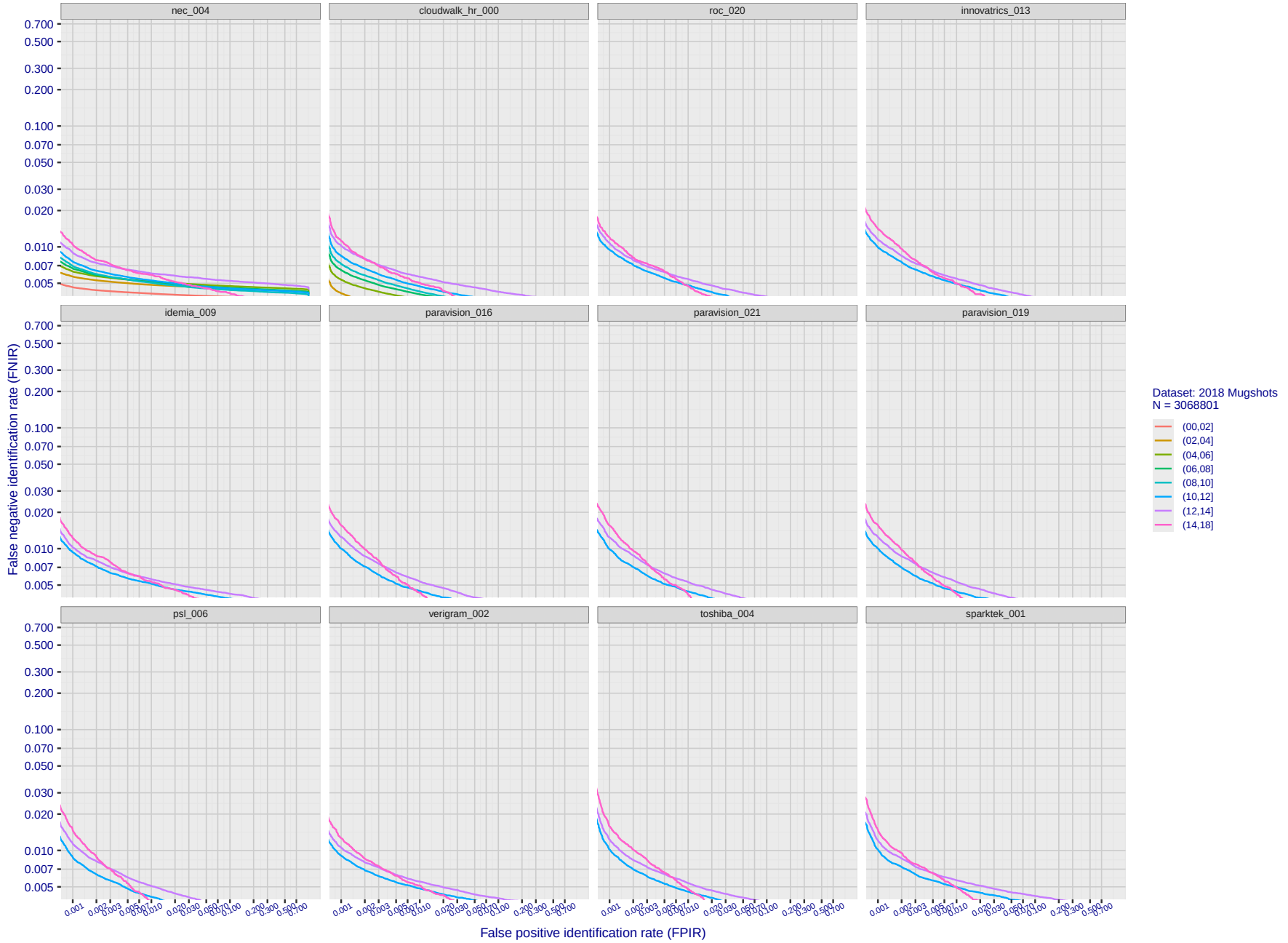

Figure 221: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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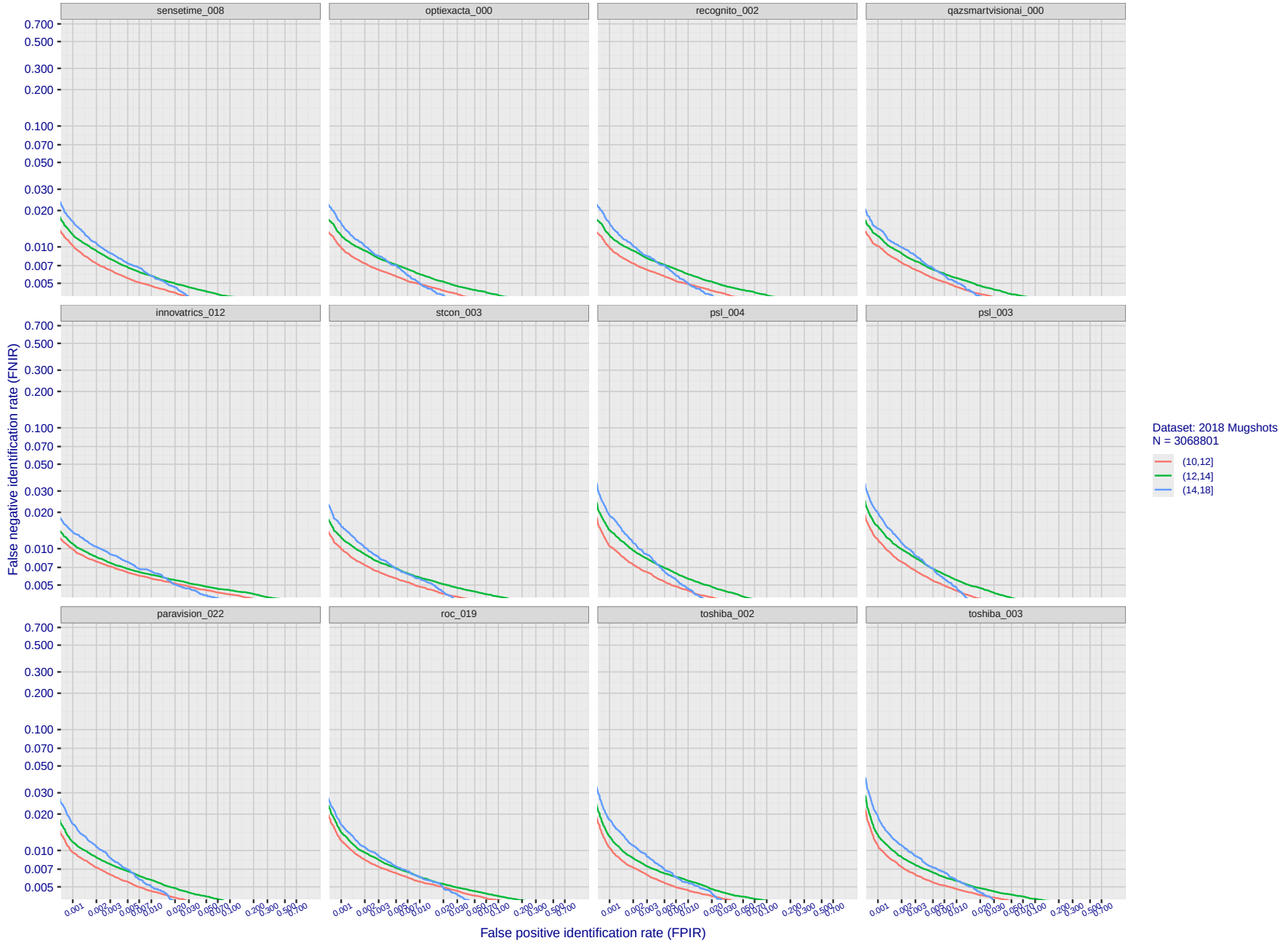

Figure 222: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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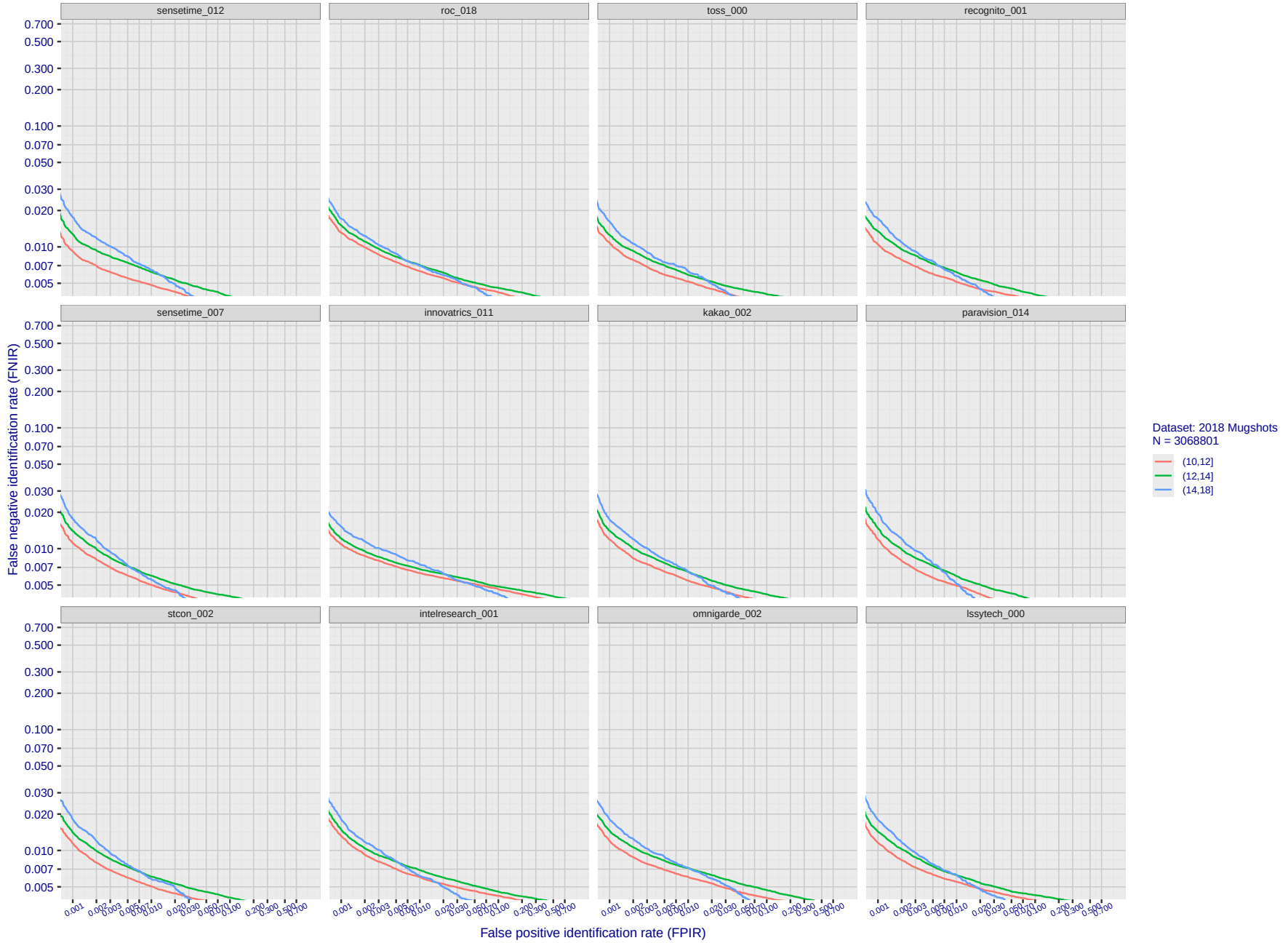

Figure 223: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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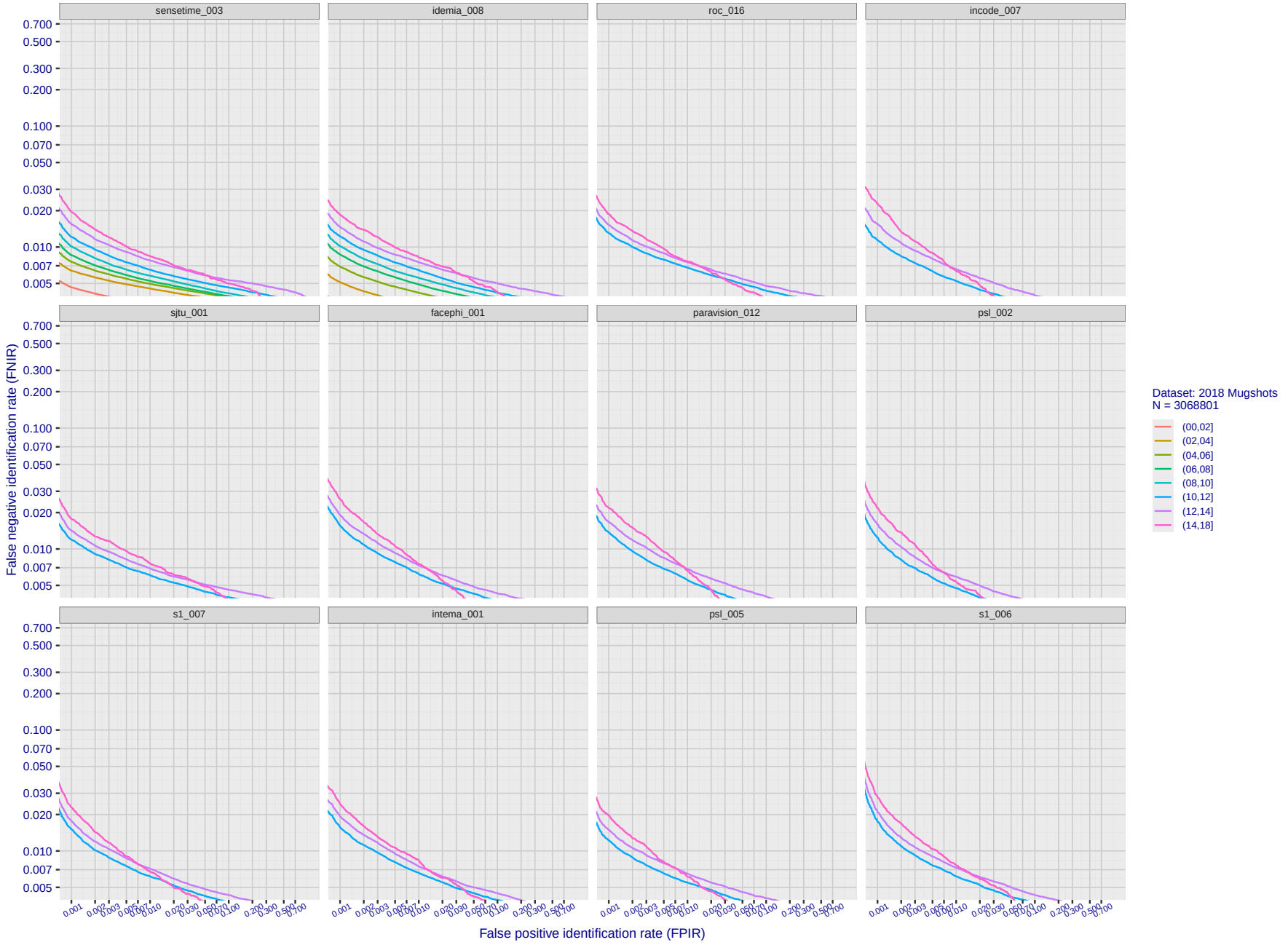

Figure 224: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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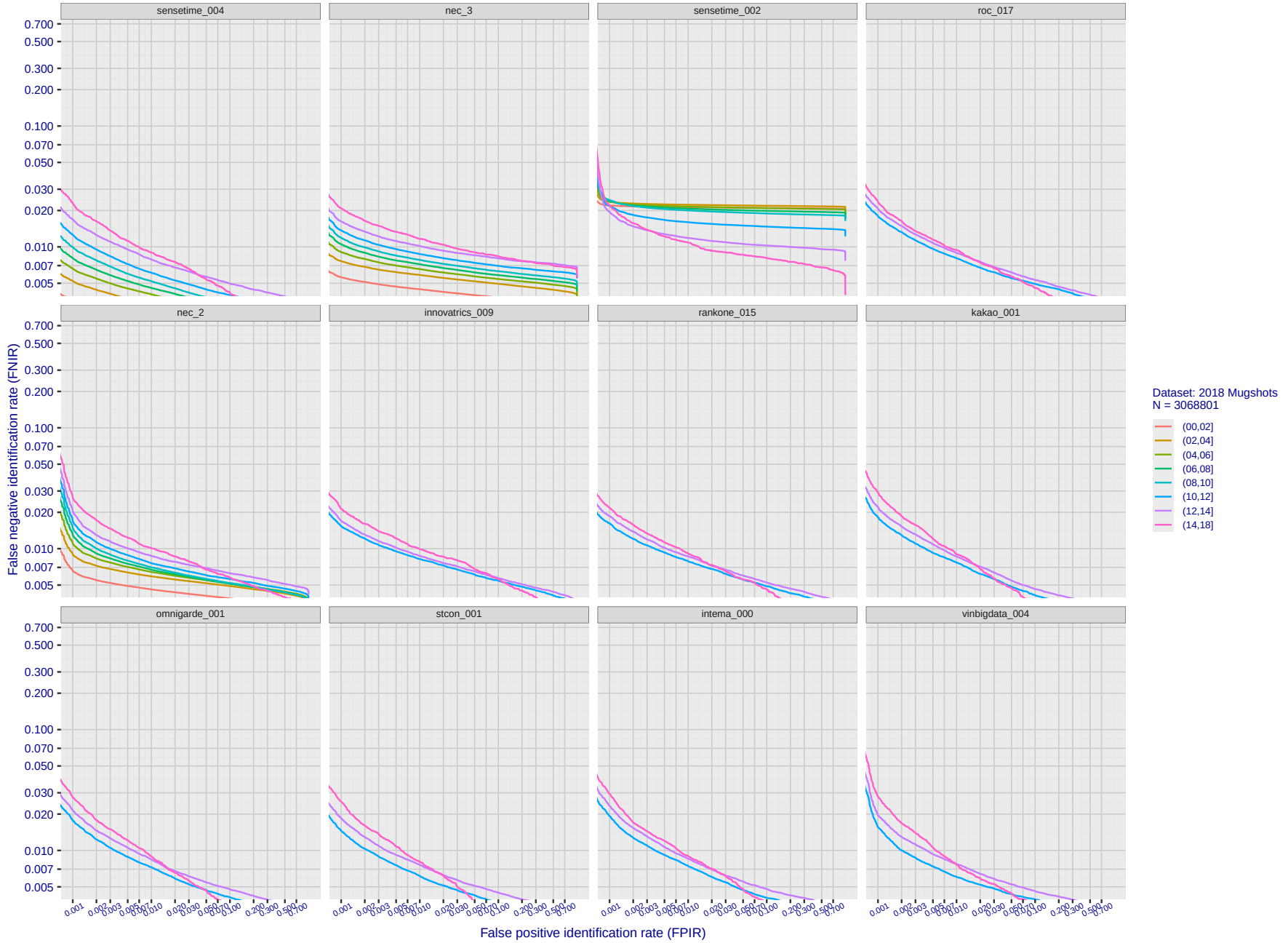

Figure 225: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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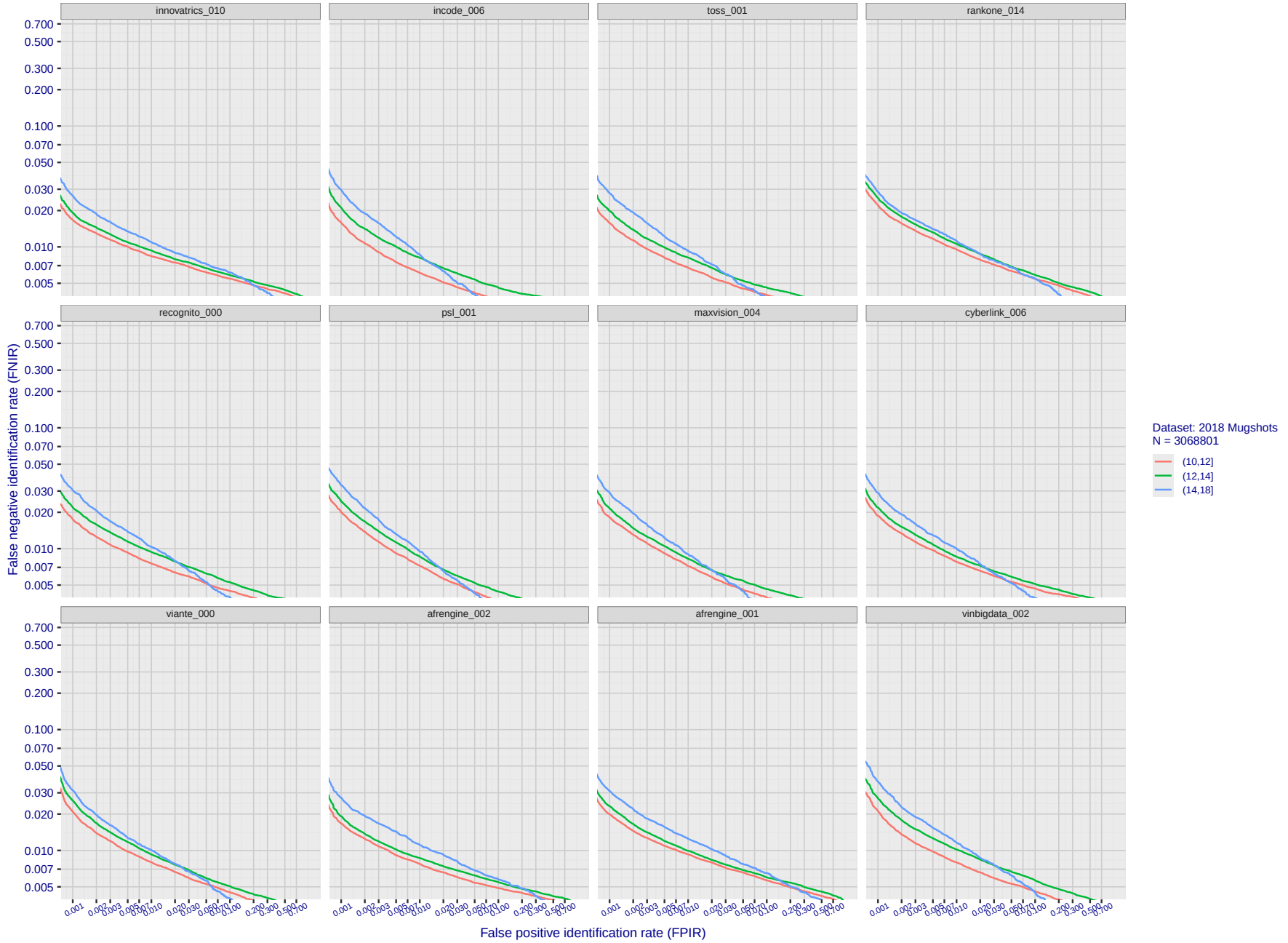

Figure 226: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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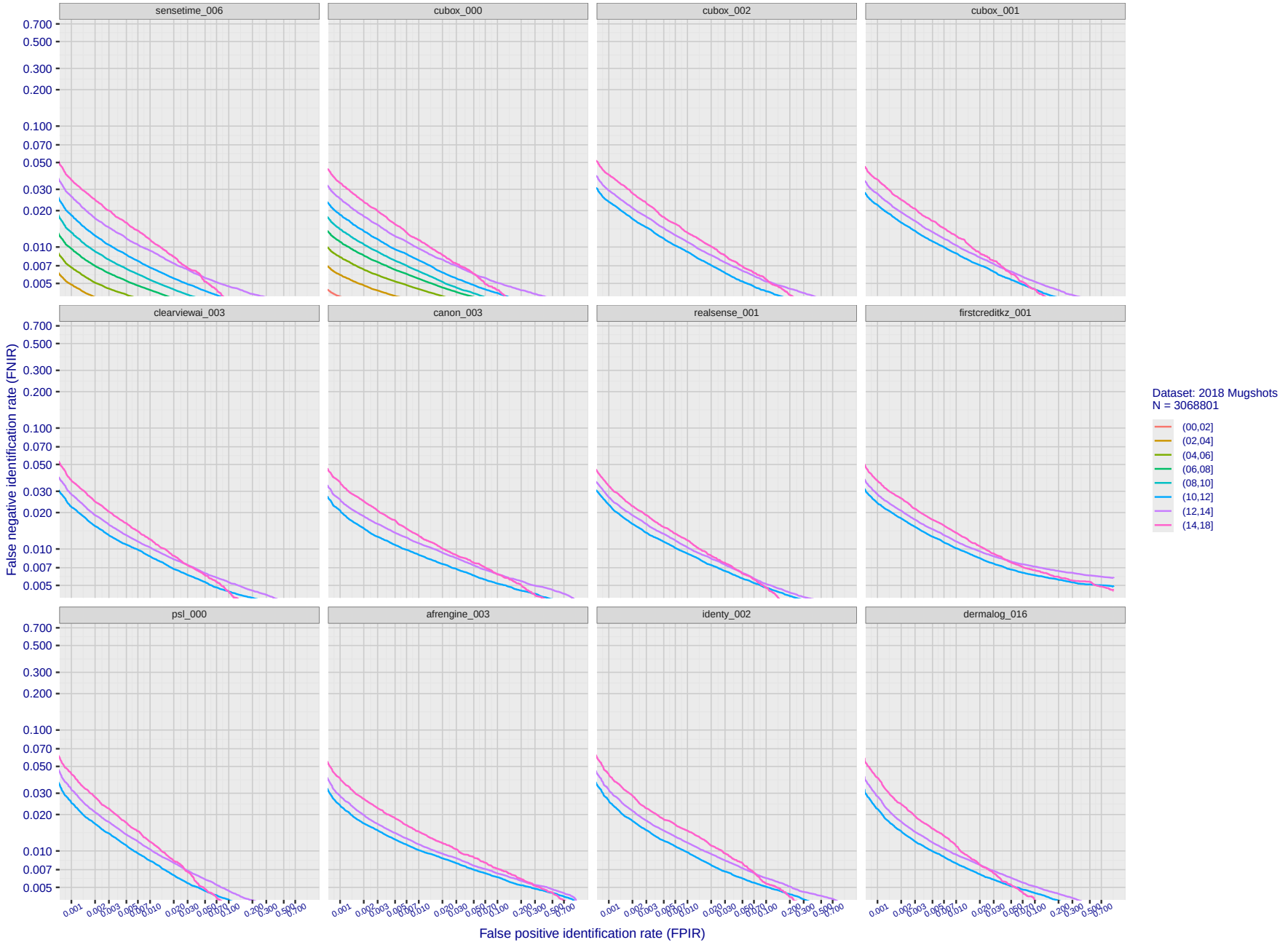

Figure 227: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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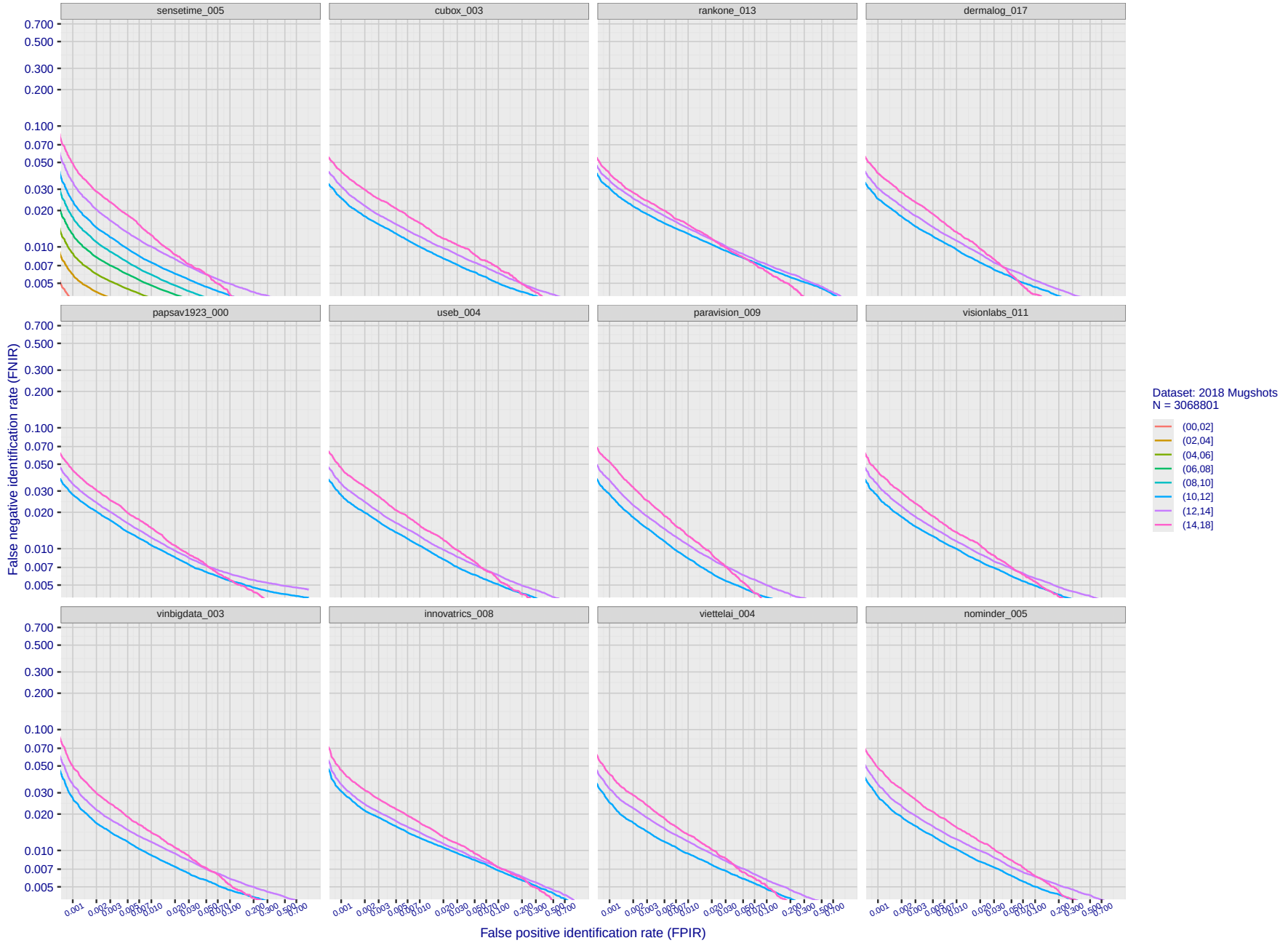

Figure 228: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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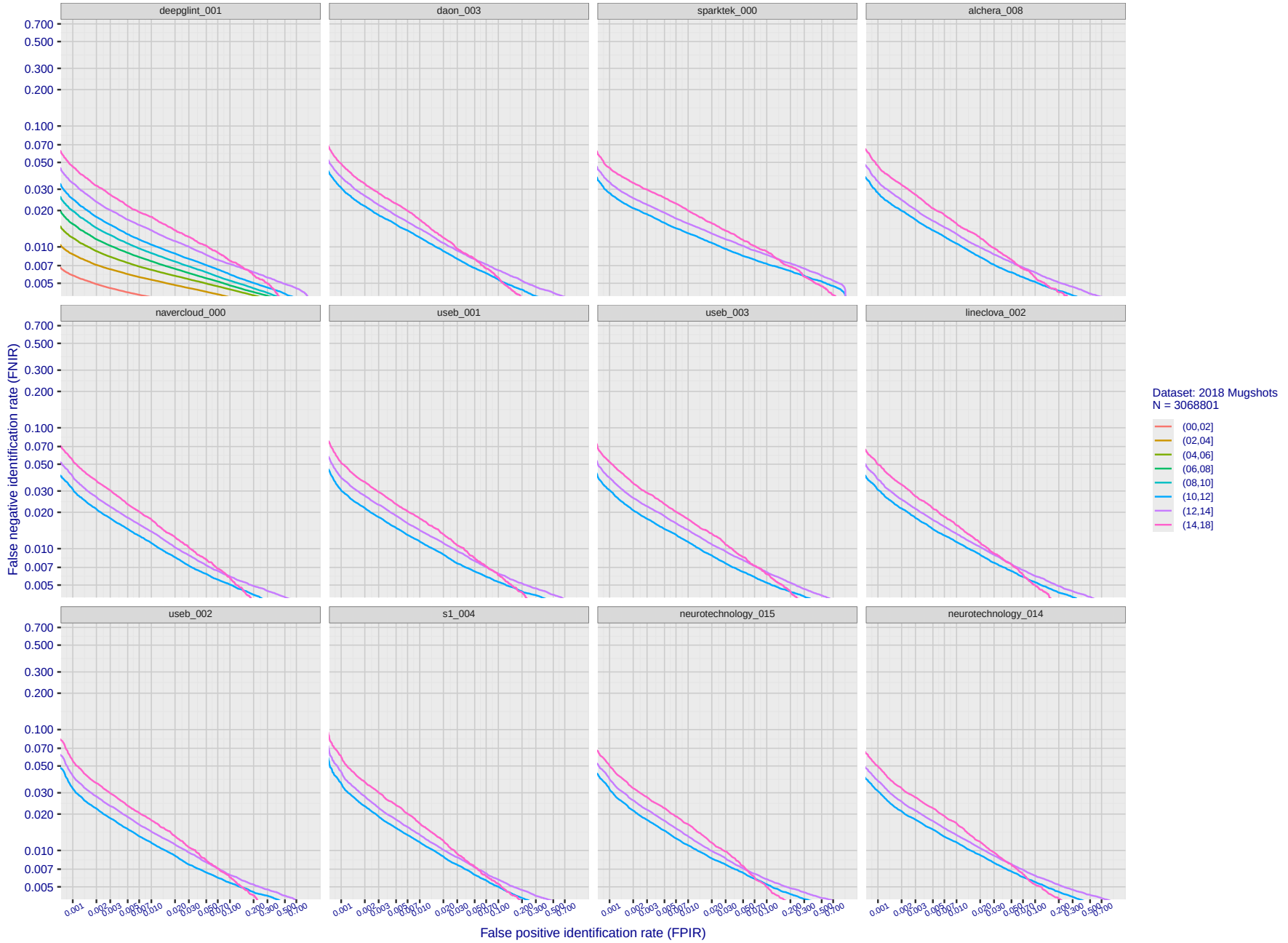

Figure 229: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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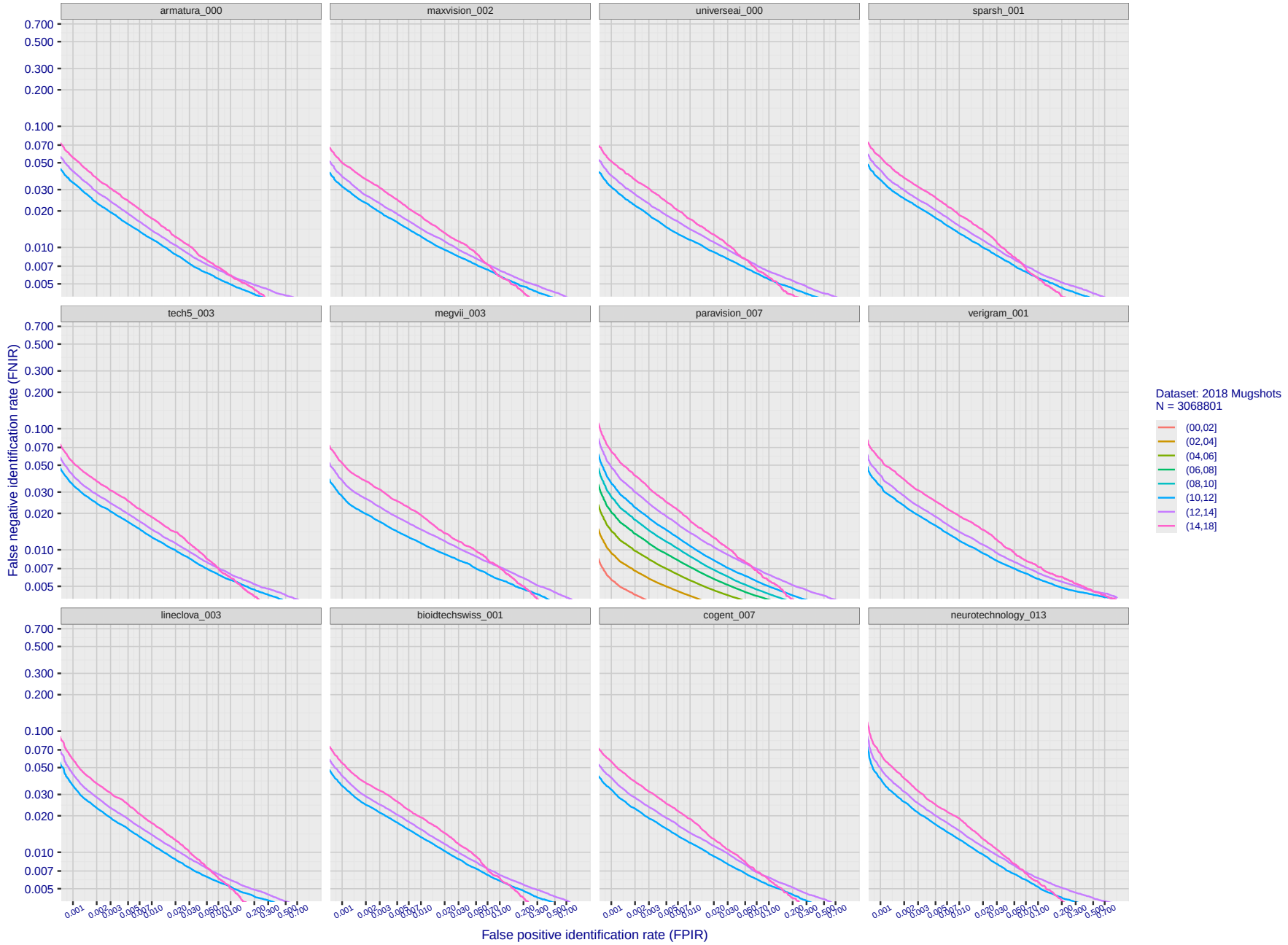

Figure 230: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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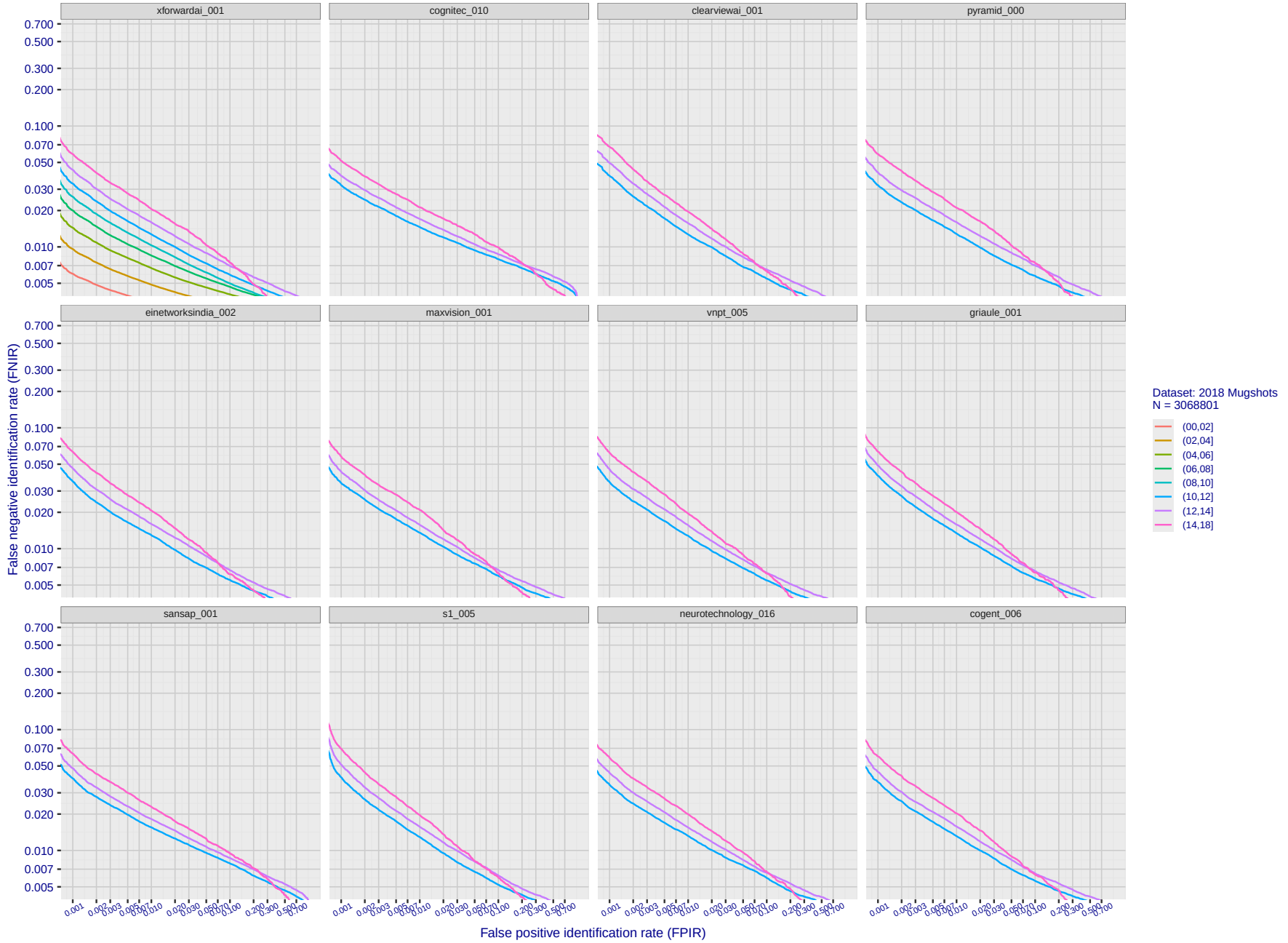

Figure 231: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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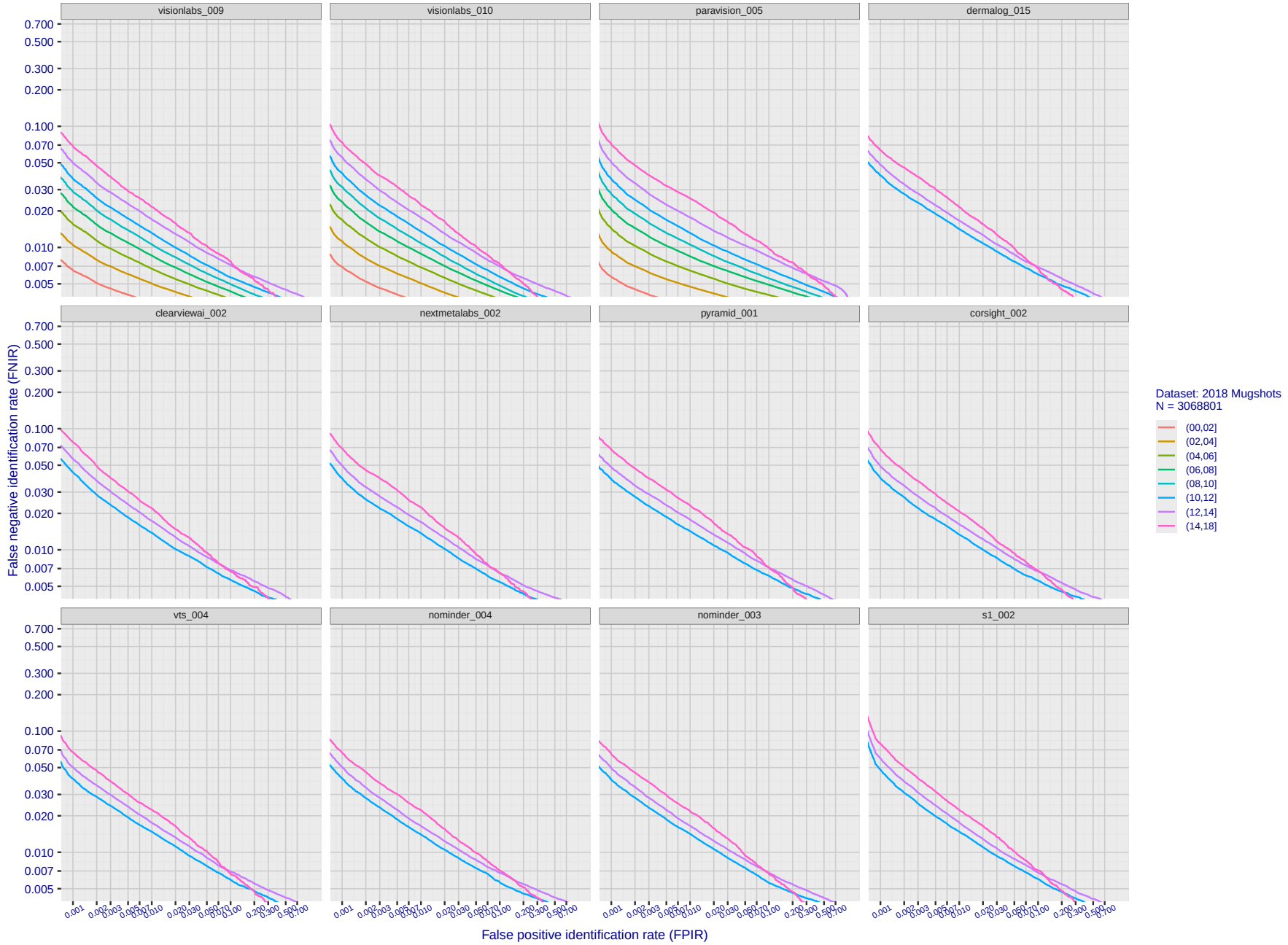

Figure 232: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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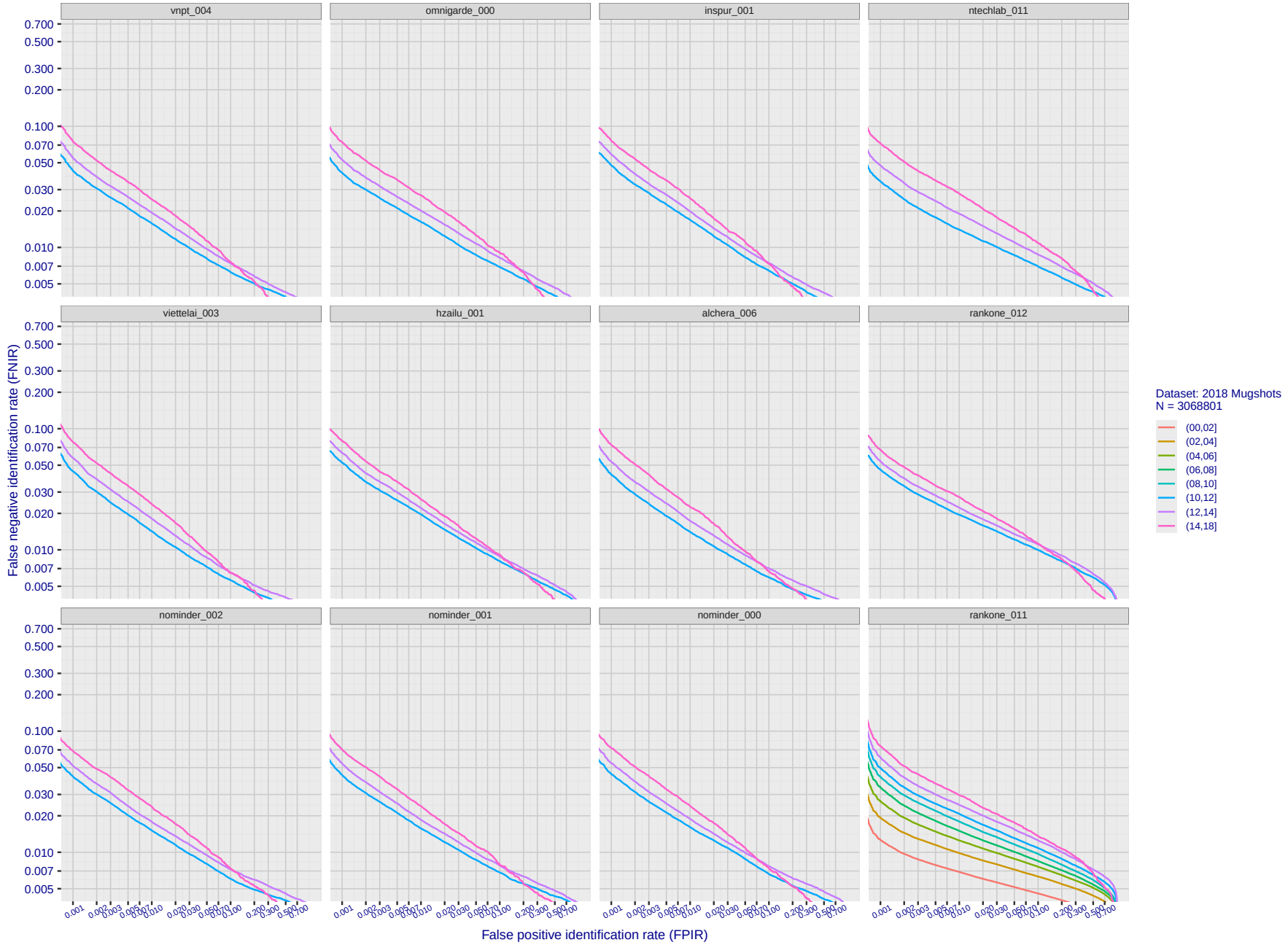

Figure 233: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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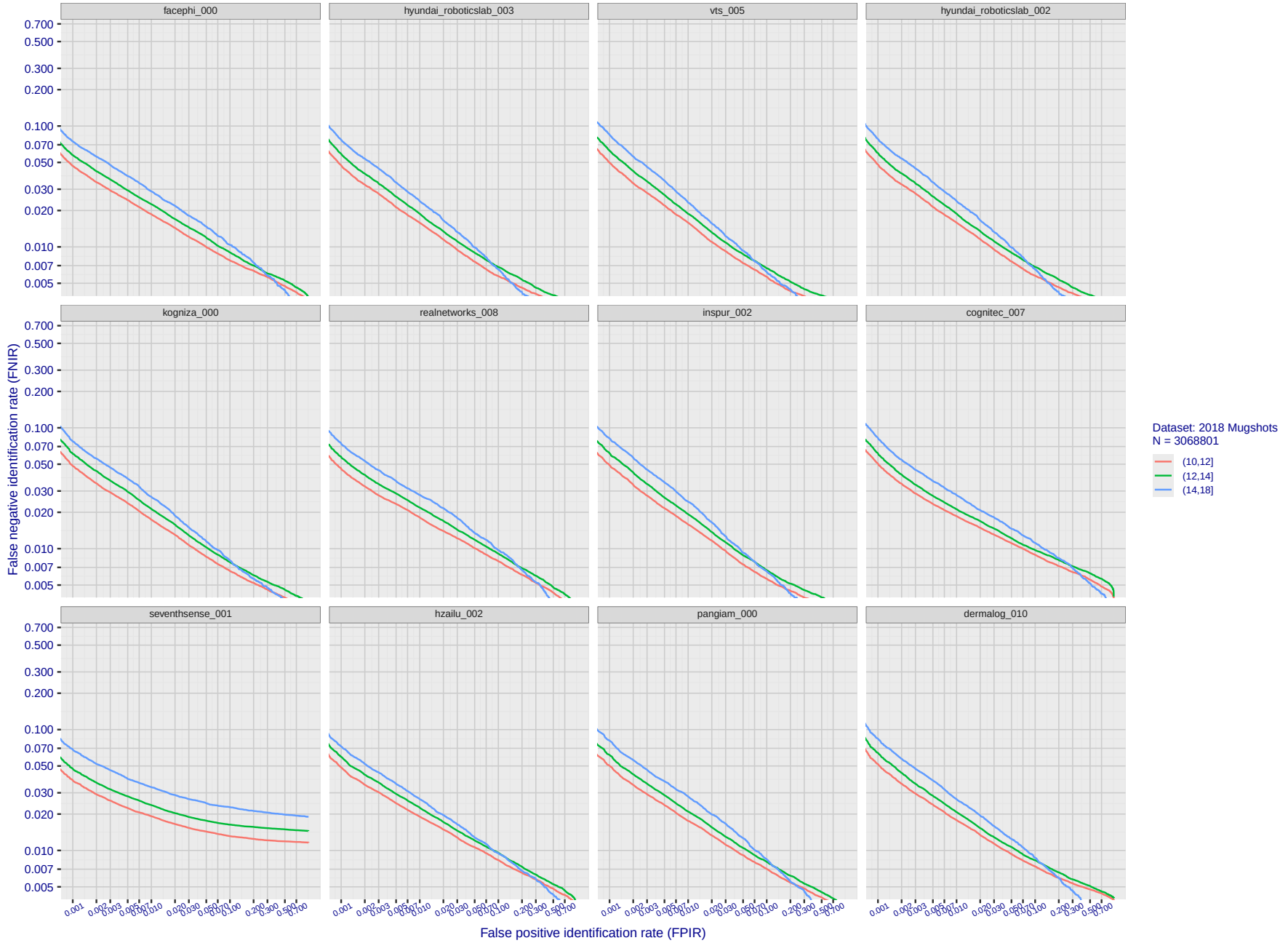

Figure 234: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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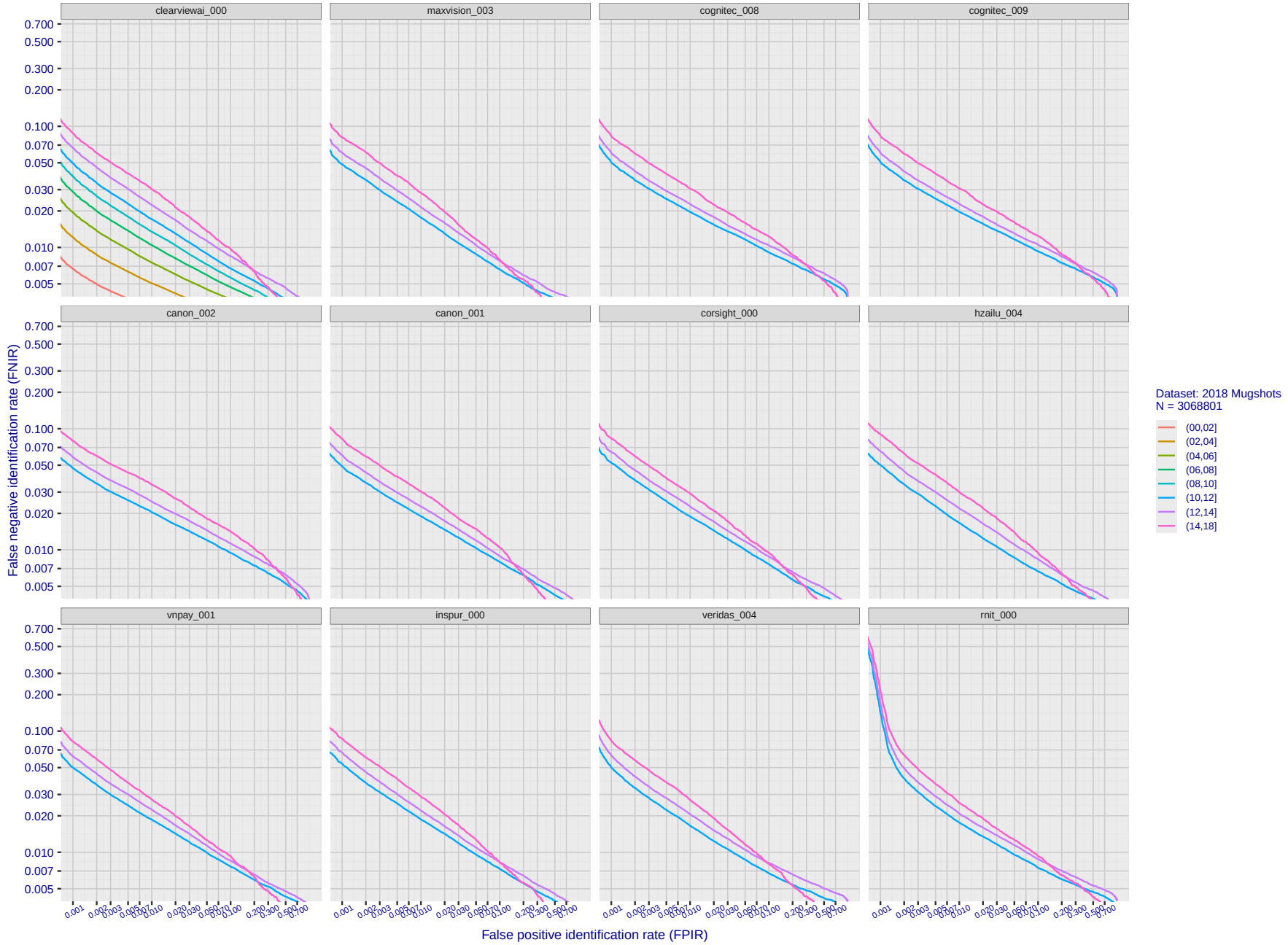

Figure 235: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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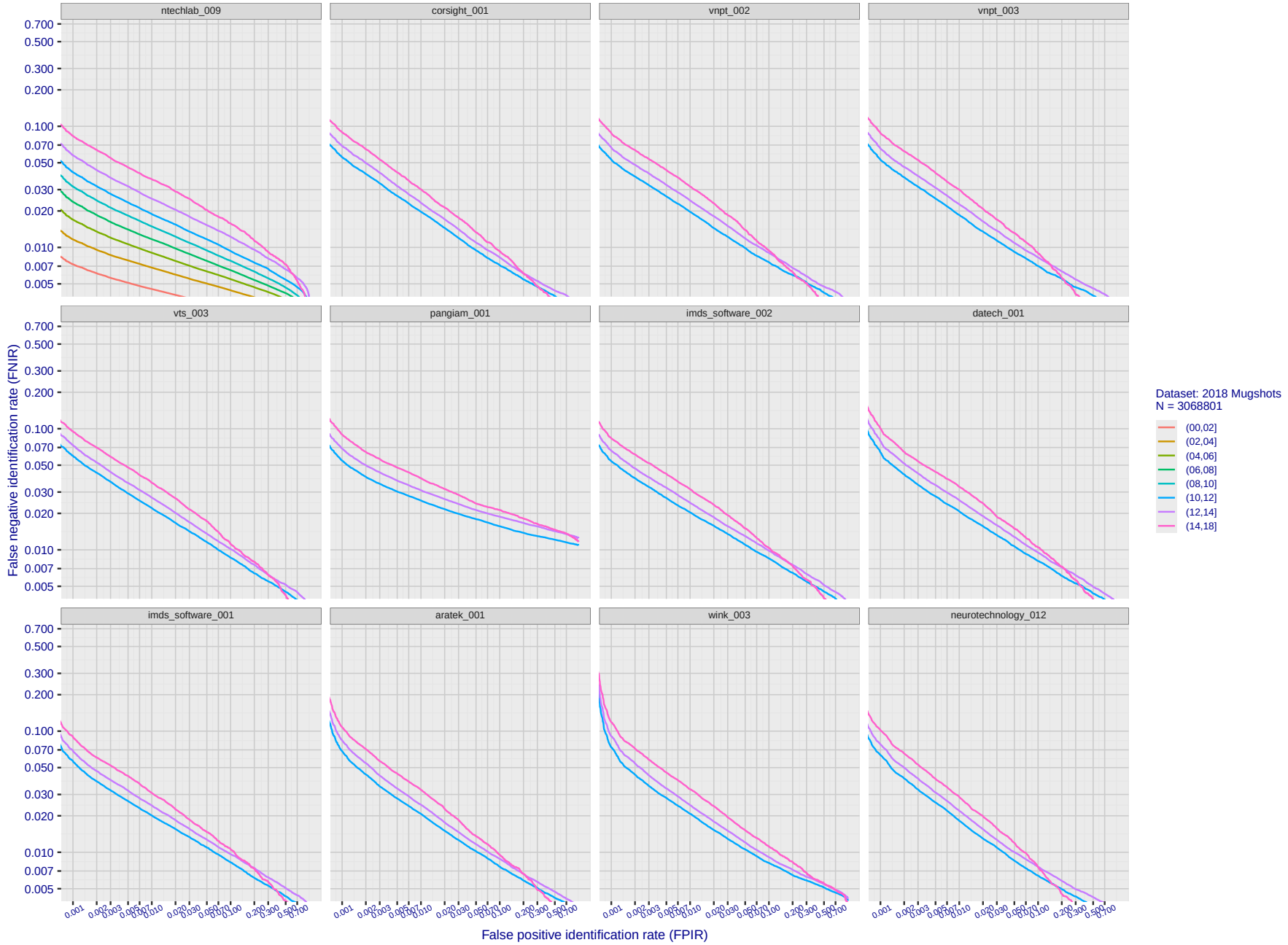

Figure 236: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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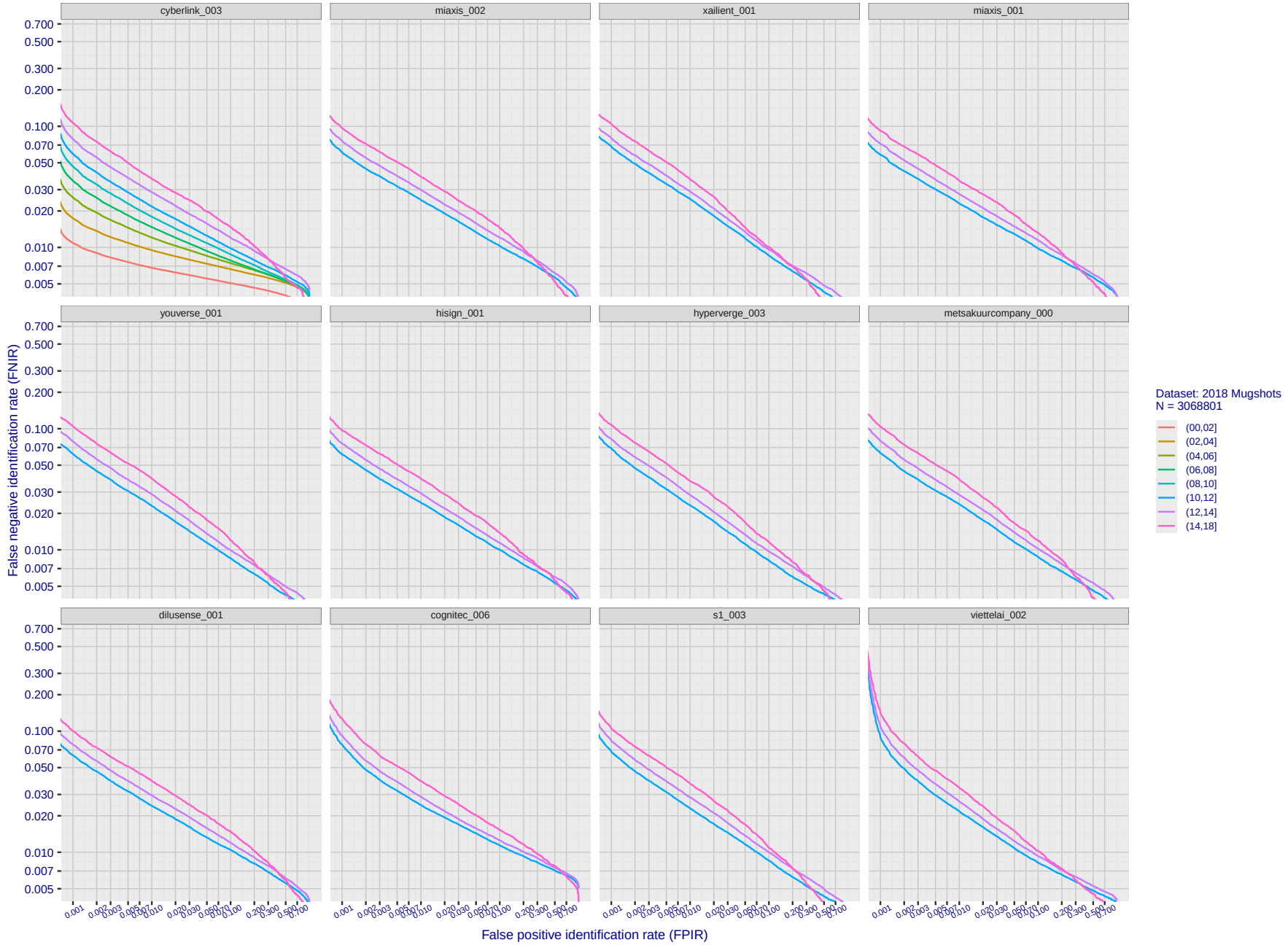

Figure 237: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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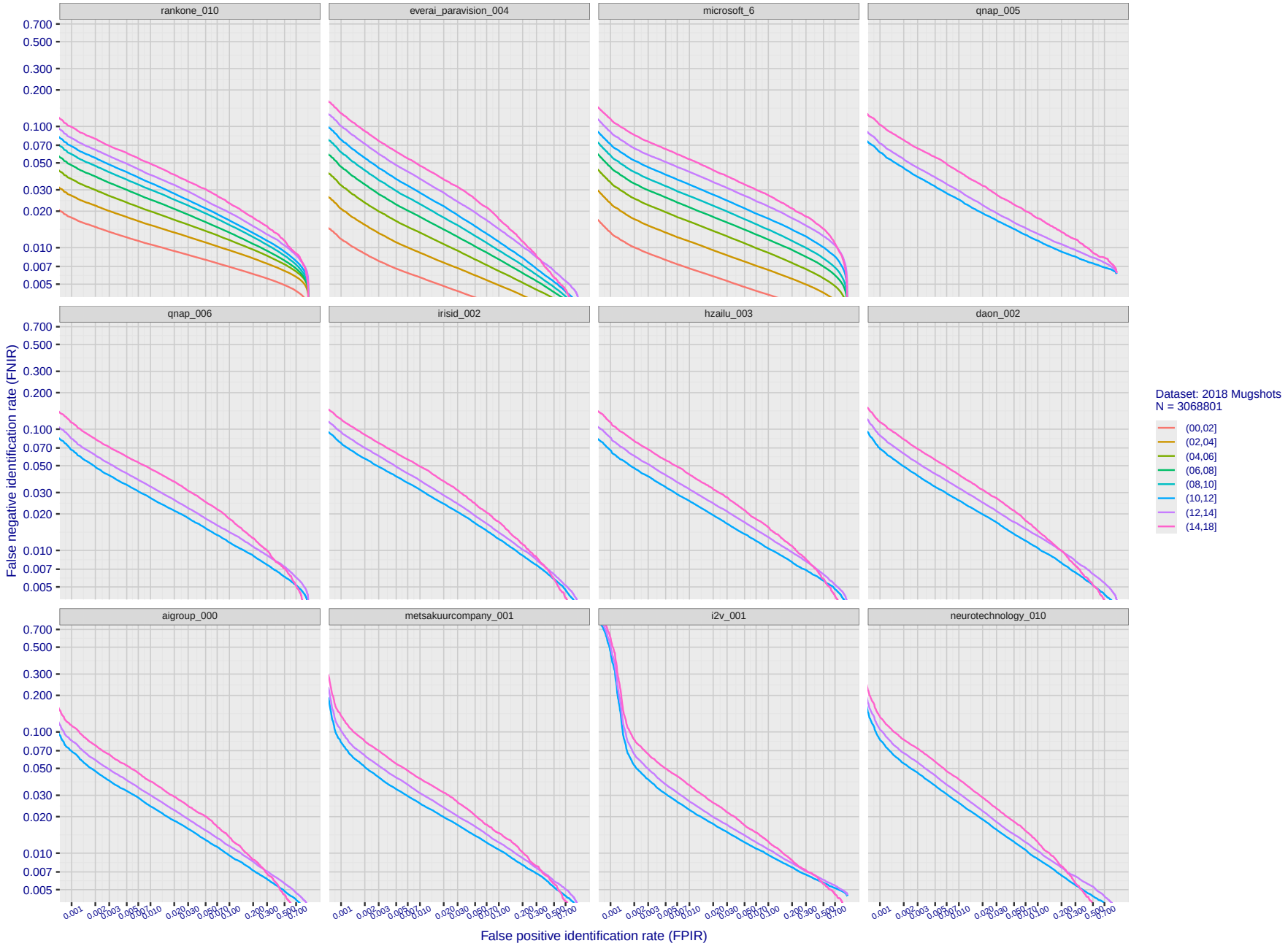

Figure 238: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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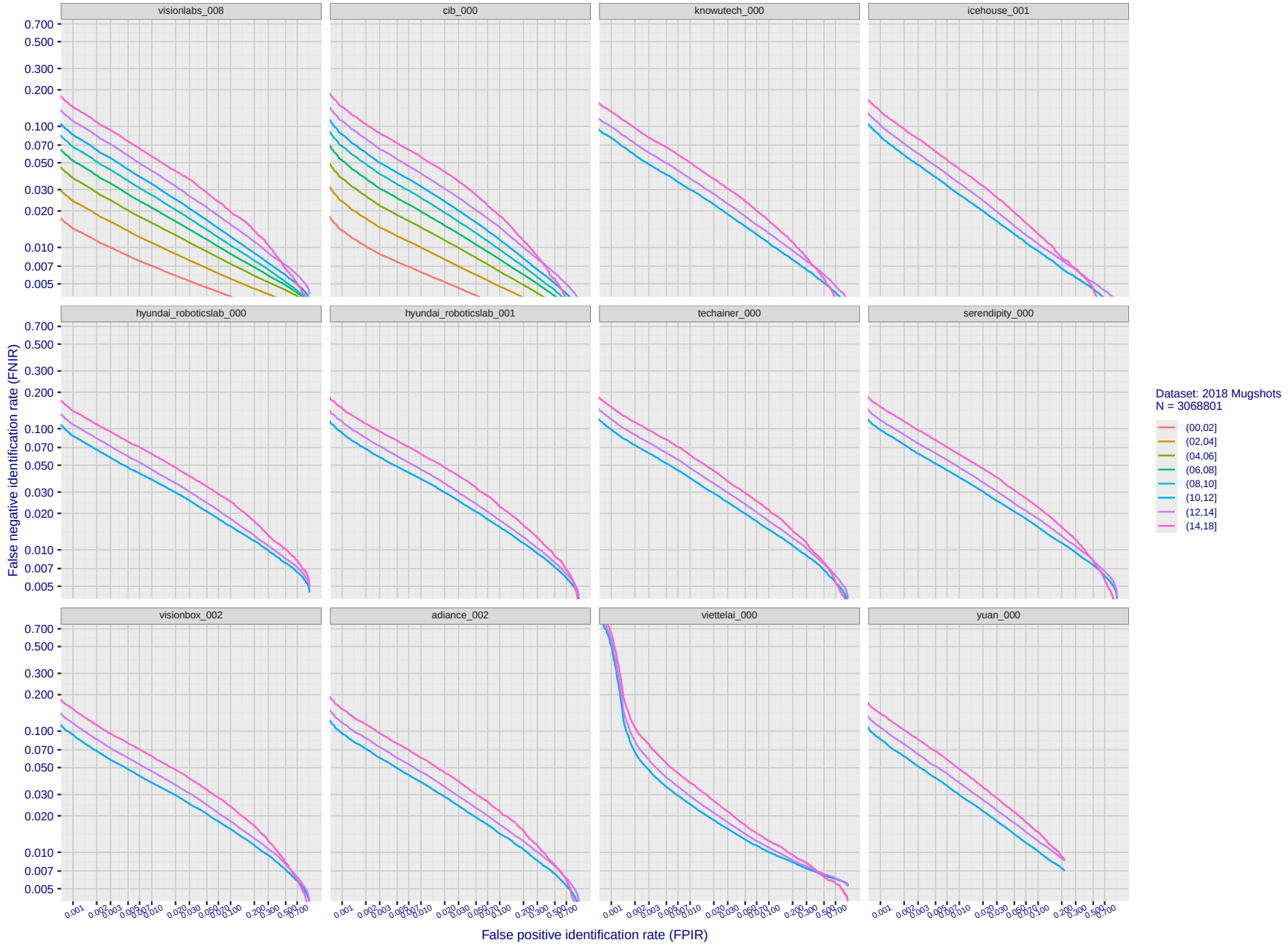

Figure 239: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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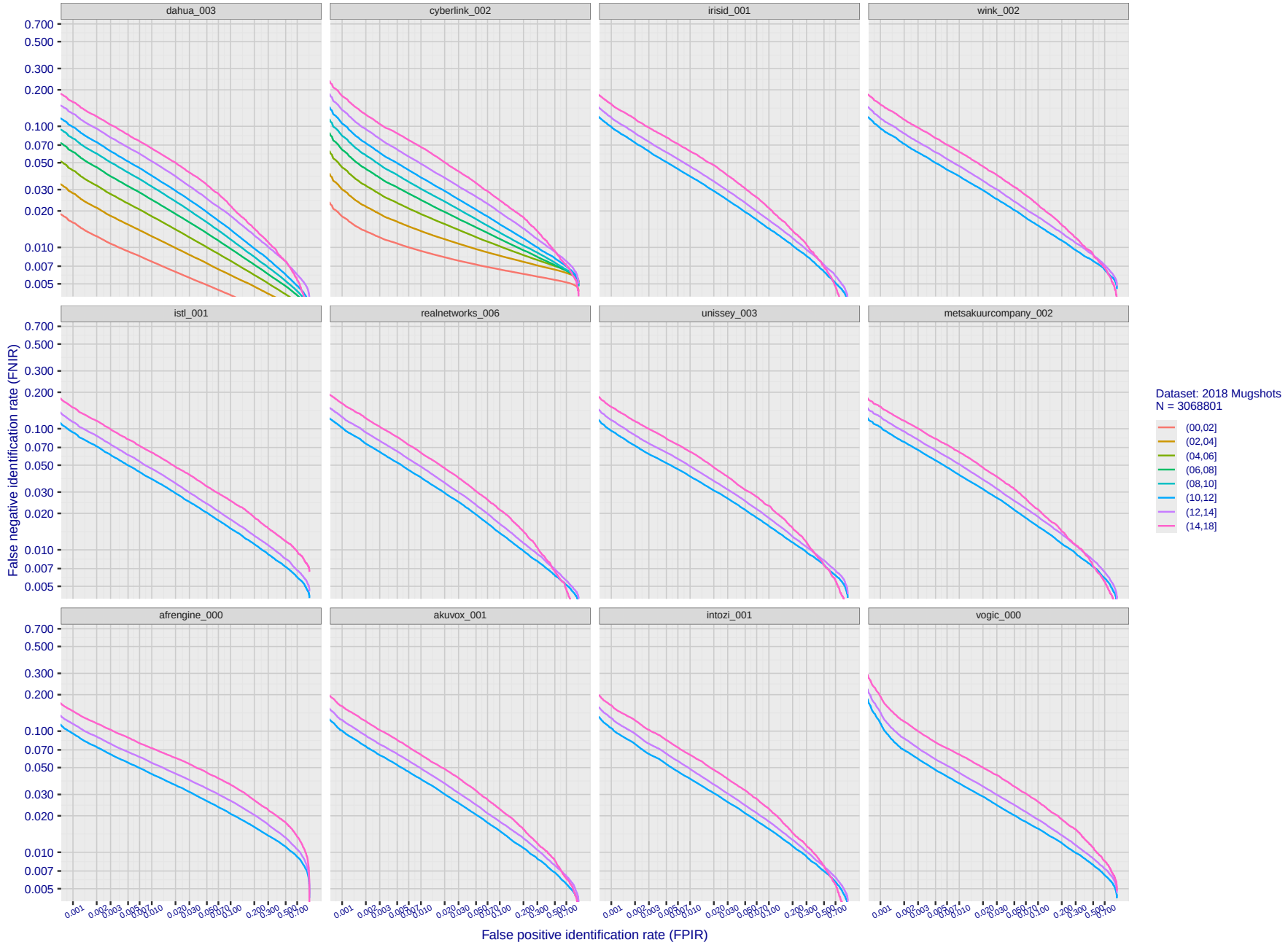

Figure 240: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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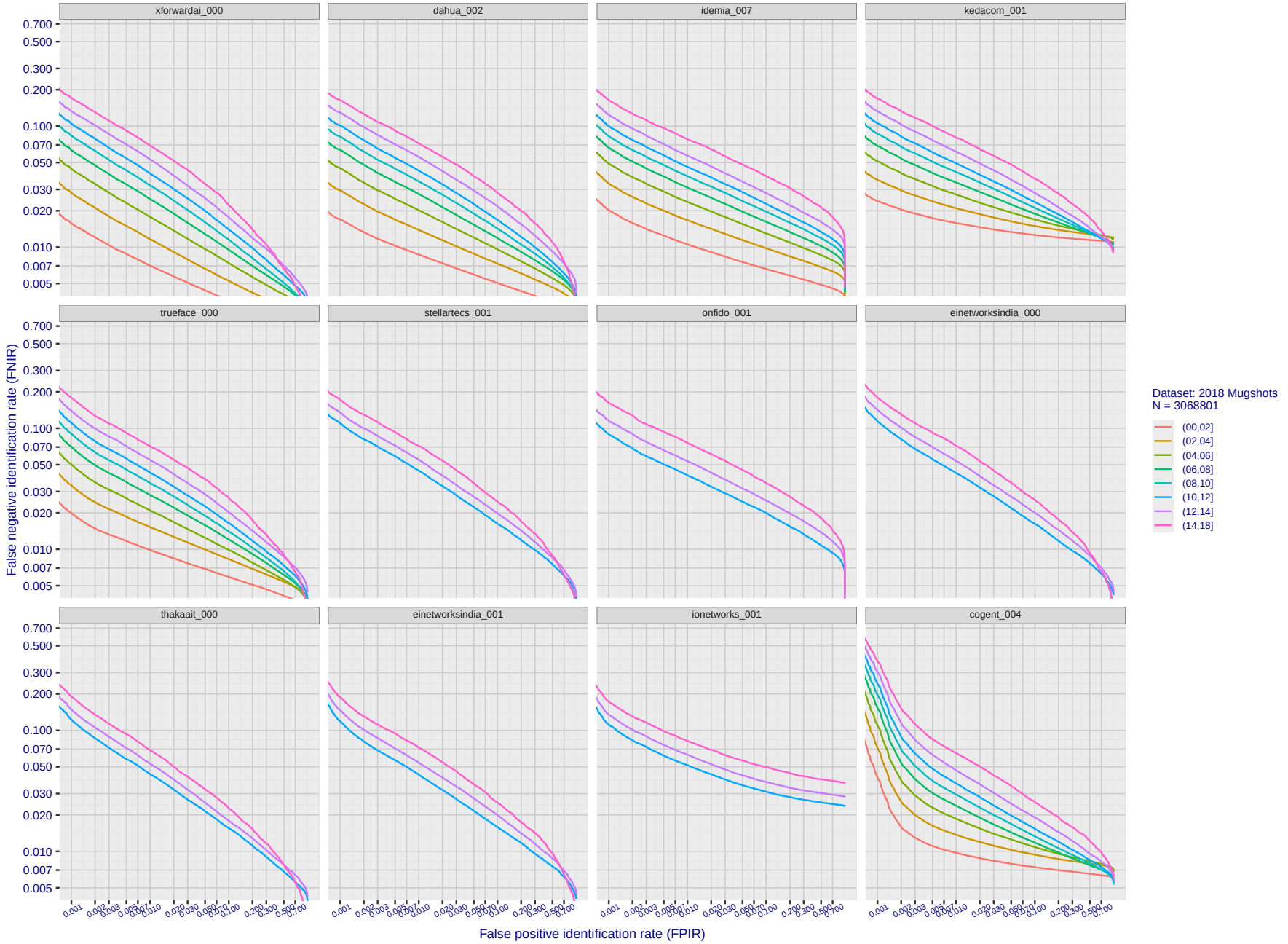

Figure 241: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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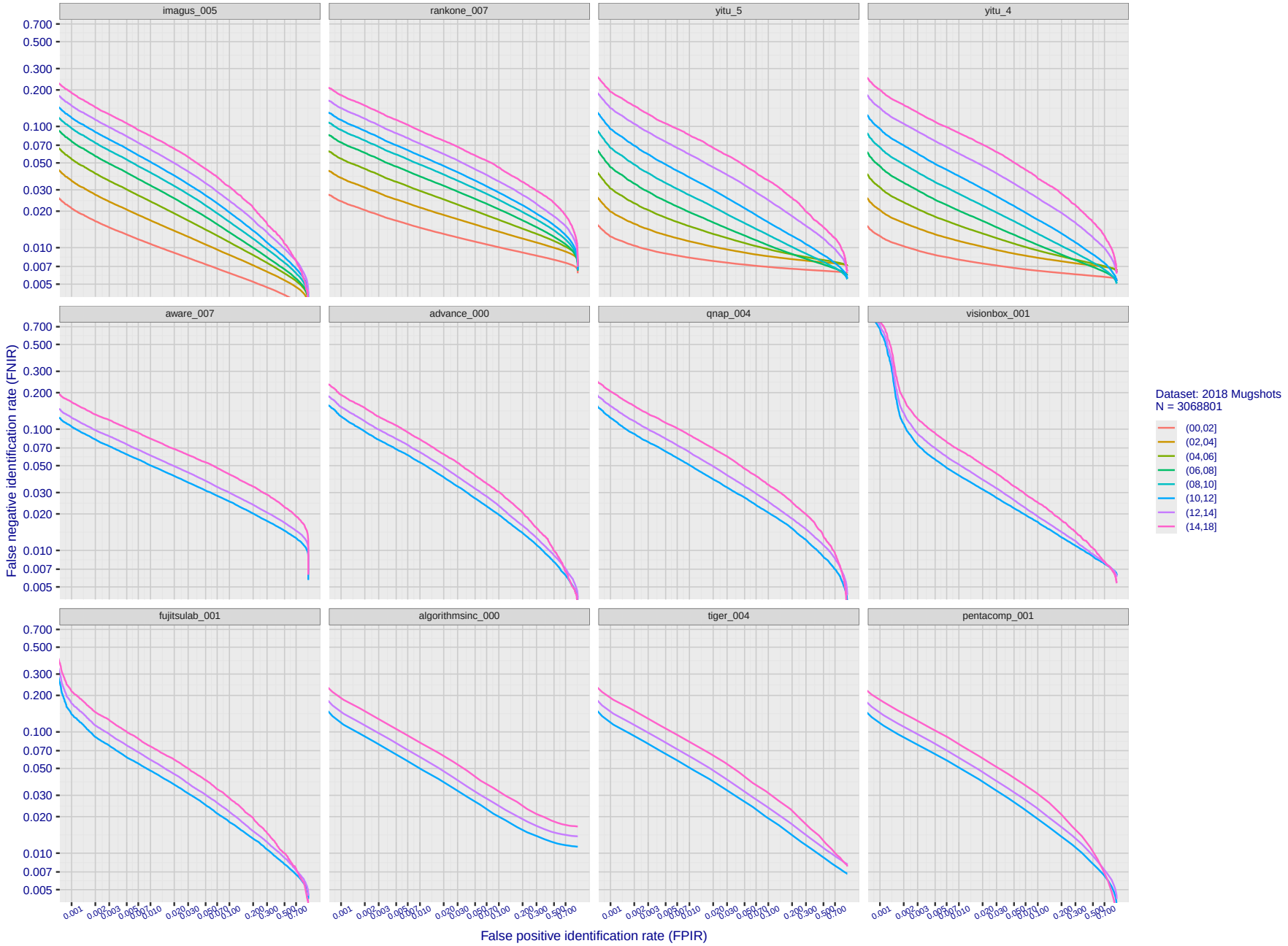

Figure 242: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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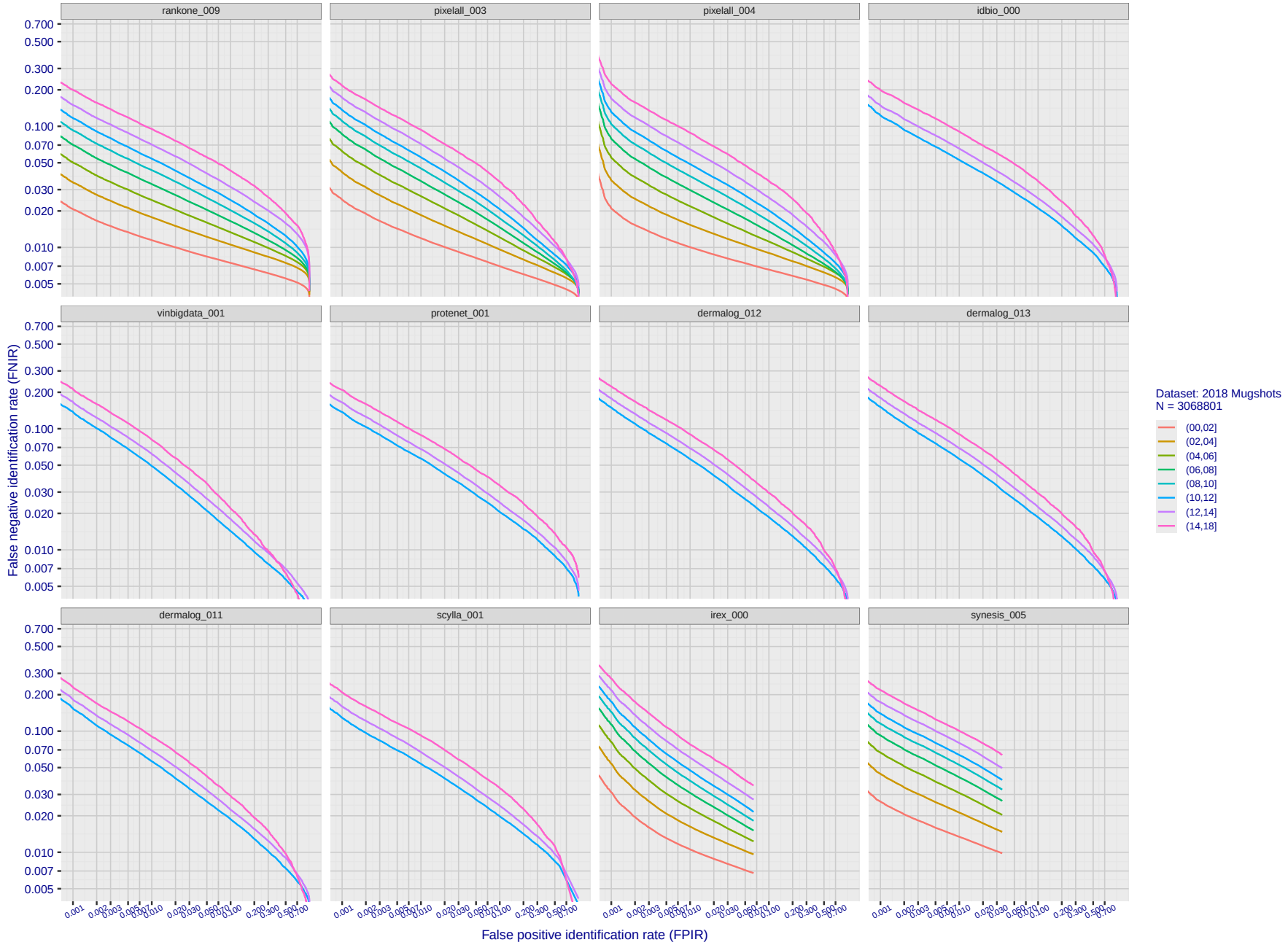

Figure 243: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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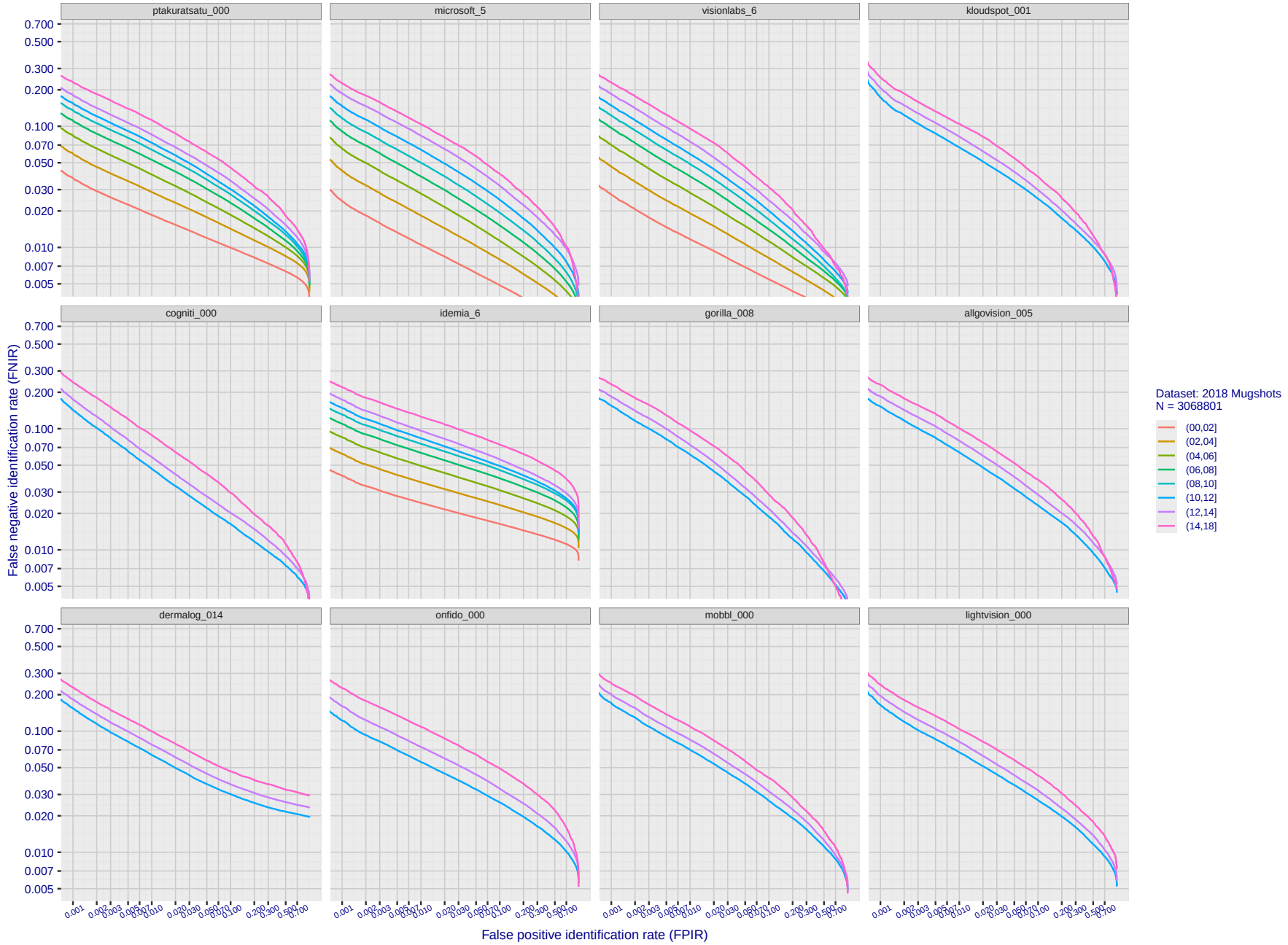

Figure 244: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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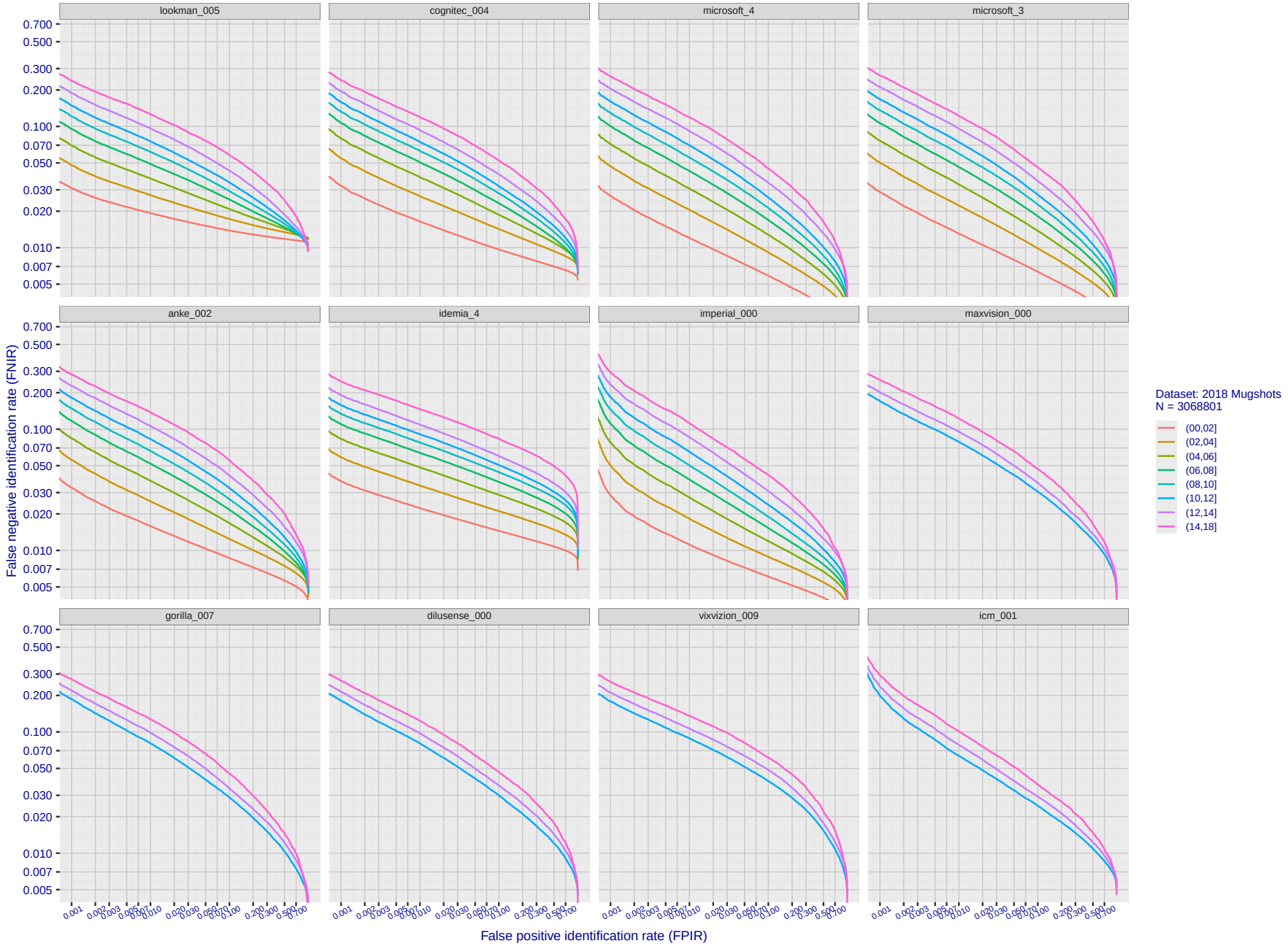

Figure 245: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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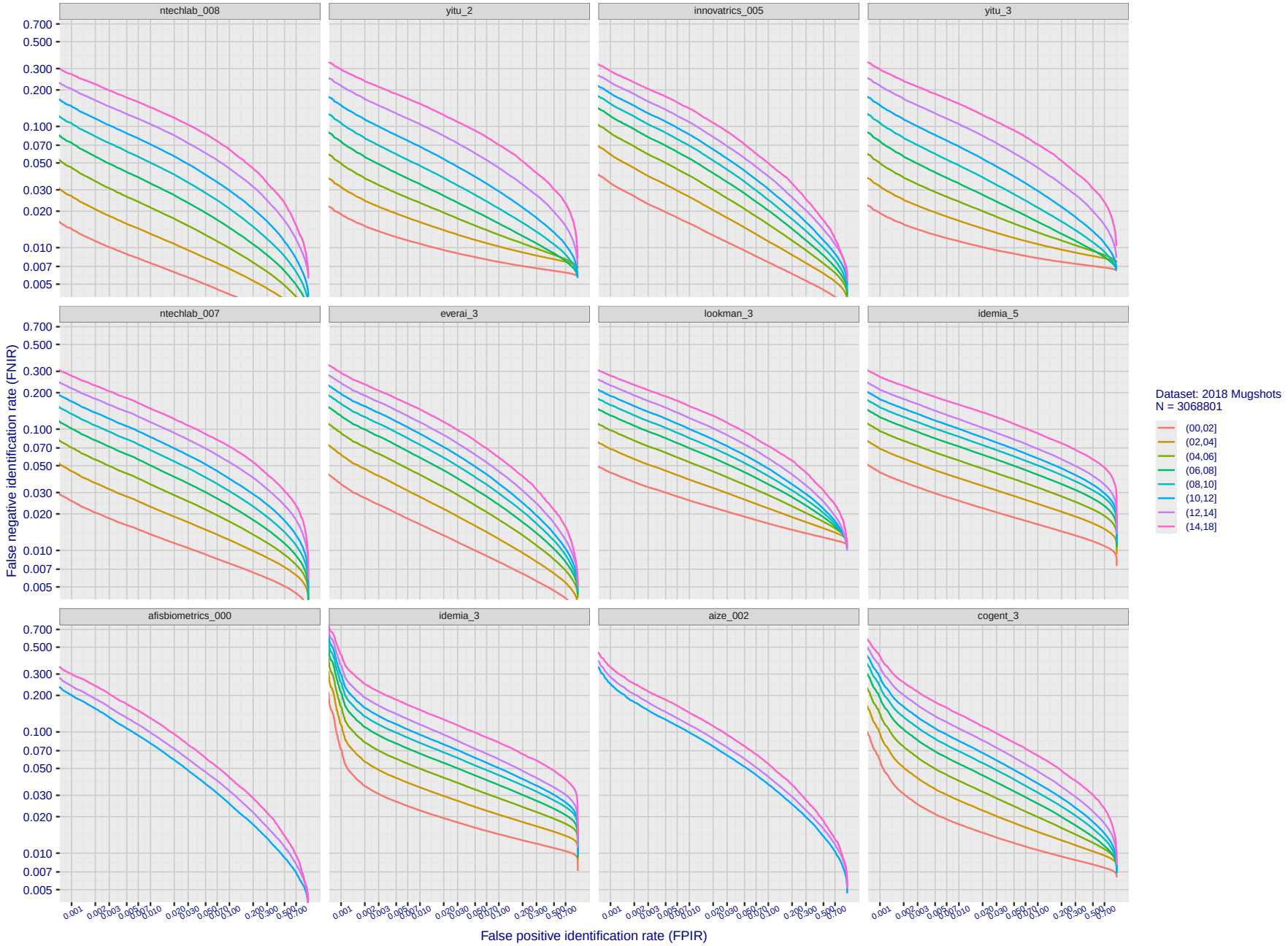

Figure 246: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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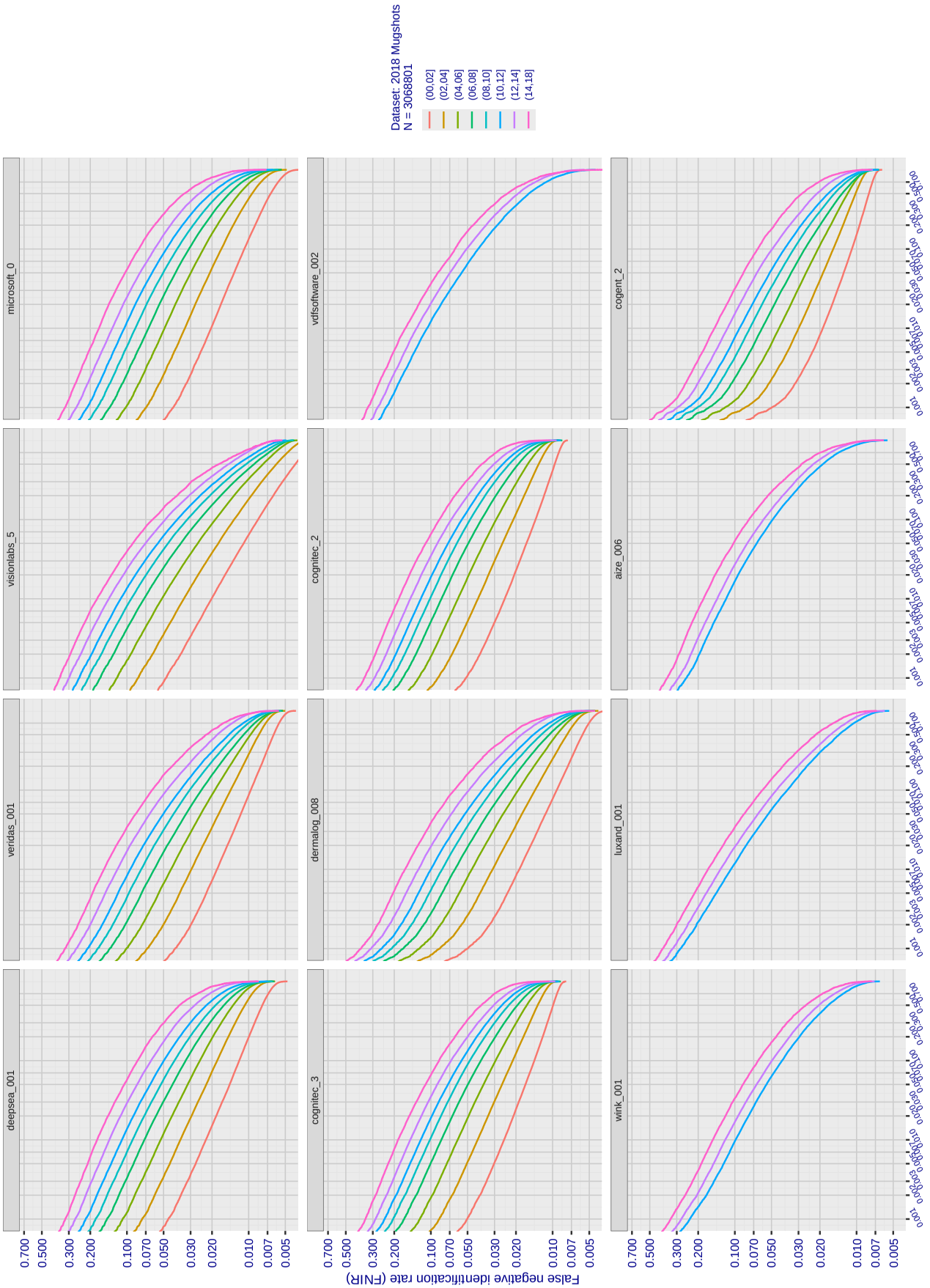

Figure 247: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed.** *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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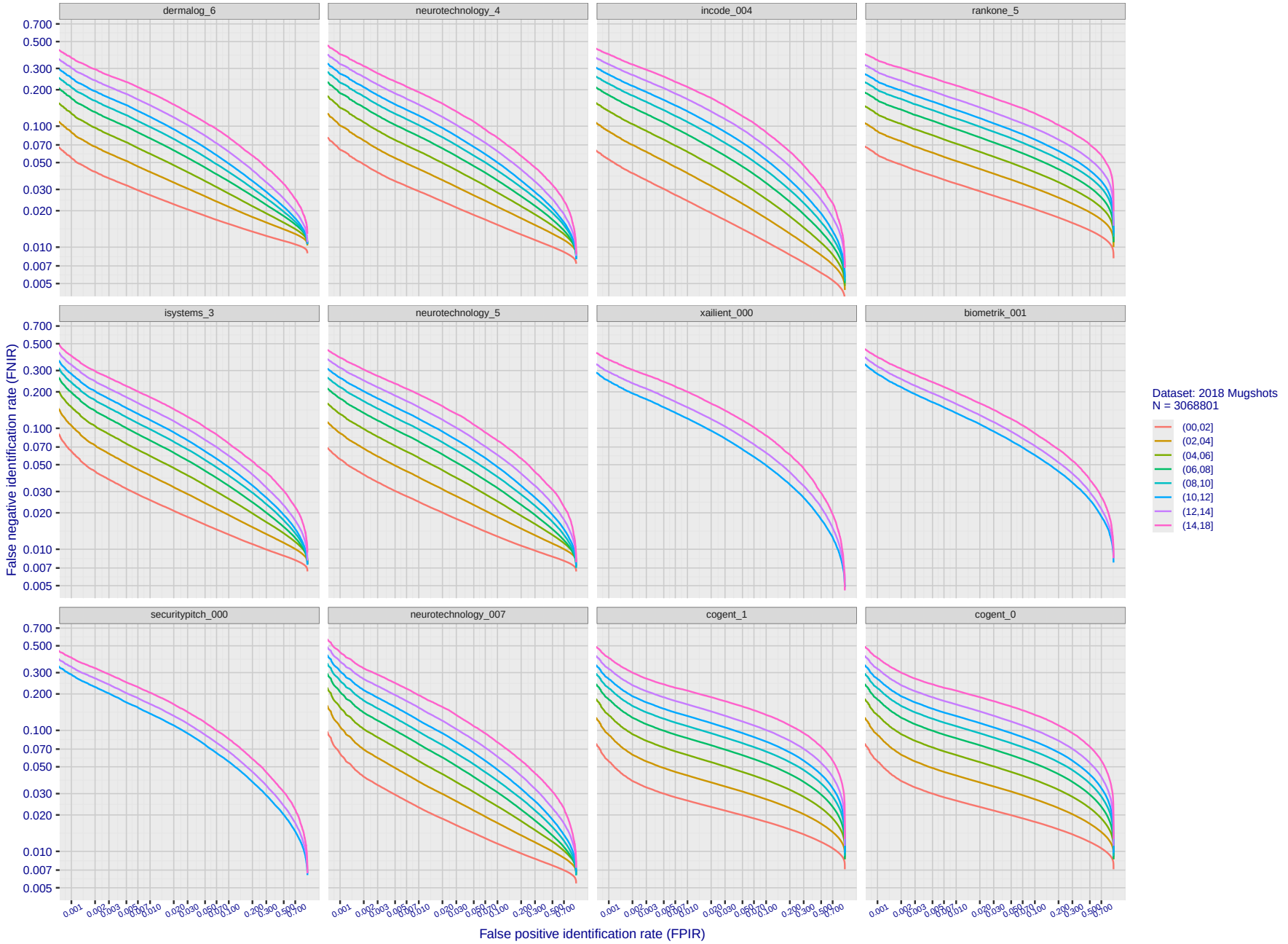

Figure 248: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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T > 0 → Identification

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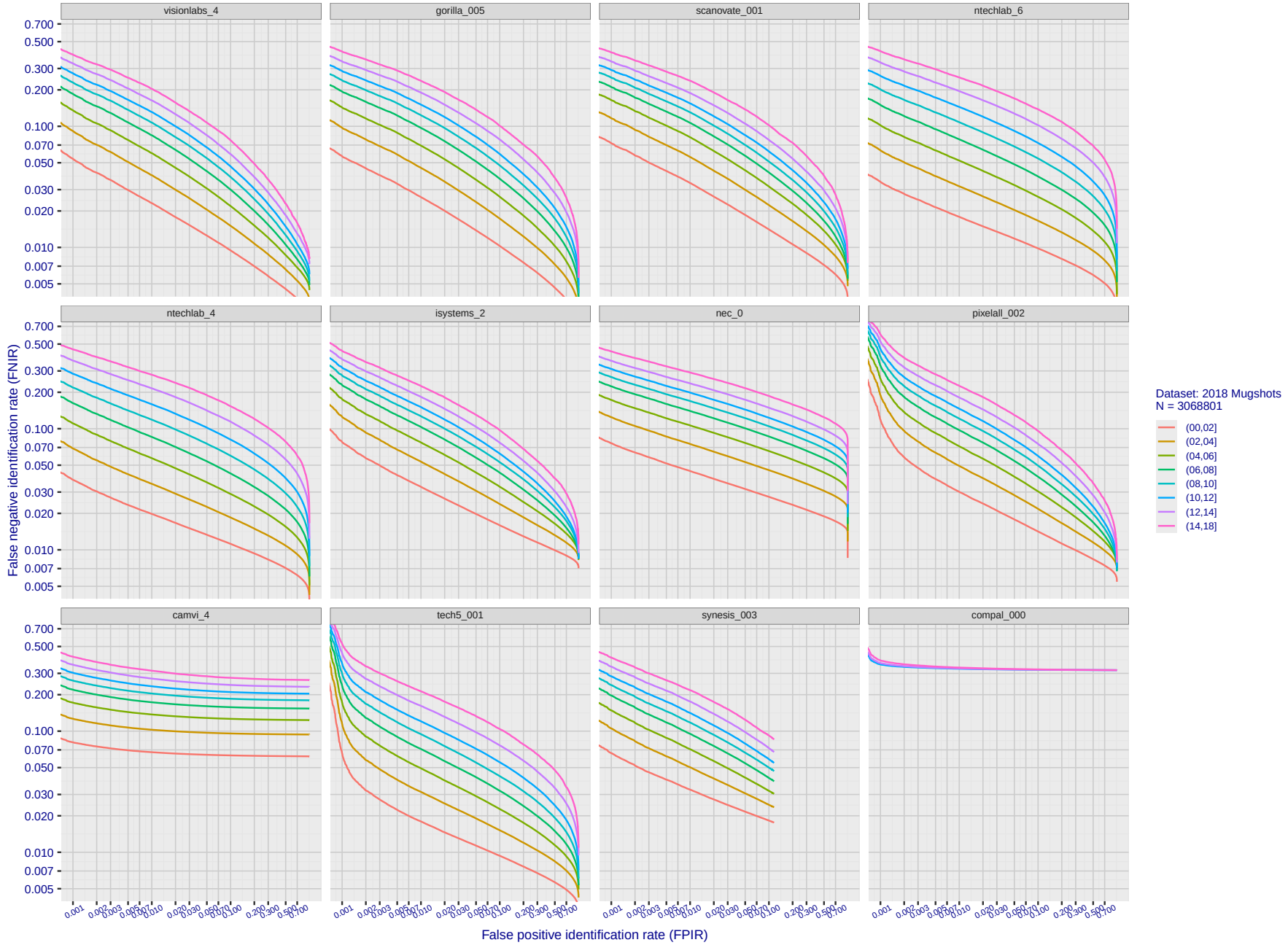

Figure 249: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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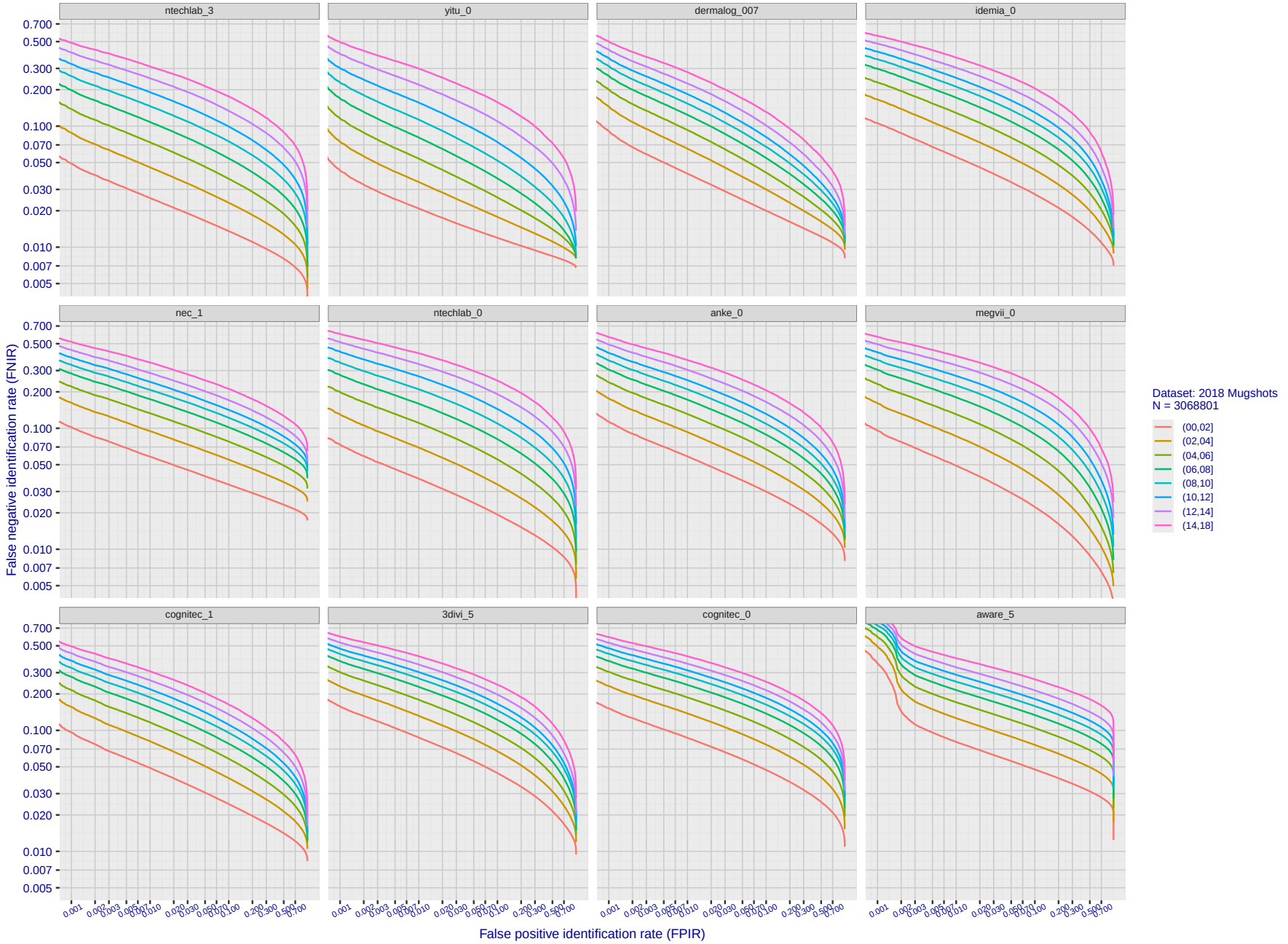

Figure 250: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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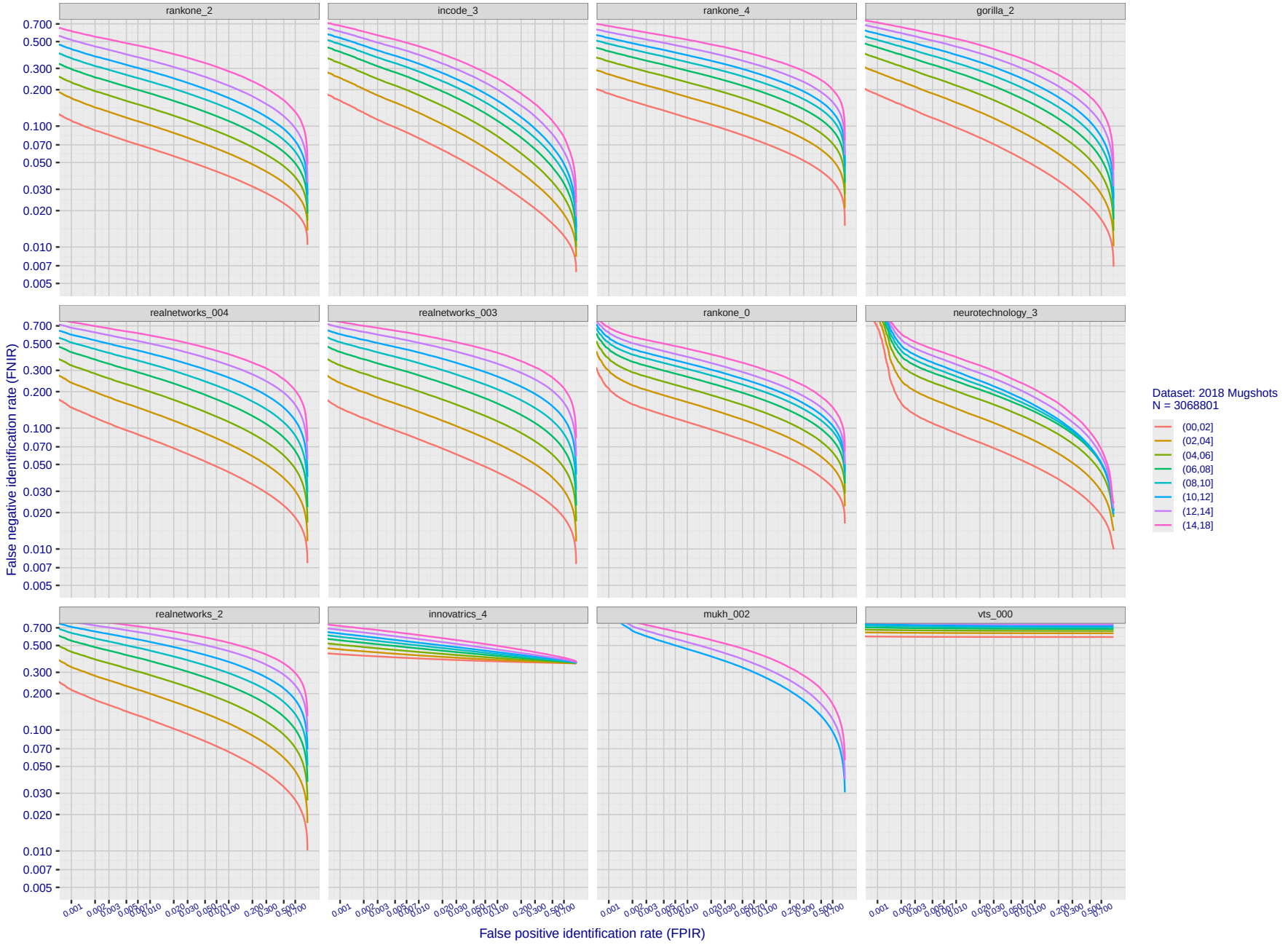

Figure 251: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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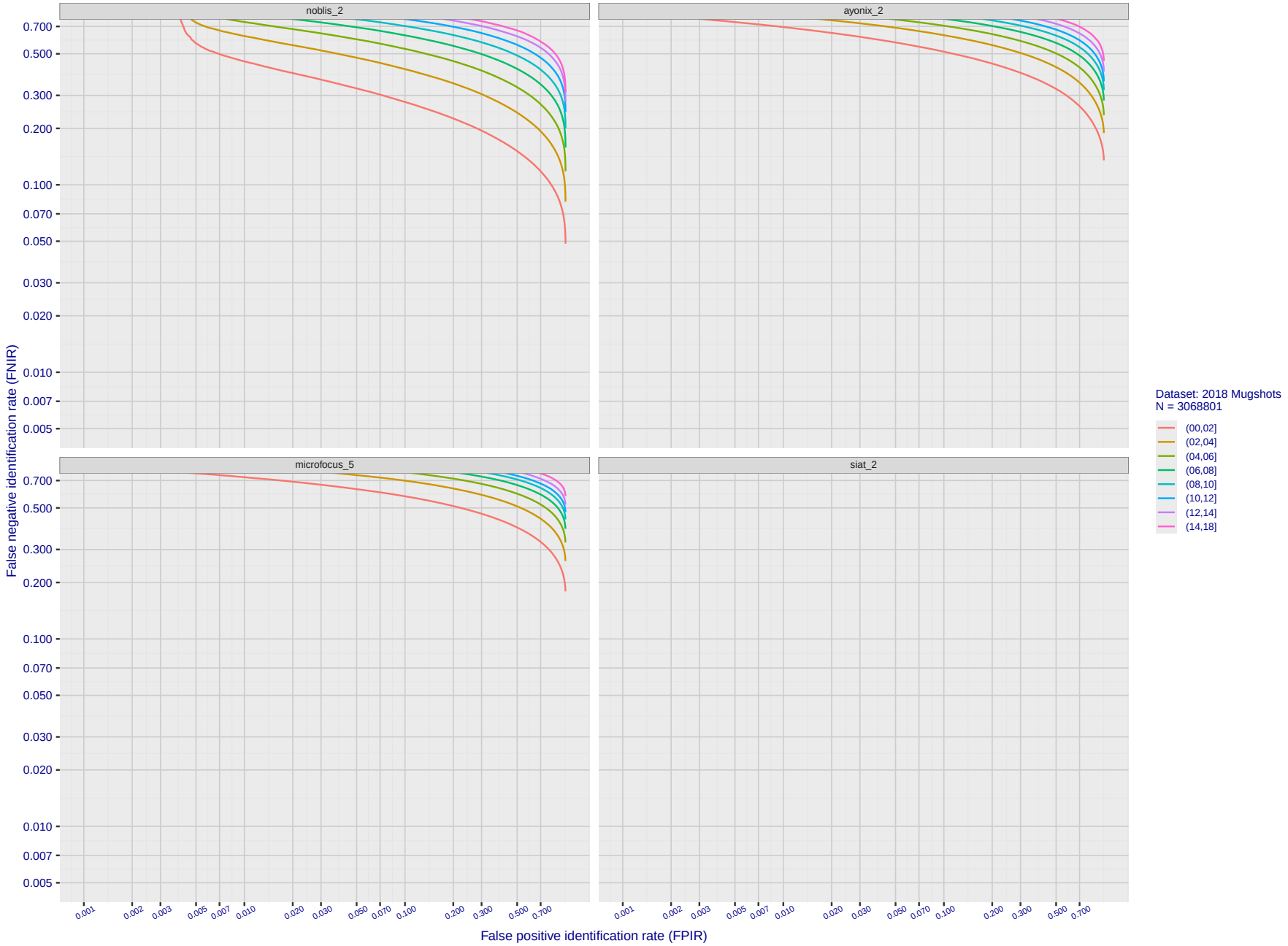

Figure 252: **[FRVT-2018 Mugshot Ageing Dataset] Identification miss rates vs. FPIR by time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Miss rates are computed over all searches noted in row 17 of Table 1 and binned by number of years between search and initial enrollment. FPIR is computed from the same FRVT 2018 non-mates noted in row 3 of Table 1 with N = 3 000 000.*

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2026/10/06 08:21:15	FNIR(N, R, T) = FPIR(N, T) =	False neg. identification rate False pos. identification rate	N = Num. enrolled subjects R = Num. candidates examined	T = Threshold	$T = 0 \rightarrow$ Investigation $T > 0 \rightarrow$ Identification

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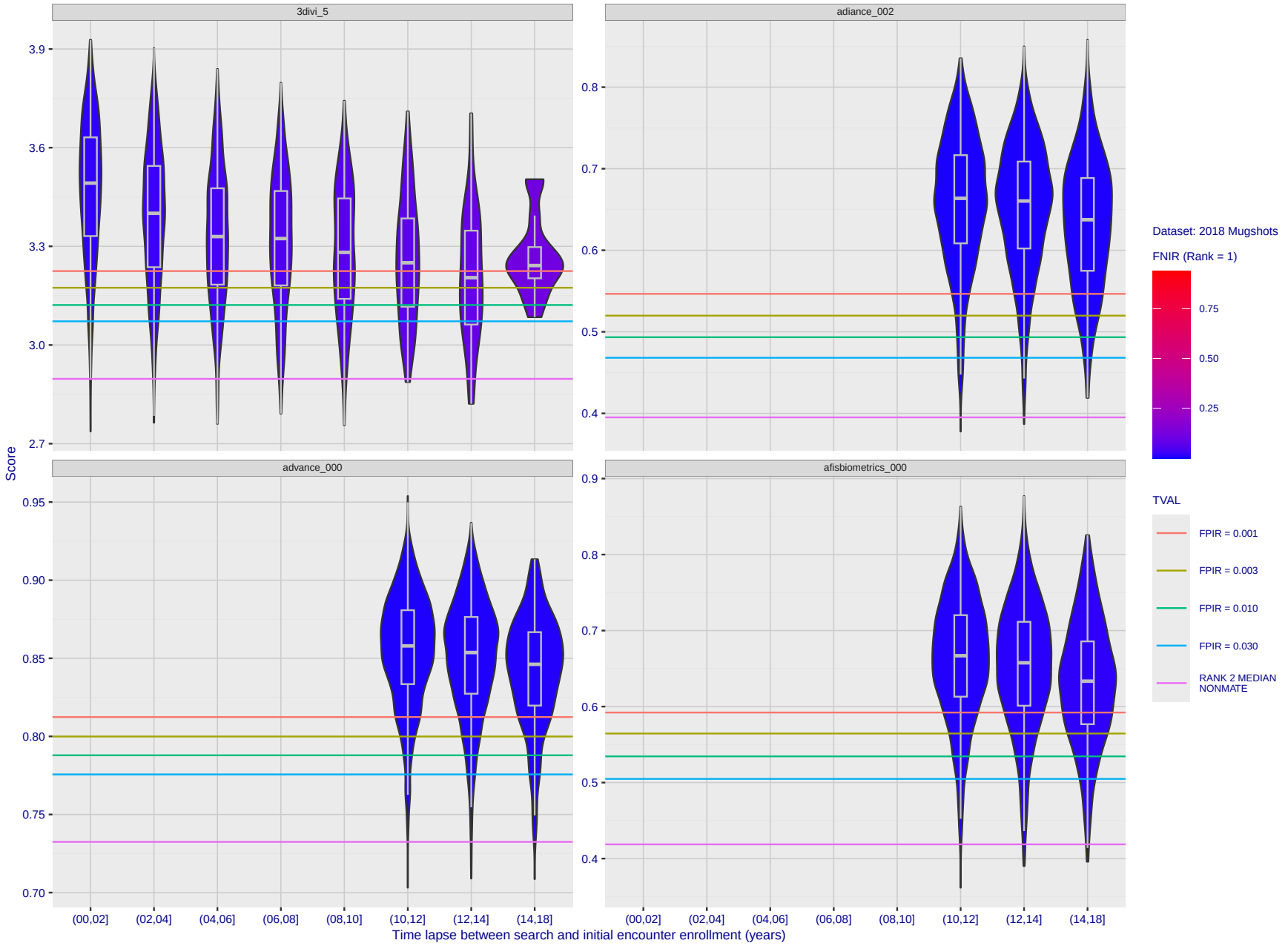

Figure 253: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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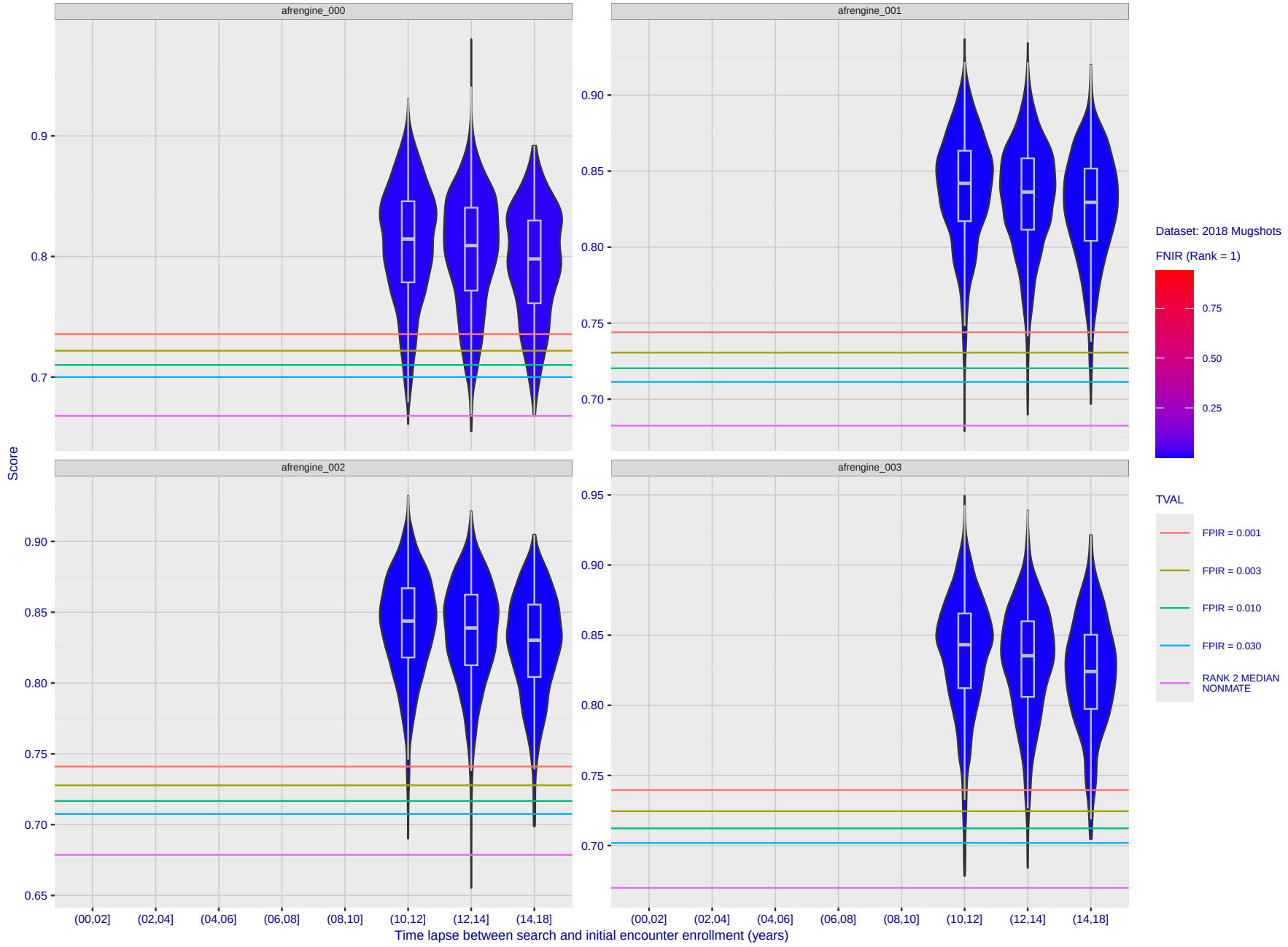

Figure 254: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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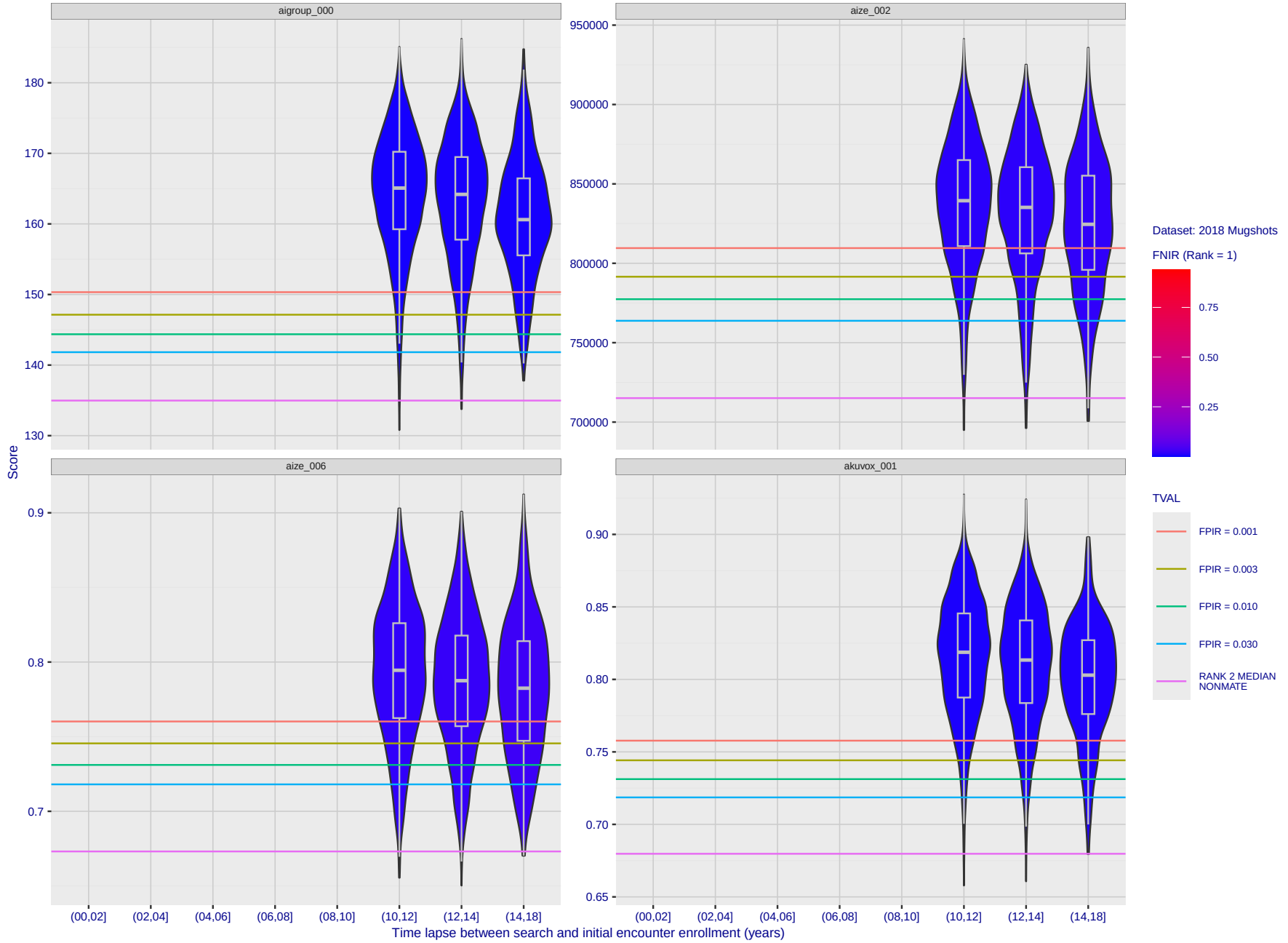

Figure 255: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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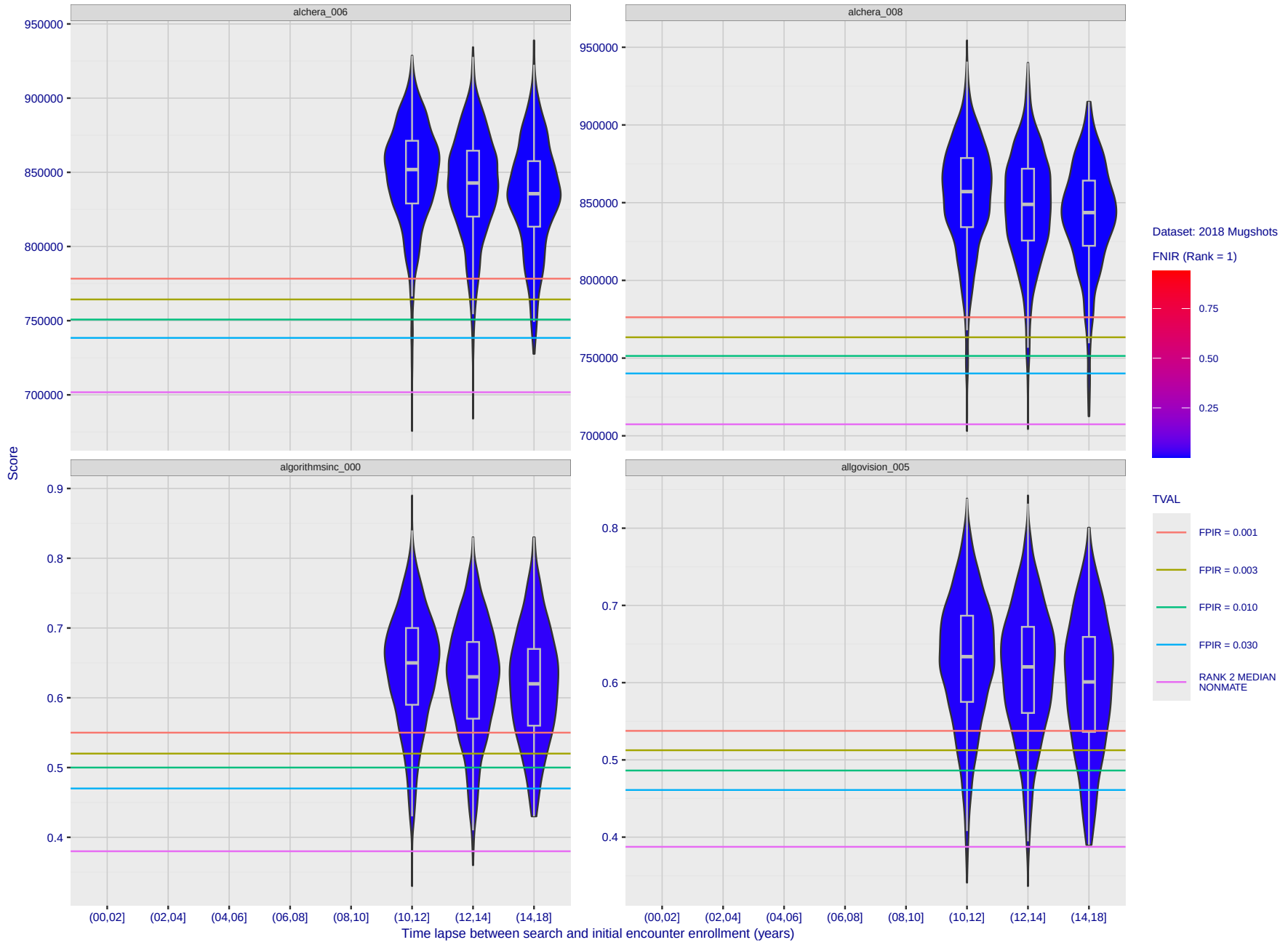

Figure 256: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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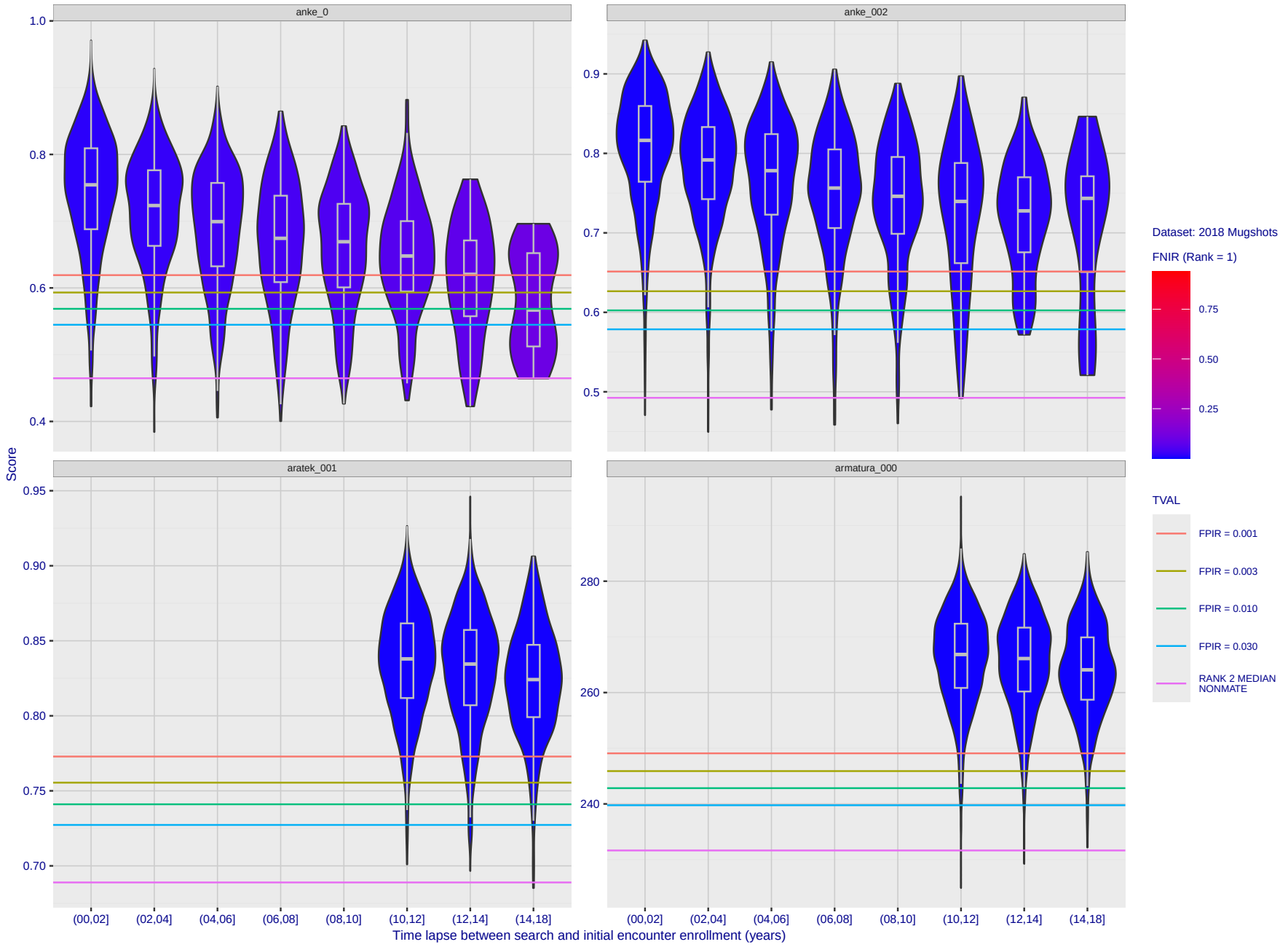

Figure 257: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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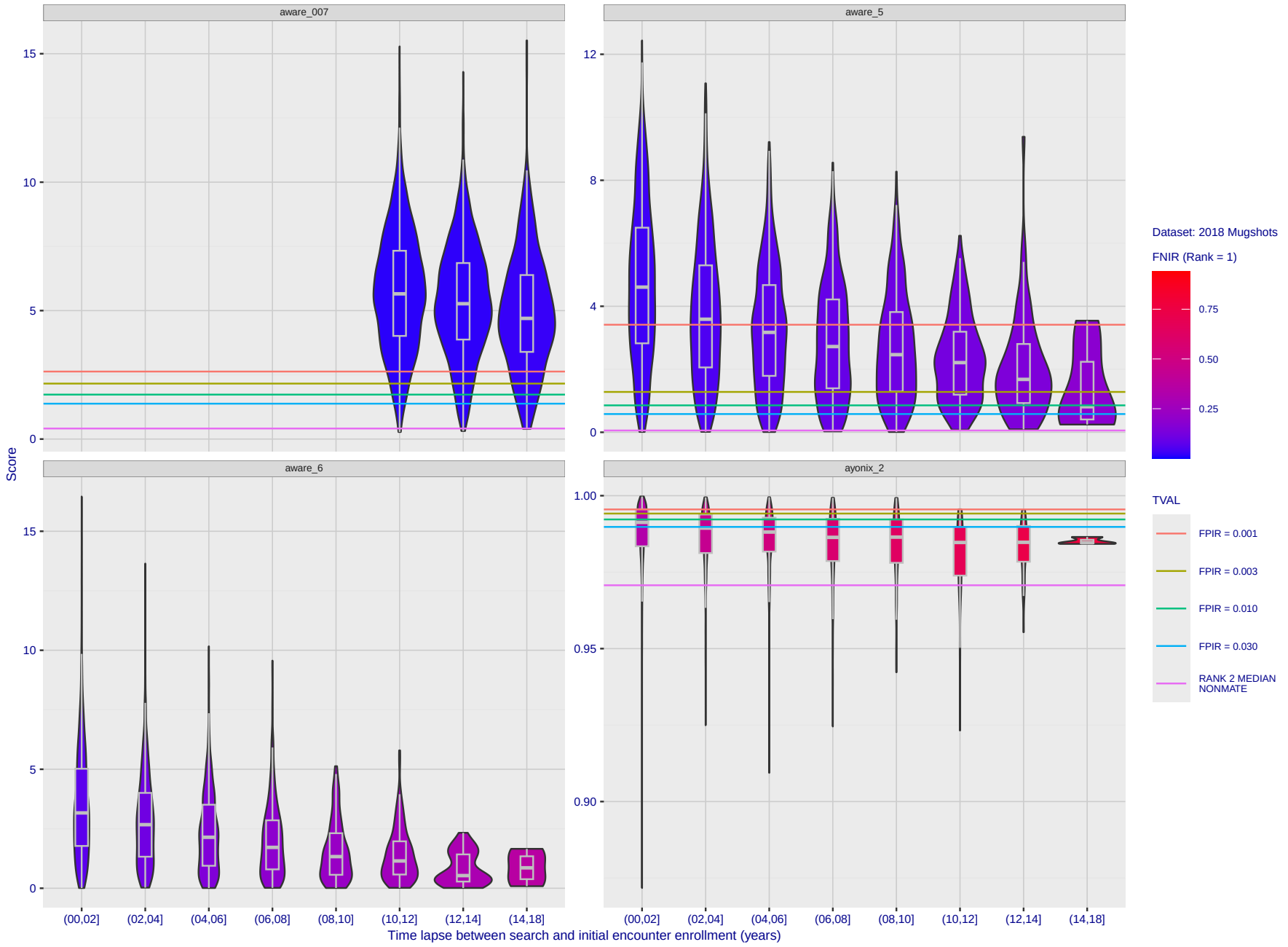

Figure 258: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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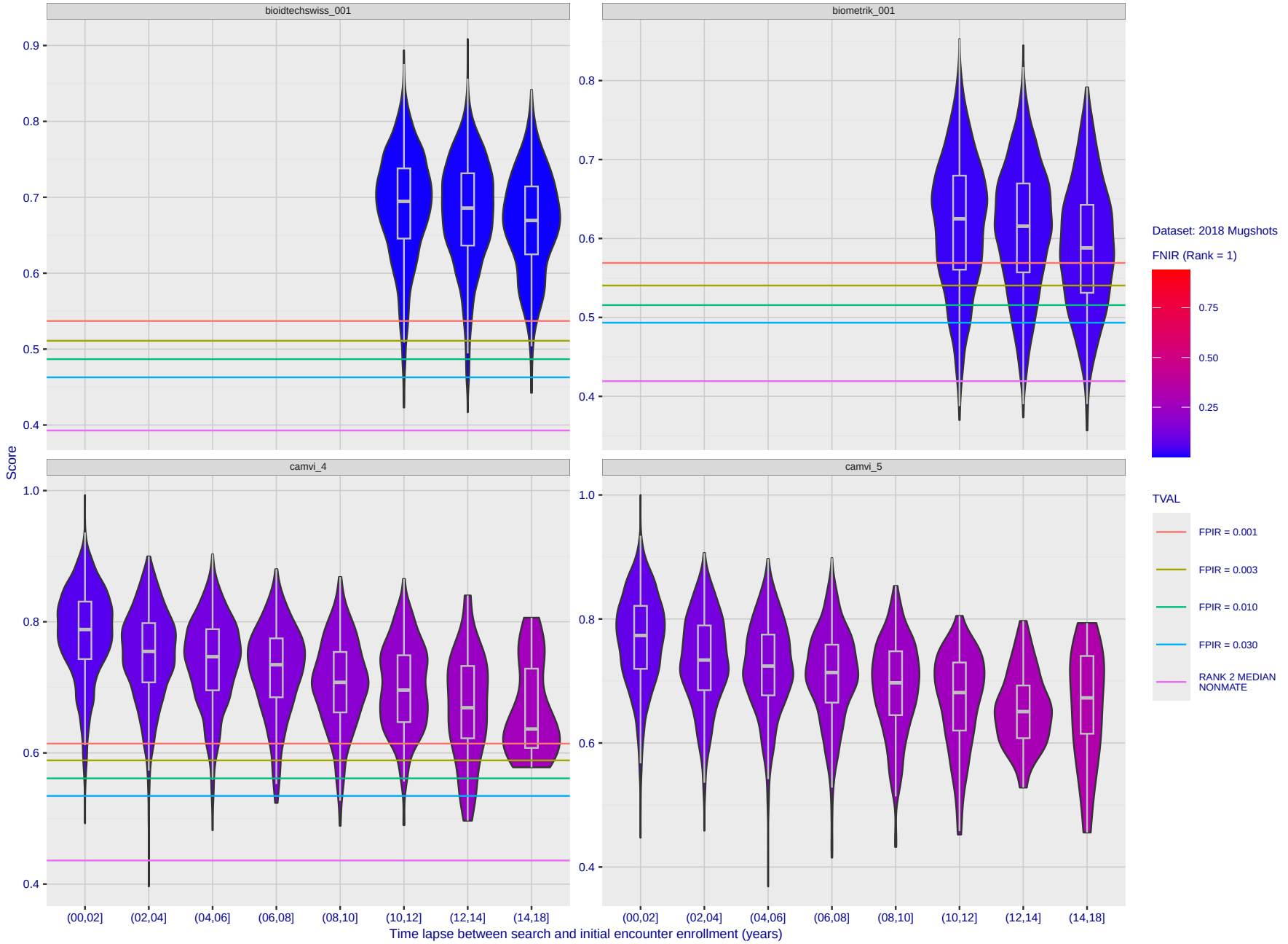

Figure 259: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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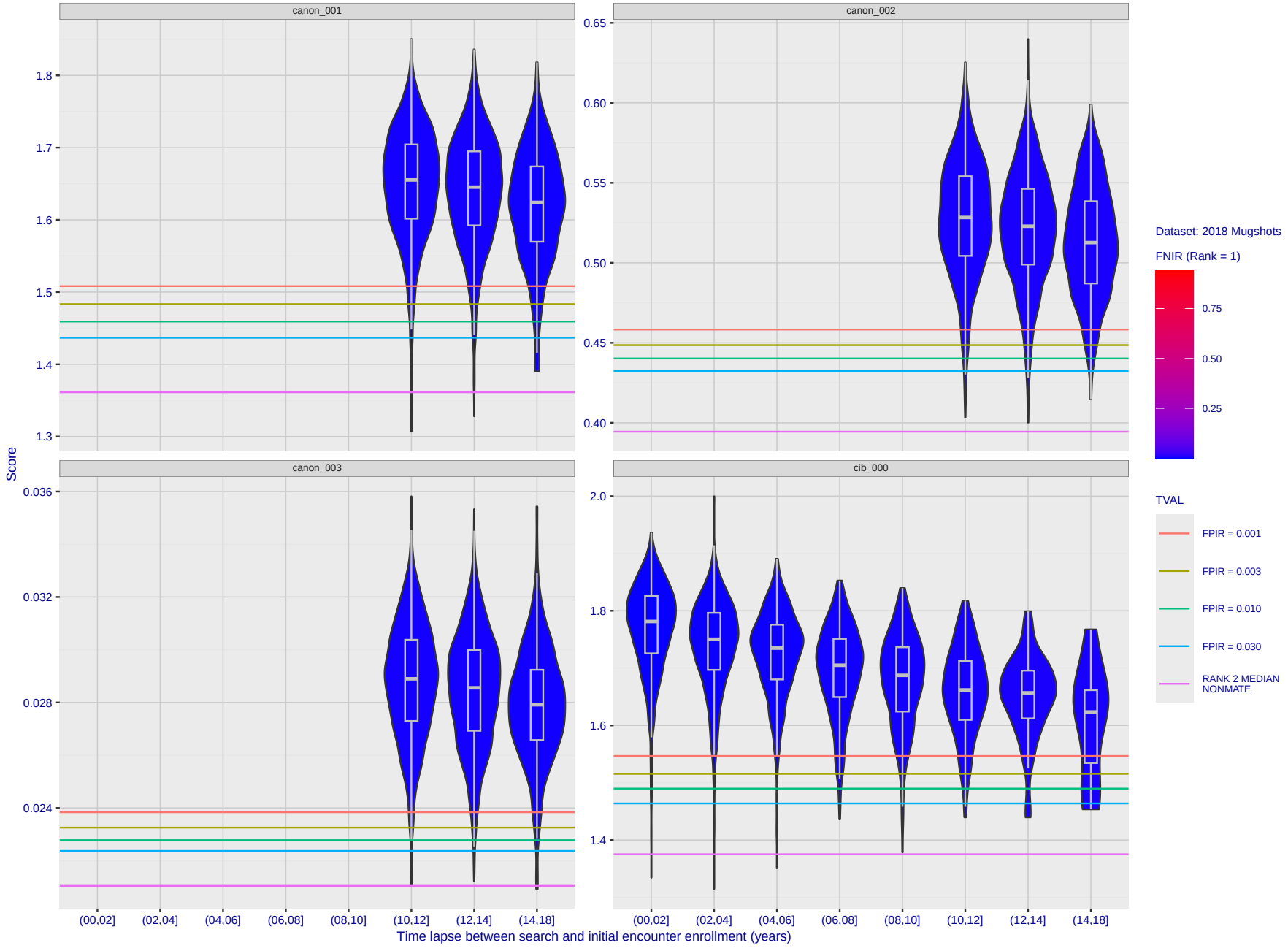

Figure 260: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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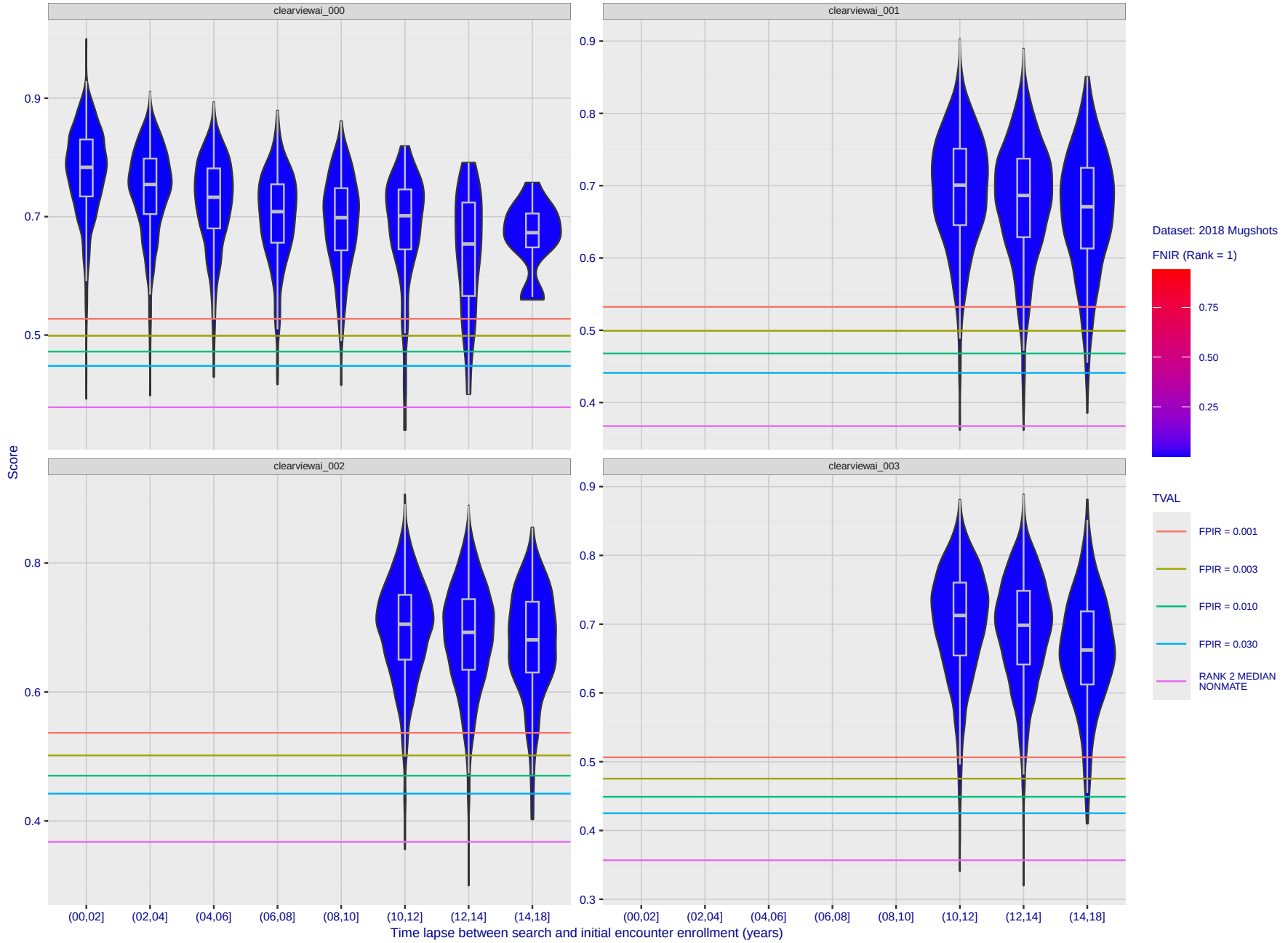

Figure 261: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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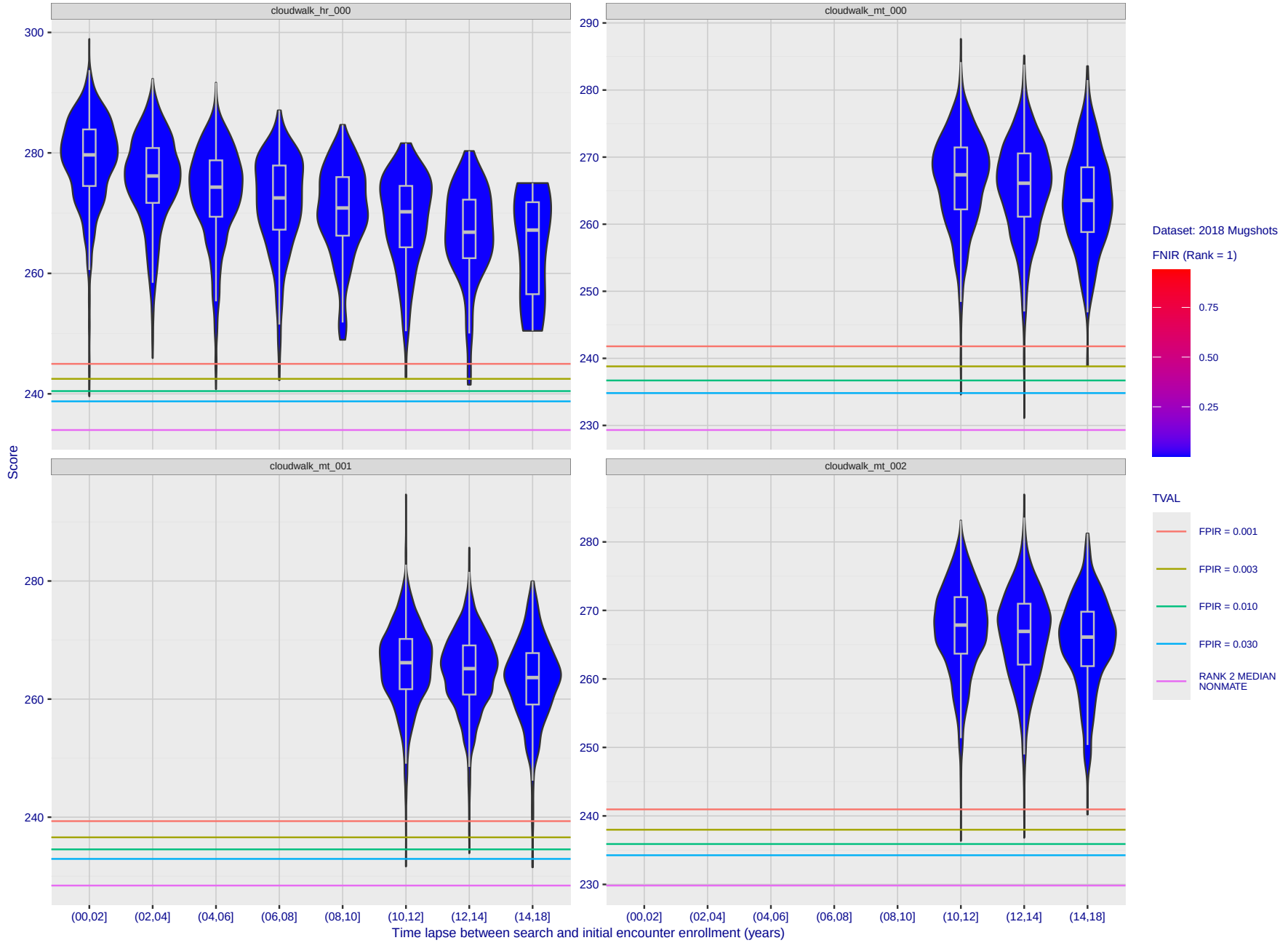

Figure 262: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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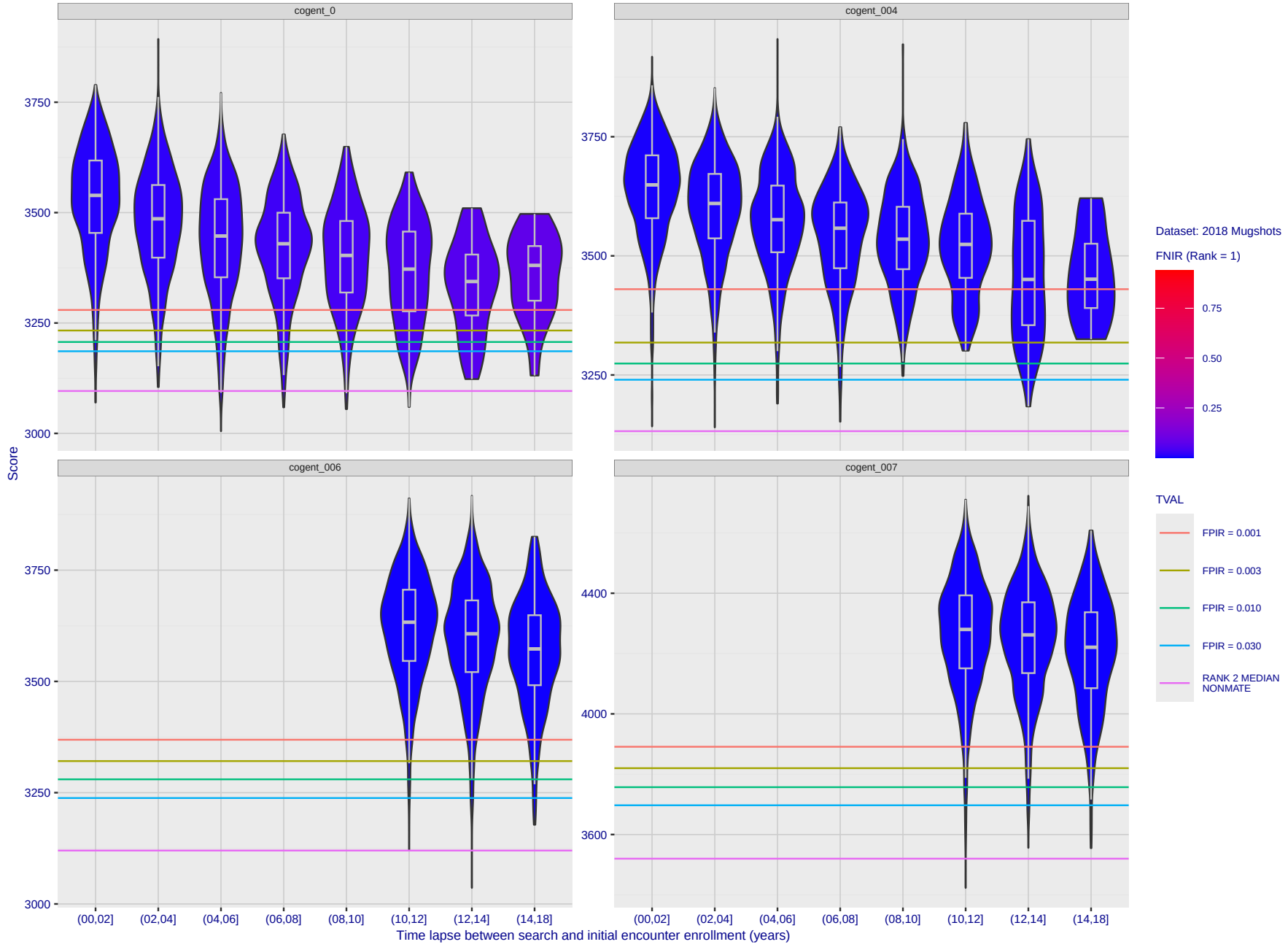

Figure 263: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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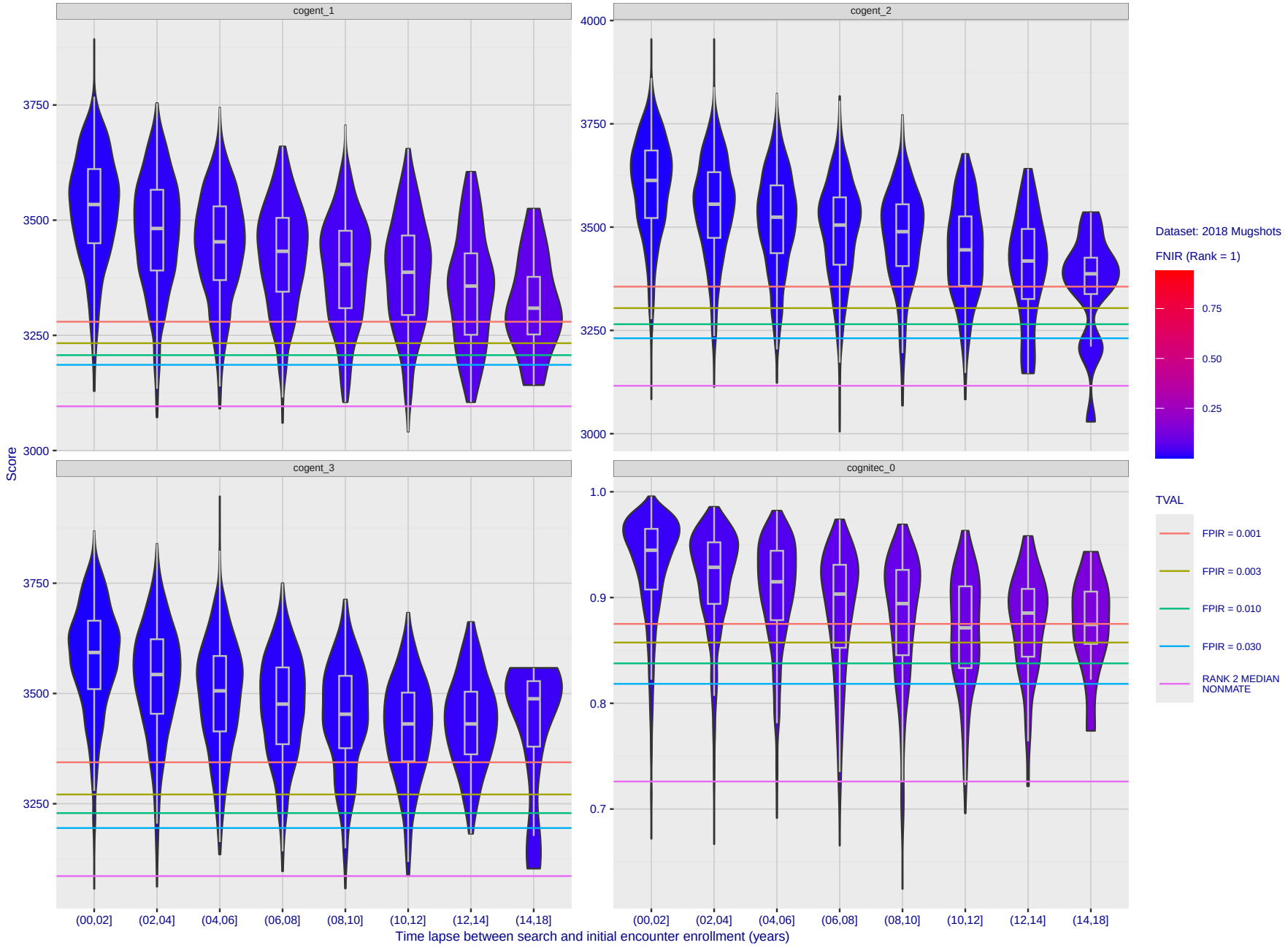

Figure 264: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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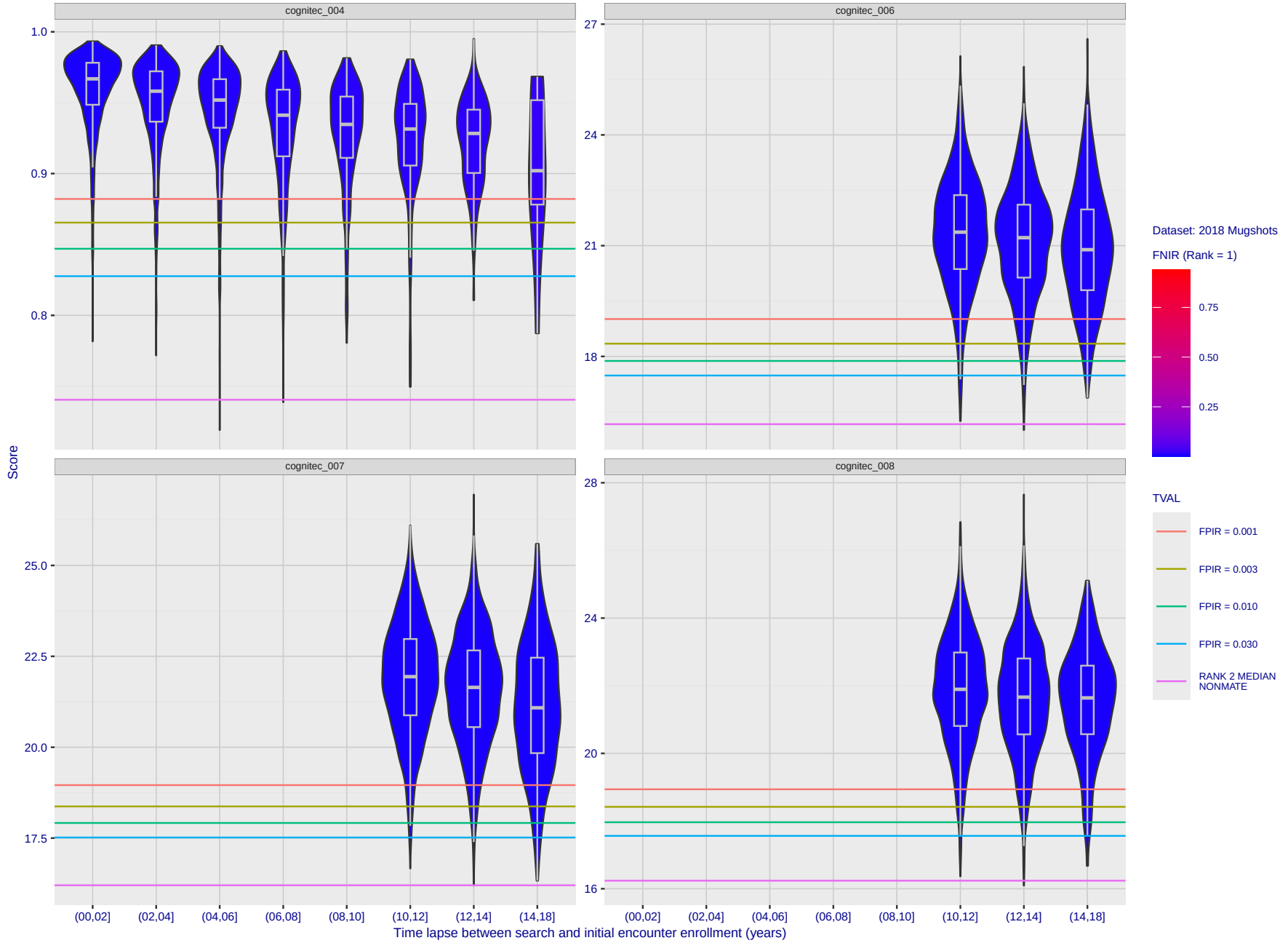

Figure 265: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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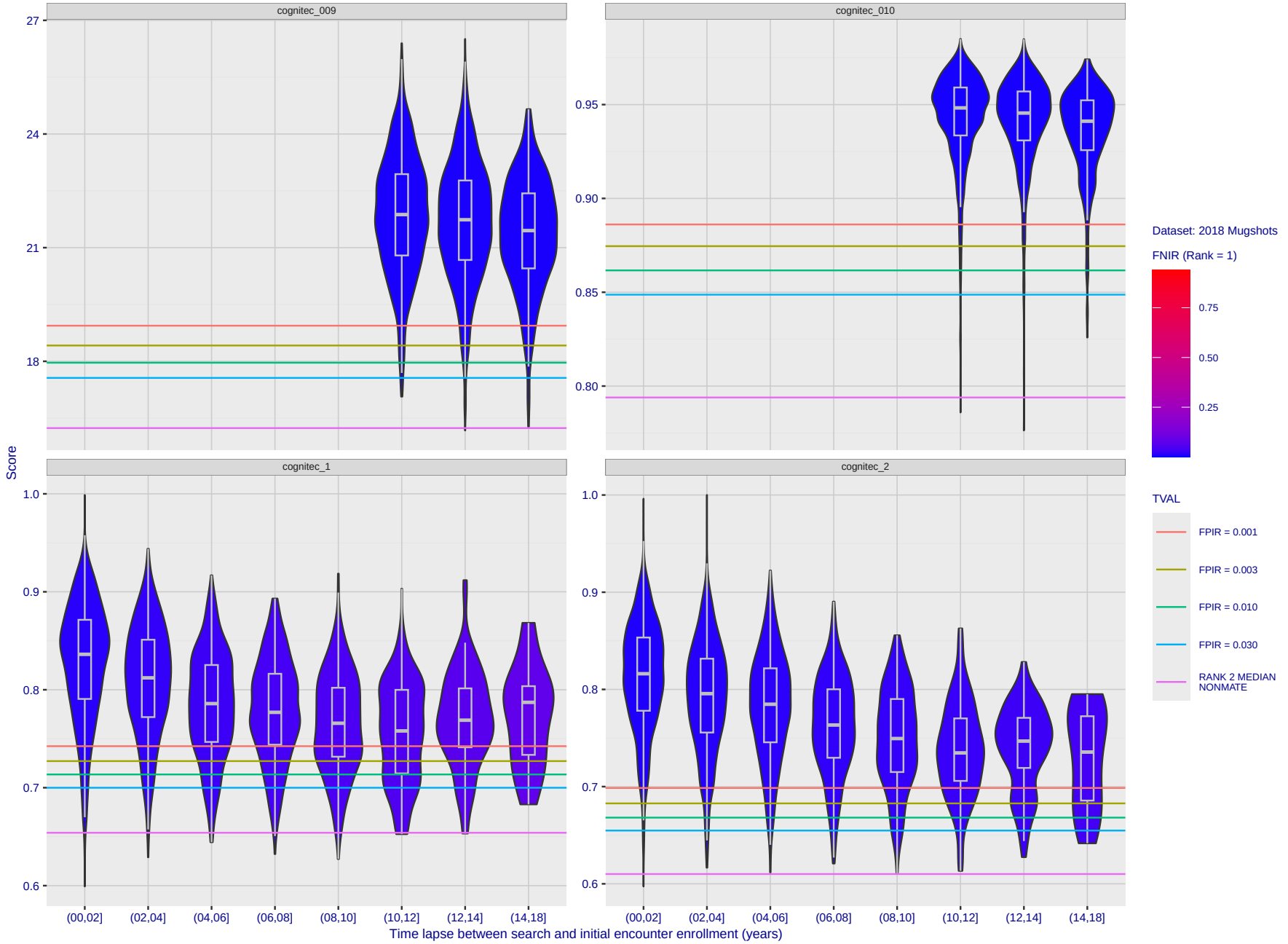

Figure 266: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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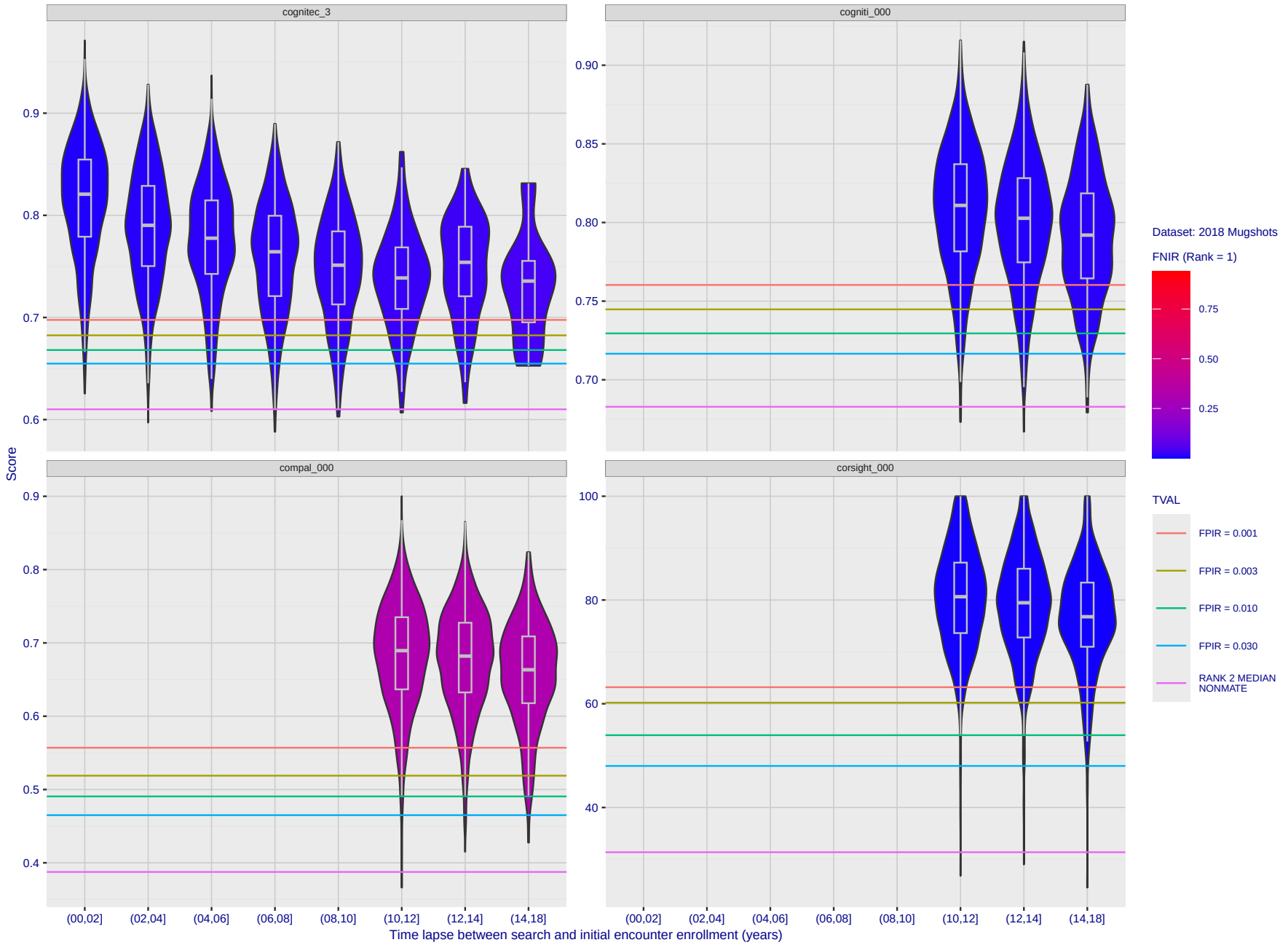

Figure 267: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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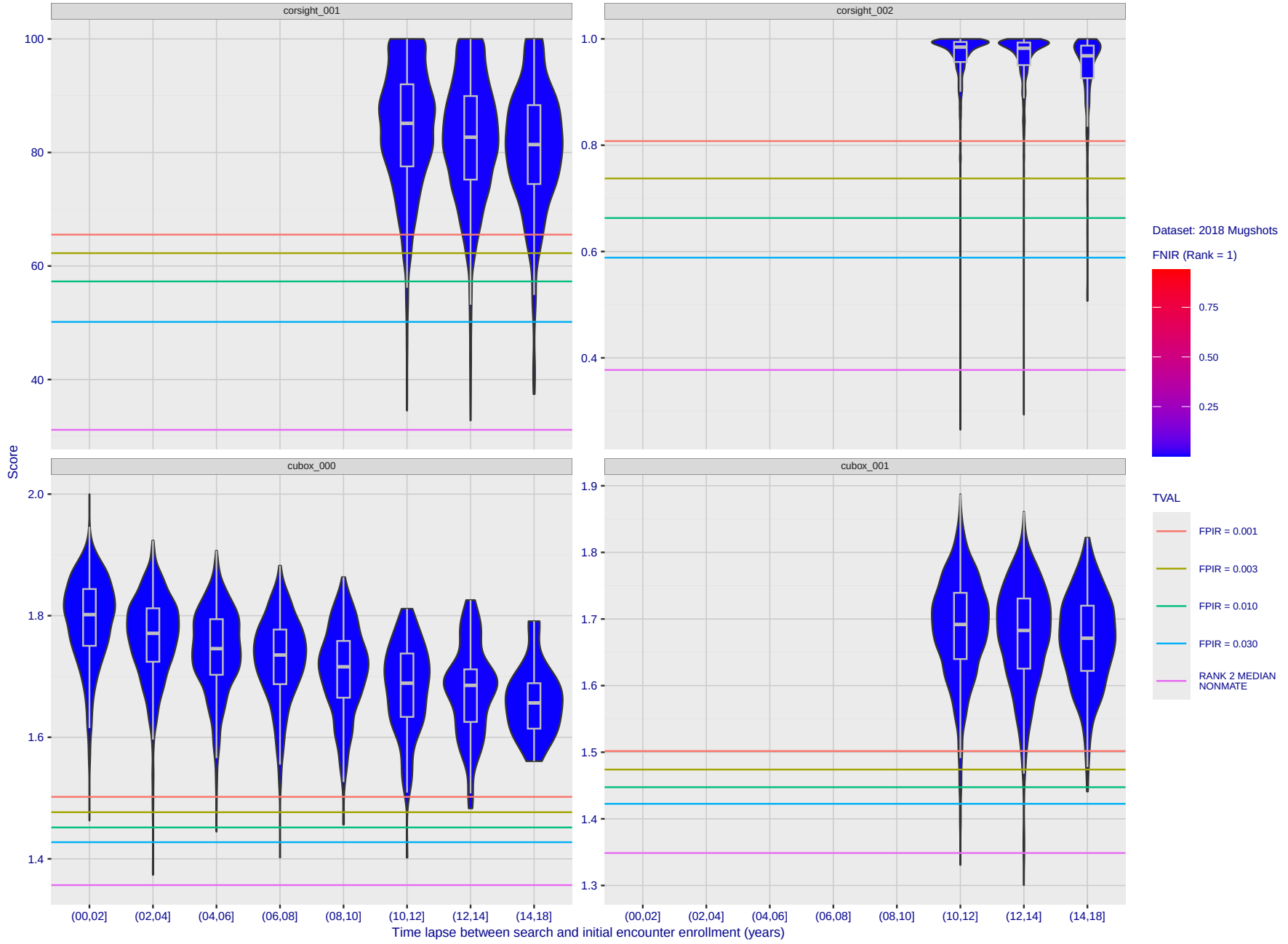

Figure 268: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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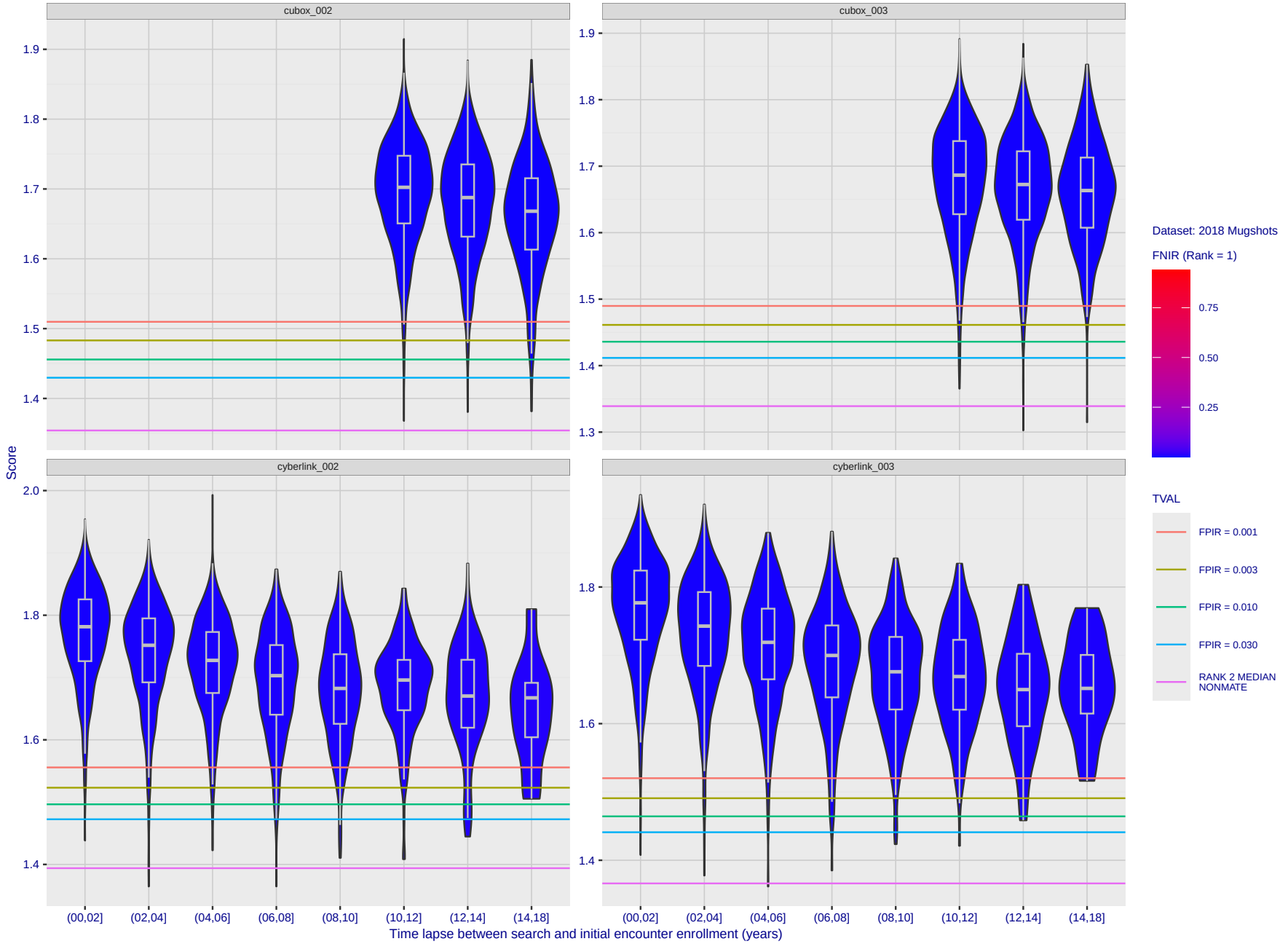

Figure 269: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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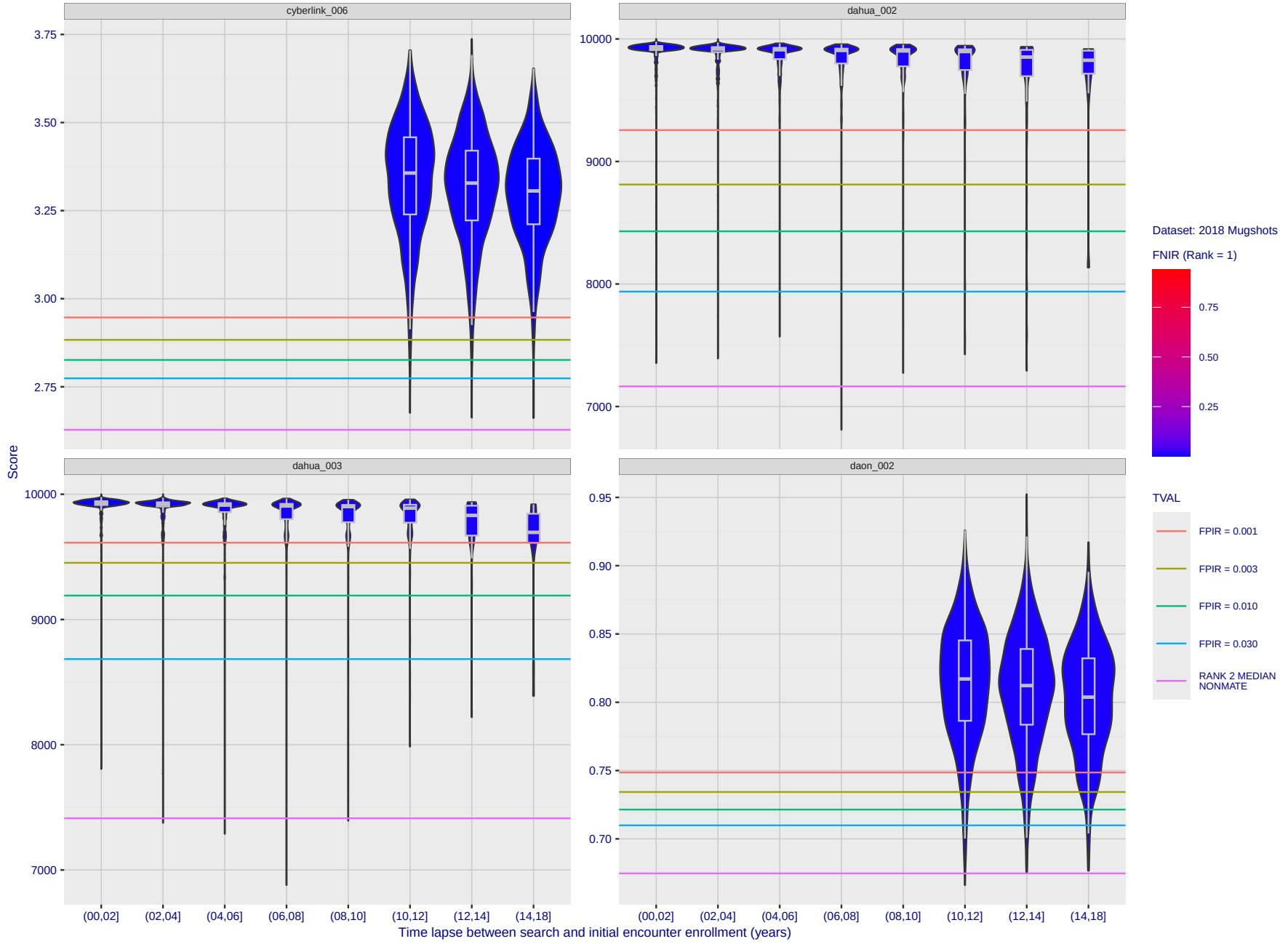

Figure 270: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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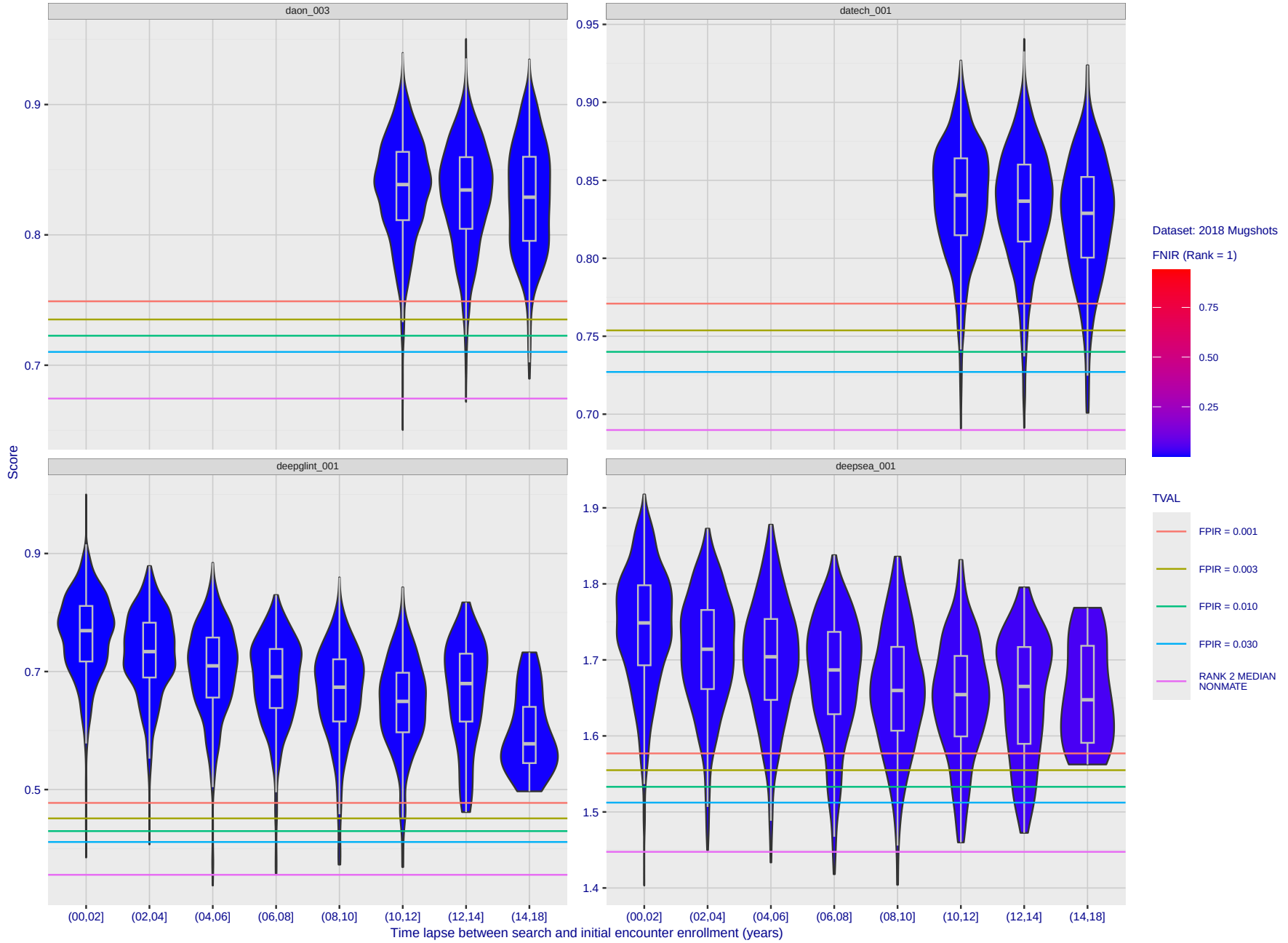

Figure 271: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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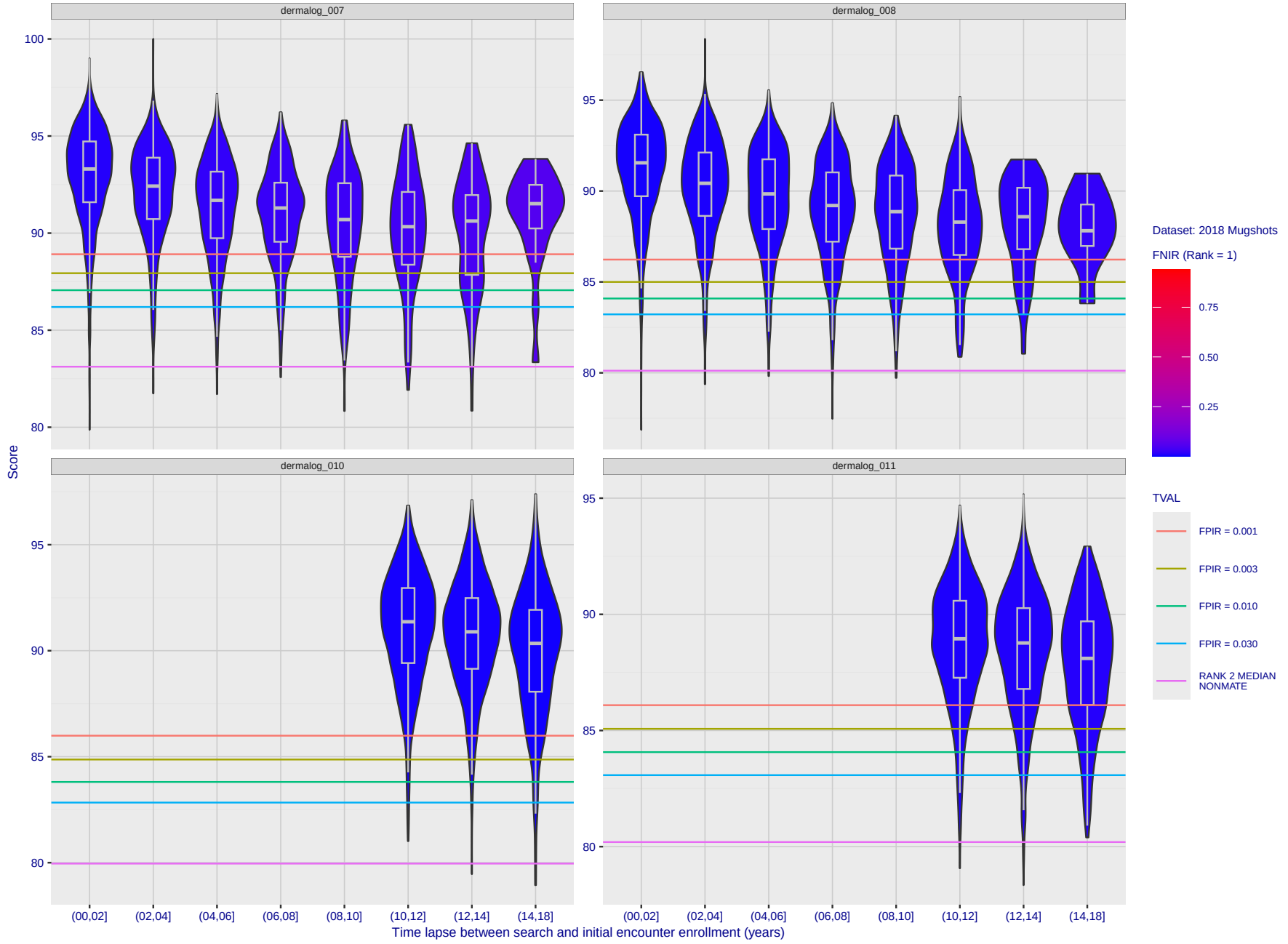

Figure 272: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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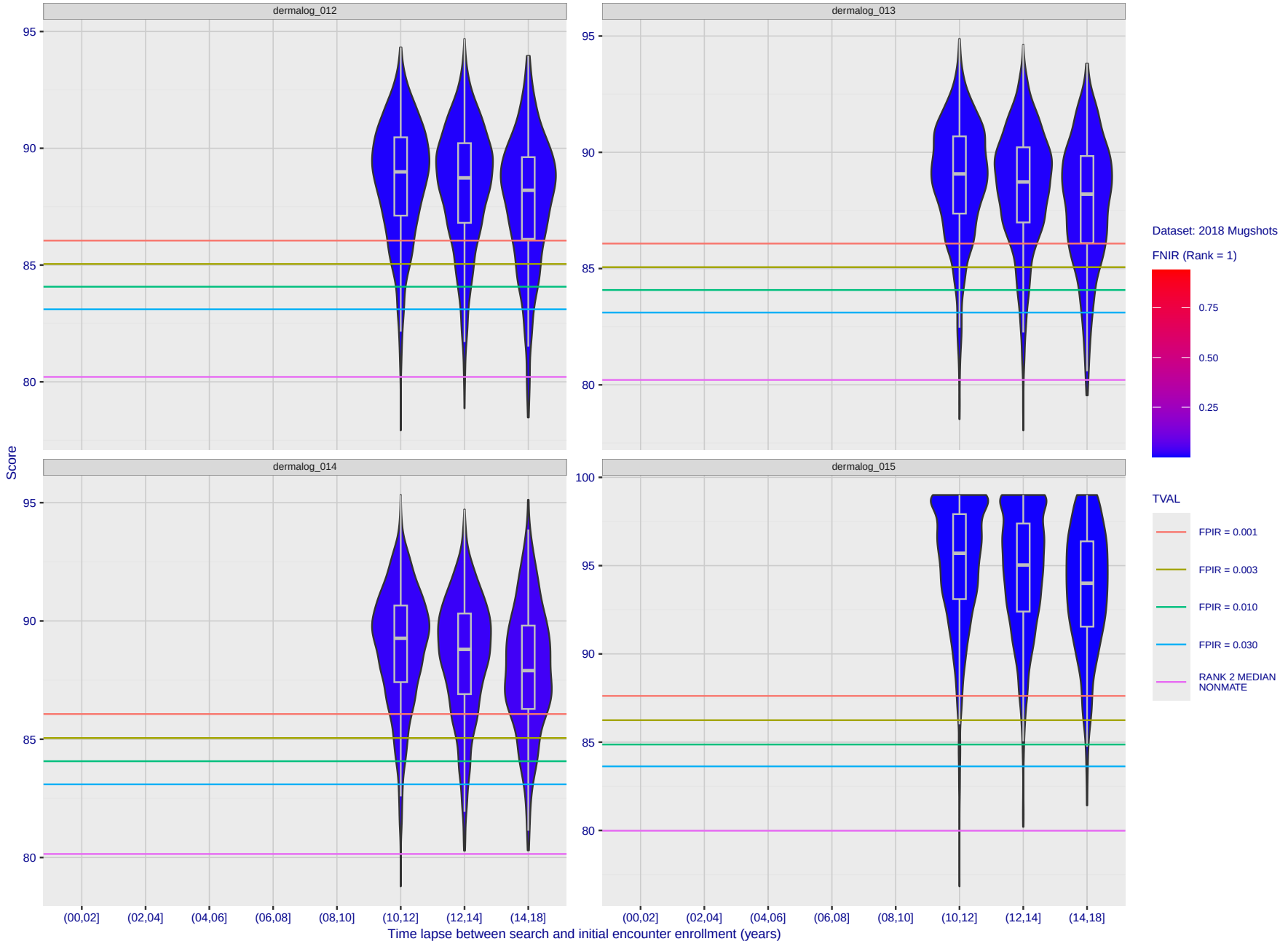

Figure 273: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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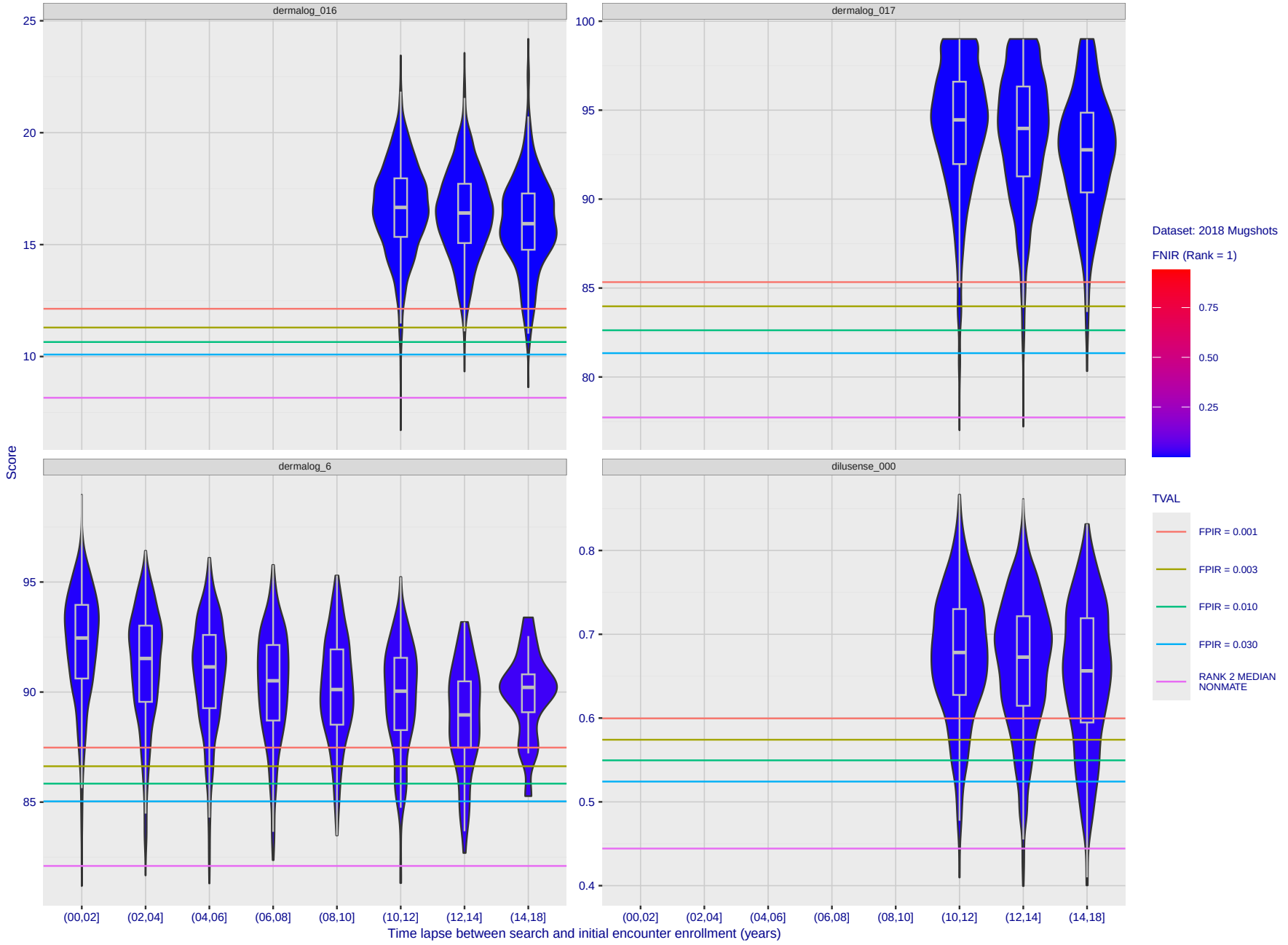

Figure 274: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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T = Threshold
T = 0 → Investigation
T > 0 → Identification

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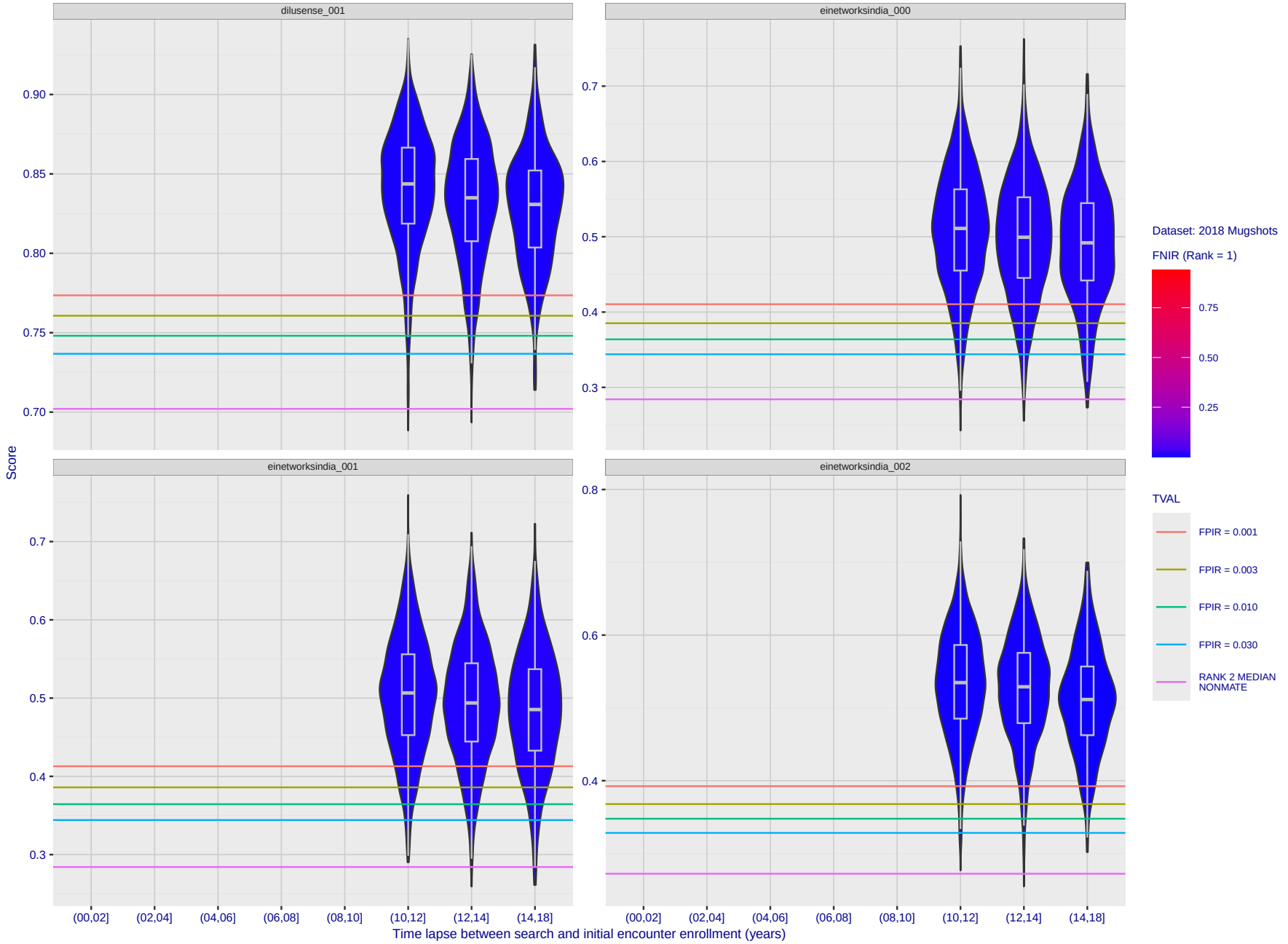

Figure 275: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
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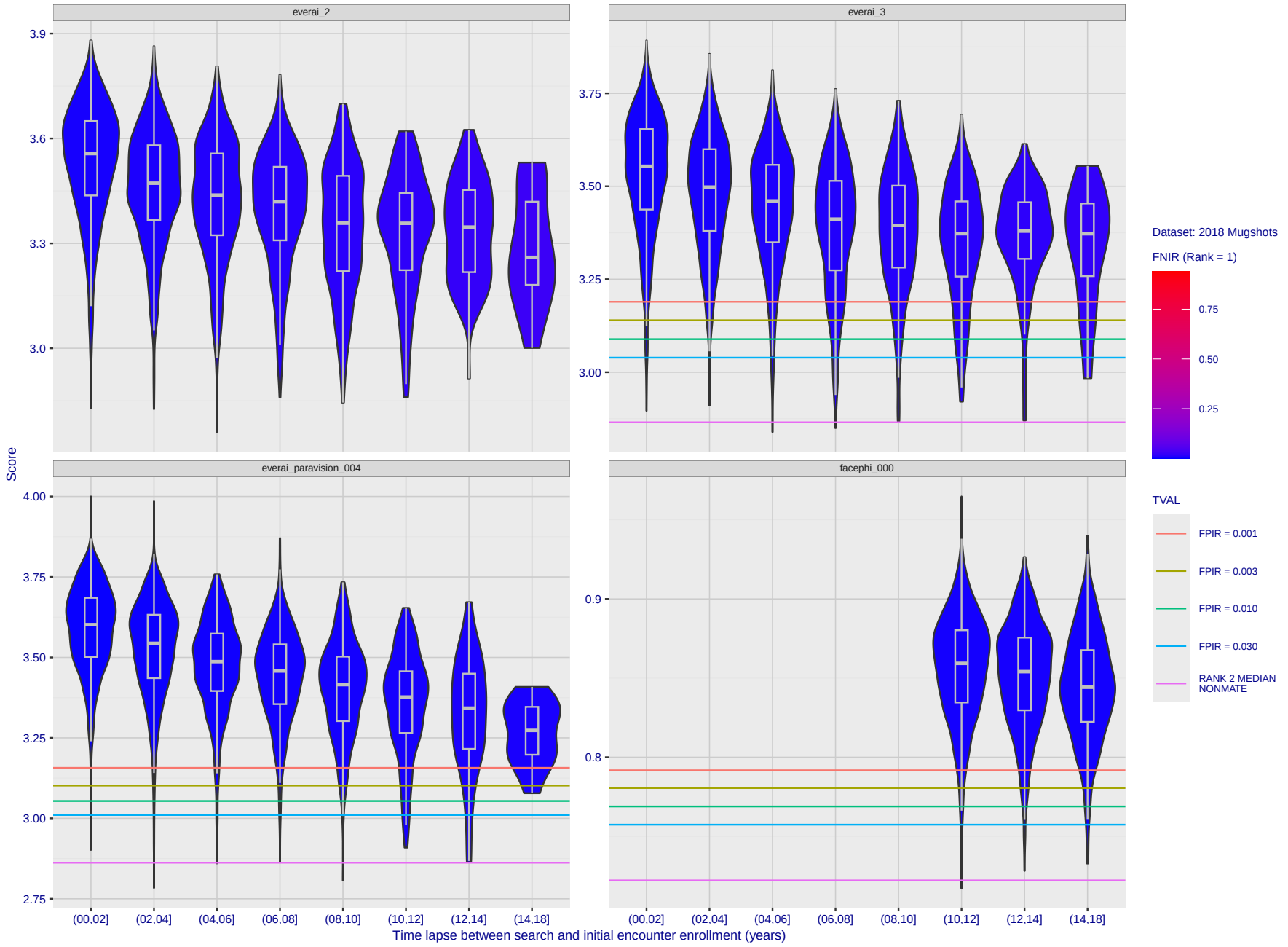

Figure 276: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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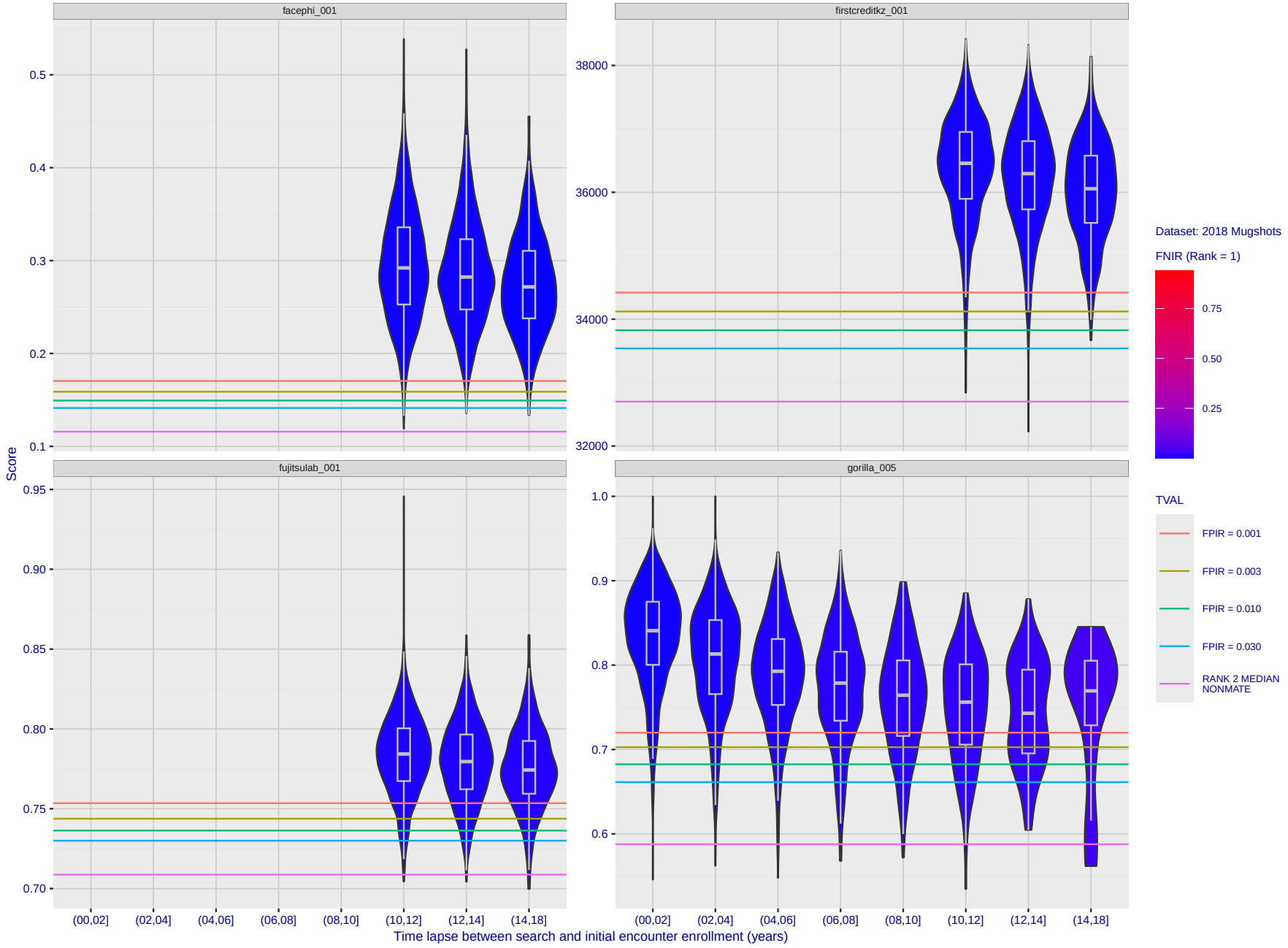

Figure 277: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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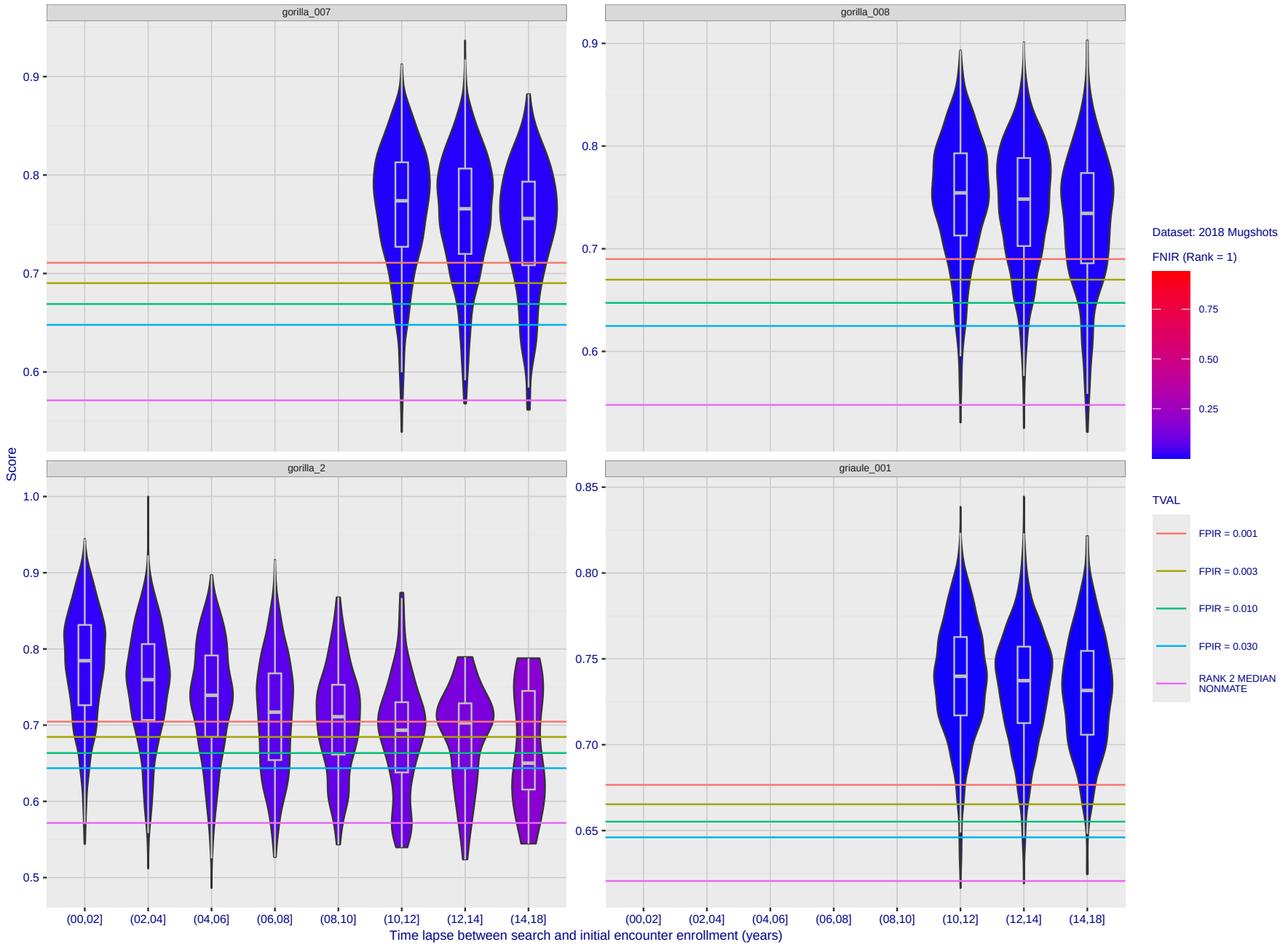

Figure 278: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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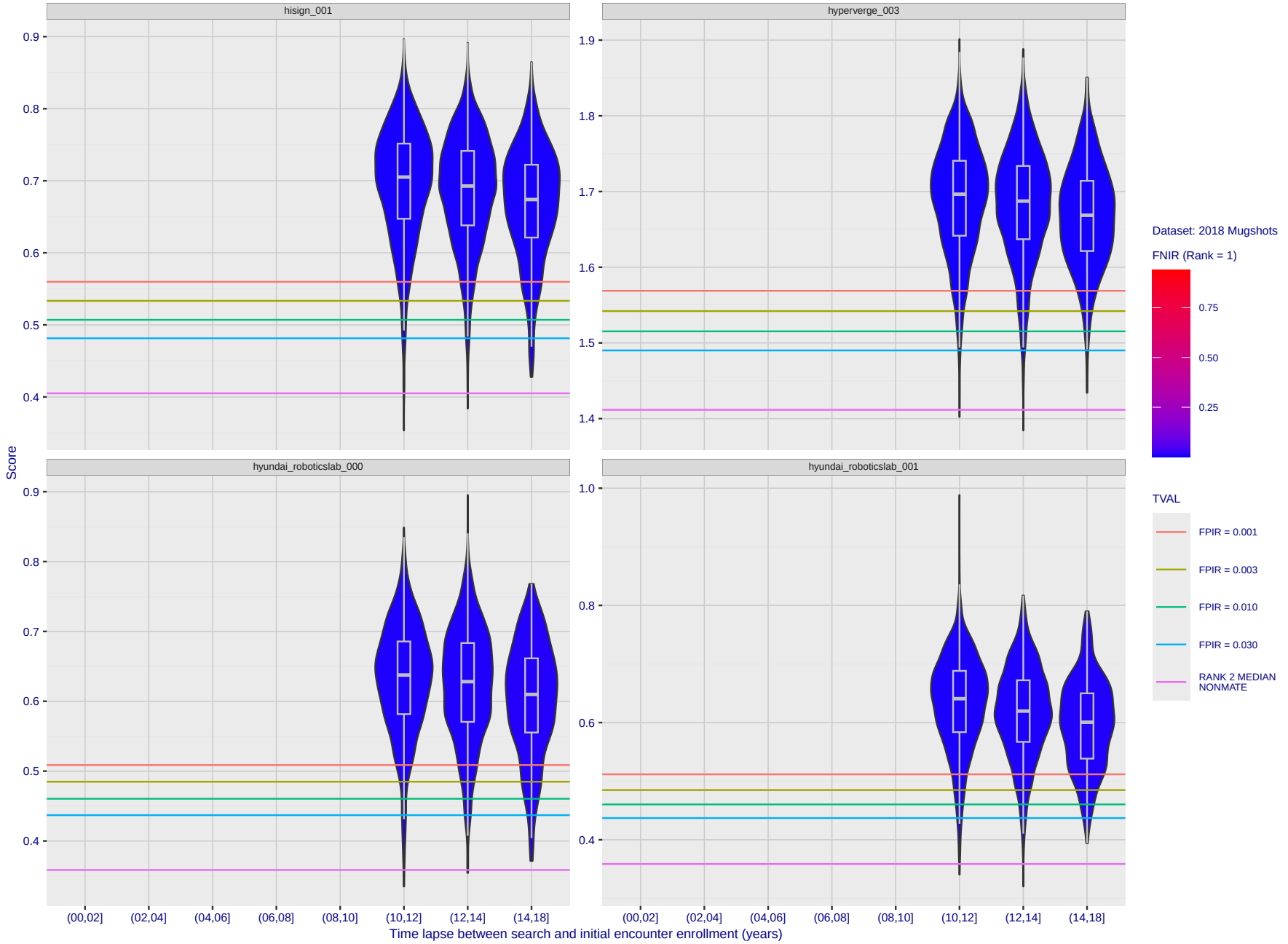

Figure 279: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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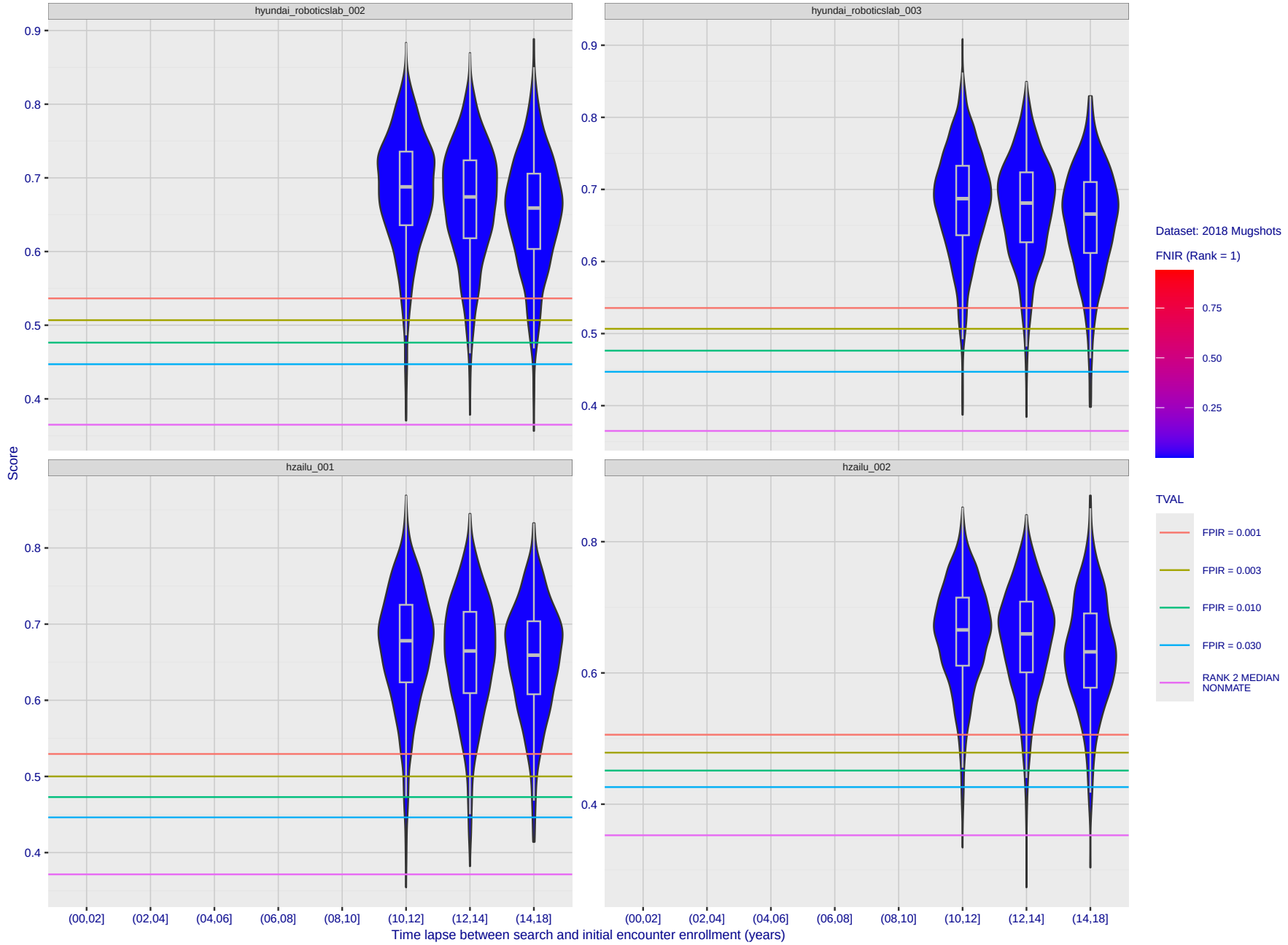

Figure 280: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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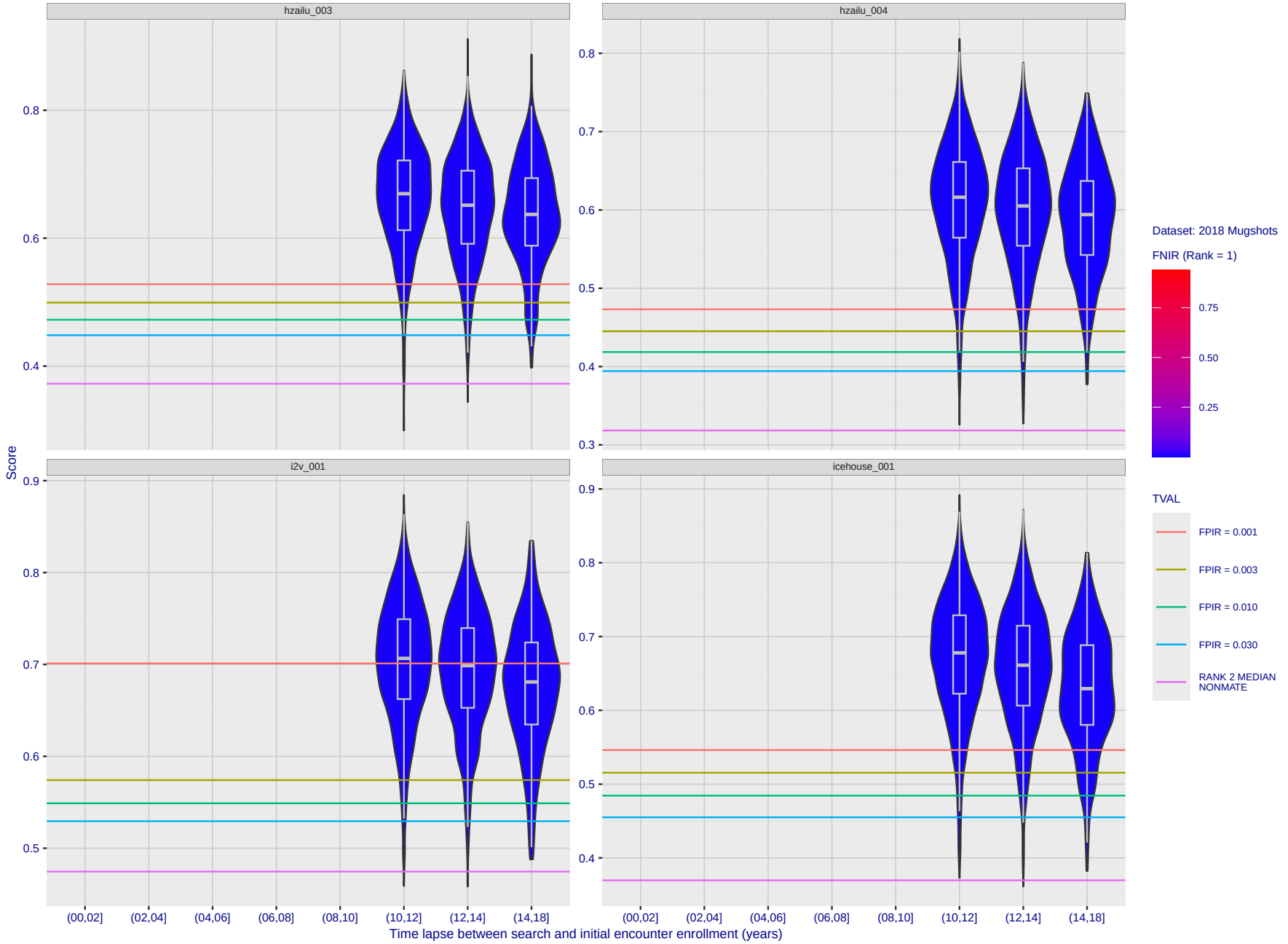

Figure 281: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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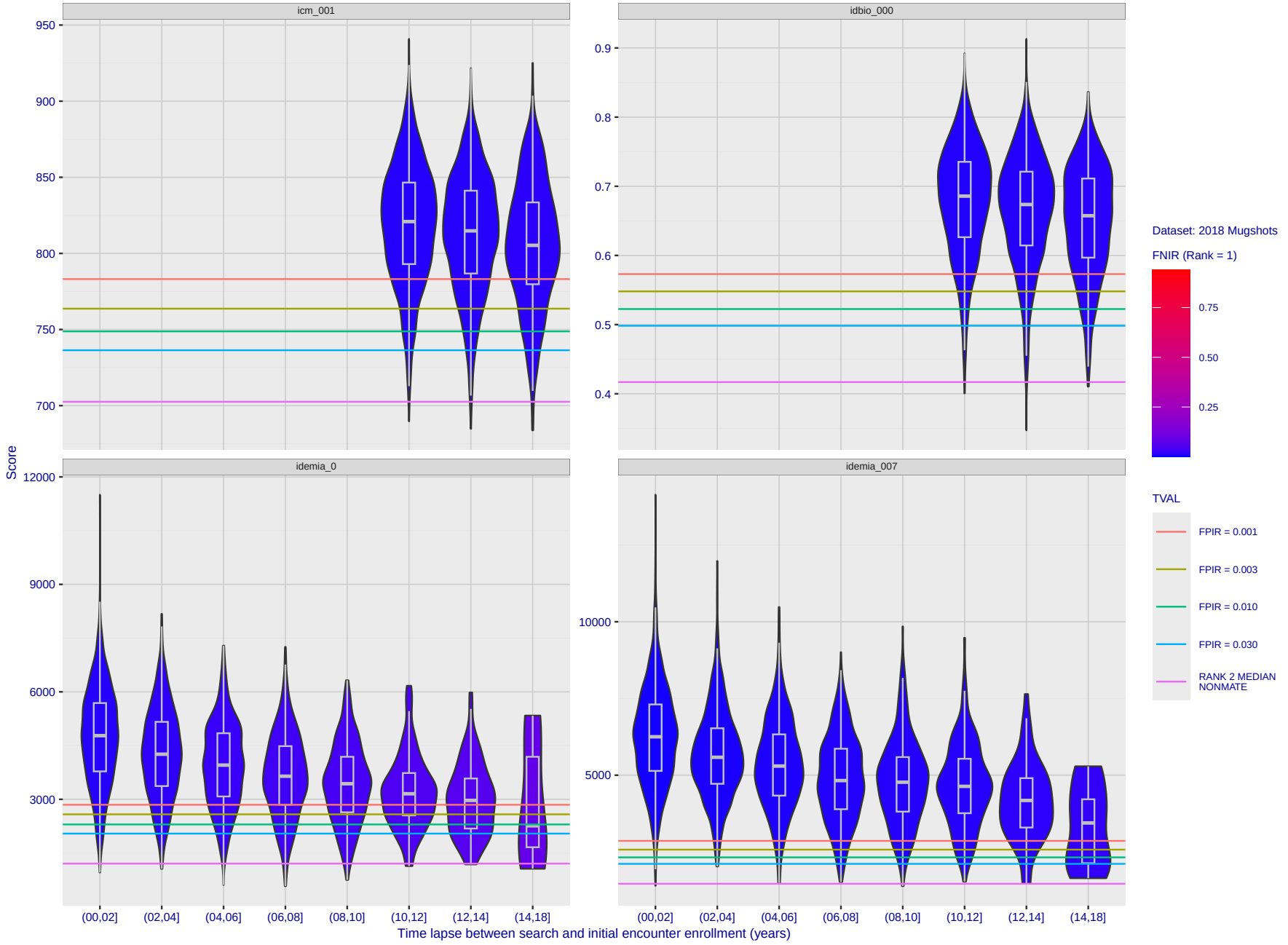

Figure 282: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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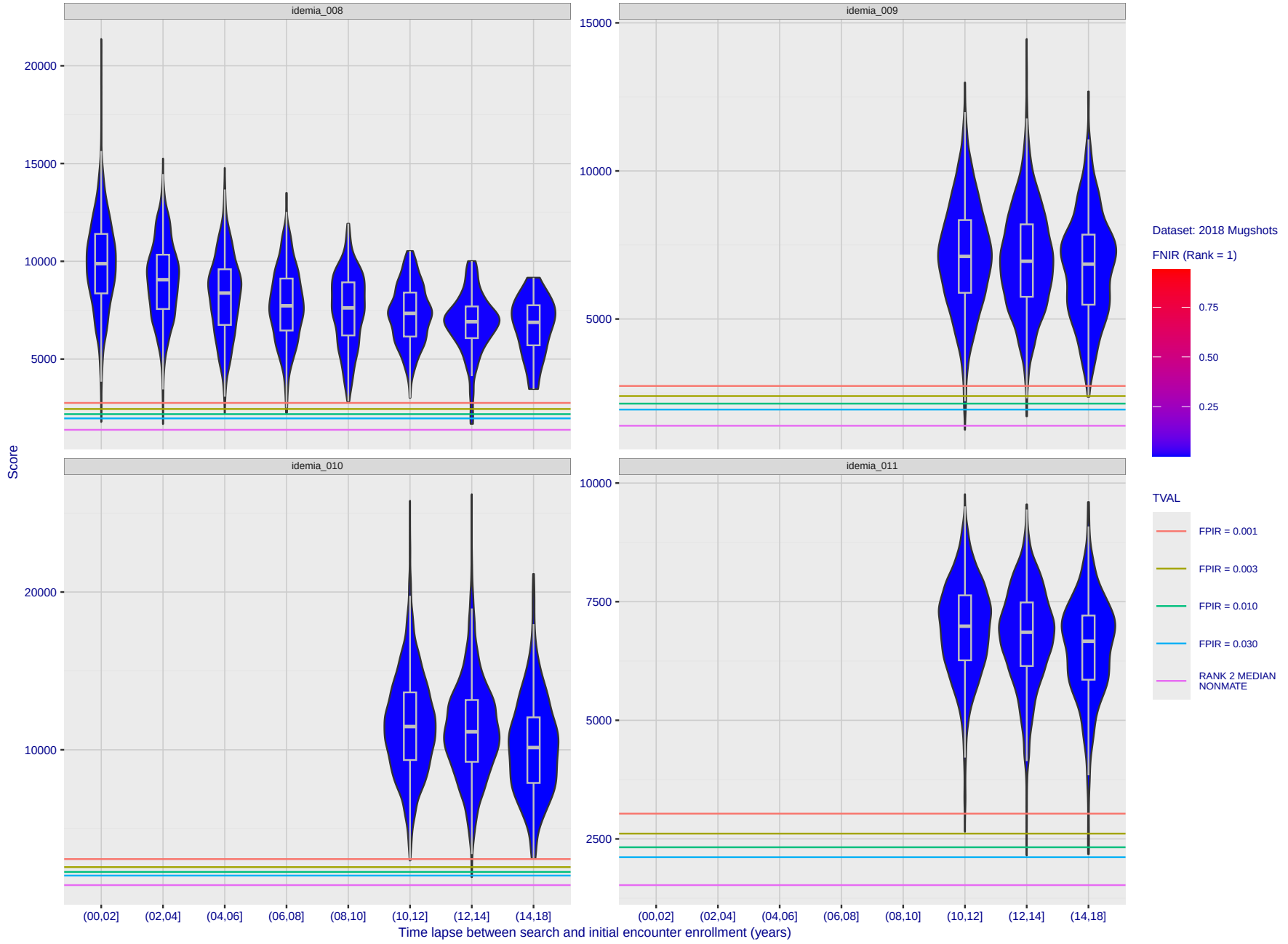

Figure 283: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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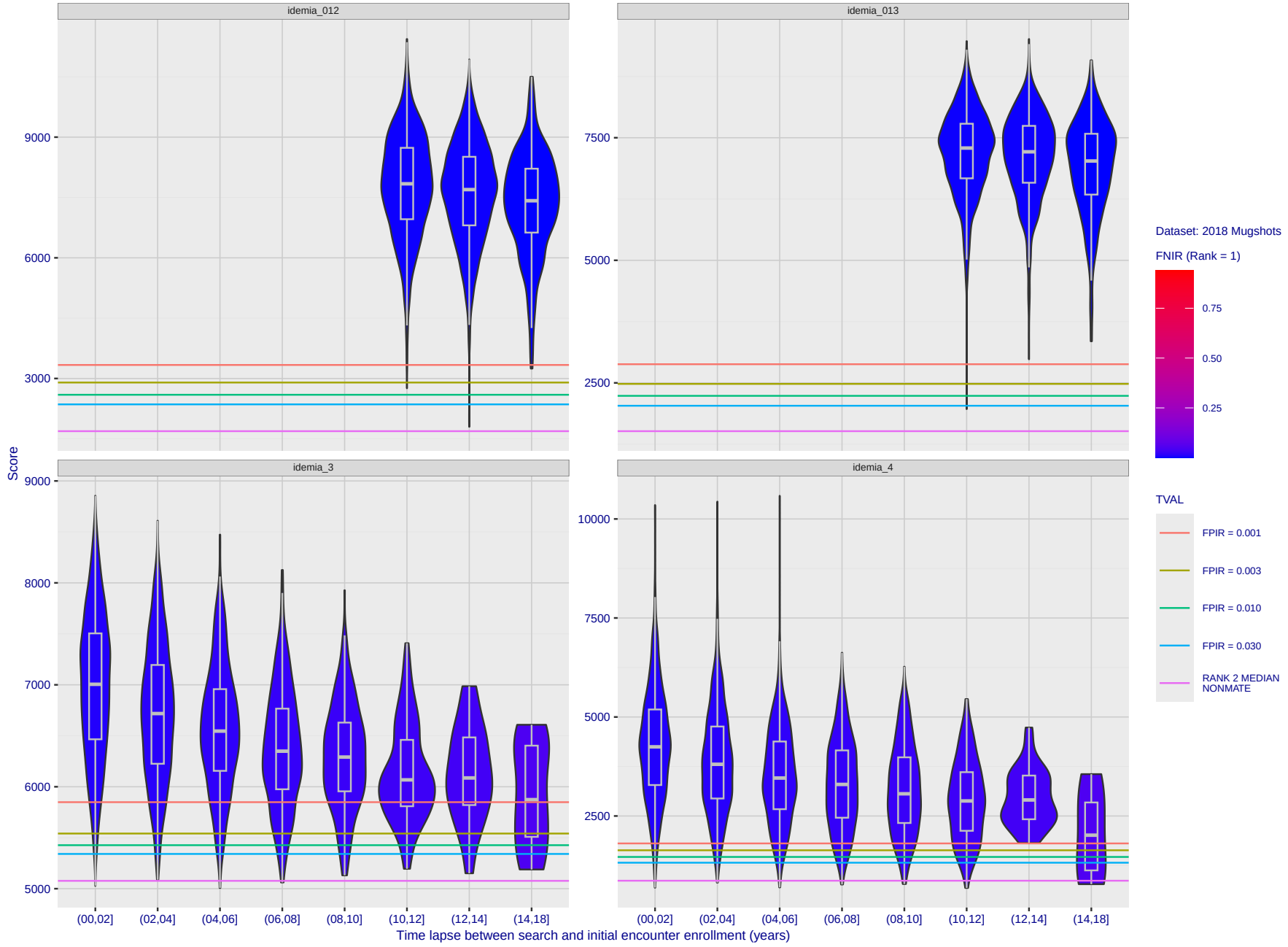

Figure 284: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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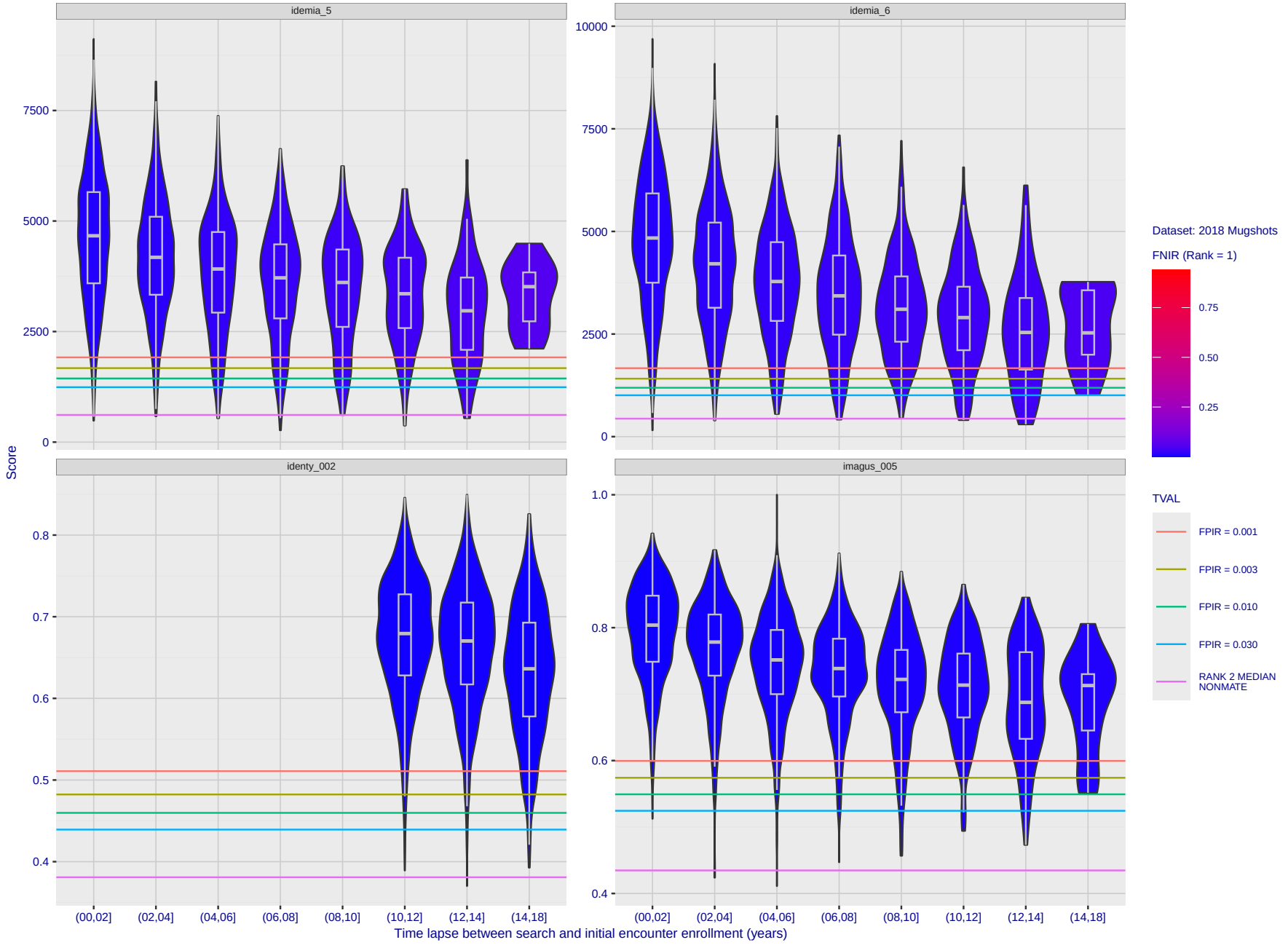

Figure 285: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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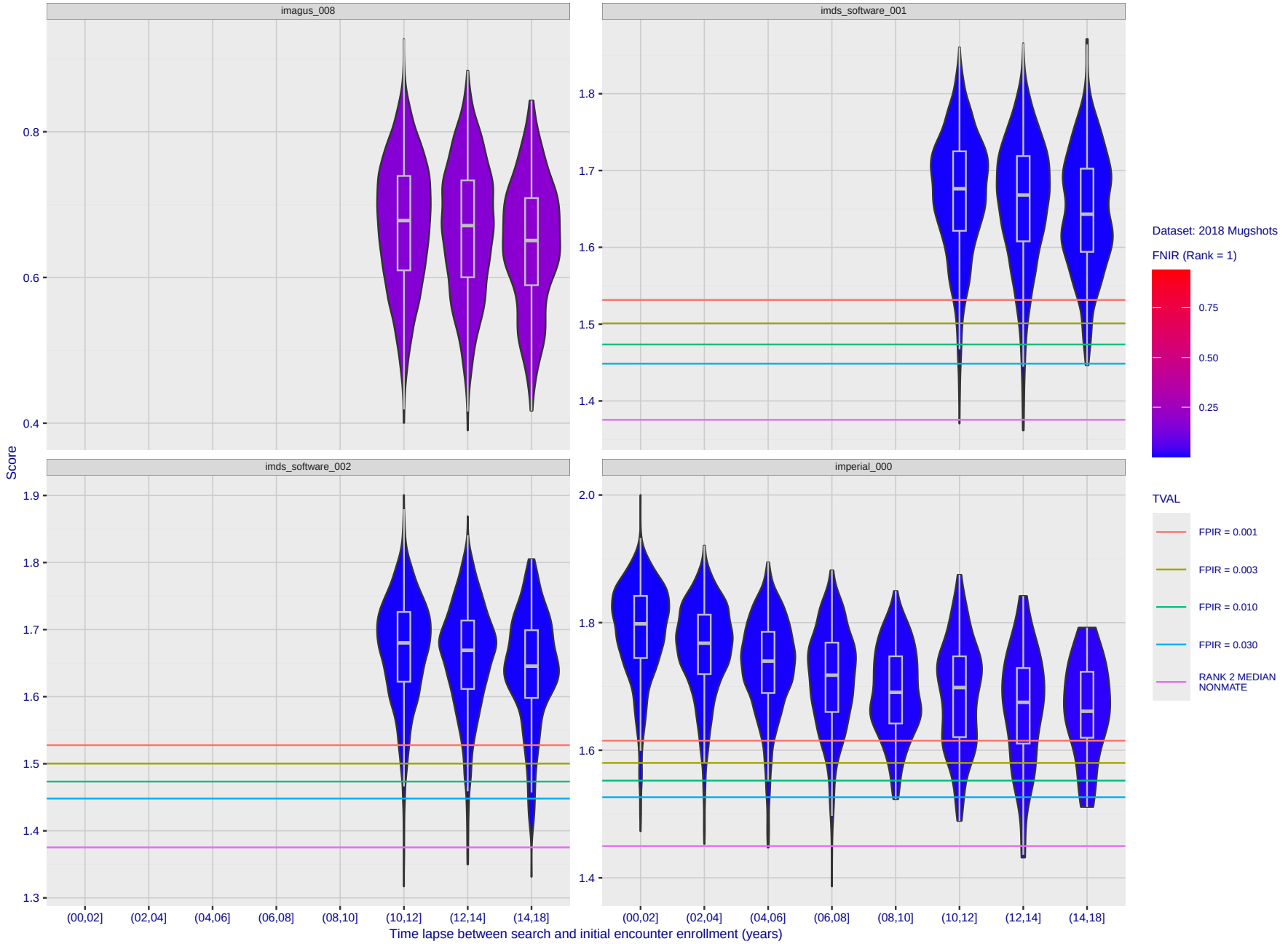

Figure 286: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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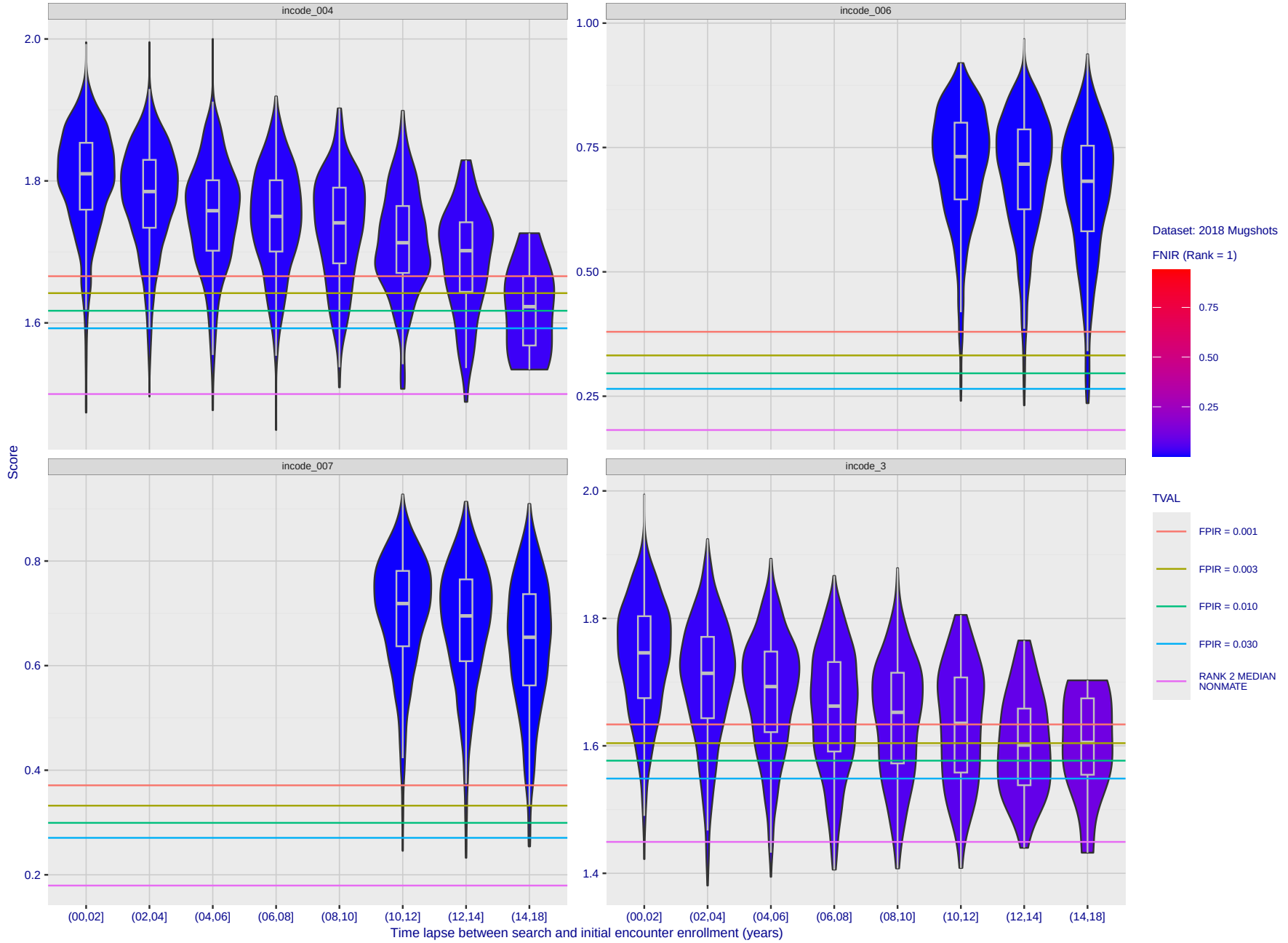

Figure 287: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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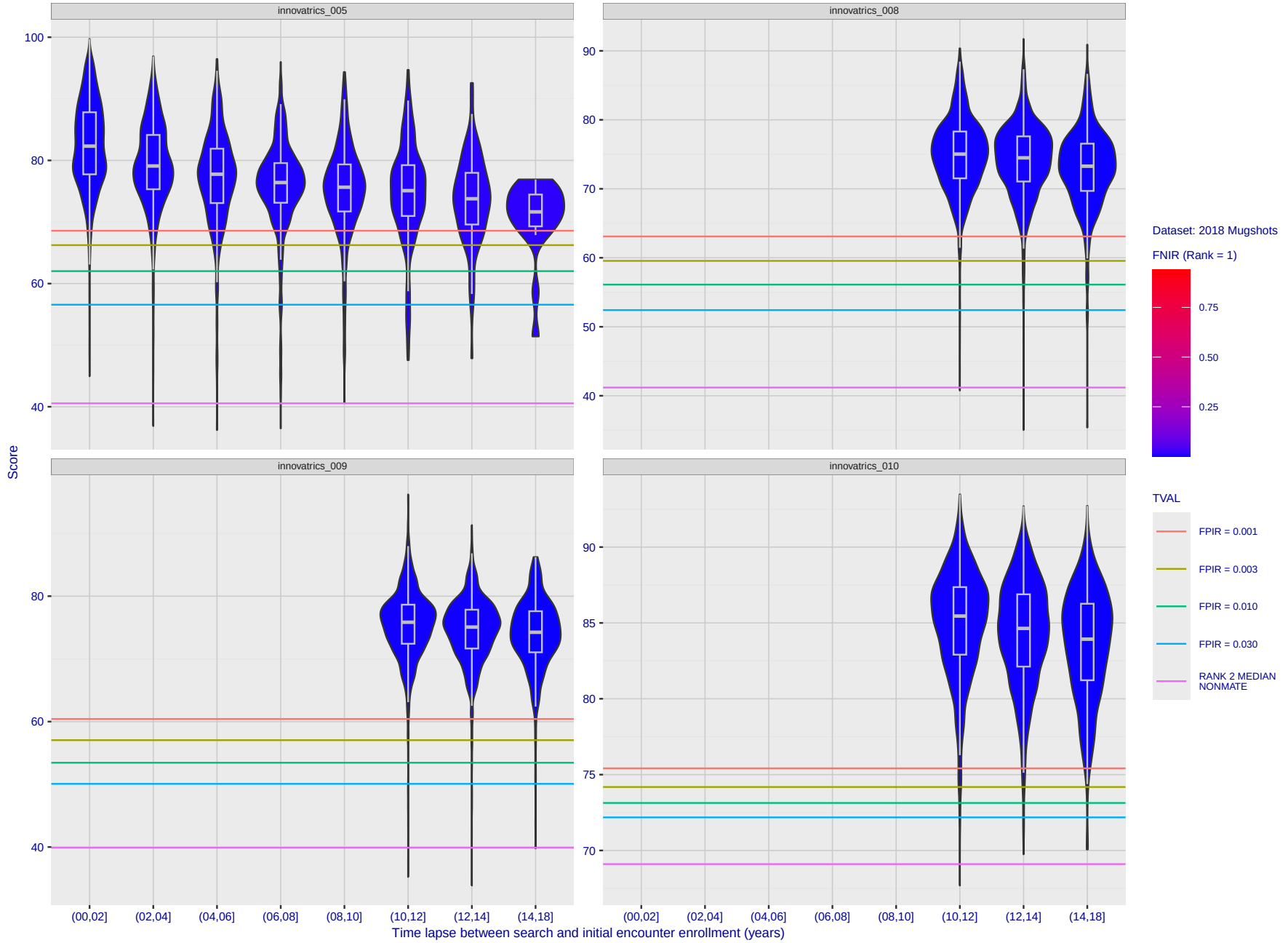

Figure 288: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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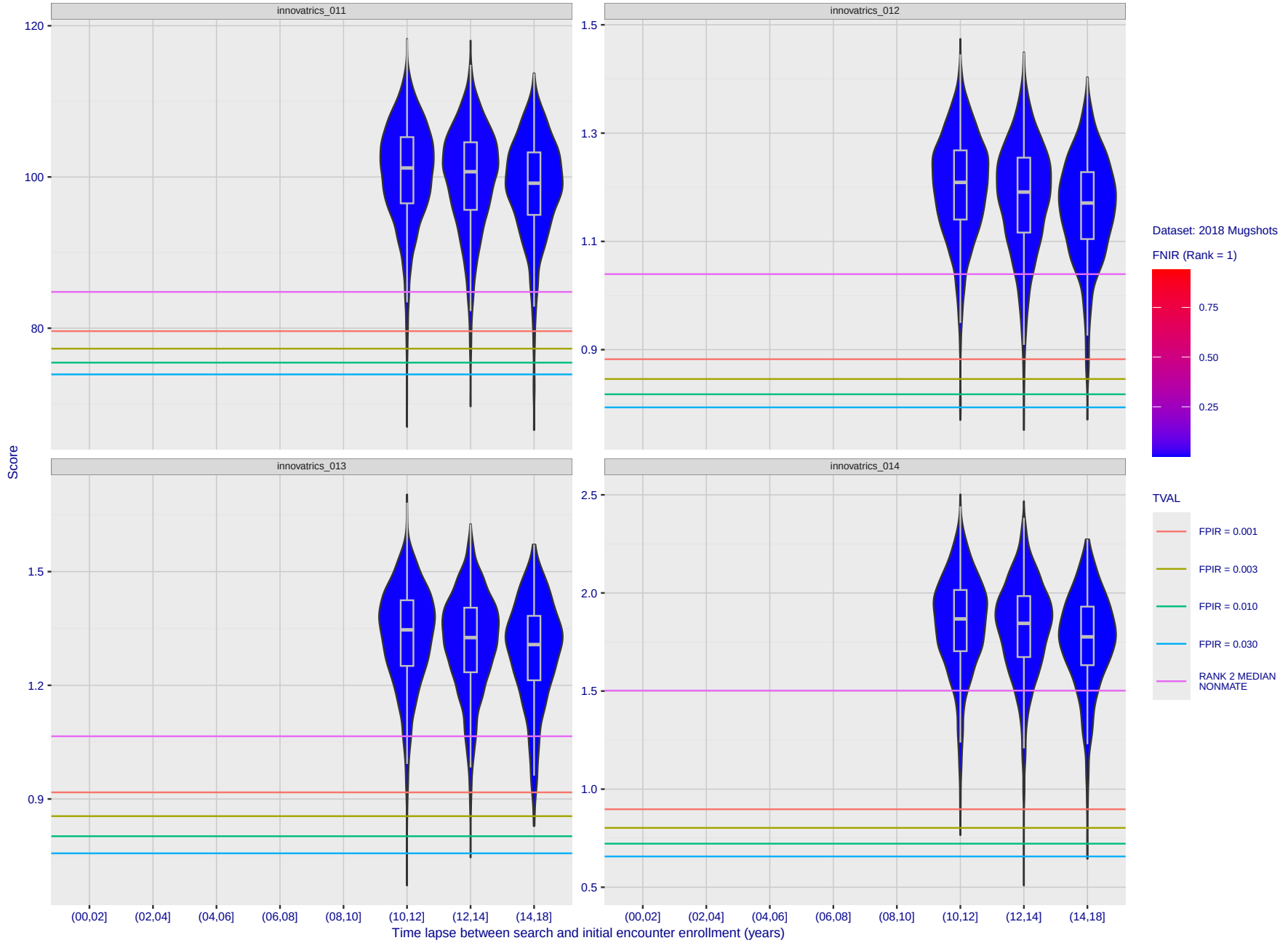

Figure 289: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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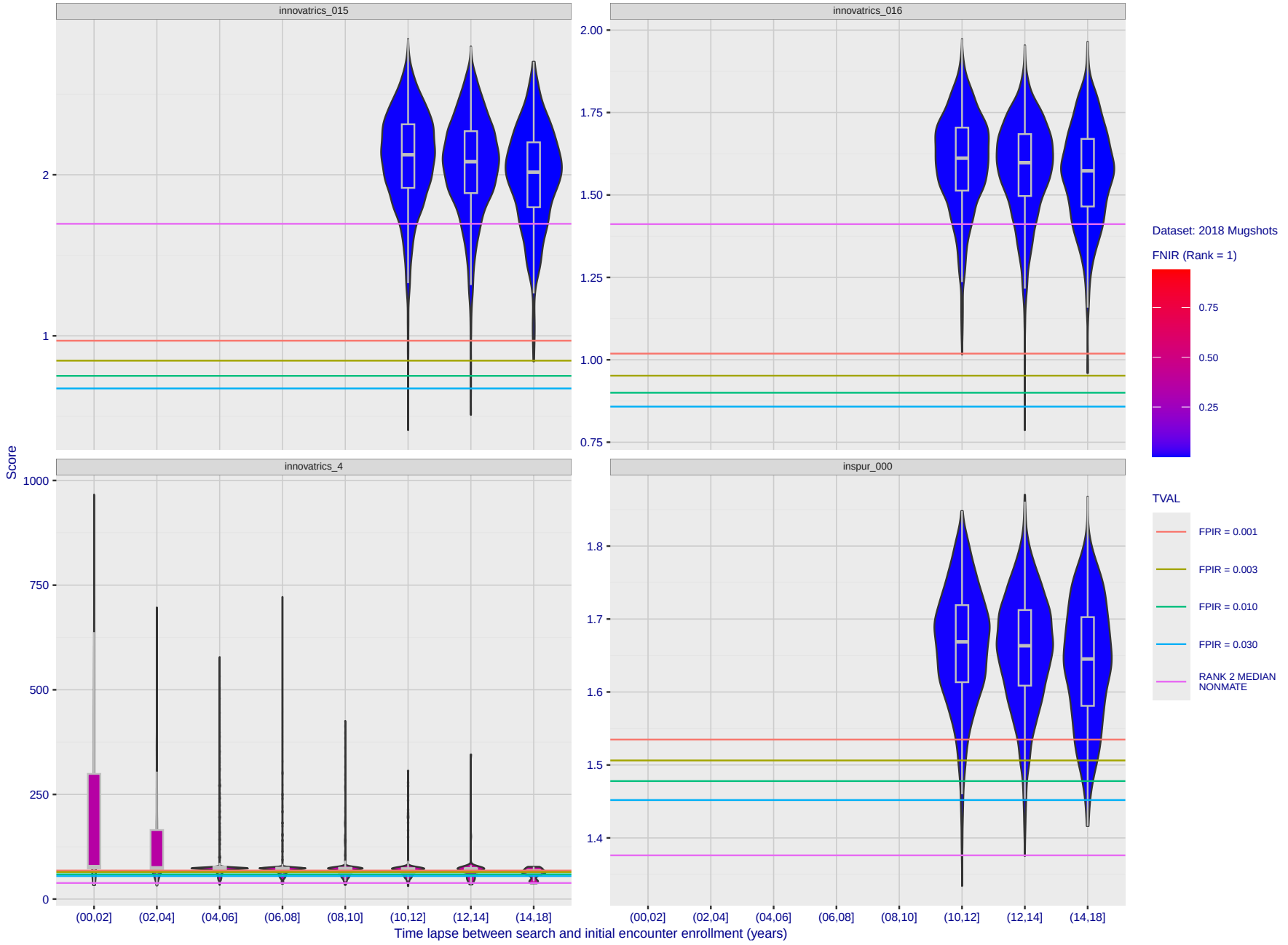

Figure 290: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate
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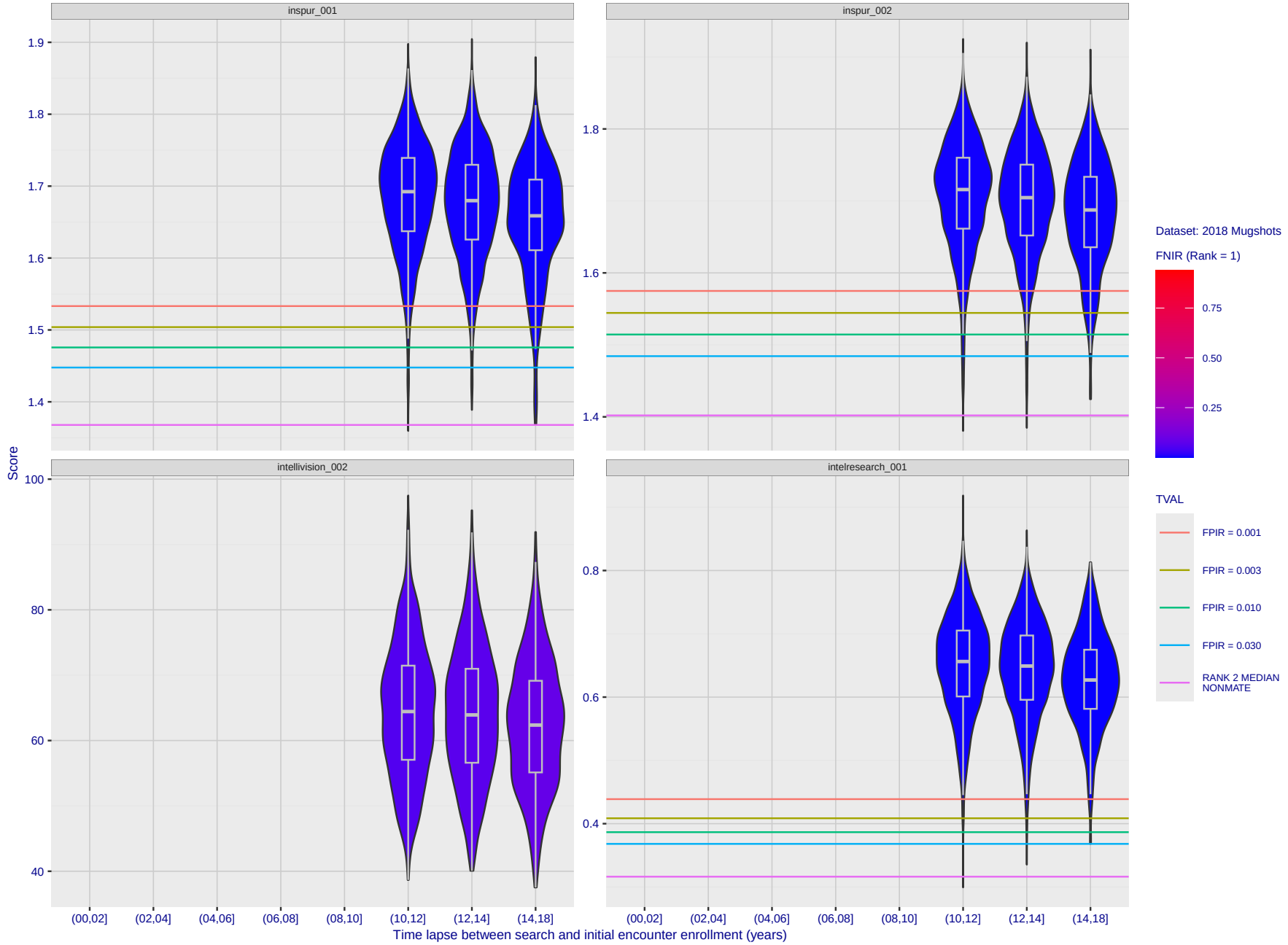

Figure 291: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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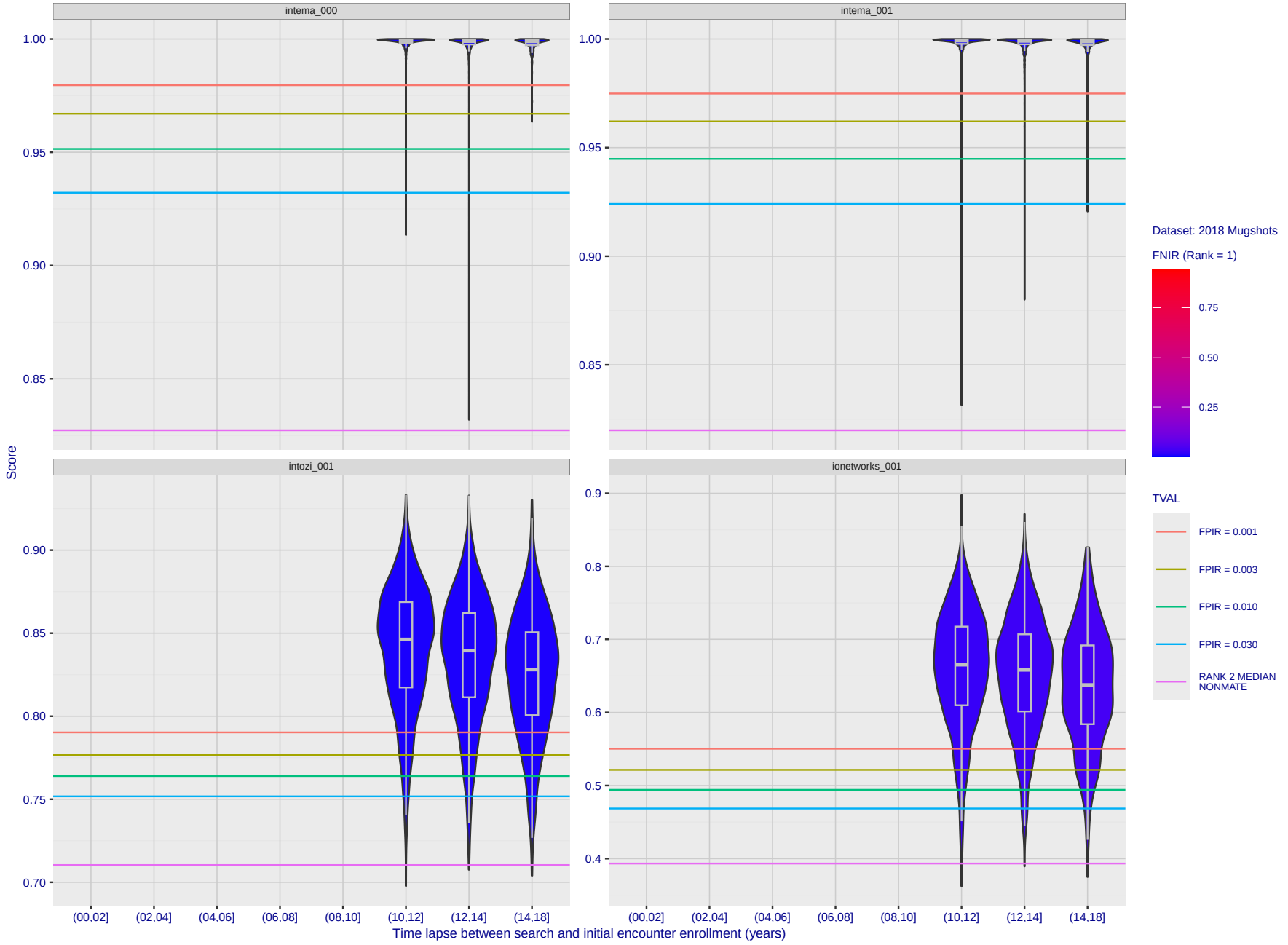

Figure 292: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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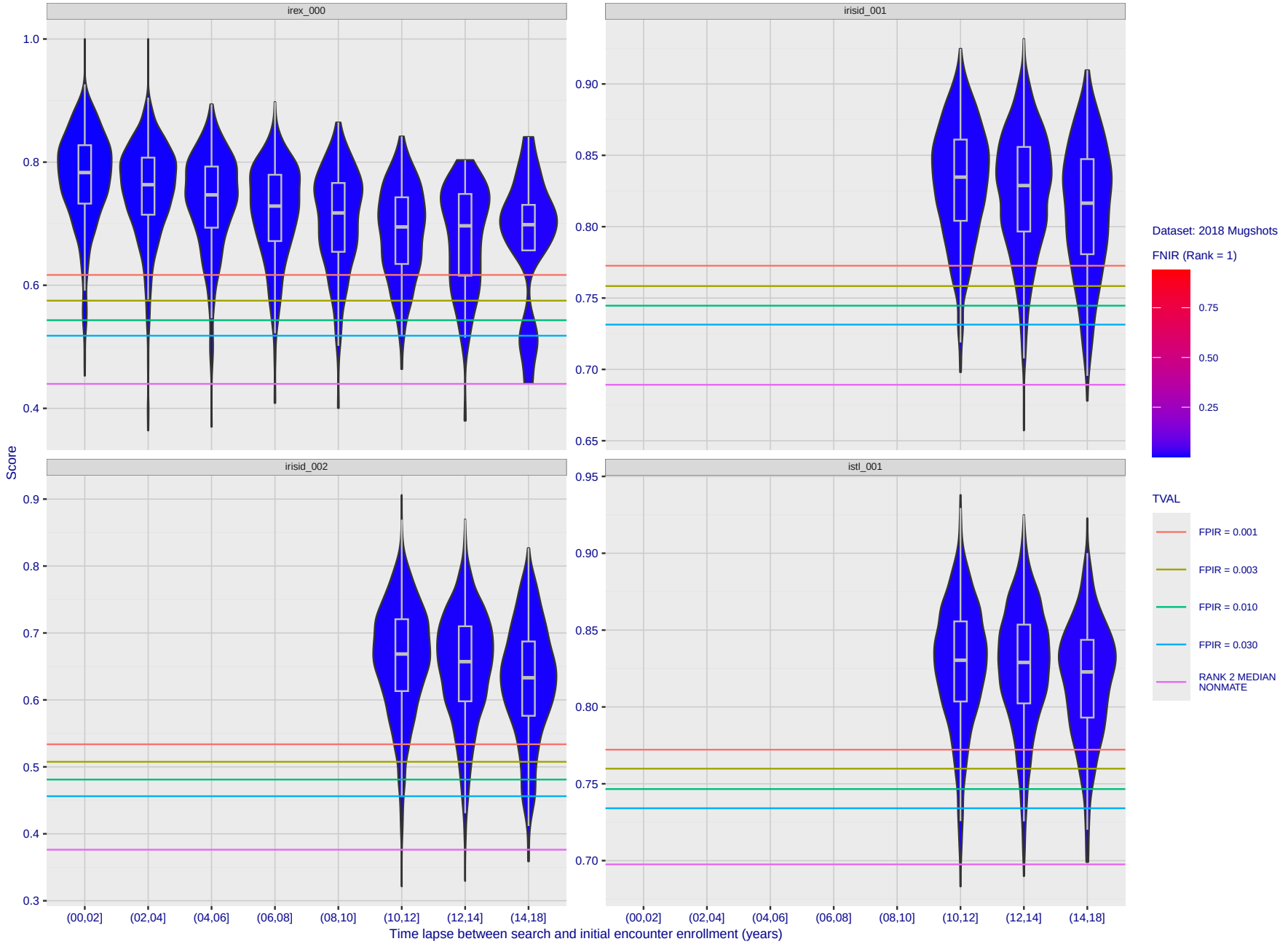

Figure 293: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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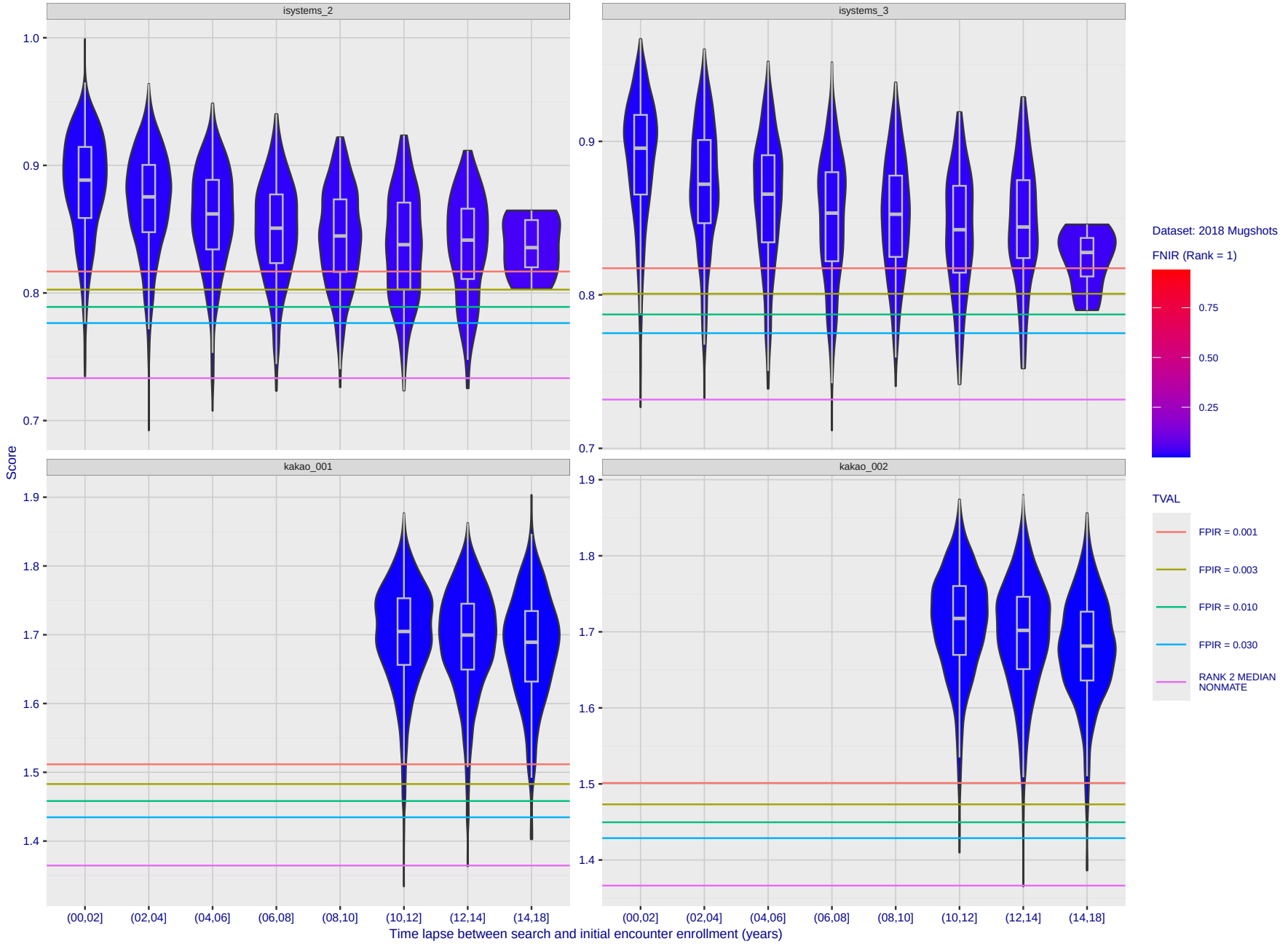

Figure 294: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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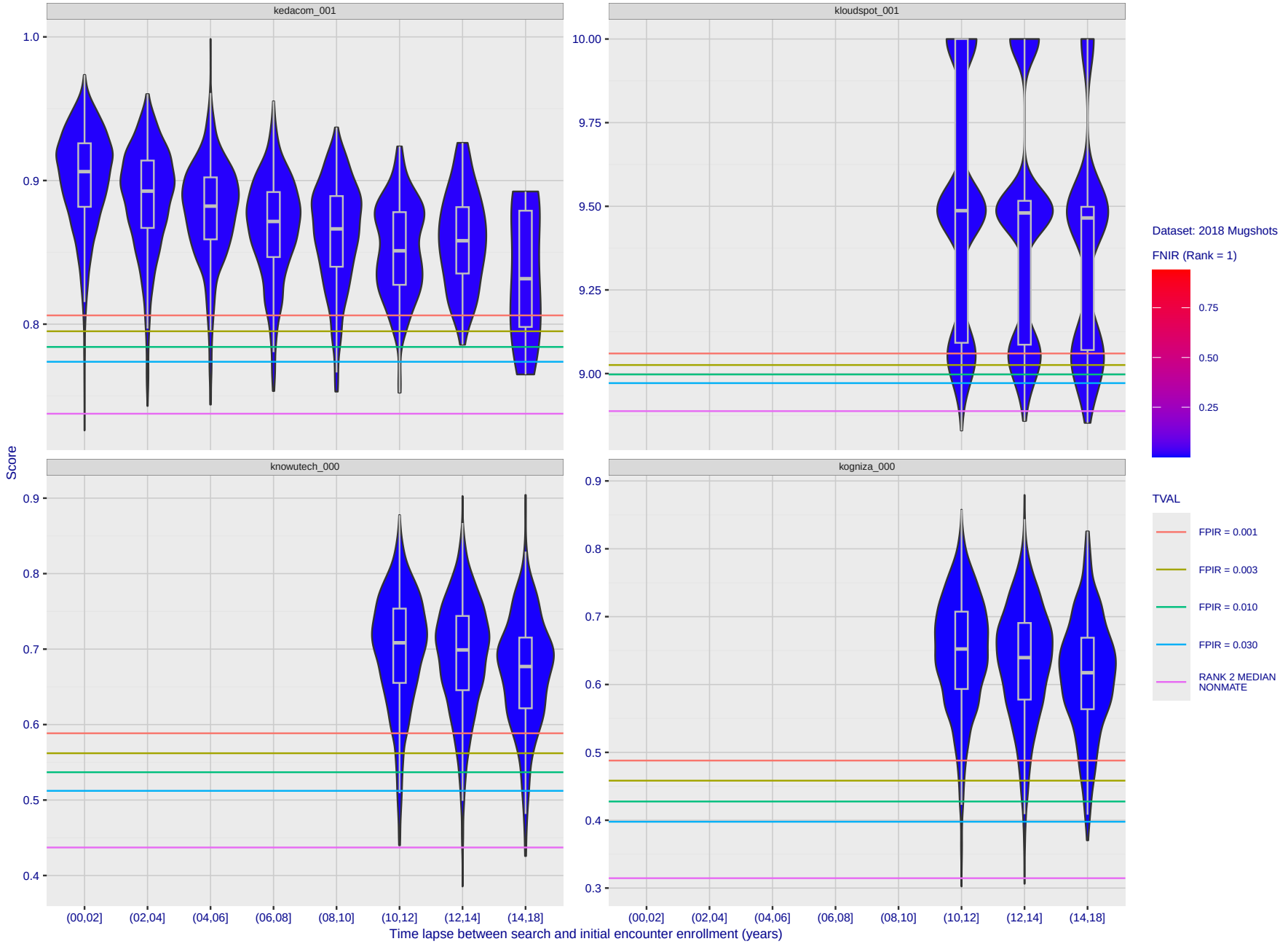

Figure 295: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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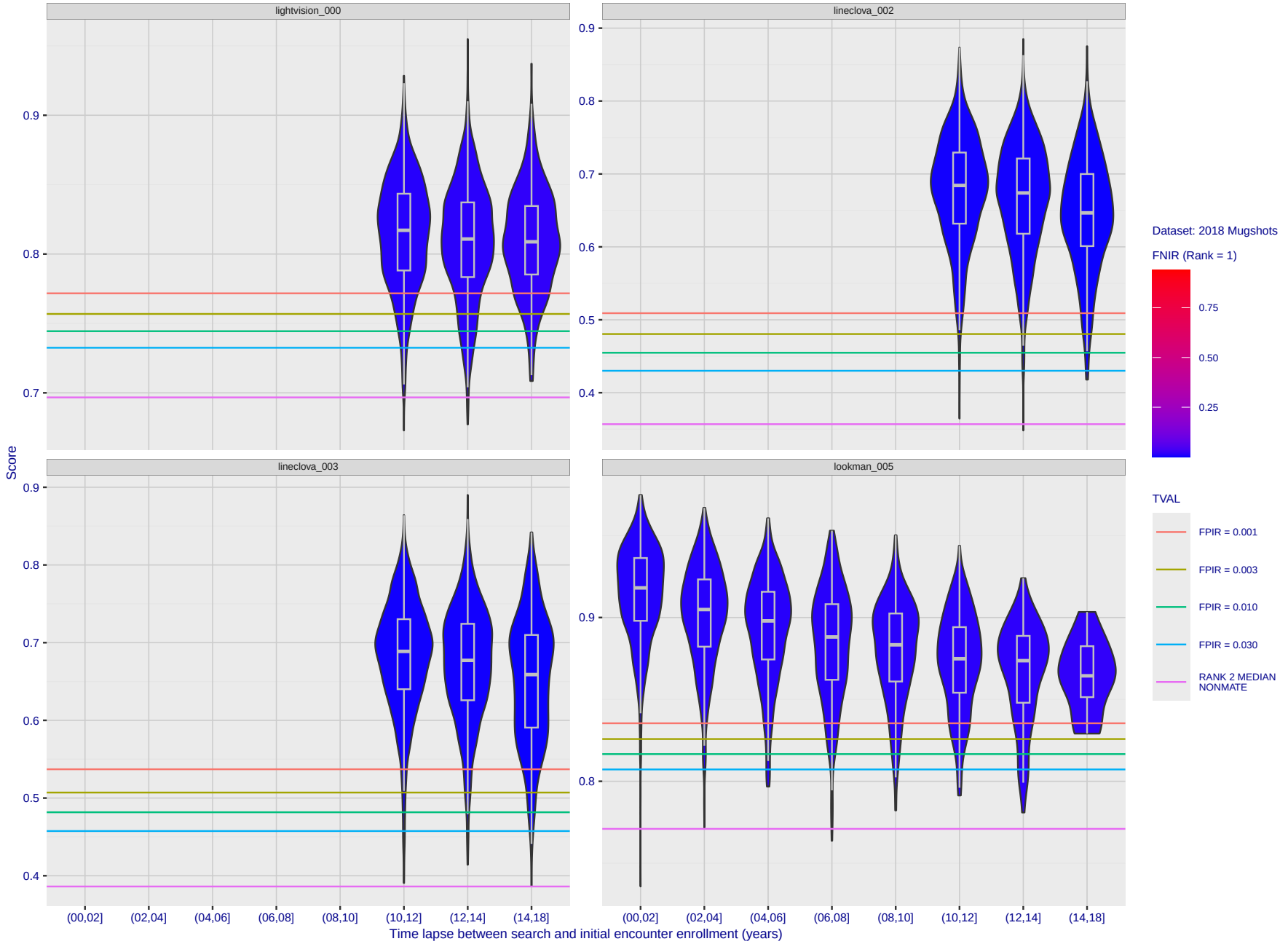

Figure 296: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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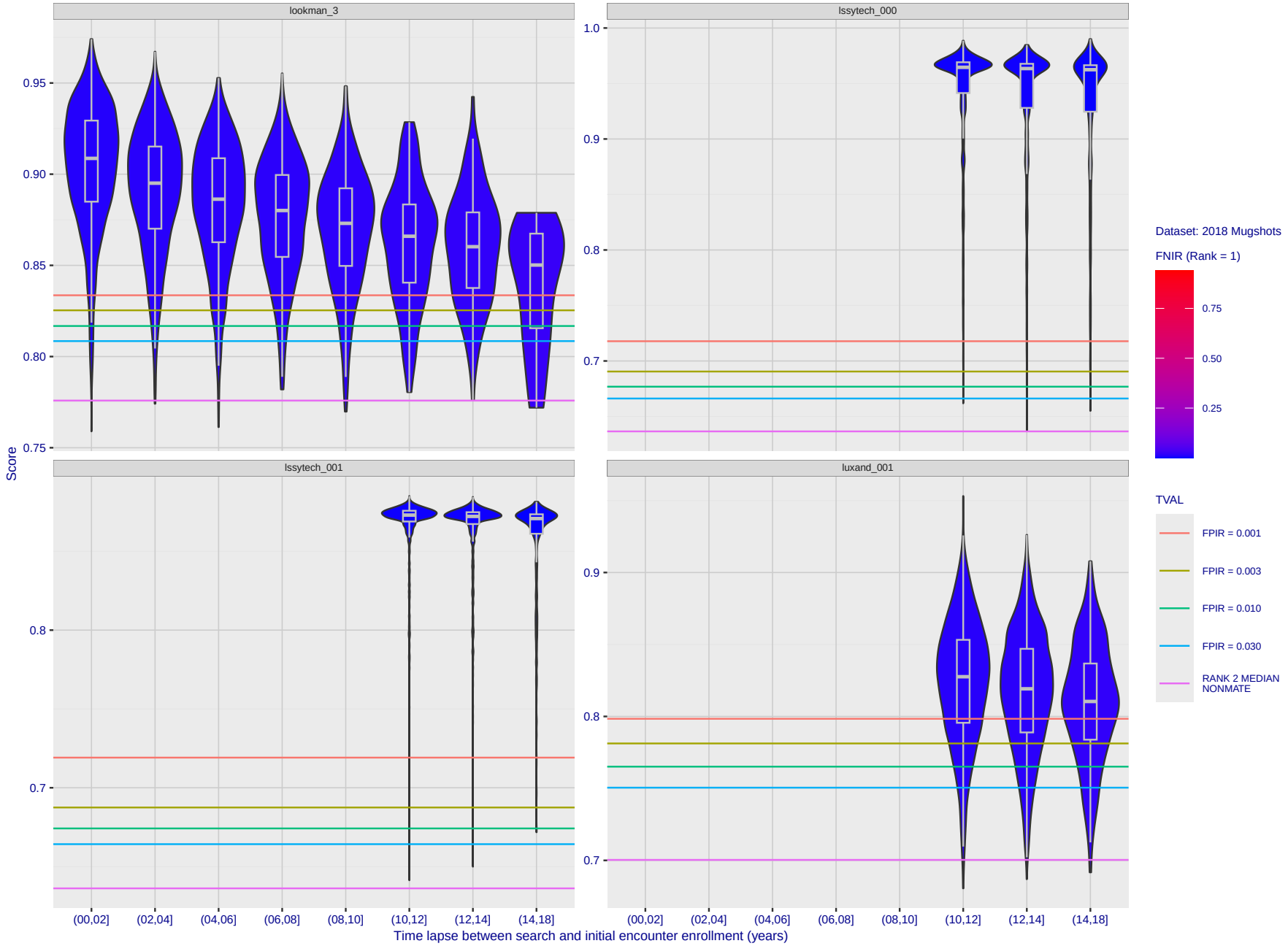

Figure 297: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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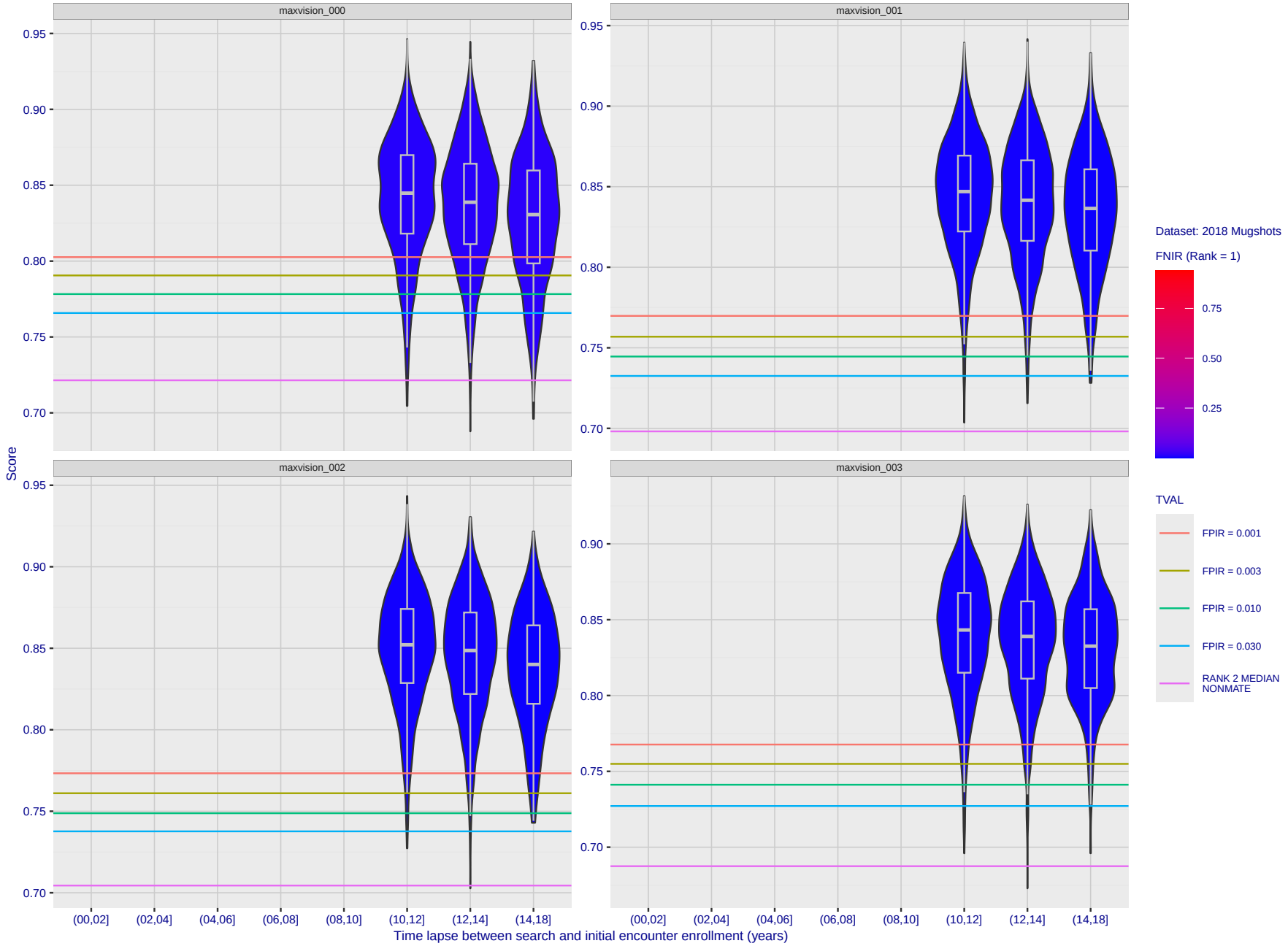

Figure 298: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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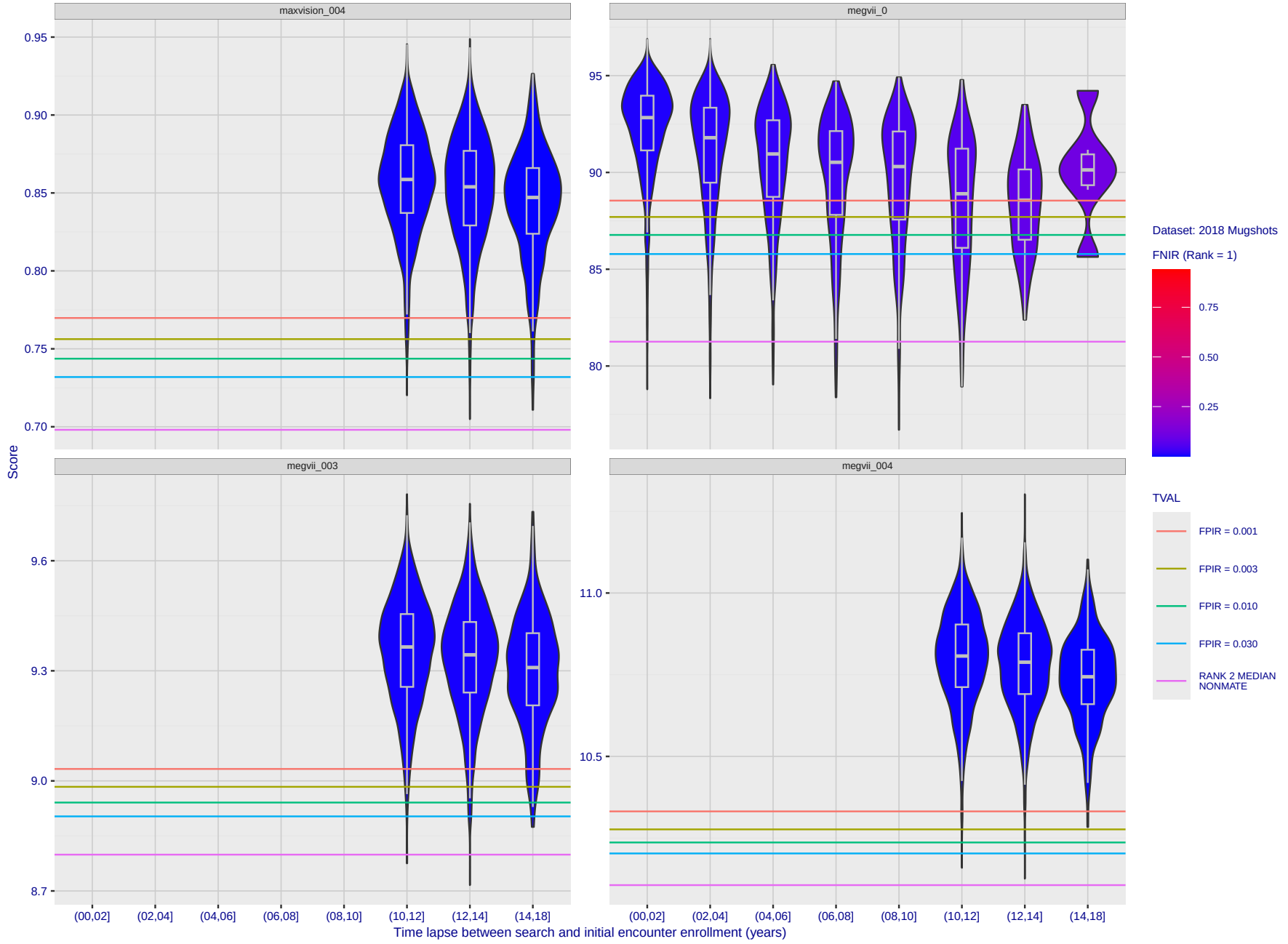

Figure 299: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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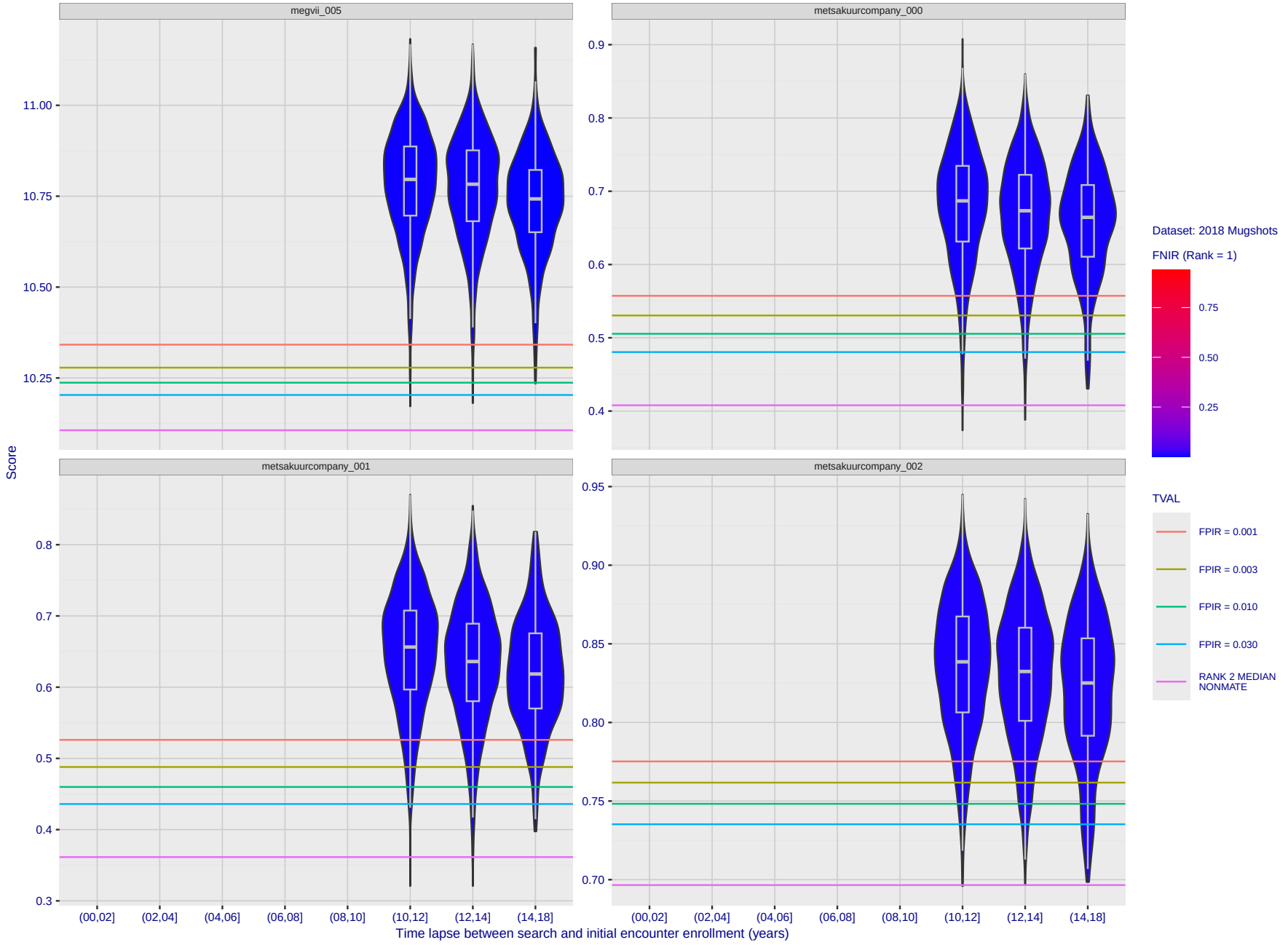

Figure 300: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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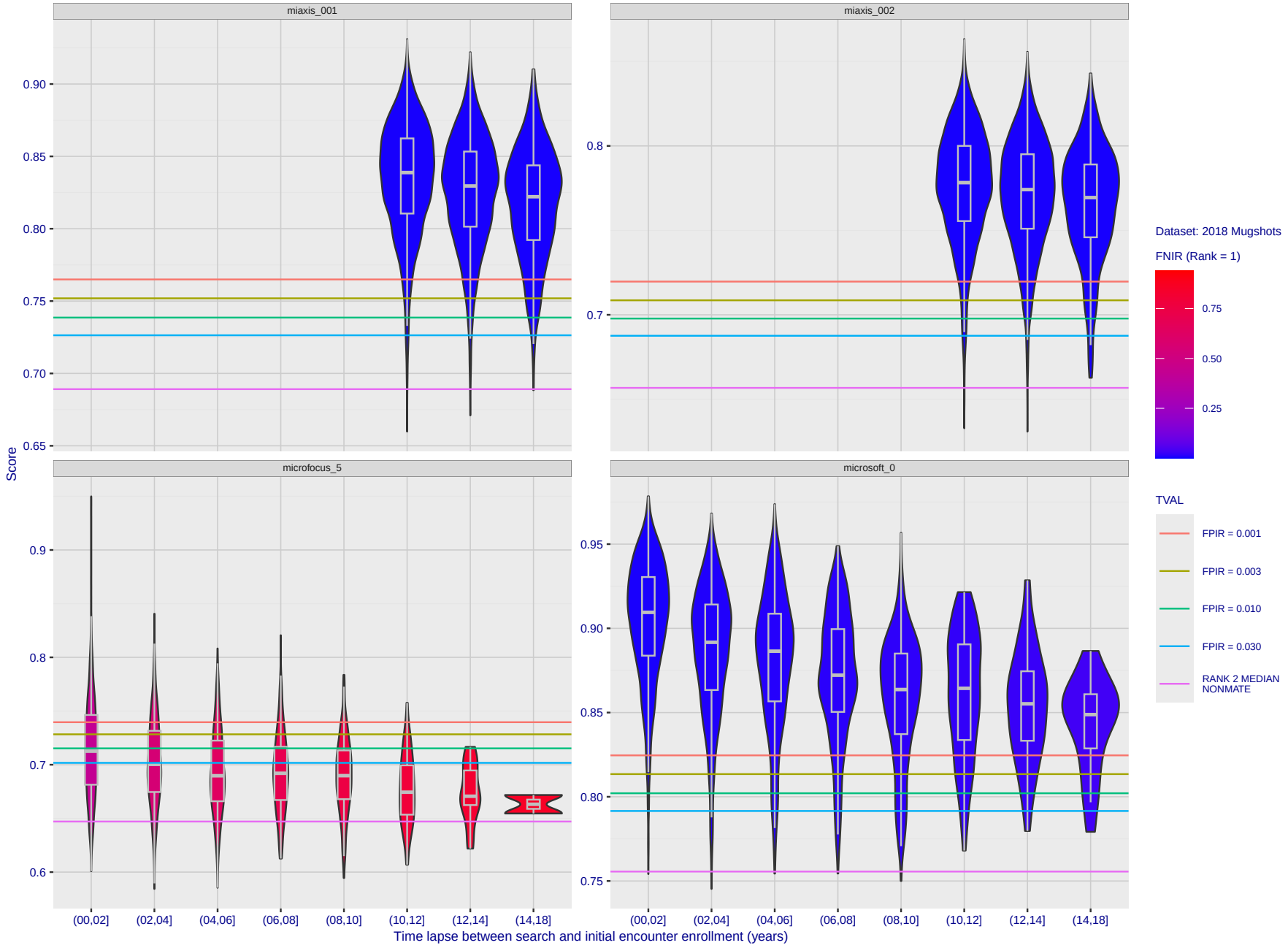

Figure 301: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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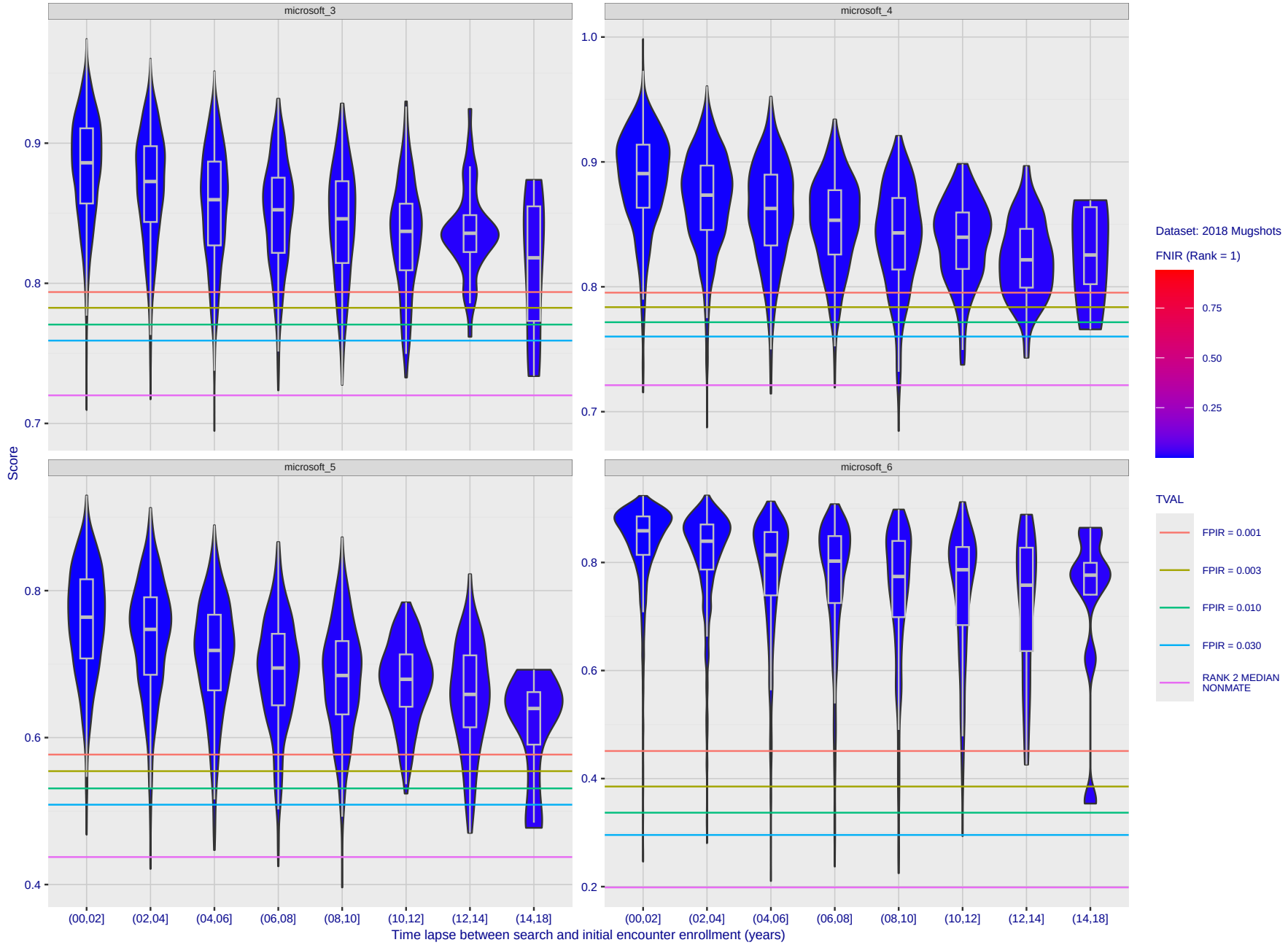

Figure 302: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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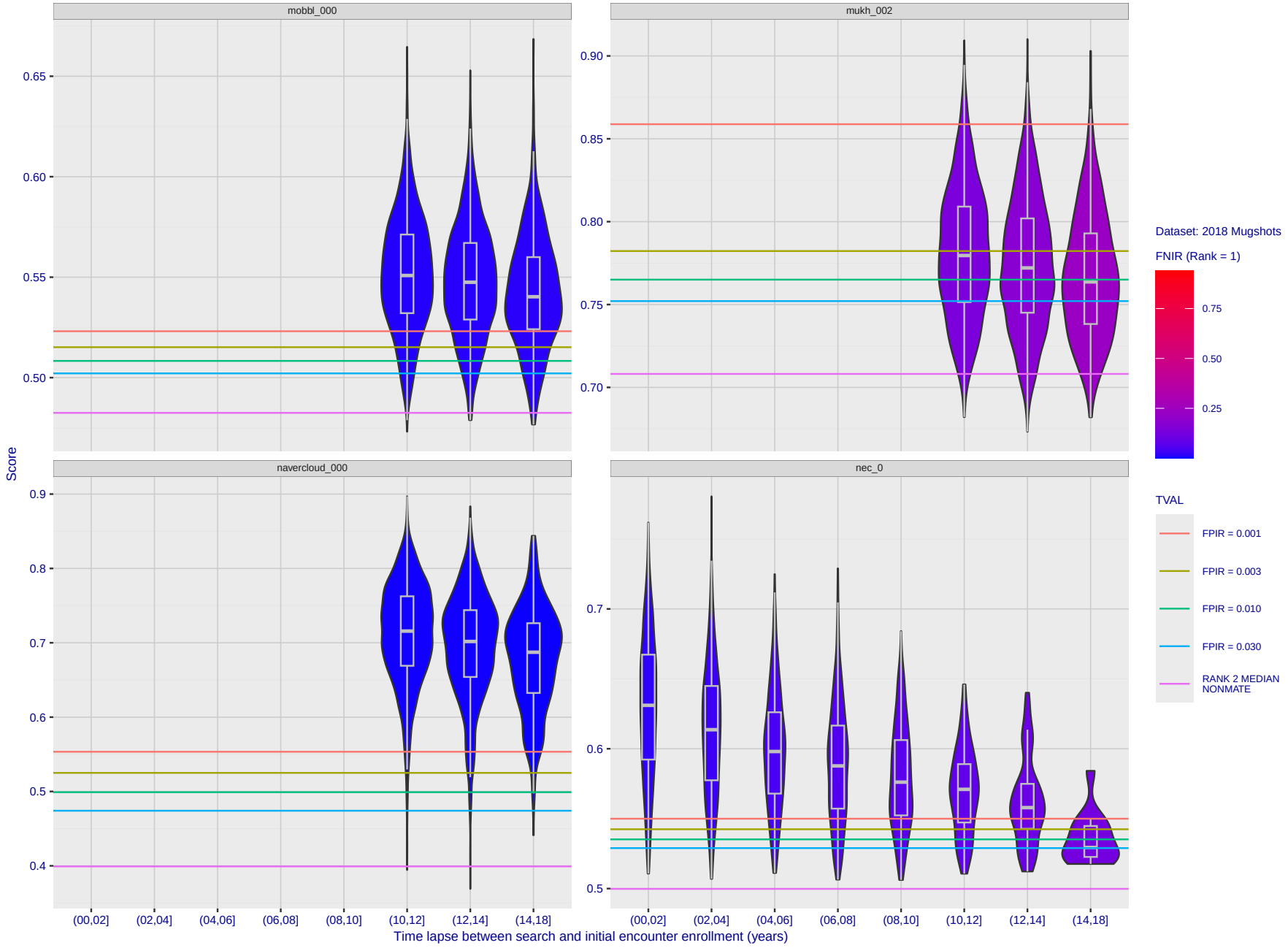

Figure 303: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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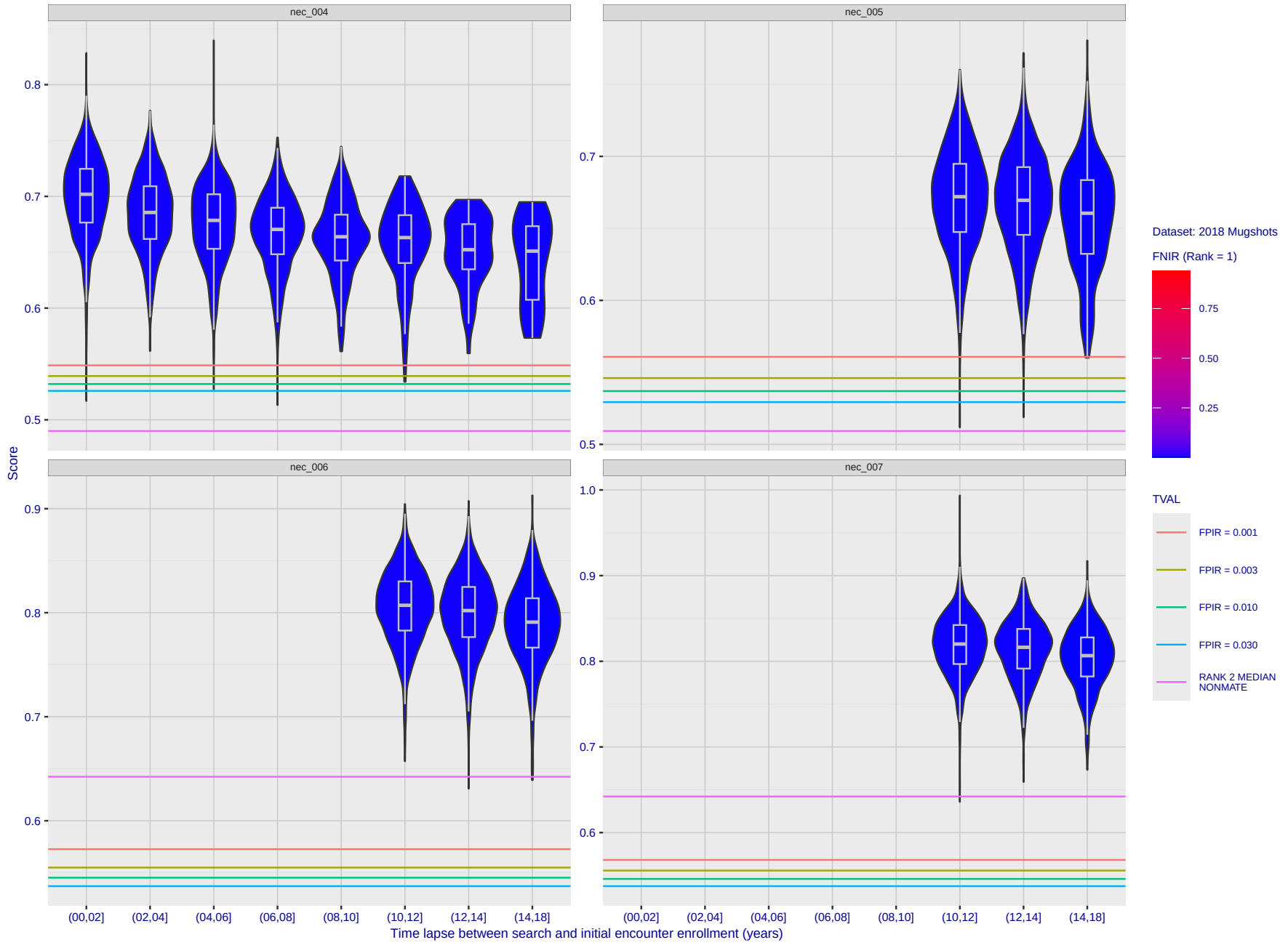

Figure 304: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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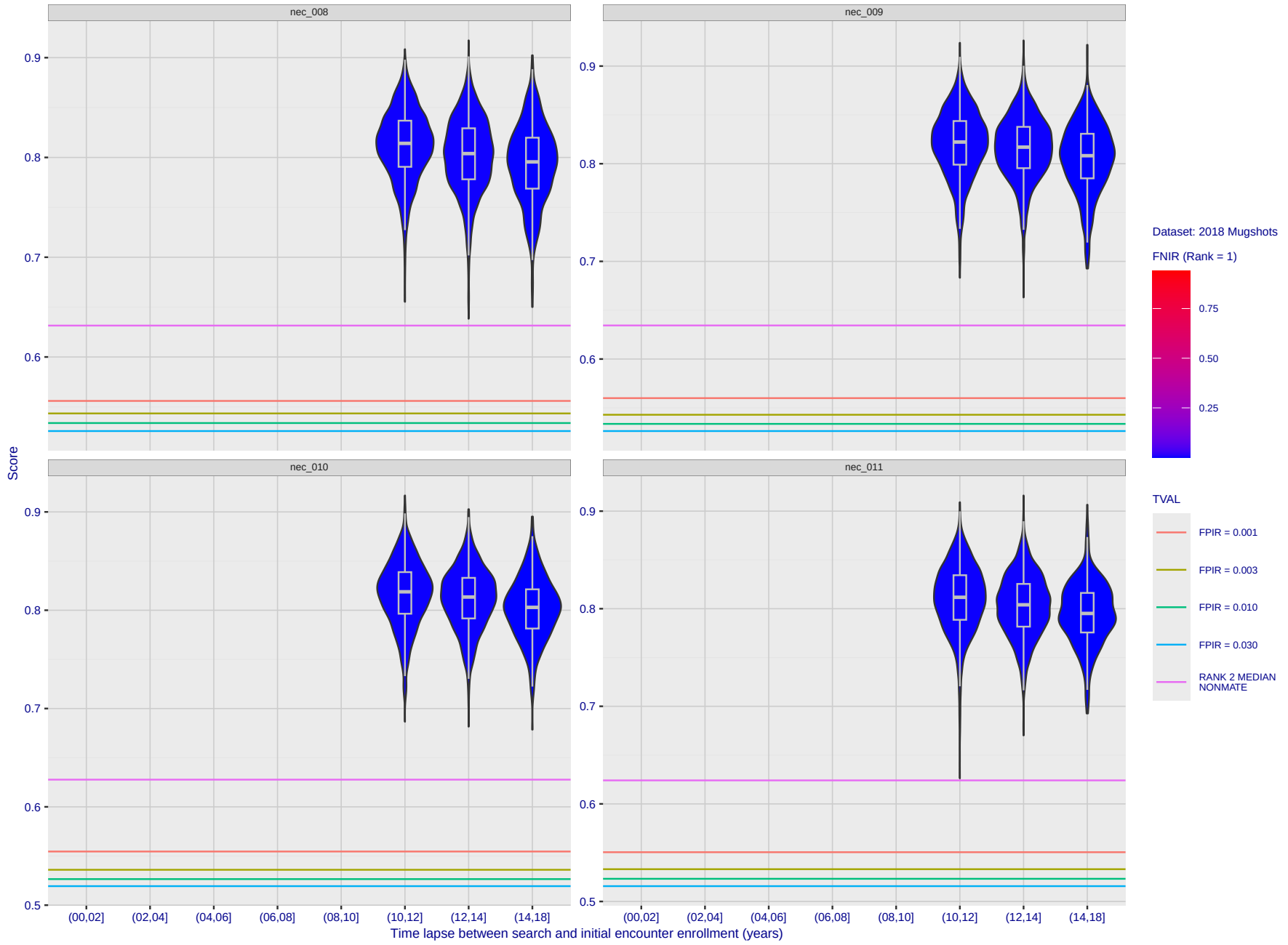

Figure 305: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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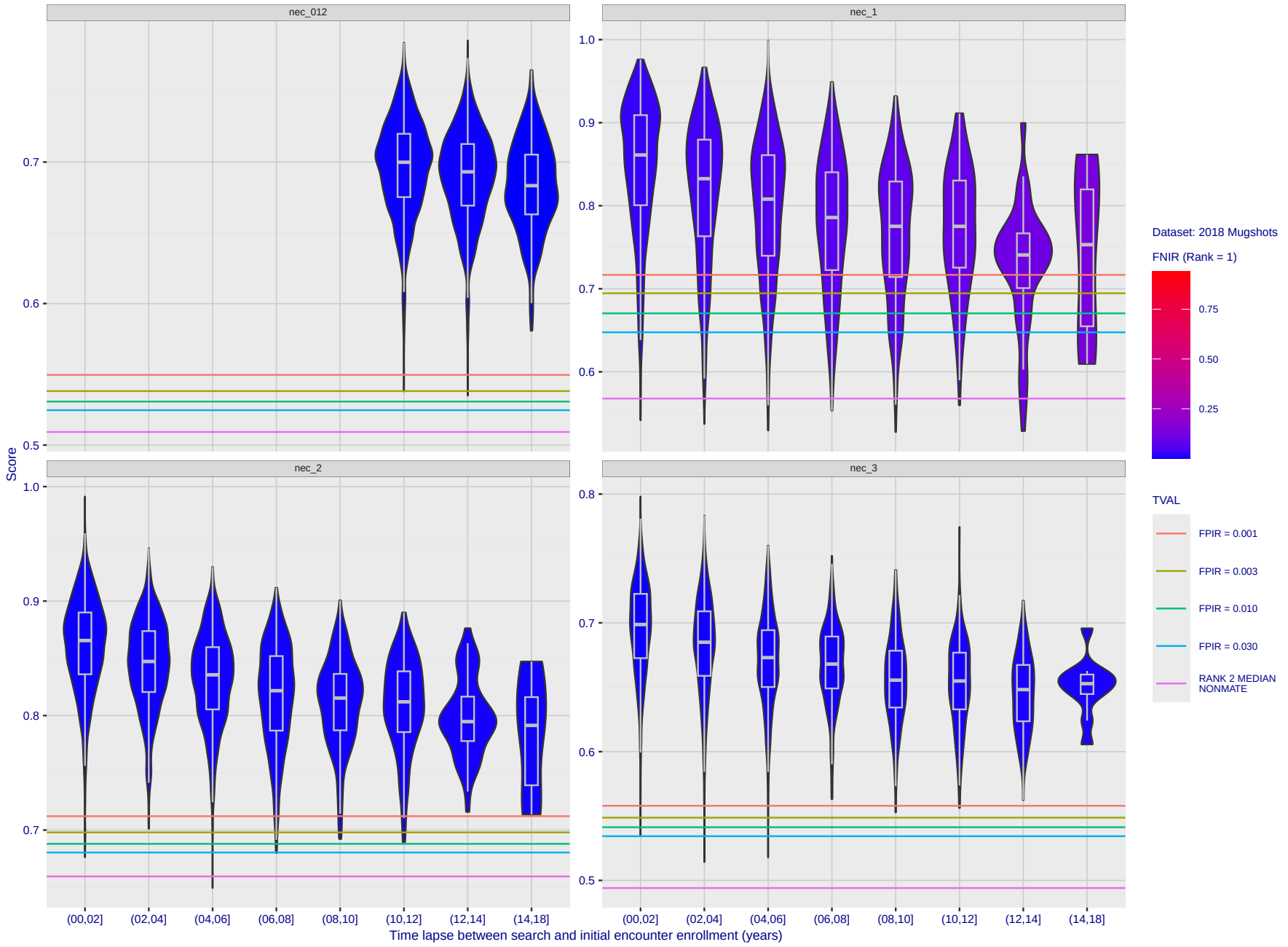

Figure 306: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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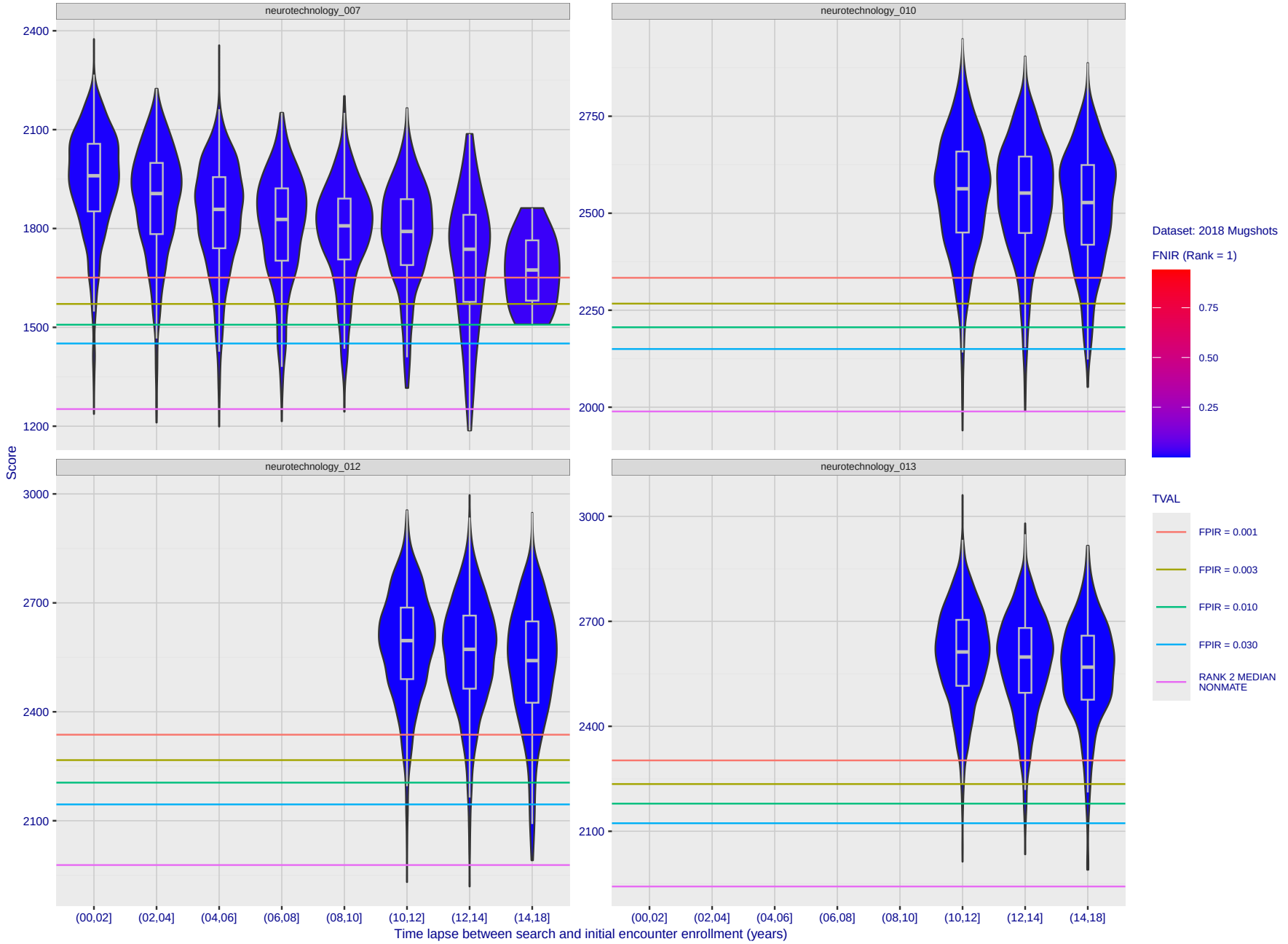

Figure 307: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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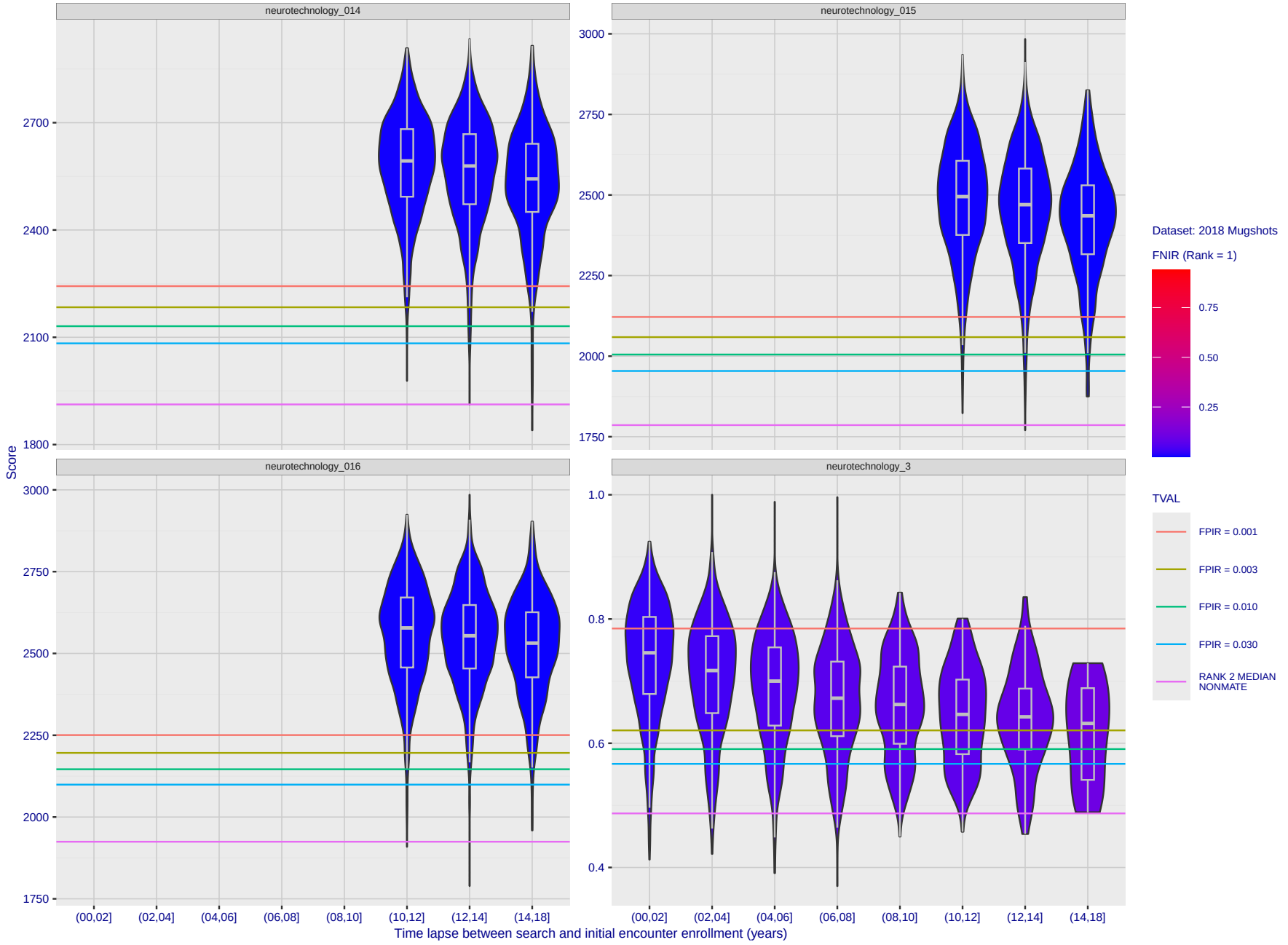

Figure 308: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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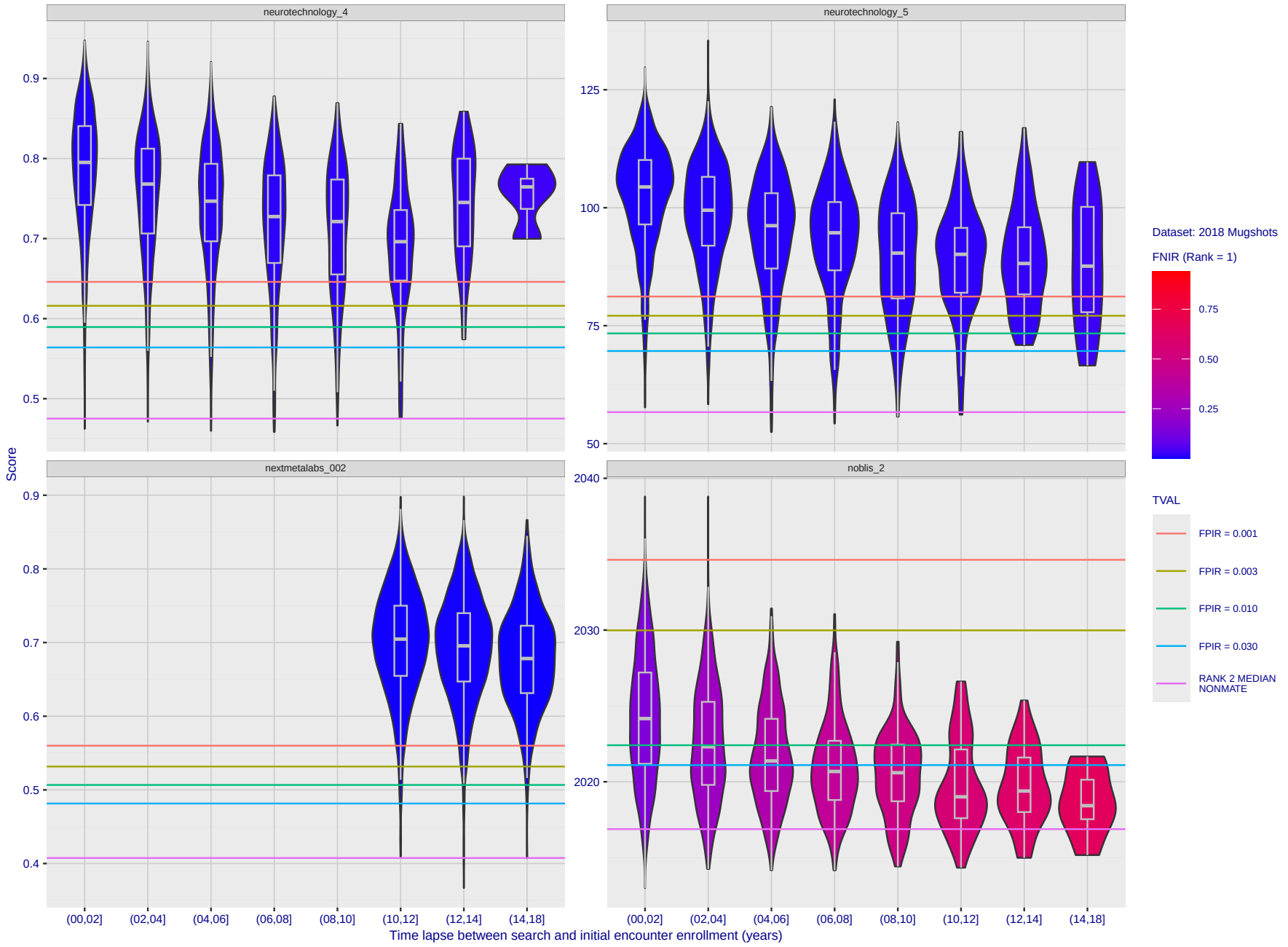

Figure 309: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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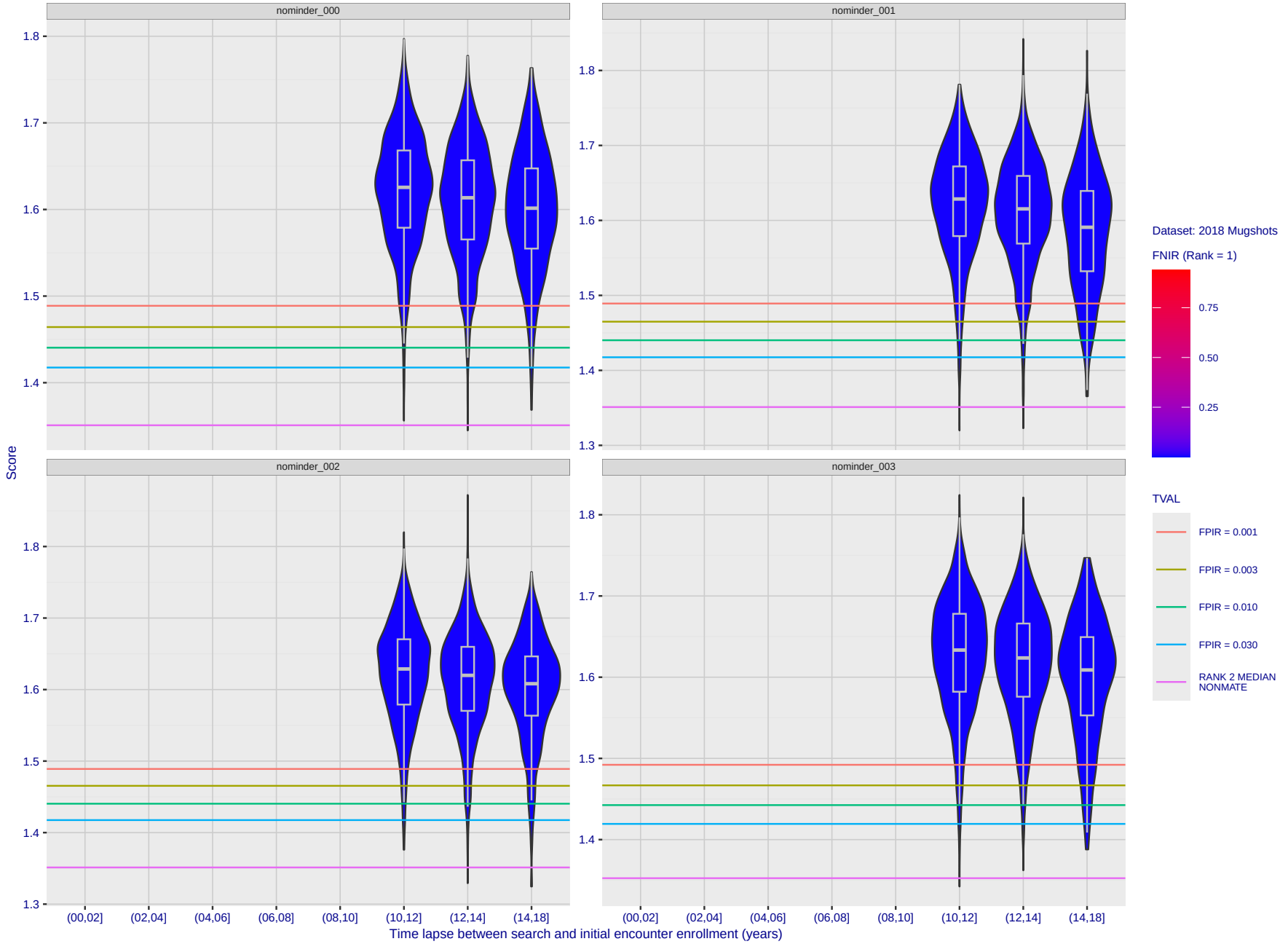

Figure 310: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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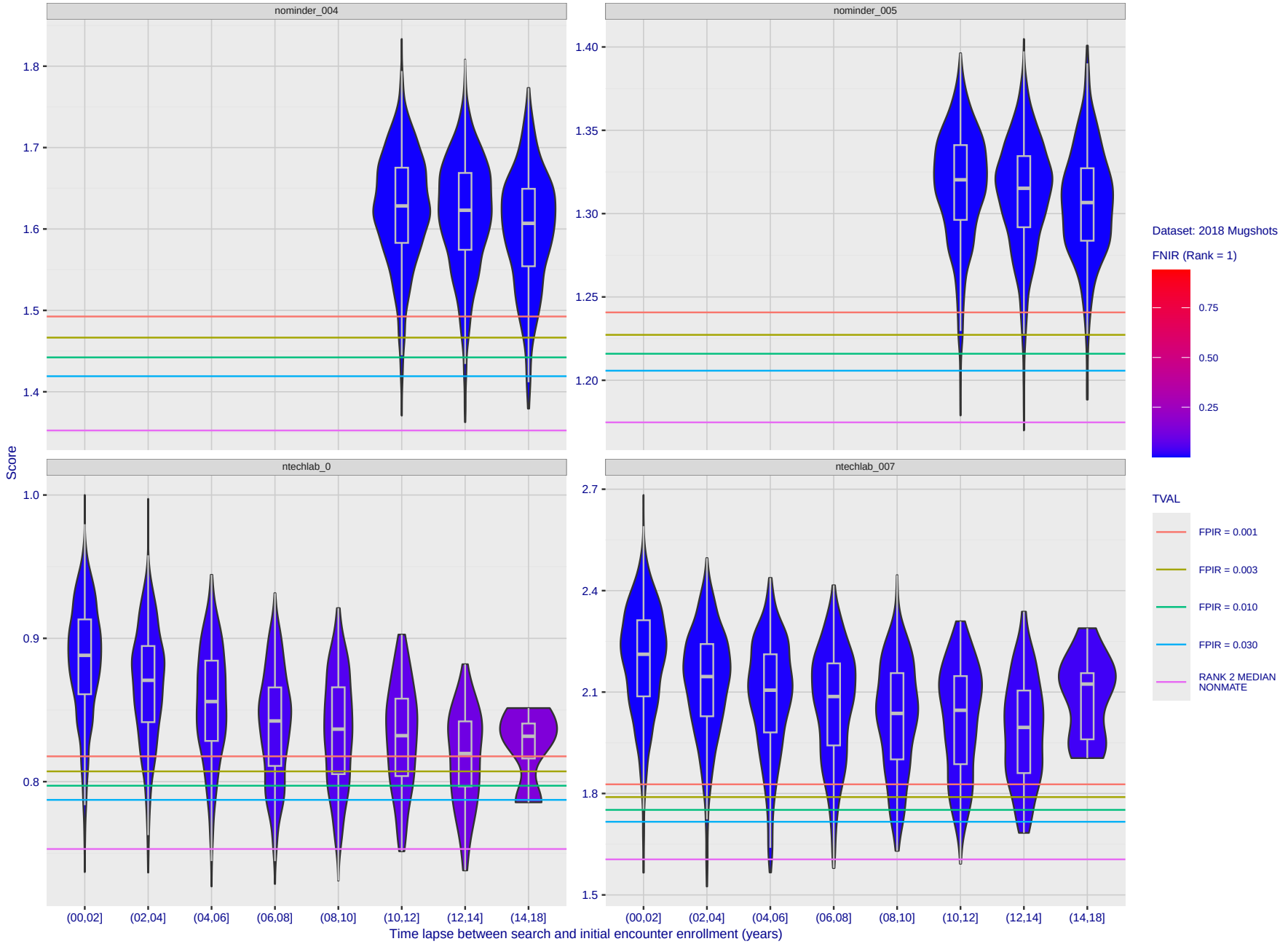

Figure 311: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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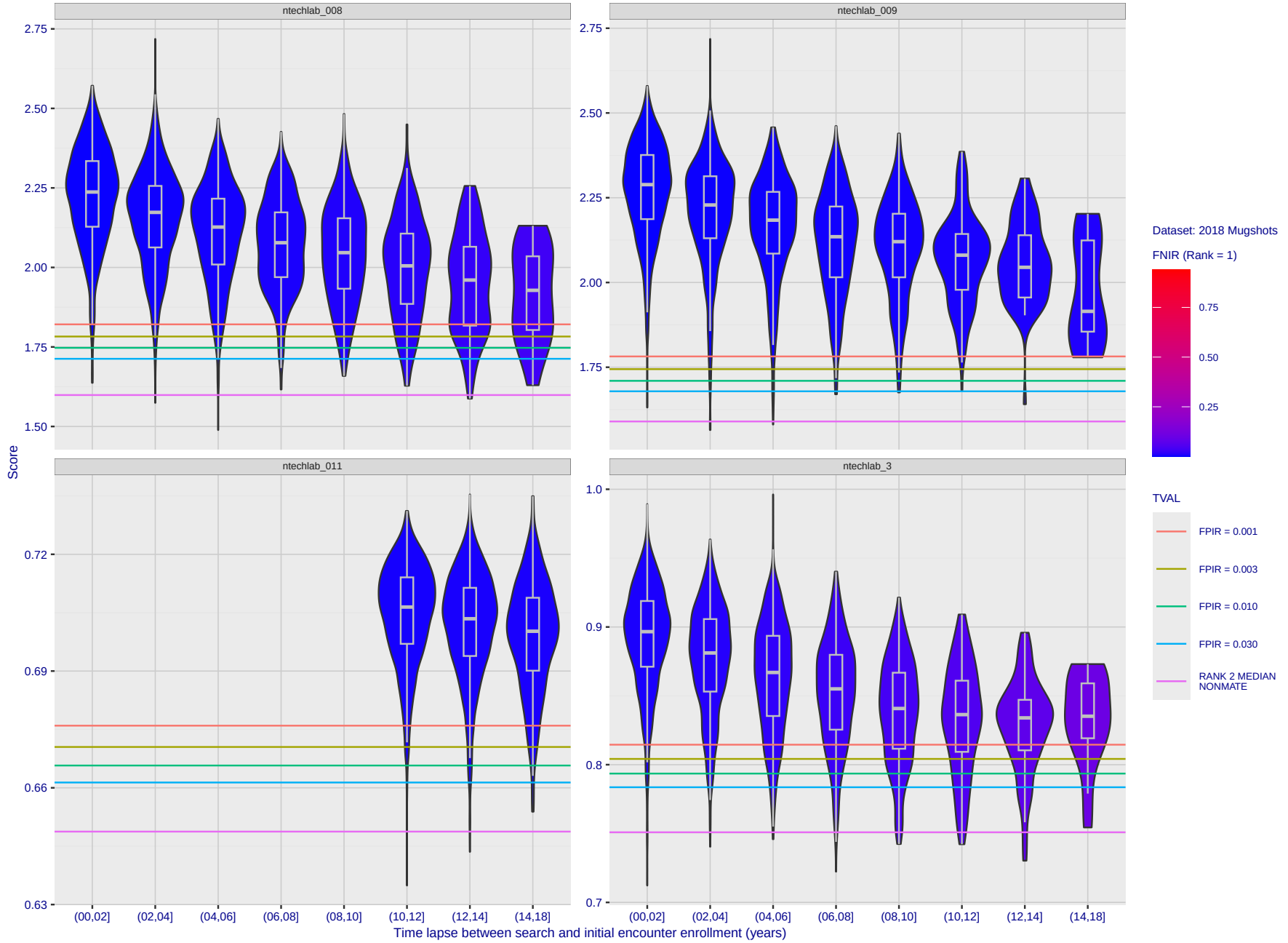

Figure 312: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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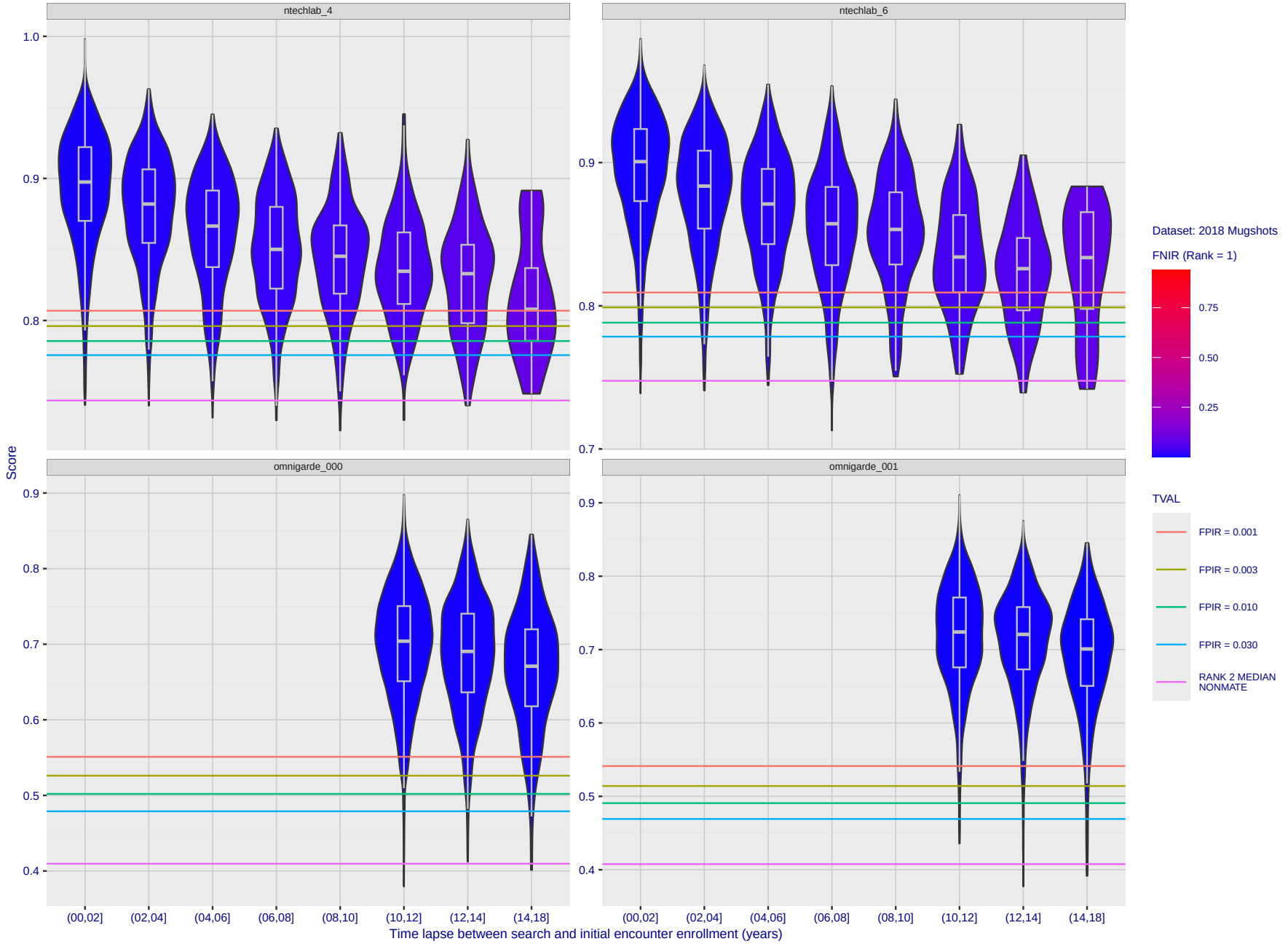

Figure 313: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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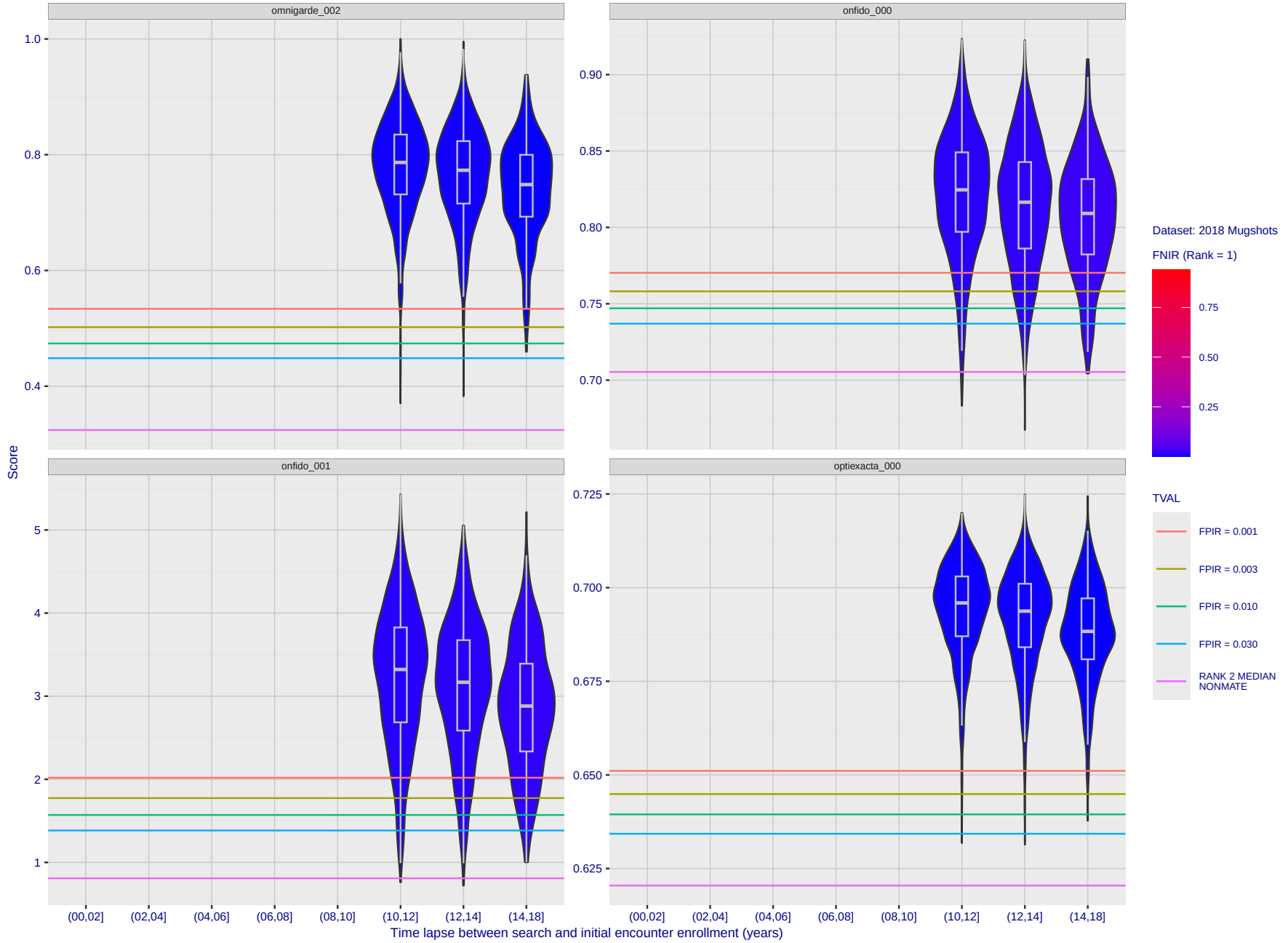

Figure 314: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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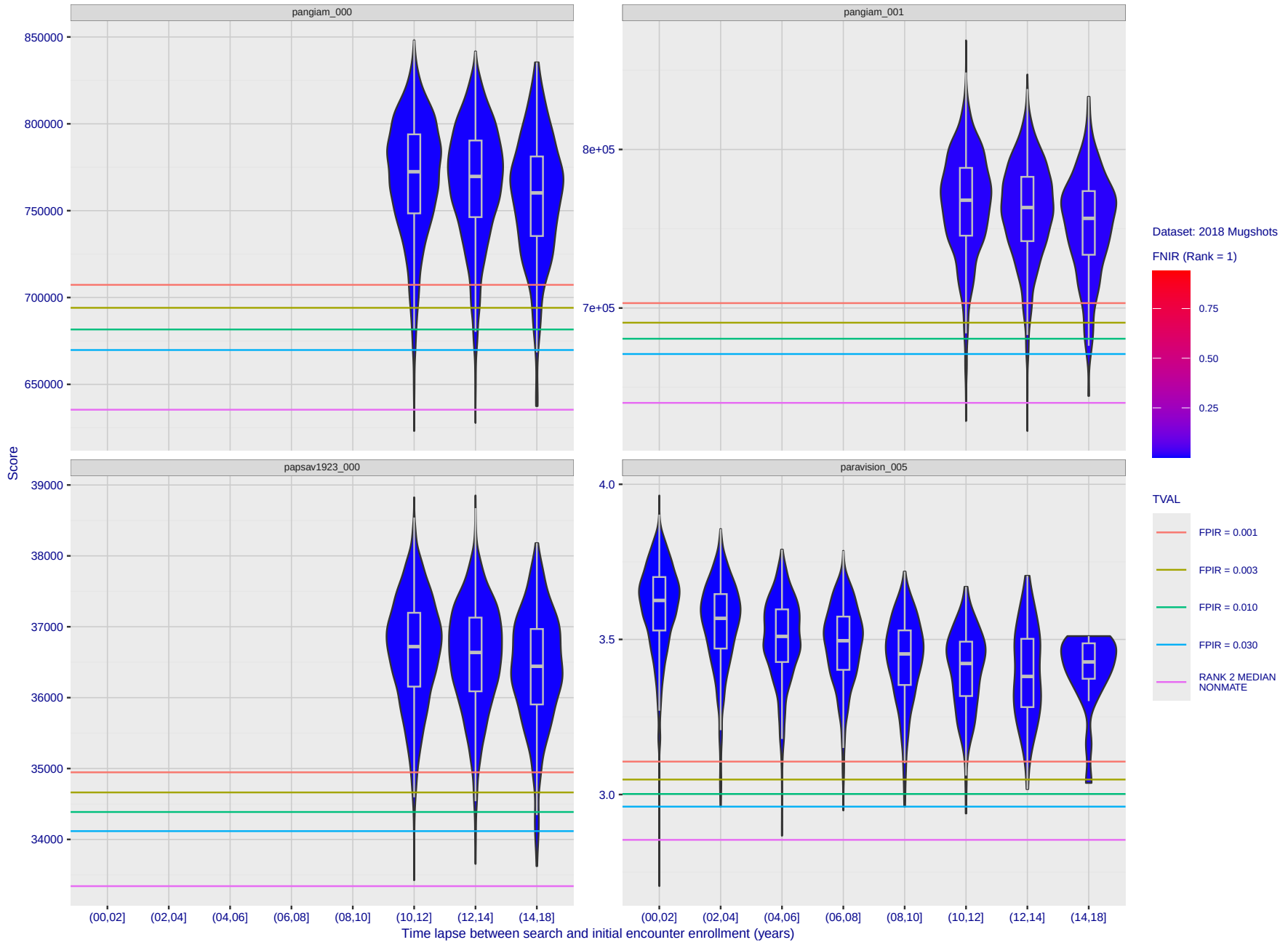

Figure 315: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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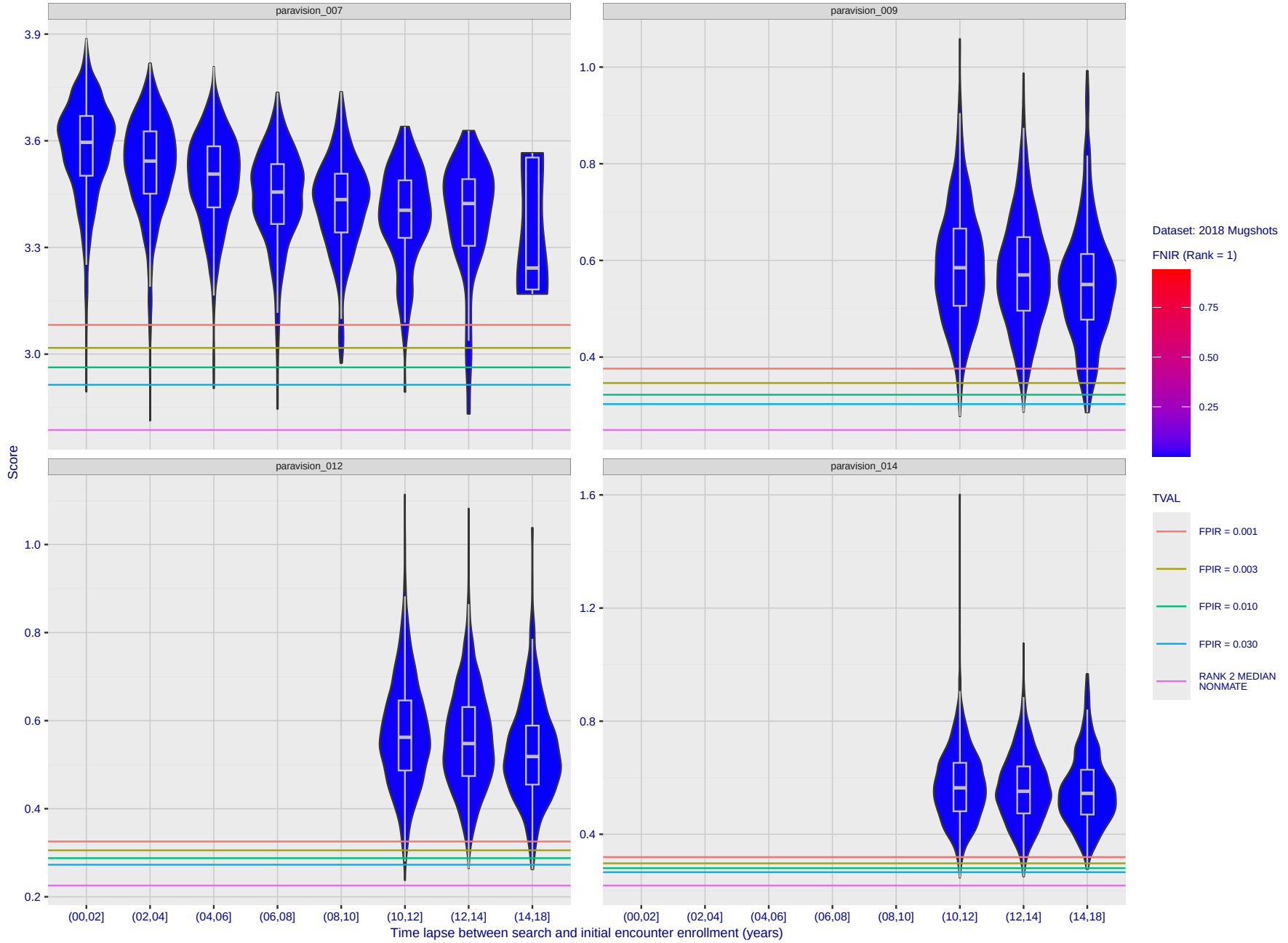

Figure 316: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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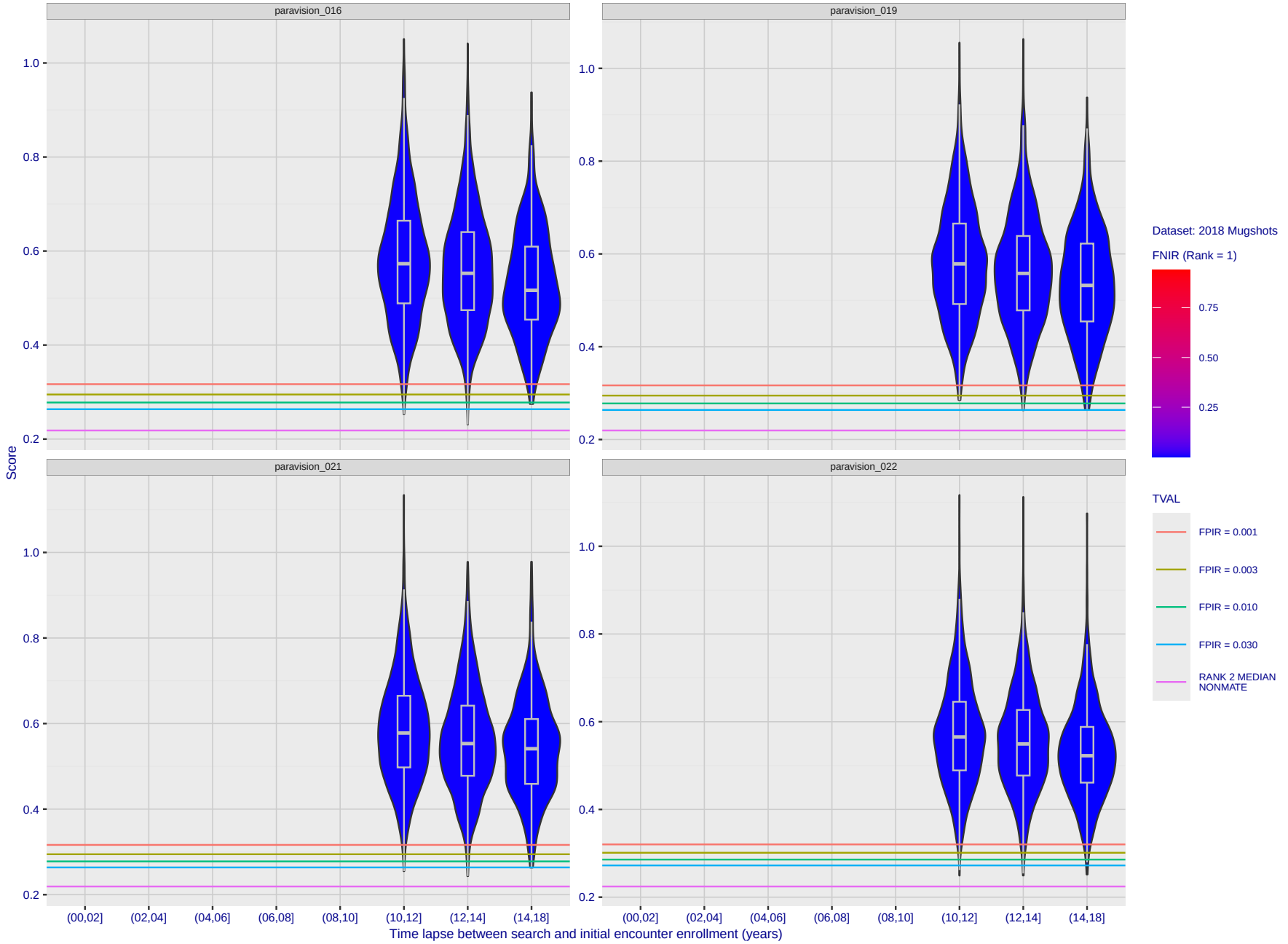

Figure 317: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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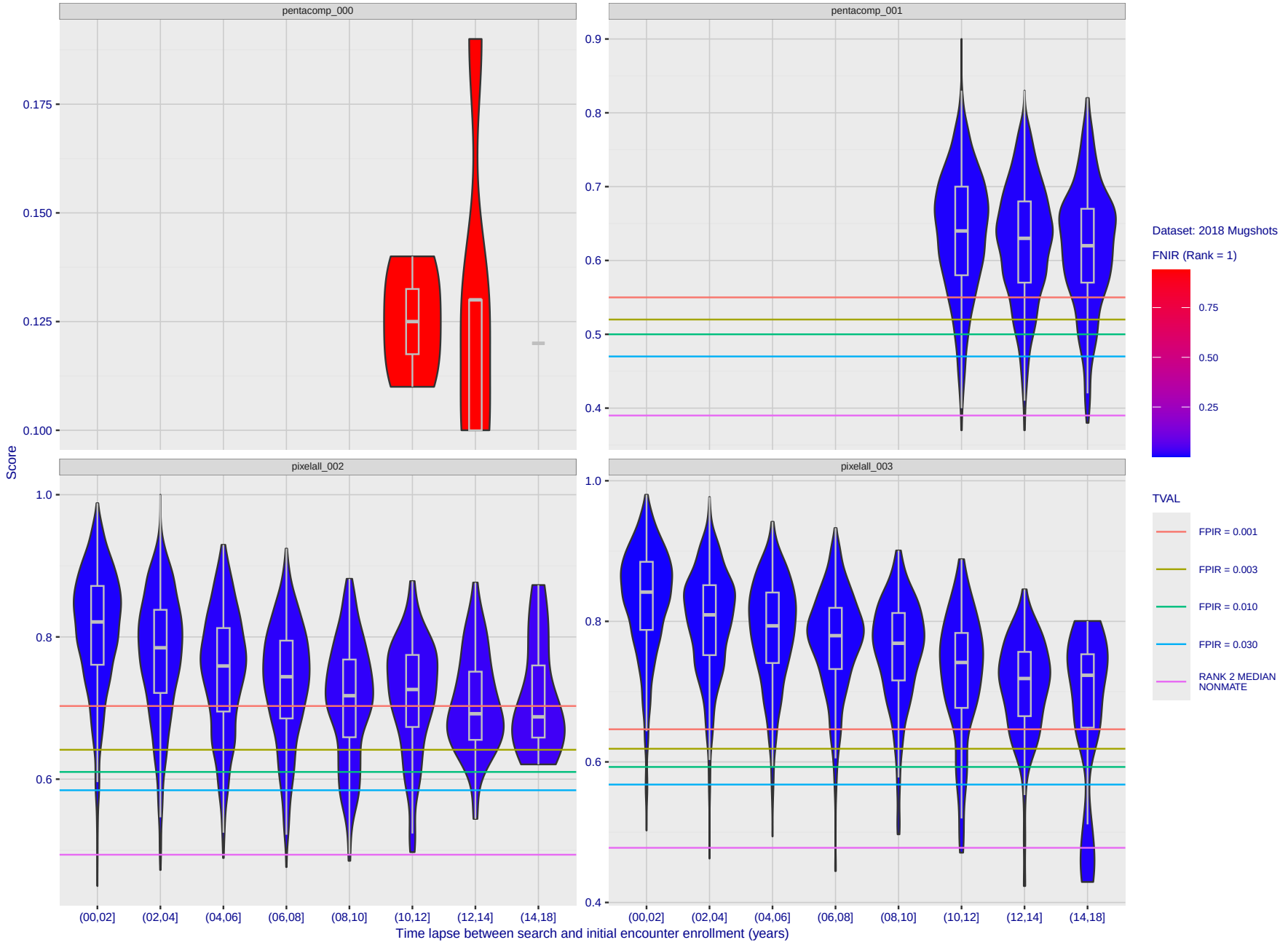

Figure 318: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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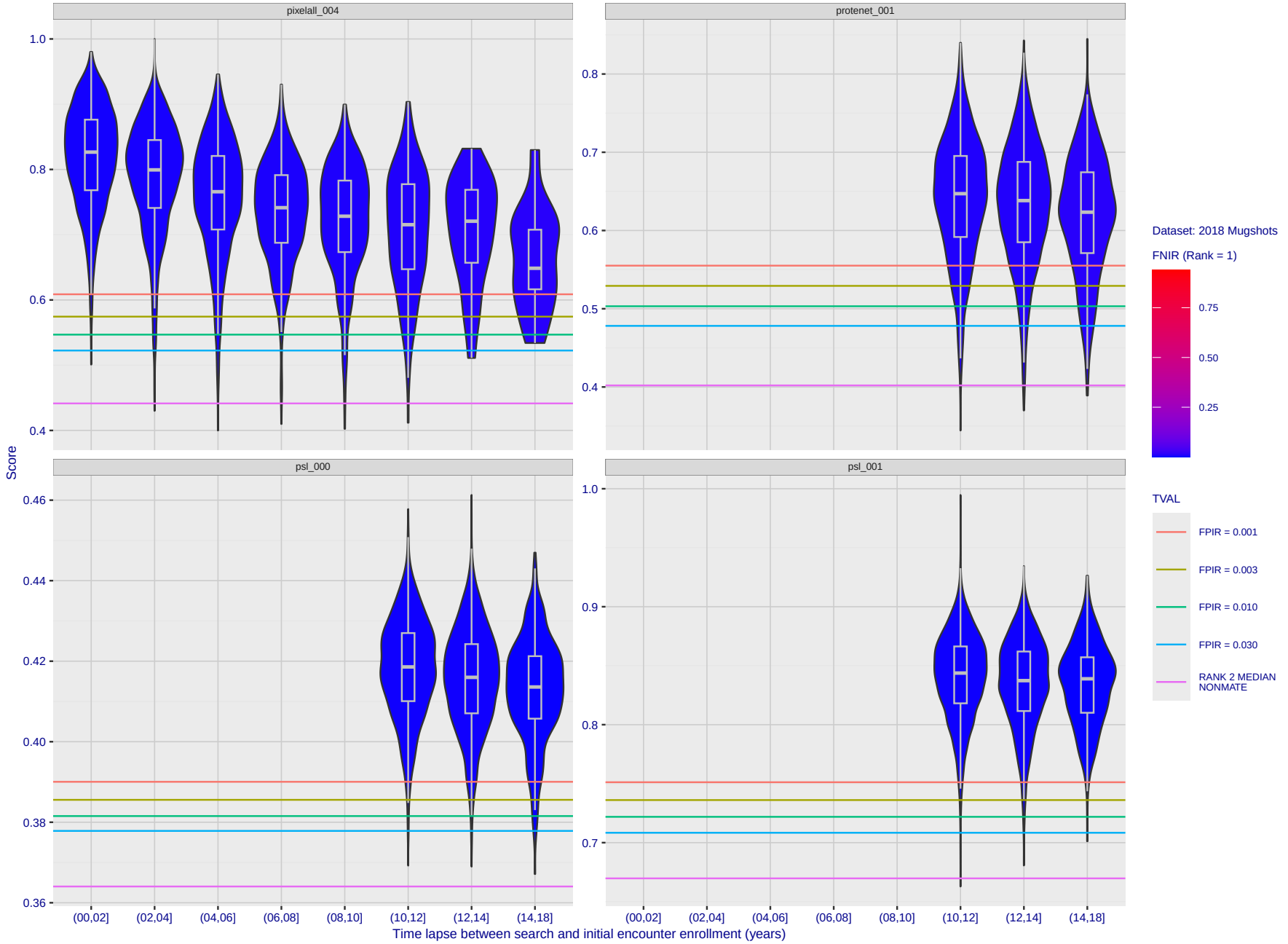

Figure 319: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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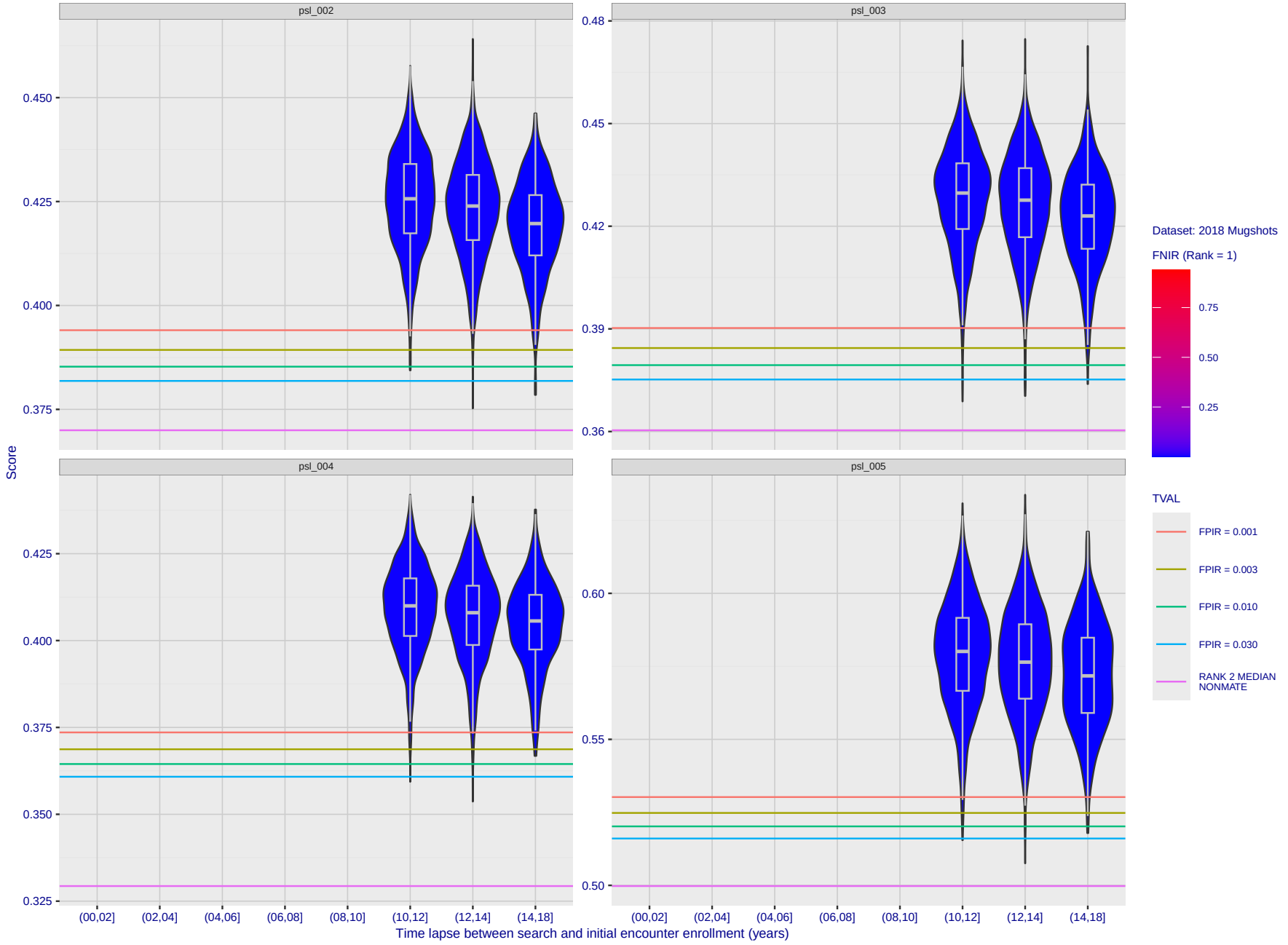

Figure 320: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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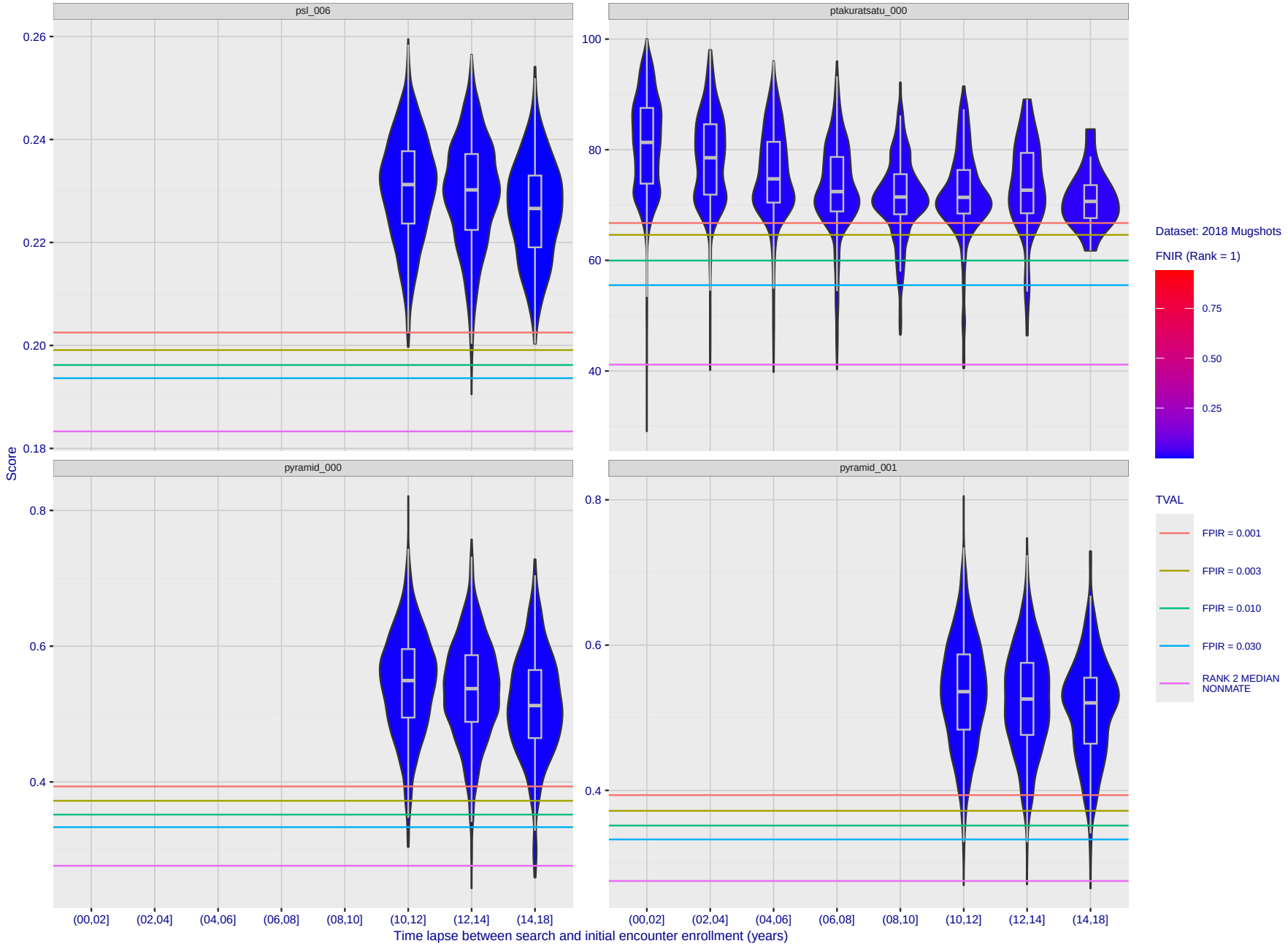

Figure 321: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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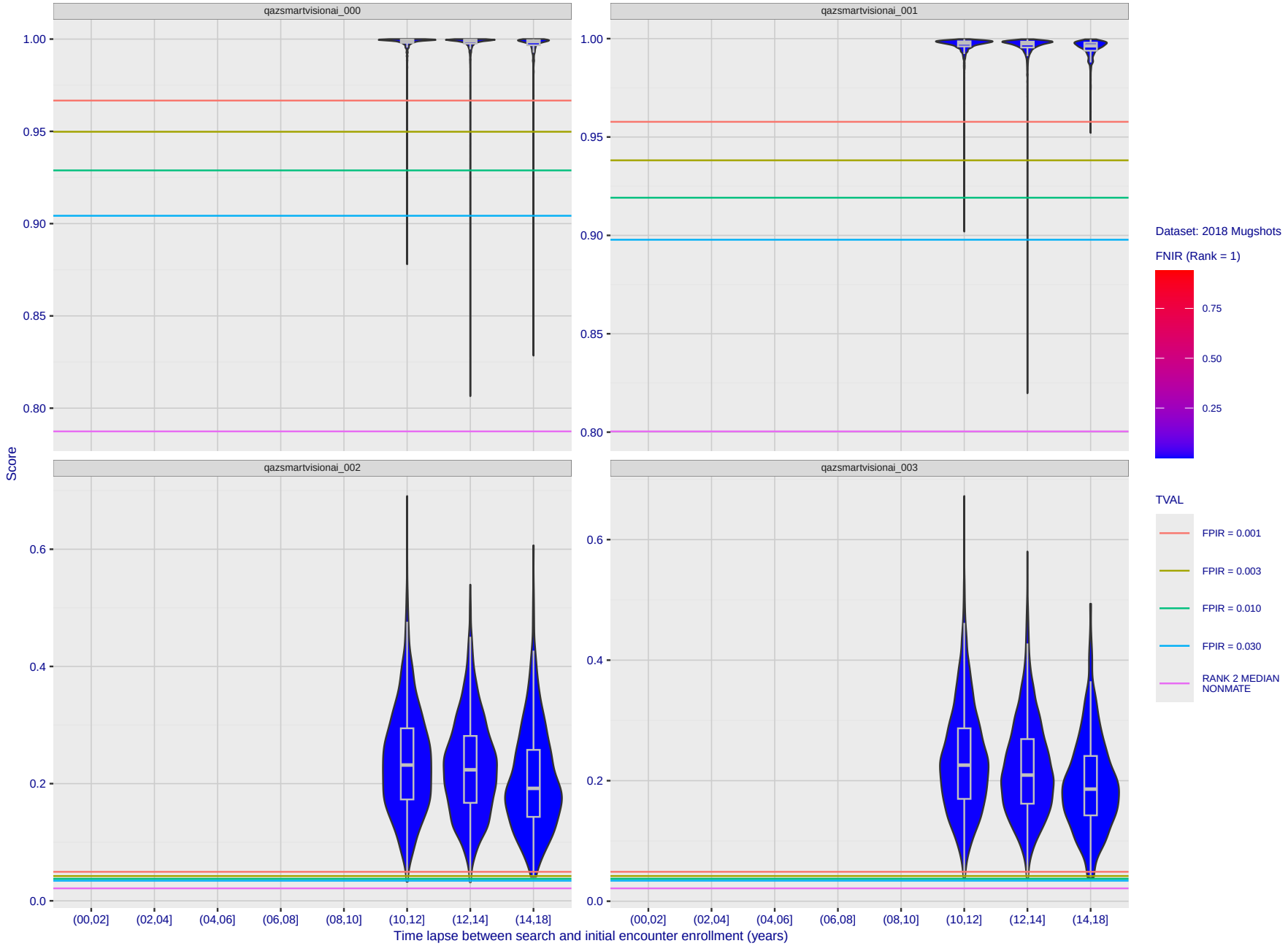

Figure 322: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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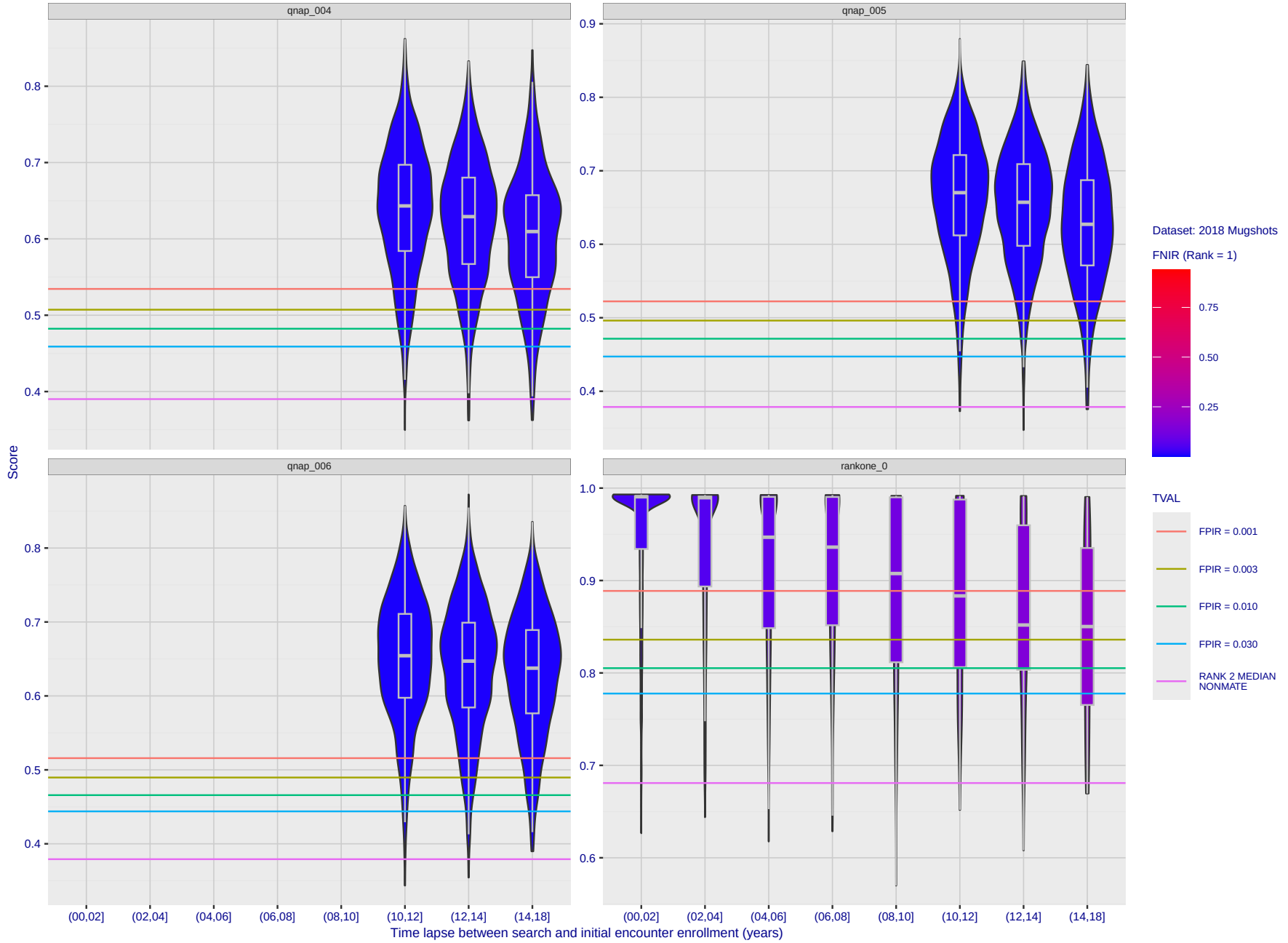

Figure 323: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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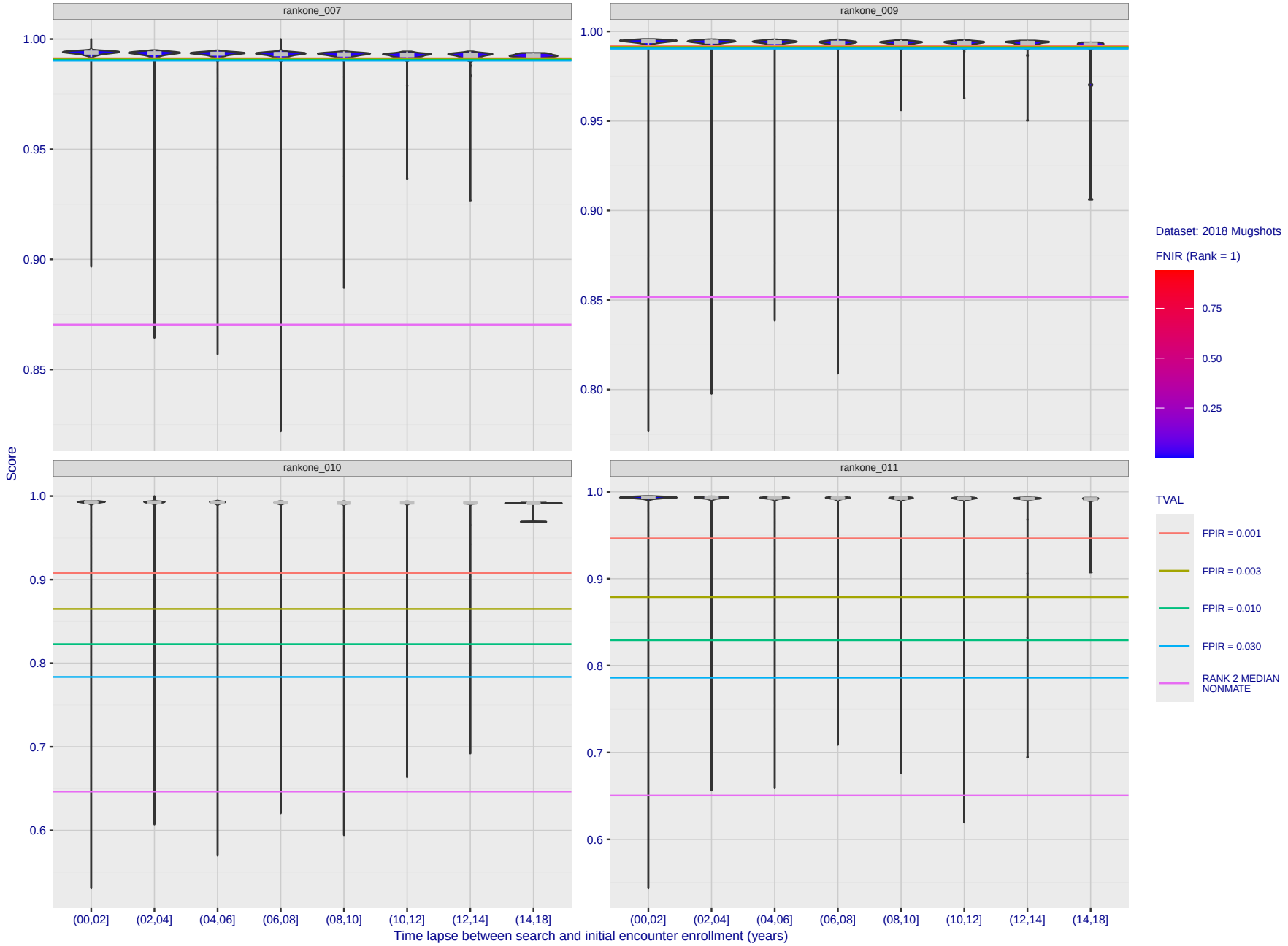

Figure 324: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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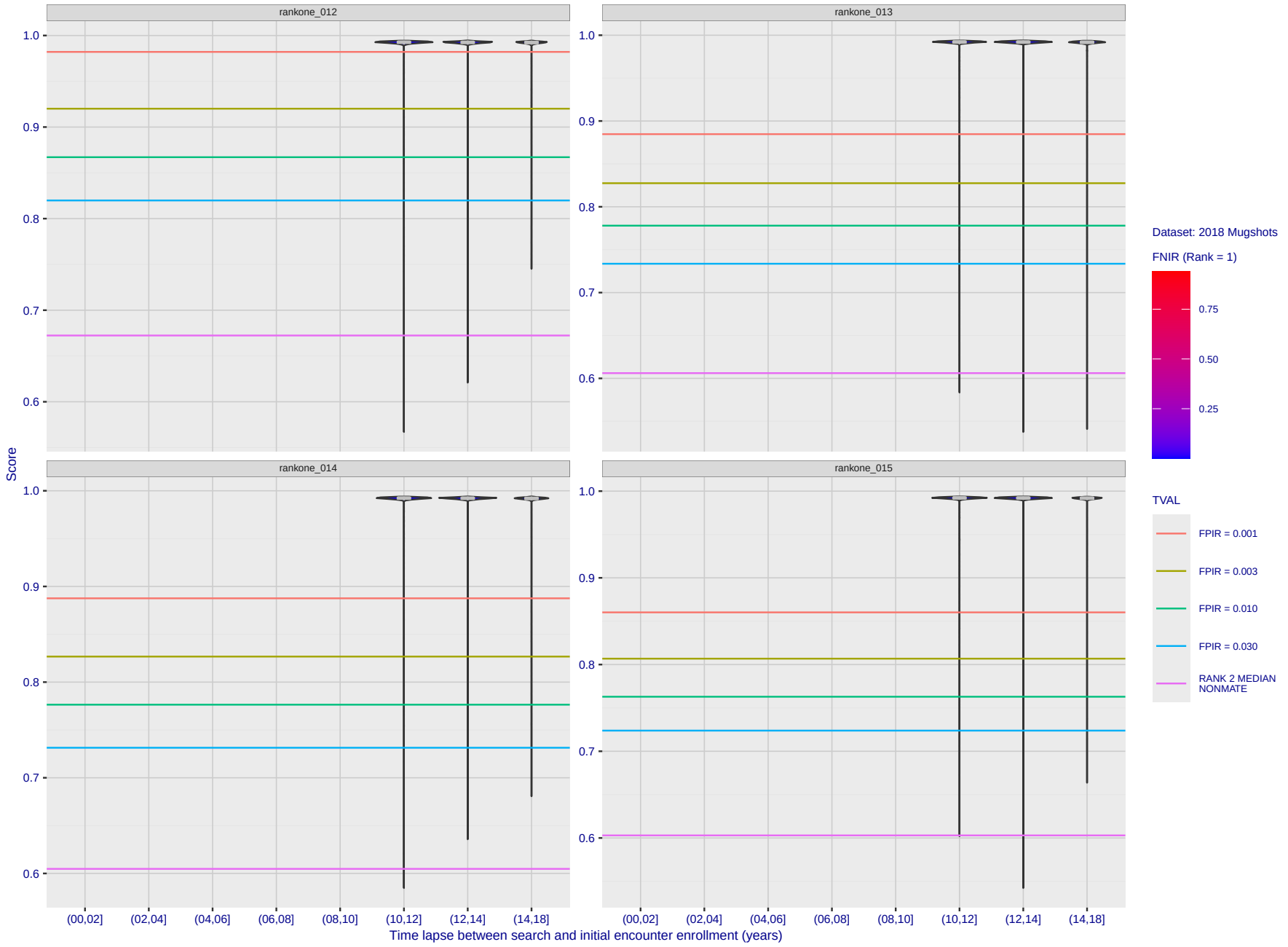

Figure 325: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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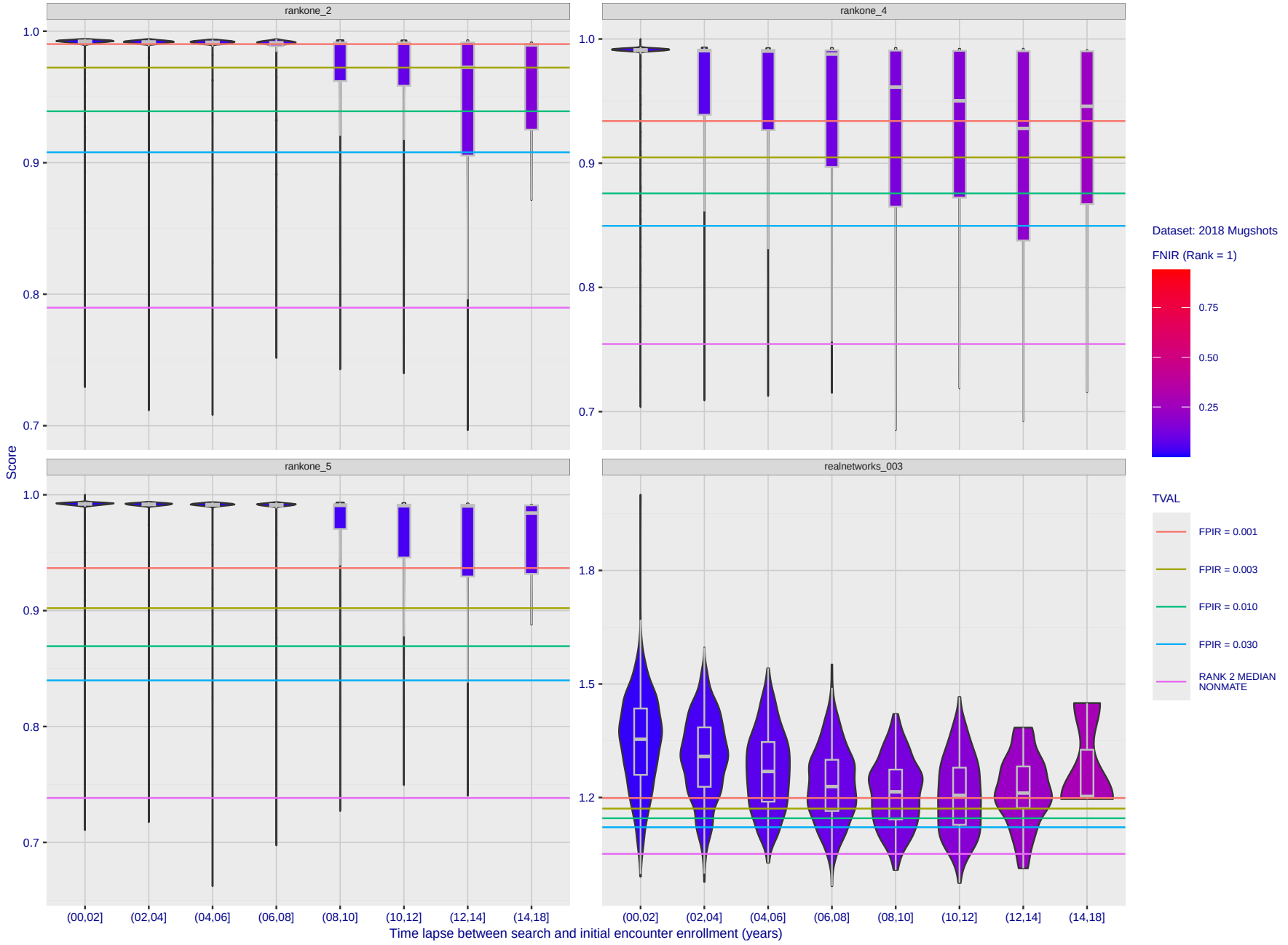

Figure 326: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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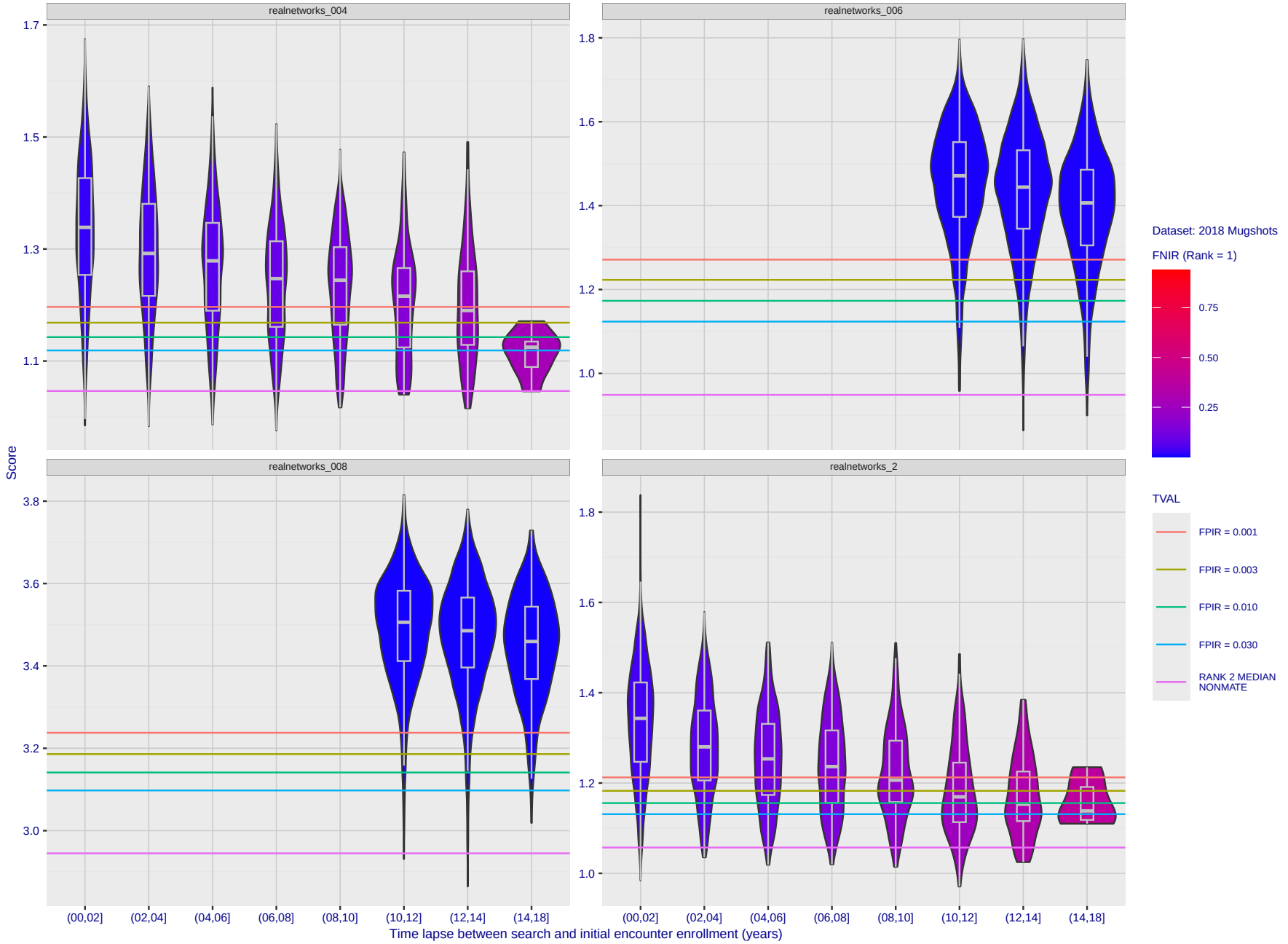

Figure 327: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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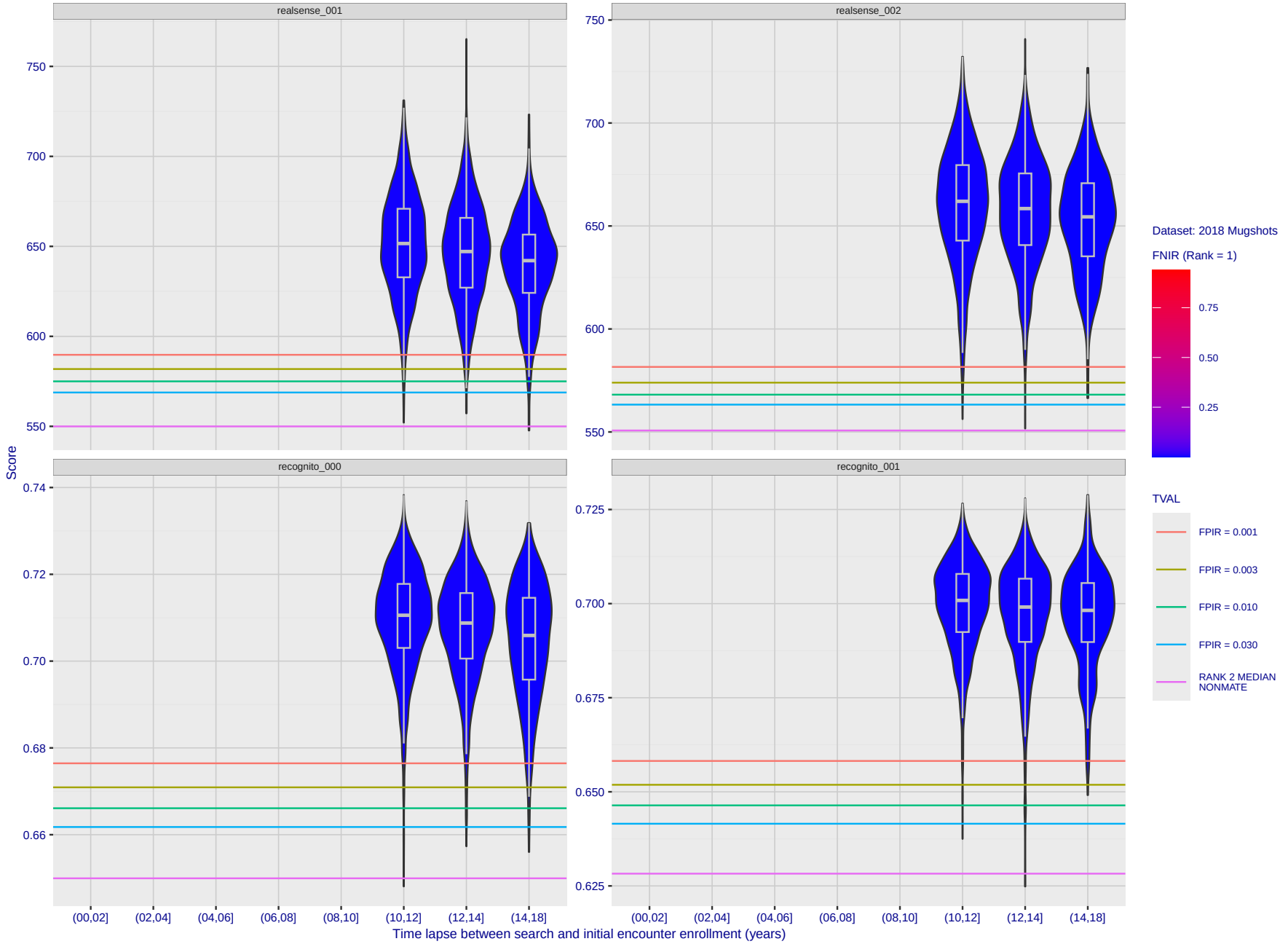

Figure 328: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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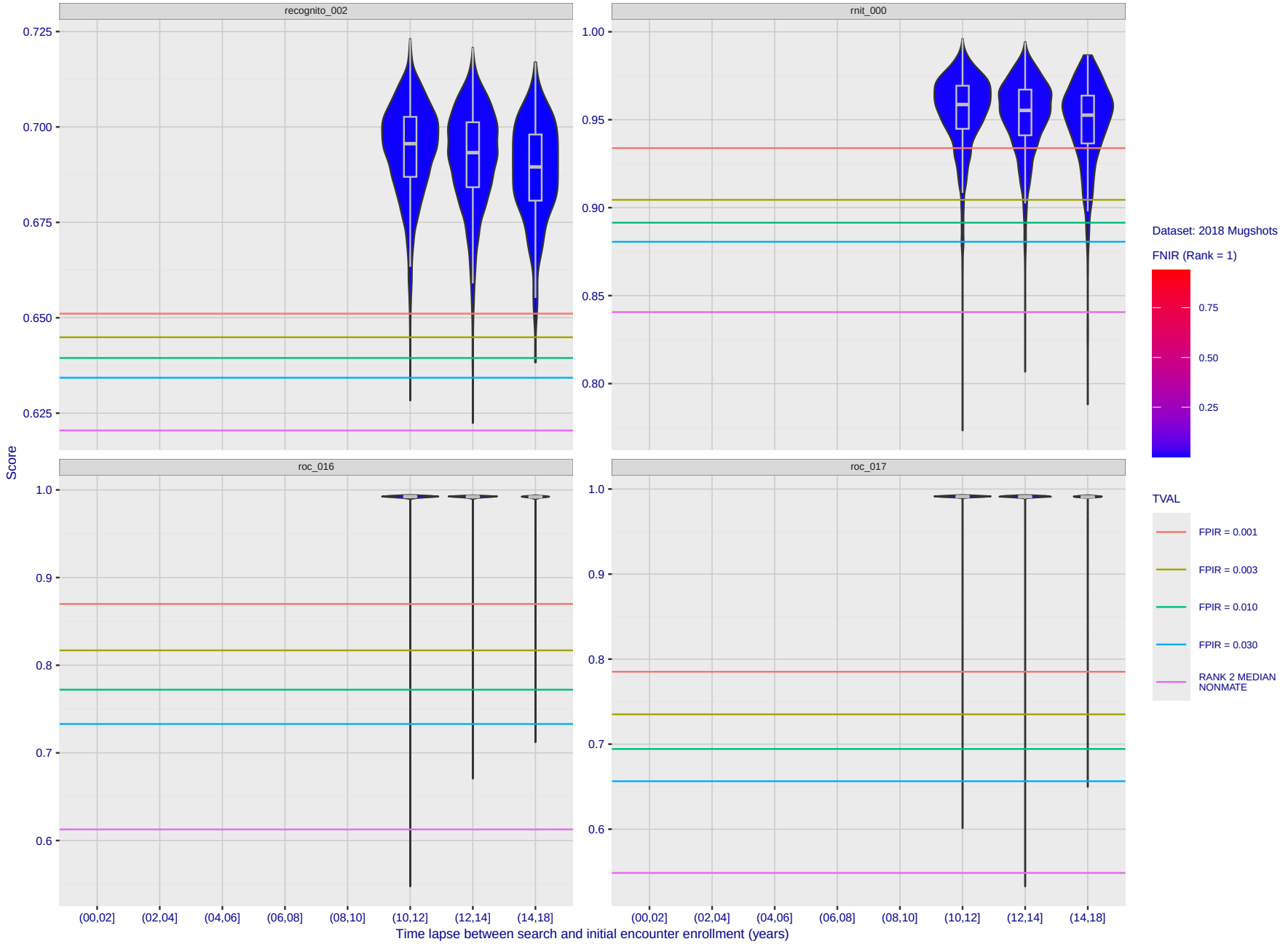

Figure 329: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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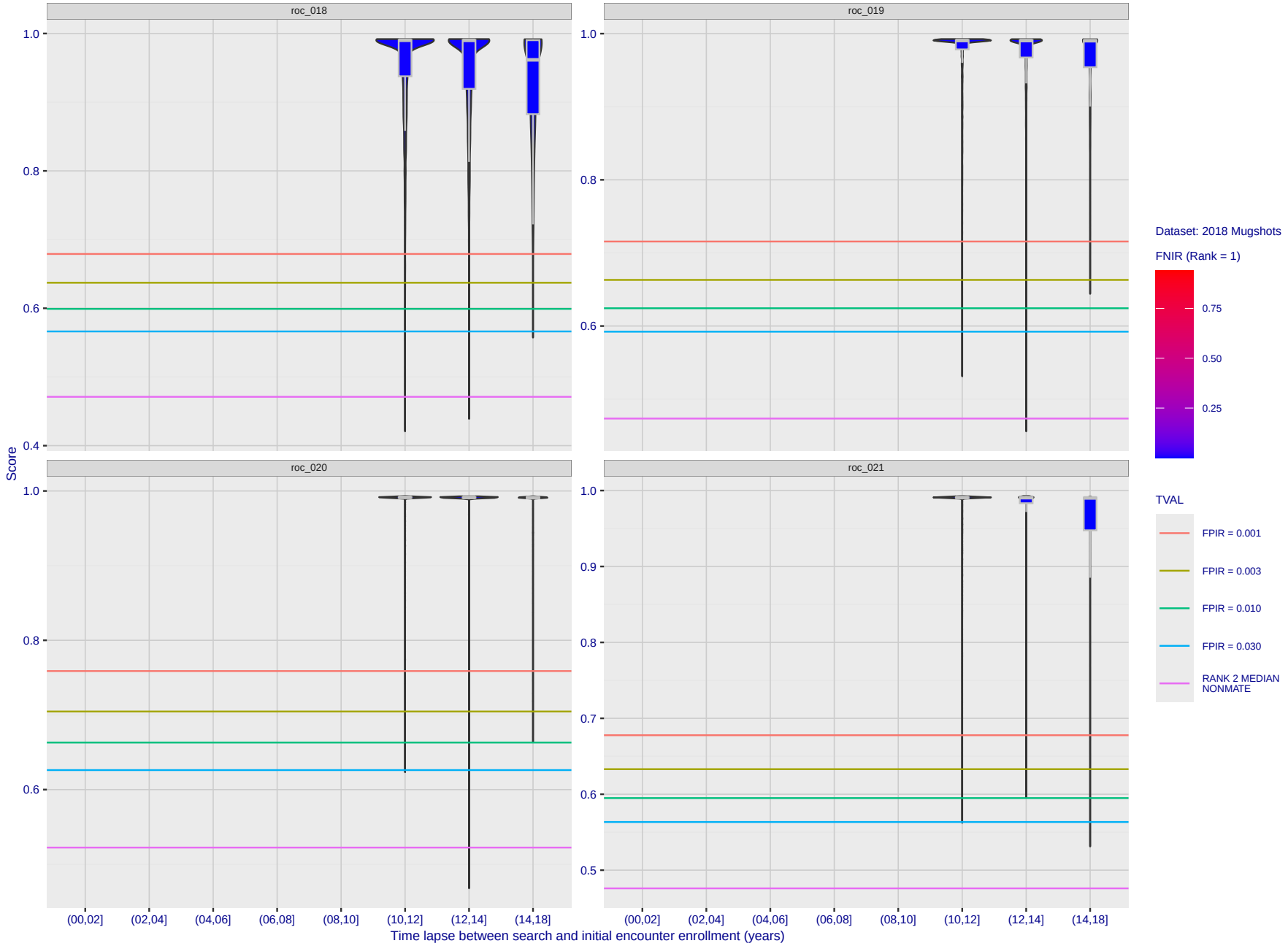

Figure 330: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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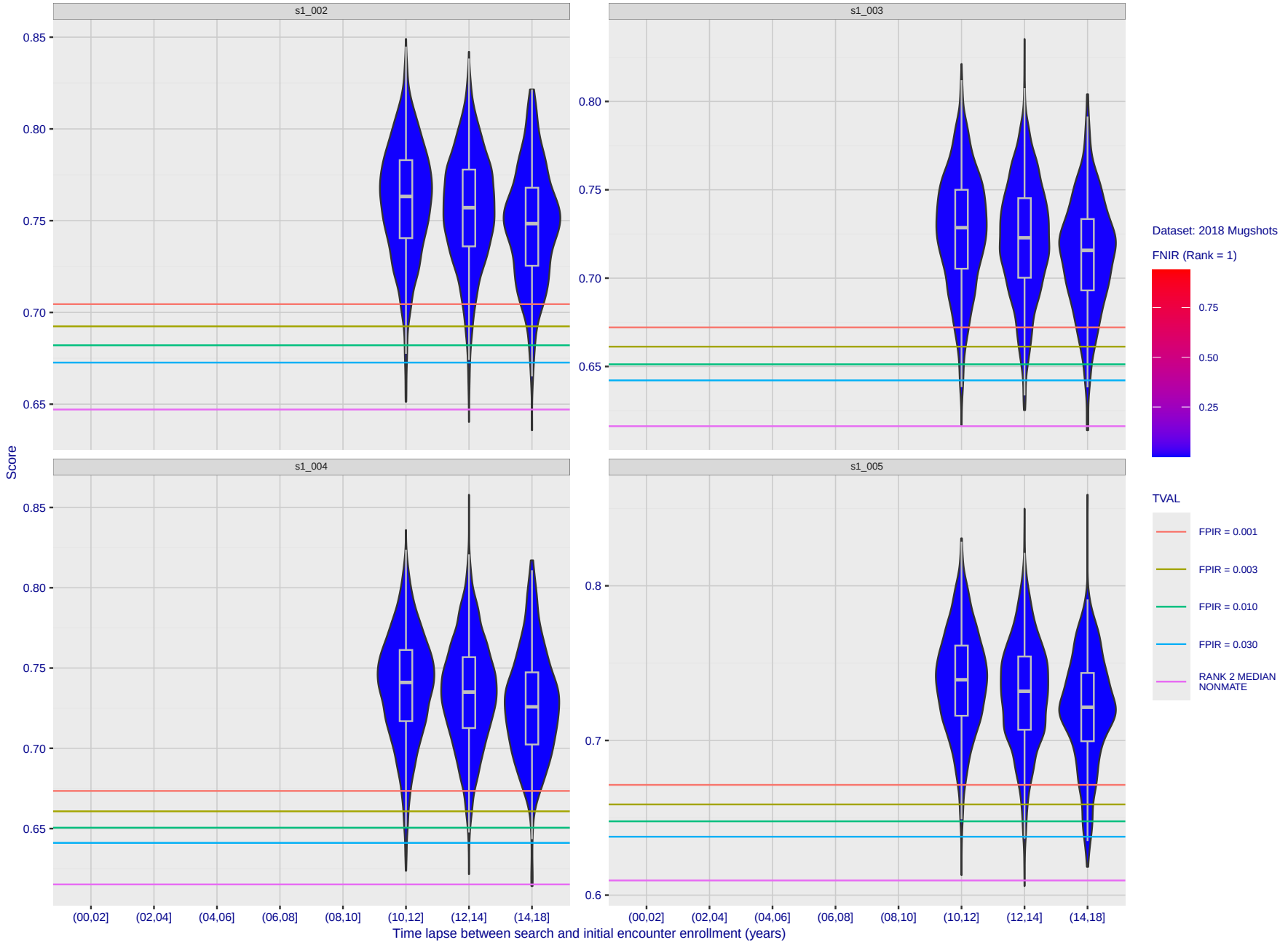

Figure 331: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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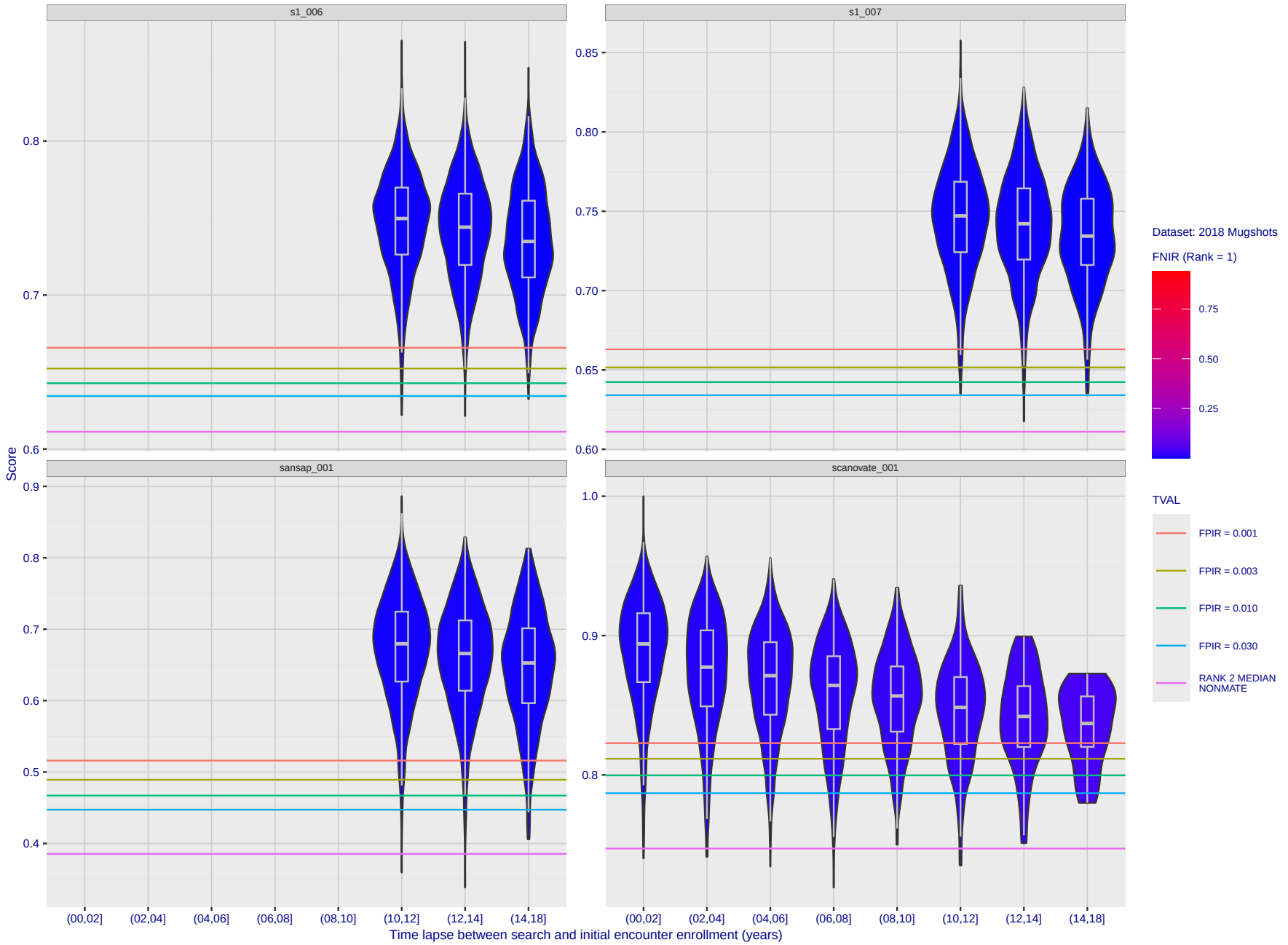

Figure 332: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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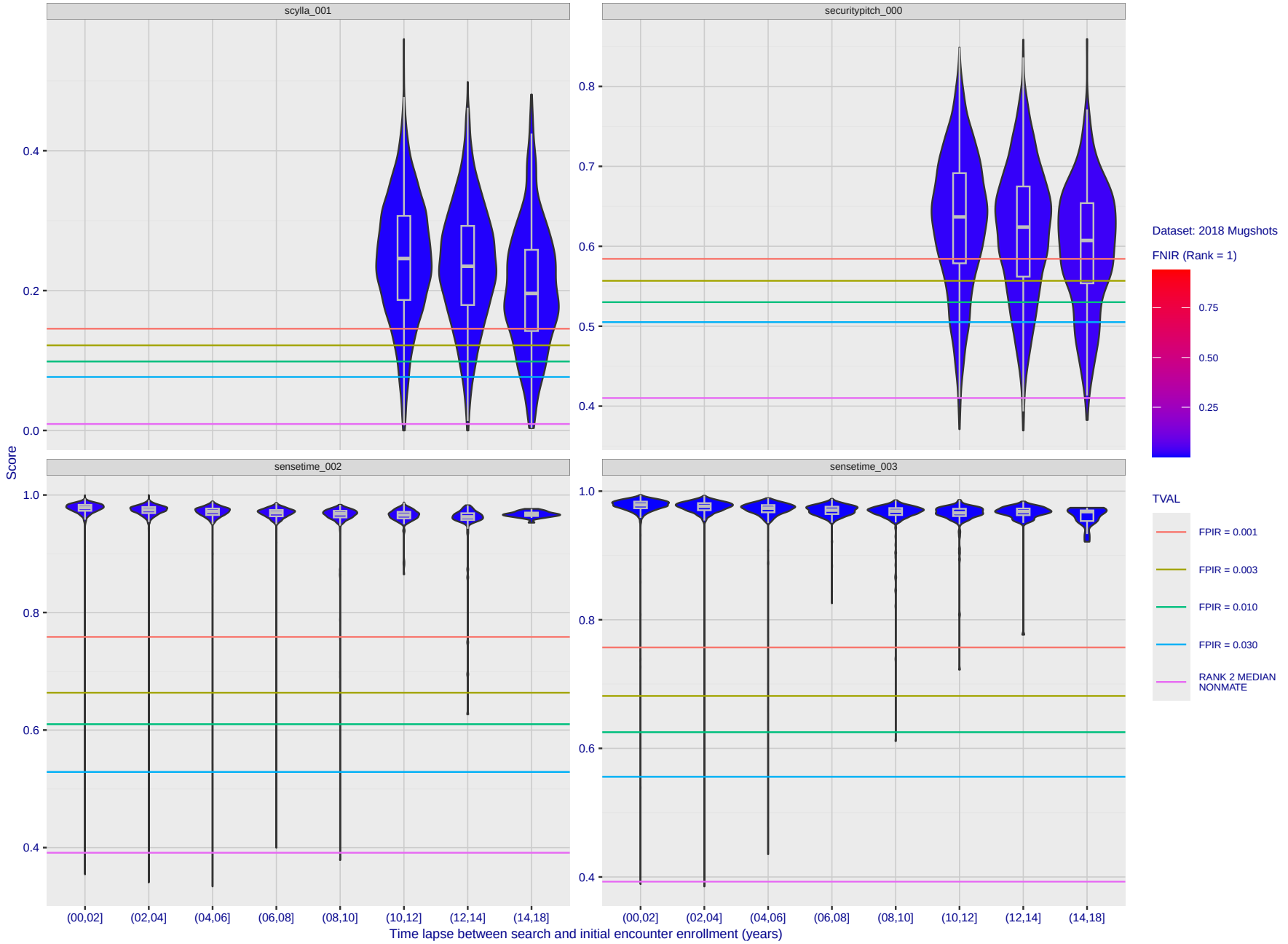

Figure 333: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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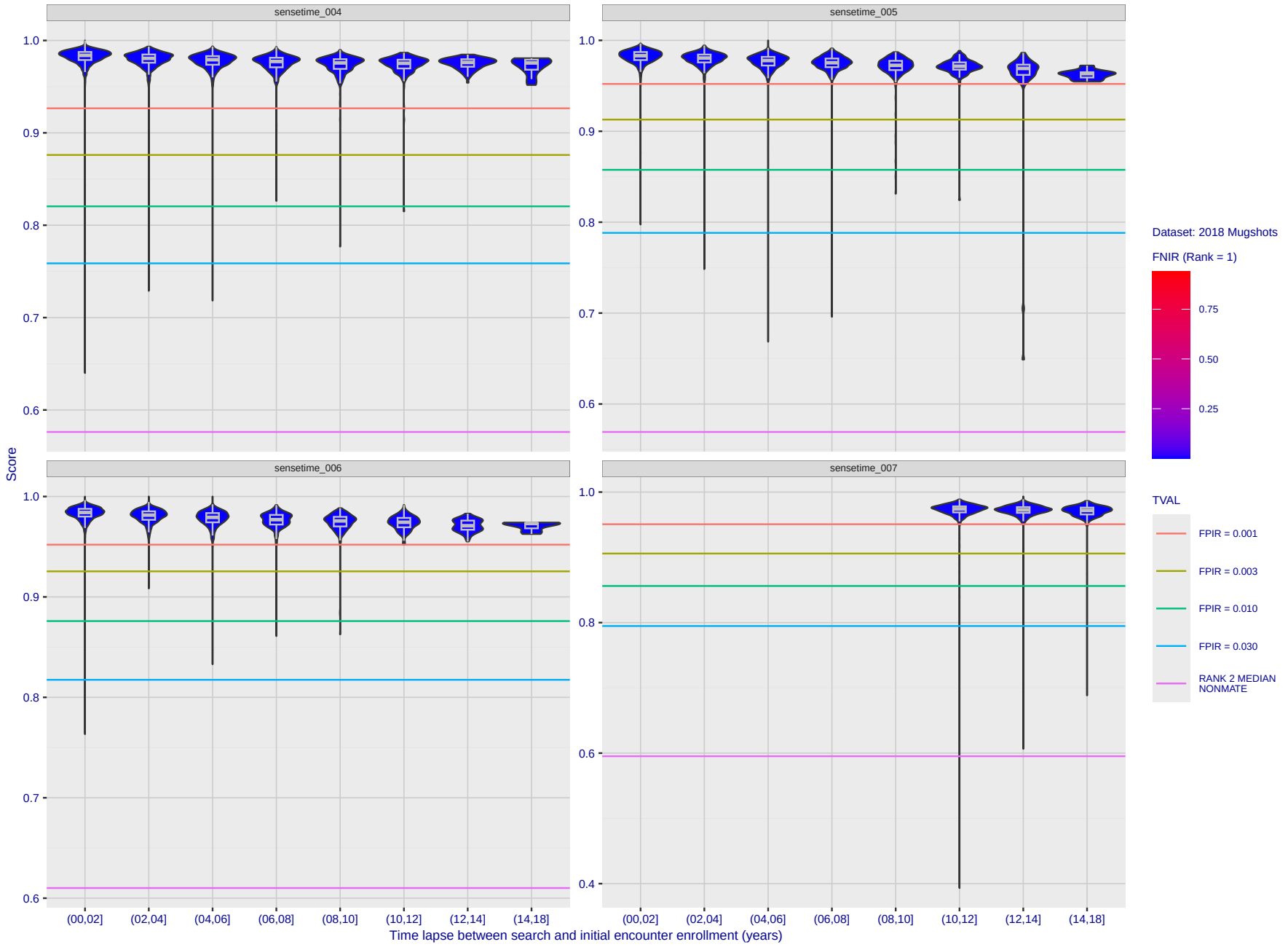

Figure 334: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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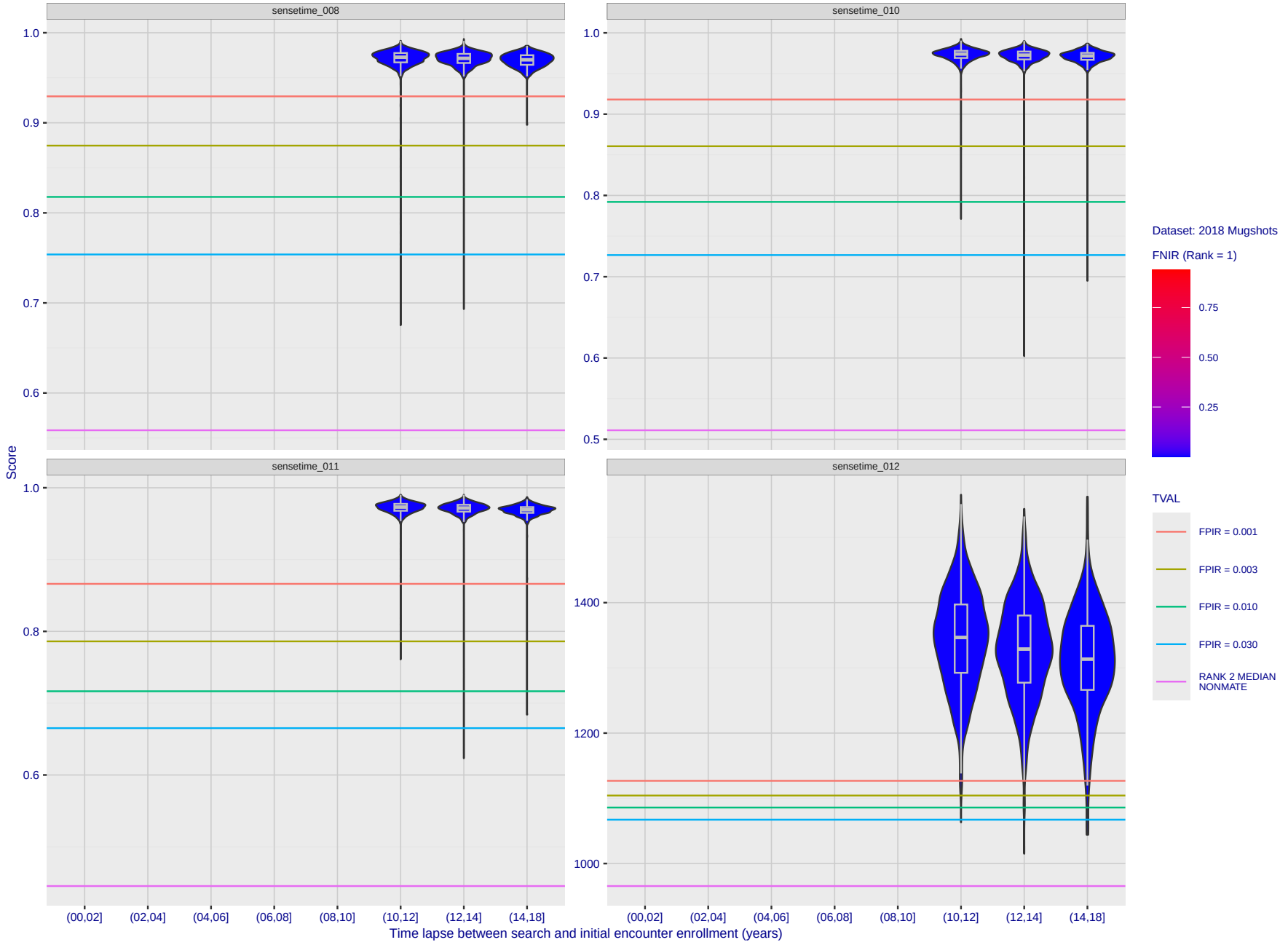

Figure 335: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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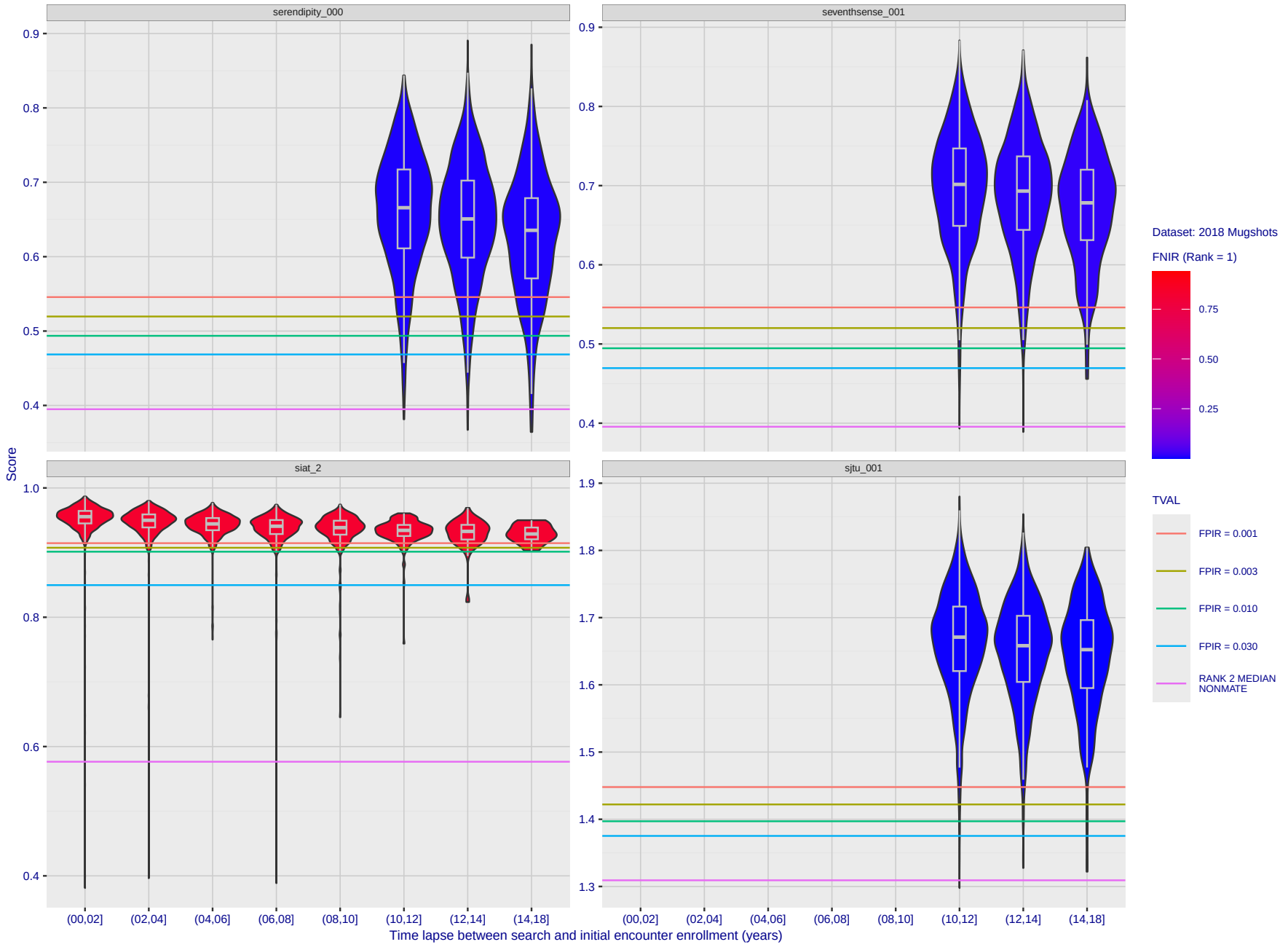

Figure 336: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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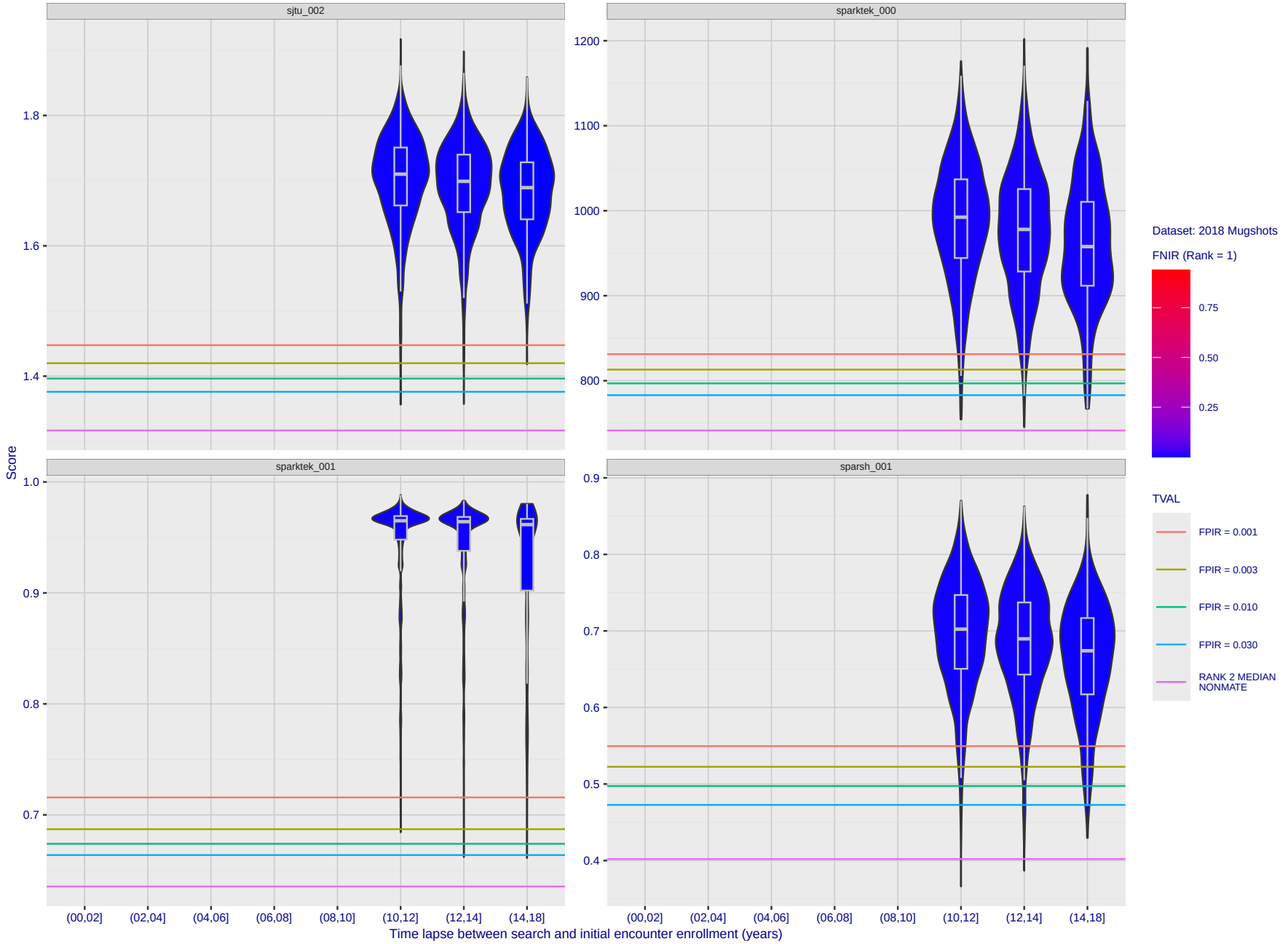

Figure 337: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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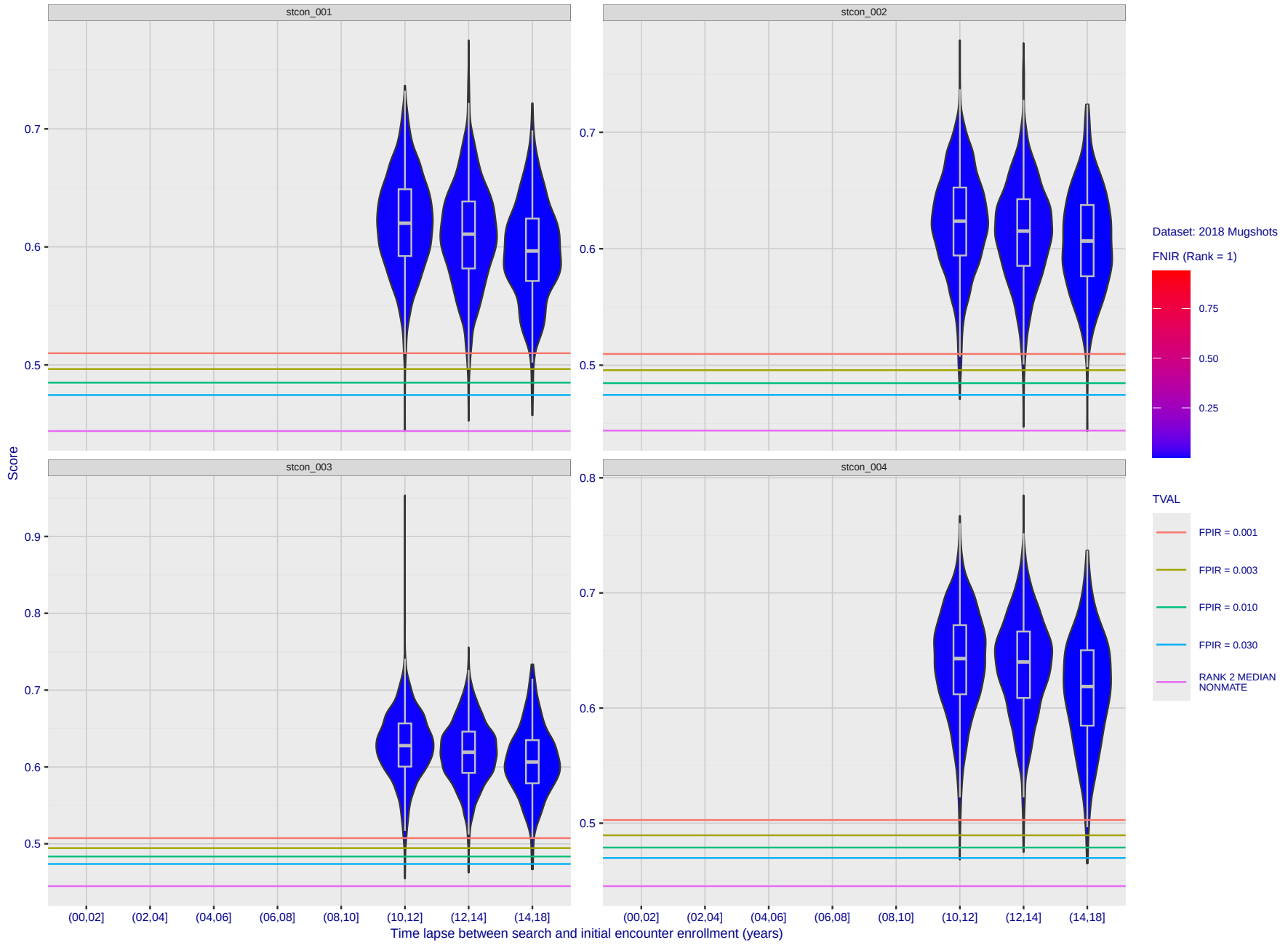

Figure 338: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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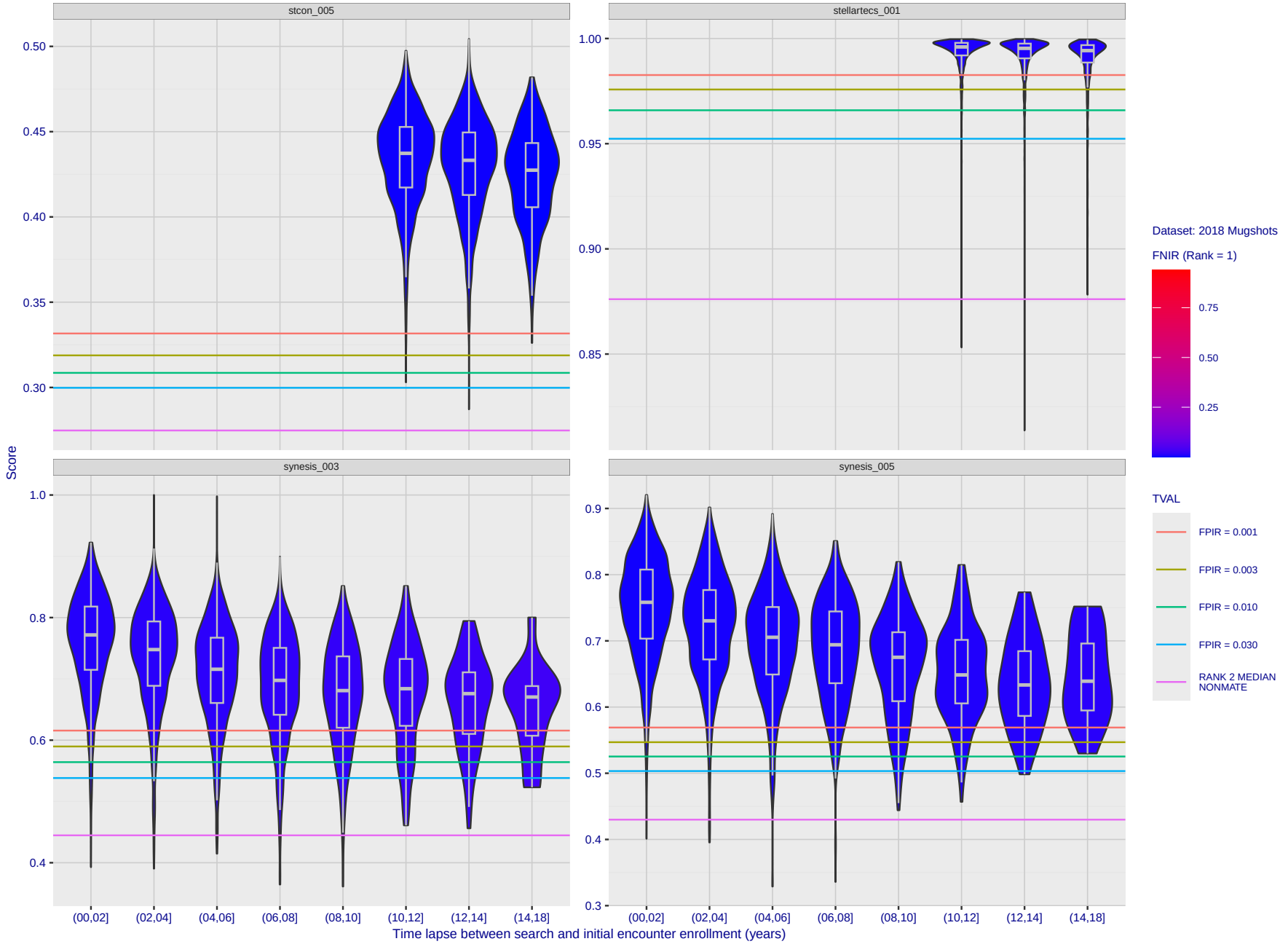

Figure 339: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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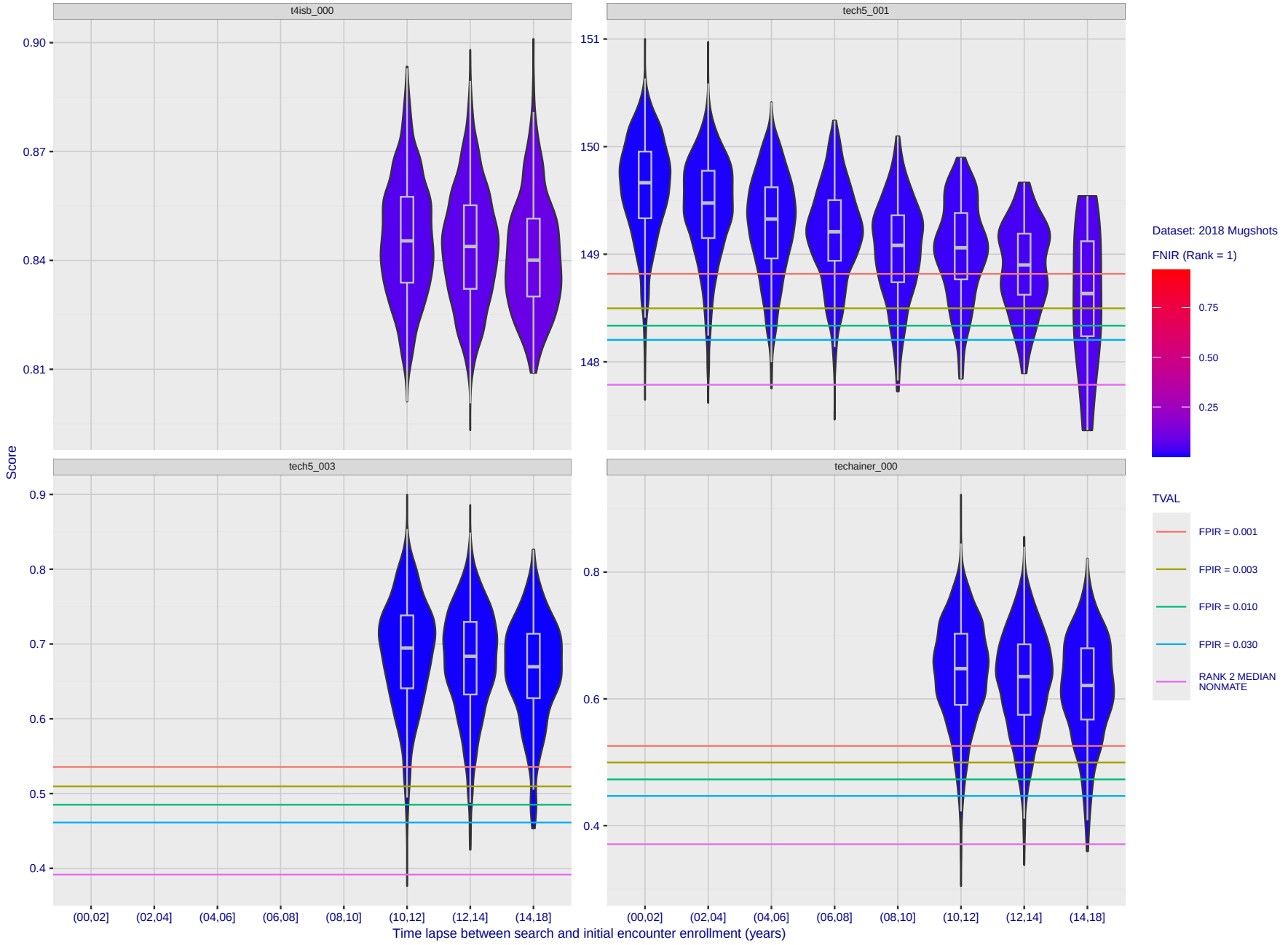

Figure 340: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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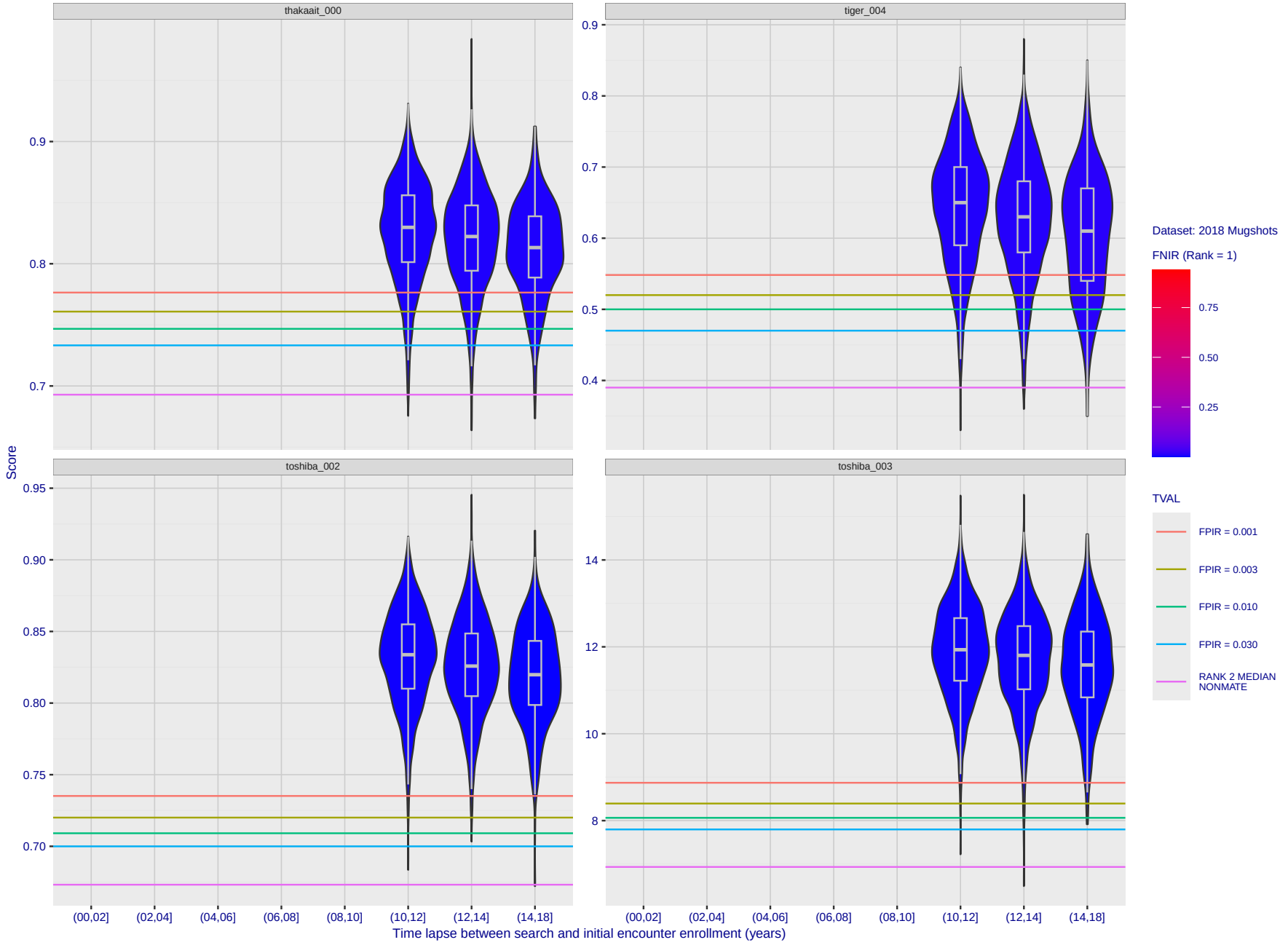

Figure 341: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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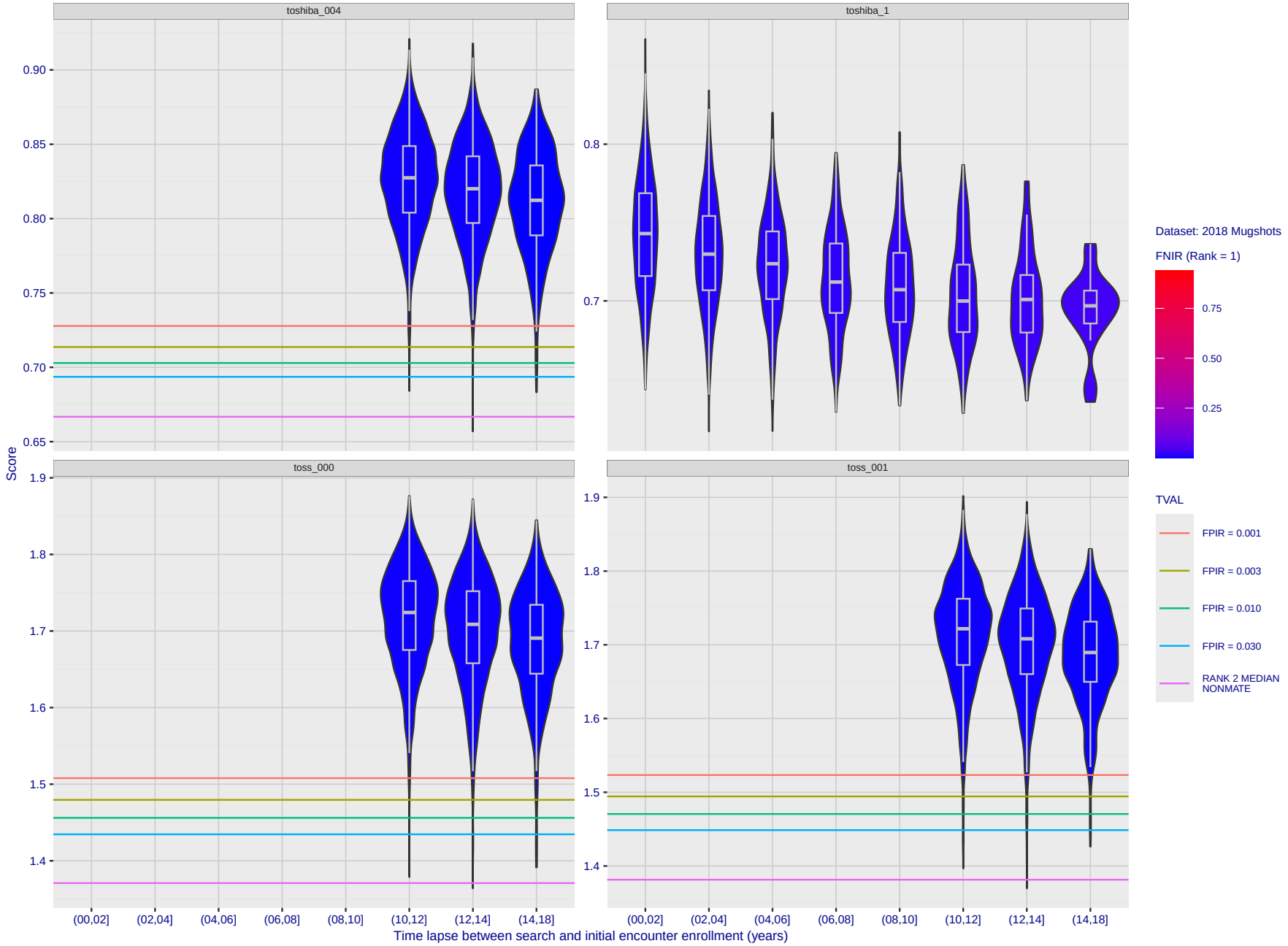

Figure 342: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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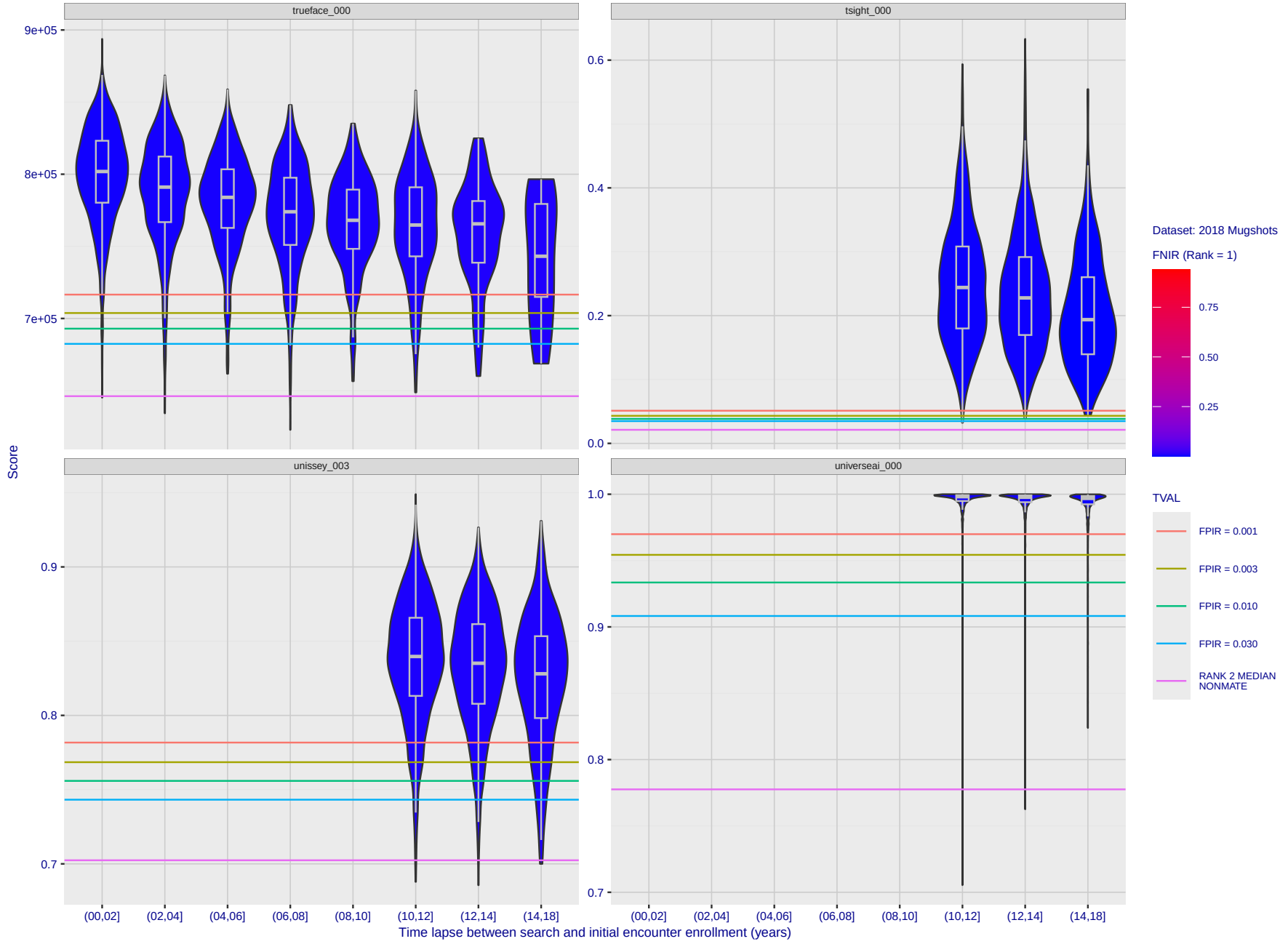

Figure 343: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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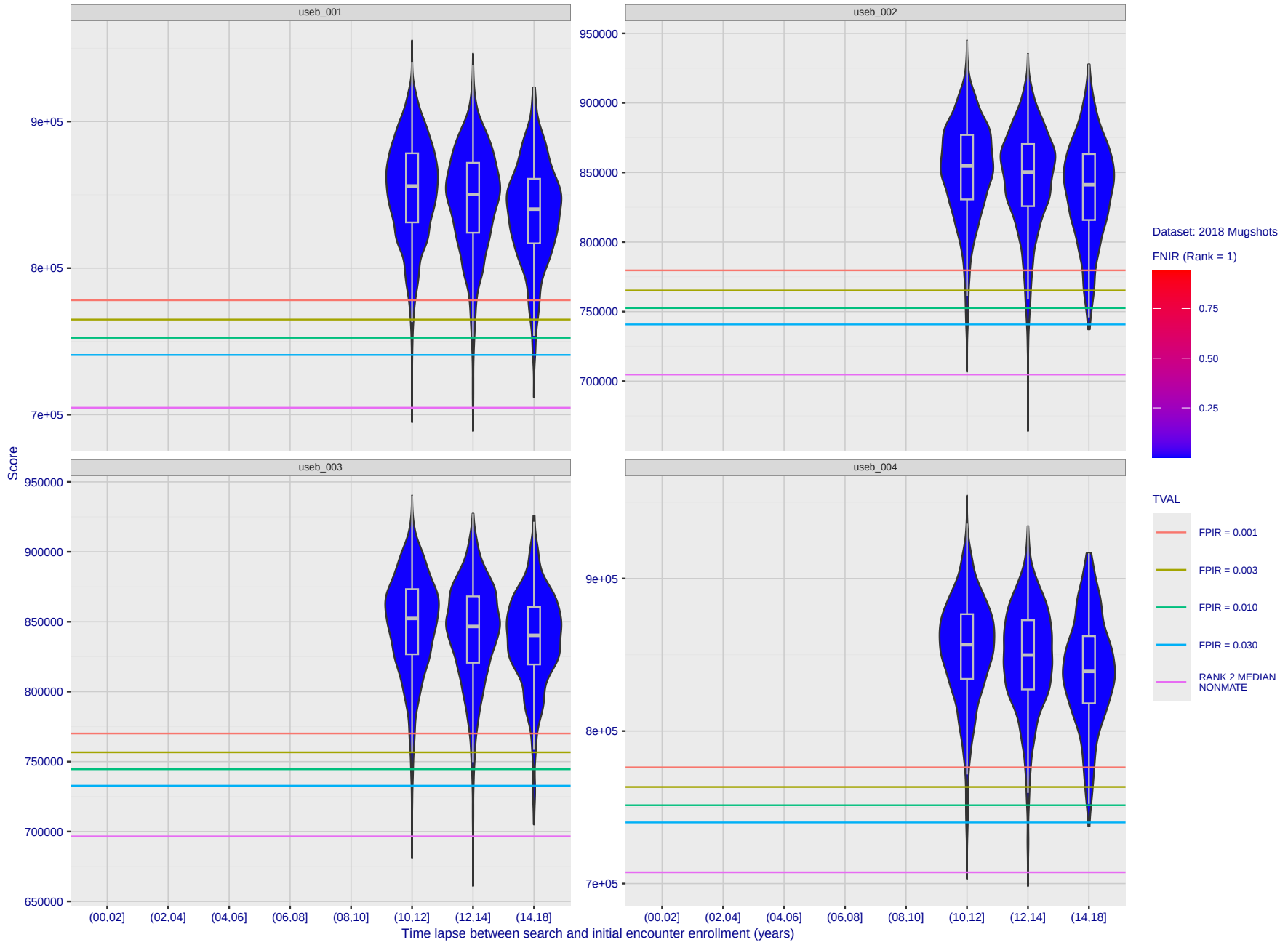

Figure 344: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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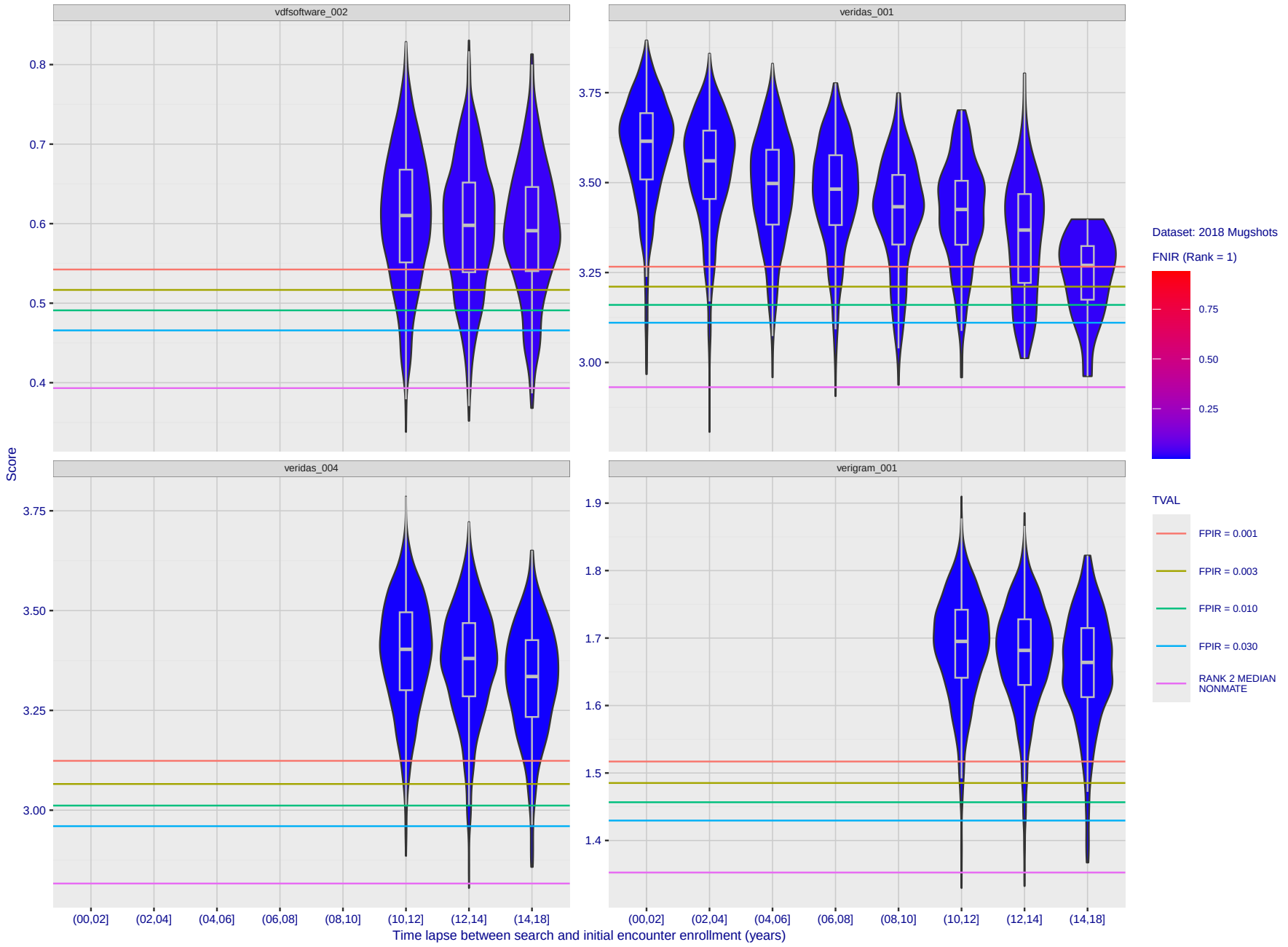

Figure 345: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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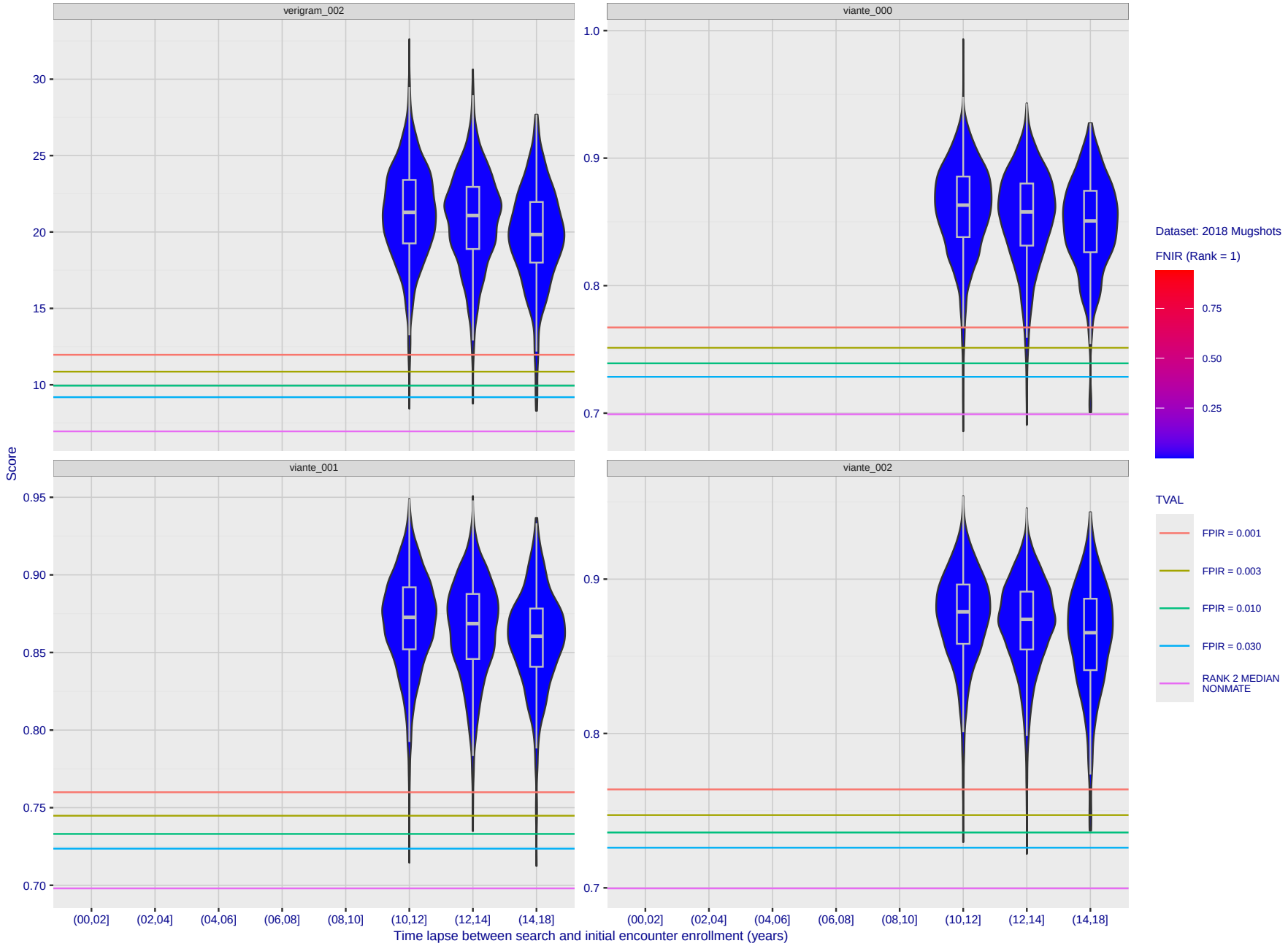

Figure 346: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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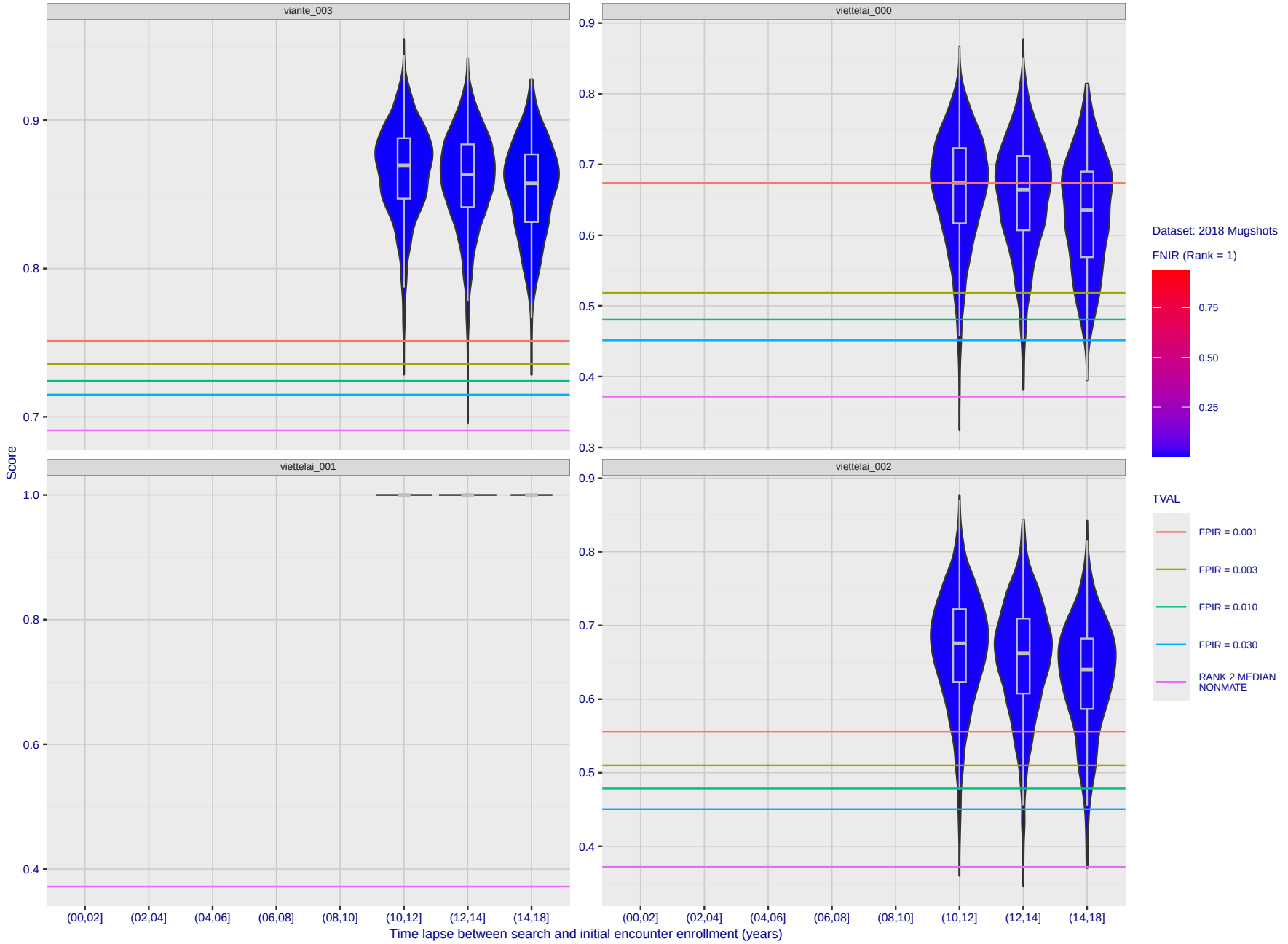

Figure 347: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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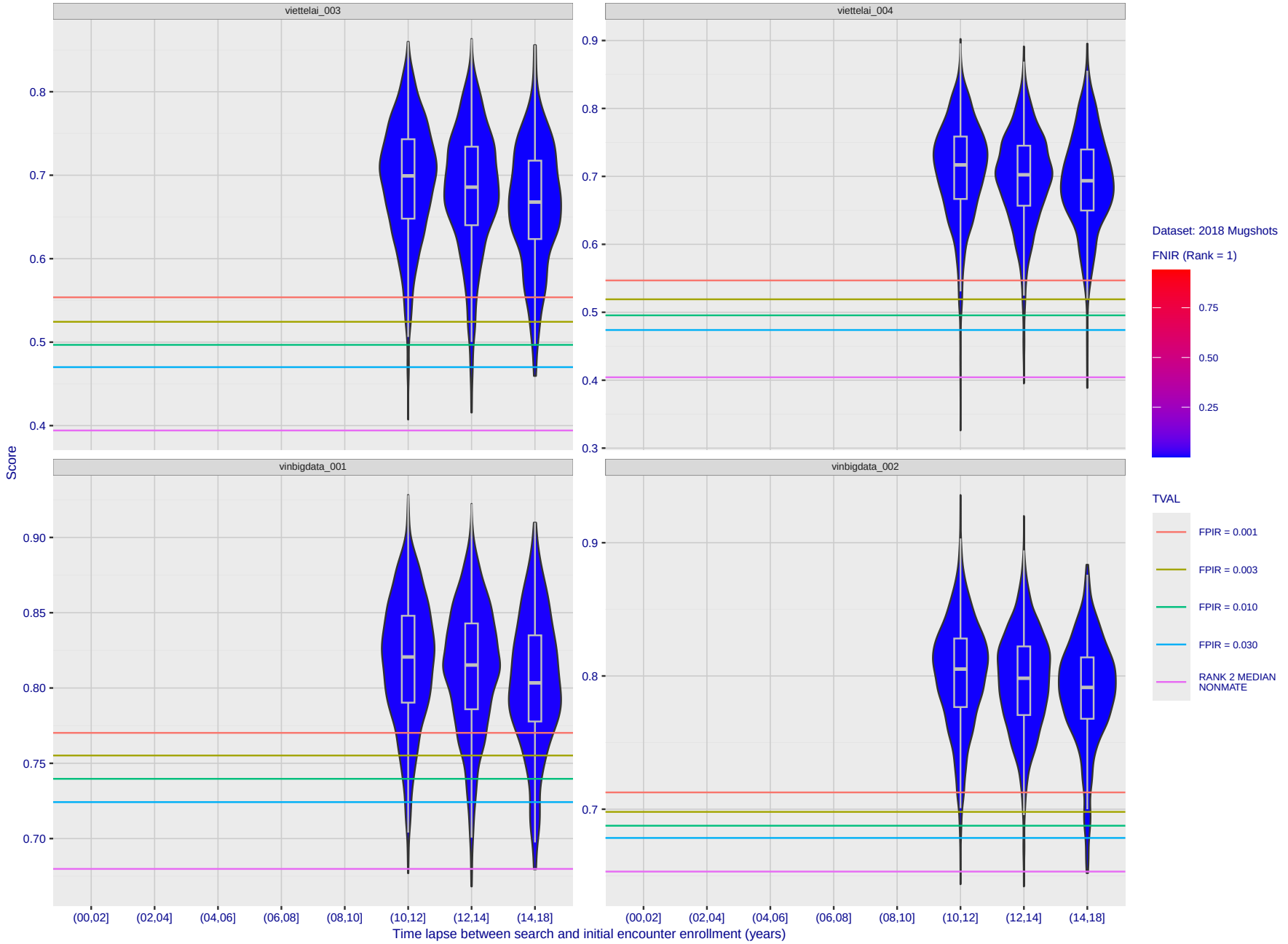

Figure 348: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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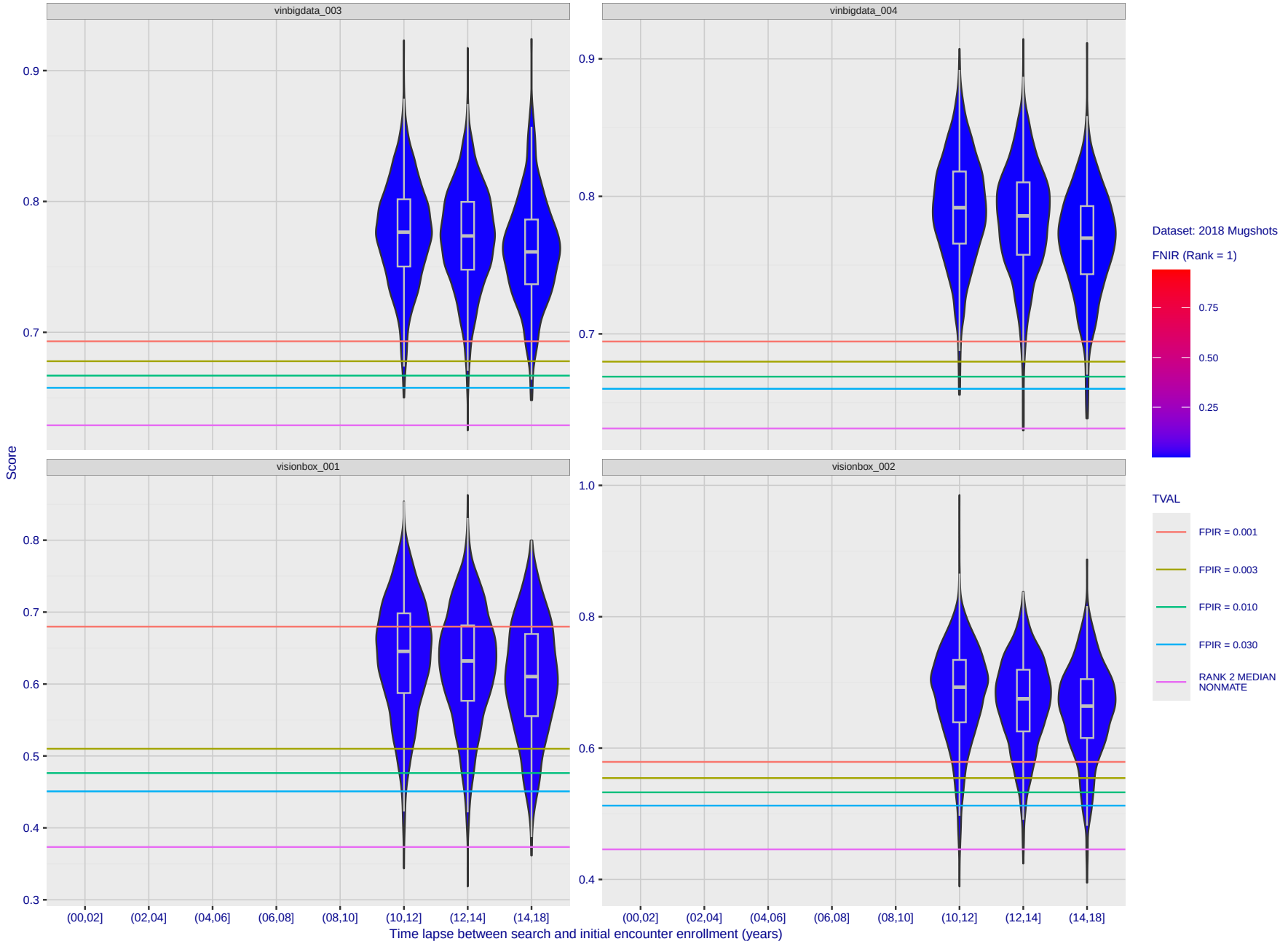

Figure 349: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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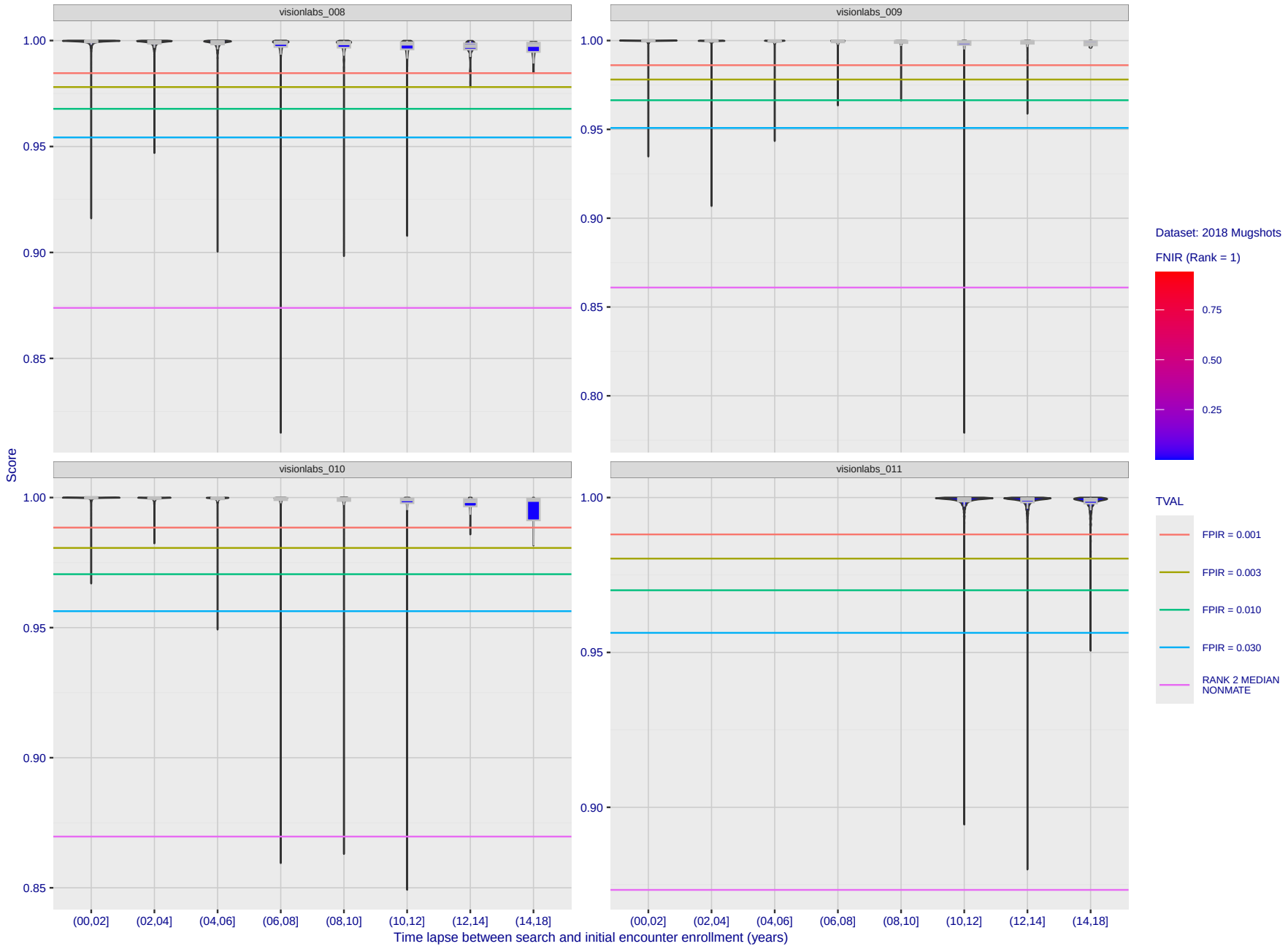

Figure 350: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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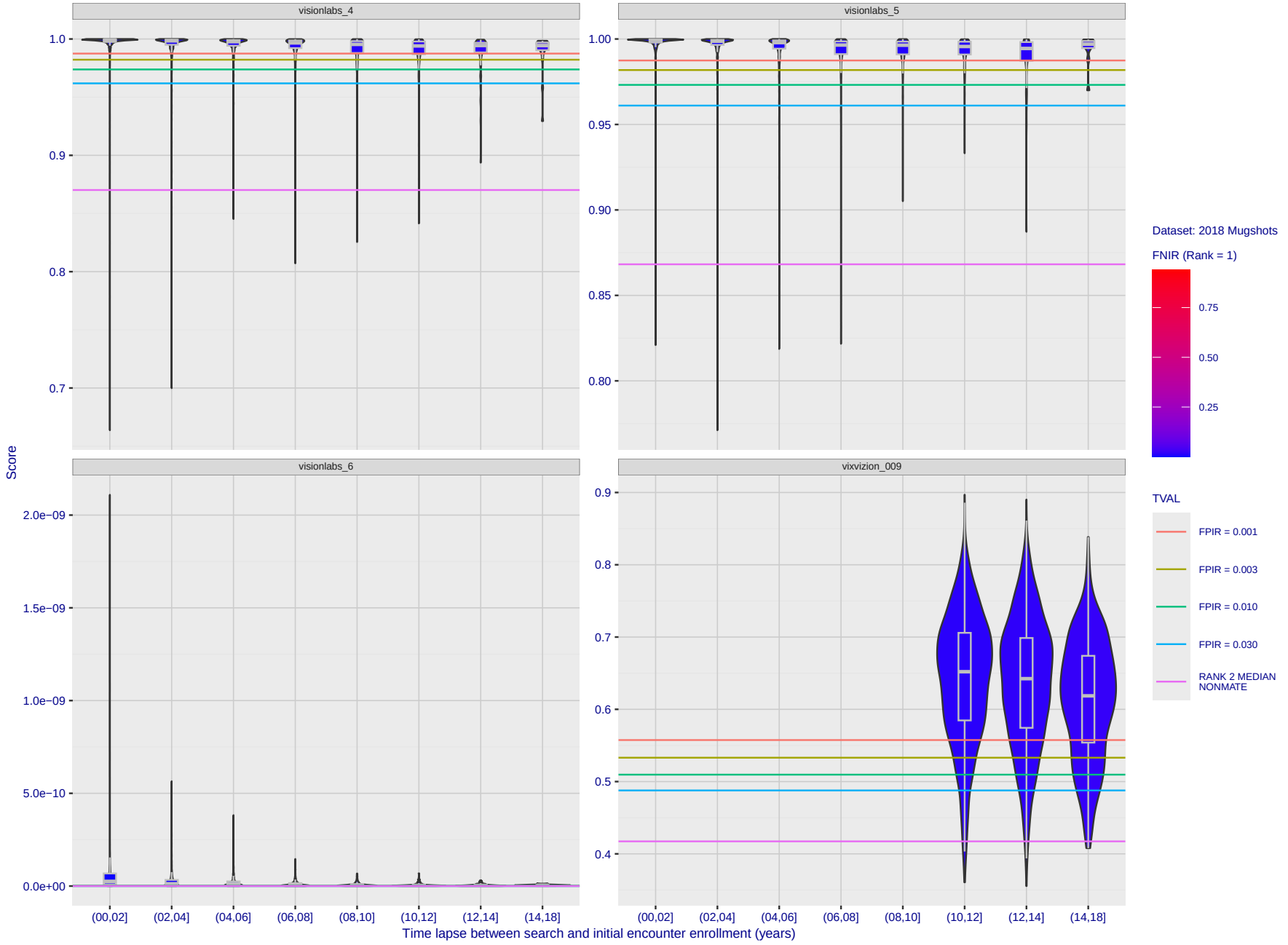

Figure 351: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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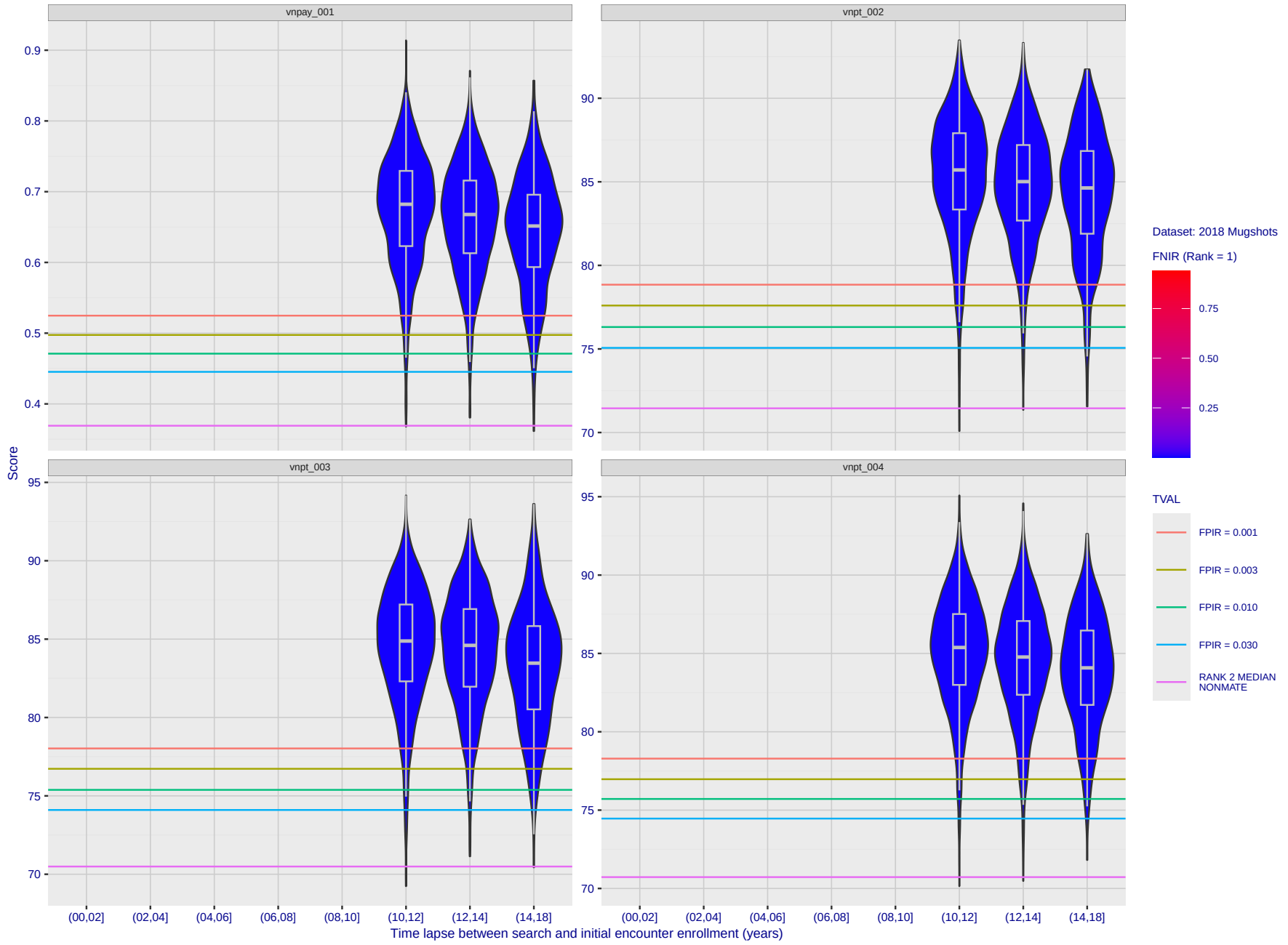

Figure 352: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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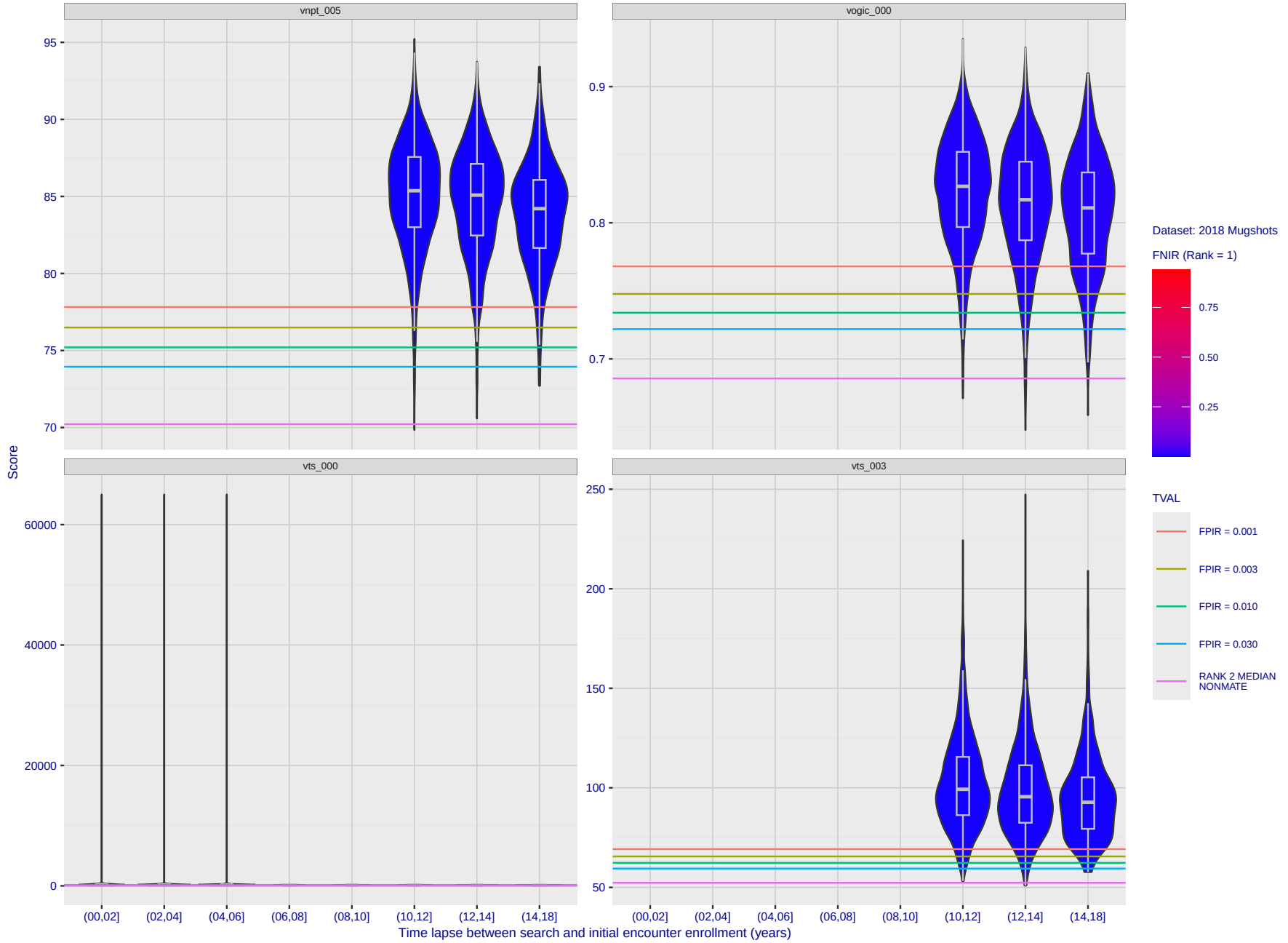

Figure 353: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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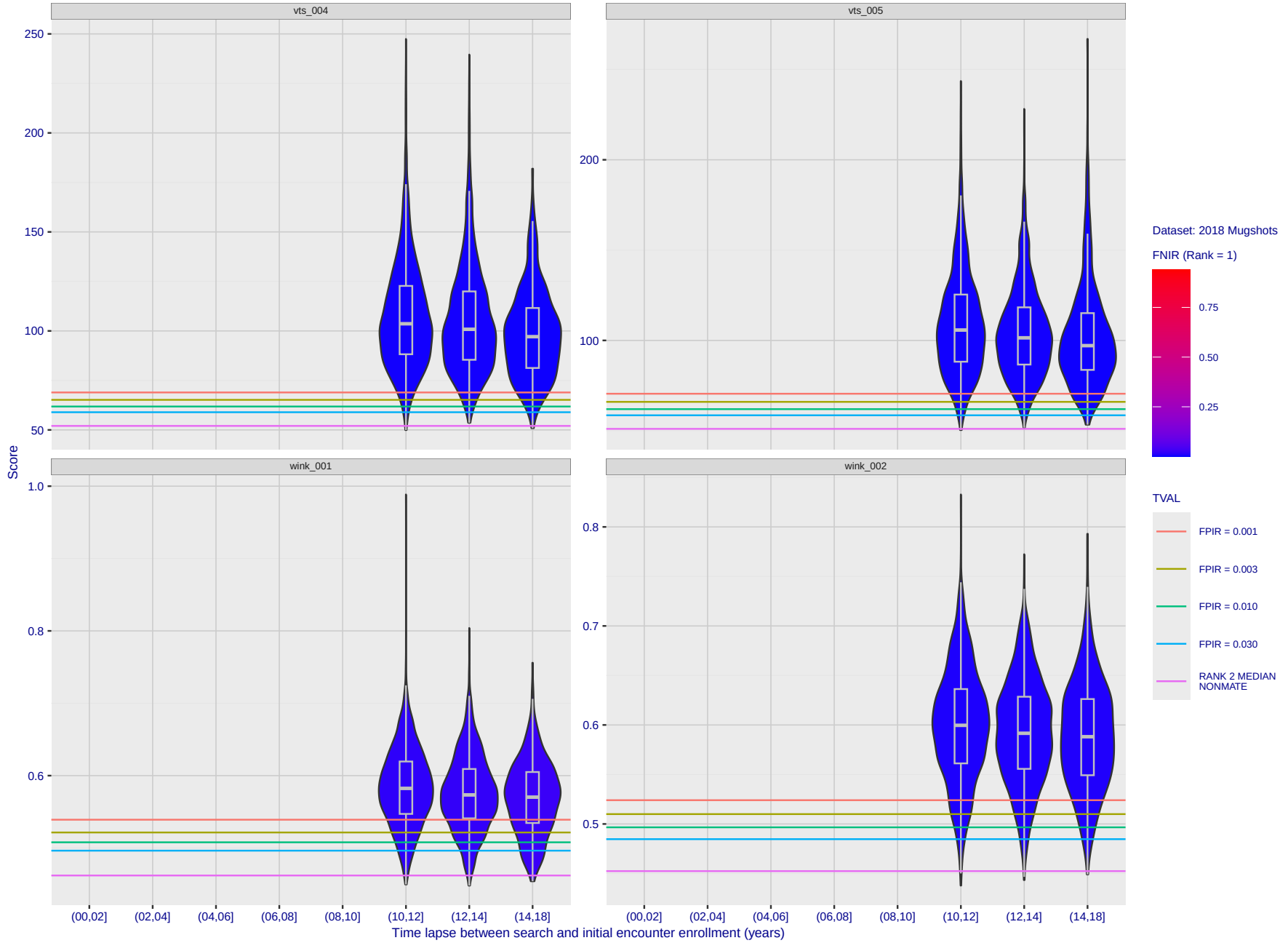

Figure 354: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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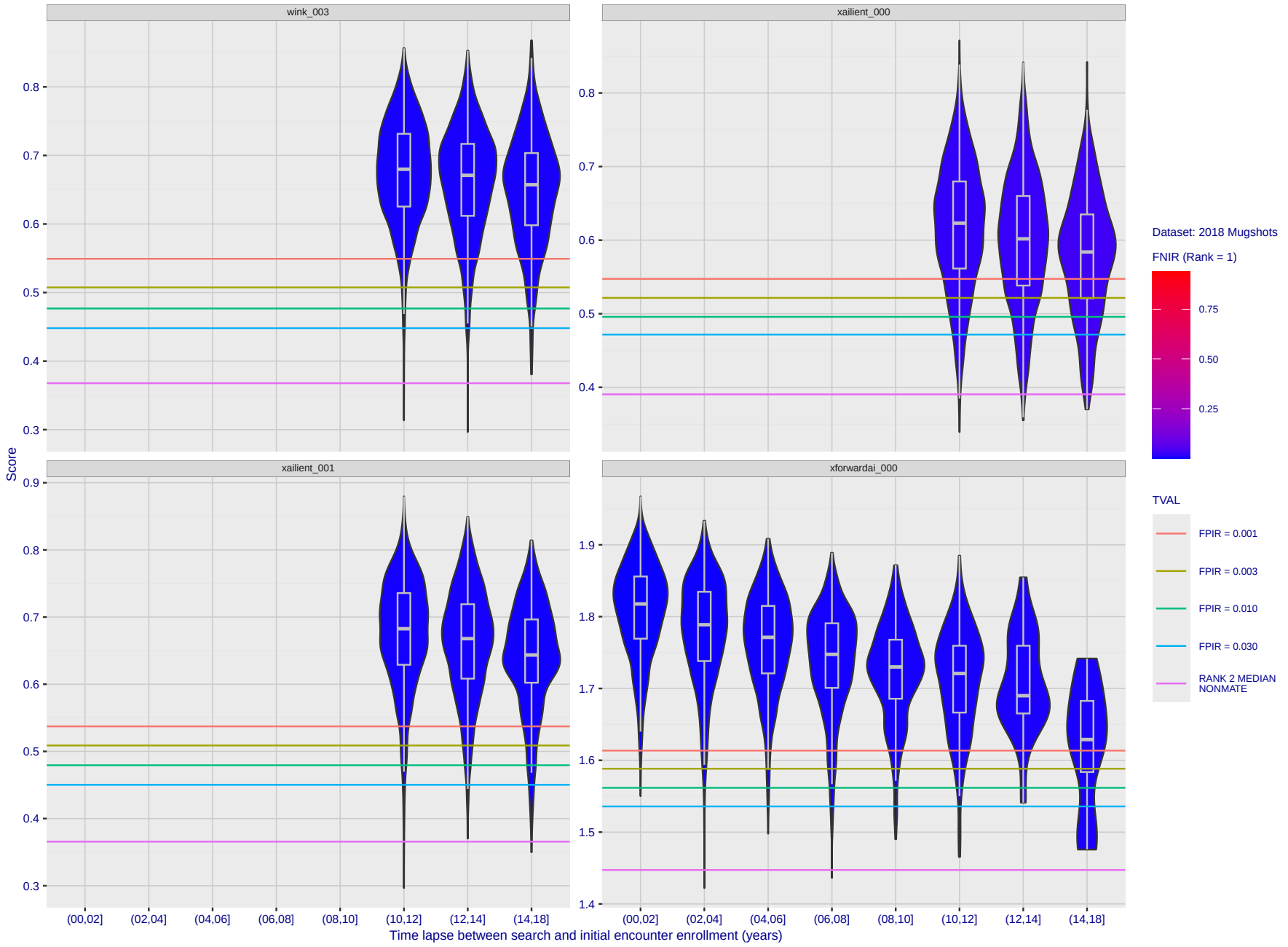

Figure 355: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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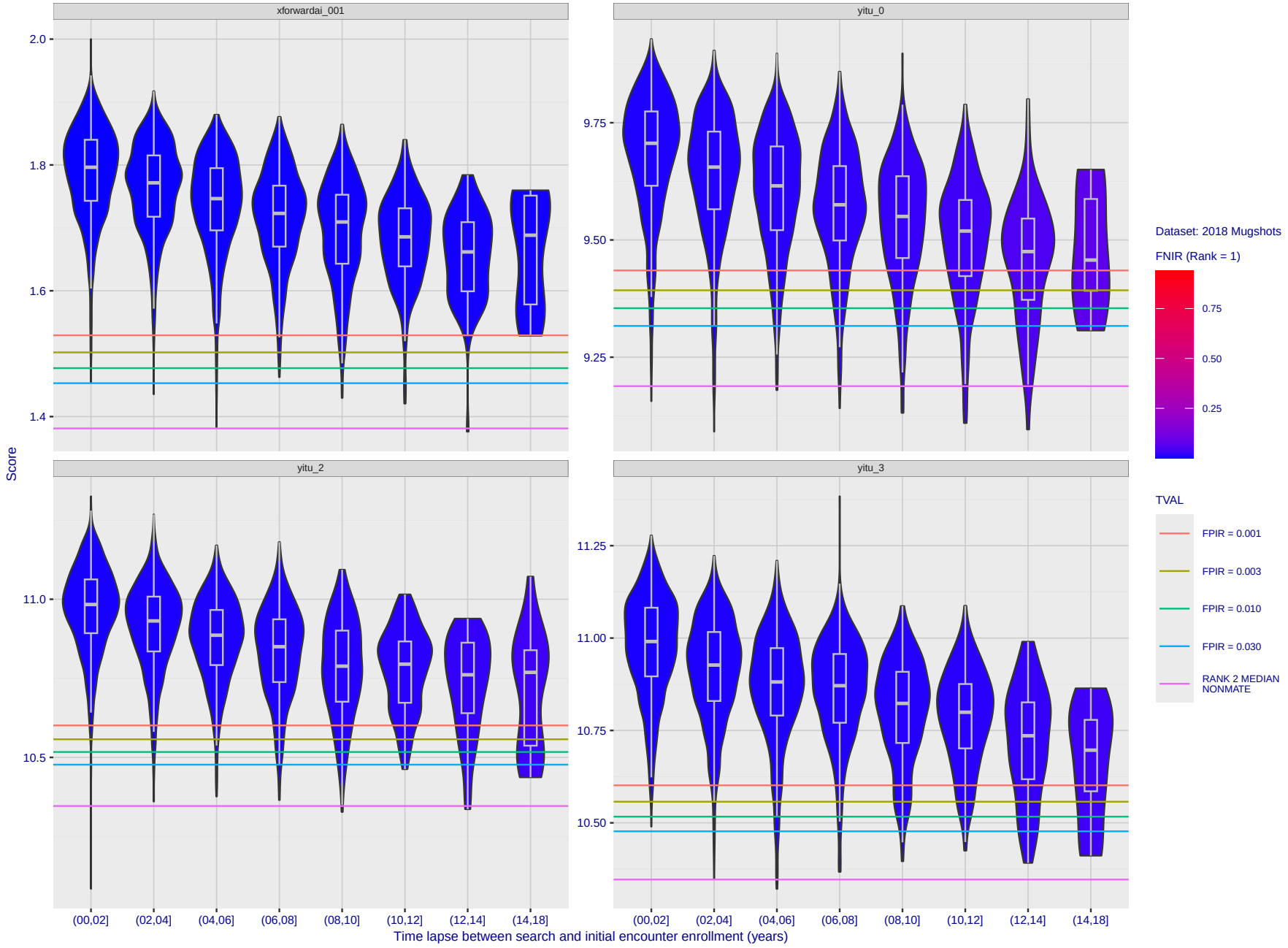

Figure 356: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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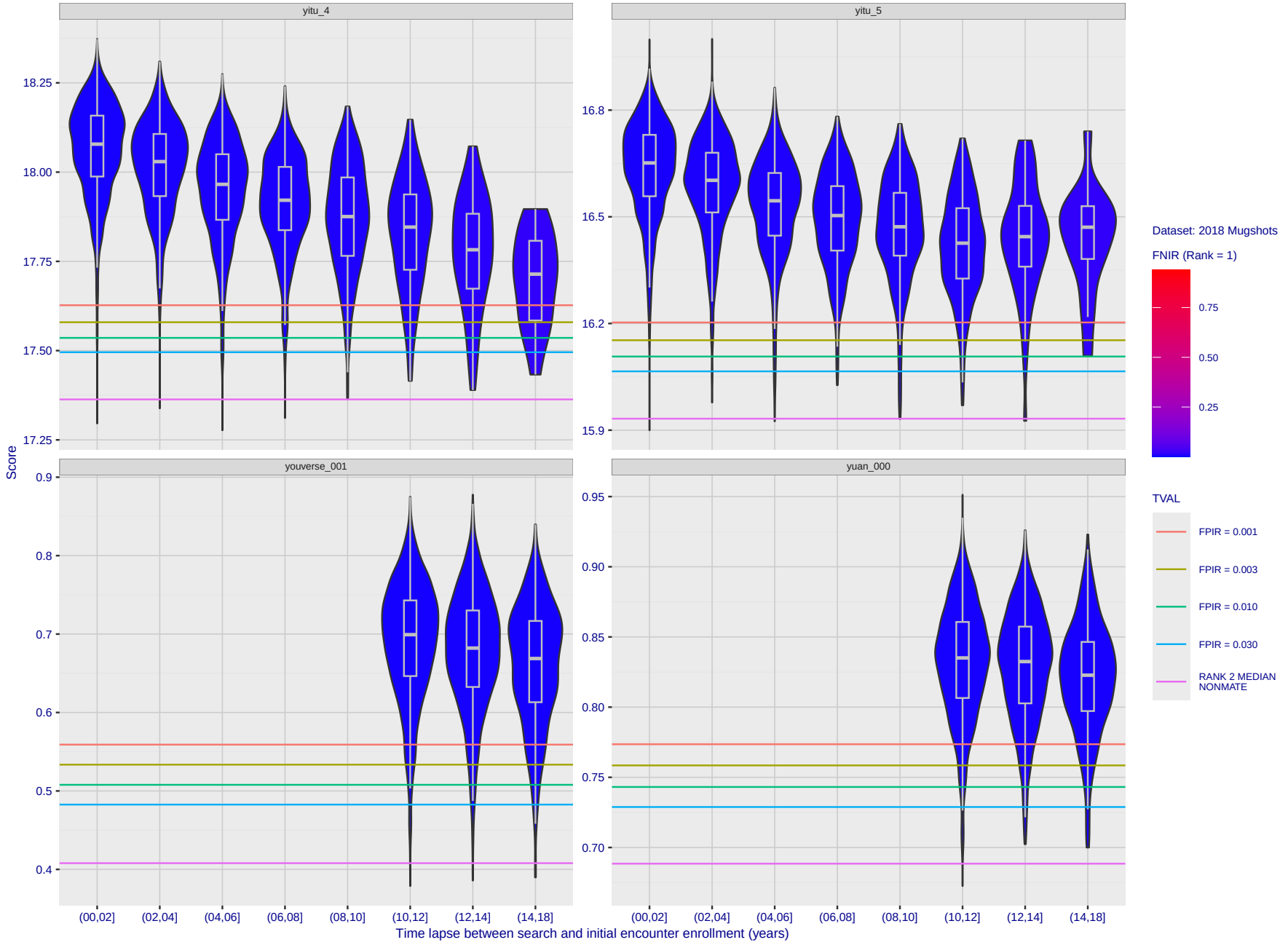

Figure 357: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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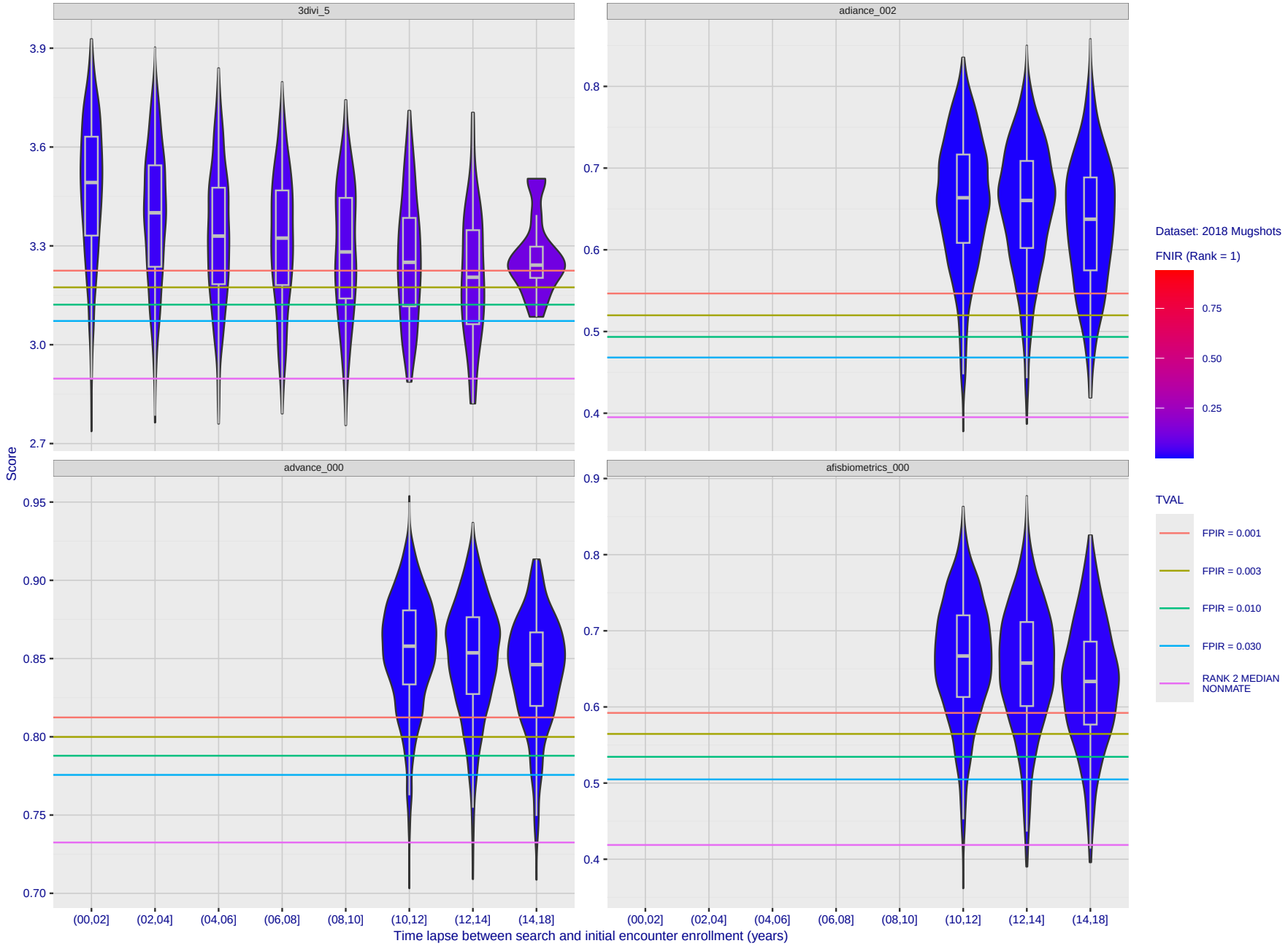

Figure 358: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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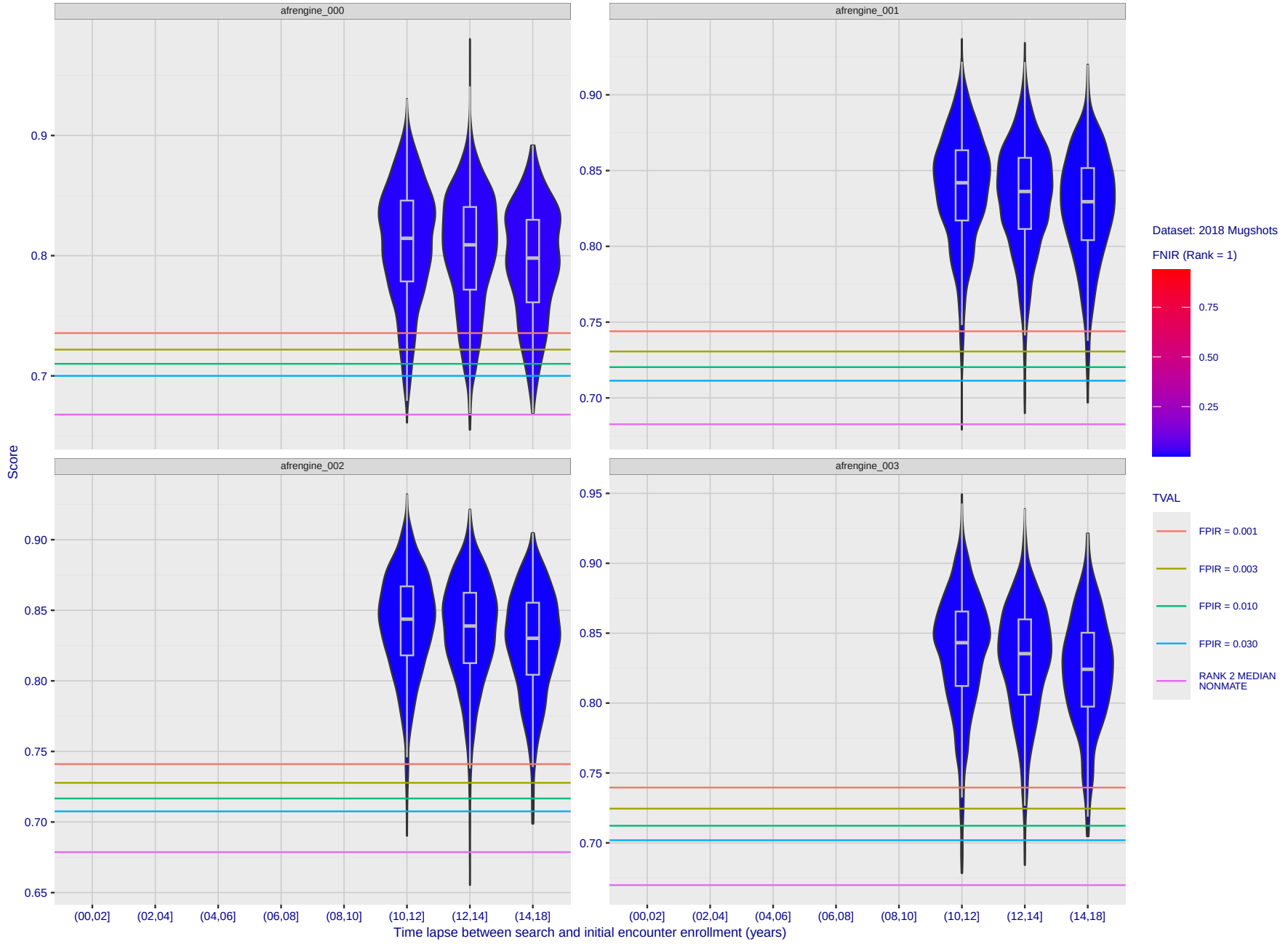

Figure 359: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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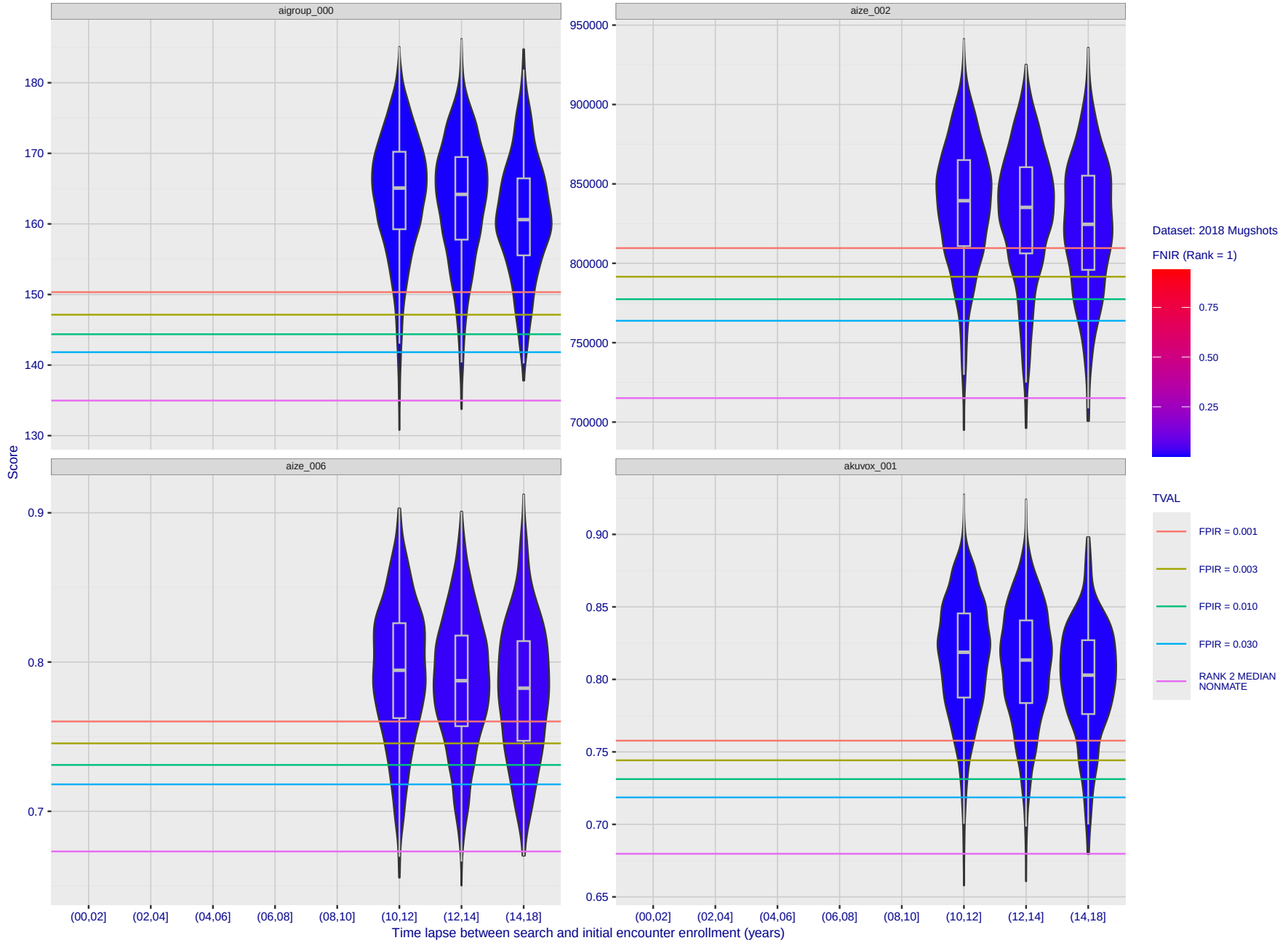

Figure 360: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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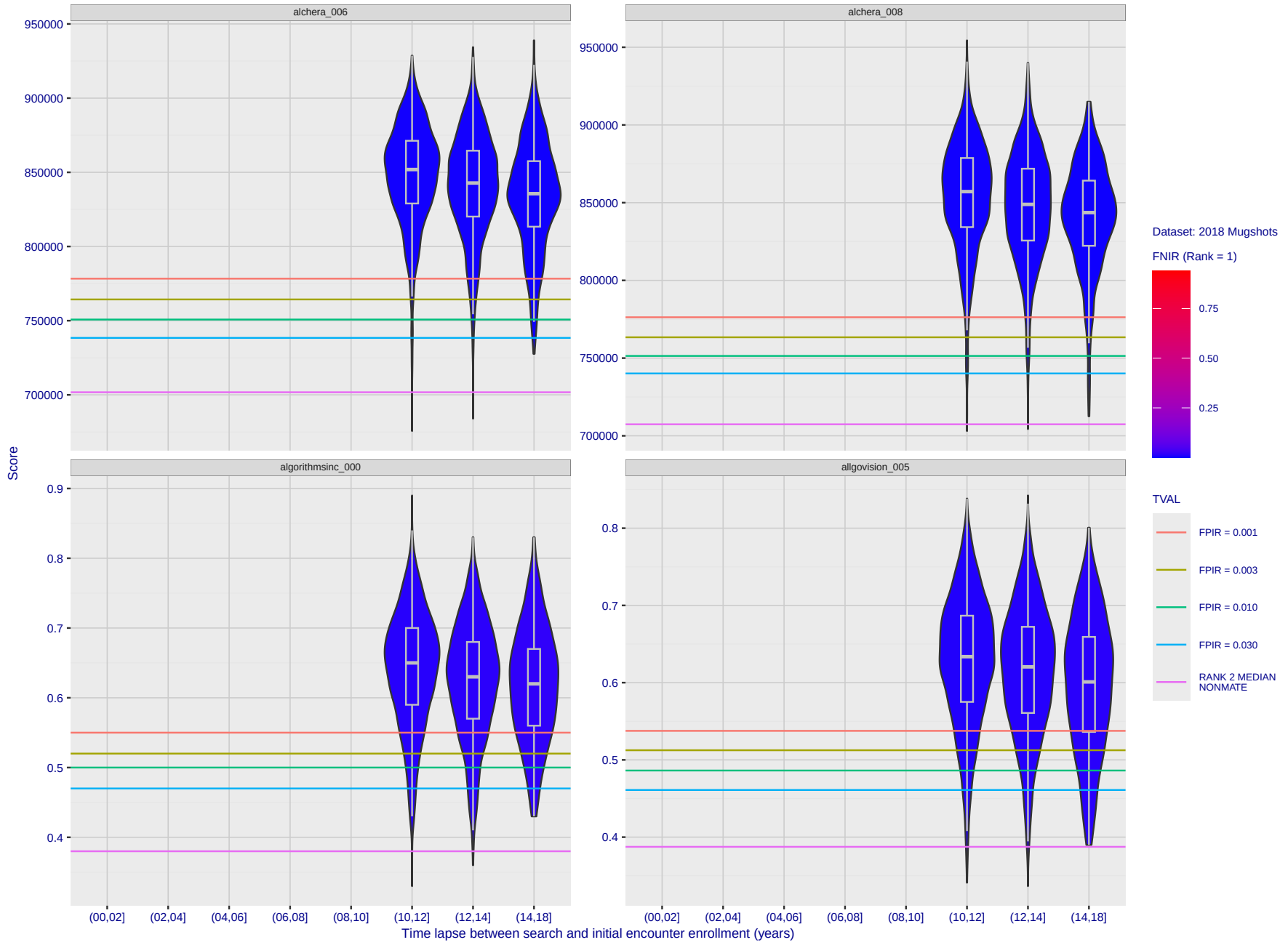

Figure 361: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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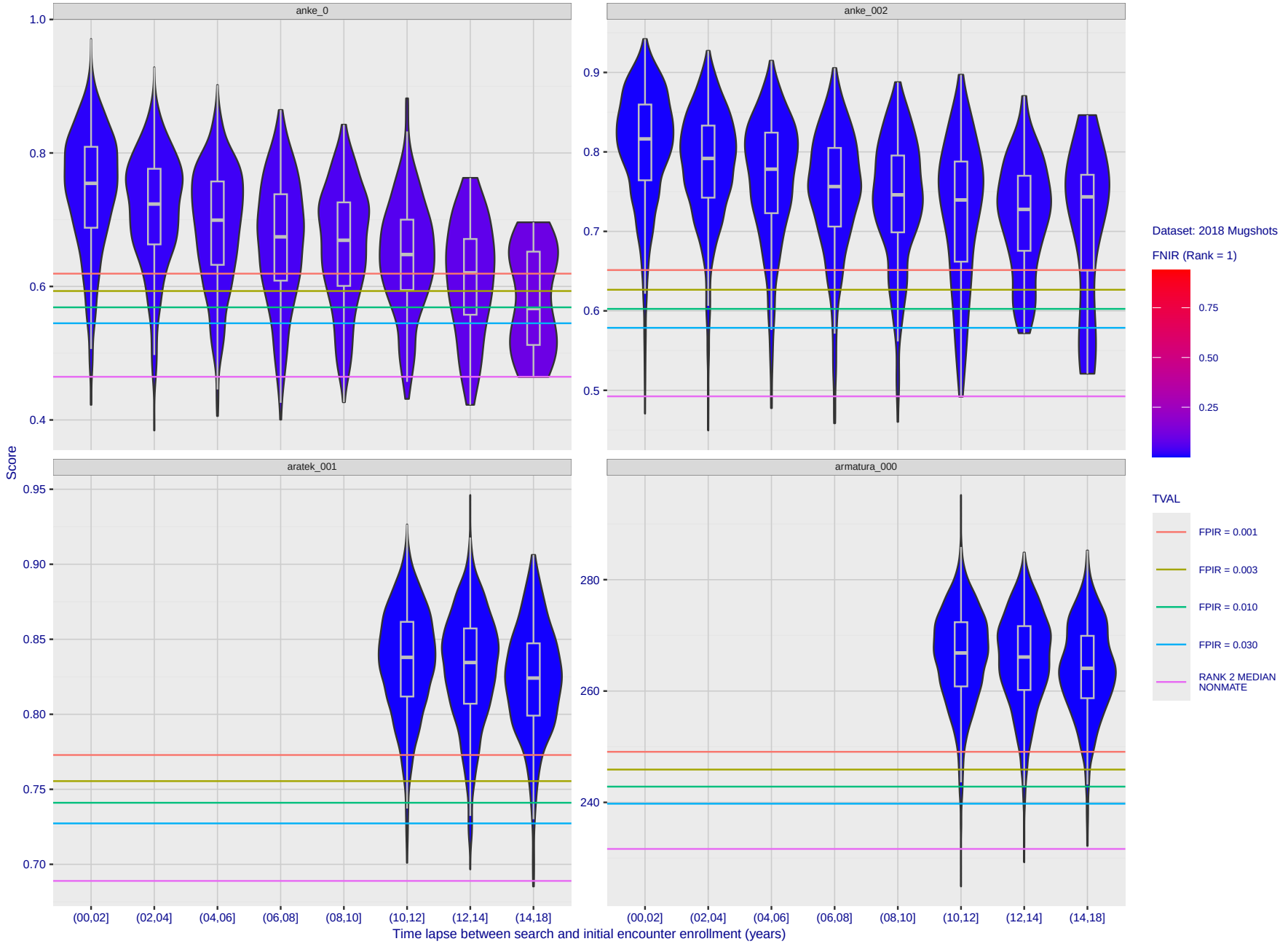

Figure 362: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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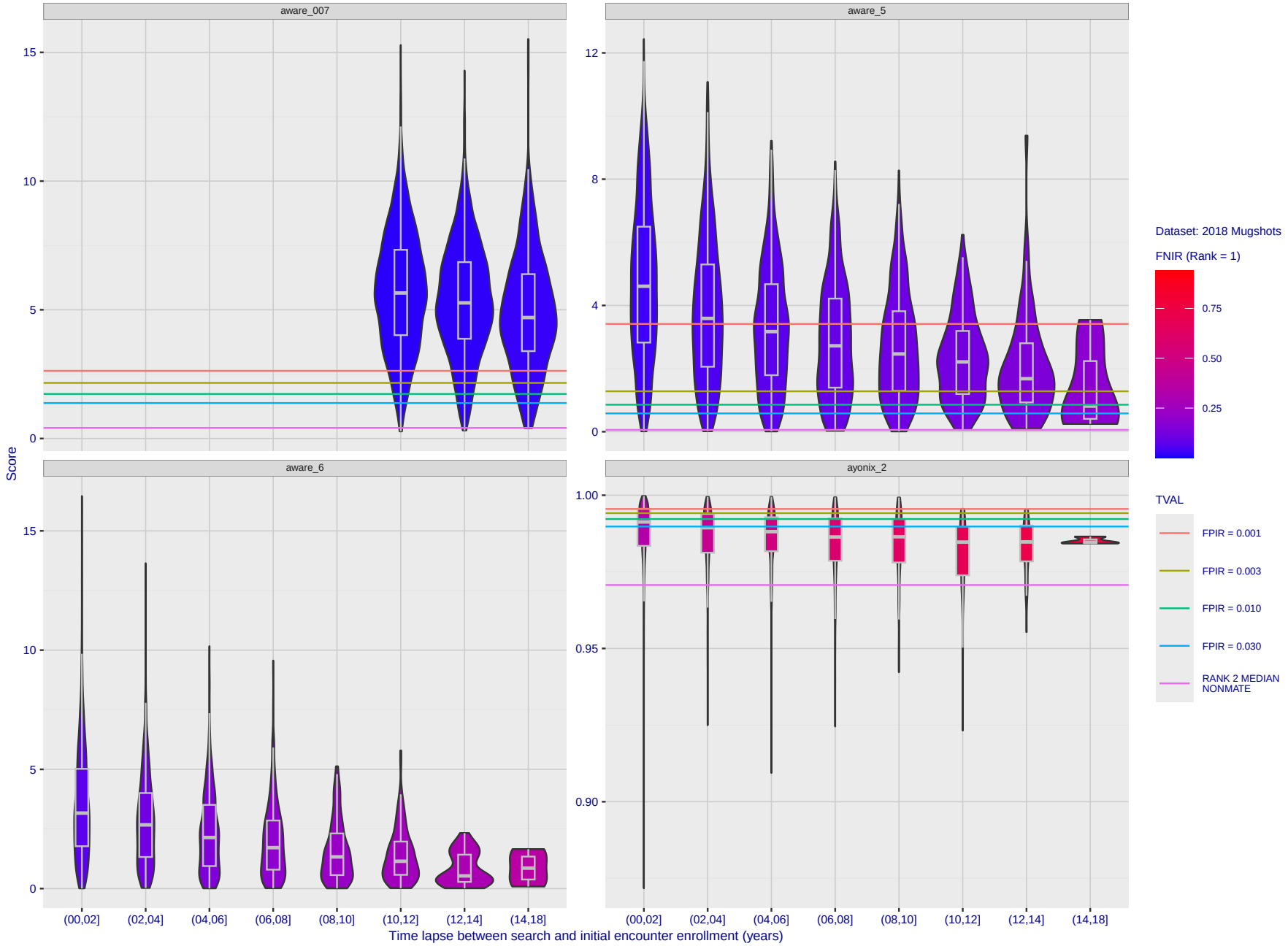

Figure 363: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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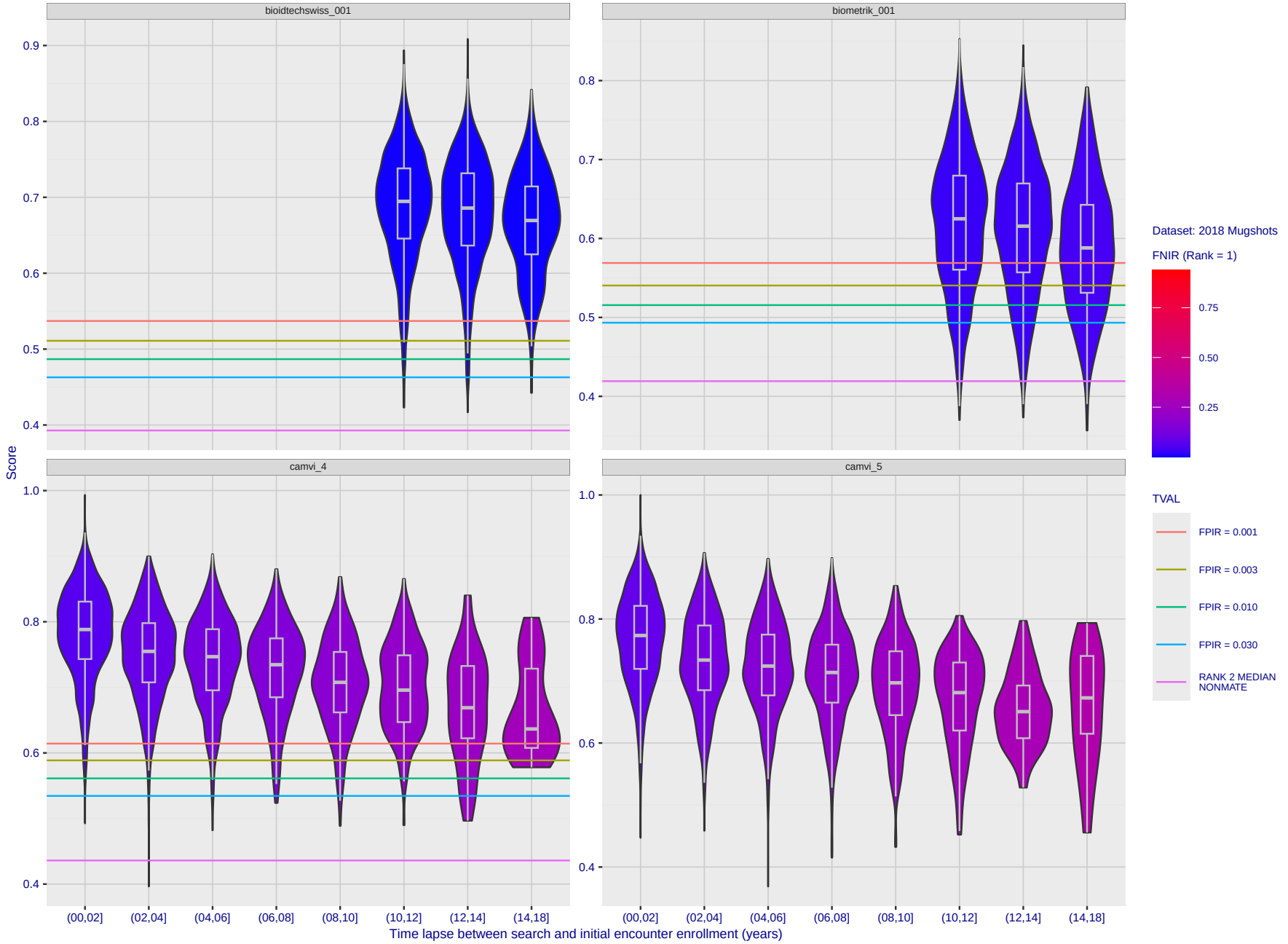

Figure 364: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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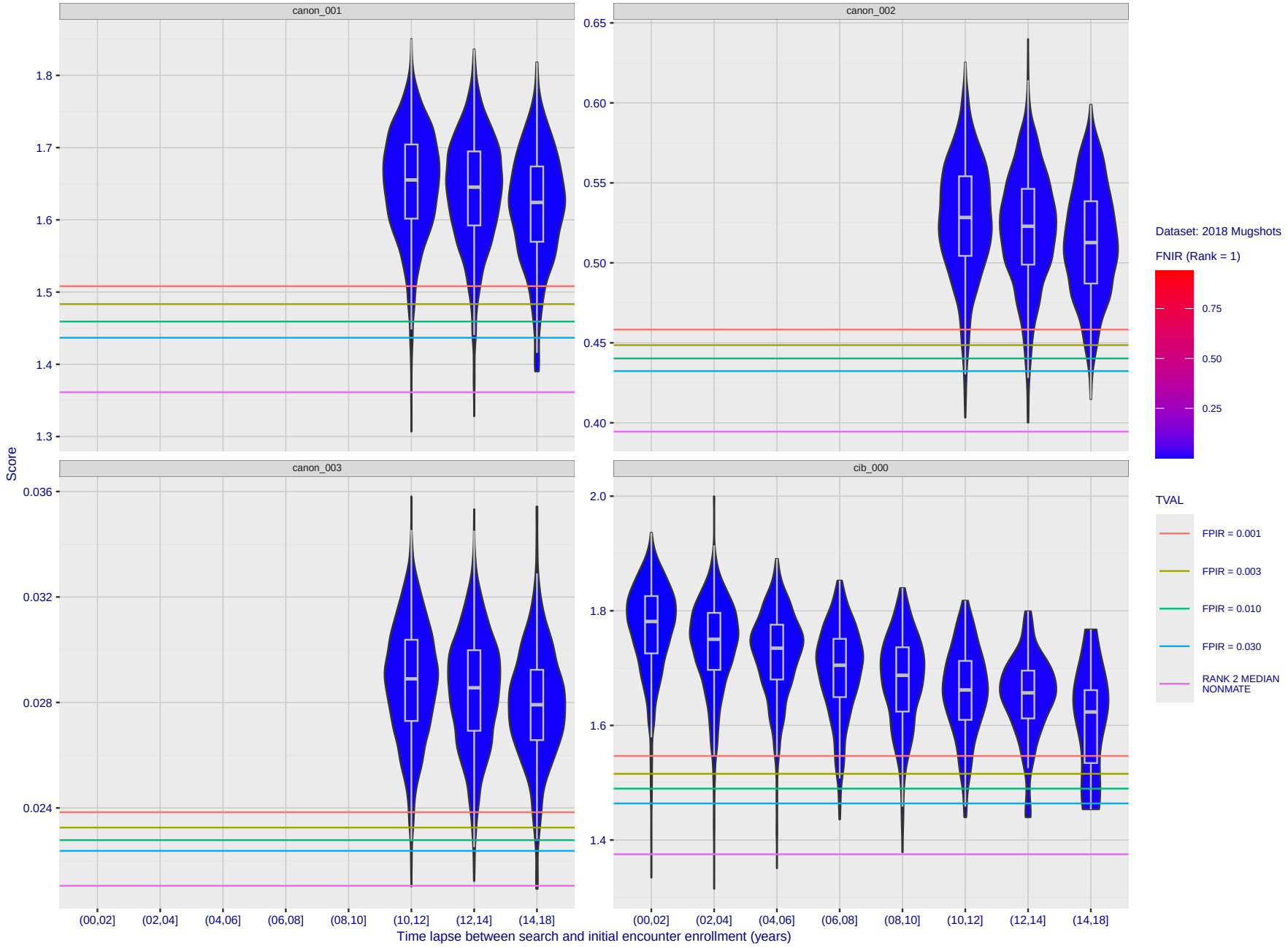

Figure 365: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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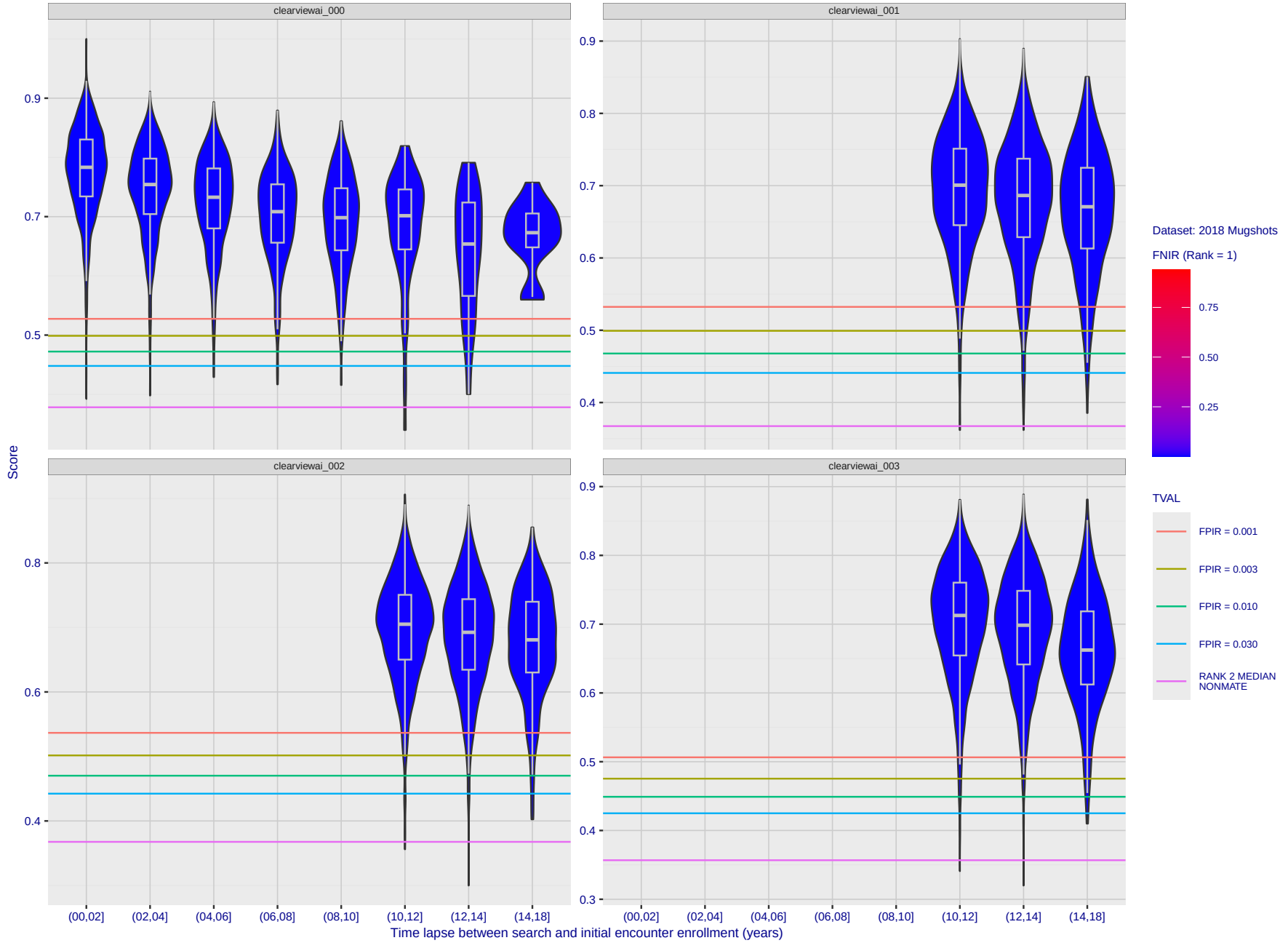

Figure 366: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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FNIR(N, R, T) = FPIR(N, T) = False neg. identification rate / False pos. identification rate
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T = Threshold
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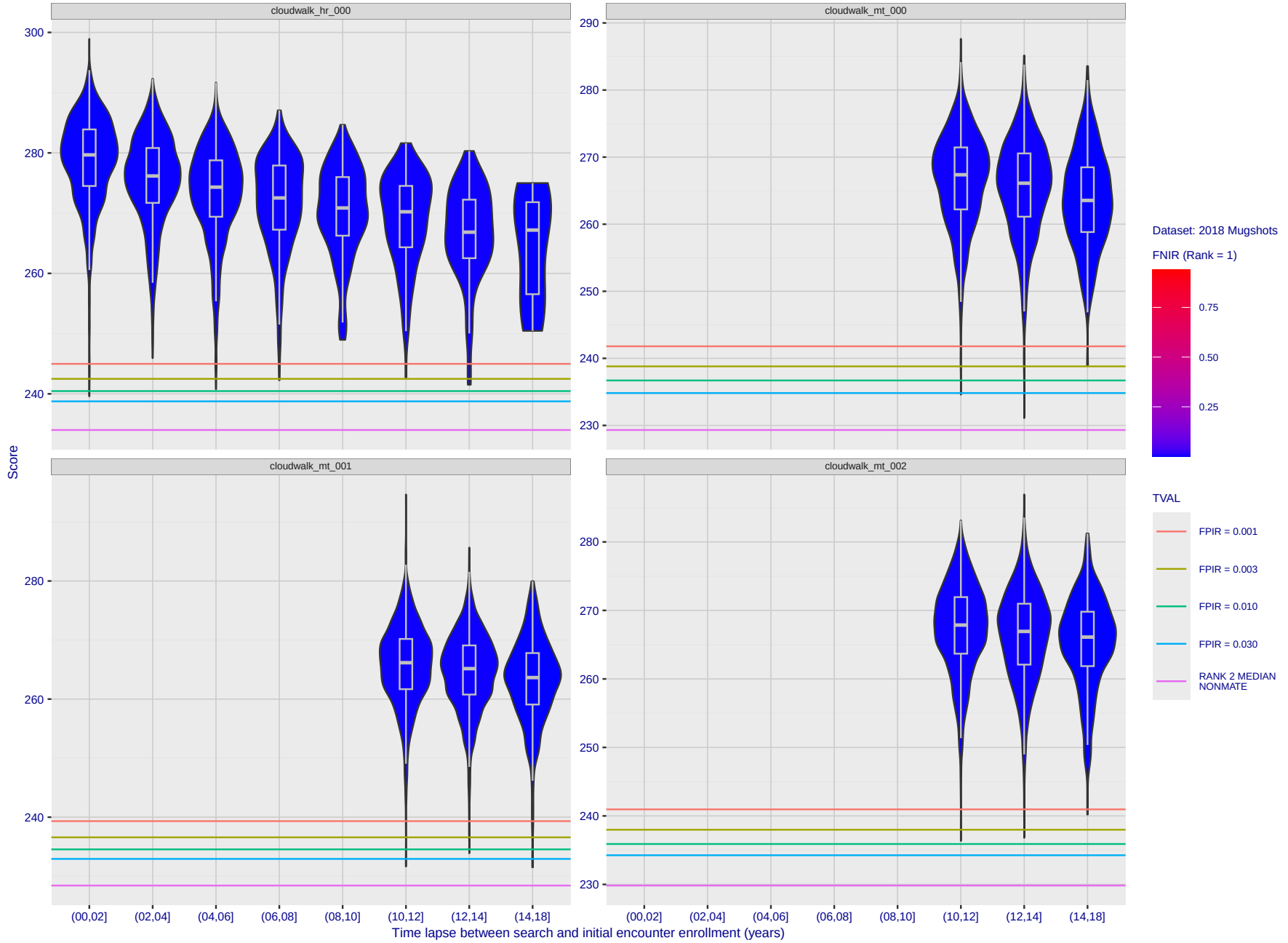

Figure 367: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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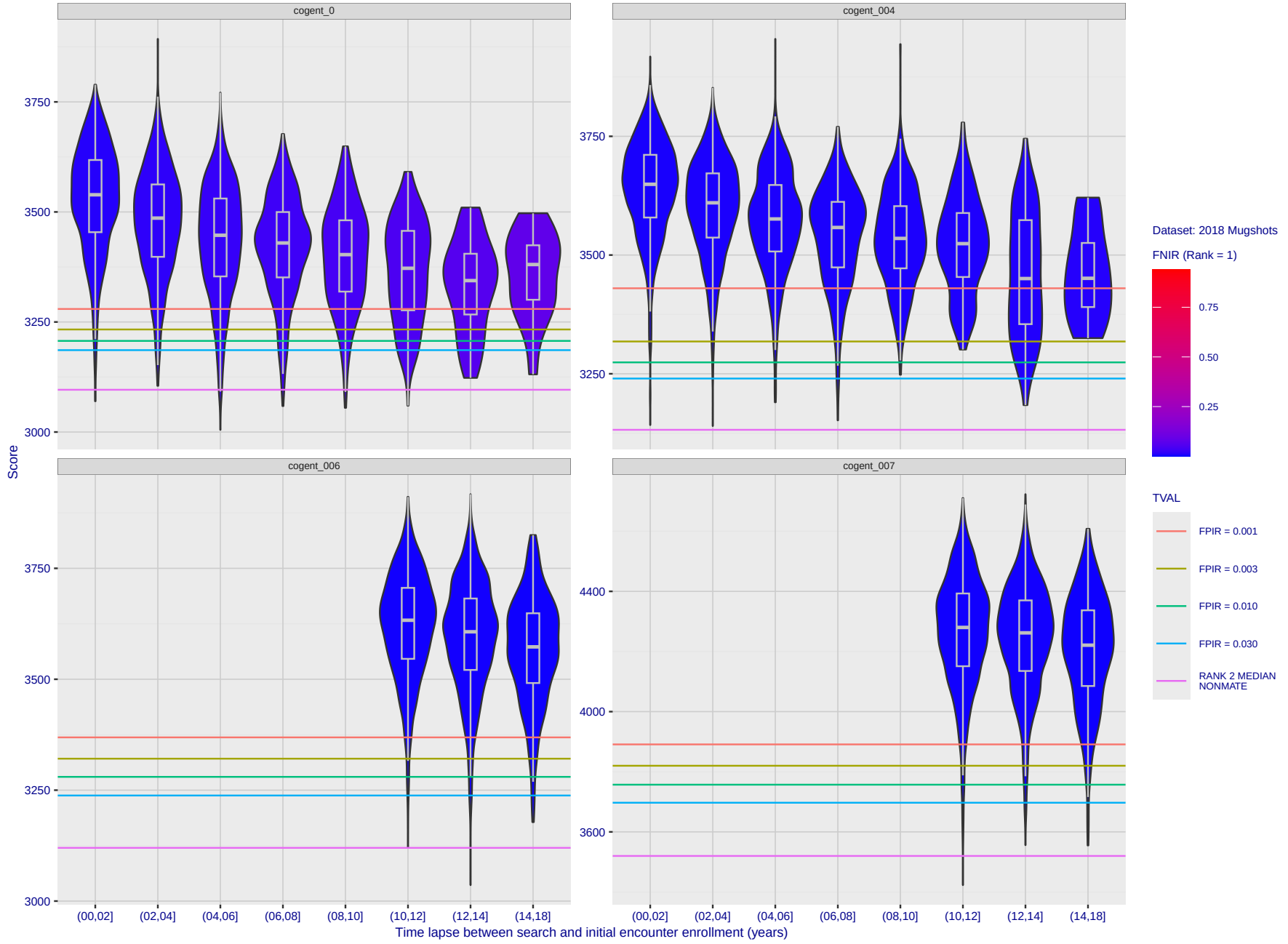

Figure 368: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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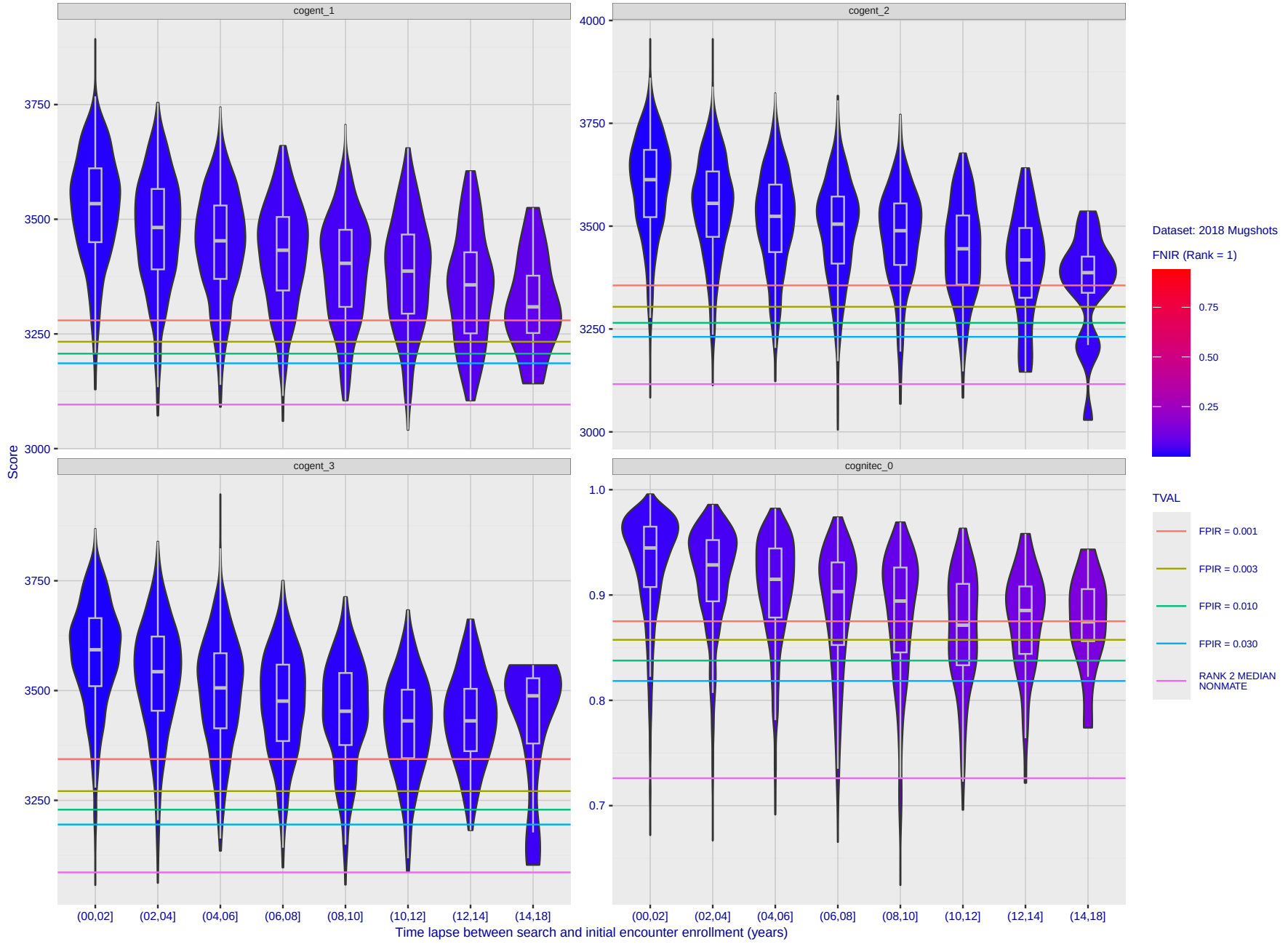

Figure 369: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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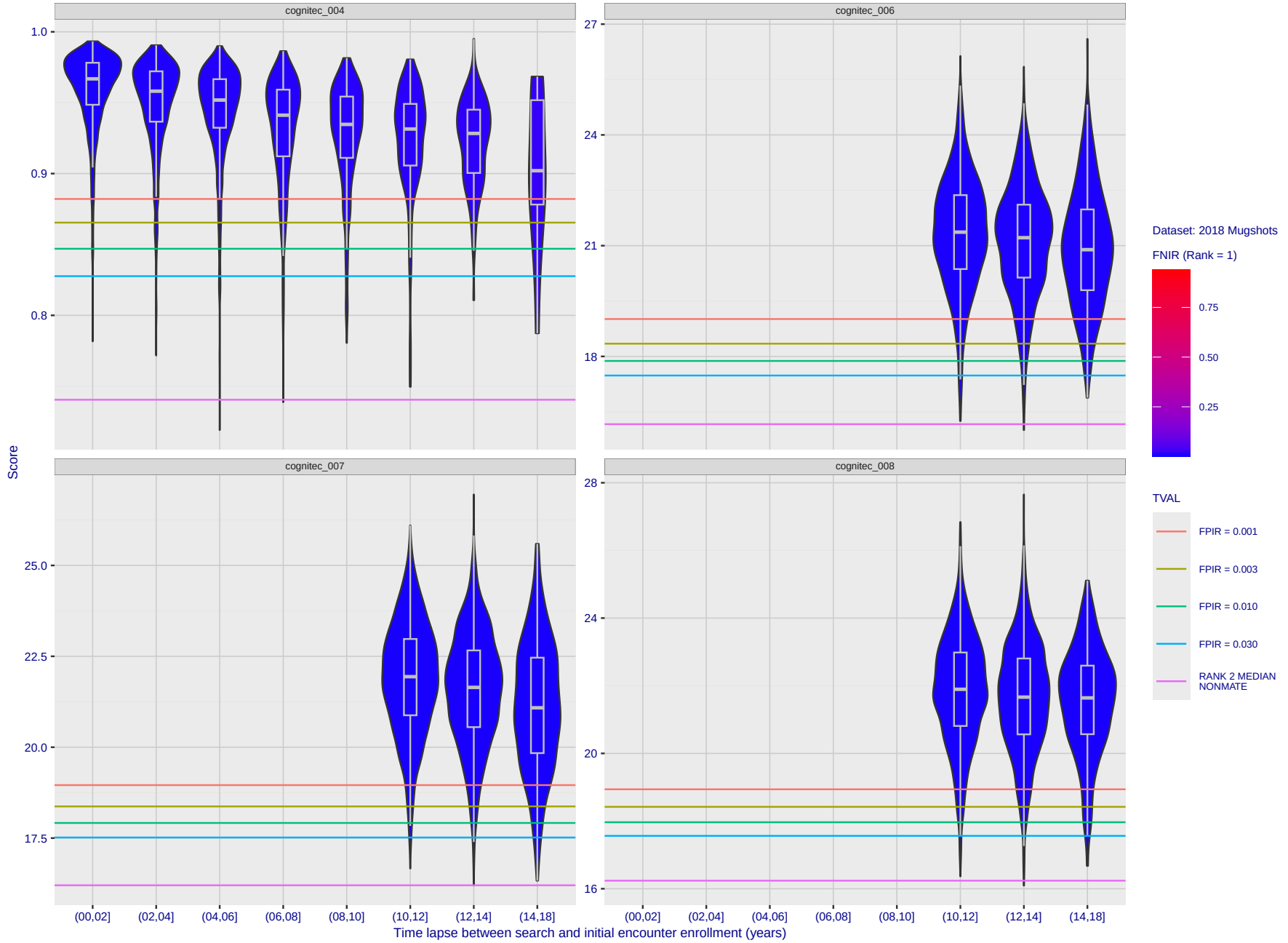

Figure 370: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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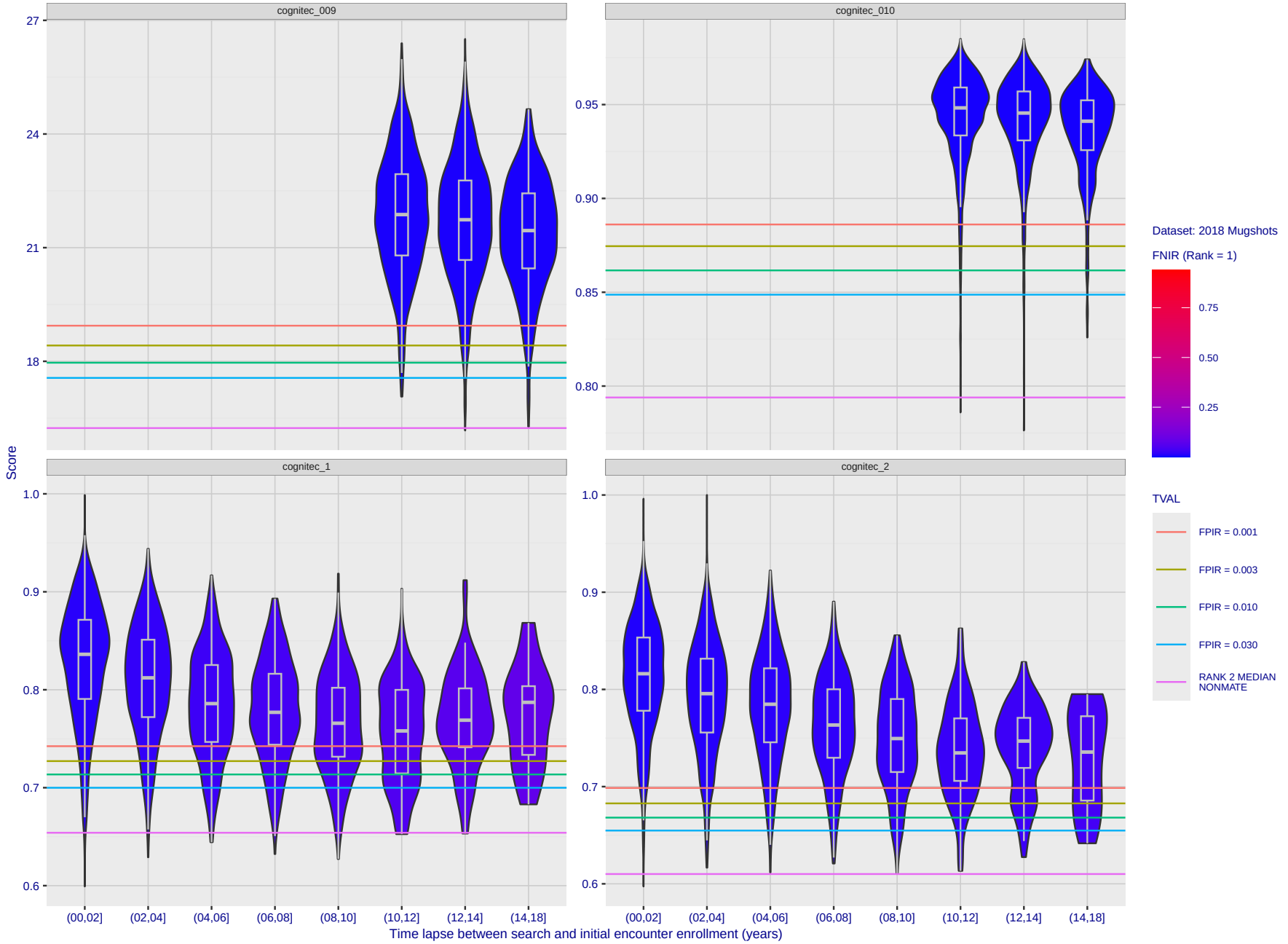

Figure 371: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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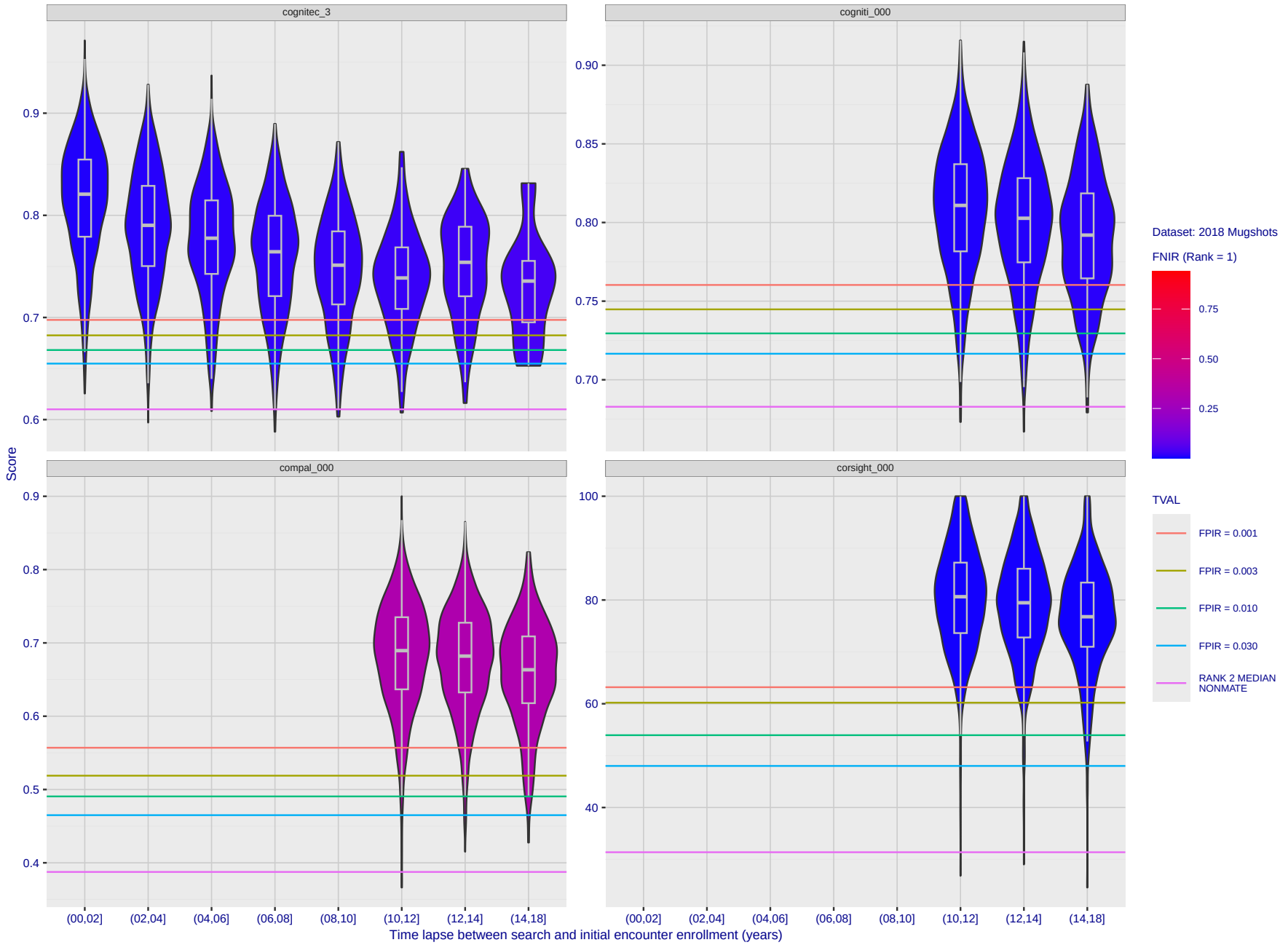

Figure 372: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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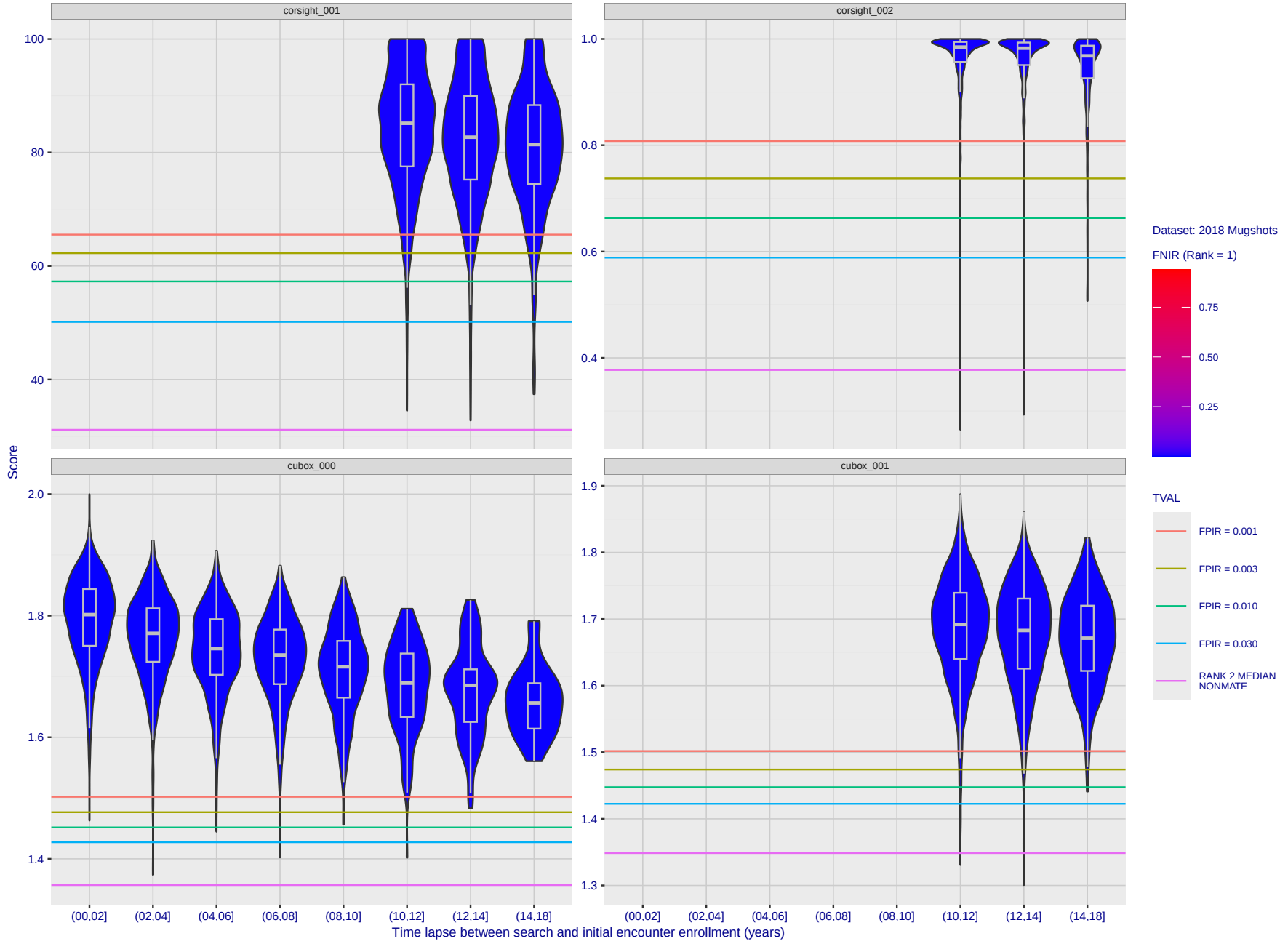

Figure 373: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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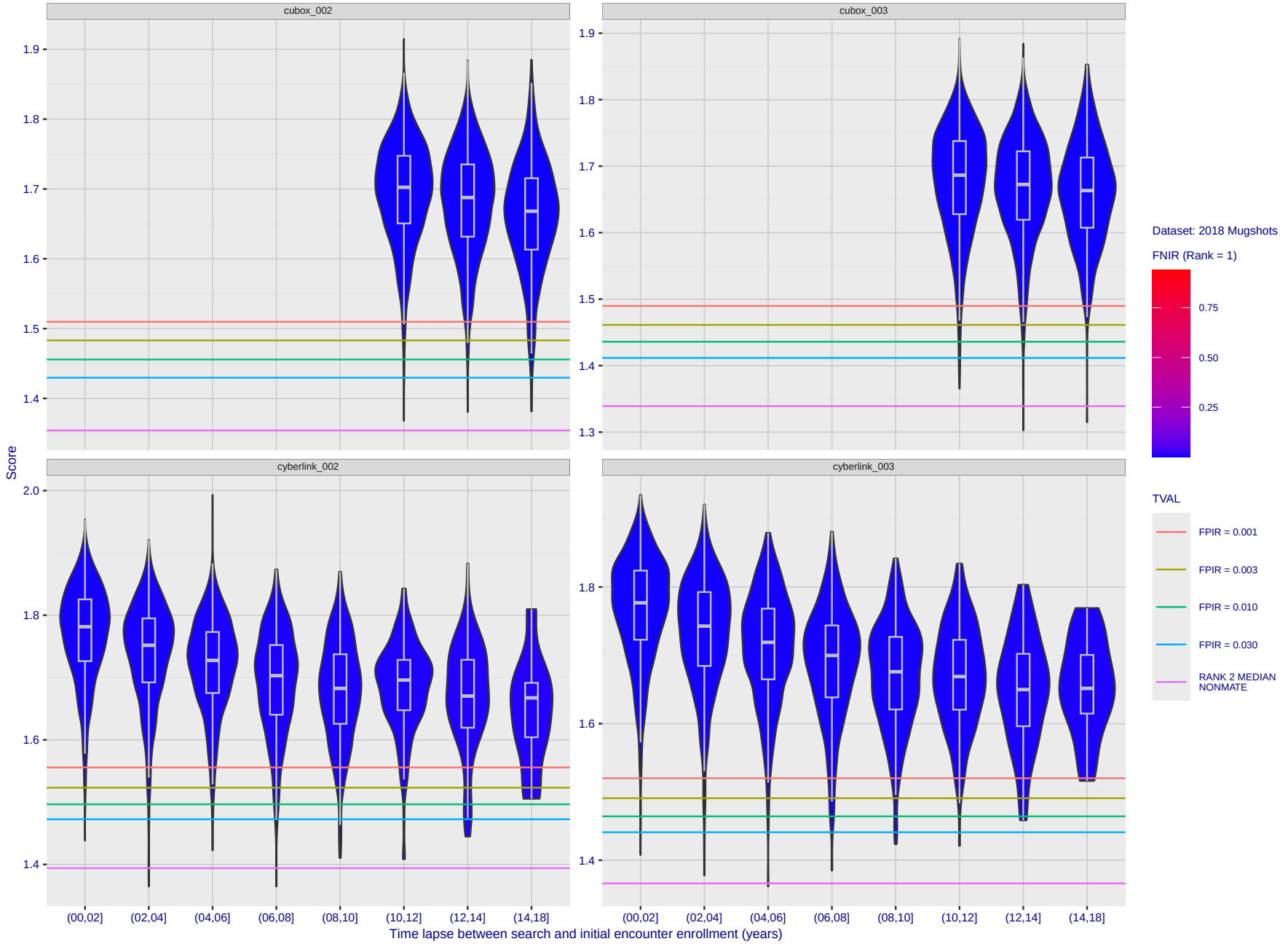

Figure 374: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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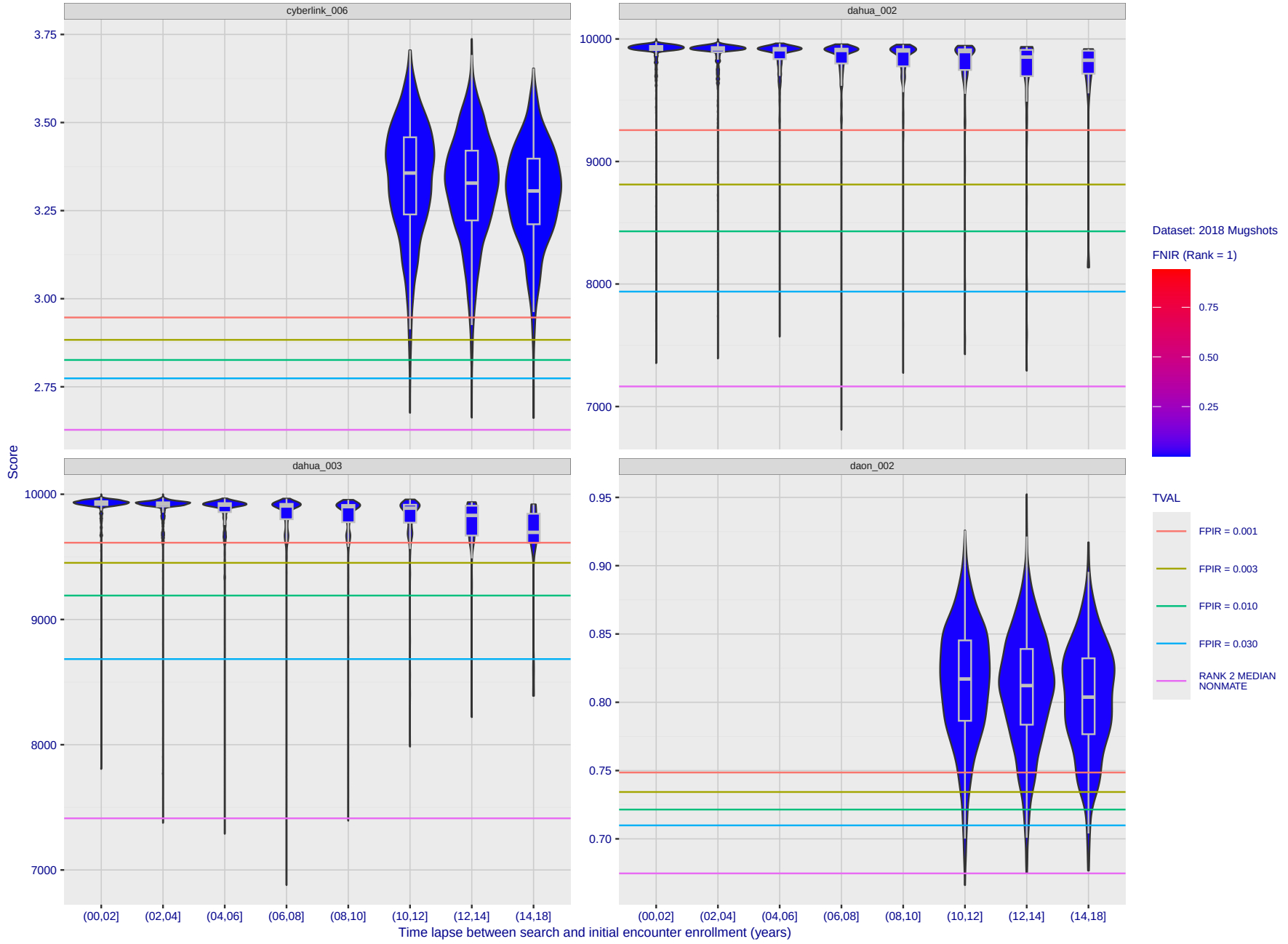

Figure 375: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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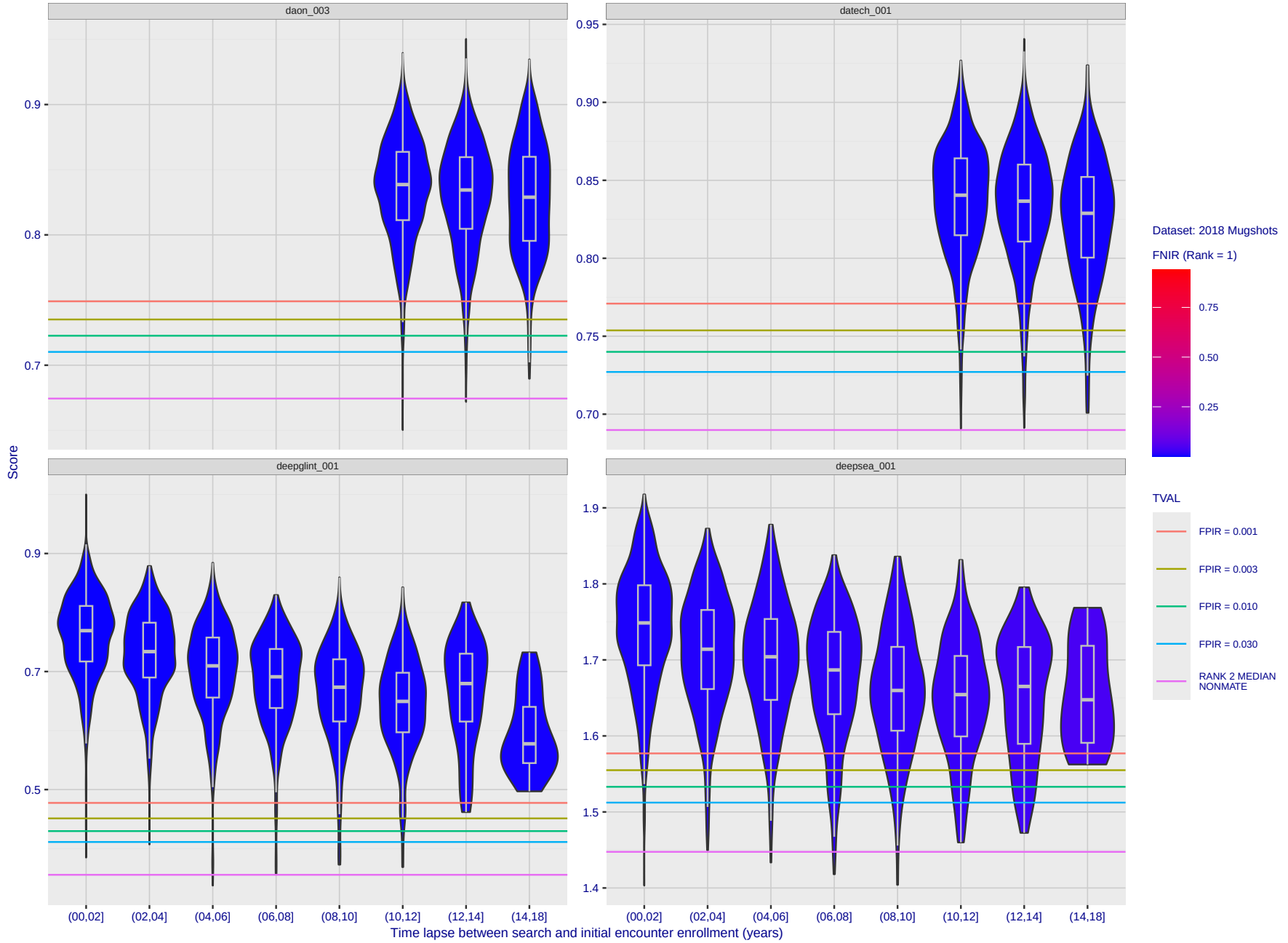

Figure 376: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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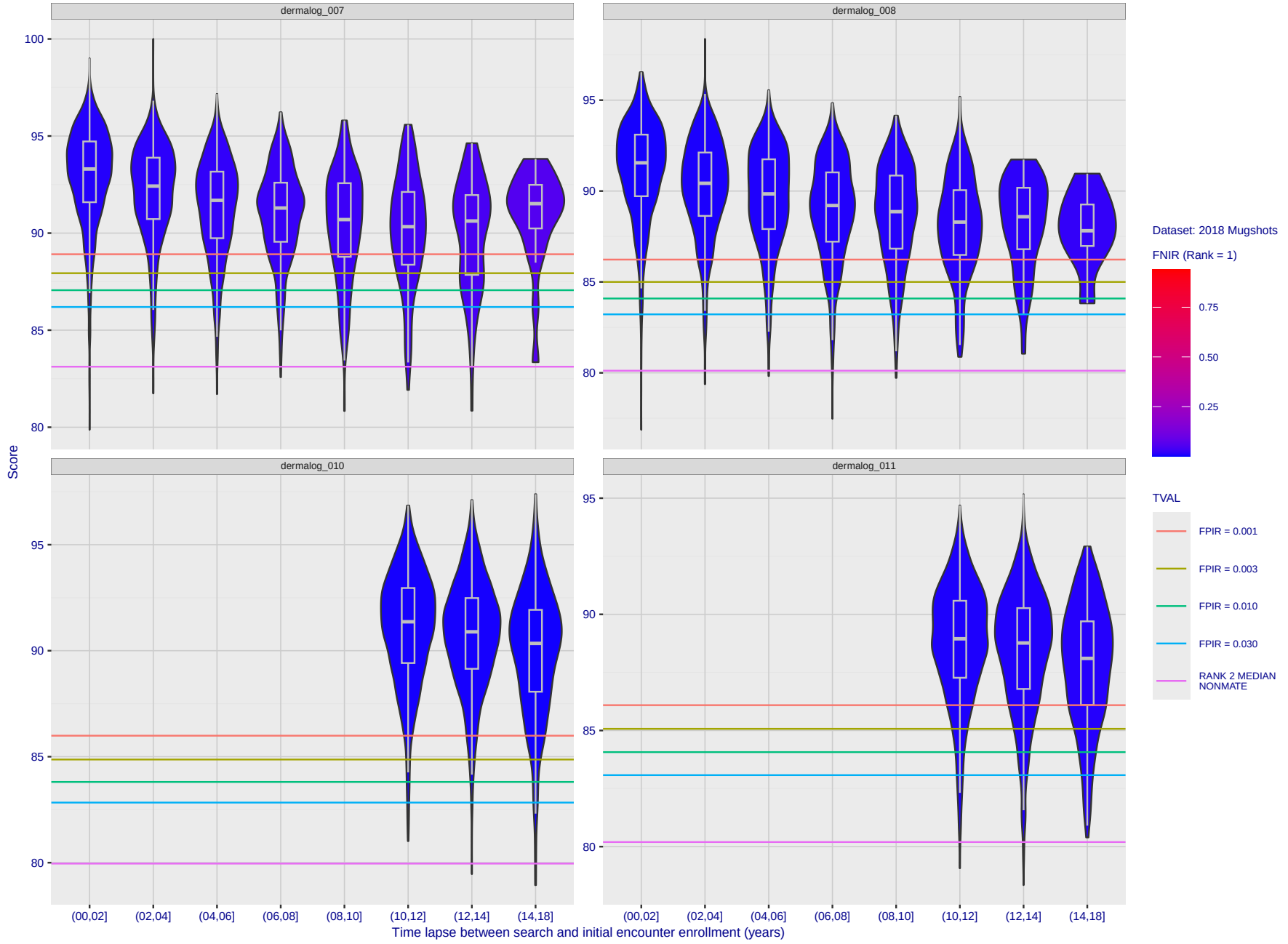

Figure 377: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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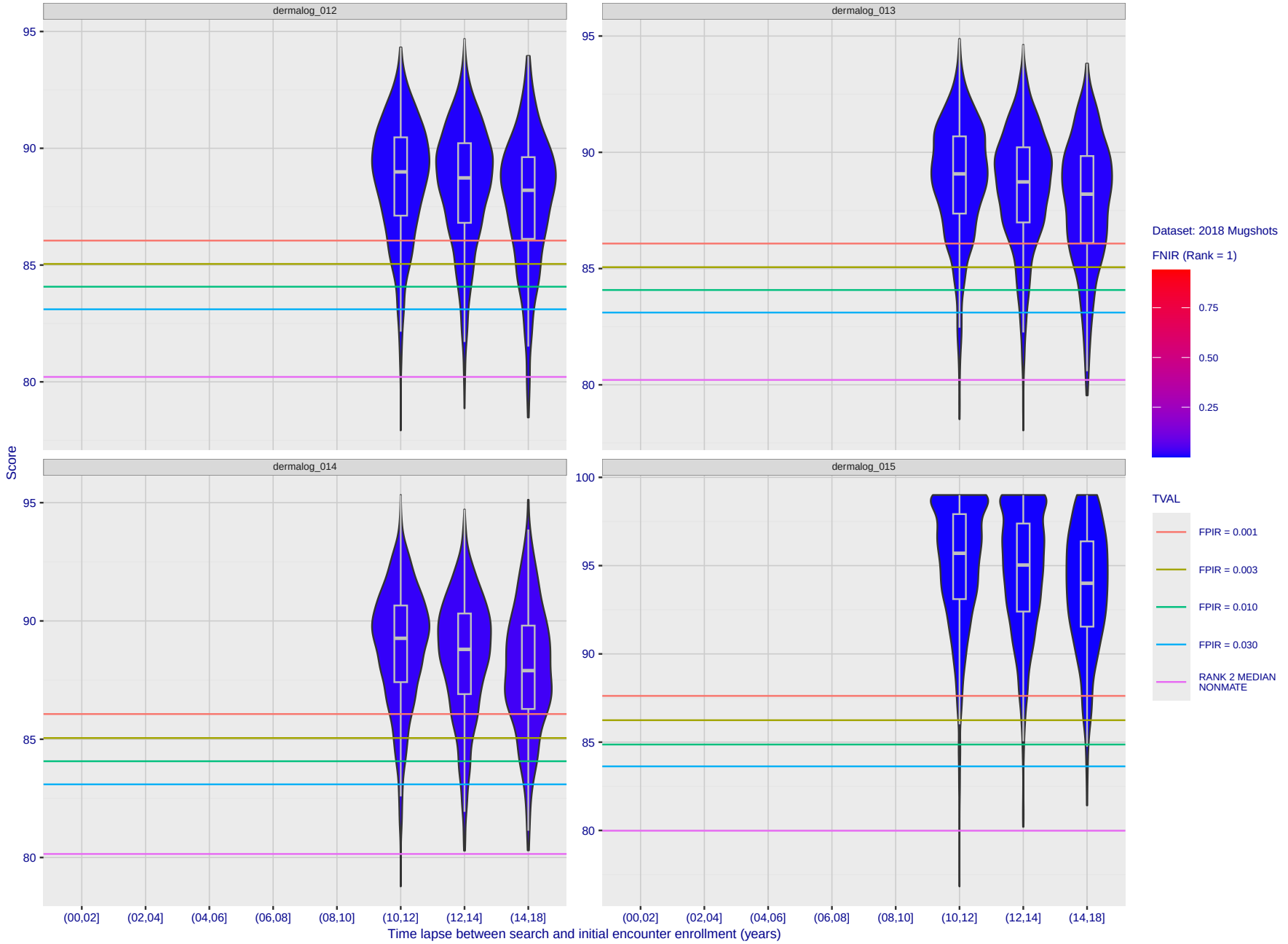

Figure 378: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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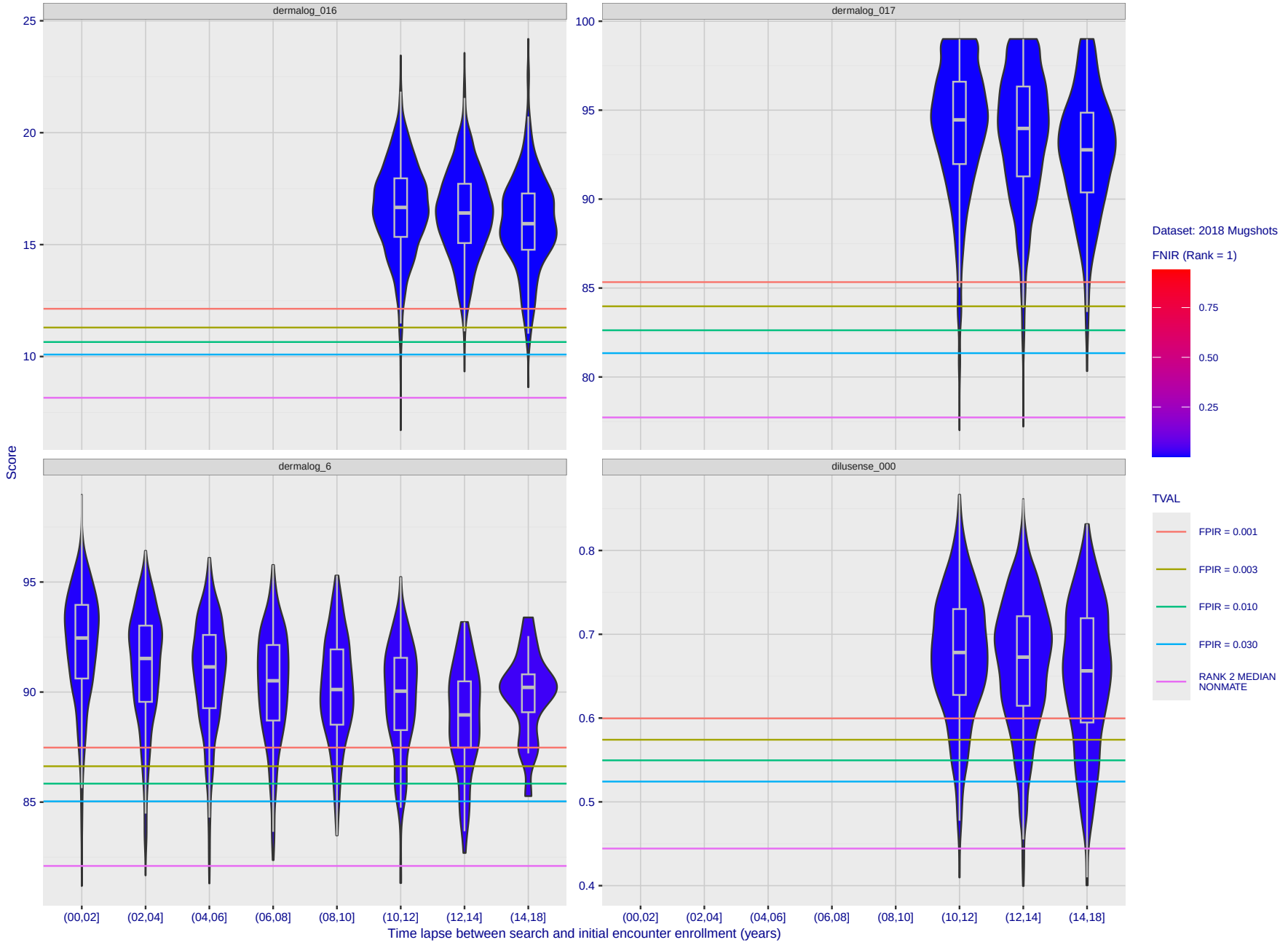

Figure 379: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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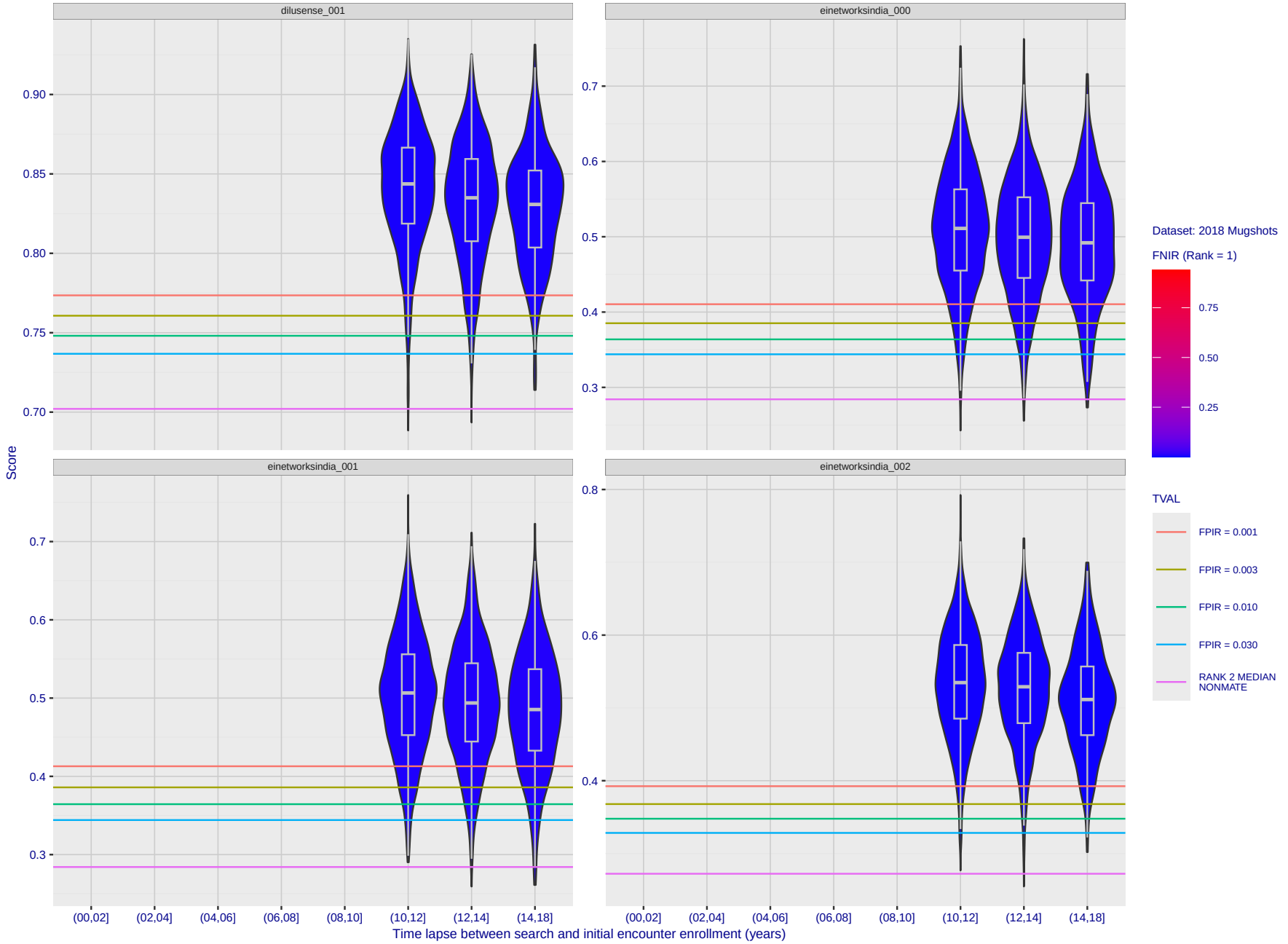

Figure 380: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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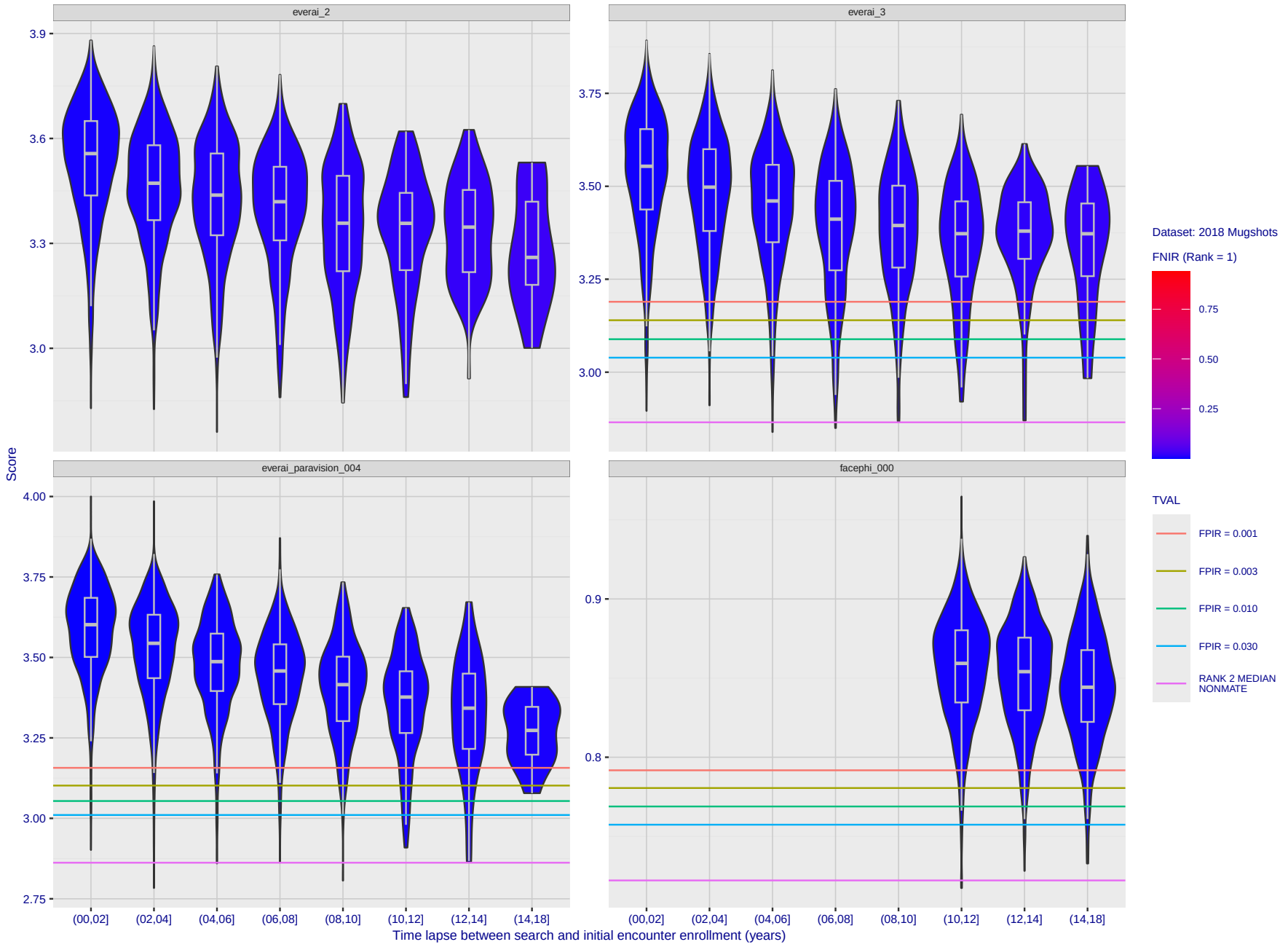

Figure 381: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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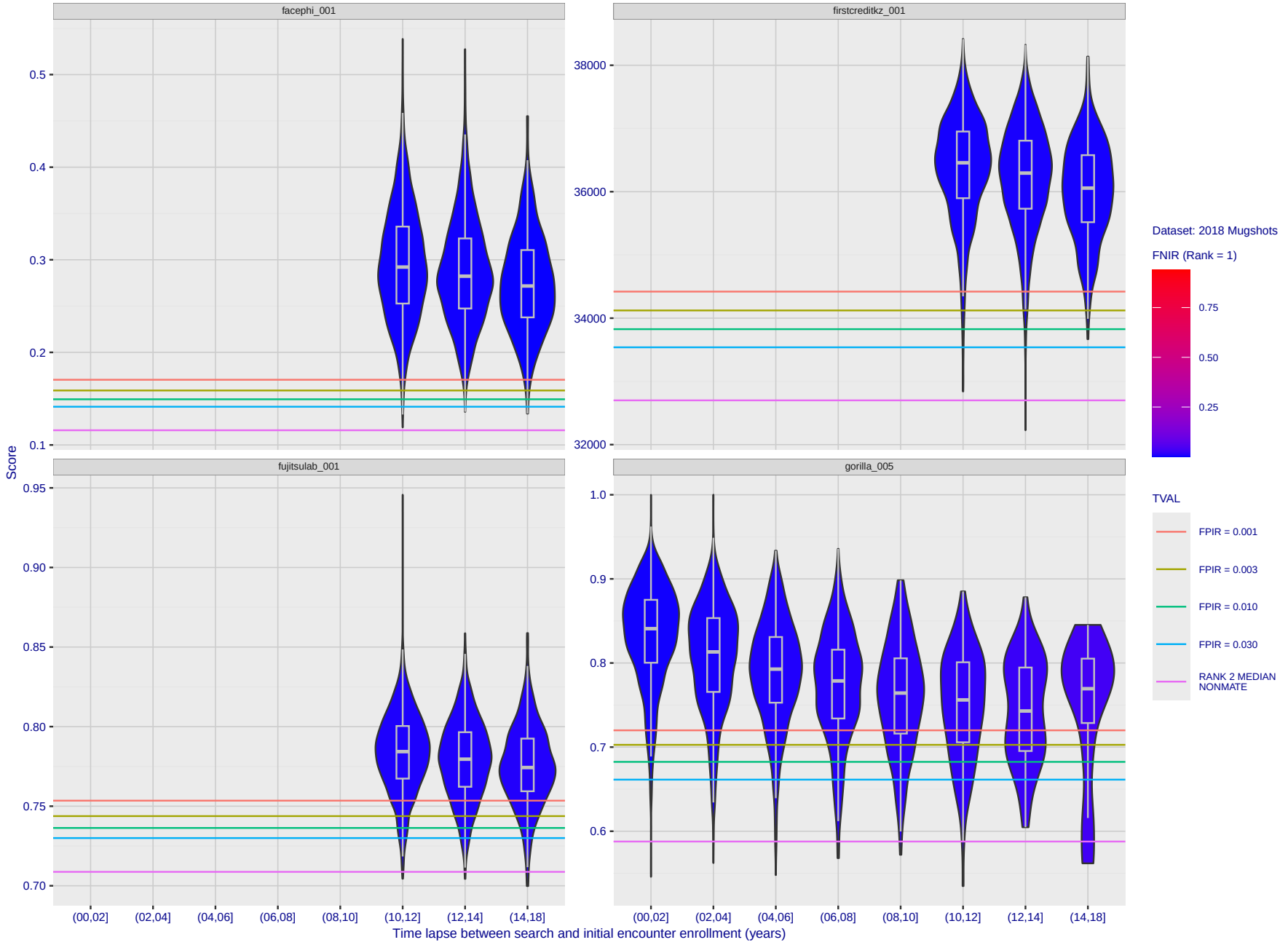

Figure 382: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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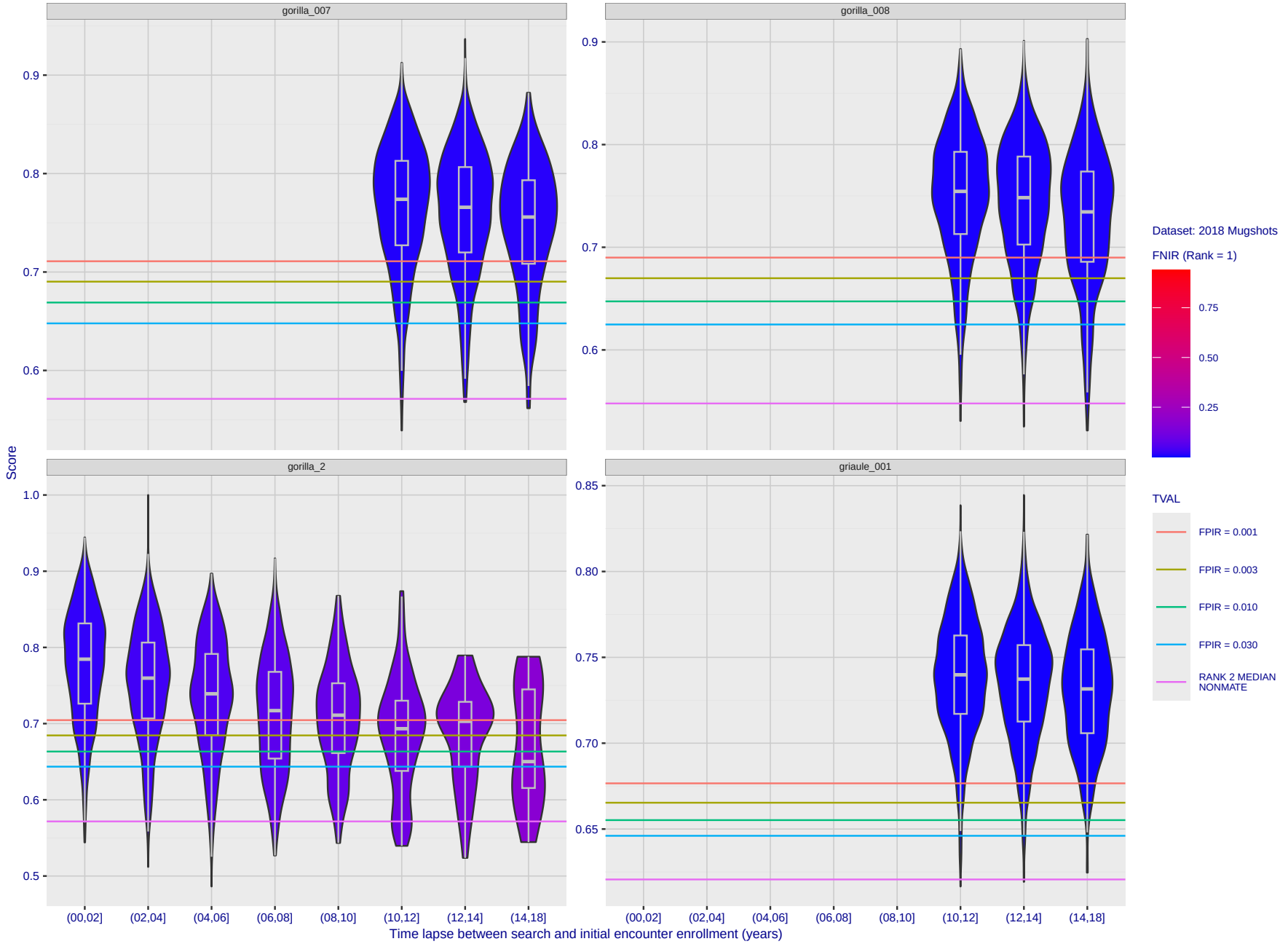

Figure 383: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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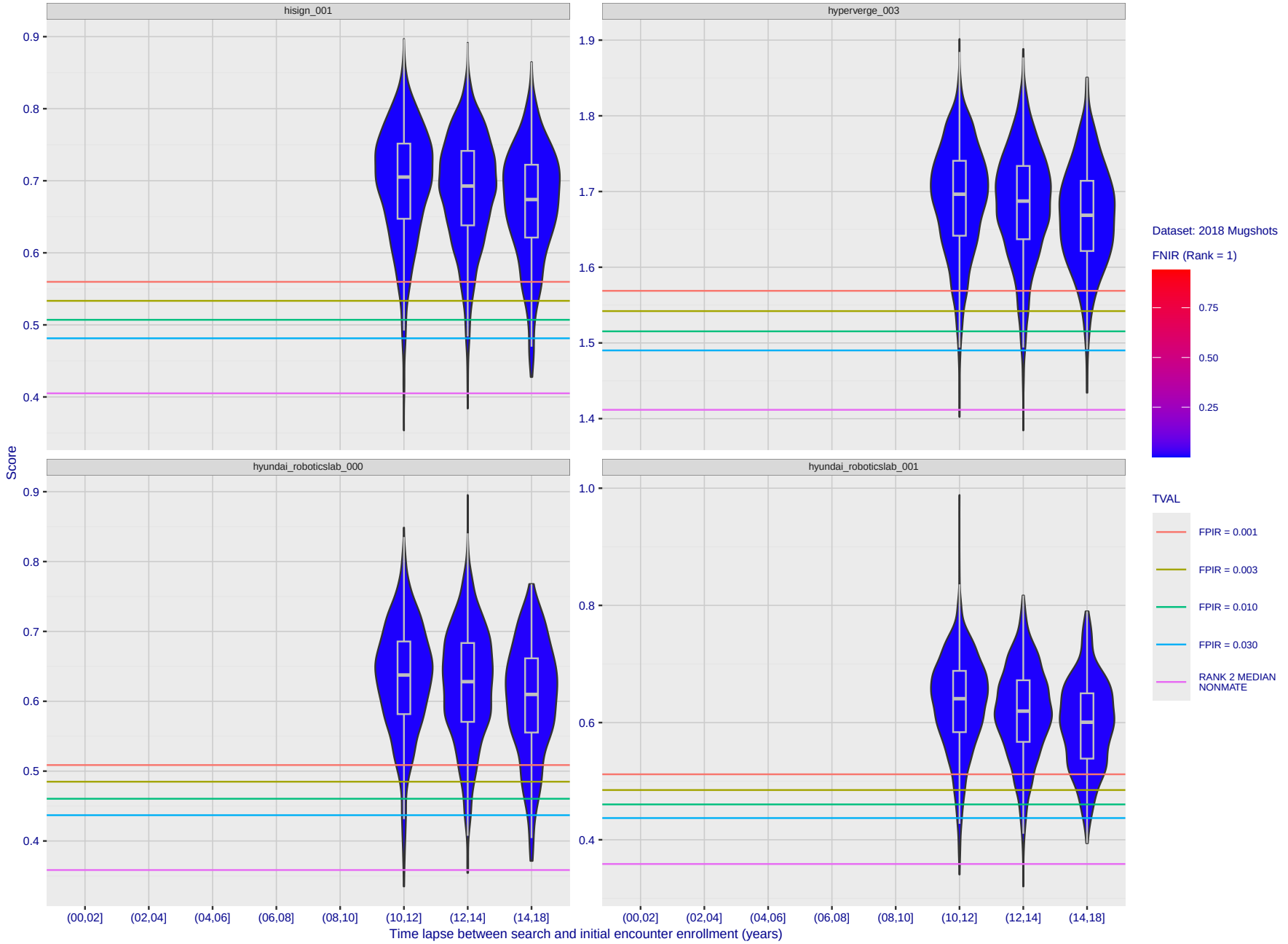

Figure 384: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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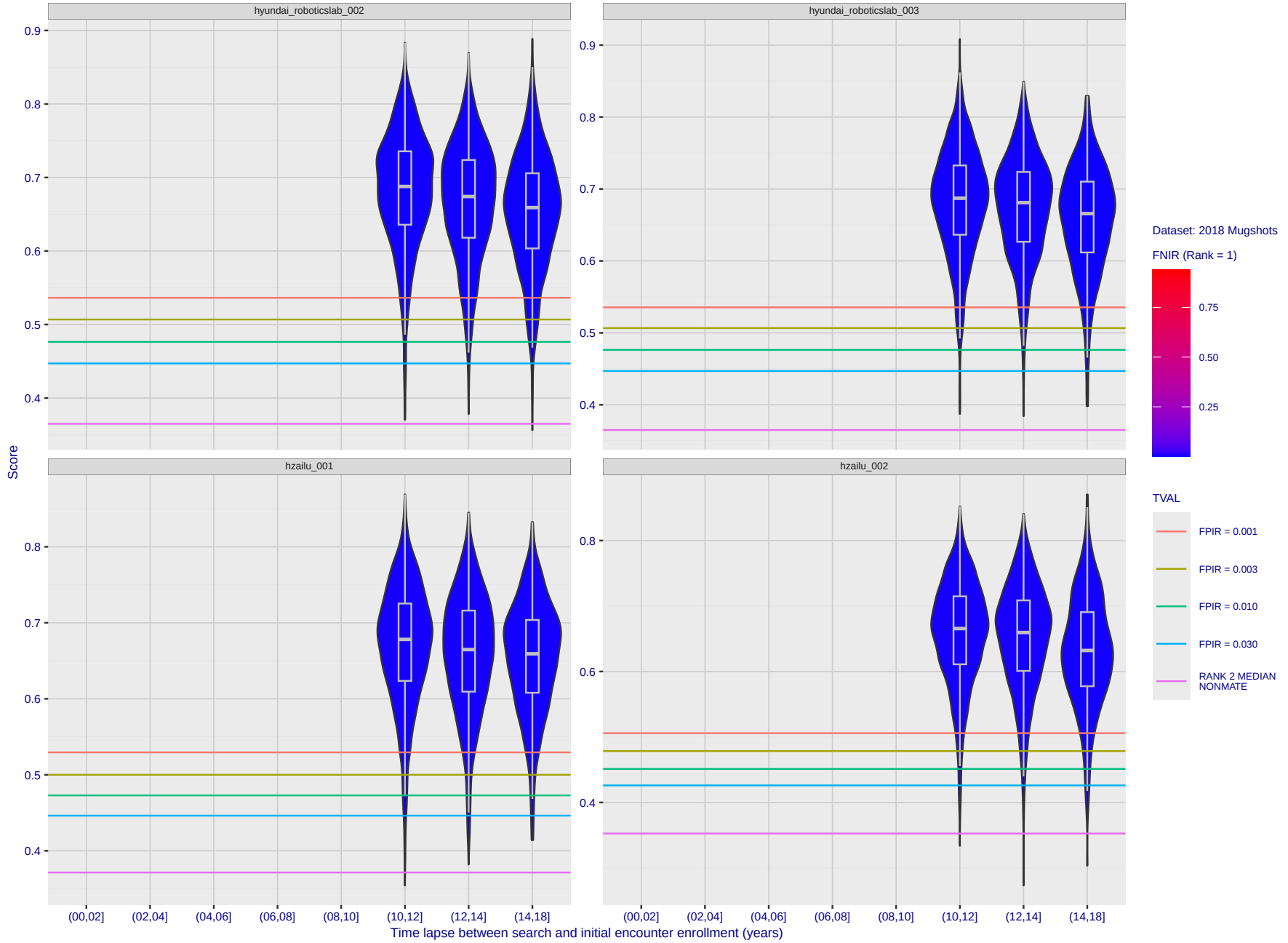

Figure 385: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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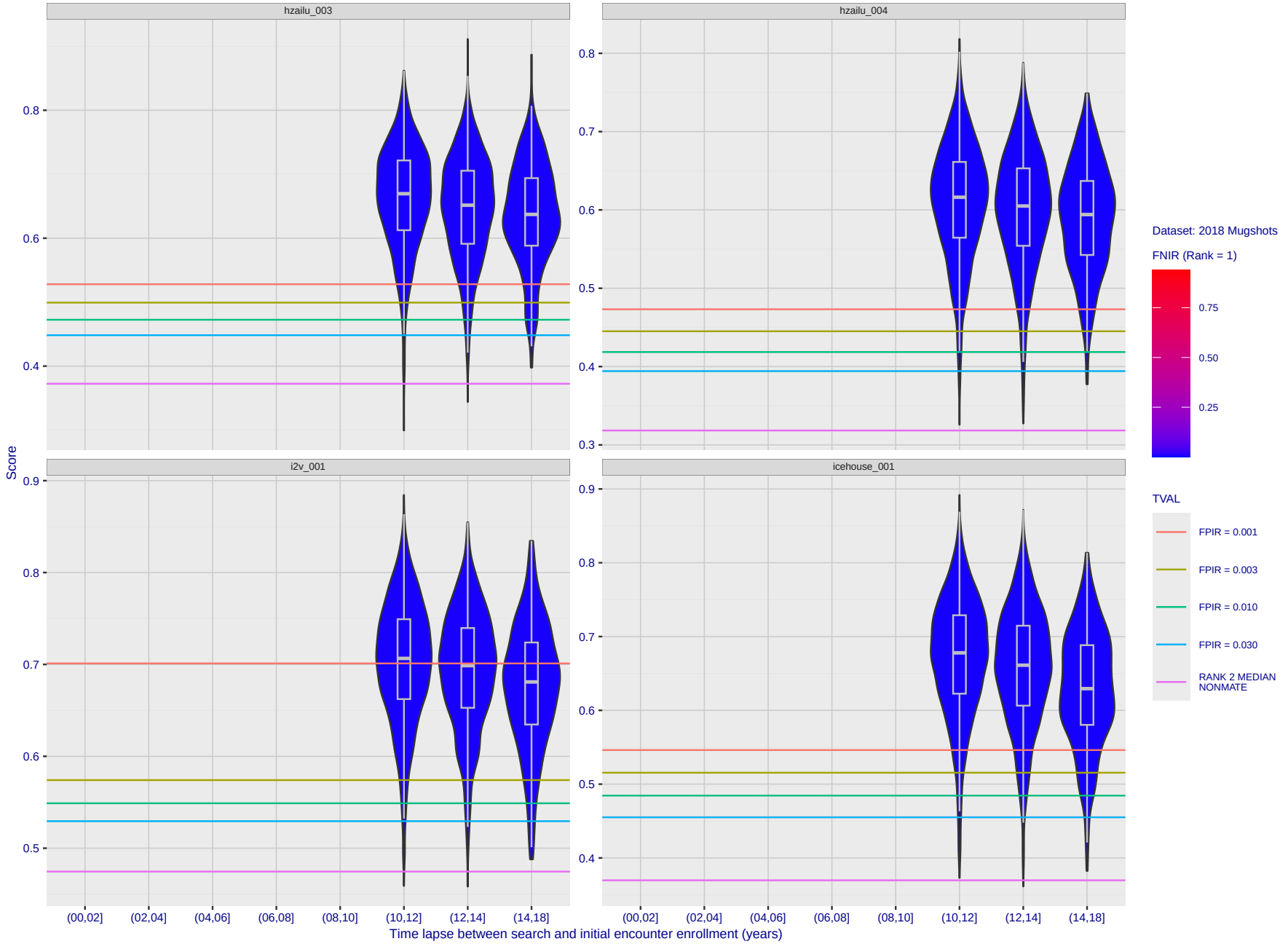

Figure 386: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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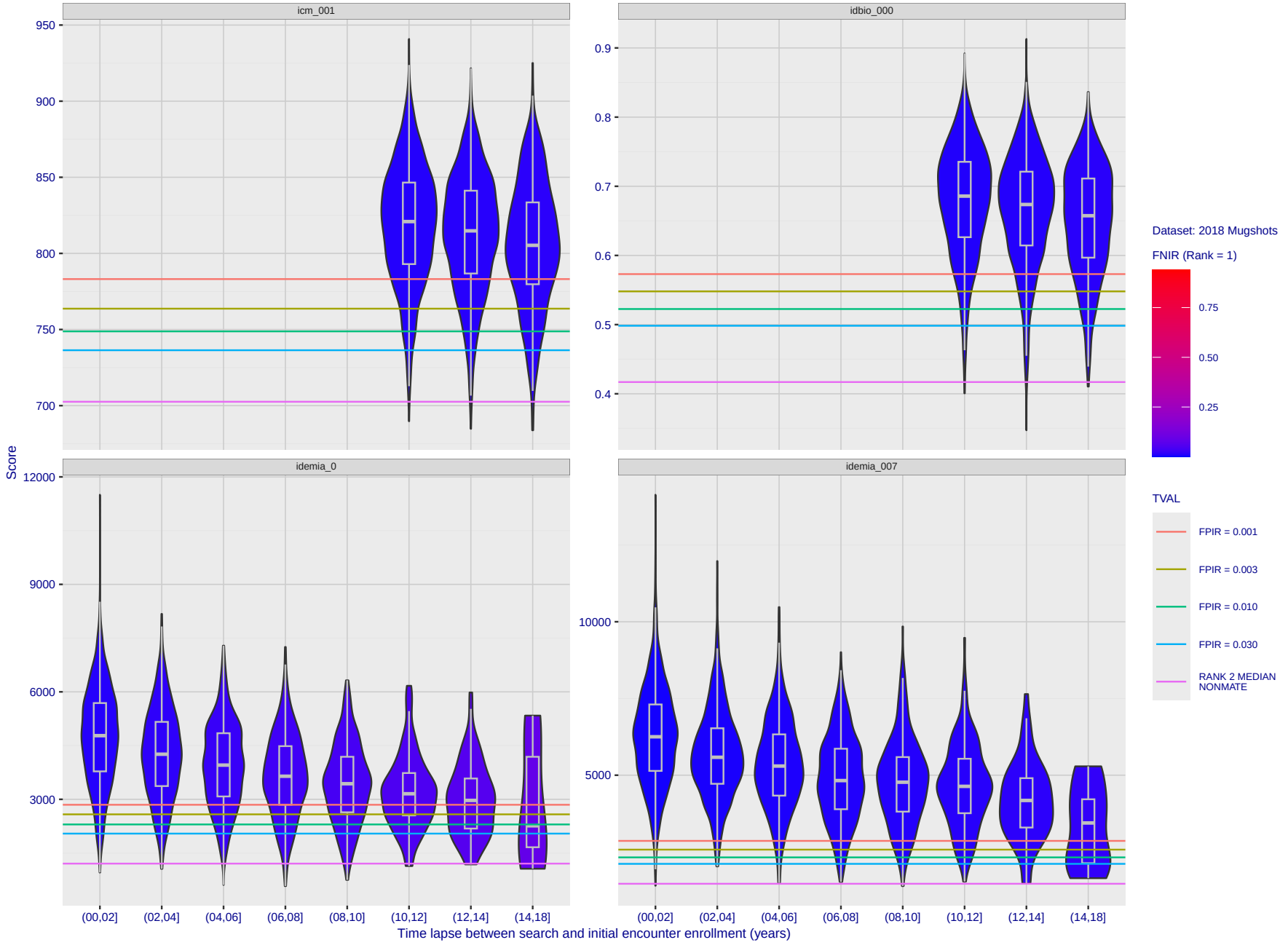

Figure 387: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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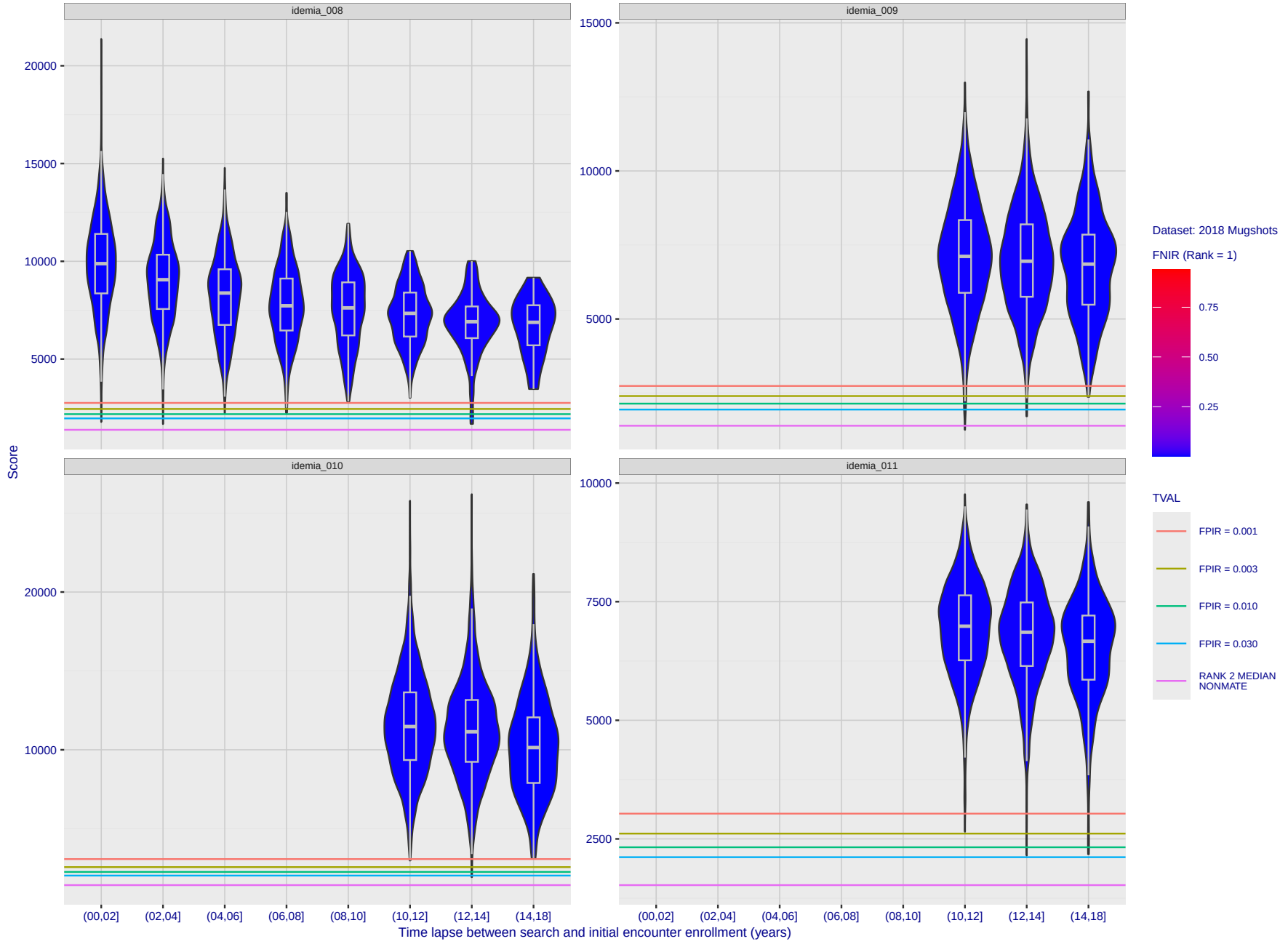

Figure 388: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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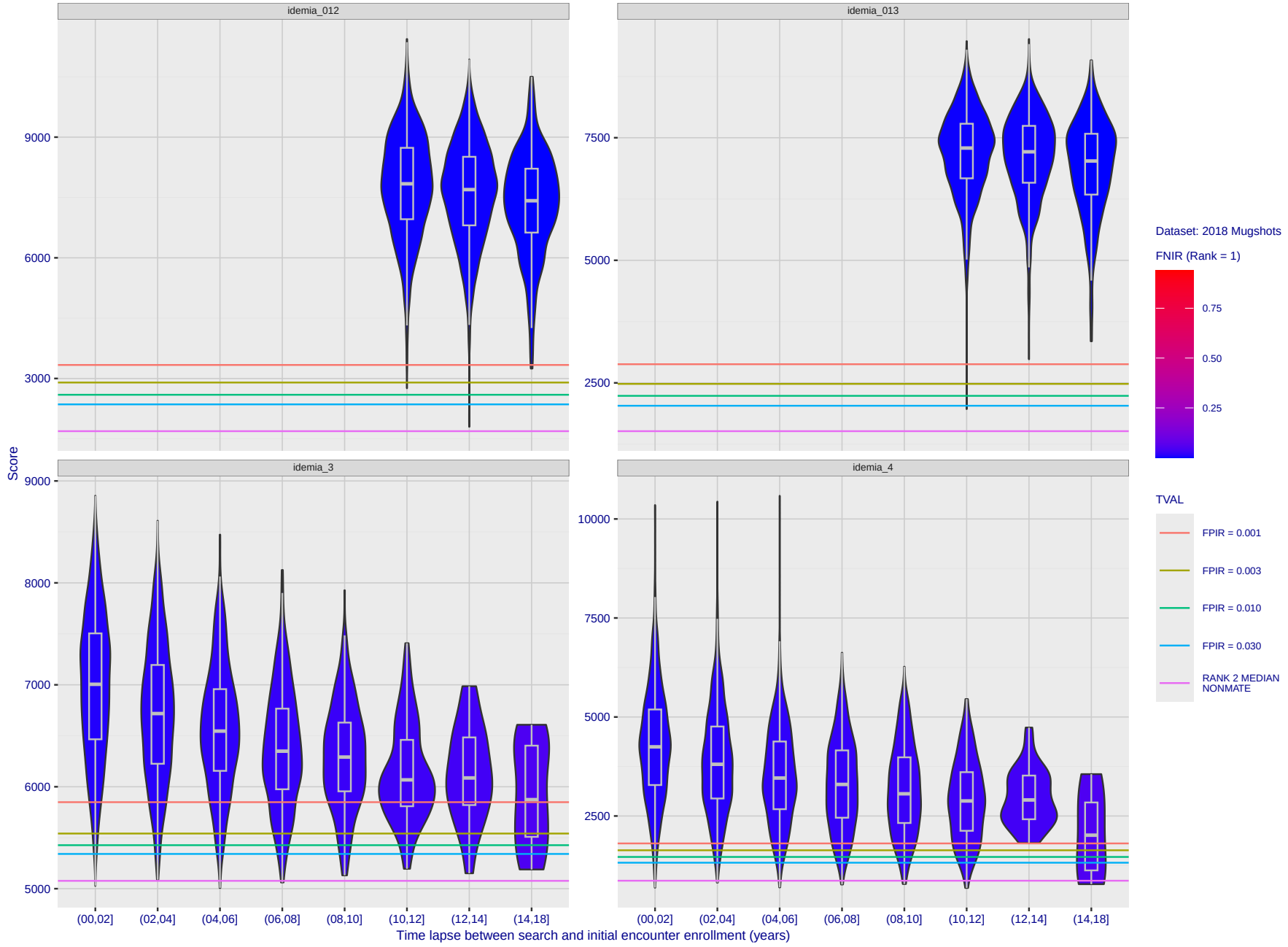

Figure 389: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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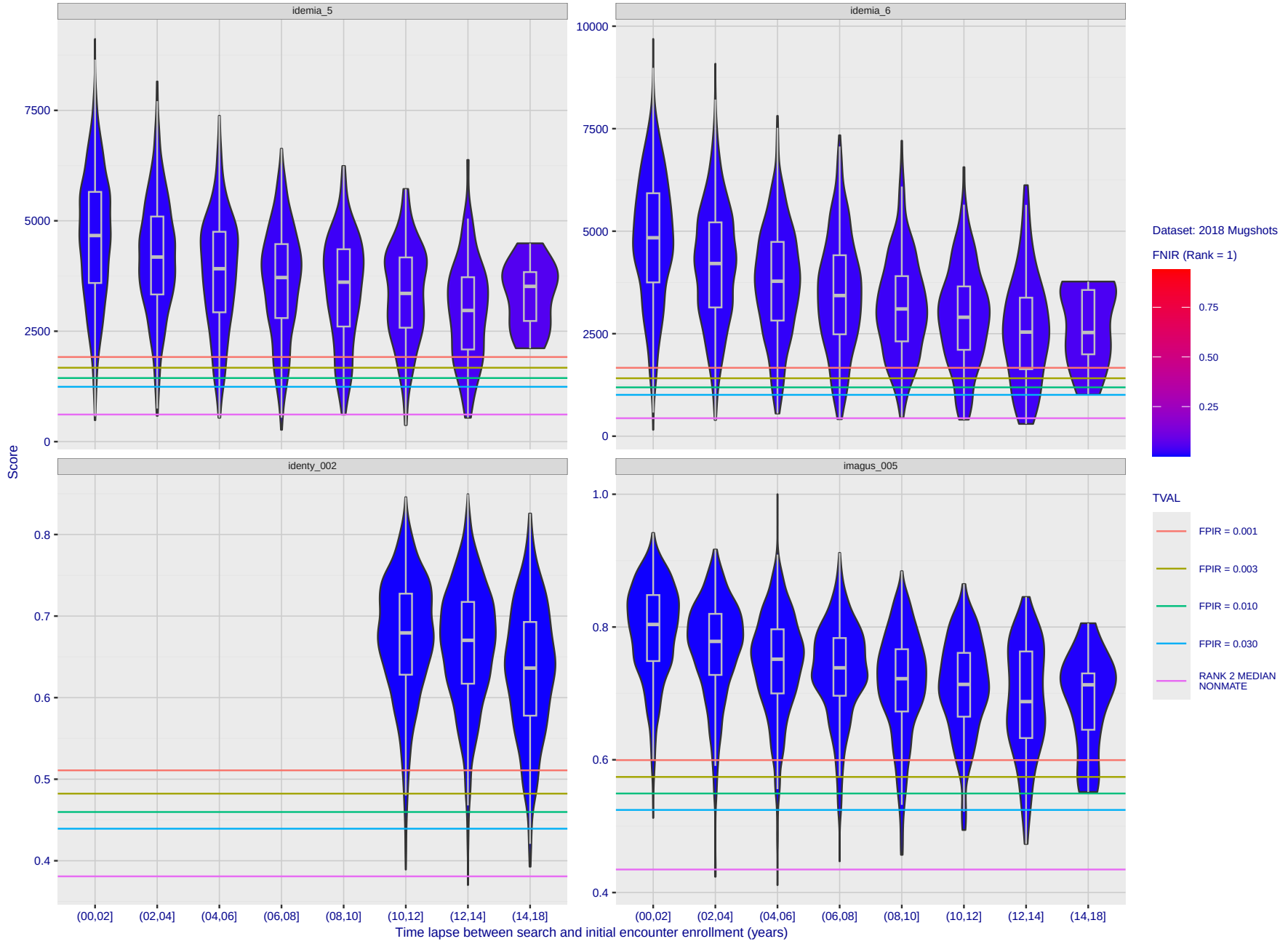

Figure 390: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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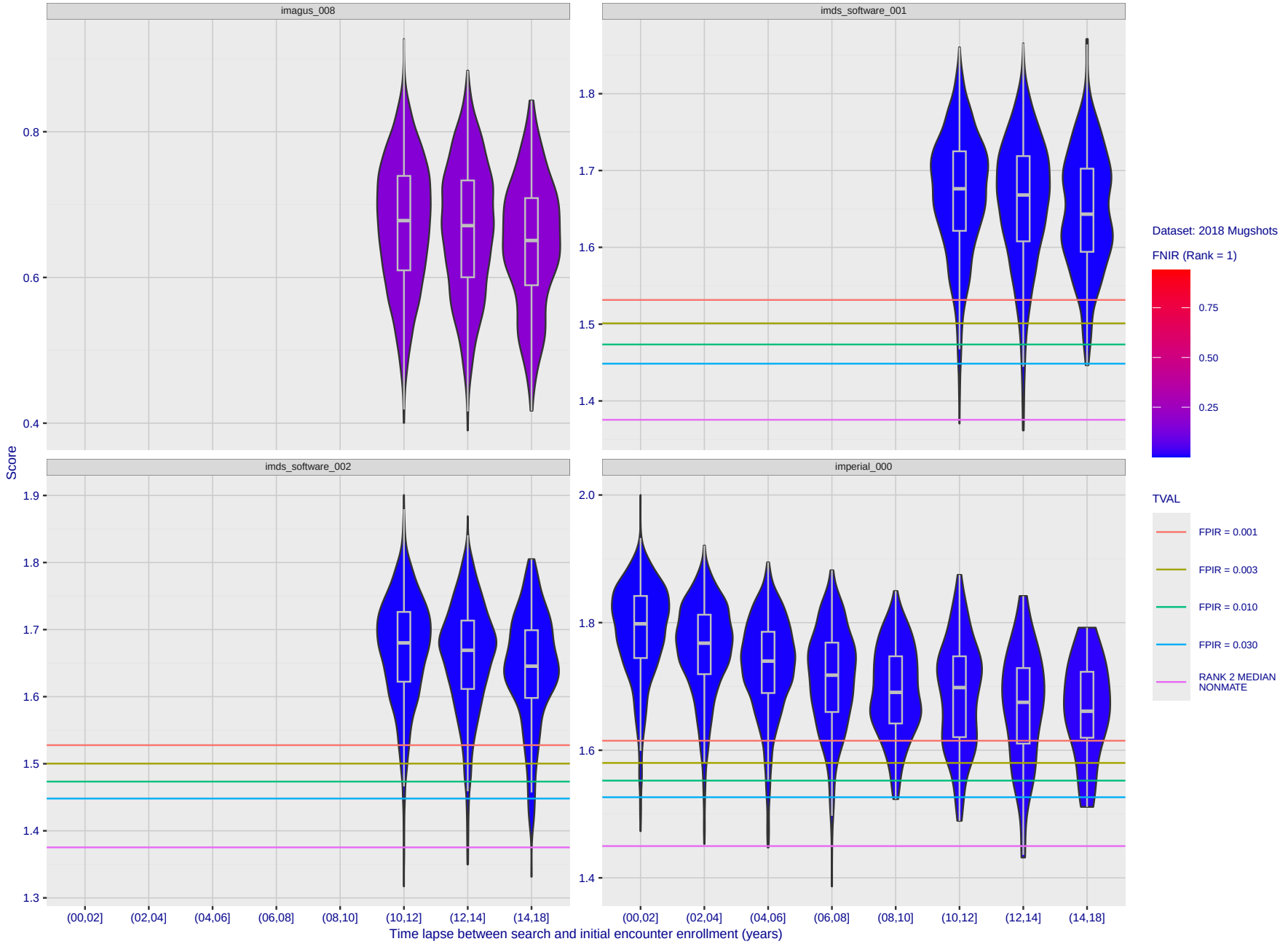

Figure 391: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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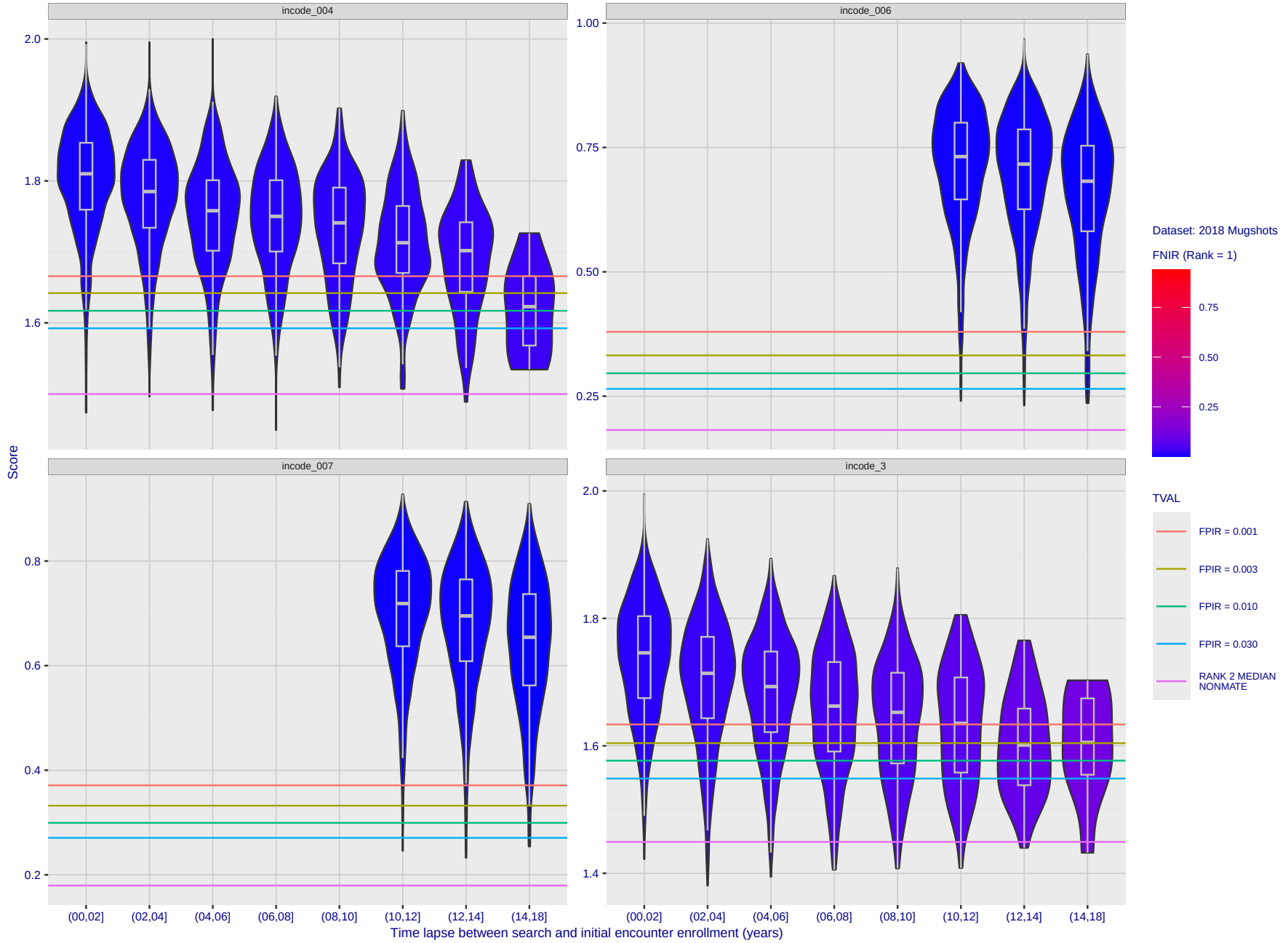

Figure 392: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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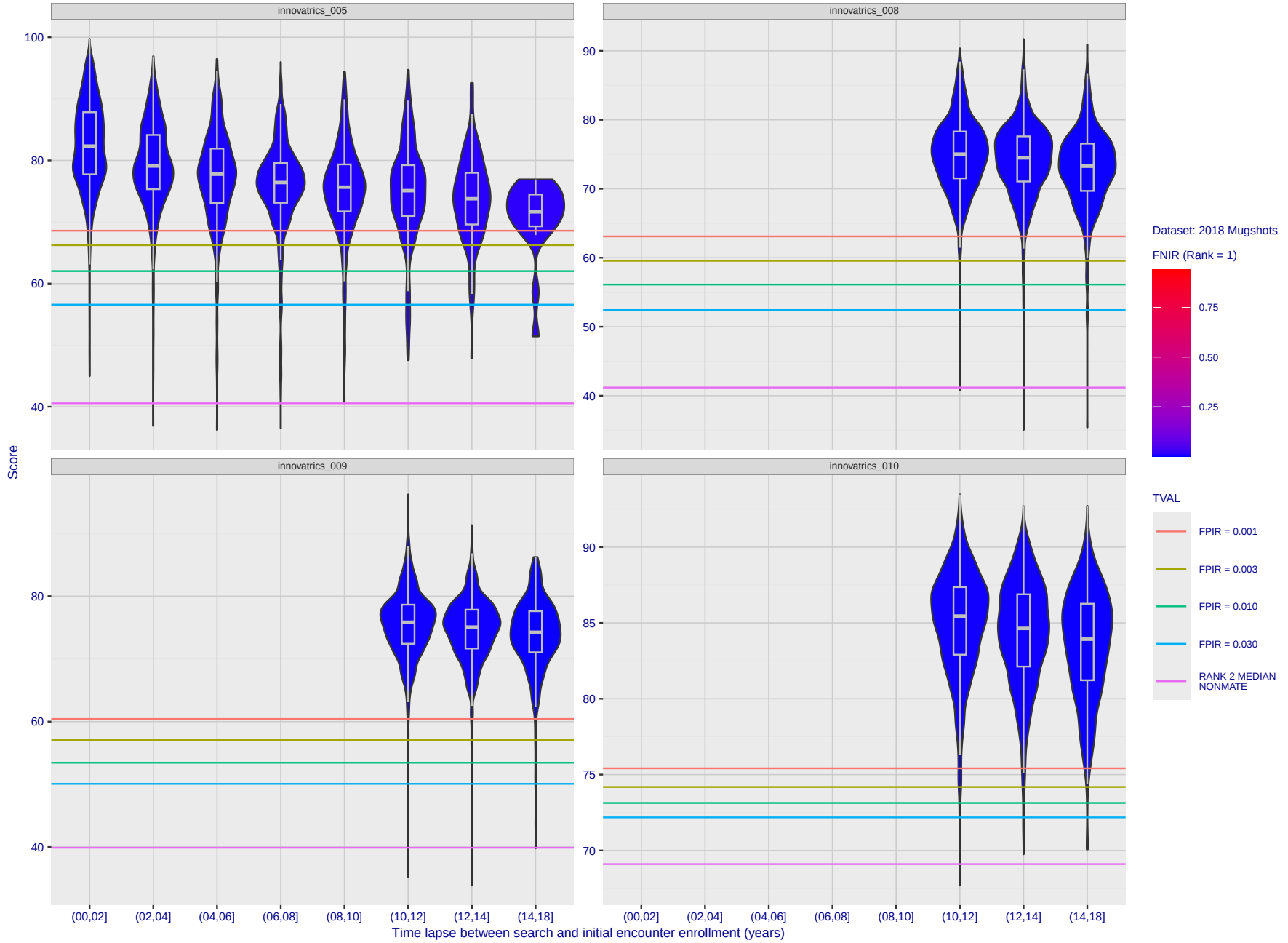

Figure 393: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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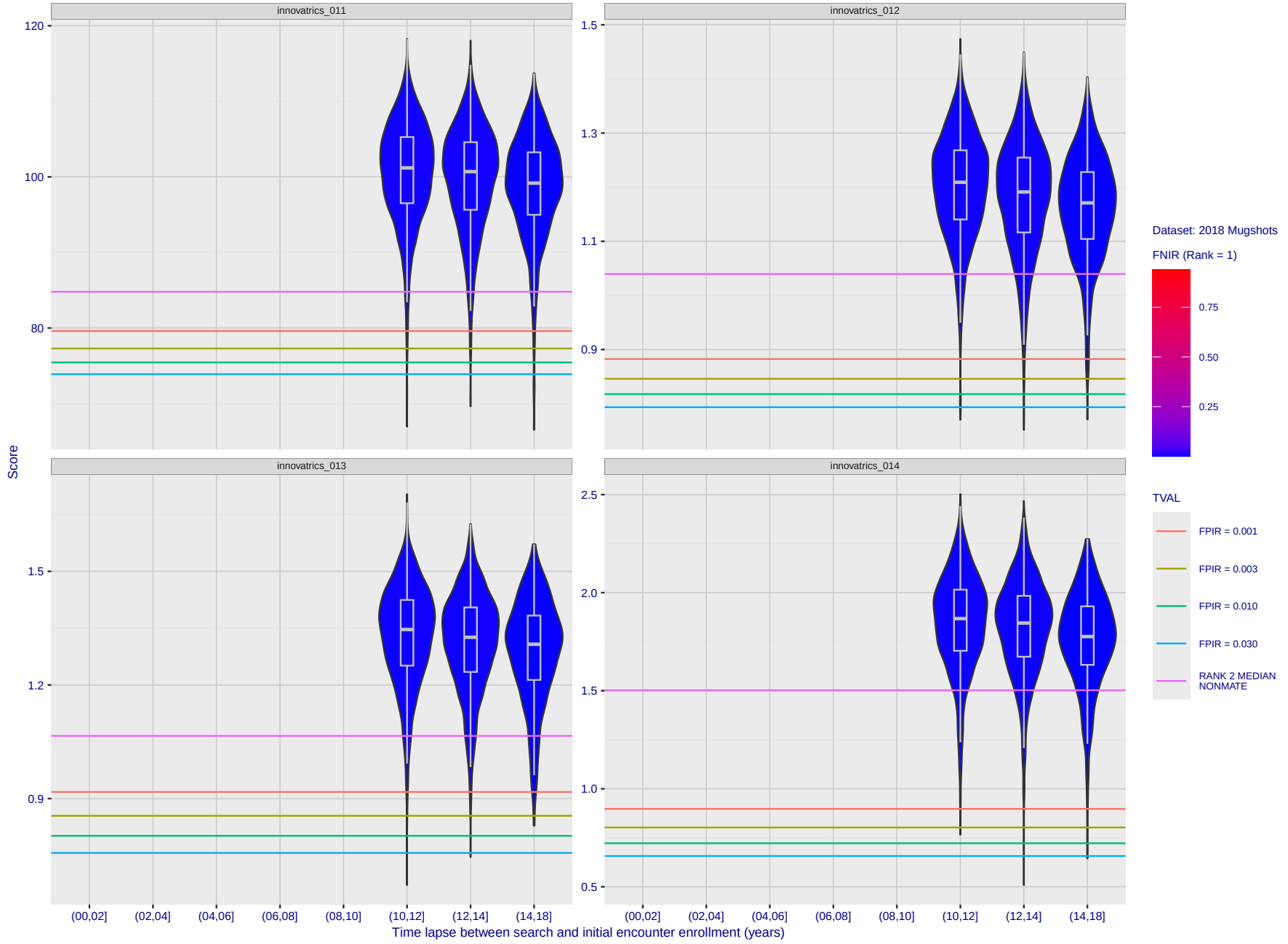

Figure 394: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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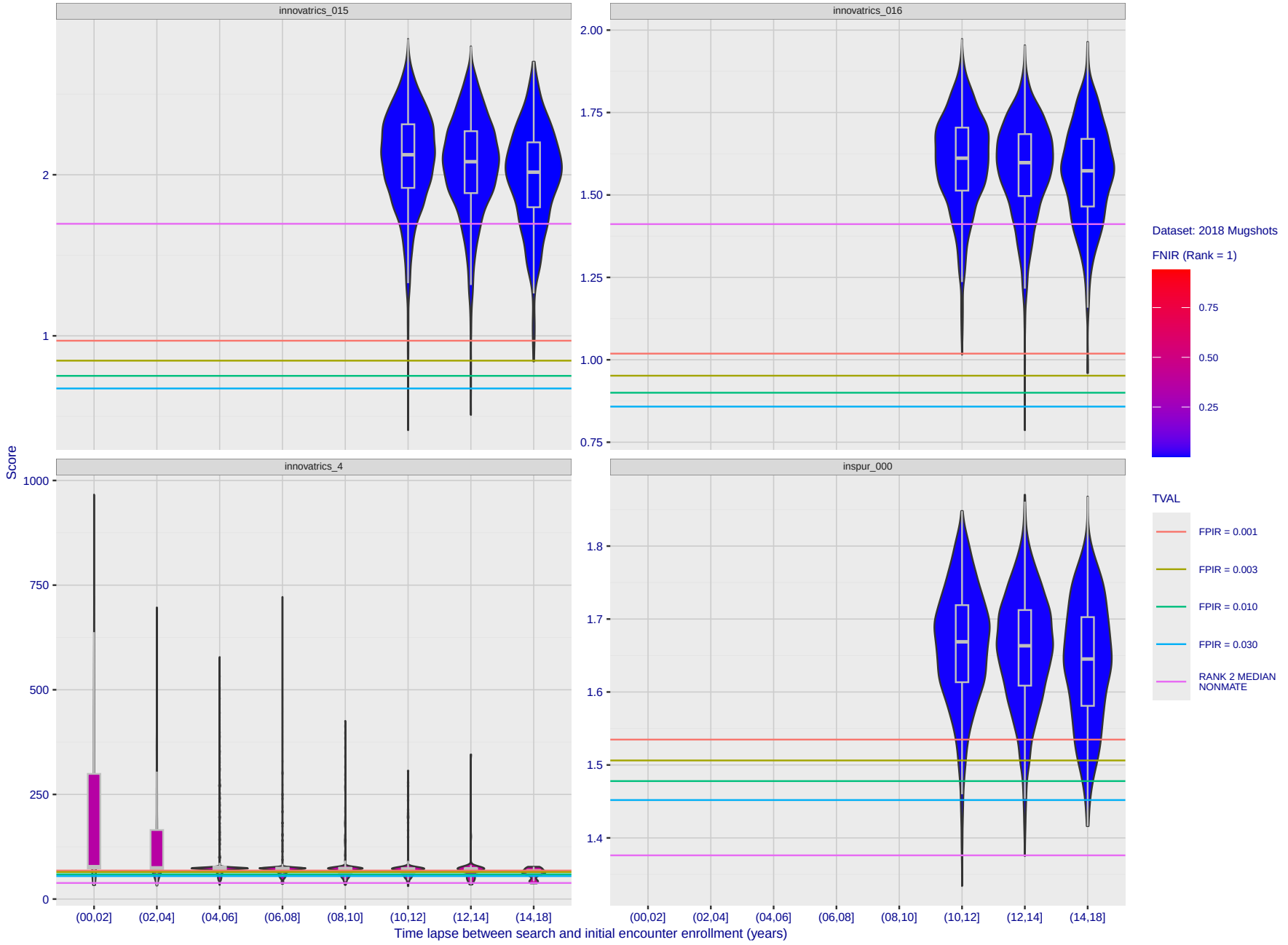

Figure 395: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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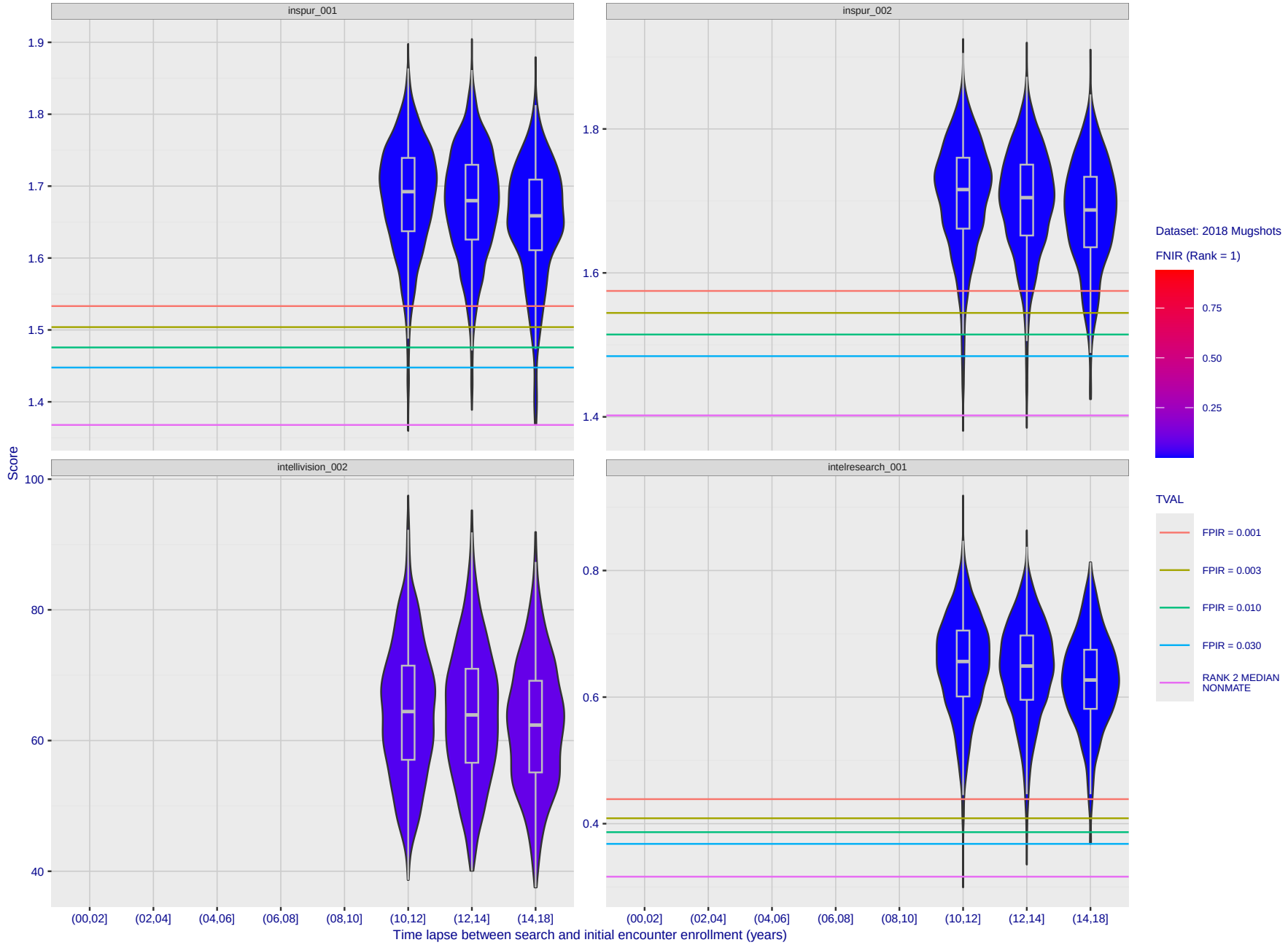

Figure 396: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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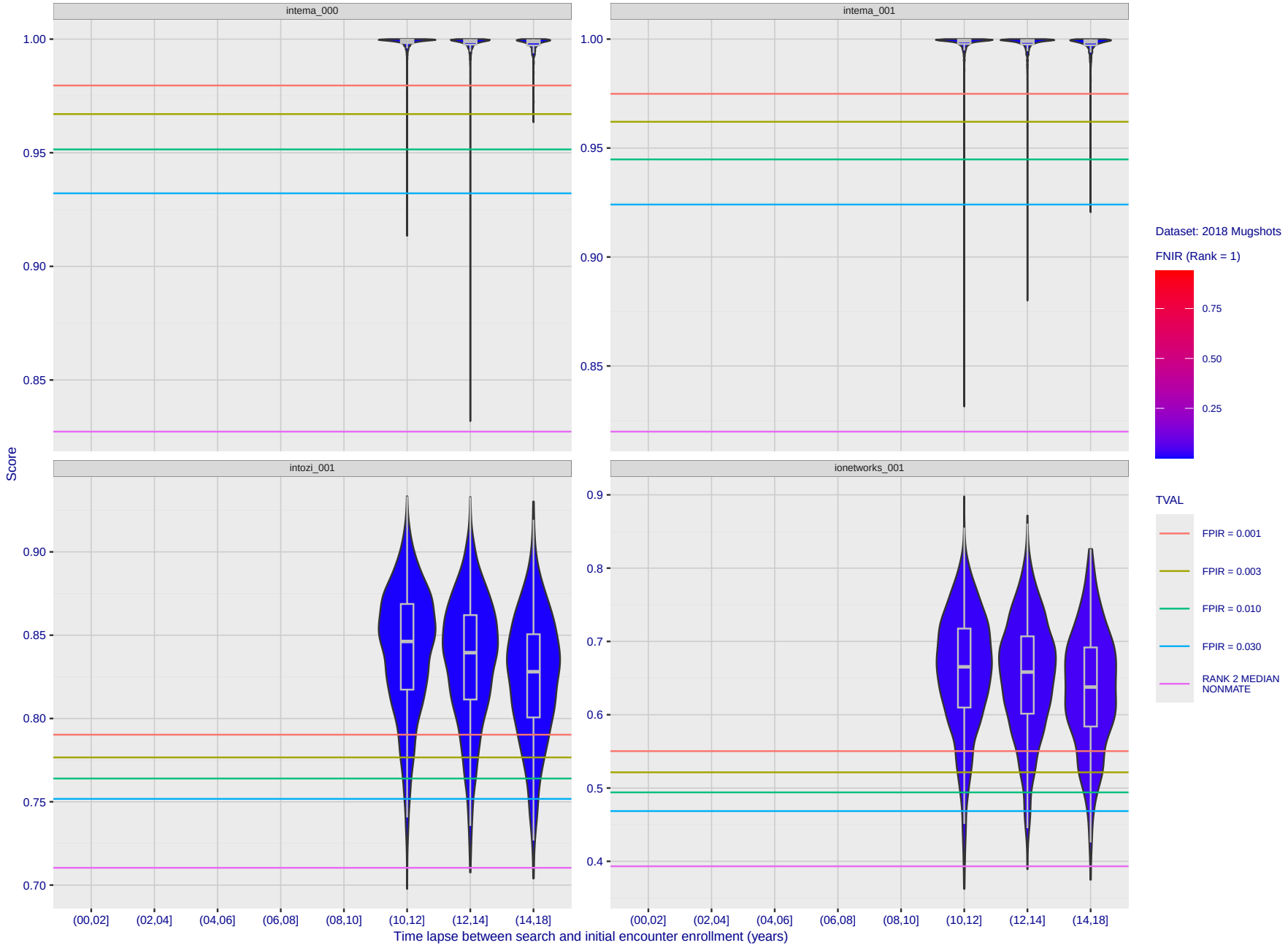

Figure 397: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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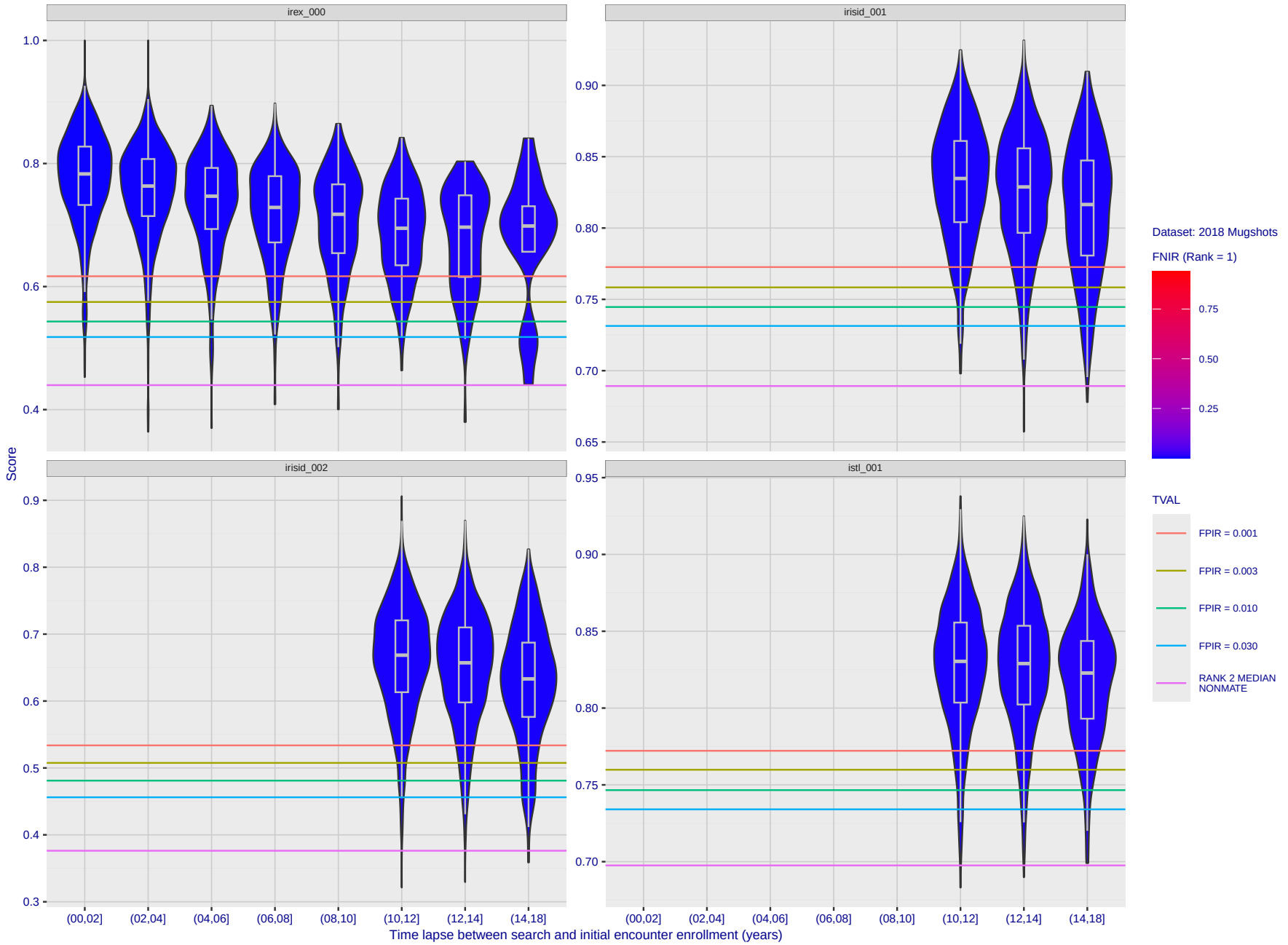

Figure 398: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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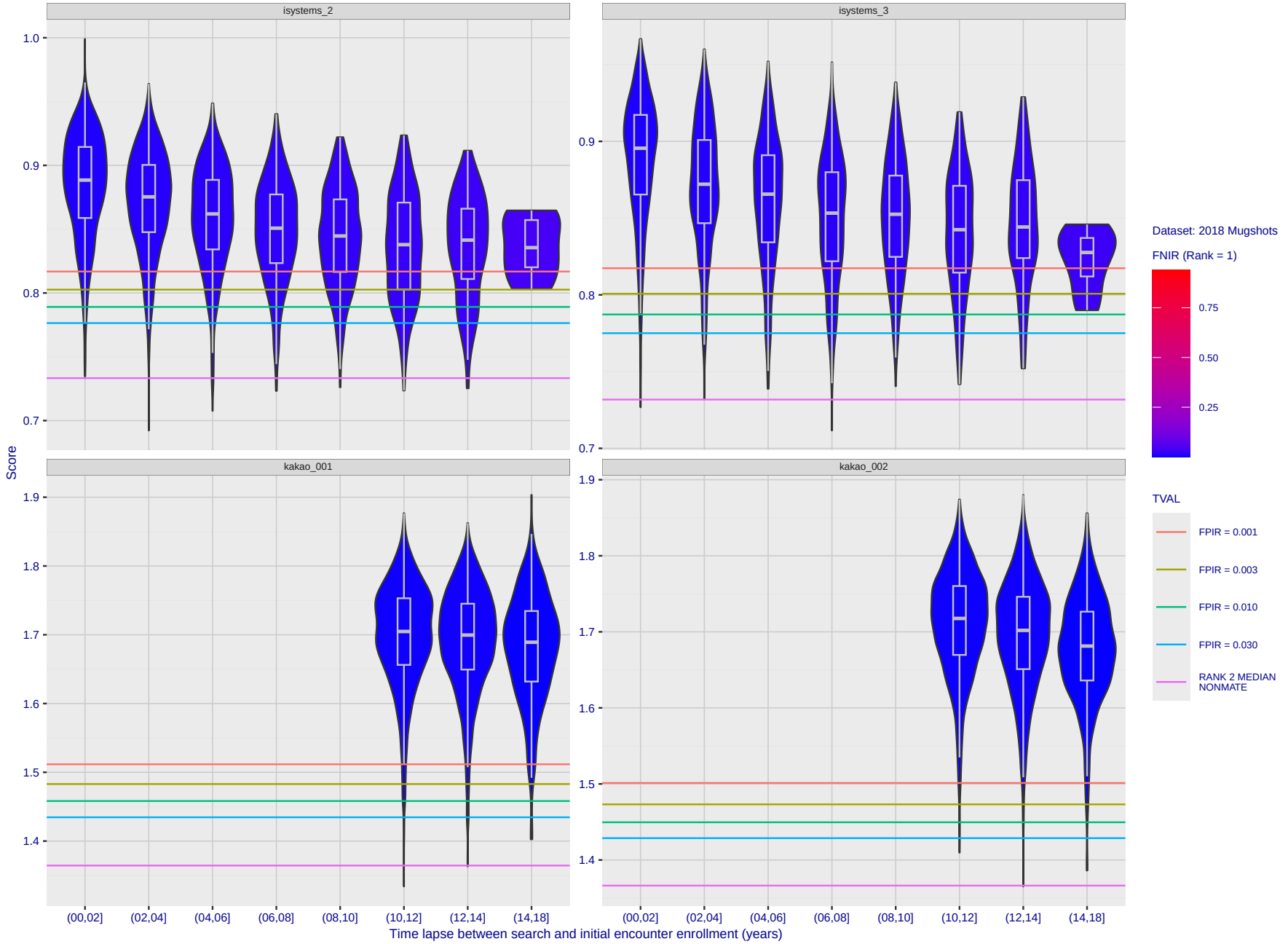

Figure 399: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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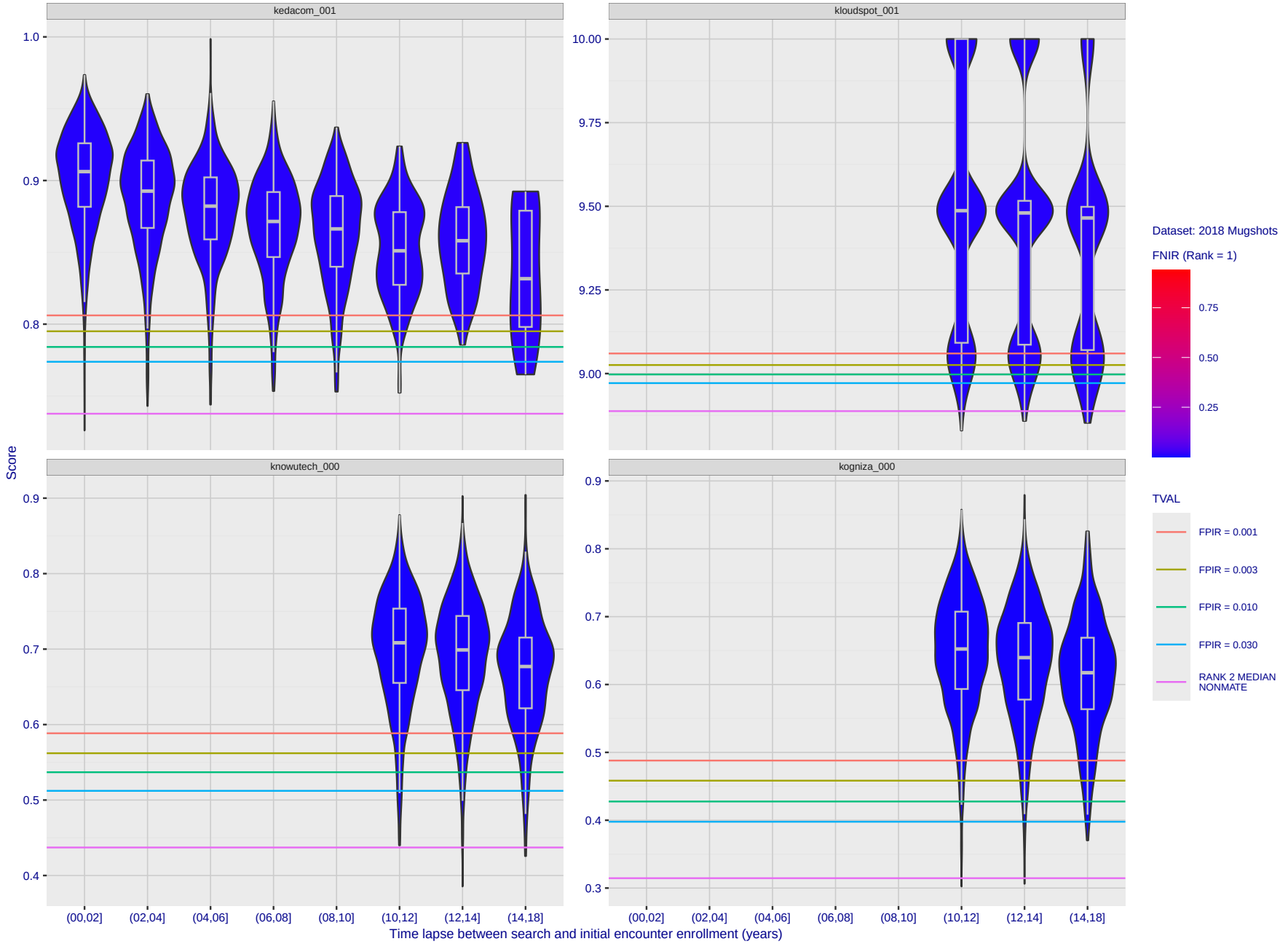

Figure 400: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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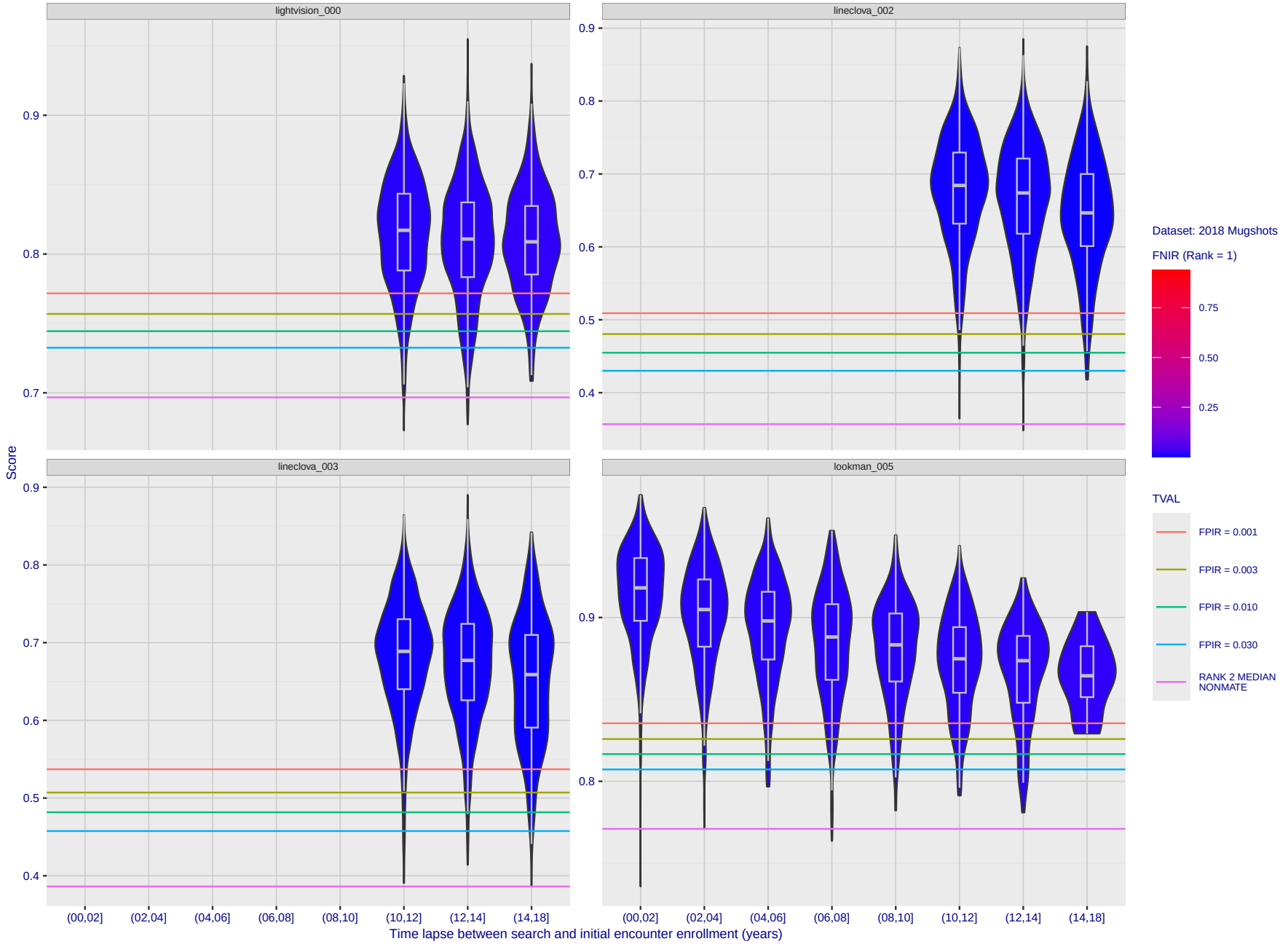

Figure 401: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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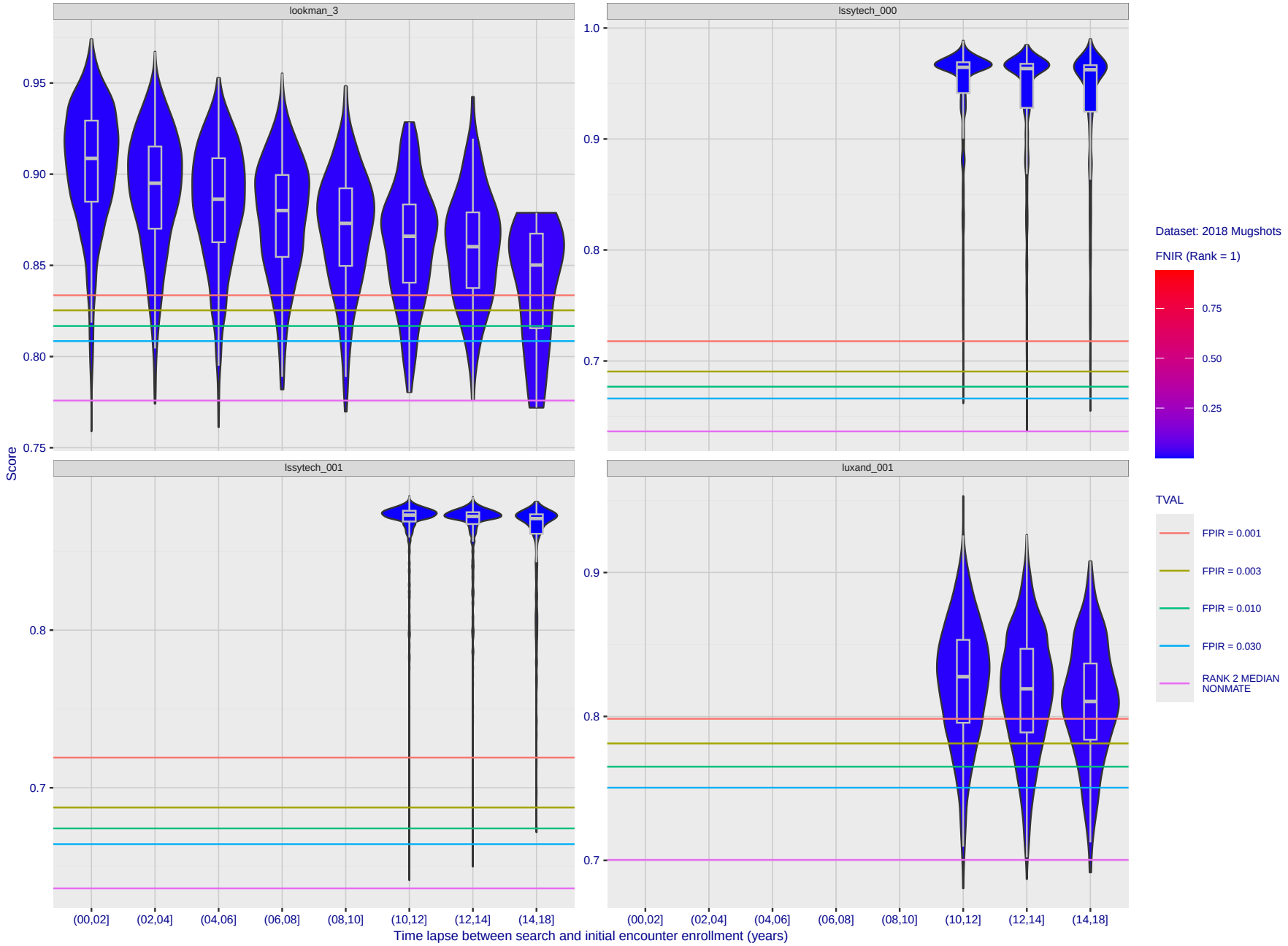

Figure 402: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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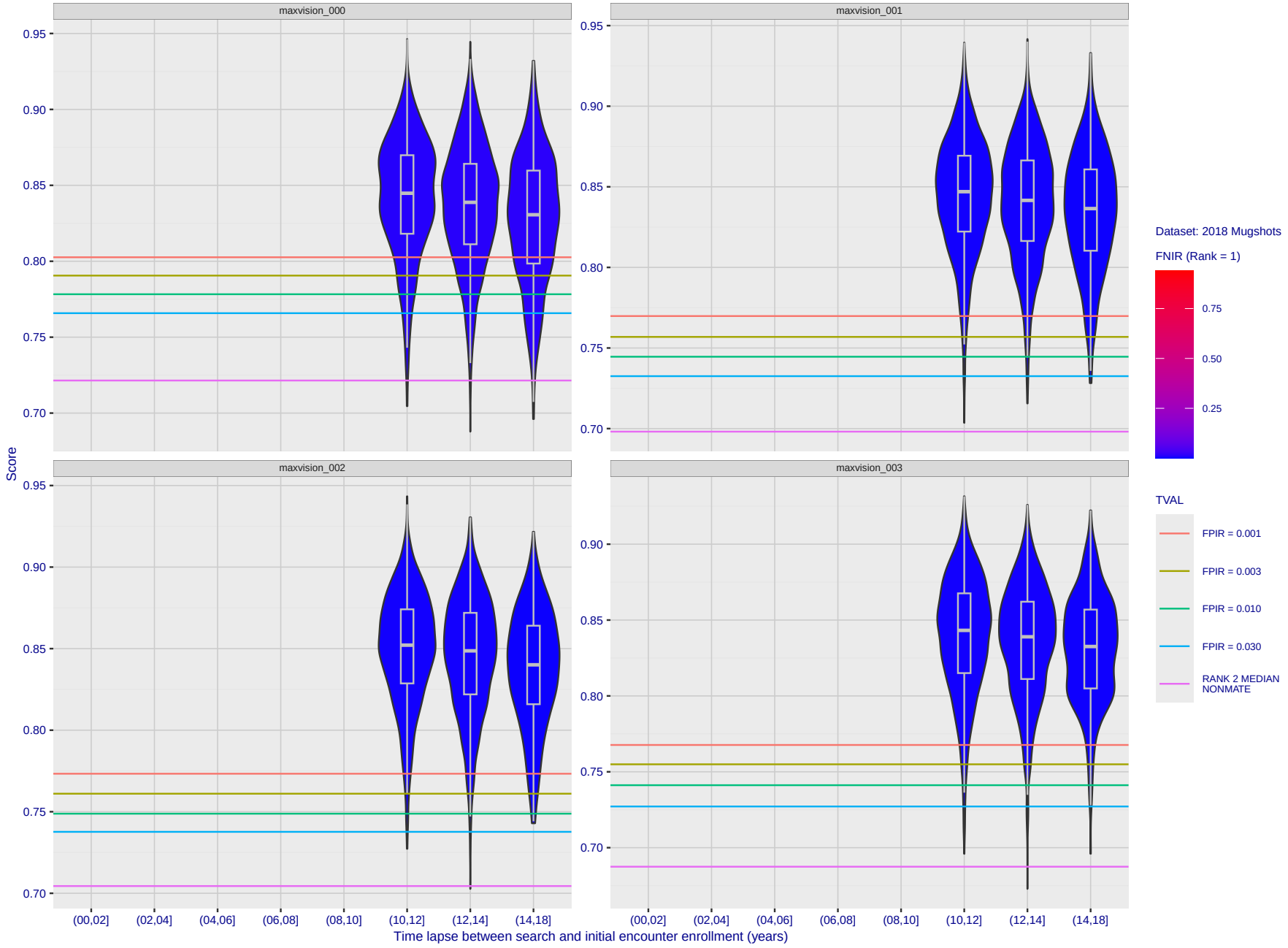

Figure 403: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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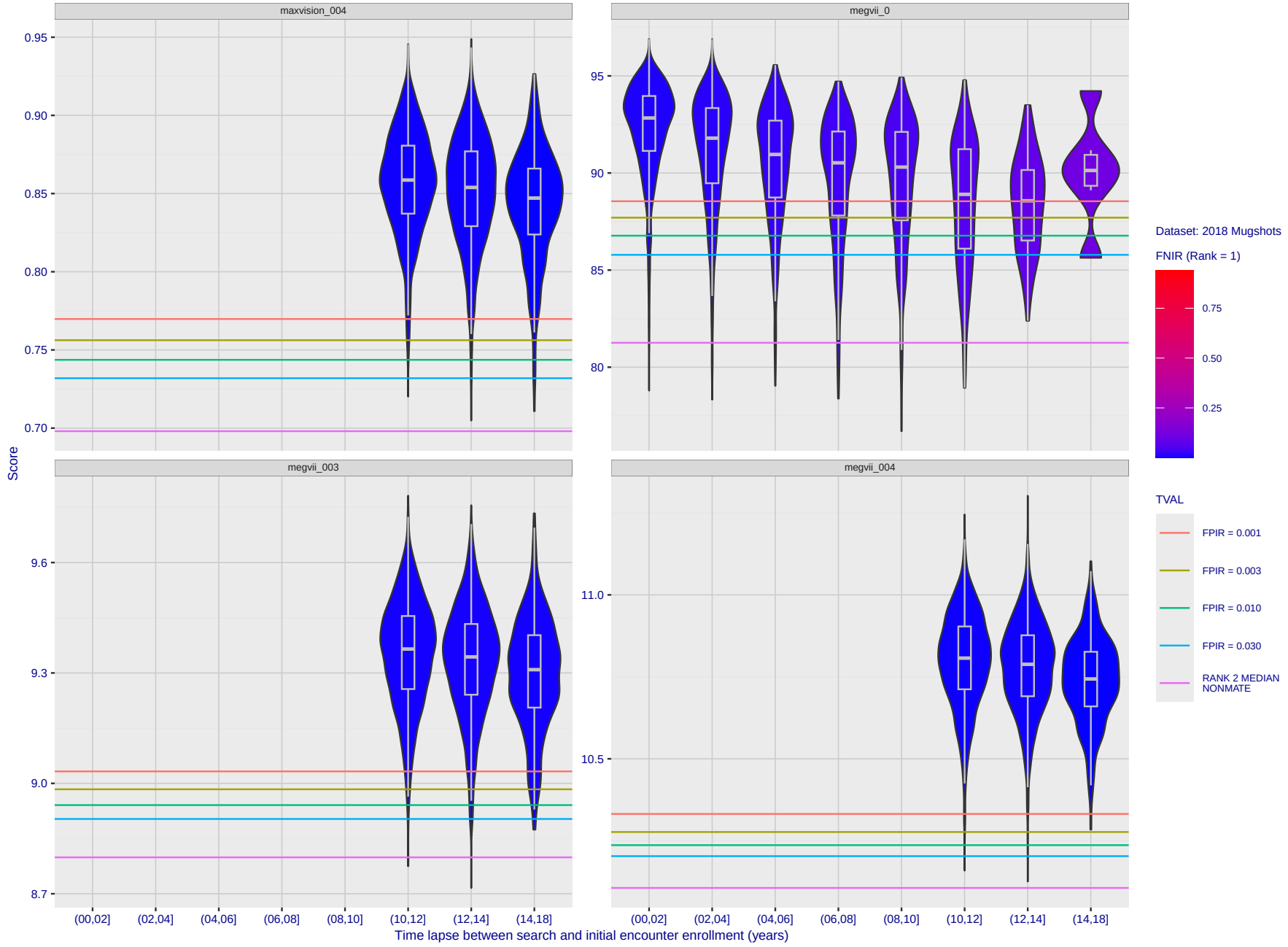

Figure 404: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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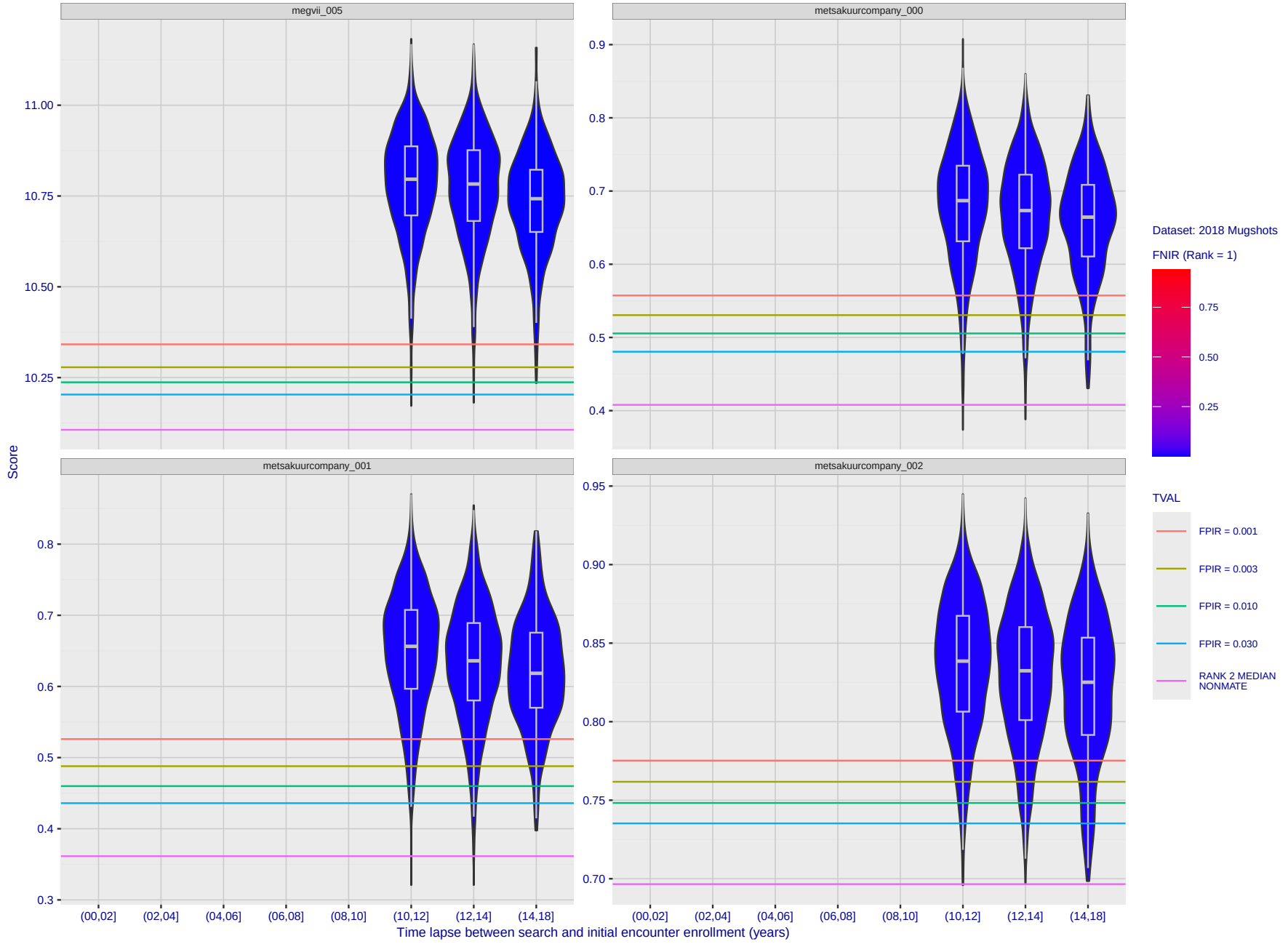

Figure 405: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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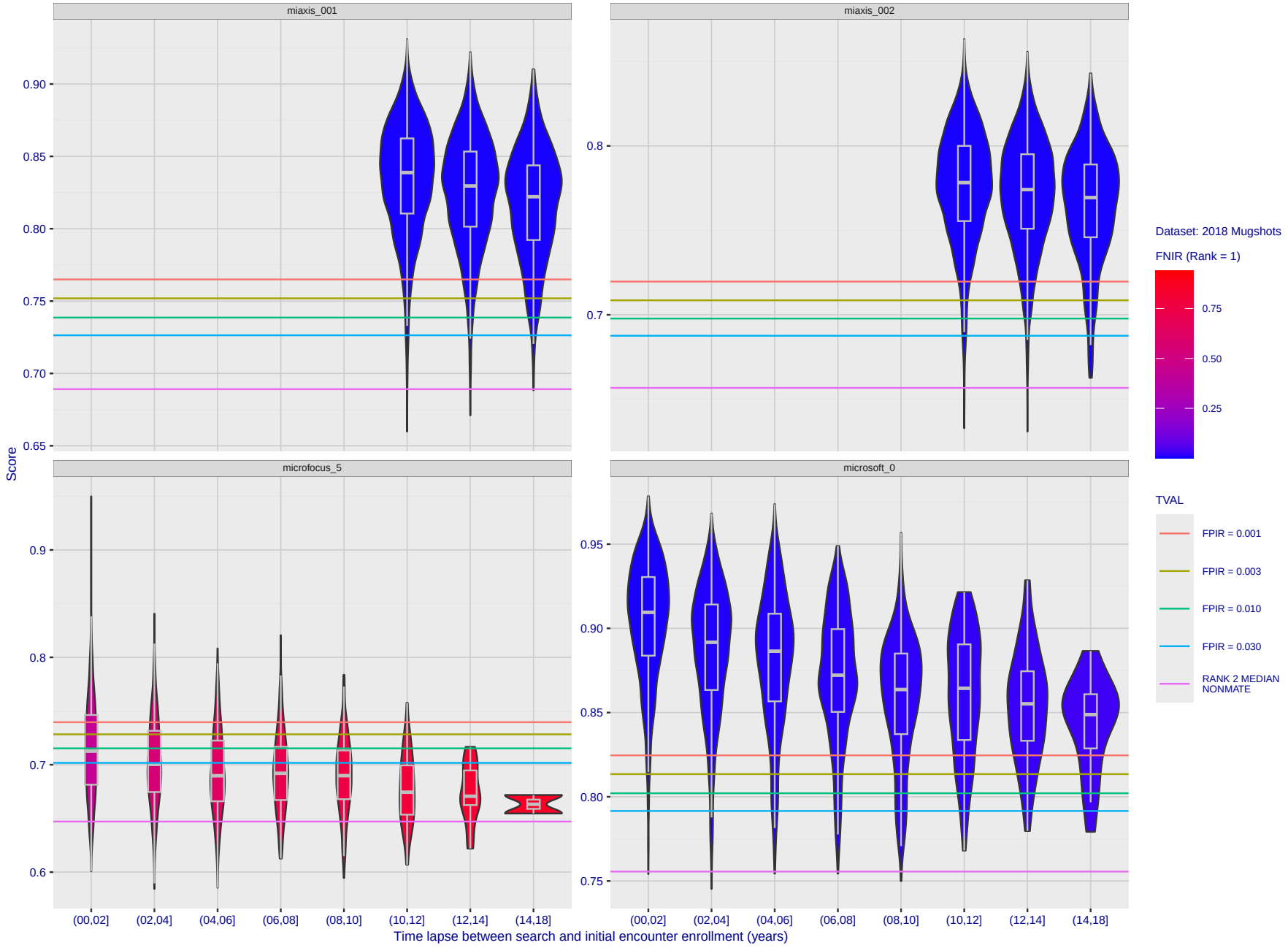

Figure 406: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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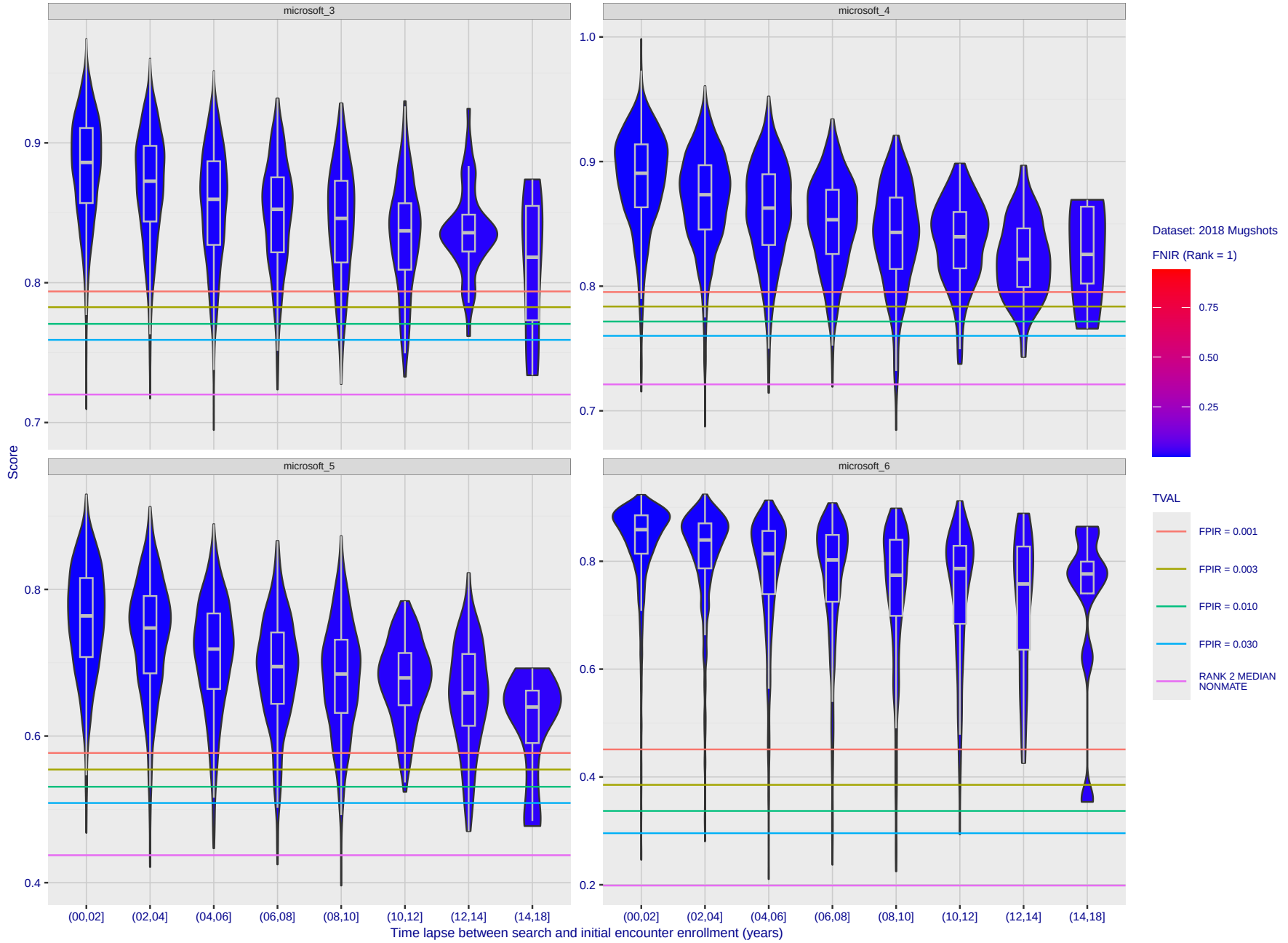

Figure 407: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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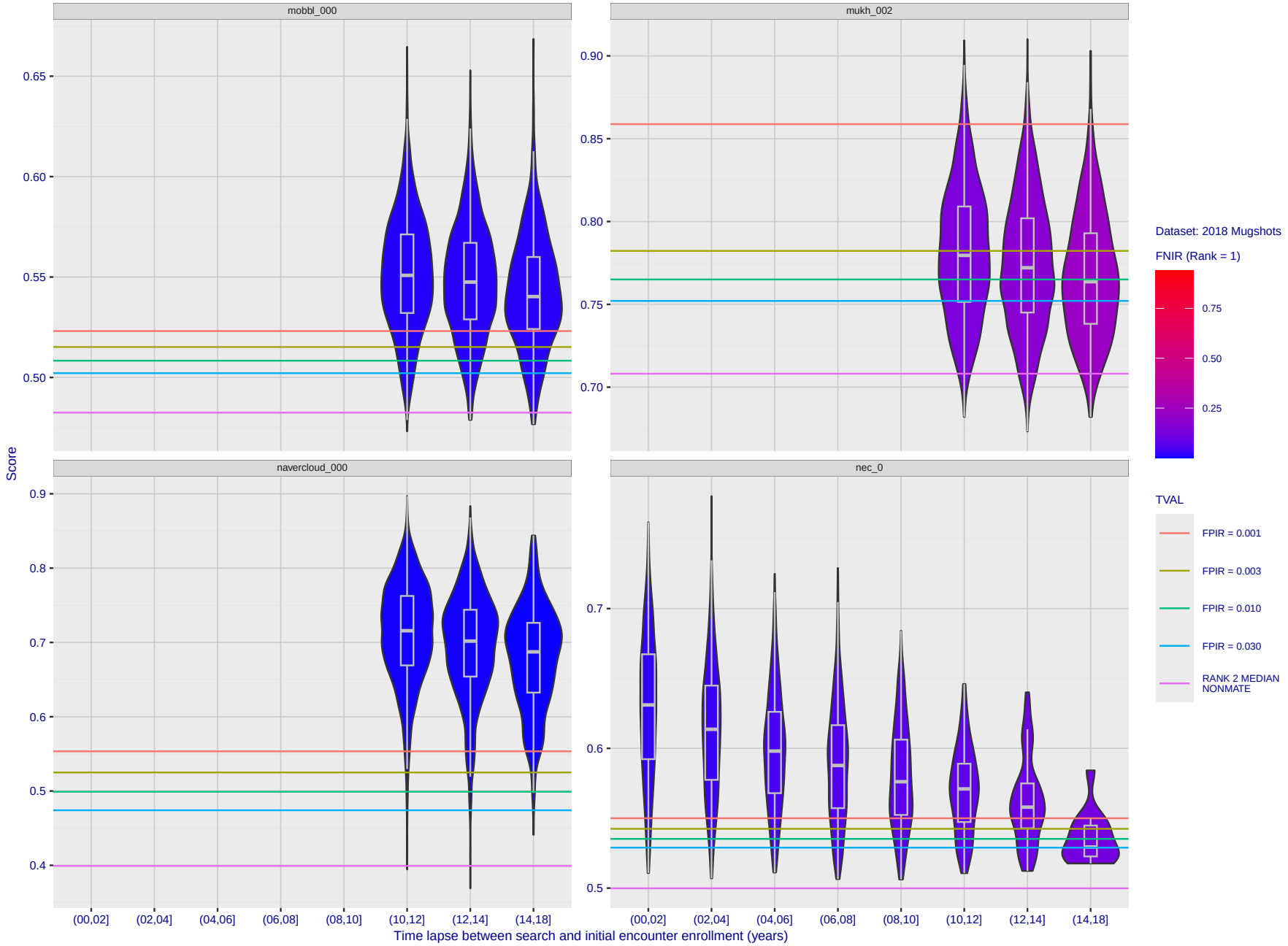

Figure 408: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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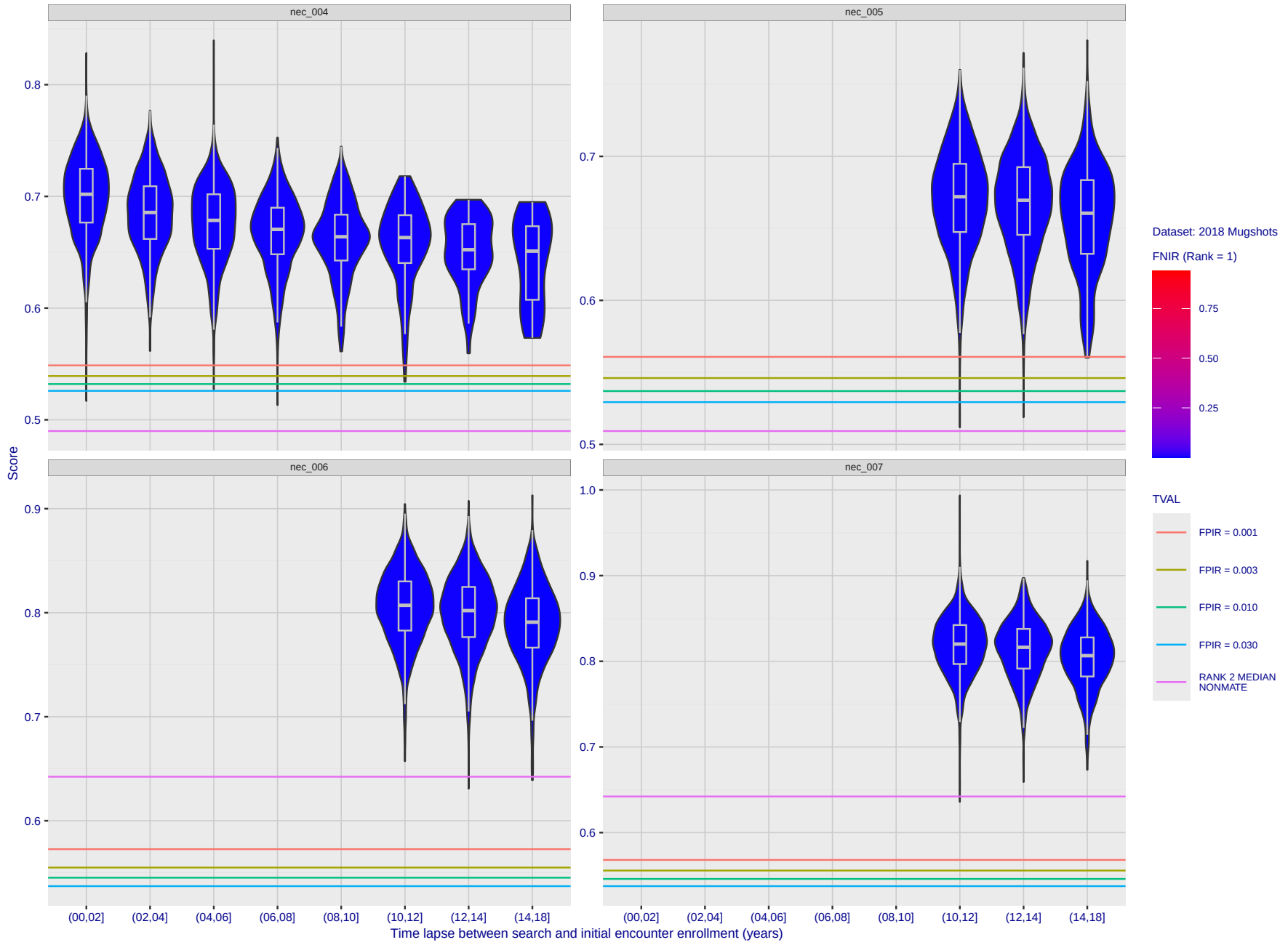

Figure 409: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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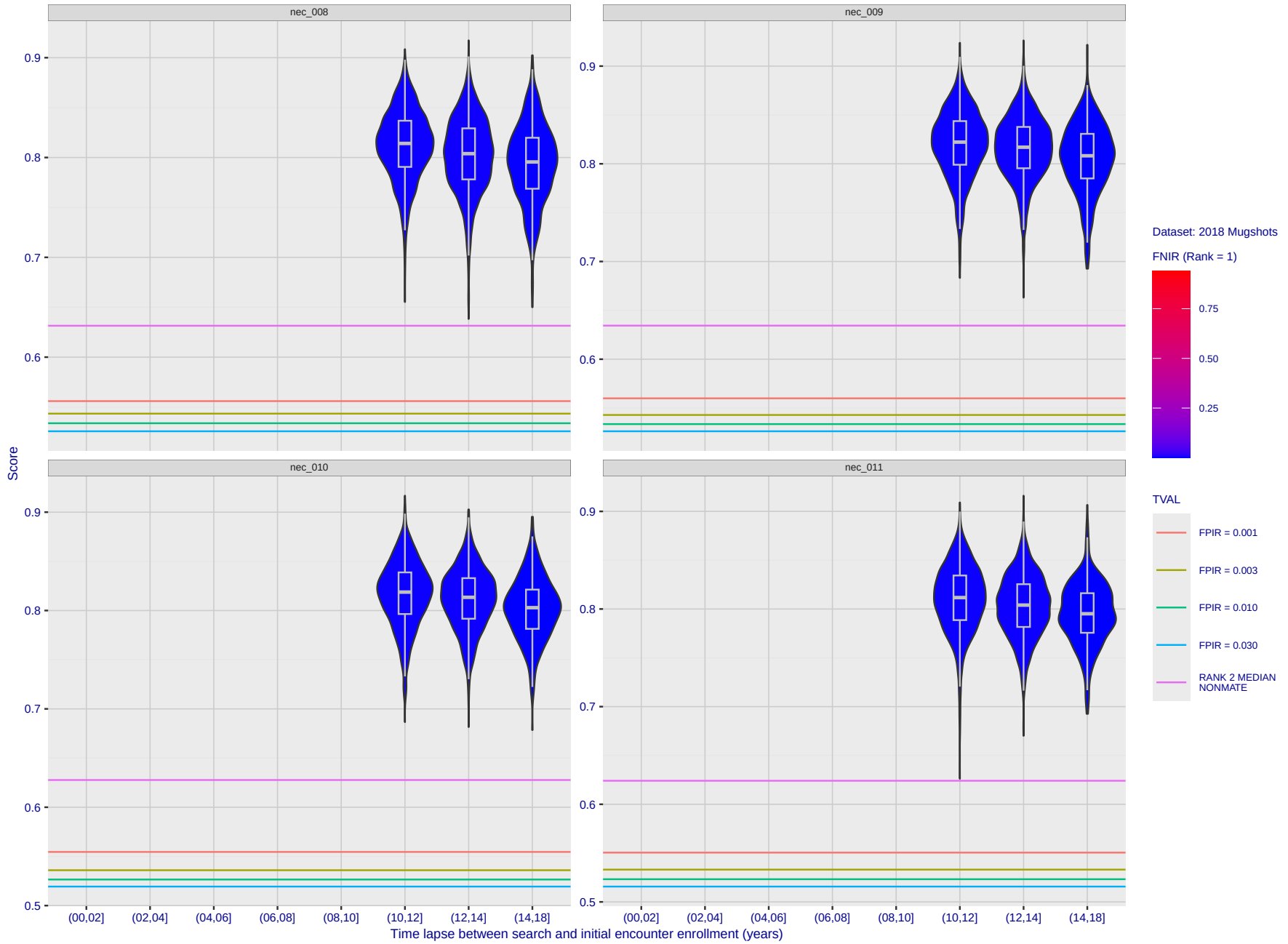

Figure 410: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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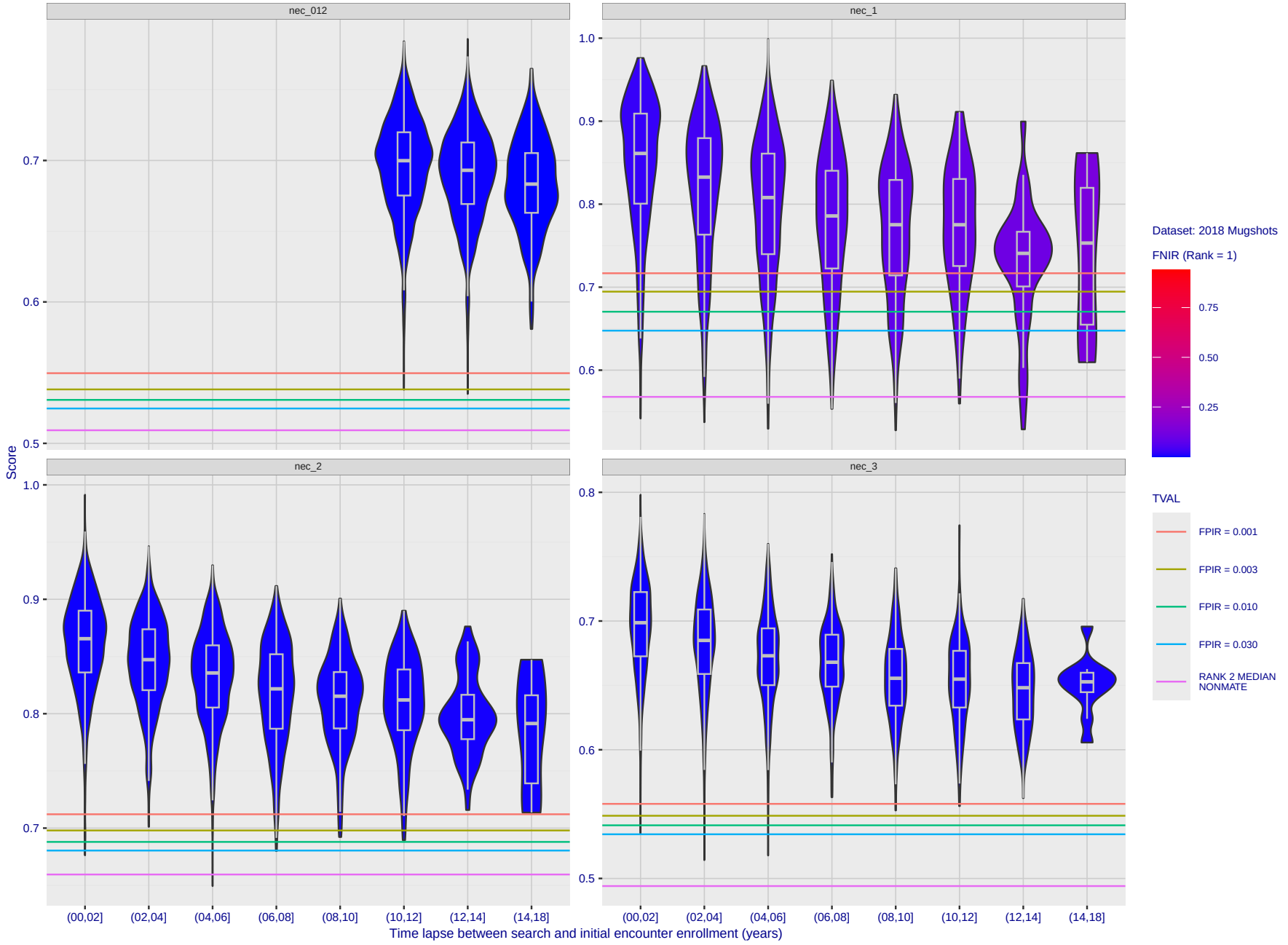

Figure 411: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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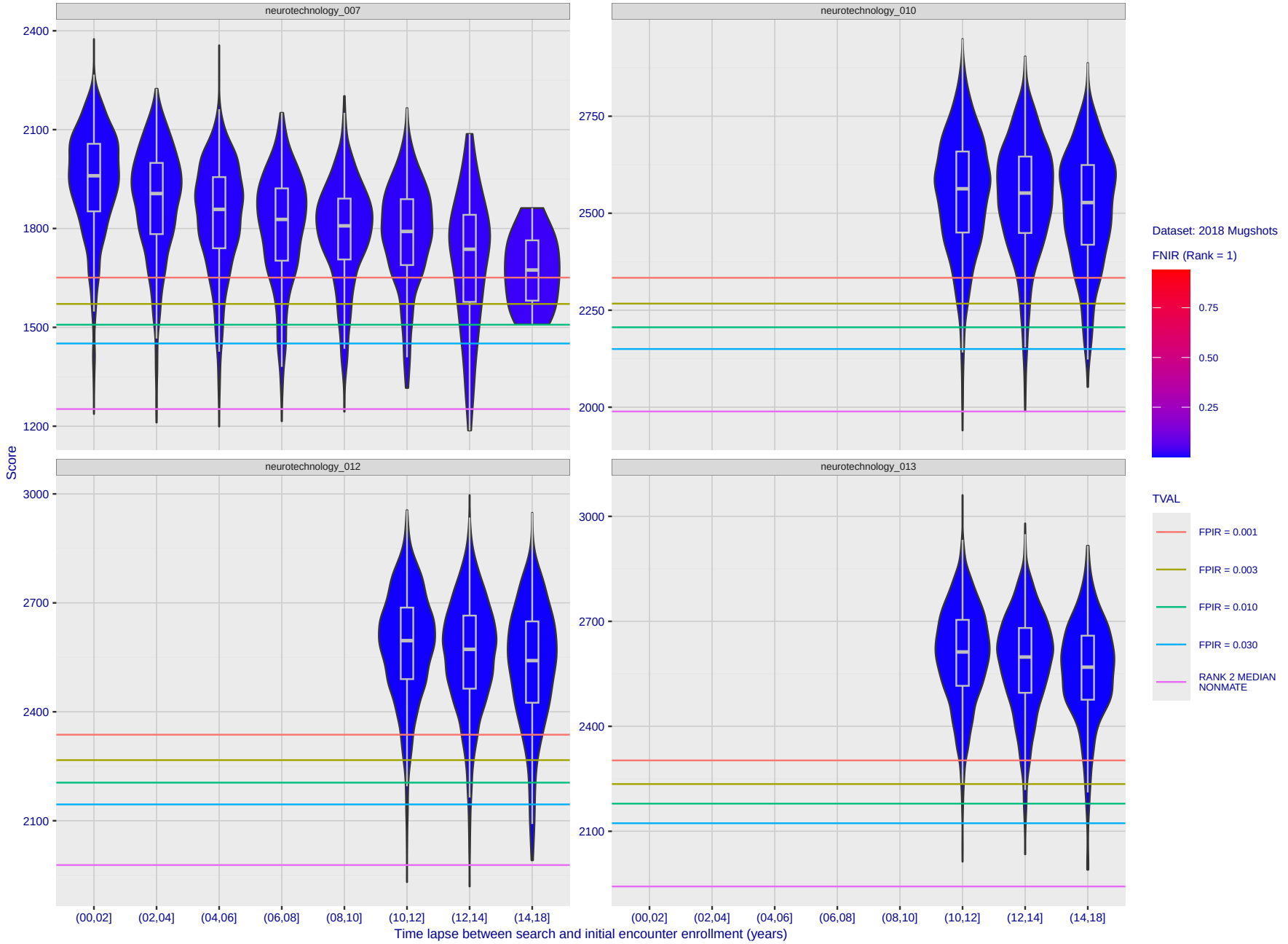

Figure 412: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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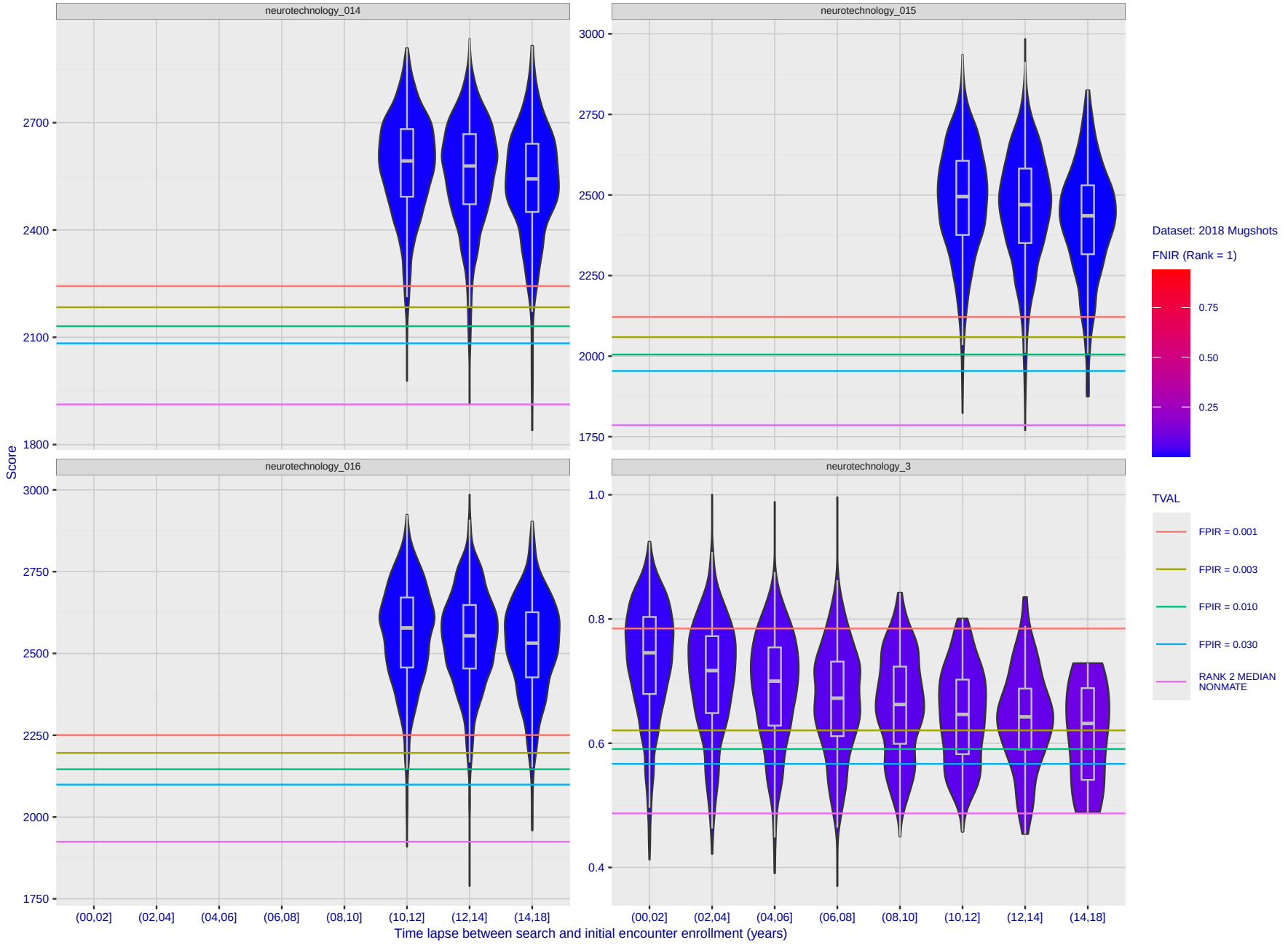

Figure 413: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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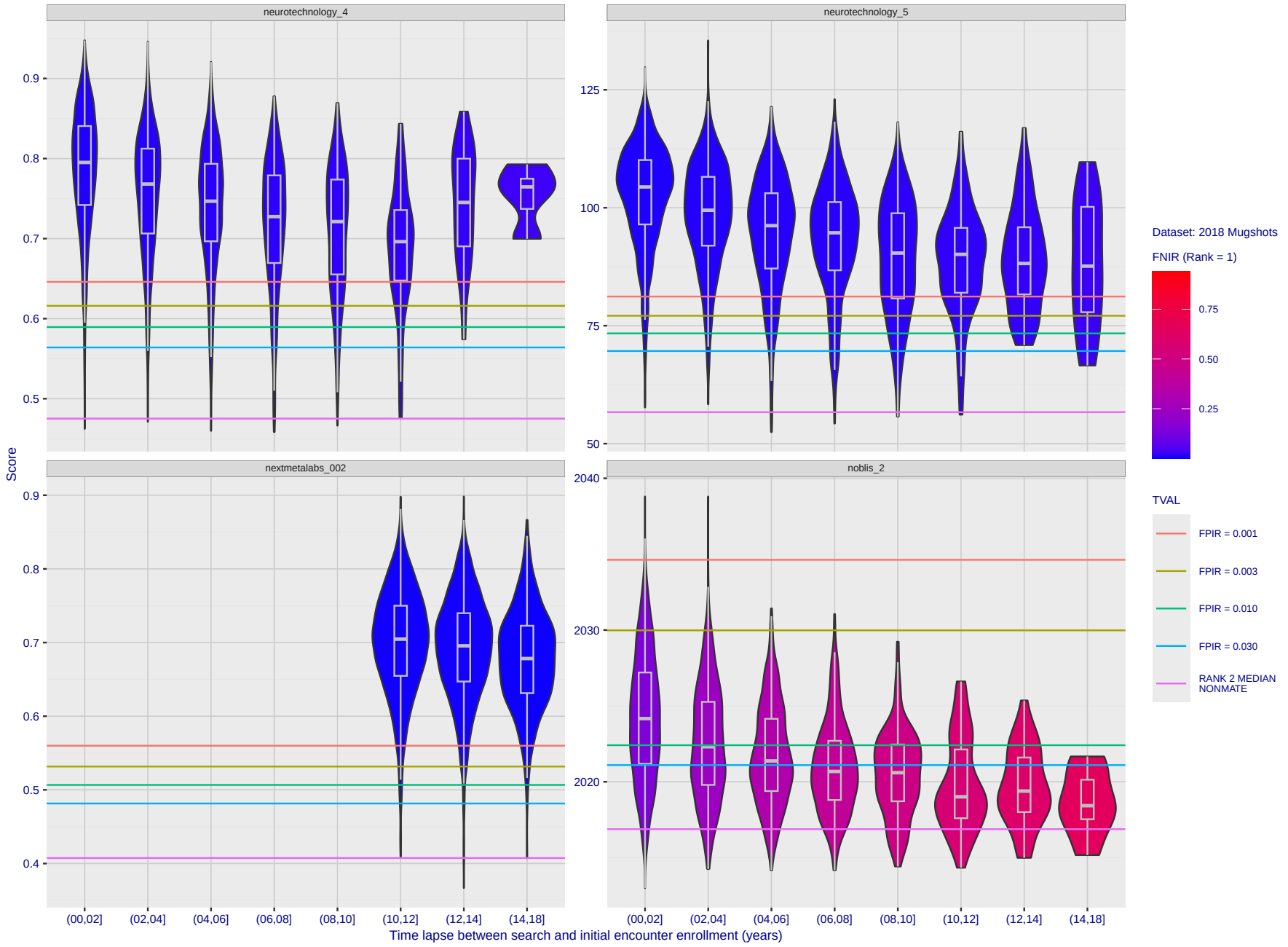

Figure 414: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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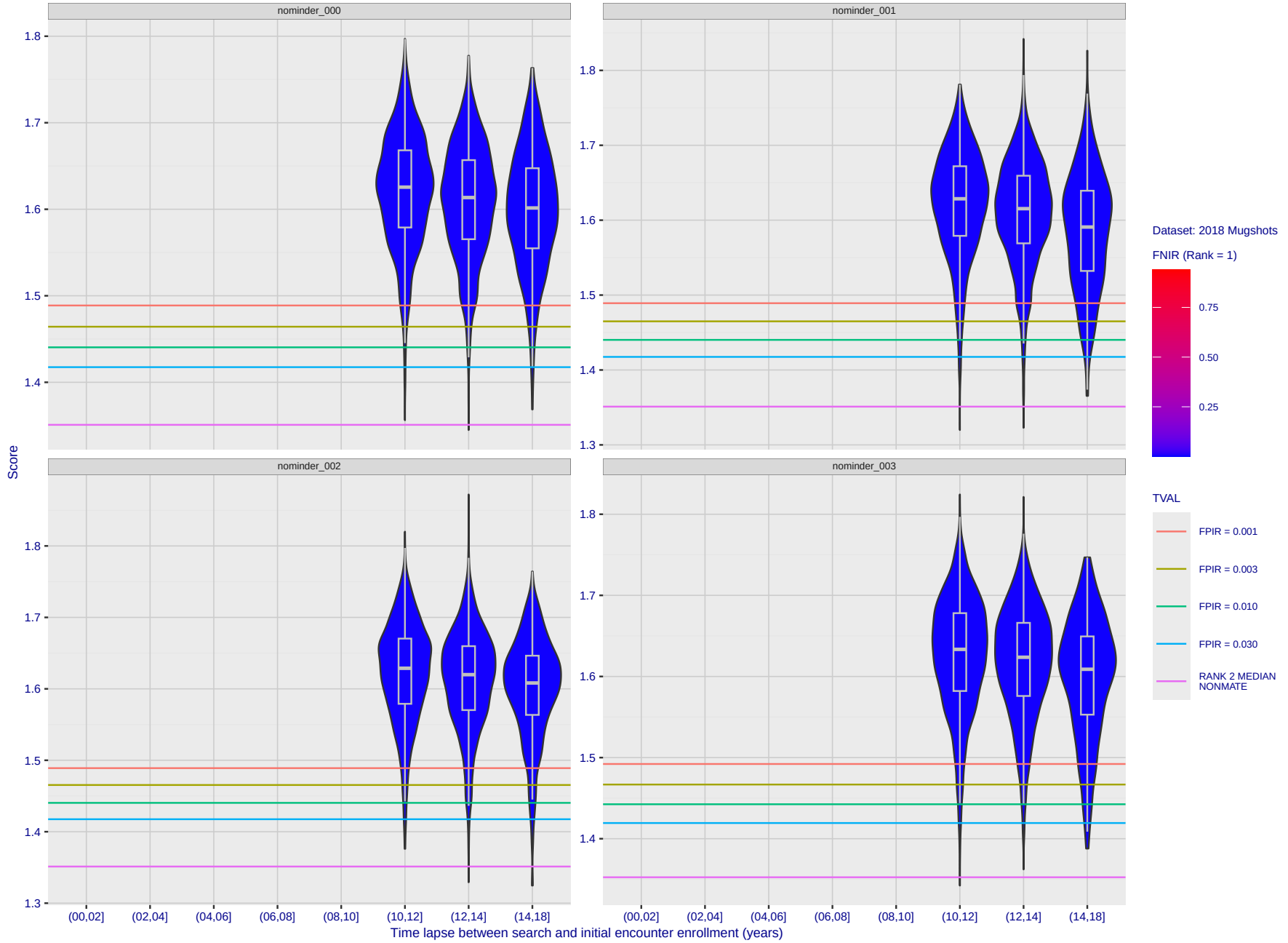

Figure 415: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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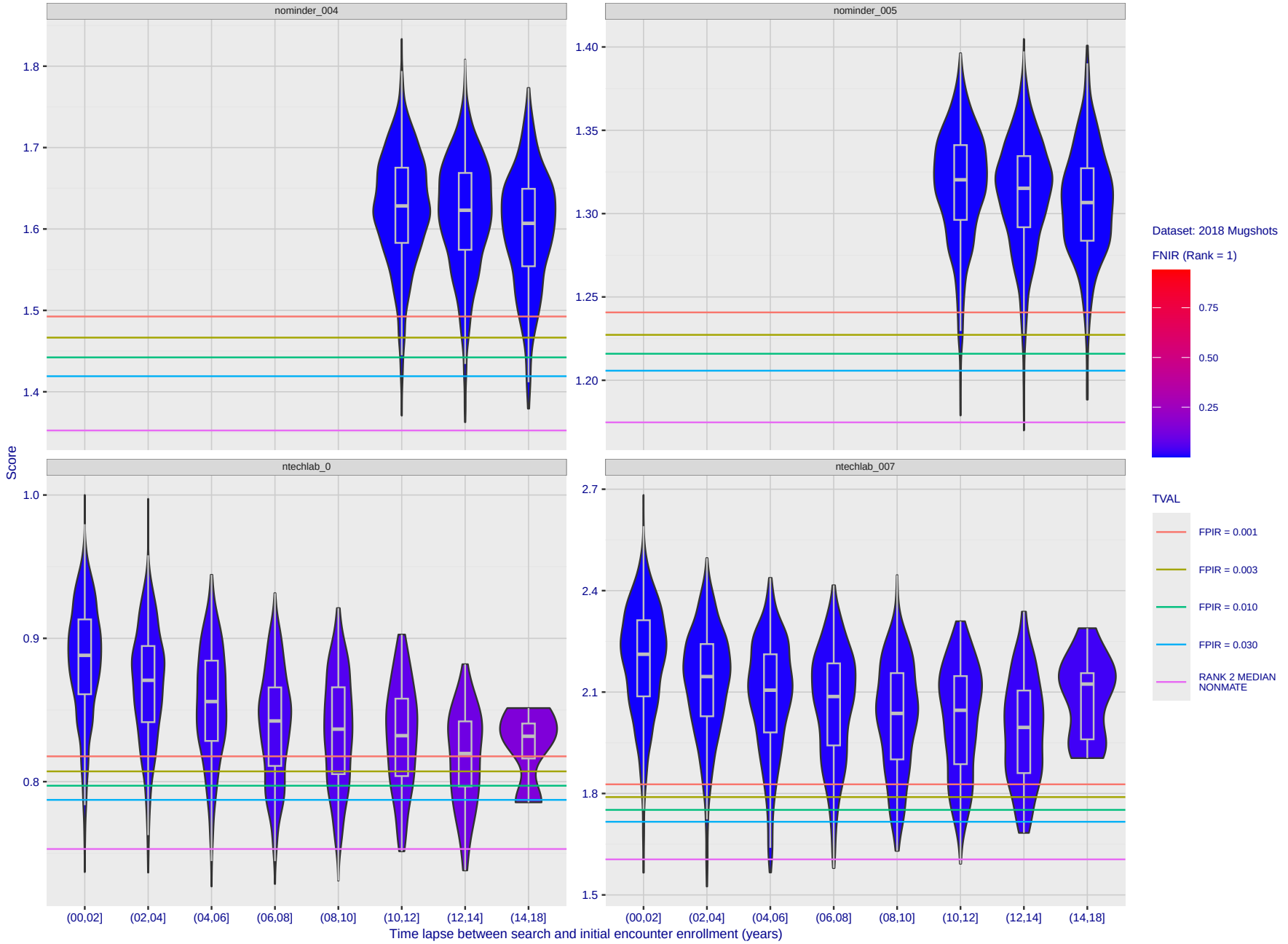

Figure 416: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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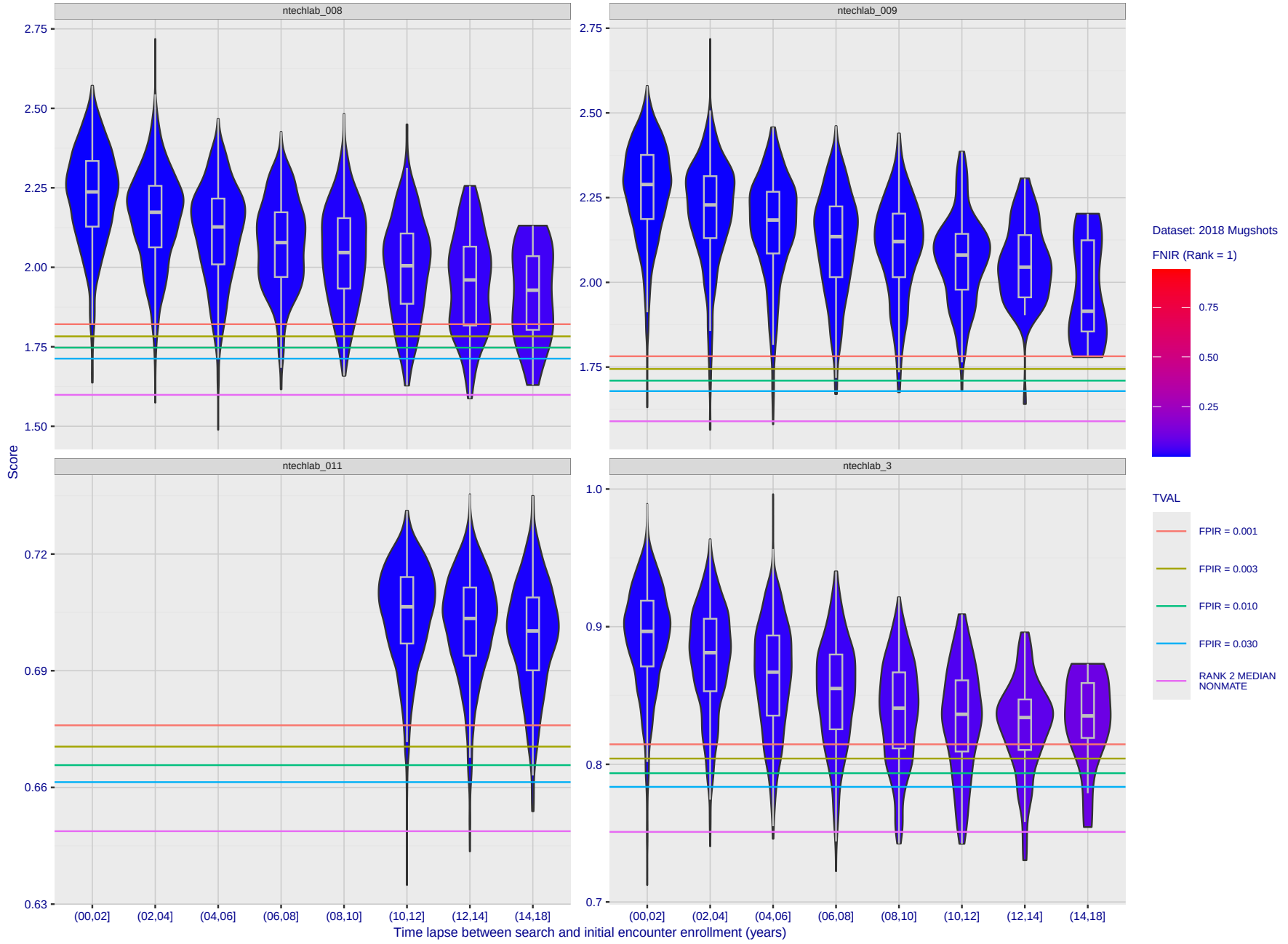

Figure 417: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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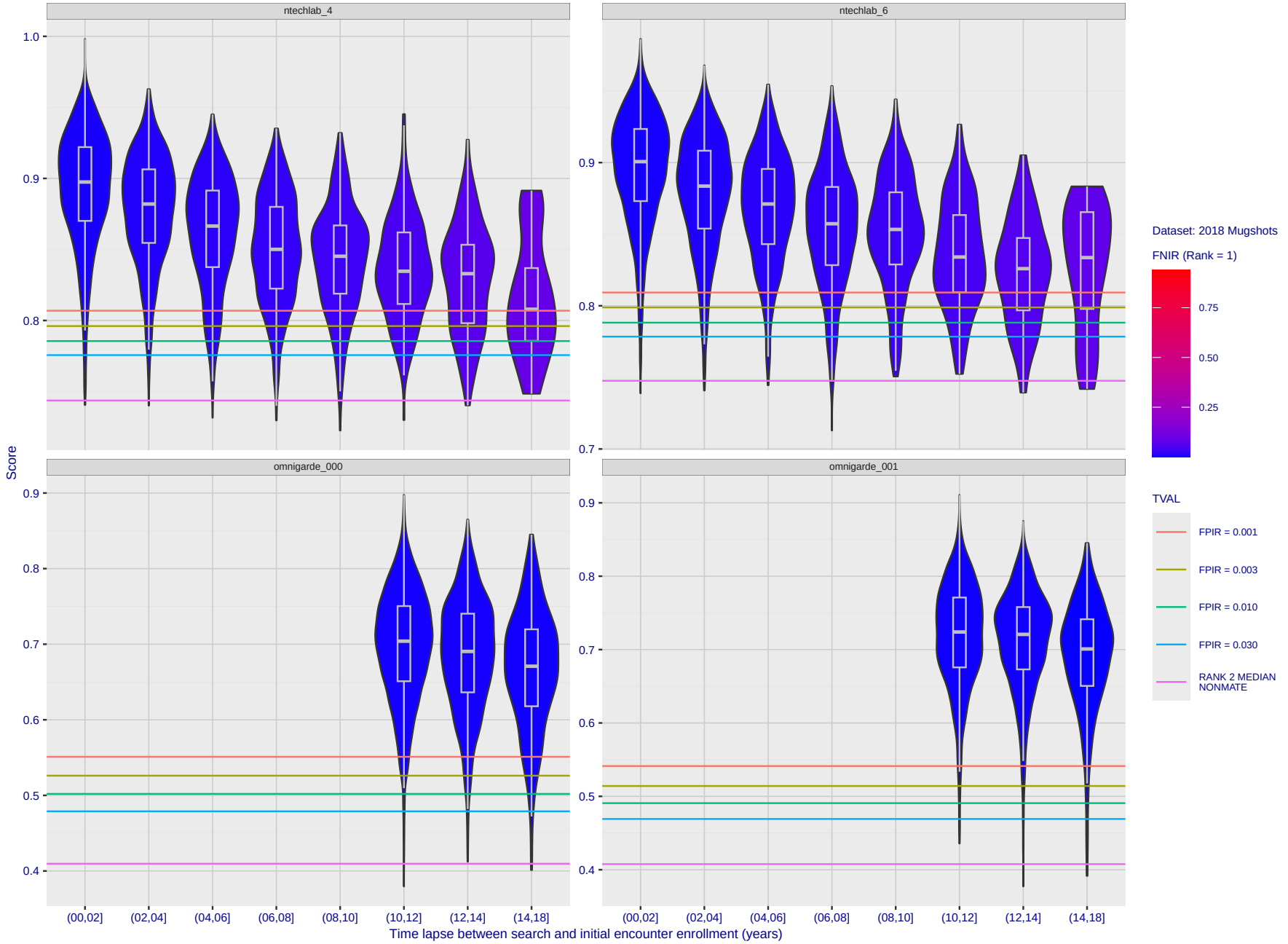

Figure 418: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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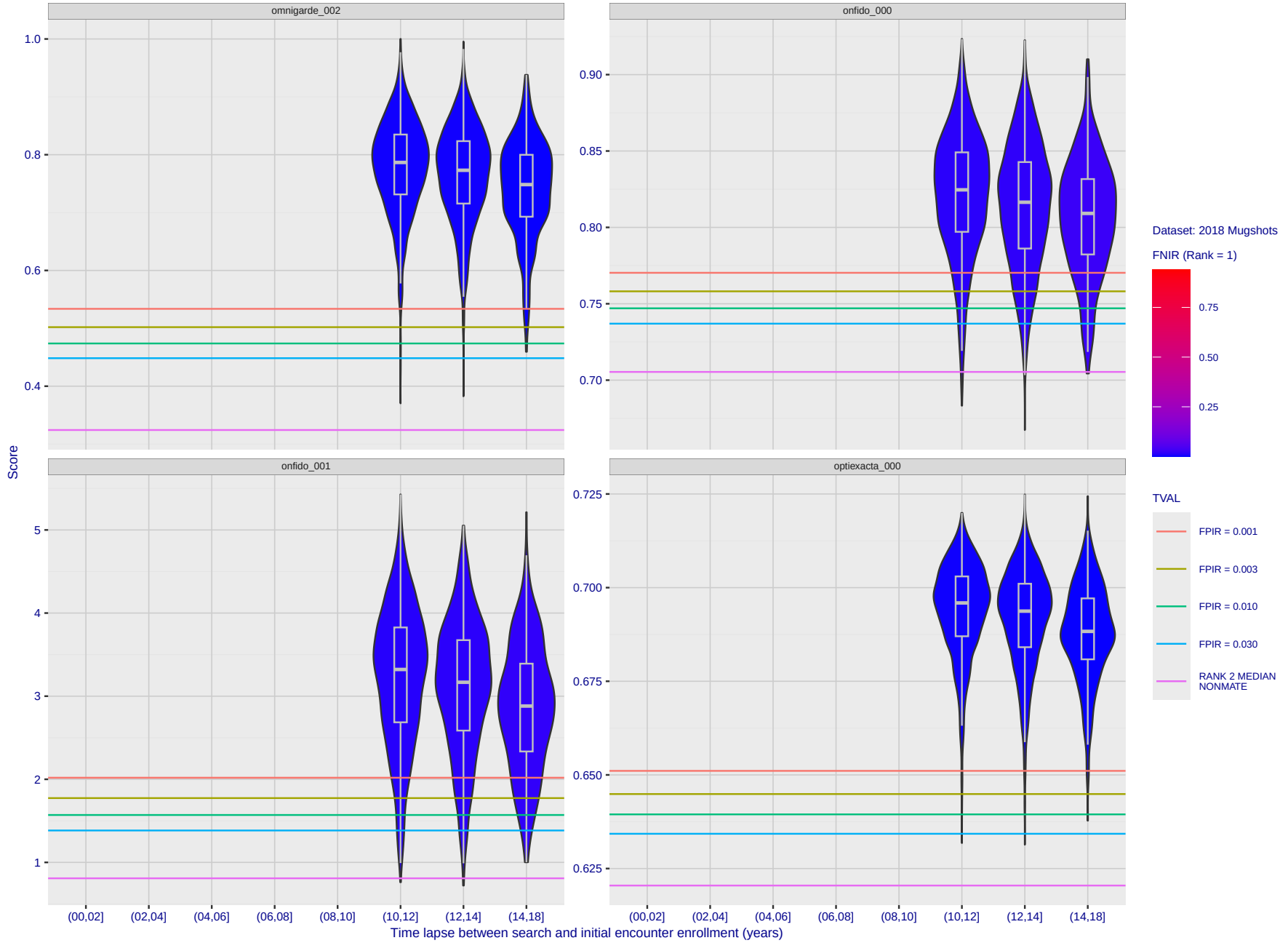

Figure 419: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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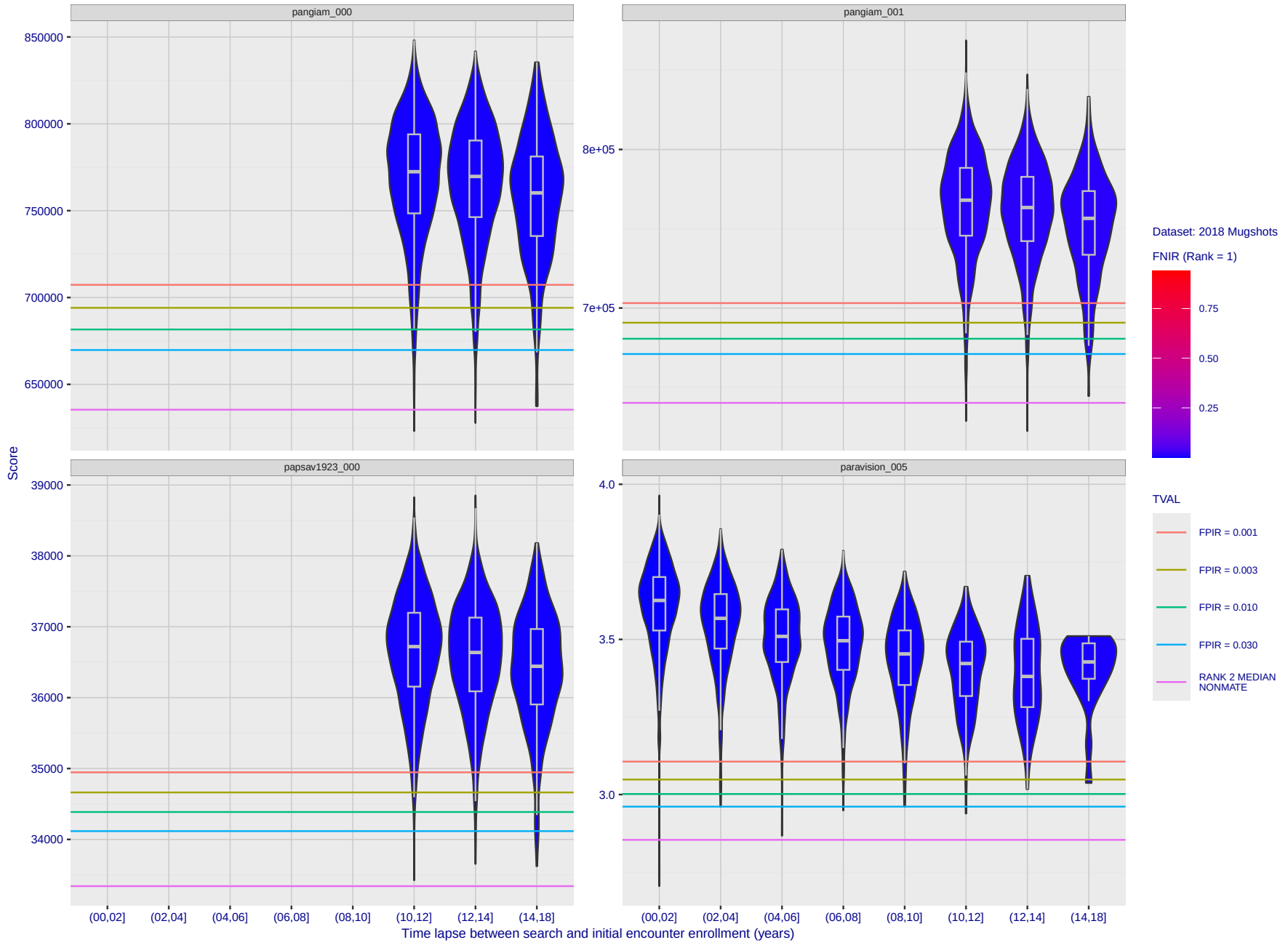

Figure 420: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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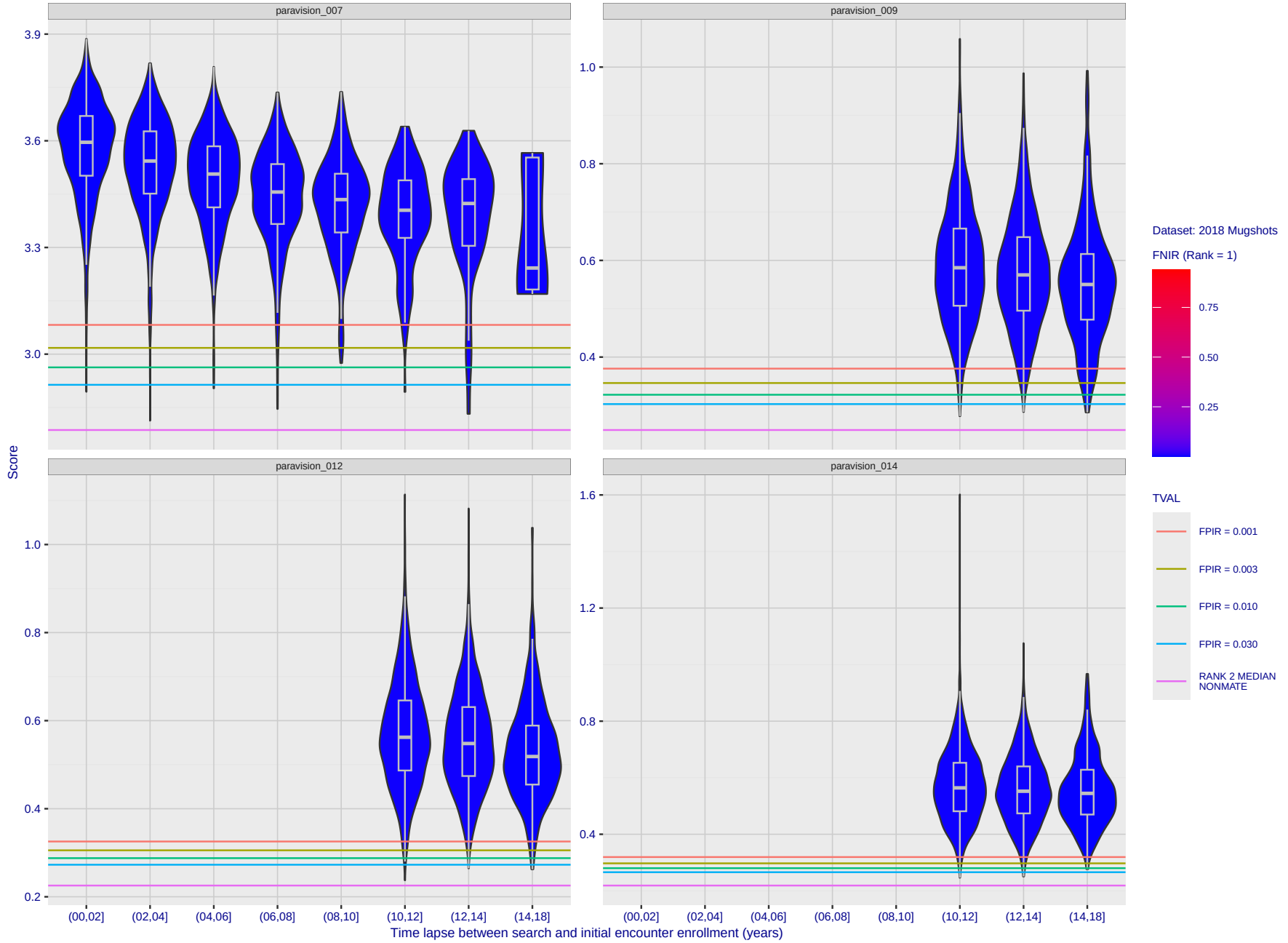

Figure 421: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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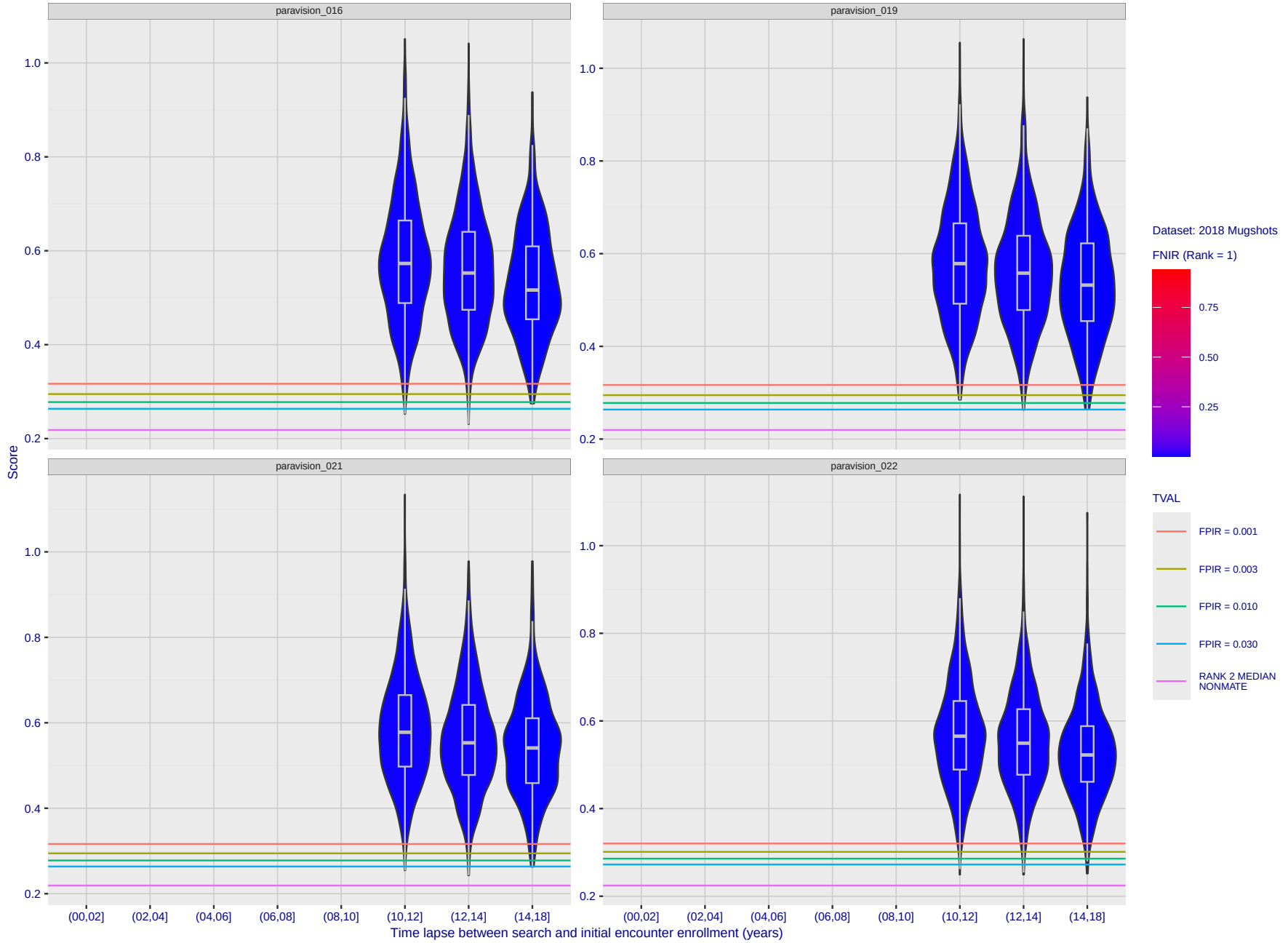

Figure 422: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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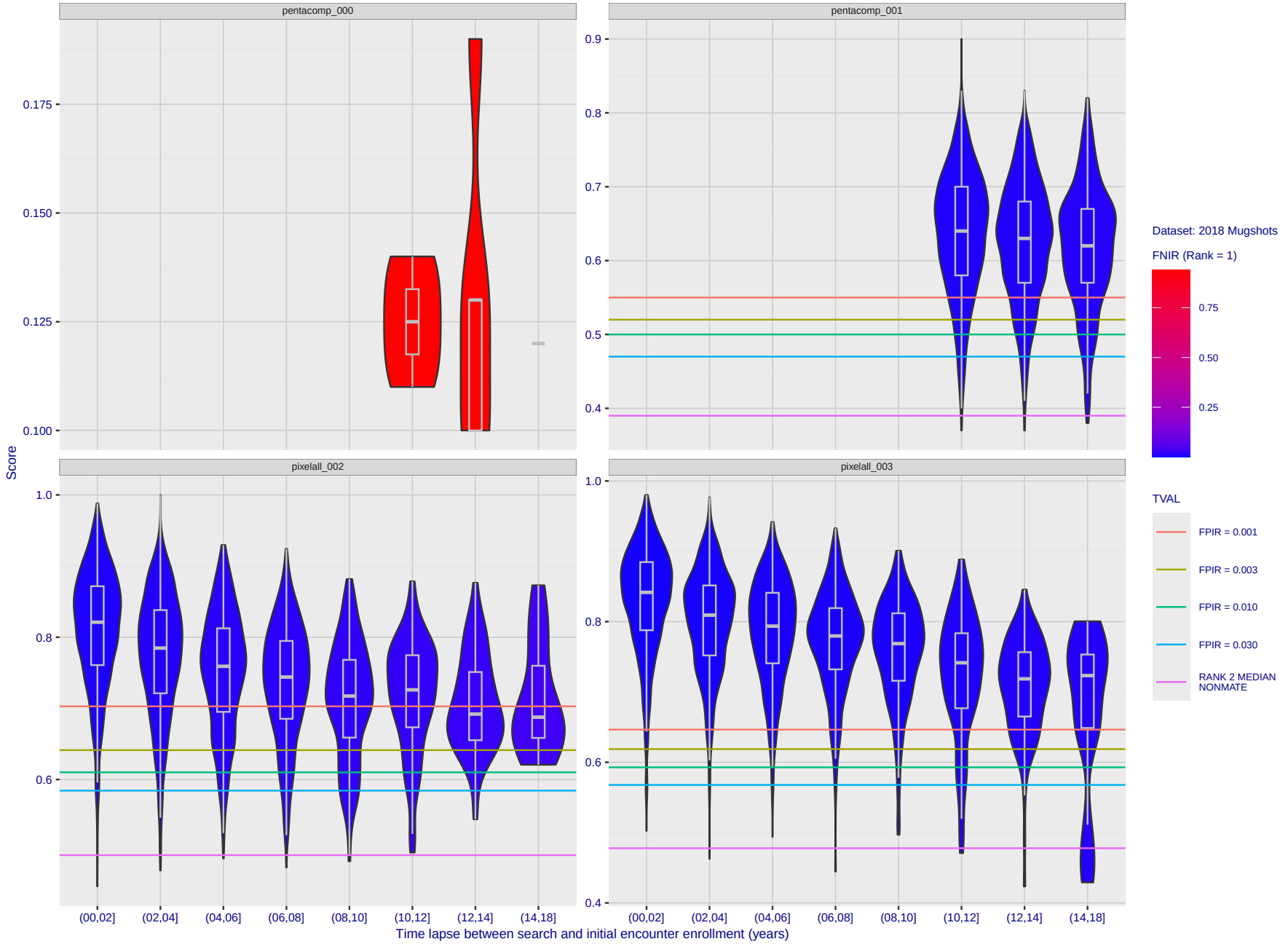

Figure 423: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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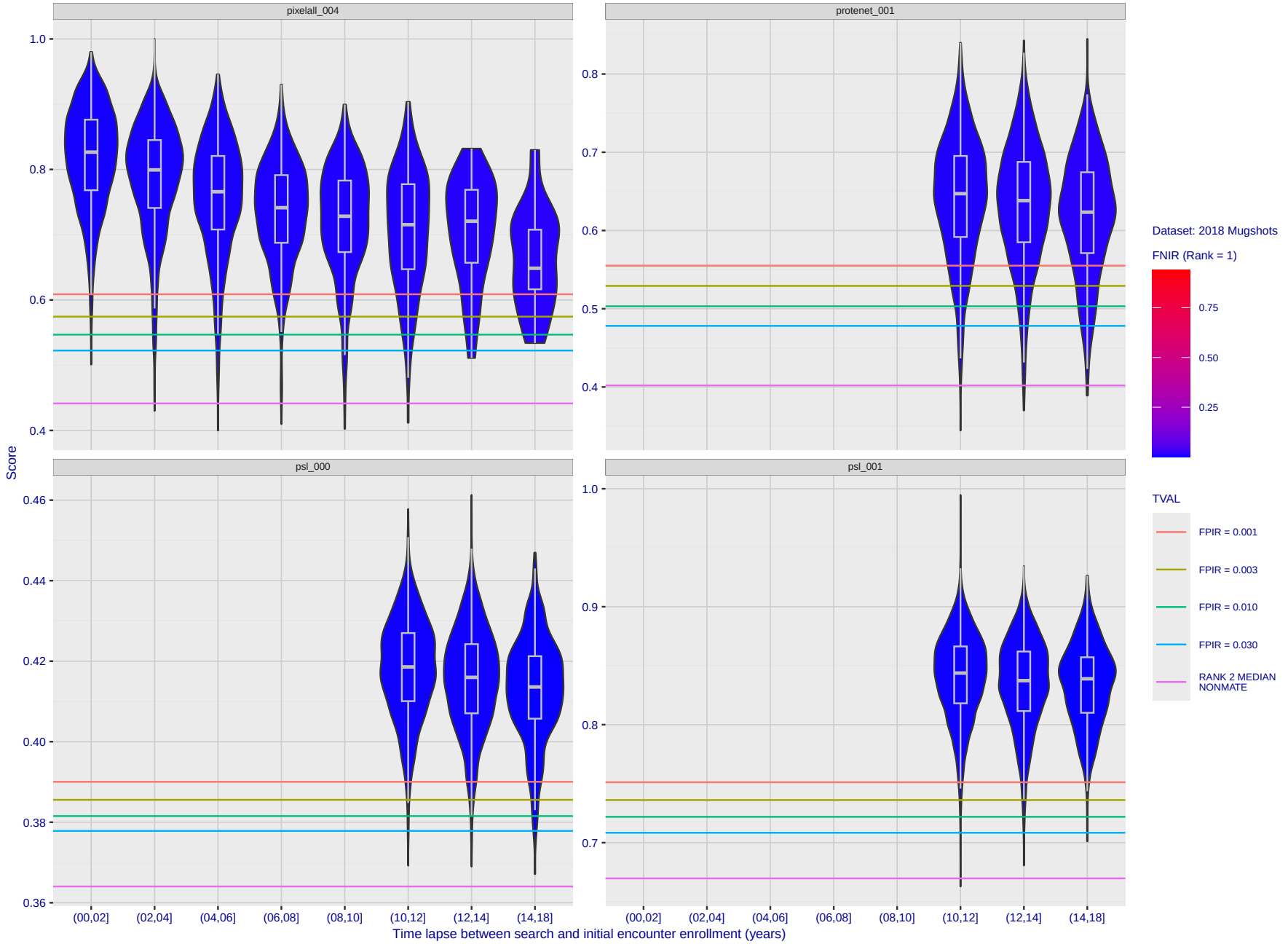

Figure 424: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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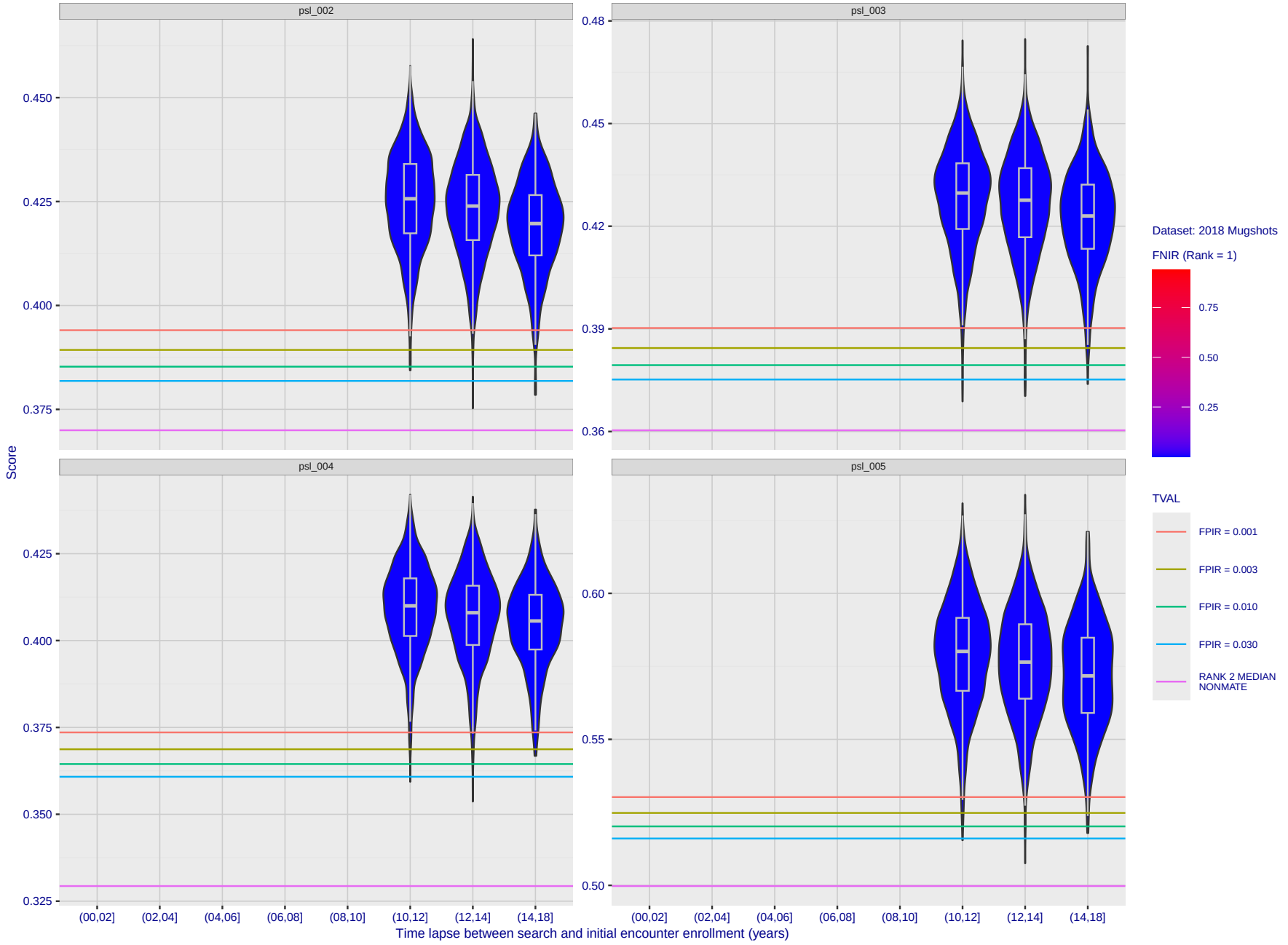

Figure 425: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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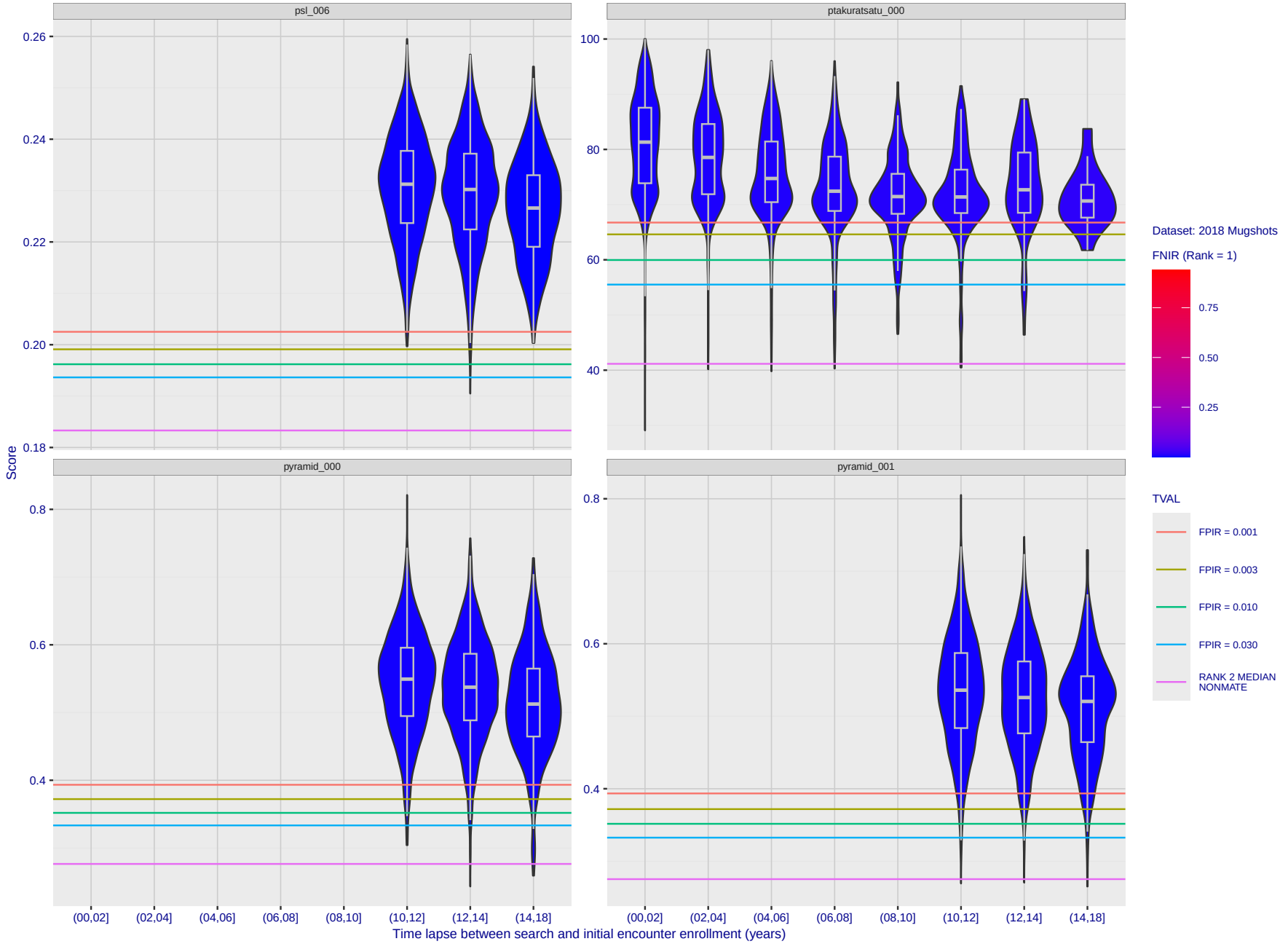

Figure 426: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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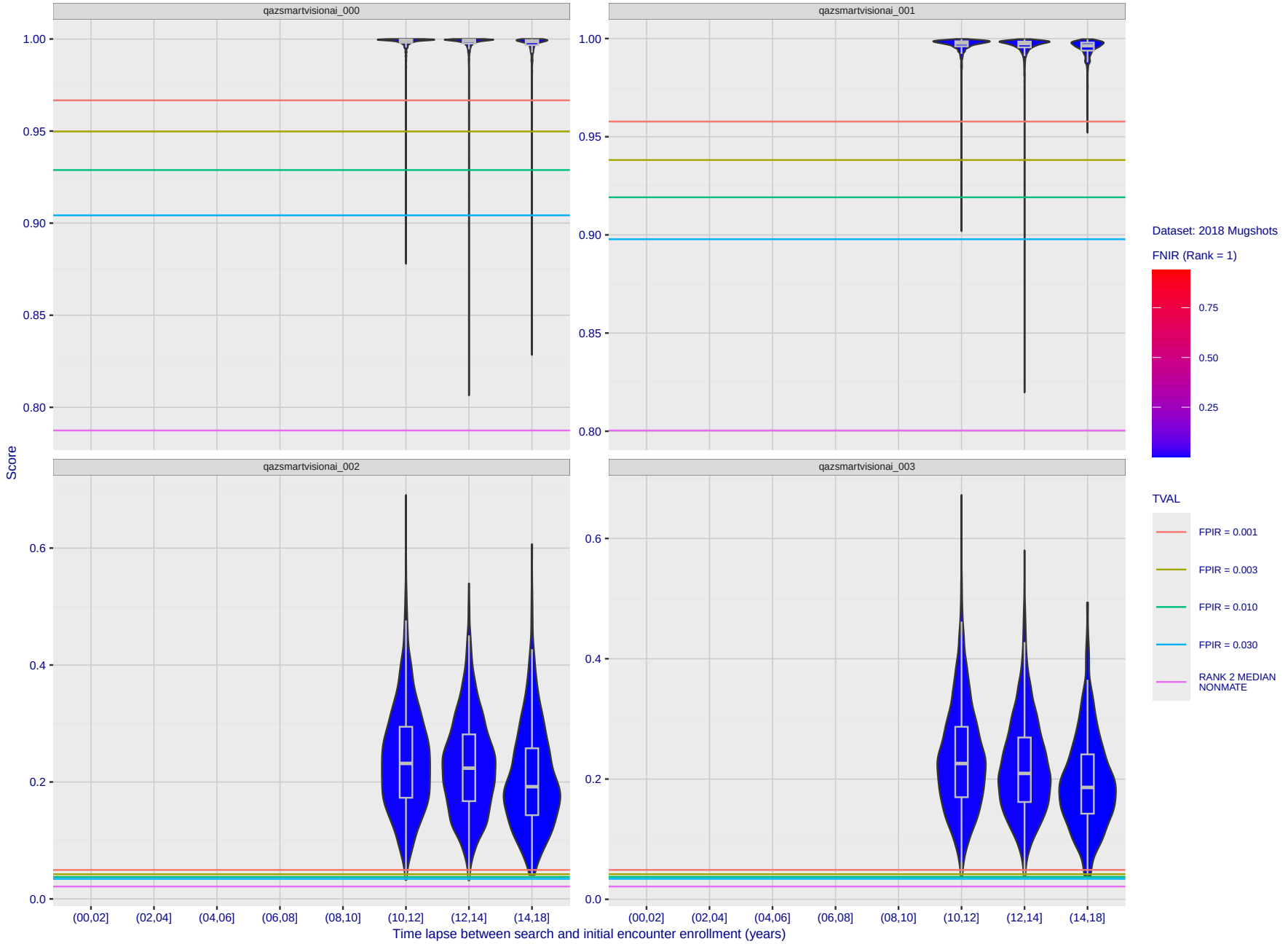

Figure 427: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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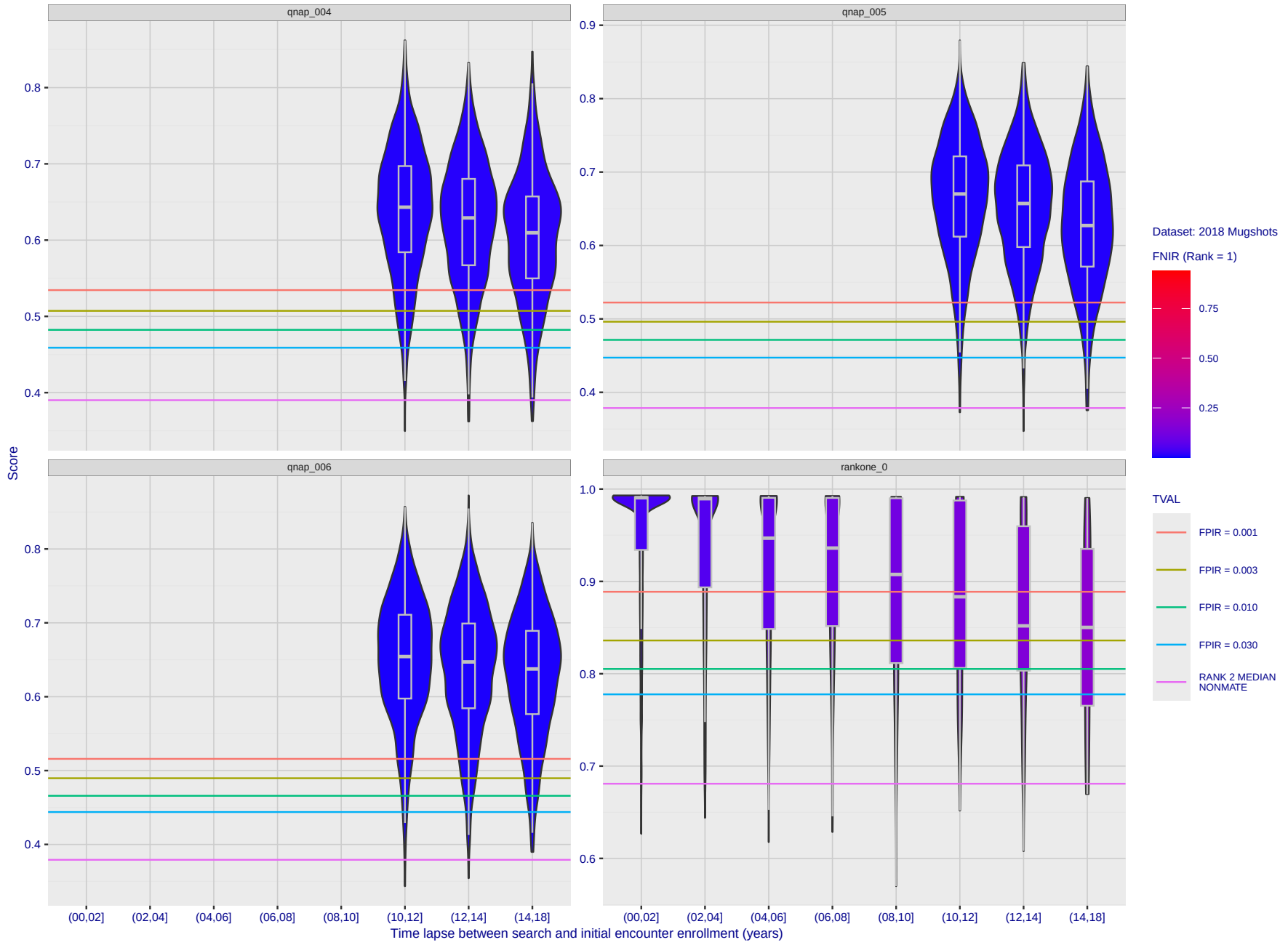

Figure 428: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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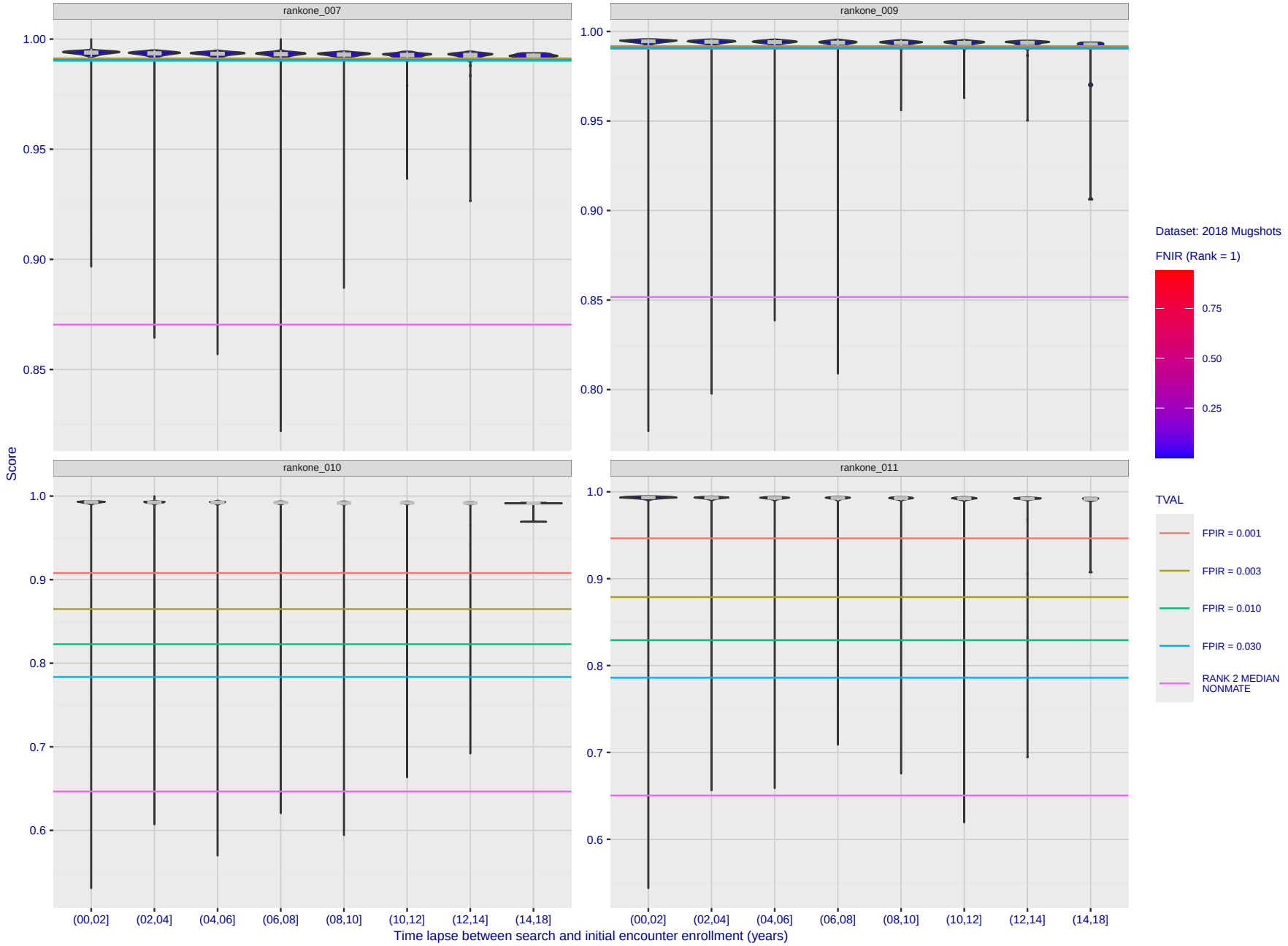

Figure 429: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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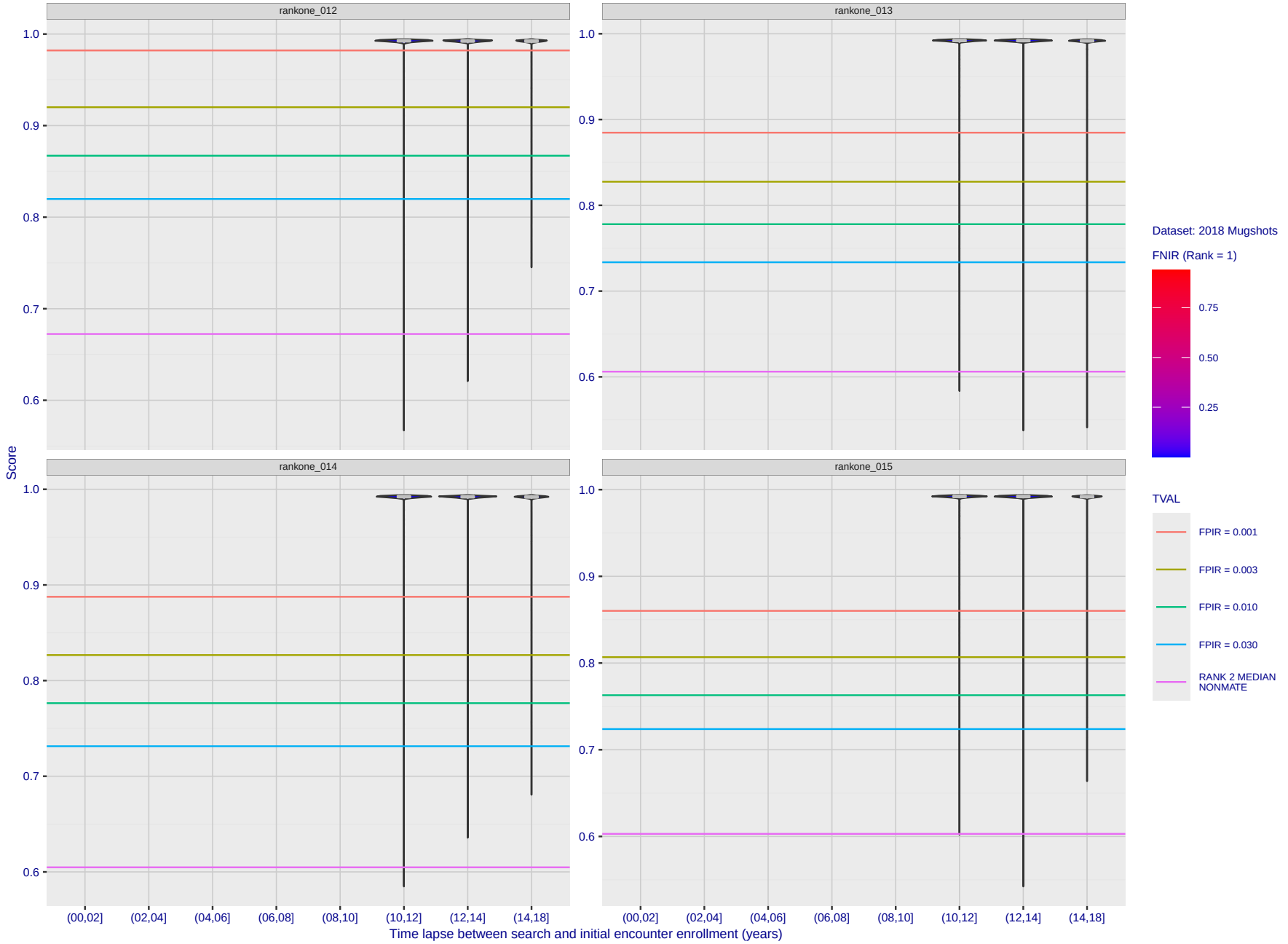

Figure 430: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
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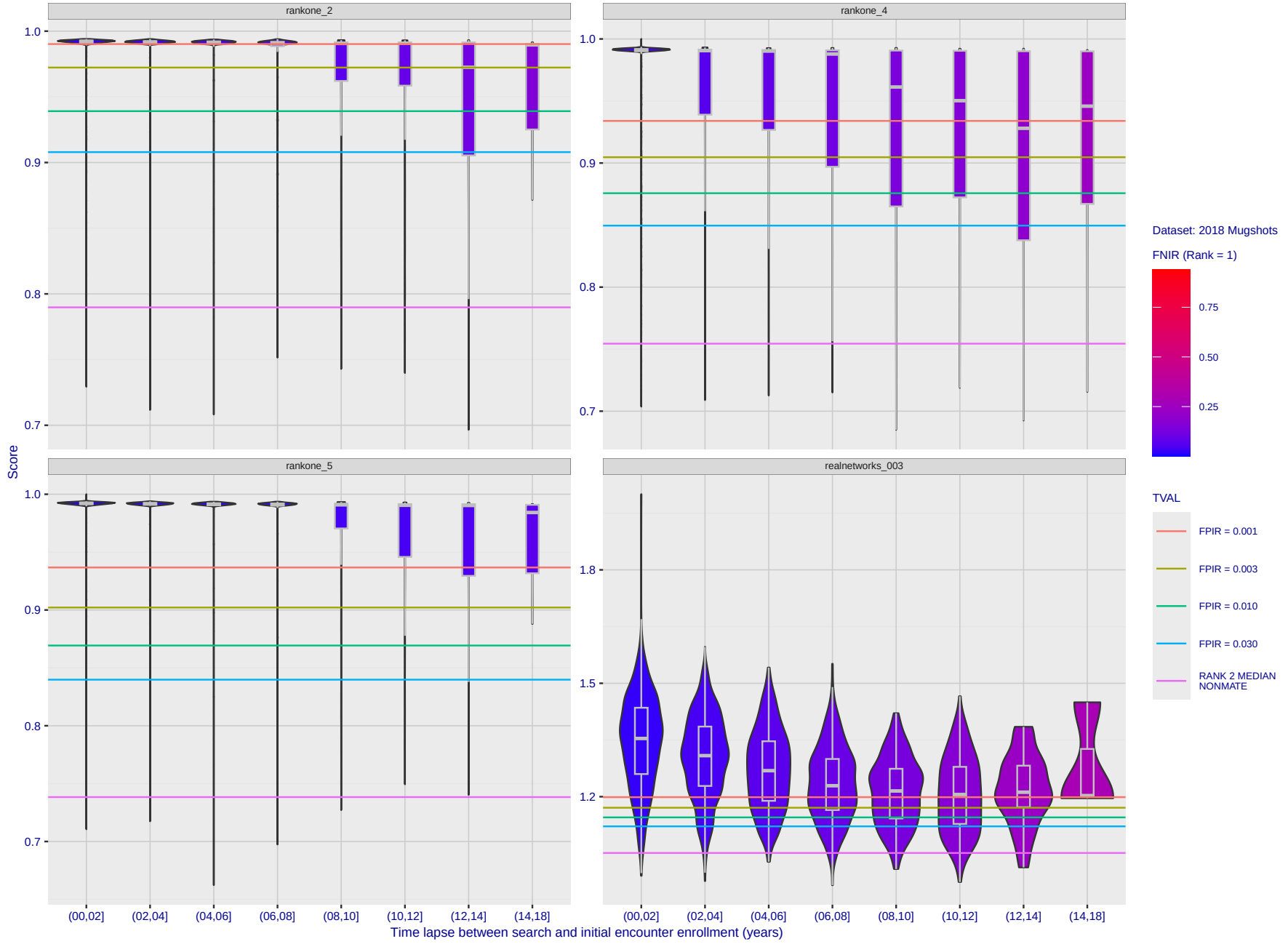

Figure 431: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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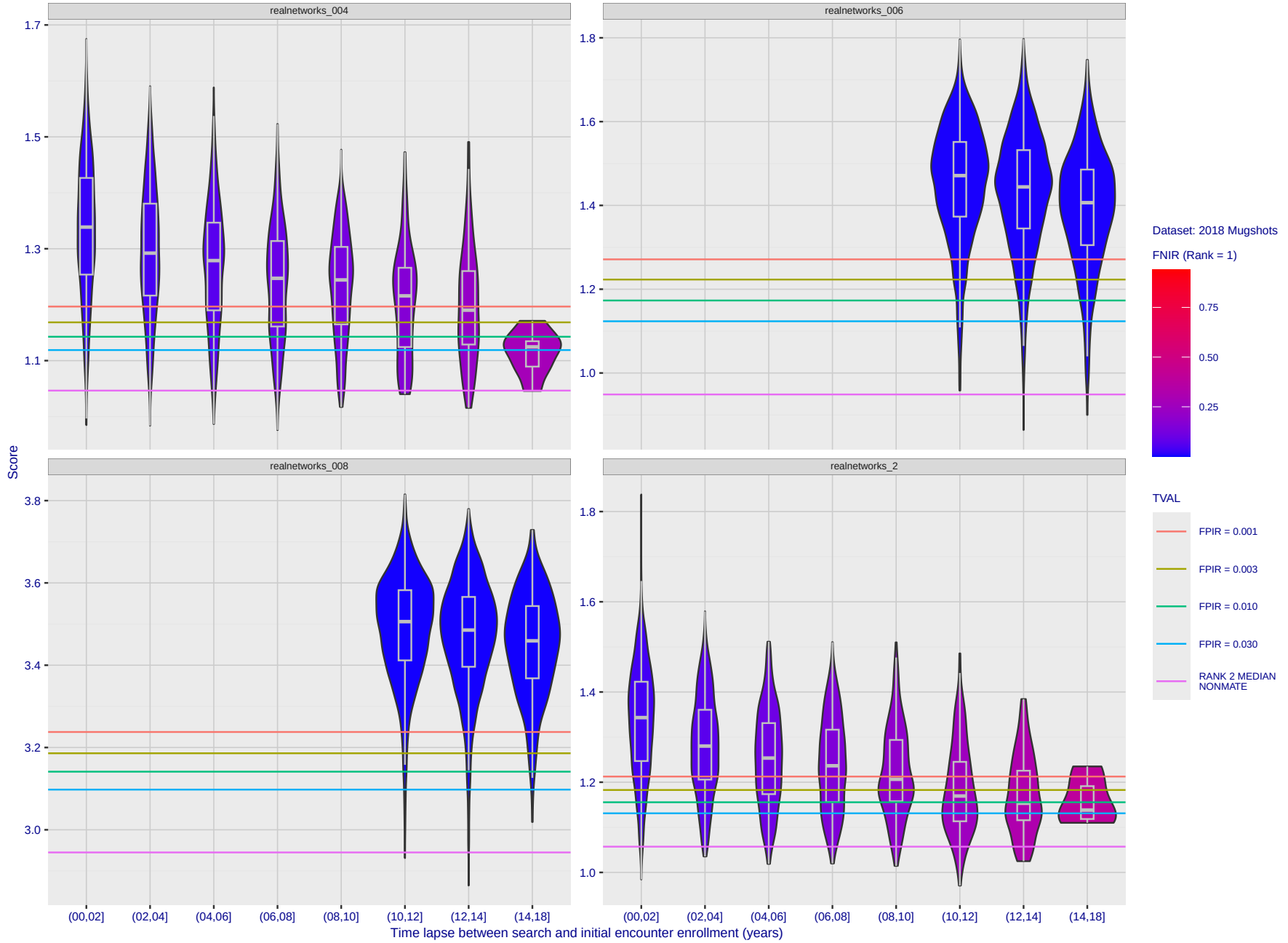

Figure 432: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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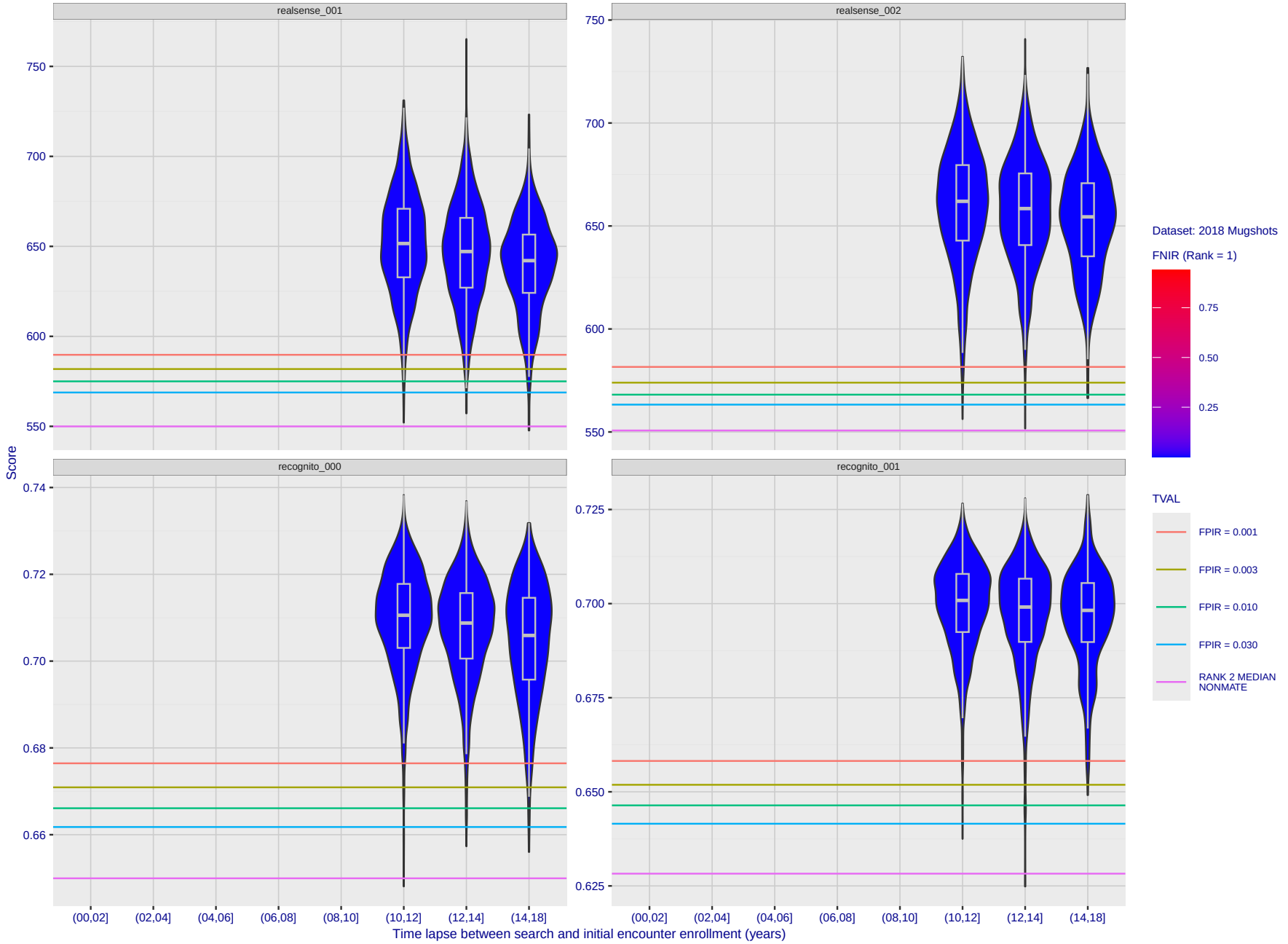

Figure 433: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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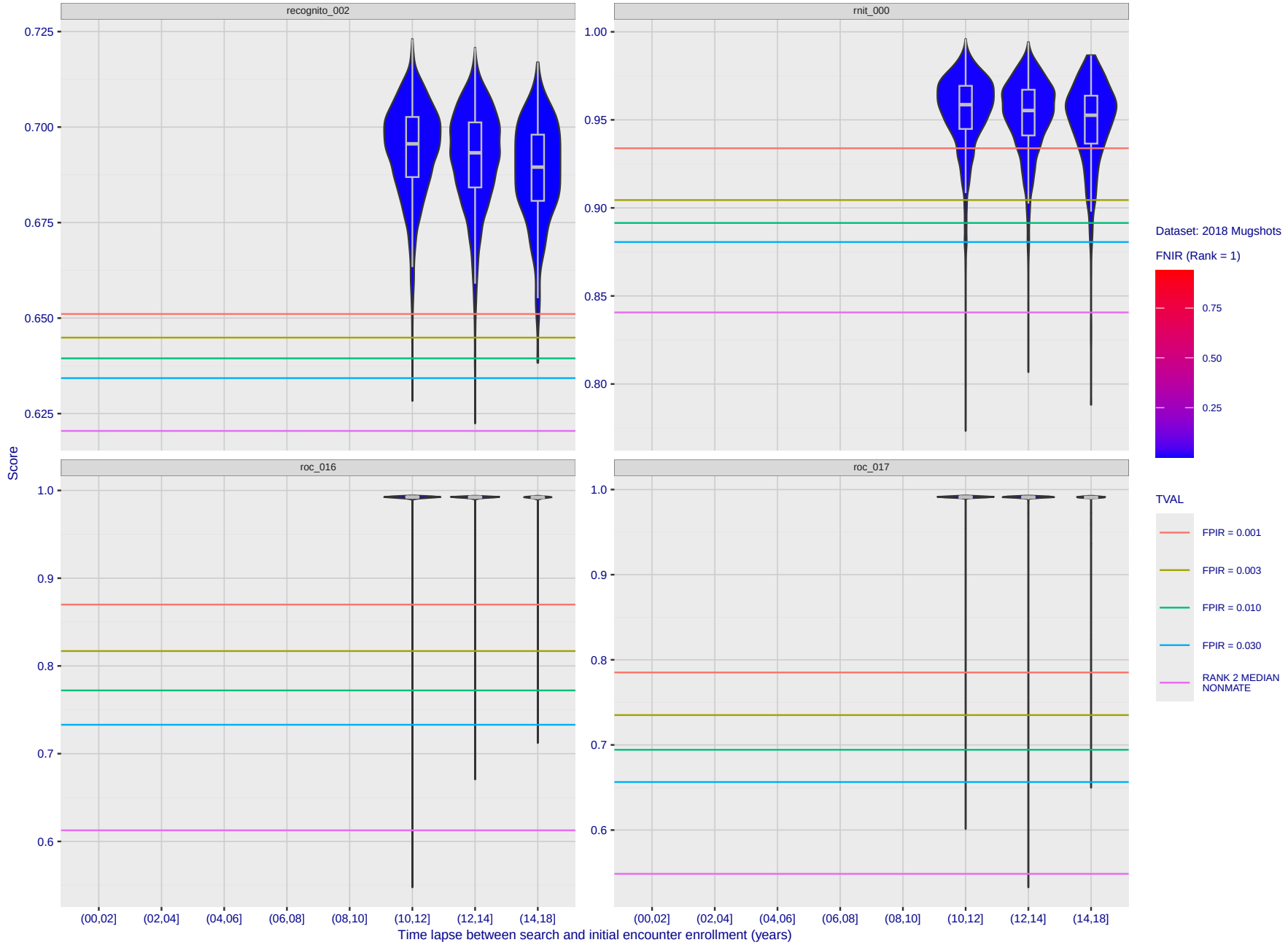

Figure 434: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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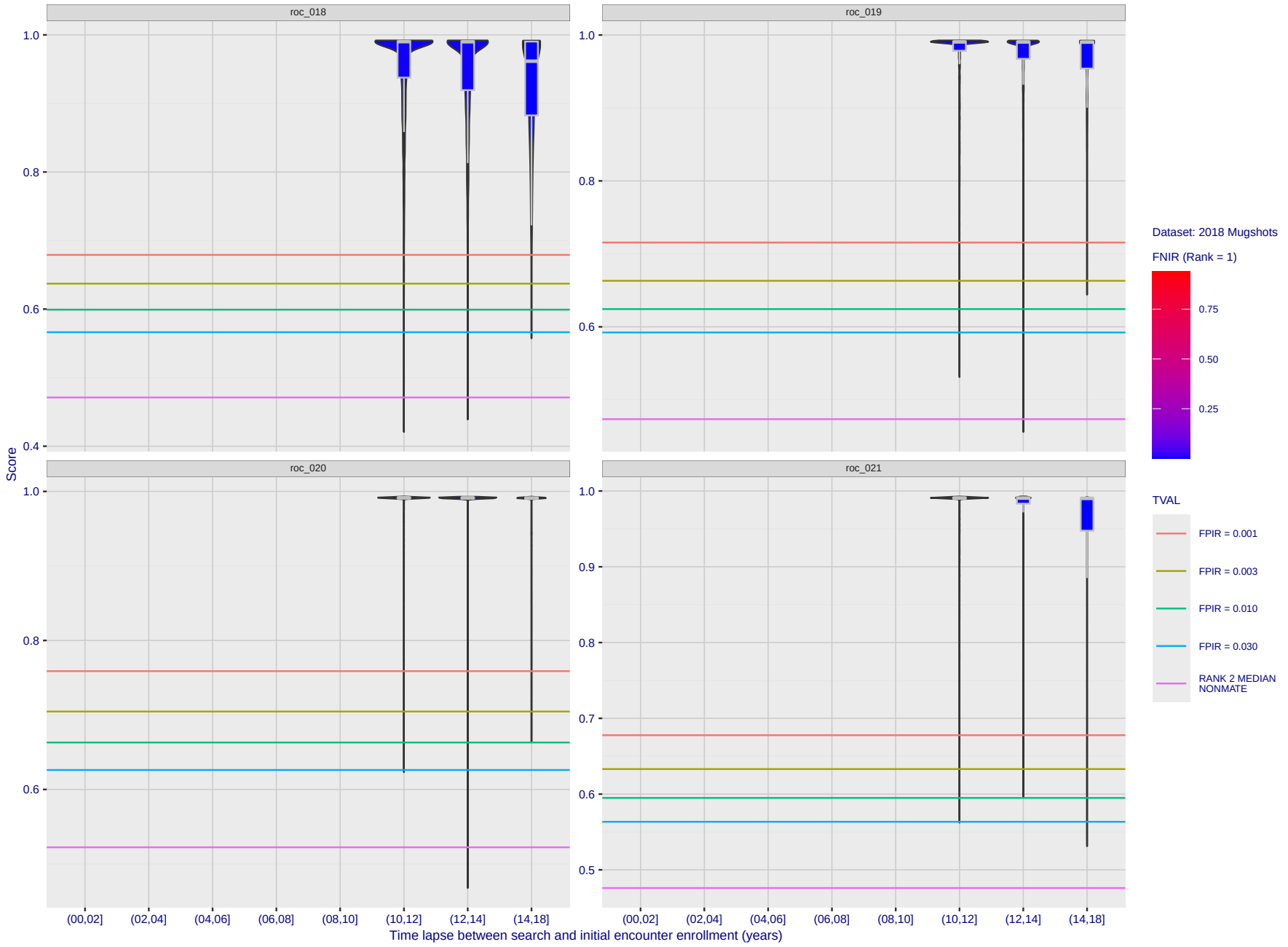

Figure 435: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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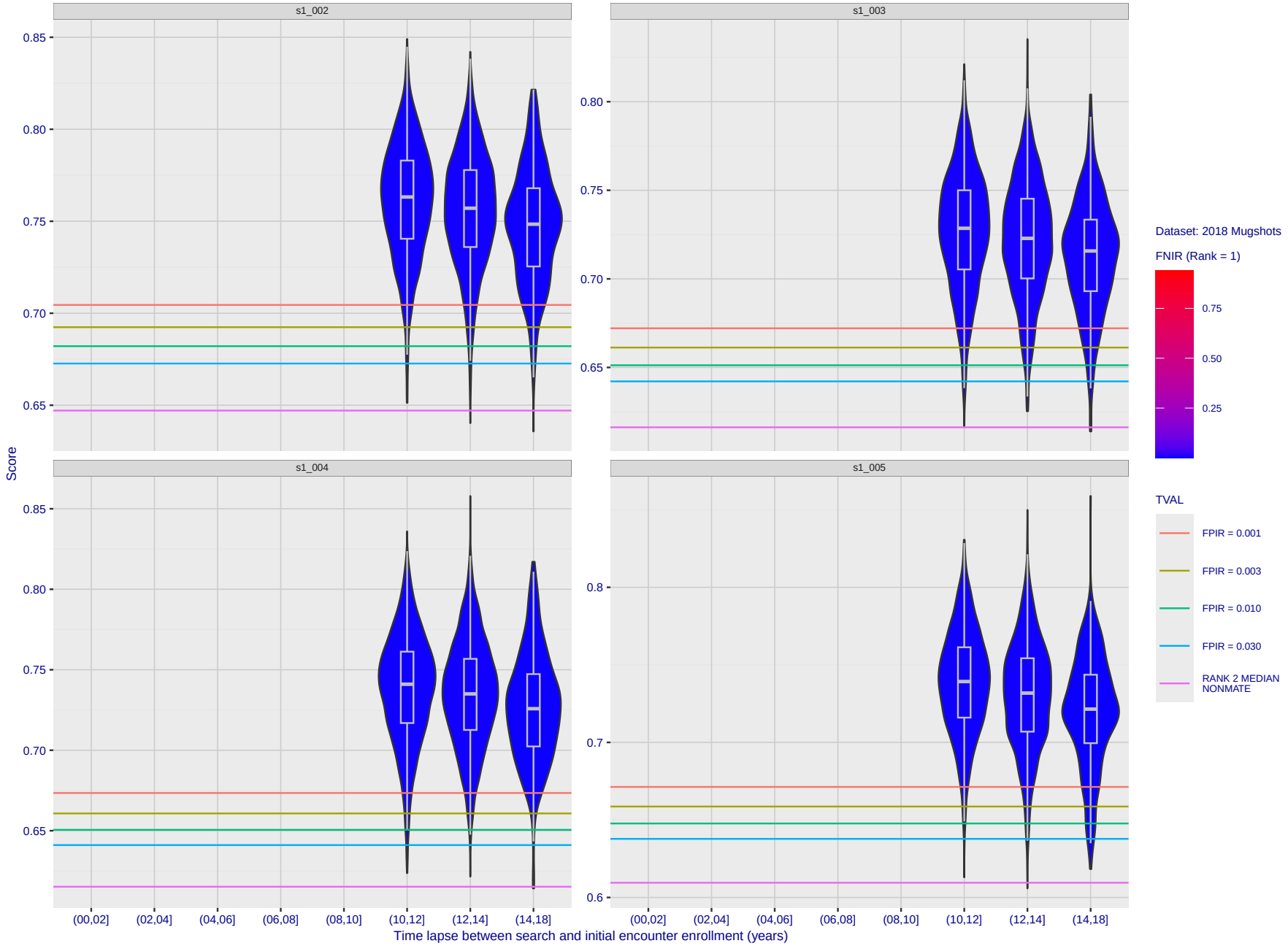

Figure 436: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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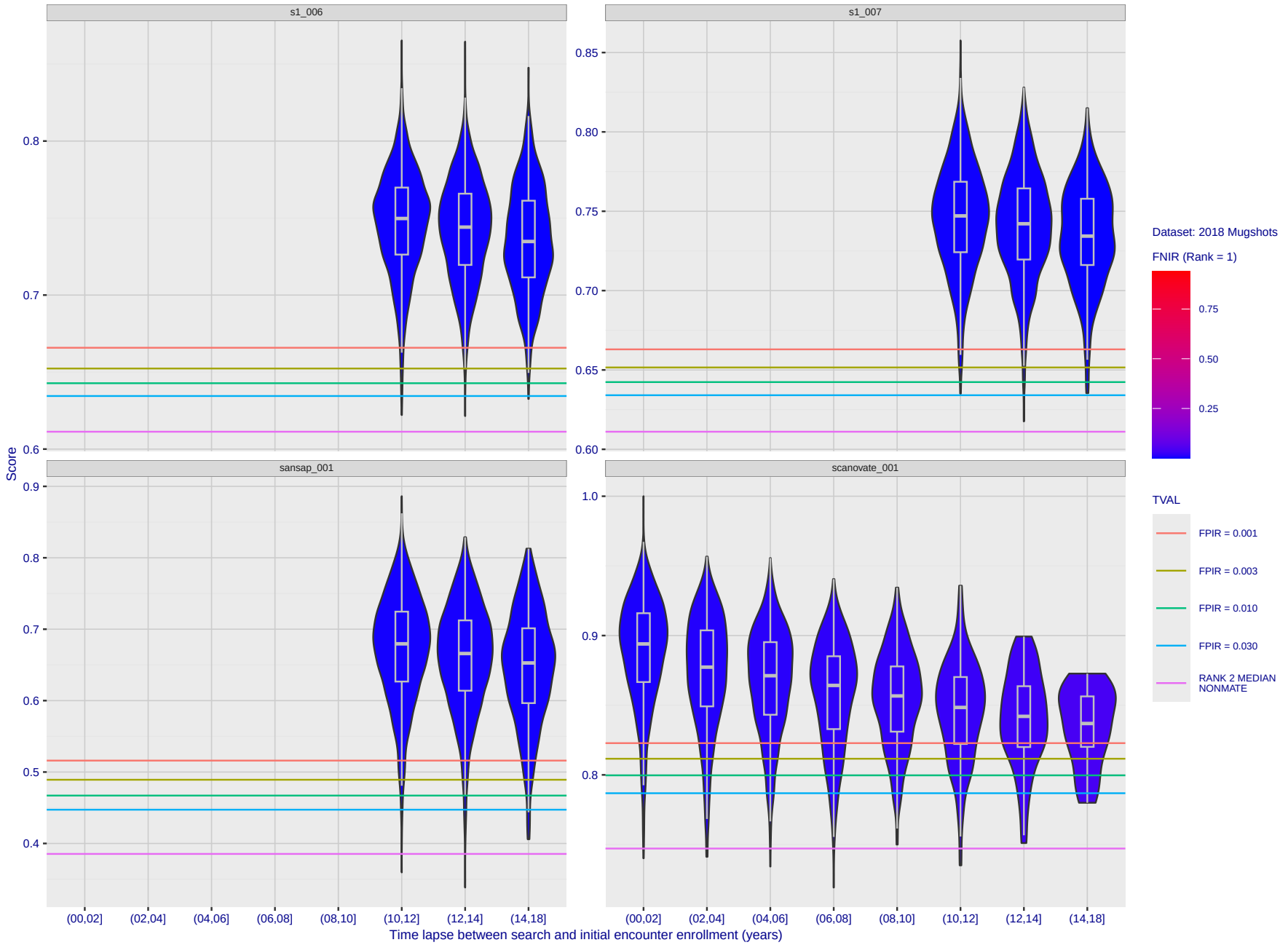

Figure 437: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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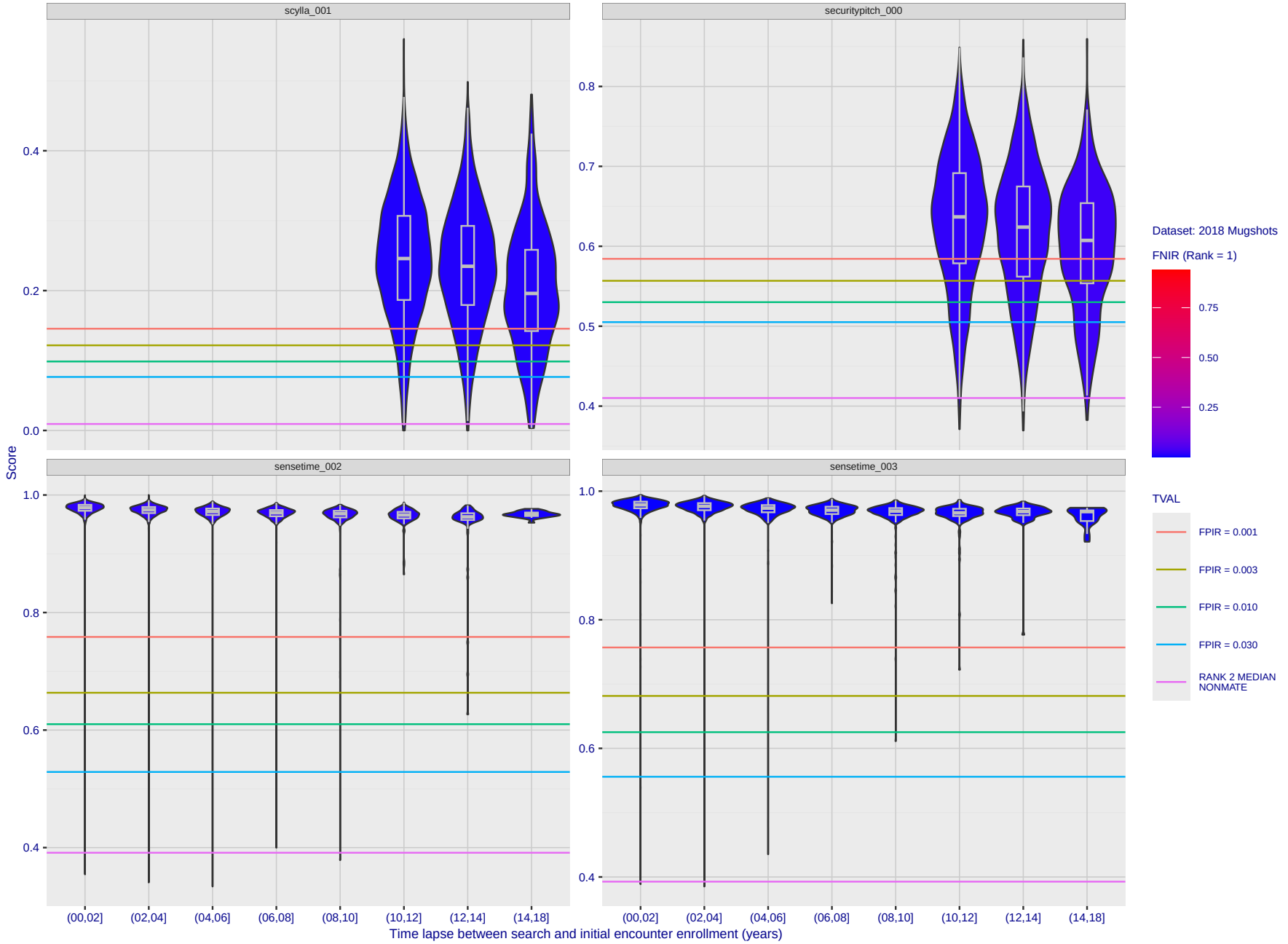

Figure 438: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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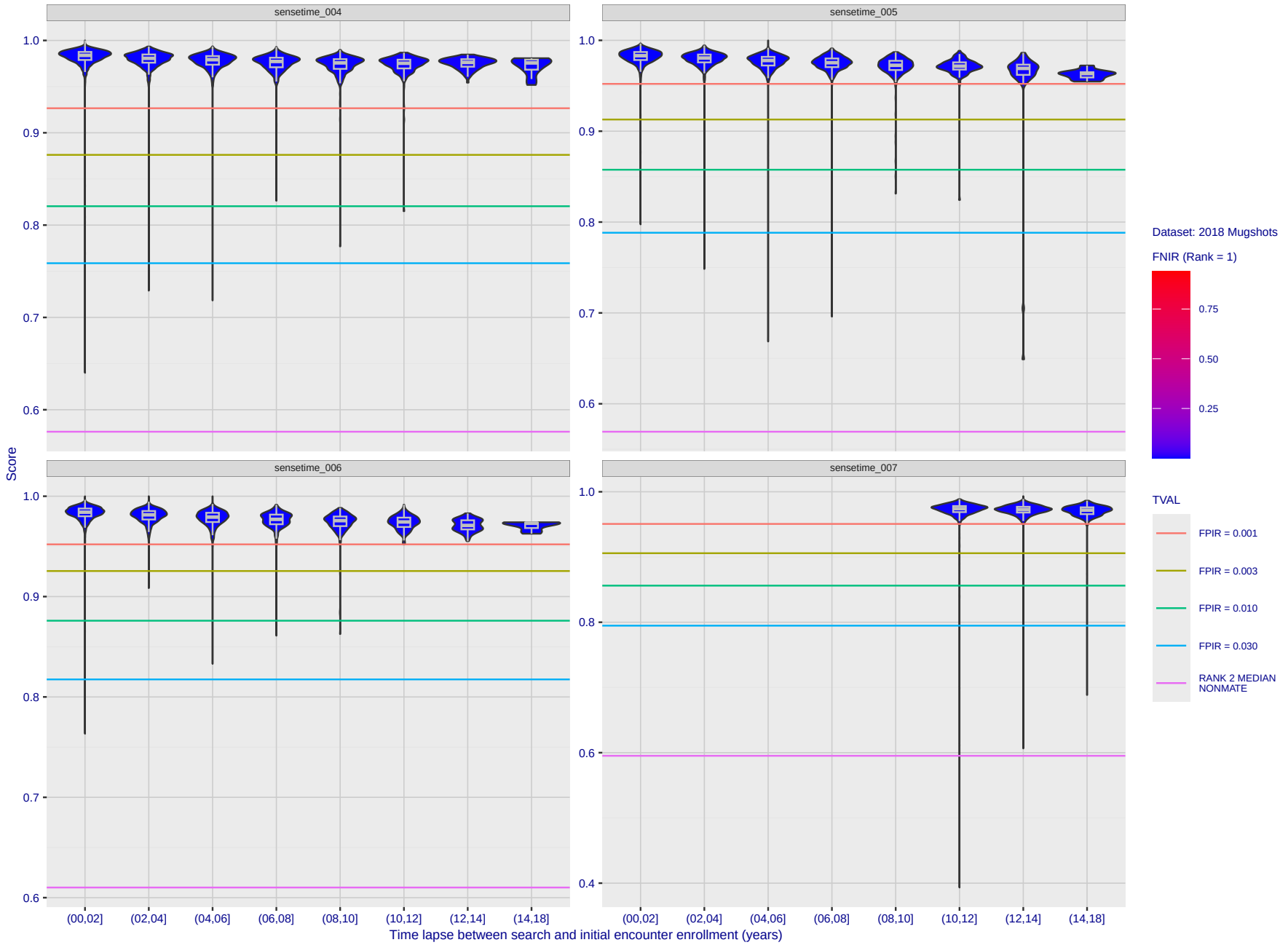

Figure 439: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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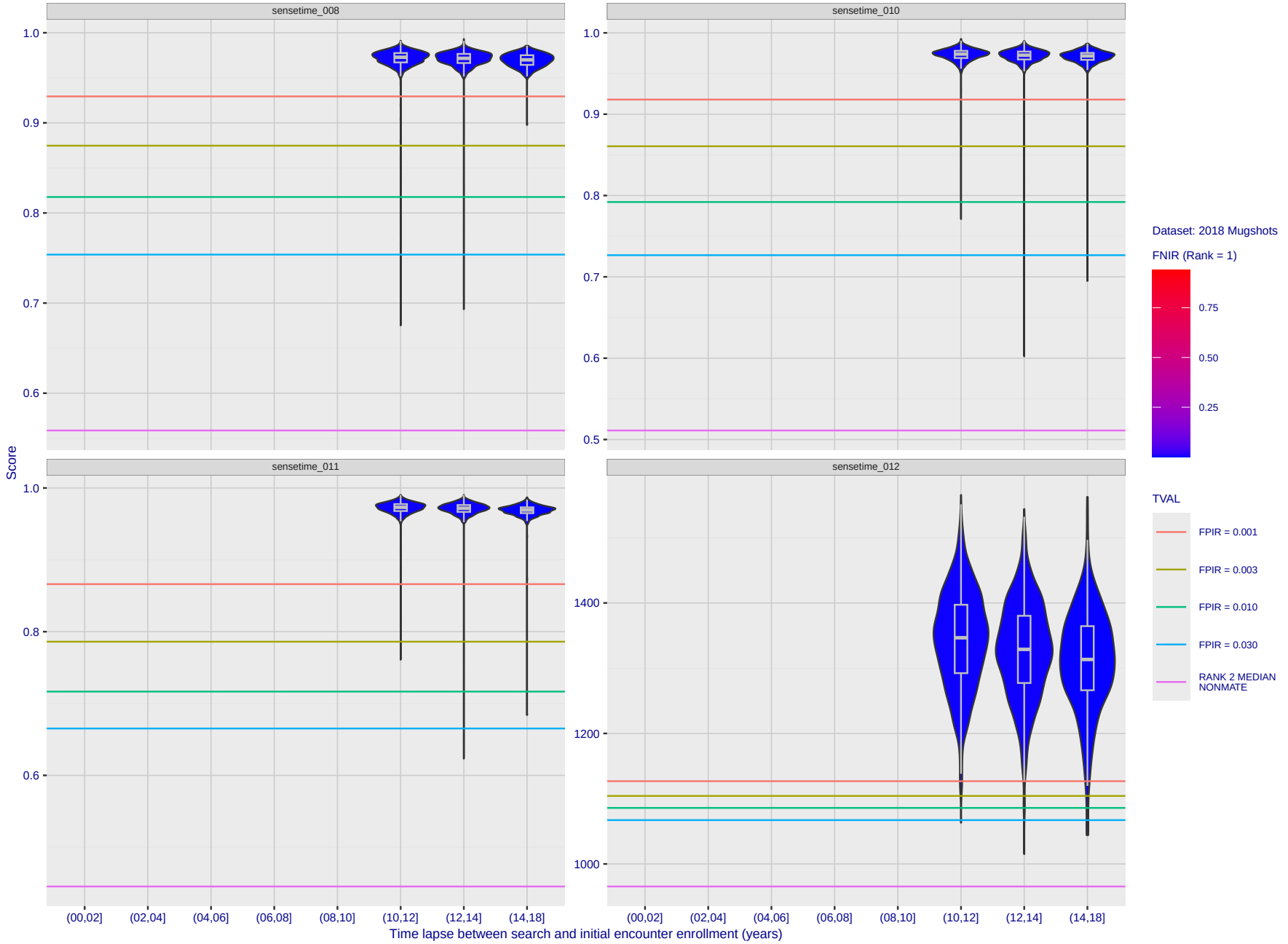

Figure 440: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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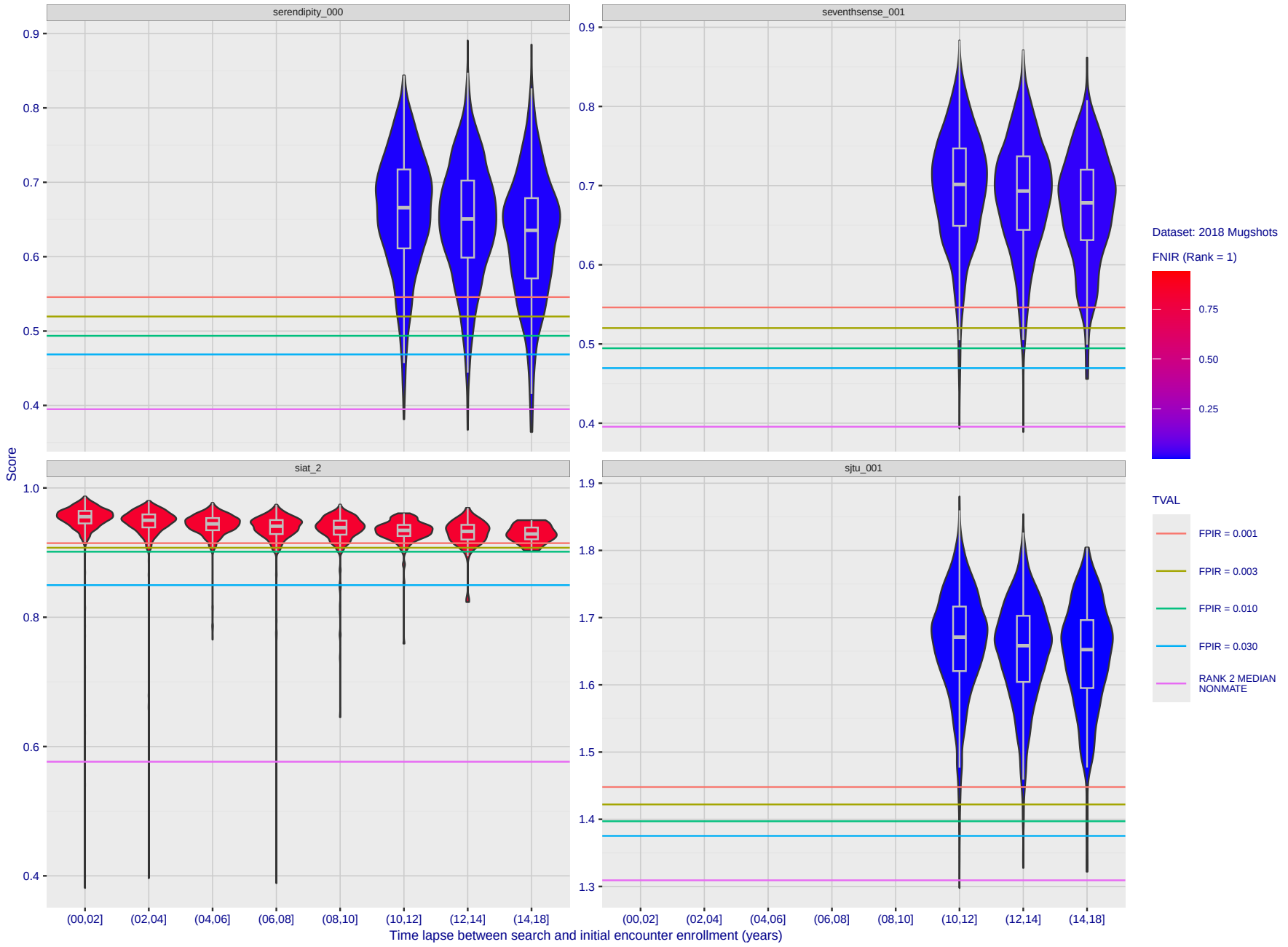

Figure 441: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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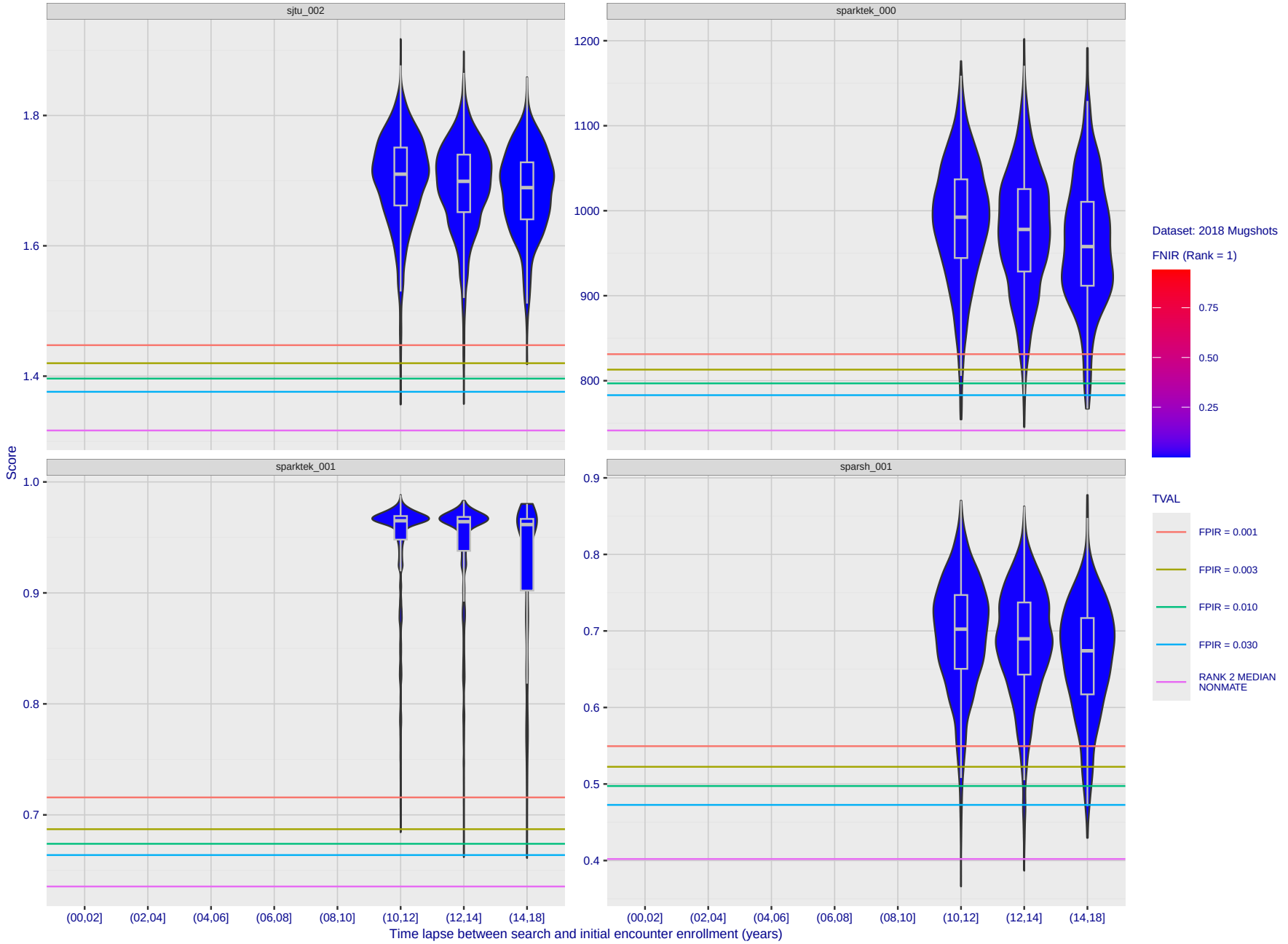

Figure 442: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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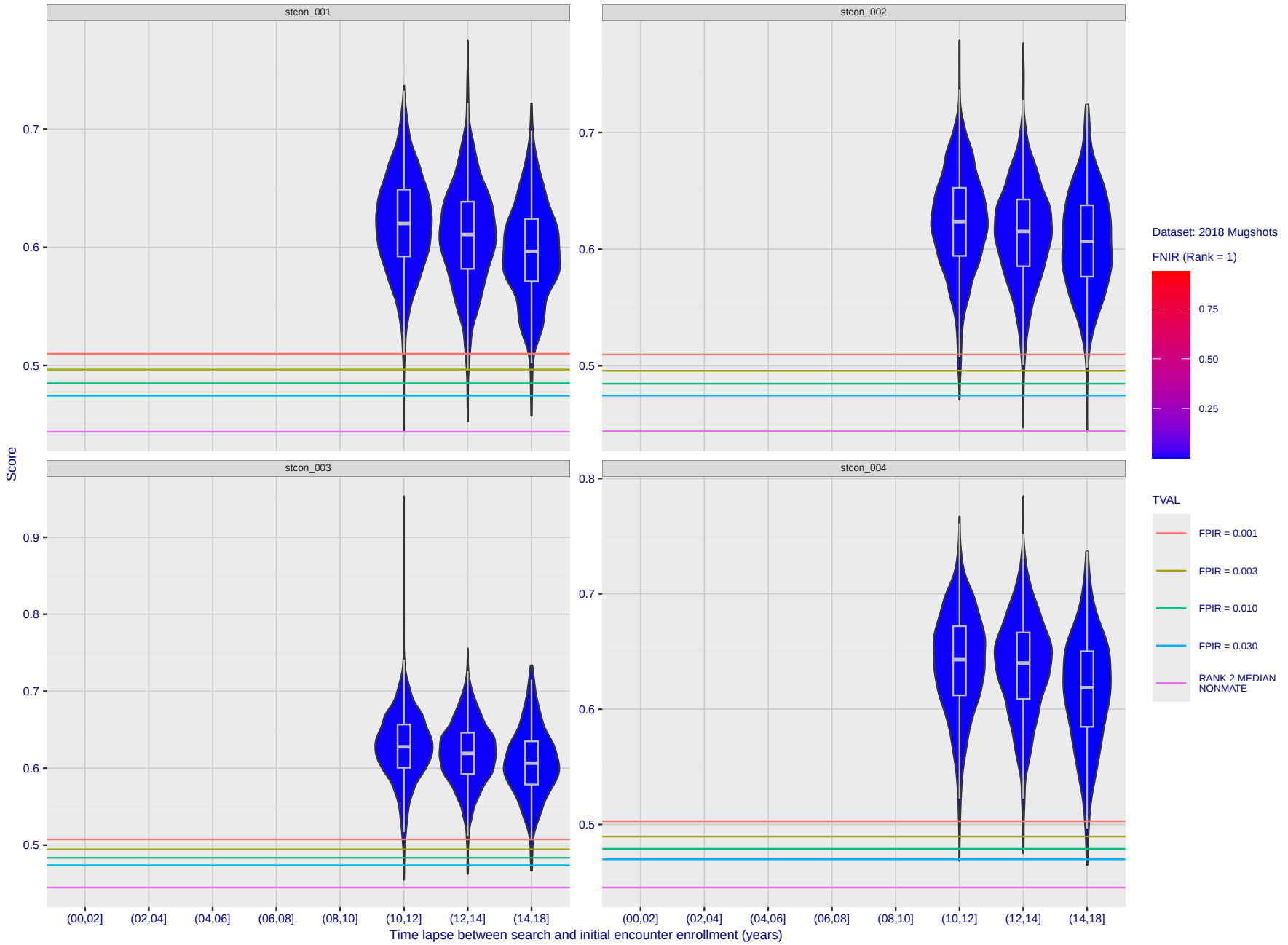

Figure 443: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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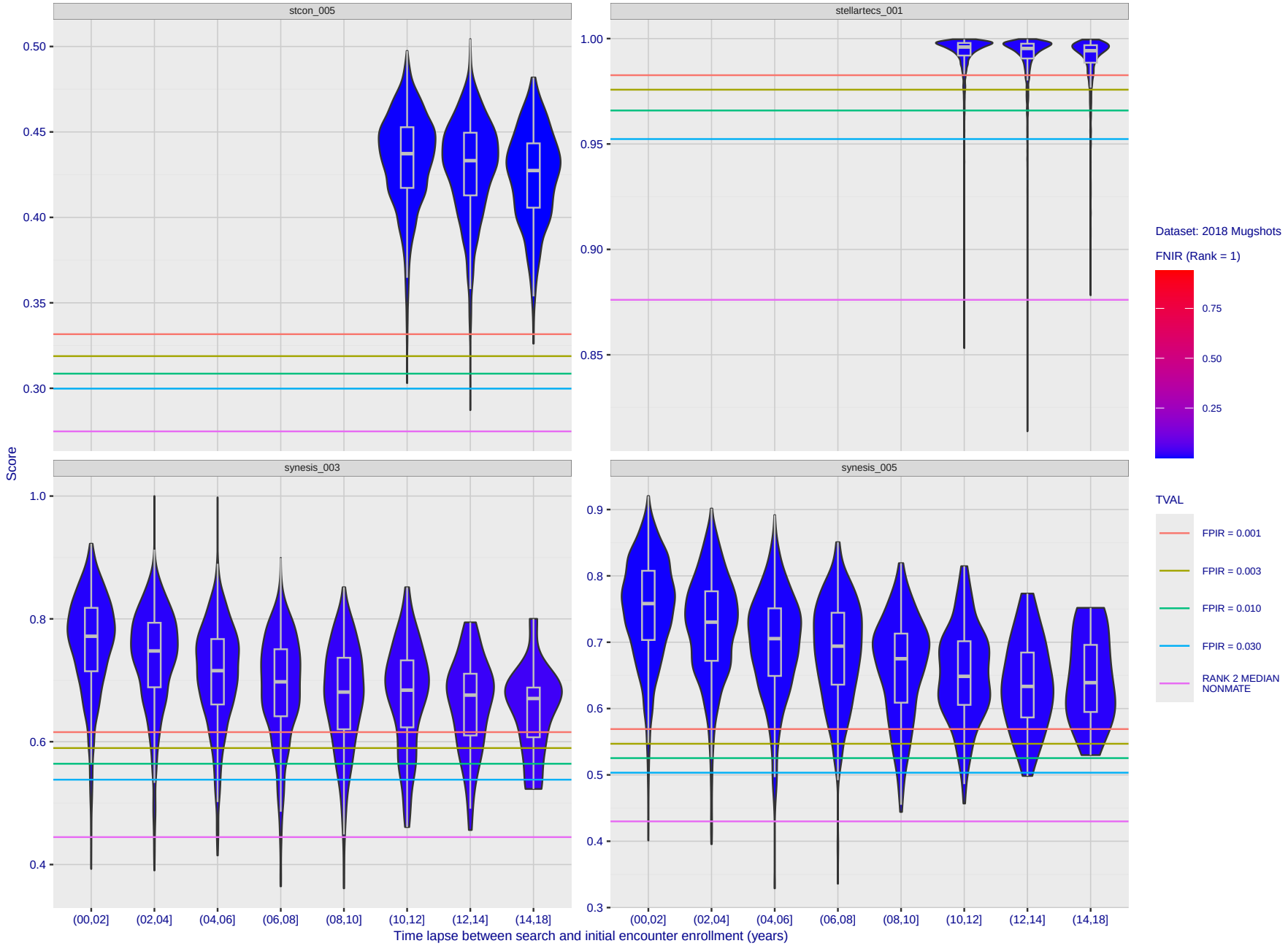

Figure 444: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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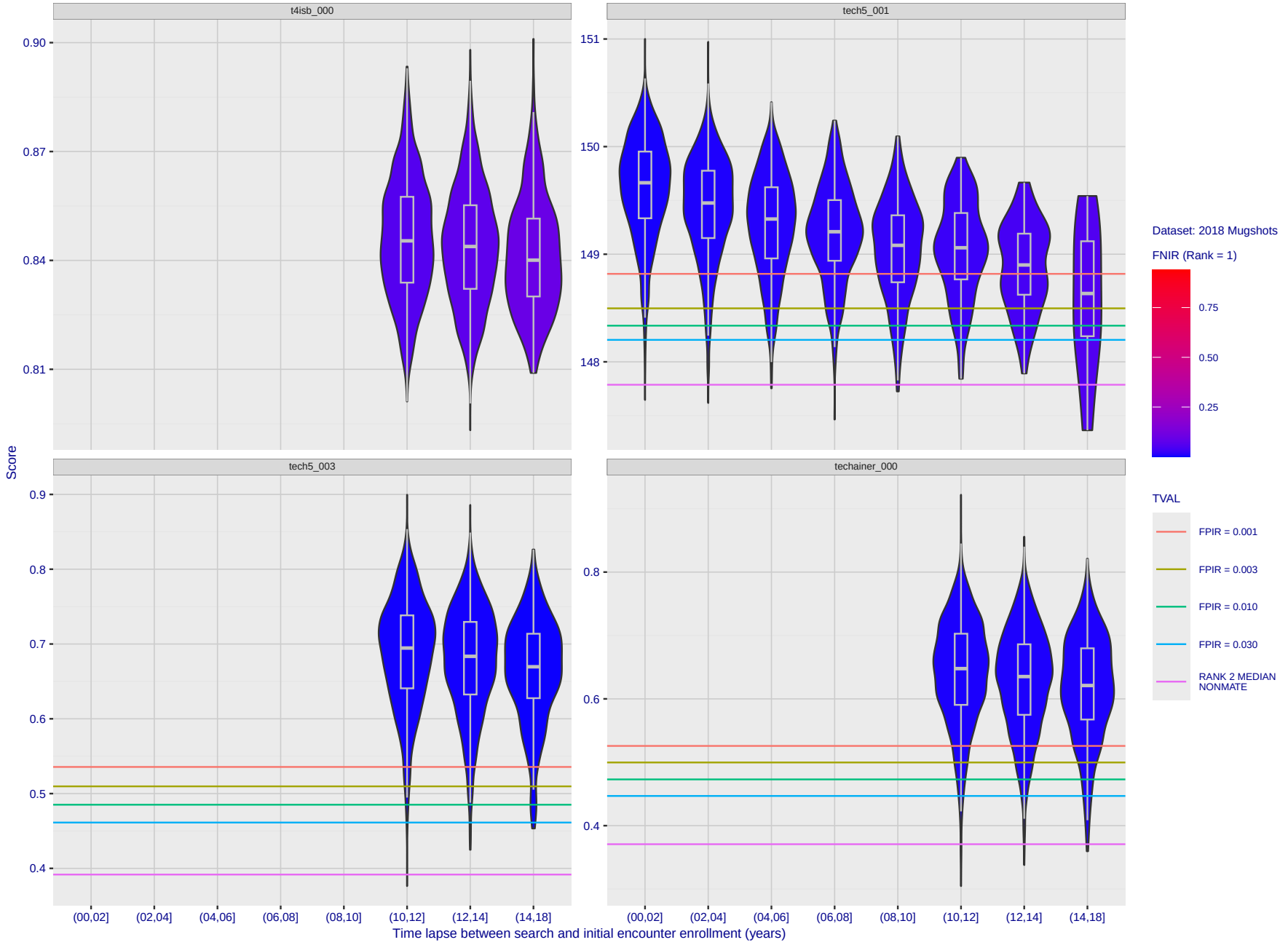

Figure 445: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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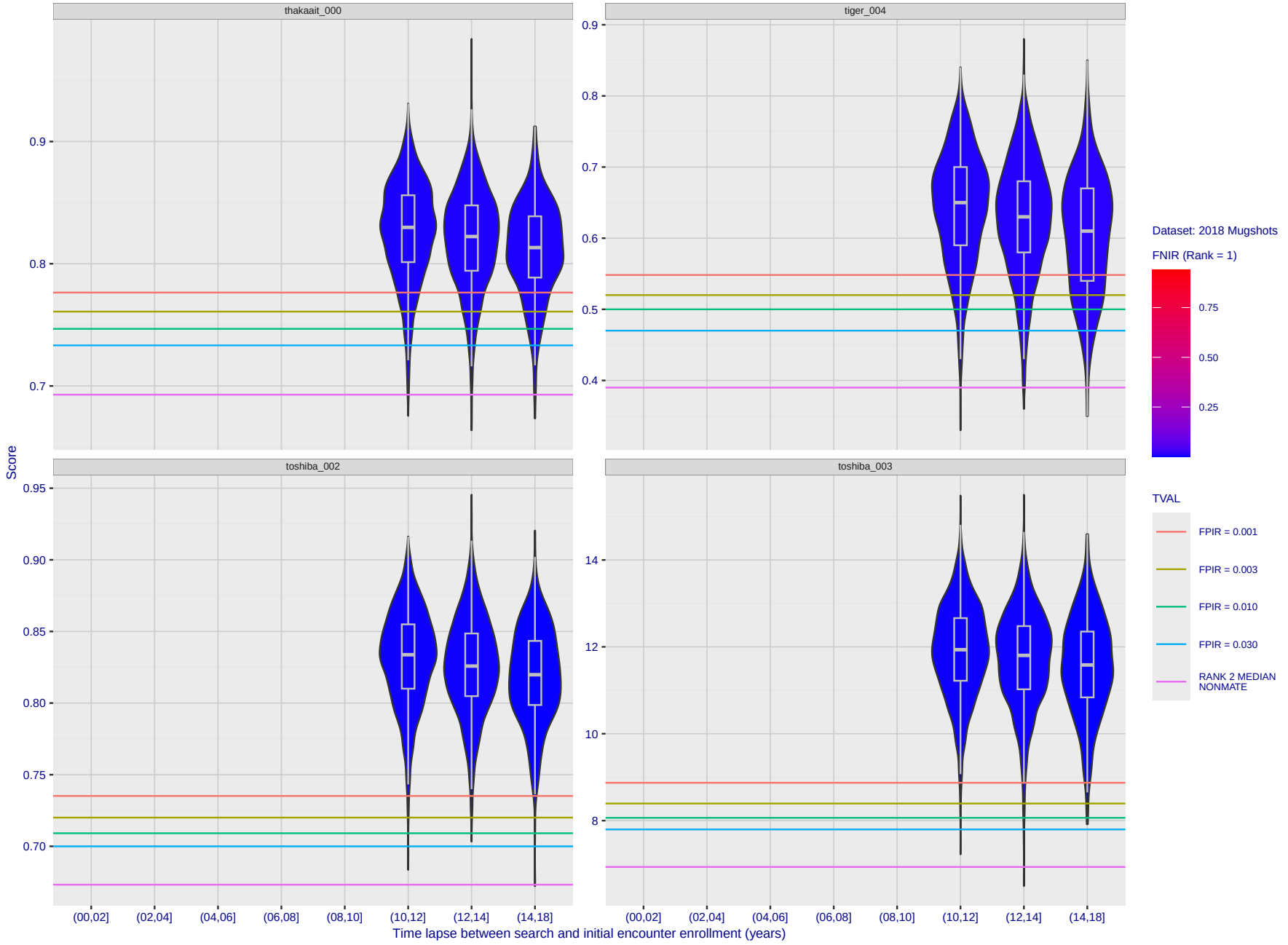

Figure 446: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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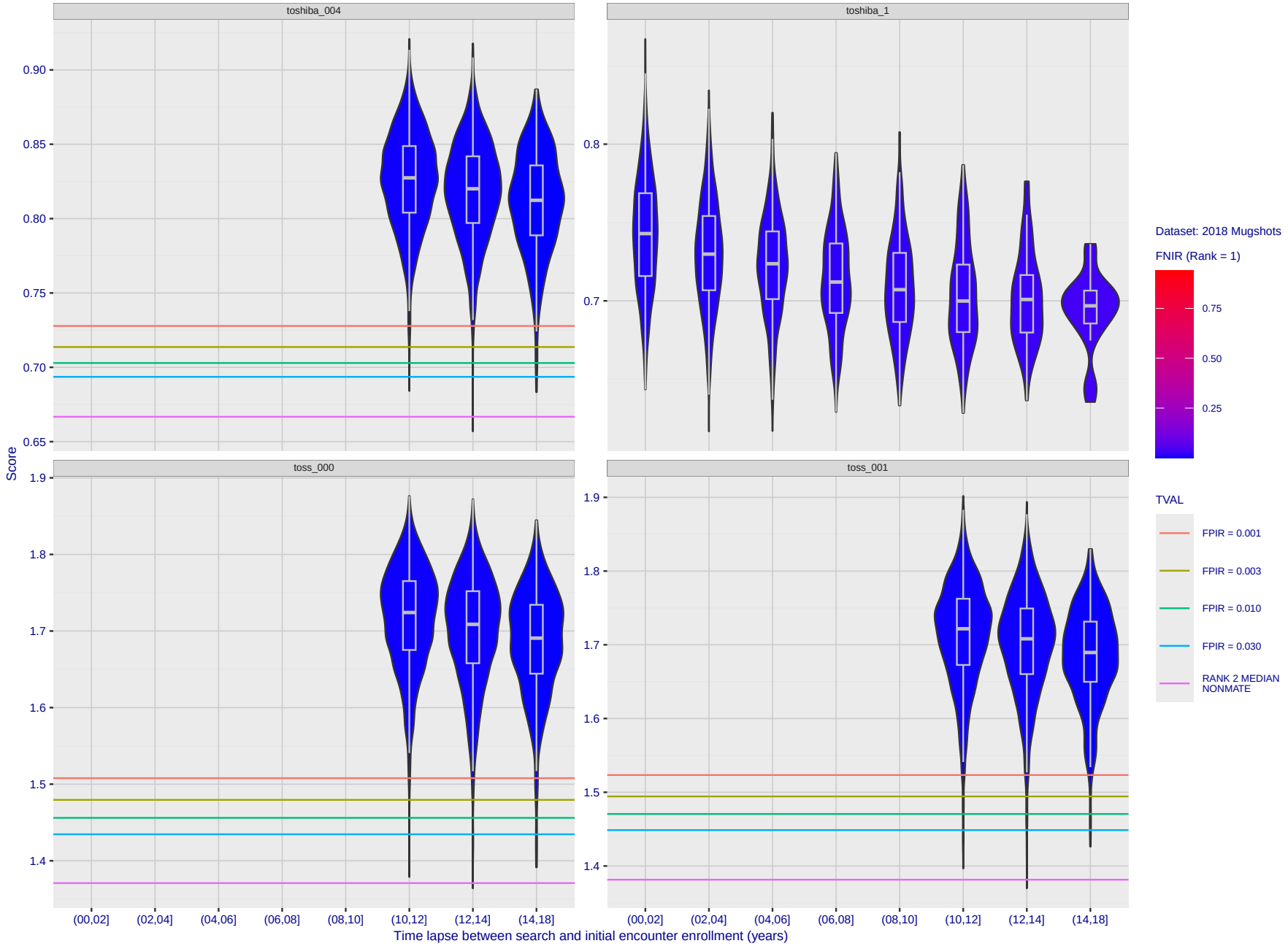

Figure 447: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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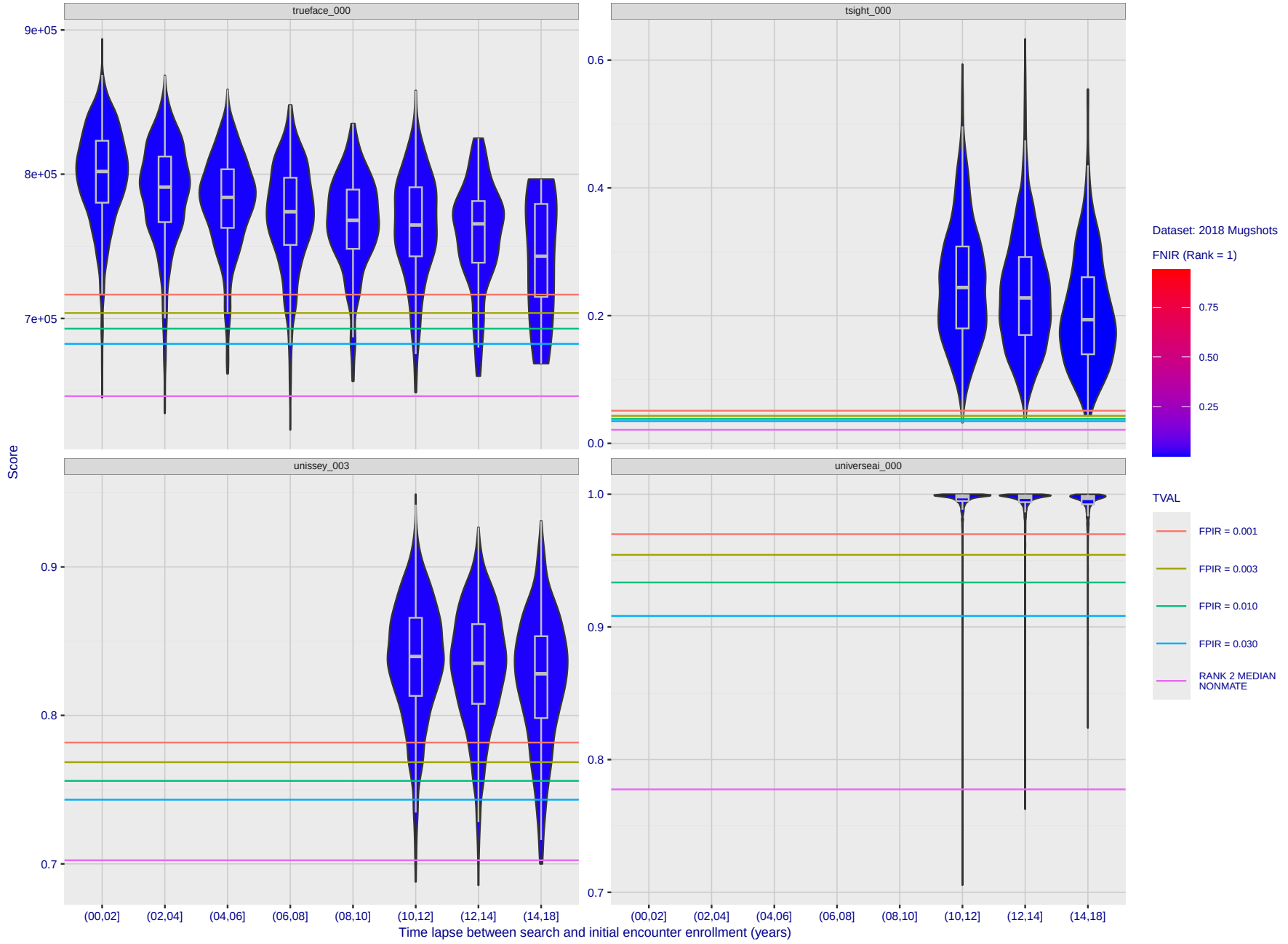

Figure 448: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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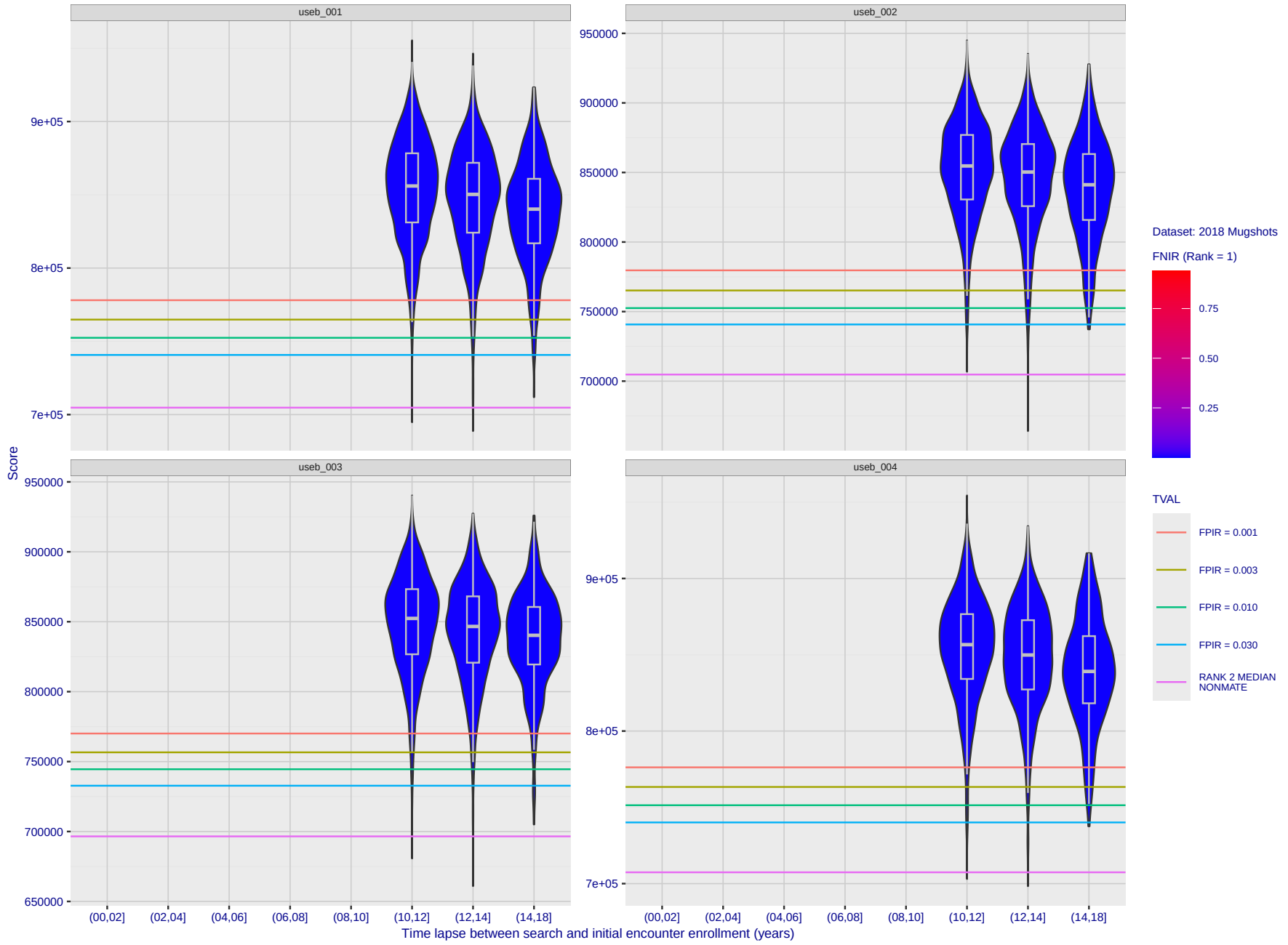

Figure 449: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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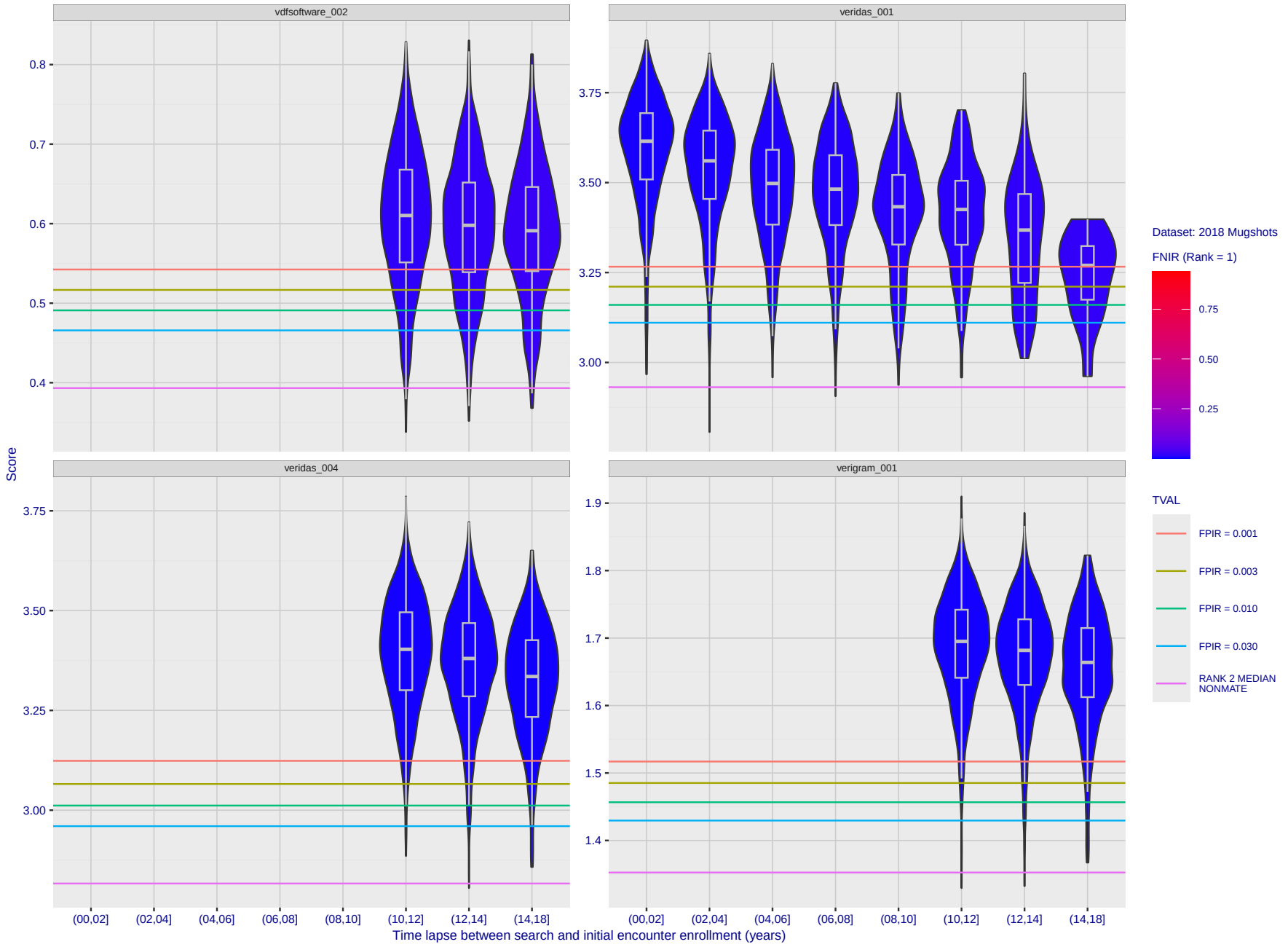

Figure 450: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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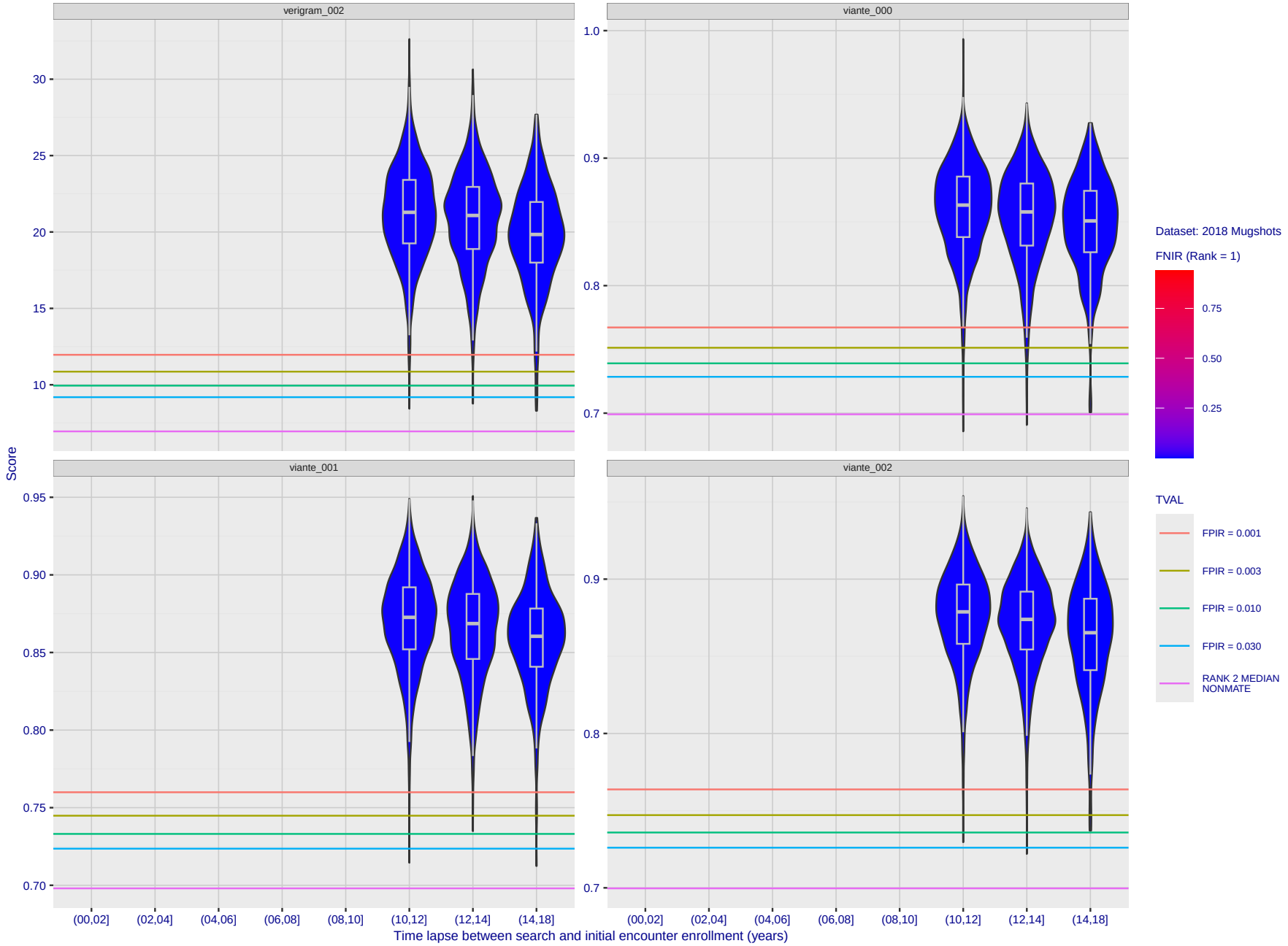

Figure 451: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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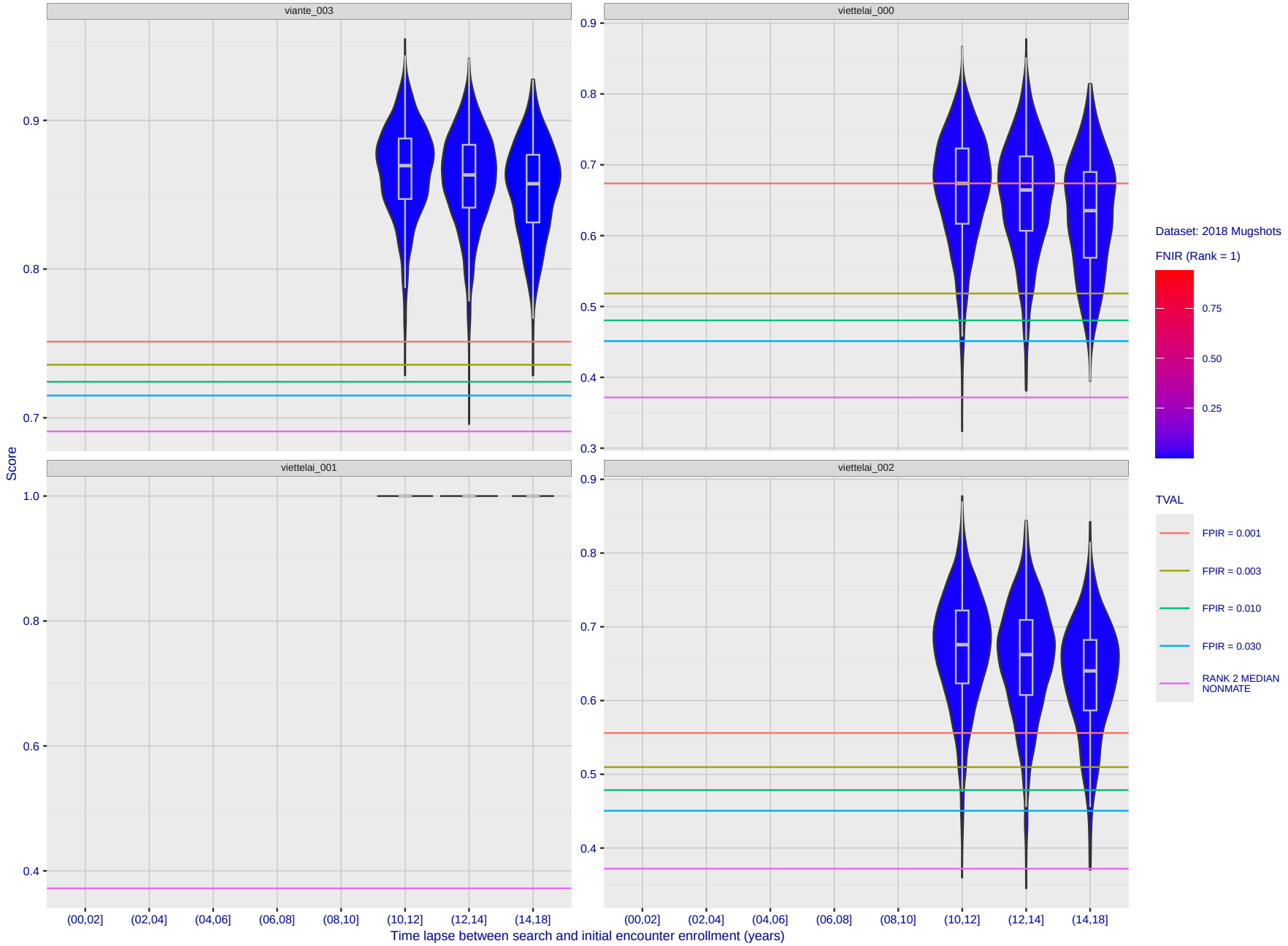

Figure 452: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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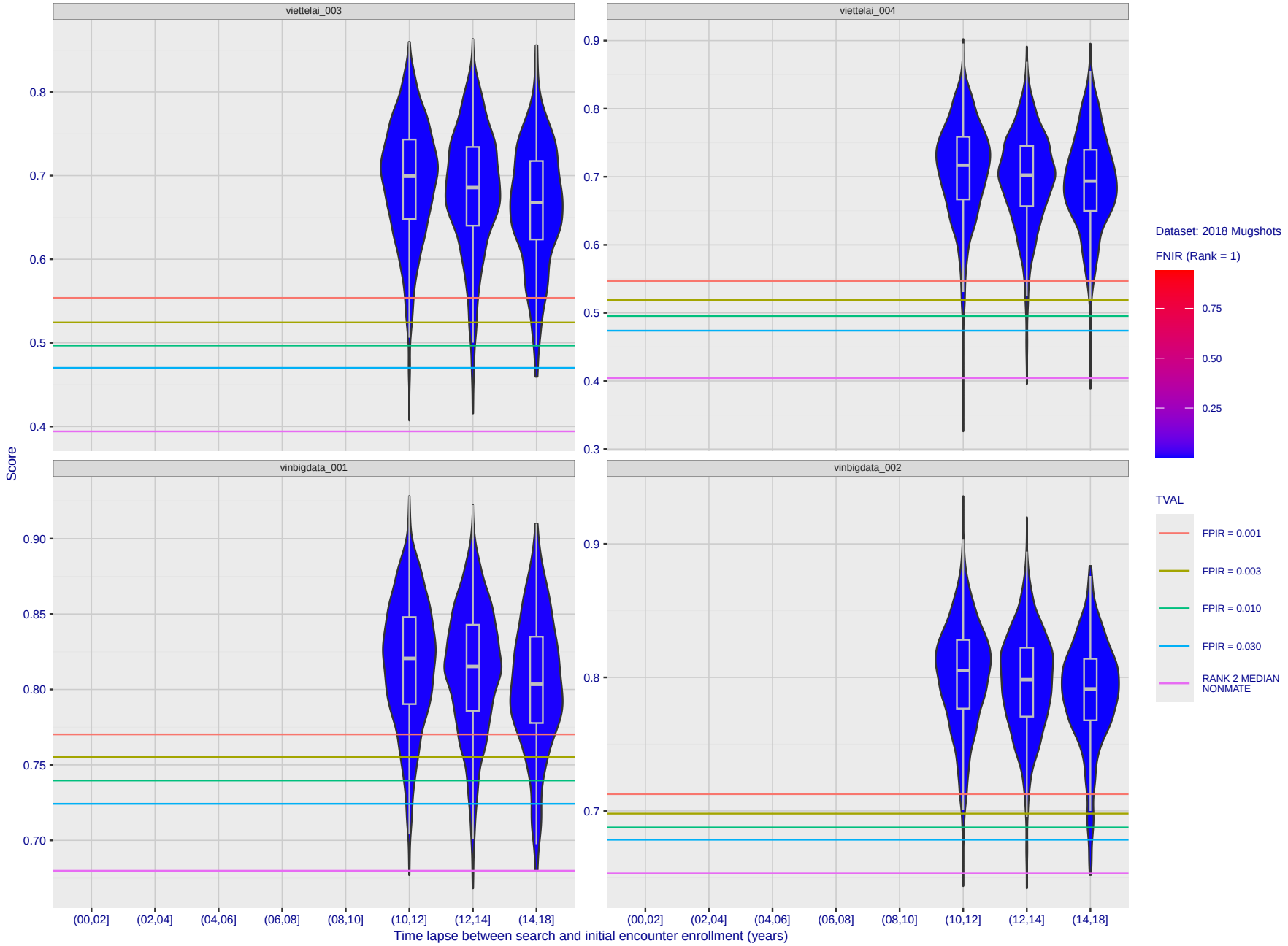

Figure 453: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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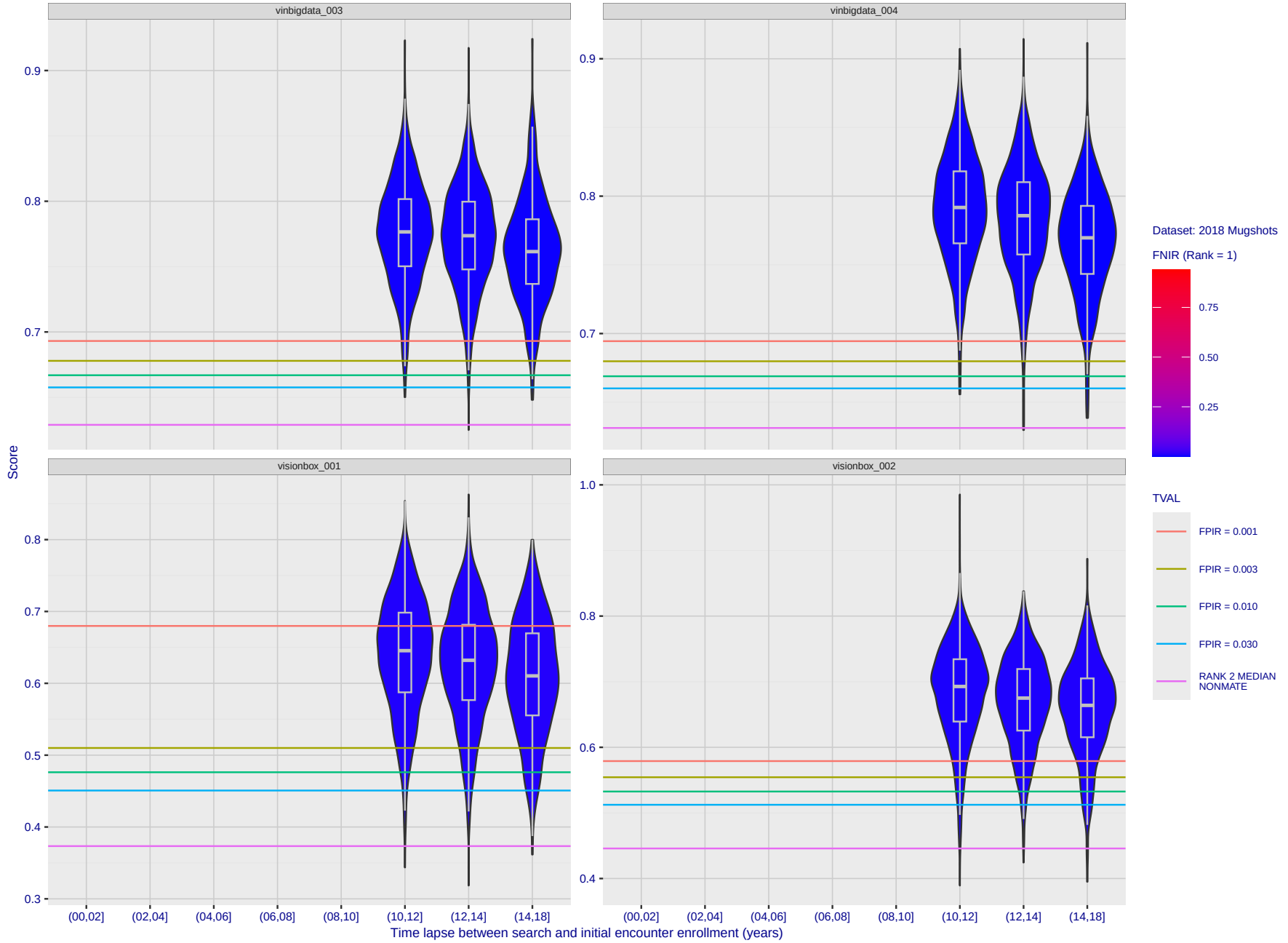

Figure 454: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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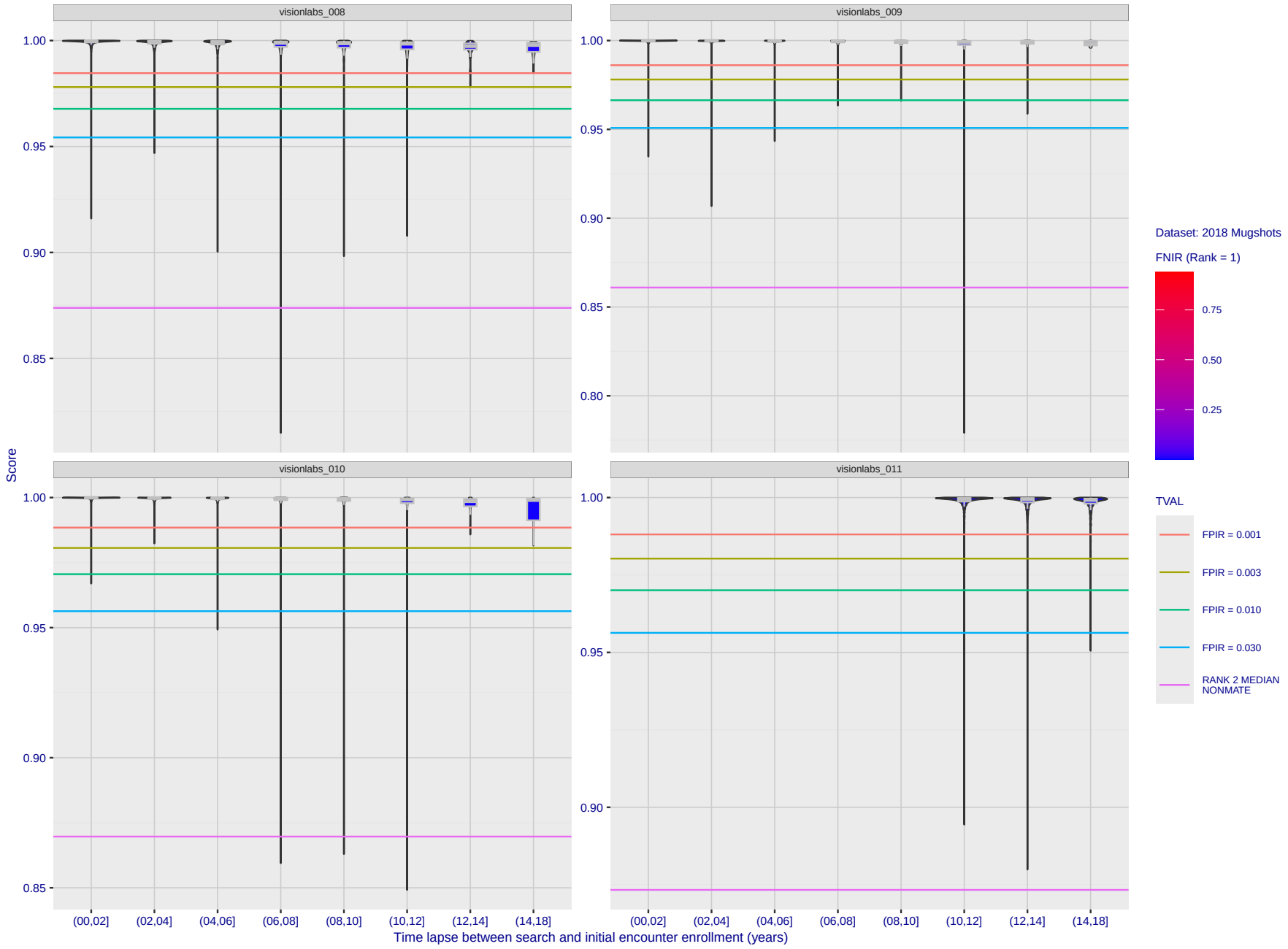

Figure 455: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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T > 0 → Identification

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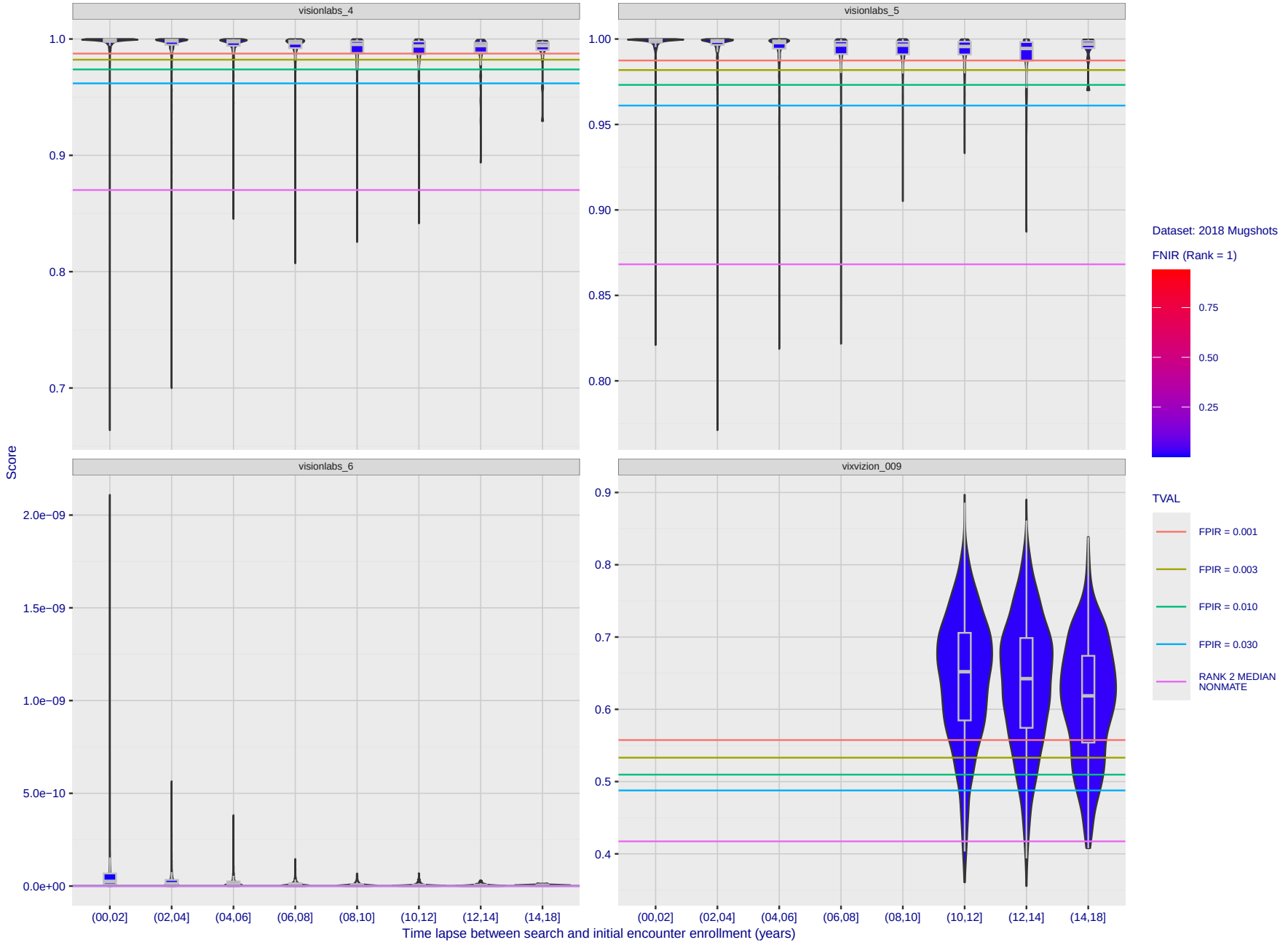

Figure 456: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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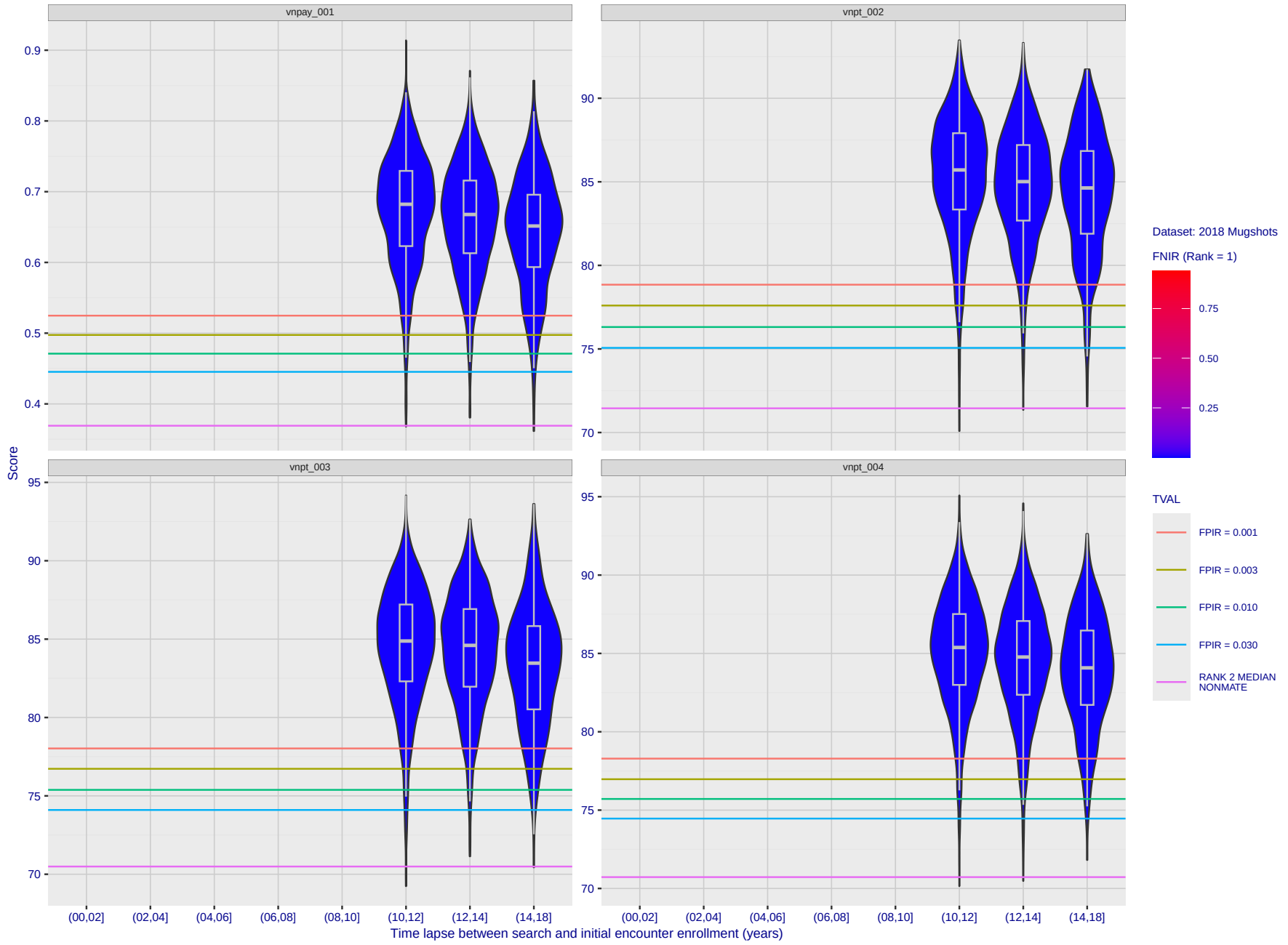

Figure 457: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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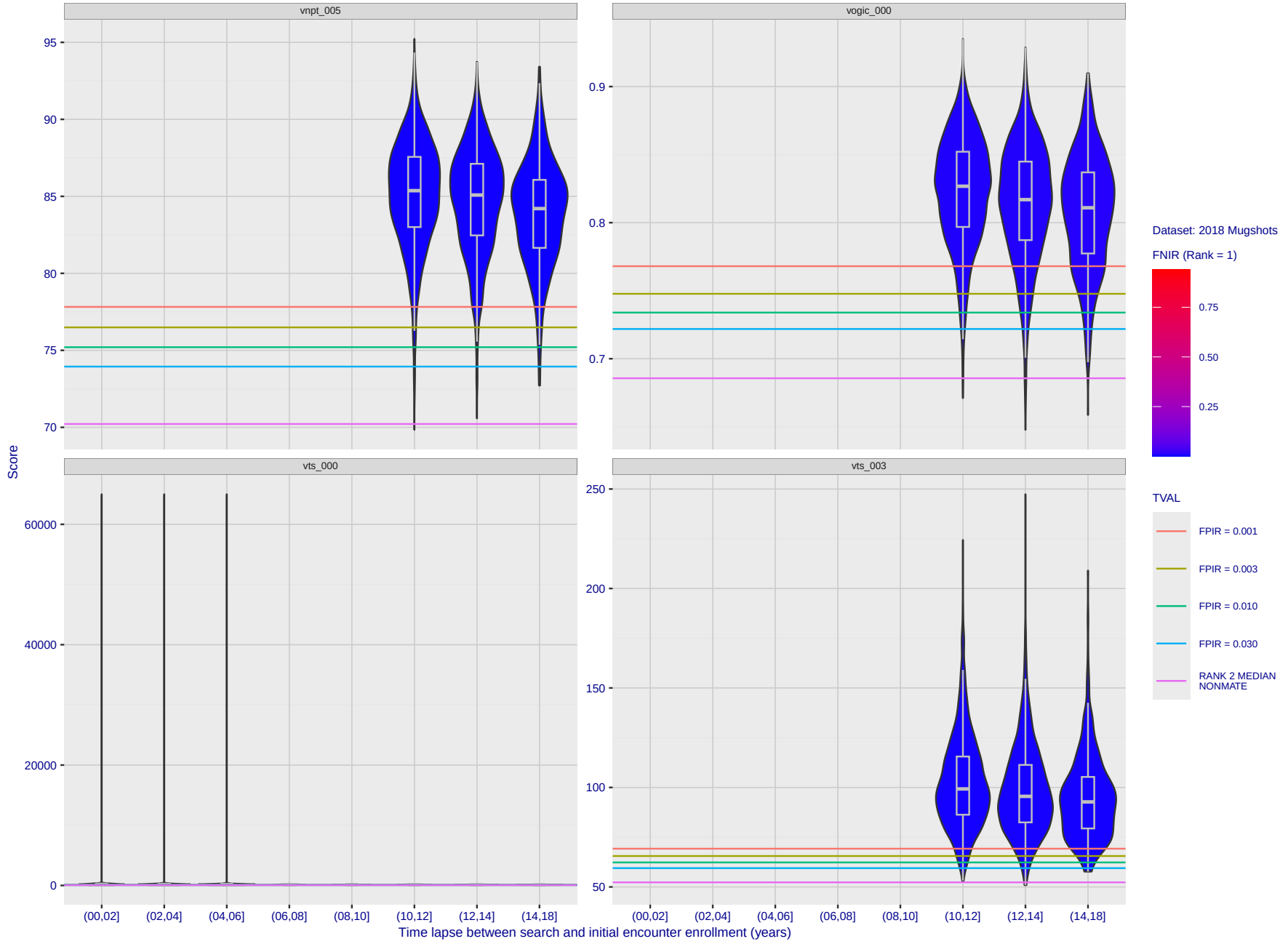

Figure 458: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

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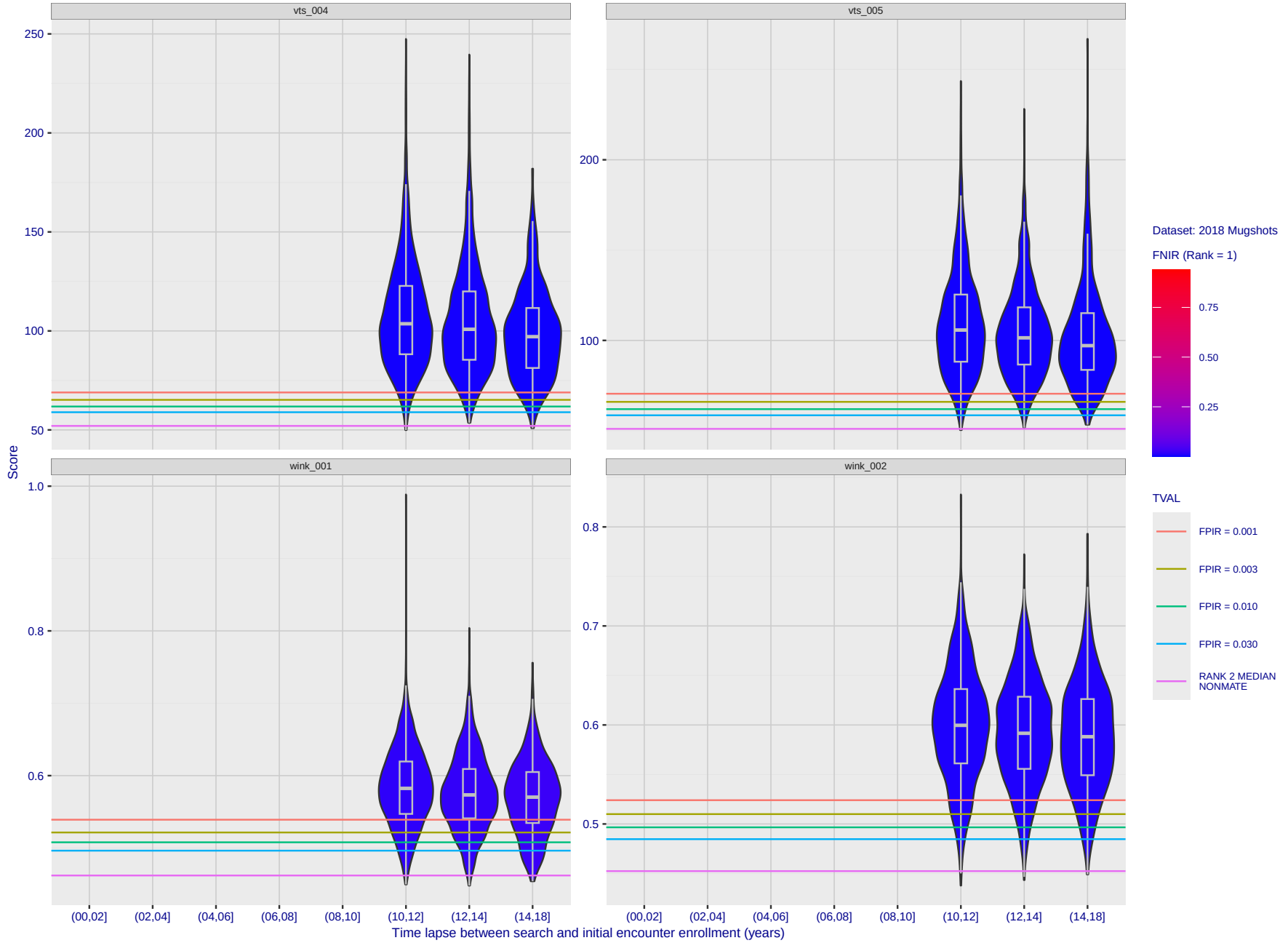

Figure 459: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
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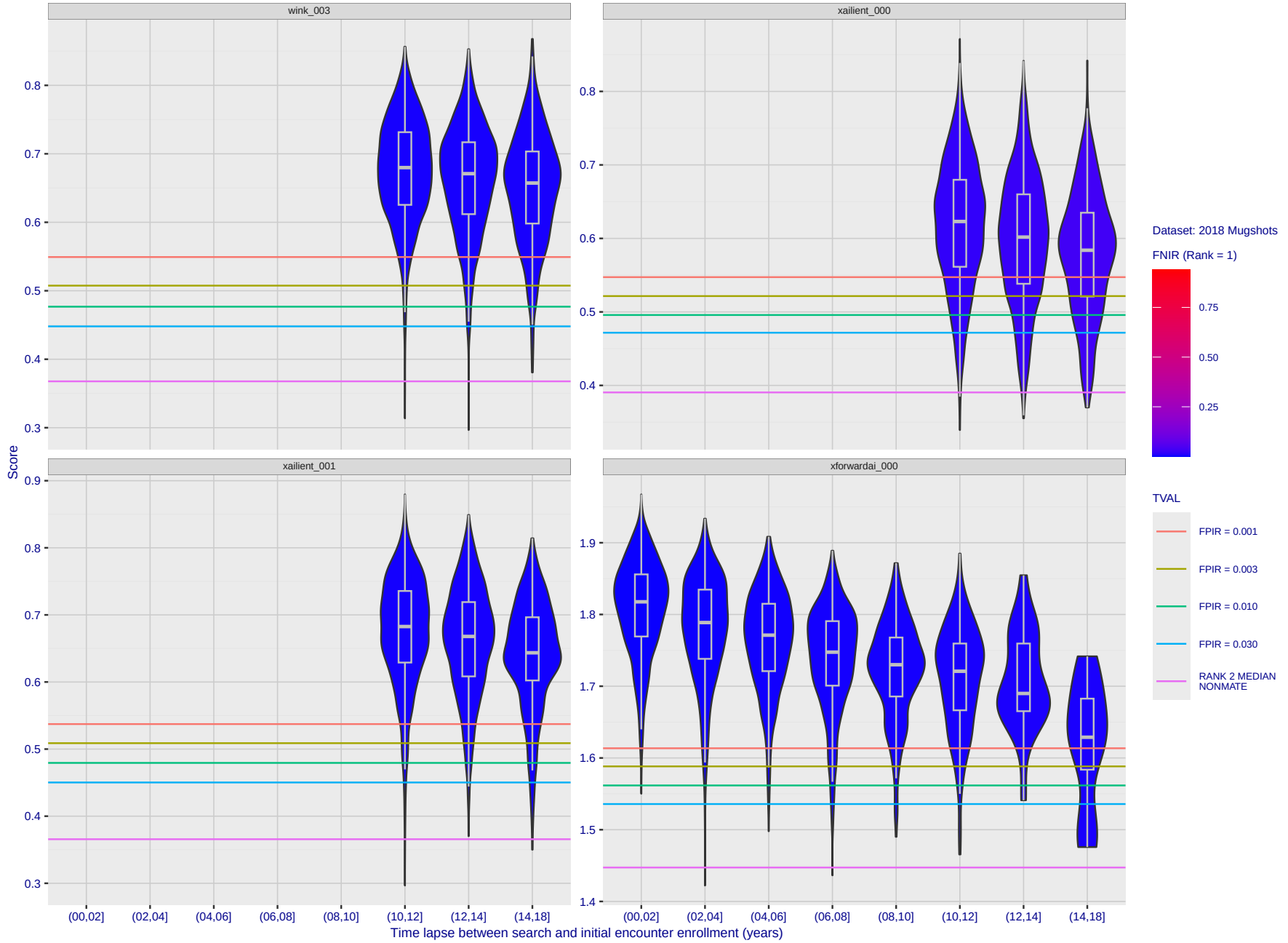

Figure 460: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
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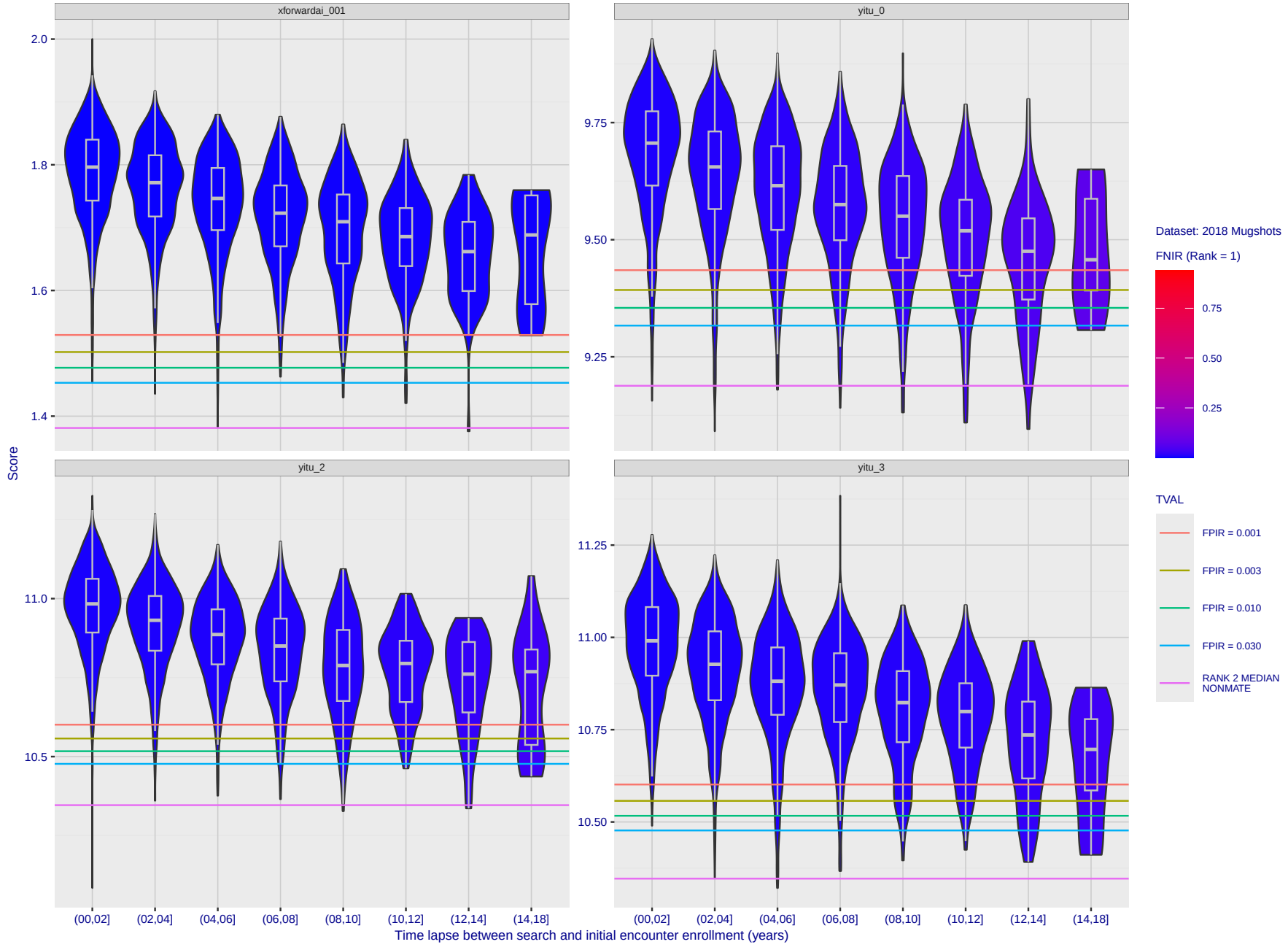

Figure 461: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
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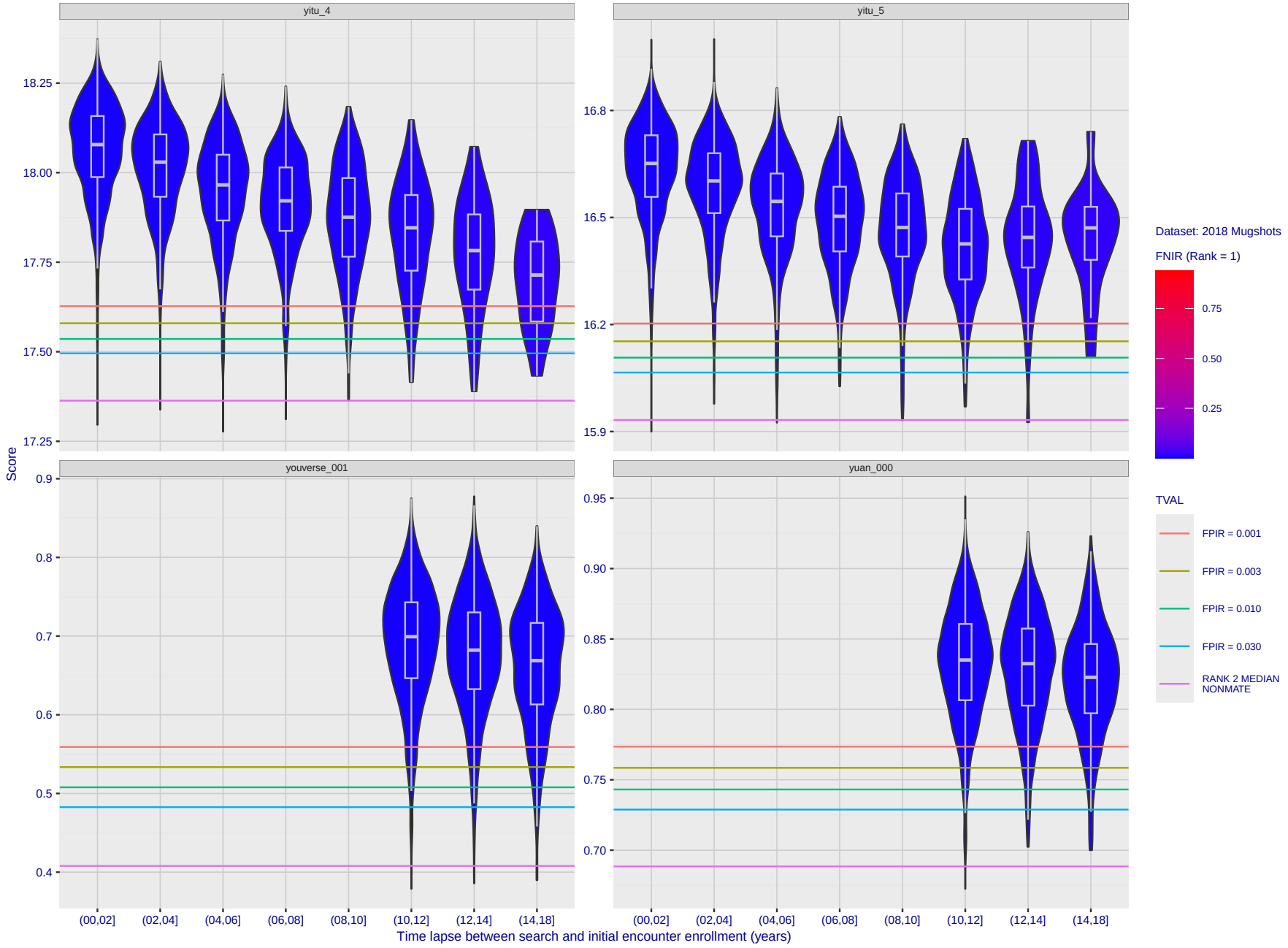

Figure 462: **[FRVT-2018 Mugshot Ageing Dataset] Native mate scores vs. time-elapsed**. *The oldest image of each individual is enrolled. Thereafter, all more recent images are searched. Mated score distributions are computed over all searches noted in row 17 of Table 1 binned by number of years between search and initial enrollment.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

Appendix C Effect of enrolling multiple images

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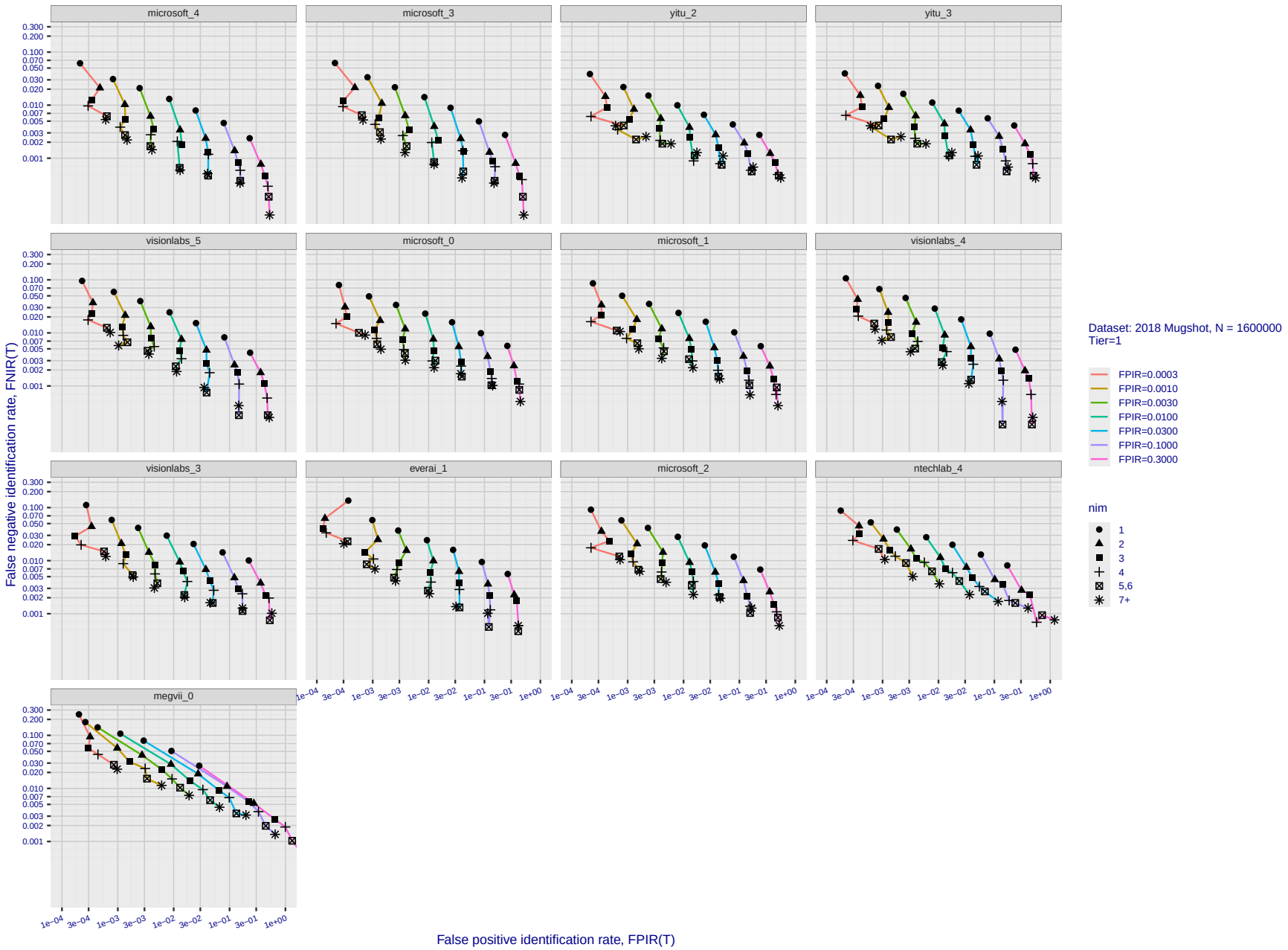

Figure 463: **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
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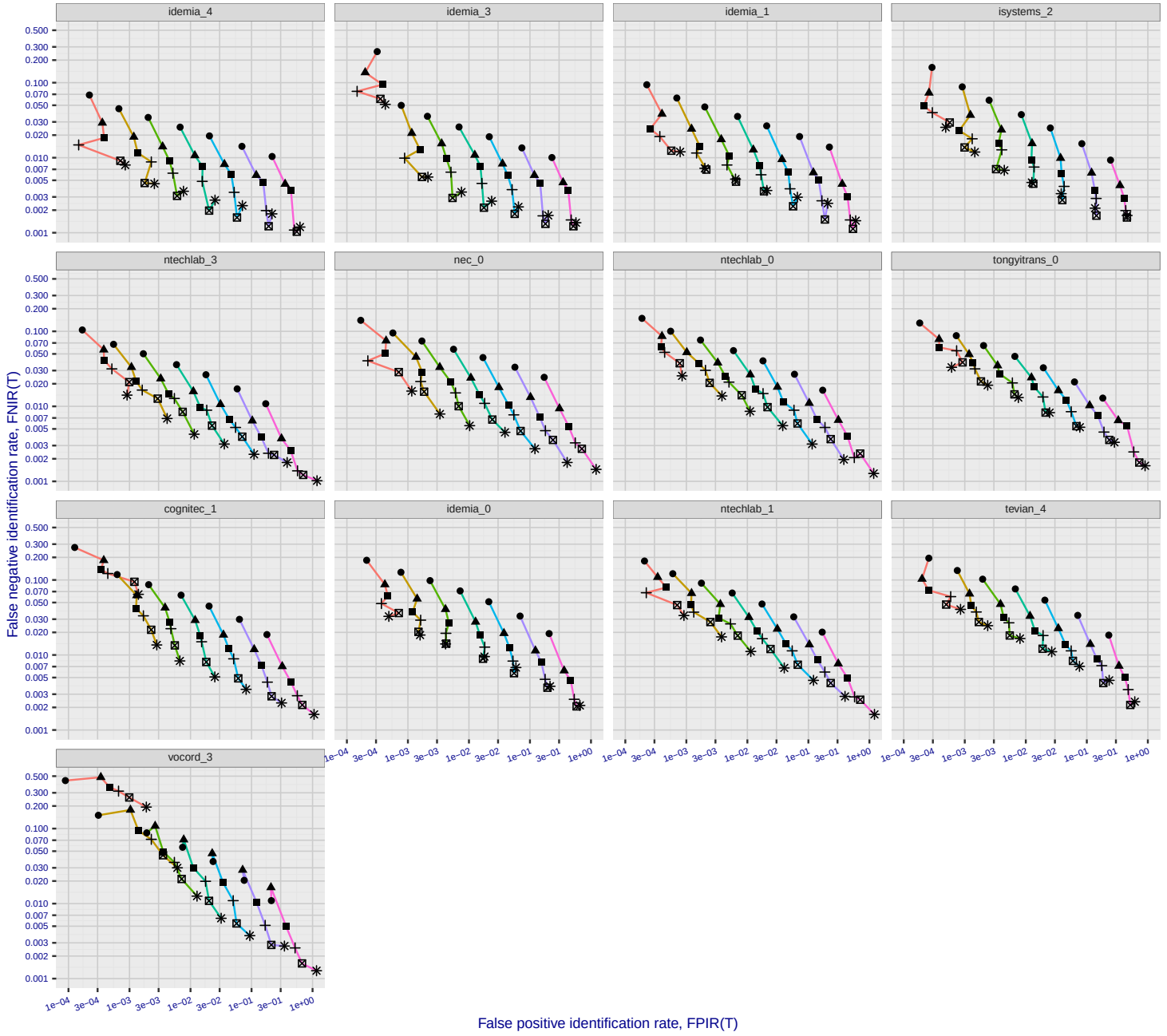

Figure 464: **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

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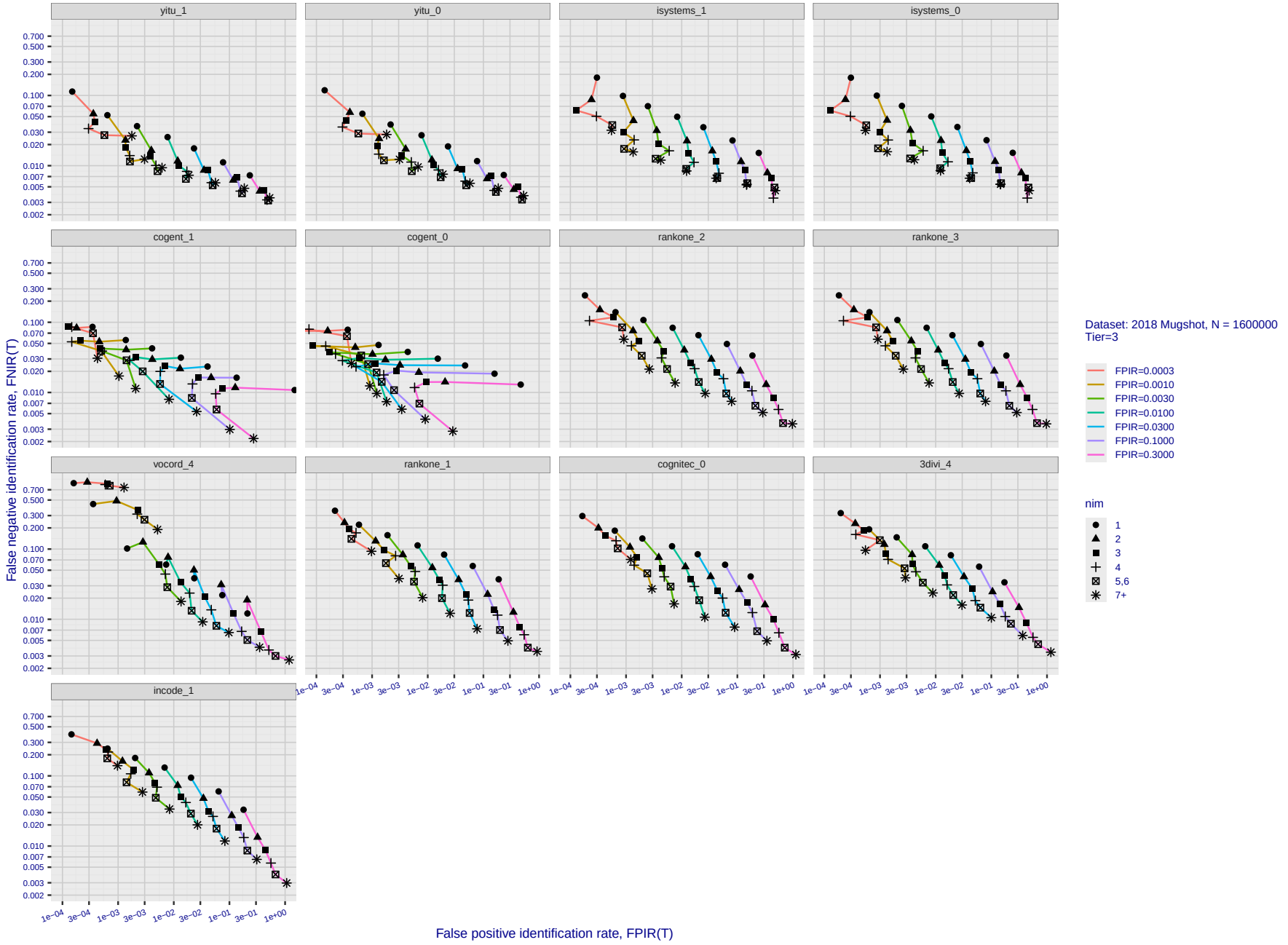

Figure 465: **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

2026/10/06 08:21:15
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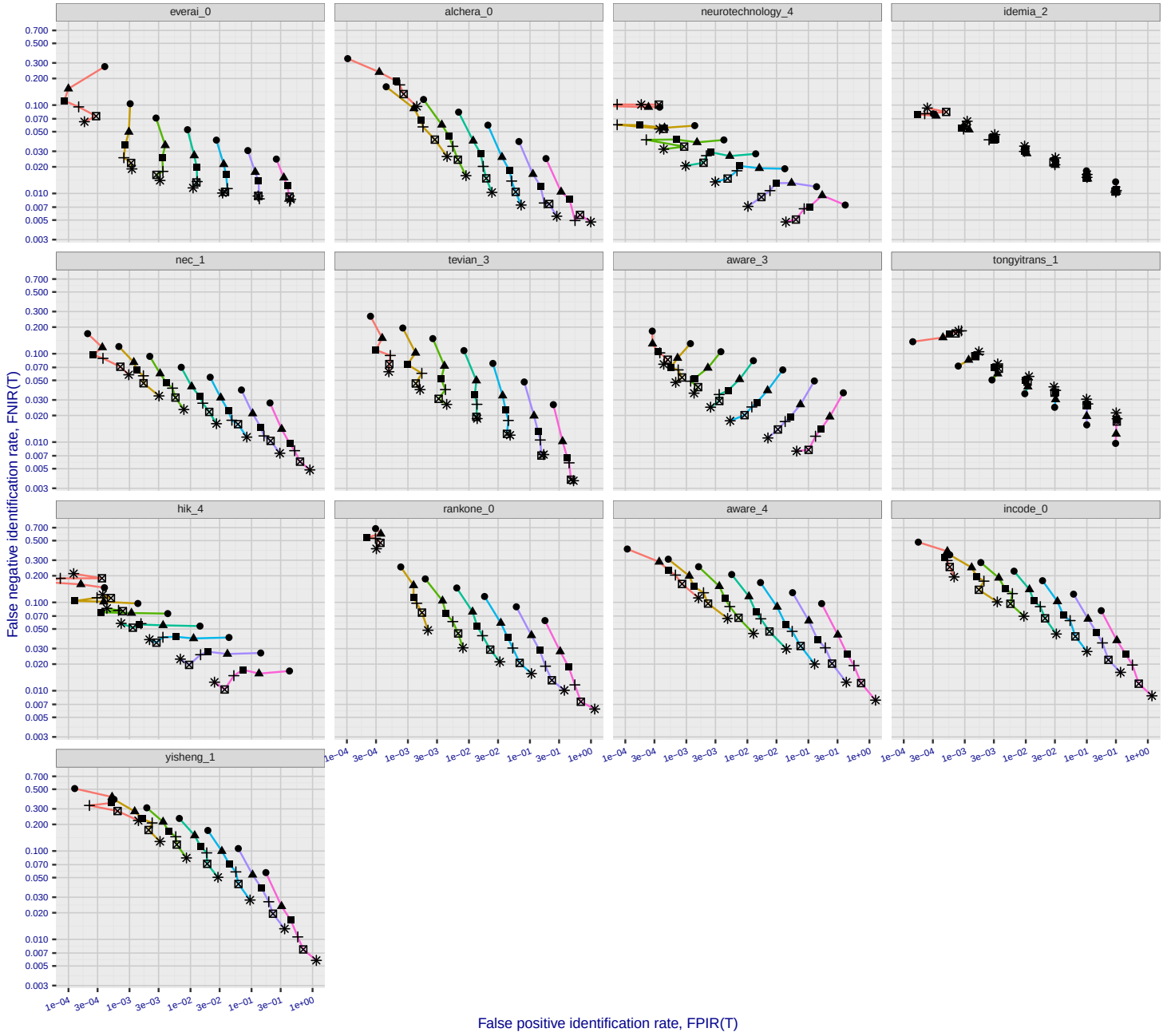

Figure 466: **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

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FNIR(N, R, T) = False neg. identification rate
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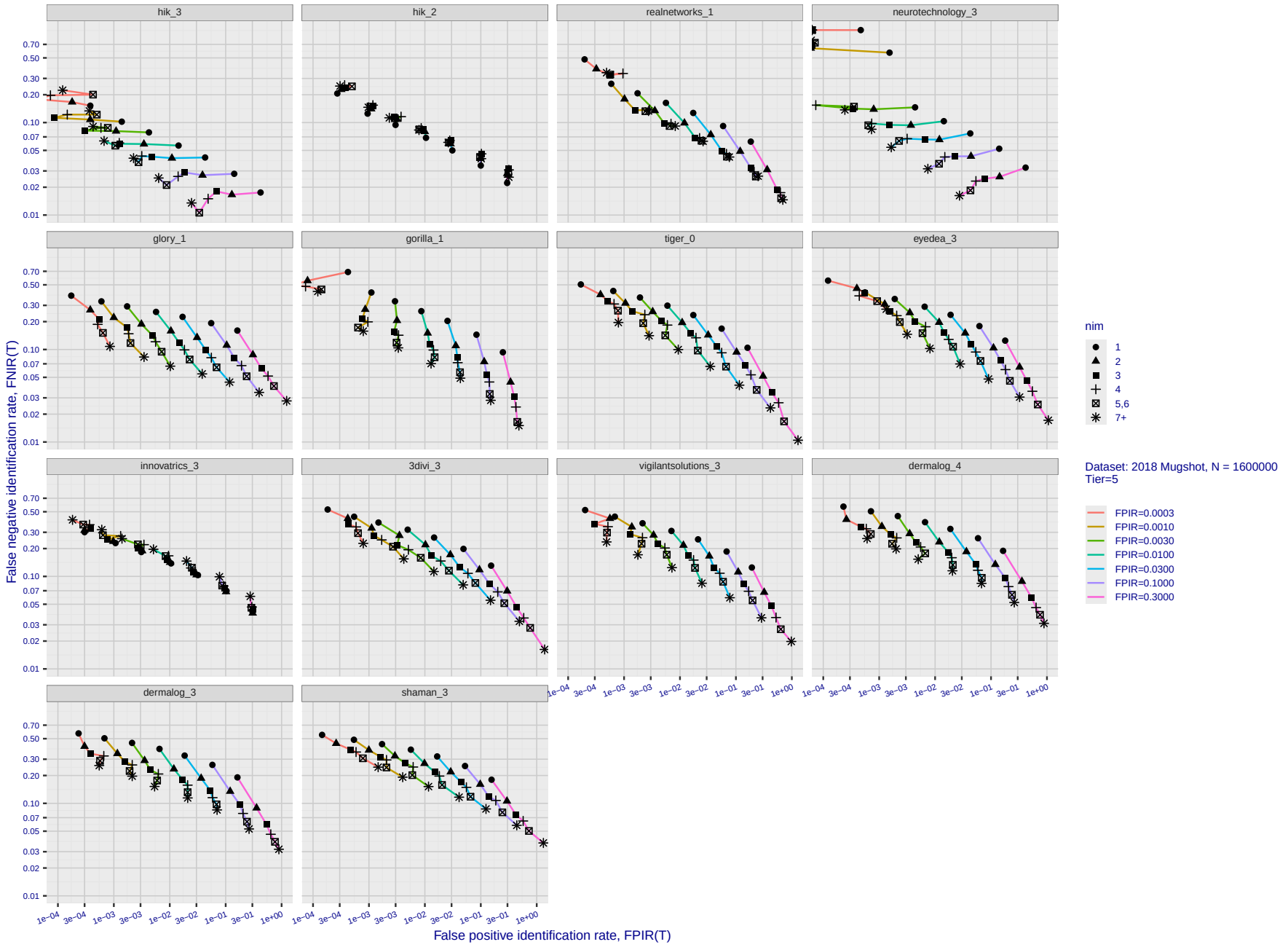

Figure 467: **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

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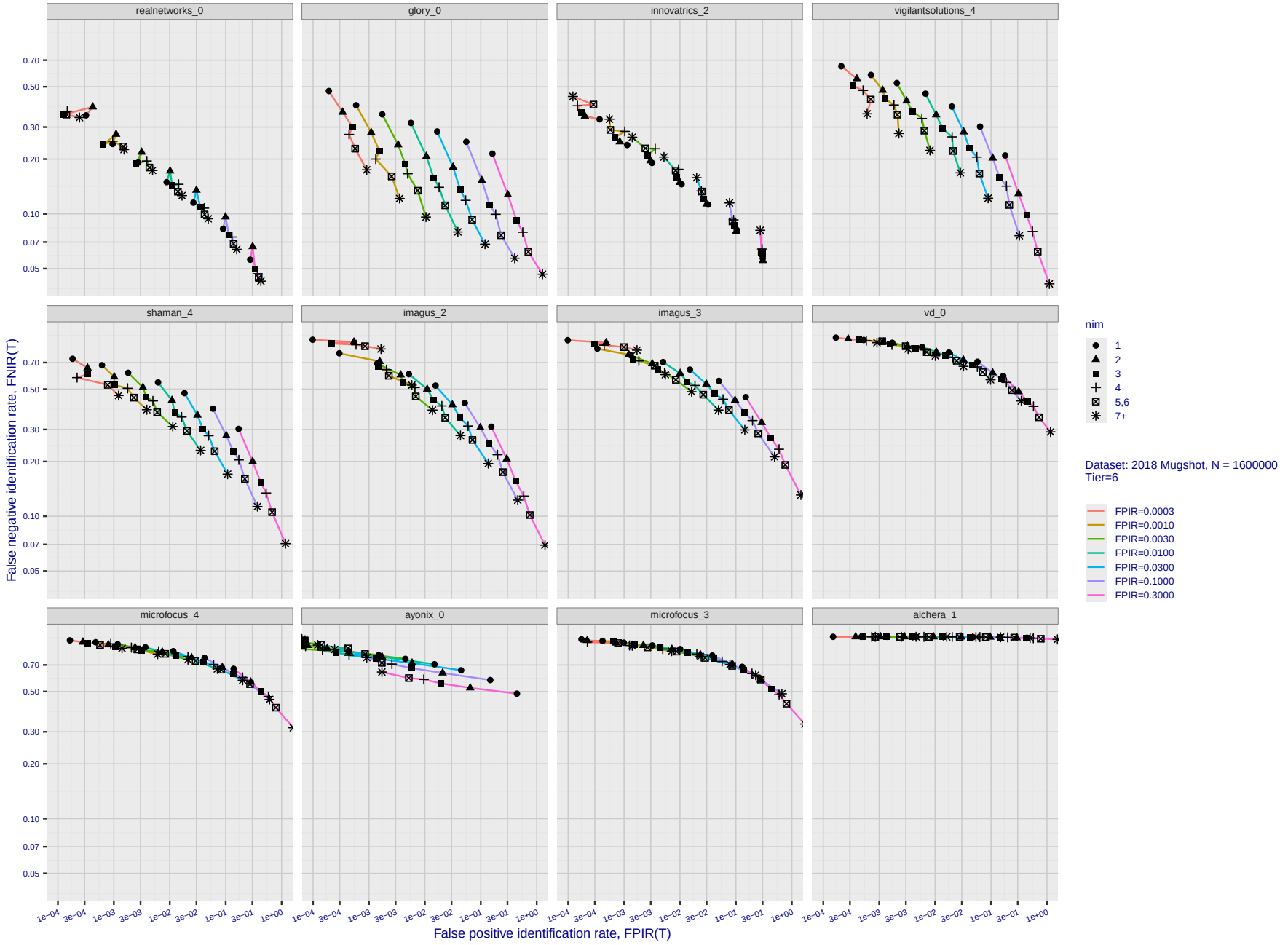

Figure 468: **[FRVT-2018 Mugshot Dataset] Effect of enrolling multiple images for each identity**. *The plot shows an identification miss rates vs. false positive rates, at seven operating thresholds. The enrolled population size is fixed. The images are enrolled with lifetime-consolidation - see section 2.3.*

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

Appendix D Accuracy with poor quality webcam images

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2026/10/06 08:21:15	FNIR(N, R, T) = FPIR(N, T) =	False neg. identification rate False pos. identification rate	N = Num. enrolled subjects R = Num. candidates examined	T = Threshold	$T = 0 \rightarrow$ Investigation $T > 0 \rightarrow$ Identification

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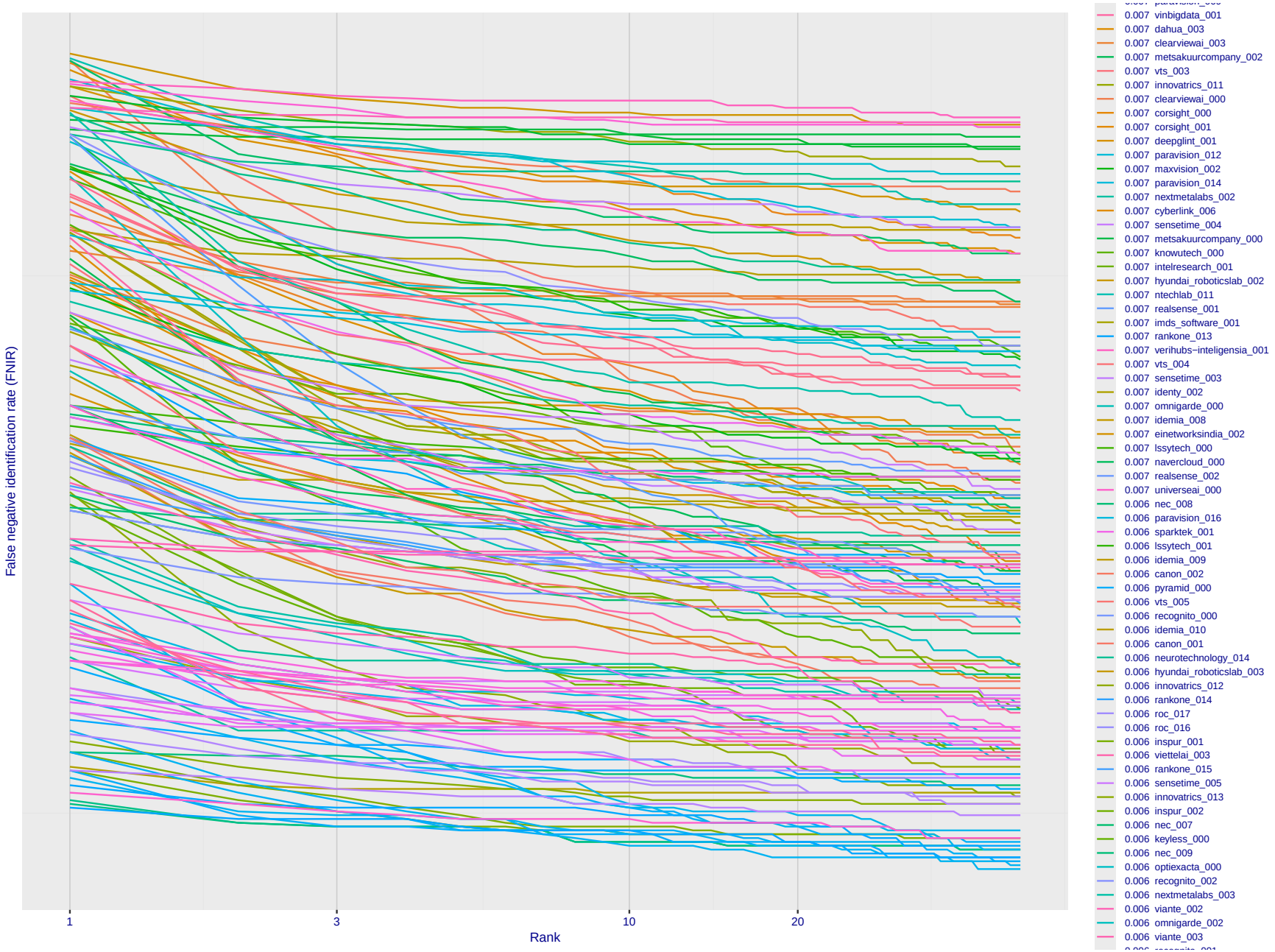

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

Figure 469: **[Webcam Dataset] Identification miss rates vs. rank**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

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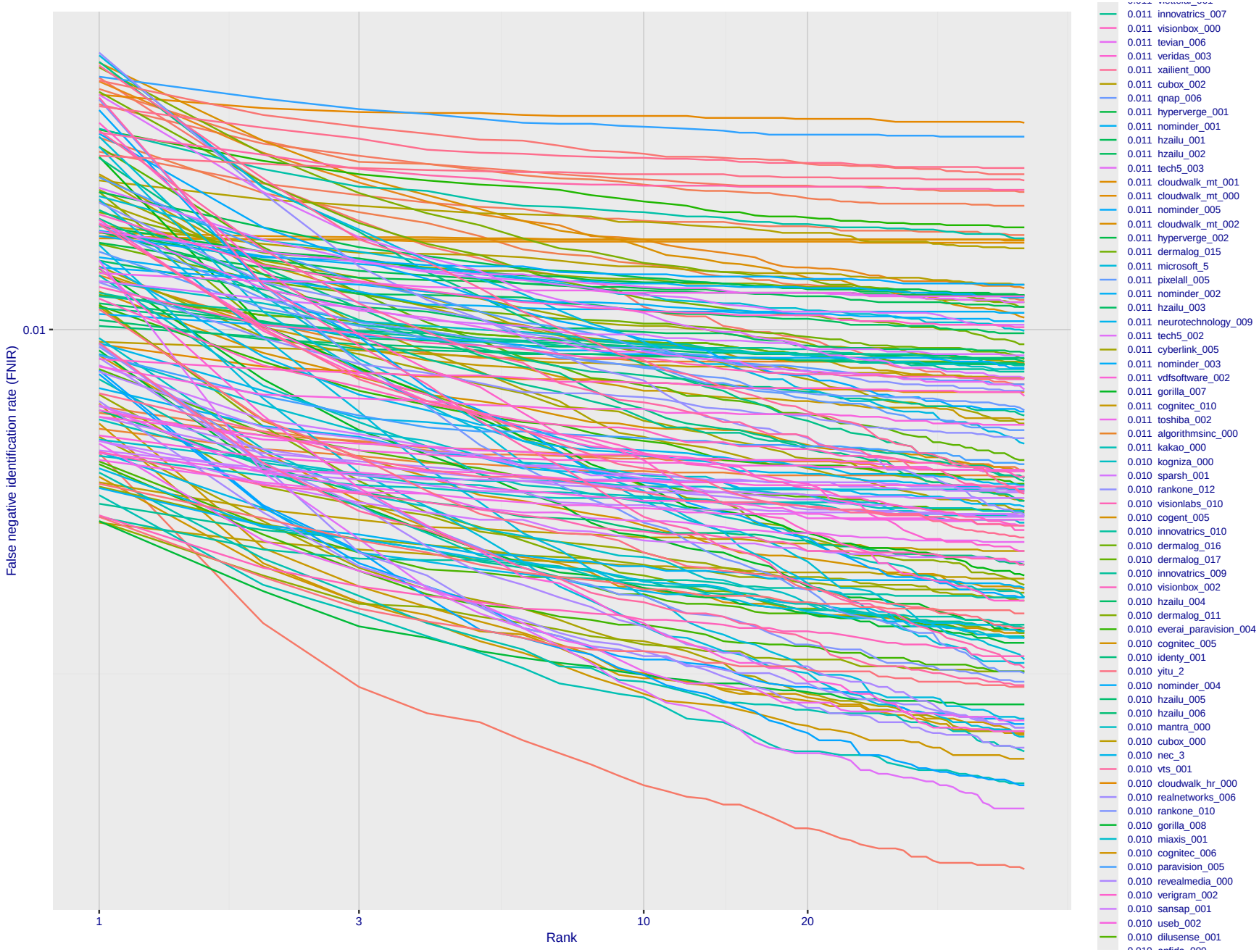

Figure 470: **[Webcam Dataset] Identification miss rates vs. rank**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

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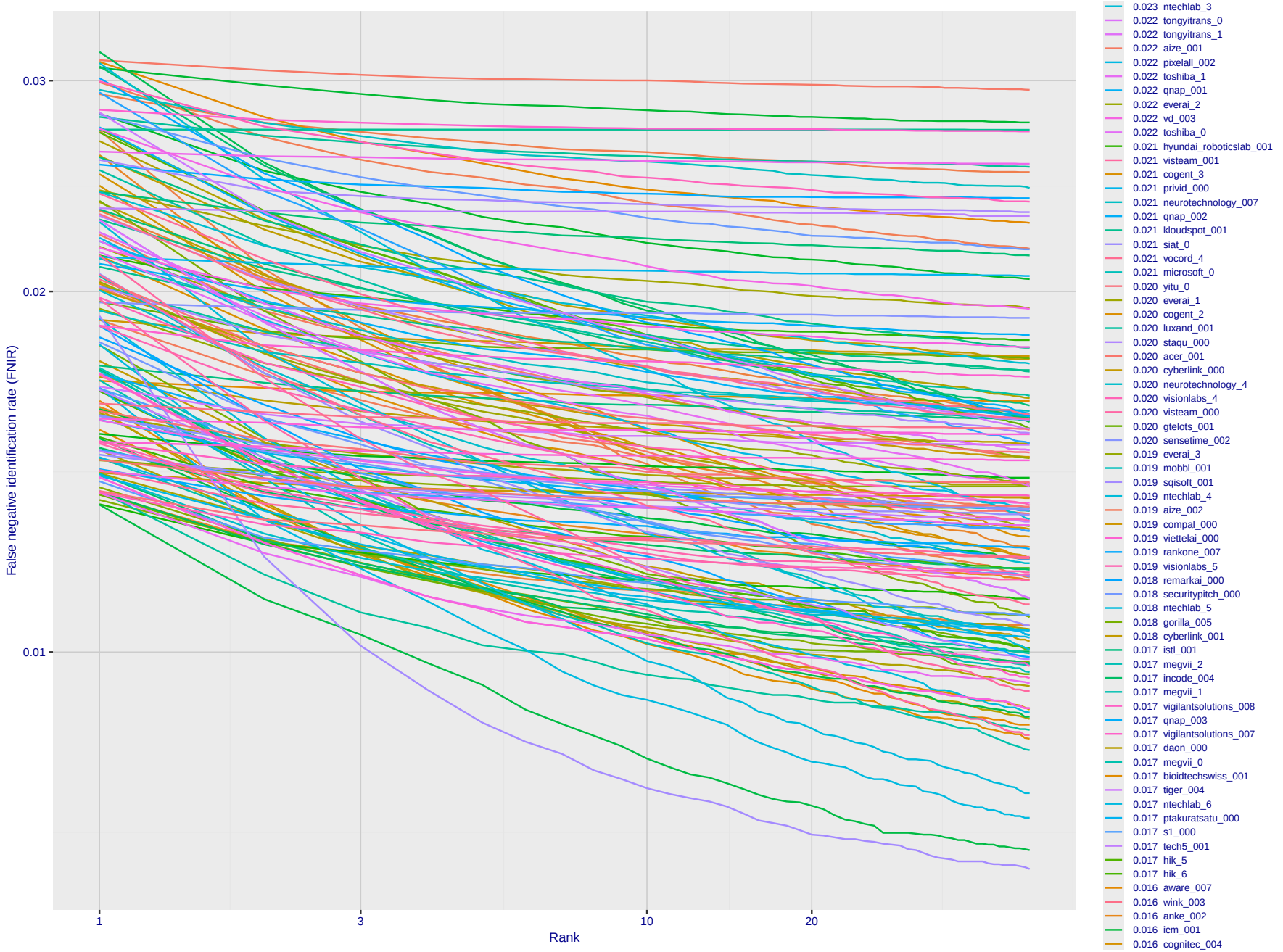

Figure 471: **[Webcam Dataset] Identification miss rates vs. rank**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
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R = Num. candidates examined
T = Threshold
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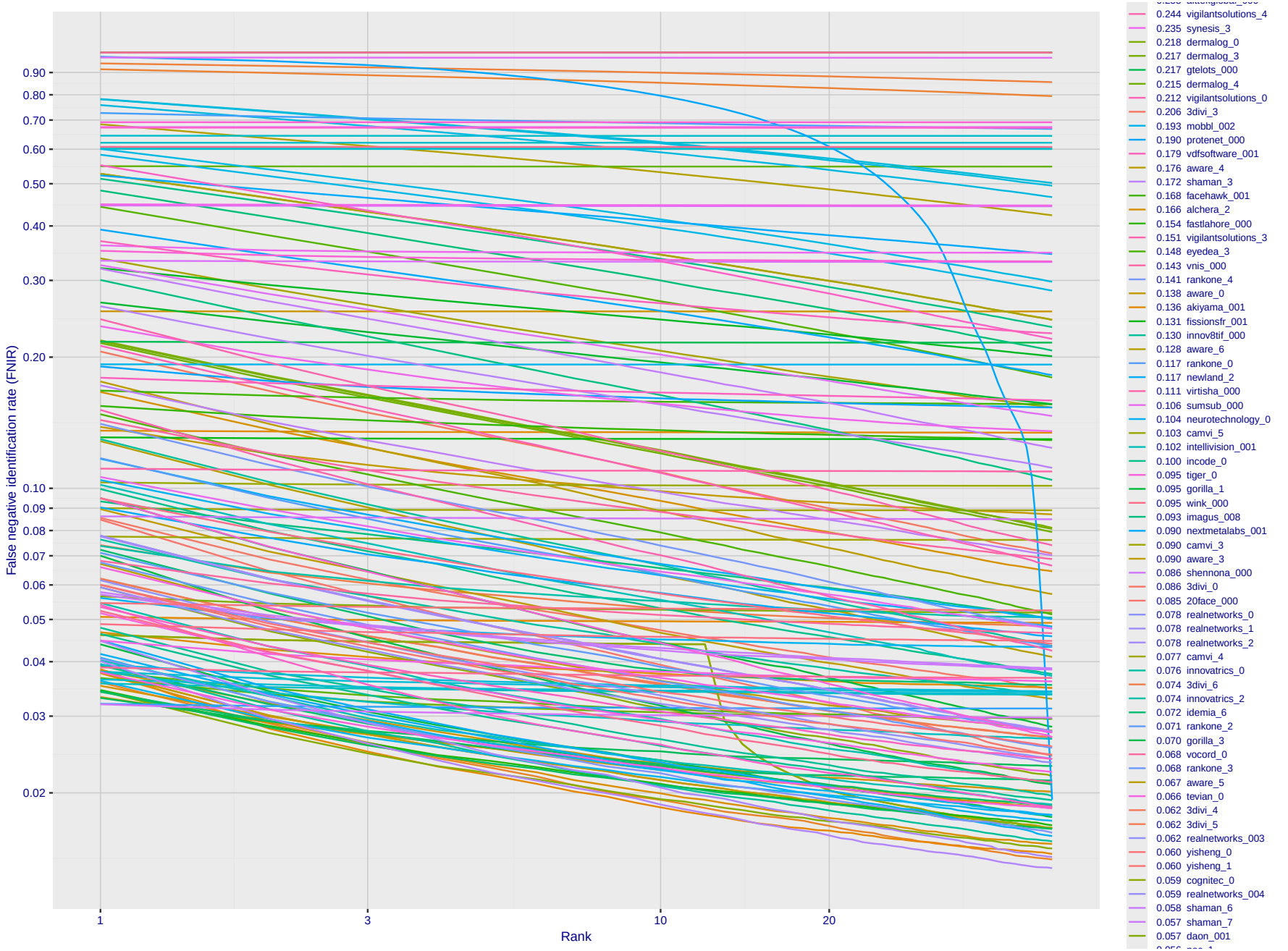

Figure 472: **[Webcam Dataset] Identification miss rates vs. rank**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

2026/10/06 08:21:15	FNIR(N, R, T) = FPIR(N, T) =	False neg. identification rate False pos. identification rate	N = Num. enrolled subjects R = Num. candidates examined	T = Threshold	$T = 0 \rightarrow$ Investigation $T > 0 \rightarrow$ Identification

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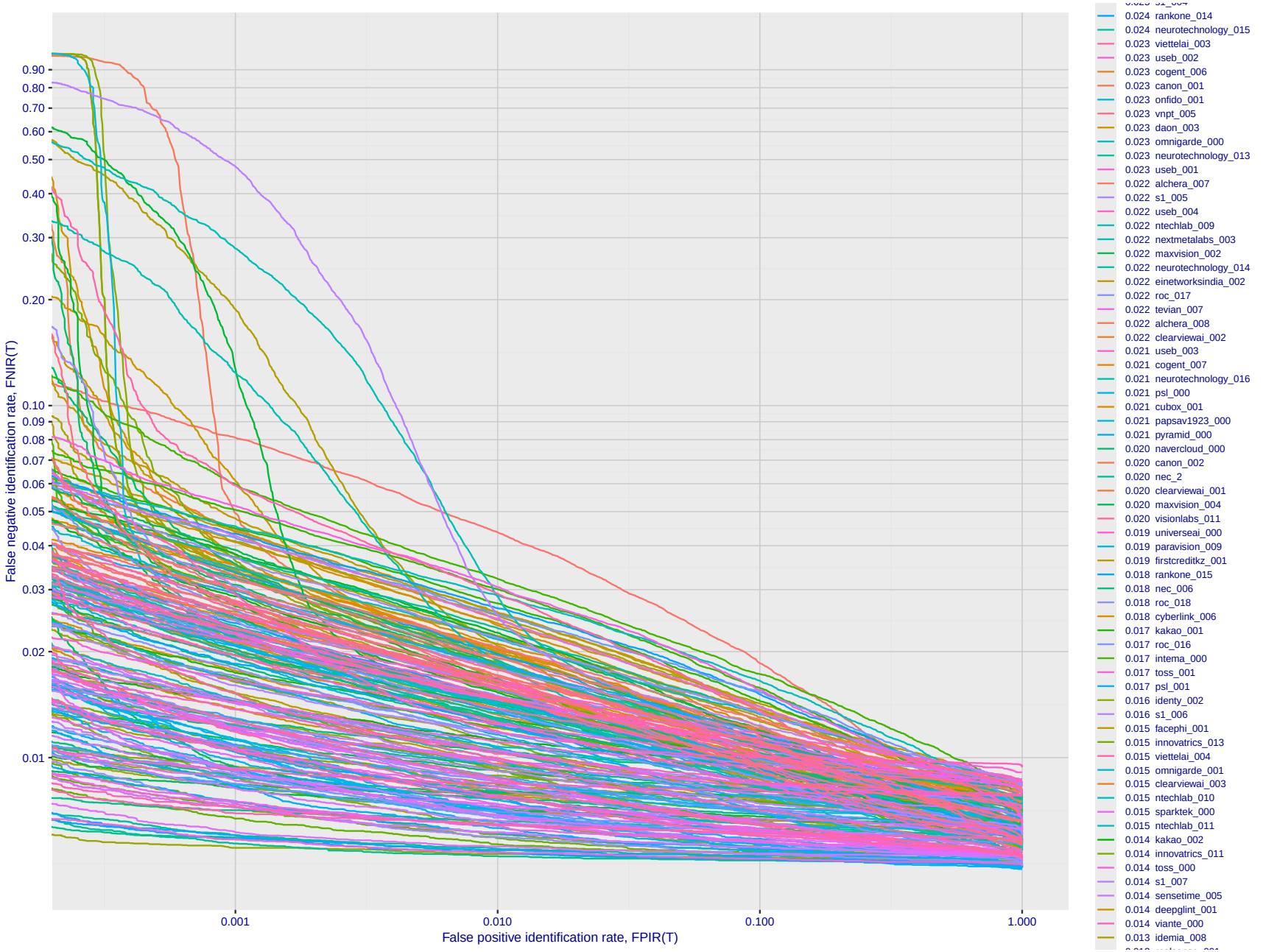

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate

N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

T = 0 → Investigation
T > 0 → Identification

Figure 473: **[Webcam Dataset] Identification miss rates vs. false positive rates**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

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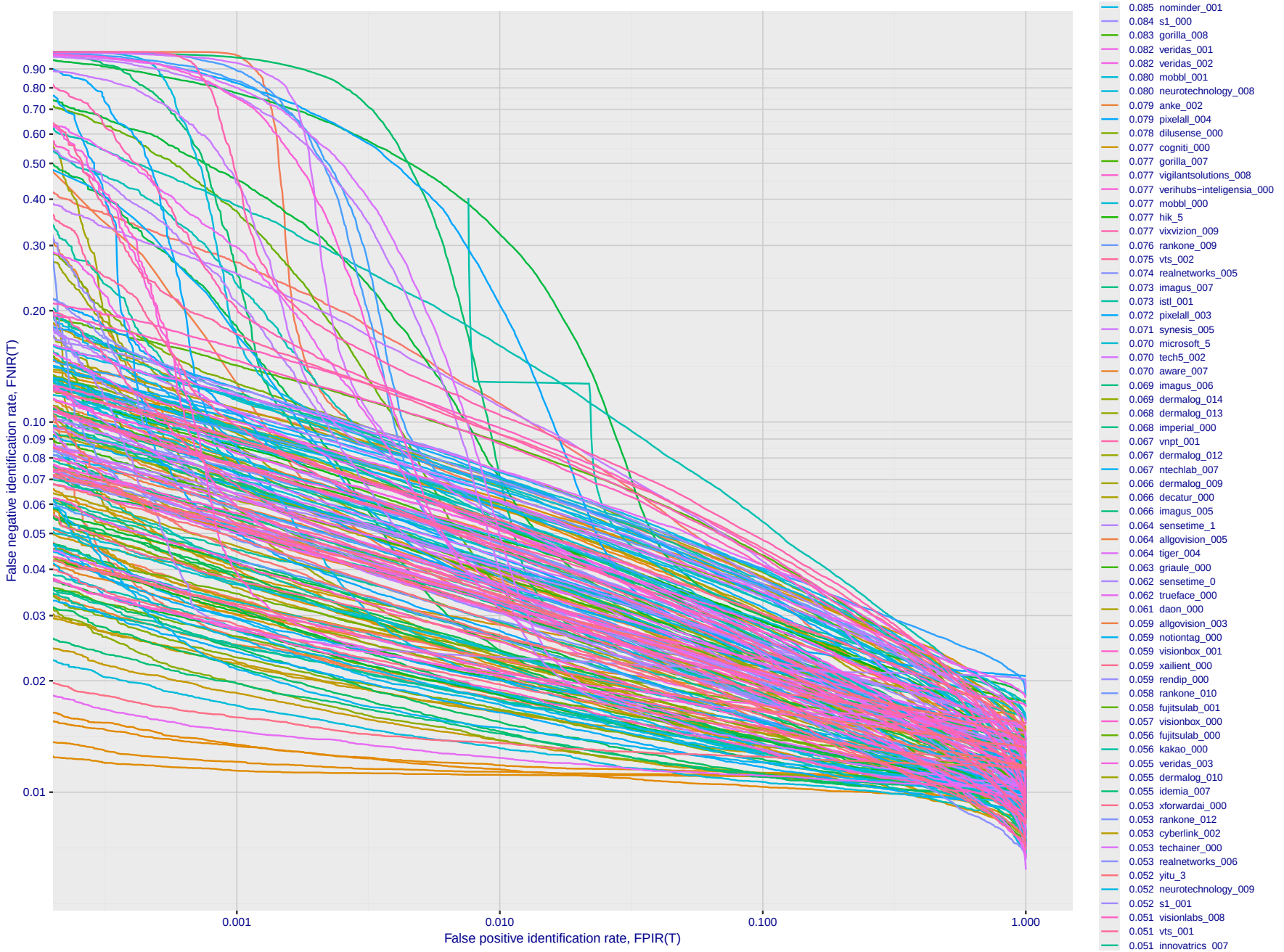

Figure 474: **[Webcam Dataset] Identification miss rates vs. false positive rates**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

2026/10/06 08:21:15

FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined

T = Threshold

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T > 0 → Identification

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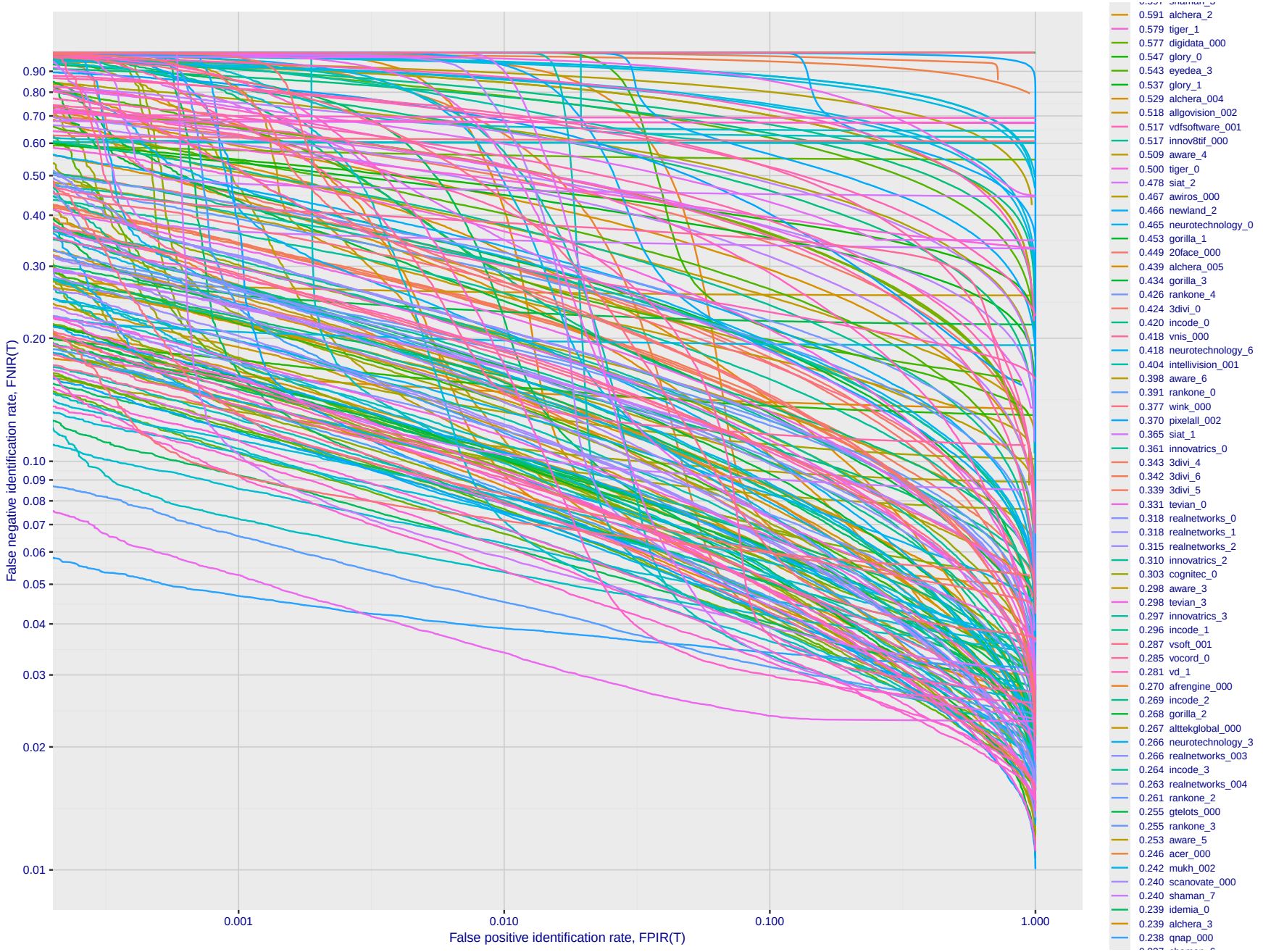

Figure 475: **[Webcam Dataset] Identification miss rates vs. false positive rates**. *The results apply to cross-domain recognition in which webcams are searched against enrolled mugshots. The FNIR values are higher than those for mugshot-mugshot identification due to low image resolution, lighting and less constrained subject pose in webcam images - see Figure 6.*

2026/10/06 08:21:15
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FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

Appendix E Accuracy for profile-view to frontal recognition

Figures 476 - 478 gives accuracy results for searching 100 000 mated and 100 000 non-mated profile-view images against the same FRVT 2018 frontal enrollment dataset, N = 1 600 000, used in the main mugshot trials. This experiment corresponds to row-13 of Table 1. An example of profile-view image is given in Figure 7.

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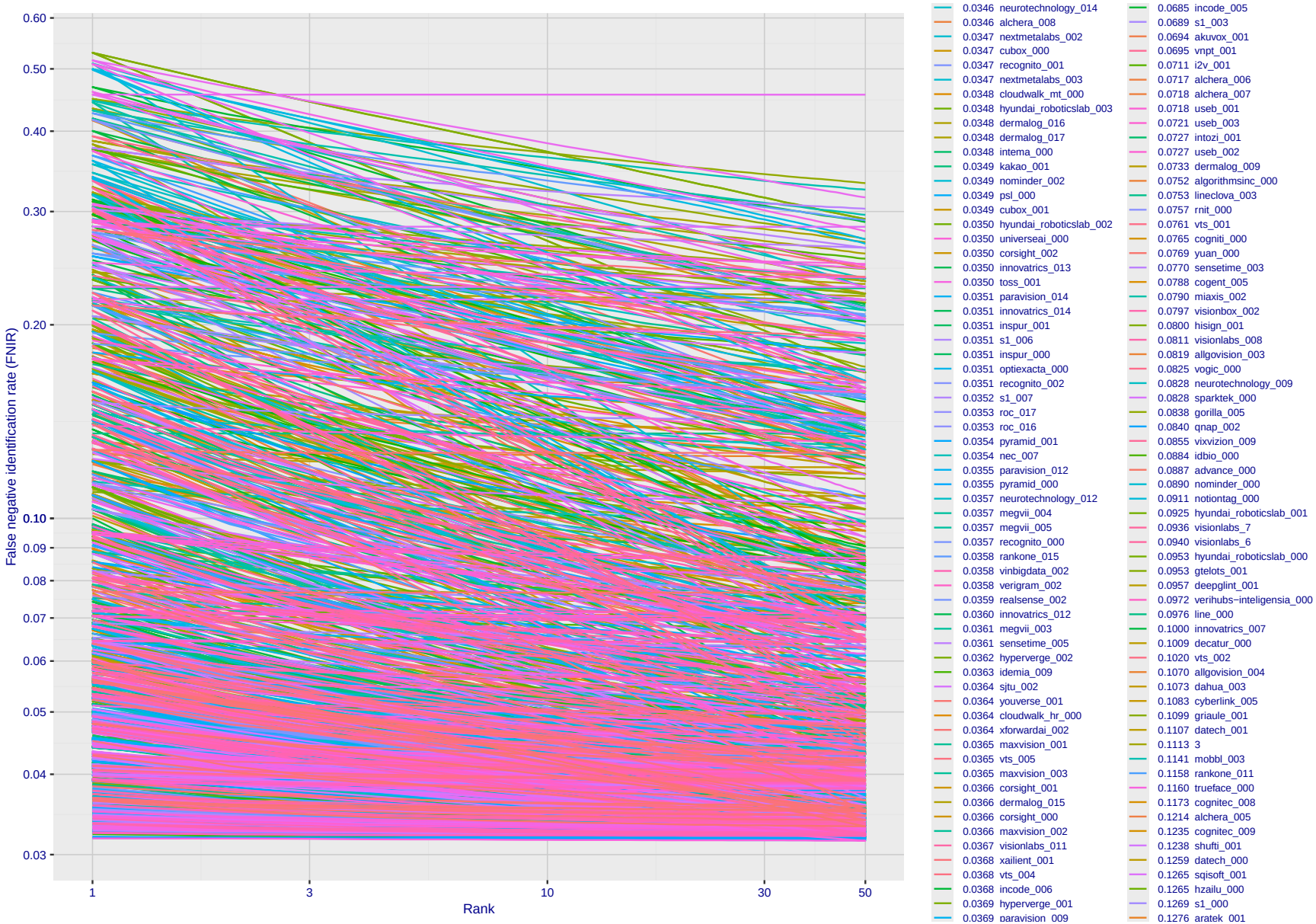

Figure 476: **[Mugshot and profile-view dataset] Rank-based accuracy**. *For some of the more accurate Phase 3 algorithms the figure plots error tradeoff characteristics for frontal and profile-view searches into an enrolled set of N = 1 600 000 frontal images. Note that some algorithms fail on profile-view images with FNIR → 1 - this evaluation did not ask developers to provide profile-view capability. Some algorithms, on the other hand, give FNIR approaching that for frontal-view searches using c. 2010 algorithms. The best result is that 91% of profile-view searches yield the correct mate at rank 1, and better than 94% in the top-50 candidates.*

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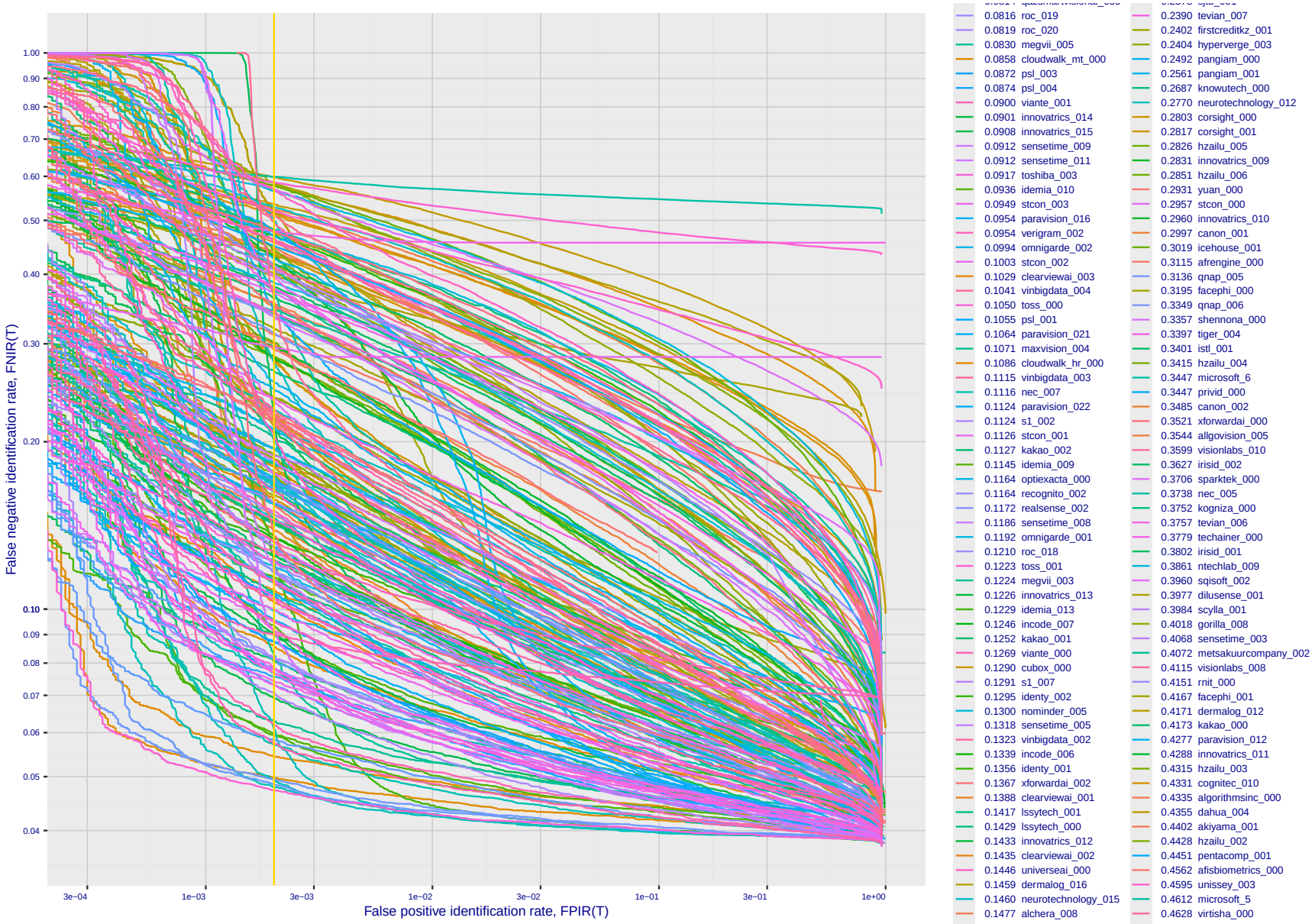

Figure 477: **[Mugshot and profile-view dataset] Threshold-based accuracy**. *For some of the more accurate Phase 3 algorithms the figure plots error tradeoff characteristics for frontal and profile-view searches into an enrolled set of N = 1 600 000 frontal images. Note that some algorithms fail on profile-view images with FNIR → 1 - this evaluation did not ask developers to provide profile-view capability. Some algorithms, on the other hand, give FNIR approaching that for frontal-view searches using c. 2010 algorithms.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
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N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

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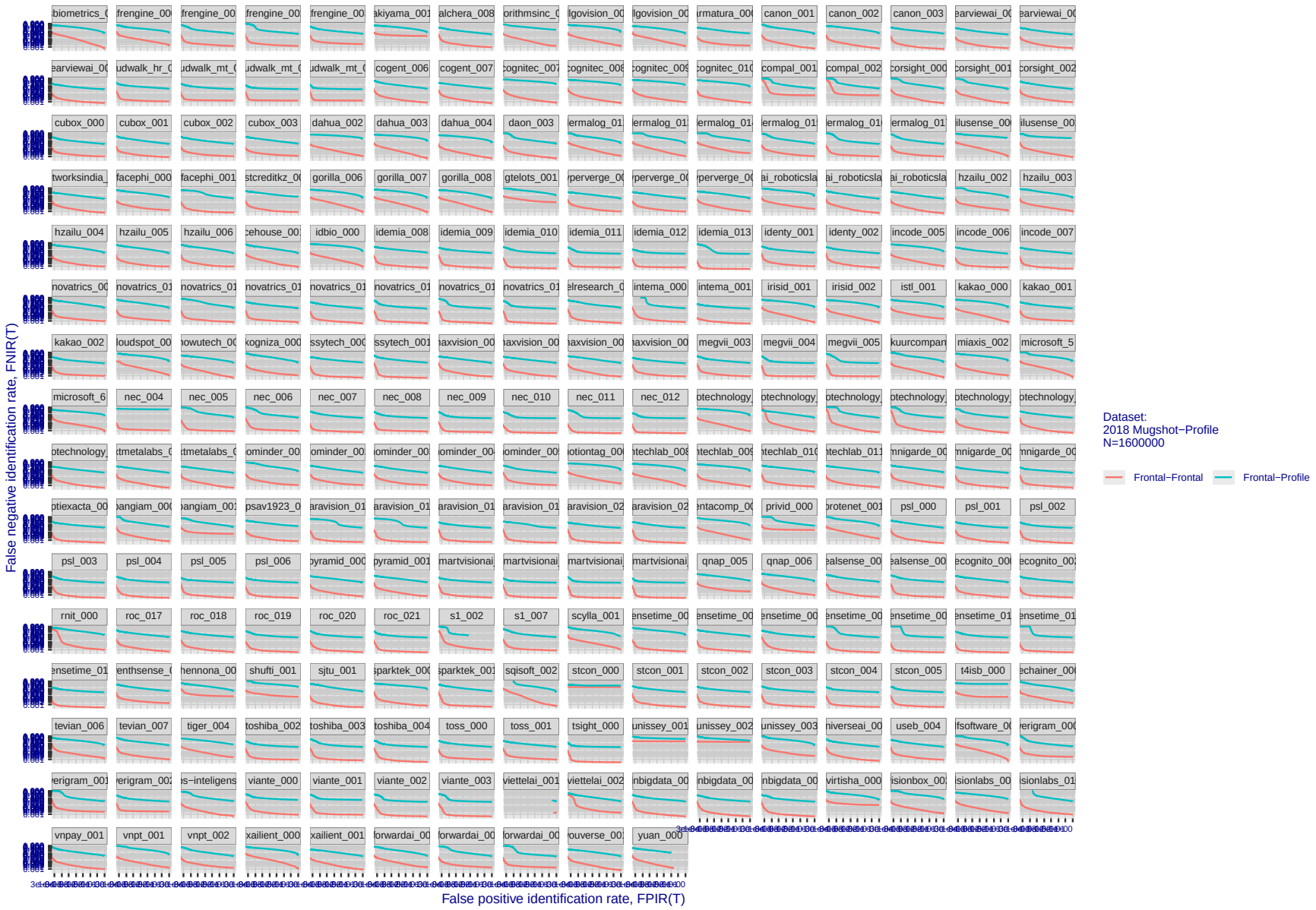

Figure 478: **[Mugshot and profile-view dataset] Speed-accuracy tradeoff**. *For some of the more accurate Phase 3 algorithms the figure plots error tradeoff characteristics for frontal and profile-view searches into an enrolled set of N = 1 600 000 frontal images. Some algorithms fail on profile-view images with FNIR → 1 - this evaluation did not ask developers to provide profile-view capability. Some algorithms, on the other hand, give FNIR approaching that for frontal-view searches using c. 2010 algorithms. Blue lines connect points of equal threshold from which it is evident that some algorithms would give markedly higher false positive outcomes if profile-view images were searched in a system configured for frontal searches. This would be a vulnerability in an access control system.*

2026/10/06 08:21:15
FNIR(N, R, T) = False neg. identification rate
FPIR(N, T) = False pos. identification rate
N = Num. enrolled subjects
R = Num. candidates examined
T = Threshold
T = 0 → Investigation
T > 0 → Identification

Appendix F Search duration

As in and prior tests, this section documents search speeds spanning three orders of magnitude. In applications where search volumes are high enough, this will have implications for hardware requirements especially for large N or when search duration is appreciably larger than the time it takes to prepare a template from the search image(s). Further, given very large (and growing) operational databases, the scalability of algorithms is important. It has been reported previously [8] that search duration can scale sublinearly with enrolled population size N. Further there has been considerable recent research on indexing, exact [13] and approximate nearest neighbor search [1,13] and fast-search [14,16].

Figure 479 charts the search duration measurements presented earlier in Tables 2 - 4.

- Most algorithms scale linearly. For those in that category, there is a wide range in speed with search durations ranging from 82 milliseconds for a 12 million gallery (for NEC-3) to more than 40 seconds (for Yitu-3, Toshiba-2) and even higher for less accurate algorithms.
- Some developers (Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs) provide algorithms whose template search durations grow approimately logarithmically i.e. $T(N)\tilde{a}\log N$ with the constant a varying between implementations. In the figure this model is fit using the point $T(1) = 0$, and $T(640\,000)$. This very sublinear behaviour affords extremely fast search times in very large galleries. One caveat for the sublinear algorithms is that their fast-search data structures can require considerable computation time - on the order of hours - for N in the millions, and this scales mildy super-linearly, i.e. $O(N^b), b > 1$. There are exceptions: the Camvi algorithms take minutes; and Innvovatrics' scale sublinearly.

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2026/10/06 08:21:15	FNIR(N, R, T) = FPIR(N, T) =	False neg. identification rate False pos. identification rate	N = Num. enrolled subjects R = Num. candidates examined	T = Threshold	$T = 0 \rightarrow$ Investigation $T > 0 \rightarrow$ Identification

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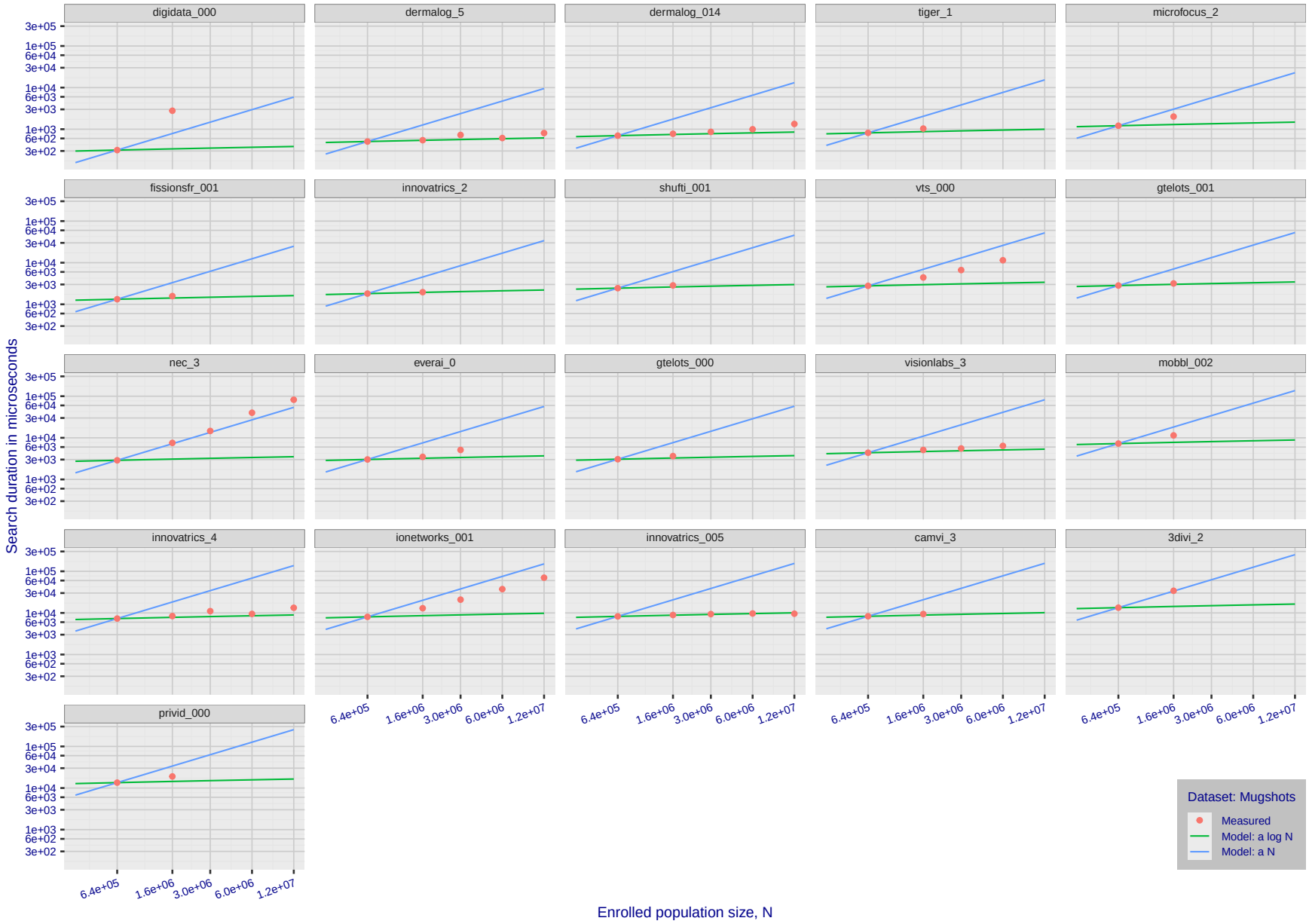

Figure 479: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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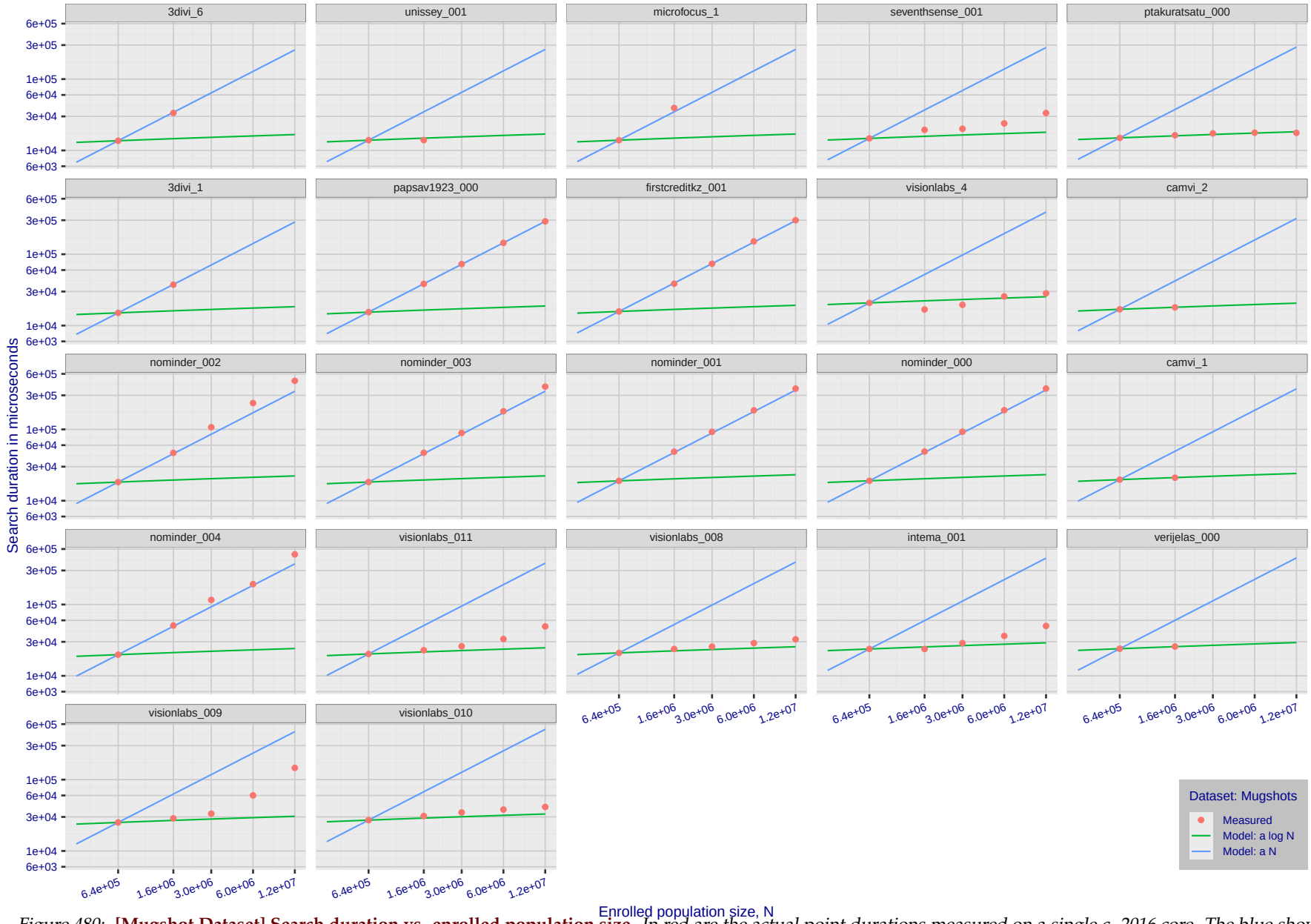

Figure 480: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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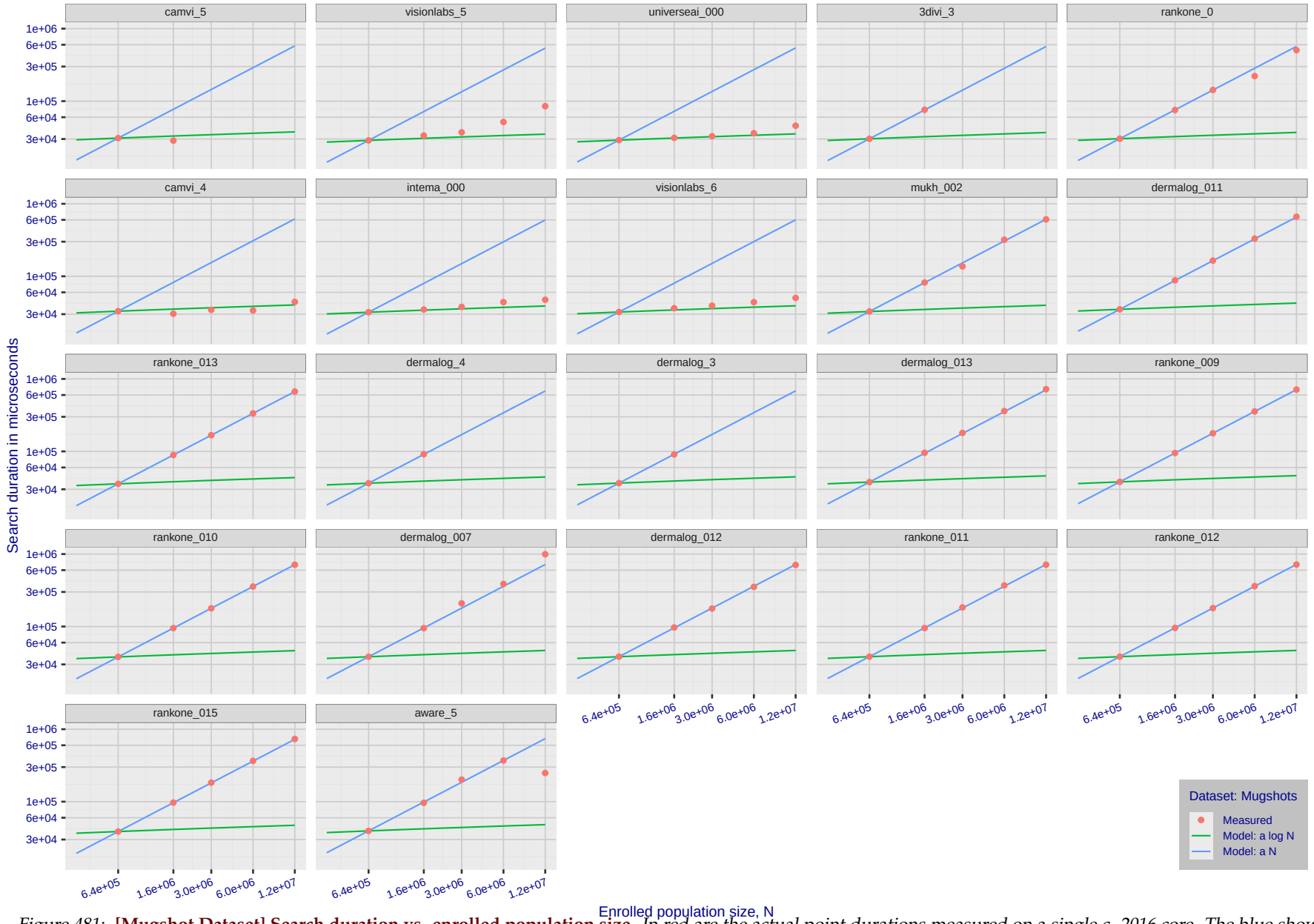

Figure 481: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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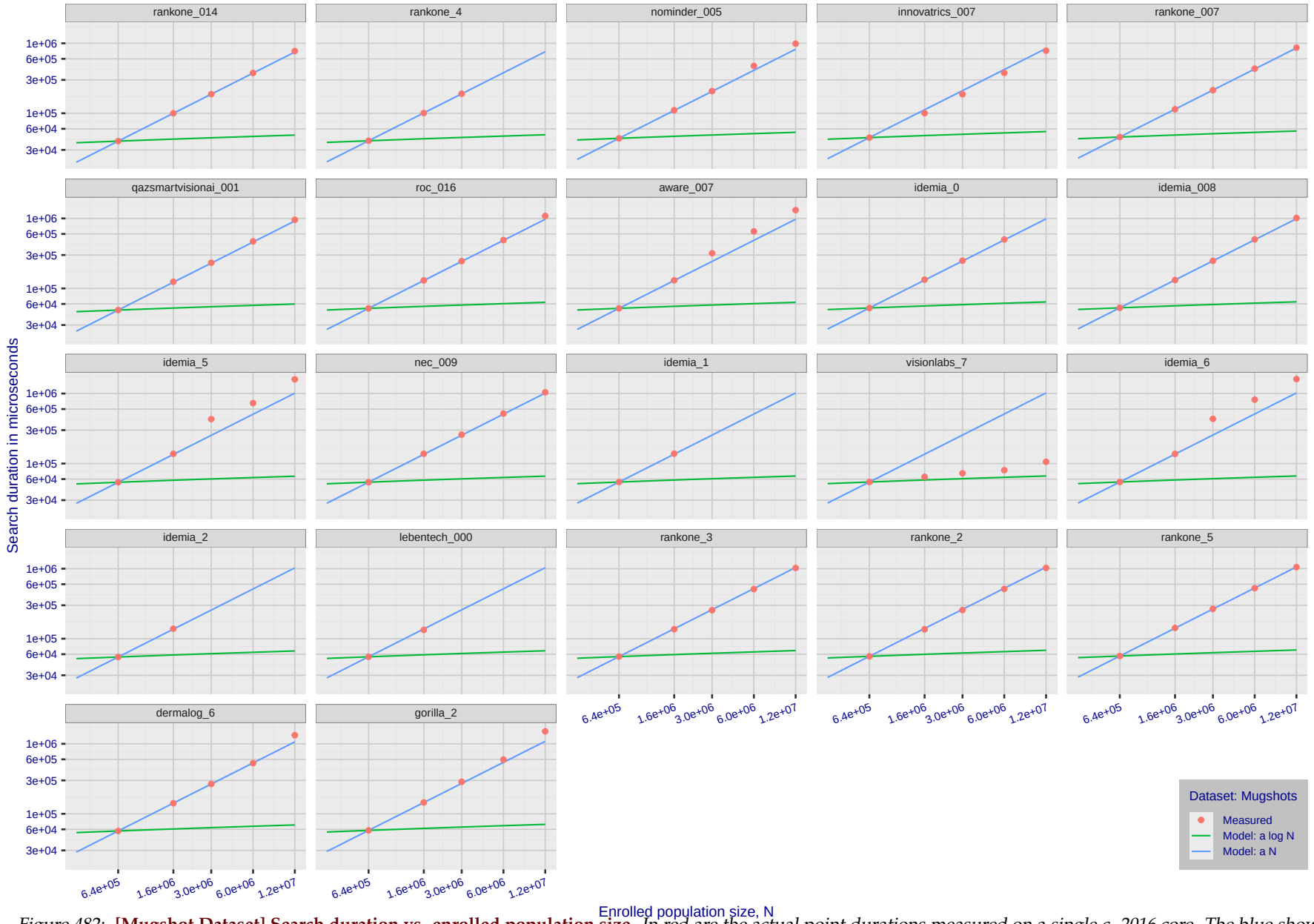

Figure 482: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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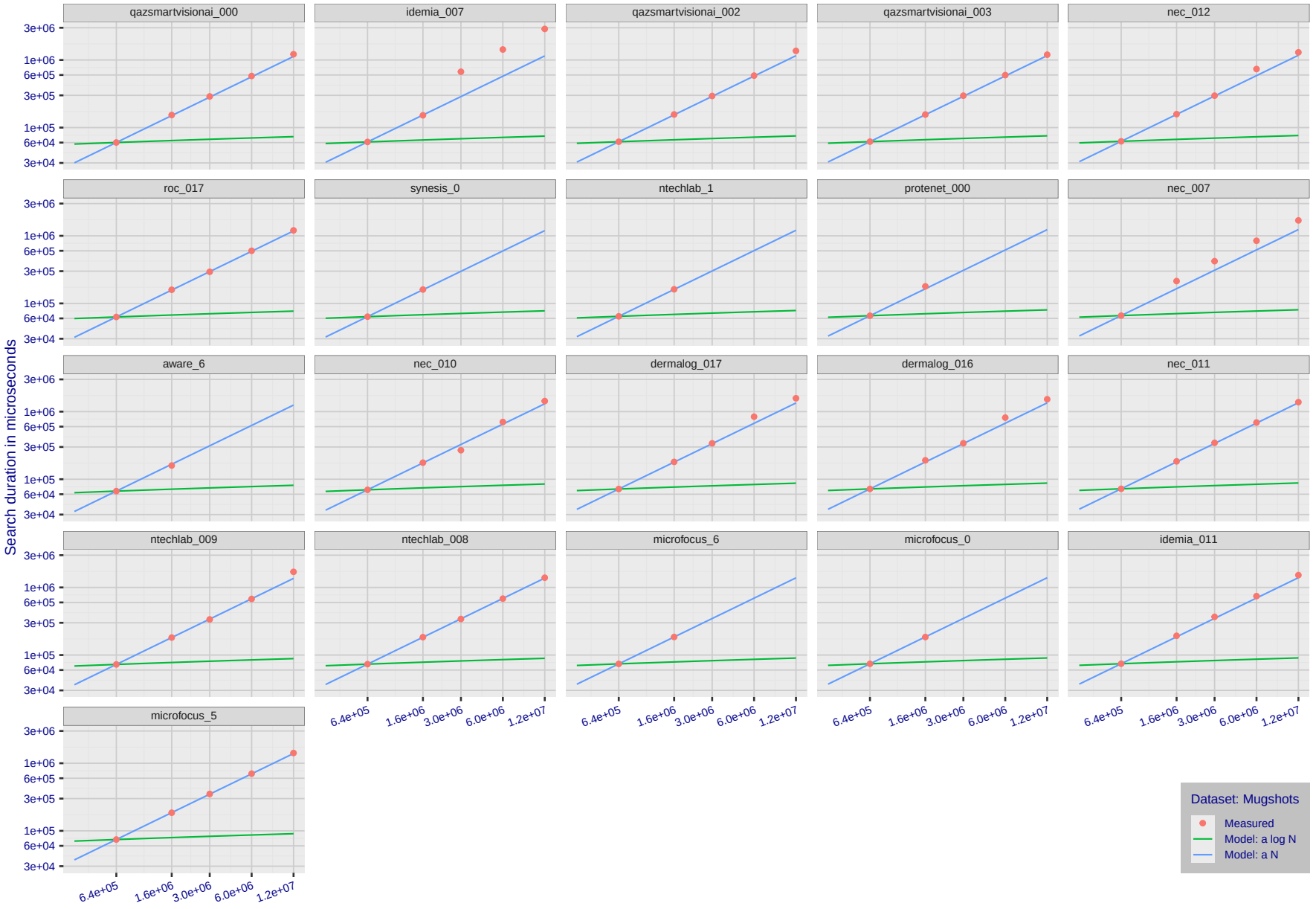

Figure 483: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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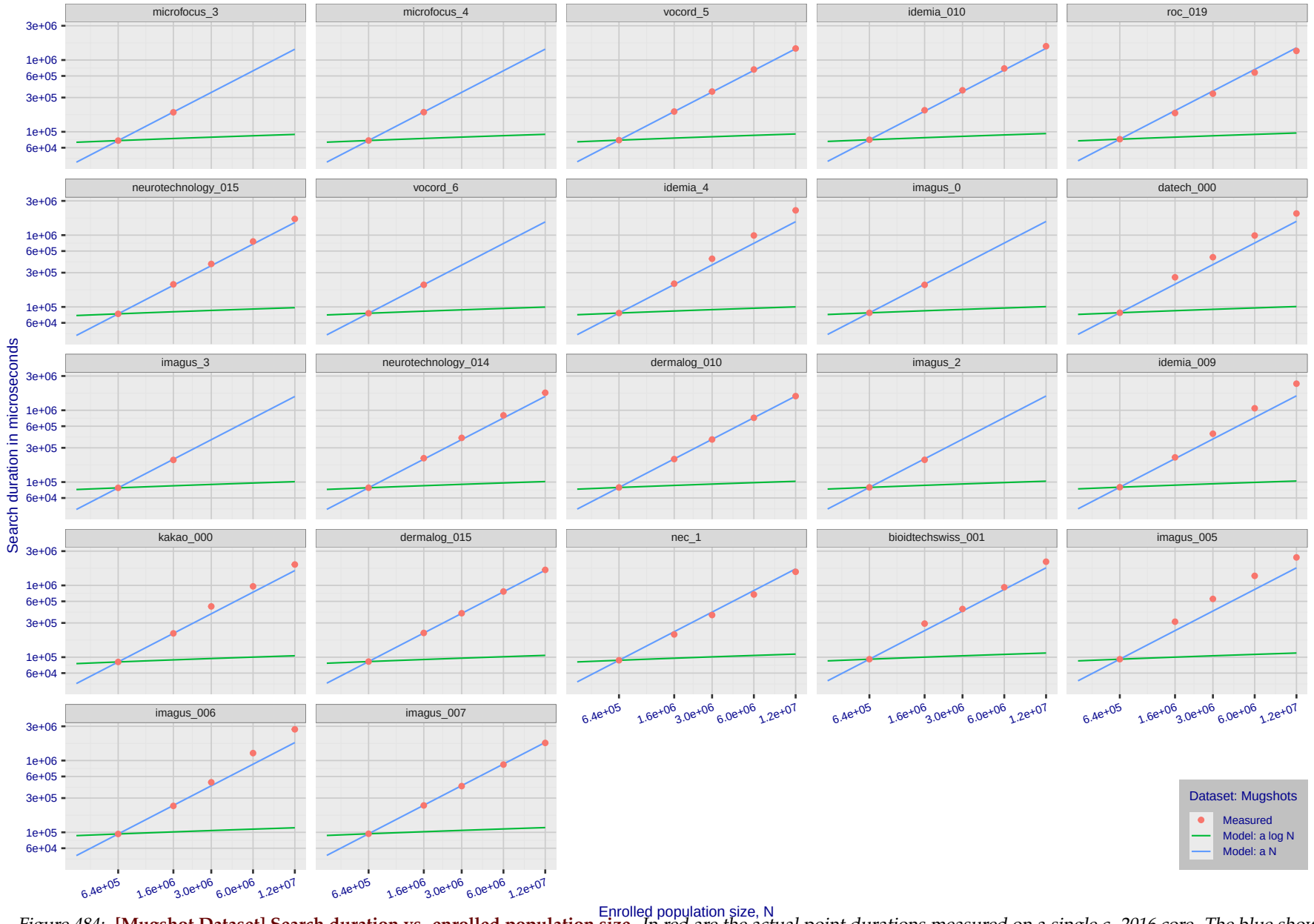

Figure 484: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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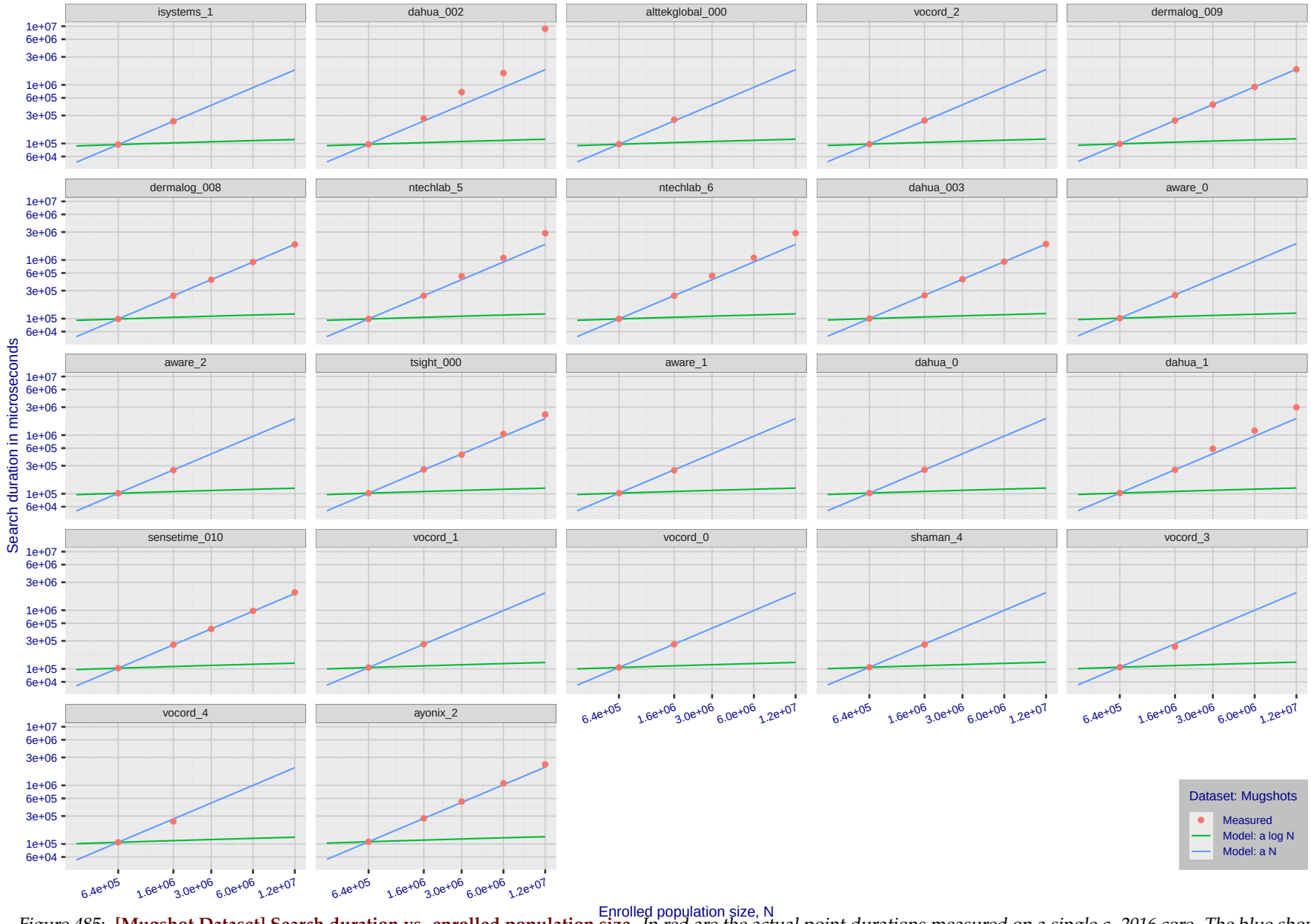

Figure 485: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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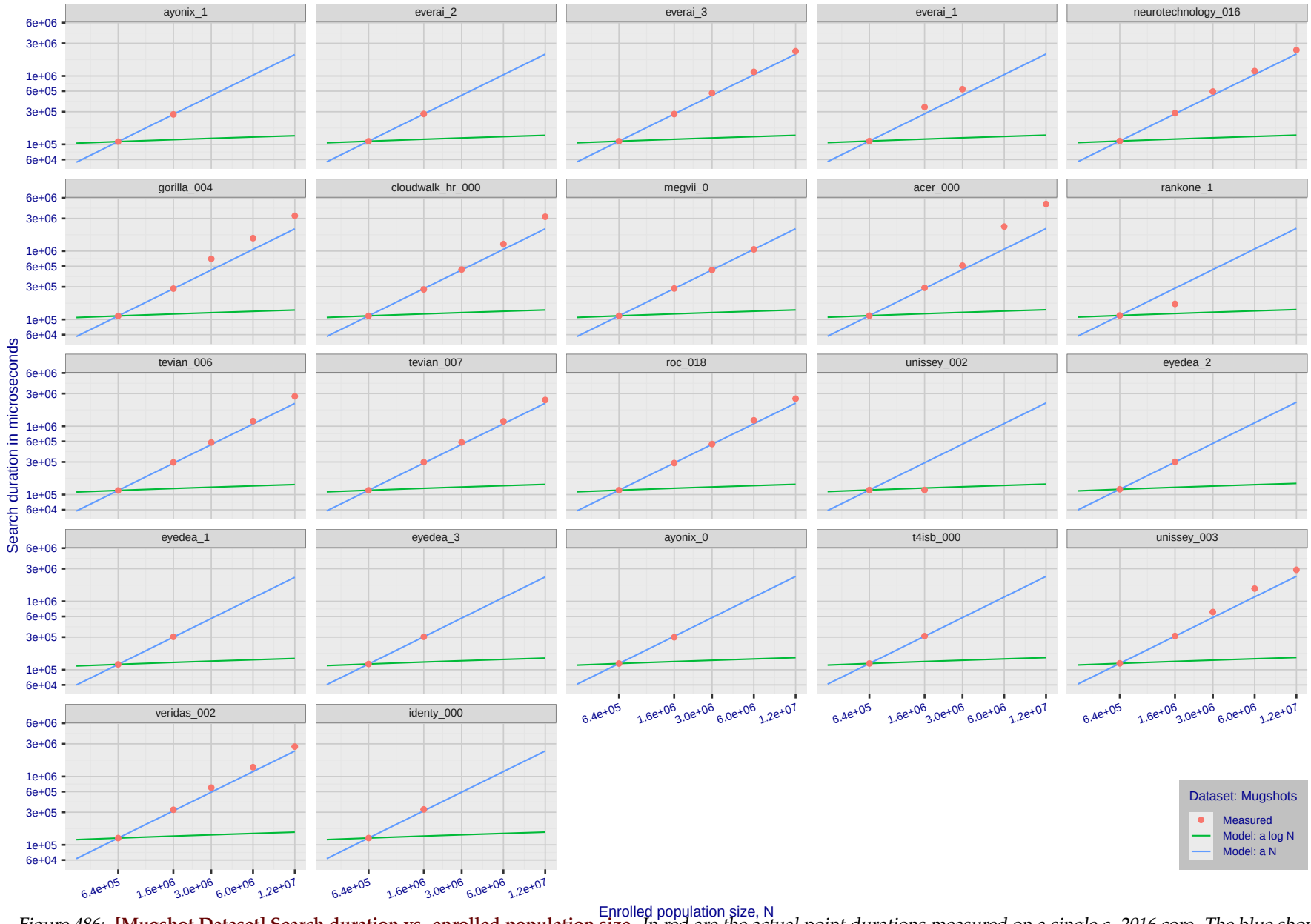

Figure 486: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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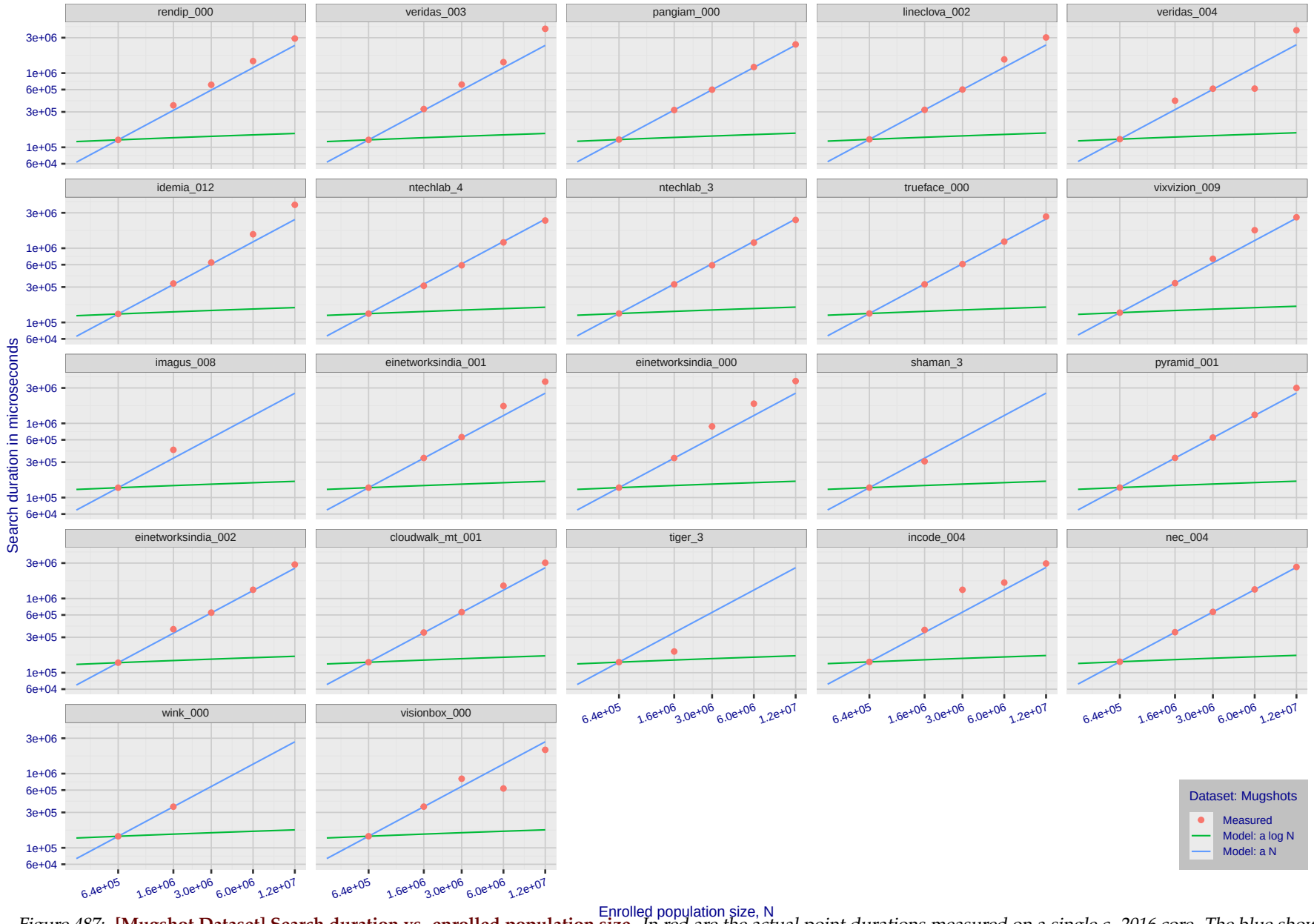

Figure 487: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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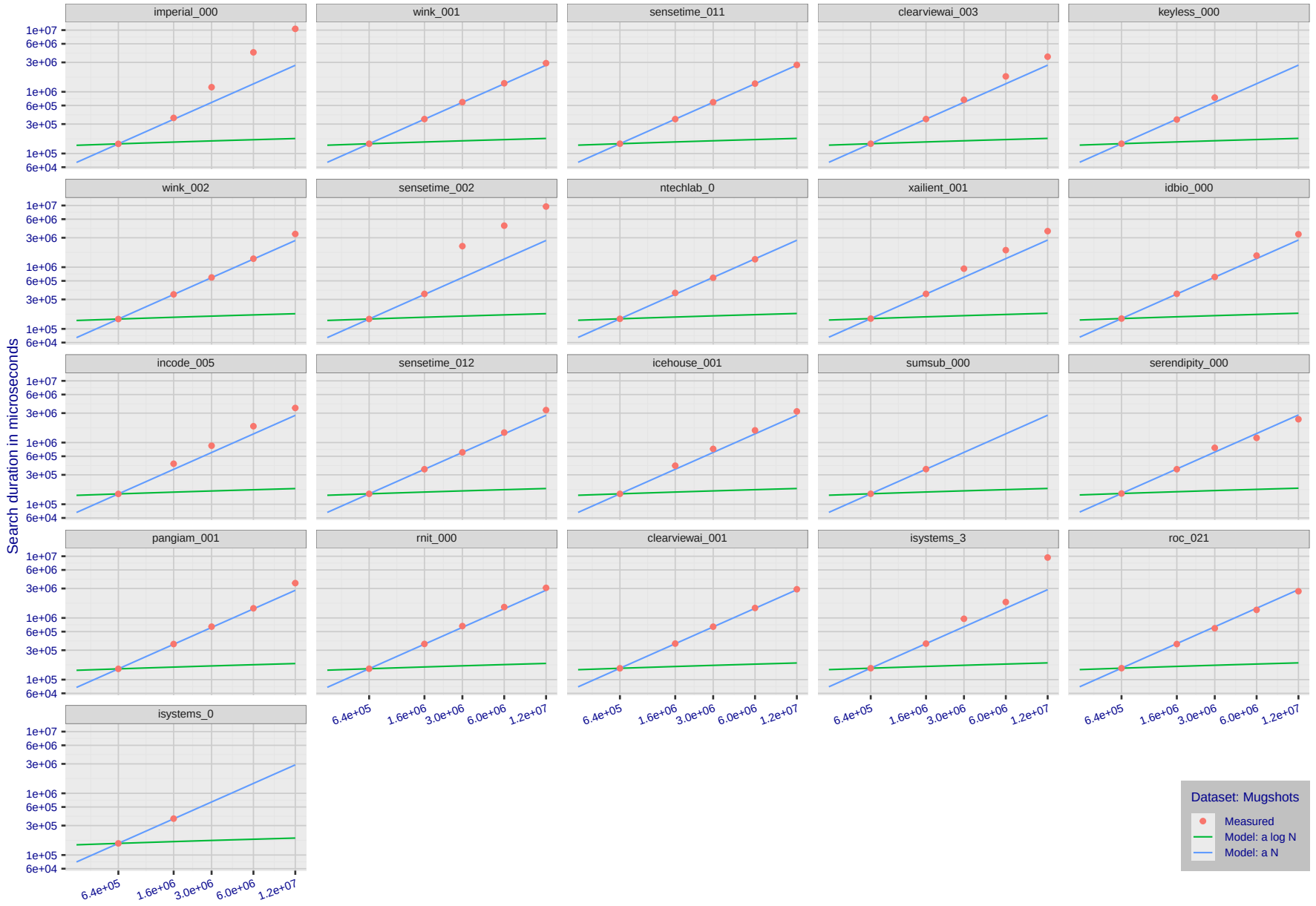

Figure 488: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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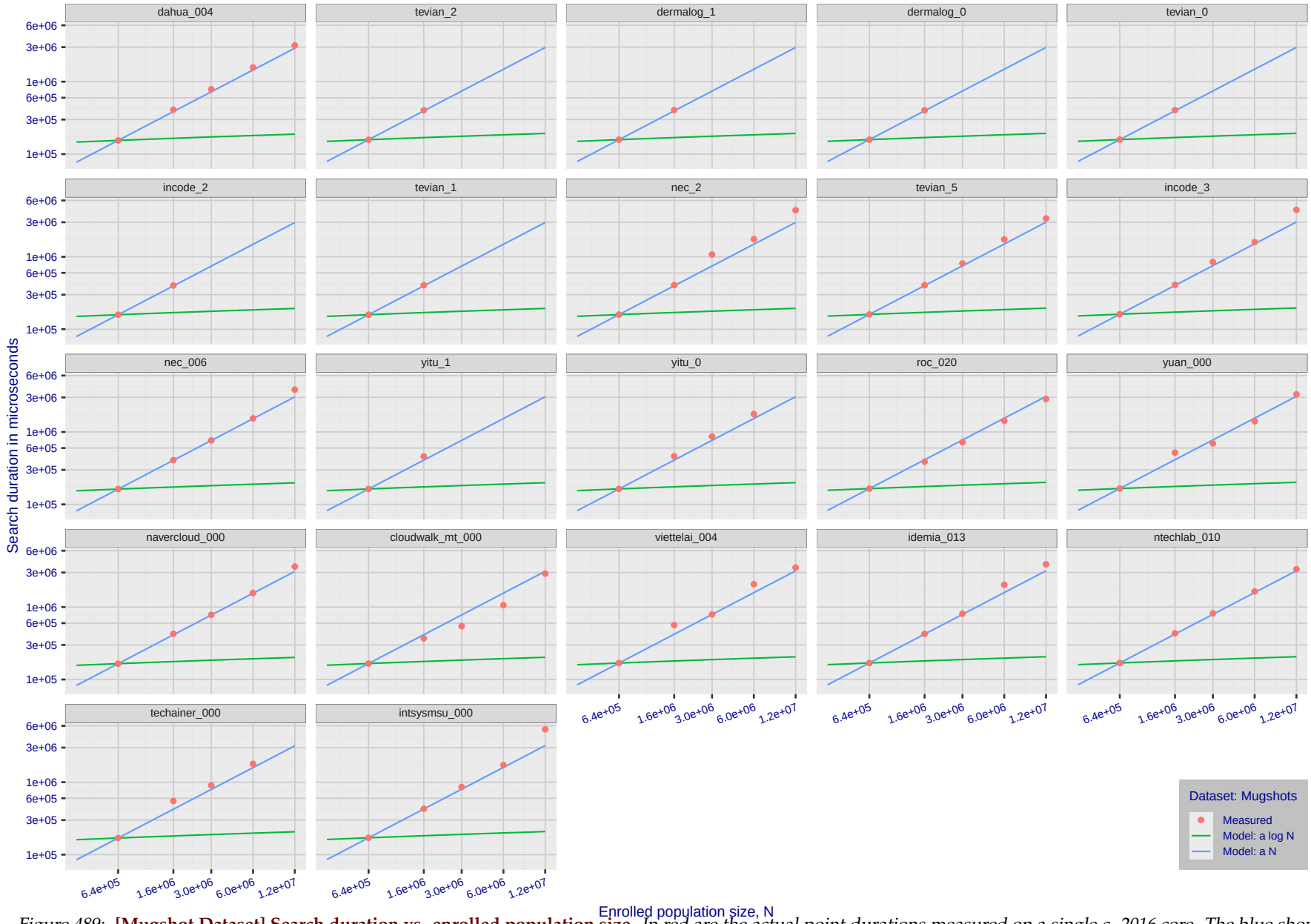

Figure 489: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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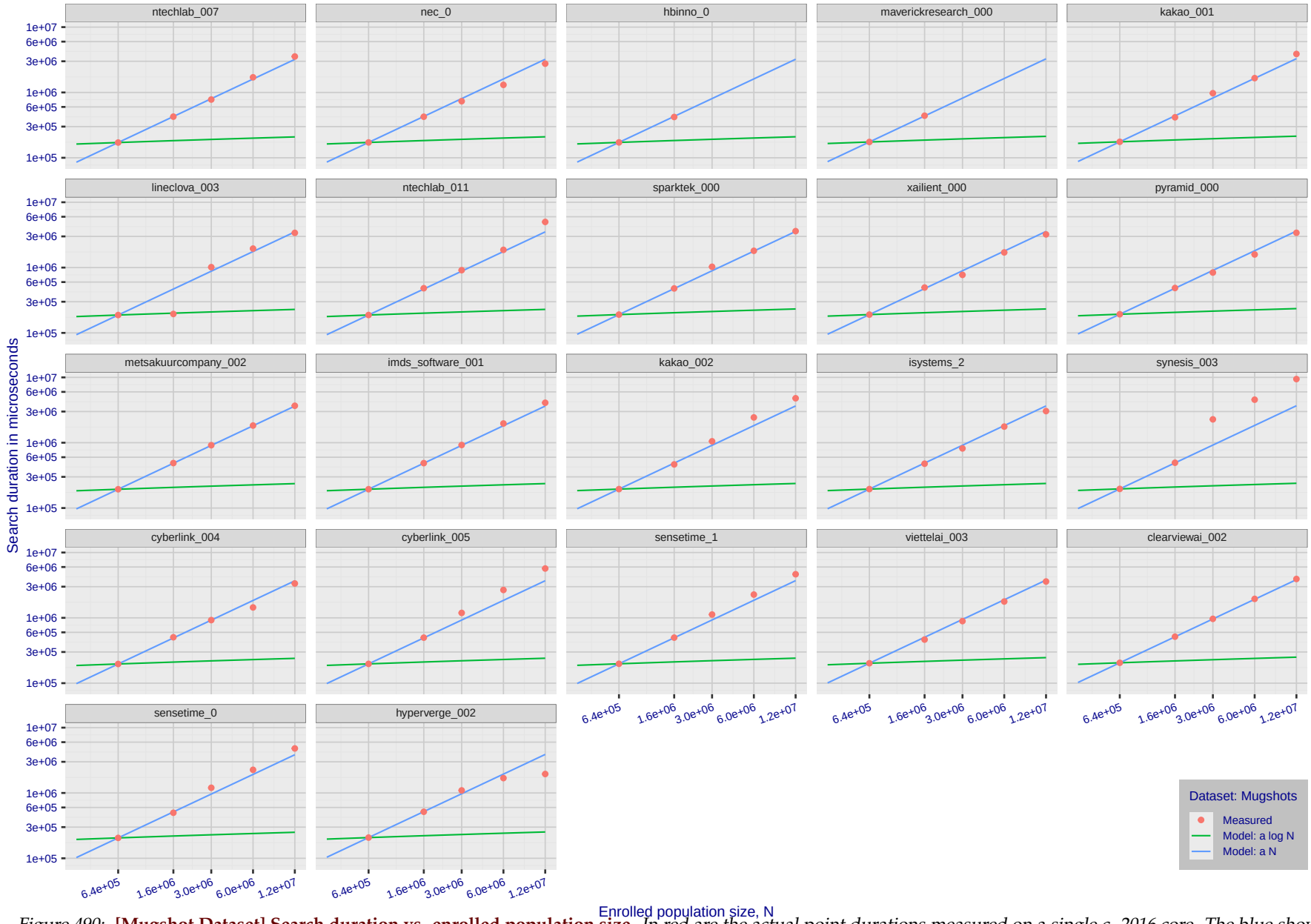

Figure 490: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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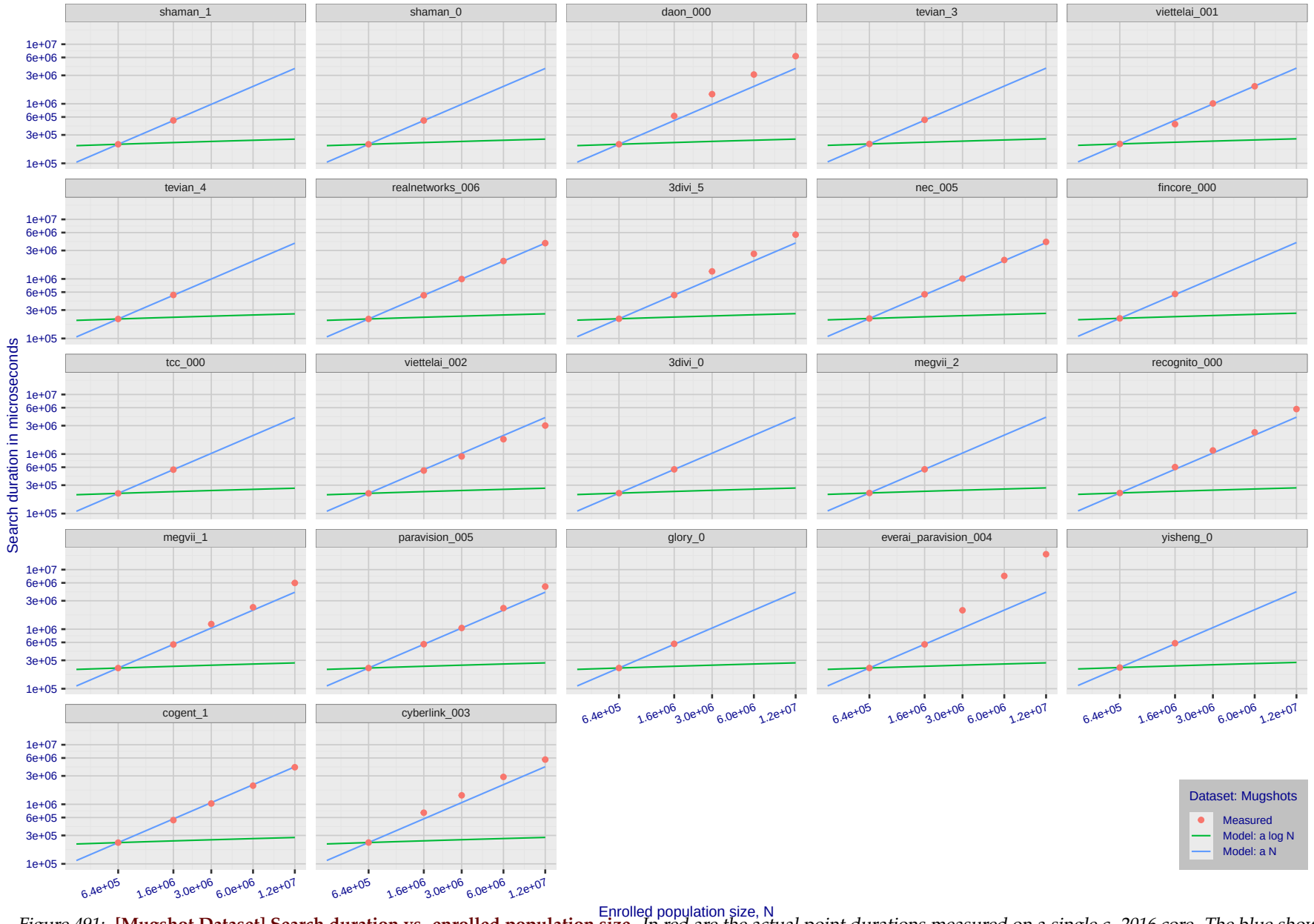

Figure 491: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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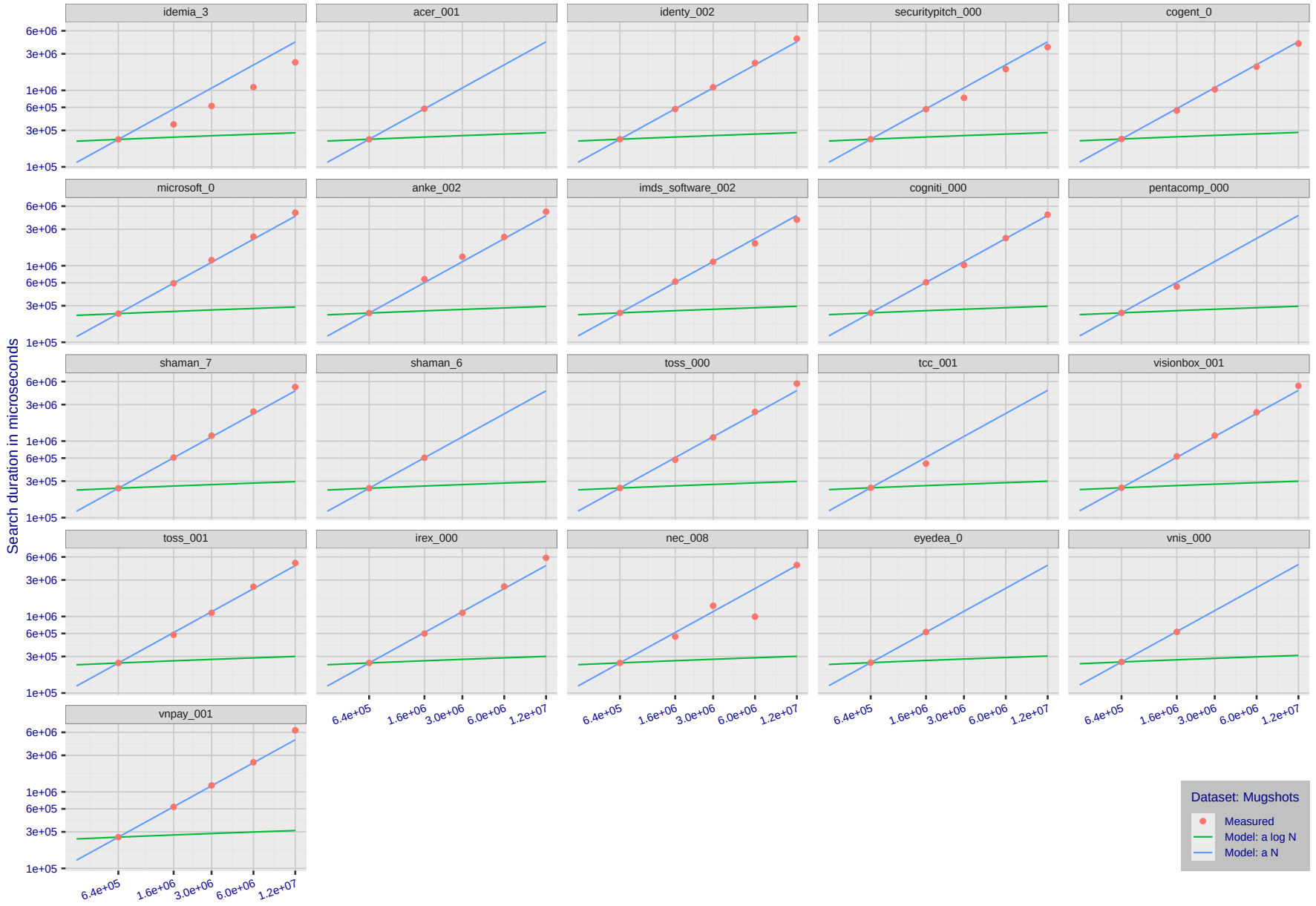

Figure 492: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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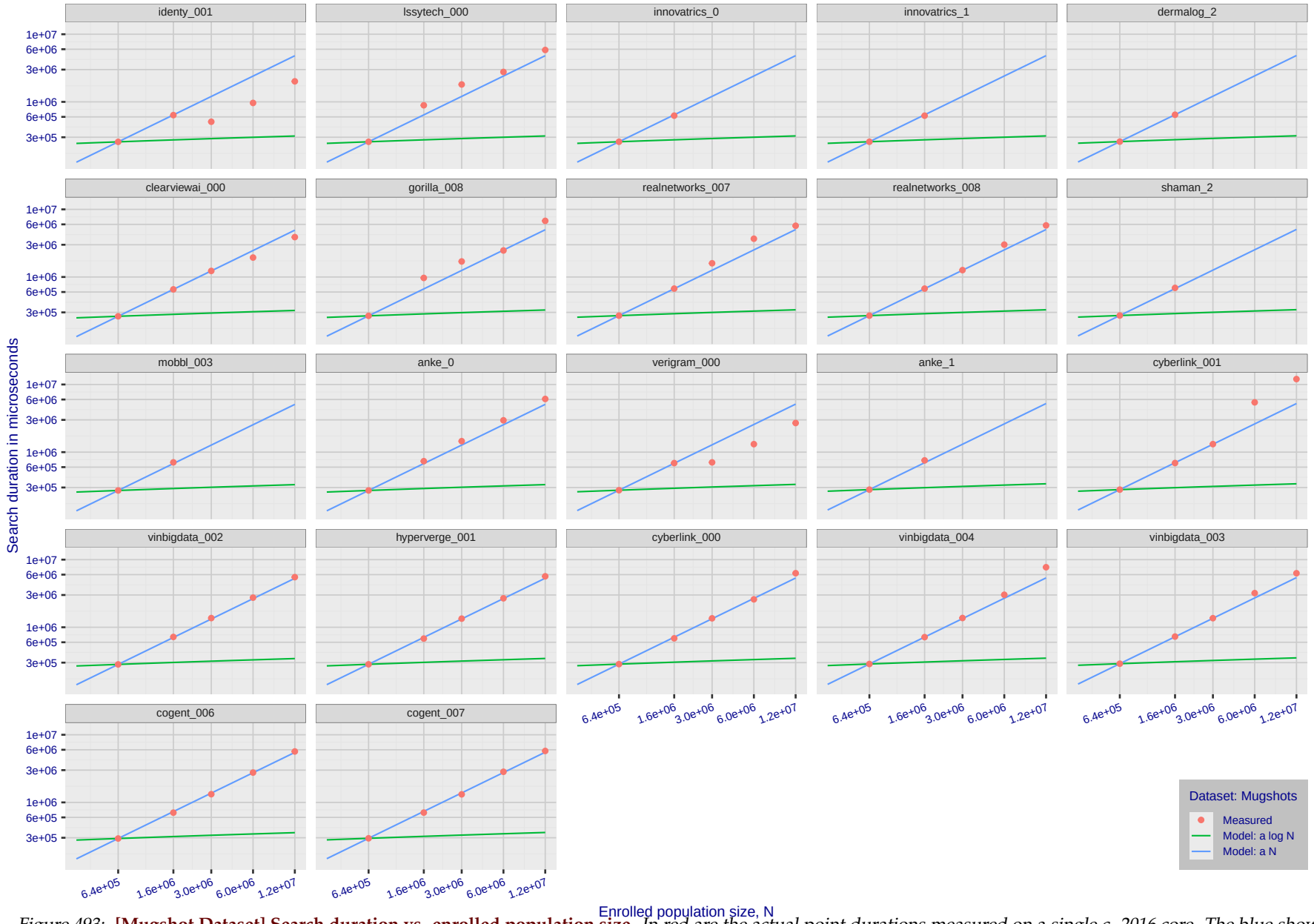

Figure 493: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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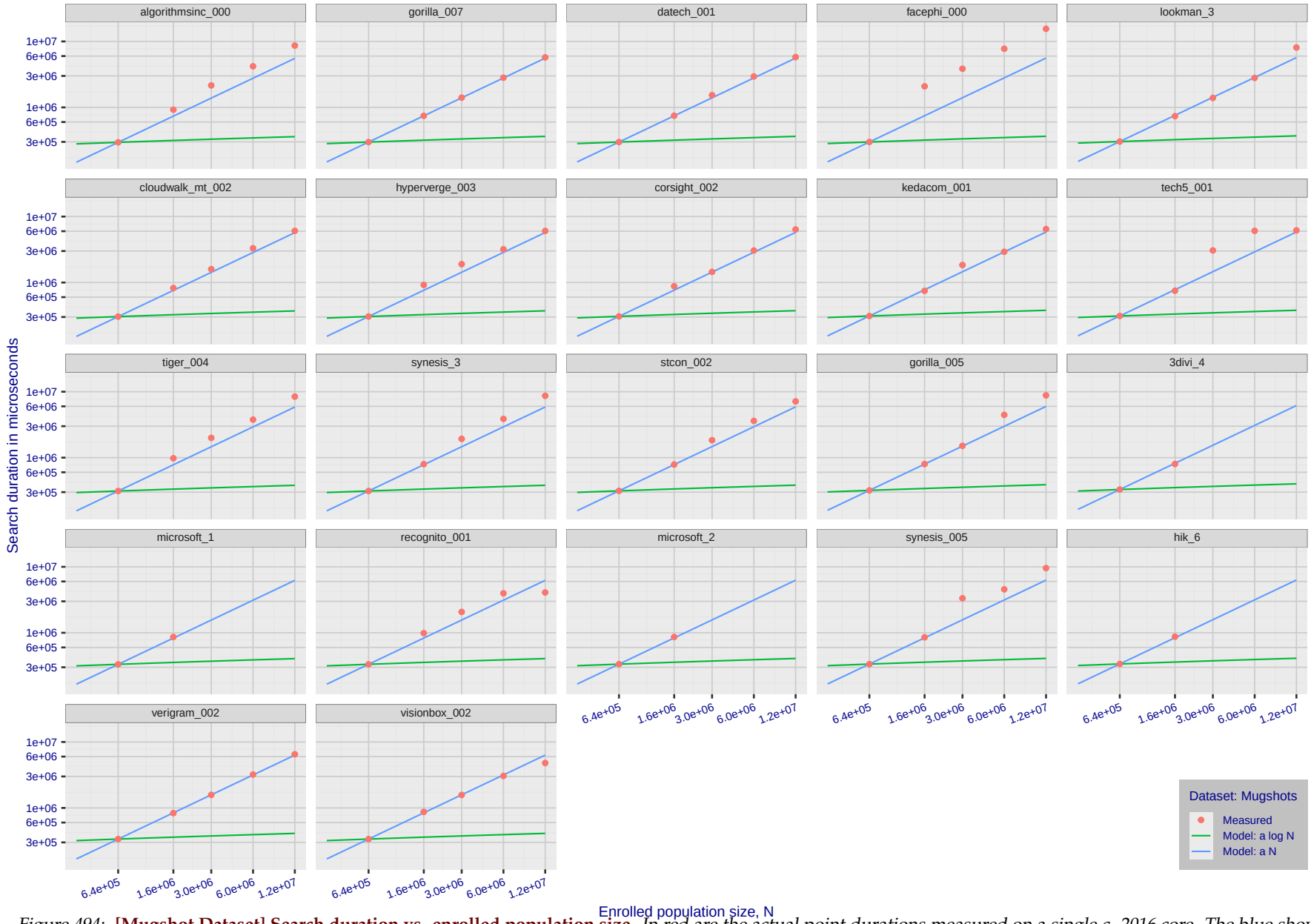

Figure 494: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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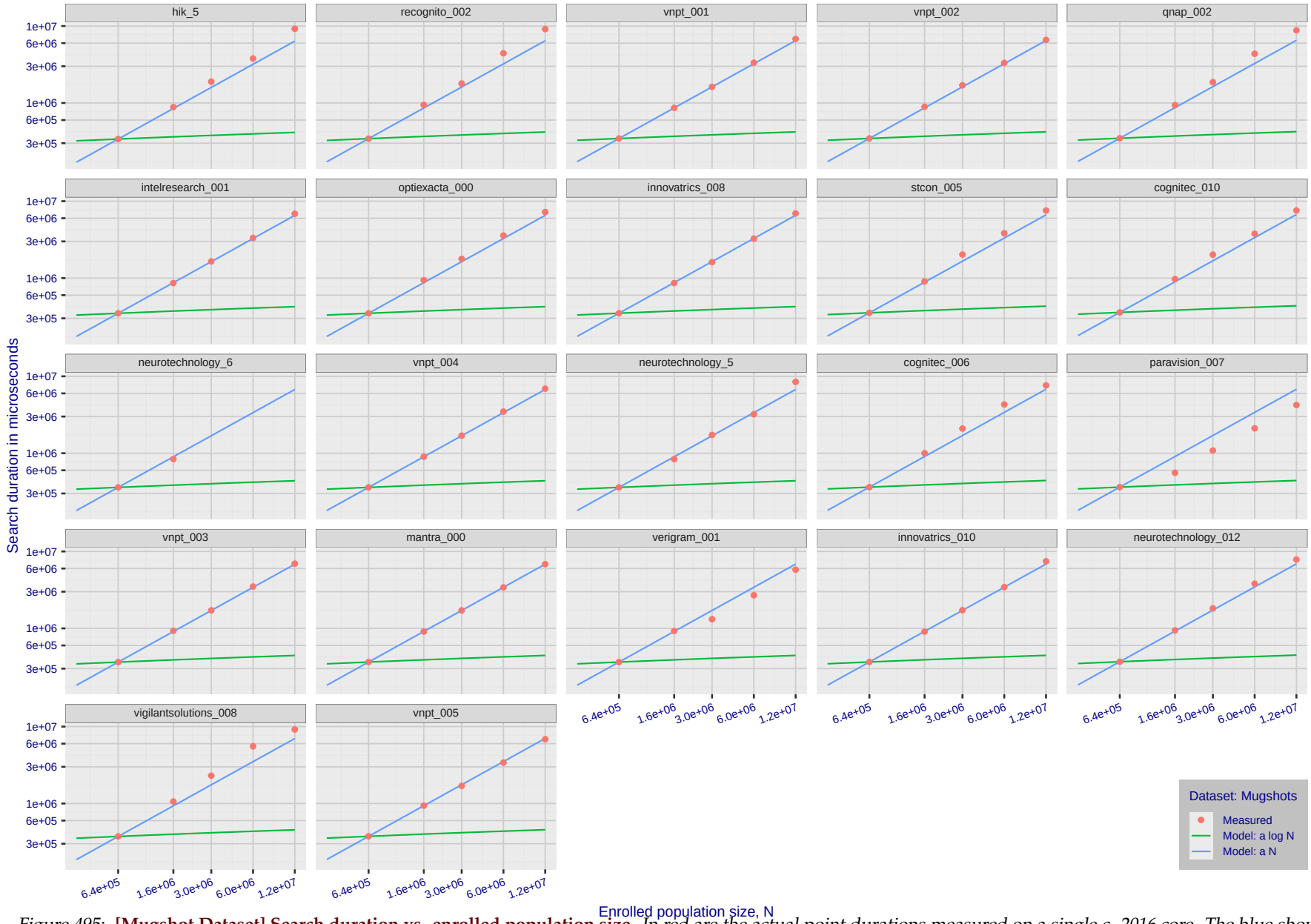

Figure 495: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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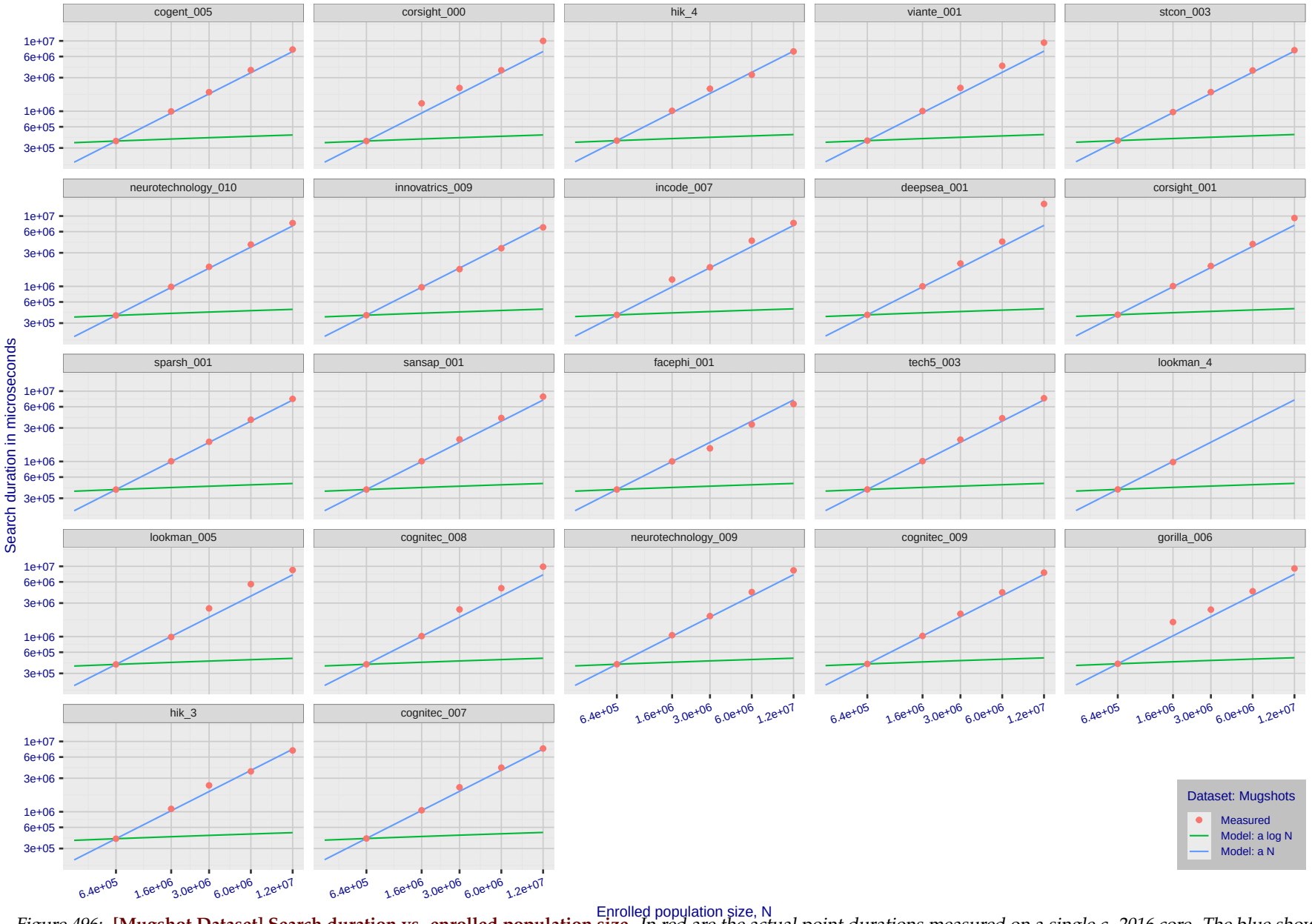

Figure 496: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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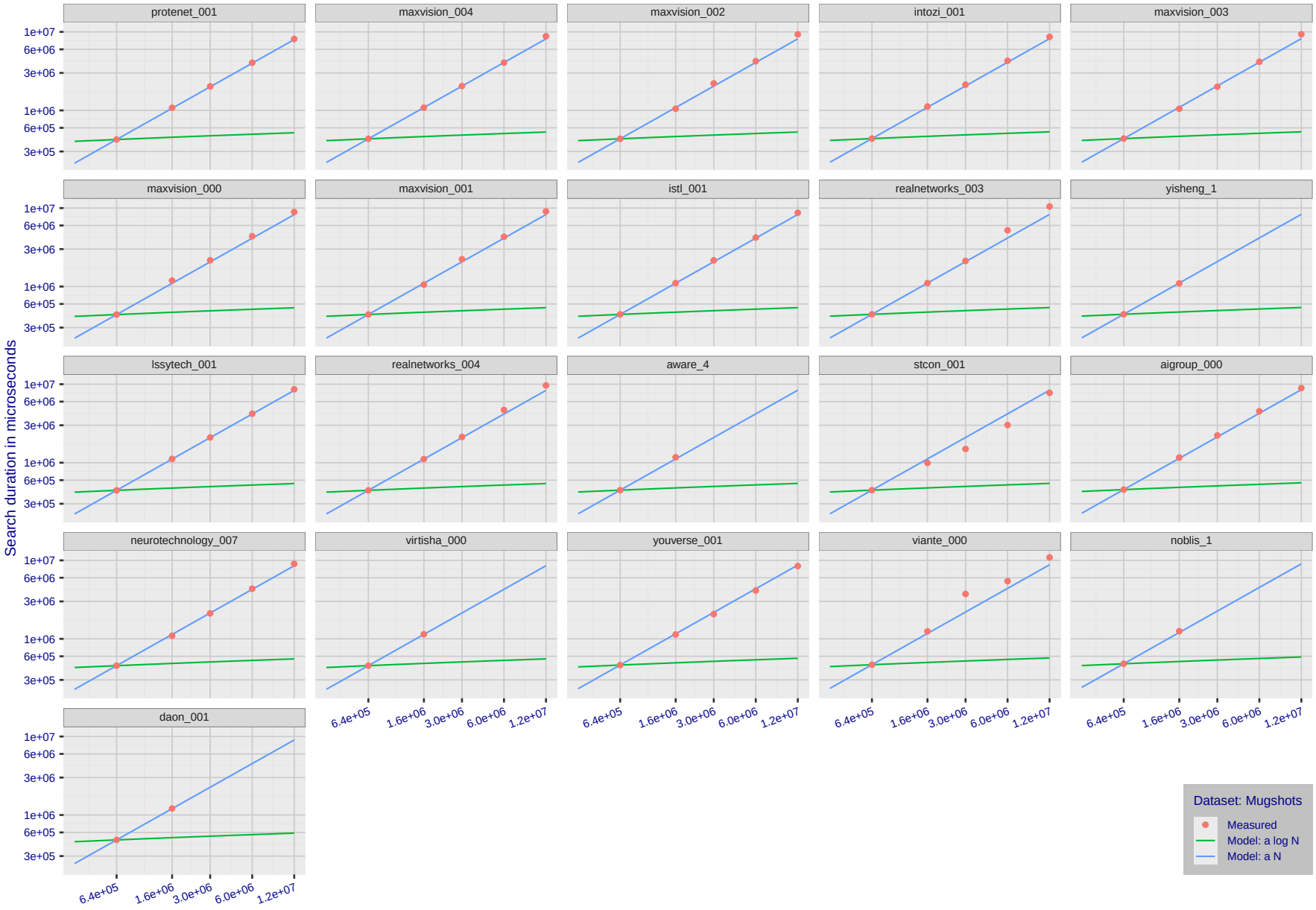

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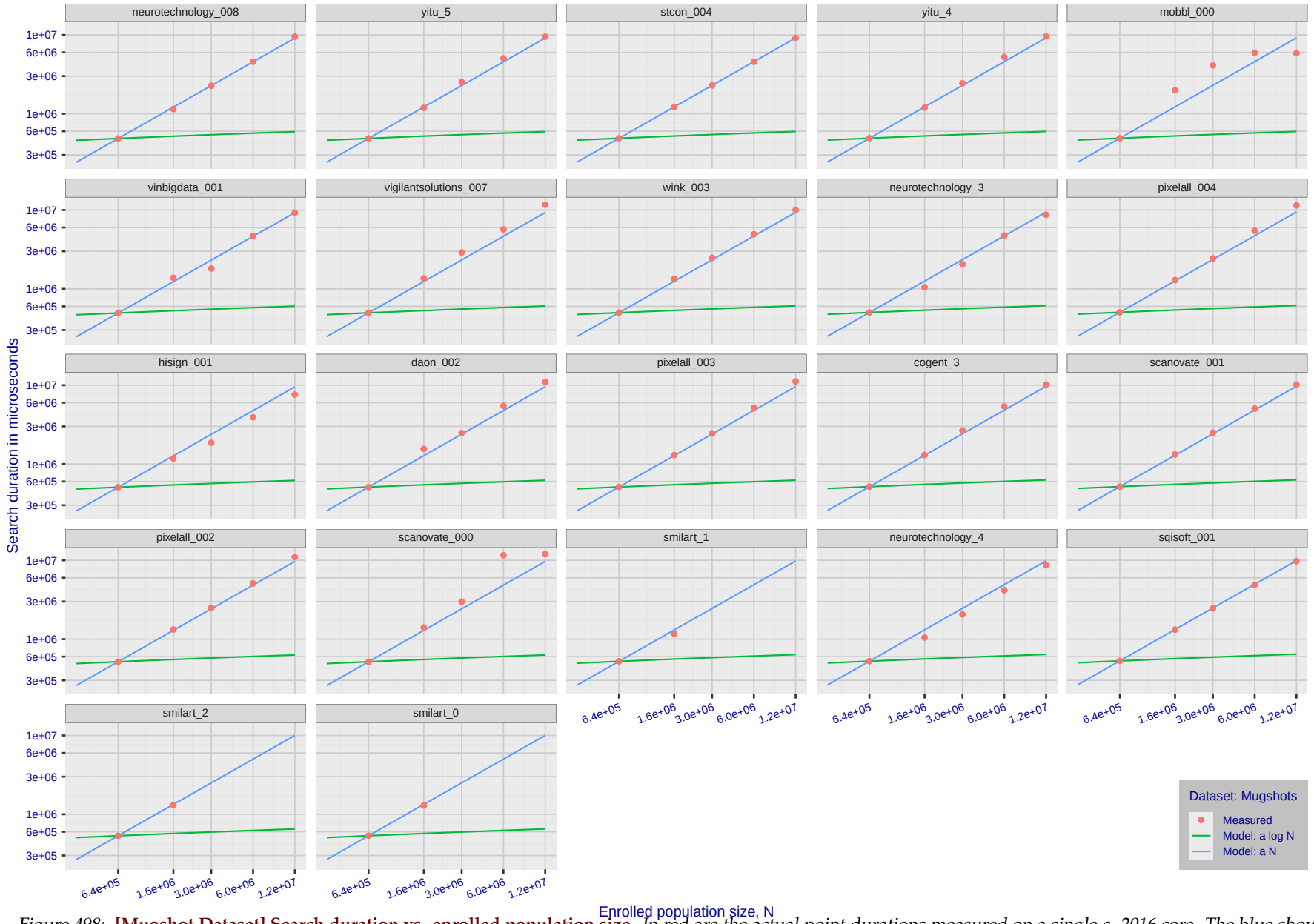

Figure 498: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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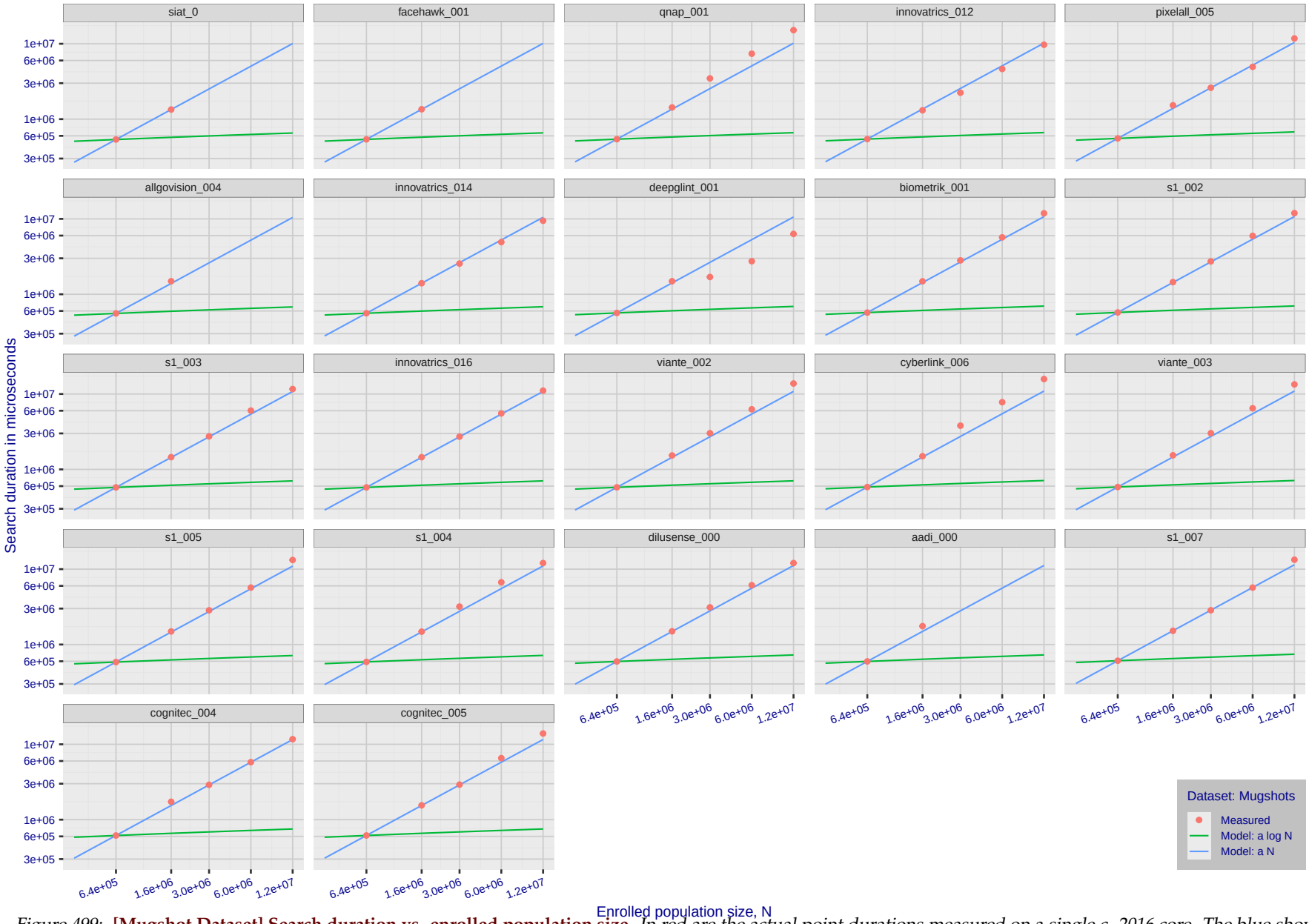

Figure 499: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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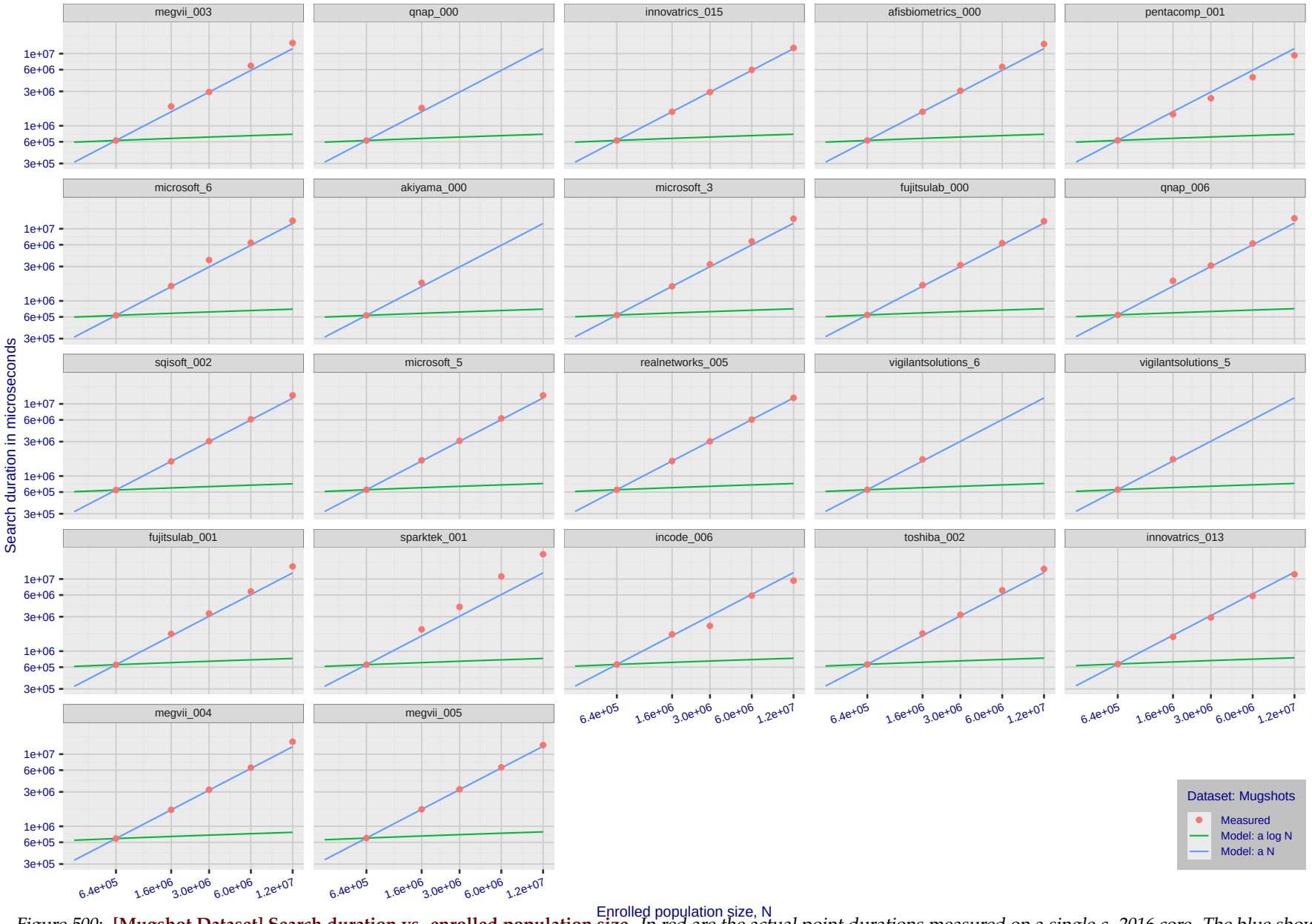

Figure 500: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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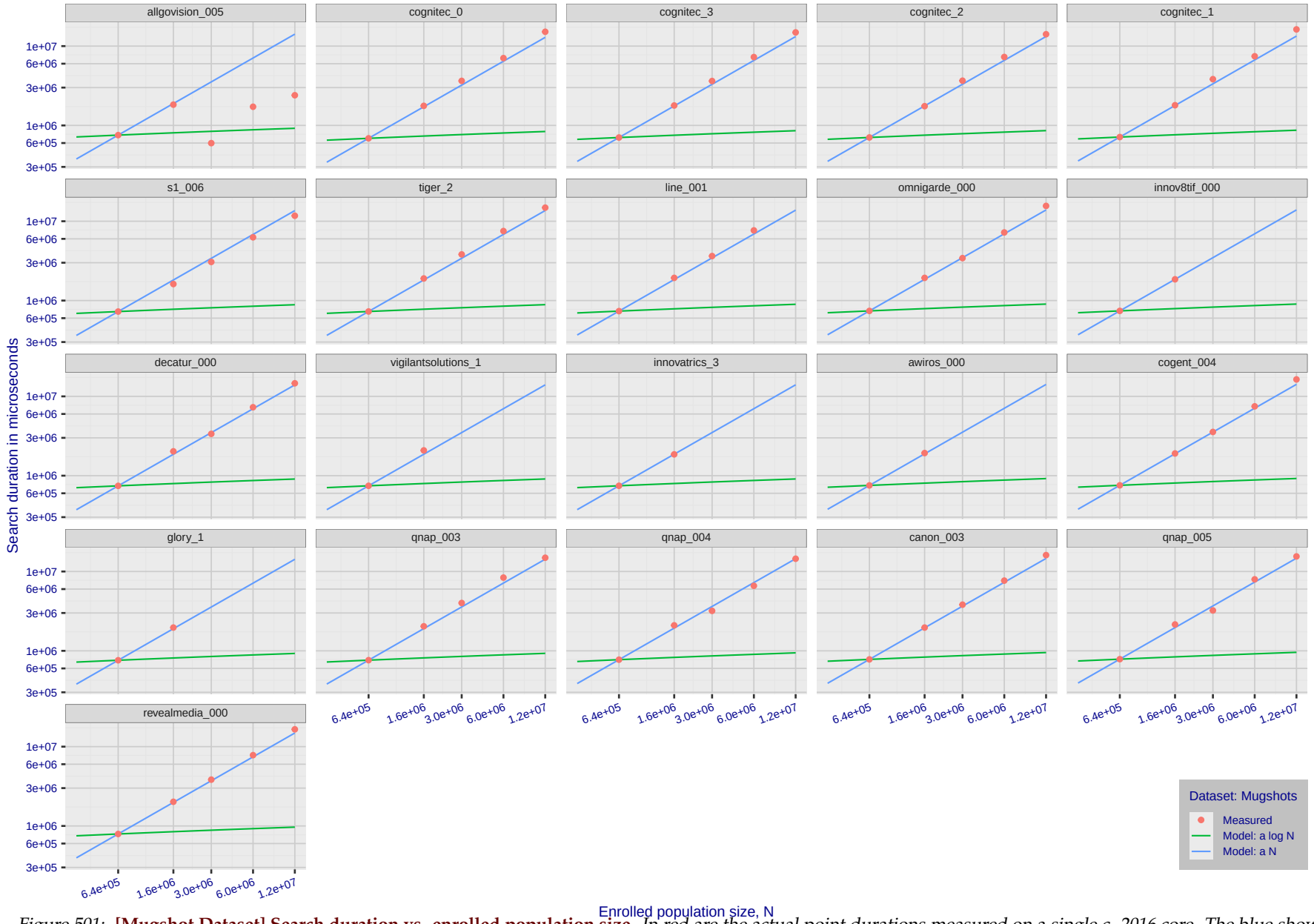

Figure 501: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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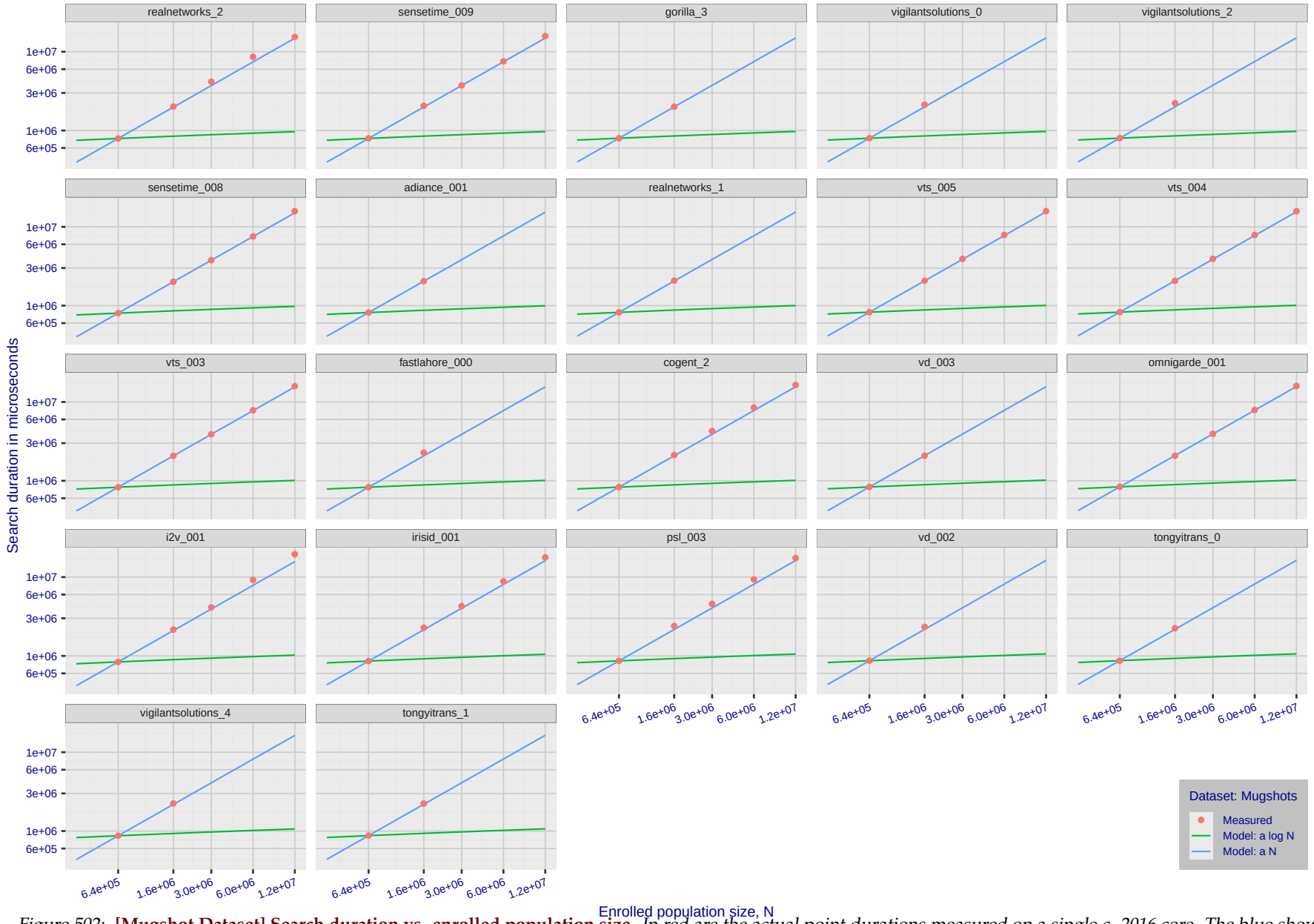

Figure 502: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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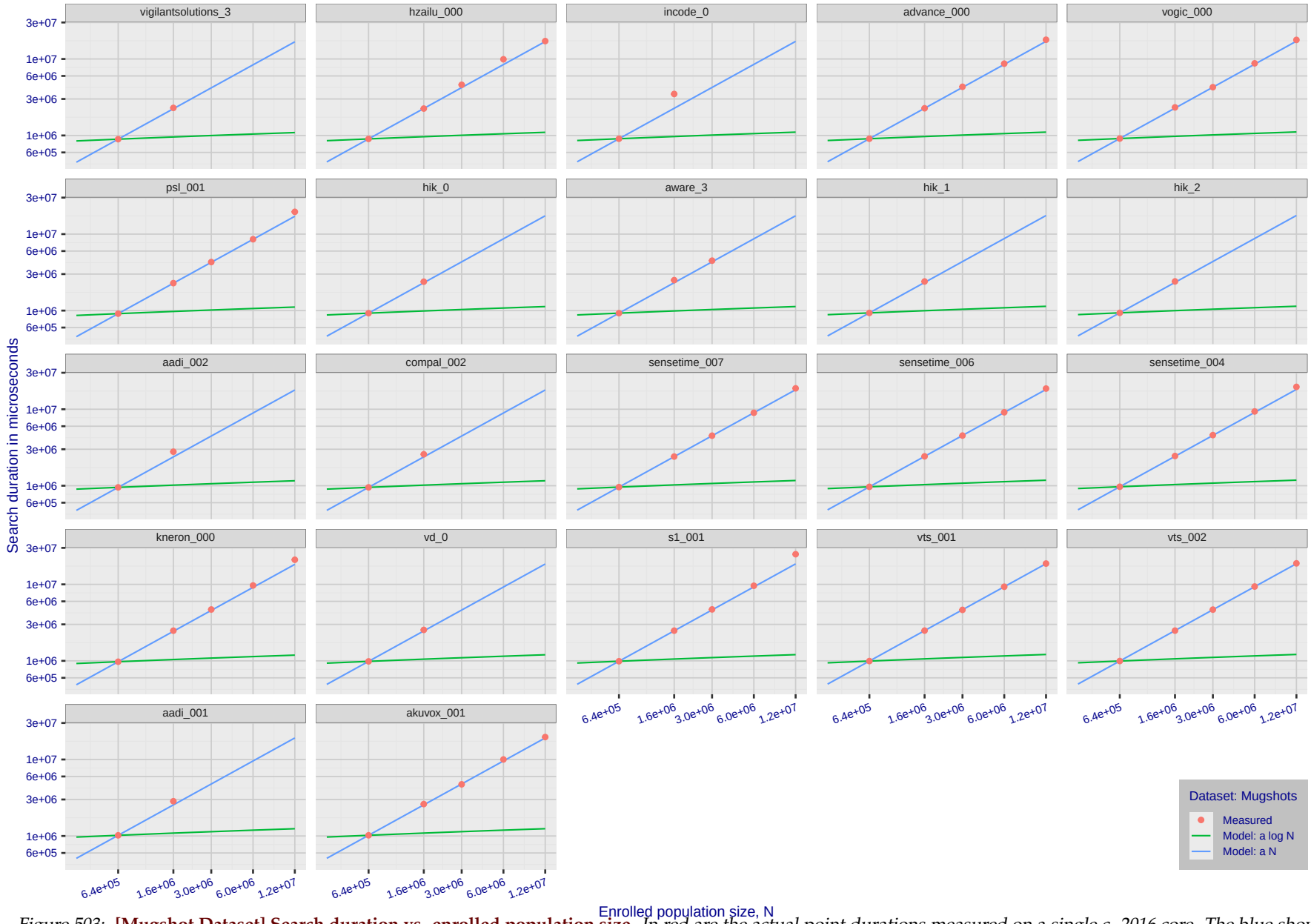

Figure 503: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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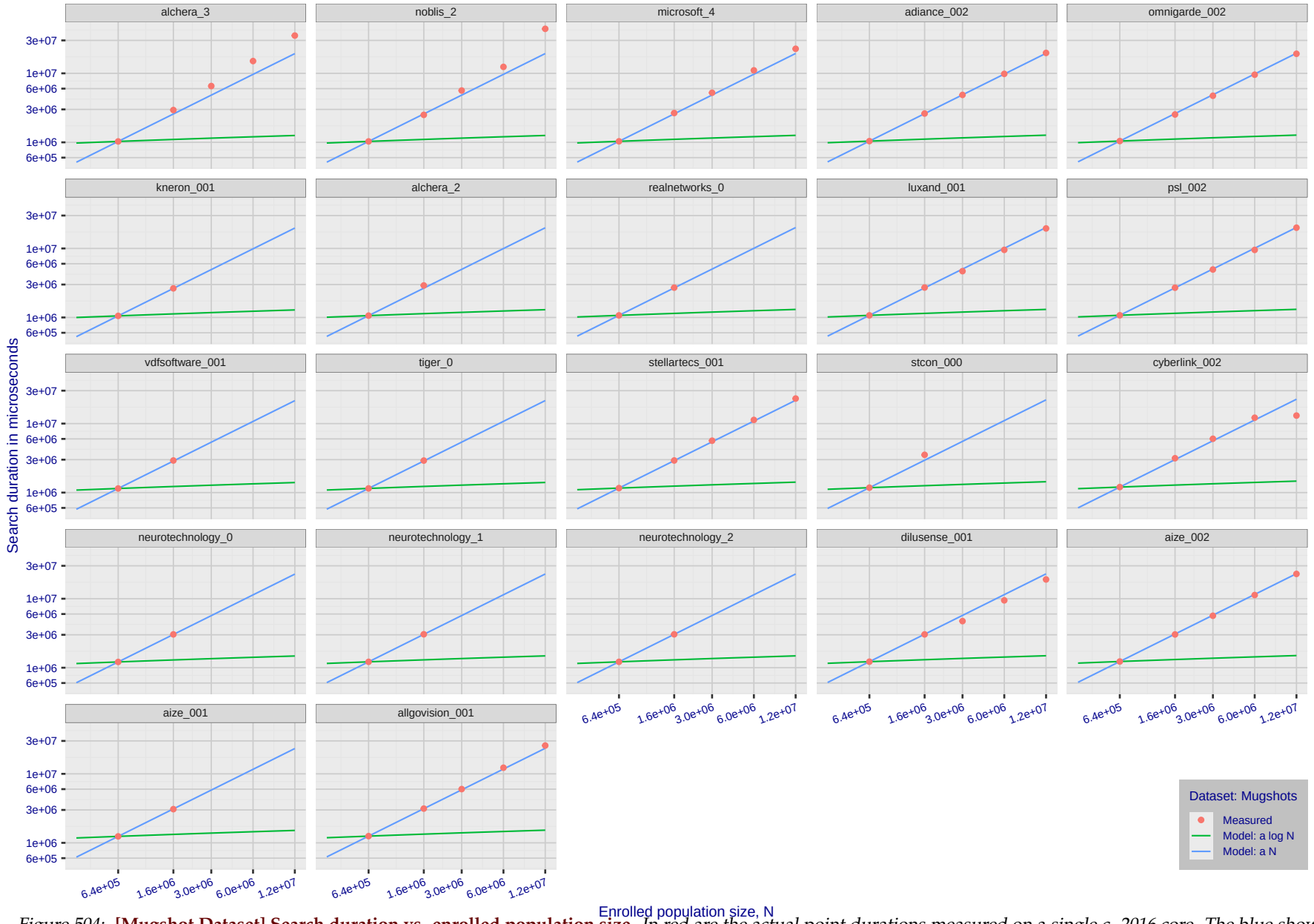

Figure 504: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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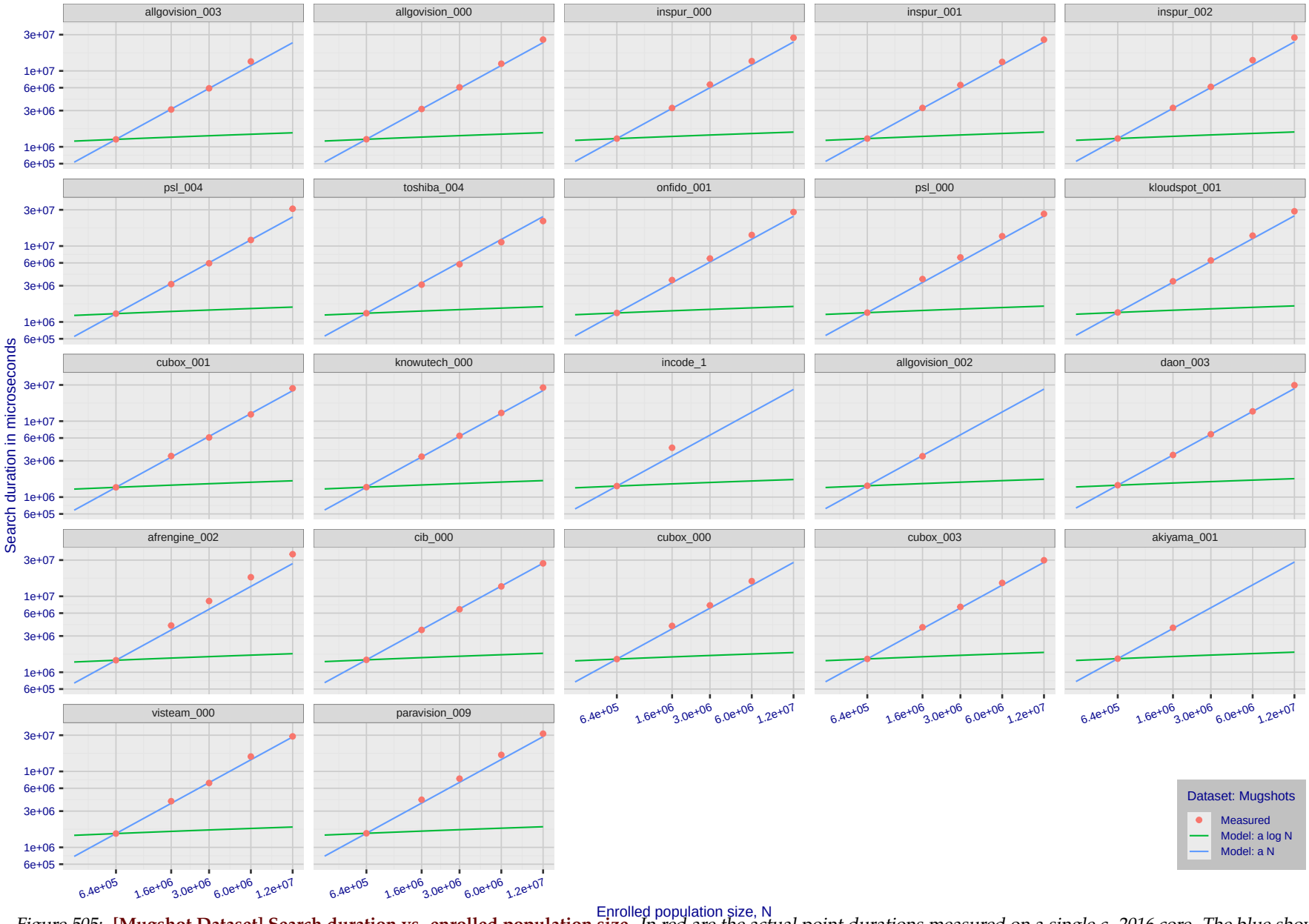

Figure 505: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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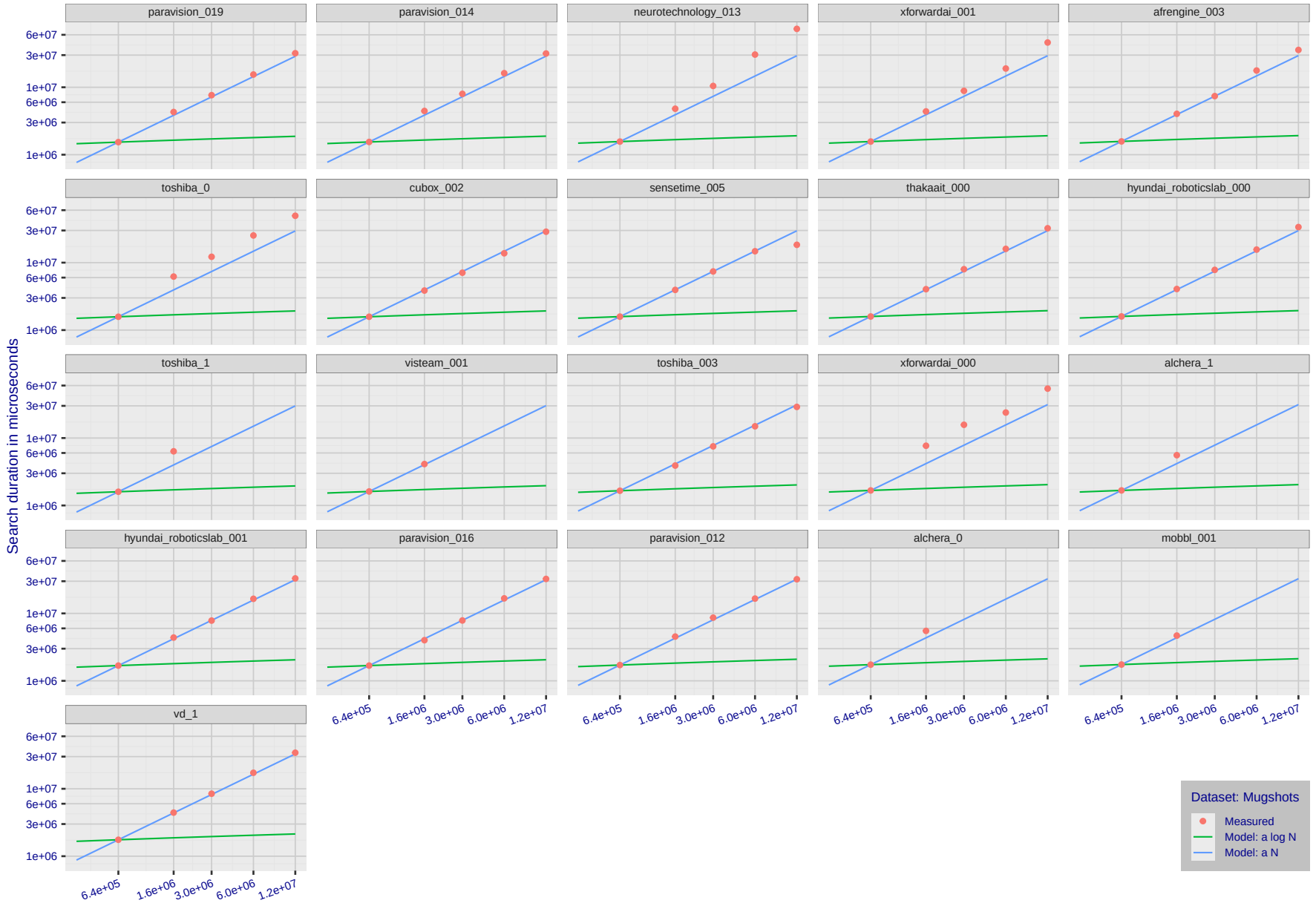

Figure 506: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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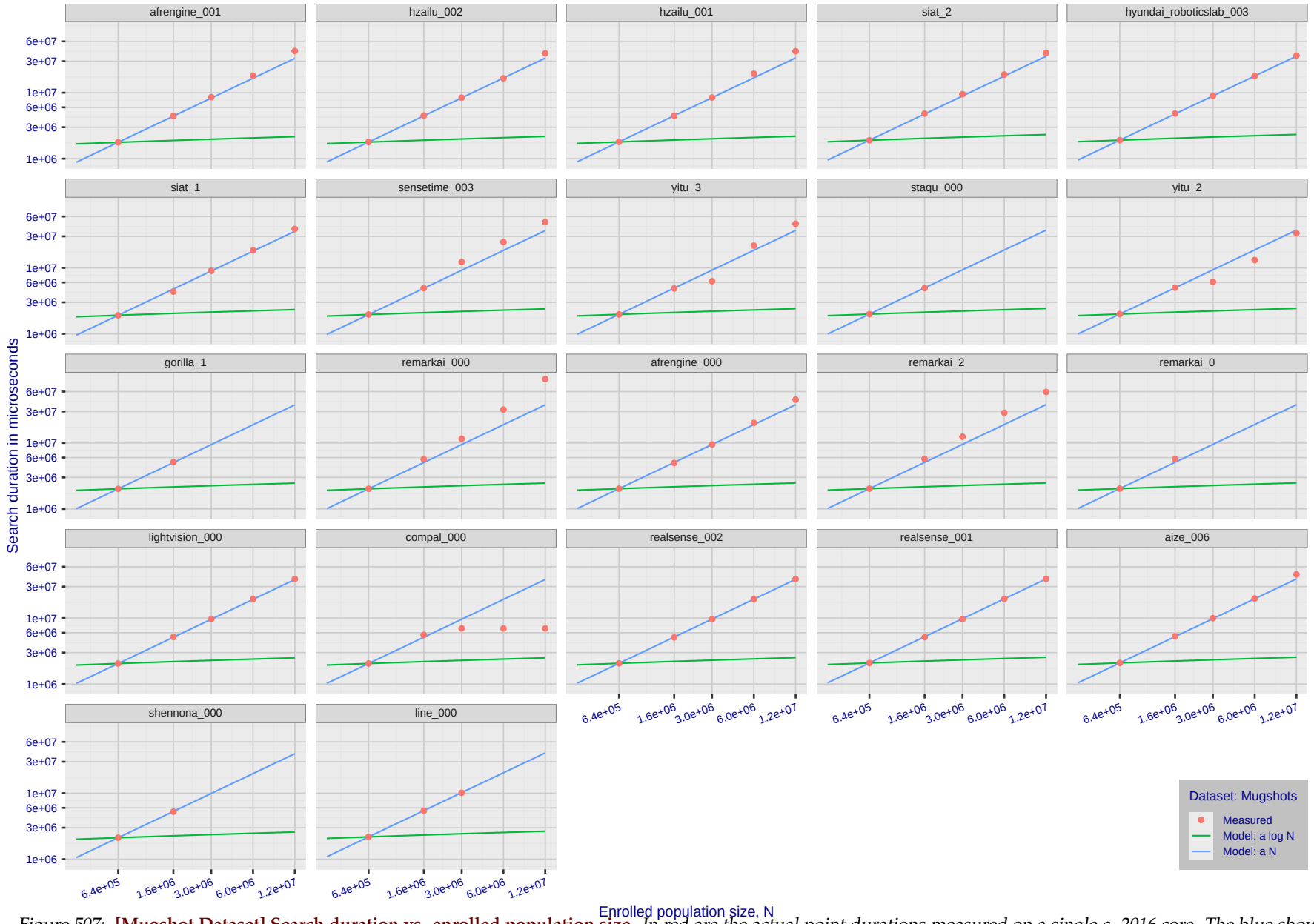

Figure 507: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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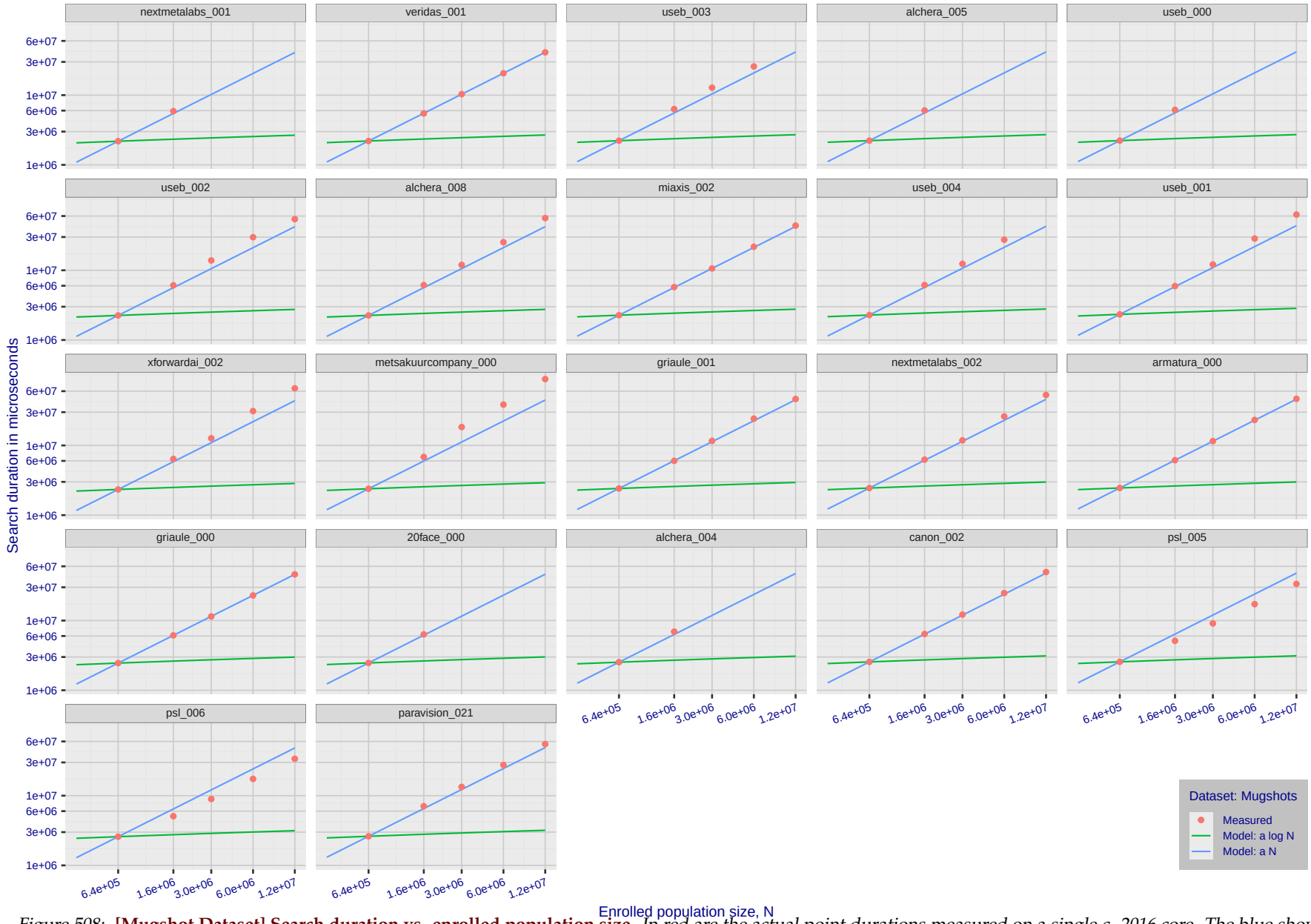

Figure 508: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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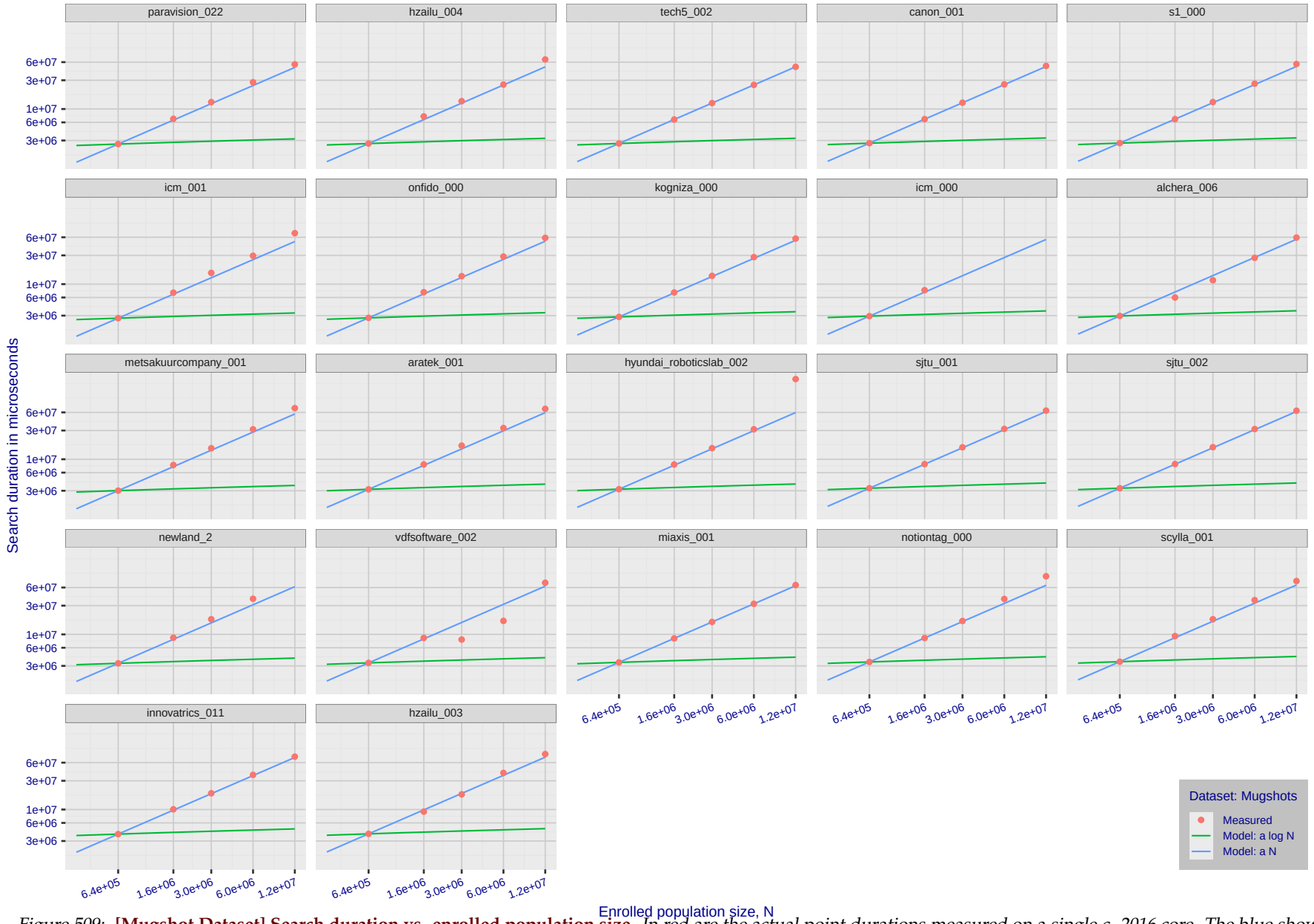

Figure 509: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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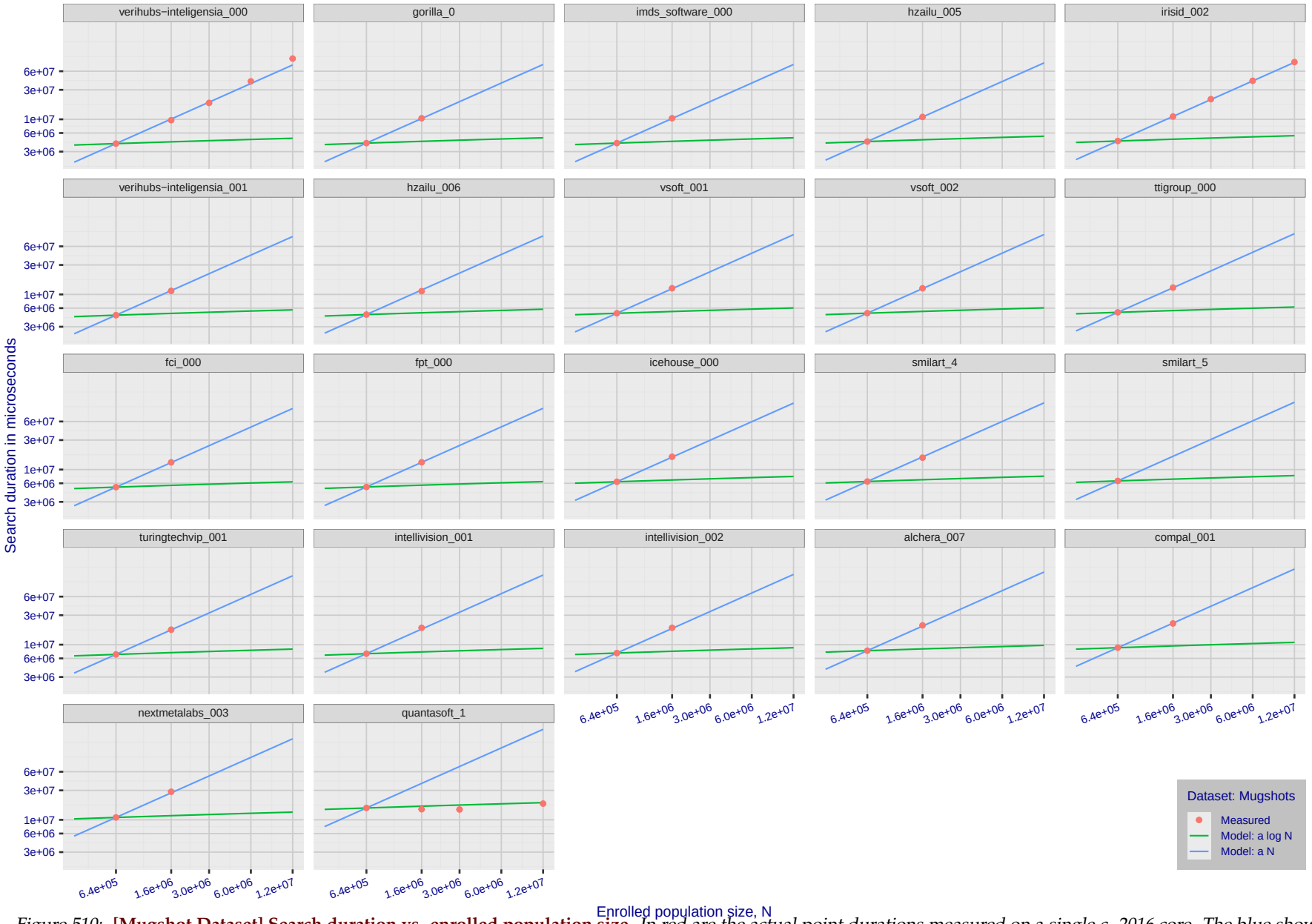

Figure 510: **[Mugshot Dataset] Search duration vs. enrolled population size**. *In red are the actual point durations measured on a single c. 2016 core. The blue shows linear growth from $N = 640\,000$. The green line shows logathmic growth from that point to $N = 1\,600\,000$. Note the sublinear growth from algorithms from Camvi, Dermalog, EverAI, Innovatrics, and Visionlabs. The tiger_1 algorithm is also sublinear, but inaccurate and inoperable at $N \geq 3000000$. This capability sometimes comes at the additional expense of converting a linear gallery data structure into whatever fast-search data structure is used. Note that search times are sometimes dominated by the template generation times shown in Table 52.*

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Appendix G Gallery Insertion Timing

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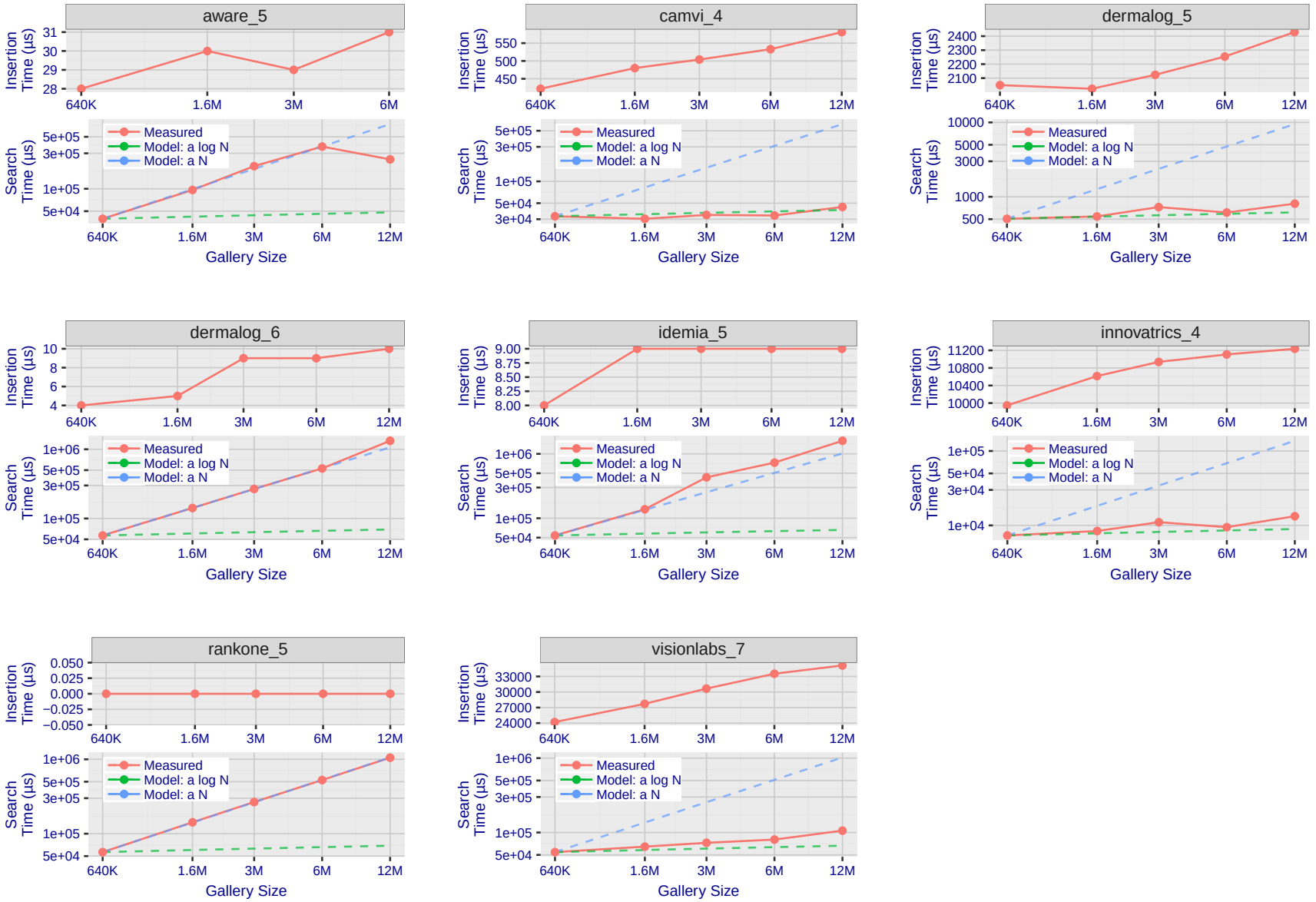

Figure 512: **[Mugshot Dataset] Gallery insertion duration vs. enrolled population size**. *This chart plots the time it takes to insert a single template into a finalized gallery, illustrated over increasing gallery sizes. For reference, search times on finalized galleries of corresponding sizes are plotted right underneath. Gallery insertion time plots were generated on algorithms that 1) successfully implemented gallery insertion with no errors and 2) that were run on galleries with N up to* 12 000 000. *Generally, only the more accurate algorithms were run on galleries with N up to* 12 000 000.

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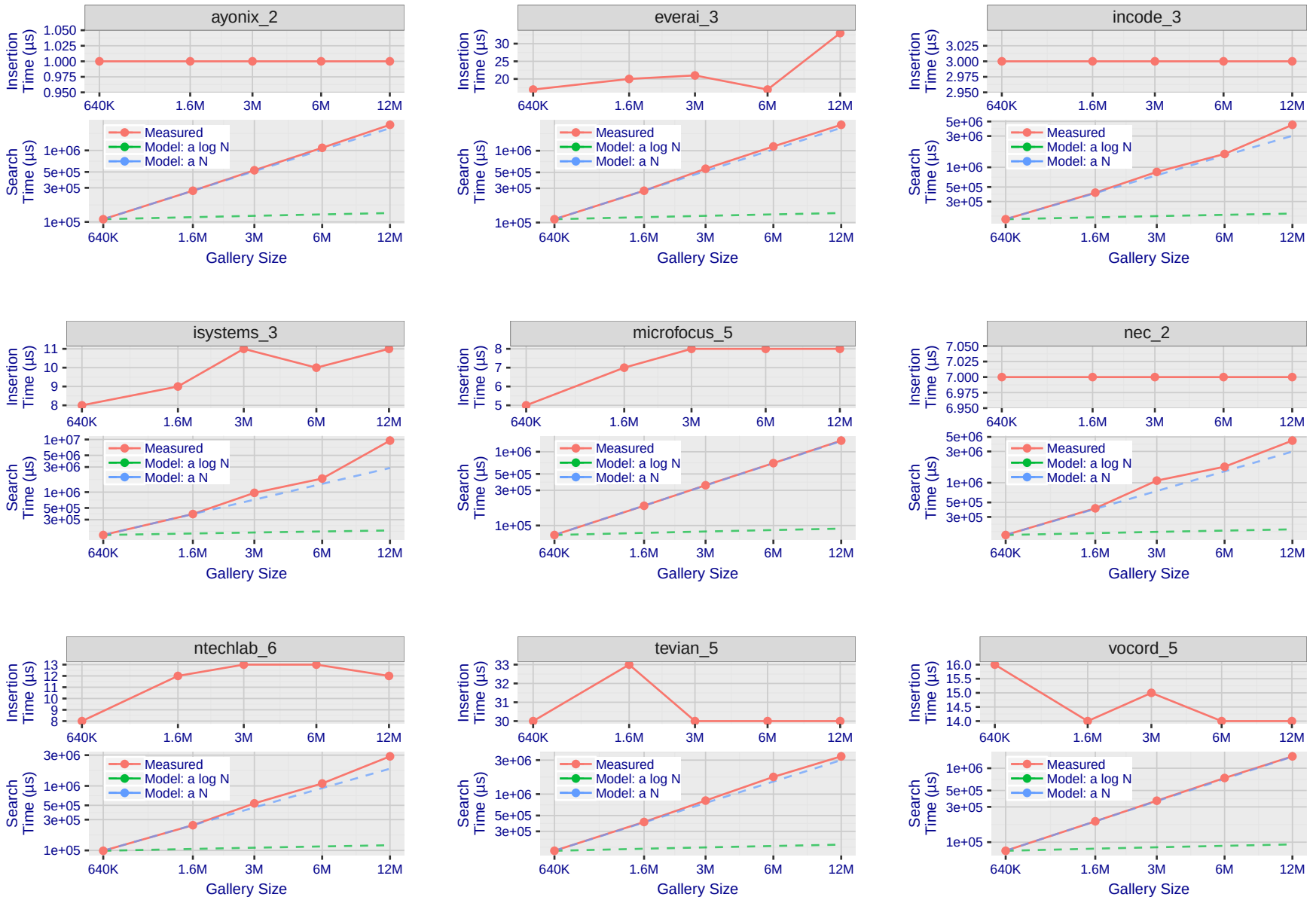

Figure 513: **[Mugshot Dataset] Gallery insertion duration vs. enrolled population size**. *This chart plots the time it takes to insert a single template into a finalized gallery, illustrated over increasing gallery sizes. For reference, search times on finalized galleries of corresponding sizes are plotted right underneath. Gallery insertion time plots were generated on algorithms that 1) successfully implemented gallery insertion with no errors and 2) that were run on galleries with N up to 12 000 000. Generally, only the more accurate algorithms were run on galleries with N up to 12 000 000.*

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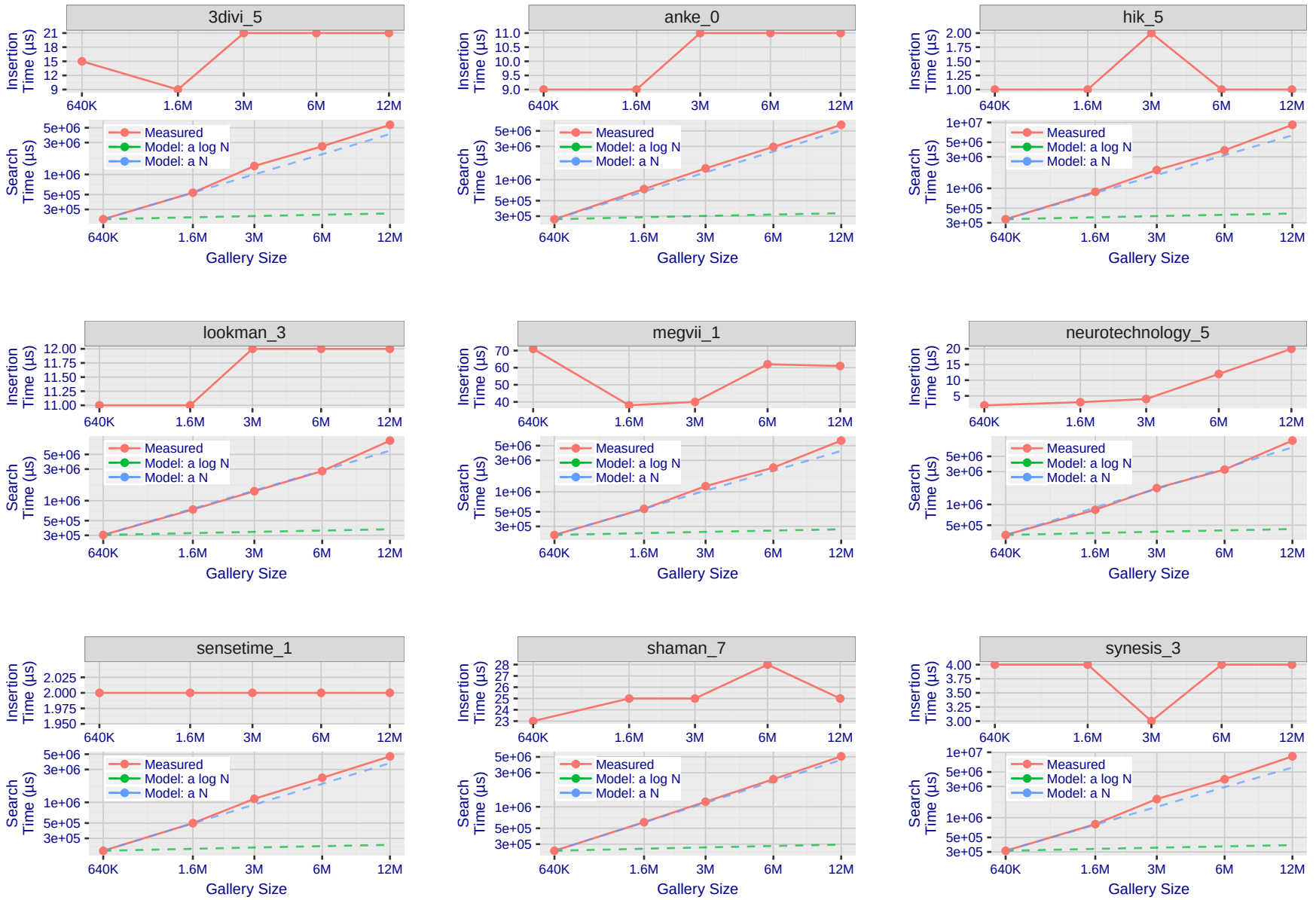

Figure 514: **[Mugshot Dataset] Gallery insertion duration vs. enrolled population size**. *This chart plots the time it takes to insert a single template into a finalized gallery, illustrated over increasing gallery sizes. For reference, search times on finalized galleries of corresponding sizes are plotted right underneath. Gallery insertion time plots were generated on algorithms that 1) successfully implemented gallery insertion with no errors and 2) that were run on galleries with N up to 12 000 000. Generally, only the more accurate algorithms were run on galleries with N up to 12 000 000.*

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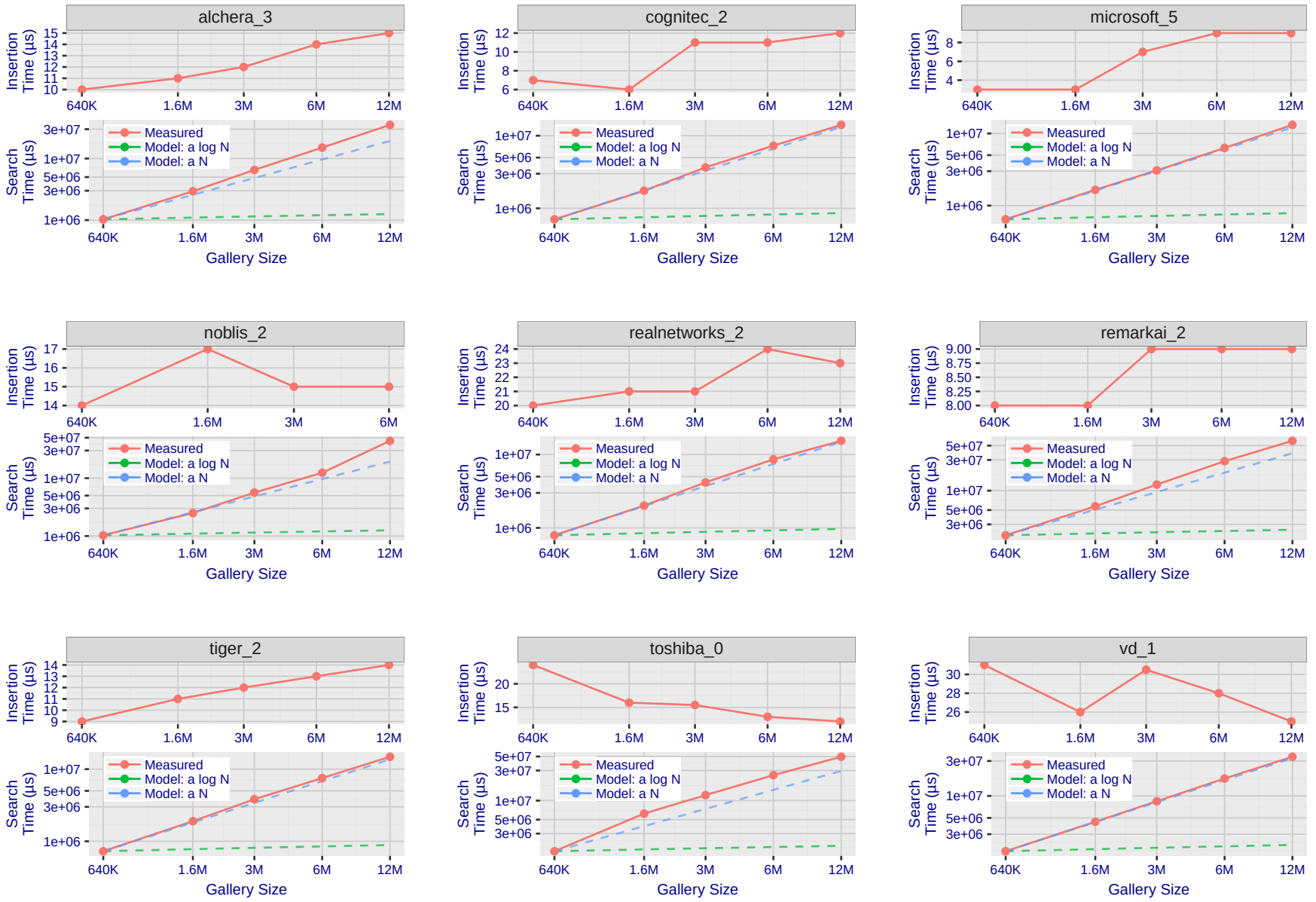

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